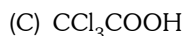
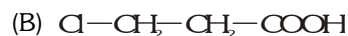
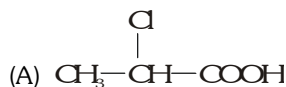


EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. Acids have much higher boiling points than isomeric esters because :-

- (A) Acids form dimers by H-Bonding
- (B) Acids are volatile in steam
- (C) Ester are non-volatile
- (D) Acids can ionise to give protons in aqueous solution

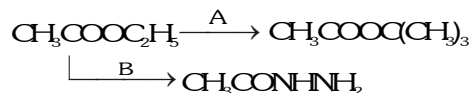
2. Which of the following acids have the lowest pK_a value :-



3. The reaction of an amide with bromine and alkali to form a primary amine is called :-

- (A) Hunsdiecker reaction
- (B) Hofmann mustard oil reaction
- (C) Hoffmann degradation of amides
- (D) Hell-Volhard-Zelinski reaction

4. The reagents A and B in the reaction sequence



are given by the set :-

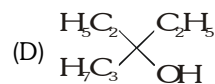
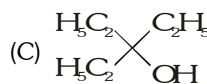
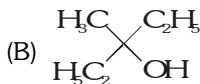
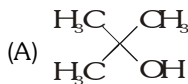
- (A) Isopropyl alcohol, hydrazine
- (B) Isopropyl alcohol, hydroxylamine
- (C) t-butyl alcohol, hydrazine
- (D) t-butyl alcohol, hydroxylamine

5. The carboxylic acids react with hydrazoic acid in presence of H_2SO_4 to form a primary amine. This reaction is called as :-

- (A) Curtius rearrangement
- (B) Lossen rearrangement
- (C) Schmidt reaction
- (D) Schotten-Boumann reaction

6. Ethyl acetate $\xrightarrow[\text{Excess}]{\text{CH}_3\text{MgBr}}$ $\xrightarrow{\text{H}_3\text{O}^+}$ P

The product P will be :-



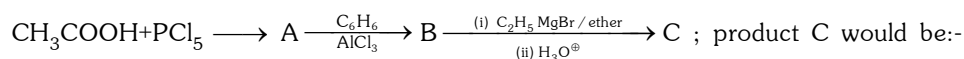
7. HVZ reaction is specific for -

- (A) Replacement of β -hydrogens
- (B) Replacement of α -hydrogens
- (C) Replacement of β - carbons
- (D) Replacement of α - carbons

8. Decreasing order of acidity of p-methoxy benzoic acid (A), p-nitrobenzoic acid (B) and benzoic acid (C) is-

- (A) B, C, A
- (B) A, B, C
- (C) C, A, B
- (D) None

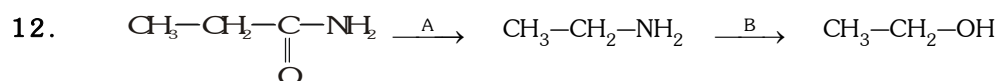
9. Benzoic acid reacts with $\text{Ca}(\text{OH})_2$. The product obtained on dry distillation gives—
 (A) Benzophenone (B) Acetophenone (C) Benzaldehyde (D) None of these
10. In a set of the given reactions, acetic acid yielded a product C.



- (A) $\text{CH}_3\text{CH}(\text{OH})\text{C}_2\text{H}_5$ (B) $\text{CH}_3\text{COC}_6\text{H}_5$ (C) $\text{CH}_3\text{CH}(\text{OH})\text{C}_6\text{H}_5$ (D) $\text{CH}_3-\overset{\text{CH}_3}{\underset{\text{OH}}{\text{C}}}-\text{C}_6\text{H}_5$

11. The correct reactivity order with the nucleophile is :

- (A) $\text{CH}_3\text{COCl} > \text{CH}_3\text{CONH}_2 > \text{CH}_3\text{COOCH}_3$
 (B) $\text{CH}_3\text{COCl} > \text{CH}_3\text{COOCH}_3 > \text{CH}_3\text{CONH}_2$
 (C) $\text{CH}_3\text{CONH}_2 > \text{CH}_3\text{COOCH}_3 > \text{CH}_3\text{COCl}$
 (D) $\text{CH}_3\text{COOCH}_3 > \text{CH}_3\text{COCl} > \text{CH}_3\text{CONH}_2$



In the above sequence A & B respectively are -

- (A) Br_2/KOH , NaOH (B) Br_2/KOH , HNO_2 (C) KMnO_4 , KOH (D) HNO_2 , Br_2 / KOH

13. Ethyl acetate on treatment with hydrazine gives -

- (A) $\text{CH}_3-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{NHNH}_2$ (B) $\text{CH}_3-\overset{\text{O}}{\underset{\text{NHNH}_2}{\text{C}}}-\text{OC}_2\text{H}_5$ (C) $\text{CH}_3-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{ONHNH}_2$ (D) $\text{CH}_3-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{NH}_2$

14. Among the following, which is the strongest acid?

- (A) $\text{CHF}_2 - \text{CH}_2 - \text{CH}_2 - \text{COOH}$ (B) $\text{CH}_3 - \text{CH}_2 - \text{CF}_2 - \text{COOH}$
 (C) $\text{CH}_2\text{F} - \text{CHF} - \text{CH}_2 - \text{COOH}$ (D) $\text{CH}_3 - \text{CF}_2 - \text{CH}_2 - \text{COOH}$

15. In the Rosenmund reduction, the catalyst used is:

- (A) Pd/BaSO_4 (B) Raney Ni (C) Sn/HCl (D) Zn/HCl

16. Which of the following carboxylic acids undergoes decarboxylation easily?

- (A) $\text{C}_6\text{H}_5\text{COCH}_2\text{COOH}$ (B) $\text{C}_6\text{H}_5\text{COCOCH}_3$ (C) $\text{C}_6\text{H}_5-\underset{\text{OH}}{\text{CH}}-\text{COOH}$ (D) $\text{C}_6\text{H}_5-\underset{\text{NH}_2}{\text{CH}}-\text{COOH}$

17. The reactivities of acid halides (I), anhydrides (II), esters (III) and amides (IV) with nucleophilic reagents follow the order

- (A) $\text{I} > \text{II} > \text{III} > \text{IV}$ (B) $\text{IV} > \text{III} > \text{II} > \text{I}$ (C) $\text{I} > \text{III} > \text{II} > \text{IV}$ (D) $\text{III} > \text{II} > \text{I} > \text{IV}$

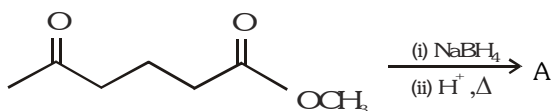
18. In the following sequence of reactions



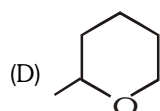
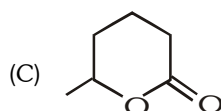
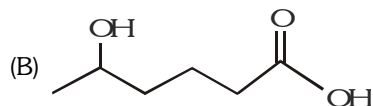
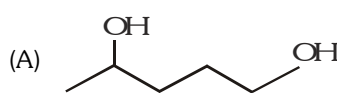
the product (B) is:

- (A) $\text{PhCOCH}_2\text{CH}_3$ (B) $\text{PhCHOHCH}_2\text{CH}_3$ (C) $\text{PhCH}_2\text{CH}_2\text{CH}_3$ (D) $\text{PhCH}=\text{CHCH}_3$

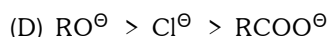
19. Consider the following reaction.



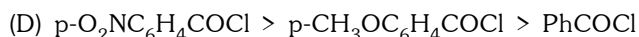
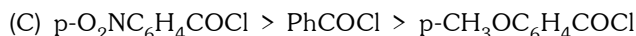
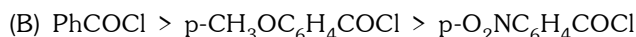
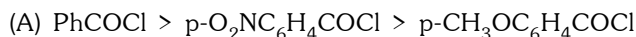
The product (A) is :



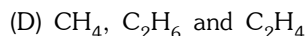
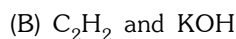
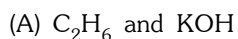
20. Which of the following orders regarding the base strength of a leaving group in a reaction of an acid derivative with a nucleophile is correct :



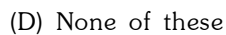
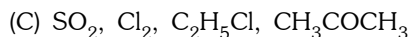
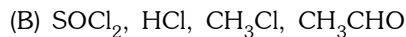
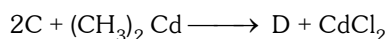
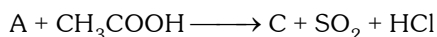
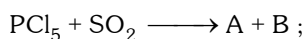
21. The correct sequence of decreasing order of reactivity of hydrolysis of acid chlorides is :



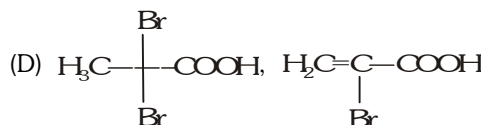
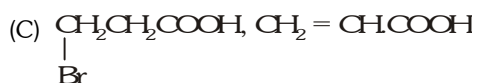
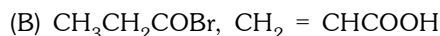
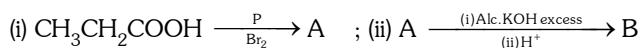
22. Kolbe electrolysis of potassium succinate gives CO_2 and :



23. In the following reaction identify compounds A, B, C and D :



24. What are A and B in the following sequence of reactions :



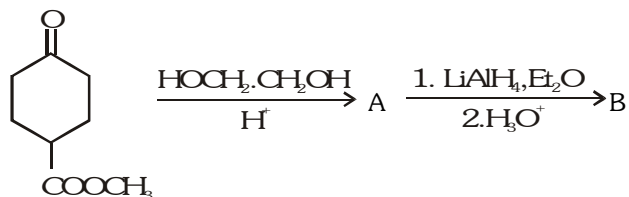
25.



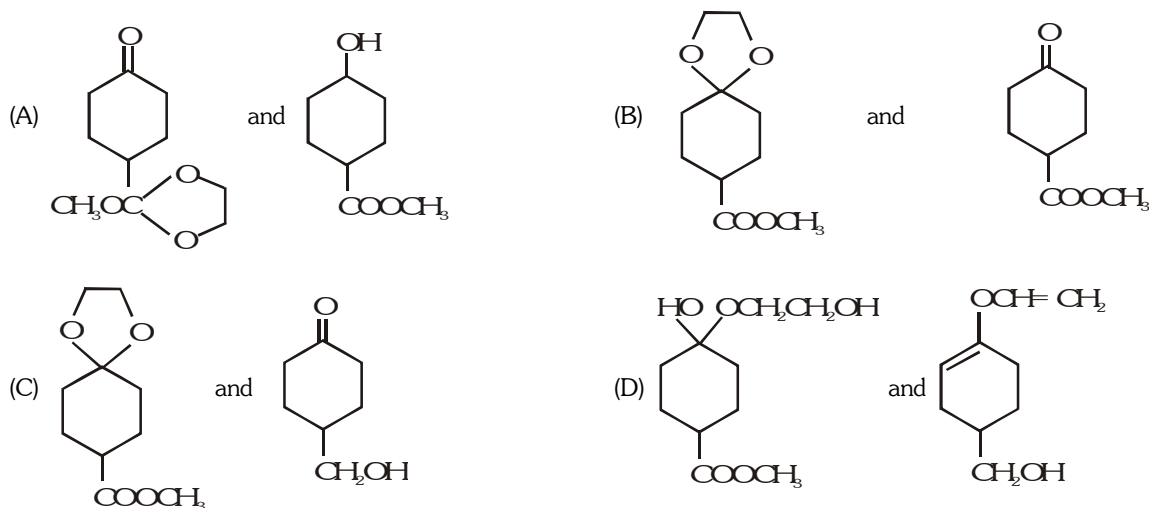
CHECK YOUR GRASP								ANSWER KEY					EXERCISE -1							
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	A	C	C	C	C	A	B	A	A	D	B	B	A	B	A	A	A	C	C	C
Que.	21	22	23	24	25															
Ans.	C	C	A	A	A															

SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)

- Which of the following reactions involve a decrease in the length of the carbon chain :-
 (A) Schmidt reaction (C) Hofmann's bromamide reaction
 (C) Hell-Volhard-Zelinski reaction (D) All of these
- Dry distillation of a mixture of calcium formate and the calcium acetate gives -
 (A) HCHO (B) CH₃CHO (C) CH₃COCH₃ (D) None
- Consider the following sequence of reactions.



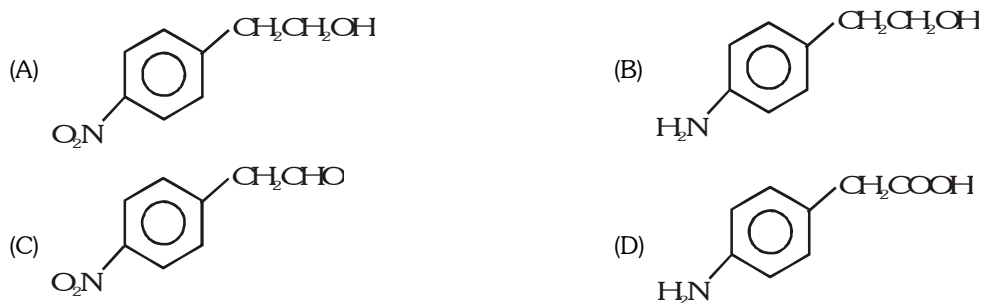
The products (A) and (B) are, respectively,



- Consider the following reaction.



The product (A) is



5. Which of the following methods are used for the conversion of carboxylic acids into acid chlorides ($\text{RCOOH} \longrightarrow \text{RCOCl}$) ?

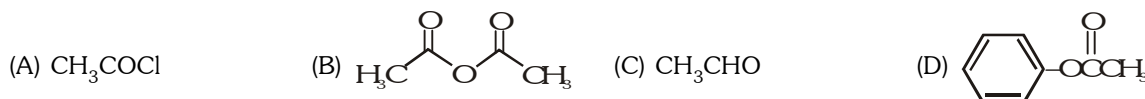
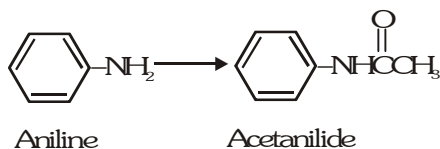


6. Which of the following statements are correct for benzoic acid ?

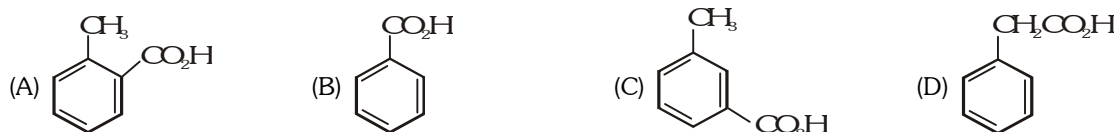
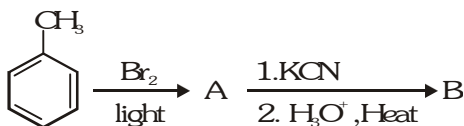
- (A) Nitration gives o-and p-nitrobenzoic acid.
 (B) Bromination ($\text{Br}_2/\text{FeBr}_3$) gives m-bromobenzoic acid.
 (C) The Friedel-Crafts reaction with $\text{CH}_3\text{COCl}/\text{AlCl}_3$ gives m-carboxyacetophenone.
 (D) The reaction with concentrated sulphuric acid gives 3-carboxybenzenesulphonic acid.

7. Which of the following compounds react with aniline to give acetanilide :

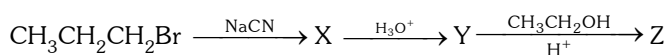
Which of the following compounds react with aniline to give acetanilide.

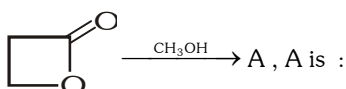


8. What is the final product (B) of this sequence :



9. What is compound Z :



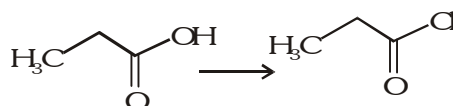
10.  A is :



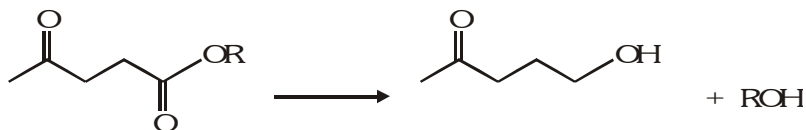
11. Which of the following orders of acid strength is correct :



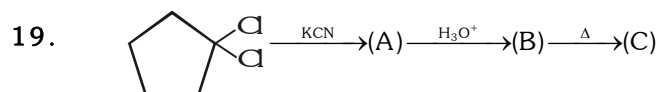
12. The relative order of esterification of alcohols is :-
(A) $1 < 2 < 3$ (B) $1 > 2 > 3$ (C) $1 > 3 > 2$ (D) $1 < 3 < 2$
13. The treatment of $\text{CH}_3\text{CH}_2\text{COOH}$ with chlorine in the presence of phosphorus gives :
(A) $\text{CH}_3\text{CH}_2\text{COCH}_3$ (B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$ (C) $\text{CH}_3\text{CH}(\text{Cl})\text{COOH}$ (D) $\text{ClCH}_2\text{CH}_2\text{COOH}$
14. Which of the following sequences of rate of alkaline hydrolysis of esters is correct :
(A) $\text{CH}_3\text{CH}_2\text{COOCH}_3 < (\text{CH}_3)_2\text{CHCOOCH}_3 < (\text{CH}_3)_3\text{CCOOCH}_3$
(B) $\text{CH}_3\text{CH}_2\text{COOCH}_3 > (\text{CH}_3)_2\text{CHCOOCH}_3 > (\text{CH}_3)_3\text{CCOOCH}_3$
(C) $\text{CH}_3\text{CH}_2\text{COOCH}_3 > (\text{CH}_3)_2\text{CHCOOCH}_3 < (\text{CH}_3)_3\text{CCOOCH}_3$
(D) $\text{CH}_3\text{CH}_2\text{COOCH}_3 < (\text{CH}_3)_2\text{CHCOOCH}_3 > (\text{CH}_3)_3\text{CCOOCH}_3$
15. Which of the following is used to perform following transformation :



- (A) SOCl_2 (B) PCl_5 (C) PCl_3 (D) SO_2Cl_2
16. Which of the following does not give iodoform :
- (A) Acetic acid (B) lactic acid (C) Acetophenone (D) propionic acid
17. One can distinguish between HCOOH and CH_3COOH with :
- (A) NaHCO_3 (B) H_2SO_4 (C) tollens reagent (D) fehling's solution
18. Which of the following reagents are involved in the following transformation ?



- (A) H_3O^+ (B) LiAlH_4 (C) Ethylene glycol (D) Acetone



Identify the correct statement(s) about the above sequence of reactions :

- (A) Compound (A) is formed through S_N reaction
- (B) Compound (C) on reduction with $LiAlH_4$ forms a product which on dehydration gives cyclohexene.
- (C) compound (A) requires two moles of hydrogen for complete reduction.
- (D) Compound (C) on Schmidt's reaction gives a product which reacts with HNO_2 to give (D) as major product.
- Compound (D) on dehydration gives cyclopentene

20. Which of the following on reduction with LiAlH_4 will give ethyl alcohol ?
 (A) $(\text{CH}_3\text{CO})_2\text{O}$ (B) CH_3COCl (C) CH_3CONH_2 (D) $\text{CH}_3\text{COOC}_2\text{H}_5$

BRAIN TEASERS						ANSWER KEY				EXERCISE -2					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	A,B	A,B,C	C	A	A,B,D	B,D	A,B,D	D	D	A	B	B	C	B	A,B,C
Que.	16	17	18	19	20										
Ans.	A,D	C,D	A,B,C	A,B,D	A,B,D										

EXERCISE-03**MISCELLANEOUS TYPE QUESTIONS****TRUE OR FALSE :**

1.  cannot undergo Claisen condensation.
2. Heating of β -hydroxy acid gives lactones.
3. Hunsdiecker reaction involves free radical in intermediate steps.
4. Benzoic acid is stronger than methanoic acid but weaker than ethanoic acid.
5. Acid halides are more reactive than acid amides towards hydrolysis.

FILL IN THE BLANKS :

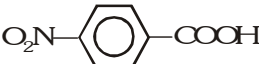
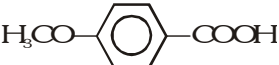
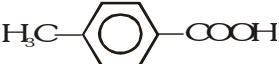
1. When sodium benzoate is heated with sodalime, it gives.....
2. pK_a and K_a of an acid are connected by the relation.....
3. Benzoic acid does not undergo Friedel-Crafts reaction due to of the benzene ring by the effect of $-\text{COOH}$ group.
4. Hell-Volhard-Zelinsky reaction involves the replacement of an atom from the alkyl group of a monocarboxylic acid by a atom.
5. Carboxylic acids may be prepared by the reaction of Grignard reagents with

MATCH THE COLUMN

1. Match the column I with column II.

Column-I (Reaction)		Column-II (Possible products)	
(A)	Arndt Eistert synthesis	(p)	Carbanion
(B)	Hunsdiecker reaction	(q)	Carbocation
(C)	Claisen condensation	(r)	Ketene
(D)	Esterification reaction	(s)	Free radical

2. Match the column I with column II.

Column-I (Acid)		Column-II (K_a)	
(A)	Benzoic acid	(p)	3.3×10^{-5}
(B)		(q)	10.2×10^{-5}
(C)		(r)	30.6×10^{-5}
(D)		(s)	6.4×10^{-5}
(E)		(t)	4.2×10^{-5}

3. Match the column I with column II.

Column-I		Column-II	
(A)	Schmidt reaction	(p)	$\text{RCOOH} \xrightarrow[\Delta]{\text{NaOH / CaO}} \text{RH}$
(B)	Curtius reaction	(q)	$\text{R-CH}_2\text{COOH} \xrightarrow[\Delta]{\text{Red P / X}_2} \text{R}-\underset{\text{X}}{\text{CH}}-\text{COOH}$
(C)	Decarboxylation	(r)	$\text{RCOCl} \xrightarrow[\Delta]{\text{NaN}_3} \text{RNH}_2$
(D)	HVZ reaction	(s)	$\text{RCOOH} \xrightarrow[\Delta]{\text{N}_3\text{H}} \text{RNH}_2$

ASSERTION & REASON QUESTION :

These questions contains, Statement-I (assertion) and Statement-II (reason).

(A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I

(B) Statement-I is True, Statement-II is True ; Statement-II is NOT a correct explanation for Statement-I

(C) Statement-I is True, Statement-II is False.

(D) Statement-I is False, Statement-II is True.

1. **Statement-I** : Unlike the $>\text{C}=\text{O}$ group of aldehydes and ketones, the $>\text{C}=\text{O}$ of $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ does not undergo nucleophilic addition reactions.

Because

Statement-II : Carboxylic acids exist as dimers due to intermolecular hydrogen bonding in aprotic medium.

2. **Statement-I** : $\text{CH}_3\text{COCH}_2\text{COOC}_2\text{H}_5$ will give iodoform test.

Because

Statement-II : It contains $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-$ group linked to a carbon atom.

3. **Statement-I** : Acetic acid does not undergo haloform reaction.

Because

Statement-II : Acetic acid has no α hydrogen.

4. **Statement-I** : Benzoic acid on nitration will give m- Nitro benzoic acid.

Because

Statement-II : $-\text{COOH}$ group will increase e^- density on meta position.

5. **Statement-I** : Acyl halide are more reactive than acid substance amide towards nucleophilic substitution.

Because

Statement-II : X^- are better leaving group than NH_2^- .

COMPREHENSION BASED QUESTIONS :

Comprehension # 1

Amides undergo hydrolysis to yield carboxylic acid plus amine on heating in either aqueous acid or aqueous base. The conditions required for amide hydrolysis are more severe than those required for the hydrolysis of esters, anhydrides or acid chlorides, but the mechanism is similar (nucleophilic acyl substitution). Nucleophilic

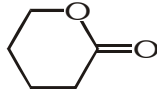
acyl substitutions involve a tetrahedral intermediate, hence these are quite different from alkyl substitution ($\text{RCH}_2\text{Br} \xrightarrow{\text{NaCN}} \text{RCH}_2\text{CN}$) which involves a pentavalent intermediate or transition state.

One of the important reactions of esters is their reaction with two equivalent of Grignard reagent to give tertiary alcohols.

- The mechanism involved during the hydrolysis of acid derivatives is :
 (A) elimination-addition (B) addition-elimination
 (C) nucleophilic addition elimination (D) electrophilic addition elimination
- Which of the following constitutes the best substrate during the acidic hydrolysis of amides ?



- For which functional derivative of carboxylic acids, acidic hydrolysis is avoided?
 (A) Acid chlorides (B) Acid amides (C) Acid anhydrides (D) Esters

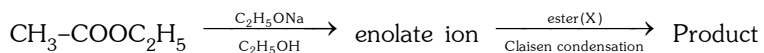
- When  is treated with two equivalent of methyl magnesium iodide and the product acidified the final product will be



Comprehension # 2

Ester gives nucleophilic addition reaction followed by elimination reaction with carbon nucleophile. When carbon nucleophile is of an ester then the reaction is known as Claisen condensation reaction. This reaction is also carried out between ester and a ketone. A successful Claisen condensation requires an ester with two α -hydrogens and an equivalent amount of base rather than a catalytic amount of base.

- Consider the given reaction



For the above reaction the most reactive ester is:

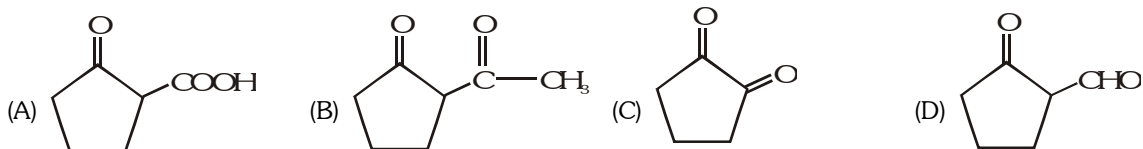


- Intramolecular Claisen condensation given by diester is known as:
 (A) Stobbe condensation (B) Dieckmann condensation
 (C) Mannich reaction (D) Reformatsky reaction

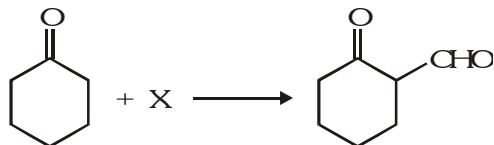
3. In the given reaction :



[X] is :



4. In the given reaction

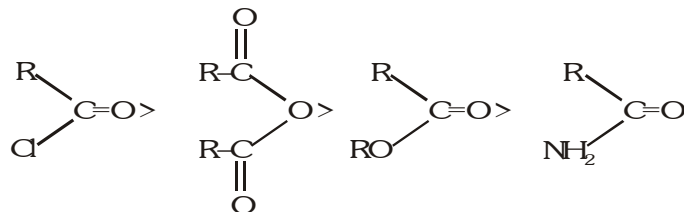


[X] is :



Comprehension # 3

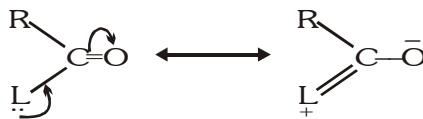
The reactivity of acid derivatives in general follows the order :



The above order of reactivity can be explained in terms of the :

(i) Basicity of leaving group (ii) Resonance effect (iii) Inductive effect

Weaker is the basic character of leaving group, more is the reactivity of acid derivative. In general, all the acid derivatives show resonance as follows:



More is the stabilization, lesser is the reactivity and *vice-versa*.

1. Which among the following anions is the most basic ?



2. Which of the most reactive acid derivative?

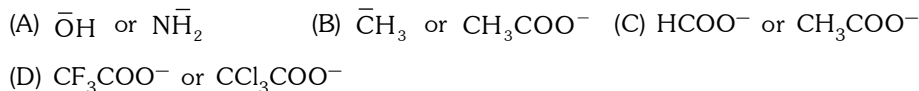


3. Which among the following ester is most reactive towards nucleophilic attack ?
 (A) $\text{CH}_3\text{COOCH}_3$ (B) HCOOCH_3 (C) $\text{CH}_3\text{CH}_2\text{COOC}_6\text{H}_5$ (D) All are equally reactive
4. Acid derivatives although contain $\text{—}\overset{\text{O}}{\parallel}{\text{C}}\text{—}$ group, yet do not undergo the usual properties of carbonyl group. It is due to:
 (A) inductive effect (B) resonance (C) eletromeric effect (D) all of these
5. Which of the following compounds will be most easily hydrolysed ?
 (A) Acid halide (B) Acid amide (C) Ester (D) Acid anhydride

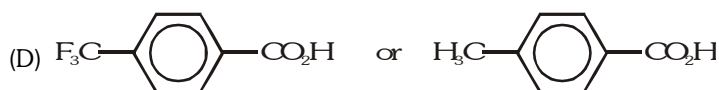
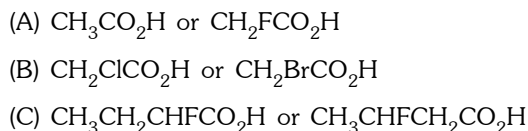
MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
<u>True / False</u> 1. T 2. F 3. T 4. F 5. T <u>Fill in the Blanks</u> 1. Benzene 2. $\text{p}K_a = -\log K_a$ 3. deactivation , electron withdrawing 4. α - Hydrogen, halogen 5. CO_2 <u>Match the Column</u> 1. (A) $\rightarrow$ r ; B $\rightarrow$ s ; (C) $\rightarrow$ p ; (D) $\rightarrow$ q 2. (A) $\rightarrow$ s ; (B) $\rightarrow$ r ; (C) $\rightarrow$ q ; (D) $\rightarrow$ p ; (E) $\rightarrow$ t 3. (A) $\rightarrow$ s ; (B) $\rightarrow$ r ; (C) $\rightarrow$ p ; (D) $\rightarrow$ q <u>Assertion - Reason Questions</u> 1. C 2. D 3. C 4. C 5. A <u>Comprehension Based Questions</u> Comprehension #1 : 1. (C) 2. (C) 3. (A) 4. (B) Comprehension #2 : 1. (B) 2. (B) 3. (A) 4. (C) Comprehension #3 : 1. (A) 2. (A) 3. (B) 4. (B) 5. (A)		

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

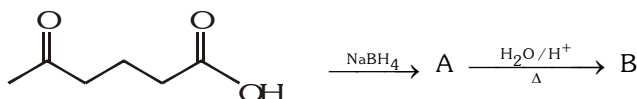
1. Carbon-oxygen bond length in formic acid are 1.23 Å and 1.36 Å but in sodium formate both the carbon-oxygen bonds have same value, i.e., 1.27 Å. Explain.
2. The second dissociation constant of fumaric acid is greater than maleic acid. Explain.
3. Which is stronger conjugate base in each pair ?



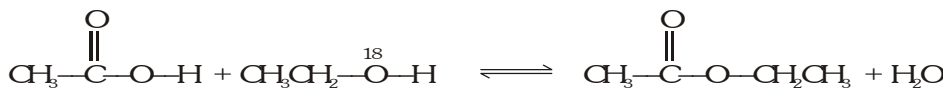
4. Which acid of each pair shown here would you expect to be stronger?



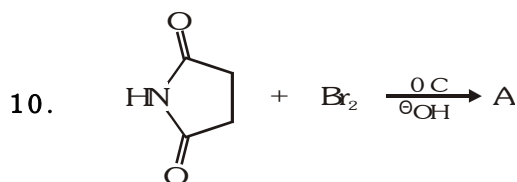
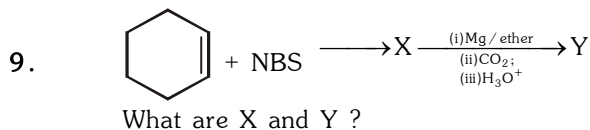
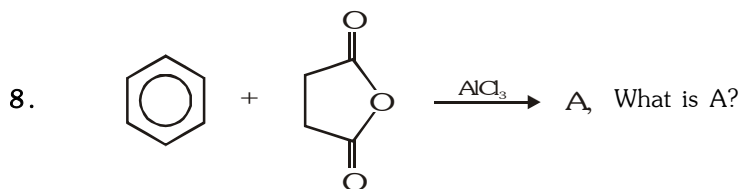
5. What are A and B in the following ?



6. In the following reaction, trace the position of isotopic O^{18} .



7. Write the reagents to carry out following conversions:

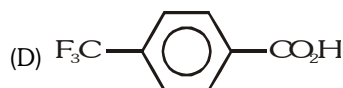
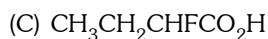


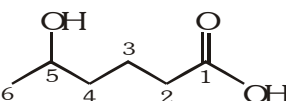
Write down the structure of A? What is the use of A?

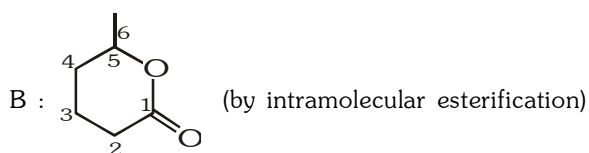
1. Formate ion shows equivalent resonating structures while formic acid does not.
 2. After the first dissociation, maleate ion is more stabilised due to intramolecular H-bonding, whereas fumarate ion does not have intramolecular H-bonding.

3. (A) NH_2^- (B) CH_3^- (C) CH_3COO^- (D) CCl_3COO^-

4. (A) $\text{CH}_2\text{FCO}_2\text{H}$ (B) $\text{CH}_2\text{ClCO}_2\text{H}$

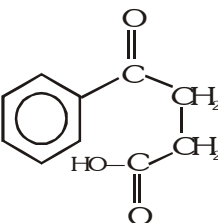


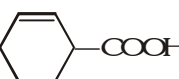
5. A :  (NaBH₄ reduces keto group)

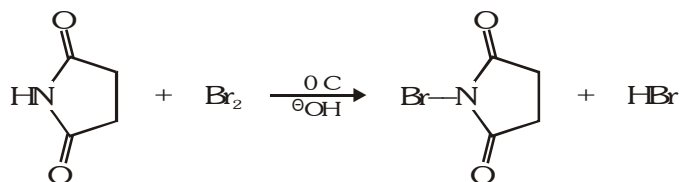


6. ¹⁸O is in ester 

7. I : LiAlH₄ II : Pd/BaSO₄ (Rosenmund)

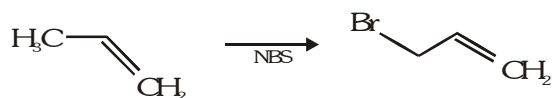
8. A is formed by Friedel-Crafts reaction A : 

9. X :  Y : 

10. 

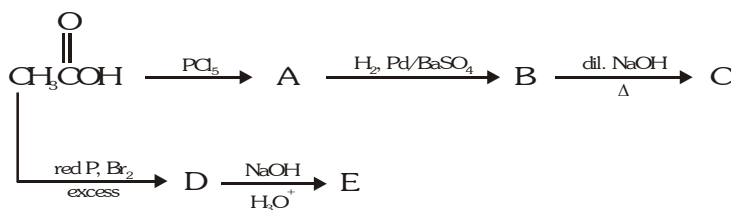
1-Bromopyrrolidine-2, 5-dione
or N-Bromosuccinimide (NBS)

It is used for brominating in allylic and benzylic hydrogen.

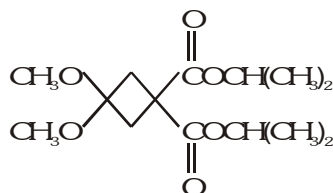


EXERCISE-04 [B]**BRAIN STORMING SUBJECTIVE EXERCISE**

1. Identify A, B, C, D and E in the following sequence of reactions:

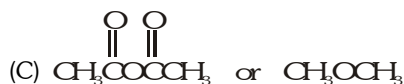
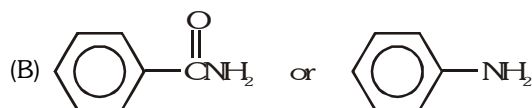


2. When the compound shown was heated in refluxing hydrochloric acid, a compound with the molecular formula $\text{A}(\text{C}_5\text{H}_6\text{O}_3)$ was isolated. Identify this product. Along with this product, three other carbon-containing substances are formed. What are they ?



What happens when $\text{A}(\text{C}_5\text{H}_6\text{O}_3)$ reacts with

- (A) HCN followed by hydrolysis
(B) soda lime/ Δ
(C) NH_2NH_2 /glycol, OH^-
(D) P/Br_2
(E) N_3H
3. CH_3COO^- (acetate ion) is more stable than $\text{C}_2\text{H}_5\text{O}^-$ (ethoxide ion). Explain.
4. Which is more reactive in each pair towards S_N reaction ?



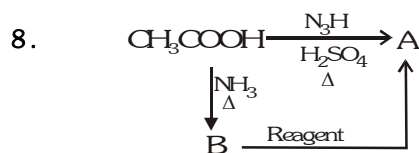
5.

6. reacts with :
 γ -Butyrolactone (ester)

- (A) NH_3 (B) LiAlH_4 (C) EtOH, H^+ (D) $\text{NaBH}_4/\text{EtOH}$

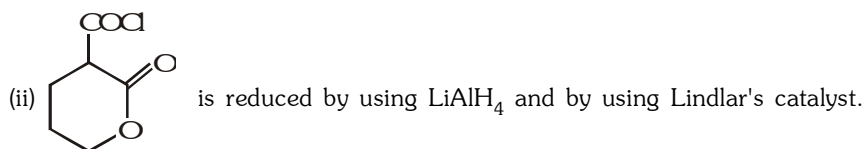
What are the product in each case ?

7. In case of aldehydes and ketones there is addition of nucleophile but in case of acyl compound there is nucleophilic substitution. Explain.

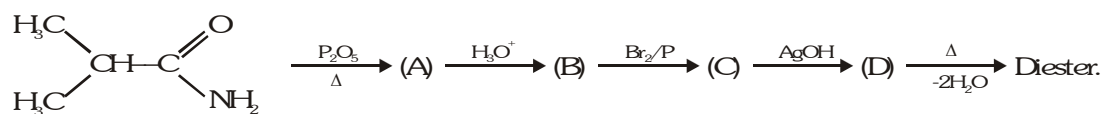


- (i) What are A & B ?
(ii) Which reagent will convert B into A ?

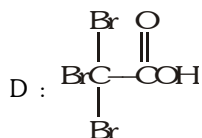
9. What happens when?



10. Complete the following sequence of reactions :

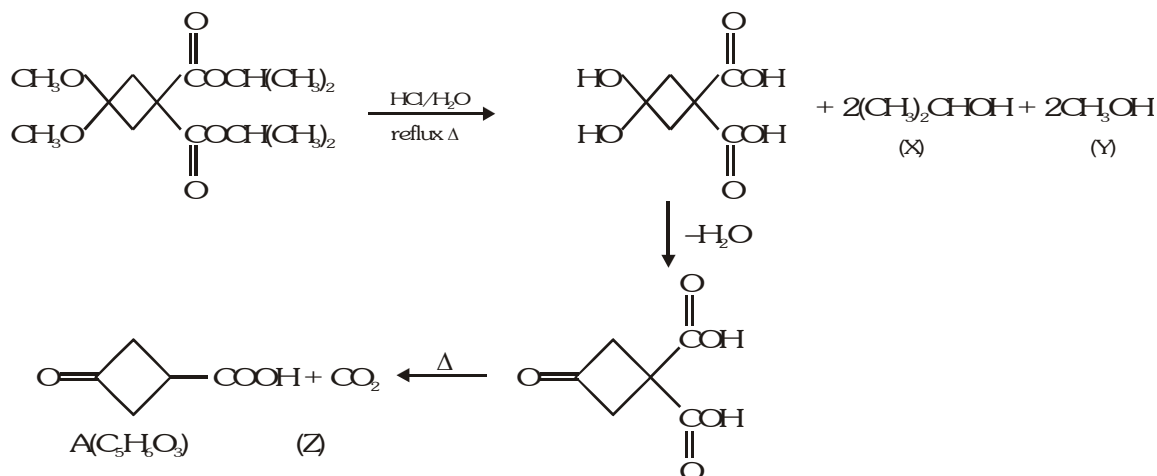


1. A : CH_3COCl B : CH_3CHO (Rosenmund)
C : $\text{CH}_3\text{CH}=\text{CHCHO}$ (aldol condensation)

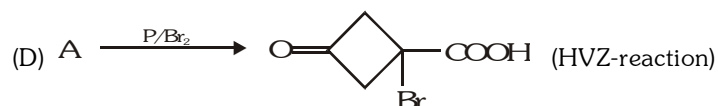
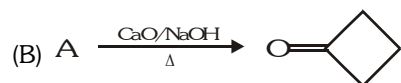
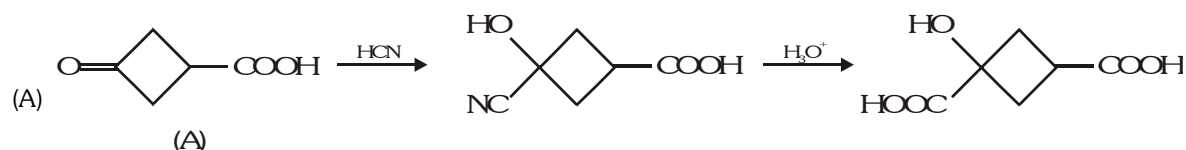


2. Given compound represents
(i) acetal (ii) ester

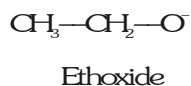
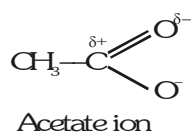
When it is subjected to acidic hydrolysis, acetal changes to carbonyl group and ester changes to acid.



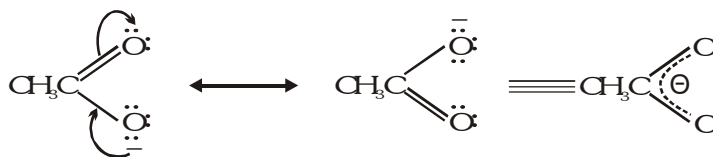
(X), (Y) and (Z) are thus, other carbon containing product along with $\text{C}_5\text{H}_6\text{O}_3$ which is the main product. $\text{A}(\text{C}_5\text{H}_6\text{O}_3)$ has (i) keto group (ii) carboxylic group



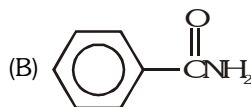
3.

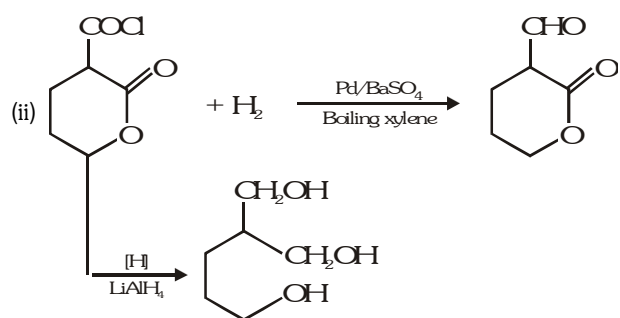


Electron delocalation, as shown by following resonance structures, causes the negative charge in acetate to be shared equally by both oxygens. This type of resonance effect is not possible in ethoxide ion.

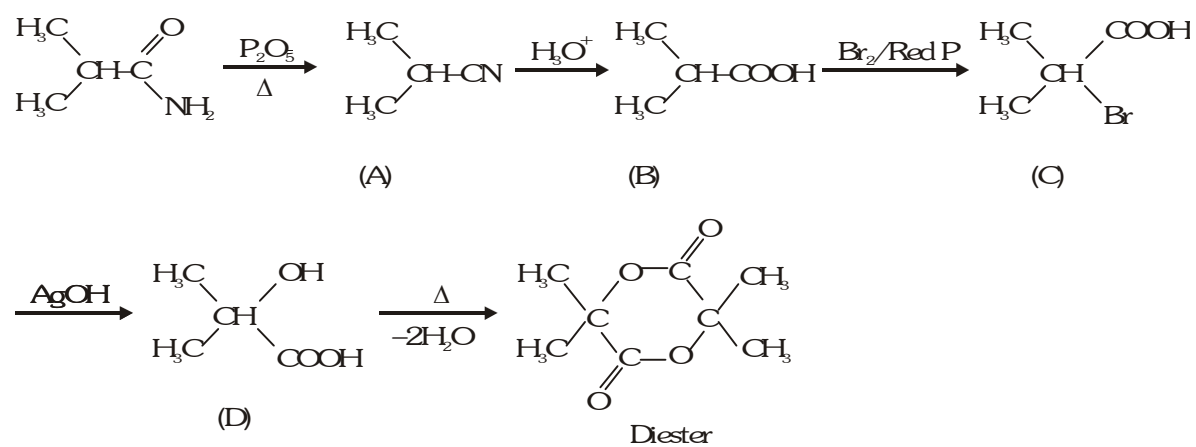


4.



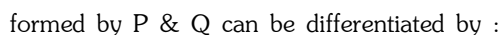


10.



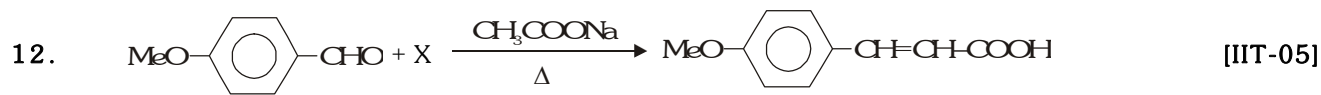
PREVIOUS YEARS QUESTIONS

- 11.



- (A) 2, 4- DNP
(B) Lucas reagent (ZnCl_2 & conc. HCl)
(C) NaHSO_3
(D) Fehlings solution

[IIT-03]



What is X ?

- (A)  (B) $\text{BrCH}_2\text{-COOH}$ (C) $\text{COOH}(\text{CH}_3\text{CO})_2\text{O}$ (D) $(\text{CH}_3\text{CO})_2\text{O}$

13. Compound (A) $\text{C}_5\text{H}_8\text{O}_2$ liberated CO_2 on reaction with sodium bicarbonate. It exists in two forms neither of which is optically active. It yielded compound (B), $\text{C}_5\text{H}_{10}\text{O}_2$ on hydrogenation. Compound (B) can be separated into enantiomorphs. Write structures of (A) and (B). [IIT-87]

14. An organic compound (A) on treatment with acetic acid in presence of sulphuric acid produces an ester (B). (A) on mild oxidation gives (C). (C) with 50% KOH followed by acidification with dilute HCl generates (A) and (D). (D) with PCl_5 followed by reaction with ammonia gives (E). (E) on dehydration produces hydrocyanic acid. Identify (A) to (E) [IIT-87]

15. Compound (A) ($\text{C}_6\text{H}_{12}\text{O}_2$) on reduction with LiAlH_4 yielded two compounds (B) and (C). The compound (B) on oxidation gave (D). 2 moles of (D) on treatment with alkali (aqueous) and subsequent heating furnished (E). The later on catalytic hydrogenation gave (C). The compound (D) was oxidized further to give (F) which was found to be monobasic acid (m.wt. 60.0). Deduce structures of (A) to (F). [IIT-90]

16. The sodium salt of a carboxylic acid, (A) was produced by passing a gas (B) into aqueous solution of caustic alkali at an elevated temperature and pressure (A) on heating in presence of sodium hydroxide followed by treatment with sulphuric acid gave a dibasic acid (C). A sample of 0.4g of (C) on combustion gave 0.08g of H_2O and 0.39g of CO_2 . The silver salt of the acid, weighing 1.0g, on ignition yielded 0.71 g of Ag as residue. Identify (A), (B) and (C). [IIT-90]

17. Two mole of an ester (A) are condensed in presence of sodium ethoxide to give a β -keto ester (B) and ethanol. On heating in an acidic solution (B) gives ethanol and β - keto acid (C). On decarboxylation (C) gives 3-pentanone. Identify (A), (B) and (C) with proper reasoning and give reactions. [IIT-92]

18. An organic compound A ($\text{C}_4\text{H}_6\text{O}_3$) on treatment with ethyl alcohol gives a carboxylic acid B and compound C. Hydrolysis of C under acidic conditions gives B and D. Oxidation of D with KMnO_4 also gives B. B on heating with $\text{Ca}(\text{OH})_2$ gives E (Molecular formula $\text{C}_3\text{H}_6\text{O}$) E does not gives Tollen's test and does not reduce Fehling solution but forms 2,4-dinitrophenylhydrazone. Identify A to E. [IIT-92]

19. An acidic compound (A), $\text{C}_4\text{H}_8\text{O}$ loses its optical activity on strong heating yielding (B). $\text{C}_4\text{H}_6\text{O}_2$ which reacts readily with KMnO_4 . (B) forms a derivative (C) with SOCl_2 , which on reaction with CH_3NH_2 gives (D). The compound (A) on oxidation with dilute chromic acid gives an unstable compound (E) which decarboxylate readily to give (F), $\text{C}_3\text{H}_6\text{O}$. The compound (F) gives a hydrocarbon (G) on treatment with amalgamated Zn and HCl. Give structures of (A) to (G) with proper reasoning. [IIT-95]

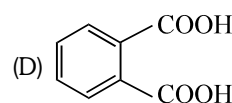
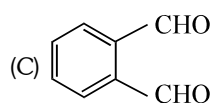
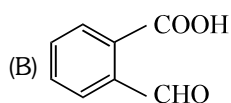
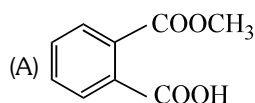
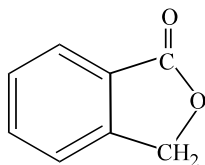
20. A liquid (X) having molecular formula $\text{C}_6\text{H}_{12}\text{O}_2$ is hydrolysed with water in presence of an acid to give a carboxylic acid (Y) and an alcohol (Z). Oxidation of (Z) with chromic acid gives (Y). What are (X), (Y) and (Z). [IIT-96]

21. Acetophenone on reaction with hydroxylamine-hydrochloride can produce two isomeric oximes. Write structures of the oximes. [IIT-97]

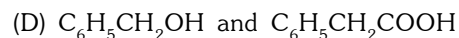
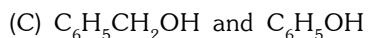
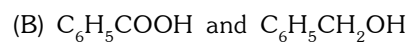
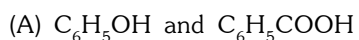
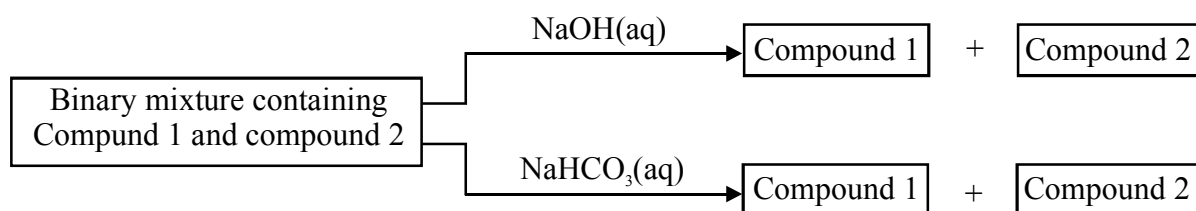
22. An organic acid (A), $\text{C}_5\text{H}_{10}\text{O}_2$ reacts with Br_2 in the presence of phosphorus to give (B). Compound (B) contains an asymmetric carbon atom and yields (C) on dehydrobromination. Compound (C) does not show geometric isomerism and on decarboxylation gives an alkene (D) which on ozonolysis gives (E) and (F). compound (E) gives a positive schiff's test but (F) does not. Give structures of (A) to (F) with reasons. [IIT-97]

23. The correct IUPAC name of C_6H_5COCl is
 (A) Benzoyl chloride (B) Benzene chloro ketone
 (C) Benzene carbonyl chloride (D) Chloro phenyl ketone [IIT 2006]

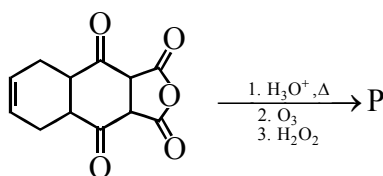
24. Which of the following reactants on reaction with conc. NaOH followed by acidification gives the following lactone as the only product? [IIT 2006]



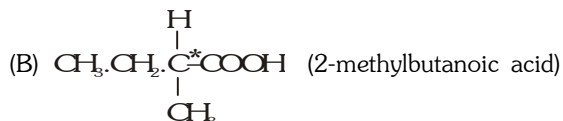
25. Identify the binary mixtures (s) that can be separated into the individual compounds, by differential extraction, as shown in the given scheme - [IIT 2012]



26. The total number of carboxylic acid groups in the product P is [JEE 2013]



1. (A) 2. (A) 3. (A) 4.(B) 5. (D) 6. (A) 7.(C) 8. (D) 9. (C)
 10. (C) 11. (D) 12. (D)



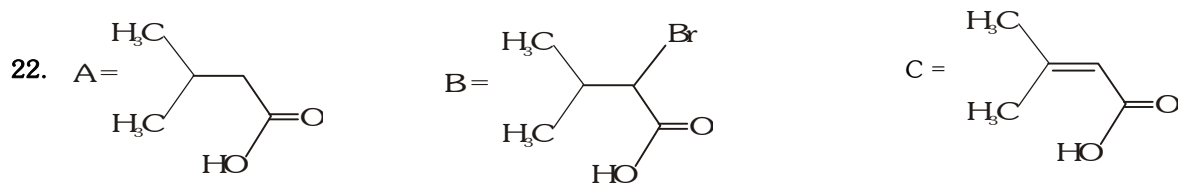
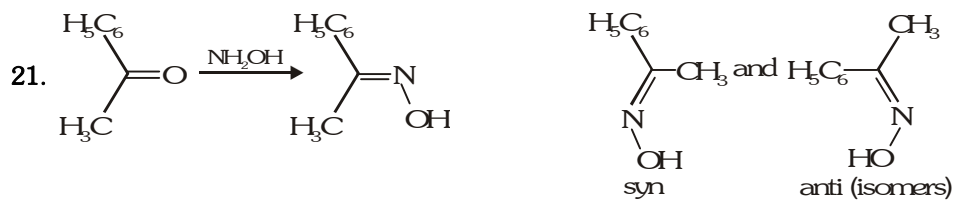
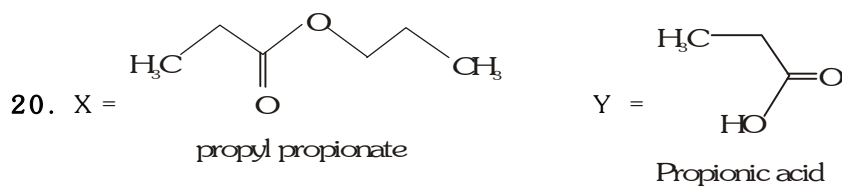
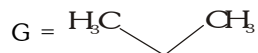
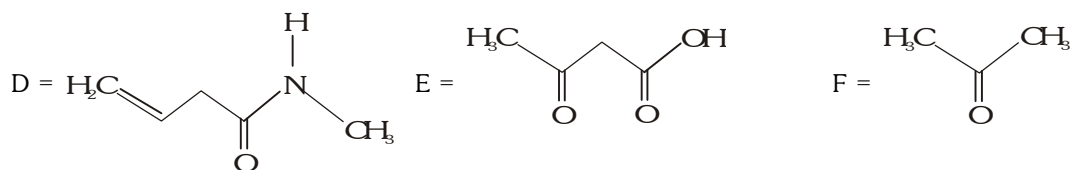
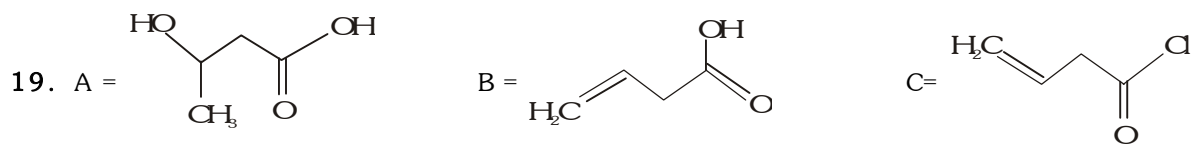
14. A = CH_3OH (Methanol)
 B = $\text{CH}_3\text{COOCH}_3$ (Methyl ethanoate)
 C = HCHO (Methanal)
 D = HCOOH (Methanoic acid)
 E = HCONH_2 (Formamide or methanamide)

15. (A) $\text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
 (B) $\text{C}_2\text{H}_5\text{OH}$
 (C) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
 (D) CH_3CHO
 (E) $\text{CH}_3\text{CH}=\text{CHCHO}$
 (F) CH_3COOH

16. (A) HCOOH ; (B) CO ; (C) $\begin{array}{c} \text{COOH} \\ | \\ \text{COOH} \end{array}$

17. (A) $\text{CH}_3\text{CH}_2\text{COOC}_2\text{H}_5$ (Ethyl propanoate)
 (B) $\begin{array}{c} \text{CH}_3\text{CH}_2\text{COCHCOOC}_2\text{H}_5 \\ | \\ \text{CH}_3 \end{array}$ Ethyl - (2-methyl-3-ketopentanoate)
 (C) $\begin{array}{c} \text{CH}_3\text{CH}_2\text{COCHCOOC}_2\text{H}_5 \\ | \\ \text{CH}_3 \end{array}$ (2-methyl-3-ketopentanoic acid)

18. (A) $(\text{CH}_3\text{CO})_2\text{O}$ (Acetic anhydride)
 (B) CH_3COOH (Ethanoic acid)
 (C) $\text{CH}_3\text{COOC}_2\text{H}_5$ (Ethyl ethanoate)
 (D) $\text{C}_2\text{H}_5\text{OH}$ (Ethanol)
 (E) CH_3COCH_3



23. C

24. C

25. B, D

26. (2)

EXERCISE # JEE MAIN
ALL QUESTIONS BASED ON
HALOGEN & OXYGEN CONTAINING ORGANIC COMPOUND

- Following reaction :
 $(\text{CH}_3)_3\text{C}-\text{Br} + \text{H}_2\text{O} \longrightarrow (\text{CH}_3)_3\text{C}-\text{OH} + \text{HBr}$
 is an example of- [AIEEE-2002]

(A) Elimination reaction (B) Free radical substitution
 (C) Nucleophilic substitution (D) Electrophilic substitution
- SN^1 reaction is easible in- [AIEEE-2002]

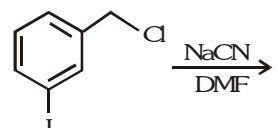
(A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + \text{KOH} \longrightarrow$ (B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + \text{KOH} \longrightarrow$
 (C) $\text{C}_6\text{H}_5\text{Cl} + \text{KOH} \longrightarrow$ (D) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{Cl} + \text{KOH} \longrightarrow$
- Bottles containing $\text{C}_6\text{H}_5\text{I}$ and $\text{C}_6\text{H}_5\text{CH}_2\text{I}$ lost their original labels. They were labelled A and B for testing. A and B were separately taken in a test tube and boiled with NaOH solution. The end solution in each tube was made acidic with dilute HNO_3 and then some AgNO_3 solution was added. Substance B gave a yellow precipitate. Which one of the following statements is true for this experiment. [AIEEE-2003]

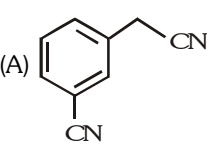
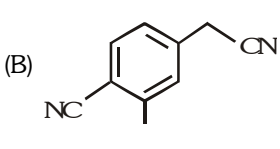
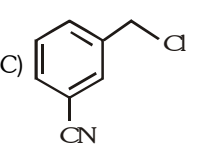
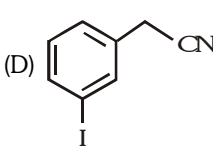
(A) A was $\text{C}_6\text{H}_5\text{I}$ (B) A was $\text{C}_6\text{H}_5\text{CH}_2\text{I}$
 (C) B was $\text{C}_6\text{H}_5\text{I}$ (D) Addition of HNO_3 was unnecessary
- The reaction of chloroform with alcoholic KOH and p-toluidine form- [AIEEE-2003]

(A) $\text{H}_3\text{C}-\text{C}_6\text{H}_4-\text{CN}$ (B) $\text{H}_3\text{C}-\text{C}_6\text{H}_4-\text{N}_2\text{Cl}$
 (C) $\text{H}_3\text{C}-\text{C}_6\text{H}_4-\text{NHCHCl}_2$ (D) $\text{H}_3\text{C}-\text{C}_6\text{H}_4-\text{NC}$
- The compound formed on heating chlorobenzene with chloral in the presence of concentrated sulphuric acid is- [AIEEE-2004]

(A) Gammaxe (B) DDT
 (C) Freon (D) Hexa chloro ethane
- Among the following the one that gives positive iodoform test upon reaction with I_2 and NaOH is- [AIEEE-2006]

(A) $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ (B) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{OH}$
 (C) $\text{H}_3\text{C}-\text{CH}(\text{CH}_3)-\text{CH}_2\text{OH}$ (D) PhCHOHCH_3
- The structure of the major product formed in the following reaction is : [AIEEE-2006]

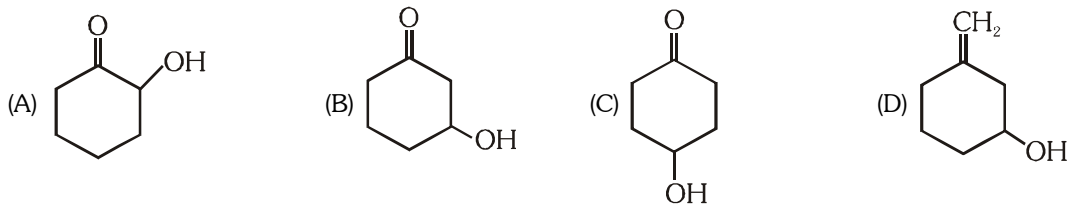


(A)  (B)  (C)  (D) 
- Which of the following on heating with aqueous KOH , produces acetaldehyde ? [AIEEE-2009]

(A) $\text{CH}_2\text{ClCH}_2\text{Cl}$ (B) CH_3CHCl_2 (C) CH_3COCl (D) $\text{CH}_3\text{CH}_2\text{Cl}$

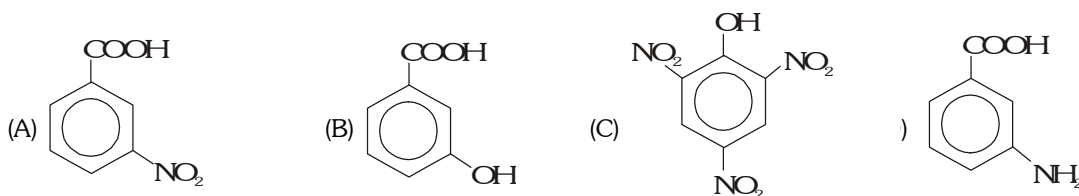
9. Maximum dehydration takes place that of -

[AIEEE-2002]



10. Picric acid is -

[AIEEE-2002]



11. An ether is more volatile than an alcohol having the same molecular formula. This is due to -

[AIEEE-2003]

- (A) Inter molecular hydrogen bonding in ethers (B) Inter molecular hydrogen bonding in alcohols
(C) Dipolar character of ethers (D) Alcohols having resonance structures

12. When $\text{CH}_2 = \text{CH}-\text{COOH}$ is reduced with LiAlH_4 , the compound obtained will be -

[AIEEE-2003]

- (A) $\text{CH}_3-\text{CH}_2-\text{CH}_2\text{OH}$ (B) $\text{CH}_3-\text{CH}_2-\text{CHO}$ (C) $\text{CH}_3-\text{CH}_2-\text{COOH}$ (D) $\text{CH}_2=\text{CH}-\text{CH}_2\text{OH}$

13. The general formula $\text{C}_n\text{H}_{2n}\text{O}_2$ represents

[AIEEE-2003]

- (A) Diols (B) Dialdehydes (C) Diketones (D) Carboxylic acids

14. Which one of the following undergoes reaction with 50% sodium hydroxide solution to give the corresponding alcohol and acid ?

[AIEEE-2004]

- (A) Phenol (B) Benzaldehyde (C) Butanal (D) Benzoic acid

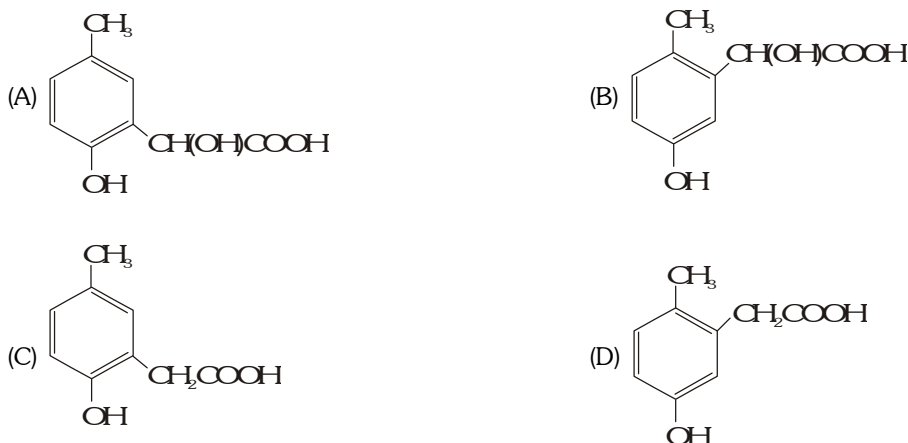
15. Among the following compounds which can be dehydrated very easily is -

[AIEEE-2004]

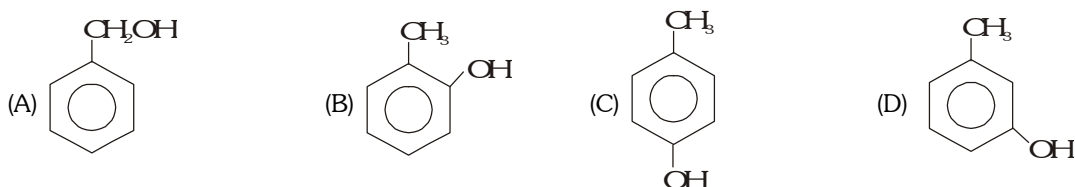


16. p-cresol reacts with chloroform in alkaline medium to give the compound A which adds hydrogen cyanide to form, the compound B. The later on acidic hydrolysis gives chiral carboxylic acid. The structure of the carboxylic acid is -

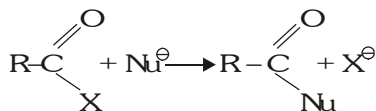
[AIEEE-2005]



17. HBr reacts with $\text{CH}_2 = \text{CH} - \text{OCH}_3$ under anhydrous conditions at room temperature to give -
 (A) BrCH_2CHO and CH_3OH (B) $\text{BrCH}_2\text{CH}_2\text{OCH}_3$ [AIEEE-2006]
 (C) $\text{H}_3\text{C}-\text{CHBr}-\text{OCH}_3$ (D) CH_3CHO and CH_3Br
18. The structure of the compound that gives tribromo derivative on treatment with bromine water is -
 [AIEEE-2006]

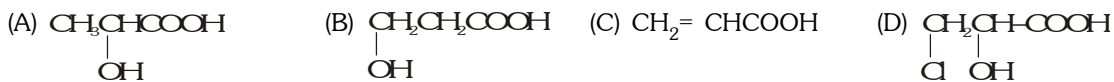


19. Phenol, when it first reacts with concentrated sulphuric acid and then with concentrated nitric acid, gives
 [AIEEE-2008]
 (A) 2, 4, 6- trinitrobenzene (B) α -nitrophenol
 (C) p-nitrophenol (D) nitrobenzene
20. Bakelite is obtained from phenol by reacting with
 [AIEEE-2008]
 (A) $(\text{CH}_2\text{OH})_2$ (B) CH_3CHO (C) CH_3COCH_3 (D) HCHO
21. The best reagent to convert pent-3-en-2-ol into pent -3-en -2-one is -
 [AIEEE-2005]
 (A) Acidic dichromate (B) Acidic permanganate
 (C) Pyridinium chloro-chromate (D) Chromic anhydride in glacial acetic acid
22. Rate of the reaction-
 [AIEEE-2005]

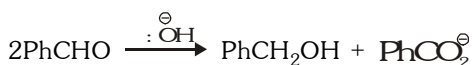


is fastest when X is -

- (A) NH_2 (B) Cl (C) OCOR (D) OC_2H_5
23. In the following sequence of reactions
 $\text{CH}_3\text{CH}_2\text{OH} \xrightarrow[\text{Ether}]{\text{P} + \text{I}_2} \text{A} \xrightarrow{\text{Mg}} \text{B} \xrightarrow{\text{HCHO}} \text{C} \xrightarrow{\text{H}_2\text{O}} \text{D}$, then compound 'D' is -
 [AIEEE-2007]
 (A) butanal (B) n-butyl alcohol (C) n-propyl alcohol (D) propanal
24. End product of the following reaction is -
 [AIEEE-2002]



25. On mixing ethyl acetate with aqueous sodium chloride, the composition of the resultant solution is -
 [AIEEE-2004]
 (A) $\text{CH}_3\text{COOC}_2\text{H}_5 + \text{NaCl}$ (B) $\text{CH}_3\text{COONa} + \text{C}_2\text{H}_5\text{OH}$
 (C) $\text{CH}_3\text{COCl} + \text{C}_2\text{H}_5\text{OH} + \text{NaOH}$ (D) $\text{CH}_3\text{Cl} + \text{C}_2\text{H}_5\text{COONa}$
26. The major product obtained on interaction of phenol with sodium hydroxide and carbon dioxide is :-
 [AIEEE-2009]
 (A) Salicylic acid (B) Phthalic acid (C) Benzoic acid (D) Salicylaldehyde
27. In Cannizzaro reaction given below :-

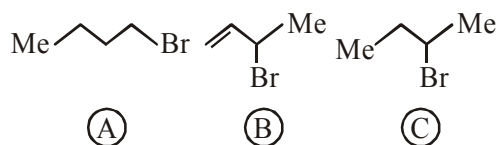


the slowest step is :-

- (A) The abstraction of proton from the carboxylic group
 (B) The deprotonation of PhCH_2OH
 (C) The attack of $:\text{OH}^\ominus$ at the carboxyl group
 (D) The transfer of hydride to the carbonyl group

[AIEEE-2009]

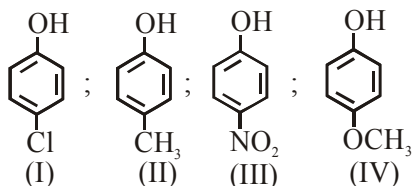
28. A liquid was mixed with ethanol and a drop of concentrated H_2SO_4 was added. A compound with a fruity smell was formed. The liquid was :- [AIEEE-2009]
 (A) CH_3COCH_3 (B) CH_3COOH (C) CH_3OH (D) HCHO
29. From amongst the following alcohols the one that would react fastest with conc. HCl and anhydrous ZnCl_2 , is :- [AIEEE-2010]
 (A) 1-Butanol (B) 2-Butanol (C) 2-Methylpropan-2-ol (D) 2-Methylpropanol
30. Sodium ethoxide has reacted with ethanoyl chloride. The compound that is produced in the above reaction is :- [AIEEE-2011]
 (A) Ethyl chloride (B) Ethyl ethanoate (C) Diethyl ether (D) 2-Butanone
31. Phenol is heated with a solution of mixture of KBr and KBrO_3 . The major product obtained in the above reaction is :- [AIEEE-2011]
 (A) 4-Bromophenol (B) 2,4,6-Tribromophenol
 (C) 2-Bromophenol (D) 3-Bromophenol
32. Trichloroacetaldehyde was subjected to Cannizzaro's reaction by using NaOH . The mixture of the products contains sodium trichloroacetate and another compound. The other compound is :- [AIEEE-2011]
 (A) 2,2,2-Trichloropropanol (B) Chloroform
 (C) 2,2,2-Trichloroethanol (D) Trichloromethanol
33. Silver Mirror test is given by which one of the following compounds ? [AIEEE-2011]
 (A) Formaldehyde (B) Benzophenone (C) Acetaldehyde (D) Acetone
34. Which of the following reagents may be used to distinguish between phenol and benzoic acid? [AIEEE-2011]
 (A) Molisch reagent (B) Neutral FeCl_3
 (C) Aqueous NaOH (D) Tollen's reagent
35. Consider the following reaction :
 $\text{C}_2\text{H}_5\text{OH} + \text{H}_2\text{SO}_4 \rightarrow \text{Produce}$
 Among the following, which one cannot be formed as a product under any conditions ? [AIEEE-2011]
 (A) Ethyl-hydrogen sulphate (B) Ethylene
 (C) Acetylene (D) Diethyl ether
36. Iodoform can be prepared from all except : [AIEEE-2012]
 (A) Isobutyl alcohol (B) Ethyl methyl ketone
 (C) Isopropyl alcohol (D) 3-Methyl-2-butanone
37. Which of the following on heating with aqueous KOH , produces acetaldehyde ? [AIEEE-2009]
 (A) $\text{CH}_2\text{ClCH}_2\text{Cl}$ (B) CH_3CHCl_2 (C) CH_3COCl (D) $\text{CH}_3\text{CH}_2\text{Cl}$
38. Consider the following bromides :- [AIEEE-2010]



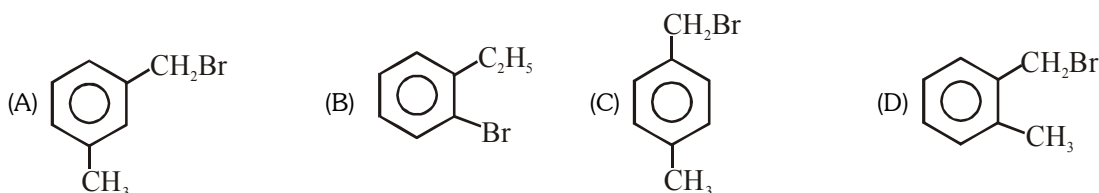
The correct order of $\text{S}_{\text{N}}1$ reactivity is

- (A) $\text{A} > \text{B} > \text{C}$ (B) $\text{B} > \text{C} > \text{A}$ (C) $\text{B} > \text{A} > \text{C}$ (D) $\text{C} > \text{B} > \text{A}$

39. A solution of (–) –1-chloro–1-phenylethane in toluene racemises slowly in the presence of a small amount of SbCl_5 , due to the formation of :- **[JEE-MAIN 2013]**
- (A) carbanion (B) Carbene (C) carbocation (D) free radical
40. Arrange the following compounds in order of decreasing acidity :



- (A) $\text{II} > \text{IV} > \text{I} > \text{III}$ (B) $\text{I} > \text{II} > \text{III} > \text{IV}$ (C) $\text{III} > \text{I} > \text{II} > \text{IV}$ (D) $\text{IV} > \text{III} > \text{I} > \text{II}$
- 41.** Compound (A), $\text{C}_8\text{H}_9\text{Br}$, gives a white precipitate when warmed with alcoholic AgNO_3 . Oxidation of (A) gives an acid (B), $\text{C}_8\text{H}_6\text{O}_4$. (B) easily forms anhydride on heating. Identify the compound (A) :

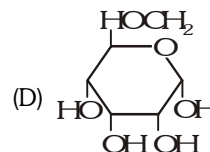
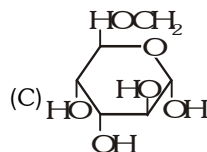
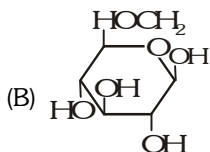
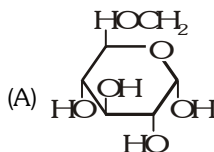


42. An organic compound A upon reacting with NH_3 gives B. On heating, B gives C. C in presence of KOH reacts with Br_2 to give $\text{CH}_3\text{CH}_2\text{NH}_2$. A is :-
- (A) CH_3COOH (B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
- (C) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{COOH}$ (D) $\text{CH}_3\text{CH}_2\text{COOH}$
43. An unknown alcohol is treated with the "Lucas reagent" to determine whether the alcohol is primary, secondary or tertiary. Which alcohol reacts fastest and by what mechanism :-
- (A) secondary alcohol by SN_1 (B) tertiary alcohol by SN_1
- (C) secondary alcohol by SN_2 (D) tertiary alcohol by SN_2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13
Ans	B	C	B	D	D	B	C	A	B	C	B	D	D
Que.	14	15	16	17	18	19	20	21	22	23	24	25	26
Ans	B	C	A	D	D	Bonus	D	D	B	C	C	B	A
Que.	27	28	29	30	31	32	33	34	35	36	37	38	39
Ans	D	B	C	B	B	C	A,C	B	C	A			
Que.	40	41	42	43									
Ans	C	D	D	B									

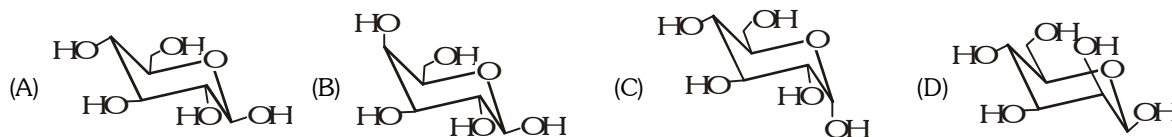
EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. The minimum number of carbon atoms that should be present in a monosaccharite is :
(A) 2 (B) 3 (C) 4 (D) 6
2. Carbohydrates that, on attempted hydrolysis, are not cleaved to smaller carbohydrates are called :
(A) oligosaccharides (B) polysaccharides (C) disaccharides (D) monosaccharides
3. The IUPAC name of the ketotriose is :
(A) 1, 3-dihydroxypropanone (B) 1-3-dihydroxy-2-oxopropane
(C) 2-oxo-1, 3-propanediol (D) dihydroxyacetone
4. The number of chiral centres in the open chain structure of glucose is :
(A) 3 (B) 4 (C) 5 (D) 6
5. Cane sugar on hydrolysis gives :
(A) glucose and galactose (B) glucose and fructose
(C) glucose only (D) fructose only
6. In D-erythrose, the configurations at C-2 and C-3 are respectively :
(A) 2R, 3R (B) 2S, 3S (C) 2R, 3S (D) 2S, 3R
7. Glucose and fructose are :
(A) diastereomers (B) anomers (C) epimers (D) functional isomers
8. Which of the following structures represents α -D-glucopyranose?

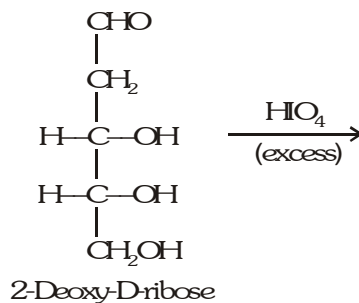


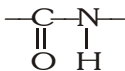
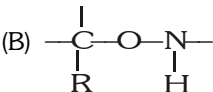
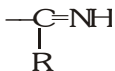
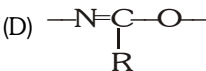
9. Sorbitol can be obtained by the reduction of -
(A) sorbose (B) glucose (C) fructose (D) all of these
10. Freshly prepared solution of cane sugar, under the influence of acid catalyst, undergoes change in optical rotation with laps of time. This phenomenon is called -
(A) mutarotation (B) inversion
(C) racemization (D) optical rotatory dispersion
11. In the formation of osazone derivatives of aldohexoses and ketohexoses, the carbon atom (s) that participate (s) in the reactions is -
(A) C-1 (B) C-2 (C) C-1 and C-2 (D) C-2 and C-3
12. Glucose gives positive silver mirror test with ammoniacal silver nitrate because it contains -
(A) hydroxyl group (B) aldehyde group
(C) ketone group (D) vicinal diol group
13. Fructose reduces fehling's solution due to the presence of -
(A) hydroxy group (B) aldehyde group
(C) ketone group (D) α -hydroxyketone group

14. In the multi-step conversion of an aldose into next higher aldose by Kiliani-Fischer synthesis, the reagent employed in the first step is -
 (A) $C_6H_5NHNH_2$ (B) NH_2OH (C) Br_2/H_2O (D) HCN
15. When methyl D-Glucopyranoside is oxidized with periodic acid, how many moles of the oxidizing agent are consumed per mole of the sugar ?
 (A) 2 (B) 3 (C) 4 (D) 5
16. β -D-glucose is represented as -

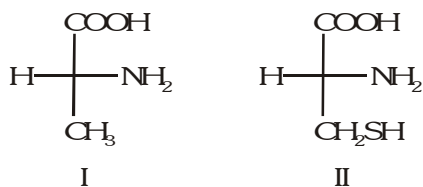


17. The number of carbon atoms and oxygen atoms present in the ring of the D-glucopyranose are respectively:
 (A) 5 and 1 (B) 4 and 1 (C) 4 and 2 (D) 3 and 2
18. Glucose is oxidized by nitric acid to give
 (A) gluconic acid (B) glucaric acid (C) glucuronic acid (D) glycolic acid
19. D-(-)- Erythrose and D-(-)- threose are separately reduced with $NaBH_4$ to give the products (A) and (B) respectively. Which of the following statements is correct about (A) and (B) ?
 (A) Both (A) and (B) will be optically active.
 (B) Both (A) and (B) will be optically inactive.
 (C) (A) will be optically inactive but (B) will be optically active
 (D) Neither (A) nor (B) possesses any asymmetric carbon atom.
20. The products expected from the reaction are :



- (A) OHCCH_2CHO , HCOOH and HCHO (B) HOOCCHOHCHOHCOOH and HCOOH
 (C) $\text{HOOCCH}_2\text{COOH}$, HCOOH and HCHO (D) HOCCOOH , HCOOH and HCHO
21. Glucose does not react with
 (A) $C_6H_5NHNH_2$ (B) NH_2OH (C) HCN (D) $NaHSO_3$
22. Which one among the following is a peptide linkage ?
 (A)  (B)  (C)  (D) 
23. The pH at which an amino acids carries no net charge is called its :
 (A) isoelectric point (B) Inversion point (C) Neutralization point (D) triple point

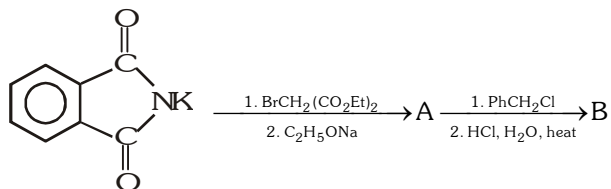
24. The configurations of the compounds I and II



are respectively

- (A) R and R (B) R and S (C) S and S (D) S and R

25. Consider the following sequence of reactions.



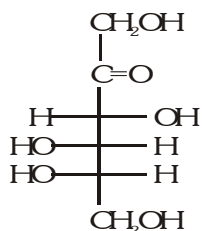
The major final product (B) is

- (A) $\text{H}_2\text{NCH}_2\text{COOH}$
- (B) $\text{PhCH}(\text{NH}_2)\text{COOH}$
- (C) $\text{PhCH}_2\text{CH}(\text{NH}_2)\text{COOH}$
- (D) $\begin{array}{c}\text{PhCH}_2\text{C}(\text{COOEt})_2 \\ | \\ \text{NH}_2\end{array}$

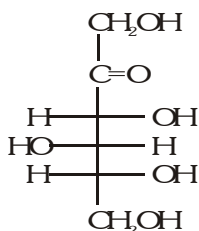
CHECK YOUR GRASP								ANSWER KEY					EXERCISE -1							
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	B	D	A	B	B	A	D	A	D	B	C	B	D	D	A	A	A	B	C	A
Que.	21	22	23	24	25															
Ans.	D	A	A	B	C															

SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)

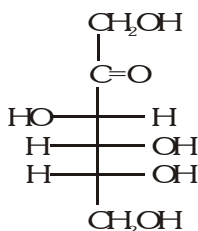
- Which of the following carbohydrates is a disaccharide?
(A) Glucose (B) Fructose (C) Maltose (D) Lactose
- α -D-Glucopyranose and β -D-glucopyranose are :
(A) Anomer (B) Epimer (C) Diastereomers (D) enantiomers
- Consider the following structures.



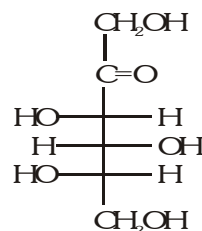
I



II



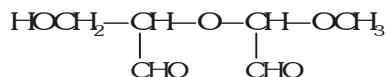
III



IV

Which of the following pairs represent D- and L-fructose respectively?

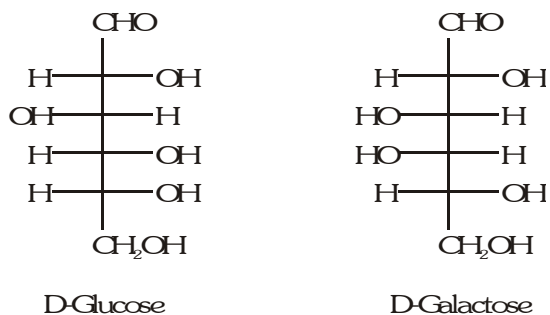
- (A) II and I (B) I and III (C) III and IV (D) II and IV
- (+) - sucrose is made up of -
(A) D-glucose and L-fructose (B) D-glucose and D-fructose
(C) D-fructose and L-glucose (D) L-fructose and L-glucose
- Glucose cannot be classified as -
(A) a carbohydrate (B) a hexose
(C) an aldose (D) an oligosaccharide
- D-Glucose and D-mannose are -
(A) anomers (B) epimers (C) diastereomers (D) enantiomers
- D-Glucose reacts with anhydrous methyl alcohol in the presence of dry HCl gas to form -
(A) 2,3,4,5,6-penta-O-methyl D-glucose (B) α -methyl D-glucopyranoside
(C) β -methyl D-glucopyranoside (D) Both (B) and (C)
- When glucose is treated with an excess of HIO_4 , the products formed are -
(A) one mole of glyoxal (CHOCHO), one mole of HCHO and three moles of HCOOH
(B) five moles of HCOOH and one mole of HCOH
(C) one mole of oxalic acid (COOHCOOH), three moles of HCOOH and one mole of HCHO
(D) six moles of HCOOH
- Methyl D-glucoside on reaction with HIO_4 consume two moles of the reagent and produces the dialdehyde
(A) and one mole of HCOOH :



The result of this reaction proves that glucose has :

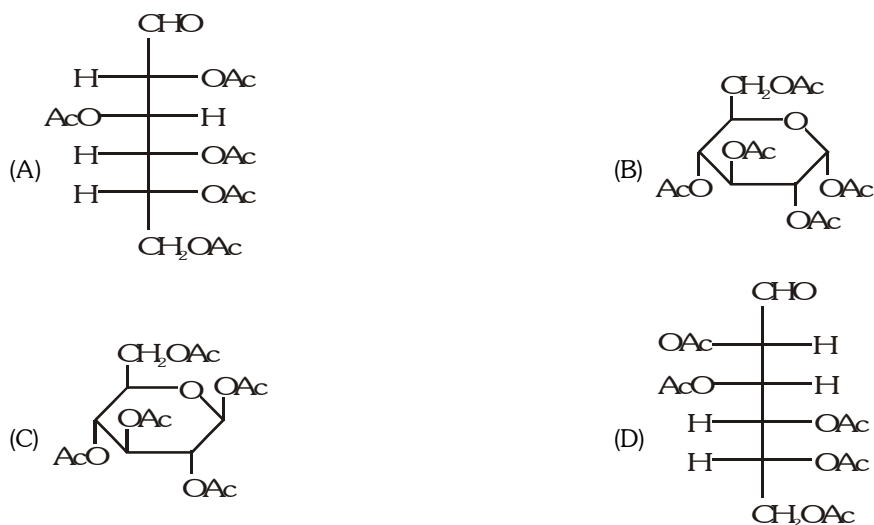
- a pyranose structure
- a furanose structure
- an open-chain structure
- a four-membered ring structure

10. Following are the structure of D-glucose and D-galactose-

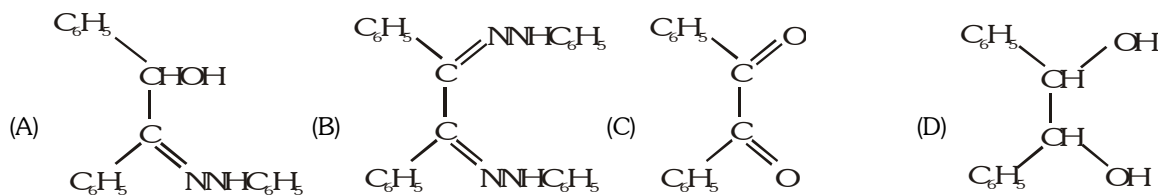
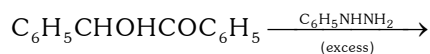


which of the statement is correct about these compounds :

- (A) They are diastereomers (B) both are components of lactose
 (C) They are C-4 epimers (D) they are enantiomers
11. D-Glucose is treated with an excess of acetic anhydride in the presence of pyridine. Identify the products formed.



12. The product formed in the reaction is :



13. (+)-Sucrose has a specific rotation of +66.5°, while D-(+)-glucose and D-(-)-fructose have specific rotations of +52.5° and -92.4° respectively. After complete hydrolysis of (+)-sucrose by dilute acid, what will be the specific rotation of the hydrolysis product, called invert sugar [a 1 : 1 mixture of D-(+)-glucose and D-(-)-fructose]?
- (A) -39.9 (B) -72.4 (C) -19.9 (D) -34.9

14. Consider the following statements :
- (A) Monosaccharides are optically active polyhydroxy carbonyl compounds.
- (B) Fructose does not reduce Fehling's solution because it is a ketose.
- (C) α -D(+)-Glucose and β -D(+)-glucose are anomers.
- (D) D-Glucose and D-mannose are C-2 epimers.
15. Consider the following statements about sucrose.
- (A) Hydrolysis of sucrose with dilute acid yields an equimolar mixture of D-glucose and D-fructose
- (B) Acid hydrolysis of sucrose is accompanied by a change in optical rotation.
- (C) In sucrose, the glycosidic linkage is between C-1 of glucose and C-2 of fructose.
- (D) Aqueous solution of sucrose exhibits mutarotation.
16. Alanine, at its isoelectric point, exists in solution as :
- (A) $\text{H}_2\text{NCHCOO}^\ominus$
|
 CH_3
- (B) $\text{H}_3\text{N}^\oplus\text{CHCOOH}$
|
 CH_3
- (C) $\text{H}_3\text{N}^\oplus\text{CHCOO}^\ominus$
|
 CH_3
- (D) $\text{H}_2\text{NCHCOOH}$
|
 CH_3
17. The pK_a of acetylsalicylic acid (aspirin) is 3.5. The pH of gastric juice in human stomach is about 2-3 and pH in the small intestine is about 8, aspirin will be -
- (A) Unionized in the small intestine and in the stomach
- (B) Completely ionized in the stomach and almost unionized in the small intestine
- (C) Ionized in the stomach and almost unionized in the small intestine
- (D) Ionised in the small intestine and almost unionised in the stomach
18. Which of the following are natural polymers ?
- (A) proteins (B) cellulose (C) Teflon (D) Natural rubber
19. Amino acids are produced on hydrolysis of :
- (A) nucleic acids (B) carbohydrates (C) fats (D) proteins
20. Which of the following do not undergo hydrolysis :
- (A) glucose (B) fructose (C) galactose (D) sucrose

BRAIN TEASERS						ANSWER KEY				EXERCISE -2					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	C, D	A,B,C	C	B	D	B,C	D	B	A	A,B,C	B,C	B	C	A,C,D	A,B,C
Que.	16	17	18	19	20										
Ans.	C	D	A,B,D	D	A,B,C										

EXERCISE-03

MISCELLANEOUS TYPE QUESTIONS

TRUE OR FALSE :

1. Fructose is an aldose.
2. Glucose and fructose are both monosaccharides.
3. Carbohydrate is a polyhydroxy compound that has an aldehydic or a ketonic functional group either free or as hemiacetal or acetal.
4. Glucose undergoes oxidation with Fehling's solution or Tollen's reagent.
5. Epimers are a pair of diastereomers that differ only in the configuration about a single carbon atom.
6. The solution having equal molecules of D-glucose and D-fructose is termed invert sugar.

FILL IN THE BLANKS :

1. A disaccharide consists of two.....joined by..... bonds.
2. A polysaccharide is a polymer of.....
3. Protein is a polymer of
4. The glycerides which contain saturated carboxylic acids are called
5. Glucose and cane sugar can be distinguished byreagent orsolution.

MATCH THE COLUMN

1. Match the column I with column II.

Column-I		Column-II	
(A)	α - and β - Glucose	(p)	Mutarotation
(B)	(+) and (-) Glucose	(q)	Enantiomers
(C)	D- and L- Notations	(r)	Anomers
(D)	α -Form $\rightleftharpoons$ open-chain form $\rightleftharpoons$ β -form	(s)	Configurational relationship

2. Match the column I with column II.

Column-I		Column-II	
(A)	Glucose	(p)	Inversion
(B)	Hydrolysis of cane sugar	(q)	α -amino acid
(C)	Zwitter ion	(r)	Protein
(D)	Peptide linkage	(s)	Monosaccharide

3. Match the column I with column II.

Column-I		Column-II	
(A)	Sucrose	(p)	Non-reducing sugar
(B)	Fructose	(q)	Polysaccharide
(C)	Maltose	(r)	Reducing sugar
(D)	Cellulose	(s)	Disaccharide

ASSERTION & REASON QUESTION :

These questions contains, Statement-I (assertion) and Statement-II (reason).

(A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I

(B) Statement-I is True, Statement-II is True ; Statement-II is NOT a correct explanation for Statement-I

(C) Statement-I is True, Statement-II is False.

(D) Statement-I is False, Statement-II is True.

1. **Statement-I** : A solution of sucrose in water is dextrorotatory but on hydrolysis in presence of H^+ , it becomes levorotatory.

Because

Statement-II : Inversion of sugar follows first order kinetics.

2. **Statement-I** : The digestion of carbohydrates does not take place in stomach.

Because

Statement-II : Enzymes like salivary amylase becomes inactive in stomach where there is acidic pH.

3. **Statement-I** : Proteins are made up of any α -amino acid.

Because

Statement-II : During denaturation, primary structure of proteins is not affected.

4. **Statement-I** : Reducing sugar give red precipitate with Fehling's solution and show mutarotation.

Because

Statement-II : During mutarotation, one pure anomer is converted into mixture of two anomers.

5. **Statement-I** : Natural rubber is a polymer of cis-Isoprene.

Because

Statement-II : Polytrans isoprene is called Gutta percha.

COMPREHENSION BASED QUESTIONS :

Comprehension # 1

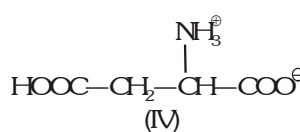
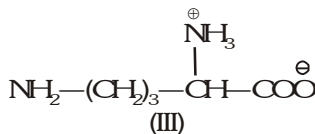
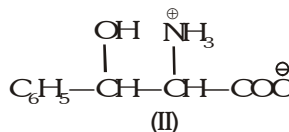
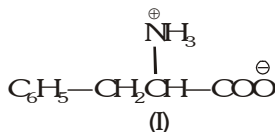
Amino acids contains as $-NH_2$ as well as $-COOH$ group. They exist as zwitter ions.



Which explain their several characteristic properties, like decomposition on heating, solubility in water, larger dipole moment. Thus in solution, amino acids may exist as dipolar ion (neutral pH), cationic (In strongly acidic solution), or anionic (in strongly basic solution).

Amino acids undergo usual reaction of the $-COOH$ group as well as $-NH_2$ group.

1. At intracellular pH (:6–7), amino acids can be divided into four types positively charged, negatively charged, hydrophobic and hydrophilic. Which is the correct classification of the following four amino acids ?



I	II	III	IV
(A) hydrophobic	+ vely charged	- vely charged	hydrophilic
(B) hydrophobic	hydrophilic	+ vely charged	- vely charged
(C) hydrophilic	hydrophobic	+ vely charged	- vely charged
(D)+ vely charged	- vely charged	hydrophobic	hydrophilic

2. Amino acid are -
- (A) As basic as a typical amine and as acidic as a carboxylic acid
 (B) Less basic than a typical amine and less acidic than a —COOH
 (C) More basic than a typical amine and more acidic than a —COOH
 (D) Nothing is certain
3. Base treatment of an amino acid usually result in the conversion of the acid to a derivative via the amino carboxylate salt. The above procedure -
- (A) Decrease the rate of electrophilic reaction of the free amino group
 (B) Decrease the rate of nucleophilic reaction of the free amino group
 (C) Enhances the rate of nucleophilic reaction of the free amino group
 (D) Enhances the rate of electrophilic reaction of the free amino group
4. Benzoylation of an amino acid can best be done by treating the amino acid with benzoyl chloride -
- (A) In presence of dil. NaOH
 (B) In presence conc. NaOH
 (C) In absence of NaOH
 (D) In presence of HCl

Comprehension # 2

Monosaccharides have —CHO (or C=O) and —OH groups, so they undergo usual oxidation and reduction. Further, monosaccharides form osazone when treated with excess of phenylhydrazine (3 equivalents). In osazone formation only the first two carbon atoms are involved. Thus monosaccharides having identical configuration on rest of C atoms except first two will form same osazone. As is the case with glucose and fructose.

A, B and C are three hexoses and form same osazone D. Compounds A to D behave as below :



1. Compound D is a osazone which can be obtained from :
- (A) Only one compound (B) Two compounds
 (C) Three compounds (D) Four compounds
2. Compound A should be :
- (A) D-glucose (B) D-fructose (C) L-glucose (D) L-fructose
3. Compound B and C, respectively, are :
- (A) D-glucose and D-mannose (B) D-mannose and D-glucose
 (C) D-glucose and L-glucose (D) D-glucose and L-mannose

Comprehension # 3

Carbohydrates are polyhydroxy aldehydes and ketones and those compounds which on hydrolysis give such compounds are also carbohydrates. The carbohydrates which are not hydrolysed are called monosaccharides. Other carbohydrates are oligosaccharides and polysaccharides. Monosaccharides with aldehydic group are called aldose and those with free ketonic groups are called ketose. Carbohydrates are optically active.

Number optical isomers = 2^n

where n = number of asymmetric carbons. Carbohydrates are mainly synthesised by plants during photosynthesis.

1. Number of optical isomers of glucose is :
(A) 8 (B) 16 (C) 32 (D) 4
 2. Fructose is :
(A) aldotriose (B) ketohexose (C) aldohexose (D) Ketotriose
 3. First member of ketose sugar is :
(A) ketotriose (B) ketotetrose (C) ketopentose (D) ketohexose
-

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
<u>True / False</u> 1. F 2. T 3. T 4. T 5. T 6. T <u>Fill in the Blanks</u> 1. monosaccharides, glycosidic 2. monosaccharides 3. α-amino acids 4. fats 5. tollen's , Fehling's <u>Match the Column</u> 1. (A) $\rightarrow$ p, r ; B $\rightarrow$ q ; (C) $\rightarrow$ s ; (D) $\rightarrow$ p 2. (A) $\rightarrow$ s ; (B) $\rightarrow$ p ; (C) $\rightarrow$ q ; (D) $\rightarrow$ r 3. (A) $\rightarrow$ p,s ; (B) $\rightarrow$ r ; (C) $\rightarrow$ p,s ; (D) $\rightarrow$ p, q <u>Assertion - Reason Questions</u> 1. B 2. A 3. B 4. A 5. B <u>Comprehension Based Questions</u> Comprehension #1 : 1. (B) 2. (B) 3. (C) 4. (A) Comprehension #2 : 1. (C) 2. (B) 3. (A) Comprehension #3 : 1. (B) 2. (B) 3. (A)		

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

- Deduce the molecular formula of glucose from the following data -
The % composition is C = 40%, H=6.7% and O=53.3%. A solution of 9.0g in 100 g of H₂O freezes at 0.93 C. (K_f of H₂O = 1.86 C/mol)
- Prove by giving one chemical reaction that glucose has hydroxyl (—OH) groups.
- Glucose, mannose and fructose give identical osazones. Explain.
- (+) Sucrose on hydrolysis gives a mixture of D-(+) glucose, $[\alpha]_D=52.7$ and D-(-) fructose = -92.4 , known as invert sugar. Calculate the rotation of mixture, i.e., invert sugar.
- Explain why, although cellulose contains so many —OH groups, it is insoluble in water ?
- The β -and α -D-glucose have different specific rotations. When either anomer is dissolved in water, their rotation change until the same fixed value results. What term is used for these changes.
- Can amino acid be isolated at its isoelectric point ?
- Why is the acidic hydrolysis of sucrose called inversion reaction ?
- Illustrate the following terms :
(a) Copolymerization (b) Vulcanization
- Explain how does 1, 3-butadiene polymerise by different routes.

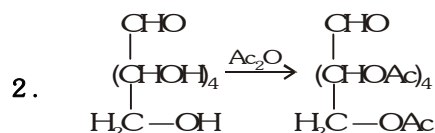
CONCEPTUAL SUBJECTIVE EXERCISE**ANSWER KEY****EXERCISE -4(A)**

- Empirical Formula is CH₂O, emp. wt. = 30, $m = \frac{\Delta T_f}{K_f}$

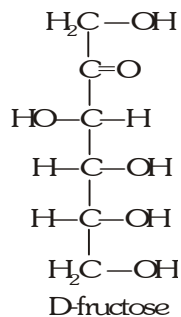
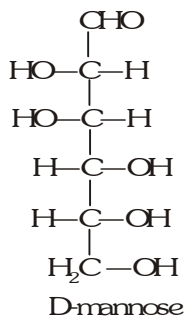
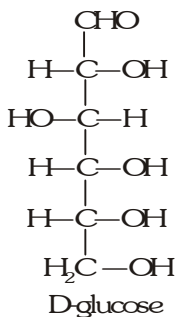
$$m = \frac{0.93}{1.86^\circ\text{C/mol}} = 0.50 \text{ mol}$$

$$\text{M.W.} = \frac{90\text{g}}{0.50 \text{ mol}} = 180 \text{ g/mol}, \quad n = \frac{180}{30} = 6$$

Hence, formula - 6 CH₂O = C₆H₁₂O₆



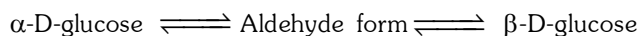
3. Only C_1 and C_2 are involved in osazone formation. Hence aldohexoses and ketohexoses which have the same configuration at C_3 , C_4 and C_5 give the same osazone.



4. We know that 1 mole of sucrose yields 1 mole of glucose and 1 mole of fructose. Hence, the specific rotation of mixture will be one half of the sum of the specific rotation of the two monosaccharides.

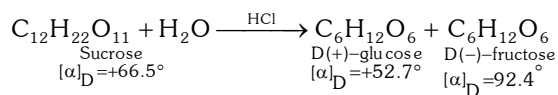
$$[\alpha]_D \text{ of invert sugar} = \frac{1}{2} [52.7 + (-92.4)] = -19.9$$

5. Cellulose does not form hydrogen bonds with water.
6. This specific rotation change is known as mutarotation. The α - and β -D-glucoses are each in equilibrium with the open chain aldehyde form



As each anomer begins to establish this equilibrium, its specific rotation value changes. When equilibrium is reached, the experimentally determined rotation remains constant. A base like NaOH catalyzes the attainment of the equilibrium.

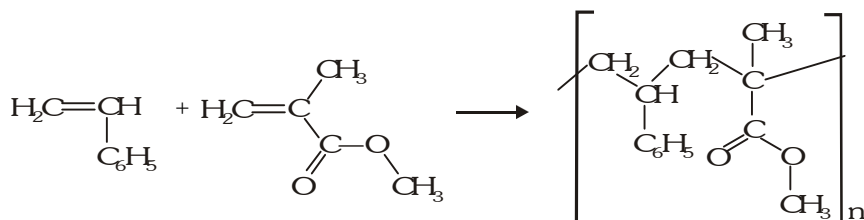
7. Yes, the solubility of an amino acid in water is minimum at its isoelectric point and thus its isolation is attempted at this pH.
8. The hydrolysis proceed as follows :



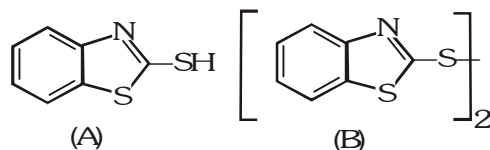
Since there is a change in the sign of rotation from Dextro to Laevo as a result of hydrolysis, it is called inversion reaction. Sucrose is often called invert sugar.

9. (a) **Copolymerization** : Polymerization of a single monomeric compound to form a homopolymer is known as homopolymerization. Evidently such a polymer is made up of identical units. In contrast, a copolymer results when two different kinds of monomers are polymerized together to give a product containing both the monomers. Such a process, known as copolymerization.

Example :

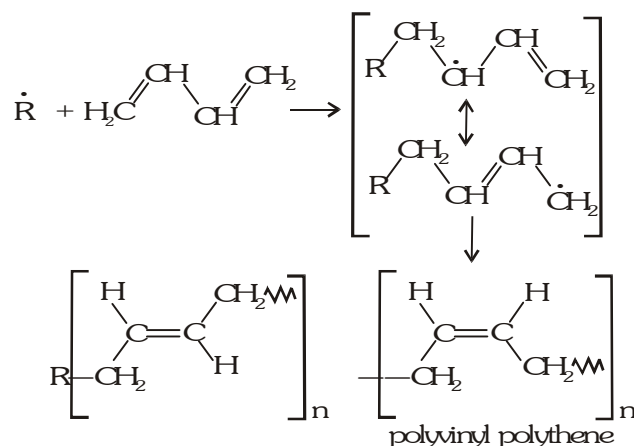


- (b) **Vulcanization** : In its original form natural rubber is not a very useful substance as it softens at moderate temperatures and hardens quickly to a brittle substance. It was subsequently discovered by Goodyear in 1839 that heating the gum rubber with sulphur produced a material of improved elasticity with much greater toughness and resistance to heat. The process is called vulcanization and is brought about with the help of vulcanization accelerators such as 2-mercaptobenzothiazole (A) and 2,2'-dithiobisbenzothiazole (B).

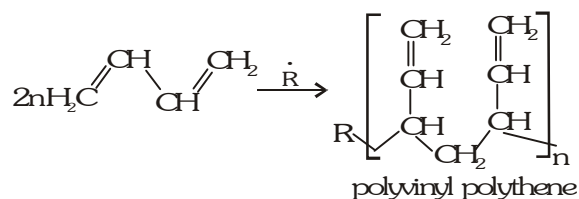


10. Butadiene is a conjugated diene and its free radical polymerisation can occur in two ways.

- (i) When the polymerization takes place at C_1 and C_4 of butadiene, an unbranched polymer results. It can exist either as trans-polybutadiene or as cis isomer or as the mixture of both.

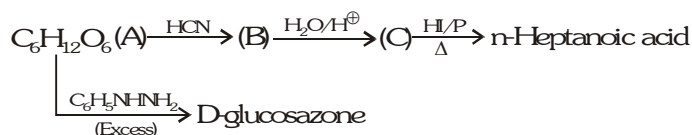


- (ii) 1,3-butadiene can also undergo polymerization at C_1 and C_2 to give polyvinyl polyethene as the product.



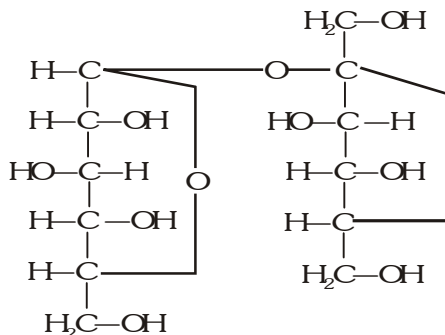
EXERCISE-04 [B]**BRAIN STORMING SUBJECTIVE EXERCISE**

1. Glucose and fructose are both reducing sugars. Why is sucrose regarded as a non-reducing sugar ?
2. (i) Name the type of linkages responsible for the formation of primary and secondary structures of proteins.
(ii) On electrolysis in acidic solution, α -amino acids migrate towards cathode while in alkaline medium, they migrate towards anode. Explain.
(iii) What are essential and non-essential amino acids. Give two example of each.
3. When the ketohexose, D(-) fructose is treated with phenyl hydrazine it produces an osazone that is identical with the one prepared from either D(+) glucose or D(+) mannose. How is the configuration of D(-) fructose related to those of D(+) glucose and D(+) mannose ?
4. Glycine exists as a dipolar ion, while anthranilic acid does not. Explain
5. Compound (A), $C_6H_{12}O_6$, is oxidized by bromine water into a monobasic acid and also reduces Tollen's reagent and Fehling's solution. It also responds to the following reactions. Identify the compound (A).

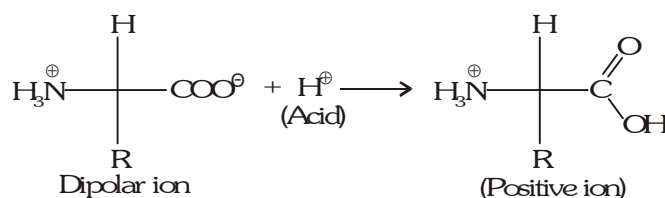


6. What is a peptide linkage ? What is the geometry and bond lengths in a peptide molecule ?
 7. Cellulose, a polysaccharide having β -D-glucoside units, has a stronger and more compact physical structure than starch which has α -D-glucose unit. Explain ?
 8. Order the following monomers with respect to their expected reactivity toward cationic polymerization and explain your answer.
 $H_2C=CHCH_3$, $H_2C=CHCl$, $H_2C=CH-C_6H_5$, $H_2C=CHCO_2CH_3$.
 9. What product would you expect to obtain from catalytic hydrogenation of natural rubber ? Would the product be syndiotactic, atactic, or isotactic ?
 10. Irradiation of poly (1, 3-butadiene), followed by addition of styrene, yields a graft copolymer that is used to make rubber soles for shoes. Draw the structure of a representative segment of this styrene- butadiene graft copolymer.
-

1. In the formation of sucrose (disaccharide), both glucose and fructose molecules are linked to each other at their aldehydic and ketonic groups respectively. Since these are not free in sucrose, it is a non-reducing sugar. This is further confirmed by the structure of sucrose.

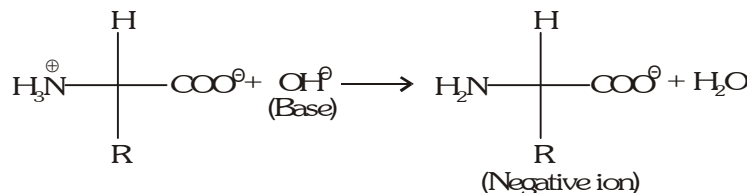


2. (a) Peptide linkages ($-\text{CO}-\text{NH}-$) are present in the primary structures of proteins while the secondary structures of proteins involve hydrogen bonding.
 (b) An α -amino acid has a dipolar structure, In acidic medium, it exists as a positive ion

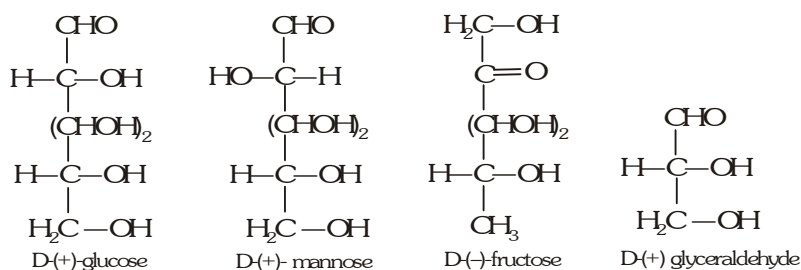


In electric field, the positive ion moves towards cathode.

In alkaline medium, the dipolar ion changes to anion and moves towards anode under the influence of applied electric field.

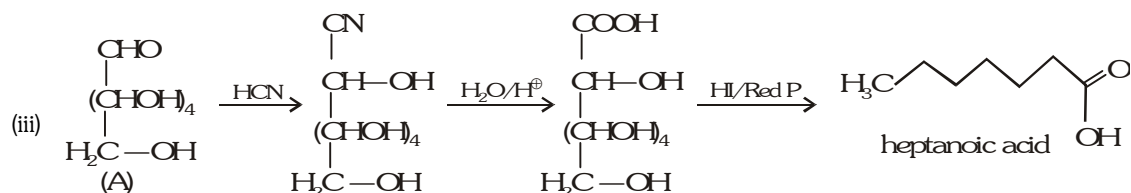
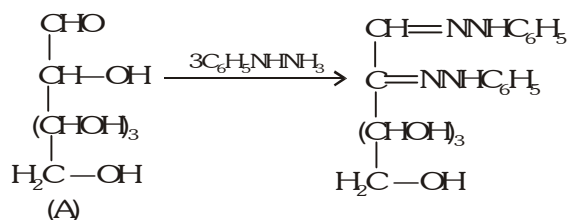


- (c) Amino acids which are not synthesized by the body are called essential amino acids. For ex., Leucine and Lysine.
 Amino acids which are synthesized by the body are known as non-essential amino acids. for ex. Glycine and Alanine.
3. Sugars forming same osazone have different configuration of C_1 and C_2 that are involved in osazone formation, the rest four carbon atoms (C_3 , C_4 , C_5 and C_6 are same) are identical. Since all belongs to D-Family hence the configuration of last but one asymmetric carbon atom would be the same.



4. In anthranilic acid, the $-\text{COOH}$ group is too weakly acidic to transfer a proton (H^+) to the weakly basic $-\text{NH}_2$ group attached to the electron-withdrawing benzene ring. When attached to an aliphatic carbon, the $-\text{NH}_2$ group is sufficiently basic to accept H^+ from $-\text{COOH}$ group.

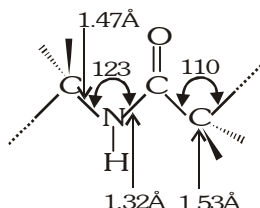
5. (i) (A) is glucose because it is readily oxidised by Br_2 water to gluconic acid. It reduces Fehling's and Tollen's reagent because it is a reducing sugar.
- (ii) With excess phenyl hydrazine it gives glucosazone.



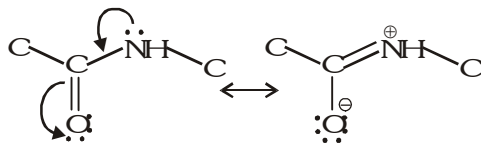
(A) is glucose.

6. Peptides are the polyamides formed by the condensation of amino group of one amino acid with the carboxylic group of the other. They are in fact secondary amides having $-\text{CO}-\text{NH}-$ linkage commonly referred to as peptide linkage.

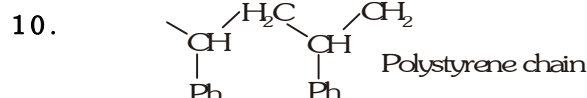
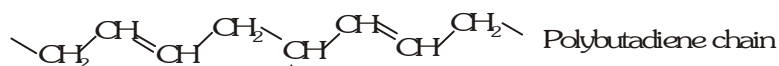
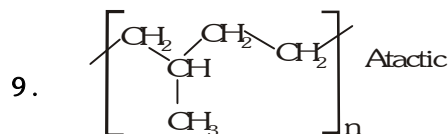
Studies on the structure determination of peptides have revealed that the amide group is flat and carbonyl and amino groups lie in one plane having H (of NH) and O (of CO) trans with respect to each other. Further the pioneering work by Linus Pauling on the X-ray studies of peptides has revealed that the C—N bond length of $-\text{CO}-\text{NH}-$ is 1.32 Å which is shorter than the usual 1.47 Å showing slight double bond character.



the double bond character of the C—N bond may be rationalized on the basis of resonance and because of this there is restricted rotation about amide bond, thus giving rise to cis, trans isomerism in which the trans isomer predominates.

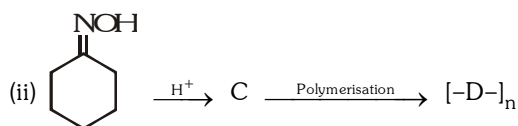


7. In cellulose the C—O bonds in β -D-glucoside units are all equatorial which would give a more stable polymeric chain (less steric interaction), whereas in starch the C—O bonds in α -D-glucoside units have to be axial if other C—O bonds are equatorial. The latter arrangement is less stable than the former because of increased steric interaction.
8. $\text{H}_2\text{C}=\text{CHCO}_2\text{CH}_3 < \text{H}_2\text{C}=\text{CHCl} < \text{H}_2\text{C}=\text{CHCH}_3 < \text{H}_2\text{C}=\text{CH}-\text{C}_6\text{H}_5$

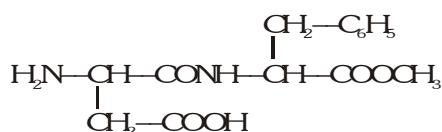


EXERCISE-05**PREVIOUS YEARS QUESTIONS**

1. The pair of compounds in which both the compounds give positive test with Tollen's reagent is :
(A) Glucose and Sucrose (B) Fructose and Sucrose [IIT-04]
(C) Acetophenone and Hexanal (D) Glucose and Fructose
2. The two forms of D-glucopyranose obtained from the solution of D-glucose are called [IIT-05]
(A) Isomers (B) Anomers
(C) Epimers (D) Enantiomers
3. Give the structure of the product in each of the following reactions. [IIT-2000]
(i) Suctose $\xrightarrow{H^+}$ A + B



4. Aspartame, an artificial sweetener, is a peptide and has the following structure : [IIT-2001]

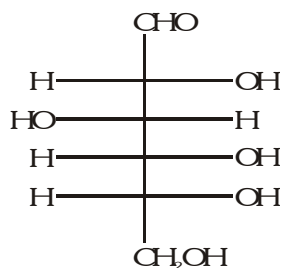


- (i) Identify the four functional groups.
(ii) Write the zwitterionic structure.
(iii) Write the structure of the amino acids obtained from the hydrolysis of aspartame.
(iv) Which of the two amino acids is more hydrophobic ?
5. Following two amino acids lysine and glutamine form dipeptide linkage. What are two possible dipeptides ?



[IIT-2003]

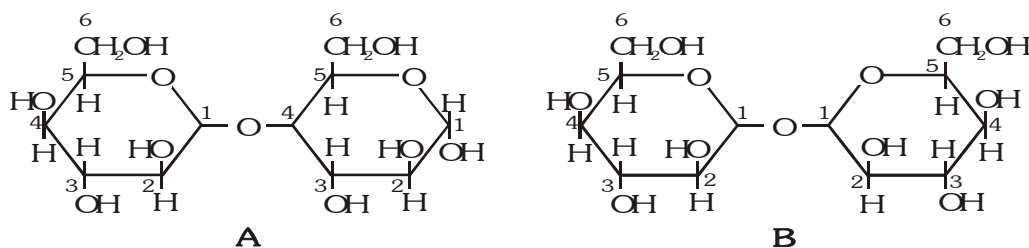
6. The Fisher projection of D-glucose is drawn below. [IIT-2004]



- (i) Draw the Fisher projection of L-glucose.
(ii) Give the reaction of L-glucose with Tollen's reagent.

7. Which of the following will reduce Tollen's reagent? Explain.

[IIT-2005]



8. Match column I with column II

[IIT-2007]

Column X

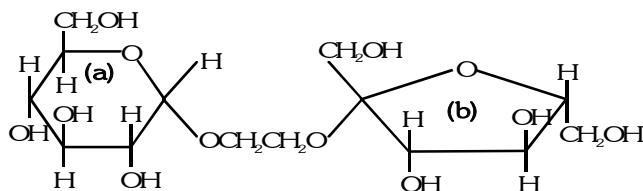
- (a) cellulose
- (b) nylon-6, 6
- (c) protein
- (d) sucrose

Column Y

- (p) natural polymer
- (q) synthetic polymer
- (r) amide linkage
- (s) glycoside linkage

9. The correct statement about the following disaccharide is -

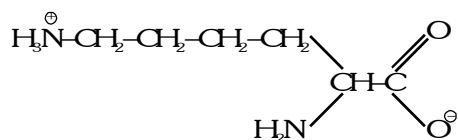
[IIT-2010]



- (A) Ring (a) is pyranose with α -glycosidic link
- (B) Ring (a) is furanose with α -glycosidic link
- (C) Ring (b) is furanose with α -glycosidic link
- (D) Ring (b) is pyranose with β -glycosidic link

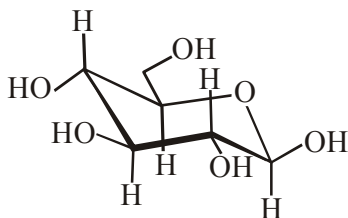
10. The total number of basic groups in the following form of lysine is :

[IIT-2010]



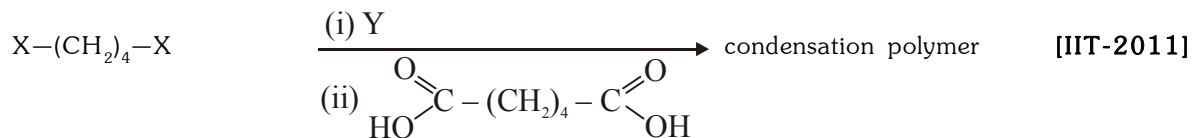
11. The following carbohydrate is

[IIT-2011]

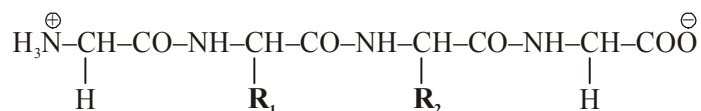


- (A) a ketohexose
- (B) an aldohexose
- (C) an α -furanose
- (D) an α -pyranose

12. The correct functional group X and the reagent/reaction conditions Y in the following scheme are

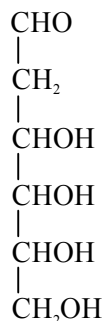


- (A) X = COOCH₃, Y = H₂/Ni/heat
 (B) X = CONH₂, Y = H₂/Ni/heat
 (C) X = CONH₂, Y = Br₂/NaOH
 (D) X = CN, Y = H₂/Ni/heat
13. The substitutes **R₁** and **R₂** for nine peptides are listed in the table given below. How many of these peptides are positively charged at pH = 7.0 ? [IIT-2012]



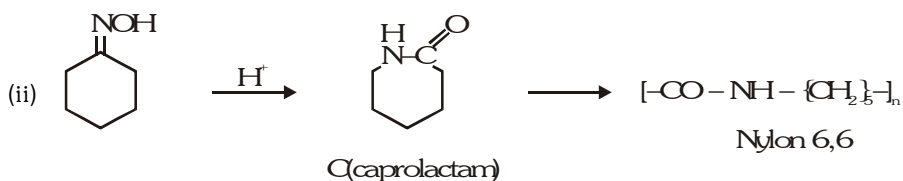
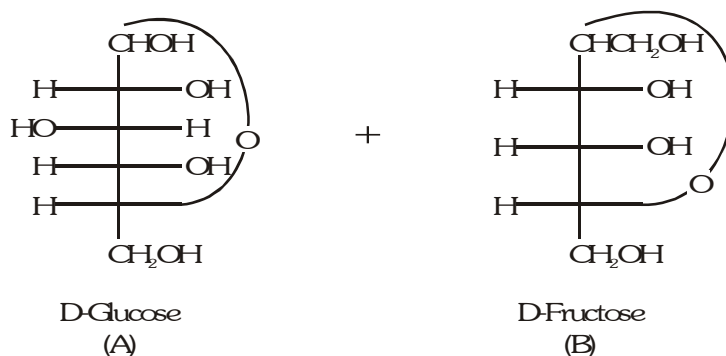
Peptide	R ₁	R ₂
I	H	H
II	H	CH ₃
III	CH ₂ COOH	H
IV	CH ₂ CONH ₂	(CH ₂) ₄ NH ₂
V	CH ₂ CONH ₂	CH ₂ CONH ₂
VI	(CH ₂) ₄ NH ₂	(CH ₂) ₄ NH ₂
VII	CH ₂ COOH	CH ₂ CONH ₂
VIII	CH ₂ OH	(CH ₂) ₄ NH ₂
IX	(CH ₂) ₄ NH ₂	CH ₃

14. When the following aldohexose exists in its d-configuration, the total number of stereoisomers in its pyranose form is - [IIT-2012]

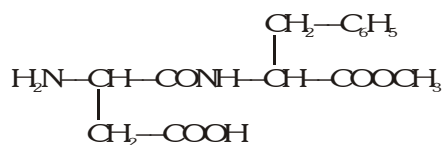


15. A tetrapeptide has -COOH group on alanine. This produces glycine (Gly), valine (Val), phenyl alanine (Phe) and alanine (Ala), on complete hydrolysis. For this tetrapeptide, the number of possible sequences (Primary structures) with -NH₂ group attached to a chiral center is. [IIT-2013]

- 1.(D) Glucose being an aldehyde responds to Tollen's test while fructose, although a ketose, undergoes rearrangement in presence of basic medium (provided by Tollen's reagent) to form glucose, which then responds to Tollen's test).
2. (B) The two isomeric forms (α - and β -) of D-glucopyranose differ in configuration only at C-1; hence these are called anomers.
3. (i) sucrose $\longrightarrow$



4.



Aspartame (Aspartamine)

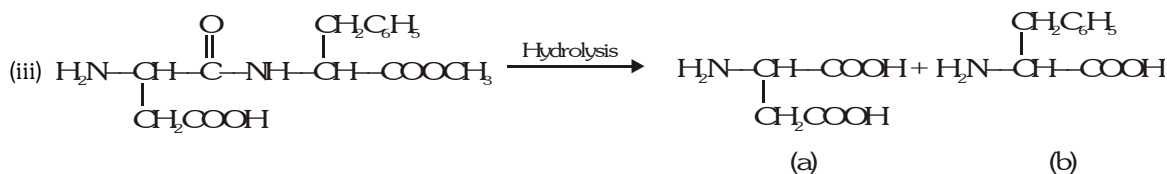
- (i) Four functional groups present in aspartamine are

(a) $-\text{NH}_2$ (Amine)(b) $-\text{COOH}$ (Carboxylic acid)

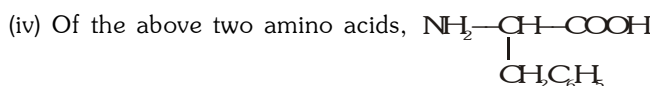
(c) (Amide)

(d) (Ester)

- (ii) Zwitterion structure is given as follows :
-

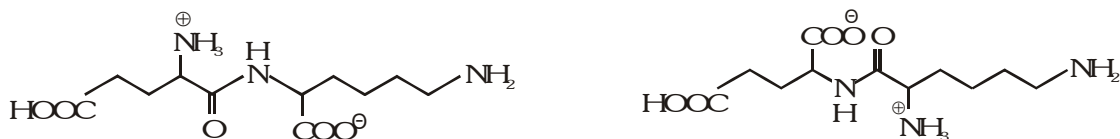


Hence on hydrolysis two amino acids (a) and (b) are obtained.

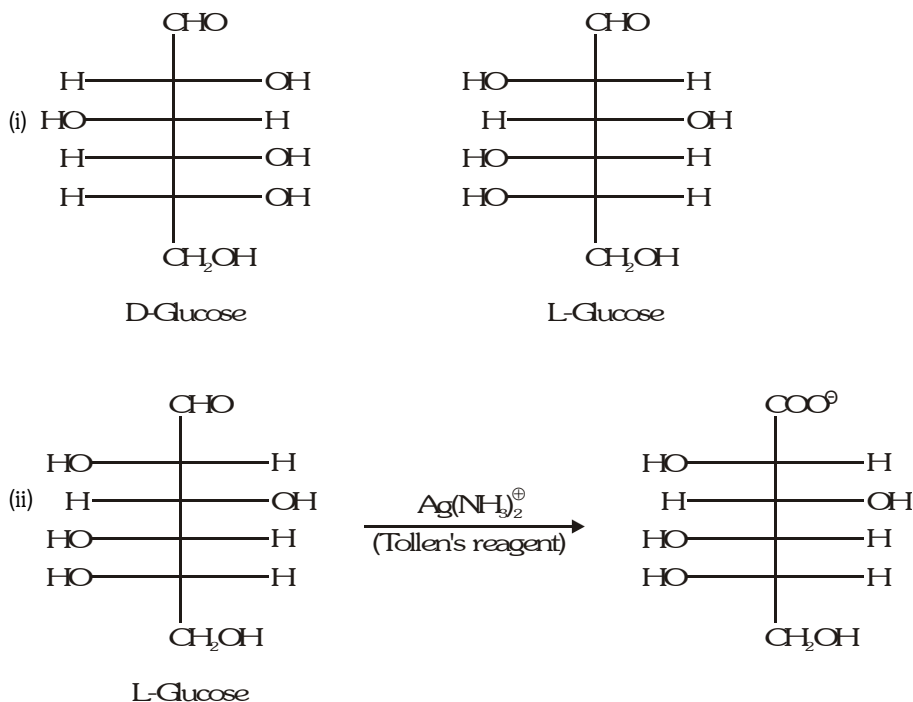


is more hydrophobic due to presence of non-polar and bulky benzyl group.

5. The structure of two possible dipeptides are



6. L-Glucose is an enantiomer of D-glucose, hence



7. In the two disaccharides structure A will be reducing sugar since both monosaccharides units are not linked through their reducing centres, while in structure B both the monosaccharide units are linked through their reducing centres, hence it will be non-reducing.

8. (A) : (p) and (s) Cellulose is a natural polymer and has a C_1-C_4 β -glycosidic linkage.
 (B) : (q) and (r) Nylon-6, 6 is a synthetic polymer of hexamethylenediamine and adipic acid and has amide linkage.
 (C) : (p) and (r) Proteins are natural polymers of α amino acids joined by amide linkages (peptide bonds).
 (D) : (s) Sucrose is a disaccharide of α -D glucose and β -D-fructose and has an α, β -glycosidic linkage.

9. (A) 12. (A,B,C,D) 13. 4 14. 8 15. 4

EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. Which of the following expressions is/are not true ?
(A) $[H^+] = [OH^-] = \sqrt{K_w}$ for a neutral solution at all temperatures.
(B) $[H^+] > \sqrt{K_w}$ & $[OH^-] < \sqrt{K_w}$ for an acidic solution
(C) $[H^+] < \sqrt{K_w}$ & $[OH^-] > \sqrt{K_w}$ for an alkaline solution
(D) $[H^+] = [OH^-] = 10^{-7} \text{ M}$ for a neutral solution at all temperatures
2. Addition of HCl will not suppress the ionization of-
(A) acetic acid (B) Benzoic acid (C) H_2S (D) Sulphuric acid
3. An acid solution of pH 6 is diluted thousand times. The pH of solution becomes approx-
(A) 6.96 (B) 6 (C) 4 (D) 9
4. pOH of H_2O is 7.0 at 298 K. If water is heated at 350 K, which of the following should be true ?
(A) pOH will decrease
(B) pOH will increase
(C) pOH will remain 7.0
(D) concentration of H^+ ions will increase but that of OH^- will decrease
5. Which of the following solution will have a pH exactly equal to 8 ?
(A) 10^{-8} M HCl solution at 25 C (B) 10^{-8} M H^+ solution at 25
(C) $2 \times 10^{-6} \text{ M}$ $Ba(OH)_2$ solution at 25 C (D) 10^{-6} M NaOH solution at 50 C
6. The number of hydrogen ions in 10 mL of a solution with pH = 13 is-
(A) 10^{13} (B) 6.023×10^8 (C) 6.023×10^{13} (D) 6.023×10^{10}
7. At 55 C autoprotolysis constant of water is 4×10^{-14} . If a given sample of water has a pH of 6.9, then it is-
(A) acidic (B) basic (C) neutral (D) explosive
8. 0.1 mol HCl is dissolved in distilled water of volume V then $V \xrightarrow{\text{lim}} \infty$ (pH)_{solution} is equal to-
(A) zero (B) 1 (C) 7 (D) 14
9. A 50 mL solution of pH = 1 is mixed with a 50 mL solution of pH = 2. then pH of the mixture will be nearly-
(A) 0.76 (B) 1.26 (C) 1.76 (D) 2.26
10. The pH of a solution obtained by mixing 50 mL of 0.4 N HCl and 50 mL of 0.2 N NaOH is-
(A) $-\log 2$ (B) $-\log 0.2$ (C) 1.0 (D) 2.0
11. The pH of a solution is 7.00. To this solution sufficient base is added to increase the pH to 12.0. The increase OH^- ion concentration is-
(A) 5 times (B) 1000 times (C) 10^5 times (D) 4 times
12. Which of the following solution will have pH close to 1.0 ?
(A) 100 mL of M/10HCl + 100 mL of M/10 NaOH
(B) 55 mL of M/10 HCl + 45 mL of M/10 NaOH
(C) 10 mL of M/10 HCl + 90 mL of M/10 NaOH
(D) 75 mL of M/5 HCl + 25 mL of M/5 NaOH.
13. The dissociation constants of two acids HA_1 and HA_2 are 3.0×10^{-4} and 1.8×10^{-5} respectively. The relative strengths of the acids will be approximately-
(A) 1 : 4 (B) 4 : 1 (C) 1 : 16 (D) 16 : 1

14. Which of the following is true-
- (A) pK_b for OH^- is -1.74 at $25^\circ C$
- (B) the equilibrium constant for the reaction between HA ($pK_a = 4$) and $NaOH$ at $25^\circ C$ will be equal to 10^{10}
- (C) the pH of a solution containing 0.1 M HCOOH ($K_a = 1.8 \times 10^{-4}$) and 0.1 M HOCN . ($K_a = 3.2 \times 10^{-4}$) will be nearly $(3 - \log 7)$.
- (D) all the above are correct.
15. Which statement/relationship is correct ?
- (A) upon hydrolysis of salt of a strong base and weak acid gives a solution with $pH < 7$
- (B) $pH = -\log \frac{1}{[H^+]}$
- (C) only at $25^\circ C$ the pH of water is 7
- (D) the value of pK_w at $25^\circ C$ is 7
16. If 50 mL of 0.2 (M) KOH is added to 40 mL of 0.5 (M) $HCOOH$. The pH of the resulting solution is ($K = 1.8 \times 10^{-4}$) :
- (A) 3.75 (B) 5.6 (C) 7.5 (D) 3.4
17. 50% neutralization of a solution of formic acid ($K_a = 2 \times 10^{-4}$) with $NaOH$ would result in a solution having a hydrogen ion concentration of-
- (A) 2×10^{-4} (B) 3.7 (C) 2.7 (D) 1.85
18. The correct order of increasing $[H_3O^+]$ in the following aqueous solution is-
- (A) $0.01\text{ M H}_2\text{S} < 0.01\text{ M H}_2\text{SO}_4 < 0.01\text{ M NaCl} < 0.01\text{ M NaNO}_2$
- (B) $0.01\text{ M NaCl} < 0.01\text{ M NaNO}_2 < 0.01\text{ M H}_2\text{S} < 0.01\text{ M H}_2\text{SO}_4$
- (C) $0.01\text{ M NaNO}_2 < 0.01\text{ M NaCl} < 0.01\text{ M H}_2\text{S} < 0.01\text{ M H}_2\text{SO}_4$
- (D) $0.01\text{ M H}_2\text{S} < 0.01\text{ M NaNO}_2 < 0.01\text{ M NaCl} < 0.01\text{ M H}_2\text{SO}_4$
19. The sodium salt of a certain weak monobasic organic acid is hydrolysed to an extent of 3% in its 0.1 M solution at $25^\circ C$. Given that the ionic product of water is 10^{-14} at this temperature, what is the dissociation constant of the acid ?
- (A) $\approx 1 \times 10^{-10}$ (B) $\approx 1 \times 10^{-9}$ (C) 3.33×10^{-9} (D) 3.33×10^{-10}
20. The correct order for the increasing extent of hydrolysis is-
- (A) $PO_4^{3-} < HPO_4^{2-} < H_2PO_4^-$ (B) $H_2PO_4^- < HPO_4^{2-} < PO_4^{3-}$
- (C) $HPO_4^{2-} < PO_4^{3-} < H_2PO_4^-$ (D) $PO_4^{3-} < H_3PO_4^- < HPO_4^{2-}$
21. The correct order of increasing pH of decimolar solution of each of the following in-
- (A) $NH_4NO_3 < NaNO_3 < NaHCO_3 < Na_2CO_3$
- (B) $NaNO_3 < NH_4NO_3 < NaHCO_3 < Na_2CO_3$
- (C) $NaNO_3 < NH_4NO_3 < Na_2CO_3 < NaHCO_3$
- (D) $Na_2CO_3 < NaHCO_3 < NaNO_3 < NH_4NO_3$
22. When 100 mL of $0.4\text{ M CH}_3\text{COOH}$ are mixed with 100 mL of 0.2 M NaOH , the $[H_3O^+]$ in the solution is approximately : [$K_a(\text{CH}_3\text{COOH}) = 1.8 \times 10^{-5}$]
- (A) 1.8×10^{-6} (B) 1.8×10^{-5} (C) 9×10^{-6} (D) 9×10^{-5}
23. A solution is $0.1\text{ M CH}_3\text{COOH}$ and $0.1\text{ M CH}_3\text{COONa}$. Which of the following will change the pH significantly, of the solution ?
- (A) addition of water
- (B) addition of CH_3COONa without change in volume
- (C) addition of CH_3COOH without change in volume
- (D) none will change the pH significantly

24. What will be the pH at the equivalence point during the titration of a 100 mL 0.2 M solution of CH_3COONa with 0.2 M solution of HCl ? $K_a = 2 \times 10^{-5}$
- (A) $3 - \log \sqrt{2}$ (B) $3 + \log \sqrt{2}$ (C) $3 - \log 2$ (D) $3 + \log 2$
25. K_a for HCN is 5×10^{-10} at 25 °C. For maintaining a constant pH of 9, the volume of 5 M KCN solution required to be added to 10 mL of 2M HCN solution is-
- (A) 4 mL (B) 7.95 mL (C) 2 mL (D) 9.3 mL
26. The pH of a solution obtained by mixing 100 mL of 0.2 M CH_3COOH with 100 mL of 0.2 M NaOH would be ($\text{p}K_a$ for $\text{CH}_3\text{COOH} = 4.74$ and $\log 2 = 0.301$)
- (A) 4.74 (B) 8.87 (C) 9.10 (D) 8.57
27. Which of the following when added to 1.0 L of 0.5 M HCl would result in maximum increase in pH?
- (A) 0.5 mol CH_3COOH (B) 1.0 mol NaCl
(C) 0.4 mol NaOH (D) 0.6 mole CH_3COONa
28. What % of the carbon in the H_2CO_3 , HCO_3^- . Buffer should be in the form of HCO_3^- so as to have a neutral solution? ($K_a = 4 \times 10^{-7}$)
- (A) 20 % (B) 40 % (C) 60 % (D) 80 %
29. Which of the following solution would have same pH?
- (A) 100 mL of 0.2 M HCl + 100 mL of 0.4 M NH_3
(B) 50 mL of 0.1 M HCl + 50 mL of 0.2 M NH_3
(C) 100 mL of 0.3 M HCl + 100 mL of 0.6 M NH_3
(D) All will have same pH
30. 50 mL of 0.1 M NaOH is added to 60 mL of 0.15 M H_3PO_4 solution (K_1 , K_2 and K_3 may be taken as 10^{-3} , 10^{-8} and 10^{-13} respectively). The pH of the mixture would be about-
- (A) 3.1 (B) 5.5 (C) 4.1 (D) 6.5
31. The solubility of a certain sparingly soluble substance MX_n is nearly 1.4×10^{-4} M. If the solubility product is 1.1×10^{-11} , what is the value of n?
- (A) 1 (B) 2 (C) 3 (D) 1.5
32. The concentration of Mg^{2+} in the solution made by mixing 10 mL of 0.25 M $\text{Mg}(\text{NO}_3)_2$ and 25 mL of 0.2 M NaF will be ($K_{sp}(\text{MgF}_2) = 8 \times 10^{-8}$)
- (A) 0.0027 M (B) 0.0714 M (C) 0.0030 M (D) 0.0060 M
33. The pH of a saturated solution of $\text{Mg}(\text{OH})_2$ in water will be ($K_{sp} = 4 \times 10^{-12}$)
- (A) $4 - \log 2$ (B) $10 - \log 2$ (C) $4 + \log 2$ (D) $10 + \log 2$
34. When NaCl is added to the reaction mixture of an oil and caustic soda, the soap is thrown out because-
- (A) NaCl is an ionic compound
(B) soap is insoluble in the presence of chloride ions
(C) the solubility product of NaCl decreases in the presence of soap
(D) the solubility product of the soap is exceeded due to the increased concentration of Na^+ ions.
35. 16.6 mg of solid silver chromate (molar mass = 332) when put into 500 mL water, silver ion and chromate ion are obtained. On adding more solid, the concentration of ions does not increase. If 100 mg of solid is put into 2 litre of water, then the amount of solid remained undissociated would be-
- (A) 33.6 mg (B) 3.36 mg (C) 66.4 mg (D) 6.64 mg
36. In which of the following solvents will AgBr has highest solubility-
- (A) 10^{-3} M NaBr (B) 10^{-3} M NH_4OH (C) pure water (D) 10^{-3} M HBr
37. In the system $\text{CaF}_2(\text{s}) \rightleftharpoons \text{Ca}^{2+}(\text{aq}) + 2\text{F}^-$ increasing the concentration of Ca^{2+} ions 4 times will cause the equilibrium concentration of F^- ions to change to times the initial value.
- (A) 4 (B) 1/2 (C) 2 (D) 1/4

38. The solubility of $\text{Fe}(\text{OH})_3$ would be maximum in-
 (A) 0.1 M NaOH (B) 0.1 M HCl (C) 0.1 M KOH (D) 0.1 M H_2SO_4
39. Arrange in increasing order of solubility of AgBr in solutions given :
 (i) 0.1 M NH_3 (ii) 0.1 M AgNO_3 (iii) 0.2 M NaBr (iv) pure water
 (A) (iii) < (ii) < (iv) < (i) (B) (iii) < (ii) < (i) < (iv)
 (C) (iii) < (ii) = (i) < (iv) (D) (ii) < (iii) < (iv) < (i)
40. The solubility product of BaCrO_4 is $2.4 \times 10^{-10} \text{ M}^2$. The maximum concentration of $\text{Ba}(\text{NO}_3)_2$ possible without precipitation in a $6 \times 10^{-4} \text{ M}$ K_2CrO_4 solution is-
 (A) $4 \times 10^{-7} \text{ M}$ (B) $1.2 \times 10^{10} \text{ M}$ (C) $6 \times 10^{-4} \text{ M}$ (D) $3 \times 10^{-4} \text{ M}$
41. At 25 °C, the solubility product values of AgCl and AgCNS are 1.7×10^{-10} and 1.0×10^{-12} respectively. When water is saturated with both solids, calculate the ratio $[\text{Cl}^-]/[\text{CNS}^-]$ and also $[\text{Ag}^+]$ in the solution.
 (A) 1.3×10^2 , $1.7 \times 10^{-5} \text{ M}$ (B) 1.7×10^2 , $1.308 \times 10^{-5} \text{ M}$
 (C) 1.3×10^4 , $1.308 \times 10^{-4} \text{ M}$ (D) 1.7×10^3 , $1.67 \times 10^{-6} \text{ M}$
42. When pure water is saturated with CaCO_3 and CaC_2O_4 , the concentration of calcium ion in the solution under equilibrium is $8.426 \times 10^{-5} \text{ M}$. If the ratio of the solubility product of CaCO_3 to that of CaC_2O_4 is 2.087, what is the solubility product of CaCO_3 in pure water ?
 (A) 4.80×10^{-8} (B) 9.60×10^{-9} (C) 9.60×10^{-8} (D) 4.80×10^{-9}
43. The solubility of Ag_2CO_3 in water at 25 °C is 1×10^{-4} mole/litre. What is its solubility in 0.1 M Na_2CO_3 solution ? Assume no hydrolysis of CO_3^{2-} ion.
 (A) 6.323×10^{-6} mole/litre (B) 4.74×10^{-5} mole/litre
 (C) 3.16×10^{-6} mole/litre (D) 5.51×10^{-5} mole/litre
44. The solubility of calcium phosphate in water is $x \text{ mol L}^{-1}$ at 25 °C. Its solubility product is equal to-
 (A) $108 x^2$ (B) $36 x^3$ (C) $36 x^5$ (D) $108 x^5$
45. The solubility product of AgCl is 1.8×10^{-10} . Precipitation of AgCl will occur only when equal volumes of solutions of-
 (A) 10^{-4} M Ag^+ and 10^{-4} M Cl^- are mixed (B) 10^{-7} M Ag^+ and 10^{-7} M Cl^- are mixed
 (C) 10^{-5} M Ag^+ and 10^{-5} M Cl^- are mixed (D) 10^{-10} M Ag^+ and 10^{-10} M Cl^- are mixed
46. The solubility of CaF_2 ($K_{\text{sp}} = 3.4 \times 10^{-11}$) in 0.1 M solution of NaF would be-
 (A) $3.4 \times 10^{-12} \text{ M}$ (B) $3.4 \times 10^{-10} \text{ M}$ (C) $3.4 \times 10^{-9} \text{ M}$ (D) $3.4 \times 10^{-13} \text{ M}$
47. The precipitate of CaF_2 ($K_{\text{sp}} = 1.7 \times 10^{-10}$) is obtained when equal volumes of the following are mixed.
 (A) $10^{-4} \text{ M Ca}^{2+} + 10^{-4} \text{ M F}^-$ (B) $10^{-2} \text{ M Ca}^{2+} + 10^{-3} \text{ M F}^-$
 (C) $10^{-5} \text{ M Ca}^{2+} + 10^{-3} \text{ M F}^-$ (D) $10^{-3} \text{ M Ca}^{2+} + 10^{-5} \text{ M F}^-$
48. Let the solubilities of AgCl in H_2O , 0.01 M CaCl_2 , 0.01 M NaCl & 0.05 M AgNO_3 be S_1 , S_2 , S_3 & S_4 respectively what is the correct relationship between these quantities. Neglect any complexation.
 (A) $S_1 > S_2 > S_3 > S_4$ (B) $S_1 > S_2 = S_3 > S_4$
 (C) $S_1 > S_3 > S_2 > S_4$ (D) $S_4 > S_2 > S_3 > S_1$
49. The solubility product $\text{Mg}(\text{OH})_2$ in water at 25 °C is $8.9 \times 10^{-13} (\text{mole dm}^{-3})^3$ while that of $\text{Al}(\text{OH})_3$ is $5 \times 10^{-33} (\text{mol dm}^{-3})^4$. If S_1 and S_2 are the solubilities of $\text{Mg}(\text{OH})_2$ and $\text{Al}(\text{OH})_3$ in water in mol dm^{-3} at 25 °C, what is the order of magnitude of the ratio, S_1/S_2 ?
 (A) 10^5 (B) 10^4 (C) 10^6 (D) 10^3
50. When HCl gas is passed through a saturated solution of common salt, pure NaCl is precipitated because-
 (A) HCl is highly ionised in solution
 (B) HCl is highly soluble in water
 (C) the solubility product of NaCl is lowered by HCl
 (D) the ionic product of $[\text{Na}^+][\text{Cl}^-]$ exceeds the solubility product of NaCl

51. A certain indicator (an organic dye) has $pK_a = 5$. For which of the following titrations may it be suitable.
- (A) acetic acid against NaOH
 (B) aniline hydrochloride against NaOH
 (C) sodium carbonate against HCl
 (D) barium hydroxide against oxalic acid
52. The pH indicators are-
- (A) salts of strong acids & strong bases
 (B) salts of weak acids & weak bases
 (C) either weak acids or weak bases
 (D) either strong acids or strong bases
53. What fraction of an indicator HIn is in the basic form at a pH of 6 if pK_a of the indicator is 5 ?
- (A) $\frac{1}{2}$ (B) $\frac{1}{11}$ (C) $\frac{10}{11}$ (D) $\frac{1}{10}$
54. An acid-base indicator which is a weak acid has a pK_a value = 5.5. At what concentration ratio of sodium acetate to acetic acid would the indicator show a colour half-way between those of its acid and conjugate base forms ? pK_a of acetic acid = 4.75
- (A) 4.93 : 1 (B) 6.3 : 1 (C) 5.62 : 1 (D) 2.37 : 1

CHECK YOUR GRASP						ANSWER KEY				EXERCISE -1					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	D	D	A	A	B	B	B	C	B	C	C	D	B	D	C
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	A	A	C	A	B	A	B	D	A	C	B	D	D	D	A
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	B	A	D	D	A	B	B	D	A	A	B	D	C	D	A
Que.	46	47	48	49	50	51	52	53	54						
Ans.	C	B	C	B	D	C	C	C	C						

EXERCISE-02**BRAIN TEASERS****SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)**

1. The conjugate acid of NH_2^- is-
(A) NH_3 (B) NH_2OH (C) NH_4^+ (D) N_2H_4
2. Out of the following, amphoteric species are
(I) HPO_3^{2-} (II) OH^- (iii) H_2PO_4^- (IV) HCO_3^-
(A) I, III, IV (B) I and III (C) III and IV (D) All
3. pH of an aqueous solution of NaCl at 85 °C should be-
(A) 7 (B) > 7 (C) < 7 (D) 0
4. 1 CC of 0.1 N HCl is added to 99 CC solution of NaCl. The pH of the resulting solution will be-
(A) 7 (B) 3 (C) 4 (D) 1
5. 10 mL of $\frac{M}{200}$ H_2SO_4 is mixed with 40 mL of $\frac{M}{200}$ H_2SO_4 . The pH of the resulting solution is-
(A) 1 (B) 2 (C) 2.3 (D) none of these
6. If pK_b for fluoride ion at 25 °C is 10.83, the ionization constant of hydrofluoric acid in water at this temperature is-
(A) 1.74×10^{-5} (B) 3.52×10^{-3} (C) 6.75×10^{-4} (D) 5.38×10^{-2}
7. If K_1 & K_2 be first and second ionisation constant of H_3PO_4 and $K_1 \gg K_2$ which is incorrect-
(A) $[\text{H}^+] = [\text{H}_2\text{PO}_4^-]$ (B) $[\text{H}^+] = \sqrt{K_1[\text{H}_3\text{PO}_4]}$
(C) $K_2 = [\text{HPO}_4^{2-}]$ (D) $[\text{H}^+] = 3[\text{PO}_4^{3-}]$
8. The degree of hydrolysis of a salt of weak acid and weak base in its 0.1 M solution is found to be 50%. If the molarity of the solution is 0.2 M, the percentage hydrolysis of the salt should be-
(A) 100 % (B) 50 % (C) 25 % (D) none of these
9. What is the percentage hydrolysis of NaCN in N/80 solution when the dissociation constant for HCN is 1.3×10^{-9} and $K_w = 1.0 \times 10^{-14}$
(A) 2.48 (B) 5.26 (C) 8.2 (D) 9.6
10. pH of 0.01 M $(\text{NH}_4)_2\text{SO}_4$ and 0.02 M NH_4OH buffer solution (pK_a of $\text{NH}_4^+ = 9.26$) is-
(A) 9.26 (B) 4.74 (C) $4.74 + \log 2$ (D) none
11. The range of most suitable indicator which should be used for titration of X^-Na^+ (0.1 M, 10 mL) with 0.1 M HCl should be (Given : $k_b(\text{X}^-) = 10^{-6}$)
(A) 2 – 3 (B) 3 – 5 (C) 6 – 8 (D) 8-10
12. When NO_2 is bubbled into water, it disproportionates completely into HNO_2 and HNO_3 .
$$2\text{NO}_2 + \text{H}_2\text{O}(\ell) \longrightarrow \text{HNO}_2(\text{aq.}) + \text{HNO}_3(\text{aq.})$$

The concentration of NO_2^- in a solution prepared by dissolving 0.05 mole of NO_2 gas in 1 litre H_2O is $\{K_a(\text{HNO}_2) = 5 \times 10^{-4}\}$
(A) $\sim 5 \times 10^{-4}$ (B) $\sim 4.8 \times 10^{-5}$
(C) $\sim 4.8 \times 10^{-3}$ (D) $\sim 2.55 \times 10^{-2}$
13. If K_{sp} for HgSO_4 is 6.4×10^{-5} , then solubility of this substance in mole per m^3 is-
(A) 8×10^{-3} (B) 6.4×10^{-5} (C) 8×10^{-6} (D) none of these
14. Which of the following is most soluble in water ?
(A) MnS ($K_{sp} = 8 \times 10^{-37}$) (B) ZnS ($K_{sp} = 7 \times 10^{-16}$)
(C) Bi_2S_3 ($K_{sp} = 1 \times 10^{-72}$) (D) $\text{Ag}_3(\text{PO}_4)$ ($K_{sp} = 1.8 \times 10^{-8}$)

15. How many moles NH_3 must be added to 2.0 litre of 0.80 M AgNO_3 in order to reduce the Ag^+ concentration to 5×10^{-8} M. K_f of $[\text{Ag}(\text{NH}_3)_2]^+ = 10^8$
 (A) 0.4 (B) 2 (C) 3.52 (D) 4
16. The solubility product of BaF_2 has the value, 1.7×10^{-4} . If V_1 mL of BaCl_2 Solution (10^{-1} M) and V_2 mL of NaF solution (10^{-2} M) are mixed, what is the ratio $V_1 : V_2$ so that a precipitate may be obtained?
 (A) 1 : 1
 (B) 2 : 1
 (C) 1 : 2
 (D) precipitate cannot be obtained for any possible ratio
17. At 18 °C, the solubility of CdS in water is 6.33×10^{-15} M. What is the concentration of Cd^+ ion in a solution of pH = 1 saturated with H_2S gas, in which concentration of $\text{H}_2\text{S} = 0.1$ M ? The product of the first and second ionization constants of H_2S is 1.1×10^{-22} at this temperature.
 (A) 6.343×10^{-8} M (B) 4.368×10^{-8} M
 (C) 4.368×10^{-9} M (D) 3.643×10^{-8} M
18. 100 mL of 0.02 M benzoic acid ($\text{pK}_a = 4.2$) is titrated using 0.02 M NaOH . pH after 50 mL and 100 mL of NaOH have been added are-
 (A) 3.50, 7 (B) 4.2, 7 (C) 4.2, 8.1 (D) 4.2, 8.25
19. What is the pH of solution made by adding 3.9 g NaNH_2 into water to make a 500 mL solution $K_b(\text{NH}_3) = 2 \times 10^{-5}$ [$\text{Na} = 23$, $\text{N} = 14$, $\text{H} = 1$]
 (A) 13.3 (B) 0.7 (C) 5.3 (D) 13.7
20. 10.2 g of acetic anhydride was added to 989.8 g of water to make a solution with a density of 1g/mL. If the k_a of acetic acid is 2×10^{-5} , the pH of the solution would be [$\text{C} = 12$, $\text{H} = 1$, $\text{O} = 16$]
 (A) 2.7 (B) 3.7 (C) 4.7 (D) 5.7
21. A well is dug in a bed of rock containing fluor spar (CaF_2). If the well contains 20000 L of water, what is the amount of F^- in it ? $K_{sp} = 4 \times 10^{-11}$
 (A) 4.3 mol (B) 6.8 mol (C) 8.6 mol (D) 13.6 mol
22. If $\text{HA} + \text{NaOH} \longrightarrow \text{NaA} + \text{H}_2\text{O}$ $\Delta H = -12$ kcal
 and $\text{HB} + \text{NaOH} \longrightarrow \text{NaB} + \text{H}_2\text{O}$ $\Delta H = -11$ kcal
 then equimolar solution of which acid has higher pH-
 (A) HA (B) HB
 (C) both have same pH (D) information insufficient
23. Aniline behaves as a weak base. When 0.1 M, 50 mL solution of aniline was mixed with 0.1 M, 25 mL solution of HCl the pH of resulting solution was 8. Then the pH of 0.01 M solution of aniliniumchloride will be ($K_w = 10^{-14}$)
 (A) 6 (B) 6.5 (C) 5 (D) 5.5
24. pH of a mixture of 1 M benzoic acid ($\text{pK}_a = 4.20$) and 1M $\text{C}_6\text{H}_5\text{COONa}$ is 4.5, what is the volume of benzoic acid required to prepare a 300 mL buffer [$\log 2 = 0.3$] ?
 (A) 200 mL (B) 150 mL (C) 100 mL (D) 50 mL
25. What is the difference in pH for 1/3 and 2/3 stages of neutralisation of 0.1 M CH_3COOH with 0.1 M NaOH .
 (A) $-2 \log 3$ (B) $2 \log(1/4)$ (C) $2 \log(2/3)$ (D) $-2 \log 2$
26. An acid HA ($k_a = 10^{-5}$) reacts with NaOH at 298 K. What would be the value of the rate constant of the reverse reaction at the same temperature if the rate constant of the forward reaction is $10^{-11} \text{ mol}^{-1} \text{ L sec}^{-1}$?
 (A) 10^{-9} (B) 10^9 (C) 10^{-5} (D) 10^{-20}

27. The pH of 1.0 M NaHSO_4 solution will be (given that K_1 and K_2 for H_2SO_4 equal to ∞ and 10^{-2} respectively)-
 (A) nearly 1.0 (B) between 2 and 3
 (C) between 1.2 and 1.8 (D) between 3 and 4
28. 0.1 millimole of CdSO_4 are present in 10 mL acid solution of 0.08 N HCl. Now H_2S is passed to precipitate all the Cd^{2+} ions. The pH of the solution after filtering off precipitate, boiling off H_2S and making the solution 100 mL by adding H_2O is-
 (A) 2 (B) 4 (C) 6 (D) 8
29. Zn salt is mixed with $(\text{NH}_4)_2\text{S}$ of molarity 0.021 M. The amount of Zn^{2+} remains unprecipitated in 12 mL of this solution would be (Given : $K_{\text{sp}} \text{ZnS} = 4.51 \times 10^{-24}$)
 (A) 1.677×10^{-22} g (B) 1.767×10^{-22} g
 (C) 2.01×10^{-23} g (D) none of these
30. The self ionisation constant for pure formic acid, $K = [\text{HCOOH}_2^+][\text{HCOO}^-]$ has been estimated as 10^{-6} at room temperature. The density of formic acid is 1.22 g/cm^3 . The percentage of formic acid molecules in pure formic acid that are converted to formate ion would be-
 (A) 0.002 % (B) 0.004 % (C) 0.006 % (D) 0.008 %
-

BRAIN TEASERS							ANSWER KEY			EXERCISE -2					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	A	C	C	B	B	C	D	B	A	A	B	A	D	D	D
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	D	D	C	A	A	C	B	C	C	D	D	A	A	A	B

EXERCISE-03

MISCELLANEOUS TYPE QUESTIONS

TRUE / FALSE

1. When a solution of a weak monoprotic acid is titrated against a strong base, at half-neutralization point,
$$\text{pH} = \frac{1}{2} \text{pK}_a$$
2. A solution of sodium acetate and ammonium acetate can act as a buffer.
3. If the solubility of the salt $\text{Li}_3\text{Na}_3(\text{AlF}_6)_2$ is x , then its solubility product would be $2916 x^8$.
4. A buffer has maximum buffer capacity when the ratio of salt to acid is 10.
5. In the presence of a common ion (incapable of forming complex ion), the solubility of salt decreases.
6. The ionic product of water changes if a few drops of acid or base are added to it.
7. When equilibrium is attained, the concentration of each of the reactants & products become equal.
8. The reaction, $\text{HCN} + \text{OH}^- \rightleftharpoons \text{CN}^- + \text{H}_2\text{O}$ is displaced to the right indicating that the acid strength of HCN is greater than water & the base strength of CN^- is greater than that of OH^- .
9. The hydroxyl ion is hydrated to give several ionic species like $\text{H}_2\text{O}_2^{-2}$, $\text{H}_3\text{O}_3^{-3}$ and $\text{H}_4\text{O}_4^{-4}$.
10. Ostwald's dilution formula is applicable to weak as well as strong electrolytes.
11. CO_2 is a Lewis base
12. Solution whether neutral, acidic or basic contain both H^+ & OH^- ions.
13. The ionic product of a saturated solution is equal to solubility product constant of its solute.
14. A Lewis base is a substance which can donate a pair of electrons ?
15. If ionic product is less than K_{sp} , no precipitation will occur.
16. A buffer has definite pH value which changes on keeping it or on diluting it.
17. A salt of strong acid with a strong base does not undergo hydrolysis
18. HCl does not act as an acid in benzene
19. Water acts as a base when ammonia is dissolved in it.
20. In the reaction, $\text{SnCl}_4 + 2\text{Cl}^- \longrightarrow [\text{SnCl}_6]^{2-}$, SnCl_4 is a Lewis acid.
21. The strength of an oxy acid increases with increase in the EN value of central atom.
22. In aqueous solution the hydronium ion is further hydrated to give species like H_5O_2^+ , H_7O_3^+ and H_9O_4^+

FILL IN THE BLANKS

1. In a mixture of weak acid and its salt, the ratio of concentration of salt to acid is increased ten fold. The pH of the solution would by unit.
2. The solubility of CH_3COOAg in water considering hydrolysis of CH_3COO^- ions would be than that ignoring the hydrolysis.
3. From an equimolar solution of Cl^- and Br^- ions, the addition of Ag^+ will selectively precipitates (K_{sp} of AgCl & AgBr are 1×10^{-10} & 1×10^{-13} respectively).
4. The solubility of AgCl in NH_3 is than the solubility in pure water because of complex ion, $[\text{Ag}(\text{NH}_3)_2]^+$ formation.
5. The hydrolytic constant K_h for the hydrolytic equilibrium
$$\text{H}_2\text{PO}_4^- + \text{H}_2\text{O} \longrightarrow \text{H}_3\text{PO}_4 + \text{OH}^-$$
 is 1.4×10^{-12}
The value of ionization constant for the $\text{H}_3\text{PO}_4 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{PO}_4^- + \text{H}_3\text{O}^+$ is
6. Given the equilibrium constants
$$\text{Cl}^- + \text{HgCl}_2^+ \rightleftharpoons \text{HgCl}_2 \quad ; \quad K_1 = 3 \times 10^6$$
$$\text{HgCl}_2 + \text{Cl}^- \rightleftharpoons \text{HgCl}_3^- \quad ; \quad K_2 = 8.9$$

The equilibrium constants for the disproportionation equilibrium.
$$2\text{HgCl}_2 \rightleftharpoons \text{HgCl}_2^+ + \text{HgCl}_3^-$$
 is-

7. If the salts M_2X , QY_2 and PZ_3 have same solubilities ($\lll 1$), their K_{sp} values are related as
8. K_a for an acid HA is 1×10^{-6} . K_b for A^- would be
9. An aqueous solution of K_2SO_4 has pH nearly equal to
10. The pH of a solution which is 0.1 M sodium acetate and 0.01 M acetic acid ($pK_a = 4.74$) would be
11. The conjugative acid of sulphate (SO_4^{2-}) is
12. The value of K_w with increase in temperature.
13. $AgCl$ is soluble in aqueous sodium chloride solution than in pure water.
14. The buffer $HCOOH/HCOONa$ will have pH than 7.
15. In the reaction $I_2 + I^- \longrightarrow I_3^-$, I_2 acts as
16. An equimolar solution of $NaNO_2$ and HNO_2 can act as a solution.
17. Larger the value of pK_a , is the acid
18. Between Na^+ & Ag^+ ion, is a stronger Lewis acid.
19. Salt of strong acids and weak base undergo hydrolysis
20. For salts of weak acid with weak bases, degree of hydrolysis is of concentration of the salt in solution.
21. The solubility of $KAl(SO_4)_2$ in terms of its solubility product is
22. The dissociation constant of NH_4OH is 1.8×10^{-5} . The hydrolysis constant of NH_4^+ ions at $25^\circ C$ would be
23. solution of $CuSO_4$ is due to the hydrolysis of ions
24. The colour of unionized form of phenolphthalein is whereas that of ionized form is
25. In general, in aqueous solution $pH + pOH =$ at all temperatures.
26. The ionization constant of water is related to ionic product by the expression
27. The smaller the value of K_a of a weak acid, is the hydrolysis constant of its conjugate base.

MATCH THE COLUMN

1. Match the effect of addition of 1 M NaOH to 100 mL 1 M CH_3COOH (in Column I) with pH (in Column II) :

Column-I		Column-II	
(A)	25 mL of NaOH	(p)	pK_a
(B)	50 mL of NaOH	(q)	$pK_a + \log 3$
(C)	75 mL of NaOH	(r)	$pK_a - \log 3$
(D)	100 mL of NaOH	(s)	$\frac{1}{2} [pK_w + pK_a - \log 2]$

2. When we titrate sodium carbonate solution (in beaker) with hydrochloric acid.

Column-I		Column-II	
(A)	At the start of titration	(p)	Buffer solution of HCO_3^- and CO_3^{2-}
(B)	Before the first equivalent point	(q)	Buffer solution of H_2CO_3 and HCO_3^-
(C)	At the first equivalent point	(r)	Amphiprotic anion, $pH = 1/2(pK_{a_1} + pK_{a_2})$
(D)	Between the first and second equivalent points	(s)	Hydrolysis of CO_3^{2-}

ASSERTION & REASON

These questions contains, Statement I (assertion) and Statement II (reason).

(A) Statement-I is true, Statement-II is true ; Statement-II is correct explanation for Statement-I.

(B) Statement-I is true, Statement-II is true ; Statement-II is NOT a correct explanation for statement-I

(C) Statement-I is true, Statement-II is false

(D) Statement-I is false, Statement-II is true

1. **Statement-I** : On dilution of a concentrated solution of CH_3COOH , the concentration of $[\text{H}^+]$ decreases

Because

Statement-II : Because increase in the volume is more than the increase in degree of ionisation

2. **Statement-I** : pH of boiling water is less than the water at 4°C

Because

Statement-II : Because density of water is maximum at 4°C

3. **Statement-I** : Solubility of BaSO_4 in $0.1\text{ M Na}_2\text{SO}_4$ is 10^{-9} M hence its K_{sp} is 10^{-18}

Because

Statement-II : Because for BaSO_4 $K_{\text{sp}} = (s)^2$

4. **Statement-I** : Aqueous solution of $\text{CH}_3\text{COONH}_4$ is found to be neutral

Because

Statement-II : because this salt does not undergo hydrolysis

5. **Statement-I** : An aqueous solution of HCl is a much better conductor of electricity than an aqueous solution of CH_3COOH of the same concentration.

Because

Statement-II : The freezing point depression and the boiling point elevation of weak electrolytes are significantly less than for strong electrolytes of the same concentration.

6. **Statement-I** : CH_3NH_3^+ CH_3NH_2 is acid base conjugate pair.

Because

Statement-II : H_3O^+ , OH^- is acid base conjugate pair.

7. **Statement-I** : The equilibrium constant for the reaction,



Because

Statement-II : This shows that CN^- is stronger base than ONO^-

COMPREHENSION BASED QUESTIONS

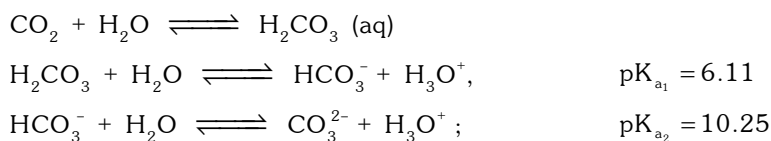
Comprehension # 1

The importance of pH maintenance in Blood

Maintenance of the pH in blood and in intracellular fluids is absolutely crucial to the processes that occur in living organisms. This primarily because the functioning of enzymes-catalysts for these processes - is sharply pH dependent. The normal pH value of blood plasma is 7.4. Severe illness or death can result from sustained variations of a few tenths of pH unit.

Among the factors that lead to a condition of acidosis, in which there is decreases in the pH of blood are heart failure, kidney failure diabetes mellitus, persistent diarrhoea or a long term high protein diet. A temporary condition acidosis may result from prolonged, intensive exercise. Alkalosis, which causes increase in pH of blood, may occur as a result of severe vomiting overbreathing or exposure to high altitudes.

Several factors are involved in the control of the pH of blood. A particularly important one is the ratio of dissolved HCO_3^- to H_2CO_3 . $\text{CO}_2(\text{g})$ is moderately soluble in water and in aqueous solution reacts only to a limited extent to produce H_2CO_3 .



In the H_2CO_3 , HCO_3^- buffer system we deal only with the first ionisation step (K_{a_1}) : H_2CO_3 is weak acid and HCO_3^- is the conjugate base (salt). CO_2 enters the blood from tissues as the by - product of metabolic reaction. In lungs, CO_2 (g) is exchanged for O_2 (g), which is transported throughout the body by the blood.

- The pH of blood stream is maintained by a proper balance of H_2CO_3 and NaHCO_3 concentration. What volume of 5 M NaHCO_3 solution should be mixed with a 10 mL sample of blood which is 2 M in H_2CO_3 in order to maintain its pH? :
(A) 40 mL (B) 38 mL (C) 50 mL (D) 78 mL
- Important diagnostic analysis in the blood is :
(A) $[\text{H}_2\text{PO}_4^-]/[\text{HPO}_4^{2-}]$ (B) $[\text{HCO}_3^-]/[\text{CO}_2]$
(C) $[\text{CO}_3^{2-}]/[\text{HCO}_3^-]$ (D) $[\text{PO}_4^{3-}]/[\text{HPO}_4^{2-}]$
- Following reaction occurs in the body :
$$\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3 \rightleftharpoons \text{H}^+ + \text{HCO}_3^-$$

If CO_2 escapes from the system :
(A) pH will decrease
(B) pH will increase
(C) $[\text{H}_2\text{CO}_3]$ remains unchanged
(D) forward reaction is promoted

Comprehension # 2

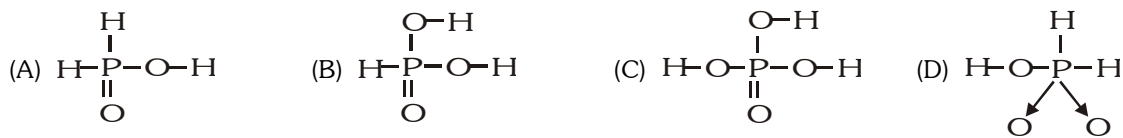
Phosphoric acid is of great importance in fertilizer production. Besides, phosphoric acid and its various salts have a number of application in metal treatment, food, detergent and toothpaste industries.

$$\text{pK}_{a_1} = 2.12, \text{pK}_{a_2} = 7.21, \text{pK}_{a_3} = 12.32$$

Small quantities of phosphoric acid are extensively used to impart the sour or tart taste to many soft drinks such as colas and root beers, in which a density of 1.00 g mL^{-1} contains 0.05% by weight of phosphoric acid.

Phosphoric acid is used as a fertiliser for agriculture and an aqueous soil digesting. $1.00 \times 10^{-3} \text{ M}$ phosphoric acid is found to have $\text{pH} = 7$. Zinc is an essential micronutrient for plant growth. Plants can absorb zinc in water soluble form only. In the given soil, zinc phosphate is only the source of zinc and phosphate ions, K_{sp} (zinc phosphate) = 9.1×10^{-33} .

- Phosphoric acid is a tribasic acid with three-step ionisation constants. Thus, its structure is :



- What is the pH of the cola assuming that the acidity of the cola arises only from phosphoric acid and second and third ionisation constants are of no importance ?
(A) 2.2 (B) 3.3 (C) 4.4 (D) 1.8

3. Molar concentration of phosphate ion in the soil with pH 7 is :
- (A) $1.2 \times 10^{-4} \text{ M}$ (B) $2.2 \times 10^{-4} \text{ M}$
 (C) $1 \times 10^{-3} \text{ M}$ (D) $1.1 \times 10^{-10} \text{ M}$
4. Concentration of $[\text{Zn}^{2+}]$ in the soil is :
- (A) $9.1 \times 10^{-5} \text{ M}$ (B) $5.7 \times 10^{-9} \text{ M}$
 (C) $4.0 \times 10^{-10} \text{ M}$ (D) $3.0 \times 10^{-6} \text{ M}$

MISCELLANEOUS TYPE QUESTION

ANSWER KEY

EXERCISE -3

● True / False

1. F	2. F	3. T	4. F
5. T	6. F	7. F	8. F
9. F	10. F	11. F	12. T
13. T	14. T	15. T	16. F
17. T	18. T	19. F	20. T
21. T	22. T		

● Fill in the Blanks

1. Increase, one	2. Greater	3. Br ⁻ ion	4. Greater
5. 7.14×10^{-3}	6. 3×10^{-6}	7. $M_2X = QY_2 > PZ_3$	8. 10^{-8}
9. 7	10. 5.74	11. HSO ₄ ⁻	12. Increases
13. Less	14. Less	15. Lewis acid	16. Buffer
17. Weaker	18. Ag ⁺	19. Cationic	20. Independent
21. $(K_{sp}/4)^{1/4}$	22. 5.556×10^{-10}	23. Acidic, Cu ⁺⁺	24. Colourless, Pink
25. pK _w (not 14 !!)	26. $k_w = k[H_2O]$	27. Greater	

● Match the Column

1. A - (r), B - (p), C - (q), D - (s)	2. A - (s), B - (p), C - (r), D - (q)
---------------------------------------	---------------------------------------

● Assertion - Reason Questions

1. A	2. B	3. D	4. C
5. B	6. C	7. A	

● Comprehension Based Questions

Comprehension #1 : 1. D	2. B	3. B	
Comprehension #2 : 1. C	2. A	3. B	4. B

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

1. Calculate
 - (i) K_a for H_2O ($K_w = 10^{-14}$)
 - (ii) K_b for $B(OH)_4^-$, $K_a(B(OH)_3) = 6 \times 10^{-10}$
 - (iii) K_a for HCN , $K_b(CN^-) = 2.5 \times 10^{-5}$
2. Calculate the ratio of degree of dissociation (α_2/α_1) when 1 M acetic acid solution is diluted to $\frac{1}{100}$ times. (Given $K_a = 1.8 \times 10^{-5}$)
3. Calculate the ratio of degree of dissociation of acetic acid and hydrocyanic acid (HCN) in 1 M their respective solution of acids (Given $K_a(CH_3COOH) = 1.8 \times 10^{-5}$; $K_a(HCN) = 6.2 \times 10^{-10}$)
4. Calculate :
 - (a) K_a for a monobasic acid whose 0.10 M solution has pH of 4.50
 - (b) K_b for a monoacidic base whose 0.1 M solution has a pH of 10.50
5. Calculate pH of following solution :
 - (a) 0.1 M HCl
 - (b) 0.1 M H_2SO_4 (50 mL) + 0.4 M HCl 50 (mL)
 - (c) 0.1 M CH_3COOH ($K_a = 1.8 \times 10^{-5}$)
 - (d) 0.1 M NH_4OH ($K_b = 1.8 \times 10^{-5}$)
 - (e) 10^{-8} M HCl
 - (f) 10^{-10} M NaOH
 - (g) 10^{-6} M CH_3COOH
 - (h) 10^{-8} M CH_3COOH
 - (i) 0.1 M HA + 0.1 M HB [$K_a(HA) = 2 \times 10^{-5}$; $K_a(HB) = 4 \times 10^{-5}$]
 - (j) Decimolar solution of Baryta ($Ba(OH)_2$), diluted 100 times.
 - (k) 10^{-3} mole of KOH dissolved in 100 L of water.
 - (l) 0.5 M HCl (25 mL) + 0.5 M NaOH (10 mL) + 40 mL H_2O
 - (m) equal volume of HCl solution (PH = 4) + 0.0019 N HCl solution
6. The value of K_w at the physiological temperature (37 °C) is 2.56×10^{-14} . What is the pH at the neutral point of water at this temperature, where there are equal number of H^+ and OH^- ?
7. Calculate the number of H^+ present in one mL of solution whose pH is 13
8. Calculate change in concentration of H^+ ion in one litre of water, when temperature changes from 298 K to 310 K. Given $K_w(298) = 10^{-14}$, $K_w(310) = 2.56 \times 10^{-14}$.
9.
 - (i) K_w for H_2O is 9.62×10^{-14} at 60 °C. What is pH of water at 60 °C.
 - (ii) What is the nature of solution at 60 °C whose.
 - (a) pH = 6.7
 - (b) pH = 6.35
10. The pH of aqueous solution of ammonia is 11.5. Find molarity of solution. $K_b(NH_4OH) = 1.8 \times 10^{-5}$.
11. The solution of weak monoprotic acid which is 0.01 M has pH = 3. Calculate K_a of weak acid.
12. Boric acid is a weak monoprotic acid. It ionizes in water as
$$B(OH)_3 + H_2O \rightleftharpoons B(OH)_4^- + H^+ : K_a = 5.9 \times 10^{-10}$$
Calculate pH of 0.3 M boric acid.
13. Calculate $[H^+]$ and $[CHCl_2COO^-]$ in a solution that is 0.01 M in HCl and 0.01 M in $CHCl_2COOH$. Take ($K_a = 2.55 \times 10^{-2}$).
14. Calculate $[H^+]$, $[CH_3COO^-]$ and $[C_7H_5O_2^-]$ in a solution that is 0.02 M in acetic acid and 0.01 M in benzoic acid. $K_a(\text{acetic}) = 1.8 \times 10^{-5}$, $K_a(\text{benzoic}) = 6.4 \times 10^{-5}$.
15. At 25 °C, the dissociation constant of HCN and HF are 4×10^{-10} and 6.7×10^{-4} . Calculate the pH of a mixture of 0.1 M HF and 0.1 M HCN.

POLYPROTIC ACIDS & BASES

16. Determine the $[S^{2-}]$ in a saturated (0.1 M) H_2S solution to which enough HCl has been added to produce a $[H^+]$ of 2×10^{-4} . $K_1 = 10^{-7}$, $K_2 = 10^{-14}$
17. Calculate $[H^+]$, $[H_2PO_4^-]$, $[HPO_4^{2-}]$ and $[PO_4^{3-}]$ in a 0.01 M solution of H_3PO_4 .
Take $K_1 = 7.225 \times 10^{-3}$, $K_2 = 6.8 \times 10^{-8}$, $K_3 = 4.5 \times 10^{-13}$.
18. Calculate the pH of a 0.1 M solution of $H_2NCH_2CH_2NH_2$; ethylenediamine (en). Determine the en H_2^{2+} concentration in the solution. K_{b1} and K_{b2} values of ethylenediamine are 8.5×10^{-5} and 7.1×10^{-8} respectively.
19. What are the concentration of H^+ , HSO_4^- , SO_4^{2-} & H_2SO_4 in a 0.20 M solution of sulphuric acid ?
Given : $H_2SO_4 \longrightarrow H^+ + HSO_4^-$; strong
 $HSO_4^- \rightleftharpoons H^+ + SO_4^{2-}$; $K_2 = 1.3 \times 10^{-2}$ M

BUFFER SOLUTION

20. Determine $[OH^-]$ of a 0.050 M solution of ammonia to which has been added sufficient NH_4Cl to make the total $[NH_4^+]$ equal to 0.100. $[K_b(NH_3) = 1.8 \times 10^{-5}]$
21. Calculate the pH of a solution prepared by mixing 50.0 mL of 0.200 M $HC_2H_3O_2$ and 50.0 mL of 0.100 M NaOH. $[K_a(CH_3COOH) = 1.8 \times 10^{-5}]$
22. A buffer of pH 9.26 is made by dissolving x moles of ammonium sulphate and 0.1 mole of ammonia into 100 mL solution. If pK_b of ammonia is 4.74, calculate value of x.
23. 50 mL of 0.1 M NaOH is added to 75 mL of 0.1 M NH_4Cl to make a basic buffer. If pK_a of NH_4^+ is 9.26, calculate pH.
24. (a) Determine the pH of a 0.2 M solution of pyridine C_5H_5N . $K_b = 1.5 \times 10^{-9}$
(b) Predict the effect of addition of pyridinium ion $C_5H_5NH^+$ on the position of the equilibrium. Will the pH be raised or lowered ?
(c) Calculate the pH of 1.0 L of 0.10 M pyridine solution to which 0.3 mol of pyridinium chloride $C_5H_5NH^+Cl^-$ has been added, assuming no change in volume.
25. A buffer solution was prepared by dissolving 0.02 mol propionic acid & 0.015 mol sodium propionate in enough water to make 1.00 L of solution. (K_a for propionic acid is 1.34×10^{-5})
(a) What is the pH of the buffer ?
(b) What would be the pH if 1.0×10^{-5} mol HCl were added to 10 mL of the buffer ?
(c) What would be the pH if 1.0×10^{-5} mol NaOH were added to 10 mL of the buffer.
(d) Also report the percent change in pH of original buffer in case (b) and (c)
26. A solution was made up of 0.01 M in chloroacetic acid $ClCH_2COOH$ and also 0.002 M in sodium chloracetate $ClCH_2COONa$. What is $[H^+]$ in the solution? $K_a = 1.5 \times 10^{-3}$.

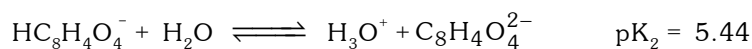
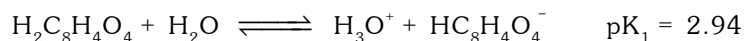
INDICATORS

27. A certain solution has a hydrogen ion concentration 4×10^{-3} M. For the indicator thymol blue, pH is 2.0 when half the indicator is in unionised form. Find the % of indicator in unionised form in the solution with $[H^+] = 4 \times 10^{-3}$ M.
28. Bromophenol blue is an indicator with a K_a value of 6×10^{-5} . What % of this indicator is in its basic form at a pH of 5 ?
29. An acid base indicator has a K_a of 3×10^{-5} . The acid form of the indicator is red & the basic form is blue. By how much must the pH change in order to change the indicator form 75% red to 75% blue ?

HYDROLYSIS

30. What is the OH^- concentration of a 0.08 M solution of CH_3COONa . $[K_a(CH_3COOH) = 1.8 \times 10^{-5}]$
31. Calculate the pH of a 2.0 M solution of NH_4Cl . $[K_b(NH_3) = 1.8 \times 10^{-5}]$

32. 0.25 M solution of pyridinium chloride $C_5H_6N^+Cl^-$ was found to have a pH of 2.699. What is K_b for pyridine, C_5H_5N ?
33. Calculate the extent of hydrolysis & the pH of 0.02 M CH_3COONH_4
 $[K_b(NH_3) = 1.8 \times 10^{-5}, K_a(CH_3COOH) = 1.8 \times 10^{-5}]$
34. Calculate the percent hydrolysis in a 0.06 M solution of KCN. $[K_a(HCN) = 6 \times 10^{-10}]$
35. Calculate the extent of hydrolysis of 0.005 M K_2CrO_4 . $[K_2 = 3.1 \times 10^{-7}$ for H_2CrO_4]
 (It is essentially strong for first ionization).
36. A 0.010 M solution of $PuO_2(NO_3)_2$ was found to have a pH of 4.0. What is the hydrolysis constant k_h for PuO_2^{2+} , and what is K_b for PUO_2OH^+ ?
37. Calculate the pH of 1.0×10^{-3} M sodium phenolate, $NaOC_6H_5$ K_a for HOC_6H_5 is 1.05×10^{-10} .
38. What is the pH of 0.1 M $NaHCO_3$? $K_1 = 4.5 \times 10^{-7}$, $K_2 = 4.5 \times 10^{-11}$ for carbonic acids.
39. Calculate pH of 0.05 M potassium hydrogen phthalate, $KHC_8H_4O_4$.



40. Calculate OH^- concentration at the equivalent point when a solution of 0.1 M acetic acid is titrated with a solution of 0.1 M NaOH. K_a for the acid = 1.9×10^{-5} .
41. The acid ionization hydrolysis constant of Zn^{2+} is 1.0×10^{-9}
 (a) Calculate the pH of a 0.001 M solution of $ZnCl_2$
 (b) What is the basic dissociation constant of $Zn(OH)^+$?

ACID BASE REACTIONS & TITRATIONS

42. Calculate the hydronium ion concentration and pH at the equivalence point in the reaction of 22.0 mL of 0.10 M acetic acid, CH_3COOH , with 22.0 mL of 0.10 M NaOH.
43. Calculate the hydronium ion concentration and the pH at the equivalence point in a titration of 50.0 mL of 0.40 M NH_3 with 0.40 M HCl.
44. In the titration of a solution of a weak acid HX with NaOH, the pH is 5.8 after 10.0 mL of NaOH solution has been added and 6.402 after 20.0 mL of NaOH has been added. What is the ionization constant of HX ?
45. The equivalent point in a titration of 40.0 mL of a solution of a weak monoprotic acid occurs when 35.0 mL of a 0.10 M NaOH solution has been added. The pH of the solution is 5.75 after the addition of 20.0 mL of NaOH solution. What is the dissociation constant of the acid ?
46. A weak base (50.0 mL) was titrated with 0.1 M HCl. The pH of the solution after the addition of acid 10.0 mL and 25.0 mL were found to be 9.84 and 9.24, respectively. Calculate K_b of the base and pH at the equivalence point.
47. A weak acid (50.0 mL) was titrated with 0.1 M NaOH. The pH values when 10.0 mL and 25.0 mL of base have been added are found to be 4.16 and 4.76, respectively. Calculate K_a of the acid and pH at the equivalence point.
48. CH_3COOH (50 mL, 0.1 M) is titrated against 0.1 M NaOH solution. Calculate the pH at the addition of 0 mL, 10 mL, 20 mL, 25 mL, 40 mL, 50 mL of NaOH. K_a of CH_3COOH is 2×10^{-5} .

SOLUBILITY & SOLUBILITY PRODUCT'S

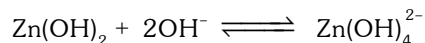
49. The values of K_{sp} for the slightly soluble salts MX and OX_2 are each equal to 4.0×10^{-18} . Which salt is more soluble ? Explain your answer fully.
50. The solubility of $PbSO_4$ in water is 0.038 g/L. Calculate the solubility product constant of $PbSO_4$.
51. Calculate the solubility of $Mg(OH)_2$ in water. $K_{sp} = 1.2 \times 10^{-11}$.

52. How many mol CuI ($K_{\text{sp}} = 5 \times 10^{-12}$) will dissolve in 1.0 L of 0.10 M NaI solution ?
53. What is the solubility (in mol/L) of Fe(OH)_3 in a solution of $\text{pH} = 8.0$? [K_{sp} for $\text{Fe(OH)}_3 = 1.0 \times 10^{-36}$]
54. Calculate the solubility of A_2X_3 in pure water, assuming that neither kind of ion reacts with water. For A_2X_3 , [$K_{\text{sp}} = 1.1 \times 10^{-23}$]
55. What mass of Pb^{2+} ion is left in solution when 50.0 mL of 0.20 M $\text{Pb(NO}_3)_2$ is added to 50.0 mL of 1.5 M NaCl ? [Given K_{sp} for $\text{PbCl}_2 = 1.7 \times 10^{-4}$]
56. A solution has a Mg^{2+} concentration of 0.0010 mol/L. will Mg(OH)_2 precipitate if the OH^- concentration of the solution is [$K_{\text{sp}} = 1.2 \times 10^{-11}$]
 (a) 10^{-5} mol/L (b) 10^{-3} mol/L ?
57. Calculate solubility of PbI_2 ($K_{\text{sp}} = 1.4 \times 10^{-8}$) in water at 25 , which is 90% dissociated.
58. Calculate the Simultaneous solubility of AgSCN and AgBr . $K_{\text{sp}}(\text{AgSCN}) = 1.1 \times 10^{-12}$, $K_{\text{sp}}(\text{AgBr}) = 5 \times 10^{-13}$.
59. Calculate F^- in a solution saturated with respect of both MgF_2 and SrF_2 . $K_{\text{sp}}(\text{MgF}_2) = 9.5 \times 10^{-9}$, $K_{\text{sp}}(\text{SrF}_2) = 4 \times 10^{-9}$.

COMPLEXATION EQUILIBRIA

60. Assuming no change in volume, calculate the minimum mass of NaCl necessary to dissolve 0.010 mol AgCl in 100 L solution. [$K_{\text{f}}(\text{AgCl}_2^-) = 3 \times 10^5$, $K_{\text{sp}}(\text{AgCl}) = 1 \times 10^{-10}$]
61. How much AgBr could dissolve in 1.0 L of 0.40 M NH_3 ? Assume that $\text{Ag(NH}_3)_2^+$ is the only complex formed. [$K_{\text{f}}(\text{Ag(NH}_3)_2^+) = 1 \times 10^8$, $K_{\text{sp}}(\text{AgBr}) = 5 \times 10^{-13}$]
62. A solution of 0.1 M Cl^- , 0.1 M Br^- and 0.1 M I^- solid AgNO_3 is gradually added to this solution. Assuming that the addition of AgNO_3 does not change the volume. Answer the following :
 (a) What conc. of Ag^+ ions will be required to start precipitation of each of the three ions.
 (b) Which ion will precipitate first
 (c) What will be the conc. of this ion when the second ion start precipitating.
 (d) What will be the conc. of both ions when the third ion start precipitating.
 Given : $K_{\text{sp}}(\text{AgCl}) = 1.7 \times 10^{-10}$, $K_{\text{sp}}(\text{AgBr}) = 5 \times 10^{-13}$
 $K_{\text{sp}}(\text{AgI}) = 8.5 \times 10^{-17}$.
63. The solubility of CaCO_3 is 7 mg/litre. Calculate the solubility product of BaCO_3 from this information and from the fact that when Na_2CO_3 is added slowly to a solution containing equimolar concentration of Ca^{+2} and Ba^{+2} , no precipitate is formed until 90% of Ba^{+2} has been precipitated as BaCO_3 .
64. A solution containing 0.10 M Zn^{2+} and 0.10 M Fe^{2+} is saturated with H_2S . What must be the H^+ concentration to separate these ions by selective precipitating ZnS . What is the smallest Zn^{2+} concentration that can be achieved without precipitating any of Fe^{2+} as FeS .
 $K_{\text{sp}}\text{ZnS} = 1.2 \times 10^{-23}$, $K_{\text{sp}}\text{FeS} = 3.7 \times 10^{-19}$.
 [Take : $[\text{H}_2\text{S}] = 0.1 \text{ M}$ and $K_1 = 1.1 \times 10^{-7}$, $K_2 = 1 \times 10^{-14}$]
65. (a) Will Mg(OH)_2 precipitate from a solution that is 0.01 M in MgCl_2 and also 0.10 M in NH_3 .
 (b) What NH_4^+ must maintained to prevent the precipitation of Mg(OH)_2 from a solution that is 0.010 MgCl_2 in 0.1 M NH_3 $K_{\text{sp}} \text{Mg(OH)}_2 = 2.5 \times 10^{-11}$, $K_{\text{b}} \text{NH}_3 = 1.8 \times 10^{-5}$
66. Calculate the solubility of AgCN in a buffer solution of $\text{pH} 3$. Neglect any complexation. Take $K_{\text{sp}}(\text{AgCN}) = 3.2 \times 10^{-16}$, $K_{\text{a}}(\text{HCN}) = 6.4 \times 10^{-10}$.
67. Silver ion forms Ag(CN)_2^- in the presence of excess CN^- . How much KCN should be added to 1 litre of a 0.0005 M Ag^+ solution in order to reduce $[\text{Ag}^+]$ to 1×10^{-19} $K_{\text{b}} \text{Ag(CN)}_2^- = 1 \times 10^{-21}$.

68. How many moles of solid NaOH must be added to 1.0 litre of H₂O in order to dissolve 0.10 mole of Zn(OH)₂ according to the equation



$$K_{\text{sp}} \text{Zn(OH)}_2 = 4.5 \times 10^{-17}, K_{\text{inst}} \text{Zn(OH)}_4^{2-} = 3.6 \times 10^{-16}.$$

CONCEPTUAL SUBJECTIVE EXERCISE			ANSWER KEY		EXERCISE-4(A)	
1. (i) 1.8 × 10 ⁻¹⁶ , (ii) 1.66 × 10 ⁻⁵ , (iii) 4 × 10 ⁻¹⁰	2. 10	3. 170.4				
4. (a) K _a = 10 ⁻⁸ , (b) K _b = 10 ⁻⁶						
5. (a) +1, (b) 0.522, (c) 2.87, (d) 11.13 (e) 6.97, (f) 6.996, (g) 6.01, (h) 6.97, (i) 2.61, (j) 11.30 (k) 9 (l) 1, (m) 3						
6. 6.795	7. 6.022 × 10 ⁷	8. 0.6 × 10 ⁻⁷				
9. (i) 6.51 ; (ii) (a) Basic, (b) Acidic						
10. 0.556 M	11. 1.11 × 10 ⁻⁴	12. 4.87				
13. [H ⁺] = 1.612 × 10 ⁻² M, [CHCl ₂ COO ⁻] = 6.126 × 10 ⁻³ M						
14. [H ⁺] = 10 ⁻³ M, [CH ₃ COO ⁻] = 3.6 × 10 ⁻⁴ M, [C ₇ H ₅ O ₂ ⁻] = 6.4 × 10 ⁻⁴ M						
15. 2.08	16. [S ²⁻] = 2.5 × 10 ⁻¹⁵					
17. [H ⁺] = [H ₂ PO ₄ ⁻] = 5.623 × 10 ⁻³ , [HPO ₄ ⁻²] = 6.8 × 10 ⁻⁸ , [PO ₄ ³⁻] = 5.441 × 10 ⁻¹⁸						
18. pH = 11.46, [enH ₂ ²⁺] = 7.1 × 10 ⁻⁸	19. 0.2216 M, 0.1884 M, 0.0116 M, 0					
20. [OH ⁻] = 9.0 × 10 ⁻⁶	21. 4.74	22. 0.05 mol				
23. 9.56	24. (a) pH = 9.239 (b) lowered (c) pH = 4.699					
25. (a) 4.7525 (b) 4.697 (c) 4.798 (d) 1.134% on acid addition 0.96 % on base addition.						
26. [H ⁺] = 2.5 × 10 ⁻³	27. [HI _n] = 28.57 %	28. 85.71% 29. ΔpH=0.954				
30. [OH ⁻] = 6.664 × 10 ⁻⁶		31. pH = 4.477 32. K _b = 6.25 × 10 ⁻¹⁰				
33. 0.56%, pH=7	34. 1.667 %	35. 0.26% 36. 10 ⁻⁶ ; 10 ⁻⁸				
37. pH = 10.43	38. 8.34	39. 4.19 40. 5.12 × 10 ⁻⁶ M				
41. (a) 6 (b) 1 × 10 ⁻⁵		42. 8.71 43. 4.98				
44. 7.94 × 10 ⁻⁷	45. K _a = 2.37 × 10 ⁻⁶	46. 1.73 × 10 ⁻⁵ , 5.27				
47. K _a = 1.73 × 10 ⁻⁵ , pH = 8.73						
48. (i) 2.85, (ii) 4.0969, (iii) 4.5229, (iv) 4.699 (v) 5.301 (vi) 8.699						
49. OX ₂ is more soluble	50. 1.6 × 10 ⁻⁸	51. 1.4 × 10 ⁻⁴ 52. [Cu ⁺] = 5 × 10 ⁻¹¹ M				
53. 10 ⁻¹⁸ M	54. 1.0 × 10 ⁻⁵ mol/lit	55. 12 mg				
56. (a) no precipitation will occur, (b) a precipitate will form	57. 1.6 × 10 ⁻³					
58. 4 × 10 ⁻⁷ mol/L AgBr, 9 × 10 ⁻⁷ mol/L AgSCN	59. [F ⁻] = 3 × 10 ⁻³ M					
60. 19.5 kg	61. 2.8 × 10 ⁻³ M					
62. (a) 8.5 × 10 ⁻¹⁶ , 5 × 10 ⁻¹² , 1.7 × 10 ⁻⁹ (b) I ⁻ (c) 1.7 × 10 ⁻⁵ (d) Br ⁻ = 2.9 × 10 ⁻⁴ , I ⁻ = 5 × 10 ⁻⁸						
63. 4.9 × 10 ⁻¹⁰	64. 5.5 × 10 ⁻³ ≤ [H ⁺] < 0.96, 3.24 × 10 ⁻⁶					
65. (a) yes (b) 0.036 M	66. s = 2.236 × 10 ⁻⁵ M	67. 3.236 × 10 ⁻³ mol 68. 1.09 mol				

EXERCISE-04 [B]**BRAIN STORMING SUBJECTIVE EXERCISE**

1. A solution of volume V contains n_1 moles of QCl and n_2 moles of RCl where QOH and ROH are two weak bases of dissociation constants k_1 and k_2 respectively. Show that the pH of the solution is given by

$$pH = \frac{1}{2} \log \left[\left(\frac{k_1 k_2}{k_w} \right) \frac{V}{(n_1 k_2 + k_1 n_2)} \right]$$

State assumptions, if any.

2. EDTA, often abbreviated as H_4Y , forms very stable complexes with almost all metal ions. Calculate the fraction of EDTA in the fully protonated form, H_4Y in a solution obtained by dissolving 0.1 mol Na_4Y in 1 lit. The acid dissociation constants of H_4Y :

$$k_1 = 1.02 \times 10^{-2}, k_2 = 2.13 \times 10^{-3}, k_3 = 6.92 \times 10^{-7}, k_4 = 5.50 \times 10^{-11} :$$

3. Calculate the solubility of solid zinc hydroxide at a pH of 5, 9 and 13. Given :



4. The standard free energy of formation for $AgCl$ at 298 K is $-109.7 \text{ kJ mole}^{-1}$. $\Delta G (Ag^+) = 77.2 \text{ kJ/mole}$, $\Delta G (Cl^-) = -131.2 \text{ kJ/mole}$. Find the solubility of $AgCl$ in 0.05 M KCl . Neglect any complication due to complexation :-

5. Salt mixture containing $Cu_3(AsO_4)_2$ ($K_{sp} = 8 \times 10^{-36}$) and $Pb_3(AsO_4)_2$ ($K_{sp} = 4.096 \times 10^{-36}$) is shaken with water. Find the concentration of metal cations in the solution at equilibrium. Neglect any hydrolysis of the dissolved ions.

6. (a) At what minimum pH will 1.0×10^{-3} mol of $Al(OH)_3$ go into 1 L solution as $[Al(OH)_4]^-$.

(b) At what minimum pH will 1.0×10^{-3} mol of $Al(OH)_3$ go into 1 L solution as Al^{3+} ?

Given : $K_{sp} [Al(OH)_3] = 5.0 \times 10^{-33}$ and for $[Al(OH)_4]^- \rightleftharpoons Al^{3+} + 4OH^-$. $K = 1.3 \times 10^{-34}$.

7. A solution contains HCl , $Cl_2HCCOOH$ & CH_3COOH at concentrations 0.09 M in HCl , 0.09 M in $Cl_2HCCOOH$ & 0.1 M in CH_3COOH , pH for the solution is 1. Ionization constant of $CH_3COOH = 10^{-5}$. What is the magnitude of K for dichloroacetic acid ?

8. A solution of ammonia bought for cleaning the windows was found to be 10% ammonia by mass, having a density of 0.935 g.mL^{-1} . What is the pH of the solution. Take K_b for protonation of ammonia = 5.5×10^{-6} .

9. The k_w of water at two different temperature is :-

T 25 C 50 C

k_w 1.08×10^{-14} 5.474×10^{-14}

Assuming that ΔH of any reaction is independent of temperature, calculate the enthalpy of neutralization of a strong acid and strong base.

10. What is the pH of a 1.0 M solution of acetic acid ? To what volume must 1 litre of the solution be diluted so that the pH of the resulting solution will be twice the original value. Given $K_a = 1.8 \times 10^{-5}$:-

11. A handbook states that the solubility of methylamine CH_3NH_2 (g) in water at 1 atm pressure at 25 C is 959 volumes of CH_3NH_2 (g) per volume of water ($pK_b = 3.39$) :-

(a) Estimate the max. pH that can be attained by dissolving methylamine in water.

(b) What molarity $NaOH$ (aq.) would be required to yield the same pH?

12. Mixture of solutions. Calculate the pH of the following solutions.
For H_3PO_4 ; $K_{a_1} = 7.5 \times 10^{-3}$, $K_{a_2} = 6.2 \times 10^{-8}$, $K_{a_3} = 10^{-12}$
- 50 mL of 0.12 M H_3PO_4 + 20 mL of 0.15 M NaOH ;
 - 50 mL of 0.12 M H_3PO_4 + 40 mL of 0.15 M NaOH ;
 - 40 mL of 0.12 M H_3PO_4 + 40 mL of 0.18 M NaOH ;
 - 40 mL of 0.10 M H_3PO_4 + 40 mL of 0.25 M NaOH.
13. Mixtured of solutions. Calculate the pH of the following solution.
Use data of above question & For H_2CO_3 ; $K_1 = 4.2 \times 10^{-7}$, $K_2 = 4.8 \times 10^{-11}$
- 40 mL of 0.050 M Na_2CO_3 + 50 mL of 0.040 M HCl ;
 - 40 mL of 0.020 M Na_3PO_4 + 40 mL of 0.040 M HCl ;
 - 50 mL of 0.10 M Na_3PO_4 + 50 mL of 0.10 M NaH_2PO_4 ;
 - 40 mL of 0.10 M H_3PO_4 + 40 mL of 0.10 M Na_3PO_4 .
14. When a 40 mL of a 0.1 M weak base is titrated with 0.16 M HCl, the pH of the solution at the end point is 5.23. What will be the pH if 15 mL of 0.12 M NaOH is added to the resulting solution :-
15. A buffer solution was prepared by dissolving 0.05 mol formic acid & 0.06 mol sodium formate in enough water to make 1.0 L of solution. K_a for formic acid is 1.80×10^{-4} .
- Calculate the pH of the solution.
 - If this solution were diluted to 10 times its volume, what would be the pH ?
 - If the solution in (b) were diluted to 10 times its volume, what would be the pH?
16. How many moles of sodium hydroxide can be added to 1.00 L of a solution 0.1 M in NH_3 & 0.1 M in NH_4Cl without changing the pOH by more than 1.00 unit ? Assume no change in volume. $K_b(\text{NH}_3) = 1.8 \times 10^{-5}$.
17. Calculate the OH^- concentration and the H_3PO_4 concentration of a solution prepared by dissolving 0.1 mol of Na_3PO_4 in sufficient water to make 1 L of solution $K_1 = 7.1 \times 10^{-3}$, $K_2 = 6.3 \times 10^{-8}$, $K_3 = 4.5 \times 10^{-13}$.
18. If 0.00050 mol NaHCO_3 is added to 1 litre of a buffered solution at pH 8.00, how much material will exist in each of the three forms H_2CO_3 , HCO_3^- and CO_3^{2-} ? For H_2CO_3 , $K_1 = 5 \times 10^{-7}$, $K_2 = 5 \times 10^{-13}$.
19. Equilibrium constant for the acid ionization of Fe^{3+} to $\text{Fe}(\text{OH})^{+2}$ and H^+ is 6.5×10^{-3} . What is the max. pH, which could be used so that at least 95% of the total Fe^{3+} in a dilute solution. exists as Fe^{3+} .
20. How much Na_2HPO_4 must be added to one litre of 0.005 M solution of NaH_2PO_4 in order to make a 1 L of the solution of pH = 6.7 ? $K_1 = 7.1 \times 10^{-3}$, $K_2 = 6.3 \times 10^{-8}$, $K_3 = 4.5 \times 10^{-13}$ for H_3PO_4 .
21. The indicator phenol red is half in the ionic form when pH is 7.2. If the ratio of the undissociated form to the ionic form is 1 : 5, find the pH of the solution. With the same pH for solution, if indicator is altered such that the ratio of undissociated form to dissociated form becomes 1 : 4, find the pH when 50% of the new indicator is in ionic form :-
22. A buffer solution, 0.080 M in Na_2PO_4 and 0.020 M in Na_3PO_4 , is prepared. The electrolytic oxidation of 1.0 mmol of the organic compound RNHOH is carried out in 100 mL of the buffer. The reaction is
- $$\text{RNHOH} + \text{H}_2\text{O} \longrightarrow \text{RNO}_2 + 4\text{H}^+ + 4\text{e}^-$$
- Calculate the approximate pH of the solution after the oxidation is complete ?
23. A solution of weak acid HA was titrated with base NaOH. The equivalence point was reached when 36.12 mL of 0.1 M NaOH has been added. Now 18.06 mL of 0.1 M HCl were added to titrated solution, the pH was found to be 4.92. What will be the pH of the solution obtained by mixing 10 mL of 0.2 M NaOH and 10 mL of 0.2 M HA :-
24. A weak base BOH was titrated against a strong acid. The pH at 1/4th equivalence point was 9.24. Enough strong base was now added (6m. eq.) to completely convert the salt. The total volume was 50 mL. Find the pH at this point :-
25. An organic monoprotic acid [0.1 M] is titrated against 0.1 M NaOH. By how much does the pH change between one fourth and three fourth stages of neutralization ? If at one third stage of neutralization, the pH is 4.45 what is the dissociation constant of the acid ? Between what stages of neutralization may the pH change by 2 units ?

26. The salt Zn(OH)_2 is involved in the following two equilibria,
 $\text{Zn(OH)}_2 (\text{s}) \rightleftharpoons \text{Zn}^{2+} (\text{aq}) + 2\text{OH}^- (\text{aq})$; $K_{\text{sp}} = 1.2 \times 10^{-17}$
 $\text{Zn(OH)}_2 (\text{s}) + 2\text{OH}^- (\text{aq}) \rightleftharpoons [\text{Zn(OH)}_4]^{2-} (\text{aq})$; $K_c = 0.13$
 Calculate the pH of solution at which solubility is minimum.
27. What is the solubility of AgCl in 0.20 M NH_3 ?
 Given : $K_{\text{sp}}(\text{AgCl}) = 1.7 \times 10^{-10} \text{ M}^2$, $K_1 = [\text{Ag(NH}_3\text{)}^+] / [\text{Ag}^+][\text{NH}_3] = 2.33 \times 10^3 \text{ M}^{-1}$ and
 $K_2 = [\text{Ag(NH}_3\text{)}_2^+] / [\text{Ag(NH}_3\text{)}^+][\text{NH}_3] = 7.14 \times 10^3 \text{ M}^{-1}$.
28. Predict whether or not AgCl will be precipitated from a solution which is 0.02 M in NaCl and 0.05 M in KAg(CN)_2 . Given $K_{\text{inst}}(\text{Ag(CN)}_2^-) = 4.0 \times 10^{-19} \text{ M}^2$ and $K_{\text{sp}}(\text{AgCl}) = 2.8 \times 10^{-10} \text{ M}^2$.
29. Equal volumes of 0.02 M AgNO_3 and 0.02 M HCN were mixed. Calculate $[\text{Ag}^+]$ at equilibrium. Take $K_a(\text{HCN}) = 9 \times 10^{-10}$, $K_{\text{sp}}(\text{AgCN}) = 4 \times 10^{-16}$:-
30. Show that solubility of a sparingly soluble salt $\text{M}^{2+}\text{A}^{2-}$ in which A^{2-} ions undergoes hydrolysis is given by :
- $$S = \sqrt{K_{\text{sp}} \left(1 + \frac{[\text{H}^+]}{K_2} + \frac{[\text{H}^+]^2}{K_1 K_2} \right)}$$
- where K_1 and K_2 are the dissociation constant of acid H_2A . K_{sp} is solubility product of MA .

BRAIN STORMING SUBJECTIVE EXERCISE		ANSWER KEY		EXERCISE-4(B)	
2.	3.82×10^{-26}	3.	10 M , $1.12 \times 10^{-6} \text{ M}$, $2 \times 10^{-4} \text{ M}$		
4.	$K_{\text{sp}} = 1.723 \times 10^{-10}$, $s = 3.446 \times 10^{-9} \text{ M}$				
5.	$[\text{Cu}^{+2}] = 8.825 \times 10^{-8}$, $[\text{Pb}^{+2}] = 7.119 \times 10^{-8}$				
6.	(a) $\text{pH} = 9.415$ (b) $\text{pH} = 4.23$	7.	$k_{a_2} = 1.25 \times 10^{-2}$		
8.	$\text{pH} = 11.74$				
9.	$\Delta H_{\text{neut}} = -51952.6 \text{ J} = -51.95 \text{ kJ/mole}$	10.	$V = 2.77 \times 10^4 \text{ L}$		
11.	(a) 13.097 (b) 0.1252				
12.	(a) 2.12 (b) 4.66 (c) 7.2 (d) 12				
13.	(a) 8.347 (b) 4.66 (c) 9.6 (d) 7.2				
14.	9.1628				
15.	(a) $\text{pH} = 3.823$ (b) $\text{pH} = 3.846$ (c) $\text{pH} = 3.9899$				
16.	0.0818 moles				
17.	$[\text{OH}^-] = 3.73 \times 10^{-2} \text{ M}$, $[\text{H}_3\text{PO}_4] = 5.93 \times 10^{-18} \text{ M}$				
18.	$[\text{H}_2\text{CO}_3] = 9.85 \times 10^{-6} \text{ M}$, $[\text{HCO}_3^-] = 4.9 \times 10^{-4} \text{ M}$, $[\text{CO}_3^{2-}] = 2.45 \times 10^{-8} \text{ M}$				
19.	$\text{pH} = 0.908$	20.	1.6 m mole	21.	$\text{pH} = 7.9$, 7.3
22.	7.81	23.	$\text{pH} = 9$	24.	$\text{pH} = 11.22$
25.	0.9542 , $\text{p}K_a = 4.751$, $\frac{1}{11}^{\text{th}}$ & $\frac{10}{11}^{\text{th}}$ stages of neutralization	26.	$\text{pH} = 9.99$, $s = 10^{-9} \text{ M}$		
27.	$9.6 \times 10^{-3} \text{ M}$	28.	precipitate will occur	29.	$[\text{Ag}^+] = 6.66 \times 10^{-5} \text{ M}$

EXERCISE - 05 [A]**JEE-[MAIN] : PREVIOUS YEAR QUESTIONS**

1. How many litres of water must be added to 1 litre of an aqueous solution of HCl with a pH of 1 to create an aqueous solution with pH of 2 ? [AIEEE-2013]
(A) 0.1 L (B) 0.9 L (C) 2.0 L (D) 9.0 L
2. The pH of a 0.1 molar solution of the acid HQ is 3. The value of the ionization constant, K_a of this acid is :- [AIEEE-2012]
(A) 1×10^{-7} (B) 3×10^{-7} (C) 1×10^{-3} (D) 1×10^{-5}
3. The K_{sp} for $\text{Cr}(\text{OH})_3$ is 1.6×10^{-30} . The molar solubility of this compound in water is :- [AIEEE-2011]
(A) $\sqrt[3]{1.6 \times 10^{-30}}$ (B) $\sqrt[4]{1.6 \times 10^{-30}}$ (C) $\sqrt[4]{1.6 \times 10^{-30} / 27}$ (D) $1.6 \times 10^{-30} / 27$
4. An acid HA ionises as
 $\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$
The pH of 1.0 M solution is 5. Its dissociation constant would be :- [AIEEE-2011]
(A) 1×10^{-10} (B) 5 (C) 5×10^{-8} (D) 1×10^{-5}
5. Solubility product of silver bromide is 5.0×10^{-13} . The quantity of potassium bromide (molar mass taken as 120 g mol^{-1}) to be added to 1 litre of 0.05 M solution of silver nitrate to start the precipitation of AgBr is :- [AIEEE-2010]
(A) $5.0 \times 10^{-8} \text{ g}$ (B) $1.2 \times 10^{-10} \text{ g}$ (C) $1.2 \times 10^{-9} \text{ g}$ (D) $6.2 \times 10^{-5} \text{ g}$
6. In aqueous solution the ionization constants for carbonic acid are
 $K_1 = 4.2 \times 10^{-7}$ and $K_2 = 4.8 \times 10^{-11}$
Select the correct statement for a saturated 0.034 M solution of the carbonic acid :- [AIEEE-2010]
(A) The concentration of H^+ is double that of CO_3^{2-}
(B) The concentration of CO_3^{2-} is 0.034 M
(C) The concentration of CO_3^{2-} is greater than that of HCO_3^-
(D) The concentrations of H^+ and HCO_3^- are approximately equal
7. At 25 °C, the solubility product of $\text{Mg}(\text{OH})_2$ is 1.0×10^{-11} . At which pH, will Mg^{2+} ions start precipitating in the form of $\text{Mg}(\text{OH})_2$ from a solution of 0.001 M Mg^{2+} ions ? [AIEEE-2010]
(A) 8 (B) 9 (C) 10 (D) 11
8. Solid $\text{Ba}(\text{NO}_3)_2$ is gradually dissolved in a $1.0 \times 10^{-4} \text{ M Na}_2\text{CO}_3$ solution. At what concentration of Ba^{2+} will a precipitate begin to form? [AIEEE-2009]
(K_{sp} for $\text{BaCO}_3 = 5.1 \times 10^{-9}$)
(A) $8.1 \times 10^{-8} \text{ M}$ (B) $8.1 \times 10^{-7} \text{ M}$ (C) $4.1 \times 10^{-5} \text{ M}$ (D) $5.1 \times 10^{-5} \text{ M}$
9. The pK_a of a weak acid, HA, is 4.80. The pK_b of a weak base, BOH, is 4.78. The pH of an aqueous solution of the corresponding salt, BA, will be - [AIEEE-2008]
(A) 9.58 (B) 4.79 (C) 7.01 (D) 9.22
10. In a saturated solution of the sparingly soluble strong electrolyte AgIO_3 (molecular mass = 283) the equilibrium which sets in is - [AIEEE-2007]
 $\text{AgIO}_3 \rightleftharpoons \text{Ag}^+_{(\text{aq})} + \text{IO}_3^-_{(\text{aq})}$
If the solubility product constant K_{sp} of AgIO_3 at a given temperature is 1.0×10^{-8} , what is the mass of AgIO_3 contained in 100 ml of its saturated solution ?
(A) $28.3 \times 10^{-2} \text{ g}$ (B) $2.83 \times 10^{-3} \text{ g}$ (C) $1.0 \times 10^{-7} \text{ g}$ (D) $1.0 \times 10^{-4} \text{ g}$

11. The solubility product of a salt having general formula MX_2 , in water is : 4×10^{-12} . The concentration of M^{2+} ions in the aqueous solution of the salt is - [AIEEE-2005]
 (A) $1.0 \times 10^{-4} \text{ M}$ (B) $2.0 \times 10^{-6} \text{ M}$ (C) $4.0 \times 10^{-10} \text{ M}$ (D) $1.6 \times 10^{-4} \text{ M}$
12. Hydrogen ion concentration in mol/L in a solution of pH = 5.4 will be - [AIEEE-2005]
 (A) 3.88×10^6 (B) 3.98×10^8 (C) 3.98×10^{-6} (D) 3.68×10^{-6}
13. The molar solubility in mol L^{-1} of a sparingly soluble salt MX_4 is 's'. The corresponding solubility product is K_{SP} . 's' is given in terms of K_{SP} by relation : [AIEEE-2004]
 (A) $s = (K_{SP} / 128)^{1/4}$ (B) $s = (128K_{SP})^{1/4}$
 (C) $s = (256K_{SP})^{1/5}$ (D) $s = (K_{SP}/256)^{1/5}$
14. The solubility of $Mg(OH)_2$ is x mole/lit. then its solubility product is- [AIEEE-2002]
 (A) x^3 (B) $5x^3$ (C) $4x^3$ (D) $2x^2$
15. The solubility in water of a sparingly soluble salt AB_2 is $1.0 \times 10^{-5} \text{ mol L}^{-1}$. Its solubility product will be [AIEEE-2003]
 (A) 1×10^{-15} (B) 1×10^{-10} (C) 4×10^{-15} (D) 4×10^{-10}

JEE-[MAIN] : PREVIOUS YEAR QUESTIONS		ANSWER KEY		EXERCISE -5[A]	
1.	(D)	2.	(D)	3.	(C)
5.	(C)	6.	(D)	7.	(C)
9.	(C)	10.	(B)	11.	(A)
13.	(D)	14.	(C)	12.	(C)
		15.	(C)		

EXERCISE - 05 [B]**JEE-[ADVANCED] : PREVIOUS YEAR QUESTIONS**

1. Which of the following statement(s) is/are correct ? [JEE 1998]
(A) the pH of 1.0×10^{-8} M solution of HCl is 8
(B) the conjugate base of H_2PO_4^- is HPO_4^{2-}
(C) autoprotolysis constant of water increases with temperature
(D) when a solution of a weak monoprotic acid is titrated against a strong base, at half - neutralization point $\text{pH} = (1/2) \text{pK}_a$.
2. A buffer solution can be prepared from a mixture of- [JEE 1999]
(A) sodium acetate and acetic acid in water
(B) sodium acetate and hydrochloric acid in water
(C) ammonia and ammonium chloride in water
(D) ammonia and sodium hydroxide in water.
3. The pH of 0.1 M solution of the following salts increases in the order- [JEE 1999]
(A) $\text{NaCl} < \text{NH}_4\text{Cl} < \text{NaCN} < \text{HCl}$
(B) $\text{HCl} < \text{NH}_4\text{Cl} < \text{NaCl} < \text{NaCN}$
(C) $\text{NaCN} < \text{NH}_4\text{Cl} < \text{NaCl} < \text{HCl}$
(D) $\text{HCl} < \text{NaCl} < \text{NaCN} < \text{NH}_4\text{Cl}$
4. An aqueous solution of 6.3 g oxalic acid dihydrate is made up to 250 mL. The volume of 0.1 N NaOH required to completely neutralise 10 mL of this solution is- [JEE 2001]
(A) 40 mL (B) 20 mL (C) 10 mL (D) 4 mL
5. For sparingly soluble salt ApBq , the relationship of its solubility product (L_s) with its solubility (S) is- [JEE 2001]
(A) $L_s = S^{p+q}$, $p^p \cdot q^q$ (B) $L_s = S^{p+q}$, $p^p \cdot q^p$ (C) $L_s = S^{pq}$, $p^p \cdot q^q$ (D) $L_s = S^{pq}$, $(p \cdot q)^{p+q}$
6. A solution which is 10^{-3} M each in Mn^{2+} , Fe^{2+} , Zn^{2+} and Hg^{2+} is treated with 10^{-16} M sulphide ion. If K_{sp} , MnS , FeS , ZnS and HgS are 10^{-15} , 10^{-23} , 10^{-20} and 10^{-54} respectively, which one will precipitate first ? [JEE 2003]
(A) FeS (B) MnS (C) HgS (D) ZnS
7. HX is a weak acid ($K_a = 10^{-5}$). It forms a salt NaX (0.1 M) on reacting with caustic soda. The degree of hydrolysis of NaX is- [JEE 2004]
(A) 0.01 % (B) 0.0001 % (C) 0.1 % (D) 0.5 %
8. CH_3NH_2 (0.1 mole, $K_b = 5 \times 10^{-4}$) is added to 0.08 moles of HCl and the solution is diluted to one litre, resulting hydrogen ion concentration is- [JEE 2005]
(A) 1.6×10^{-11} (B) 8×10^{-11} (C) 5×10^{-5} (D) 2×10^{-2}

SUBJECTIVES :

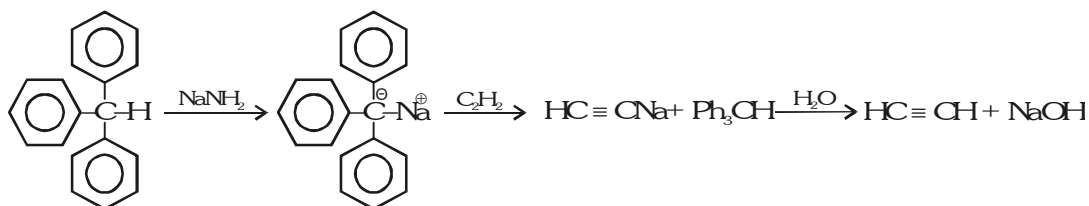
9. Given : $\text{Ag}(\text{NH}_3)_2^+ \rightleftharpoons \text{Ag}^+ + 2 \text{NH}_3$, $K_c = 6.2 \times 10^{-8}$ & K_{sp} of $\text{AgCl} = 1.8 \times 10^{-10}$ at 298 K. Calculate the concentration of the complex in 1.0 M aqueous ammonia. [JEE 1998]
10. What will be the resultant pH when 200 mL of an aqueous solution of HCl (pH = 2.0) is mixed with 300 mL of an aqueous solution of NaOH (pH = 12.0) ? [JEE 1998]
11. The solubility of $\text{Pb}(\text{OH})_2$ in water is 6.7×10^{-6} M. Calculate the solubility of $\text{Pb}(\text{OH})_2$ in a buffer solution of pH = 8. [JEE 1999]
12. The average concentration of SO_2 in the atmosphere over a city on a certain day is 10 ppm, when the average temperature is 298 K. Given that the solubility of SO_2 in water at 298 K is $1.3653 \text{ moles litre}^{-1}$ and the pK_a of H_2SO_3 is 1.92, estimate the pH of rain on that day. [JEE 2000]
13. 500 mL of 0.2 M aqueous solution of acetic acid is mixed with 500 mL of 0.2 M HCl at 25 °C.
(a) Calculate the degree of dissociation of acetic acid in the resulting solution and pH of the solution.
(b) If 6g of NaOH is added to the above solution, determine final pH. Assume there is no change in volume on mixing. K_a of acetic acid is 1.75×10^{-5} M. [JEE 2002]
14. Will the pH of water be same at 4 °C and 25 °C ? Explain. [JEE 2003]
15. 0.1 M of HA is titrated with 0.1 M NaOH, calculate the pH at end point.
Given $K_a(\text{HA}) = 5 \times 10^{-6}$ and $\alpha \ll 1$. [JEE 2004]
16. CH_3NH_2 (0.1 mole, $K_b = 5 \times 10^{-4}$) is added to 0.08 moles of HCl and the solution is diluted to one litre, resulting hydrogen ion concentration is
(A) 1.6×10^{-11} (B) 8×10^{-11} (C) 5×10^{-5} (D) 2×10^{-2} [JEE 2005]
17. The species present in solution when CO_2 is dissolved in water :
(A) CO_2 , H_2CO_3 , HCO_3^- , CO_3^{2-} (B) H_2CO_3 , CO_3^{2-} [JEE 2006]
(C) CO_3^{2-} , HCO_3^- (D) CO_2 , H_2CO_3
18. Solubility product constants (K_{sp}) of salts of types MX, MX_2 and M_3X at temperature 'T' are 4.0×10^{-8} , 3.2×10^{-14} and 2.7×10^{-15} , respectively. Solubilities (mol dm^{-3}) of the salts at temperature 'T' are in the order :
(A) $\text{MX} > \text{MX}_2 > \text{M}_3\text{X}$ (B) $\text{M}_3\text{X} > \text{MX}_2 > \text{MX}$ [JEE 2008]
(C) $\text{MX}_2 > \text{M}_3\text{X} > \text{MX}$ (D) $\text{MX} > \text{M}_3\text{X} > \text{MX}_2$
19. 2.5 mL of $\frac{2}{5}$ M weak monoacidic base ($K_b = 1 \times 10^{-12}$ at 25 °C) is titrated with $\frac{2}{15}$ M HCl in water at 25 °C.
The concentration of H^+ at equivalence point is [JEE 2008]
($K_w = 1 \times 10^{-14}$ at 25 °C)
(A) 3.7×10^{-13} M (B) 3.2×10^{-7} M (C) 3.2×10^{-2} M (D) 2.7×10^{-2} M

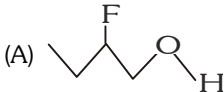
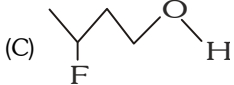
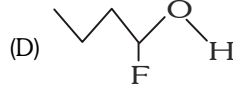
20. The dissociation constant of a substituted benzoic acid at 25°C is 1.0×10^{-4} . The pH of a 0.01 M solution of its sodium salt is [JEE 2009]
21. In 1 L saturated solution of AgCl [$K_{sp}(\text{AgCl}) = 1.6 \times 10^{-10}$], 0.1 mol of CuCl [$K_{sp}(\text{CuCl}) = 1.0 \times 10^{-6}$] is added. The resultant concentration of Ag^+ in the solution is 1.6×10^{-x} . The value of 'x' is. [JEE 2011]
22. The K_{sp} of Ag_2CrO_4 is 1.1×10^{-12} at 298 K. The solubility (in mol/L) of Ag_2CrO_4 in a 0.1 M AgNO_3 solution is [JEE 2013]
 (A) 1.1×10^{-11} *(B) 1.1×10^{-10} (C) 1.1×10^{-12} (D) 1.1×10^{-9}
23. The initial rate of hydrolysis of methyl acetate (1M) by a weak acid (HA, 1M) is $1/100^{\text{th}}$ of that of a strong acid (HX, 1M), at 25°C. The K_a of HA is [JEE 2013]
 *(A) 1×10^{-4} (B) 1×10^{-5} (C) 1×10^{-6} (D) 1×10^{-3}

JEE-[ADVANCED] : PREVIOUS YEAR QUESTIONS		EXERCISE -5[B]	
1.	B,C	2.	A,B,C
3.	B	4.	A
5.	A	6.	C
7.	A	8.	B
9.	$[\text{Ag}(\text{NH}_3)_2^+] = 0.0539$	10.	pH = 11.3010
11.	$s = 1.203 \times 10^{-3} \text{ M}$	12.	0.4948
13.	(a) 0.0175 %, pH = 1, (b) 4.75	14.	NO
15.	pH = 9	16.	B
17.	A	18.	D
19.	D	20.	8
21.	7	22.	A
23.	A		

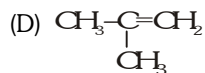
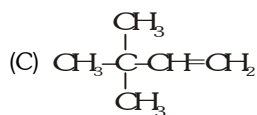
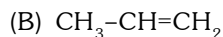
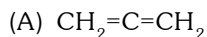
EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

- Carbanion is a :-
(A) Base (B) Nucleophile (C) Both (A) and (B) (D) None
- Electrophile is :
(A) H_2O (B) NH_3 (C) AlCl_3 (D) $\text{C}_2\text{H}_5\text{NH}_2$
- Which of the following species has a pyramidal shape-
(A) CH_3^+ (B) BF_3 (C) CH_3^- (D) $\cdot\text{CH}_3$
- Increasing order (least basic first) of basic strength is shown by the set
(A) CINH_2 , NH_3 , CH_3NH_2 (B) CINH_2 , CH_3NH_2 , NH_3
(C) NH_3 , CINH_2 , CH_3NH_2 (D) CH_3NH_2 , CINH_2 , NH_3
- The decreasing order of acid strength indicated by the following sequence of reaction is :-



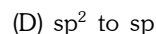
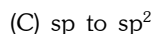
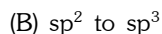
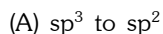
- (A) $\text{NH}_3 > \text{Ph}_3\text{CH} > \text{C}_2\text{H}_2 > \text{H}_2\text{O}$ (B) $\text{H}_2\text{O} > \text{HC}\equiv\text{CH} > \text{Ph}_3\text{CH} > \text{NH}_3$
(C) $\text{HC}\equiv\text{CH} > \text{H}_2\text{O} > \text{Ph}_3\text{CH} > \text{NH}_3$ (D) $\text{Ph}_3\text{CH} > \text{HC}\equiv\text{CH} > \text{H}_2\text{O} > \text{NH}_3$
- In which of the following compounds is hydroxylic proton the most acidic -
(A)  (B)  (C)  (D) 
 - Correct arrangement of the following acids in correct K_a order is :-
(I) $\text{H}_3\text{N}^{\oplus}-\text{CH}_2-\text{COOH}$ (II) $\text{NC}-\text{CH}_2-\text{COOH}$ (III) $\text{H}_3\text{C}-\text{CH}_2-\text{COOH}$ (IV) $\text{OOC}^{\ominus}-\text{CH}_2-\text{COOH}$
(A) I > II > III > IV (B) II > I > III > IV (C) I > III > II > IV (D) IV > III > II > I
 - Which of the following orders of acid strength is correct :-
(A) $\text{RCOOH} > \text{ROH} > \text{HOH} > \text{HC}\equiv\text{CH}$ (B) $\text{RCOOH} > \text{HOH} > \text{ROH} > \text{HC}\equiv\text{CH}$
(C) $\text{RCOOH} > \text{HOH} > \text{HC}\equiv\text{CH} > \text{ROH}$ (D) $\text{RCOOH} > \text{HC}\equiv\text{CH} > \text{HOH} > \text{ROH}$
 - Which of them is false for order of -I effect
(A) $-\text{F} > -\text{Cl} > -\text{Br} > \text{I}$ (B) $-\text{NR}_3^{\oplus} > -\text{NH}_3^{\oplus} > -\text{NO}_2$
(C) $-\text{OCH}_3 > -\text{OH} > -\text{NH}_2$ (D)  > $-\text{C}\equiv\text{CH} > \text{H}$
 - The order of basicity of halides is :
(A) $\text{Cl}^- < \text{Br}^- < \text{I}^- < \text{F}^-$ (B) $\text{F}^- < \text{I}^- < \text{Br}^- < \text{Cl}^-$
(C) $\text{I}^- < \text{Br}^- < \text{Cl}^- < \text{F}^-$ (D) $\text{Cl}^- < \text{F}^- < \text{I}^- < \text{Br}^-$

11. Which of the following molecule has longest C=C bond length -

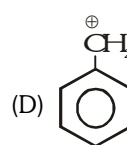
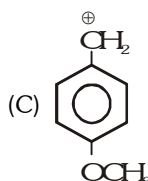
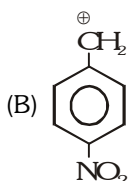
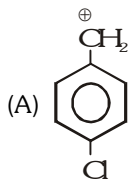


12. In the reaction $\text{CH}_3\text{CN} \xrightarrow[\text{heat}]{\text{H}_3\text{O}^+} \text{CH}_3\text{COOH}$

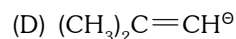
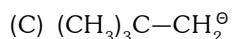
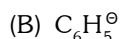
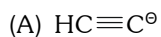
the hybridization state of the functional carbon changes from



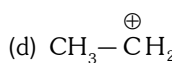
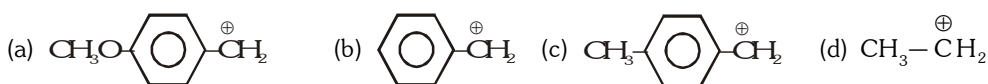
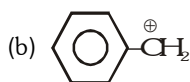
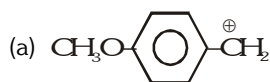
13. Most stable carbocation is :-



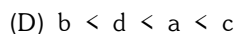
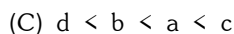
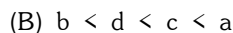
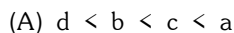
14. Most stable carbanion is :-



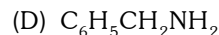
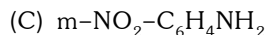
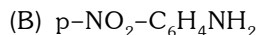
15. Consider the following carbocations



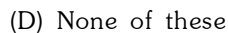
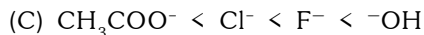
The relative stabilities of these carbocations are such that :-



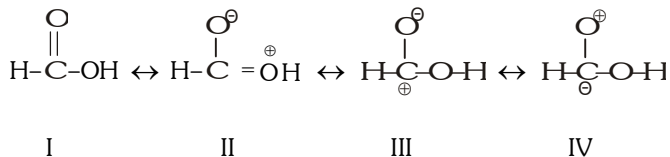
16. Among the following, the strongest base is :-



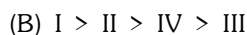
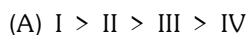
17. The increasing order of base strength of Cl^- , CH_3COO^- , OH^- and F^- is :



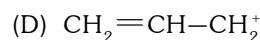
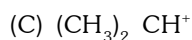
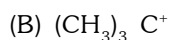
18. Formic acid is considered as a hybrid of the four structures



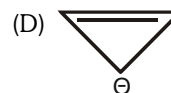
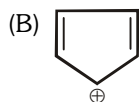
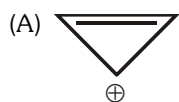
Which of the following order is correct for the stability of the four contributing structures.



19. Which of the following carbonium ion is most stable



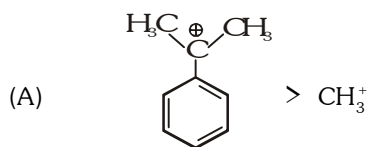
20. Among the following the aromatic compound is



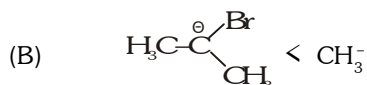
21. Match List- I with List- II and select the correct answer using the codes given below -

List-I (Stability)

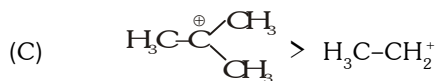
List -II (Reason)



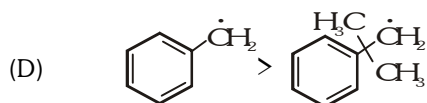
(1) Inductive effect



(2) Resonance



(3) Hyperconjugation and resonance



(4) Hyperconjugation and inductive effect

Codes :

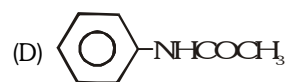
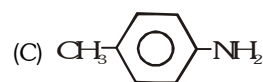
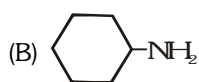
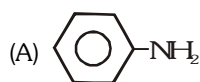
(A) A - 2 ; B - 3 ; C - 4 ; D - 1

(B) A - 3 ; B - 1 ; C - 4 ; D - 2

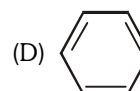
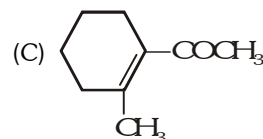
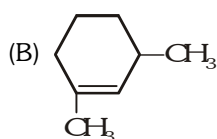
(C) A - 4 ; B - 3 ; C - 1 ; D - 2

(D) A - 3 ; B - 4 ; C - 2 ; D - 1

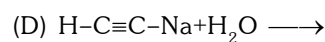
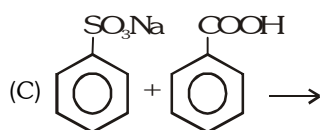
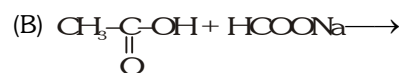
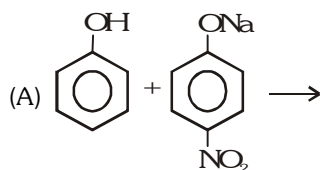
22. Which of the following compounds is the strongest base -

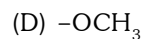
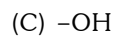
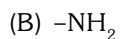
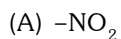
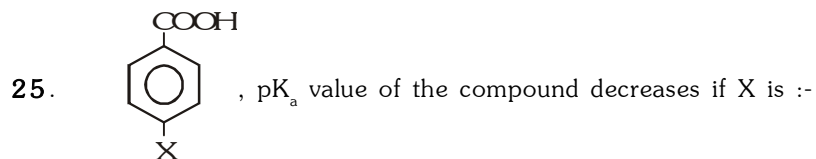


23. In which of the following molecule all the effect namely inductive, mesomeric & hyperconjugation operate:

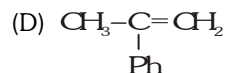
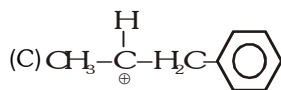
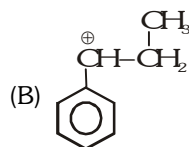
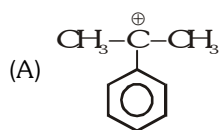
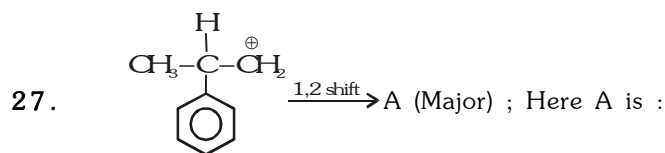
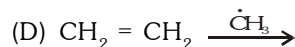
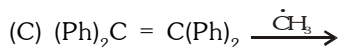
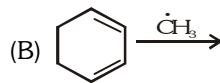
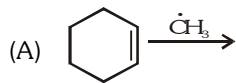


24. Which of the following reaction is possible :

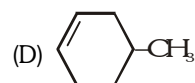
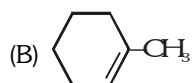
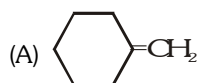




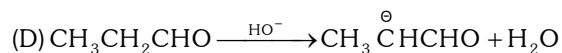
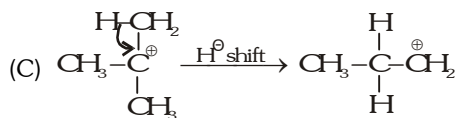
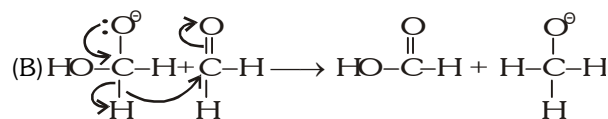
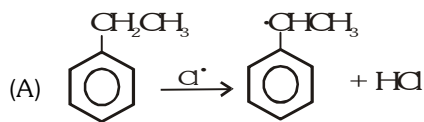
26. Which one of the following results most stable carbon free radical :



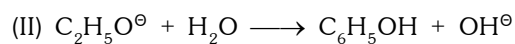
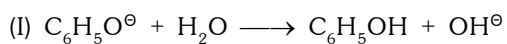
28.  ; On dehydration with conc. H_2SO_4 forms predominantly :



29. Which of the following is incorrect :



30. The K_a values of alcohols, water and phenol are of order 10^{-17} , 10^{-14} and 10^{-10} respectively. Which of the following reactions is possible :



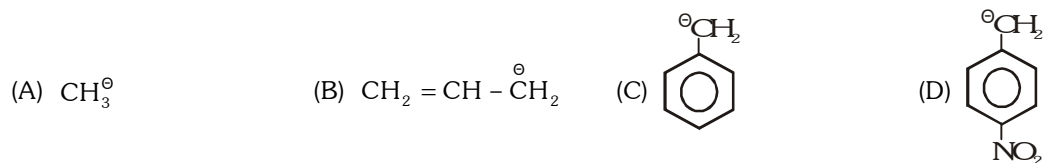
(A) Both I and II

(B) only II

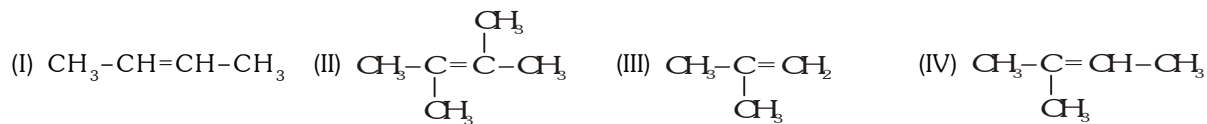
(C) only I

(D) neither of two

31. Most stable carbanion is :



32. Stability of :



in the increasing order is :

(A) I < III < IV < II (B) I < II < III < IV (C) I < IV < III < I (D) II < III < IV < I

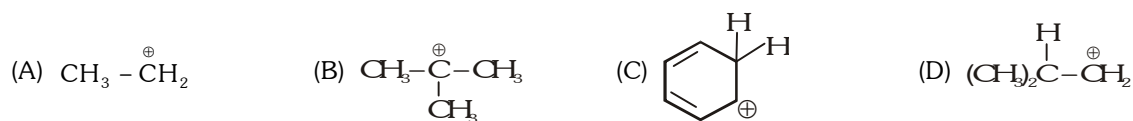
33.  $\xrightarrow{\text{H}_2\text{O}}$ [A] ; the product (A) is :



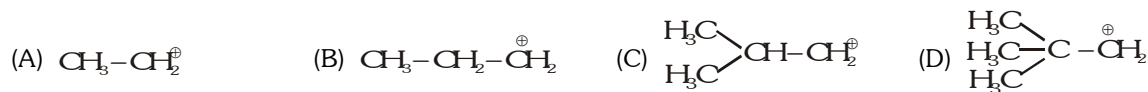
34. Spin multiplicity of Triplet nitrene is :

(A) 1 (B) 2 (C) 3 (D) 4

35. The maximum probability of proton loss is in the case of :



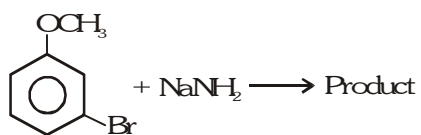
36. Select the least stable one :



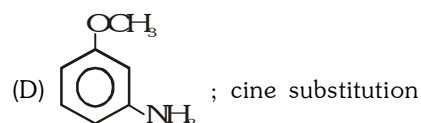
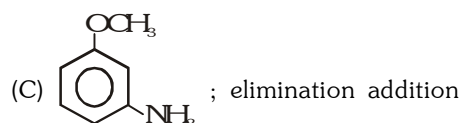
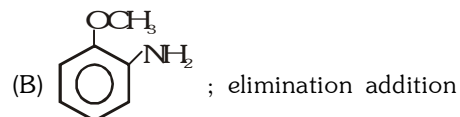
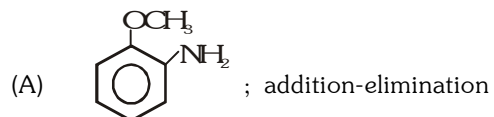
37. Which of the following will react fastest with conc. HCl :



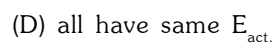
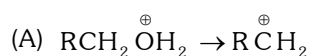
38. Consider the following reaction :



The product (P) and reaction (R) are :



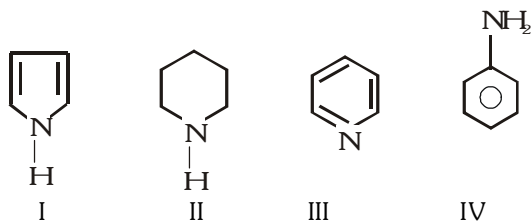
39. Energy of activation is lower for which reaction :



CHECK YOUR GRASP						ANSWER KEY				EXERCISE -1					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	C	C	C	A	B	D	A	B	D	C	D	C	C	A	A
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	D	A	A	A	A	B	B	C	D	A	C	A	B	C	B
Que.	31	32	33	34	35	36	37	38	39						
Ans.	D	A	B	C	C	D	B	C	C						

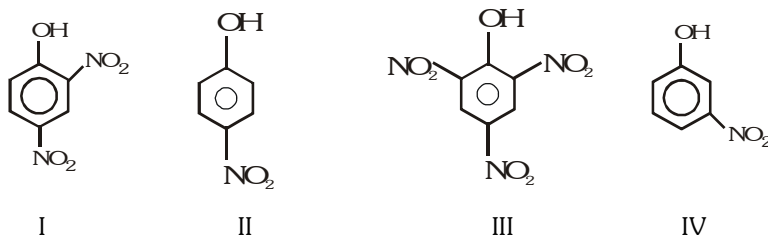
SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)

1. The correct order of decreasing basic strength is :



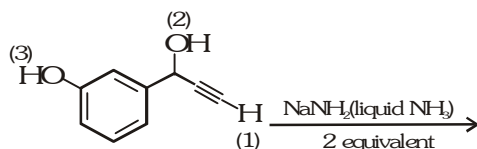
- (A) I > II > III > IV (B) II > III > I > IV (C) II > IV > I > III (D) II > III > IV > I

2. The correct order of increasing dissociation constant of the following compound is :-



- (A) II < IV < I < III (B) IV < III < I < II (C) IV < II < I < III (D) IV < I < II < III

3. In the following arrange the H (numbered) for their ease of displacement during acid base reaction :

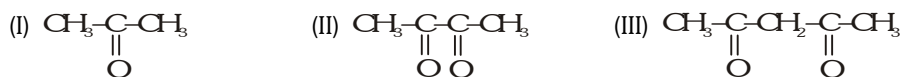


- (A) 1 > 2 > 3 (B) 3 > 2 > 1 (C) 3 > 1 > 2 (D) 2 > 3 > 1

4. Which one of the following is the strongest base in aqueous medium :-

- (A) (C₂H₅)₃N (B) C₂H₅NH₂ (C) NH₃ (D) (C₂H₅)₂NH

5. The correct order of acid strength of the following compound is -

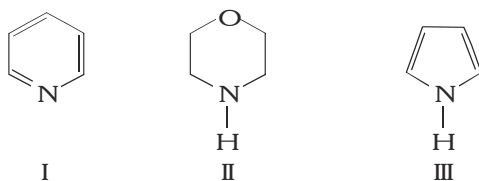


- (A) III > II > I (B) III > I > II (C) II > I > III (D) I > II > III

6. Which of the following shows the correct order of decreasing basicity in aqueous medium -

- (A) (CH₃)₃N > (CH₃)₂NH > CH₃NH₂ > NH₃ (B) (CH₃)₂NH > (CH₃)₃N > CH₃NH₂ > NH₃
(C) (CH₃)₂NH > CH₃NH₂ > (CH₃)₃N > NH₃ (D) (CH₃)₂NH > CH₃NH₂ > NH₃ > (CH₃)₃N

7. For the compounds



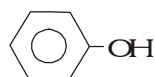
the order of basicity is -

- (A) III > II > I (B) II > III > I (C) I > II > III (D) II > I > III

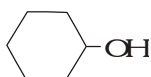
8. Which of the following shows the correct order of decreasing acidity-

- (A) $\text{PhCO}_2\text{H} > \text{PhSO}_3\text{H} > \text{PhCH}_2\text{OH} > \text{PhOH}$
 (B) $\text{PhSO}_3\text{H} > \text{PhOH} > \text{PhCO}_2\text{H} > \text{PhCH}_2\text{OH}$
 (C) $\text{PhCO}_2\text{H} > \text{PhOH} > \text{PhCH}_2\text{OH} > \text{PhSO}_3\text{H}$
 (D) $\text{PhSO}_3\text{H} > \text{PhCO}_2\text{H} > \text{PhOH} > \text{PhCH}_2\text{OH}$

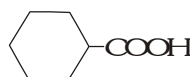
9. Give the correct order of increasing acidity of the following compounds-



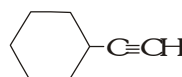
I



II



III

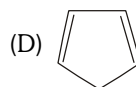


IV

- (A) $\text{II} < \text{I} < \text{III} < \text{IV}$ (B) $\text{IV} < \text{II} < \text{I} < \text{III}$ (C) $\text{I} < \text{II} < \text{IV} < \text{III}$ (D) $\text{IV} < \text{I} < \text{II} < \text{III}$

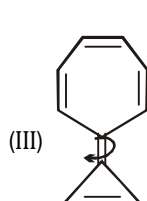
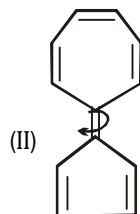
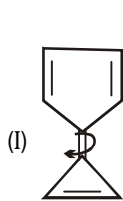
10. Which of the following is the most acidic

- (A) $\text{CH}_2 = \text{CH}_2$ (B) $\text{HC} \equiv \text{CH}$



11. Which of the following substituents will increase the acidity of phenol -

- (A) $-\text{NO}_2$ (B) $-\text{CN}$ (C) $-\text{CHO}$ (D) $-\text{CH}_3$

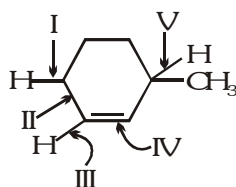


12.

Compare carbon-carbon bond rotation across I, II, III.

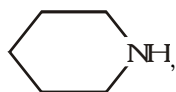
- (A) $\text{I} > \text{II} > \text{III}$ (B) $\text{I} > \text{III} > \text{II}$ (C) $\text{II} > \text{I} > \text{III}$ (D) $\text{II} > \text{III} > \text{I}$

13. Which of the following s-bonds participate in hyperconjugation :

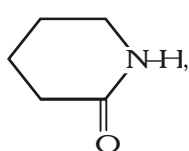


- (A) I and II (B) I and V (C) I and V (D) III and IV

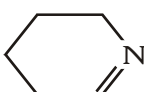
14.



(x)



(y)

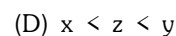
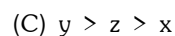
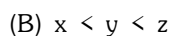
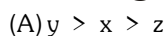
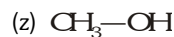
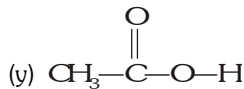
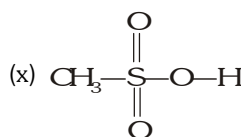


(z)

The correct order of decreasing basic strengths of x, y and z is :

- (A) $x > y > z$ (B) $x > z > y$ (C) $y > x > z$ (D) $y > z > x$

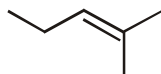
15. Set the following in increasing order of their pK_a values:



16. Rank the following alkenes in order of decreasing heats of hydrogenation (largest first):



(I)



(II)



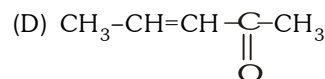
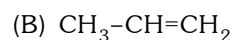
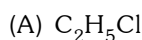
(III)



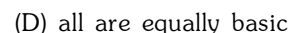
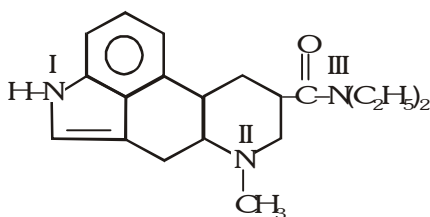
(IV)



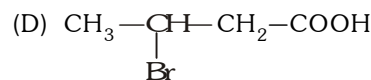
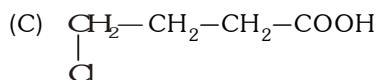
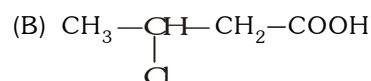
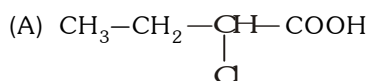
17. Which of the following molecules has all the effects : inductive, mesomeric and Baker Nathan effect ?



18. Which nitrogen in LSD (Lysergic acid diethylamide) is more basic :

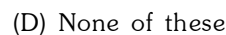
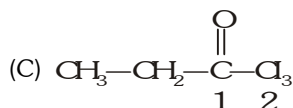
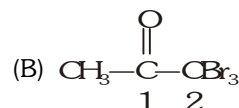
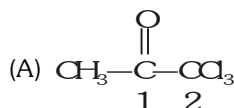


19. Which of the following substituted carboxylic acids has the highest K_a value :

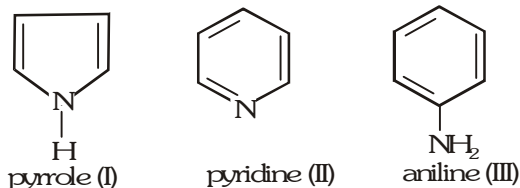


SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)

20. Which of the following compounds yield most stable carbanion after rupture of (C_1-C_2) bond :



21.

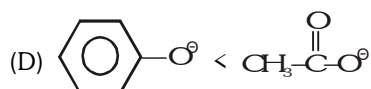
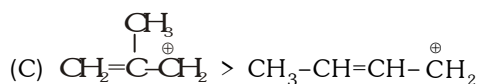
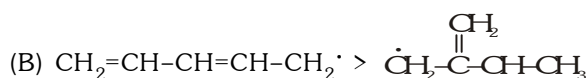
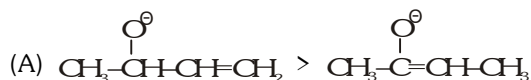


Which is/are correct statements ?

- (A) I is more basic than II (B) II is more basic than I and III
(C) III is more basic than II (D) all are aromatic bases

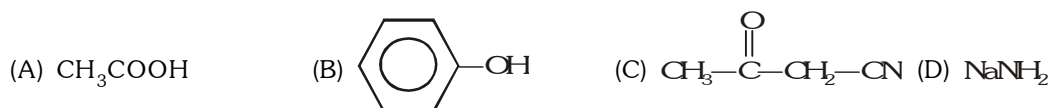
22.

Following is true for stability between the two structures :



23.

Which have acidic hydrogen :



24.

C—C and C=C bond lengths are unequal in :

- (A) Benzene (B) 1,3 buta-di-ene
(C) 1, 3 cyclohexa-di-ene (D) None

25.

The acid strength of substituted carboxylic acids is known to be dependent on the nature and position of the substituent. In the following examples, an attempt has been made to arrange the acids in order of acid strength, the strongest first. One of the series is incorrect-which one :

- (A) $\text{CH}_3\cdot\text{CH}_2\text{CH}(\text{Cl})\cdot\text{COOH} > \text{CH}_3\cdot\text{CH}(\text{Cl})\cdot\text{CH}_2\text{COOH} > \text{CH}_3\cdot\text{CH}_2\cdot\text{CH}_2\cdot\text{COOH}$
(B) $\text{NO}_2\cdot\text{CH}_2\cdot\text{COOH} > \text{HOCH}_2\cdot\text{COOH} > \text{CH}_3\cdot\text{COOH}$
(C) $\text{Cl}_3\text{C}\cdot\text{COOH} > \text{BrCH}_2\cdot\text{COOH} > \text{FCH}_2\cdot\text{COOH}$
(D) $\text{CH}_3\cdot\text{COOH} > \text{CH}_3\cdot\text{CH}_2\cdot\text{COOH} > (\text{CH}_3)_3\text{C}\cdot\text{COOH}$

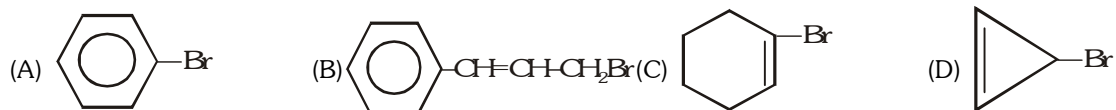
26.

Which species is not aromatic :



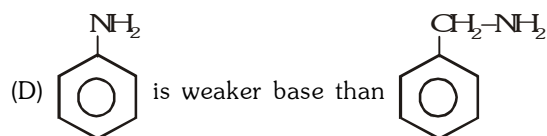
27. Select the correct option :
- (A) carbonic acid is weaker acid than acetic acid
 (B) the boiling points of acids are higher than corresponding alcohols
 (C) chloroacetic acid is stronger acid than acetic acid
 (D) phenol is more acidic than ethanol

28. Which reacts with AgNO_3 to give ppt. :

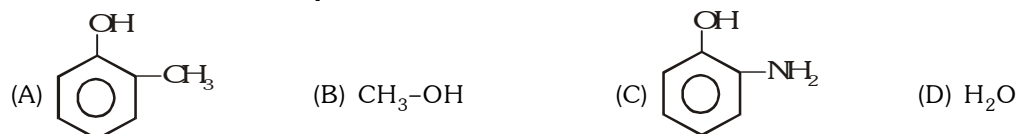


29. Which statement is true :

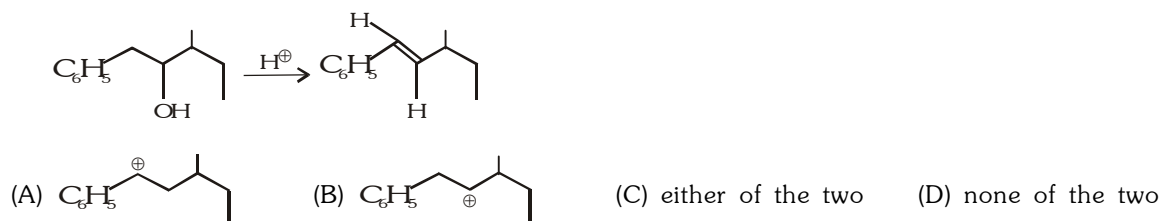
- (A) $\ddot{\text{N}}\text{H}_2$ is stronger base than $\ddot{\text{O}}\text{H}$
 (B) $\text{NH}_2\text{-OH}$ is less basic than NH_3
 (C) $\text{CH}_3\text{-NH}_2$ is weaker base than $\text{N}(\text{CH}_3)_3$ in (aq) medium



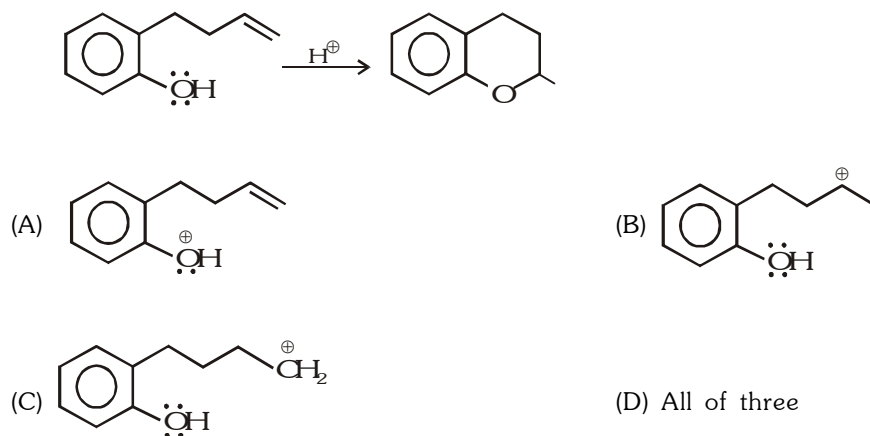
30. Which is less acidic than phenol :

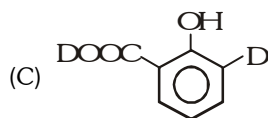
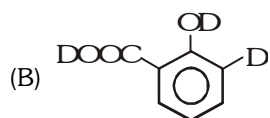
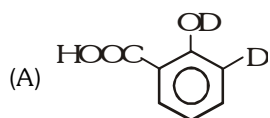


31. The precursor carbocation to the product in the following reaction is :



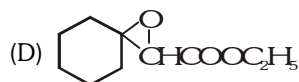
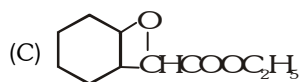
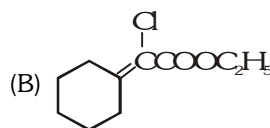
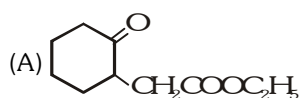
32. Which of the following ion is formed in the following reaction :



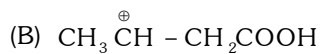
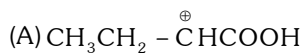
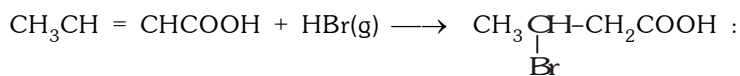


(D) Reaction not possible

34. Predict the nature of A in the following reaction :

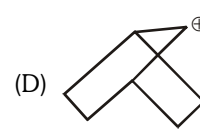
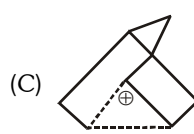
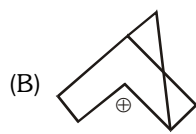
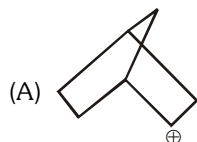
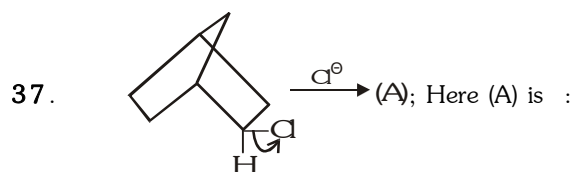
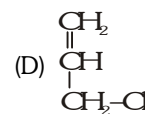
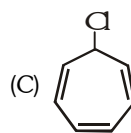
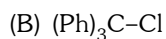
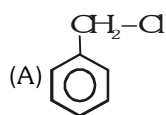


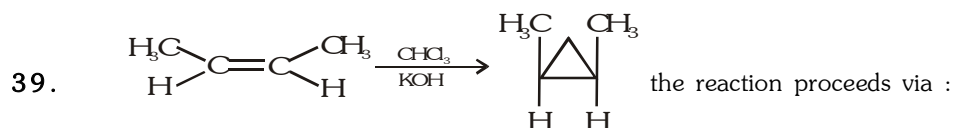
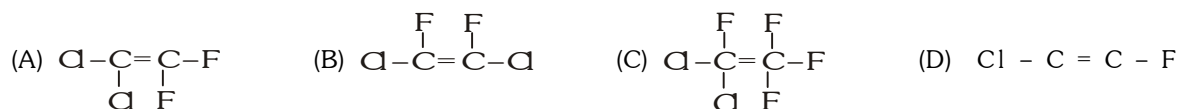
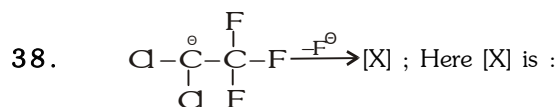
35. Which of the following intermediate is most likely to be formed during addition of HBr on crotonic acid :



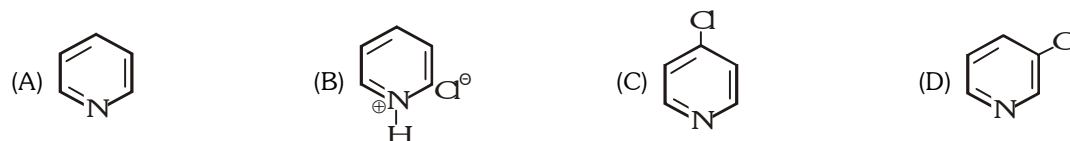
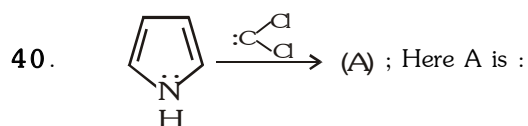
(D) none of these

36. Which will give most stable cation upon strong heating :

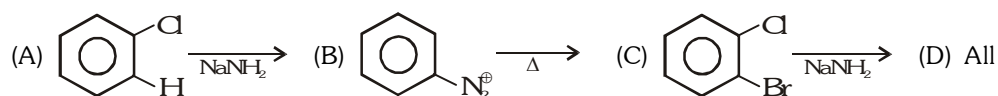




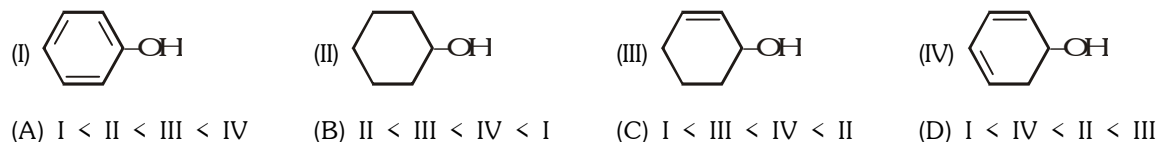
- (A) carbene (B) Carbon free radical (C) Carbocation (D) Carbanion



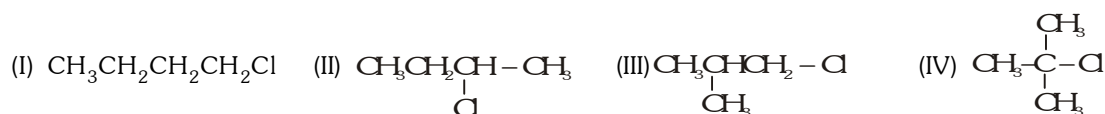
41. Which of the following can give benzyne :



42. Dehydration of the following in increasing order is :



43. Increasing tendency for $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ reaction is :

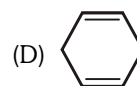
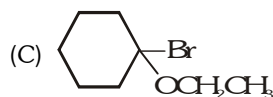
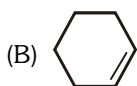
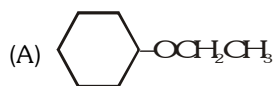


- (A) $\text{S}_{\text{N}}1$; I < III < II < IV (B) $\text{S}_{\text{N}}2$; IV < II < III < I
 (C) (A) and (B) both are correct (D) Both incorrect

44. Cyclopentyl ethyl ether can be obtained by :



45. Major product in the following reaction is :



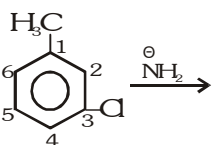
46. Mechanism of insertion reaction is of :

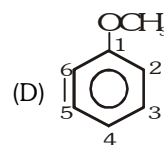
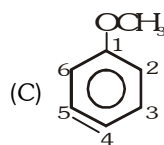
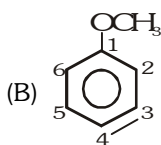
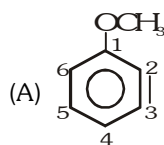
(A) One step for triplet carbene

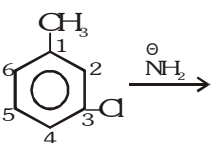
(B) Two step for singlet carbene

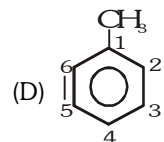
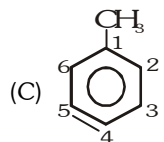
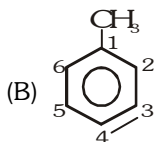
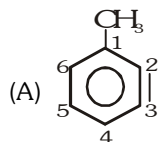
(C) One step for singlet carbene

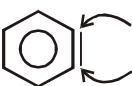
(D) Two step for both

47. . The most probable structure of intermediate P is :



48. . The most probable structure of intermediate A is :



49. Hybridisation of arrow headed 'C' is 

(A) $sp^2 sp^3$

(B) $sp^3 sp^3$

(C) $sp^2 sp^2$

(D) $sp^3 sp^3 d$

50. Which of the following is most strongest electrophile :

(A) singlet $\text{H}-\ddot{\text{N}}\cdot$

(B) triplet $\text{H}-\ddot{\text{N}}\cdot$

(C) singlet $\text{C} \equiv \text{C} \cdot$

(D) triplet $\text{C} \equiv \text{C} \cdot$

51. In benzyne two external unshared electronic orbital :

(A) do not overlap to form a π bond

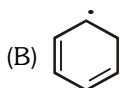
(B) can overlap to form a complete π bond

(C) poorly overlap to form a very weak bond

(D) contribute in benzene ring

52. Which of the following is aromatic intermediate :

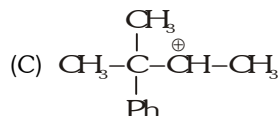
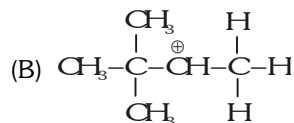
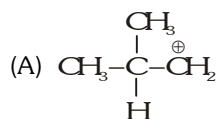
(A) carbene



(C) Benzyne

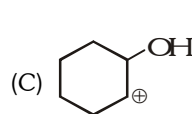
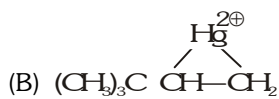
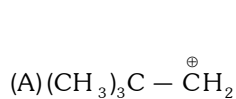


53. In which of the following case 'H' shift is more preferable than 'CH₃' shift :



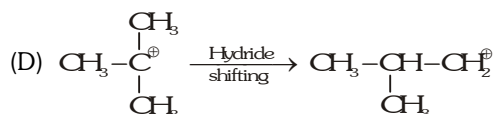
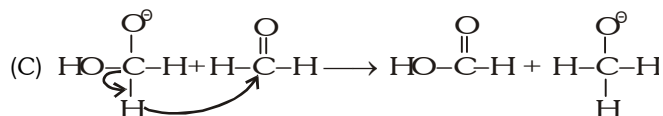
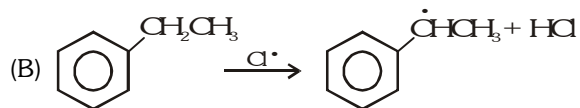
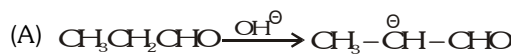
(D) all

54. Which of the following will not give 1, 2, shift :

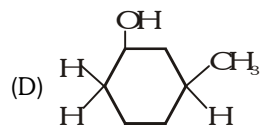
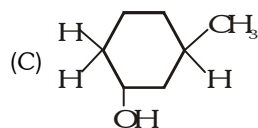
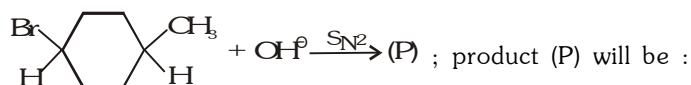


(D) all

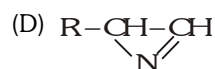
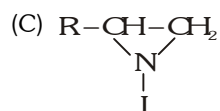
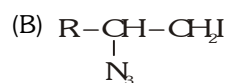
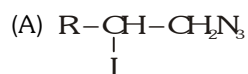
55. Which of the following reactions is wrong :

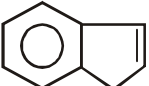


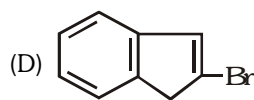
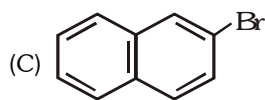
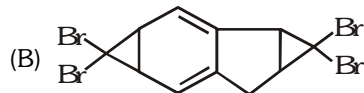
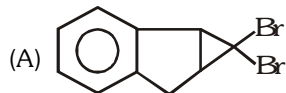
56. Methyl benzylic carbonium ion is the most stable one, hence it will react fastest consider the following reaction :

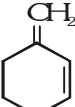


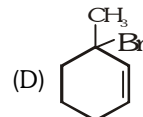
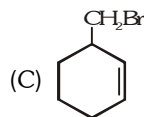
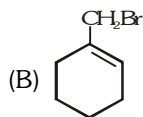
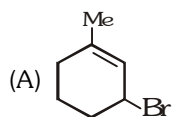
57. Consider the following reaction -
 $R-CH=CH_2 + IN_3 \longrightarrow A$, A is :

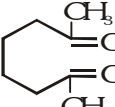


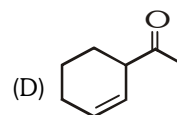
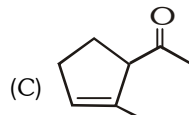
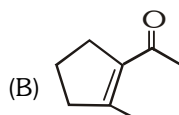
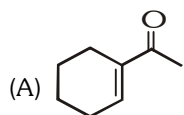
58.  + $\text{CHBr}_3 \xrightarrow{1. \text{BuOK}}$ product ; product will be :



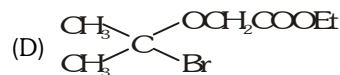
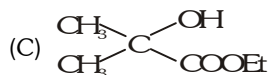
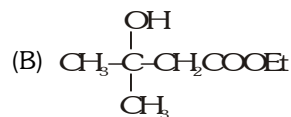
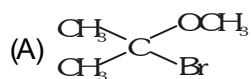
59.  $\xrightarrow{\text{HBr}}$ major product will be :

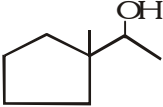


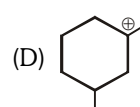
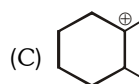
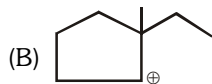
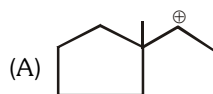
60.  $\xrightarrow[\Delta]{\text{OH}^\ominus}$ Product ; major product of the reaction can be :



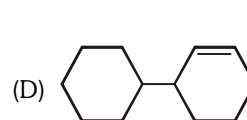
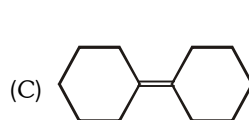
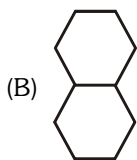
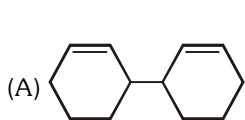
61. $\text{CH}_3\text{COCH}_3 + \text{BrCH}_2\text{COOEt} \xrightarrow[\text{(ii) H}_2\text{O/H}^\oplus]{\text{(i) Zn}}$ (A) ; the product (A) is :



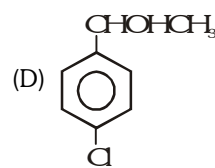
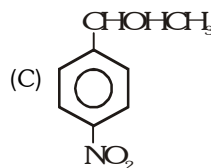
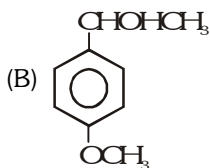
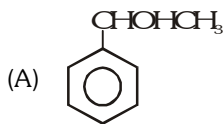
62. Which carbocation is more likely to be formed in the dehydration of  $\xrightarrow{H^+}$:



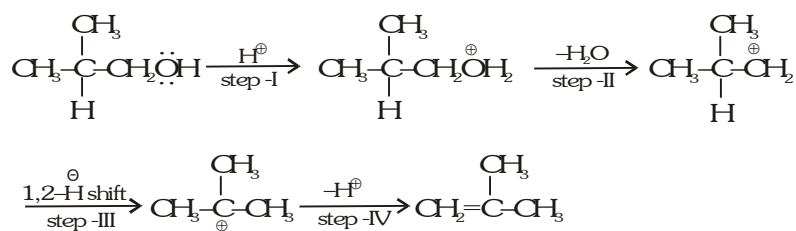
63. The most probable structure for (P) in the following reaction is :



64. Which of the following alcohols is dehydrated most readily with conc. H_2SO_4 ?



65. Dehydration of alcohols by conc. H_2SO_4 takes place according to following steps :



the slowest and fastest steps in the above reaction are :

- (A) step I is slowest, while III is fastest
 (B) step II is slowest, while III is fastest
 (C) step II is slowest, while IV is fastest
 (D) all step proceed at equal rate

EXERCISE-03**MISCELLANEOUS TYPE QUESTIONS****TRUE OR FALSE :**

1. In benzene, carbon uses all the three p-orbitals for hybridization.
2. All the carbon atoms in bicyclbutane are sp^2 -hybridized.
3. Allene ($CH_2=C=CH_2$) is a planar molecule.
4. The greater stability of trans-but-2-ene over cis-but-2-ene can be explained on the basis of hyperconjugation effect.
5. In a two step reaction, the rate determining step has the lowest energy of activation.

FILL IN THE BLANKS :

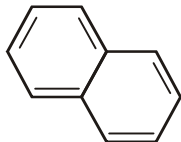
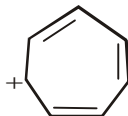
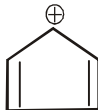
1. Out of benzene, ethylene and acetylene, the carbon-carbon bond is longest in
2. The cyclopentadienyl cation is while cyclopentadienyl anion is
3. The type of delocalization involving sigma bond orbitals with π bond or vacant orbital is called
4. Hydroperoxide ion is a stronger but weaker than hydroxide ion.
5. Catalytic hydrogenation of cis-2, 3-diphenyl-2-butene gives.....

MATCH THE COLUMN

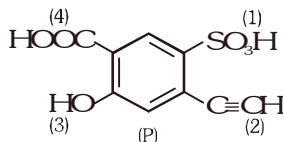
1. Match the column I with column II.

Column-I		Column-II	
(A)	$-\text{NO}_2$	(p)	- m effect
(B)	$-\text{O}^-$	(q)	+ m effect
(C)	$-\text{O}-\text{CH}_3$	(r)	+ I effect
(D)	$-\text{C} \equiv \text{N}$	(s)	- I effect

2. Match the column I with column II.

Column-I		Column-II	
(A)		(p)	Aromatic
(B)		(q)	Non-aromatic
(C)		(r)	Anti-aromatic
(D)		(s)	Aliphatic

3. Observe the following compound P and match the column I with column II.



Column-I		Column-II	
(A)	Strongest acidic group in P	(p)	1
(B)	Weakest acidic group in P	(q)	2
(C)	Number of Intra molecular H-Bonding in P	(r)	3
(D)	Number of mole of hydrogen gas that is liberated on reaction of 'P' with excess of sodium metal if one mole of P is used	(s)	4

ASSERTION & REASON QUESTION :

These questions contains, Statement-I (assertion) and Statement-II (reason).

- (A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I
 (B) Statement-I is True, Statement-II is True ; Statement-II is NOT a correct explanation for Statement-I
 (C) Statement-I is True, Statement-II is False.
 (D) Statement-I is False, Statement-II is True.

1. **Statement-I** : Greater the s-character of the hybrid orbital, the smaller is its size.

Because

Statement-II : Bond formed by the overlap of sp^3-sp^3 hybrid orbital is the longest while the bond formed by the overlap of $sp-sp$ hybrid orbital is shortest.

2. **Statement-I** : $Me_3\overset{+}{C}$ is more stable than $Me_2\overset{+}{CH}$ and $Me_2\overset{+}{CH}$ is more stable than the $Me\overset{+}{CH_2}$.

Because

Statement-II : Greater the number of hyperconjugative structures, more is the stability of carbocation.

3. **Statement-I** : Cyclopentadienyl anion is much more stable than allyl anion.

Because

Statement-II : Cyclopentadienyl anion is aromatic in nature.

4. **Statement-I** : CH_3OCH_3 and C_2H_5OH have comparable molecular masses but boiling point of C_2H_5OH is higher than dimethyl ether.

Because

Statement-II : C_2H_5OH forms intermolecular H-bonding while CH_3OCH_3 forms intramolecular H-bonding.

5. **Statement-I** : In the compound, $\overset{IV}{CH_3}-\overset{III}{CH_2}-\overset{II}{C}\equiv\overset{I}{N}$, the most electroegative carbon is II.

Because

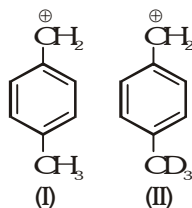
Statement-II : Carbon atom II has more s-character.

6. **Statement-I** : pK_{a1} of fumaric acid is more than maleic acid.

Because

Statement-II : Conjugate base of fumaric acid is stabilised by intramolecular H-bonding.

7. **Statement-I** : Carbocation (II) is more stable than carbocation (I)



Because

Statement-II : Carbocation (II) has more positive inductive effect of $-CD_3$ group as compared to $-CH_3$ group.

8. **Statement-I** : Carbon-oxygen bonds are of equal length in acetate ion.

Because

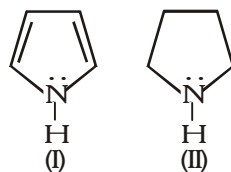
Statement-II : Bond length decreases with the multiplicity of bond between two atoms.

9. **Statement-I** : The potential energy barrier for rotation about C=C bond in 2-butene is much higher than that in ethylene.

Because

Statement-II : Hyperconjugation effect decreases the double bond character.

10. **Statement-I** : Pyrrolidine (II) is more basic than pyrrole (I).



Because

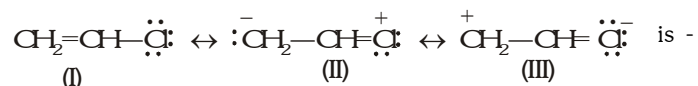
Statement-II : Protonated pyrrole has resonance stabilization of positive charge in aromatic ring.

COMPREHENSION BASED QUESTIONS :

Comprehension # 1

The intramolecular delocalisation of π and /non-bonding electrons without any change in the position of atoms is called resonance. Delocalisation may occur in conjugated system involving carbon atom and atom other than the carbon. Delocalisation makes system stable. More is the number of resonating structures, more is the stability of the system. A conjugated structure is least stable when a higher electronegative atom has positive charge and when identical charges are present on adjacent atoms.

1. The decreasing order of stability of the following resonating structures :

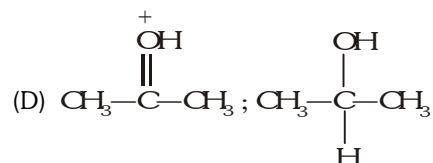
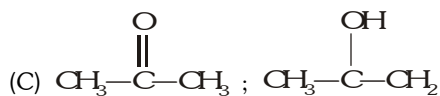
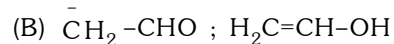
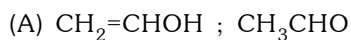


- (A) I > II > III (B) II > III > I (C) III > II > I (D) I > III > II

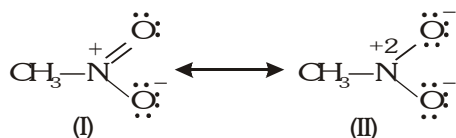
2. If A is $PhCH_2^+$ and B is $CH_2=CH-CH_2^+$, the greater number of resonating structure is of -

- (A) A (B) B (C) both A and B (D) None of these

3. Which of the following pairs represent resonance ?



4. Examine the structures I and II for nitromethane and choose the statement correctly :



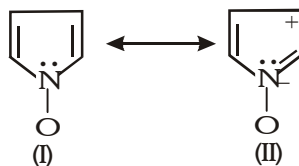
(A) II is less important because electrons have shifted to oxygen

(B) II is less important because nitrogen has sextet of electrons

(C) II is acceptable and important structure

(D) none of these

5. Examine the following two structures for pyrrole and choose the correct statement given below :



(A) II is not an acceptable resonating structure because carbonium ions is less stable than nitride ion

(B) II is not an acceptable resonating structure because there is charge separation

(C) II is not an acceptable resonating structure because nitrogen has ten valance electrons

(D) II is an acceptable resonating structure

6. Delocalization of electrons increases molecular stability because :

(A) Potential energy of the molecule decreases

(B) Electron-nuclei attraction decreases

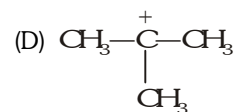
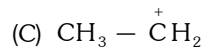
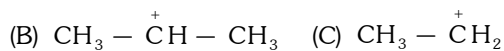
(C) Both (A) and (B)

(D) Electron-electron repulsion increases

Comprehension # 2

Carbocation is a group of atoms with positively charged carbon atom having six electrons in the valence shell after sharing. Carbocations are formed in the heterolysis of a bond and are planar species. Stability of carbocation is determined by inductive effect, hyperconjugation and resonance effect. Greater the number of contributing structures, more is the stability of a Carbocation. Electron releasing groups (+I effect) increases the stability of a carbocation whereas the electron withdrawing groups (-I effect) have an opposite effect.

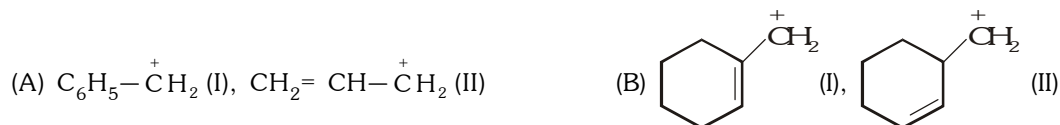
1. Which of the following is most stable carbocation ?

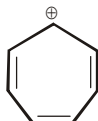


2. The most stable carbocation among the following :



3. In which of the following cases, the carbocation (I) is less stable than the carbocation (II) ?



4. The most stable and the least stable carbocation among  (I), $CH_2=CH-\overset{+}{C}H_2$ (II), $C_6H_5-\overset{+}{C}H_2$ (III)

and $CH_3-\overset{+}{C}H-CH_3$ (IV) are respectively :

(A) II, I (B) III, IV (C) I, II (D) I, IV

5. Most stable carbocation is formed by the heterolysis of :

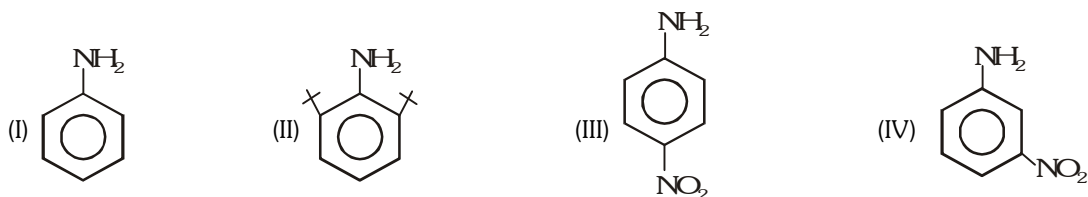
(A) $(CH_3)_3CBr$ (B) $(C_6H_5)_3CBr$ (C) $(C_6H_5)_2CHBr$ (D) $C_6H_5CH_2Br$

Comprehension # 3

The most important condition for resonance to occur is that the involved atoms in resonating structure must be coplanar or nearly coplanar for maximum delocalisation. If this condition does not fulfil, involved orbitals cannot be parallel to each other and as a consequence delocalisation cannot occur. Bulky groups present on adjacent atoms inhibit the of orbitals. This phenomenon is known as steric inhibition of resonance. Steric inhibition of resonance has profound effect on

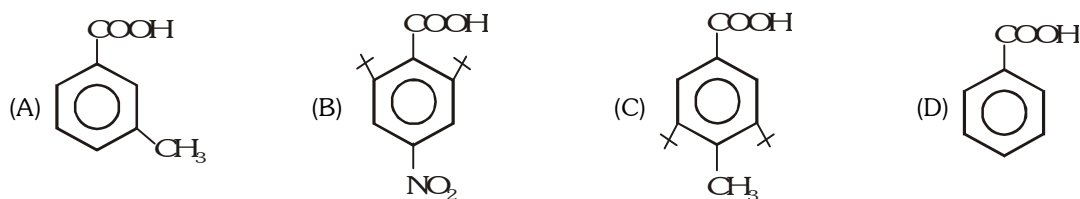
(1) Physical properties (2) Acidity and basicity (3) Reactivity of organic compounds

1. Arrange the following in the increasing order of basicity :

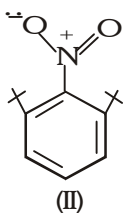
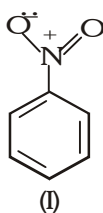


(A) I > II > III > IV (B) IV > III > II > I (C) II > I > IV > III (D) I > IV > III > II

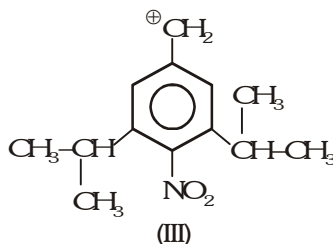
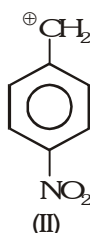
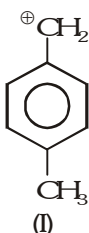
2. Which of the following is most acidic :



3. Consider the following two structures and choose the correct statements -



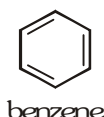
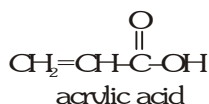
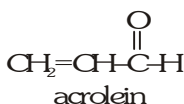
- (A) carbon-nitrogen bond length structure I is greater than that in structure II
 (B) carbon-nitrogen bond length in structure I is less than in structure II
 (C) carbon-nitrogen bond length in both structure is same
 (D) It can not be compared
4. Arrange the following carbocation in the increasing order of stability :



- (A) I < II < III (B) II < III < I (C) III < II < I (D) III < I < II

Comprehension # 4

1-Butene (A) and 1,3-Butadiene (B) differ not only in the number of π bonds, but (B) also has σ and π bonds at alternate positions. This type of the system is called conjugate system. Following are some of the conjugate system.



in such systems, π electron shifting takes place consecutively giving permanent polarity on the chain. This type of π -electron shift in the conjugate systems is called mesomeric effect or resonance.

Rules for resonance forms :

- Individual resonance forms are imaginary, not real
- Resonance forms differ only in the placement of their π electrons or nonbonding electrons.
- Different resonance forms of a substance don't have to be equivalent.
- Resonance forms must be valid Lewis structures and obey normal rules of valency.
- The resonance hybrid is more stable than any individual resonance.

Rules for stability :

- Structures with more covalent bonds are more stable than other structures.
- Structures in which all of the atoms have a complete valence shell of electrons (i.e. the noble gas structure) are especially stable and make large contribution to the hybrid.
- Stability is decreased by an increase in charge separation.
- Structure that carry negative charge on a more electronegative atom and positive charge on less electronegative atom are comparatively more stable.

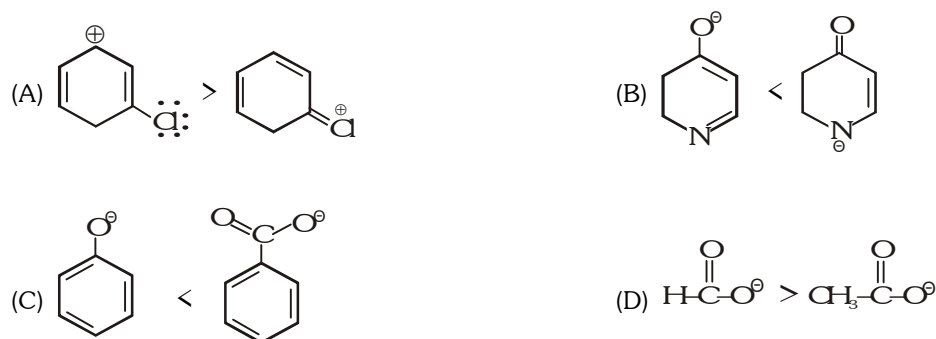
1. In which of the following compound, resonance is not possible :



2. Which of the following statement is incorrect -

- (A) In a resonance hybrid all the molecules are the same. A resonance hybrid cannot be expressed by any single structure.
 (B) The value of the resonance energy of any resonance hybrid is not an absolute value
 (C) The energy of hybrid structures is always less than that of any resonating structure
 (D) Only one individual resonating structure explains all characteristics of the molecule

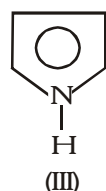
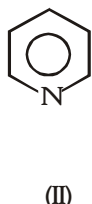
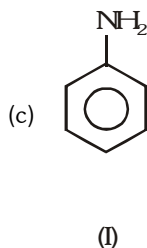
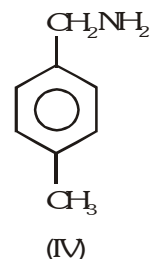
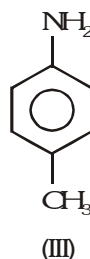
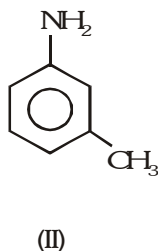
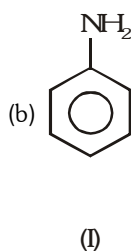
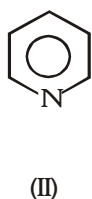
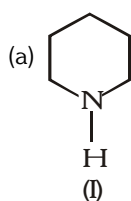
3. Choose the correct option about stability of resonating structure :



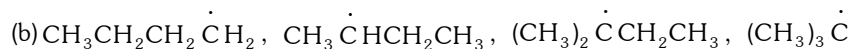
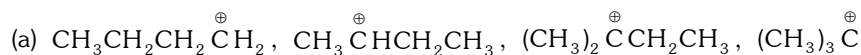
MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
• <u>True / False</u>		
1. F	2. F	3. T
4. F	5. F	
• <u>Fill in the Blanks</u>		
1. benzene	2. antiaromatic, aromatic	3. hyperconjugation
5. 2,3-diphenylbutane (optically inactive, meso)	4. nucleophile, base	
• <u>Match the Column</u>		
1. (A) → p,s ; B → q,r ; (C) → q,s ; (D) → p,s	2. (A) → q ; (B) → p ; (C) → p ; (D) → r	
3. (A) → p ; (B) → q ; (C) → p ; (D) → q		
• <u>Assertion - Reason Questions</u>		
1. B	2. A	3. A
4. C	5. A	
6. C	7. D	8. B
9. D	10. C	
• <u>Comprehension Based Questions</u>		
Comprehension #1 : 1. (A)	2. (A)	3. (B)
4. (B)	5. (C)	6. (C)
Comprehension #2 : 1. (D)	2. (C)	3. (C)
4. (D)	5. (B)	
Comprehension #3 : 1. (C)	2. (B)	3. (B)
4. (B)		
Comprehension #4 : 1. (A)	2. (D)	3. (C,D)

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

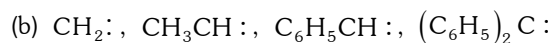
1. Rank the following amines in increasing basic nature :



2. Rank the following intermediates according to the stability (most stable first).



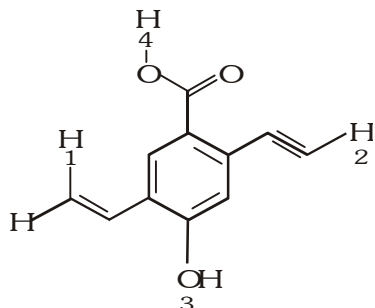
3. Rank the following intermediates according to the stability (most stable first).



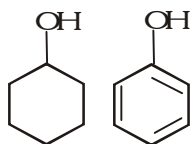
4. Show why the phenolate ion, $\text{C}_6\text{H}_5\text{O}^-$, has a greater resonance stabilization than phenol, $\text{C}_6\text{H}_5\text{OH}$.

5. Is phenol a stronger acid than acetic acid. Explain your answer with proper reasoning.

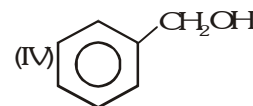
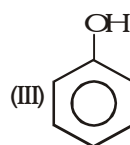
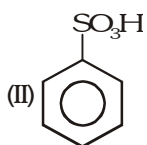
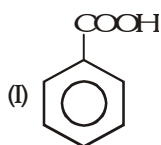
6. Arrange all these numbered H in order of their decreasing acidic strength.



7. Between $\begin{array}{c} | & | \\ \text{---C} & \text{---C---} \end{array} \ddot{\text{F}}$ and $\begin{array}{c} | & | \\ \text{---C} & \text{---C---} \end{array} \ddot{\text{Cl}}$ which lone pair- p conjugation will be more preferred.
8. Which is more acidic and why ?



9. Arrange the following compound in the correct order of acidity :

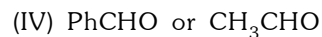
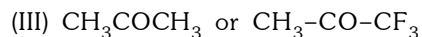
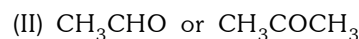
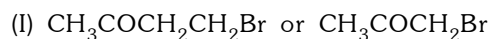


10. Why a cation like is not possible.

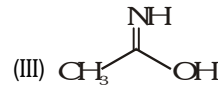
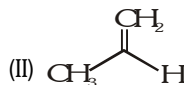
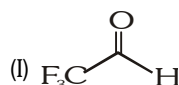
11. (A) Product.

12. $\text{CH}_2=\text{CHCHO} + \text{C}_2\text{H}_5\text{MgBr} \xrightarrow{\text{H}_3\text{O}^+} \text{P}$; Here P is :

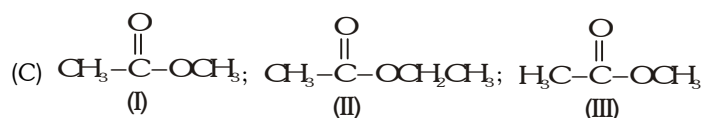
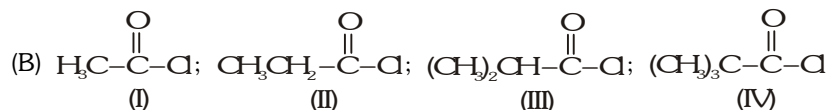
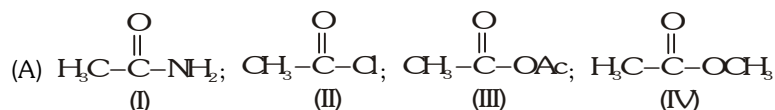
13. Within each set, select the compound which is more reactive in nucleophilic addition at carbonyl carbon :



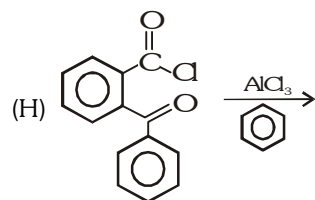
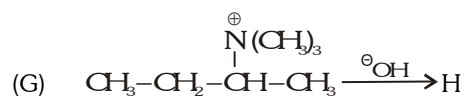
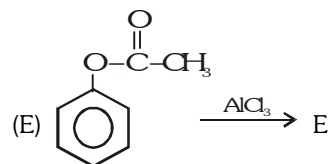
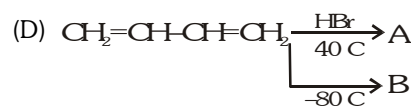
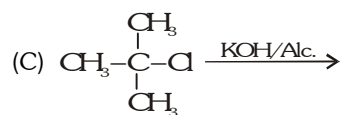
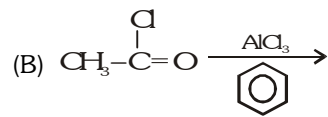
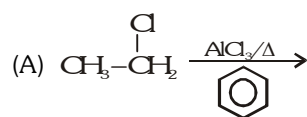
14. Rank of the following compounds in increasing order of electrophilicity :



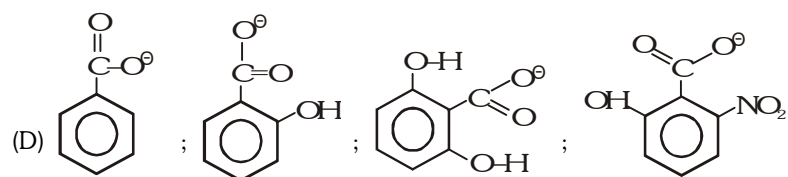
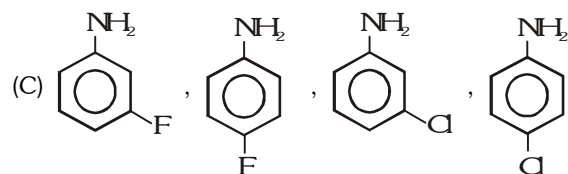
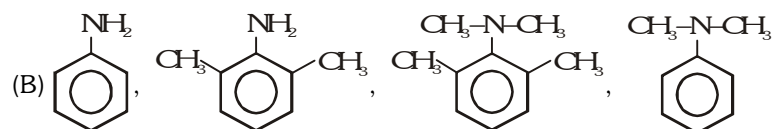
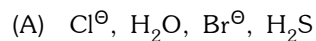
15. Arrange the followings in increasing order of reactivity towards nucleophile :



16. Suggest the product of the reactions :



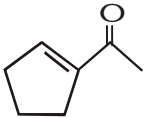
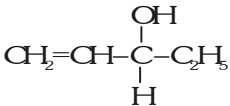
17. Arrange the following in increasing order of basic character :



18. Heat of hydrogenation of $\text{CH}_2=\text{CH}_2$ is greater than $\text{CH}_3-\text{CH}=\text{CH}_2$. Why ?

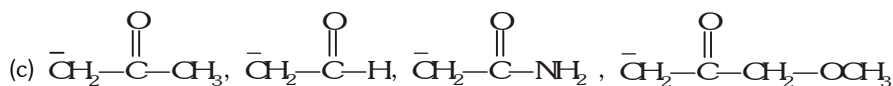
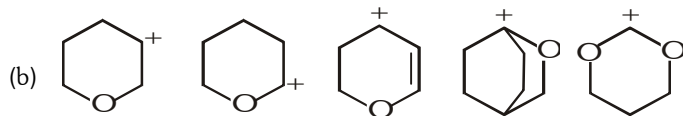
19. In the case of $\left[\text{CH}_3-\text{CH}_2-\overset{\oplus\text{N}(\text{CH}_3)_3}{\underset{|}{\text{CH}}}-\text{CH}_3 \right] \text{OH}^-$, less stable 1-Butene is obtained. Why ?

20. When carbene attacks cis 2-butene singlet gives cis product while triplet gives trans product. Why ?

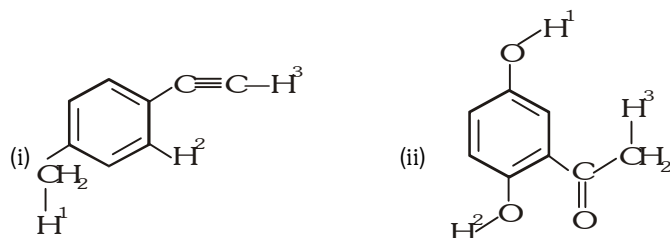
1. (a) $I > III > II > IV$ (b) $IV > III > II > I$ (c) $II > I > III$
2. (a) $(CH_3)_3\overset{\oplus}{C} > (CH_3)_2\overset{\oplus}{C}CH_2CH_3 > CH_3\overset{\oplus}{C}HCH_2CH_3 > CH_3CH_2CH_2\overset{\oplus}{C}H_2$
 (b) $(CH_3)_3\dot{C} > (CH_3)_2\dot{C}CH_2CH_3 > CH_3\dot{C}HCH_2CH_3 > CH_3CH_2CH_2\dot{C}H_2$
3. (a) $CH_3CH_2CH_2\bar{C}H_2 > CH_3\bar{C}HCH_2CH_3 > (CH_3)_2\bar{C}CH_2CH_3 > (CH_3)_3\bar{C}$
 (b) $(C_6H_5)_2C: > C_6H_5CH: > CH_3CH: > CH_2:$
4. Free charge has more reactivity than bound charge.
5. No
 Anion of acetic acid is more stabilised by resonance than phenoxide ion.
6. $4 > 3 > 2 > 1$
7. In $\begin{array}{c} | \quad | \\ -C=C-\ddot{F} \end{array}$, lone pair - p conjugation will be more due to more overlapping of orbitals involved in conjugation.
8. Phenol is more acidic as after loss of H^+ , phenoxide ion is Resonance stabilised. No such resonance is seen in conjugate base of cyclohexanol.
9. $II > I > III > IV$
10. Due to the non-planar structure.
11. 
12. 

EXERCISE-04 [B]**BRAIN STORMING SUBJECTIVE EXERCISE**

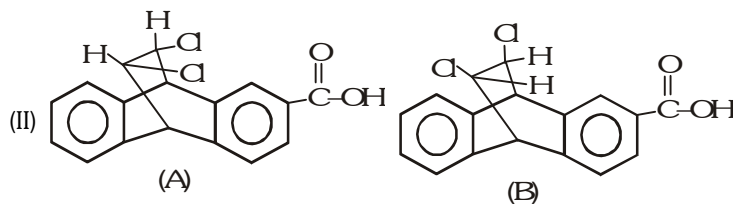
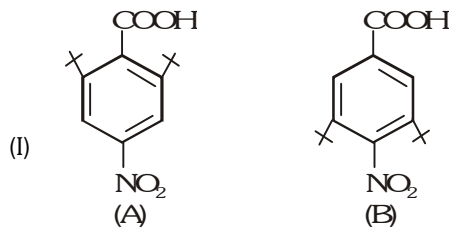
1. Rank the following sets of intermediates in increasing order of their stability giving appropriate reasons for your choice.



2. For the following compounds, arrange the labelled proton in increasing order of their ease of deprotonation :



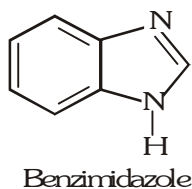
3. Benzene sulphonic acid is a stronger acid than benzoic acid, explain.
4. Which is a stronger acid, A or B and why ?



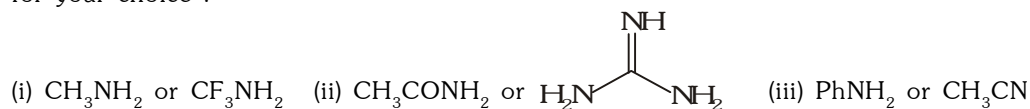
5. Discuss the following observations :

- (a) C-Cl bond in vinyl chloride is stronger than in chloroethane.
- (b) Carbon-carbon bond length in ethene is shorter than in $\text{CH}_2=\text{CHOCH}_3$.
- (c) CH_3SH is stronger acid than CH_3OH .
- (d) $\text{CH}_3\text{CH}_2\text{NH}_2$ is stronger base than $\text{CH}_2=\text{CHNH}_2$.

6. Discuss the basic strength of two nitrogens in benzimidazole.

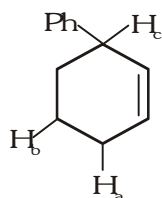


7. In each of the following pair of compounds, which is more basic in aqueous solution ? Give an explanation for your choice :



8. Answer the following questions :

- (i) Which of the indicated H is abstracted rapidly by bromine radical and why ?



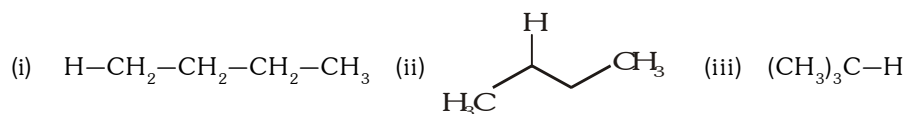
- (ii) One of the indicated proton H_a or H_b , is approximately 10^{30} times more acidic than other, which is more acidic and why ?



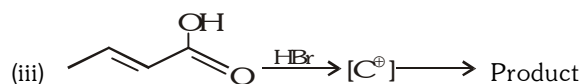
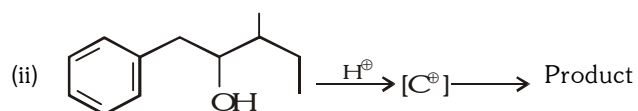
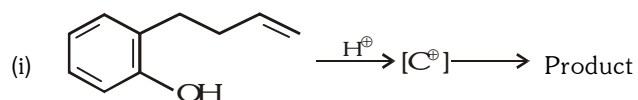
9. Explain the following

- (i) $\text{NC}-\text{CH}_2-\text{CN}$ is a stronger acid than $\text{CH}_3-\text{CO}-\text{C}_2\text{H}_5$.
 (ii) $\text{CH}_3-\text{CO}-\text{C}_2\text{H}_5$ is more acidic than $\text{CH}_3-\text{CH}=\text{CH}_2$.
 (iii) $\text{CH}_3-\text{CH}=\text{CH}-\text{OH}$ is more acidic than $\text{CH}_2=\text{CH}-\text{CH}_2-\text{OH}$.

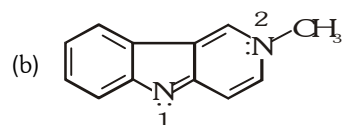
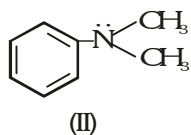
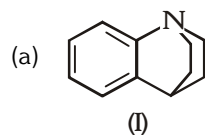
10. Compare the bond dissociation energies of C-H bonds in the following



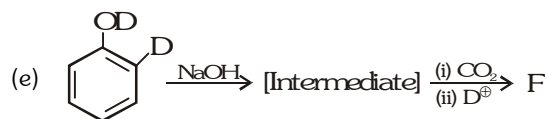
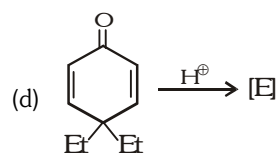
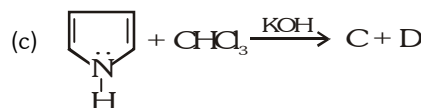
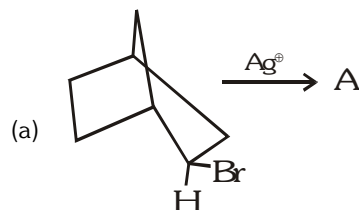
11. The most stable intermediary carbocation and major product in following reactions :



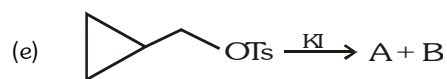
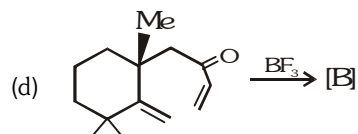
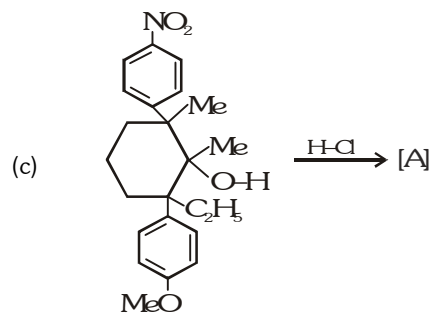
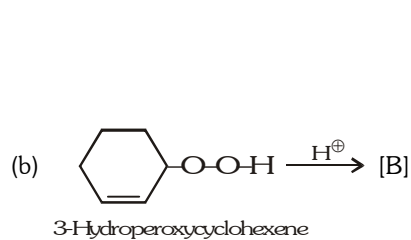
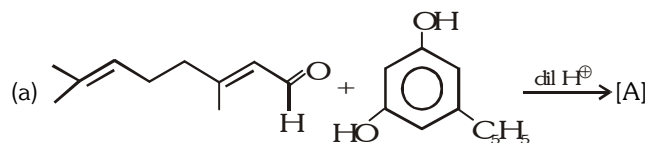
12. Which one of the following is more basic and why ?



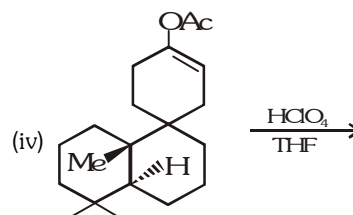
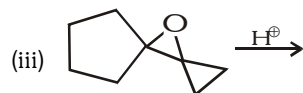
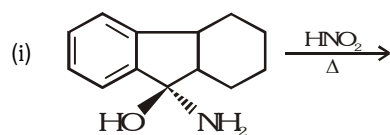
13. Find the product in following reaction :



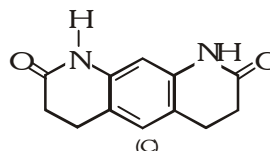
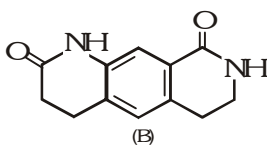
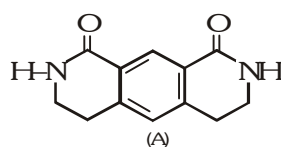
14. Write the mechanism of following reaction :



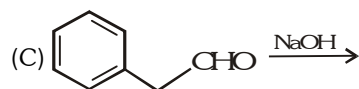
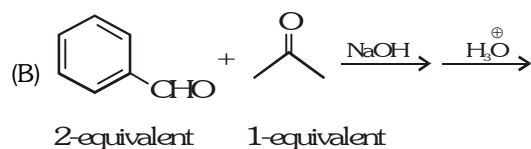
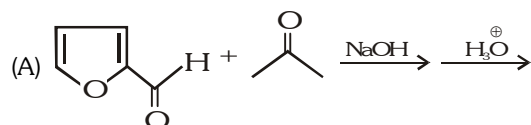
15. Find the major product in the reaction the following :



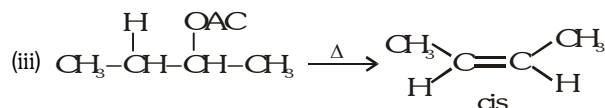
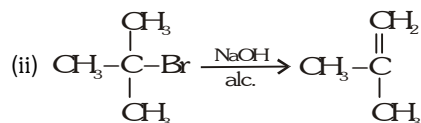
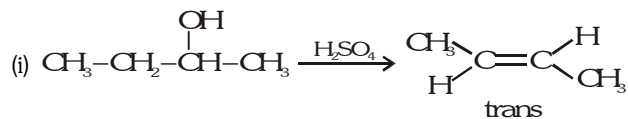
16. Decreasing order of E.S.R.

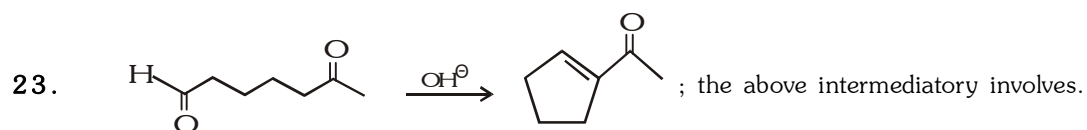
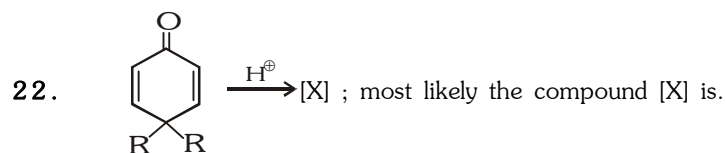
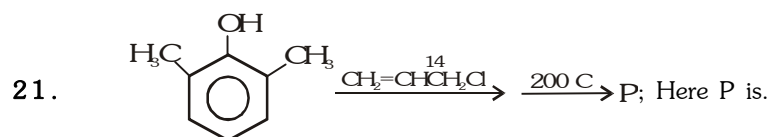
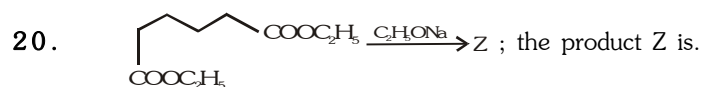
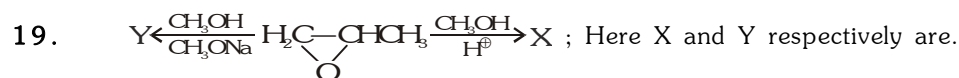
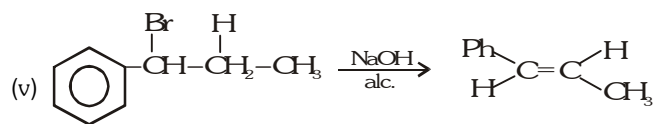
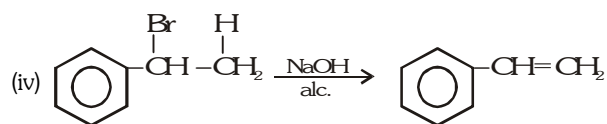


17. Complete the following equations :

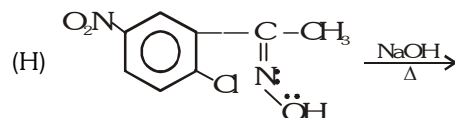
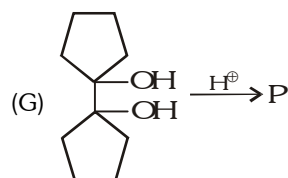
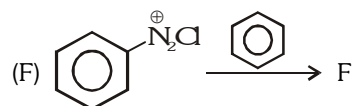
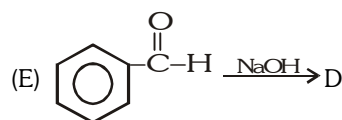
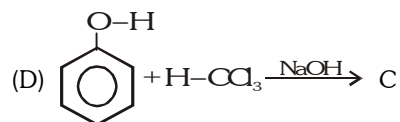
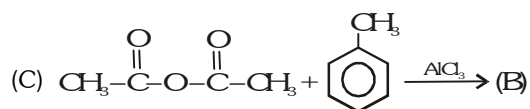
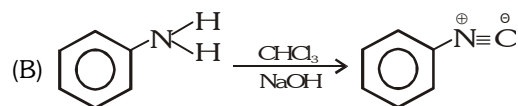
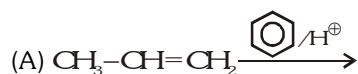


18. Suggest the mechanism of the following reaction :

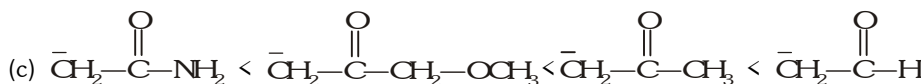
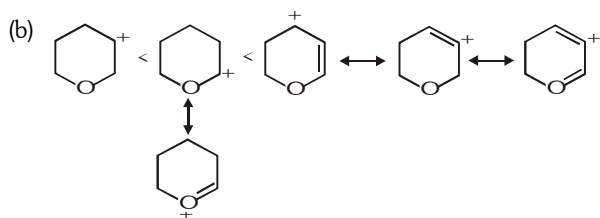




24. Suggest the product of the reactions :



1. (a) $\text{O}_2\text{NCH}_2\text{CH}_3 < \text{CH}_2\text{CH}_3 < \text{CH}_3\text{CH}_2\text{CH}_3 < \text{C}_6\text{H}_5\text{CH}_2\text{CH}_3 < (\text{C}_6\text{H}_5)_2\text{CH}_2$



2. (i) $2 < 1 < 3$ (ii) $3 < 1 < 2$

3. Conjugate base of benzene sulphonic acid has greater resonance stabilization than benzoate ion, hence benzene sulphonic acid is stronger acid than benzoic acid. PhSO_3^- has three equivalent resonance structure, so more stable. but PhCOO^- has only two equivalent resonance structure, so less stable.

4. B is stronger base than A. Here basic strength is controlled by $-\text{R}$ effect of $-\text{NO}_2$. In B, due to steric inhibition to resonance of $-\text{NO}_2$ by two bulky tertiary butyl group, it is not decreasing basic strength by $-\text{R}$ effect, hence B is stronger base than A.

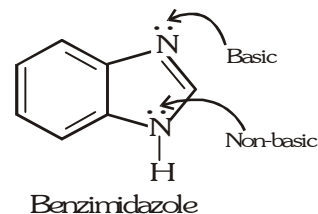
5. (a) Due to resonance in vinyl chloride, $\text{C}-\text{Cl}$ bond acquire some double bond character while in chloroethane, $\text{C}-\text{Cl}$ bond is pure, single sigma covalent bond.

(b) Due to resonance in methyl vinyl ether, bond order of $\text{C}-\text{C}$ is slightly reduced from two while in ethene $\text{C}-\text{C}$ bond order is exactly 2.

(c) Larger size of sulphur compared to oxygen gives greater acidity to CH_3SH .

(d) Lone-pair of electron in vinyl amine is involved partly in resonance delocalization and less available for donation to a Lewis acid, hence a weaker Lewis base.

6. Lone pair of the basic nitrogen is fully available with nitrogen, i.e. not involved in delocalization while lone pair of non-basic nitrogen is part of aromatic delocalization.



7. (i) CH_3NH_2 , (ii) $\text{H}_2\text{N}-\text{C}(\text{NH}_2)=\text{NH}$ (iii) PhNH_2

8. (i) $\text{H}_c > \text{H}_a > \text{H}_b$ (ii) H_a : Conjugate base is aromatic

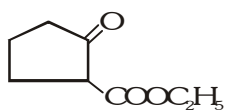
9. Acidic strength $\propto$ stability of anion formed after removal of $\text{H}^+ \propto$ number of e^- withdrawing groups.

10. $\text{III} > \text{II} > \text{I}$

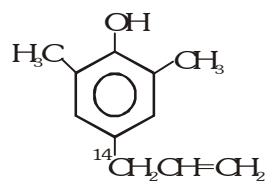
$$\text{B.D.E.} \propto \frac{1}{\text{stability of intermediate formed}}$$

19. X : $\text{HOCH}_2-\text{CH}(\text{OCH}_3)-\text{CH}_3$ and Y : $\text{CH}_2=\text{CH}(\text{OCH}_3)-\text{CH}_3$

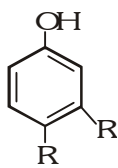
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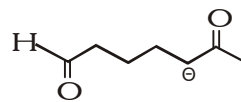
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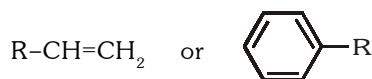


23.



EXERCISE-05(A)**PREVIOUS YEARS QUESTIONS**

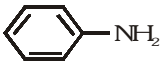
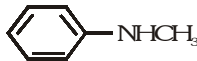
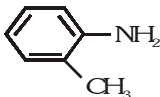
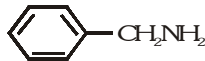
1. In the following benzyl/allyl system



[AIEEE 2002]

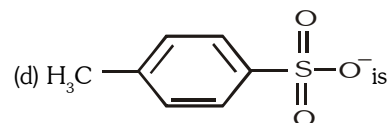
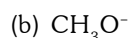
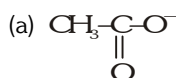
(R is alkyl group)

decreasing order of inductive effect is-

- (1) $(CH_3)_3C- > (CH_3)_2CH- > CH_3CH_2-$
(2) $CH_3-CH_2- > (CH_3)_2CH- > (CH_3)_3C-$
(3) $(CH_3)_2CH- > CH_3CH_2- > (CH_3)_3CH-$
(4) None of these
2. In the anion $HCOO^-$ the two carbon-oxygen bonds are found to be of equal length. What is the reason for it- [AIEEE 2003]
- (1) Electronic orbits of carbon atoms are hybridised
(2) The $C=O$ bond is weaker than the $C-O$ bond
(3) The anion $HCOO^-$ has two resonating structure
(4) The anion is obtained by removal of a proton from the acid molecule
3. The correct order of increasing basic nature for the bases NH_3 , CH_3NH_2 and $(CH_3)_2NH$ is-[AIEEE 2003]
- (1) $CH_3NH_2 < NH_3 < (CH_3)_2NH$
(2) $(CH_3)_2NH < NH_3 < CH_3NH_2$
(3) $NH_3 < CH_3NH_2 < (CH_3)_2NH$
(4) $CH_3NH_2 < (CH_3)_2NH < NH$
4. Consider the acidity of the carboxylic acids- [AIEEE 2004]
- (i) $PhCOOH$ (ii) $o-NO_2C_6H_4COOH$
(iii) $p-NO_2C_6H_4COOH$ (iv) $m-NO_2C_6H_4COOH$
- which of the following order is correct for acidic strength-
- (1) $i > ii > iii > iv$ (2) $ii > iv > iii > i$
(3) $ii > iv > i > iii$ (4) $ii > iii > iv > i$
5. Which of the following is the strongest base- [AIEEE 2004]
- (1)  (2)  (3)  (4) 
6. Rate of the reaction $R-\overset{\overset{O}{\parallel}}{C}-Z + Nu^- \rightarrow R-\overset{\overset{O}{\parallel}}{C}-Nu + Z^-$ is fastest when Z is [AIEEE 2004]
- (1) $OCOCH_3$ (2) NH_2
(3) OC_2H_5 (4) Cl
7. Which one of the following does not have sp^2 hybridised carbon [AIEEE 2004]
- (1) Acetamide (2) Acetic acid
(3) Acetonitrile (4) Acetone
8. Due to the presence of an unpaired electron, free radicals are [AIEEE-2005]
- (1) Chemically inactive (2) Chemically reactive
(3) Cations (4) Anions

9. The decreasing order of nucleophilicity among the nucleophiles

[AIEEE-2005]



(1) (d), (c), (b), (a)

(2) (a), (b), (c), (d)

(3) (c), (b), (a), (d)

(4) (b), (c), (a), (d)

10. Tertiary alkyl halides are practically inert to substitution by SN^2 mechanism because of [AIEEE-2005]

(1) unstability

(2) insolubility

(3) steric hinderance

(4) inductive effect

11. Among the following acids which has the lowest pK_a value- [AIEEE-2005]

(1) $\text{CH}_3\text{CH}_2\text{COOH}$

(2) $(\text{CH}_3)_2\text{CHCOOH}$

(3) HCOOH

(4) CH_3COOH

12. Amongst the following the most basic compound is- [AIEEE-2005]

(1) p-nitro aniline

(2) Acetanilide

(3) Aniline

(4) Benzylamine

13. The increasing order of the rate of HCN addition to compounds A-D is [AIEEE-2006]

(A) HCHO

(B) CH_3COCH_3

(C) PhCOCH_3

(D) PhCOPh

(1) $\text{D} < \text{B} < \text{C} < \text{A}$

(2) $\text{D} < \text{C} < \text{B} < \text{A}$

(3) $\text{C} < \text{D} < \text{B} < \text{A}$

(4) $\text{A} < \text{B} < \text{C} < \text{D}$

14. The increasing order of stability of the following free radicals is [AIEEE-2006]

(1) $(\text{C}_6\text{H}_5)_3\dot{\text{C}} < (\text{C}_6\text{H}_5)_2\dot{\text{C}}\text{H} < (\text{CH}_3)_3\dot{\text{C}} < (\text{CH}_3)_2\dot{\text{C}}\text{H}$

(2) $(\text{C}_6\text{H}_5)_2\dot{\text{C}}\text{H} < (\text{C}_6\text{H}_5)_3\dot{\text{C}} < (\text{CH}_3)_3\dot{\text{C}} < (\text{CH}_3)_2\dot{\text{C}}\text{H}$

(3) $(\text{CH}_3)_2\dot{\text{C}}\text{H} < (\text{CH}_3)_3\dot{\text{C}} < (\text{C}_6\text{H}_5)_3\dot{\text{C}} < (\text{C}_6\text{H}_5)_2\dot{\text{C}}\text{H}$

(4) $(\text{CH}_3)_2\dot{\text{C}}\text{H} < (\text{CH}_3)_3\dot{\text{C}} < (\text{C}_6\text{H}_5)_2\dot{\text{C}}\text{H} < (\text{C}_6\text{H}_5)_3\dot{\text{C}}$

15. $\text{CH}_3\text{Br} + \text{Nu}^- \longrightarrow \text{CH}_3 - \text{Nu} + \text{Br}^-$

The decreasing order of the rate of the above reaction with nucleophiles (Nu^-) A to D is [AIEEE-2006]

$[\text{Nu}^- = (\text{A}) \text{PhO}^-, (\text{B}) \text{AcO}^-, (\text{C}) \text{HO}^-, (\text{D}) \text{CH}_3\text{O}^-]$

(1) $\text{D} > \text{C} > \text{B} > \text{A}$

(2) $\text{A} > \text{B} > \text{C} > \text{D}$

(3) $\text{B} > \text{D} > \text{C} > \text{A}$

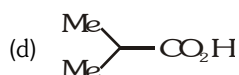
(4) $\text{D} > \text{C} > \text{A} > \text{B}$

16. The correct order of increasing acid strength of the compounds is [AIEEE-2006]

(a) $\text{CH}_3\text{CO}_2\text{H}$

(b) $\text{MeOCH}_2\text{CO}_2\text{H}$

(c) $\text{CF}_3\text{CO}_2\text{H}$

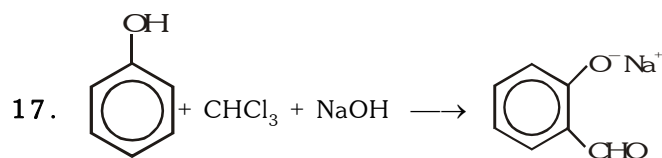


(1) $\text{d} < \text{a} < \text{c} < \text{b}$

(2) $\text{d} < \text{a} < \text{b} < \text{c}$

(3) $\text{a} < \text{d} < \text{c} < \text{b}$

(4) $\text{b} < \text{d} < \text{a} < \text{c}$



The electrophile involved in the above reaction is

[AIEEE-2006]

- (1) dichlorocarbene (:CCl₂)
- (2) trichloromethyl anion (CCl₃)[⊖]
- (3) formyl cation (CHO)[⊕]
- (4) dichloromethyl cation (CHCl₂)[⊕]

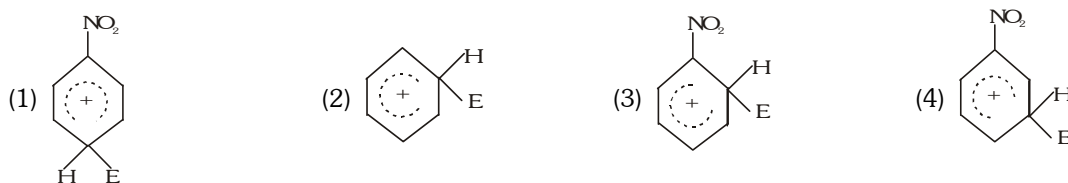
18. Which one of the following is the strongest base in aqueous solution ?

[AIEEE-2007]

- (1) Trimethylamine
- (2) Aniline
- (3) Dimethylamine
- (4) Methylamine

19. The electrophile, E[⊕] attacks the benzene ring to generate the intermediate σ-complex. Of the following, which σ-complex is of lowest energy?

[AIEEE-2008]



20. Arrange the carbanions, (CH₃)₃C[⊖], CCl₃[⊖], (CH₃)₂CH[⊖], C₆H₅CH₂[⊖], in order of their decreasing stability :-

[AIEEE-2009]

- (1) CCl₃[⊖] > C₆H₅CH₂[⊖] > (CH₃)₂CH[⊖] > (CH₃)₃C[⊖]
- (2) (CH₃)₃C[⊖] > (CH₃)₂CH[⊖] > C₆H₅CH₂[⊖] > CCl₃[⊖]
- (3) C₆H₅CH₂[⊖] > CCl₃[⊖] > (CH₃)₃C[⊖] > (CH₃)₂CH[⊖]
- (4) (CH₃)₂CH[⊖] > CCl₃[⊖] > C₆H₅CH₂[⊖] > (CH₃)₃C[⊖]

21. The correct order of increasing basicity of the given conjugate base (R=CH₃) is :-

[AIEEE-2010]

- (1) RCOO[⊖] < HC≡C[⊖] < NH₂[⊖] < R[⊖]
- (2) RCOO[⊖] < HC≡C[⊖] < R[⊖] < NH₂[⊖]
- (3) R[⊖] < HC≡C[⊖] < RCOO[⊖] < NH₂[⊖]
- (4) RCOO[⊖] < NH₂[⊖] < HC≡C[⊖] < R[⊖]

22. The strongest acid amongst the following compounds is ?

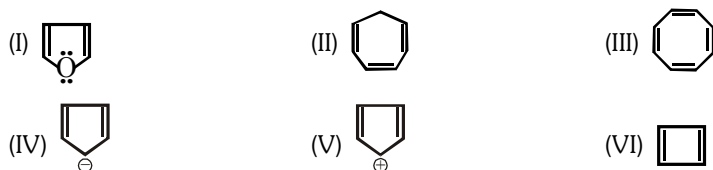
[AIEEE-2011]

- (1) CH₃CH₂CH(Cl)CO₂H
- (2) ClCH₂CH₂CH₂COOH
- (3) CH₃COOH
- (4) HCOOH

23. Ortho-Nitrophenol is less soluble in water than p- and m- Nitrophenols because :- [AIEEE-2012]

- (1) Melting point of o-Nitrophenol is lower than those of m- and p- isomers
- (2) o-Nitrophenol is more volatile in steam than those of m- and p- isomers
- (3) o-Nitrophenol shows Intramolecular H-bonding
- (4) o-Nitrophenol shows Intermolecular H-bonding

24. Which of the following compounds are antiaromatic :- [AIEEE-2012(Online)]



- (1) (III) and (VI)
- (2) (II) and (V)
- (3) (I) and (V)
- (4) (V) and (VI)

25. In the following compounds : [AIEEE-2012(Online)]



the order of basicity is as follows :

- (1) IV > III > II > I
- (2) II > III > I > IV
- (3) I > III > II > IV
- (4) III > I > II > IV

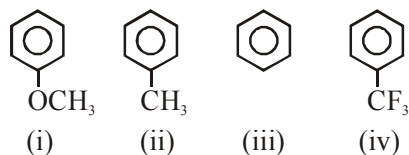
26. Dipole moment is shown by :- [AIEEE-2012(Online)]

- (1) trans-2, 3-dichloro- 2-butene
- (2) 1, 2-dichlorobenzene
- (3) 1, 4-dichlorobenzene
- (4) trans-1, 2-dinitroethene

27. Among the following chloro-compound having the lowest dipole moment is :- [AIEEE-2012(Online)]



28. In the below mentioned compounds the decreasing order of reactivity towards electrophilic substitution is : [AIEEE-2012(Online)]



- (1) (iii) > (i) > (iv) > (ii)
- (2) (iv) > (i) > (ii) > (iii)
- (3) (ii) > (iii) > (i) > (iv)
- (4) (i) > (ii) > (iii) > (iv)

29. Among the following the molecule with the lowest dipole moment is :- [AIEEE-2012(Online)]

- (1) CHCl_3
- (2) CH_2Cl_2
- (3) CCl_4
- (4) CH_3Cl

30. The most basic compound among the following is :- [AIEEE-2012(Online)]

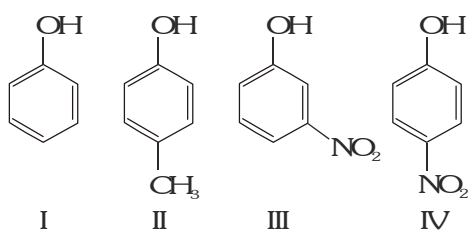
- (1) Acetanilide
- (2) Benzylamine
- (3) p-Nitro aniline
- (4) Aniline

EXERCISE-05(B)**PREVIOUS YEARS QUESTIONS**

1. Which one of the following has the smallest heat of hydrogenation per mole ? [IIT-93]
- (A) 1-Butene (B) trans-2-Butene
(C) cis-2-Butene (D) 1, 3-Butadiene

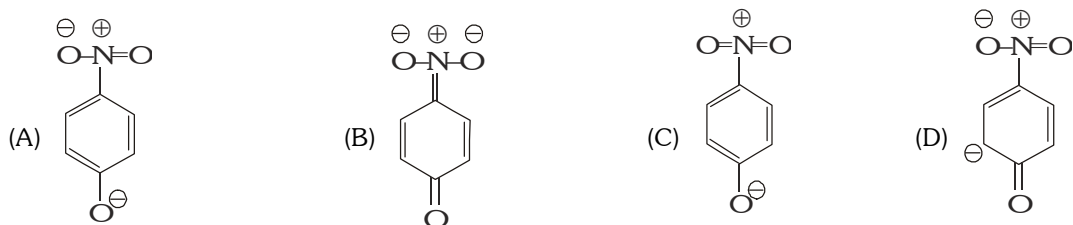
2. What is the decreasing order of strength of bases $\text{OH}^\ominus$, $\text{NH}_2^\ominus$, $\text{HC}\equiv\text{C}^\ominus$, $\text{CH}_3\text{-CH}_2^\ominus$ [IIT-93]
- (A) $\text{CH}_3\text{-CH}_2^\ominus > \text{NH}_2^\ominus > \text{HC}\equiv\text{C}^\ominus > \text{OH}^\ominus$ (B) $\text{HC}\equiv\text{C}^\ominus > \text{CH}_3\text{-CH}_2^\ominus > \text{NH}_2^\ominus > \text{OH}^\ominus$
(C) $\text{OH}^\ominus > \text{NH}_2^\ominus > \text{HC}\equiv\text{C}^\ominus > \text{CH}_3\text{-CH}_2^\ominus$ (D) $\text{NH}_2^\ominus > \text{HC}\equiv\text{C}^\ominus > \text{OH}^\ominus > \text{CH}_3\text{-CH}_2^\ominus$

3. In the following compounds [IIT-96]



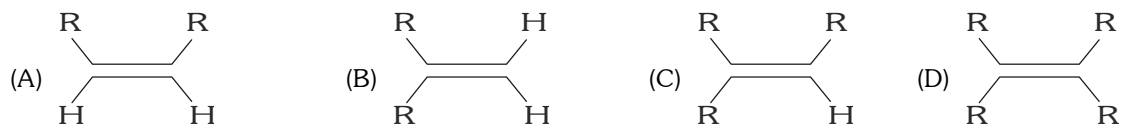
The order of acidity is -

- (A) III > IV > I > II (B) I > IV > III > II
(C) II > I > III > IV (D) IV > III > I > II
4. The formation of cyanohydrin from a ketone is an examples of - [IIT-98]
- (A) Electrophilic addition (B) Nucleophilic addition
(C) Nucleophilic substitution (D) Electrophilic substitution
5. The most unlikely representation of resonance structure of p-nitrophenoxide ion is - [IIT-99]

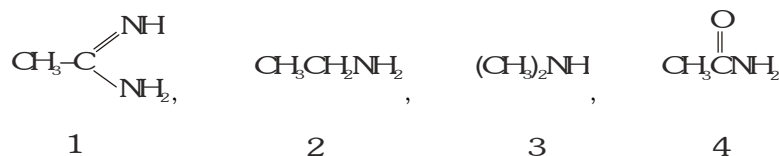


6. An aromatic molecule will not [IIT-99]
- (A) have $4n$ π electrons (B) have $(4n + 2)\pi$ electrons
(C) be planar (D) be cyclic
7. Amongst the following, the most basic compound is - [IIT-2000]
- (A) $\text{C}_6\text{H}_5\text{NH}_2$ (B) $\text{p-NO}_2\text{-C}_6\text{H}_4\text{NH}_2$
(C) $\text{m-NO}_2\text{-C}_6\text{H}_4\text{NH}_2$ (D) $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$

8. Which of the following alkenes will react fastest with H_2 under catalytic hydrogenation conditions [IIT-2001]



9. The correct order of basicities of the following compounds is : [IIT-2001]

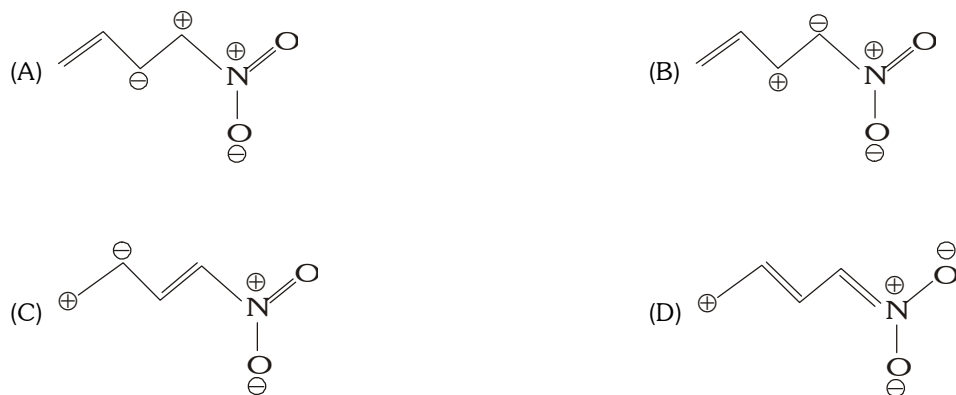


- (A) $2 > 1 > 3 > 4$ (B) $1 > 3 > 2 > 4$ (C) $3 > 1 > 2 > 4$ (D) $1 > 2 > 3 > 4$ [IIT-2005]

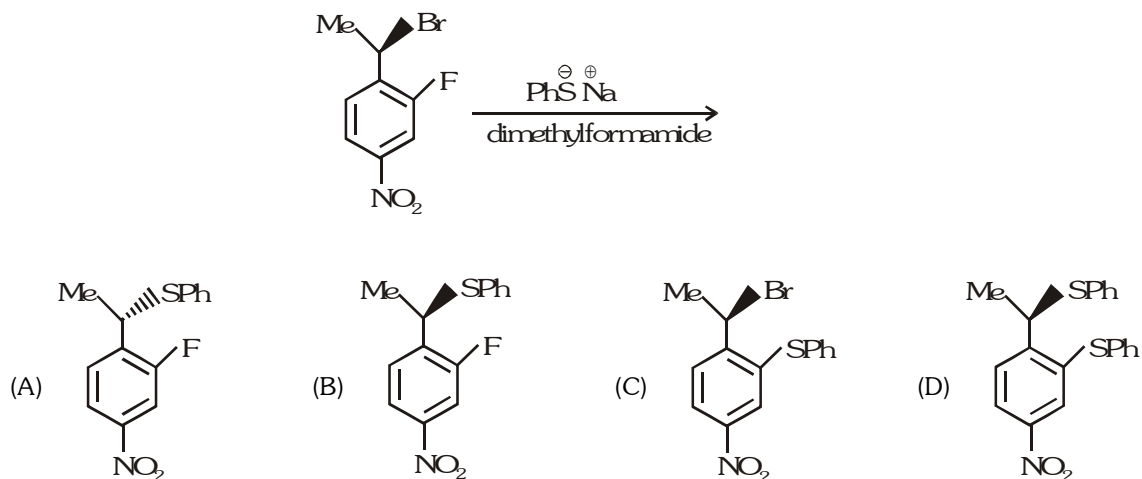
10. Which of the following is least stable :



11. Among the following, the least stable resonance structure is - [IIT-2007]

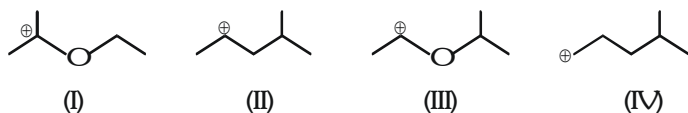


12. The major product of the following reaction is : [IIT-2008]



13. The correct stability order for the following species is :

[IIT-2008]



- (A) II > IV > I > III (B) I > II > III > IV (C) II > I > IV > III (D) I > III > II > IV

14. **Statement-I** : p-Hydroxybenzoic acid has a lower boiling point than o-hydroxybenzoic acid.

Because

Statement-II : o-Hydroxybenzoic acid has intramolecular hydrogen bonding.

[IIT 2003]

- (A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I
 (B) Statement-I is True, Statement-II is True ; Statement-II is NOT a correct explanation for Statement-I
 (C) Statement-I is True, Statement-II is False.
 (D) Statement-I is False, Statement-II is True.

15. **Statement-I** : Bromobenzene upon reaction with Br_2/Fe gives 1, 4-dibromobenzene as the major product.

Because

Statement-II : In bromobenzene, the inductive effect of the bromo group is more dominant than the mesomeric effect in directing the incoming electrophile.

[IIT 2008]

- (A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I
 (B) Statement-I is True, Statement-II is True ; Statement-II is NOT a correct explanation for Statement-I
 (C) Statement-I is True, Statement-II is False.
 (D) Statement-I is False, Statement-II is True.

16. Out of anhydrous AlCl_3 and hydrous AlCl_3 which is more soluble in diethyl ether ? Explain with reason.

[IIT 2003]

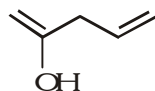
17. Match K_a values with suitable acid :

[IIT 2003]

K_a		Acid	
(A)	3.3×10^{-5}	(p)	
(B)	4.2×10^{-5}	(q)	
(C)	6.3×10^{-5}	(r)	
(D)	6.4×10^{-5}	(s)	
(E)	30.6×10^{-5}	(t)	

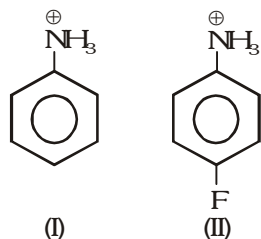
18. Give resonating structures of following compound.

[IIT 2003]



19. Which of the following is more acidic and why ?

[IIT 2004]



20. The compound that does NOT liberate CO_2 , on treatment with aqueous sodium bicarbonate solution, is - [JEE ADVANCED 2013]

- (A) Benzoic acid (B) Benzenesulphonic acid
(C) Salicylic acid (D) Carbolic acid (phenol)

21. The hyperconjugative stabilities of tert-butyl cation and 2-butene, respectively, are due to

[JEE ADVANCED 2013]

- (A) $\sigma \rightarrow p$ (empty) and $\sigma \rightarrow \pi^*$ electron delocalisations
(B) $\sigma \rightarrow \sigma^*$ and $\sigma \rightarrow \pi$ electron delocalisations
(C) $\sigma \rightarrow p$ (filled) and $\sigma \rightarrow \pi$ electron delocalisations
(D) p (filled) $\rightarrow \sigma^*$ and $\sigma \rightarrow \pi^*$ electron delocalisations

22. The total number of lone-pairs of electrons in melamine is

[JEE ADVANCED 2013]

PREVIOUS YEARS QUESTIONS				ANSWER KEY			EXERCISE -5(B)
1. (D)	2. (A)	3. (D)	4.(B)	5. (C)	6. (A)	7.(D)	8. (A)
9. (B)	10. (D)	11. (A)	12. (A)	13. (D)	14. (D)	15. (C)	
16. Anhydrous AlCl_3 is more soluble.							
17. A-(s) ; B-(q) ; C-(p) ; D-(r) ; E-(t) ;							
19. II is more acidic.		20. (D)		21. (A)		22. (6)	

HYDROGEN, POLYMER, CHEMISTRY IN EVERYDAY LIFE & ENVIRONMENTAL CHEMISTRY

DIHYDROGEN

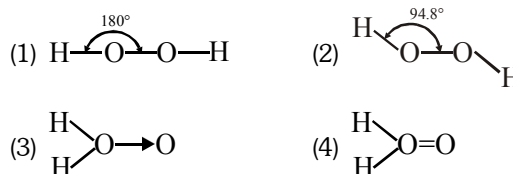
- The sum number of neutrons and protons in one of the isotopes of hydrogen is :-
(1) 3 (2) 4 (3) 5 (4) 6
- The catalyst used in Bosch process of manufacture of H_2 is :-
(1) Finely divided Ni (2) V_2O_5
(3) Pb (4) $Fe_2O_3 + Cr_2O_3$
- The most abundant isotope of hydrogen is :-
(1) Tritium (2) Deuterium
(3) Protium (4) Para hydrogen
- The n/p ratio for ${}_1H^1$ is :-
(1) 1 (2) 2 (3) 3 (4) Zero
- Ordinary hydrogen at room temperature is a mixture of :-
(1) 75% o-Hydrogen + 25% p-Hydrogen
(2) 25% o-Hydrogen + 75% p-Hydrogen
(3) 50% o-Hydrogen + 50% p-Hydrogen
(4) 1% o-Hydrogen + 99% p-Hydrogen
- In all its properties, hydrogen resembles :-
(1) Alkali metals only
(2) Halogens only
(3) Both alkali metals and halogens
(4) Neither alkali metals nor halogens
- Hydrogen is :-
(1) Electropositive
(2) Electronegative
(3) Both electropositive as well as electro-negative
(4) Neither electropositive nor electronegative
- Which of the following will not produce hydrogen gas :-
(1) Reaction between Fe and dil. HCl
(2) Reaction between Zn and conc. H_2SO_4
(3) Reaction between Zn and NaOH
(4) Electrolysis of NaCl in Nelson's cell
- Para hydrogen is :-
(1) Less stable than ortho hydrogen
(2) More stable than ortho hydrogen
(3) As stable as ortho hydrogen
(4) None of these

WATER (H_2O)

- Both temporary and permanent hardness in water is removed by :-
(1) Boiling (2) Filtration
(3) Distillation (4) Decantation
- Both temporary and permanent hardness is removed on boiling water with :-
(1) $Ca(OH)_2$ (2) Na_2CO_3
(3) $CaCO_3$ (4) CaO
- Temporary hardness is caused due to the presence of :-
(1) $CaSO_4$ (2) $CaCl_2$
(3) $CaCO_3$ (4) $Ca(HCO_3)_2$
- High boiling point of water is due to :-
(1) Its high specific heat
(2) Hydrogen bonding
(3) High dielectric constant
(4) Low dissociation constant

HYDROGEN PEROXIDE (H_2O_2)

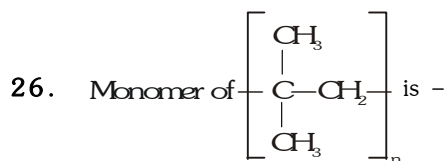
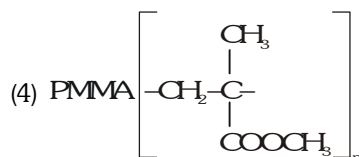
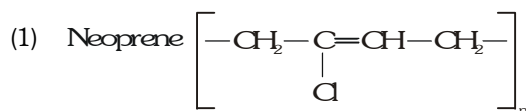
- Hydrogen peroxide is not :-
(1) A reducing agent (2) An oxidising agent
(3) A dehydrating agent (4) A bleaching agent
- The bleaching properties of H_2O_2 are due to its :-
(1) Reducing properties (2) Oxidising properties
(3) Unstable nature (4) Acidic nature
- Hydrogen peroxide has a :-
(1) Linear structure
(2) Pyramidal structure
(3) Closed book type structure
(4) Half open book type structure
- Hydrogen peroxide is a :-
(1) Liquid (2) Gas
(3) Solid (4) Semi-solid
- Which of the following is a true structure of H_2O_2



19. Decomposition of H_2O_2 is retarded by :-
 (1) Acetanilide (2) MnO_2
 (3) Zinc (4) Finely divided metals
20. H_2O_2 is :-
 (1) An oxidising agent
 (2) Both oxidising and reducing agent
 (3) Reducing agent
 (4) None of the above
21. H_2O_2 is :-
 (1) Diamagnetic (2) Paramagnetic
 (3) Ferromagnetic (4) None of these
22. The hybridisation of the orbitals of oxygen in H_2O_2 is :-
 (1) sp^3d (2) sp (3) sp^2 (4) sp^3
23. H_2O_2 is always stored in black bottles because :-
 (1) It is highly unstable
 (2) Its enthalpy of decomposition is high
 (3) It undergoes autooxidation on prolonged standing
 (4) None of these

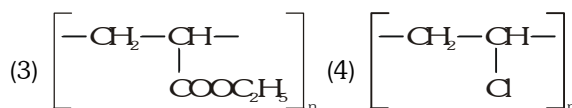
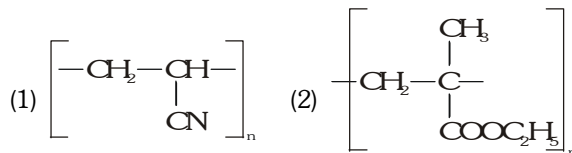
POLYMER

24. $\text{CF}_2=\text{CF}_2$ is a monomer of -
 (1) Teflon (2) Orlon
 (3) Polythene (4) Nylon-6
25. Which of the following is not correctly matched -



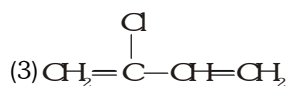
- (1) 2-methyl propene (2) Styrene
 (3) Propylene (4) Ethene

27. Acrylon is a hard, horny and a high melting material. Which of the following represents its structure -



28. Which one of the following monomers gives the polymer neoprene on polymerization -

- (1) $\text{CH}_2=\text{CHCl}$ (2) $\text{CCl}_2=\text{CCl}_2$



- (4) $\text{CF}_2=\text{CF}_2$

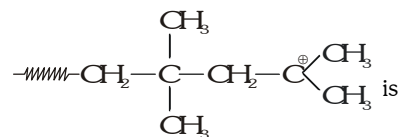
29. Which of the following is a biodegradable polymer

- (1) Cellulose (2) Polythene
 (3) Polyvinyl chloride (4) Nylon-6

30. Which one of the following is a chain growth polymer

- (1) Nucleic acid (2) Polystyrene
 (3) Protein (4) Starch

31. The monomer of the polymer -



- (1) $\text{CH}_3\text{CH}=\text{CHCH}_3$ (2) $\text{CH}_3\text{CH}=\text{CH}_2$
 (3) $(\text{CH}_3)_2\text{C}=\text{C}(\text{CH}_3)_2$ (4) $\text{H}_2\text{C}=\text{C}(\text{CH}_3)_2$

32. Which one of the following polymers is prepared by condensation polymerization

- (1) Styrene (2) Nylon-66
 (3) Teflon (4) Rubber

CHEMISTRY IN EVERYDAY LIFE

33. An antipyretic is –
(1) Quinine (2) Paracetamol
(3) Luminal (4) Piperazine
34. Medicine which is an antibiotic is –
(1) Ampicillin (2) Aspirin
(3) Chloroquine (4) None of these
35. Alizarin belongs to the class of –
(1) Vat dyes (2) Mordant dyes
(3) Substantive dyes (4) Reactive dyes
36. Which of the following is a basic dye –
(1) Alizarin (2) Phthalein
(3) Aniline yellow (4) Orange-I
37. Diazo coupling is useful to prepare some –
(1) Pesticides (2) Dyes
(3) Proteins (4) Vitamins
38. Which of the following is an azo dye –
(1) Methyl orange (2) Phenolphthalein
(3) Malachite green (4) Methylene blue
39. Paracetamol is a/an –
(1) Both antipyretic and analgesic
(2) Analgesic (3) Antipyretic
(4) Antimalarial
40. Which of the following compounds is aspirin –
(1) Methyl salicylate (2) Acetylsalicylic acid
(3) Phenyl salicylate (4) Salicylic acid
41. Sulpha drugs are derivatives of –
(1) Benzene sulphononic acid (2) Sulphanillic acid
(3) Sulphanilamide (4) p - aminobenzoic acid
42. Which of the following is a natural dye –
(1) Phenolphthalein (2) Alizarin
(3) Martius yellow (4) Malachite green

ENVIRONMENTAL CHEMISTRY

- 43.** The term biosphere is used for the zone of the earth where life exists
- (1) On the lithosphere
 - (2) In the hydrosphere
 - (3) In the lithosphere and hydrosphere
 - (4) In the lithosphere, hydrosphere and atmosphere

44. Biosphere is
 - (1) In which individual interact to each other
 - (2) By which life originated
 - (3) The name of a bird
 - (4) Organic compound by which life diminishes
45. Which is not a renewable source
 - (1) Forest
 - (2) Coal
 - (3) Water
 - (4) Forest organism
46. Noosphere is synonyms of
 - (1) Environment
 - (2) Atmosphere
 - (3) Hydrosphere
 - (4) Stratosphere
47. When biosphere turns into human dominated environment it is called
 - (1) Noosphere
 - (2) Troposphere
 - (3) Mesosphere
 - (4) Man sphere
48. The living organisms on or around the earth constitute
 - (1) Biome
 - (2) Biosphere
 - (3) Community
 - (4) Biocoenosis
49. Biosphere refers to
 - (1) Plants of the world
 - (2) Special plants
 - (3) Area occupied by living beings
 - (4) Plants of a particular area
50. What is the correct sequence of atmospheric layers starting from earth
 - (1) Stratosphere troposphere, mesosphere, thermosphere
 - (2) Troposphere, stratosphere, mesosphere, thermosphere
 - (3) Mesosphere, troposphere, stratosphere, thermosphere
 - (4) Thermosphere, mesosphere, stratosphere, troposphere
51. On earth all living organisms constitute
 - (1) Community
 - (2) Biome
 - (3) Association
 - (4) Biosphere
52. A biosphere is composed of
 - (1) Living organisms
 - (2) Living organisms + Lithosphere
 - (3) Living organisms + lithosphere + atmosphere
 - (4) Living organisms + lithosphere + atmosphere + hydrosphere

CHECK YOUR GRASP					ANSWER KEY					EXERCISE-I					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans	1	4	3	4	1	3	3	2	1	3	2	4	2	3	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans	4	1	2	1	2	1	4	3	1	3	1	1	3	1	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans	4	2	2	1	2	3	2	1	1	2	3	2	4	1	2
Que.	46	47	48	49	50	51	52								
Ans	1	1	2	3	2	4	4								

CHECK YOUR GRASP					ANSWER KEY					EXERCISE-I					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans	1	4	3	4	1	3	3	2	1	3	2	4	2	3	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans	4	1	2	1	2	1	4	3	1	3	1	1	3	1	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans	4	2	2	1	2	3	2	1	1	2	3	2	4	1	2
Que.	46	47	48	49	50	51	52								
Ans	1	1	2	3	2	4	4								

CHECK YOUR GRASP					ANSWER KEY					EXERCISE-I					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans	1	4	3	4	1	3	3	2	1	3	2	4	2	3	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans	4	1	2	1	2	1	4	3	1	3	1	1	3	1	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans	4	2	2	1	2	3	2	1	1	2	3	2	4	1	2
Que.	46	47	48	49	50	51	52								
Ans	1	1	2	3	2	4	4								

HYDROGEN, POLYMER, CHEMISTRY IN EVERYDAY LIFE & ENVIRONMENTAL CHEMISTRY

1. The function of enzymes in the living system is to –
 - (1) Transport oxygen
 - (2) Provide immunity
 - (3) Catalyse biochemical reaction
 - (4) Provide energy
2. Enzymes are :-
 - (1) Carbohydrates
 - (2) Nucleic acids
 - (3) Globular proteins
 - (4) Fibrous proteins
3. The sex hormone which controls the development and maintenance of pregnancy is :-
 - (1) Cortisone
 - (2) Thyroxine
 - (3) Progesterone
 - (4) Estrone
4. Vitamin E is also called :-
 - (1) Cyanocobalamin
 - (2) tocopherol
 - (3) Lactoflavin
 - (4) Ascorbic acid
5. The deficiency of vitamin K causes :-
 - (1) Haemorrhage
 - (2) Lengthening time of blood clotting
 - (3) Inflammation of tongue
 - (4) Both (1) and (2)
6. The function of haemoglobin is to :-
 - (1) Help in muscular movement
 - (2) Store oxygen until it is needed for energy reproduction
 - (3) Transport oxygen from lungs to various tissues through blood stream
 - (4) Catalyse biochemical processes
7. Which one among the following is a thermosetting plastic -
 - (1) PVC
 - (2) PVA
 - (3) Bakelite
 - (4) Perspex
8. The basis on the mode of their formation, the polymers can be classified -
 - (1) As addition polymers only
 - (2) As condensation polymers only
 - (3) As copolymers
 - (4) Both as addition and condensation polymers
9. Which of the following is not a polymer -
 - (1) Gun cotton
 - (2) Perspex
 - (3) Shellac (eg. lac shellac)
 - (4) Wax (eg. bees wax)
10. Among the following a natural polymer is -
 - (1) Cellulose
 - (2) PVC
 - (3) Teflon
 - (4) Polyethylene
11. Which is a naturally occurring polymer-
 - (1) Polythene
 - (2) PVC
 - (3) Acetic acid
 - (4) Protein
12. Which one of the following is a linear polymer-
 - (1) Amylopectin
 - (2) Glycogen
 - (3) Starch
 - (4) Amylose
13. Natural rubber is which type of polymer-
 - (1) Condensation polymer
 - (2) Addition polymer
 - (3) Co-ordination polymer
 - (4) None of these
14. Polyethylene is -
 - (1) Random copolymer
 - (2) Homo polymer
 - (3) Alternate copolymer
 - (4) Crosslinked copolymer
15. Nylon is not a -
 - (1) Condensation polymer
 - (2) Polyamide
 - (3) Copolymer
 - (4) Homopolymer
16. Bakelites are -
 - (1) Rubber
 - (2) Rayon
 - (3) Resins
 - (4) Plasticisers
17. Which of the following is a basic dye –
 - (1) Alizarin
 - (2) Malachite green
 - (3) Indigo
 - (4) Orange-I
18. The compound used to fix a dye to the fabric is called –
 - (1) Mordant
 - (2) Lake
 - (3) Bleaching agent
 - (4) Oxidising agent
19. Aspirin is called –
 - (1) Pyretic
 - (2) Antiseptic
 - (3) Antibiotic
 - (4) Antipyretic
20. Which of the following is an antidiabetic drug –
 - (1) Insulin
 - (2) Penicillin
 - (3) Chloroquine
 - (4) Aspirin

21. 2-Acetoxybenzoic acid is called –
 (1) Antiseptic (2) Aspirin
 (3) Antibiotic (4) Mordant dye
22. Match List-I with List-II and select the correct answer using the codes given below the lists –
 List I List II
 I. Iodoform A. Anaesthetic
 II. Methyl salicylate B. Antiseptic
 III. Diethyl ether C. Insecticide
 IV. Hexachlorocyclohexane D. Detergent
 E. Pain Balm
 (1) I-B, II-E, III-C, IV-D
 (2) I-D, II-B, III-A, IV-C
 (3) I-B, II-E, III-A, IV-C
 (4) I-C, II-A, III-D IV-B
23. Arsenic drugs are mainly used in the treatment of–
 (1) Jaundice (2) Typhoid
 (3) Syphilis (4) Cholera
24. Aspirin is an acetylation product of –
 (1) p-dihydroxybenzene
 (2) o-hydroxybenzoic acid
 (3) o-dihydroxy benzene
 (4) m-hydroxybenzoic acid
25. An example of vat dye is –
 (1) Indigo (2) Alizarin
 (3) Malachite green (4) Martius yellow
26. Which of the following is an azo dye –
 (1) Orange - I (2) Malachite green
 (3) Indigo (4) Martius yellow
27. Which of the following is the non conventional source of energy
 (1) Coal
 (2) Petroleum
 (3) Electricity from nuclear power plants
 (4) Solar radiations
28. The population of India is 15% of the world but its annual energy consumption is only
 (1) 0.2% (2) 2.0%
 (3) 10% (4) 25%
29. Petroleum resource is
 (1) Renewable
 (2) Non renewable
 (3) Synthetic & biodegradable
 (4) Infinite & unconventional
30. The main aim of plant conservation is -
 (1) To conserve the necessary ecological activities and life supporting systems
 (2) To conserve species diversity and range of genetic material
 (3) Both the above
 (4) None of the above
31. Environmental laning organisation is
 (1) CSIR (2) CEPHERI
 (3) ICAR (4) NEERI
32. Which will not cause any atmospheric pollution
 (1) Hydrogen (2) Sulphur dioxide
 (3) Carbon dioxed (4) Carbon monoxide
33. Which of the following is the main factor of water pollution
 (1) Smoke (2) Industrial waste
 (3) Detergent (4) Ammonia
34. Main air pollutant among the following is
 (1) CO (2) CO₂ (3) N₂ (4) Sulphur
35. Which is more important for water pollution
 (1) Sound (2) SO₂
 (3) Salts of arsenic (4) Sewage
36. Which of the following atmospheric pollutants is not produced by the exhaust of motor vehicle in Delhi
 (1) SO₂ (2) Hydrocarbon gases
 (3) Fly ash (4) CO
37. Riboflavin is the chemical name of :-
 (1) Vitamin B₁ (2) Vitamin B₂
 (3) Vitamin B₆ (4) Vitamin B complex
38. Calorific value is in the order :-
 (1) Fats > Carbohydrates > Proteins
 (2) Carbohydrates > Fats > Proteins
 (3) Proteins > Carbohydrates > Fats
 (4) Fats > Proteins > Carbohydrates

39. Which of the following is a step growth polymer-
 (1) Polyisoprene (2) Polythene
 (3) Nylon (4) Polyacrylonitrile
40. An example of chain growth polymer is
 (1) Nylon -66 (2) Bakelite
 (3) Terylene (4) Teflon
41. Which of the following is not an example of natural polymer-
 (1) Wool (2) Silk
 (3) Leather (4) Nylon
42. Natural rubber is a -
 (1) Polyester (2) Polyamide
 (3) Polyisoprene (4) Polysaccharide
43. Which of the following is not a synthetic polymer -
 (1) Polyethylene (2) PVC
 (3) Nylon (4) Cellophane
44. Which of the following is not correct regarding terylene -
 (1) Step -growth polymer
 (2) Synthetic fibre
 (3) Condensation polymer
 (4) Thermosetting plastic
45. When heated with zinc chloride, lactides form a linear polymer which may be -
 (1) Polystyrene (2) Polyamide
 (3) Polyester (4) Polythene
46. The catalyst used for the polymerisation of olefins is -
 (1) Ziegler Natta catalyst (2) Wilkinson's catalyst
 (3) Pd-catalyst (4) Zeise's salt catalyst
47. PVC is prepared by the polymerisation of -
 (1) Ethylene (2) 1-chloropropene
 (3) Propene (4) 1-chloroethene
48. Acrylonitrile forms -
 (1) Terylene (2) Orlon
 (3) PVC (4) Bakelite
49. The synthetic polymer which resembles natural rubber is -
 (1) Neoprene (2) Chloroprene
 (3) Glyptal (4) Nylon
50. Chloramphenicol is an -
 (1) Analgesic (2) Anaesthetic
 (3) Antibiotic (4) Antiseptic
51. Detergents are prepared by the action of H_2SO_4 followed by neutralization by starting with-
 (1) Cholesterol (2) Lauryl alcohol
 (3) Cyclohexanol (4) p-Nitrophenol
52. 2, 4, 6 - trinitrophenol is a/an -
 (1) Acid dye (2) Basic dye
 (3) Azo dye (4) Vat dye
53. Substances which bring body temperature down are known as -
 (1) Antipyretics (2) Analgin
 (3) Antibiotics (4) Hypnotics
54. The indicator used in the titration of a strong acid and a strong base is -
 (1) Phenolphthalein (2) Methyl Orange
 (3) Alizarin yellow (4) Red litmus
55. The drug given during hypertension is -
 (1) Streptomycin (2) Chloroxylenol
 (3) Equanil (4) Aspirin
56. One of the most widely used drug in medicine is
 (1) Methyl salicylate (2) Ethyl salicylate
 (3) Acetylsalicylic acid (4) o-hydroxybenzoic acid
57. Which of the following is known as broad spectrum antibiotic -
 (1) Streptomycin (2) Ampicillin
 (3) Chloramphenicol (4) Penicillin
58. Phenol is used as -
 (1) An antiseptic (2) A disinfectant
 (3) Both (1) and (2) (4) None of these
59. The antiseptic action of Dettol is due to -
 (1) Chloro benzene (2) Chloroxylenol
 (3) Chloroquine (4) Chloramphenicol

60. Pollution can be controlled by
 (1) Sewage treatment
 (2) Checking atomic blasts
 (3) Manufacturing electrically operated vehicles
 (4) All the above
61. If water pollution continues at its present rate, it will eventually
 (1) Stop water cycle
 (2) Prevent precipitation
 (3) Make oxygen molecules unavailable to water plants.
 (4) Make nitrate molecules unavailable to water plants.
62. In cities like Bombay and Calcutta the major air pollutants are
 (1) Ozone
 (2) Carbon monoxide and oxides of Sulphur
 (3) Hydrocarbons and not air
 (4) Algal spores and marsh gas
63. Recent reports of acid rains in industrial cities are due to the effect of atmospheric pollution by
 (1) Excessive release of NO_2 and SO_2 by burning of fossil fuels.
 (2) Excessive release of CO_2 by burning of fuel like wood and charcoal, cutting of forests and increased animal population.
 (3) Excessive release of NH_3 by industrial plants and coal gas.
 (4) Excessive release of CO in atmosphere by incomplete combustion of coal, charcoal and other carbonaceous fuels in scarcity of oxygen,
64. Pollution is a change in physical, chemical or biological characters of our land and water that may be
 (1) Desirable and harmful to human
 (2) Desirable and useful to human
 (3) Undesirable and harmful to human
 (4) Undesirable and useful to human
65. Which is the greatest air pollutant these days
 (1) Factories (2) Motor vehicles
 (3) Domestic appliances (4) animals
66. Removal of the soil by the action of wind and water is known as
 (1) Erosion (2) Fossilization
 (3) Leaching (4) Calcification
67. Acid rain occurs due to atmospheric pollution of
 (1) SO_2 (2) NH_3
 (3) CO_2 (4) N_2O
68. Photochemical smog was first observed in -
 (1) London (2) Los Angeles
 (3) Paris (4) Tokyo
69. An increase in CO_2 concentration in the atmosphere will result in
 (1) Adverse effects of natural vegetation
 (2) Global warming
 (3) Temperature decrease in global atmosphere
 (4) Genetic disorders in plants and animals
70. Calgon is an industrial name given to :-
 (1) Normal sodium phosphate
 (2) Sodium meta-aluminate
 (3) Sodium hexametaphosphate
 (4) Hydrated sodium aluminium silicate
71. When the same amount of zinc is treated separately with excess of sulphuric acid and excess of sodium hydroxide, the ratio of volumes of hydrogen evolved is :-
 (1) 1 : 1 (2) 1 : 2 (3) 2 : 1 (4) 9 : 4
72. Permutit is :-
 (1) Hydrated sodium aluminium silicate
 (2) Sodium hexametaphosphate
 (3) Sodium silicate
 (4) Sodium meta-aluminate
73. Ortho and Para hydrogen differ :-
 (1) In the number of protons
 (2) In the molecular mass
 (3) In the nature of spins of protons
 (4) In the nature of spins of electrons
74. In Bosch's process which gas is utilised for the production of hydrogen :-
 (1) Producer gas (2) Water gas
 (3) Coal gas (4) Natural gas

75. The gas used in the hydrogenation of oils in presence of nickel as a catalyst is :-
(1) Methane (2) Ethane
(3) Ozone (4) Hydrogen

76. Water softening by Clarke's process uses :-
(1) Calcium bicarbonate (2) Sodium bicarbonate
(3) Potash alum (4) Calcium hydroxide

77. Which of the following produces hydrolith with dihydrogen :-
(1) Mg (2) Al (3) Cu (4) Ca

78. The lightest gas is :-
(1) Nitrogen (2) Helium
(3) Oxygen (4) Hydrogen

79. The ratio of electron, proton and neutron in tritium is :-
(1) 1 : 1 : 1 (2) 1 : 1 : 2 (3) 2 : 1 : 1 (4) 1 : 2 : 1

80. The nuclei of tritium (H^3) atom would contain neutrons :-
(1) 1 (2) 2 (3) 3 (4) 4

81. The adsorption of hydrogen by metals is called :-
(1) Dehydrogenation (2) Hydrogenation
(3) Occlusion (4) Adsorption

82. Heavy water (D_2O) is :-
(1) A product of oxygen and hydrogen
(2) Ordinary water containing dissolved salts of heavy metals
(3) Water of mineral springs
(4) Water produced by repeated distillation and condensation

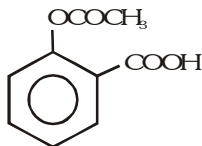
83. Ionic hydrides are usually :-
(1) Good electrically conductors when solid
(2) Easily reduced
(3) Good reducing agents
(4) Liquid at room temperature

BRAIN TEASERS					ANSWER KEY					EXERCISE-II					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans	3	3	3	2	4	3	3	4	4	1	4	4	2	2	4
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans	3	2	1	4	1	2	3	3	2	1	1	4	2	2	3
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans	4	1	2	1	3	3	1	1	3	4	4	3	4	4	1
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans	1	4	2	1	3	2	1	1	1	3	1	3	3	2	4
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans	3	2	1	3	2	1	1	2	2	3	1	1	3	2	4
Que.	76	77	78	79	80	81	82	83							
Ans	4	4	4	2	2	3	4	3							

HYDROGEN, POLYMER, CHEMISTRY IN EVERYDAY LIFE & ENVIRONMENTAL CHEMISTRY

- Which hydride is an ionic hydride :- [AIIMS 1985]
(1) NH_3 (2) H_2S (3) $\text{TiH}_{1.73}$ (4) NaH
- The reaction, $\text{H}_2\text{S} + \text{H}_2\text{O}_2 \longrightarrow \text{S} + 2\text{H}_2\text{O}$ manifests: [MLNR 1987]
(1) Acidic nature of H_2O_2
(2) Alkaline nature of H_2O_2
(3) Oxidising nature of H_2O
(4) Reducing nature of H_2O_2
- Heavy water has found application in atomic reactor as :- [MLNR 1988]
(1) Coolant
(2) Moderator
(3) Both coolant and moderator
(4) Neither coolant nor moderator
- Calgon (a water softener) is :- [CBSE 1989]
(1) $\text{Na}_2[\text{Na}_4(\text{PO}_3)_6]$ (2) $\text{Na}_4[\text{Na}_2(\text{PO}_3)_6]$
(3) $\text{Na}_2[\text{Na}_4(\text{PO}_4)_6]$ (4) $\text{Na}_4[\text{Na}_2(\text{PO}_4)_6]$
- Hydrogen peroxide works as :- [CPMT 1990]
(1) An oxidant only
(2) A reductant only
(3) An acid only
(4) An oxidant, a reductant and an acid
- The hair dyes available in the market generally contain two bottles, one containing the dye and the other hydrogen peroxide. Before applying the dye, the two solutions are mixed. the hydrogen peroxide [NCERT 1990]
(1) Is added to dilute the solution of the dye
(2) Oxidises the dye to give the desired colour
(3) Reduces the dye to give the the desired colour
(4) Acidifies the solution of the dye
- When zeolite (hydrated sodium aluminium silicate) is treated with hard water, the sodium ions are exchanged with :- [IIT 1990]
(1) H^+ ions (2) Ca^{2+} ions
(3) SO_4^{2-} ions (4) OH^- ions
- Hydrogen peroxide is now generally prepared on industrial scale by the :- [Roorkee 1992]
(1) Action of H_2SO_4 on barium peroxide
(2) Action of H_2SO_4 on sodium peroxide
(3) Electrolysis of 50% H_2SO_4
(4) Burning hydrogen in excess of oxygen
- The hardness of water is due to.....metal ions [BHU 1992]
(1) Ca^{2+} and Na^+ (2) Mg^{2+} and K^+
(3) Ca^{2+} and Mg^{2+} (4) Zn^{2+} and Ba^{2+}
- Which of the following statement is correct :- [BHU 1997]
(1) Hydrogen has same ionisation potential as sodium
(2) H has same electronegativity as halogens
(3) It will not be liberated at anode
(4) H has oxidation state + 1, zero and - 1
- The formula of heavy water is :- [CPMT 1991 ; AFMC 1997]
(1) H_2O^{18} (2) D_2O (3) T_2O (4) H_2O^{17}
- Polyphosphates are used as water softening agent because they :- [IIT 2002]
(1) Form soluble complexes with anionic species
(2) Precipitate anionic species
(3) Form soluble complexes with cationic species
(4) Precipitate cationic species.
- Which one of the following processes will produce hard water :- [AIEEE 2003]
(1) Saturation of water with CaSO_4
(2) Addition of Na_2SO_4 to water
(3) Saturation of water with CaCO_3
(4) Saturation of water with MgCO_3
- In an organic compound of molar mass 108gmol^{-1} C, H and N atoms are present in 9 : 1 : 35 by weight. Molecular formula can be- [AIEEE - 2002]
(1) $\text{C}_6\text{H}_8\text{N}_2$ (2) $\text{C}_7\text{H}_{10}\text{N}$
(3) $\text{C}_5\text{H}_6\text{N}_3$ (4) $\text{C}_4\text{H}_{18}\text{N}_3$

15. Compound A given below is :



[AIEEE - 2002]

- (1) Antiseptic (2) Antibiotic
(3) Analgesic (4) Pesticide
16. Monomers are converted to polymer by-
(1) Hydrolysis of monomer [AIEEE - 2002]
(2) Condensation reaction between monomers
(3) protonation of monomers
(4) none of the above
17. Nylon threads are made of- [AIEEE - 2003]
(1) polyvinyl polymer
(2) polyester polymer
(3) polyamide
(4) polyethylene polymer
18. Which of the following could act as a propellant for rockets- [AIEEE - 2003]
(1) Liquid hydrogen + liquid nitrogen
(2) Liquid oxygen + liquid argon
(3) Liquid hydrogen + liquid oxygen
(4) Liquid nitrogen + liquid oxygen
19. Identify the correct statement regarding enzymes- [AIEEE - 2004]
(1) Enzymes are specific biological catalysts that can normally function at very high temperatures ($T \approx 1000\text{ K}$)
(2) Enzymes are normally heterogeneous catalysts that are very specific in their action
(3) Enzymes are specific biological catalysts that can not be poisoned
(4) Enzymes are specific biological catalysts that possess well defined active sites
20. Insulin production and its action in human body are responsible for the level of diabetes. This compound belongs to which of the following categories [AIEEE - 2004]
(1) A coenzyme (2) A hormone
(3) An enzyme (4) An antibiotic

21. The ammonia evolved from the treatment of 0.30g of an organic compound for the estimation of nitrogen was passed in 100 mL of 0.1 M sulphuric acid. The excess of acid required 20 mL of 0.5 M sodium hydroxide solution for complete neutralization. The organic compound is-

[AIEEE - 2004]

- (1) acetamid (2) benzamide
(3) urea (4) thiourea
22. Which one of the following types of drugs reduces fever- [AIEEE - 2005]
(1) Tranquilizer (2) Antibiotic
(3) Antipyretic (4) Analgesic
23. Which of the following is a polyamide [AIEEE - 2005]
(1) Bakelite (2) Terylene
(3) Nylon-66 (4) Teflon
24. Which of the following is fully fluorinated polymer- [AIEEE - 2005]
(1) PVC (2) Thiokol
(3) Teflon (4) Neoprene
25. An organic compound having molecular mass 60 is found to contain C = 20%, H = 6.67 % and N = 46.67 % while rest is oxygen on heating it gives NH_3 along with a solid residue. The solid residue gives violet colour with alkaline copper sulphate solution. The compound is- [AIEEE - 2005]
(1) $\text{CH}_3\text{CH}_2\text{CONH}_2$ (2) $(\text{NH}_2)_2\text{CO}$
(3) CH_3CONH_2 (4) CH_3NCO
26. Regular use of which of the following fertilisers increases the acidity of soil ? [AIEEE - 2007]
(1) Potassium nitrate
(2) Urea
(3) Superphosphate of lime
(4) Ammonium sulphate
27. Identify the wrong statement in the following [AIEEE-2008]
(1) Chlorofluorocarbons are responsible for ozone layer depletion
(2) Greenhouse effect is responsible for global warming
(3) Ozone layer does not permit infrared radiation from the sun to reach the earth
(4) Acid rain is mostly because of oxides of nitrogen and sulphur

28. Buna-N synthetic rubber is a copolymer of :-

[AIEEE-2009]

- (1) $\text{H}_2\text{C} = \text{CH} - \text{CN}$ and $\text{H}_2\text{C} = \text{CH} - \text{CH} = \text{CH}_2$

- (2) $\text{H}_2\text{C} = \text{CH} - \text{CN}$ and $\text{H}_2\text{C} = \text{CH} - \underset{\text{CH}_3}{\text{C}} = \text{CH}_2$

- (3) $\text{H}_2\text{C}=\text{CH}-\overset{\text{Cl}}{\underset{|}{\text{C}}}=\text{CH}_2$ and $\text{H}_2\text{C}=\text{CH}-\text{CH}=\text{CH}_2$

- (4) $\text{H}_2\text{C} = \text{CH} - \text{CH} = \text{CH}_2$ and $\text{H}_5\text{C}_6 - \text{CH} = \text{CH}_2$

29. The two functional groups present in a typical carbohydrate are :- **[AIEEE-2009]**

- (1) $\text{>C} = \text{O}$ and $-\text{OH}$

- (2) -OH and -CHO

- (3) $-\text{OH}$ and $-\text{COOH}$

- (4) $-\text{CHO}$ and $-\text{COOH}$

30. 29.5 mg of an organic compound containing nitrogen was digested according to Kjeldahl's method and the evolved ammonia was absorbed in 20 mL of 0.1 M HCl solution. The excess of the acid required 15 mL of 0.1 M NaOH solution for complete neutralization. The percentage of nitrogen in the compound is :- **[AIEEE-2010]**

- (1) 29.5 (2) 59.0

- (3) 47.4 (4) 23.7

31. The polymer containing strong intermolecular forces e.g. hydrogen bonding, is :- **[AIEEE-2010]**

- (1) natural rubber (2) teflon

- (3) nylon 6, 6 (4) polystyrene

32. Biurest test is not given by :- [AIEEE-2010]

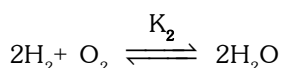
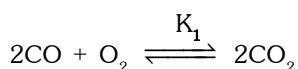
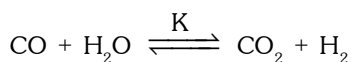
- (1) proteins (2) carbohydrates

- (3) polypeptides (4) urea

PREVIOUS YEAR QUESTIONS ANSWER KEY EXERCISE-III															
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans	4	3	3	1	4	2	2	3	3	4	2	3	1	1	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans	2	3	3	4	2	3	3	3	3	2	4	3	1	2	4
Que.	31	32													
Ans	3	2													

EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

- In a reaction $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$ degree of dissociation is 30%. If initial moles of PCl_5 is one then total moles at equilibrium is
(A) 1.3 (B) 0.7 (C) 1.6 (D) 1.0
- For reaction $\text{HI} \rightleftharpoons \frac{1}{2} \text{H}_2 + \frac{1}{2} \text{I}_2$ value of K_c is $1/8$ then value of K_c for $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$.
(A) $\frac{1}{64}$ (B) 64 (C) $\frac{1}{8}$ (D) 8
- In a equilibrium reaction $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$
 $\Delta H = -3000$ calories,
which factor favours dissociation of HI :-
(A) Low temp. (B) High Pressure (C) High temp. (D) Low pressure.
- $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$
If temp. of following equilibrium reaction increase then -
(A) Shift Right side (B) Shift left side (C) Unchanged (D) Nothing say.
- $\text{C}(\text{s}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{CO}(\text{g})$
by increasing pressure following equilibrium
(A) Unaffected (B) Proceed in backward direction
(C) Proceed in forward direction (D) Unfixed
- Unit of equilibrium constant K_c for following homogenous reaction :-
 $4\text{NH}_3 + 5\text{O}_2 \rightleftharpoons 4\text{NO} + 6\text{H}_2\text{O}$ is
(A) $(\text{Conc}^n)^{-1}$ (B) $(\text{Conc}^n)^{+1}$ (C) $(\text{Conc}^n)^{+10}$ (D) Have no unit
- Which of the following factor shifted the reaction $\text{PCl}_3 + \text{Cl}_2 \rightleftharpoons \text{PCl}_5$ at left side.
(A) Adding PCl_5 (B) Increase pressure (C) Constant temp. (D) Catalyst.
- In which of the following process reaction is fastest complete.
(A) $K = 10$ (B) $K = 1$ (C) $K = 10^3$ (D) $K = 10^{-2}$
- At 298 K equilibrium constant K_1 and K_2 of
following reaction $\text{SO}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightleftharpoons \text{SO}_3(\text{g})$ ---- (1)
 $2\text{SO}_3(\text{g}) \rightleftharpoons 2\text{SO}_2(\text{g}) + \text{O}_2(\text{g})$ ----- (2)
The relation between K_1 and K_2 is
(A) $K_1 = K_2$ (B) $K_2 = K_1^2$ (C) $K_2 = 1/K_1^2$ (D) $K_2 = 1/K_1$
- In the following reaction $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ at constant temp. rate of backward reaction is increase by :
(A) Inert gas mixed at constant volume (B) Cl_2 gas mixed at constant volume
(C) Inert gas mixed at constant pressure (D) PCl_5 mixed in constant volume.
- Some gaseous equilibrium are following :



then find out the relation between equilibrium constants :-

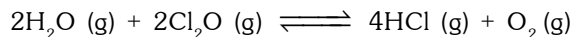
- (A) $K = K_1 K_2$ (B) $K = (K_1 K_2)^2$ (C) $K = (K_1 K_2)^{-1/2}$ (D) $K = (K_1 / K_2)^{1/2}$

12. For the equilibrium process $x + y \rightleftharpoons xy$. If the concⁿ. of x and y is doubled, then equilibrium constant.
 (A) Become twice (B) Become half (C) Unchanged (D) Become thrice
13. Two moles of PCl_5 were heated in closed vessel of 2 litre capacity, at equilibrium 40% of PCl_5 was dissociated into PCl_3 and Cl_2 . The value of equilibrium constant is :
 (A) .267 (B) .53 (C) 2.67 (D) 5.3
14. The reaction $\text{A(g)} + \text{B(g)} \rightleftharpoons 2\text{C(g)}$ is occur by mixing of 3 moles of A and 1 mole of B in one litre Container. if α of B is $\frac{1}{3}$, then K_c for this reaction is :-
 (A) 0.12 (B) 0.25 (C) 0.50 (D) 0.75
15. Reaction $2\text{BaO}_2(\text{s}) \rightleftharpoons 2\text{BaO}(\text{s}) + \text{O}_2(\text{g})$; $\Delta H = + \text{ve}$. At equilibrium condition, Pressure of O_2 is depends on :-
 (A) Increase mass of BaO_2 (B) Increase mass of BaO
 (C) Increase temp. at Eq^m. (D) Increase mass of BaO_2 and BaO both
16. Consider the following equilibrium in a closed container $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$. At a fixed temperature, the volume of the reaction container is halved. For this change, which of the following statements held true regarding the equilibrium constant (K_p) and degree of dissociation (α) :-
 (A) Neither K_p nor α changes (B) Both K_p and α - changes
 (C) K_p changes, but α does not change (D) K_p does not change, but α - changes
17. $\text{C(s)} + \text{CO}_2(\text{g}) \rightleftharpoons 2\text{CO(g)}$
 According to above reaction, partial Pressure of CO_2 & CO are 4 & 8 respectively then find out K_p of the above reaction :-
 (A) 6 (B) 2 (C) 16 (D) 32
18. For the reaction, $\text{A+B} \rightleftharpoons \text{C + D}$, $K_c = 9$. If A and B are taken in equal amounts, then amount of C at equilibrium is :-
 (A) 1 (B) 0.25 (C) 0.75 (D) None of these
19. At equilibrium 500mL vessel contains 1.5 M of each A, B, C, D. If 0.5M of C and D expelled out then what would be the K_c :-
 (A) 1 (B) $\frac{1}{9}$ (C) $\frac{4}{9}$ (D) $\frac{5}{9}$
20. The following equilibrium are given
 $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ ----- K_1
 $\text{N}_2 + \text{O}_2 \rightleftharpoons 2\text{NO}$ ----- K_2
 $\text{H}_2 + \frac{1}{2}\text{O}_2 \rightleftharpoons \text{H}_2\text{O}$ ----- K_3
 The equilibrium constant of the reaction
 $2\text{NH}_3 + \frac{5}{2}\text{O}_2 \rightleftharpoons 2\text{NO} + 3\text{H}_2\text{O}$, in terms of K_1 , K_2 and K_3 is :
 (A) $\frac{K_1 K_2}{K_3}$ (B) $\frac{K_1 K_3^2}{K_2}$ (C) $\frac{K_2 K_3^3}{K_1}$ (D) $K_1 K_2 K_3$
21. The reaction $\text{A + B} \rightleftharpoons \text{C + D}$ is studied in a one litre vessel at 250 C. The initial concentration of A was 3n and that of B was n. When equilibrium was attained, equilibrium concentration of C was found to be equal to the equilibrium concentration of B. What is the concentration of D at equilibrium :
 (A) n/2 (B) (3n - 1/2) (C) (n - n/3) (D) n

22. In a reversible reaction $A \xrightleftharpoons[k_2]{k_1} B$, the initial concentration of A and B are a and b in moles per litre and the equilibrium concentration are (a - x) and (b + x) respectively ; express x in terms of k_1 , k_2 , a and b :

(A) $\frac{k_1 a - k_2 b}{k_1 + k_2}$ (B) $\frac{k_1 a - k_2 b}{k_1 - k_2}$ (C) $\frac{k_1 a - k_2 b}{k_1 k_2}$ (D) $\frac{k_1 a + k_2 b}{k_1 + k_2}$

23. The value of K_p for the reaction



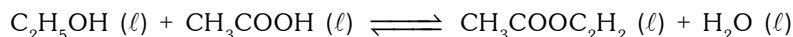
is 0.03 atm at 427 C, when the partial pressure are expressed in atmosphere then the value of K_c for the same reaction is :

(A) 5.23×10^{-4} (B) 7.34×10^{-4} (C) 3.2×10^{-3} (D) 5.43×10^{-5}

24. The equilibrium constant of the reaction $SO_2(g) + \frac{1}{2} O_2(g) \rightleftharpoons SO_3(g)$ is $4 \times 10^{-3} \text{ atm}^{-1/2}$. The equilibrium constant of the reaction $2 SO_3(g) \rightleftharpoons 2 SO_2(g) + O_2(g)$ would be :

(A) 250 atm (B) $4 \times 10^3 \text{ atm}$ (C) $0.25 \times 10^4 \text{ atm}$ (D) $6.25 \times 10^4 \text{ atm}$

25. When alcohol (C_2H_5OH) and acetic acid are mixed together in equimolar ratio at 27 C, 33% is converted into ester. Then the K_c for the equilibrium :



(A) 4 (B) 1/4 (C) 9 (D) 1/9

26. 2 moles each of SO_3 , CO, SO_2 and CO_2 is taken in a one lit. vessel. If K_c for $SO_3 + CO \rightleftharpoons SO_2 + CO_2$ is 1/9 then :

(A) total no. of moles at equilibrium are less than 8

(B) $n(SO_3) + n(CO_2) = 4$

(C) $[n(SO_2)/n(CO)] < 1$

(D) both (B) and (C)

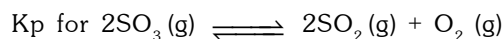
27. An equilibrium mixture in a vessel of capacity 100 litre contain 1 mol N_2 , 2 mol O_2 and 3 mol NO. No. of moles of O_2 to be added so that at new equilibrium the conc. of NO is found to be 0.04 mol/lit :

(A) (101/18) (B) (101/9) (C) (202/9) (D) None of these

28. Ammonia gas at 15 atm is introduced in a rigid vessel at 300 K. At equilibrium the total pressure of the vessel is found to be 40.11 atm at 300 C. The degree of dissociation of NH_3 will be :

(A) 0.6 (B) 0.4 (C) unpredictable (D) none of these

29. The degree of dissociation of SO_3 is α at equilibrium pressure P_0 :



(A) $[(P_0 \alpha^3)/2(1 - \alpha)^3]$ (B) $[(P_0 \alpha^3)/(2 + \alpha)(1 - \alpha)^2]$ (C) $[(P_0 \alpha^2)/2(1 - \alpha)^2]$ (D) none of these

30. For the reaction $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$ at a given temperature the equilibrium amount of $CO_2(g)$ can be increased by :

(A) adding a suitable catalyst

(B) adding an inert gas

(C) decreasing the volume of container

(D) increasing the amount of CO (g)

31. For the reaction :



The forward reaction at constant temperature is favoured by

(A) introducing an inert gas at constant volume (B) introducing chlorine gas at constant volume

(C) introducing an inert gas at constant pressure (D) increasing the volume of the container

(E) introducing PCl_5 at constant volume

32. Given the following reaction at equilibrium $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$. Some inert gas at constant pressure is added to the system. Predict which of the following facts will be affected :
- (A) more $\text{NH}_3(\text{g})$ is produced (B) less $\text{NH}_3(\text{g})$ is produced
(C) no affect on the equilibrium (D) K_p of the reaction is decreased
33. For an equilibrium $\text{H}_2\text{O}(\text{s}) \rightleftharpoons \text{H}_2\text{O}(\text{l})$ which of the following statement is true :
- (A) the pressure changes do not affect the equilibrium
(B) more of ice melts if pressure on the system is increased
(C) more of liquid freezes if pressure on the system is increase
(D) the pressure changes may increase or decrease the degree of advancement of the reaction depending upon the temperature of the system.
34. When a bottle of cold drink is opened, the gas comes out with a fizzle due to :
- (A) decrease in temperature (B) increase in pressure
(C) decrease in pressure suddenly which results in decrease of solubility of CO_2 gas in water
(D) none
35. The equilibrium, $\text{SO}_2\text{Cl}_2(\text{g}) \rightleftharpoons \text{SO}_2(\text{g}) + \text{Cl}_2(\text{g})$ is attained at 25 C in a closed container and an inert gas, helium, is introduced. Which of the following statements are correct :
- (A) concentrations of SO_2 , Cl_2 and SO_2Cl_2 are changed
(B) no effect on equilibrium
(C) concentration of SO_2 is reduced
(D) K_p of reaction is increasing
36. For the reaction $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$
 $K_c = 66.9$ at 350 C and $K_c = 50.0$ at 448 C. The reaction has :
- (A) $\Delta H = +ve$ (B) $\Delta H = -ve$
(C) $\Delta H = \text{Zero}$ (D) $\Delta H = \text{Not found the signs}$
37. Variation of $\log_{10} K$ with $\frac{1}{T}$ is shown by the following graph in which straight line is at 45°, hence ΔH is :
- (A) +4.606 cal (B) -4.606 cal
(C) 2 cal (D) - 2cal



CHECK YOUR GRASP						ANSWER KEY				EXERCISE -1					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	A	B	C	B	B	B	A	C	C	B	D	C	A	B	C
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	D	C	C	A	C	A	A	A	D	B	D	A	B	B	D
Que.	31	32	33	34	35	36	37								
Ans.	C,D,E	B	B	C	B	B	B								

EXERCISE-02**BRAIN TEASERS****SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)**

1. Consider following reaction in equilibrium with equilibrium concentration 0.01 M of every species
- (I) $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$
- (II) $2\text{HI}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{I}_2(\text{g})$
- (III) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
- Extent of the reactions taking place is :
- (A) $\text{I} > \text{II} > \text{III}$ (B) $\text{I} < \text{II} < \text{III}$ (C) $\text{II} < \text{III} < \text{I}$ (D) $\text{III} < \text{I} < \text{II}$
2. For the reaction $3\text{A}(\text{g}) + \text{B}(\text{g}) \rightleftharpoons 2\text{C}(\text{g})$ at a given temperature, $K_c = 9.0$. What must be the volume of the flask, if a mixture of 2.0 mol each of A, B and C exist in equilibrium?
- (A) 6 L (B) 9 L (C) 36 L (D) None of these
3. Sulphide ion in alkaline solution reacts with solid sulphur to form polysulphide ions having formulae S_2^{2-} , S_3^{2-} , S_4^{2-} and so on. The equilibrium constant for the formation of S_2^{2-} is 12 (K_1) & for the formation of S_3^{2-} is 132 (K_2), both from S and S^{2-} . What is the equilibrium constant for the formation of S_3^{2-} from S_2^{2-} and S ?
- (A) 11 (B) 12 (C) 132 (D) None of these
4. For the following gases equilibrium $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$
- K_p is found to be equal to K_c . This is attained when :
- (A) 0 C (B) 273 K (C) 1 K (D) 12.19 K
5. 1 mole N_2 and 3 mol H_2 are placed in a closed container at a pressure of 4 atm. The pressure falls to 3 atm at the same temperature when the following equilibrium is attained
- $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$. The equilibrium constant K_p for dissociation of NH_3 is :
- (A) $\frac{1}{0.5} \times (1.5)^3 \text{ atm}^{-2}$ (B) $0.5 (1.5)^3 \text{ atm}^2$ (C) $\frac{0.5 \times (1.5)^3}{3 \times 3} \text{ atm}^2$ (D) $\frac{3 \times 3}{0.5 \times (1.5)^3} \text{ atm}^{-2}$
6. One mole of $\text{N}_2\text{O}_4(\text{g})$ at 300 K is left in a closed container under one atm. It is heated to 600 K when 20% by mass of $\text{N}_2\text{O}_4(\text{g})$ decomposes to $\text{NO}_2(\text{g})$. The resultant pressure is :
- (A) 1.2 atm (B) 2.4 atm (C) 2.0 atm (D) 1.0 atm
7. For the reaction : $2\text{HI}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{I}_2(\text{g})$, the degree of dissociation (α) of HI (g) is related to equilibrium constant K_p by the expression :
- (A) $\frac{1 + 2\sqrt{K_p}}{2}$ (B) $\sqrt{\frac{1 + 2K_p}{2}}$ (C) $\sqrt{\frac{2K_p}{1 + 2K_p}}$ (D) $\frac{2\sqrt{K_p}}{1 + 2\sqrt{K_p}}$
8. The vapour density of N_2O_4 at a certain temperature is 30. What is the % dissociation of N_2O_4 at this temperature :
- (A) 53.3% (B) 106.6% (C) 26.7% (D) None
9. For the reaction $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$, the forward reaction at constant temperature is favoured by :
- (A) introducing an inert gas at constant volume
- (B) introducing chlorine gas at constant volume
- (C) introducing an inert gas at constant pressure
- (D) introducing PCl_5 at constant volume

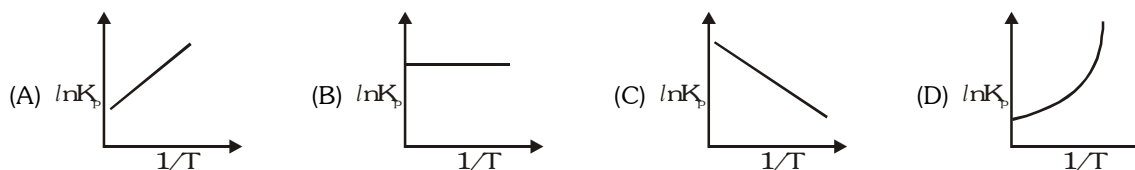
10. When N_2O_5 is heated at temp. T, it dissociates as $\text{N}_2\text{O}_5 \rightleftharpoons \text{N}_2\text{O}_3 + \text{O}_2$, $K_c = 2.5$. At the same time N_2O_3 also decomposes as : $\text{N}_2\text{O}_3 \rightleftharpoons \text{N}_2\text{O} + \text{O}_2$. If initially 4.0 moles of N_2O_5 are taken in 1.0 litre flask and allowed to attain equilibrium, concentration of O_2 was formed to be 2.5 M. Equilibrium concentration of N_2O is :
- (A) 1.0 (B) 1.5 (C) 2.166 (D) 0.334

11. Densities of diamond and graphite are 3.5 and 2.3 g/mL.

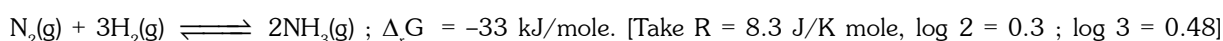


favourable conditions for formation of diamond are :

- (A) high pressure and low temperature
 (B) low pressure and high temperature
 (C) high pressure and high temperature
 (D) low pressure and low temperature
12. When NaNO_3 is heated in a closed vessel, oxygen is liberated and NaNO_2 is left behind. At equilibrium:
- (A) addition of NaNO_2 favours reverse reaction
 (B) addition of NaNO_3 favours forward reaction
 (C) increasing temperature favours forward reaction
 (D) increasing pressure favours reverse reaction
13. The equilibrium $\text{SO}_2\text{Cl}_2(\text{g}) \rightleftharpoons \text{SO}_2(\text{g}) + \text{Cl}_2(\text{g})$ is attained at 25 C in a closed rigid container when an inert gas, helium is introduced. Which of the following statement is / are correct :
- (A) concentrations of SO_2 , Cl_2 and SO_2Cl_2 do not change
 (B) more chlorine is formed
 (C) concentration of SO_2 is reduced
 (D) more SO_2Cl_2 is formed
14. For the gas phase reaction, $\text{C}_2\text{H}_4 + \text{H}_2 \rightleftharpoons \text{C}_2\text{H}_6$ ($\Delta H = -32.7 \text{ kcal}$), carried out in a closed vessel, the equilibrium moles of C_2H_4 can be increased by :
- (A) increasing the temperature (B) decreasing the pressure
 (C) removing some H_2 (D) adding some C_2H_6
15. An exothermic reaction is represented by the graph :



16. The correct relationship between free energy change in a reaction and the corresponding equilibrium constant K is :
- (A) $-\Delta G = RT \ln K$ (B) $\Delta G = RT \ln K$ (C) $-\Delta G = RT \ln K$ (D) $\Delta G = RT \ln K$
17. The value of ΔG_f° of gaseous mercury is 31 kJ/mole. At what external pressure mercury start boiling 25 C. [R = 8.3 J/K mole]
- (A) $10^{-5.44}$ (B) $10^{-12.5}$ (C) $10^{-6.52}$ (D) $10^{-3.12}$
18. What is $\Delta_r G$ (kJ/mole) for synthesis of ammonia at 298 K at following sets of partial pressure :



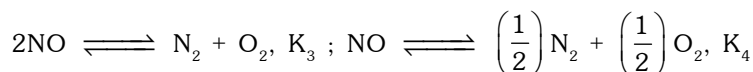
Gas	N_2	H_2	NH_3
Pressure (atm)	1	3	0.02
(A) +6.5	(B) -6.5	(C) +60.5	(D) -60.5

19. In a 7.0 L evacuated chamber, 0.50 mol H_2 and 0.50 mol I_2 react at 427 C.

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$. At the given temperature, $K_c = 49$ for the reaction.

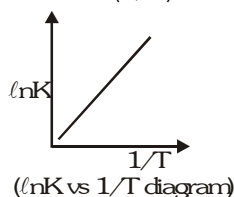
- (i) What is the value of K_p ?
 (A) 7 (B) 49 (C) 24.5 (D) None
- (ii) What is the total pressure (atm) in the chamber ?
 (A) 83.14 (B) 831.4 (C) 8.21 (D) None
- (iii) How many moles of the iodine remain unreacted at equilibrium ?
 (A) 0.388 (B) 0.112 (C) 0.25 (D) 0.125
- (iv) What is the partial pressure (atm) of HI in the equilibrium mixture?
 (A) 6.385 (B) 12.77 (C) 40.768 (D) 646.58

20. $N_2 + O_2 \rightleftharpoons 2NO$, K_1 ; $\left(\frac{1}{2}\right)N_2 + \left(\frac{1}{2}\right)O_2 \rightleftharpoons NO$, K_2

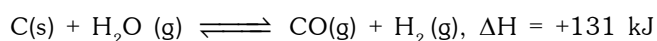


Correct relation between K_1 , K_2 , K_3 and K_4 is :

- (A) $K_1 K_3 = 1$ (B) $\sqrt{K_1} \times K_4 = 1$ (C) $\sqrt{K_3} \times K_2 = 1$ (D) None
21. The equation, $a = \frac{D-d}{(n-1)d}$ is correctly matched for :
 (A) $A \rightleftharpoons nB/2 + nC/3$ (B) $A \rightleftharpoons nB/3 + (2n/3)C$
 (C) $A \rightarrow (n/2)B + (n/4)C$ (D) $A \rightleftharpoons (n/2)B + C$
22. Variation of equilibrium constant K for the reaction ; $2A(s) + B(g) \rightleftharpoons C(g) + 2D(g)$ is plotted against absolute temperature T in figure as - $\ln K$ Vs $(1/T)$:



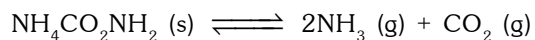
- (A) the forward reaction is exothermic (B) the forward reaction is endothermic
 (C) the slope of line is proportional to ΔH (D) adding 'A' favours forward reaction
 (E) removing C favours forward reaction
23. The equilibrium of which of the following reactions will not be disturbed by the addition of an inert gas at constant volume?
 (A) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ (B) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$
 (C) $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$ (D) $C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$
24. An industrial fuel, 'water gas', which consists of a mixture of H_2 and CO can be made by passing steam over red-hot carbon. The reaction is :



The yield of CO and H_2 at equilibrium would be shifted to the product side by

- (A) raising the relative pressure of the steam
 (B) adding hot carbon
 (C) raising the temperature
 (D) reducing the volume of the system

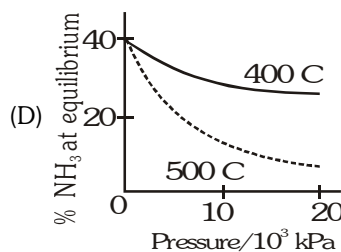
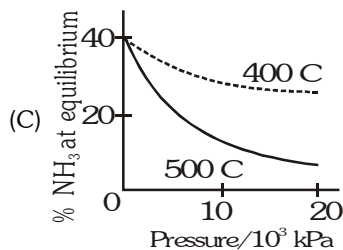
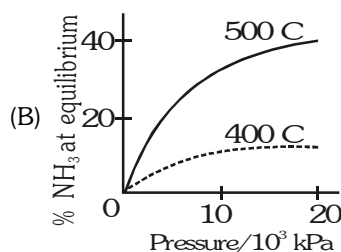
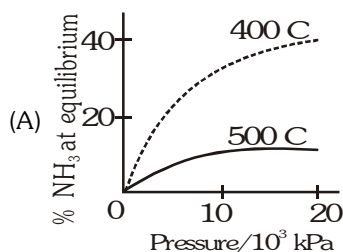
25. The dissociation of ammonium carbamate may be represented by the equation :



ΔH° for the forward reaction is negative. The equilibrium will shift from right to left if there is

- (A) a decrease in pressure
- (B) an increase in temperature
- (C) an increase in the concentration of ammonia
- (D) an increase in the concentration of carbon dioxide

26. The percentage of ammonia obtainable, if equilibrium were to be established during the Haber process, is plotted against the operating pressure for two temperatures 400 C and 500 C. Which of the following correctly represents the two graphs ?



BRAIN TEASERS							ANSWER KEY				EXERCISE -2				
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	B	A	A	D	B	B	D	A	C,D	D	C	C,D	A	A,B,C,D	A
Que.	16	17	18	19(i)	(ii)	(iii)	(iv)	20	21	22	23	24	25	26	
Ans.	A	A	D	B	C	B	A	A,B,C	B	A,C,E	A,B,C,D	A,C	B,C,D	A	

EXERCISE-03

MISCELLANEOUS TYPE QUESTIONS

TRUE / FALSE

1. Van't Hoff's equation gives the quantitative relation between change in value of K with change in temperature.
2. The larger value of K indicates that the product is more stable relative to reactants.
3. The value of equilibrium constant changes with change in the initial concentration of the reactants.
4. Extent of a reaction can always be increased by increasing the temperature.
5. K_p is related to K_c as $K_p = K_c (RT)^{\Delta n}$.
6. Introduction of inert gas in a gaseous reaction ($\Delta n_g \neq 0$) at equilibrium keeping pressure constant has no effect on equilibrium state.
7. For the reaction, $N_2O_4(g) \rightleftharpoons 2NO_2(g)$, $K_p = K_c(RT)$.
8. For a reaction the value of Q greater than K indicates that the net reaction is proceeding in backward direction.
9. Solubilities of all solids in water increase with increase in temperature.
10. Dissolution of all gases in water is accompanied by evolution of heat.
11. For the reaction, $N_2 + 3H_2 \rightleftharpoons 2NH_3$, the equilibrium expression may be written as $K = \frac{[NH_3]^2}{[N_2][H_2]^3}$.
12. For the reaction, $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$, $K_p = p_{CO_2}$.
13. A catalyst increases the value of the equilibrium constant for a reaction.
14. If concentration quotient of reaction is less than K , the net reaction is proceeding in the backward direction.
15. In case of endothermic reaction, the equilibrium shifts in backward direction on increasing the temperature.
16. The value of K increases with increase in pressure.
17. For the reaction, $H_2 + I_2 \rightleftharpoons 2HI$, the equilibrium constant, K is dimensionless.
18. The reaction $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, $\Delta H = -X \text{ kJ}$, is favoured by high pressure and high temperature.
19. A very high value of K indicates that at equilibrium most of the reactants are converted into products.
20. The value of K for the reaction, $N_2 + 3H_2 \rightleftharpoons 2NH_3$, can be increased by applying high pressure or by using a catalyst.

FILL IN THE BLANKS

1. K for the reaction $2A + B \rightleftharpoons 2C$ is 1.5×10^{12} . This indicates that at equilibrium the concentration of would be maximum.
2. The reaction $N_2 + O_2 \rightleftharpoons 2NO - \text{Heat}$, would be favoured by temperature.
3. K for the reaction $X_2 + Y_2 \rightleftharpoons 2XY$ is 100. K for this reaction $XY \rightleftharpoons \frac{1}{2}X_2 + \frac{1}{2}Y_2$ would be
4. Compared to K for the dissociation, $2H_2S \rightleftharpoons 2H^+ + 2HS^-$, then K' for the $H^+ + HS^- \rightleftharpoons H_2S$ would have
5. The equilibrium constant for a reaction decreases with increase in temperature, the reaction must be
6. For the reaction, $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$, K_p and K_c are related as
7. For the reaction, $N_2O_4(g) \rightleftharpoons 2NO_2(g)$, at equilibrium, increase in pressure shifts the equilibrium in direction.

8. ΔG is related to K by the relation
9. Vant Hoff's equation is
10. When the reaction is at equilibrium, the value of ΔG is
11. Dimensions of equilibrium constant, K_c for the reaction $2\text{NH}_3 \rightleftharpoons \text{N}_2 + 3\text{H}_2$, are
12. The value of K for a reaction can be changed by changing
13. The law of mass action was proposed by
14. The degree of dissociation of PCl_5 [$\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$], with increase in pressure at equilibrium.
15. If concentration quotient, Q is greater than K_c , the net reaction is taking place in direction.
16. The reaction, $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ would be favoured by pressure.
17. K_p is related to K_c as
18. Solubility of a gas in water with increase in temperature.
19. Introduction of inert gas at constant volume to a gaseous reaction at equilibrium results in formation of product.
20. The product is more stable than reactants in reaction having K.

MATCH THE COLUMN

1.	Column-I (Reactions)	Column-II (Favourable conditions)
(A)	Oxidation of nitrogen $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) + 180.5 \text{ kJ} \rightleftharpoons 2\text{NO}(\text{g})$	(p) Addition of inert gas at constant pressure
(B)	Dissociation of $\text{N}_2\text{O}_4(\text{g})$ $\text{N}_2\text{O}_4(\text{g}) + 57.2 \text{ kJ} \rightleftharpoons 2\text{NO}_2(\text{g})$	(q) Decrease in pressure
(C)	Oxidation of $\text{NH}_3(\text{g})$ $4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightleftharpoons 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{g}) + 905.6 \text{ kJ}$	(r) Decrease in temperature
(D)	Formation of $\text{NO}_2(\text{g})$ $\text{NO}(\text{g}) + \text{O}_3(\text{g}) \rightleftharpoons \text{NO}_2(\text{g}) + \text{O}_2(\text{g}) + 200 \text{ kJ}$	(s) Increase in temperature

2.	Column-I (Reaction)	Column-II (If α is negligible w.r.t. 1)
(A)	$2\text{X}(\text{g}) \rightleftharpoons \text{Y}(\text{g}) + \text{Z}(\text{g})$	(p) $\alpha = 2 \sqrt{K_c}$
(B)	$\text{X}(\text{g}) \rightleftharpoons \text{Y}(\text{g}) + \text{Z}(\text{g})$	(q) $\alpha = 3 \sqrt{K_c}$
(C)	$3\text{X}(\text{g}) \rightleftharpoons \text{Y}(\text{g}) + \text{Z}(\text{g})$	(r) $\alpha = (2K_c)^{1/3}$
(D)	$2\text{X}(\text{g}) \rightleftharpoons \text{Y}(\text{g}) + 2\text{Z}(\text{g})$	(s) $\alpha = \sqrt{K_c}$

ASSERTION & REASON

These questions contains, Statement I (assertion) and Statement II (reason).

(A) Statement-I is true, Statement-II is true ; Statement-II is correct explanation for Statement-I.

(B) Statement-I is true, Statement-II is true ; Statement-II is NOT a correct explanation for statement-I

(C) Statement-I is true, Statement-II is false

(D) Statement-I is false, Statement-II is true

1. **Statement-I** : The melting point of ice decreases with increase of pressure.

Because

Statement-II : Ice contracts on melting.

2. **Statement-I** : The equilibrium of $A(g) \rightleftharpoons B(g) + C(g)$ is not affected by changing the volume.

Because

Statement-II : K_c for the reaction does not depend on volume of the container.

3. **Statement-I** : For the reaction $A(g) \rightleftharpoons B(g) + C(g)$, $K_p = 1$ atm. If we start with equal moles of all gases at 9 atm of initial pressure, then at equilibrium partial pressure of A increases.

Because

Statement-II : Reaction quotient $Q_p > K_p$ hence equilibrium shifts in backward direction.

4. **Statement-I** : For a reaction at equilibrium, the Gibb's free energy of reaction is minimum at constant temp. and pressure.

Because

Statement-II : The Gibb's free energy of both reactants and products increases and become equal at equilibrium.

5. **Statement-I** : Equilibrium constant for the reverse reaction is the inverse of the equilibrium constant for the reaction in the forward direction.

Because

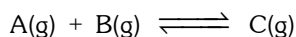
Statement-II : Equilibrium constant depends upon the way in which the reaction is written.

6. **Statement-I** : For the reaction $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ if the volume of vessel is reduced to half of its original volume, equilibrium concentration of all gases will be doubled.

Because

Statement-II : According to Le Chatelier's principle, reaction shifts in a direction that tends to minimized the effect of the stress.

7. **Statement-I** : For the reaction at certain temperature



there will be no effect by addition of inert gas at constant volume.

Because

Statement-II : Molar concentration of all gases remains constant.

8. **Statement-I** : The catalyst does not alter the equilibrium constant.

Because

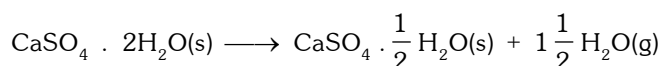
Statement-II : For the catalysed reaction and uncatalysed reaction ΔH remains same and equilibrium constant depends on ΔH .

COMPREHENSION BASED QUESTIONS

Comprehension # 1

On July, 1, 2000, the combined tunnel and bridge connecting Denmark and Sweden was officially opened. It consists of a tunnel from Copenhagen to an artificial Island and a bridge from the island to Malmö in Sweden. The major construction materials employed are concrete and steel. This problem deals with chemical reactions relating to production and degradation of such materials.

Concrete is produced from a mixture of cement, water, sand and small stones. Cement consists primarily of calcium silicates and calcium aluminates formed by heating and grinding of clay and limestone. In the later steps of cement production may lead to formation of unwanted hemihydrate, $\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}$. Consider the following reaction :



The following thermodynamic data apply at 25 C, standard pressure : 1.00 bar :

Compound	H / (kJ mol ⁻¹) (ΔH_f)	S / (JK ⁻¹ mol ⁻¹)
$\text{CaSO}_4 \cdot 2\text{H}_2\text{O(s)}$	-2021.0	194.0
$\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O(s)}$	-1575.0	130.5
$\text{H}_2\text{O(g)}$	-241.8	188.6

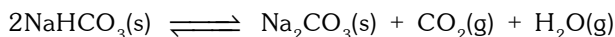
Gas constant ; R = 8.314 J mol⁻¹ K⁻¹

- ΔH for the transformation of 1.00 kg of $\text{CaSO}_4 \cdot 2\text{H}_2\text{O(s)}$ to $\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O(s)}$ is :
(A) +446 kJ (B) +484 kJ (C) -446 kJ (D) -484 kJ
- Equilibrium pressure (in bar) of water vapour in a closed vessel containing $\text{CaSO}_4 \cdot 2\text{H}_2\text{O(s)}$, $\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O(s)}$ and $\text{H}_2\text{O(g)}$ at 25 C is :
(A) 7.35 $\times 10^{-4}$ bar (B) 2.15 $\times 10^{-4}$ bar (C) 8.10 $\times 10^{-3}$ bar (D) 7.00 $\times 10^{-4}$ bar
- Temperature at which the equilibrium water vapour pressure is 1.00 bar.
(A) 107 C (B) 380 C (C) 215 C (D) 240 C

Comprehension # 2

Questions are based on the manufacture of Na_2CO_3 by Solvay process :

In the manufacture of $\text{Na}_2\text{CO}_3(\text{s})$ by Solvay process, $\text{NaHCO}_3(\text{s})$ is decomposed by heating :



$$K_p = 0.23 \text{ at } 100 \text{ C} \quad \Delta H = 136 \text{ kJ}$$

- If a sample of $\text{NaHCO}_3(\text{s})$ is brought to a temperature of 100 C in a closed container total gas pressure at equilibrium is :
(A) 0.96 atm (B) 0.23 atm
(C) 0.48 atm (D) 0.46 atm

2. A mixture of 1.00 mol each of $\text{NaHCO}_3(\text{s})$ and $\text{Na}_2\text{CO}_3(\text{s})$ is introduced into a 2.5 L flask in which $P_{\text{CO}_2} = 2.10 \text{ atm}$ and $P_{\text{H}_2\text{O}} = 0.94 \text{ atm}$. When equilibrium is established at 100°C , then partial pressure of :
- (A) $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{g})$ will be greater than their initial pressure
 (B) $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{g})$ will be less than their initial pressure
 (C) $\text{CO}_2(\text{g})$ will be larger and that of $\text{H}_2\text{O}(\text{g})$ will be less than their initial pressure
 (D) $\text{H}_2\text{O}(\text{g})$ will be larger and that of $\text{CO}_2(\text{g})$ will be less than their initial pressure

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
True / False 1. T 2. T 3. F 4. F 5. T 6. F 7. T 8. T 9. F 10. T 11. T 12. T 13. F 14. F 15. F 16. F 17. T 18. F 19. T 20. F Fill in the Blanks 1. C 2. high 3. $\frac{1}{10}$ 4. $\frac{1}{\sqrt{K}}$ 5. exothermic 6. $K_p = K_c (RT)$ 7. backward 8. $\Delta G = -RT \ln K$ 9. $\log \frac{K_2}{K_1} = \frac{\Delta H^\circ}{2.303R} \left[\frac{T_2 - T_1}{T_2 T_1} \right]$ 10. zero 11. $\text{mol}^2 \text{L}^{-2}$ 12. temperature 13. Guldberg and Waage 14. decreases 15. backward 16. high 17. $K_p = K_c (RT)^{\Delta n}$ 18. decreases 19. same amount of 20. large value of Match the Column 1. A - (s), B - (p,q,s), C - (p,q,r), D - (r) 2. A - (p), B - (s), C - (q), D - (r) Assertion - Reason Questions 1. A 2. D 3. A 4. C 5. A 6. B 7. A 8. A Comprehension Based Questions Comprehension #1 : 1. B 2. C 3. A Comprehension #2 : 1. A 2. B		

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE****Reaction quotient and equilibrium constant**

1. The initial concentration or pressure of reactants and products are given for each of the following systems. Calculate the reaction quotient and determine the directions in which each system will shift to reach equilibrium.
- (a) $2\text{NH}_3(\text{g}) \rightleftharpoons \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$ $K = 17$
[NH₃] = 0.20 M ; [N₂] = 1.00 M ; [H₂] = 1.00 M
- (b) $2\text{NH}_3(\text{g}) \rightleftharpoons \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$ $K_p = 6.8 \times 10^4 \text{ atm}^2$
Initial pressure : NH₃ = 3.0 atm ; N₂ = 2.0 atm ; H₂ = 1.0 atm
- (c) $2\text{SO}_3(\text{g}) \rightleftharpoons 2\text{SO}_2(\text{g}) + \text{O}_2(\text{g})$ $K = 0.230 \text{ atm}$
[SO₃] = 0.00 M ; [SO₂] = 1.00 M ; [O₂] = 1.00 M
- (d) $2\text{SO}_3(\text{g}) \rightleftharpoons 2\text{SO}_2(\text{g}) + \text{O}_2(\text{g})$ $K_p = 16.5 \text{ atm}$
Initial pressure : SO₃ = 1.0 atm ; SO₂ = 1.0 atm ; O₂ = 1.0 atm
- (e) $2\text{NO}(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons 2\text{NOCl}(\text{g})$ $K = 4.6 \times 10^4$
[NO] = 1.00 M ; [Cl₂] = 1.00 M ; [NOCl] = 0 M
2. Among the solubility rules, the statement that all chlorides are soluble except Hg₂Cl₂, AgCl, PbCl₂, and CuCl.
- (a) Write the expression for the equilibrium constant for the reaction represented by the equation
 $\text{AgCl}(\text{s}) \rightleftharpoons \text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq})$
Is K greater than 1, less than 1, or about equal to 1? Explain your answer.
- (b) Write the expression for the equilibrium constant for the reaction represented by the equation
 $\text{Pb}^{2+}(\text{aq}) + 2\text{Cl}^-(\text{aq}) \rightleftharpoons \text{PbCl}_2(\text{s})$
Is K greater than 1, less than 1, or about equal to 1? Explain your answer.
3. Benzene is one of the compounds used as octane enhancers in unleaded gasoline. It is manufactured by the catalytic conversion of acetylene to benzene.
- $$3\text{C}_2\text{H}_2 \longrightarrow \text{C}_6\text{H}_6$$
- Would this reaction be most useful commercially if K were about 0.01, about 1, or about 10? Explain your answer.
4. For which of the following reactions will the equilibrium mixture contain an appreciable concentration of both reactants and products?
- (a) $\text{Cl}_2(\text{g}) \rightleftharpoons 2\text{Cl}(\text{g})$; $K_c = 6.4 \times 10^{-39}$
- (b) $\text{Cl}_2(\text{g}) + 2\text{NO}(\text{g}) \rightleftharpoons 2\text{NOCl}(\text{g})$; $K_c = 3.7 \times 10^8$
- (c) $\text{Cl}_2(\text{g}) + 2\text{NO}_2(\text{g}) \rightleftharpoons 2\text{NO}_2\text{Cl}(\text{g})$; $K_c = 1.8$
5. The value of K_c for the reaction $3\text{O}_2(\text{g}) \rightleftharpoons 2\text{O}_3(\text{g})$ is 1.7×10^{-56} at 25 C. Do you expect pure air at 25 C to contain much O₃ (ozone) when O₂ and O₃ are in equilibrium ? If the equilibrium concentration of O₂ in air at 25 C is $8 \times 10^{-3} \text{ M}$, what is the equilibrium concentration of O₃.

6. At 1400 K, $K_c = 2.5 \times 10^{-3}$ for the reaction $\text{CH}_4(\text{g}) + 2\text{H}_2\text{S} \rightleftharpoons \text{CS}_2(\text{g}) + 4\text{H}_2(\text{g})$. A 10.0 L reaction vessel at 1400 K contains 2.0 mol of CH_4 , 3.0 mol of CS_2 , 3.0 mol of H_2 and 4.0 mol of H_2S . Is this reaction mixture at equilibrium? If not, in which direction does the reaction proceed to reach equilibrium?
7. An equilibrium mixture of N_2 , H_2 and NH_3 at 700 K contains 0.036 M N_2 and 0.15 M H_2 . At this temperature, K_c for the reaction $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ is 0.29. What is the concentration of NH_3 ?
8. The air pollutant NO is produced in automobile engines from the high temperature reaction $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$; $K_c = 1.7 \times 10^{-3}$ at 2300 K. If the initial concentrations of N_2 and O_2 at 2300 K are both 1.40 M, what are the concentration of NO, N_2 and O_2 when the reaction mixture reaches equilibrium?
9. At a certain temperature, the reaction $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ has an equilibrium constant $K_c = 5.8 \times 10^{-2}$. Calculate the equilibrium concentrations of PCl_5 , PCl_3 and Cl_2 if only PCl_5 is present initially, at a concentration of 0.160 M.
10. At 700 K, $K_p = 0.140$ for the reaction $\text{ClF}_3(\text{g}) \rightleftharpoons \text{ClF}(\text{g}) + \text{F}_2(\text{g})$. Calculate the equilibrium partial pressure of ClF_3 , ClF and F_2 if only ClF_3 is present initially; at a partial pressure of 1.47 atm.

Homogeneous equilibria degree of dissociation, vapour density and equilibrium constant

11. The degree of dissociation of N_2O_4 into NO_2 at 1.5 atmosphere and 40 C is 0.25. Calculate its K_p at 40 C. Also report degree of dissociation at 10 atmospheric pressure at same temperature.
12. At 46 C, K_p for the reaction $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ is 0.667 atm. Compute the percent dissociation of N_2O_4 at 46 C at a total pressure of 380 Torr.
13. When 36.8 g $\text{N}_2\text{O}_4(\text{g})$ is introduced into a 1.0 - litre flask at 27 C. The following equilibrium reaction occurs :

$$\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g}) ; K_p = 0.1642 \text{ atm.}$$
 - (a) Calculate K_c of the equilibrium reaction.
 - (b) What are the number of moles of N_2O_4 and NO_2 at equilibrium?
 - (c) What is the total gas pressure in the flask at equilibrium?
 - (d) What is the percent dissociation of N_2O_4 ?
14. At some temperature and under a pressure of 4 atm, PCl_5 is 10% dissociated. Calculate the pressure at which PCl_5 will be 20% dissociated, temperature remaining same.
15. In a mixture of N_2 and H_2 in the ratio of 1 : 3 at 64 atmospheric pressure and 300 C, the percentage of ammonia under equilibrium is 33.33 by volume. Calculate the equilibrium constant of the reaction using the equation.

$$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$$
16. The system $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$ maintained in a closed vessel at 60 C & pressure of 5 atm has an average (i.e. observed) molecular weight of 69, calculate K_p . At what pressure at the same temperature would the observed molecular weight be (230/3)?
17. The vapour density of N_2O_4 at a certain temperature is 30. Calculate the percentage dissociation of N_2O_4 at this temperature. $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$.
18. In the esterification $\text{C}_2\text{H}_5\text{OH}(\ell) + \text{CH}_3\text{COOH}(\ell) \rightleftharpoons \text{CH}_3\text{COOC}_2\text{H}_5(\ell) + \text{H}_2\text{O}(\ell)$ an equimolar mixture of alcohol and acid taken initially yields under equilibrium, the water with mole fraction = 0.333. Calculate the equilibrium constant.

Heterogeneous equilibrium

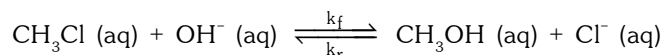
19. Solid Ammonium carbamate dissociates as : $\text{NH}_2\text{COONH}_4 (\text{s}) \rightleftharpoons 2\text{NH}_3 (\text{g}) + \text{CO}_2 (\text{g})$. In a closed vessel solid ammonium carbamate is in equilibrium with its dissociation products. At equilibrium, ammonia is added such that the partial pressure of NH_3 at new equilibrium now equals the original total pressure. Calculate the ratio of total pressure at new equilibrium to that of original total pressure.
20. A sample of $\text{CaCO}_3 (\text{s})$ is introduced into a sealed container of volume 0.821 litre & heated to 1000 K until equilibrium is reached. The equilibrium constant for the reaction $\text{CaCO}_3 (\text{s}) \rightleftharpoons \text{CaO} (\text{s}) + \text{CO}_2 (\text{g})$ is 4×10^{-2} atm at this temperature. Calculate the mass of CaO present at equilibrium.
21. Anhydrous calcium chloride is often used as a desiccant. In the presence of excess of CaCl_2 , the amount of the water taken up is governed by $K_p = 6.4 \times 10^{85}$ for the following reaction at room temperature, $\text{CaCl}_2 (\text{s}) + 6\text{H}_2\text{O} (\text{g}) \rightleftharpoons \text{CaCl}_2 \cdot 6\text{H}_2\text{O} (\text{s})$. What is the equilibrium vapour pressure of water in a closed vessel that contains $\text{CaCl}_2 (\text{s})$?
22. 20.0 grams of $\text{CaCO}_3 (\text{s})$ were placed in a closed vessel, heated & maintained at 727 C under equilibrium $\text{CaCO}_3 (\text{s}) \rightleftharpoons \text{CaO} (\text{s}) + \text{CO}_2 (\text{g})$ and it is found that 75% of CaCO_3 was decomposed. What is the value of K_p ? The volume of the container was 15 litres.

Changes in concentration at equilibrium Le Chatelier's principle

23. Suggest four ways in which the concentration of hydrazine, N_2H_4 , could be increased in an equilibrium described by the equation
- $$\text{N}_2 (\text{g}) + 2\text{H}_2 (\text{g}) \rightleftharpoons \text{N}_2\text{H}_4 (\text{g}) \quad \Delta H = 95 \text{ kJ}$$
24. How will an increase in temperature and increase in pressure affect each of the following equilibria?
- (a) $2\text{NH}_3 (\text{g}) \rightleftharpoons \text{N}_2 (\text{g}) + 3\text{H}_2 (\text{g}) \quad \Delta H = 92 \text{ kJ}$
- (b) $\text{N}_2 (\text{g}) + \text{O}_2 (\text{g}) \rightleftharpoons 2\text{NO} (\text{g}) \quad \Delta H = 181 \text{ kJ}$
- (c) $2\text{O}_3 (\text{g}) \rightleftharpoons 3 \text{O}_2 (\text{g}) \quad \Delta H = -285 \text{ kJ}$
- (d) $\text{CaO} (\text{s}) + \text{CO}_2 (\text{g}) \rightleftharpoons \text{CaCO}_3 (\text{s}) \quad \Delta H = -176 \text{ kJ}$
25. (a) Water gas, a mixture of H_2 and CO , is an important industrial fuel produced by the reaction of steam with red-hot coke, essentially pure carbon. Write the expression for the equilibrium constant for the reversible reaction.
- $$\text{C} (\text{s}) + \text{H}_2\text{O} (\text{g}) \rightleftharpoons \text{CO} (\text{g}) + \text{H}_2 (\text{g}) \quad \Delta H = 131.30 \text{ kJ}$$
- (b) Assume that equilibrium has been established and predict how the concentration of each reactant and product will differ at a new equilibrium if (1) more C is added. (2) H_2O is removed. (3) CO is added (4) the pressure on the system is increased. (5) the temperature of the system is increased.
26. Ammonia is weak base that reacts with water according to the equation
- $$\text{NH}_3 (\text{aq}) + \text{H}_2\text{O} (\text{l}) \rightleftharpoons \text{NH}_4^+ (\text{aq}) + \text{OH}^- (\text{aq})$$
- Will any of the following increase the percent of ammonia that is converted to the ammonium ion in water?
- (a) Addition of NaOH.
- (b) Addition of HCl.
- (c) Addition of NH_4Cl .
27. Suggest two ways in which equilibrium concentration of Ag^+ can be reduced in a solution of Na^+ , Cl^- , Ag^+ and NO_3^- , in contact with solid AgCl .
- $$\text{Na}^+ (\text{aq}) + \text{Cl}^- (\text{aq}) + \text{Ag}^+ (\text{aq}) + \text{NO}_3^- (\text{aq}) \rightleftharpoons \text{AgCl} (\text{s}) + \text{Na}^+ (\text{aq}) + \text{NO}_3^- (\text{aq}) \quad \Delta H = -65.9 \text{ kJ}$$

Kinetics and equilibrium constant

28. Consider a general single-step reaction of the type $A + B \rightleftharpoons C$. Show that the equilibrium constant is equal to the ratio of the rate constant for the forward and reverse reaction, $K_c = K_f/K_r$.
29. Consider the reaction of chloromethane with OH^- in aqueous solution

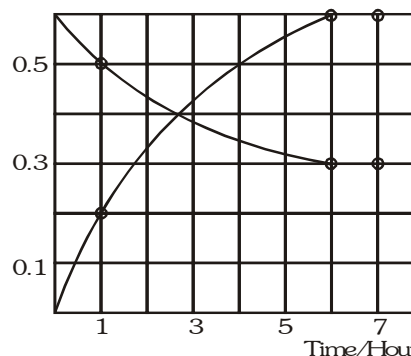


At 25 C, the rate constant for the forward reaction is $6 \times 10^{-6} \text{ M}^{-1} \text{ s}^{-1}$, and the equilibrium constant K_c is 1×10^{16} . Calculate the rate constant for the reverse reaction at 25 C.

30. The progress of the reaction

$A \rightleftharpoons nB$ with time, is presented in figure, Determine

- the value of n.
- the equilibrium constant K.
- the initial rate of conversion of A.



Temperature dependence of equilibrium constant

31. Listed in the table are forward and reverse rate constants for the reaction $2\text{NO (g)} \rightleftharpoons \text{N}_2 \text{ (g)} + \text{O}_2 \text{ (g)}$

Temperature (K)	$k_f \text{ (M}^{-1}\text{s}^{-1}\text{)}$	$k_r \text{ (M}^{-1}\text{s}^{-1}\text{)}$
1400	0.29	1.1×10^{-6}
1500	1.3	1.4×10^{-5}

Is the reaction endothermic or exothermic?

32. Rate of disappearance of the reactant A at two different temperature is given by $A \rightleftharpoons B$

$$\frac{-d[A]}{dt} = (2 \times 10^{-2} \text{ s}^{-1}) [A] - 4 \times 10^{-3} \text{ s}^{-1} [B] ; 300 \text{ K}$$

$$\frac{-d[A]}{dt} = (4 \times 10^{-2} \text{ s}^{-1}) [A] - 16 \times 10^{-4} \text{ s}^{-1} [B] ; 400 \text{ K}$$

Calculate heat of reaction in the given temperature range. When equilibrium is set up.

33. The K_p for reaction $A + B \rightleftharpoons C + D$ is 1.34 at 60 C and 6.64 at 100 C. Determine the free energy change of this reaction at each temperature and ΔH for the reaction over this range of temperature?

Equilibrium expressions and equilibrium constants

34. If $K_c = 7.5 \times 10^{-9}$ at 1000 K for the reaction $\text{N}_2 \text{ (g)} + \text{O}_2 \text{ (g)} \rightleftharpoons 2\text{NO (g)}$, what is K_c at 1000 K for the reaction $2\text{NO (g)} \rightleftharpoons \text{N}_2 \text{ (g)} + \text{O}_2 \text{ (g)}$?
35. A sample of HI ($9.30 \times 10^{-3} \text{ mol}$) was placed in an empty 2.00 L container at 1000 K. After equilibrium was reached, the concentration of I_2 was $6.29 \times 10^{-4} \text{ M}$. Calculate the value of K_c at 1000 K for the reaction $\text{H}_2 \text{ (g)} + \text{I}_2 \text{ (g)} \rightleftharpoons 2\text{HI (g)}$.
36. The vapour pressure of water at 25 C is 0.0313 atm. Calculate the values of K_p and K_c at 25 C for the equilibrium $\text{H}_2\text{O (l)} \rightleftharpoons \text{H}_2\text{O (g)}$.

37. For each of the following equilibria, write the equilibrium constant expression for K_c . Where appropriate, also write the equilibrium constant expression for K_p .
- (a) $\text{Fe}_2\text{O}_3 (\text{s}) + 3\text{CO} (\text{g}) \rightleftharpoons 2\text{Fe} (\ell) + 3\text{CO}_2 (\text{g})$
- (b) $4\text{Fe} (\text{s}) + 3\text{O}_2 (\text{g}) \rightleftharpoons 2\text{Fe}_2\text{O}_3 (\text{s})$
- (c) $\text{BaSO}_4 (\text{s}) \rightleftharpoons \text{BaO} (\text{s}) + \text{SO}_3 (\text{g})$
- (d) $\text{BaSO}_4 (\text{s}) \rightleftharpoons \text{Ba}^{2+} (\text{aq}) + \text{SO}_4^{2-} (\text{aq})$

General problems

38. When 0.5 mol of N_2O_4 is placed in a 4.00 L reaction vessel and heated at 400 K, 79.3 % of the N_2O_4 decomposes to NO_2 .
Calculate K_c and K_p at 400 K for the reaction $\text{N}_2\text{O}_4 (\text{g}) \rightleftharpoons 2\text{NO}_2 (\text{g})$.
39. At 100 K, then value of K_c for the reaction $\text{C}(\text{s}) + \text{H}_2\text{O} (\text{g}) \rightleftharpoons \text{CO} (\text{g}) + \text{H}_2 (\text{g})$ is 3.0×10^{-2} . Calculate equilibrium concentrations of H_2O , CO_2 and H_2 in the reaction mixture obtained by heating 6.0 mole of steam and an excess of solid carbon in a 5.0 L container. What is the molar composition of the equilibrium mixture?
40. When 1.0 mol of PCl_5 is introduced into a 5.0 L container at 500 K, 78.5 % of the PCl_5 dissociates to given an equilibrium mixture of PCl_5 , PCl_3 and Cl_2 .
$$\text{PCl}_5 (\text{g}) \rightleftharpoons \text{PCl}_3 (\text{g}) + \text{Cl}_2 (\text{g})$$
- (a) Calculate the values of K_c and K_p .
- (b) If the initial concentrations in a particular mixture of reactants and products are $[\text{PCl}_5] = 0.5 \text{ M}$, $[\text{PCl}_3] = 0.15 \text{ M}$, and $[\text{Cl}_2] = 0.6 \text{ M}$, in which direction does the reaction proceed to reach equilibrium? What are the concentrations when the mixture reaches equilibrium?

Thermodynamic and equilibrium constant

41. α -D-Glucose undergoes mutarotation to β -D-Glucose in aqueous solution. If at 298 K there is 60% conversion. Calculate ΔG of the reaction.
$$\alpha\text{-D-Glucose} \rightleftharpoons \beta\text{-D-Glucose}$$
42. For the reaction at 298 K
$$\text{A} (\text{g}) + \text{B} (\text{g}) \rightleftharpoons \text{C} (\text{g}) + \text{D} (\text{g})$$

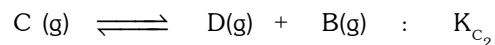
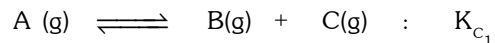
 $\Delta H = -29.8 \text{ kcal}$; $\Delta S = -0.1 \text{ kcal/K}$
Calculate ΔG and K .
43. The equilibrium constant of the reaction $2\text{C}_3\text{H}_6 (\text{g}) \rightleftharpoons \text{C}_2\text{H}_4 (\text{g}) + \text{C}_4\text{H}_8 (\text{g})$ is found to fit the expression
$$\ln K = -1.04 - \frac{1088}{T}$$

Calculate the standard reaction enthalpy and entropy at 400 K.
44. PCl_5 dissociates according to the reaction $\text{PCl}_5 (\text{g}) \rightleftharpoons \text{PCl}_3 (\text{g}) + \text{Cl}_2 (\text{g})$. At 523 K, $K_p = 1.78 \text{ atm}$. Find the density of the equilibrium mixture at a total pressure of 1 atm.
45. The following data for the equilibrium composition of the reaction
$$2\text{Na} (\text{g}) \rightleftharpoons \text{Na}_2 (\text{g})$$

at 1.013 MP_a pressure and 1482.53 K have been obtained.
mass % Na (monomer gas) = 71.3
mass % Na_2 (dimer gas) = 28.7
Calculate the equilibrium constant K_p .

46. A certain gas A polymerizes to a small extent at a given temperature & pressure, $nA \rightleftharpoons A_n$. Show that the gas obeys the approx equation $\frac{PV}{RT} = \left[1 - \frac{(n-1)K_c}{V^{n-1}} \right]$ where $K_c = \frac{[A_n]}{[A]^n}$ & V is the volume of the container. Assume that initially one mole of A was taken in the container.

47. When 1 mole of A (g) is introduced in a closed rigid 1 litre vessel maintained at constant temperature the following equilibria are established.

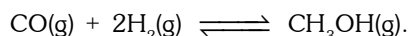


The pressure at equilibrium is twice the initial pressure. Calculate the value of $\frac{K_{c_2}}{K_{c_1}}$ if $\frac{[C]_{eq}}{[B]_{eq}} = \frac{1}{5}$

CONCEPTUAL	SUBJECTIVE EXERCISE	ANSWER KEY	EXERCISE-4(A)
1.	(a) 25, shifts left, (b) 0.22, shifts right, (c) ∞ , shifts left, (d) 1, shifts right, (e) 0, shifts right		
2.	(a) $K = [\text{Ag}^+][\text{Cl}^-]$ is less than 1. AgCl is insoluble thus the concentration of ions are much less than 1 M (b) $K = 1/[\text{Pb}^{2+}][\text{Cl}^-]^2$ is greater than one because PbCl_2 is insoluble and formation of the solid will reduce the concentration of ions to a low level.		
3.	K about 10	4. c	5. $\sim 9 \times 10^{-32} \text{ mol/L}$
6.	The reaction is not in equilibrium because $Q_c > K_c$. The reaction will proceed from right to left to reach equilibrium.		
7.	$5.9 \times 10^{-3} \text{ M}$	8. $[\text{NO}] = 0.056 \text{ M}$, $[\text{N}_2] = [\text{O}_2] = 1.37 \text{ M}$	
9.	$[\text{PCl}_3] = [\text{Cl}_2] = 0.071 \text{ M}$, $[\text{PCl}_5] = 0.089 \text{ M}$	10. $P_{\text{ClF}} = P_{\text{F}_2} = 0.389 \text{ atm}$, $P_{\text{ClF}_3} = 1.08 \text{ atm}$	
11.	$K_p = 0.4$, $\alpha \sim 0.1$	12. 50%	
13.	(a) $6.667 \times 10^{-3} \text{ mol L}^{-1}$; (b) $n(\text{N}_2\text{O}_4) = 0.374 \text{ mol}$; $n(\text{NO}_2) = 0.052 \text{ mol}$; (c) 10.49 atm (d) 6.44%		
14.	0.97 atm	15. $K_p = 1.3 \times 10^{-3} \text{ atm}^{-2}$	
16.	$K_p = 2.5 \text{ atm}$, $P = 15 \text{ atm}$	17. 53.33%	18. $K = 4$
19.			31/27
20.	22.4 mg	21. $P_{\text{H}_2\text{O}} = 5 \times 10^{-15} \text{ atm}$	22. 0.821 atm
23.	add N_2 , add H_2 , increase the pressure, heat the reaction		
24.	(a) shift right, shift left, (b) shift right, no effect, (c) shift left, shift left, (d) shift left, shift right		
25.	(a) $K = [\text{CO}][\text{H}_2] / [\text{H}_2\text{O}]$; (b) in each of the following cases the mass of carbon will change, but its concentration (activity) will not change.		
1.	$[\text{H}_2\text{O}]$ no change, $[\text{CO}]$ no change, $[\text{H}_2]$ no change ;	2. $[\text{H}_2\text{O}]$ decrease, $[\text{CO}]$ decrease, $[\text{H}_2]$ decrease ;	
3.	$[\text{H}_2\text{O}]$ increase, $[\text{CO}]$ increase, $[\text{H}_2]$ decrease ;	4. $[\text{H}_2\text{O}]$ increase, $[\text{CO}]$ decrease, $[\text{H}_2]$ decrease ;	
5.	$[\text{H}_2\text{O}]$ decrease, $[\text{CO}]$ increase, $[\text{H}_2]$ increase ;		
26.	b	27. Add NaCl or some other salt that produces Cl^- in the solution. Cool the solution	
28.	$k_f [\text{A}][\text{B}] = k_r [\text{C}]$; $\frac{k_f}{k_r} = \frac{[\text{C}]}{[\text{A}][\text{B}]} = K_c$	29. 6×10^{-22}	30. (i) 2 ; (ii) 1.2 mol/L ; (iii) 0.1 moles/hr
31.	The reaction is exothermic	32. 16.06 kJ	
33.	-810 J/mol ; -5872 J/mol and 41.3 kJ/mol		34. 1.3×10^8
35.	29.0	36. $K_p = 0.0313 \text{ atm}$, $K_c = 1.28 \times 10^{-3}$	
37.	(a) $K_c = \frac{[\text{CO}_2]^3}{[\text{CO}]^3}$, $K_p = \frac{(P_{\text{CO}_2})^3}{(P_{\text{CO}})^3}$, (b) $K_c = \frac{1}{[\text{O}_2]^3}$, $K_p = \frac{1}{(P_{\text{O}_2})^3}$; (c) $K_c = [\text{SO}_3]$, $K_p = P_{\text{SO}_3}$ (d) $K_c = [\text{Ba}^{2+}][\text{SO}_4^{2-}]$		
38.	$K_c = 1.51$, $K_p = 49.6$	39. $[\text{CO}] = [\text{H}_2] = 0.18 \text{ M}$; $[\text{H}_2\text{O}] = 1.02 \text{ M}$	
40.	(a) $K_c = 0.573$ and $K_p = 23.5$; (b) to the right, $[\text{PCl}_5] = 0.365 \text{ M}$; $[\text{PCl}_3] = 0.285 \text{ M}$; $[\text{Cl}_2] = 0.735 \text{ M}$		
41.	-1.005 kJ/mol	42. $\Delta G = 0$; $K = 1$	
43.	$\Delta H = 9.04 \text{ kJ/mol}$; $\Delta S = -8.64 \text{ J / mol}^{-1} \text{ K}^{-1}$	44. 2.71 g/L	
45.	$P_{\text{Na}} = 0.843 \text{ M Pa}$, $P_{\text{Na}_2} = 0.170 \text{ M Pa}$, $k_p = 0.293$	46. To be proved	47. 4

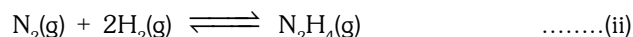
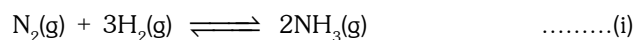
EXERCISE-04 [B]**BRAIN STORMING SUBJECTIVE EXERCISE**

1. 0.15 mole of CO taken in a 2.5 litre flask is maintained at 750 K along with a catalyst so that the following reaction can take place ;



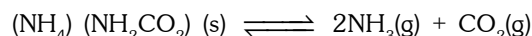
Hydrogen is introduced until the total pressure of the system is 8.5 atm at equilibrium and 0.08 mole of methanol is formed. Calculate :

- (i) K_p & K_c ;
(ii) the final pressure if the same amount of CO and H_2 as before are used, but with no catalyst so that the reaction does not take place.
2. In a vessel, two equilibrium are simultaneously established at the same temperature as follows :



Initially the vessel contains N_2 and H_2 in the molar ratio of 9 : 13. The equilibrium pressure is $7P_0$, in which pressure due to ammonia is P_0 and due to hydrogen is $2P_0$. Find the values of equilibrium constant (K_p 's) for both the reactions.

3. The decomposition of solid ammonium carbamate, $(\text{NH}_4)(\text{NH}_2\text{CO}_2)$, to gaseous ammonia and carbon dioxide is an endothermic reaction.

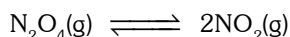


- (a) When solid $(\text{NH}_4)(\text{NH}_2\text{CO}_2)$ is introduced into an evacuated flask at 25 C, the total pressure of gas at equilibrium is 0.116 atm. What is the value of K_p at 25 C ?
(b) Given that the decomposition reaction is at equilibrium, how would the following changes affect the total quantity of NH_3 in the flask once equilibrium is re-established?
(i) Adding CO_2
(ii) Adding $(\text{NH}_4)(\text{NH}_2\text{CO}_2)$
(iii) Removing CO_2
(iv) Increasing the total volume
(v) Adding neon
(vi) Increasing the temperature

4. A container contains three gases. A, B and C in equilibrium $\text{A} \rightleftharpoons 2\text{B} + \text{C}$

At equilibrium the concentration of A was 3 M, and of B was 4 M. On doubling the volume of container, the new equilibrium concentration of B was 3M. Calculate K_c and initial equilibrium concentration of C.

5. The density of an equilibrium mixture of N_2O_4 and NO_2 at 1 atm and 346 K is 1.8 g/L. Calculate K_c for the reaction.



6. At 90 C, the following equilibrium is established :



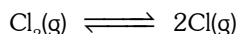
If 0.2 mol of hydrogen and 1.0 mol of sulphur are heated to 90 C in a 1.0 litre vessel, what will be the partial pressure of H_2S at equilibrium ?

7. At 817 C, K_p for the reaction between pure CO_2 and excess hot graphite to form 2CO(g) is 10 atm.

- (a) What is the analysis of the gases at equilibrium at 817 C & a total pressure of 4.0 atm ? What is the partial pressure of CO_2 at equilibrium ?
(b) At what total pressure will the gas mixture analyze 6%, CO_2 by volume ?

8. For the reaction $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$, equilibrium mixture contains NO_2 at $P = 1.1$ atm & N_2O_4 at $P = 0.28$ atm at 350 K. The volume of the container is doubled. Calculate the equilibrium pressures of the two gases when the system reaches new equilibrium.

9. The degree of dissociation of HI at a particular temperature is 0.8. Find the volume of 1.5 M sodium thiosulphate solution required to react completely with the iodine present at equilibrium in acidic conditions, when 0.135 mol each of H_2 and I_2 are heated at 440 K in a closed vessel of capacity 2.0 L.
10. A mixture of hydrogen & iodine in the mole ratio 1.5 : 1 is maintained at 450 C. After the attainment of equilibrium $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, it is found on analysis that the mole ratio of I_2 to HI is 1 : 18. Calculate the equilibrium constant & the number of moles of each species present under equilibrium, if initially, 127 g of iodine were taken.
11. The equilibrium constant for the reaction $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$ is 7.3 at 450 C & 1 atm pressure. The initially concentration of water gas [CO & H_2 in equimolar ratio] & steam are 2 moles & 5 moles respectively. Find the number of moles of CO , H_2 , CO_2 & H_2O (vapour) at equilibrium.
12. At 1200 C, the following equilibrium is established between chlorine atoms & molecule.



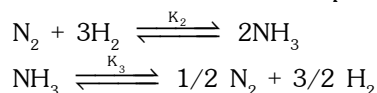
The composition of equilibrium mixture may be determined by measuring the rate of effusion of the mixture through a pin hold. It is found that at 1200 C and 1 atm pressure the mixture effuses 1.16 times as fast as krypton effuses under the same condition. Calculate the equilibrium constant K_c .

13. Two solids X and Y dissociate into gaseous products at a certain temperature as follows :
 $X(s) \rightleftharpoons A(g) + C(g)$, and $Y(s) \rightleftharpoons B(g) + C(g)$. At a given temperature, pressure over excess solid X is 40 mm and total pressure over solid Y is 60 mm. Calculate :
 (a) the values of K_p for two reactions (in mm)
 (b) the ratio of moles of A and B in the vapour state over a mixture of X and Y.
 (c) the total pressure of gases over a mixture of X and Y.
14. SO_3 decomposes at a temperature of 1000 K and at a total pressure of 1.642 atm. At equilibrium, the density of mixture is found to be 1.28 g/L in a vessel of 90 litres. Find the degree of dissociation of SO_3 for $SO_3 \rightleftharpoons SO_2 + 1/2O_2$.
15. The density of an equilibrium mixture of N_2O_4 and NO_2 at 101.32 KP_a is 3.62 $g\ dm^{-3}$ at 288 K and 1.84 $g\ dm^{-3}$ at 348 K. What is the heat of the reaction for $N_2O_4 \rightleftharpoons 2NO_2(g)$.
16. The equilibrium constant for the following reaction at 1395 K.



Calculate the value of K for the reaction : $H_2 + CO_2 \rightleftharpoons CO + H_2O$

17. A saturated solution of iodine in water contains 0.33 g I_2 /L. More than this can dissolve in a KI solution because of the equilibrium : $I_2(aq) + I^-(aq) \rightleftharpoons I_3^-(aq)$. A 0.10 M KI solution (0.10 MI^-) actually dissolves 12.5 g of iodine/L, most of which is converted to I_3^- . Assuming that the concentration of I_2 in all saturated solutions is the same, calculate the equilibrium constant for the above reaction. What is the effect of adding water to a clear saturated of I_2 in the KI solution ?
18. A mixture of N_2 & H_2 are in equilibrium at 600 K at a total pressure of 80 atm. If the initial ratio of N_2 and H_2 are 3 : 1 and at equilibrium NH_3 is 10% by volume, calculate K_p of reaction at given temperature.
19. ΔG (298 K) for the reaction $1/2N_2 + 3H_2 \xrightleftharpoons{K_1} NH_3$ is $-16.5\ kJ\ mol^{-1}$. Find the equilibrium constant (K_1) at 25 C. What will be the equilibrium constants K_2 and K_3 for the following reactions :



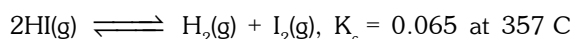
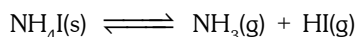
20. When NO & NO_2 are mixed, the following equilibria are readily obtained ;



In an experiment when NO & NO_2 are mixed in the ratio of 1 : 2, the total final pressure was 5.05 atm & the partial pressure of N_2O_4 was 1.7 atm. Calculate :

- (a) the equilibrium partial pressure of NO . (b) K_p for $NO + NO_2 \rightleftharpoons N_2O_3$

21. Solid NH_4I on rapid heating in a closed vessel at 357 C develops a constant pressure of 275 mm Hg owing to partial decomposition of NH_4I into NH_3 and HI but the pressure gradually increases further (when the excess solid residue remains in the vessel) owing to the dissociation of HI. Calculate the final pressure developed at equilibrium.



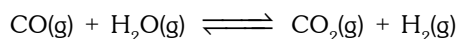
22. For the reaction $\text{SO}_2\text{(g)} + 1/2\text{O}_2\text{(g)} \rightleftharpoons \text{SO}_3\text{(g)}$ $\Delta H_{298} = -98.32 \text{ kJ/mole}$,

$\Delta S_{298} = -95.0 \text{ J/K/mole}$. Find the K_p for this reaction at 298 K.

23. Given are the following standard free energies of formation at 298 K.

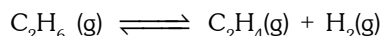
	CO(g)	$\text{CO}_2\text{(g)}$	$\text{H}_2\text{O(g)}$	$\text{H}_2\text{O(l)}$
$\Delta_f G / \text{kJ mole}^{-1}$	-137.17	-394.36	-228.57	-237.13

(a) Find $\Delta_r G$ and the standard equilibrium constant K_p^0 at 298 K for the reaction



(b) If CO , CO_2 and H_2 are mixed so that the partial pressure of each is 101.325 kPa and the mixture is brought into contact with excess of liquid water, what will be the partial pressure of each gas when equilibrium is attained at 298 K. The volume available to the gases is constant.

24. For the reaction



K_p^0 is 0.05 and $\Delta_r G$ is 22.384 kJ mol⁻¹ at 900 K. If an initial mixture comprising 20 mol of C_2H_6 and 80 mol of N_2 (inert) is passed over a dehydrogenation catalyst at 900 K, what is the equilibrium percentage composition of the effluent gas mixture? The total pressure is kept at 0.5 bar. Given : $\Delta_r S = 135.143 \text{ J K}^{-1} \text{ mol}^{-1}$ at 300 K. Calculate $\Delta_r G$ at 300 K. (Assume $\Delta_r C_p = 0$).

BRAIN STORMING SUBJECTIVE EXERCISE	ANSWER KEY	EXERCISE-4(B)
1. (i) $k_c = 187.85 \text{ mol}^{-2} \text{ lit}^2$, $k_p = 0.05 \text{ atm}^{-2}$ (ii) $P = 12.438 \text{ atm}$		
2. $K_{P_1} = \frac{1}{20P_0^2}$, $K_{P_2} = \frac{3}{20P_0}$		
3. (a) 2.31×10^{-4} (b) (i) decrease (ii) no change (iii) increase (iv) increase (v) no change (vi) increase		
4. 28.8	5. 12.8	6. 0.379 atm
7. (i) $p(\text{CO}_2) = 0.938 \text{ atm}$ (ii) $P_{\text{Total}} = 0.68 \text{ atm}$	8. $P_{\text{NO}_2} = 0.64 \text{ atm}$, $P_{\text{N}_2\text{O}_4} = 0.095 \text{ atm}$	
9. $V = 144 \text{ mL}$	10. $k_c = 54$, $n_{\text{HI}} = 0.9 \text{ mol}$, $n_{\text{I}_2} = 0.05 \text{ mol}$, $n_{\text{H}_2} = 0.3 \text{ mol}$	
11. $n_{\text{CO}_2} = 0.938$, $n_{\text{H}_2} = 1.938$, $n_{\text{CO}} = 0.062$, $n_{\text{H}_2\text{O}} = 4.062$	12. 6.3×10^{-4}	
13. (a) 400 mm^2 , 900 mm^2 (b) $4 : 9$ (c) 72.11 mmHg	14. $\alpha = 0.5$	
15. $\Delta_r H = 75.68 \text{ kJ mol}^{-1}$	16. $k = 2.58$	
17. $K = 707.2$ backward reaction is favoured	18. 1.032×10^{-3}	
19. $K_1 = 779.4$, $K_2 = 6.074 \times 10^5$; $K_3 = 1.283 \times 10^{-3}$		
20. (a) 1.05 atm (b) 3.43 atm^{-1}	21. 337 mmHg	
22. $K_p = 1.862 \times 10^{12} \text{ atm}^{-1/2}$		
23. (a) $\Delta G = -28.62 \text{ kJ/mole}$ (b) $P_{\text{CO}_2} = 202.44 \text{ kPa}$, $P_{\text{H}_2\text{O}} = 3.16 \text{ kPa}$, $P_{\text{CO}} = 0.2 \text{ kPa}$, $K_p = 1.039 \times 10^5$		
24. 103.47 kJ/mol		

EXERCISE - 05 [A]**JEE-[MAINS] : PREVIOUS YEAR QUESTIONS**

1. Reaction $\text{CO}_{(g)} + \frac{1}{2} \text{O}_{2(g)} \rightleftharpoons \text{CO}_{2(g)}$. The value of $\frac{K_p}{K_c}$ is - [AIEEE-2002]
- (A) $\frac{1}{RT}$ (B) $\sqrt{RT}$ (C) $\frac{1}{\sqrt{RT}}$ (D) RT
2. One of the following equilibrium is not affected by change in volume of the flask - [AIEEE-2002]
- (A) $\text{PCl}_5(g) \rightleftharpoons \text{PCl}_3(g) + \text{Cl}_2(g)$ (B) $\text{N}_2(g) + 3\text{H}_2(g) \rightleftharpoons 2\text{NH}_3(g)$
(C) $\text{N}_2(g) + \text{O}_2 \rightleftharpoons 2\text{NO}(g)$ (D) $\text{SO}_2\text{Cl}_2(g) \rightleftharpoons \text{SO}_2(g) + \text{Cl}_2(g)$
3. For the reaction equilibrium,
 $\text{N}_2\text{O}_4(g) \rightleftharpoons 2\text{NO}_2(g)$ the concentration of N_2O_4 and NO_2 at equilibrium are 4.8×10^{-2} and $1.2 \times 10^{-2} \text{ mol L}^{-1}$ respectively. The value of K_c for the reaction is- [AIEEE-2003]
- (A) $3 \times 10^{-3} \text{ mol L}^{-1}$ (B) $3 \times 10^3 \text{ mol L}^{-1}$ (C) $3.3 \times 10^2 \text{ mol L}^{-1}$ (D) $3 \times 10^{-1} \text{ mol L}^{-1}$
4. Consider the reaction equilibrium [AIEEE-2003]
 $2\text{SO}_{2(g)} + \text{O}_{2(g)} \rightleftharpoons 2\text{SO}_{3(g)} ; \Delta H = -198 \text{ kJ}$
On the basis of Le-Chatelier's principle, the condition favourable for the forward reaction is -
(A) Lowering the temperature and increasing the pressure
(B) Any value of temperature as well as pressure
(C) Lowering of temperature as well as pressure
(D) Increasing temperature as well as pressure
5. What is the equilibrium expression for the reaction $\text{P}_{4(s)} + 5\text{O}_{2(g)} \rightleftharpoons \text{P}_4\text{O}_{10(s)}$? [AIEEE-2004]
- (A) $K_c = [\text{P}_4\text{O}_{10}] / [\text{P}_4] [\text{O}_2]^5$ (B) $K_c = [\text{P}_4\text{O}_{10}] / 5 [\text{P}_4] [\text{O}_2]$
(C) $K_c = [\text{O}_2]^5$ (D) $K_c = 1 / [\text{O}_2]^5$
6. For the reaction $\text{CO}_{(g)} + \text{Cl}_{2(g)} \rightleftharpoons \text{COCl}_{2(g)}$ the $\frac{K_p}{K_c}$ is equal to - [AIEEE-2004]
- (A) $\frac{1}{RT}$ (B) RT (C) $\sqrt{RT}$ (D) 1.0
7. The equilibrium constant for the reaction $\text{N}_{2(g)} + \text{O}_{2(g)} \rightleftharpoons 2\text{NO}_{(g)}$ at temperature T is 4×10^{-4} . The value of K_c for the reaction $\text{NO}_{(g)} \rightleftharpoons \frac{1}{2} \text{N}_{2(g)} + \frac{1}{2} \text{O}_{2(g)}$ [AIEEE-2004]
- (A) 2.5×10^2 (B) 50 (C) 4×10^{-4} (D) 0.02
8. For the reaction $2\text{NO}_{2(g)} \rightleftharpoons 2\text{NO}_{(g)} + \text{O}_{2(g)}$, ($K_c = 1.8 \times 10^{-6}$ at 184°C) ($R = 0.831 \text{ kJ}(\text{mol.K})$) When K_p and K_c are compared at 184°C it is found that [AIEEE-2005]
- (A) K_p is less than K_c
(B) K_p is greater than K_c
(C) Whether K_p is greater than, less than or equal to K_c depends upon the total gas pressure
(D) $K_p = K_c$
9. The exothermic formation of ClF_3 is represented by the equation [AIEEE-2005]
 $\text{Cl}_{2(g)} + 3\text{F}_{2(g)} \rightleftharpoons 2\text{ClF}_{3(g)} ; \Delta H_f = -329 \text{ kJ}$
Which of the following will increase the quantity of ClF_3 in an equilibrium mixture of Cl_2 , F_2 and ClF_3 ?
- (A) Removing Cl_2 (B) Increasing the temperature
(C) Adding F_2 (D) Increasing the volume of the container

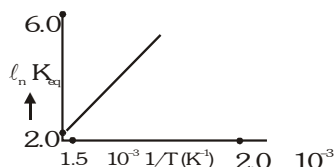
10. A schematic plot of $\ln K_{eq}$ versus inverse of temperature for a reaction is shown below. The reaction must be [AIEEE-2005]

(A) endothermic

(B) exothermic

(C) highly spontaneous at ordinary temperature

(D) one with negligible enthalpy change



11. Phosphorus pentachloride dissociates as follows, in a closed reaction vessel, [AIEEE-2006]



If total pressure at equilibrium of the reaction mixture is P and degree of dissociation of PCl_5 is x, the partial pressure of PCl_3 will be-

(A) $\left(\frac{2x}{1-x}\right)P$

(B) $\left(\frac{x}{x-1}\right)P$

(C) $\left(\frac{x}{1-x}\right)P$

(D) $\left(\frac{x}{x+1}\right)P$

12. The equilibrium constant for the reaction $\text{SO}_3(\text{g}) \rightleftharpoons \text{SO}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g})$ is $K_C = 4.9 \times 10^{-2}$. The value of K_C for the reaction [AIEEE-2006]



(A) 2.40×10^{-3}

(B) 9.8×10^{-2}

(C) 4.9×10^{-2}

(D) 416

13. The equilibrium constants K_{p1} and K_{p2} for the reaction $\text{X} \rightleftharpoons 2\text{Y}$ and $\text{Z} \rightleftharpoons \text{P} + \text{Q}$, respectively are in the ratio of 1 : 9. If the degree of dissociation of X and Z be equal then the ratio of total pressure at these equilibria is [AIEEE-2008]

(A) 1 : 36

(B) 1 : 1

(C) 1 : 3

(D) 1 : 9

14. A vessel at 1000 K contains CO_2 with a pressure of 0.5 atm. Some of the CO_2 is converted into CO on the addition of graphite. If the total pressure at equilibrium is 0.8 atm, the value of K is :- [AIEEE-2011]

(A) 0.3 atm

(B) 0.18 atm

(C) 1.8 atm

(D) 3 atm

15. The equilibrium constant (K_C) for the reaction $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{NO}(\text{g})$ at temperature T is 4×10^{-4} . The value of K_C for the reaction, $\text{NO}(\text{g}) \longrightarrow \frac{1}{2}\text{N}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g})$ at the same temperature is :- [AIEEE-2012]

(A) 50.0

(B) 0.02

(C) 2.5×10^2

(D) 4×10^{-4}

JEE-[MAIN] : PREVIOUS YEAR QUESTIONS

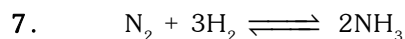
ANSWER KEY

EXERCISE -5[A]

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans	C	C	A	A	D	A	B	B	C	B	D	D	A	C	A

EXERCISE - 05 [B]**JEE-[ADVANCED] : PREVIOUS YEAR QUESTIONS**

1. For the reaction $\text{CO (g)} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 \text{ (g)} + \text{H}_2 \text{ (g)}$ at a given temperature the equilibrium amount of $\text{CO}_2 \text{ (g)}$ can be increased by : [JEE 1998]
- (A) adding a suitable catalyst
(B) adding an inert gas
(C) decreasing the volume of the container
(D) increasing the amount of CO (g)
2. For the reaction, $\text{N}_2\text{O}_5 \text{ (g)} = 2 \text{ NO}_2 \text{ (g)} + 0.5 \text{ O}_2 \text{ (g)}$, calculate the mole fraction of $\text{N}_2\text{O}_5 \text{ (g)}$ decomposed at a constant volume & temperature, if the initial pressure is 600 mm Hg & the pressure at anytime is 960 mm Hg. Assume ideal gas behaviour. [JEE 1998]
3. The degree of dissociation is 0.4 at 400 K & 1.0 atm for the gaseous reaction $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2 \text{ (g)}$. Assuming ideal behaviour of all gases. Calculate the density of equilibrium mixture at 400 K & 1.0 atm pressure. [JEE 1999]
4. When 3.06 g of solid NH_4HS is introduced into a two litre evacuated flask at 27 C, 30% of the solid decomposes into gaseous ammonia and hydrogen sulphide. [JEE 2000]
- (i) Calculate K_c & K_p for the reaction at 27 C.
(ii) What would happen to the equilibrium when more solid NH_4HS is introduced into the flask?
5. When 1-pentyne (A) is treated with 4N alcoholic KOH at 175 C, it is converted slowly into an equilibrium mixture of 1.3% 1-pentyne (A), 95.2% 2-pentyne (B) & 3.5 % of 1,2-pentadiene (C). The equilibrium was maintained at 175 C. Calculate ΔG for the following equilibria.
- $\text{B} = \text{A} \qquad \Delta G_1 = ?$
 $\text{B} = \text{C} \qquad \Delta G_2 = ?$
- From the calculated value of ΔG_1 & ΔG_2 indicate the order of stability of A, B & C. Write a reasonable reaction mechanism sharing all intermediate leading to A, B & C. [JEE 2001]
6. $\text{N}_2\text{O}_4 \text{ (g)} \rightleftharpoons 2\text{NO}_2 \text{ (g)}$
- This reaction is carried out at 298 K and 20 bar. 5 mol each of N_2O_4 and NO_2 are taken initially :
- Given : $\Delta G_{\text{N}_2\text{O}_4} = 100 \text{ kJ mol}^{-1}$; $\Delta G_{\text{NO}_2} = 50 \text{ kJ mol}^{-1}$
- (i) Find ΔG for reaction at 298 K under given condition.
(ii) Find the direction in which the reaction proceeds to achieve equilibrium. [JEE 2004]



Which is correct statement if N_2 is added at equilibrium condition?

(A) The equilibrium will shift to forward direction because according to II law of thermodynamics the entropy must increases in the direction of spontaneous reaction.

(B) The condition for equilibrium is $G_{\text{N}_2} + 3G_{\text{H}_2} \rightleftharpoons 2 G_{\text{NH}_3}$ where G is Gibbs free energy per mole of the gaseous species measured at that partial pressure. The condition of equilibrium is unaffected by the use of catalyst, which increases the rate of both the forward and backward reactions to the same extent.

(C) The catalyst will increase the rate of forward reaction by α and that of backward reaction by β .

(D) Catalyst will not alter the rate of either of the reaction. [JEE 2006]

8. The value of $\log_{10} K$ for a reaction $\text{A} \rightleftharpoons \text{B}$ is : [JEE 2007]

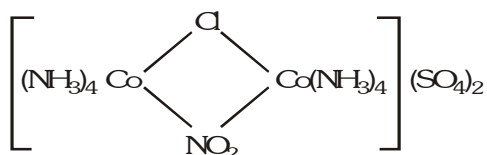
(Given : $\Delta_r H_{298\text{K}} = -54.07 \text{ kJ mol}^{-1}$, $\Delta_r S_{298\text{K}} = 10 \text{ JK}^{-1} \text{ mol}^{-1}$ and

$$R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}; 2.303 \times 8.314 \times 298 = 5705)$$

JEE-[ADVANCE] : PREVIOUS YEAR QUESTIONS		ANSWER KEY	EXERCISE -5[B]
1. D	2. Fraction decomposed = 0.4	3. 4.54 g dm^{-3}	
4. (i) $K_c = 8.1 \times 10^{-5} \text{ mol}^2 \text{ L}^2$; $K_p = 4.91 \times 10^{-2} \text{ atm}^2$,(ii) No effect ;			
5. 15991 J mol^{-1} , 12304 J mol^{-1} ; $B > C > A$			
6. (i) $5.705 \times 10^3 \text{ J mol}^{-1}$			
(ii) Since initial Gibbs free energy change of the reaction is positive, so the reverse reaction will take place.			
7. B			
8. B			
	$\begin{aligned} \Delta G &= \Delta H - T\Delta S \\ &= -54.07 \times 1000 - 298 \times 10 \\ &= -54070 - 2980 = -57050 \\ \Delta G &= -2.303 RT \log_{10} K \\ -57050 &= -2.303 \times 298 \times 8.314 \log_{10} K \\ &= -5705 \log_{10} K \\ \log_{10} K &= 10 \end{aligned}$		

SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)

- In the complex ion $[\text{Fe}(\text{EDTA})]^-$ the co-ordination number and oxidation state of central metal ion is –
 (A) C. N. = 6, O. N. = +3 (B) C. N. = 1, O. N. = -1
 (C) C. N. = 4, O. N. = +2 (D) C. N. = 3, O. N. = +3
- The IUPAC name of the complex $[\text{CrCl}_2(\text{H}_2\text{O})_4]\text{NO}_3$ is –
 (A) Dichlorotetraaqua chromium(III) nitrate (B) Tetraaquadichlorochromium(III) nitrate
 (C) Chromiumtetraaquadichloro nitrate (D) Dichlorotetraaqua chromium nitrate
- Amongst TiF_6^{2-} , CoF_6^{3-} , Cu_2Cl_2 and NiCl_4^{2-} (At. No. of Ti = 22, Co = 27, Cu = 29, Ni = 28)
 The colourless species are –
 (A) CoF_6^{3-} and NiCl_4^{2-} (B) TiF_6^{2-} and CoF_6^{3-} (C) Cu_2Cl_2 and NiCl_4^{2-} (D) TiF_6^{2-} and Cu_2Cl_2
- Which of the following complex can not exhibit geometrical isomerism –
 (A) $[\text{Pt}(\text{NH}_3)_2\text{Cl NO}_2]$ (B) $[\text{Pt}(\text{gly})_2]$ (C) $[\text{Cu}(\text{en})_2]^+$ (D) $[\text{Pt}(\text{H}_2\text{O})(\text{NH}_3)\text{BrCl}]$
- $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ absorbs orange light and the transmitted complementary colour will be –
 (A) Green (B) Yellow (C) Blue (D) Violet
- AgCl precipitate dissolves in NH_3 due to the formation of –
 (A) $[\text{Ag}(\text{NH}_3)_2]\text{OH}$ (B) $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$
 (C) $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$ (D) $[\text{Ag}(\text{NH}_3)_2]\text{OH}$
- Which one of the following compounds will exhibit linkage isomerism –
 (A) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ (B) $[\text{Co}(\text{NH}_3)_2\text{NO}_2]\text{Cl}_2$ (C) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ (D) $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$
- A magnetic moment of 1.73 BM will be shown by one among of the following compounds –
 (A) $[\text{Cu}(\text{NH}_3)_4]^{2+}$ (B) $[\text{Ni}(\text{CN})_4]^{2-}$ (C) TiCl_4 (D) $[\text{CoCl}_6]^{-3}$
- Give the name of the complex compound $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ according to IUPAC system –
 (A) Potassium Ferric oxalate (B) Potassium trioxalatoiron(III)
 (C) Potassium trioxalatoferrate(III) (D) Tripotassium trioxalatoferrate(III)
- Give the IUPAC name of the complex compound $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Br}](\text{NO}_3)_2$:-
 (A) Bromoaquatetraaminocobalt(III) nitrate (B) Bromoaquotetraaminocobalt(III) nitrate
 (C) Bromoaquatetraamminecobalt(III) nitrate (D) Tetraammineaquabromocobalt(III) nitrate
- Name the following compound according to IUPAC system



- Octamine- μ -chloro- μ -nitrodicobalt(III) sulphate
- Octaamine- μ -chloro- μ -nitrodicobalt(III) sulphate
- Octaamine- μ -nitro- μ -chlorodicobalt(III) sulphate
- None of the above

12. Chemical Name of "Turn bull's blue" is –
 (A) Ferrous ferricyanide (B) Ferri ferrocyanide (C) Potassium cyanide (D) Potassium ferricyanide
13. Hexafluorocobaltate(III) ion is found to be high spin complex, the probable hybrid state of cobalt in it, is–
 (A) dsp^2 (B) d^2sp^3 (C) sp^3d^2 (D) sp^3d
14. In the complex $[Ni(H_2O)_2(NH_3)_4]^{+2}$ the magnetic moment (μ) of Ni is –
 (A) Zero (B) 2.83 BM (C) 1.73 BM (D) 3.87 BM
15. Which of the following system has maximum number of unpaired electrons –
 (A) d^5 (Octahedral, low spin) (B) d^8 (Tetrahedral)
 (C) d^6 (Octahedral, low spin) (D) d^3 (Octahedral)
16. Nessler's reagent is –
 (A) K_2HgI_4 (B) $K_2HgI_4 + KOH$ (C) $K_2HgI_2 + KOH$ (D) $K_2HgI_4 + Hg$
17. A blue colouration is not obtained when –
 (A) NH_4OH is added to $CuSO_4$ (B) $CuSO_4$ solution reacts with $K_4[Fe(CN)_6]$
 (C) $FeCl_3$ reacts with $K_4[Fe(CN)_6]$ (D) Anhydrous white $CuSO_4$ is dissolved in water
18. One among the following complex ions will not show optical activity –
 (A) $[Pt (Br)(Cl)(I)(NO_2)(C_2H_5N)(NH_3)]$ (B) $Cis-[Co(en)_2Cl_2]^+$
 (C) $[Co(en)(NH_3)_2Cl_2]^+$ (D) $[Cr(NH_3)_4Cl_2]$
19. A Planar Complex (Mabcd) gives –
 (A) Two Optical isomer (B) Two geometrical isomer
 (C) Three optical isomer (D) Three geometrical isomers
20. A complex shown below can exhibit –
 (A) Geometrical isomerism only
 (B) Optical isomerism only
 (C) Both Optical and geometrical isomerism
 (D) None
-
21. $[Co(NH_3)_4Cl_2]NO_2$ and $[Co(NH_3)_4Cl NO_2]Cl$ are –
 (A) Geometrical isomers (B) Optical isomers
 (C) Linkage isomers (D) Ionisation isomers
22. Theoritically the No. of geometrical isomers expected for octahedral complex $[Mabcdef]$ is –
 (A) Zero (B) 30 (C) 15 (D) 9
23. Which of the following has two geometrical isomers, and is also a non ionisable complex –
 (A) $PtCl_4 \cdot 2NH_3$ (B) $PtCl_4 \cdot 3NH_3$ (C) $PtCl_4 \cdot 4NH_3$ (D) $PtCl_4 \cdot 6NH_3$
24. $Na_2S_2O_3 \cdot 5H_2O$ is used in photography to –
 (A) Reduce $AgBr$ to metallic Ag
 (B) Remove reduced Ag
 (C) Remove undecomposed $AgBr$ as a soluble complex
 (D) Converts metallic Ag to silver salt

25. Oxidation state of Ag in $\text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2]$ is –
 (A) + 2 (B) – 2 (C) 0 (D) +1
26. Out of $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Ni}(\text{CN})_4]^{-2}$ and $[\text{Ni}(\text{CO})_4]$
 (A) All have identical geometry.
 (B) All are paramagnetic.
 (C) All are diamagnetic.
 (D) $[\text{Fe}(\text{CN})_6]^{4-}$ is diamagnetic but $[\text{Ni}(\text{CN})_4]^{-2}$ and $[\text{Ni}(\text{CO})_4]$ are paramagnetic
27. Which gives only 25% mole of chloride as AgCl, when reacts with excess AgNO_3 –
 (A) $\text{PtCl}_2 \cdot 4\text{NH}_3$ (B) $\text{PtCl}_4 \cdot 5\text{NH}_3$ (C) $\text{PtCl}_4 \cdot 4\text{NH}_3$ (D) $\text{PtCl}_4 \cdot 3\text{NH}_3$
28. Which of the following compound is paramagnetic –
 (A) Tetracyanonickelate(II) ion (B) Tetraamminezinc(II) ion
 (C) Hexamine chromium(III) ion (D) Diammine silver(I) ion
29. The complexes $[\text{Co}(\text{NO}_2)(\text{NH}_3)_5] \text{Cl}_2$ and $[\text{Co}(\text{ONO})(\text{NH}_3)_5] \text{Cl}_2$ are the examples of
 (A) Co-ordination isomerism (B) Ionisation isomerism
 (C) Geometrical isomerism (D) Linkage isomerism
30. The complex $[\text{Mn}(\text{CN})_6]^{4-}$ is –
 (A) High spin complex (B) Diamagnetic ion
 (C) Having magnetic moment 1.73 BM (D) Outer orbital complex
31. Amongst the following, identify the species with an atom in +6 oxidation state –
 (A) MnO_4^- (B) $\text{Cr}(\text{CN})_6^{3-}$ (C) NiF_6^{2-} (D) CrO_2Cl_2
32. Which of the following statement is incorrect about $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$ –
 (A) It produce in brown ring test for nitrates (B) Oxidation state of Fe is +1
 (C) It exhibits geometrical isomerism (D) Charge on NO is +1
33. Which is/are organometallic compound –
 (I) Grignard reagent (II) Sodium methoxide (III) Sodium acetylide
 (A) II (B) I, III (C) I, II (D) I
34. Which of the following is most likely formula of platinum complex, if 1/4 of total chlorine of the compound is precipitated by adding AgNO_3 to its aqueous solution
 (A) $\text{PtCl}_4 \cdot 6\text{H}_2\text{O}$ (B) $\text{PtCl}_4 \cdot 5\text{H}_2\text{O}$ (C) $\text{PtCl}_2 \cdot 2\text{H}_2\text{O}$ (D) $\text{PtCl}_4 \cdot 3\text{H}_2\text{O}$
35. A complex compound of cobalt has molecular formula containing five NH_3 molecules, one nitro group and two chlorine atom for one cobalt atom. One mole of this compound produces three moles of ion in aqueous solution. On reaction with excess of AgNO_3 solution two moles of AgCl get precipitated. The Ionic formula of the compound.
 (A) $[\text{Co}(\text{NH}_3)_4 \text{NO}_2 \text{Cl}] \text{NH}_3\text{Cl}$ (B) $[\text{Co}(\text{NH}_3)_5\text{Cl}] \text{ClNO}_2$
 (C) $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ (D) None of these

36. Which of the following will give maximum number of isomers –
 (A) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ (B) $[\text{Ni}(\text{en})(\text{NH}_3)_4]^{+2}$ (C) $[\text{Ni}(\text{C}_2\text{O}_4)(\text{en})_2]^0$ (D) $[\text{Cr}(\text{SCN})_2(\text{NH}_3)_4]^+$
37. In the complexes $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Fe}(\text{en})_3]^{3+}$, $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ and $[\text{FeCl}_6]^{3-}$, more stability is shown by
 (A) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ (B) $[\text{Fe}(\text{en})_3]^{3+}$ (C) $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ (D) $[\text{FeCl}_6]^{3-}$
38. Both geometrical and optical isomerism are shown by –
 (A) $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ (B) $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$ (C) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ (D) $[\text{Cr}(\text{ox})_3]^{3-}$
39. An example for a double salt is
 (A) Cuprammonium sulphate (B) Mohr's salt
 (C) Potassium ferricyanide (D) Cobalthexammine chloride
40. One mole of the complex compound $\text{Co}(\text{NH}_3)_5\text{Cl}_3$, gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with two moles of AgNO_3 solution to yield two moles of AgCl (s). The structure of the complex is –
 (A) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3] \cdot 2\text{NH}_3$ (B) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2] \text{Cl} \cdot \text{NH}_3$
 (C) $[\text{Co}(\text{NH}_3)_4\text{Cl}] \text{Cl}_2 \cdot \text{NH}_3$ (D) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
41. What are the geometric shape and the oxidation number of the copper atom, respectively, for the complex ion, $[\text{Cu}(\text{NH}_3)_4(\text{OH})_2]^{2+}$
 (A) Tetrahedral; + 2 (B) Square planar; – 2 (C) Linear; + 3 (D) Octahedral; + 2
42. What is the relationship between the following two linear complex ions ?
 $[\text{Cl} - \text{Ag} - \text{SCN}]^{1-}$ $[\text{SCN} - \text{Ag} - \text{Cl}]^{1-}$
 The complex ions are –
 (A) Linkage isomers (B) Coordination isomers (C) Geometric isomers (D) Optical isomers
43. Which of the following common shapes (I–IV) can never exist as geometric isomers, regardless of the identity of the ligands –
 (I) Linear (II) Square planar (III) Tetrahedral (IV) Octahedral
 (A) I only (B) I and II (C) I and III (D) II and IV
44. Synergic bonding involves –
 (A) The transference of electrons from ligands to metal
 (B) The transference of electrons from filled metal orbitals to anti-bonding orbitals of ligands
 (C) Both (A) and (B)
 (D) None of these

ANSWER KEY																				
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	A	B	D	C	C	C	B	A	C	D	D	A	C	B	D	B	B	D	D	C
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	D	C	A	C	D	C	D	C	D	C	D	C	D	D	C	D	B	A	B	D
Que.	41	42	43	44																
Ans.	D	A	C	C																

SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)

- A complex cation is formed by Pt (in some oxidation state) with ligands (in proper number so that coordination number of Pt becomes six). Which of the following can be its correct IUPAC name –
 (A) Diammineethylenediaminedithiocyanato-S-platinum(II)
 (B) Diammineethylenediaminedithiocyanato-S-platinate(IV) ion
 (C) Diammineethylenediaminedithiocyanato-S-platinum(IV) ion
 (D) Diamminebis (ethylenediamine) dithiocyanato-S-platinum(IV) ion
- In which of the following configuration (s) the value of 'spin only' magnetic moment is 2.84 BM for octahedral complex
 (A) d^4 (in strong field ligand) (B) d^2 (in weak field ligand)
 (C) d^3 (in weak as well as in strong field ligand) (D) d^5 (in strong field ligand)
 (E) d^8 (in weak field ligand)
- In the crystal field of the complex $[\text{Fe}(\text{Cl})(\text{CN})_4(\text{O}_2)]^{4-}$, the electronic configuration of metal is found to be t_{2g}^6, e_g^0 then which of the following is/are true about this complex ion –
 (A) It is a paramagnetic complex
 (B) O – O bond length will be more than found in O_2 molecule
 (C) Its IUPAC name will be chlorotetracyanosuperoxidoferrate(II) ion.
 (D) It will show geometrical as well as optical isomerism
- The solubility of AgBr in hypo solution is due to the formation of –
 (A) Ag_2SO_3 (B) $\text{Ag}_2\text{S}_2\text{O}_3$ (C) $[\text{Ag}(\text{S}_2\text{O}_3)]^-$ (D) $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$
- Photographic films or plates have.....as an essential ingredient –
 (A) Silver oxide (B) Silver bromide (C) Silver thio sulphate (D) Silver nitrate
- Silver halides are used in photography because they are –
 (A) Photosensitive (B) Soluble in hypo solution
 (C) Soluble in NH_4OH (D) Insoluble in acids
- The compound which shows paramagnetism is –
 (A) $[\text{Cu}(\text{NH}_3)_4]\text{Cl}_2$ (B) $\text{Fe}(\text{CO})_5$ (C) NO (D) NO_2
- Magnetic moment of x^{n+} is $\sqrt{24}$ B.M. Hence No. of unpaired electron and value of 'n' respectively. (Atomic number = 26)
 (A) 4, 3 (B) 3, 5 (C) 4, 2 (D) 4, 1
- Among the following compound that is both paramagnetic and coloured is –
 (A) $\text{K}_2\text{Cr}_2\text{O}_7$ (B) $(\text{NH}_4)_2[\text{TiCl}_6]$ (C) VOSO_4 (D) $\text{K}_3[\text{Cu}(\text{CN})_4]$
- The total number of possible isomer for the complex compound : $[\text{Cu}(\text{NH}_3)_4][\text{PtCl}_4]$
 (A) 3 (B) 6 (C) 5 (D) 4
- The image on an exposed and developed photography film is due to –
 (A) AgBr (B) $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3+}$ (C) Ag (D) Ag_2O
- The chloro-bis (ethylenediamine) nitrocobalt(III) ion is –
 (A) $[\text{Co}(\text{NO}_2)_2(\text{en})_2\text{Cl}]^+$ (B) $[\text{CoCl}(\text{NO}_2)_2(\text{en})_2]^+$
 (C) $[\text{Co}(\text{NO}_2)\text{Cl}(\text{en})_2]^+$ (D) $[\text{Co}(\text{en})\text{Cl}_2(\text{NO}_2)_2]^-$

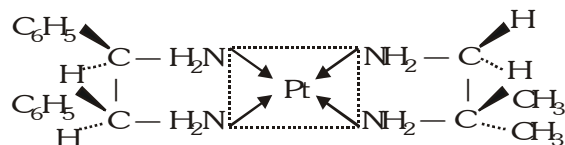
13. Zeigler natta catalyst is –
 (A) Pt/PtO (B) $\text{Al}(\text{C}_2\text{H}_5)_3 + \text{TiCl}_4$
 (C) $\text{K}[\text{PtCl}_3(\eta^2 - \text{C}_2\text{H}_4)]$ (D) Pt/Rh
14. Which of the following set of isomer isomerism is/are correct –
 (A) $\text{Cis} - [\text{Co}(\text{gly})_2\text{Cl}_2]^-$ – optical isomerism (B) $[\text{Zn}(\text{NH}_3)_3\text{Cl}]^+$ – Geometrical isomerism
 (C) $[\text{Fe}(\text{H}_2\text{O})_6]\text{Cl}_3$ – Hydrate isomerism (D) $[\text{Co}(\text{en})_2(\text{NCS})_2]\text{Cl}$ – Linkage isomerism
15. In $\text{Na}_2 [\text{Fe}(\text{CN})_5\text{NO}]$ sodium nitroprusside –
 (A) Oxidation state of Fe is +2 (B) This has NO^+ as ligand
 (C) It is paramagnetic (D) None is correct
16. Which of the following statement is correct –
 (A) $[\text{Cu}(\text{NH}_3)_6]^{2+}$ is a colourless ion (B) $[\text{Ni}(\text{CN})_4]^{2-}$ ion has tetrahedral shape
 (C) $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$ ion is blue coloured (D) Nickel dimethylglyoxime is red in colour
17. Which of the following have square planar geometry –
 (a) $[\text{NiCl}_4]^{-2}$ (b) $[\text{Cu}(\text{NH}_3)_4]^{+2}$ (c) $[\text{Ni}(\text{CO})_4]$ (d) ClF_4^-
 (A) b, c and d (B) a, b and c (C) b and d (D) All
18. Pick up the incorrect statement –
 (A) Cisplatin is a complex of platinum (B) Vitamin B_{12} is a complex of cobalt
 (C) Chlorophyll is a complex of Manganese (D) Haemoglobin is a complex of iron
19. Which of the following pair of molecule have identical shape –
 (A) $[\text{NiCl}_4]^{-2}$ and XeF_4 (B) $[\text{Zn}(\text{H}_2\text{O})_4]^{+2}$ and SiCl_4
 (C) $[\text{Fe}(\text{CO})_5]$ and XeOF_4 (D) $[\text{Ag}(\text{NH}_3)_2]^+$ and SF_2
20. Hypo is used in photography because it is –
 (A) A strong reducing agent (B) A strong oxidising agent
 (C) A strong Complexing agent (D) Photo sensitive Compound
21. The kind of isomerism exhibited by $[\text{Rh}(\text{en})_2\text{Cl}_2][\text{Ir}(\text{en})\text{Cl}_4]$ and $[\text{Rh}(\text{en})_3][\text{IrCl}_6]$ is –
 (A) Linkage (B) Co-ordination (C) Ligand (D) Ionisation
22. Identify the complex which are expected to be colourless –
 (A) $[\text{Ti}(\text{NO}_3)_4]^0$ (B) $[\text{Cu}(\text{NCCH}_3)_4]^+$ (C) $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ (D) K_3VF_6
23. Which of following organometallic compound is σ and π bonded –
 (A) $[\text{Fe}(\eta^5 - \text{C}_5\text{H}_5)_2]$ (B) $\text{K}[\text{PtCl}_3(\eta^2 - \text{C}_2\text{H}_4)]$ (C) $[\text{Co}(\text{CO})_5\text{NH}_3]^{+2}$ (D) $\text{Fe}(\text{CH}_3)_3$
24. Which statement is incorrect –
 (A) $[\text{Ni}(\text{CO})_4]$ - Tetrahedral, paramagnetic (B) $[\text{Ni}(\text{CN})_4]^{2-}$ - Square planar, diamagnetic
 (C) $[\text{Ni}(\text{CO})_4]$ - Tetrahedral, diamagnetic (D) $[\text{NiCl}_4]^{-2}$ - Tetrahedral, paramagnetic
25. A complex $\text{K}_n [\text{MnF}_6]$ has magnetic moment 4.9 BM what will be the oxidation state of Mn and the value of n ?
 (A) Mn(II), n = 4 (B) Mn(III) ; n = 3 (C) Mn(IV) ; n = 2 (D) Mn(V) ; n = 1
26. In $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$, the number of covalent bonds & coordinate bonds respectively –
 (A) 3, 6 (B) 6, 6 (C) 8, 9 (D) 18, 6

27. CuSO_4 when reacts with KCN forms x, which is insoluble in water. x is soluble in excess of KCN, due to formation of –
 (A) $\text{K}_2[\text{Cu}(\text{CN})_4]$ (B) $\text{K}_3[\text{Cu}(\text{CN})_4]$ (C) CuCN_2 (D) $\text{Cu}[\text{KCu}(\text{CN})_4]$
28. A square planar complex is formed by hybridization of which atomic orbitals –
 (A) s, p_x , p_y , d_{yz} (B) s, p_x , p_y , $d_{x^2 - y^2}$ (C) s, p_x , p_y , d_{z^2} (D) s, p_x , p_y , d_{xy}
29. When AgNO_3 is added to a solution of $\text{Co}(\text{NH}_3)_5 \text{Cl}_3$, the precipitate of AgCl shows two ionized chloride ions. This means –
 (A) Only two chlorine atoms satisfy primary valency and one secondary valency
 (B) One chlorine atom satisfies primary as well as secondary valency
 (C) Three chlorine atoms satisfy primary valency
 (D) Three chlorine atoms satisfy secondary valency.
30. Which of the following compound is not coloured ?
 (A) $\text{Na}_2[\text{CuCl}_4]$ (B) $\text{Na}_2[\text{CdCl}_4]$ (C) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ (D) $\text{K}_3[\text{Fe}(\text{CN})_6]$
31. Colourless species is :
 (A) VCl_3 (B) VOSO_4 (C) Na_3VO_4 (D) $[\text{Ni}(\text{H}_2\text{O})_6] \text{SO}_4 \cdot \text{H}_2\text{O}$
32. The coordination number of a central metal atom in a complex is determined by –
 (A) The number of ligands around a metal ion bonded by sigma and pi-bonds both
 (B) The number of ligands around a metal ion bonded by pi-bonds
 (C) The number of ligands around a metal ion bonded by sigma bonds
 (D) The number of only anionic ligands bonded to the metal ion
33. The correct order of magnetic moments (spin only values in B.M.) among is –
 (A) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-}$ (B) $[\text{MnCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-}$
 (C) $[\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-}$ (D) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-}$
34. Among the following series of transition metal ions, the one where all metal ions have $3d^2$ electronic configuration is –
 (A) Ti^{2+} , V^{3+} , Cr^{4+} , Mn^{5+} (B) Ti^{2+} , V^{2+} , Cr^{3+} , Mn^{4+}
 (C) Ti^+ , V^{4+} , Cr^{6+} , Mn^{7+} (D) Ti^{4+} , V^{3+} , Cr^{2+} , Mn^{3+}
35. CN^- is a strong field ligand. This is due to the fact that:
 (A) It forms high spin complexes with metal species (B) It gives negative charge
 (C) It is a pseudohalide (D) It can accept electrons from metal species
36. Which of the following coordination compounds would not exhibit optical isomerism ?
 (A) tris-(ethylenediamine)cobalt(III) bromide
 (B) pentaamminenitrocobalt(III) iodide
 (C) diamminedichloroplatinum(II)
 (D) trans-dicyanobis (ethylenediamine) chromium(III) chloride
37. Among $[\text{Ni}(\text{CO})_4]$, $[\text{Ni}(\text{CN})_4]^{2-}$, $[\text{NiCl}_4]^{2-}$ species, the hybridization states at the Ni atom are, respectively (Atomic Number of Ni = 28)–
 (A) dsp^2 , sp^3 , sp^3 (B) sp^3 , dsp^2 , dsp^2
 (C) sp^3 , dsp^2 , sp^3 (D) sp^3 , sp^3 , dsp^2

38. Which one of the following is expected to not exhibit optical isomerism ?
(en = ethylenediamine)
(A) *cis* - [Pt(NH₃)₂Cl₂] (B) *cis* - [Co(en)₂Cl₂] (C) *trans* - [Co(en)₂Cl₂] (D) *trans* - [Pt(NH₃)₂Cl₂]
39. An aqueous solution of CoCl₂ on addition of excess of concentrated HCl turns blue due to formations of :
(A) [Co(H₂O)₄Cl₂] (B) [Co(H₂O)₂Cl₄]²⁻ (C) [CoCl₄]²⁻ (D) [Co(H₂O)₂Cl₂]
40. In which of the following pairs both the complexes show optical isomerism ?
(A) *cis*-[Cr(C₂O₄)₂Cl₂]³⁻, *cis*-[Co(NH₃)₄Cl₂]
(B) [Co(en)₃] Cl₃, *cis*- [Co(en)₂Cl₂]Cl
(C) [PtCl(dien)]Cl, [NiCl₂ Br₂]²⁻
(D) [Co(NO₃)₃(NH₃)₃], *cis* -[Pt(en)₂Cl₂]
41. The correct order for the wavelength of absorption in the visible region is :
(A) [Ni(NO₂)₆]⁴⁻ < [Ni(NH₃)₆]²⁺ < [Ni(H₂O)₆]²⁺ (B) [Ni(NO₂)₆]⁴⁻ < [Ni(H₂O)₆]²⁺ < [Ni(NH₃)₆]²⁺
(C) [Ni(H₂O)₆]²⁺ < [Ni(NH₃)₆]²⁺ < [Ni(NO₂)₆]⁴⁻ (D) [Ni(NH₃)₆]²⁺ < [Ni(H₂O)₆]²⁺ < [Ni(NO₂)₆]⁴⁻
42. Which one of the following cyano complexes would exhibit the lowest value of magnetic moment ?
(A) [Cr(CN)₆]³⁻ (B) [Mn(CN)₆]³⁻ (C) [Fe(CN)₆]³⁻ (D) [Co(CN)₆]³⁻
43. The oxidation state of Mo in its oxo-complex species [Mo₂O₄(C₂H₄)₂(H₂O)₂]²⁻ is -
(A) + 2 (B) + 3 (C) + 4 (D) + 5
44. The correct order of hybridisation of the central atom in the following species :
NH₃, [PtCl₄]²⁻, PCl₅ and BCl₃ is
(A) dsp², dsp³, sp², sp³ (B) sp³, dsp², dsp³, sp²
(C) dsp², sp², sp³, dsp³ (D) dsp², sp³, sp², sp³
45. [Cr(H₂O)₆] Cl₃ (at. no. of Cr=24) has a magnetic moment of 3.83 B.M. The correct distribution of 3d electrons in the Chromium of the complex is -
(A) 3d_{xy}¹, 3d_{yz}¹, 3d_{z²}¹ (B) (3d_{x²-y²})¹, 3d_{z²}¹, 3d_{xz}¹
(C) 3d_{xz}¹, (3d_{x²-y²})¹, 3d_{yz}¹ (D) 3d_{xy}¹, 3d_{yz}¹, 3d_{xz}¹
46. [Co (NH₃)₄ (NO₂)₂]Cl exhibits
(A) linkage isomerism, geometrical isomerism and optical isomerism
(B) linkage isomerism, ionization isomerism and optical isomerism
(C) linkage isomerism, ionization isomerism and geometrical isomerism
(D) ionization isomerism, geometrical isomerism and optical isomerism
47. The pair in which both species have same magnetic moment (spin only value) is -
(A) [Cr(H₂O)₆]²⁺, [CoCl₄]²⁻ (B) [Cr(H₂O)₆]²⁺, [Fe(H₂O)₆]²⁺
(C) [Mn(H₂O)₆]²⁺, [Cr(H₂O)₆]²⁺ (D) [CoCl₄]²⁻, [Fe(H₂O)₆]²⁺
48. Among the following the species having same geometry for central atom are
(i) XeF₄ (ii) SF₄ (iii) [NiCl₄]²⁻ (iv) [PdCl₄]²⁻
(A) (i) and (iv) (B) (i), (iii) and (iv) (C) (ii) and (iii) (D) (iii) and (iv)

49. For $[\text{FeF}_6]^{3-}$ and $[\text{Fe}(\text{CN})_6]^{3-}$ magnetic moment of the fluoride complex is expected to be –
- The same as the magnetic moment of the cyanide complex
 - Larger than the magnetic moment of the cyanide complex because there are more unpaired electrons in the fluoride complex
 - Smaller than the magnetic moment of the cyanide complex because there are more unpaired electrons in the fluoride complex
 - Larger than the magnetic moment of the cyanide complex because there are few unpaired electrons in the fluoride complex
50. Coordination compounds which contain cyanide (CN^-) ligands tend to be yellow where as coordination compounds which contain water (H_2O) ligands tend to be blue or green because –
- The complexes diffract light at different angles
 - Cyanide is a weak-field ligand and water is a strong-field ligand
 - Cyanide is a strong-field ligand and water is a weak-field ligand
 - Cyanide compounds absorb yellow light and water compounds absorb blue or green light
51. Which of the following complexes is not a chelate –
- bis (dimethylglyoximate) nickel(II)
 - Potassium ethylenediaminetetrathiocyanato chromate (III)
 - pantamminecarbonatocobalt (III) nitrate
 - Trans-diglycinatoplatinum (II)
52. The stability constants of the complexes formed by a metal ion M^{2+} with NH_3 , CN^- , H_2O are of the order of 10^{15} , 10^{27} , 10^{11} respectively. Then –
- NH_3 is the strongest ligand
 - CN^- is the strongest ligand
 - These values cannot predict the strength of the ligand
 - All the ligands are equally strong
53. $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ a blue coloured complex. Average oxidation number of Fe in $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ is 'a' oxidation state of central iron atom 'b' oxidation state of counter iron atom 'c', a,b,c are respectively –
- $+\frac{5}{2}$, + 2 , + 3
 - $+\frac{5}{2}$, + 3 , + 2
 - $+\frac{18}{7}$, + 2 , + 3
 - $+\frac{18}{7}$, + 3 , + 2
54. Which of the following is correct about Tetraamminedithiocyanato-S-cobalt(III) tris(oxalato)cobaltate(III)?
- Formula of the complex is $[\text{Co}(\text{NH}_3)_4(\text{SCN})_2][\text{Co}(\text{ox})_3]$
 - It is a chelating complex and show linkage isomerism
 - It shows optical isomerism
 - It show geometrical isomerism

55. Select the correct statement about given square planar complex.



- (A) It has no geometrical isomer
 (B) It is optically active because it does not have plane of symmetry
 (C) It is optically inactive because square planar complex have plane of symmetry
 (D) It is optically active because it has symmetric carbon.
56. Total number of stereo isomers of $[\text{Co}(\text{acac})_2\text{BrCl}]^-$ are:
 (A) 4 (B) 3 (C) 6 (D) 2
57. Match list-I (Species) with List-II (Hybrid orbitals used by the central atom in their formation) and select the correct answer:

Column-I		Column-II	
(a)	$\text{Ni}(\text{CN})_5^{3-}$	(1)	sp^3
(b)	CuCl_5^{3-}	(2)	dsp^2
(c)	AuCl_4^-	(3)	$\text{sp}^3\text{d}_{z^2}$
(d)	ClO_4^-	(4)	$\text{d}_{x^2-y^2}\text{sp}^3$

Codes :

	a	b	c	d
(A)	1	3	2	4
(B)	3	4	2	1
(C)	4	2	1	3
(D)	4	3	2	1

58. Which of the following is correct IUPAC name of any complex compound?
 (A) Tris (acetylacetonato)iron(III) chloride
 (B) Hexachloroplatinum(IV) tetraamminedicyanoplatinate(IV)
 (C) Amminebromochloromethylamineplatinum(II)
 (D) Cis-dichloro(ethylenediamine)platinum(II)
59. Which of the following statement(s) is (are) correct?
 (A) The complexes $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ differ in the magnetic properties
 (B) The complexes $[\text{NiCl}_4]^{2-}$ and $[\text{NiCN}_4]^{2-}$ differ in the geometry
 (C) The complexes $[\text{NiCl}_4]^{2-}$ and $[\text{NiCN}_4]^{2-}$ differ in primary valencies of nickel
 (D) The complexes $[\text{NiCl}_4]^{2-}$ and $[\text{NiCN}_4]^{2-}$ differ in the state of hybridization of nickel.

60. Which of the following statement(s) is/are correct with reference to Fe^{2+} and Fe^{3+} ions?
- (1) Fe^{3+} gives brown colour with potassium ferricyanide
(2) Fe^{2+} gives blue colour with potassium ferricyanide
(3) Fe^{3+} gives red colour with potassium thiocyanate
(4) Fe^{2+} gives brown colour with ammonium thiocyanate
- (A) 1, 2 (B) 1, 4 (C) 1, 2, 3 (D) All of these
61. 0.001 mol of $\text{Co}(\text{NH}_3)_5(\text{NO}_3)(\text{SO}_4)$ was passed through a cation exchanger and the acid coming out of its required 20 ml of 0.1 M NaOH for neutralisation. Hence, the complex is -
- (A) $[\text{CoSO}_4(\text{NH}_3)_5]\text{NO}_3$ (B) $[\text{CoNO}_3(\text{NH}_3)_5]\text{SO}_4$
(C) $[\text{Co}(\text{NH}_3)_5](\text{SO}_4)(\text{NO}_3)$ (D) None of these
62. The IUPAC name of the red coloured complex $[\text{Fe}(\text{C}_4\text{H}_7\text{O}_2\text{N}_2)_2]$ obtained from the reaction of Fe^{2+} and dimethyl glyoxime -
- (A) bis(dimethyl glyoxime) ferrate(II)
(B) bis (dimethyl glyoximato) iron(II)
(C) bis (2, 3-butanediol dioximato) iron(II)
(D) bis (2, 3-butanedione dioximato) iron(II)
63. An ion M^{2+} , forms the complexes $[\text{M}(\text{H}_2\text{O})_6]^{2+}$, $[\text{M}(\text{en})_3]^{2+}$ and $[\text{MBr}_6]^{4-}$, Match the complex with the appropriate colour -
- (A) Green, blue and red (B) Blue, red and green
(C) Green, red and blue (D) Red, blue and green

ANSWER KEY

ANSWER KEY																				
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	C	A,B,E	A,B,C	D	B	A	A,C,D	C	C	D	C	C	B	A,C,D	A,B	D	C	C	B	C
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	B	A,B	C	A	B	D	B	B	B,C	B	C	C	C	A	D	B,C,D	C	A,C,D	C	B
Que.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	A	D	B	B	D	C	B	A	B	C	C	B	C	B,C,D	B	B	D	C	A,B,D	C
Que.	61	62	63																	
Ans.	B	B	B																	

EXERCISE-03**MISCELLANEOUS TYPE QUESTIONS****TRUE / FALSE**

- Coordination number and oxidation state of a metal means the same thing.
- Stability of coordination compounds increases with increase in charge density of the metal ions.
- $\text{Fe}(\text{CO})_5$ has trigonal bipyramidal geometry.
- $[\text{NiCl}_4]^{2-}$ is diamagnetic in nature.
- Metal Carbonyls are organometallic compounds.
- Both optical and geometrical isomerisms are shown by $[\text{Co}(\text{en})_2 \text{Cl}_2]^+$.
- The complex $[\text{Co}(\text{NH}_3)_3 \text{Cl}_3]$ is ionizable,
- The complex ion trans-dichlorobis(ethylenediamine)rhodium(III) is optically active.

FILL IN THE BLANKS

- A Solution of potassium ferrocyanide contains ions.
- EDTA^{-4} is a ligand.
- The secondary valency of metal is equal to and is
- The coordination number and oxidation number of cobalt in $[\text{Co}(\text{edta})]^{-1}$ are and respectively.
- The total number of electrons on the central metal atom/ion including those gained by bonding is called
- $[\text{Co}(\text{NH}_3)_5 \text{SO}_4]\text{Br}$ and $[\text{Co}(\text{NH}_3)_5 \text{Br}]\text{SO}_4$ show isomerism.
- One molecule of $[\text{Pt}(\text{NH}_3)_6]\text{Cl}_4$ gives ions in solution and requires of AgNO_3 for complete precipitation of chloride ions.
- $[\text{Pt}(\text{NH}_3)(\text{NH}_2\text{OH})(\text{py})(\text{NO}_2)]\text{NO}_3$ has shape and has geometrical isomers.

MATCH THE COLUMN

- | Column-I | | Column-II | |
|----------|-----------------------------------|-----------|--------------|
| (A) | $[\text{Ni}(\text{CN})_4]^{2-}$ | (p) | Paramagnetic |
| (B) | $[\text{NiCl}_4]^{2-}$ | (q) | Diamagnetic |
| (C) | $[\text{MnCl}_4]^{2-}$ | (r) | Tetrahedral |
| (D) | $[\text{Cu}(\text{NH}_3)_4]^{+2}$ | (s) | SQ. planar |
- | Column-I | | Column-II | |
|----------|--|-----------|------------------------------|
| (A) | Sodium nitroprusside | (p) | $\mu = 0 \text{ BM}$ |
| (B) | Brown ring complex | (q) | Octahedral |
| (C) | Complex of Ag formed during its extraction | (r) | $\mu = \sqrt{15} \text{ BM}$ |
| (D) | Potassium ferrocyanide | (s) | NO^+ ligand |
| | | (t) | cynide ligand is present |

3.	Column-I	Column-II
	Co-ordination compound (A) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Br}$ (B) $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$ (C) $[\text{Co}(\text{en})_2(\text{NO}_2)\text{Cl}]\text{SCN}$ (D) $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{F})_6]$	Type of isomerism (p) Optical isomerism (q) Ionization isomerism (r) Coordination isomerism (s) Geometrical isomerism (t) Linkage isomerism

ASSERTION & REASON QUESTIONS

These questions contains, Statement-I (assertion) and Statement-II (reason).

(A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I

(B) Statement-I is True, Statement-II is True ; Statement-II is NOT a correct explanation for Statement-I

(C) Statement-I is True, Statement-II is False.

(D) Statement-I is False, Statement-II is True.

- Statement-I** : $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ is a double salt compound.
Because
Statement-II : It ionises to give a complex ion.
- Statement-I** : $[\text{Fe}(\text{CO})_5]$ is diamagnetic complex.
Because
Statement-II : In the given complex oxidation state of Iron is zero.
- Statement-I** : $[\text{Ni}(\text{CN})_4]^{-2}$ has zero unpaired electron while that of $[\text{NiCl}_4]^{-2}$ has two unpaired e^-
Because
Statement-II : $[\text{Ni}(\text{CN})_4]^{-2}$ has strong crystal field while $[\text{NiCl}_4]^{-2}$ has weak crystal field
- Statement-I** : $\text{K}_2[\text{PtCl}_6]$ gives white ppt when reacts with AgNO_3
Because
Statement-II : Chloride ion in the complex is non-ionisable.
- Statement-I** : Trans $[\text{CoCl}_2(\text{en})_2]^+$ is optically inactive.
Because
Statement-II : It has plane of symmetry.
- Statement-I** : Cis - $[\text{Fe}(\text{en})_2\text{Cl}_2]^+$ can form racemic mixture.
Because
Statement-II : Cis - $[\text{Fe}(\text{en})_2\text{Cl}_2]^+$ is optically active square planar complex .
- Statement-I** : Potassium ferrocyanide is diamagnetic whereas potassium ferricyanide is paramagnetic.
Because
Statement-II : Crystal field splitting in ferrocyanide ion is greater than that of ferricyanide ion.
- Statement-I** : $[\text{Co}(\text{NO}_2)_3(\text{NH}_3)_3]$ does not show optical isomerism.
Because
Statement-II : It has plane of symmetry.
- Statement-I** : C-C bond length in Zeise's salt is same as ethylene.
Because
Statement-II : Double bond is shorter as compared to single bond.
- Statement-I** : Hydrazine is a neutral ligand.
Because
Statement-II : It has two N as donor atoms and behaves as a chelating ligand.

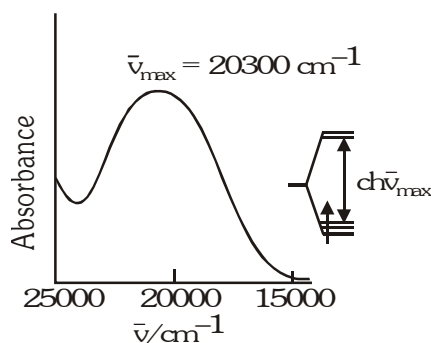
11. **Statement-I** : $[\text{Ti}(\text{H}_2\text{O})_6]^{4+}$ is coloured while $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$ is colourless.
Because
Statement-II : d-d transition is not possible in $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$.
12. **Statement-I** : EAN of Fe in ferrocene is 36.
Because
Statement-II : 6π e^- are co-ordinated by each cyclopentadienyl ring with central metal ion.

COMPREHENSION BASED QUESTIONS

Comprehension # 1

With the help of CFT number of unpaired electron in a compound can be calculated and we can calculate its paramagnetic moment (due to spin only), by the formula:

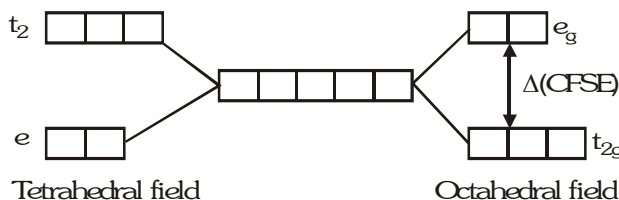
$\mu = \sqrt{n(n+2)}$ Bohr magneton (BM), where 'n' is the number of unpaired electron in the complex. For spectral analysis the separation between t_{2g} and e_g orbitals, called ligand field splitting parameter Δ_0 (for octahedral complexes) should be known to us, which can be easily calculated by observing the absorption spectrum of one e^- complex figure shows the optical absorption spectrum of the d^1 hexaaquatitanium (III) ion $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$. The CFT assigns the first absorption maximum at $20,300 \text{ cm}^{-1}$ to the transition $e_g \leftarrow t_{2g}$. For multielectronic (d^2 to d^{10}) system, the calculation of Δ_0 by absorption spectrum is not that easy as the absorption spectrum will also be affected by electron-electron repulsions.



- The crystal field stabilization energy (CFSE) for complex given in the passage, $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ will be (in kJ/mol) –
 (A) 243 kJ/mole (B) 97 kJ/mole (C) 194 kJ/mole (D) 143 kJ/mole
- The complex for which the calculation of crystal field splitting can be most easily done, by knowing its absorption spectrum, will be –
 (A) $[\text{TiCl}_6]^{2-}$ (B) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ (C) $[\text{Ti}(\text{CN})_6]^{3-}$ (D) $[\text{CoF}_6]^{3-}$
- The magnetic moments of following, arranged in increasing order will be (atomic number of Co = 27)
 (1) Co^{3+} (octahedral complex with a strong field ligand)
 (2) Co^{3+} (octahedral complex with a weak field ligand)
 (3) Co^{2+} (tetrahedral complex)
 (4) Co^{2+} (square planar complex)
 (A) $1 > 2 > 3 > 4$ (B) $2 > 3 > 4 > 1$ (C) $3 > 2 > 4 > 1$ (D) $2 > 4 > 3 > 1$

Comprehension#2

When degenerate d-orbitals of an isolated atom/ion come under influence of magnetic field of ligands, the degeneracy is lost. The two set $t_{2g}(d_{xy}, d_{yz}, d_{xz})$ and $e_g(d_{z^2}, d_{x^2-y^2})$ are either stabilized or destabilized depending upon the nature of magnetic field. It can be expressed diagrammatically as:



Value of CFSE depends upon nature of ligand and a spectrochemical series has been made experimentally, for tetrahedral complexes, Δ is about $4/9$ times to Δ_0 (CFSE for octahedral complex). This energy lies in visible region and i.e., why electronic transition are responsible for colour. Such transitions are not possible with d^0 and d^{10} configuration.

- The CFSE for $[\text{CoCl}_6]^{4-}$ complex is 18000 cm^{-1} . The Δ for $[\text{CoCl}_4]^{2-}$ will be -
(A) 18000 cm^{-1} (B) 16000 cm^{-1} (C) 8000 cm^{-1} (D) 2000 cm^{-1}
- The d-orbitals, which are stabilised in an octahedral magnetic field, are -
(A) d_{xy} and d_{z^2} (B) $d_{x^2-y^2}$ and d_{z^2} (C) d_{xy} , d_{xz} and d_{yz} (D) d_{z^2} only
- For an octahedral complex, which of the following d-electron configuration will give maximum CFSE?
(A) High spin d^6 (B) Low spin d^4 (C) Low spin d^5 (D) High spin d^7
- $\text{Ti}_{(\text{aq})}^{3+}$ is purple while $\text{Ti}_{(\text{aq})}^{4+}$ is colourless because -
(A) There is no crystal field effect in Ti^{4+}
(B) There energy difference between t_{2g} and e_g of Ti^{4+} is quite high and does not fall in the visible region
(C) Ti^{4+} has d^0 configuration.
(D) Ti^{4+} is very small in comparison to Ti^{3+} and hence does not absorb any radiation.
- Crystal field stabilization energy for $[\text{CoF}_6]^{3-}$ in terms of parameter Dq is - ($\Delta = 10Dq$)
(A) 4 (B) 6 (C) 12 (D) 24

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
True / False 1.F 2.T 3.T 4.F 5.T 6.T 7.F 8.F Fill in the Blanks 1. Five 2. hexadentate 3.Co-ordination no. & satisfied by ligand 4. Six & +3 5. EAN 6. Ionization 7. Five & Four 8. Square planer, 3 Match the Column 1. (A)-q,s (B) p,r (C)-p,r (d)-p,s 2. (A)-p,q,s,t (B)-q,r,s (C)-p,t (D)-p,q,t 3. (A)-q,s (B)-p,s (C)-p,q,s,t (D)-r Assertion - Reason Questions 1. C 2. B 3. A 4. D 5. A 6. C 7. C 8. A 9. D 10. C 11. D 12. A Comprehension Based Questions Comprehension #1 : 1. B 2. C 3. B Comprehension #2 : 1. C 2. C 3. C 4. C 5. A		

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

1. Draw the structure of
 - (a) Cis-dichlorotetracyano-chromate(III),
 - (b) Mer-triamminetrichlorocobalt(III)
 - (c) Fac-triaquatritritito-N-cobalt(III)
2. Combination of Pt(IV), NH_3 , Cl^- and K^+ results in the formation of seven complexes and one such complex is $[\text{Pt}(\text{NH}_3)_6]\text{Cl}_4$.
 - (i) Write the formula of the other six members of series.
 - (ii) Name these complexes according IUPAC system of nomenclature.
 - (iii) Which will have highest molar conductivity?
 - (iv) Which of these is non-ionic?
 - (v) What is the coordination number and oxidation state of Pt in these complexes.
3. What type of isomers are the following :
 - (i) $[(\text{CO})_5\text{MnSCN}]$ and $[(\text{CO})_5\text{MnNCS}]$,
 - (ii) $[\text{Co}(\text{en})_3][\text{Cr}(\text{CN})_6]$ and $[\text{Cr}(\text{en})_3][\text{Co}(\text{CN})_6]$
 - (iii) $[\text{Co}(\text{NH}_3)_5\text{NO}_3]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{NO}_3$
4. How many geometrical isomer are possible :
 - (a) $[\text{Co}(\text{NH}_3)_2\text{Cl}_4]^-$, octahedral?
 - (b) $[\text{AuCl}_2\text{Br}_2]^-$, square planar?
 - (c) $[\text{CoCl}_2\text{Br}_2]^{2-}$, tetrahedral?
5. What are the types of hybridization involved in the following geometrical shapes of the complex?
 - (a) Square planar,
 - (b) Tetrahedral,
 - (c) Octahedral
6. On the basis of CFT, explain the following, giving appropriate reasons for your answer:
 - (1) The magnetic moment of $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ ion is 5.92 B.M. and that of $[\text{Fe}(\text{CN})_6]^{3-}$ ion is 1.73 B.M.
 - (2) $[\text{Fe}(\text{CN})_6]^{3-}$ ion is weakly paramagnetic while $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ ion strongly paramagnetic
 - (3) Complexes of Co(III) like $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Co}(\text{en})_3]^{3+}$ and $[\text{Co}(\text{NO}_2)_6]^{3-}$ are diamagnetic while $[\text{CoF}_6]^{3-}$ and $[\text{Co}(\text{H}_2\text{O})_3\text{F}_3]$ are paramagnetic.
 - (4) $[\text{Co}(\text{CN})_6]^{4-}$ ion is paramagnetic while $[\text{Co}(\text{CN})_6]^{3-}$ ion is diamagnetic, although both the ions have strong ligands.
7. Which of the following pairs of complex ions has higher value of Δ_0 and why?
 - (a) $[\text{Co}(\text{NH}_3)_6]^{3+}$ and $[\text{Rh}(\text{NH}_3)_6]^{3+}$
 - (b) $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{4-}$ and $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$
 - (c) $[\text{Cr}(\text{en})_3]^{3+}$ and $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$
8. Discuss the geometry of $[\text{HgI}_3]^-$ ion on the basis of valence bond theory.
9. Calculate CFSE of the following complexes: (i) $[\text{CoF}_6]^{3-}$ (ii) $[\text{Fe}(\text{CN})_6]^{4-}$ (iii) $[\text{Cu}(\text{NH}_3)_6]^{2+}$
10. Calculate CFSE values for the following system:
 - (i) d^1 - octahedral (ii) d^1 - tetrahedral (iii) d^5 - low spin octahedral (iv) d^5 - high spin octahedral

11. Write the formula of the following compounds :

- (i) di- μ -carbonyl octacarbonyl diiron(0)
- (ii) Ammonium aquapentafluoronickelate(IV)
- (iii) Tetrammineaquabromocobalt(III) chloride
- (iv) Sodium dithiosulphatoargentate(I)
- (v) Bromodichloriodopalladate(II) ion

12. Explain how $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ and $[\text{Pt}(\text{NH}_3)_6]\text{Cl}_4$ will differ in their electrolytic conductance, Give the hybridisation state of Pt in these compounds. (At. No. of Pt is 78)

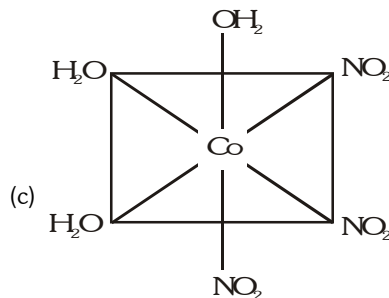
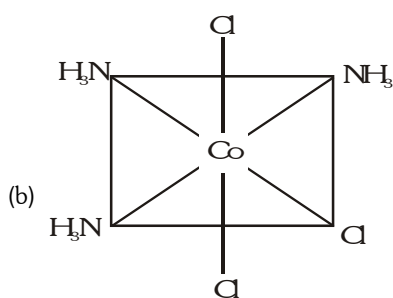
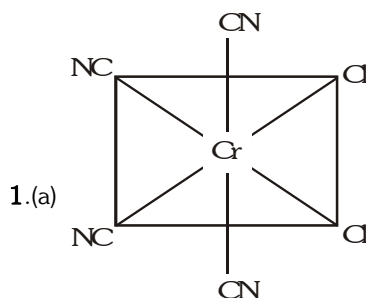
13. A coordination compound has the formula $\text{CoCl}_3 \cdot 4\text{NH}_3$. It does not liberate ammonia but precipitates Chloride ion as AgCl . Give the IUPAC name of the complex and write its structural formula.

14. What is the coordination entity formed when excess of aqueous KCN is added to an aqueous solution of copper sulphate ? Why is it that no precipitate of copper sulphide is obtained when $\text{H}_2\text{S}(\text{g})$ is passed through this solution ?

CONCEPTUAL SUBJECTIVE EXERCISE

ANSWER KEY

EXERCISE -4(A)



2. (i) $[\text{Pt}(\text{NH}_3)_5\text{Cl}]\text{Cl}_3$
 $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}_2$
 $[\text{Pt}(\text{NH}_3)_3\text{Cl}_3]\text{Cl}^-$
 $[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$
 $\text{K}^+[\text{Pt}(\text{NH}_3)\text{Cl}_5]$
 $\text{K}_2[\text{PtCl}_6]$

(iii) $[\text{Pt}(\text{NH}_3)_6]\text{Cl}_4$

(iv) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$

(v) 6 and 4

3. (i) Linkage (ii) coordination (iii) ionization

4. (a) two (b) two (c) 0

5. (a) dsp^2 (b) sp^3 (c) sp^3d^2 / d^2sp^3

EXERCISE-04 [B]

BRAIN STORMING SUBJECTIVE EXERCISE

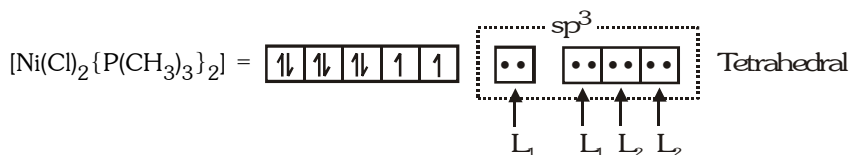
- Draw all the possible isomers of the following complexes –
 $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]^{2+}$, $[\text{Pt}(\text{gly})_2]_0$, $[\text{Pt}\{\text{P}(\text{C}_2\text{H}_5)_3\}_2\text{Cl}_2]$, $[\text{Cr}(\text{NH}_3)(\text{OH})_2\text{Cl}_3]^{2-}$, $[\text{Pt}(\text{NH}_3)(\text{NH}_2\text{OH})(\text{py})(\text{NO}_2)]^+$,
 $[\text{Co}(\text{en})\text{Cl}_2\text{Br}_2]^-$, $[\text{Rh}(\text{py})_3\text{Cl}_3]_0$, $[\text{Co}(\text{en})\text{Cl}_3\text{Br}]^-$
- Arrange the following pairs of complex ions on the basis of their crystal field splitting parameters (Δ_0) :
 (a) $[\text{V}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{V}(\text{H}_2\text{O})_6]^{3+}$ (b) $[\text{Fe}(\text{CN})_6]^{4-}$ and $[\text{Os}(\text{CN})_6]^{4-}$
- A complex of the type $[\text{M}(\text{AA})_2\text{X}_2]$ is known to be optically active. What does this indicate about the structure of the complex? Give one example of such complex.
- A complex of the type $[\text{M}(\text{AB})_2]$ is known to be optically active. What does this indicate about the structure of the complex? Give one example of such complex.
- $[\text{Ni}(\text{Cl})_2(\text{P}(\text{CH}_3)_3)_2]$ is a paramagnetic complex of Ni(II). Analogous Pd(II) complex is diamagnetic. How many geometrical isomers will be possible for Ni(II) and Pd(II) complexes? Also explain their magnetic behaviour.
- On the basis of CFT give the electronic configuration of Rh^{2+} ion in an octahedral complex for which $\Delta_0 > P$.
- Two compounds have the empirical formula, $\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3$. In aqueous solution one of these compounds does not conduct electricity while the other does. Write the possible structures of these two compounds.
- Compute CFSE (in terms of Δ_0) values for d^7 (octahedral) and d^7 (tetrahedral) ions in strong field.

BRAIN STORMING SUBJECTIVE EXERCISE

ANSWER KEY

EXERCISE -4(B)

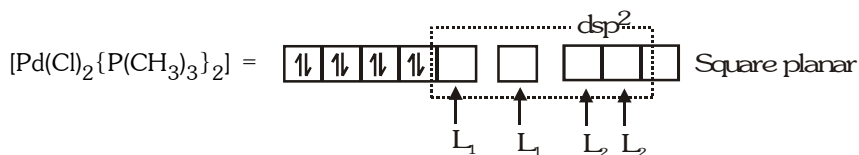
5. (Hint)- Ni(II) complex is paramagnetic, so it has unpaired electrons while complex of Pd(II) is diamagnetic without any unpaired electron. In both Ni(II) and Pd(II), there is d^8 configuration. In Ni(II), value of crystal field splitting energy (Δ) is less than in Pd(II). So in Ni(II) pairing is less favoured while in Pd(II), all electrons are paired because unpairing is unfavoured owing to high value of CFSE. Thus the geometry and hybridization in two complexes is explained as follows:



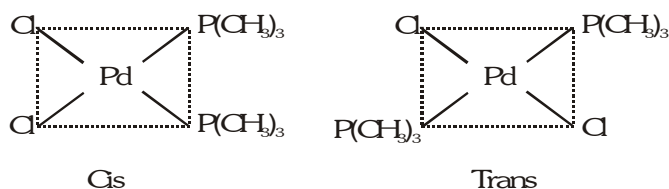
Pd(II) - C.F.S.E. > P.E. irrespective of ligand

Ni(II) - C.F.S.E < P.E. for W.F.L. & C.F.S.E. > P.E. for S.F.L.

because complex is tetrahedral so only one structure will be possible



So it is complex of $\text{M}_{a_2b_2}$ type and will exist as cis and trans isomer. a,b = monodentate ligands.



EXERCISE-05 [A]**(JEE-MAIN)**

1. In $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$, the isomerism shown is [AIEEE-2002]
(A) Ligand (B) Optical (C) Geometrical (D) Ionization
2. In the complexes $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Fe}(\text{SCN})_6]^{3-}$, $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ and $[\text{FeCl}_6]^{3-}$, more stability is shown by [AIEEE-2002]
(A) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ (B) $[\text{Fe}(\text{SCN})_6]^{3-}$ (C) $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ (D) $[\text{FeCl}_6]^{3-}$
3. One mole of the complex compound $\text{Co}(\text{NH}_3)_5\text{Cl}_3$, gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with two moles of AgNO_3 solution to yield two moles of AgCl (s). The structure of the complex is [AIEEE-2003]
(A) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3] \cdot 2\text{NH}_3$ (B) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2] \text{Cl} \cdot \text{NH}_3$
(C) $[\text{Co}(\text{NH}_3)_4\text{Cl}] \text{Cl}_2 \cdot \text{NH}_3$ (D) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
4. In the coordination compound $\text{K}_4[\text{Ni}(\text{CN})_4]$, the oxidation state of nickel is – [AIEEE-2003]
(A) 0 (B) +1 (C) +2 (D) –1
5. The number of 3d-electrons remained in Fe^{2+} (At.no. of Fe = 26) ion is – [AIEEE-2003]
(A) 4 (B) 5 (C) 6 (D) 3
6. Ammonia forms the complex ion $[\text{Cu}(\text{NH}_3)_4]^{2+}$ with copper ions in alkaline solutions but not in acidic solution. What is the reason for it :- [AIEEE-2003]
(A) In acidic solutions hydration protects copper ions
(B) In acidic solutions protons coordinate with ammonia molecules forming NH_4^+ ions and NH_3 molecules are not available
(C) In alkaline solutions insoluble $\text{Cu}(\text{OH})_2$ is precipitated which is soluble in excess of any alkali
(D) Copper hydroxide is an amphoteric substance
7. Among the properties (a) reducing (b) oxidising (c) complexing, the set of properties shown by CN^- ion towards metal species is :- [AIEEE-2004]
(A) c, a (B) b, c (C) a, b (D) a, b, c
8. The coordination number of a central metal atom in a complex is determined by :- [AIEEE-2004]
(A) The number of ligands around a metal ion bonded by sigma and pi-bonds both
(B) The number of ligands around a metal ion bonded by pi-bonds
(C) The number of ligands around a metal ion bonded by sigma bonds
(D) The number of only anionic ligands bonded to the metal ion
9. Which one of the following complexes is an outer orbital complex :- [AIEEE-2004]
(A) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (B) $[\text{Mn}(\text{CN})_6]^{4-}$ (C) $[\text{Fe}(\text{CN})_6]^{4-}$ (D) $[\text{Ni}(\text{NH}_3)_6]^{2+}$
(Atomic nos.: Mn=25 ; Fe=26 ; Co=27 ; Ni = 28)
10. Coordination compounds have great importance in biological systems. In this context which of the following statements is incorrect ? [AIEEE-2004]
(A) Cyanocobalamin is vitamin B_{12} and contains cobalt
(B) Haemoglobin is the red pigment of blood and contains iron
(C) Chlorophylls are green pigments in plants and contain calcium
(D) Carboxypeptidase - A is an enzyme and contains zinc

11. The correct order of magnetic moments (spin only values in B.M.) among is :-[AIEEE-2004]
 (A) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-}$ (B) $[\text{MnCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-}$
 (C) $[\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-}$ (D) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-}$
 (Atomic nos. : Mn = 25, Fe = 26, Co = 27)
12. For octahedral complex, the value of the 'spin only' magnetic moment for one of the following configurations is 2.84 BM. The correct one is [AIEEE-2005]
 (A) d^4 (in strong ligand field) (B) d^4 (in weak ligand field)
 (C) d^3 (in weak as well as in strong field) (D) d^5 (in strong ligand field)
13. The IUPAC name for the complex $[\text{Co}(\text{NO}_2)(\text{NH}_3)_5]\text{Cl}_2$ is [AIEEE-2006]
 (A) pentaammine nitrito-N- cobalt (II) chloride (B) pentaammine nitrito-N- cobalt (III) chloride
 (C) nitrito-N- pentaamminecobalt (III) chloride (D) nitrito-N- pentaamminecobalt (II) chloride
14. Nickel ($Z=28$) combines with a uninegative monodentate ligand X^- to form a paramagnetic complex $[\text{NiX}_4]^{2-}$. The number of unpaired electron in the nickel and geometry of this complex ion are, respectively. [AIEEE-2006]
 (A) one, square planar (B) two, square planar (C) one, tetrahedral (D) two, tetrahedral
15. In $\text{Fe}(\text{CO})_5$, the Fe-C bond possesses [AIEEE-2006]
 (A) ionic character (B) σ - character only (C) π -character only (D) both σ and π character
16. How many EDTA (ethylenediaminetetraacetic acid) molecules are required to make an octahedral complex with a Ca^{2+} ion ? [AIEEE-2006]
 (A) One (B) Two (C) Six (D) Three
17. The "spin-only" magnetic moment [in units of Bohr magneton, (μ_B)] of Ni^{2+} in aqueous solution would be (At. No. Ni= 28) [AIEEE-2006]
 (A) 0 (B) 1.73 (C) 2.84 (D) 4.90
18. Which one of the following has a square planar geometry :- (Co = 27, Ni = 28, Fe=26, Pt = 78) [AIEEE-2007]
 (A) $[\text{CoCl}_4]^{2-}$ (B) $[\text{FeCl}_4]^{2-}$ (C) $[\text{NiCl}_4]^{2-}$ (D) $[\text{PtCl}_4]^{2-}$
19. The coordination number and the oxidation state of the element 'E' in the complex $[\text{E}(\text{en})_2(\text{C}_2\text{O}_4^{2-})]\text{NO}_2^\ominus$ (where en) is ethylene diamine) are, respectively [AIEEE-2008]
 (A) 6 and 2 (B) 4 and 2 (C) 4 and 3 (D) 6 and 3
20. In which of the following octahedral complexes of Co (at. no. 27), will the magnitude of Δ_o be the highest ? [AIEEE-2008]
 (A) $[\text{Co}(\text{CN})_6]^{3-}$ (B) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ (C) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ (D) $[\text{Co}(\text{NH}_3)_6]^{3+}$
21. Which of the following pairs represents linkage isomers ? [AIEEE-2009]
 (A) $[\text{Co}(\text{NH}_3)_5\text{NO}_3]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{NO}_3$
 (B) $[\text{PtCl}_2(\text{NH}_3)_4]\text{Br}_2$ and $[\text{PtBr}_2(\text{NH}_3)_4]\text{Cl}_2$
 (C) $[\text{Cu}(\text{NH}_3)_4][\text{PtCl}_4]$ and $[\text{Pt}(\text{NH}_3)_4][\text{CuCl}_4]$
 (D) $[\text{Pd}(\text{PPh}_3)_2(\text{NCS})_2]$ and $[\text{Pd}(\text{PPh}_3)_2(\text{SCN})_2]$

22. Which of the following has an optical isomer? [AIEEE-2009]
 (A) $[\text{Co}(\text{H}_2\text{O})_4(\text{en})]^{3+}$ (B) $[\text{Co}(\text{en})_2(\text{NH}_3)_2]^{3+}$
 (C) $[\text{Co}(\text{NH}_3)_3\text{Cl}]^+$ (D) $[\text{Co}(\text{en})(\text{NH}_3)_2]^{2+}$
23. Which one of the following has an optical isomer? [AIEEE-2010]
 (A) $[\text{Zn}(\text{en})_2]^{2+}$ (B) $[\text{Zn}(\text{en})(\text{NH}_3)_2]^{2+}$ (C) $[\text{Co}(\text{en})_3]^{3+}$ (D) $[\text{Co}(\text{H}_2\text{O})_4(\text{en})]^{3+}$
 (en = ethylenediamine)
24. A solution containing 2.675 g of $\text{CoCl}_3 \cdot 6\text{NH}_3$ (molar mass = 267.5 g mol^{-1}) is passed through a cation exchanger. The chloride ions obtained in solution were treated with excess of AgNO_3 to give 4.78 g of AgCl (molar mass = 143.5 g mol^{-1}). The formula of the complex is :- [AIEEE-2010]
 (At. mass of Ag = 108 u)
 (A) $[\text{CoCl}(\text{NH}_3)_5]\text{Cl}_2$ (B) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (C) $[\text{CoCl}_2(\text{NH}_3)_4]\text{Cl}$ (D) $[\text{CoCl}_3(\text{NH}_3)_3]$
25. Which of the following facts about the complex $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ is wrong? [AIEEE-2011]
 (A) The complex is an outer orbital complex
 (B) The complex gives white precipitate with silver nitrate solution
 (C) The complex involves d^2sp^3 hybridisation and is octahedral in shape
 (D) The complex is paramagnetic
26. The magnetic moment (spin only) of $[\text{NiCl}_4]^{2-}$ is :- [AIEEE-2011]
 (A) 2.82 BM (B) 1.41 BM (C) 1.82 BM (D) 5.46 BM
27. Among the ligands NH_3 , en, CN^- and CO the correct order of their increasing field strength, is :- [AIEEE-2011]
 (A) $\text{CO} < \text{NH}_3 < \text{en} < \text{CN}^-$ (B) $\text{NH}_3 < \text{en} < \text{CN}^- < \text{CO}$
 (C) $\text{CN}^- < \text{NH}_3 < \text{CO} < \text{en}$ (D) $\text{en} < \text{CN}^- < \text{NH}_3 < \text{CO}$
28. Which one of the following complex ions has geometrical isomers? [AIEEE-2011]
 (A) $[\text{Co}(\text{en})_3]^{3+}$ (B) $[\text{Ni}(\text{NH}_3)_5\text{Br}]^+$ (C) $[\text{Co}(\text{NH}_3)_2(\text{en})_2]^{3+}$ (D) $[\text{Cr}(\text{NH}_3)_4(\text{en})]^{3+}$
29. Which among the following will be named as dibromidobis (ethylene diamine) chromium (III) bromide? [AIEEE-2012]
 (A) $[\text{Cr}(\text{en})\text{Br}_2]\text{Br}$ (B) $[\text{Cr}(\text{en})_3]\text{Br}_3$ (C) $[\text{Cr}(\text{en})_2\text{Br}_2]\text{Br}$ (D) $[\text{Cr}(\text{en})\text{Br}_4]^-$
30. Which of the following complex species is not expected to exhibit optical isomerism? [J-MAIN-2013]
 (A) $[\text{Co}(\text{en})_3]^{3+}$ (B) $[\text{Co}(\text{en})_2\text{Cl}_2]^+$
 (C) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ (D) $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^+$

JEE-MAIN				ANSWER KEY				EXERCISE -5[A]		
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	D	A	C	B	A	C	D	C
Que.	11	12	13	14	15	16	17	18	19	20
	C	A	B	D	D	A	C	D	D	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	B	C	B	A	A	B	C	C	C

MCQ's WITH ONE CORRECT ANSWER

- Which of the following is an organometallic compound? [1997; 1M]
 (A) Lithium methoxide (B) Lithium acetate
 (C) Lithium dimethylamide (D) Methyl lithium
- The geometry of Ni(CO)_4 and $\text{Ni(PPh}_3)_2\text{Cl}_2$ are - [1999; 2M]
 (A) Both square planar (B) Tetrahedral and square planar, respectively
 (C) Both tetrahedral (D) Square planar and tetrahedral, respectively.
- Amongst the following, identify the species with an atom in +6 oxidation state - [2000; 2M]
 (A) MnO_4^- (B) Cr(CN)_6^{3-} (C) NiF_6^{2-} (D) CrO_2Cl_2
- The complex ion which has no 'd'-electrons in the central metal atom is - [2001; 2M]
 (A) $[\text{MnO}_4]^-$ (B) $[\text{Co(NH}_3)_6]^{3+}$ (C) $[\text{Fe(CN)}_6]^{3-}$ (D) $[\text{Cr(H}_2\text{O)}_6]^{3+}$
- Mixture of (X) = 0.02 mole of $[\text{Co(NH}_3)_5\text{SO}_4]\text{Br}$ and 0.02 mole of $[\text{Co(NH}_3)_5\text{Br}]\text{SO}_4$ was prepared in 2 litre of solution [IIT-2003]
 1 litre of mixture [X] + excess $\text{AgNO}_3 \longrightarrow [\text{Y}]$
 1 litre of mixture [X] + excess $\text{BaCl}_2 \longrightarrow [\text{Z}]$
 Number of moles of [Y] and [Z] are -
 (A) 0.01, 0.01 (B) 0.02, 0.01 (C) 0.01, 0.02 (D) 0.02, 0.02
- The pair of compounds having metals in their highest oxidation state is - [IIT-2004]
 (A) MnO_2 , FeCl_3 (B) $[\text{MnO}_4]^-$, CrO_2Cl_2 (C) $[\text{Fe(CN)}_6]^{3-}$, $[\text{Co(CN)}_6]^{3-}$ (D) $[\text{NiCl}_4]^{2-}$, $[\text{CoCl}_4]^-$
- The compound having tetrahedral geometry is - [IIT-2004]
 (A) $[\text{Ni(CN)}_4]^{2-}$ (B) $[\text{Pd(CN)}_4]^{2-}$ (C) $[\text{PdCl}_4]^{2-}$ (D) $[\text{NiCl}_4]^{2-}$
- Spin only magnetic moment of the compound $\text{Hg[Co(SCN)}_4]$ is - [IIT-2004]
 (A) $\sqrt{3}$ (B) $\sqrt{15}$ (C) $\sqrt{24}$ (D) $\sqrt{8}$
- Which kind of isomerism is exhibited by octahedral $\text{Co(NH}_3)_4\text{Br}_2\text{Cl}$? [IIT-2005]
 (A) Geometrical and ionisation (B) Geometrical and optical
 (C) Optical and ionisation (D) Geometrical only
- Which pair of compounds is expected to show similar colour in aqueous medium? [IIT-2005]
 (A) FeCl_3 and CuCl_2 (B) VOCl_2 and CuCl_2 (C) VOCl_2 and FeCl_2 (D) FeCl_2 and MnCl_2
- CuSO_4 decolourises on addition of excess KCN, the product is - [IIT-JEE-2006]
 (A) $[\text{Cu(CN)}_4]^{2-}$ (B) $[\text{Cu(CN)}_4]^{3-}$ (C) Cu(CN)_2 (D) CuCN

COMPREHENSION (Q.12 to 14)

[IIT-JEE-2006]

The co-ordination number of Ni^{+2} is = 4

$\text{NiCl}_2 + \text{KCN (excess)} \longrightarrow \text{A (cyno complex)}$

$\text{NiCl}_2 + \text{Conc. HCl (excess)} \longrightarrow \text{B (Chloro complex)}$

12. The IUPAC name of A and B are –

- (A) Potassium tetracyanonickelate(II), potassium tetrachloronickelate(II),
- (B) Tetracyanopotassiumnickelate(II), tetrachloro potassiumnickelate(II),
- (C) Tetracyanonickel(II), tetrachloronickel(II)
- (D) Potassiumtetracyanonickel(II), Potassiumtetrachloronickel(II)

13. Predict the magnetic nature of A and B –

- (A) Both are diamagnetic
- (B) A is diamagnetic and B is paramagnetic with one unpaired electron
- (C) A is diiamagnetic and B is paramagnetic with two unpaired electrons
- (D) both are paramagnetic

14. The hybridisation of A and B are –

- (A) dsp^2 , sp^3
- (B) sp^3 , sp^3
- (C) dsp^2 , dsp^2
- (D) sp^3d^2 , d^2sp^3

15. If the bond length of CO bond in carbon monoxide is 1.128 Å., then what is the value of CO bond length in Fe(CO)_5 ? [IIT-2006]

- (A) 1.15 Å
- (B) 1.128 Å
- (C) 1.72 Å
- (D) 1.118 Å

16. Among the following metal carbonyls, the C – O bond order is lowest in – [IIT-2007]

- (A) $[\text{Mn(CO)}_6]^+$
- (B) $[\text{Fe(CO)}_5]$
- (C) $[\text{Cr(CO)}_6]$
- (D) $[\text{V(CO)}_6]^-$

17. Sodium-fusion extract, obtained from aniline, on treatment with iron(II) sulphate and H_2SO_4 in presence of air gives a Prussian blue precipitate. The blue colour is due to the formation of: [IIT-2007]

- (A) $\text{Fe}_4[\text{Fe(CN)}_6]_3$
- (B) $\text{Fe}_3[\text{Fe(CN)}_6]_2$
- (C) $\text{Fe}_4[\text{Fe(CN)}_6]_2$
- (D) $\text{Fe}_3[\text{Fe(CN)}_6]_3$

18. Match the complexes in Column-I with their properties listed in Column-II (matrix match type) [IIT-2007]

Column-I		Column-II	
(A)	$[\text{Co(NH}_3)_4(\text{H}_2\text{O})_2]\text{Cl}_2$	(p)	geometrical isomers
(B)	$[\text{Pt(NH}_3)_2\text{Cl}_2]$	(q)	paramagnetic
(C)	$[\text{Co(H}_2\text{O)}_5\text{Cl}]\text{Cl}$	(r)	diamagnetic
(D)	$[\text{Ni(H}_2\text{O)}_6]\text{Cl}_2$	(s)	metal ion with +2 oxidation state

19. Among the following, the coloured compound is :- [IIT-2008]

- (A) CuCl
- (B) $\text{K}_3[\text{Cu(CN)}_4]$
- (C) CuF_2
- (D) $[\text{Cu(NCCH}_3)_4]\text{BF}_4$

20. Both $[\text{Ni(CO)}_4]$ and $[\text{Ni(CN)}_4]^{2-}$ are diamagnetic. The hybridisations of nickel in these complexes, respectively, are [IIT-2008]

- (A) sp^3 , sp^3
- (B) sp^3 , dsp^2
- (C) dsp^2 , sp^3
- (D) dsp^2 , dsp^2

21. The IUPAC name of $[\text{Ni}(\text{NH}_3)_4][\text{NiCl}_4]$ is - [IIT-2008]
 (A) Tetrachloronickel(II) - tetraamminenickel(II) (B) Tetraamminenickel(II) - tetrachloronickel(II)
 (C) Tetraamminenickel(II) - tetrachloronickelate(II) (D) Tetrachloronickel(II) - tetraamminenickelate(0)
22. The spin only magnetic moment value (in Bohr magneton units) of $\text{Cr}(\text{CO})_6$ is : [IIT 2009]
 (A) 0 (B) 2.84 (C) 4.90 (D) 5.92
23. The compound(s) that exhibit(s) geometrical isomerism is (are) : [IIT 2009]
 (A) $[\text{Pt}(\text{en})\text{Cl}_2]$ (B) $[\text{Pt}(\text{en})_2\text{Cl}_2]$ (C) $[\text{Pt}(\text{en})_2\text{Cl}_2]\text{Cl}_2$ (D) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$
24. The correct structure of ethylenediaminetetraacetic acid (EDTA) is - [IIT 2010]
 (A) (B)
 (C) (D)
25. The ionization isomer of $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}(\text{NO}_2)]\text{Cl}$ is - [IIT 2010]
 (A) $[\text{Cr}(\text{H}_2\text{O})_4(\text{O}_2\text{N})]\text{Cl}_2$ (B) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2](\text{NO}_2)$ (C) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}(\text{ONO})]\text{Cl}$ (D) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2(\text{NO}_2)]\cdot\text{H}_2\text{O}$
26. The complex showing a spin-only magnetic moment of 2.82 B.M. is - [IIT 2010]
 (A) $\text{Ni}(\text{CO})_4$ (B) $[\text{NiCl}_4]^{2-}$ (C) $\text{Ni}(\text{PPh}_3)_4$ (D) $[\text{Ni}(\text{CN})_4]^{2-}$
27. Geometrical shapes of the complexes formed by the reaction of Ni^{2+} with Cl^- , CN^- and H_2O respectively, are - [JEE 2011]
 (A) octahedral, tetrahedral and square planar (B) tetrahedral, square planar and octahedral
 (C) square planar, tetrahedral and octahedral (D) octahedral, square planar and octahedral
28. Among the following complexes (K-P) [JEE 2011]
 $\text{K}_3[\text{Fe}(\text{CN})_6]$ (K), $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (L), $\text{Na}_3[\text{Co}(\text{oxalate})_3]$ (M), $[\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2$ (N),
 $\text{K}_2[\text{Pt}(\text{CN})_4]$ (O) and $[\text{Zn}(\text{H}_2\text{O})_6](\text{NO}_3)_2$ (P)
 The diamagnetic complex are -
 (A) K, L, M, N (B) K, M, O, P (C) L, M, O, P (D) L, M, N, O
29. The volume (in mL) of 0.1M AgNO_3 required for complete precipitation of chloride ions present in 30 mL of 0.01M solution of $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2$, as silver chloride is close to. [JEE 2011]
30. As per IUPAC nomenclature, the name of the complex $[\text{Co}(\text{H}_2\text{O})_4(\text{NH}_3)_2]\text{Cl}_3$ is : [JEE 2012]
 (A) Tetraaquadiaminecobalt(III) chloride
 (B) Tetraaquadiammincobalt(III) chloride
 (C) Diaminetetraaquacobalt(III) chloride
 (D) Diamminetetraaquacobalt(III) chloride
31. The colour of light absorbed by an aqueous solution of CuSO_4 is - [JEE 2012]
 (A) orange-red (B) blue-green (C) yellow (D) violet
32. $\text{NiCl}_2\{\text{P}(\text{C}_2\text{H}_5)_2(\text{C}_6\text{H}_5)\}_2$ exhibits temperature dependent magnetic behavior (paramagnetic/diamagnetic). The coordination geometries of Ni^{2+} in the paramagnetic and diamagnetic states are respectively : [JEE 2012]
 (A) tetrahedral and tetrahedral (B) square planar and square planar
 (C) tetrahedral and square planar (D) square planar and tetrahedral

33. Consider the following complex ions P, Q and R ,
 $P = [FeF_6]^{3-}$, $Q = [V(H_2O)_6]^{2+}$ and $R = [Fe(H_2O)_6]^{2+}$
 The correct order of the complex ions, according to their spin-only magnetic moment values (in B.M.) is - [JEE 2013]
 (A) $R < Q < P$ (B) $Q < R < P$ (C) $R < P < Q$ (D) $Q < P < R$
34. $EDTA^{4-}$ is ethylenediaminetetraacetate ion. The total number of N-Co-O bond angles in $[Co(EDTA)]^{-1}$ complex ion is [JEE 2013]
35. The pair(s) of coordination complex/ion exhibiting the same kind of isomerism is(are) - [JEE 2013]
 (A) $[Cr(NH_3)_5Cl]Cl_2$ and $[Cr(NH_3)_4Cl_2]Cl$ (B) $[Co(NH_3)_4Cl_2]^+$ and $[Pt(NH_3)_2(H_2O)Cl]^+$
 (C) $[CoBr_2Cl_2]^{2-}$ and $[PtBr_2Cl_2]^{2-}$ (D) $[Pt(NH_3)_3(NO_3)]Cl$ and $[Pt(NH_3)_3Cl]Br$
36. **Statement-I** : $[Fe(H_2O)_5NO]SO_4$ is paramagnetic. [IIT-2008]
Because
Statement-II : The Fe in $[Fe(H_2O)_5NO]SO_4$ has three unpaired electrons.
 (A) Statement-I is True, Statement-II is True; Statement-II is a correct explanation for Statement-I
 (B) Statement-I is True, Statement-II is True; Statement-II is **NOT** a correct explanation for Statement-I
 (C) Statement-I is True, Statement-II is False
 (D) Statement-I is False, Statement-II is True
37. **Statement-I** : The geometrical isomers of the complex $[M(NH_3)_4Cl_2]$ are optically inactive. [IIT-2008]
Because
Statement-II : Both geometrical isomers of the complex $[M(NH_3)_4Cl_2]$ possess axis of symmetry.
 (A) Statement-I is True, Statement-II is True; Statement-II is a correct explanation for Statement-I
 (B) Statement-I is True, Statement-II is True; Statement-II is **NOT** a correct explanation for Statement-I
 (C) Statement-I is True, Statement-II is False
 (D) Statement-I is False, Statement-II is True
38. **Statement-I** : Zn^{2+} is diamagnetic [1998; 2M]
Because
Statement-II : The electrons are lost from 4s-orbital to form Zn^{2+} .
 (A) Statement-I is True, Statement-II is True; Statement-II is a correct explanation for Statement-I
 (B) Statement-I is True, Statement-II is True; Statement-II is **NOT** a correct explanation for Statement-I
 (C) Statement-I is True, Statement-II is False
 (D) Statement-I is False, Statement-II is True
39. A, B and C are three complexes of chromium(III) with the empirical formula $H_{12}O_6Cl_3Cr$. All the three complexes have water and chloride ion as ligands. Complex A does not react with concentrated H_2SO_4 , whereas complexes B and C lose 6.75% and 13.5% of their original mass, respectively , on treatment with concentrated H_2SO_4 , identify A, B and C. [1999; 6M]
40. Draw the structures of $[Co(NH_3)_6]^{3+}$, $[Ni(CN)_4]^{2-}$ and $[Ni(CO)_4]$. Write the hybridisation of atomic orbital of the transition metal in each case. [2000 main; 4M]
41. A metal complex having composition $Cr(NH_3)_4Cl_2Br$ has been isolated in two forms A and B. The form A reacts with $AgNO_3$ to give a white precipitate readily soluble in dilute aqueous ammonia, whereas B gives a pale yellow precipitate soluble in concentrated ammonia. Write the formula of A and B and state the hybridization of chromium in each. Calculate their magnetic moments (spin-only value) [2001 main; 5M]
42. Deduce the structures of $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ considering the hybridization of the metal ion. Calculate the magnetic moment (spin only) of the species. [2002 main; 5M]
43. Write the IUPAC nomenclature of the given complex along with its hybridization and structure.
 $K_2[Cr(NO)(NH_3)(CN)_4]$, ($\mu = 1.73$ B.M. spin only) [2003 main; 4M]
- Integer Type Question :** [IIT-2010]
44. Total number of geometrical isomers for the complex $[RhCl(CO)(PPh_3)(NH_3)]$ is -

● Objective Questions

- 1.D 2.C 3.D 4.A 5.A 6.B 7.D 8.B 9.A 10.B
 11.B 12.A 13.C 14.A 15.A 16.D 17.A 19.C 20.B 21.C
 22. A 23. C,D 24. C 25.B 26.B 27.B 28.C 29.6 30.D 31.A
 32. C 33.B 34.8 35.B,D

● Match the Column

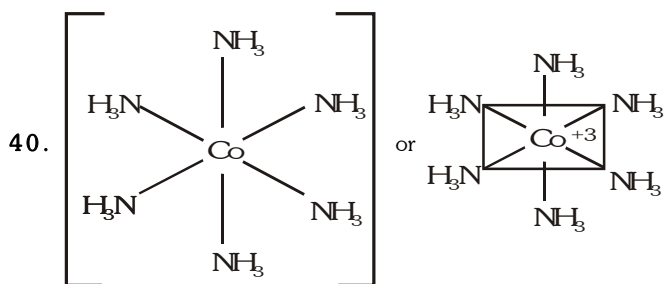
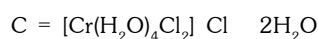
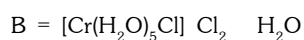
18. (A) $\rightarrow$ p,q,s ; (B) $\rightarrow$ p,r,s ; (C) $\rightarrow$ q,s ; (D) $\rightarrow$ q,s

● Assertion - Reason Questions

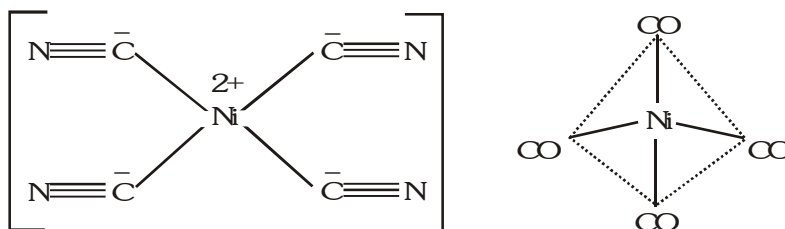
36. A 37. B 38. B

● Subjective Questions

39. A = $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$

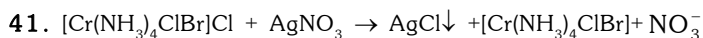


Structure of $[\text{Co}(\text{NH}_3)_6]^{3+}$ is octahedral, d^2sp^3 hybridization



Structure of $[\text{Ni}(\text{CN})_4]^{2-}$ ion is square planar, dsp^2 hybridization

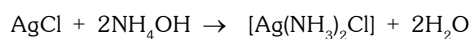
Structure of $\text{Ni}(\text{CO})_4$ is tetrahedral, sp^3 hybridization



form 'A'

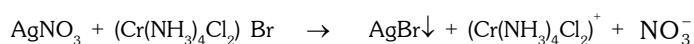
white ppt.

These precipitates of AgCl are soluble in NH_4OH due to formation of complex salt.

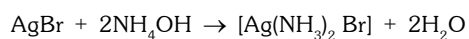


white ppt.

complex salt

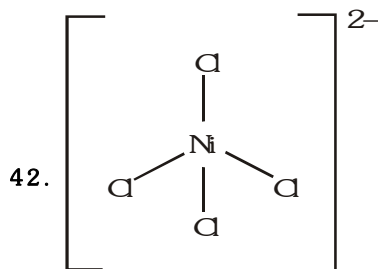


pale yellow ppt

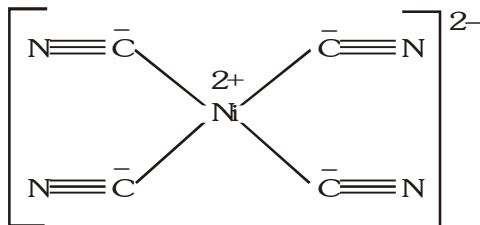


Pale yellow ppt.

Magnetic moment (μ) = 3.872 B.M.

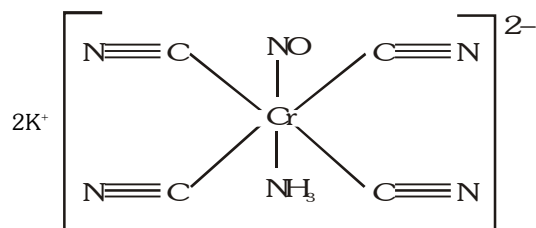


Structure of $[\text{NiCl}_4]^{2-}$ is tetrahedral, sp^3 hybridization, magnetic moment = 2.82 B.M.



Structure of $[\text{Ni}(\text{CN})_4]^{2-}$ is square planar, dsp^2 -hybridization, magnetic moment = 0

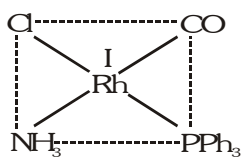
43. IUPAC name of $\text{K}_2[\text{Cr}(\text{NO})(\text{NH}_3)(\text{CN})_4]$ is potassium ammine tetra cyano nitrosyl chromate (I)



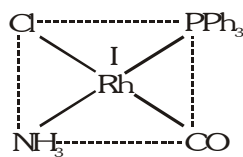
Structure of $\text{K}_2[\text{Cr}(\text{NO})(\text{NH}_3)(\text{CN})_4]$ is octahedral, d^2sp^3 -hybridization

44. (3) $[\text{RhCl}(\text{CO})(\text{PPh}_3)(\text{NH}_3)]$

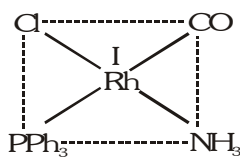
dsp^2 , square planar, total 3 geometrical isomer.



(1)



(2)



(3)

HINTS :

Do Yourself I : 1. (a) co-ordination compound ; (b) Double salt ; (c) co-ordination compound

(d) Double salt ; (e) Double salt ; (f) co-ordination compound ; (g) Double salt

Do Yourself II : 1. (a) $[\text{Pt}(\text{NH}_3)_3\text{Cl}_2\text{Br}]^+$; (b) +1 ; (c) 6 ; (d) 6 ; (e) +4 ; (f) 2 ; (g) 1 ; (h) Yes ; (i) No ;

(j) one mole of AgCl & No AgBr ; (k) No

Do Yourself III : (i) Potassium hexanitrito-Ncobaltate(III), (ii) Sodium pentacyanonitrosoniumferrate(I),

(iii) tetrachloronickelate(II) ion., (iv) pentaamminechlororuthenium(III) ion.,

(v) Tris(ethylenediamine)iron(III) ion, (vi) Bis(glycinato)nickel(II).

Do Yourself IV : 1. (i) $[\text{Pt}(\text{NH}_3)_6]\text{Cl}_6$; (ii) $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}_2$; (iii) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$

Do Yourself V : 1. (i) 36 ; (ii) 34 ; (iii) 86 ; (iv) 50

Do Yourself VI : 1. (i) sp^3 , tetrahedral, 5.92 ; (ii) d^2sp^3 octahedral, 1.73 ; (iii) sp^3d^2 octahedral, 5.92

2. (i) inner orbital complex ; (ii) outer orbital complex ;

(iii) inner [exception : (i) incase of Co^{+3} ligand $\text{C}_2\text{O}_4^{2-}$ behave as a S.F.L. ;

(ii) in case of Fe^{+2} , Mn^{+2} ligand NH_3 behave like a W.F.L.] ; (iv) outer

Do Yourself VII : 1. (i) $t_{2g}^5 e_g^0$, C.F.S.E = $-2.0 \Delta_0 + 2P$; (ii) $t_{2g}^3 e_g^2$, C.F.S.E = 0 ; (iii) $t_{2g}^3 e_g^1$, C.F.S.E = $-0.6 \Delta_0$;

(iv) $t_{2g}^6 e_g^0$, C.F.S.E = $-2.4 \Delta_0 + 3P$.

2. Less interaction between metal and ligands.

3. $t_{2g}^6 e_g^0$ - octahedral, $t_{2g}^4 e_g^2$ - octahedral, $t_{2g}^4 e_g^2$ - octahedral, $t_{2g}^5 e_g^0$ - octahedral.

Do Yourself VIII : 1. (i) 4.89BM ; (ii) 2.82BM ; (iii) 5.92BM ; (iv) 5.92BM ; (v) 4.89BM ; (vi) 2.82BM ;

(vii) 3.46 BM; (viii) 3.46 BM

Do Yourself IX : 1. Due to different crystal field splitting energy of complexes.

2. $[\text{Ni}(\text{NO}_2)_6]^{4+} > [\text{Ni}(\text{NH}_3)_6]^{2+} > [\text{Ni}(\text{H}_2\text{O})_6]^{2+}$

Do Yourself X : 1. Both produce different ions in aqueous solution.

2. (i) G.I. = None ; (ii) G.I. = 2

3. Draw the mirror image of the complexes.

4. See the chart given in theory.

5. G.I. = 3, O.I. = 0

6. (i) G.I., O.I. ; (ii) O.I. ; (iii) Linkage ; (iv) G.I.

EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. If the pressure of a gas contained in a closed vessel is increased by 0.4 % when heated by 1 C its initial temperature must be :
(A) 250 K (B) 250 C (C) 25 C (D) 25 K
2. If a mixture containing 3 moles of hydrogen and 1 mole of nitrogen is converted completely into ammonia, the ratio of initial and final volume under the same temperature and pressure would be :
(A) 3 : 1 (B) 1 : 3 (C) 2 : 1 (D) 1 : 2
3. SO_2 at STP contained in a flask was replaced by O_2 under identical conditions of pressure, temperature and volume. Then the weight of O_2 will be _____ of SO_2 :
(A) half (B) one fourth (C) twice (D) four times
4. According to Charle's law :
(A) $\left(\frac{dV}{dT}\right)_P = k$ (constant) (B) $\left(\frac{dV}{dT}\right)_P = P$ (C) $\left(\frac{dV}{dT}\right)_P = V$ (D) $\left(\frac{dV}{dT}\right)_P = T$
5. A sample of gas at 35 C & 1 atm pressure occupies a volume of 3.75 litres. At what temp. should the gas be kept if it is required to reduce the volume to 3 litres at the same pressure :
(A) -26.6 C (B) 0 C (C) 3.98 C (D) 28 C
6. Equal weights of methane and hydrogen are mixed in an empty container at 25 C. The fraction of total pressure exerted by hydrogen is :
(A) 1/2 (B) 8/9 (C) 16/19 (D) 1/9
7. The best vacuum so far attained in laboratory is 10^{-10} mm of Hg. The number of molecules of gas remain per cm^3 at 20 C in this vacuum is :
(A) 3.29×10^4 molecules (B) 3.29×10^5 molecules (C) 3.29×10^6 molecules (D) 3.29×10^7 molecules
8. A hydrocarbon contains 10.5 g of carbon per gm of H. One litre vapours of hydrocarbon at 127 C and 1 atm pressure weighs 2.8 g. The molecular formula of hydrocarbon is :
(A) C_6H_8 (B) C_7H_8 (C) C_5H_{12} (D) C_8H_4
9. A 0.5 dm^3 flask contains gas A and 1 dm^3 flask contains gas B at the same temperature. If density of A = 3 g/dm^3 and that of B = 1.5 g/dm^3 and the molar mass of A = 1/2 of B, the ratio of pressure exerted by gases is :
(A) $\frac{P_A}{P_B} = 2$ (B) $\frac{P_A}{P_B} = 1$ (C) $\frac{P_A}{P_B} = 4$ (D) $\frac{P_A}{P_B} = 3$
10. 120 g of an ideal gas of molecular weight 40 are confined to a volume of 20 litre at 400 K, then the pressure of gas is :
(A) 490 atm (B) 4.92 atm (C) 2236 atm (D) 22.4 atm
11. A cylinder contains acetylene gas at 27 C and 4.05 M Pa. The pressure in the cylinder after half the mass of gas is used up and temperature has fallen to 12 C will be :
(A) 4.05 M Pa (B) 2.025 M Pa (C) 3.84 M Pa (D) 1.92 M Pa
12. The weight of 350 mL of a diatomic gas at 0 C and 2 atm pressure is 1 g. The weight in g of one atom at NTP is :
(A) $2.64 \times 10^{-23} \text{ g}$ (B) $2.64 \times 10^{-22} \text{ g}$ (C) $5.28 \times 10^{-23} \text{ g}$ (D) $0.82 \times 10^{-22} \text{ g}$
13. Oxygen is present in a litre flask at a pressure of 7.6×10^{-10} mm of Hg. The number of oxygen molecules in the flask at 0 C is :
(A) 2.7×10^9 molecules (B) 2.7×10^{10} molecules (C) 2.7×10^{11} molecules (D) 2.7×10^{12} molecules

14. Assuming that O_2 molecule is spherical in shape with radius $2 \text{ \AA}$, the percentage of the volume of O_2 molecules to the total volume of gas at S.T.P. is :
 (A) 0.09 % (B) 0.9 % (C) 0.009 % (D) 0.045 %
15. The r.m.s. velocity of hydrogen at 27°C , $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$ is :
 (A) 1.934 m/s (B) 19.34 m/s (C) 193.4 m/s (D) 1934 m/s
16. Temperature at which r.m.s. speed of O_2 is equal to that of neon at 300 K is :
 (A) 280 K (B) 480 K (C) 680 K (D) 180 K
17. The most probable velocity of a neutron at 20°C is nearby :
 (A) 220 m/s (B) 2200 m/s (C) 22200 m/s (D) 22 m/s
18. The R.M.S. speed of the molecules of a gas of density 4 kg m^{-3} and pressure $1.2 \times 10^5 \text{ N m}^{-2}$ is :
 (A) 120 m s^{-1} (B) 300 m s^{-1} (C) 600 m s^{-1} (D) 900 m s^{-1}
19. The mass of molecule A is twice that of molecule B. The root mean square velocity of molecule A is twice that of molecule B. If two containers of equal volume have same number of molecules, the ratio of pressure P_A/P_B will be :
 (A) 8 : 1 (B) 1 : 8 (C) 4 : 1 (D) 1 : 4
20. The R.M.S. velocity of a gas whose each molecule weighs 10^{-12} g and at temperature 27°C is :
 (A) 0.70 cm/s (B) 0.35 cm/s (C) 0.35 m/s (D) 0.70 m/s
21. The average speed of an ideal gas molecule at 27°C is 0.3 m sec^{-1} . The average speed at 927°C :
 (A) 0.15 m sec^{-1} (B) 0.6 m sec^{-1} (C) 1.2 m sec^{-1} (D) 0.6 cm sec^{-1}
22. The temperature at which CO_2 has the same R.M.S. speed to that of O_2 at S.T.P. is/are :
 (A) 375.38 K (B) 102.38°C (C) 275.38 K (D) 202.38°C
23. The temperature at which the most probable speed of CO_2 molecules be twice as that of 50°C is :
 (A) 200°C (B) 1292 K (C) 100°C (D) 646 K
24. What is the total translational and rotational energy of 1 mole of oxygen at 300 K .
 $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$:
 (A) 6235.5 J (B) 623.25 J (C) 62.325 J (D) 6.2325 J
25. The kinetic energy of N molecules of O_2 is x joule at -123°C . Another sample of O_2 at 27°C has a kinetic energy of 2x. The later sample contains _____ molecules of O_2 :
 (A) N (B) $N/2$ (C) 2 N (D) 3 N
26. The average kinetic energy in joules of molecules in 8.0 gm of methane at 27°C is :
 (A) $6.21 \times 10^{-20} \text{ J/molecule}$ (B) $6.21 \times 10^{-21} \text{ J/molecule}$
 (C) $6.21 \times 10^{-22} \text{ J/molecule}$ (D) $3.1 \times 10^{-22} \text{ J/molecule}$
27. The ratio of rates of diffusion of CO_2 and SO_2 at the same pressure and temperature is :
 (A) $4 : \sqrt{11}$ (B) 11 : 4 (C) 1 : 4 (D) 1 : 6
28. 20 L of SO_2 diffuses through a porous partition in 60 seconds. Volume of O_2 diffuse under similar conditions in 30 seconds will be :
 (A) 12.14 L (B) 14.14 L (C) 18.14 L (D) 28.14 L
29. Three footballs are respectively filled with nitrogen, hydrogen and helium. If the leaking of the gas occurs with time from the filling hole, then the ratio of the rate of leaking of gases ($r_{N_2} : r_{H_2} : r_{He}$) from three footballs (in equal time interval) is :
 (A) $(1 : \sqrt{14} : \sqrt{7})$ (B) $(\sqrt{14} : \sqrt{7} : 1)$ (C) $(\sqrt{7} : 1 : \sqrt{14})$ (D) $(1 : \sqrt{7} : \sqrt{14})$

30. NH_3 & SO_2 gases are being prepared in two corners of a laboratory. The gas that will be detected first in the middle of the laboratory is :
 (A) NH_3 (B) SO_2 (C) both at the same time (D) can't determine
31. Consider an ideal gas contained in a vessel. If the intermolecular interaction suddenly begins to act, which of the following will happen :
 (A) the observed pressure decreases (B) the observed pressure increases
 (C) the observed pressure remains same (D) none of these
32. A real gas obeying Vander Waals equation will resemble ideal gas, if the :
 (A) constants a & b are small (B) a is large & b is small
 (C) a is small & b is large (D) constant a & b are large
33. Calculate the compressibility factor for CO_2 , if one mole of it occupies 0.4 litre at 300 K and 40 atm. Comment on the result :
 (A) 0.40, CO_2 is more compressible than ideal gas
 (B) 0.65, CO_2 is more compressible than ideal gas
 (C) 0.55, CO_2 is more compressible than ideal gas
 (D) 0.62, CO_2 is more compressible than ideal gas
34. Calculate the radius of He atoms if its Vander Waal's constant 'b' is 24 ml mol^{-1} :
 (Note ml = cubic centimeter)
 (A) 1.355 Å (B) 1.314 Å (C) 1.255 Å (D) 0.355 Å
35. The critical constant for water are 374 C, 218 atm and 0.0566 litre mol^{-1} . Calculate a & b.
 (A) a = 1.095 litre² atm mol^{-2} , b = 0.0185 litre mol^{-1}
 (B) a = 1.92 litre² atm mol^{-2} , b = 0.185 litre mol^{-1}
 (C) a = 2.095 litre² atm mol^{-2} , b = 0.0189 litre mol^{-1}
 (D) a = 2.95 litre² atm mol^{-2} , b = 0.1185 litre mol^{-1}
36. 10 mL of gaseous hydrocarbon on combustion gives 40 mL of CO_2 & 50 mL of H_2O . The hydrocarbon is :
 (A) C_4H_6 (B) C_4H_8 (C) C_8H_{10} (D) C_4H_{10}
37. The volume of oxygen required for complete oxidation of 2 litre of methane at NTP is :
 (A) 12.25 L (B) 4 L (C) 1 L (D) 3 L
38. LPG is a mixture of n-butane & iso-butane. The volume of oxygen needed to burn 1 kg of LPG at NTP would be :
 (A) 2240 L (B) 2510 L (C) 1000 L (D) 500 L
39. Stronger inter-molecular forces exist in :
 (1) gases (2) liquids (C) amorphous solids (D) crystalline solids
40. Association of molecules in water is due to :
 (A) covalent bonding (B) hydrogen bonding (C) ionic bonding (D) van der Waals' forces
41. Which of the following statements is wrong ?
 (A) Evaporation is a spontaneous process
 (B) Evaporation is a surface phenomenon
 (C) Vapour pressure decreases with increase of temperature
 (D) The vapour pressure of a solution is always less than the vapour pressure of a pure solvent.
42. Normal boiling point of a liquid is that temperature at which vapour pressure of the liquid is equal to :
 (A) zero (B) 380 mm of Hg (C) 760 mm of Hg (D) 100 mm of Hg

43. Water boils at lower temperature on high altitudes because :
 (A) atmospheric pressure is low there (B) atmospheric pressure is high there
 (C) water is weakly hydrogen bonded there (D) water in pure form is found there
44. When a student was given viscometer, the liquid was sucked with difficulty; the liquid may be :
 (A) benzene (B) toluene (C) water (D) glycerine
45. Mark the statement which is correct ?
 (A) Surface tension of a liquid increases with temperature
 (B) Addition of chemicals reduces the surface tension of a liquid
 (C) Stalagmometer is used for measuring viscosity of the liquid
 (D) Viscosity of the liquid does not depend on intermolecular forces
46. With the increasing molecular mass of a liquid, the viscosity :
 (A) decreases (B) increases (C) no effect (D) all wrong
47. The viscosity of a which liquid is the maximum ?
 (A) water (B) glycol (C) acetone (D) ethanol
48. The rise of a liquid in a capillary tube is due to :
 (A) viscosity (B) osmosis (C) diffusion (D) surface tension
49. With increase in temperature, the fluidity of liquids ;
 (A) increases (B) decreases (C) remains constant
 (D) may increase or decrease
50. If η_1 and η_2 are the coefficients of viscosity of two liquids, d_1 and d_2 their densities and t_1 and t_2 the flow times in Ostwald viscometer, then :
 (A) $\frac{\eta_1}{\eta_2} = \frac{d_1 t_2}{d_2 t_1}$ (B) $\frac{\eta_1}{\eta_2} = \frac{d_2 t_2}{d_1 t_1}$ (C) $\frac{\eta_1}{\eta_2} = \frac{d_1 t_1}{d_2 t_2}$ (D) $\frac{\eta_1}{\eta_2} = \frac{d_2 t_1}{d_1 t_2}$
51. Which of the following expressions regarding the unit of coefficient of viscosity is not true ?
 (A) dyne cm⁻² sec (B) dyne cm² sec⁻¹ (C) Nm⁻² sec (D) 1 poise = 10⁻¹ Nm⁻² sec
52. The boiling point of water, ethyl alcohol and diethyl ether are 100 °C, 78.5 °C and 34.6 °C respectively. The intermolecular forces will be in the order of :
 (A) water > ethyl alcohol > diethyl ether (B) ethyl > alcohol > water > diethyl ether
 (C) diethyl ether > ethyl alcohol > water (D) diethyl ether > water > ethyl alcohol
53. Which one is the amorphous solid ?
 (A) diamond (B) graphite (C) common salt (D) glass
54. Viscosity of a liquid is increased by :
 (A) increase in temperature. (B) decrease in molecular size.
 (C) increase in molecular size. (D) none of the above.
55. Which of the following statements is correct if the intermolecular forces in liquids A, B and C are in the order A < B < C ?
 (A) B evaporates more readily than A (B) B evaporates less readily than C.
 (C) A and B evaporate at the same rate. (D) A evaporates more readily than C.
56. The critical temperature of water is higher than that of O₂ because the H₂O molecule has :
 (A) fewer electrons than O₂ (B) two covalent bonds.
 (C) V-shape (D) dipole moment.

CHECK YOUR GRASP						ANSWER KEY				EXERCISE -1					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	A	C	A	A	A	B	C	B	C	B	D	A	B	A	D
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	B	B	B	A	B	B	A	B	A	A	B	A	B	A	A
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	A	A	B	A	C	D	B	B	D	B	C	C	A	D	B
Que.	46	47	48	49	50	51	52	53	54	55	56				
Ans.	B	B	B	A	C	C	A	D	C	D	D				

BRAIN TEASERS

1. Consider the following statements :

$$PV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

c : has the dimension of molar volume

(A) a and b (B) a and c (C) b and c (D) a, b and c

- Gas Y : 8.0 0.011

$$c : T_c(X) < T_c(Y)$$

(A) a alone (B) a and b (C) a,b and c (D) b and c

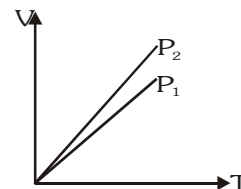
- $$\left[p + \frac{a}{V^2} \right] V = RT$$

(A) $\left(1 - \frac{a}{RTV}\right)$ (B) $\left(1 - \frac{RTV}{a}\right)$ (C) $\left(1 + \frac{a}{RTV}\right)$ (D) $\left(1 + \frac{RTV}{a}\right)$

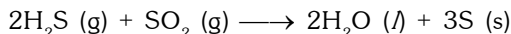
- (D) $a(\text{NH}_3) < a(\text{N}_2)$ but $b(\text{NH}_3) > b(\text{N}_2)$

- (A) $PV = nRT - \frac{n^2 a}{V}$ (B) $PV = nRT + nbP$ (C) $PV = nRT$ (D) $P = \frac{nRT}{V - b}$

6. Compressibility factor for H_2 behaving as real gas is :
- (A) 1 (B) $\left(1 - \frac{a}{RTV}\right)$ (C) $\left(1 + \frac{Pb}{RT}\right)$ (D) $\frac{RTV}{(1-a)}$
7. Equal masses of methane and oxygen are mixed in an empty container at 25 C. The fraction of the total pressure exerted by oxygen is :
- (A) 1/3 (B) 1/2 (C) 2/3 (D) (1/3) (273/298)
8. The temperature at which a real gas obeys the ideal gas laws over a wide range of pressure is :
- (A) critical temperature (B) Boyle temperature (C) boiling temperature (D) reduced temperature
9. Helium atom is two times heavier than a hydrogen molecule. At 298 K, the average kinetic energy of a helium atom is :
- (A) two times that of a hydrogen molecules (B) same as that of a hydrogen molecules
(C) four times that of a hydrogen molecules (D) half that of a hydrogen molecules
10. In vander Waal's equation of state for a non-ideal gas the term that accounts for intermolecular forces is :
- (A) $(V - b)$ (B) $\left(p + \frac{a}{V^2}\right)$ (C) RT (D) $(RT)^{-1}$
11. The compressibility of a gas is less than unity at STP. Therefore :
- (A) $V_m > 22.4 \text{ L}$ (B) $V_m < 22.4 \text{ L}$ (C) $V_m = 22.4 \text{ L}$ (D) $V_m \geq 44.8 \text{ L}$
12. If two moles of an ideal gas at 546 K occupies a volume of 44.8 litres, the pressure must be :
- (A) 2 atm (B) 3 atm (C) 4 atm (D) 1 atm
13. At STP the order of root mean square velocity of molecules of H_2 , N_2 , O_2 and HBr is :
- (A) $H_2 > N_2 > O_2 > HBr$ (B) $HBr > O_2 > N_2 > H_2$ (C) $HBr > H_2 > O_2 > N_2$ (D) $N_2 > O_2 > H_2 > HBr$
14. The density of a gas at 27 C and 1 atm is d. Pressure remaining constant at which of the following temperatures will its density become 0.75 d?
- (A) 20 C (B) 30 C (C) 400 K (D) 300 K
15. At 27 C the ratio of rms velocities of ozone to oxygen is :
- (A) $\sqrt{3/5}$ (B) $\sqrt{4/3}$ (C) $\sqrt{2/3}$ (D) 0.25
16. A real gas most closely approaches the behaviour of an ideal gas at :
- (A) 15 atm and 200 K (B) 1 atm and 273 K
(C) 0.5 atm and 500 K (D) 15 atm and 500 K
17. The temperature at which the second virial coefficient of a real gas is zero is called :
- (A) Critical temperature (B) Boyle's temperature (C) Boiling point (D) none of these
18. V vs T curves at constant P_1 and P_2 for an ideal gas are shown in figure. Which is correct :
- (A) $P_1 > P_2$
(B) $P_1 < P_2$
(C) $P_1 = P_2$
(D) all



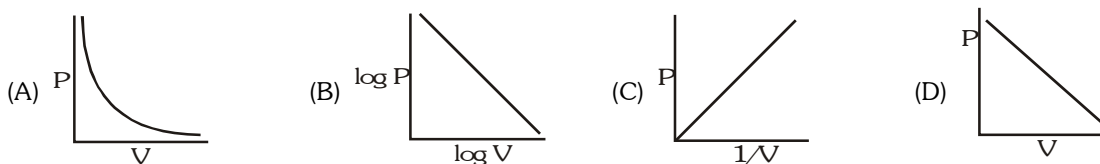
19. At STP, 2.8 litres of hydrogen sulphide were mixed with 1.6 litres of sulphur dioxide and the reaction occurred according to the equation :



Which of the following shows the volume of the gas remaining after the reaction?

- (A) 0.2 litres of SO_2 (g) (B) 0.4 litres of H_2 (g)
 (C) 1.2 litres of H_2S (g) (D) 1.2 litres of SO_2 (g)
20. The rates of diffusion of SO_3 , CO_2 , PCl_3 and SO_2 are in the following order :
- (A) $\text{PCl}_3 > \text{SO}_3 > \text{SO}_2 > \text{CO}_2$ (B) $\text{CO}_2 > \text{SO}_2 > \text{PCl}_3 > \text{SO}_3$
 (C) $\text{SO}_2 > \text{SO}_3 > \text{PCl}_3 > \text{CO}_2$ (D) $\text{CO}_2 > \text{SO}_2 > \text{SO}_3 > \text{PCl}_3$

21. Which of the following curves does not represent Boyle's law :



22. A closed vessel contains equal number of nitrogen and oxygen molecules at pressure of P mm. If nitrogen is removed from the system, then the pressure will be :

- (A) P (B) 2P (C) P/2 (D) P^2

23. The ratio of average molecular kinetic energy of PCl_5 to that of O_2 both at 300 K is :

- (A) 1 : 1 (B) 7 : 2 (C) 176 : 1 (D) 2 : 7

24. The correct order of normal boiling points of O_2 , N_2 , NH_3 and CH_4 for whom the values of vander waals constant 'a' are 1.360, 1.390, 4.170 and 2.253 $\text{L}^2\cdot\text{atm}\cdot\text{mol}^{-2}$ respectively, is :

- (A) $\text{O}_2 < \text{N}_2 < \text{NH}_3 < \text{CH}_4$ (B) $\text{O}_2 < \text{N}_2 < \text{CH}_4 < \text{NH}_3$
 (C) $\text{NH}_3 < \text{CH}_4 < \text{N}_2 < \text{O}_2$ (D) $\text{NH}_3 < \text{CH}_4 < \text{O}_2 < \text{N}_2$

25. A molecule in a cube of length ℓ has a velocity v_x in the x direction. The number of collisions with one face per second for this molecule will be given by which one of the following :

- (A) v_x/ℓ (B) $v_x/2\ell$ (C) $\frac{1}{2} \ell$ (D) $N m v^2/3\ell$

26. The rate of diffusion of methane at a given temperature is twice that of gas X. The molecule weight of X is :

- (A) 64.0 (B) 32.0 (C) 4.0 (D) 8.0

27. For a fixed volume of a gas the pressure increases when the temperature increases. This is due to :

- (A) increase in the average molecular speed (B) increase in the rate of collisions
 (C) increase in the molecular attraction (D) decrease in the mean free path

28. The ratio, $\frac{\text{rms velocity of SO}_2}{\text{rms velocity of He}}$, of sulphur dioxide and helium gases at 30 C is equal to :

- (A) 4 (B) 0.25 (C) 0.10 (D) 8

29. A certain volume of argon gas (Mol Wt. 40) requires 45 s to effuse through a hole at a certain pressure and temperature. The same volume of another gas of unknown molecular weight requires 60 s to pass through the same hole under the same conditions of temperature and pressure. The molecular weight of the gas is :

(A) 53 (B) 35 (C) 71 (D) 120

30. On the surface of the earth at 1 atm pressure, a balloon filled with H_2 gas occupies 500 mL. This volume $5/6$ of its maximum capacity. The balloon is left in air. It starts rising. The height above which the balloon will burst if temperature of the atmosphere remains constant and the pressure decreases 1 mm for every 100 cm rise of height is :

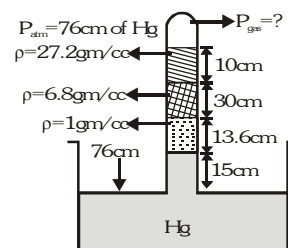
(A) 120 m (B) 136.67 m (C) 126.67 m (D) 100 m

31. A chemist has synthesized a greenish yellow gaseous compound of chlorine and oxygen and finds that its density is 7.71 g/L at 36 C and 2.88 atm. Then the molecular formula of the compound will be :

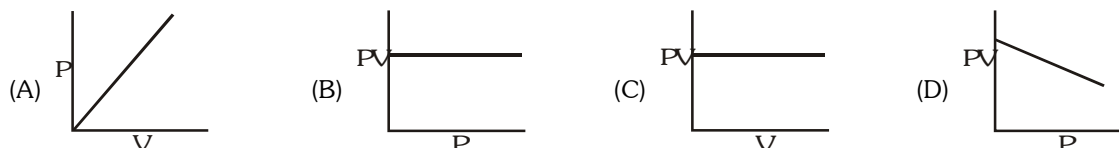
(A) ClO_3 (B) ClO_2 (C) ClO (D) Cl_2O_2

32. In the following arrangement the pressure of the confined gas will be :

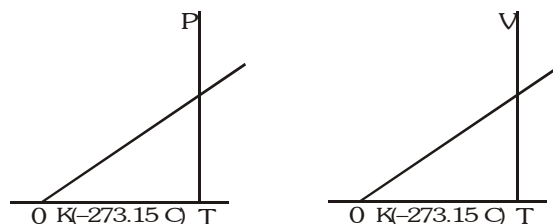
(A) 15 cm of Hg
(B) 25 cm of Hg
(C) 35 cm of Hg
(D) 45 cm of Hg



33. Which of the following graphs represent Boyle's law :



34. What conclusions would you draw from the following graphs :

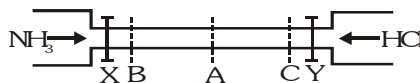


- (A) As the temperature is reduced, the volume as well as pressure increase.
(B) As the temperature is reduced, the volume becomes zero and the pressure reaches infinity.
(C) As the temperature is reduced, the volume as well as the pressure decrease.
(D) A point is reached where theoretically, the volume as well as the pressure become zero.

35. Which of the following statements are correct :

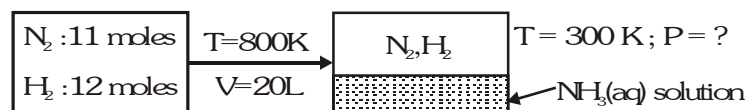
- (A) Helium diffuses at a rate 8.65 times as much as CO does
(B) Helium escapes at a rate 2.65 times as fast as CO does.
(C) Helium escapes at a rate 4 times as fast as CO_2 does.
(D) Helium escapes at a rate 4 times as fast as SO_2 does.

36. When a gas is expanded at constant temperature :
- (A) the pressure decreases
 (B) the kinetic energy of the molecules remains the same
 (C) the kinetic energy of the molecules decreases
 (D) the number of molecules of the gas decreases
37. Which of the following are correct statements :
- (A) vander waals constant 'a' is a measure of attractive force
 (B) vander waals constant 'b' is also called co-volume or excluded volume
 (C) 'b' is expressed in L mol^{-1}
 (D) 'a' is expressed in $\text{atm L}^2 \text{mol}^{-2}$
38. Which of the following statements are incorrect :
- (A) Molar volume of every gas at STP is 22.4 L
 (B) Under critical states compressibility factor is 1
 (C) All gases will have equal value of average KE at a given temperature
 (D) At absolute zero, KE is $\frac{3}{2} R$.
39. At what temperature will the total KE of 0.3 mol of He be the same as the total KE of 0.40 mol of Ar at 400 K :
- (A) 533 K (B) 400 K (C) 346 K (D) 300 K
40. If molecules of the gas are spherical of radius $1 \text{ \AA}$, the volume occupied by the molecules in 1 mol of a gas is :
- (A) 22400 mL (B) 22.4 L (C) 2.52 mL (D) 4.22 mL
41. See the figure :



The valves of X and Y are opened simultaneously. The white fumes of NH_4Cl will first form at :

- (A) A (B) B (C) C (D) A, B and C simultaneously
42. Calculate γ (ratio of C_p and C_v) for triatomic linear gas at high temperature. Assume that the contribution of vibrational degree of freedom is 75% :
- (A) 1.222 (B) 1.121 (C) 1.18 (D) 1.33
43. 11 moles N_2 and 12 moles of H_2 mixture reacted in 20 litre vessel at 800 K. After equilibrium was reached, 6 mole of H_2 was present. 3.58 litre of liquid water is injected in equilibrium mixture and resultant gaseous mixture suddenly cooled to 300 K. What is the final pressure of gaseous mixture ? Negative vapour pressure of liquid solution. Assume (i) all NH_3 dissolved in water (ii) no change in volume of liquid (iii) no reaction of N_2 and H_2 at 300 K :



Initial condition

- (A) 18.47 atm (B) 60 atm (C) 22.5 atm (D) 45 atm

44. Two closed vessel A and B of equal volume containing air at pressure P_1 and temperature T_1 are connected to each other through a narrow open tube. If the temperature of one is now maintained at T_1 and other at T_2 (where $T_1 > T_2$) then that what will be the final pressure ?
- (A) $\frac{T_1}{2P_1T_2}$ (B) $\frac{2P_1T_2}{T_1 + T_2}$ (C) $\frac{2P_1T_2}{T_1 - T_2}$ (D) $\frac{2P_1}{T_1 + T_2}$
45. A balloon containing 1 mole air at 1 atm initially is filled further with air till pressure increases to 4 atm. The initial diameter of the balloon is 1 m and the pressure at each stage is proportion to diameter of the balloon. How many no. of moles of air added to change the pressure from 1 atm to 4 atm :
- (A) 80 (B) 257 (C) 255 (D) 256
46. What is the density of wet air with 75% relative humidity at 1 atm and 300 K? Given : vapour pressure of H_2O is 30 torr and average molar mass of air is 29 g mol^{-1} :
- (A) 1.614 g/L (B) 0.96 g/L (C) 1.06 g/L (D) 1.164 g/L
47. Calculate minimum number of balloons each of volume 82.1 L required to lift a mass of 1 kg to a height of 831 m. Given : molar mass of air = 29 g/mol, temperature is constant at 290 K and mass of each balloons is 40 g. [Use $e^{-0.1} = 0.9$, pressure at sea level = 1 atm, acceleration due to gravity (g) = 10 m/s^2] :
- (A) 10 (B) 20 (C) 25 (D) 50
48. If a real gas following equation $P(V - nb) = nRT$, at low pressure then find the intercept and slope of graph between $\frac{d}{P}$ v/s P are respectively :
- (A) $\frac{MR}{T}, \frac{M(RT)^2}{b}$ (B) $\frac{M}{RT}, -\frac{Mb}{(RT)^2}$ (C) $\frac{Mb}{RT}, -\frac{M}{(RT)^2}$ (D) $\frac{RT}{M}, -\frac{b}{M(RT)^2}$

[illegible]

EXERCISE-03

MISCELLANEOUS TYPE QUESTIONS

TRUE / FALSE

1. The volume occupied by 32 g of oxygen is greater than that occupied by 16 g of methane, both being at the same T and P.
2. A real gas can be liquefied if its temperature is greater than its critical temperature.
3. Kinetic energy of gaseous molecules is zero at 0 C.
4. The term $V_m - b$ in vander waals equation represents the available volume where molecules of the gas can move.
5. The average speed of a gas varies linearly with increases of temperature.
6. A mixture of ideal gaseous is cooled upto liquid helium temperature (4.22 K) to form an ideal solution.
7. For a vander waals gas $V_c = 3b$, $P_c = a/27b^2$, $T_c = 8a/27bR$. Numerically the compressibility factor of a vander waals gas at the critical points is $Z = 0.375$.

FILL IN THE BLANKS

1. The value of PV for 5.6 L of an ideal gas is RT at STP.
2. The density of an ideal gas with increase in temperature of the gas, provided the pressure remains constant.
3. The unit of vander waals constant 'a' in SI units is
4. 0.5 L of a certain gas at STP weighs 0.58 g. Its molar mass isg mol⁻¹.
5. The root means square speed of methane gas will be that of helium gas both having the same temperature and pressure.
6.
 - (i) The rate of diffusion of a gas is proportional to or square root of molecular mass.
 - (ii) The total energy of one mole of gas (ideal monoatomic) at 27 C is calorie.
 - (iii) Equal masses of SO₂ and O₂ are kept in a vessel at 27 C. The total pressure is 2 atm. The partial pressure of SO₂ is
7.
 - (i) There is no effect of on the motion of gas molecules.
 - (ii) If the density of a gas at 27 C and 1 atmospheric pressure is 1.8 g lit⁻¹, its molecular mass is
 - (iii) If the speed of a molecule at 27 C is 0.25 metre sec⁻¹, its speed at 927 C will be
 - (iv) A steel vessel of capacity 22.4 litre contains 2 g of hydrogen, 8g of oxygen and 22 g of carbon dioxide at a temperature of 0 C. The total pressure of the gas is
8. A gas has a volume of 580 cm³ at a certain pressure. If its pressure is increased by 0.96 atm, its volume becomes 100 cm³. The new pressure of the gas is

MATCH THE COLUMN

1.

Column-I		Column-II	
(A)	$P_1 V_1 = P_2 V_2 = P_3 V_3 = \dots\dots\dots$	(p)	Dalton's law of partial pressures at constant temperature.
(B)	$\frac{V_1}{T_1} = \frac{V_2}{T_2} = \frac{V_3}{T_3} = \dots\dots\dots$ at constant pressure.	(q)	Kinetic equation of ideal gases.
(C)	$r \propto \sqrt{\frac{1}{d}}$	(r)	22.4 litre
(D)	$P = P_1 + P_2 + P_3 + \dots\dots\dots$	(s)	Isotherm
(E)	$(V - b) \left(P + \frac{a}{V^2} \right) = RT$	(t)	Isobar
(F)	R/N	(u)	Charles' law
(G)	Molar volume	(v)	Graham's law
(H)	$PV = \frac{1}{3} mnc^2$	(w)	Boyle's law
(I)	Graph between P and V at constant temperature.	(x)	Equation for real gases.
(J)	Graph between V and T at constant pressure.	(y)	Boltzmann's constant

ASSERTION & REASON

These questions contains, Statement I (assertion) and Statement II (reason).

(A) Statement-I is true, Statement-II is true ; Statement-II is correct explanation for Statement-I.

(B) Statement-I is true, Statement-II is true ; Statement-II is NOT a correct explanation for statement-I

(C) Statement-I is true, Statement-II is false

(D) Statement-I is false, Statement-II is true

1. **Statement-I** : Plot of P Vs. $1/V$ (volume) is a straight line.

Because

Statement-II : Pressure is directly proportional to volume.

2. **Statement-I** : Absolute zero is a theoretically possible temperature at which the volume of the gas becomes zero.

Because

Statement-II : The total kinetic energy of the molecule is zero at this temperature.

3. **Statement-I** : Pressure exerted by a mixture of reacting gases is equal to the sum of their partial pressures.

Because

Statement-II : Reacting gases react to form a new gas having pressure equal to the sum of both.

4. **Statement-I** : Critical temperature of the gas is the temperature at which it occupies 22.4 L of volume.

Because

Statement-II : Molar volume of every gas at NTP is 22.4 L.

5. **Statement-I** : Excluded volume or co-volume equals to $(v-nb)$ for n moles.

Because

Statement-II : Co-volume depends on the effective size of gas molecules.

6. **Statement-I** : Carbondioxide has greater value of root mean square velocity u_{rms} than carbon monoxide.

Because

Statement-II : u_{rms} is directly proportional to molar mass.

7. **Statement-I** : Most probable velocity is the velocity possessed by maximum fraction of molecules at the same temperature.
Because
Statement-II : On collision, more and more molecules acquire higher speed at the same temperature.
8. **Statement-I** : The effusion rate of oxygen is smaller than that of nitrogen.
Because
Statement-II : Molecular size of nitrogen is smaller than that of oxygen.

COMPREHENSION BASED QUESTIONS

Comprehension # 1

The rate of change of pressure (p) of a gas at constant temperature and constant external pressure due to effusion of gas from a vessel of constant volume is related to rate of change of number of molecules present by

$$\frac{dp}{dt} = \frac{kT}{V} \frac{dN}{dt}$$

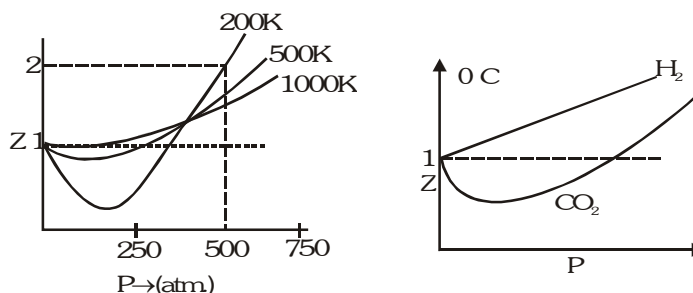
where k = Boltzmann constant, T = temperature, V = volume of vessel & N = No. of molecules and

$$\frac{dN}{dt} = \frac{-pA_0}{(2\pi mkT)^{1/2}}, \text{ where } A_0 = \text{area of orifice and } m = \text{mass of molecule.}$$

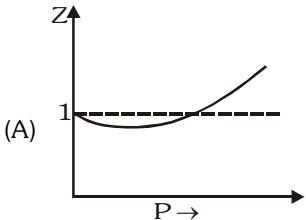
- Time required for pressure inside vessel to reduce to $1/e$ of its initial value in ($\ln e = 1$)
 (A) $\left(\frac{2\pi m}{kT}\right)^{1/2} \frac{V}{A_0}$ (B) $\left(\frac{kT}{2\pi m}\right)^{1/2} \frac{V}{A_0}$ (C) $\left(\frac{2\pi mkT}{A_0}\right)^{1/2}$ (D) $\frac{2\pi m}{kT} \frac{V}{A_0}$
- If the gas inside the vessel had molecular weight 9 times the gas in previous example and area of orifice was doubled and temperature maintained at $4T$, time required for pressure to fall to $1/e$ times of its initial value would be (t = answer of previous option)
 (A) $1.33 t$ (B) $4.24 t$ (C) $0.75 t$ (D) $1.125 t$
- The incorrect statement (s) is/are.
 [I] Pressure will not fall to zero in infinite time.
 [II] Time required for pressure to decrease to half its initial value is independent of initial pressure.
 [III] The relations given above are true for real gases also.
 (A) I (B) II (C) III (D) I and III

Comprehension # 2

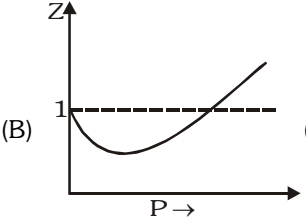
Sketch shows the plot of Z v/s P for a hypothetical gas for one mole at three distinct temperature.



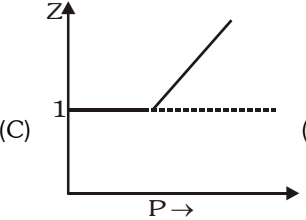
Boyle's temperature is the temperature at which a gas shows ideal behaviour over a pressure range in the low pressure region. Boyle's temperature (T_b) = $\frac{a}{Rb}$. If a plot is obtained at temperatures well below Boyle's temperature then the curve will show negative deviation, in low pressure region and positive deviation in the high pressure region. Near critical temperature the curve is more likely as CO_2 and the temperature well above critical temperature curve is more like H_2 at 0°C as shown above. At high pressure suppose all the constant temperature curve varies linearly with pressure according to the following equation $Z = 1 + \frac{Pb}{RT}$ ($R = 2 \text{ cal mol}^{-1} \text{ K}^{-1}$)

1. Which of the following is correct :
- (A) $\frac{a}{b} < 0.4 \text{ k cal mol}^{-1}$ (B) $0.4 \text{ k cal mol}^{-1} < \frac{a}{b} < 2 \text{ k cal mol}^{-1}$
 (C) $\frac{a}{b} > 0.4 \text{ k cal mol}^{-1}$ (D) $\frac{a}{b} = 1 \text{ K cal mol}^{-1}$
2. For 500 K plot value of Z changes from 2 to 2.2 if pressure is varied from 1000 atm to 1200 atm (high pressure) then the value of $\frac{b}{RT}$ will be :-
 (A) 10^{-3} atm^{-1} (B) $2 \cdot 10^{-3} \text{ atm}^{-1}$ (C) $5 \cdot 10^{-4} \text{ atm}^{-1}$ (D) 10^{-4} atm^{-1}
3. As shown in the figure at 200 K and 500 atm value of compressibility factor is 2 (approx). Then volume of the gas at this point will be :-
 (A) 0.01 L (B) 0.09 L (C) 0.065 L (D) 0.657 L
4. Plot at Boyle's temperature for the gas will be :
- 

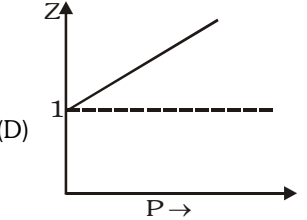
(A)



(B)



(C)



(D)
5. In very high pressure region if Z v/s P is plotted at 1200 K for the above gas then it will have greatest slope.
 (A) true (B) false
 (C) can't say (D) not related to the paragraph

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
True / False 1. F 2. F 3. F 4. T 5. F 6. F 7. T Fill in the Blanks 1. 0.25 2. decreases 3. $\text{Pa m}^6 \text{ mol}^{-2}$ 4. 26 5. One half 6. (i) inversely, square root of density, (ii) 900, (iii) $2/3$ 7. w (i) gravity (ii) 44.33 (iii) 0.5 m/s (iv) 1.75 atm 8. 1.16 atm Match the Column 1. A - (w), B - (u), C - (v), D - (p), E - (x), F - (y), G - (r), H - (q), I - (s), J - (t) Assertion - Reason Questions 1. C 2. B 3. E 4. D 5. D 6. E 7. C 8. C Comprehension Based Questions Comprehension #1 : 1. (A) 2. (C) 3. (C) Comprehension #2 : 1. (B) 2. (A) 3. (C) 4. (C) 5. (B)		

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

1. 3.6 g of an ideal gas was injected into a bulb of internal volume of 8 L at pressure P atm and temp T K. The bulb was then placed in a thermostat maintained at $(T + 15)^\circ\text{K}$. 0.6 g of the gas was let off to keep the original pressure. Find P and T if mol weight of is 44.
2. A toy balloon originally held 1.0 g of He gas and had a radius 10 cm. During the night, 0.25 g of the gas effused from the balloon. Assuming ideal gas behaviour, under these constant P and T conditions, what was the radius of the balloon the next morning.
3. If a scuba diver is to remain submerged for 1 hr, what pressure must be applied to force sufficient air into the tank to be used. Assume 0.5 dm^3 of air per breath at standard atmospheric pressure, a respiration rate of 38 breaths per minute, and a tank capacity of 30 dm^3 .
4. While resting, the average human male use 0.2 dm^3 of O_2 per hour at STP for each kg of body mass. Assume that all this O_2 is used to produce energy by oxidising glucose in the body. What is the mass of glucose required per hour by a resting male having mass 60 kg. What volume, at STP of CO_2 would be produced.
5. 12 g N_2 , 4 g H_2 and 9 g O_2 are put into a one litre container at 27°C . What is the total pressure.
6. $1.0 \times 10^{-2}\text{ kg}$ of hydrogen and $6.4 \times 10^{-2}\text{ kg}$ of oxygen are contained in a $10 \times 10^{-3}\text{ m}^3$ flask at 473 K. Calculate the total pressure of the mixture. If a spark ignites the mixture, What will be the final pressure.
7. At room temp, NH_3 gas at one atm & HCl gas at "P" atm are allowed to effuse through identical pin holes to the opposite ends of a glass tube 1 m long & uniform cross-section. A white deposit is observed at a distance of 60 cm from the HCl end. What is "P".
8. A gas mixture contains equal number of molecules of N_2 and SF_6 , some of it is passed through a gaseous effusion apparatus. Calculate how many molecules of N_2 are present in the product gas for every 100 molecules of SF_6 .
9. Two gases NO and O_2 were introduced at the two ends of a one metre long tube simultaneously (tube of uniform cross-section). At what distance from NO gas end, Brown fumes will be seen.
10. At 20°C two balloons of equal volume and porosity are filled to a pressure of 2 atm, one with 14 kg N_2 & other with 1 kg H_2 . The N_2 balloon leaks to a pressure of $\frac{1}{2}$ atm in one hour. How long will it take for H_2 balloon to leaks to a pressure of $\frac{1}{2}$ atm.
11. Pure O_2 diffuses through an aperture in 224 sec, where as mixture of O_2 and another gas containing 80% O_2 diffuses from the same in 234 sec. What is molecular weight of the gas.
12. A space capsule is filled with neon gas 1.00 atm and 290 K. The gas effuses through a pin - hole into outer space at such a rate that the pressure drops by 0.3 torr/sec.
 - (a) If the capsule were filled with ammonia at the same temperature and pressure, what would be the rate of pressure drop.
 - (b) If the capsule were filled with 30.0 mol % helium, 20.0 mol% oxygen & 50.0 mol% nitrogen at a total pressure of 1.00 atm & a temp of 290 K, what would be the corresponding rate of pressure drop.
13. Show that the height at which the atmospheric pressure is reduced to half its value is given by $h = \frac{0.693RT}{Mg}$.
14. Calculate the pressure of a barometer on an aeroplane which is at an altitude of 10 Km. Assume the pressure to be 101.325 kPa at sea level & the mean temperature 243 K. Use the average molar mass of air (80% N_2 , 20% O_2)

15. An iron cylinder contains helium at a pressure of 250 kPa and 27 C. The cylinder can withstand a pressure of 1×10^6 Pa. The room in which cylinder is placed catches fire. Predict whether the cylinder will blow up before it melts or not. [melting point of cylinder = 1800 K]
16. Determine the molar mass of a gas if its pressure is to fall to one-half of its value in a vertical distance of one meter at 298 K.
17. The time taken for a given volume of gas E to effuse through a hole is 75 sec. Under identical conditions the same volume of a mix of CO & N₂ (containing 40% of N₂ by volume) effused in 70 sec. Calculate
 - (i) the relative mol mass of E, and
 - (ii) the RMS velocity (in ms⁻¹ units) of E at 0 C.
18. At what temperature in C, the U_{rms} of SO₂ is equal to the average velocity of O₂ at 27 C.
19. The density of CO at 273 K and 1 atm is 1.2504 kg m⁻³. Calculate (a) root mean square speed (b) the average speed and (c) most probable speed.
20. Calculate the temperature values at which the molecules of the first two members of the homologous series C_nH_{2n+2} will have the same rms speed as CO₂ gas at 770 K. The normal b.p. of n-butane is 273 K. Assuming ideal gas behaviour of n-butane upto this temperature, calculate the mean velocity and the most probable velocity of its molecules at this temperature.
21. Calculate the temperature at which the root mean square velocity, average velocity and most probable velocity of oxygen gas are all equal to 1500 ms⁻¹.
22. Calculate the fraction of N₂ molecules at 101.325 kPa and 300 K whose speeds are in the range of $u_{mp} - 0.005 u_{mp}$ to $u_{mp} + 0.005 u_{mp}$.
23. What is the ratio of the number of molecules having speeds in the range of $2u_{mp}$ and $2u_{mp} + du$ to the number of molecules having speeds in the range of u_{mp} and $u_{mp} + du$?
24. The density of mercury is 13.6 g/cm³. Estimate the b value.
25. Calculate the pressure exerted by 22 g of carbon dioxide in 0.5 dm³ at 298. 15 K using :
 - (a) the ideal gas law and
 - (b) vander waals equation. Given :

$$[a = 363.76 \text{ kPa dm}^6 \text{ mol}^{-2} \quad \text{and} \quad b = 42.67 \text{ cm}^3 \text{ mol}^{-1}]$$

26. The compressibility factor for N₂ at -50 C and 800 atm pressure is 1.95 and at 100 C and 200 atm, it is 1.10. A certain mass of nitrogen occupied one litre at -50 C and 800 atm. Calculate the volume occupied by the same quantity of N₂ at 100 C and 200 atm.
27. At 273.15 K and under a pressure of 10.1325 MPa, the compressibility factor of O₂ is 0.927. Calculate the mass of O₂ necessary to fill a gas cylinder of 100 dm³ capacity under the given conditions.
28. The vander waals constant for O₂ are $a = 1.36 \text{ atm L}^2 \text{ mol}^{-2}$ and $b = 0.0318 \text{ L mol}^{-1}$. Calculate the temperature at which O₂ gas behaves, ideally for longer range of pressure.
29. The vander waals constants for gases A,B and C are as follows :

Gas	$a/\text{dm}^6 \text{ kPa mol}^{-2}$	$b/\text{dm}^3 \text{ mol}^{-1}$
A	405.3	0.027
B	1215.9	0.030
C	607.95	0.032

Which gas has (i) the highest critical temperature , (ii) the largest molecular volume, and (iii) most ideal behaviour around STP?

30. A commercial cylinder contains 6.91 m³ of O₂ at 15.18 MPa and 21 C. The critical constants for O₂ are T_c = -118.4 C, P_c = 50.1 atm. Determine the reduced pressure and reduced temperature for O₂ under these conditions.

31. Show that at low densities, the vander waals equation

$$\left(p + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

and the Dieterici's equation $P(V_m - b) = RT \exp(-a/RTV_m)$ given essential the same value of p .

32. Calculate from the vander waal's equation, the temperature at which 192 g of SO_2 would occupy a vol. of 10 dm^3 at 15 atm pressure. [$a = 6.7 \text{ atm L}^2 \text{ mol}^{-2}$, $b = 0.0564 \text{ L mol}^{-1}$]
33. Calculate the pressure of 15 mol neon at 30 C in a 12 litre container using
- the ideal gas equation
 - the vander waals equation
- [$a = 0.2107 \text{ atm L}^2 \text{ mol}^{-2}$, $b = 0.0171 \text{ L mol}^{-1}$]
34. What will be the temperature difference needed in a hot air balloon to lift 1.0 kg of mass? Assume that the volume of balloons is 100 m^3 , the temperature of ambient air is 25 C, the pressure is 1 bar, and air is an ideal gas with an average molar mass of 29 g mol^{-1} (hot and cold both).
35. One mole of a non linear triatomic gas is heated in a closed rigid container from 500 C to 1500 C. Calculate the amount of energy required if vibrational degree of freedom become effective only above 1000 C.

CONCEPTUAL	SUBJECTIVE	EXERCISE	ANSWER KEY	EXERCISE-4(A)
1. $P = 0.062 \text{ atm}$, $T = 75 \text{ K}$	2. 9.08 cm	3. $3.8 \times 10^3 \text{ kPa}$		
4. 16.07 g , 12 dm^3	5. 66.74 atm			
6. $P_{\text{total}} = 27.54 \times 10^5 \text{ N/m}^2$, $P_{\text{final}} = 19.66 \times 10^5 \text{ N/m}^2$	7. 2.19 atm			
8. 228	9. 50.8 cm	10. 16 min		
11. 46.6	12. (a) 0.32 Torr/sec	(b) 0.29 Torr/sec		
13.	14. 25.027 kPa	15. Yes		
16. $175.133 \text{ kg mol}^{-1}$	17. 32.14 g/mol , 460.28 m/s			
18. 236 C	19. $U_{\text{RMS}} = 493 \text{ m/s}$, $U_{\text{mp}} = 403 \text{ m/s}$, $U_{\text{av}} = 454.4 \text{ m/s}$			
20. 280 K , 525 K , $3.157 \times 10^2 \text{ m/s}$, $2.798 \times 10^2 \text{ m/s}$				
21. $R_{\text{RMS}} = 2886 \text{ K}$, $T_{\text{av}} = 3399 \text{ K}$, $T_{\text{mp}} = 4330 \text{ K}$	22. 8.303×10^{-3}			
23. 0.199	24. 58.997 cm^3			
25. (a) $2.479 \times 10^3 \text{ kPa}$ (b) 2225.55 kPa	26. 3.77 L			
27. 15.40 kg	28. 521 K	29. (i) B (ii) C (iii) A		
30. $P_r = 2.99$, $T_r = 1.90$	32. 350.5 C	33. (i) 31.1 atm (ii) 31.4 atm		
34. 2.53 C	35. 4500 RJ			

EXERCISE-04 [B]**BRAIN STORMING SUBJECTIVE EXERCISE**

1. The respiration of a suspension of yeast cells was measured by determining the decrease in pressure of the gas above the cell suspension. The apparatus was arranged so that the gas was confined to a constant volume, 16 cm^3 and the entire pressure change was caused by uptake of oxygen by the cells. The pressure was measured in a monometer, the fluid of which had density of 1.034 g/cm^3 . The entire apparatus was immersed in a thermostat at 37°C . In a 30 minute observation period the fluid in the open side of the manometer dropped 37 mm. Neglecting the solubility of oxygen in the yeast suspension, compute the rate oxygen consumption by the cells in mm^3 of O_2 (STP) per hour.
2. In a basal metabolism measurement timed at 6 minutes, a patient exhaled 52.5 L of air measured over water at 20°C . The vapour pressure of water at 20°C is 17.5 torr. The barometric pressure was 750 torr. The exhaled air analyzed 16.75 volume % oxygen and the inhaled air 20.32 volume % oxygen. Both on a dry basis neglecting any solubility of the gases in water and any difference in the total volumes of inhaled and exhaled air, calculate the rate of oxygen consumption by the patient in mL (STP) per minute.
3. The temperature and the relative humidity of air are 20°C and 80% on a certain day. Find the fraction of the mass of water vapour that will condense if the temperature falls to 5°C . Saturation vapour pressures at 20°C and 5°C are 17.5 mm and 6.5 mm of Hg respectively.
4. 6.0 g of He having average velocity $4 \times 10^2 \text{ ms}^{-1}$ is mixed with 12.0 g of Ne^{20} having the same average velocity. What is the average kinetic energy per mole in the mixture?
5. Molar volume of He at 10.1325 MPa and 273 K is 0.011075 times its molar volume at 101.325 kPa. Calculate radius of He atom assuming negligible 'a'.
6. The Virial equation for ethane gas is given by $PV = RT + BP$. At 0°C , $B = -0.1814 \text{ L/mol}$. Calculate volume of one mole of ethane at 10 atm, and 'a'.
7. Pressure of He gas confined in a steel chamber drops from 4.0 to 1.0 atmosphere in 4.0 hours due to diffusion through a pin-hole in the steel chamber. If an equimolar mixture of He and methane gas at 20 atmosphere and the same temperature are confined in the same chamber, what will be the partial pressure of He and methane after 1.0 hour. Assume rate of diffusion to be linear function of gas pressure and inverse function of square root of molar masses.
8. A one litre flask containing NH_3 (g) at 2.0 atmosphere and 300 K is connected to another 800 mL flask containing HCl (g) at 8.0 atmosphere and 300 K by means of a narrow tube of negligible volume and gases were allowed to react quantitatively as :
$$\text{NH}_3 (\text{g}) + \text{HCl} (\text{g}) \longrightarrow \text{NH}_4\text{Cl} (\text{s}) ; \Delta H = - 43 \text{ kJ/mol}$$
If heat capacity of HCl (g) C_v is $20 \text{ JK}^{-1} \text{ mol}^{-1}$, determine final pressure inside the flask assuming negligible heat capacity of flask and negligible volume of solid NH_4Cl .
9. Calculate the value of σ , λ , Z_1 and Z_{11} for nitrogen molecules at 25°C and at pressure of 10^{-3} mm Hg . Given that b for nitrogen is $39.1 \text{ cm}^3 \text{ mol}^{-1}$.
10. The mean free path of the molecule of a certain gas at 300 K is $2.6 \times 10^{-5} \text{ m}$. The collision diameter of the molecule is 0.26 nm. Calculate
 - (a) pressure of the gas, and
 - (b) number of molecules per unit volume of the gas.
11. There are two vessels of same consisting same no of moles of two different gases at same temperature. One of the gas is CH_4 & the other is unknown X. Assuming that all the molecules of X are under random motion whereas in CH_4 except one all are stationary. Calculate Z_1 for X in terms of Z_1 of CH_4 . Given that the collision diameter for both gases are same & $(U_{\text{rms}})_x = \frac{1}{\sqrt{6}} (U_{\text{av}})_{\text{CH}_4}$.

12. A mixture of CH_4 & O_2 is used as an optimal fuel if O_2 is present in thrice the amount required theoretically for combustion of CH_4 . Calculate number of effusions steps required to convert a mixture containing 1 part of CH_4 in 193 parts mixture (parts by volume). If calorific value (heat evolved when 1 mole is burnt) of CH_4 is 100 cal/mole & if after each effusion 90% of CH_4 is collected. find out what initial mole of each gas in initial mixture required for producing 1000 cal of energy after processing.

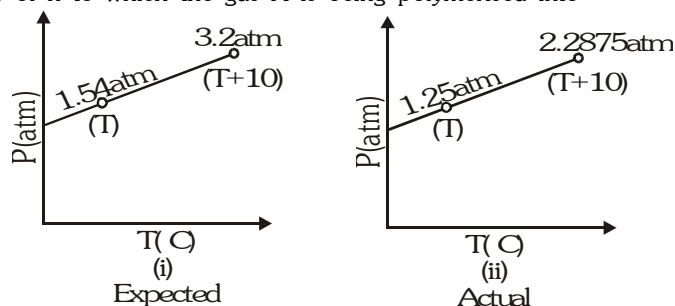
[Given $(0.9)^5 = 0.6$]

13. A closed vessel of known volume containing known amount of ideal gaseous substance 'A' was observed for variation of pressure with temperature. The expected graph was to be like as in (i) However actual observations revealed the graph to be like. (ii) The deviation was attributed to polymerisation of gas molecules as $n\text{A}(\text{g}) \rightleftharpoons \text{A}_n(\text{g})$. If it is known that the above reaction given only 50% yield.

(a) Calculate the ratio of $\frac{n_{\text{experiment}}}{n_{\text{theoretical}}}$ (where $n_{\text{exp.}}$ = Total no. of gaseous mole actually present

$n_{\text{theoretical}}$ = Total no. of mole original taken)

(b) Find the value of n to which the gas A is being polymerised into



14. During one of his adventure, Chacha choudhary got trapped in an underground cave which was sealed two hundred year back. The air inside the cave was poisonous, having some amount of carbon monoxide in addition to O_2 and N_2 . Sabu, being huge could not enter into the cave, so in order to save chacha choudhary he started sucking the poisonous air out of the cave by mouth. Each time, he filled his lunge with cave air and exhaled it out in the surroundings. In the mean time fresh air from surrounding effused into the cave till the pressure was again one atmosphere. Each time sabu sucked out some air, the pressure in the cave dropped to half of its initial value of one atmosphere.

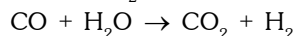
If the initial sample of air from the cave contain 5% by volume CO.

If the safe level of CO in the atmosphere is less than 0.001% by volume how many times does Sabu need to such out air in order to save Chacha choudhary.

15. A closed vertical cylinder is divided into two parts by a frictionless piston, each part contains 1 mole of air. At 27°C the volume of the upper part is 4 times than that of the lower part. Calculate the temperature when volume of the upper part will be three times that of the lower part.

16. A water gas mixture has the composition by volume of 50% H_2 , 40% CO and 5% CO_2 .

(i) Calculate the volume in litres at STP of the mixture which on treatment with excess steam will contain 5 litres of H_2 . The stoichiometry for the water gas shift reaction is



(ii) Find the density of the water gas mixture in kg/m^3 .

(iii) Calculate the moles of the absorbants KOH, $\text{Ca}(\text{OH})_2$ and ethanolamine.

$\text{HO} - \text{CH}_2 - \text{CH}_2 - \text{NH}_2$ required respectively to collect the CO_2 gas obtained.

17. A gas present in a container connected to frictionless, weightless piston operating always at one atmosphere pressure such that it permits flow of gas outside (with no adding of gas). The graph of n vs T (Kelvin) was plotted & was found to be a straight line with co-ordinates of extremen points as (300, 2) & (200, 3). Calculate :

(i) relationship between n & T

(ii) relationship between V & T

(iii) Maxima or minima value of ' V '

18. Find the critical constant (P_c , V_c and T_c) in terms of A and B, also find compressibility factor (z) for the following equation of state.

$$PV = RT - \frac{A}{V} + \frac{2B}{V^2}$$

where A and B are constant, P = pressure and V = molar volume.

19. Calculate the volume occupied by 14.0 g N_2 at 200 K and 8.21 atm pressure if $\frac{P_c V_c}{RT_c} = \frac{3}{8}$ and $\frac{P_r V_r}{T_r} = 2.2$.

BRAIN STORMING SUBJECTIVE EXERCISE		ANSWER KEY		EXERCISE-4(B)
1. 100 mm ³ /hr	2. 280 mL/min	3. 0.51	4. 807.84 J	
5. 134 pm	6. 0.918, a = 3.77 bar L ² mol ⁻²	7. $P_{He} = 7.07$ atm, $P_{CH_4} = 8.4$ atm		
8. 10.3 atmosphere	9. 314 pm, 7.015 cm, 6742 s ⁻¹ , 1.09 10 ¹⁷ cm ⁻³ s ⁻¹			
10. (a) 1.281 10 ²³ m ⁻³	(b) 5.306 10 ² Pa	11. $\frac{2\sqrt{2}}{3\sqrt{\pi}} Z_1$		
12. 10 Steps, 27.78 mol CH ₄ , 5333.3 mol O ₂		13. (a) 0.625, (b) 4	14. 13	
15. 421.9 K				
16. (i) 5.263 L, (ii) 0.7kg/m ³ , (iii) KOH = 0.2348 moles, Ca(OH) ₂ = 0.1174 moles, ethanolamine = 0.2348 moles				
17. $n = \frac{-T}{100} + 5$, $V = \frac{-RT^2}{100} + 5RT$, 51.3125 ℓ				
18. $V_c = \frac{6B}{A}$, $T_c = \frac{A^2}{6RB}$, $P_c = \frac{A^3}{108B^2}$, compressibility factor = $\frac{P_c V_c}{RT_c} = \frac{1}{3}$			19. 0.825 L	

EXERCISE-05 (A)**PREVIOUS YEARS QUESTIONS**

1. The no. of moles per litre in the equation $PV = nRT$ is expressed by : [AIEEE-02]
(1) $\frac{P}{RT}$ (2) $\frac{PV}{RT}$ (3) $\frac{RT}{PV}$ (4) None
2. The correct value of R is - [AIEEE-02]
(1) $R = 0.082$ litre atm (2) $R = 8.314 \times 10^7$ erg $^{-1}K^{-1} \text{ mol}^{-1}$
(3) $R = 2K^{-1} \text{ mol}^{-1}$ (4) None
3. According to the kinetic theory of gases, in an ideal gas, between two successive collisions a gas molecule travels - [AIEEE-03]
(1) In a straight line path (2) with an accelerated velocity
(3) In a circular path (4) In a wavy path
4. What volume of hydrogen gas, at 273K and 1 atm, pressure will be consumed in obtaining 21.6g of elemental boron (atomic mass = 10.8) from the reduction of boron trichloride by hydrogen ? [AIEEE-03]
(1) 44.8 L (2) 22.4 L (3) 89.6 L (4) 67.2 L
5. As the temperature is raised from 20 C to 40 C, the average kinetic energy of neon atoms changes by factor of which of the following ? [AIEEE-04]
(1) $1/2$ (2) $\sqrt{(313/293)}$ (3) $313/293$ (4) 2
6. In vanderwaals equation of state of the gas law, the constant 'b' is a measure of : [AIEEE-04]
(1) intermolecular repulsions (2) intermolecular attraction
(3) volume occupied by the molecules (4) intermolecular collisions per unit volume
7. An ideal gas expands in volume from $1 \times 10^{-3} \text{ m}^3$ to $1 \times 10^{-2} \text{ m}^3$ at 300 K against a constant pressure of $1 \times 10^5 \text{ Nm}^{-2}$. The work done is - [AIEEE-04]
(1) - 900 J (2) - 900 K (3) 2710 kJ (4) 900 kJ
8. For which of the following parameters the structural isomers $\text{C}_2\text{H}_5\text{OH}$ and CH_3OCH_3 would be expected to have the same values ? (Assume ideal behavior) [AIEEE-05]
(1) Heat of vaporization
(2) vapour pressure at the same temperature
(3) Boiling points
(4) Gaseous densities at the same temperature and pressure
9. Equal masses of methane and oxygen are mixed in an empty container at 25 C. The fraction of the total pressure exerted by oxygen is - [AIEEE-06]
(1) $2/3$ (2) $\frac{1}{3} \times \frac{273}{298}$ (3) $\frac{1}{3}$ (4) $\frac{1}{2}$

PREVIOUS YEAR QUESTIONS**ANSWER KEY****EXERCISE-05(A)**

Que.	1	2	3	4	5	6	7	8	9
Ans	1	2	1	4	3	3	1	4	3

EXERCISE-05 (B)**PREVIOUS YEARS QUESTIONS**

1. A mixture of ideal gases is cooled upto liquid He temperature (4.22 K) to form an ideal solution. Is this statement **true** or **false**. Justify your answer in not more than two lines. [JEE 1996]
2. The ratio between the r.m.s. velocity of H_2 at 50 K and that of O_2 at 800 K is. [JEE 1996]
(A) 4 (B) 2 (C) 1 (D) $1/4$
3. X ml H_2 gas effuses through a hole in a container in 5 sec. The time taken for the effusion of the same volume of the gas specified below under identical conditions is [JEE 1996]
(A) 10 sec, He (B) 20 sec, O_2 (C) 25 sec, CO (D) 55 sec, CO_2
4. One mole of N_2O_4 (g) at 300 K is kept in a closed container under one atm. It is heated to 600 K when 20% by mass of N_2O_4 (g) decomposes to NO_2 (g). The resultant pressure is. [JEE 1996]
(A) 1.2 atm (B) 2.4 atm (C) 2.0 atm (D) 1.0 atm
5. The absolute temperature of an ideal gas is to/than the average kinetic energy of the gas molecules. [JEE 1997]
6. One way of writing the equation of state for real gases is,
$$P\bar{V} = RT \left[1 + \frac{B}{V} + \dots \right] \text{ where B is constant.}$$

Derive an approximate expression for 'B' in terms of Vander Waals constant 'a' & 'b'. [JEE 1997]
7. Calculate the total pressure in a 10 litre cylinder which contains 0.4 g He, 1.6 g oxygen and 1.4 g nitrogen at 27 C. Also calculate the partial pressure of He gas in the cylinder. Assume ideal behaviour for gases. [JEE 1997]
8. According to Graham's law, at a given temperature the ratio of the rates of diffusion $\frac{r_A}{r_B}$ of gases A and B is given by. [JEE 1997]
(A) $\frac{P_A}{P_B} \left(\frac{M_A}{M_B} \right)^{1/2}$ (B) $\left(\frac{M_A}{M_B} \right) \left(\frac{P_A}{P_B} \right)^{1/2}$ (C) $\frac{P_A}{P_B} \left(\frac{M_B}{M_A} \right)^{1/2}$ (D) $\frac{M_A}{M_B} \left(\frac{P_B}{P_A} \right)^{1/2}$
9. An evacuated glass vessel weighs 50.0 g when empty, 148.0 g when filled with a liquid of density 0.98 g /mL and 50.5 g when filled with an ideal gas at 760 mm Hg at 300 K. Determine the molecular weight of the gas. [JEE 1998]
10. Using Vander Waals equation, calculate the constant "a" when 2 moles of a gas confined in a 4 litre flask exerts a pressure of 11.0 atm at a temperature of 300 K. The value of "b" is 0.05 litre mol^{-1} . [JEE 1998]
11. The pressure exerted by 12 g of an ideal gas at temperature t C in a vessel of volume V is one atm. When the temperature is increased by 10 degrees at the same volume, the pressure increases by 10%. Calculate the temperature 't' and volume 'V'. [molecular weight of gas = 120] [JEE 1999]
12. One mole of N_2 gas at 0.8 atm takes 38 sec to diffuse through a pin hole, whereas one mole of an unknown compound of Xenon with F at 1.6 atm takes 57 sec to diffuse through the same hole. Calculate the molecular formula of the compound. [JEE 1999]
13. A gas will approach ideal behaviour at : [JEE 1999]
(A) low temperature and low pressure (B) low temperature and high pressure
(C) low pressure and high temperature (D) high temperature and high pressure
14. The compressibility of a gas is less than unity at STP. Therefore : [JEE 2000]
(A) $V_m > 22.4$ L (B) $V_m < 22.4$ L (C) $V_m = 22.4$ L (D) $V_m = 44.8$ L

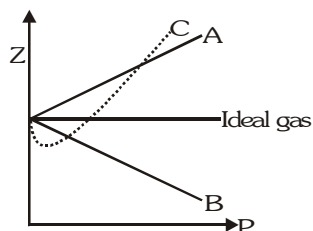
15. The r.m.s. velocity of hydrogen is $\sqrt{7}$ times the r.m.s. velocity of nitrogen. If T is the temperature of the gas : [JEE 2000]
- (A) $T(\text{H}_2) = T(\text{N}_2)$ (B) $T(\text{H}_2) > T(\text{N}_2)$ (C) $T(\text{H}_2) < T(\text{N}_2)$ (D) $T(\text{H}_2) = \sqrt{7} T(\text{N}_2)$
16. The pressure of a fixed amount of an ideal gas is proportional to its temperature. Frequency of collision and their impact both increase in proportion to the square root of temperature. True/False.[JEE 2000]
17. Calculate the pressure exerted by one mole of CO_2 gas at 273 K, if the Vander Waals constant $a = 3.592 \text{ dm}^6 \text{ atm mol}^{-2}$. Assume that the volume occupied by CO_2 molecules is negligible. [JEE 2000]
18. The root mean square velocity of an ideal gas at constant pressure varies with density as : [JEE 2001]
- (A) d^2 (B) d (C) $d^{1/2}$ (D) $1/d^{1/2}$
19. The compression factor (compressibility factor) for one mole of a vander Waals gas at 0 C and 100 atmosphere pressure is found to be 0.5. Assuming that the volume of a gas molecule is negligible, calculate the vander waals constant 'a'. [JEE 2001]
20. Which one of the following V, T plots represents the behaviour of one mole of an ideal gas at one atm? [JEE 2002]
- (A)

(B)

(C)

(D)
21. The density of the vapour of a substance at 1 atm pressure and 500 K is 0.36 Kg m^{-3} . The vapour effuse through a small hole at a rate of 1.33 times faster than oxygen under the same condition.[JEE 2002]
- (a) Determine
- (i) mol. wt. ; (ii) molar volume ; (iii) compressibility factor (z) of the vapour
- (iv) which forces among the gas molecules are dominating the attractive or the repulsive.
- (b) If the vapour behaves ideally at 1000 K, determine the average translational K.E. of a molecule.
22. The average velocity of gas molecules is 400 m/sec. Calculate its (rms) velocity at the same temperature.[JEE 2003]
23. C_v value of He is always $3R/2$ but C_v value of H_2 is $3R/2$ at low temperature and $5R/2$ at moderate temperature and more than $5R/2$ at higher temperature explain in two to three lines. [JEE 2003]
24. Positive deviation from ideal behaviour takes place because of. [JEE 2003]
- (A) molecular interaction between atoms and $\frac{PV}{nRT} > 1$
- (B) molecular interaction between atoms and $\frac{PV}{nRT} < 1$
- (C) finite size of atoms and $\frac{PV}{nRT} > 1$
- (D) finite size of atoms and $\frac{PV}{nRT} < 1$
25. For a real gas obeying vander Waal's equation a graph is plotted between PV_m (y-axis) and P(x-axis) where V_m is molar volume. Find y-intercept of the graph. [JEE 2005]
26. The ratio of the rate of diffusion of helium and methane under identical condition of pressure and temperature will be : [JEE 2005]
- (A) 4 (B) 2 (C) 1 (D) 0.5

27.



$$\text{where } Z = \frac{PV}{nRT},$$

a = Vander Waal's constant for pressure correction

b = Vander Waal's constant for volume correction

Pick the only incorrect statement

(A) for gas A, if $a = 0$, the compressibility factor is directly proportional to pressure

(B) for gas B, if $b = 0$, the compressibility factor is directly proportional to pressure

(C) for gas C, $a \neq 0$, $b \neq 0$, it can be used to calculate a and b by giving lowest P value and its intercept with $Z = 1$.

(D) slope for all three gases at high pressure (not shown in graph) is positive.

[JEE 2006]

PREVIOUS YEARS QUESTIONS		ANSWER KEY		EXERCISE -5(B)
1. Yest it is false statement	2. C	3. B	4. B	
5. directly proportional	6. $B = \left(b - \frac{a}{RT}\right)$	7. 0.492 atm, 0.246 atm		
8. C	9. 123	10. 6.46 atm L ² mol ⁻²	11. -173 C, 0.82 L	
12. XeF ₆	13. C	14. B	15. C	
16. both statements are correct	17. 0.99 atm	18. D		
19. 1.2544 atm L ² mol ⁻²	20. C			
21. (a) (i) 18.1 g/mol (ii) 50.25 L mol ⁻¹ (iii) 1.224 (iv) repulsive	(b) 2.07 10 ⁻²⁰ J	22. 434.17 m/sec		
23. Since H ₂ is diatomic and He is monoatomic degree of freedom for mono is 3 and only translational but for diatomic, vibrational and rotational are also to be considered.				
24. C	25. RT	26. B	27. B	

EXERCISE-01

CHECK YOUR GRASP

SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)

1. A white crystalline solid A on boiling with caustic soda solution gave a gas B which when passed through an alkaline solution of potassium mercuric iodide gave a brown ppt. The substance A on heating gave a gas C which rekindled a glowing splinter but did not give brown fumes with nitric oxide. The gas B is ?
(A) H_2S (B) NH_3 (C) HCl (D) CO_2
2. The gas C (in the above question) is -
(A) N_2O (B) O_2 (C) NO (D) O_3
3. A metal oxide is yellow when hot and white when cold. The metal oxide is -
(A) ZnO (B) CuO (C) PbO (D) Al_2O_3
4. When a salt is heated with dil. H_2SO_4 and KMnO_4 solution, the pink colour of KMnO_4 is discharged, the mixture may contain ?
(A) Sulphite (B) Carbonate (C) Nitrate (D) Bicarbonate
5. Sulphur dioxide may be recognised by its ?
(A) Characteristic pungent smell of burning sulphur
(B) Ability to turn dichromate paper green
(C) Ability to decolourize acidified KMnO_4 solution
(D) All
6. Chromyl chloride vapours are dissolved in water and solution is treated with acetic acid and lead acetate solution is added, then-
(A) The solution will remain colourless (B) The solution will become dark green
(C) A yellow solution will be obtained (D) A yellow precipitate will be obtained
7. In a combination of NO_3^- , Br^- and I^- present in a mixture, Br^- and I^- interfere in the ring test for NO_3^- . These are removed by adding a solution of -
(A) AgNO_3 (B) Ag_2SO_4 (C) Ag_2CO_3 (D) None of these
8. A dark green bead in borax bead test indicates the presence of -
(A) Cr^{3+} (B) Mn^{2+} (C) Co^{2+} (D) Ni^{2+}
9. Lead has been placed in group 1st and 2nd because -
(A) It shows the valency one and two (B) It forms insoluble PbCl_2
(C) It forms lead sulphide (D) Its chloride is partly soluble in water
10. The group reagent for third group is NH_4OH in presence of -
(A) $(\text{NH}_4)_2\text{CO}_3$ (B) NaCl (C) $(\text{NH}_4)_2\text{SO}_4$ (D) NH_4Cl
11. What would you observe if you add with shaking excess of dilute NaOH solution to an aqueous solution of AlCl_3 ?
(A) A permanent white ppt. is formed
(B) No change at first, but a white ppt. is formed on standing
(C) A white ppt. is formed which later dissolves
(D) A green ppt. which turns red on standing in air
12. BaCl_2 solution gives a white ppt. with a solution of an acid radical which dissolves in dil. HCl with the evolution of a colourless, pungent smelling gas. The acid radical may be ?
(A) SO_4^{2-} (B) S^{2-} (C) SO_3^{2-} (D) CO_3^{2-}

13. $K_3Co(NO_2)_6$ is known as -
 (A) Fischer's salt (B) Thenard's blue (C) Rinman's green (D) Blue vitriol
14. In the precipitation of the iron group in qualitative analysis, ammonium chloride is added before adding ammonium hydroxide to -
 (A) Decrease concentration of OH^- ions (B) Prevent interference by phosphate ions
 (C) Increase concentration of Cl^- ions (D) Increase concentration of NH_4^+ ions
15. Potassium ferrocyanide is used in the detection of -
 (A) Cu^{2+} ion (B) Fe^{3+} ions (C) Both (D) None
16. The acidic solution of a salt produced a deep blue colour with starch iodide solution. The salt may be -
 (A) Chloride (B) Nitrite (C) Acetate (D) Bromide
17. Which of the following pairs of ions would be expected to form precipitate when dilute solutions are mixed?
 (A) Na^+ , SO_4^{2-} (B) NH_4^+ , CO_3^{2-} (C) Na^+ , S_2^{2-} (D) Fe^{3+} , PO_4^{3-}
18. Nessler's reagent is ?
 (A) K_2HgI_4 (B) $K_2HgI_4 + KOH$ (C) $K_2HgI_2 + KOH$ (D) $K_2HgI_4 + KI$
19. When bismuth chloride is poured into a large volume of water the white precipitate produced is ?
 (A) $Bi(OH)_3$ (B) Bi_2O_3 (C) $BiOCl$ (D) Bi_2OCl_3
20. $Fe(OH)_3$ can be separated from $Al(OH)_3$ by addition of ?
 (A) Dil. HCl (B) NaCl solution
 (C) NaOH solution (D) NH_4Cl and NH_4OH
21. Mark the compound which turns black with NH_4OH ?
 (A) Lead chloride (B) Mercurous chloride (C) Mercuric chloride (D) Silver chloride
22. If NaOH is added to an aqueous solution of zinc ions a white precipitate appears and on adding excess NaOH, the precipitate ?
 (A) Cationic part (B) Anionic part
 (C) Both in cationic and anionic parts (D) There is no zinc ion in the solution.
23. A substance on treatment with dil. H_2SO_4 liberates a colourless gas which produces (i) turbidity with baryta water and (ii) turns acidified dichromate solution green. The reaction indicates the presence of ?
 (A) CO_3^{2-} (B) S^{2-} (C) SO_3^{2-} (D) NO_2^-
24. An aqueous solution of colourless metal sulphate M, gives a white ppt. with NH_4OH . This was soluble in excess of NH_4OH . On passing H_2S through this solution a white ppt. is formed. The metal M in the salt is ?
 (A) Ca (B) Ba (C) Al (D) Zn
25. The salt used for performing 'bead' test in qualitative inorganic analysis is ?
 (A) $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24 H_2O$ (B) $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$
 (C) $Na(NH_4)HPO_4 \cdot 4H_2O$ (D) $CaSO_4 \cdot 2H_2O$
26. A chloride dissolves appreciably in cold water. When placed on a platinum wire in Bunsen flame no distinctive colour is noticed, the cation would be ?
 (A) Mg^{2+} (B) Ba^{2+} (C) Pb^{2+} (D) Ca^{2+}
27. Which is not dissolved by dil. HCl ?
 (A) ZnS (B) MnS (C) $BaSO_3$ (D) $BaSO_4$

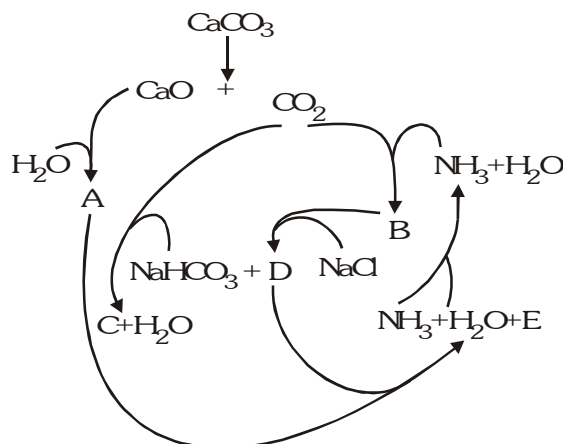
28. The brown ring test for NO_2^- and NO_3^- is due to the formation of complex ion with formula -
 (A) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ (B) $[\text{Fe}(\text{NO})(\text{CN})_5]^{2-}$
 (C) $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{2+}$ (D) $[\text{Fe}(\text{H}_2\text{O})(\text{NO})_5]^{2+}$
29. Which of the following metal sulphides has maximum solubility in water -
 (A) $\text{HgS } K_{\text{sp}} = 10^{-54}$ (B) $\text{CdS } K_{\text{sp}} = 10^{-30}$
 (C) $\text{FeS } K_{\text{sp}} = 10^{-20}$ (D) $\text{ZnS } K_{\text{sp}} = 10^{-22}$
30. Which one of the following statement is correct -
 (A) From a mixed precipitate of AgCl and AgI , ammonia solution dissolves only AgCl
 (B) Ferric ions give a deep green precipitate on adding potassium ferrocyanide solution
 (C) On boiling a solution having K^+ , Ca^{2+} and HCO_3^- ions we get a precipitate of $\text{K}_2\text{Ca}(\text{CO}_3)_2$
 (D) Manganese salts give a violet borax bead test in the reducing flame
31. When H_2S is passed through Hg_2^{2+} , we get -
 (A) HgS (B) $\text{HgS} + \text{Hg}_2\text{S}$ (C) $\text{HgS} + \text{Hg}$ (D) Hg_2S
32. Potassium chromate solution is added to an aqueous solution of a metal chloride. The precipitate thus obtained are insoluble in acetic acid. These are subjected to flame test, the colour of the flame is -
 (A) Lilac (B) Apple green (C) Crinison red (D) Golden yellow
33. Sometimes yellow turbidity appears while passing H_2S gas even in the absence of II group radicals. This is because of -
 (A) Sulphur is present in the mixture as impurity
 (B) IV group radicals are precipitated as sulphides
 (C) The oxidation of H_2S gas by some acid radicals
 (D) III group radicals are precipitated as hydroxides
34. A metal salt solution gives a yellow ppt with silver nitrate. The ppt dissolves in dil. nitric acid as well as in ammonium hydroxide. The solution contains -
 (A) Bromide (B) Iodide (C) Phosphate (D) Chromate
35. A blue colouration is not obtained when -
 (A) Ammonium hydroxide dissolves in copper sulphate
 (B) Copper sulphate solution reacts with $\text{K}_4[\text{Fe}(\text{CN})_6]$
 (C) Ferric chloride reacts with sodium ferrocyanide
 (D) Anhydrous white CuSO_4 is dissolved in water
36. A pale green crystalline metal salt of M dissolves freely in water. On standing it gives a brown ppt on addition of aqueous NaOH . The metal salt solution also gives a black ppt on bubbling H_2S in basic medium. An aqueous solution of the metal salt decolourizes the pink colour of the permanganate solution. The metal in the metal salt solution is -
 (A) Copper (B) Aluminium (C) Lead (D) Iron

CHECK YOUR GRASP								ANSWER KEY					EXERCISE -1							
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	B	A	A	A	D	D	B	A	D	D	C	C	A	A	C	B	D	B	C	C
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36				
Ans.	B	B	C	D	C	A	D	C	C	A	C	B	C	C	B	D				

EXERCISE-02**BRAIN TEASERS****SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)**

1. Identify the incorrect reaction (s) -
- (A) $\text{K}_2\text{Cr}_2\text{O}_7 + 4\text{NaCl} + 3\text{H}_2\text{SO}_4 \text{ (conc.)} \longrightarrow 2\text{CrO}_2\text{Cl}_2 + 2\text{Na}_2\text{SO}_4 + \text{K}_2\text{SO}_4 + 3\text{H}_2\text{O}$
- (B) $\text{K}_2\text{Cr}_2\text{O}_7 + 6\text{KI} + 7\text{H}_2\text{SO}_4 \text{ (conc.)} \longrightarrow 3\text{I}_2 + \text{Cr}_2(\text{SO}_4)_3 + 4\text{K}_2\text{SO}_4 + 7\text{H}_2\text{O}$
- (C) $\text{K}_2\text{Cr}_2\text{O}_7 + 4\text{AgCl} + 3\text{H}_2\text{SO}_4 \text{ (conc.)} \longrightarrow 2\text{CrO}_2\text{Cl}_2 + 2\text{Ag}_2\text{SO}_4 + \text{K}_2\text{SO}_4 + 3\text{H}_2\text{O}$
- (D) $\text{MnO}_2 + \text{NaCl} + 2\text{H}_2\text{SO}_4 \text{ (conc.)} \longrightarrow \text{NaHSO}_4 + \text{MnSO}_4 + \text{HCl} + \text{H}_2\text{O} + 1/2\text{O}_2$
2. Which of the following statement (s) is (are) true -
- (A) Cu^{2+} salts form soluble complex with excess KCN
- (B) Cu^{2+} salts form soluble complex with aqueous ammonia
- (C) Cu^{2+} salts form soluble complex with KI
- (D) A piece of iron or zinc when placed in Cu^{2+} salt solution, precipitates copper
3. A mixture of two white substances was dissolved in water. On passing Cl_2 gas through the solution a deep brown colour is developed. Addition of BaCl_2 solution to the original solution gives a white ppt. Addition of a large amount of NaOH solution to the original solution gives a white ppt, whose suspension in water is used as an ant-acid, the mixture gives golden yellow colour flame. What is the chemical composition of the mixture -
- (A) NaBr & MgSO_4 (B) NaBr & CaSO_4 (C) NaBr & $\text{Al}_2(\text{SO}_4)_3$ (D) NaBr & ZnSO_4
4. $\text{KBr} + \text{MnO}_2 + \text{H}_2\text{SO}_4 \text{ (conc.)} \longrightarrow \text{KHSO}_4 + \text{MnSO}_4 + \text{H}_2\text{O} + [\text{X}]$ (unbalanced equation) -
- (A) X turns starch paper orange red
- (B) X with AgNO_3 solution gives a pale yellow ppt which is completely soluble in excess ammonium hydroxide
- (C) X produces violet colour in organic layer in KI solution
- (D) X is liberated when a mixture of KBr, $\text{K}_2\text{Cr}_2\text{O}_7$ and conc. H_2SO_4 is heated
5. An inorganic Lewis acid [X] gives gelatinous white ppt. With NH_4OH in presence of NH_4Cl . [X] will respond to which of the following characteristics -
- (A) X fumes in moist air
- (B) X on heating with solid $\text{K}_2\text{Cr}_2\text{O}_7$ and conc. H_2SO_4 gives deep red or orange red fumes
- (C) X on addition of excess NaOH gives white ppt
- (D) X on heating with Na_2CO_3 and cobalt nitrate gives a blue bead in oxidising flame
6. SO_2 and CO_2 both turn lime water (A) milky, SO_2 also turns $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$ (B) green while O_2 is soluble in pyrogallol (C) turning it black. These gases are to be detected in order by using these reagents. The order is -
- (A) (A), (B), (C) (B) (B), (C), (A) (C) (B), (A), (C) (D) (A), (C), (B)
7. Three test tubes A, B, C contain Pb^{2+} , Hg_2^{2+} and Ag^+ (but unknown). To each aqueous solution NaOH is added in excess. Following changes occur -
- A : Black ppt B : Brown ppt C : White ppt but dissolves in excess of NaOH
- A, B and C contain respectively :
- (A) Pb^{2+} , Hg_2^{2+} and Ag^+ (B) Hg_2^{2+} , Ag^+ , Pb^{2+}
- (C) Ag^+ , Pb^{2+} , Hg_2^{2+} (D) Ag^+ , Hg_2^{2+} and Pb^{2+}

8. Of the following oxides, all are soluble in NaOH(aq) except -
 (A) ZnO (B) Al₂O₃ (C) Fe₂O₃ (D) SnO₂
9. The solvay process can be represented by the following scheme



In the above process, the correct options are :

- (A) A = Ca(OH)₂ (B) B = NH₄HCO₃ (C) E = CaCl₂ (D) C = NaHCO₃
10. Reddish brown gas is obtained when the following are treated with conc. H₂SO₄ :
 (A) Br⁻ (B) NO₃⁻ (C) NO₂⁻ (D) I⁻
11. KI solution identifies -
 (A) Hg₂²⁺ (B) Pb²⁺ (C) Ag⁺ (D) Cu²⁺
12. The only cations present in a slightly acidic solution are Fe³⁺, Zn²⁺ and Cu²⁺. The reagent that when added in excess to this solution would identify and separate Fe³⁺ in one step is -
 (A) 2 M HCl (B) 6 M NH₃ (C) 6 M NaOH (D) H₂S gas
13. A test-tube containing a nitrate and another containing a bromide and MnO₂ are treated with conc. H₂SO₄. The brown fumes evolved are passed in water. The water will be coloured by -
 (A) The nitrate (B) The bromide (C) Both (D) None of the two
14. Production of a green edged flame on igniting the vapours evolved by heating a given inorganic salt with a few ml of ethanol and conc. H₂SO₄ indicates the presence of -
 (A) Tartrate (B) Oxalate (C) Acetate (D) Borate
15. When CS₂ layer containing both Br₂ and I₂ is shaken with excess of Cl₂ water, the violet colour due to I₂ disappears and orange colour due to Br₂ appears. The disappearance of violet colour is due to the formation of -
 (A) I₃⁻ (B) HIO₃ (C) ICl₂ (D) I⁻
16. Which one among the following pairs of ions cannot be separated by H₂S in dilute hydrochloric acid-
 (A) Bi³⁺, Sn⁴⁺ (B) Al³⁺, Hg²⁺ (C) Zn²⁺, Cu²⁺ (D) Ni²⁺, Cu²⁺
17. Yellow ammonium sulphide solution is a suitable reagent for the separation of -
 (A) HgS and PbS (B) PbS and Bi₂S₃ (C) Bi₂S₃ and CuS (D) CdS and As₂S₃
18. What product is formed by mixing the solution of K₄[Fe(CN)₆] with the solution of FeCl₃ -
 (A) Ferro-ferricyanide (B) Ferric-ferrocyanide
 (C) Ferri-ferricyanide (D) None

19. Which of the following will not give positive chromyl chloride test-
- (A) Copper chloride, CuCl_2 (B) Mercuric chloride, HgCl_2
(C) Mercurous chloride, HgCl_2 (D) Zinc chloride, ZnCl_2
20. When chlorine water is added to an aqueous solution of potassium halide in presence of chloroform, a violet colour is obtained. On adding more of chlorine water, the violet colour disappears, and a colourless solution is obtained. This test confirms the presence of the following in aqueous solution -
- (A) Iodide (B) Bromide (C) Chloride (D) Iodide and bromide
21. Which compound does not dissolve in hot dilute HNO_3 -
- (A) HgS (B) PbS (C) CuS (D) CdS
22. An aqueous solution of FeSO_4 , $\text{Al}_2(\text{SO}_4)_3$ and chrome alum is heated with excess of Na_2O_2 and filtered. The materials obtained are -
- (A) A colourless filtrate and a green residue (B) A yellow filtrate and a green residue
(C) A yellow filtrate and a brown residue (D) A green filtrate and a brown residue
23. Three separate samples of a solution of a single salt gave these results. One formed a white precipitate with excess ammonia solution, one formed a white precipitate with dil. NaCl solution and one formed a black precipitate with H_2S . The salt could be -
- (A) AgNO_3 (B) $\text{Pb}(\text{NO}_3)_2$ (C) $\text{Hg}(\text{NO}_3)_2$ (D) MnSO_4
24. A white sodium salt dissolves readily in water to give a solution which is neutral to litmus. When silver nitrate solution is added to the solution, a white precipitate is obtained which does not dissolve in dil. HNO_3 . The anion could be -
- (A) CO_3^{2-} (B) Cl^- (C) SO_4^{2-} (D) S^{2-}
25. In the separation of Cu^{2+} and Cd^{2+} in 2nd group qualitative analysis of cations, tetrammine copper (II) sulphate and tetrammine cadmium (II) sulphate react with KCN to form the corresponding cyano complexes. Which one of the following pairs of the complexes and their relative stability enables the separation of Cu^{2+} and Cd^{2+} :
- (A) $\text{K}_3[\text{Cu}(\text{CN})_4]$ more stable and $\text{K}_2[\text{Cd}(\text{CN})_4]$ less stable
(B) $\text{K}_2[\text{Cu}(\text{CN})_4]$ less stable and $\text{K}_2[\text{Cd}(\text{CN})_4]$ more stable
(C) $\text{K}_2[\text{Cu}(\text{CN})_4]$ more stable and $\text{K}_2[\text{Cd}(\text{CN})_4]$ less stable
(D) $\text{K}_3[\text{Cu}(\text{CN})_4]$ less stable and $\text{K}_2[\text{Cd}(\text{CN})_4]$ more stable
26. When $\text{K}_2\text{Cr}_2\text{O}_7$ crystals are heated with conc. HCl , the gas evolved is -
- (A) O_2 (B) Cl_2 (C) CrO_2Cl_2 (D) HCl
27. On the addition of a solution containing CrO_4^{2-} ions to the solution of Ba^{2+} , Sr^{2+} and Ca^{2+} ions, the ppt obtained first will be of -
- (A) CaCrO_4 (B) SrCrO_4 (C) BaCrO_4 (D) A mixture of all the three
28. A mixture of chlorides of copper, cadmium, chromium, iron and aluminium was dissolved in water acidified with HCl and hydrogen sulphide gas was passed for sufficient time. It was filtered, boiled and a few drops of nitric acid were added while boiling. To this solution ammonium chloride and sodium hydroxide were added in excess and filtered. The filtrate shall give test for -
- (A) Sodium and iron (B) Sodium, chromium and aluminium
(C) Aluminium and iron (D) Sodium, iron, cadmium and aluminium

29. A salt on treatment with dil. HCl gives a pungent smelling gas and a yellow precipitate. The salt gives green flame when tested. The solution gives a yellow precipitate with potassium chromate. The salt is-
- (A) NiSO_4 (B) BaS_2O_3 (C) PbS_2O_3 (D) CuSO_4
30. CrO_3 dissolves in aqueous NaOH to give -
- (A) $\text{Cr}_2\text{O}_7^{2-}$ (B) CrO_4^{2-} (C) Cr(OH)_3 (D) Cr(OH)_2
31. A mixture of two salts is not water soluble but dissolves completely in dil HCl to form a colourless solution. The mixture could be -
- (A) AgNO_3 and KBr (B) BaCO_3 and ZnS
 (C) FeCl_3 and CaCO_3 (D) $\text{Mn(NO}_3)_2$ and MgSO_4
32. Which of the following combinations in an aqueous medium will give a red colour or precipitate -
- (A) $\text{Fe}^{3+} + \text{SCN}^-$ (B) $\text{Fe}^{2+} + [\text{Fe(CN)}_6]^{3-}$
 (C) $\text{Ni}^{2+} + \text{dimethylglyoxime} + \text{NH}_3$ solution (D) $\text{Cu}^{2+} + [\text{Fe(CN)}_6]^{4-}$
33. Acidic $\text{K}_2\text{Cr}_2\text{O}_7$ reacts with H_2S to produce-
- (A) Cr^{6+} ions (B) Cr^{3+} ions (C) SO_2 (D) S
34. A yellow precipitate is obtained when -
- (A) Lead acetate solution is treated with K_2CrO_4
 (B) $\text{Pb(NO}_3)_2$ solution is treated with K_2CrO_4
 (C) AgNO_3 solution treated with KI
 (D) H_2S is passed through a solution of CdSO_4
35. Which of the following ions can be separated by using NH_4Cl and NH_4OH -
- (A) Fe^{3+} and Cr^{3+} (B) Cr^{3+} and Co^{2+} (C) Cr^{3+} and Al^{3+} (D) Al^{3+} and Ba^{2+}
36. $\text{Al}_2(\text{SO}_4)_3 + \text{NH}_4\text{OH} \longrightarrow \text{X}$
 Select the correct statement (s) about compound X :
- (A) X is a white coloured compound (B) X is insoluble in excess of NH_4OH
 (C) X is soluble in NaOH (D) X can be used as an antacid
37. Which of the following cations cannot be separated by adding NH_4Cl , NH_4OH and $(\text{NH}_4)_2\text{CO}_3$ to their solution ?
- (A) Ca^{2+} and Sr^{2+} (B) Ba^{2+} and Sr^{2+} (C) Ba^{2+} and Mg^{2+} (D) Ca^{2+} and Ba^{2+}
38. On being heated, which of the following substances will give a gas that turns limewater milky ?
- (A) Na_2CO_3 (B) ZnCO_3 (C) ZnSO_3 (D) MgCO_3
39. On being heated, which of the following substances will give a white sublimate ?
- (A) NH_4Cl (B) HgCl_2 (C) AgCl (D) Hg_2Cl_2
40. On being strongly heated, which of the following substances will leave a black residue ?
- (A) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (B) ZnCO_3 (C) PbCO_3 (D) MnSO_4
41. Which of the following cations will turn a borax bead green in an oxidising flame ?
- (A) Fe^{2+} (B) Mn^{2+} (C) Cr^{3+} (D) Cu^{2+}
42. Which of the following cations will turn a borax bead blue in an oxidising flame ?
- (A) Fe^{3+} (B) Fe^{2+} (C) Co^{2+} (D) Cu^{2+}
43. Which of the following substances are blue ?
- (A) $\text{Fe(BO}_2)_2$ (B) CoAl_2O_4 (C) $\text{Co(BO}_2)_2$ (D) NaCoPO_4

44. On reaction with dilute H_2SO_4 , which of the following salts will give out a gas that turns an acidified dichromate paper green ?
- (A) Na_2CO_3 (B) Na_2S (C) ZnSO_3 (D) FeS
45. Which of the following ions can be separated by using dilute HCl ?
- (A) Ag^+ and Cu^{2+} (B) Ag^+ and Hg_2^{2+} (C) Hg_2^{2+} and Cd^{2+} (D) Ag^+ and Al^{3+}
46. Which of the following ions can be separated by using H_2S in the presence of dilute HCl ?
- (A) Cu^{2+} and Co^{2+} (B) Pb^{2+} and Ni^{2+} (C) Hg^{2+} and Cu^{2+} (D) Cu^{2+} and Bi^{3+}
47. Which of the following ions can be separated by using NH_4Cl and NH_4OH ?
- (A) Fe^{3+} and Cr^{3+} (B) Cr^{3+} and Co^{2+} (C) Cr^{3+} and Al^{3+} (D) Al^{3+} and Ba^{2+}
48. Which of the following mixtures of ions in solution can be separated by using an NH_3 solution ?
- (A) Hg_2^{2+} and Ag^+ (B) Bi^{3+} and Cu^{2+} (C) Ag^+ and Pb^{2+} (D) Cu^{2+} and Cd^{2+}
49. Which of the following mixtures of ions in solution can be separated by using an NaOH solution ?
- (A) Fe^{3+} and Pb^{2+} (B) Pb^{2+} and Sn^{2+} (C) Zn^{2+} and Sn^{2+} (D) Al^{3+} and Cu^{2+}
50. Which of the following mixtures of ions in solution can be separated by using dilute H_2SO_4 ?
- (A) Zn^{2+} and Pb^{2+} (B) Ba^{2+} and Pb^{2+} (C) Mn^{2+} and Sr^{2+} (D) Sr^{2+} and Ba^{2+}

Brain Teasers						Answer Key				Exercise -2					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	C,D	A,B,D	A	A,C,D	A,B,D,	C	B	C	A,B,C,	A,B,C	B,C	B	B	D	B
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	A	D	B	B	A	A	B	B	B	A	B	C	B	B	B
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	B	A,C	B,D	A,B,C,D	B,D	A,B,C,D	A,B,D	B,C,D	A,B,D	A,D	A,C	C,D	B,C,D	B,C,D	A,C,D
Que.	46	47	48	49	50										
Ans.	A,B	B,D	A,B,C	A,D	A,C										

EXERCISE-03**MISCELLANEOUS TYPE QUESTIONS****TRUE / FALSE**

1. In the ring test for NO_3^- , removal of Br^- and I^- is done by adding AgNO_3 solution.
2. CO_3^{2-} & HCO_3^- of sodium both produce precipitate with MgCl_2 aq.
3. NO_3^- solution produce brown ring with FeSO_4 & dil. H_2SO_4 .
4. HgCl_2 gives not chromyl chloride test.
5. Mn^{+2} when boiled with $\text{Na}_2\text{S}_2\text{O}_8/\text{H}^+$ it produce purple coloration.
6. Alkaline Solution of sodium nitro pruside produce volatile coloration with $\text{H}_2\text{S}_{(\text{g})}$.

FILL IN THE BLANKS

1. A solution of salt in HCl when diluted with water turns milky. It indicates the presence of
2. In group III, the basic radicals are precipitated as their
3. The solubility product of hydroxide of Fe^{2+} is than that of Fe^{3+} .
4. Group IV basic radicals are precipitated as from medium.
5. NaNO_3 when treated with Zn dust & NaOH solution it produce gas.
6. NH_4NO_3 on heating gives Solid substance.
7. Mix of NaI (s) + $\text{K}_2\text{Cr}_2\text{O}_7$ (s) + conc. H_2SO_4 . When heated in a test tube dark vapours evolve is
8. $\text{Cr}_2(\text{SO}_4)_3$ solution produce colour with Na_2O (excess) and colour with Na_2O_2 (excess.)

MATCH THE COLUMN

1. Match the following

Column-I		Column-II	
(A)	CrCl_3 (aq)	(p)	Produce ppt with excess of NaOH
(B)	CuSO_4 (aq)	(q)	Produce coloured Solution with excess of ammonia
(C)	$(\text{NH}_4)_2\text{CO}_3$ (aq)	(r)	Produce gases product when heated with KOH (aq)
(D)	AgNO_3 (aq)	(s)	Produce gas with dil. H_2SO_4

2. Match the following

Column-I		Column-II	
(A)	$\text{Fe}(\text{SCN})_3 + \text{KF}$ (aq) excess	(p)	Produce coloured product (s)
(B)	$\text{CrO}_2\text{Cl}_2 + \text{NaOH}$ (aq)	(q)	diamagnetic product
(C)	$\text{Ni}^{+2} + \text{dmg} \xrightarrow[\text{CH}_3\text{COONa}]{\text{CH}_3\text{COOH}}$	(r)	Hydrogen bonded product
(D)	$\text{Na}_2\text{SO}_3 + \text{Cr}_2\text{O}_7^{2-} \xrightarrow{\text{H}^+}$	(s)	Tetrahedral geometry around metal

Note :- dmg = dimethyl glyoxime

3. Match the following

Column-I		Column-II	
(A)	$\text{H}_3\text{P}_3\text{O}_9$	(p)	S-O-S bond is present
(B)	$\text{H}_2\text{S}_2\text{O}_7$	(q)	Di-basic acid
(C)	$\text{H}_2\text{S}_4\text{O}_6$	(r)	P-O-P bond is present
(D)	$\text{H}_4\text{P}_2\text{O}_5$	(s)	Central atom (S or P) in maximum oxidation state

4. Match the following

Column-I		Column-II	
(A)	Soluble in a concentrated NH_3 solution	(p)	Ag_2S
(B)	Soluble in excess KCN solution	(q)	$\text{Cu}(\text{OH})_2$
(C)	Soluble in excess hypo solution	(r)	AgBr
(D)	Soluble in conc. HCl	(s)	AgCl

5. Match the following

Column-I		Column-II	
(A)	Colourless gas evolved on addition of dil. H_2SO_4	(p)	$\text{S}_2\text{O}_3^{2-}$
(B)	White ppt. on addition of AgNO_3	(q)	S^{2-}
(C)	Black ppt. obtained when HgCl_2 is added in little amount	(r)	NO_2^-
(D)	The ppt. obtained on addition of AgNO_3 followed by NH_3 solution	(s)	CH_3CO_2^-

ASSERTION & REASON

These questions contains, Statement I (assertion) and Statement II (reason).

(A) Statement-I is true, Statement-II is true ; Statement-II is correct explanation for Statement-I.

(B) Statement-I is true, Statement-II is true ; Statement-II is NOT a correct explanation for statement-I

(C) Statement-I is true, Statement-II is false

(D) Statement-I is false, Statement-II is true

1. **Statement -I** : Borax bead test is applicable only to coloured salts.

Because

Statement -II : In borax bead test, coloured salts are decomposed to give colored metal metaborates.

2. **Statement -I** : First group basic radicals are precipitated as thier chlorides.

Because

Statement -II : The solubility product of these chlorides are more than the solubility product of other basic radical chlorides.

3. **Statement -I** : Cu^{2+} and Cd^{2+} are separated from each other by first adding KCN solution and then passing H_2S gas.
Because
Statement -II : KCN reduces Cu^{2+} to Cu^+ and forms a complex with it.
4. **Statement -I** : CaSO_4 dissolves in $(\text{NH}_4)_2\text{SO}_4$ solution.
Because
Statement -II : CaSO_4 forms a soluble complex of $(\text{NH}_4)_2[\text{Ca}(\text{SO}_4)_2]$
5. **Statement -I** : Nessler's reagent gives a brown precipitate with NH_3 .
Because
Statement -II : NH_4OH gives a brown precipitate with Fe^{3+} .
6. **Statement -I** : NH_4Cl is added in III group basic radicals to suppress the ionisation of NH_4OH .
Because
Statement -II : In the presence of high concentration of OH^- ions, basic radicals of other groups will also get precipitated in III group.

COMPREHENSION BASED QUESTIONS

Comprehension # 1

A chemist opened a cupboard to find four bottles containing water solutions, each of which has lost its label. Bottles 1, 2, 3 contained colourless solutions, whilst Bottle 4 contained a blue solution. The labels from the bottles were lying scattered on the floor of the cupboard. They were

Copper (II) sulphate

Sodium carbonate

Lead nitrate

hydrochloric acid

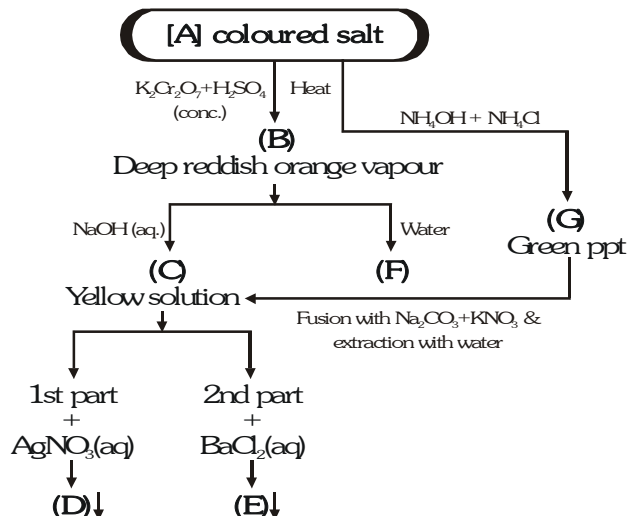
By mixing samples of the contents of the bottles, in pairs, the chemist made the following observations :

- | | | |
|-------|---------------------|------------------------|
| (i) | Bottle 1 + Bottle 2 | white precipitate |
| (ii) | Bottle 1 + Bottle 3 | white precipitate |
| (iii) | Bottle 1 + Bottle 4 | white precipitate |
| (iv) | Bottle 2 + Bottle 3 | colourless gas evolved |
| (v) | Bottle 2 + Bottle 4 | no visible reaction |
| (vi) | Bottle 3 + Bottle 4 | blue precipitate |

- Chemical formula of white precipitate in observation (i) is :
 (A) CuCl_2 (B) PbCl_2
 (C) PbCO_3 (D) CuSO_3
- Colourless solution present in Bottle-1 is -
 (A) CuSO_4 (B) HCl
 (C) $\text{Pb}(\text{NO}_3)_2$ (D) Na_2CO_3
- Nature of gas evolved in observation (iv) is -
 (A) Acidic (B) Neutral (C) Basic (D) Amphoteric
- Chemical formula of white ppt. formed in observation (iii) is :
 (A) PbCl_2 (B) PbCO_3 (C) CuCO_3 (D) PbSO_4

Comprehension # 2

Read the following comprehension carefully and answer the following questions.



- The colour of the ppt (D) & (E) are :
 (A) white & yellow (B) yellow
 (C) brick red & yellow (D) yellow and brick red
- Yellow solution (C) is an important laboratory reagent and is used in the estimation of :
 (A) Pb^{2+} (B) Fe^{3+} (C) Cd^{2+} (D) None of these
- The compound (A) is :
 (A) CrCl_3 (B) CrBr_3 (C) $\text{Cr}(\text{CH}_3\text{COO})_3$ (D) $\text{Cr}(\text{NO}_3)_3$
- $[\text{A}] (\text{s}) + \text{MnO}_2 + \text{H}_2\text{SO}_4 (\text{conc.}) \longrightarrow \text{X}$ Greenish yellow gas.
 Select the correct choice for [X] :
 (A) It gives yellow ppt. with AgNO_3 (B) It liberates I_2 from KI solution
 (C) It turns starch paper orange red (D) It turns titan yellow solution red

Comprehension # 3

Three metal ions x^{+2} , y^{+2} , z^{+2} are identify in qualitative analysis. Nitrates of x^{+2} , y^{+2} , z^{+2} dissolve in three seprate test tubes and gives following observation.

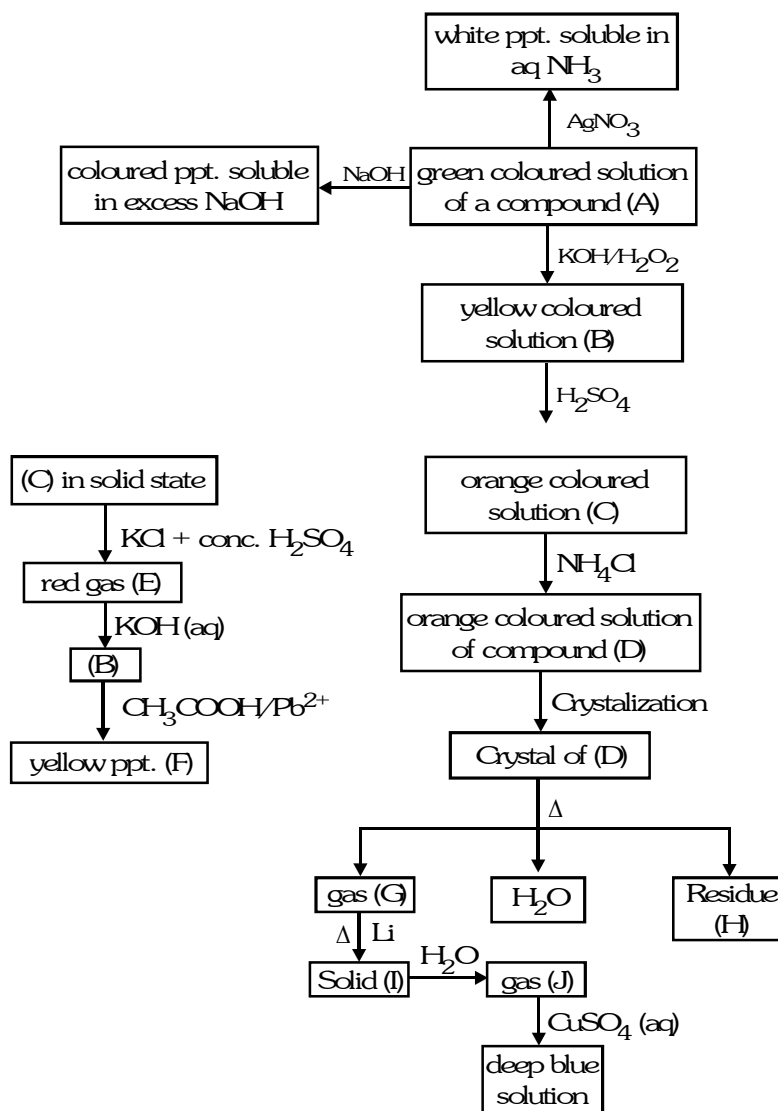
- All solution produce carbonate precipitate with $(\text{NH}_4)_2\text{CO}_3$
- Only one produce white ppt on addition of NaCl .
- Out of 3 cations two produce sulphide ppt.
- Sulphide of y^{+2} is not produce by $\text{H}_2\text{S}/\text{H}^+$ but produce when H_2S is passed in basic medium.
- Only y^{+2} produce soluble sulphate
- x^{+2} gives no ppt with dil NH_4OH .

- Select in correct statment :
 (A) y^{+2} not produce precipitate with Ist group reagent in salt analysis
 (B) y^{+2} not produce ppt with 2nd group reagent in salt analysis
 (C) z^{+2} produce ppt with IInd group reagent in salt analysis
 (D) z^{+2} is not produce ppt with Ist group reagent in salt analysis

2. Select order of K_{sp} of sulphide of x^{+2} , y^{+2} , z^{+2} -
- (A) $x_s > y_s > z_s$ (B) $x_s > z_s > y_s$
 (C) $y_s > z_s > x_s$ (D) $z_s > y_s > x_s$
3. Select correct about xCO_3 , yCO_3 , zCO_3 -
- (A) All are soluble in dil. H_2SO_4 (B) All are soluble in dil HCl
 (C) None is soluble in dil. H_2SO_4 (D) Except ZCO_3 all are soluble in dil. HCl
4. (i) $x^{+2} + H_2S \xrightarrow{NH_4OH}$ (ii) $x^{+2} + NaOH (dil) \longrightarrow$
 (iii) $x^{+2} + Na_2CO_3 \longrightarrow$
- Precipitate is obtain in
- (A) Reaction (i), (ii), (iii) (B) Only in reaction (iii)
 (C) Only in reaction (i) and (ii) (D) Only in reaction (ii)

Comprehension # 4

Read the following short write up and answer subsequent questions based on observations (A) to (J).



- Compound A and B are respectively :
 (A) FeCl_2 ; FeCl_3 (B) CuCl_2 ; $2\text{H}_2\text{O}$; $[\text{CuCl}_4]^{-2}$
 (C) CrCl_3 ; K_2CrO_4 (D) NiCl_2 ; NiCl_3
- Gas (J) is also produced by :
 (i) heating NH_4NO_3 (ii) heating NH_4NO_2
 (iii) heating NH_4Cl (iv) Reaction of NH_4Cl and Ca(OH)_2
 (A) (i) and (iii) (B) (i) and (ii) (C) (i) and (iv) (D) (iii) and (iv)
- Select the incorrect reaction :
 (A) (C) in solid state + KBr + conc. $\text{H}_2\text{SO}_4 \longrightarrow$ Red gas
 (B) (C) in solid state + KCl + conc. $\text{H}_2\text{SO}_4 \longrightarrow$ Red gas
 (C) (C) in solid state + FeCl_3 + conc. $\text{H}_2\text{SO}_4 \longrightarrow$ Red gas
 (D) (C) in solid state + HgCl_2 + conc. $\text{H}_2\text{SO}_4 \longrightarrow$ Red gas

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
<u>True / False</u> 1. F 2. F 3. F 4. T 5. T 6. T <u>Fill in the Blanks</u> 1. Bi^{3+} 2. Hydroxides 3. Higher 4. Sulphides, Ammoniacal 5. NH_3 6. None 7. I_2 8. Green, Yellow <u>Match the Column</u> 1. (A) - q (B) - p, q (C) - r, s (D) - p 2. (A) - q (B) - p, q, s (C) - p, q, r (d) - p, q, s 3. (A) - p, r (B) - p, q (C) - p, r (D) - p, q, s 4. (A) - q, r, s (B) - p, q, r, s (C) - q, r, s (d) - q, s 5. (A) - p, q, s (B) - p, r, s (C) - q, (d) - q <u>Assertion - Reason Questions</u> 1. A 2. C 3. B 4. A 5. C 6. A <u>Comprehension Based Questions</u> Comprehension #1 : 1. B 2. C 3. A 4. D Comprehension #2 : 1. C 2. A 3. A 4. B Comprehension #3 : 1. D 2. A 3. D 4. B Comprehension #4 : 1. C 2. D 3. D		

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

1. Colourless salt (A) + NaOH (excess) $\xrightarrow{\Delta}$ gas (B) giving white fumes with HCl + alkaline solution (C)
- (C) + Zn $\longrightarrow$ gas (B)
- (A) $\xrightarrow{\Delta}$ gas (D) + liquid (E)
- D, E both triatomic
- identify (A, B, C, D) and (E).
2. Complete and balance the following reactions :
- (A) $\text{Cu} + \text{HNO}_3 \text{ (dil)} \longrightarrow \text{NO} + \dots + \dots$
- (B) $\text{Pb}(\text{NO}_3)_2 \xrightarrow{\text{heat}} \text{PbO} + \dots + \dots$
- (C) $\text{CuSO}_4 + \text{NH}_4\text{OH (excess)} \longrightarrow \dots$
- (D) $\text{AgCl} + \text{NH}_4\text{OH} \longrightarrow \dots$
3. Aqua-regia dissolves gold. Write reaction.
4. What happens when -
- (i) Hydrogen sulphide is bubbled through an aqueous solution of sulphur dioxide
- (ii) Hydrogen sulphide is passed through acidified ferric chloride solution.
- (iii) Sulphur is boiled with caustic soda solution.
5. Sodium salt (A) of a dibasic acid $\xrightarrow{\text{HCl}}$ gas (B) and clear solution of gas (B) turns $\text{K}_2\text{Cr}_2\text{O}_7$ to green and also lime water milky. identify (A) and (B).
6. To a solution containing Ca^{2+} , Ag^+ , Cu^{2+} and K^+ , 2M HCl is added when a white precipitate (A) is obtained. After filtration H_2S is passed through the filtrate, a black ppt. (B) is formed. On removing (B) by filtration, it gave a white ppt. (C) with Na_2CO_3 solution. Identify (A), (B) and (C).
7. An aqueous solution of a gas (X) gives the following reactions :
- (i) It decolourises an acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution.
- (ii) On boiling with H_2O_2 and cooling it and then adding an aqueous solution of BaCl_2 , a ppt. insoluble in dil. HCl, is produced.
- (iii) On passing H_2S in the solution, white turbidity is formed.
- Identify (X) and give chemical reactions of sets (i) to (iii).
8. A solution containing several unknown cations is treated with dil. HCl and a ppt. forms. The ppt. is filtered and the filtrate at pH 1.0 is treated with H_2S , no ppt. forms. At pH 8.0 H_2S causes the formation of a ppt., the filtrate from which gives no ppt. on treatment with Na_2CO_3 . Which group of cations are present in the original solution ?
9. The aqueous solution of an inorganic compound (X) yielded a white precipitate when treated with dil HNO_3 and AgNO_3 . Another sample of the solution of (X) when treated with NaOH gave a white precipitate first which dissolved in excess of NaOH yielding a colorless solution. When H_2S gas was passed through that solution a white precipitate was obtained. Identify the compound (X) and give the reactions.

10. An orange coloured solid (A) is soluble in water and gives a gas (B) and green coloured solid (C) on heating. The compound (A) gives a gas (D) when reacts with NaOH and solution turns yellow. The gas (D) turns red litmus blue. Identify the compounds (A) to (D) and explain the reaction.
11. A compound (X) on heating with an excess of NaOH solution gives a gas (Y) which gives white fumes on exposure to HCl. Heating is continued to expel the gas completely. The resultant alkaline solution again liberates the same gas Y when heated with Zn powder. However, the compound (X) when heated alone does not give nitrogen. Identify the compounds X and Y.
12. A salt reacts with NaOH to form a green coloured ppt. (X) which is soluble in excess of NaOH. (X) on heating gives a green powder (Y). (Y) on fusion with NaOH in air gives a yellow coloured solution (Z). Identify the compound X, Y & Z.
13. Identify the inorganic salt A whose aqueous solution gives following reactions.
- Pale yellow precipitate with AgNO_3 solution, insoluble in dil HNO_3
 - White precipitate with NH_4OH and also with NaOH solution. However the precipitate does not dissolve in excess of NH_4OH but soluble in excess of NaOH.
14. Two species (A) and (B) exists in equilibrium at pH of about 4 and can be interconverted by changing the pH. Acidified solution of (B) is orange, and on adding H_2O_2 it forms deep blue colour due to the formation of compound (C), this blue colour fades away gradually. Further acidified solution of (B) on reaction with NaCl gives orange red fumes due to the formation of (D). Identify (A), (B), (C) & (D).
15. A compound X does not give N_2 on heating. Its aqueous solution when heated with caustic soda liberate a gas Y which turns red litmus blue. Heating of alkaline solution of X is continued to expell the gas Y completely. However residual solution again liberates the gas Y when heated with Zinc powder. Identify X and Y.
16. An aqueous solution of a gas (X) shows the following reactions.
- It turns red litmus blue
 - When added in excess to a CuSO_4 solution, a deep blue colour is obtained
 - On addition of FeCl_3 solution a brown precipitate soluble in dilute HNO_3 is obtained.
- Identify (X) and give equations for the reactions at step (ii) and (iii)
17. Complete the following by identifying (A) to (F).
- $\text{CuSO}_4 \cdot 5\text{H}_2\text{O} \xrightarrow{100^\circ\text{C}} (\text{A}) \xrightarrow{230^\circ\text{C}} (\text{B}) \xrightarrow{800^\circ\text{C}} (\text{C}) + (\text{D})$
 - $\text{AgNO}_3 \xrightarrow{\text{Red hot}} (\text{E}) + (\text{F}) + \text{O}_2$
18. Identify (A), (B), (C) & (D) and give their chemical formulae :
- $(\text{A}) + \text{NaOH} \xrightarrow{\text{Heat}} \text{NaCl} + \text{NH}_3 + \text{H}_2\text{O}$
 - $\text{NH}_3 + \text{CO}_2 + \text{H}_2\text{O} \longrightarrow (\text{B})$
 - $(\text{B}) + \text{NaCl} \longrightarrow (\text{C}) + \text{NH}_4\text{Cl}$
 - $(\text{C}) \xrightarrow{\text{Heat}} \text{Na}_2\text{CO}_3 + \text{H}_2\text{O} + (\text{D})$
19. A certain metal (A) is boiled in dilute HNO_3 to give a salt (B) and an oxide of nitrogen (C). An aqueous solution of (B) with brine gives a precipitate (D) which is soluble in NH_4OH . On adding aqueous solution of (B) to hypo solution, a white precipitate (E) is obtained. (E) turns black on standing. Identify (A) to (E).

20. Calcium burns in nitrogen to produce a white powder which dissolves in sufficient water to produce a gas (A) and an alkaline solution. The solution on exposure to air produces a thin solid layer of (B) on the surface. Identify the compounds (A) and (B).
21. Gradual addition of KI solution to $\text{Bi}(\text{NO}_3)_3$ solution initially produces a dark brown precipitate which dissolves in excess of KI to give a clear yellow solution. Write equations for the above reactions.

CONCEPTUAL	SUBJECTIVE	EXERCISE	ANSWER KEY	EXERCISE -4(A)
1. (A) - NH_4NO_3	(B) - NH_3	(C) - $(\text{NaNO}_3 + \text{NaOH})$	(D) - N_2O	(E) - H_2O
2. (A) - $\text{Cu}(\text{NO}_3)_2, \text{H}_2\text{O}$	(B) - NO_2, O_2	(C) - $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4 + \text{H}_2\text{O}$	(D) - $\text{Ag}(\text{NH}_3)_2\text{Cl} + \text{H}_2\text{O}$	
3. $\text{Au} + 3[\text{Cl}] \longrightarrow \text{AuCl}_3 \xrightarrow{\text{HCl}} \text{H}[\text{AuCl}_4]$				
4. (i) - $\text{S}, \text{H}_2\text{O}$	(ii) - $\text{FeCl}_2, \text{HCl}, \text{H}_2\text{O}$	(iii) - Na_2S_5		
5. (A) - Na_2SO_3	(B) - SO_2			
6. (A) - AgCl ,	(B) - CuS (Black ppt.)	(C) - CaCO_3		
7. $\text{X} \rightarrow \text{SO}_2$	8. Ions of group I and III are present.			
9. $\text{X} \rightarrow \text{ZnCl}_2$	10. (A) - $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$,	(B) - N_2 ,	(C) - Cr_2O_3 ,	(D) - 2NH_3
11. $\text{X} \rightarrow \text{NH}_4\text{NO}_3$, Y - NH_3	12. $\text{X} \rightarrow \text{Cr}(\text{OH})_3$, Y $\rightarrow \text{Cr}_2\text{O}_3$, Z $\rightarrow \text{Na}_2\text{CrO}_4$			
13. A - AlBr_3	14. (A) - CrO_4^{-2} , (B) - $\text{Cr}_2\text{O}_7^{-2}$, (C) - CrO_5 , (D) - CrO_2Cl_2			
15. X - NH_4NO_3 , Y - NH_3	16. $\text{X} \rightarrow \text{NH}_3$			
17. (A) - $\text{CuSO}_4 \cdot \text{H}_2\text{O}$,	(B) - CuSO_4 ,	(C) - CuO ,	(D) - SO_3 ,	(E) - Ag , (F) - NO_2
18. (A) - NH_4Cl ,	(B) - NH_4HCO_3 ,	(C) - NaHCO_3 ,	(D) - CO_2	
19. (A) - Ag ,	(B) - AgNO_3 ,	(C) - NO ,	(D) - AgCl	(E) - $\text{Ag}_2\text{S}_2\text{O}_3$
20. (A) - NH_3 ,	(B) - CaCO_3			
21. $\text{Bi}(\text{NO}_3)_3 + 3\text{KI} \longrightarrow \text{BiI}_3 \downarrow + 3\text{KNO}_3$,	$\text{BiI}_3 + \text{KI} \longrightarrow \text{K}[\text{BiI}_4]$			

EXERCISE-04 [B]

BRAIN STORMING SUBJECTIVE EXERCISE

1. A mixture of two white substances is soluble in water. This solution gives brown colour gas on passing chlorine gas. Another sample of solution gives white precipitate with BaCl_2 which is insoluble in concentrated HCl . The original solution of the mixture gives white precipitate with large excess of NaOH solution whose suspension is used as an antacid. After filtering off this precipitate, the filtrate was boiled with excess NaOH . solution gave a yellowish precipitate on adding NaClO_4 . One of the compound of the mixture forms alum. Identify the mixture.
2. An inorganic compound (A), transparent like glass is a strong reducing agent. Its hydrolysis in water gives a white turbidity (B). Aqueous solution of (A) gives white precipitate (C) with NaOH (aq) which is soluble in excess NaOH . (A) reduces auric chloride to produce purple of cassius. (A) also reduces I_2 and gives chromyl chloride test. Identify A, B, C & write balance reaction.
3. A unknown inorganic compound (X) gave the following reaction :
 - (i) on heating 'X' gave a residue, oxygen and oxide of nitrogen.
 - (ii) Addition of acetic acid and $\text{K}_2\text{Cr}_2\text{O}_7$ to its aqueous solution give a yellow precipitate.
 - (iii) Addition of NaOH to its aqueous solution first forms a white precipitate, Dissolve in the excess of the reagent.Identify the compound (X) and write balanced equation for step (i), (ii) & (iii).
4. A solution of white solid (A) gave white precipitate (B) with water. On treatment with HCl , the ppt. B produced A. The solution of A gives black precipitate (C) on reacting with sodium stannite and NaOH . The compound A gives a colourless gas (D) with concentrated H_2SO_4 . The gas is soluble in water and its aqueous solution produce with it precipitate with $\text{Hg}_2(\text{NO}_3)_2$ but no precipitate with $\text{Hg}(\text{NO}_3)_2$. Identify (A) to (D) and write the chemical reactions involved.
5. A mixture of three gases A, B and C is passed first into acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution when A is absorbed turnign the solution green. The remainder of the gas is passed through excess of lime water which turns milky resulting in the absorption of B. The residual gas C is absorbed by alkaline pyrogallol solution. However the original mixture does not turn lead acetate paper black. Identify A, B & C (Give necessary equations).
6. An unknown inorganic compound (X) gave the following reactions.
 - (i) The compound (X) on heating gave a residue, Oxygen and oxide of nitrogen.
 - (ii) An aqueos solution of compound (X) on addition to tap water gave a turbidity which did not dissolved in HNO_3
 - (iii) The turbidity dissolved in NH_4OH .Identify the compound (X) and give equations for the reactions (i), (ii) and (iii)
7. An unknown inorganic compound (X) loses its water of crystallisation. On heating its aqueous solution gives the following reaction :
 - (i) It gives a white turbidity with dilute HCl solution.
 - (ii) It decolourises a solution of iodine in KI .
 - (iii) It gives a white precipitate with AgNO_3 solution which turns black on standing.Identify compound (X) and give chemical equations for the reactions at step (i), (ii) & (iii).

8. A certain inorganic compound (A) on heating loses water of crystallisation. On further heating a blackish brown powder (B) and two oxides of sulphur (C & D) are obtained. The powder (B) on boiling with HCl gives a yellow solution (E). When H_2S is passed in (E) a white turbidity (F) and an apple green solution (G) is obtained. The solution (E) on treatment with thiocyanate ion gives blood red compound (H). Identify (A) to (H).
9. A black coloured compound (A) on reaction with dilute H_2SO_4 gives a gas (B) which on passing in a solution of an acid (C) gives a white turbidity (D). Gas (B) when passed in an acidified solution of a compound (E) gives a precipitate (F) soluble in dil HNO_3 . After boiling this solution when an excess of NH_4OH is added, a blue coloured compound (G) is formed. To this solution on addition of acetic acid and aqueous $\text{K}_4[\text{Fe}(\text{CN})_6]$ a chocolate precipitate (H) is obtained. On addition of an aqueous solution of BaCl_2 to an aqueous solution of (E), a white precipitate insoluble in HNO_3 is obtained. Identify (A) to (H).
10. On the basis of following reaction, Identify (A), (B), (C) & (D) and write down their chemical formulae ?
- (i) $(\text{A})_{\text{aqueous}} + \text{Zn} \xrightarrow{\text{heat}} (\text{B})_{\text{gas}}$
- (ii) $(\text{A})_{\text{aqueous}} + (\text{C}) \xrightarrow{\text{heat}} \text{PH}_3_{\text{gas}}$
- (iii) $(\text{A})_{\text{aqueous}} + \text{NH}_4\text{Cl} \xrightarrow{\text{heat}} (\text{D})_{\text{gas}}$
11. An aqueous solution of an unknown compound (X) gives the following reactions.
- (i) It gives brown precipitate with alkaline KMnO_4 solution
- (ii) It forms HCl & evolved O_2 when reacts with Cl_2 gas.
- (iii) It liberates I_2 from an acidified KI solution.
- (iv) It gives orange yellow colour with acidified titanous sulphate solution.
- Identify (X) and give the chemical equations for the reactions (i), (ii) & (iii).
12. An aqueous solution of inorganic compound (X) gives following reactions.
- (i) With an aqueous solution of BaCl_2 a precipitate insoluble in dilute HCl is obtained.
- (ii) Addition of excess of KI gives a brown appearance which turns white on addition of excess of hypo.
- (iii) With an aqueous solution of $\text{K}_4\text{Fe}(\text{CN})_6$ a chocolate coloured precipitate is obtained.
- Identify (X) and give equations for the reactions for (i), (ii) & (iii) observations.
13. An inorganic compound (X) gives brick red flame on performing the flame test. This also give the following tests :
- (i) Smell of chlorine when placed in moist air.
- (ii) If KI & CH_3COOH are added to its suspension in water, a brown colour is obtained. Identify (X) and write down equations for reactions at step (i) and (ii).
14. Two solid laboratory reagents (A) and (B) give following reactions :
- Compound : (A)
- (i) On strongly heating it gives two oxides of sulphur.
- (ii) On adding aqueous NaOH solution to its aqueous solution, a dirty green precipitate is obtained which starts turning brown on exposure to air.

Compound : (B)

(i) It imparts green colour to flame.

(ii) Its solution doesn't give precipitate on passing H_2S

(iii) When it is heated with $\text{K}_2\text{Cr}_2\text{O}_7$ & conc. H_2SO_4 , a red gas is evolved. The gas when passed in aqueous NaOH solution turns it yellow.

Identify (A) to (B) and give chemical reactions.

15. The gas liberated on heating a mixture of two salts with NaOH, give a reddish brown precipitate with an alkaline solution of K_2HgI_4 . The aqueous solution of the mixture on treatment with BaCl_2 gives a white precipitate which is sparingly soluble in concentrated HCl. On heating the mixture with $\text{K}_2\text{Cr}_2\text{O}_7$ and concentrated H_2SO_4 , red vapour (A) are produced. The aqueous solution of the mixture gives a deep blue ppt (B) with potassium ferricyanide solution. Identify the radicals.

16. When 16.8 g of white solid X was heated 4.4 g of acid gas : (A) that turned lime water milky was driven off together with 1.8 g of a gas (B) which condensed to a colourless liquid the solid that remained (Y) dissolved in water to give an alkaline solution, which with excess of BaCl_2 solution gave a white precipitate (Z). The precipitate effervesces with acid giving off CO_2 gas. Identify the compound A, B & Y and write the chemical equations for the thermal decomposition of X.

17. A metal chloride (X) shows the following reactions :

(i) When H_2S is passed in an acidified aqueous solution of (X) a black ppt is obtained.

(ii) The precipitate obtained in step (i) is not soluble in yellow ammonium sulphide.

(iii) When a solution of stannous chloride is added to an aqueous solution of (X), a white precipitate is obtained which turns grey on addition of more of stannous chloride.

(iv) When an aqueous solution of KI is added to an aqueous solution of (X), a red precipitate is obtained which dissolves on addition of excess of KI.

Identify (X) and write down the equations for the reaction at steps (i), (iii) & (iv)

18. On mixing the aqueous solutions of compounds (A) and (B), an insoluble compound (C) is produced along with another water soluble compound (D). Compound (A) on heating gives brown NO_2 gas with a cracking noise. An aqueous solution of compound (A) gives black ppt. With H_2S gas. Compound (A) also gives white ppt. with dil. HCl which is soluble in hot water and reappears on cooling. The hot water extract of compound (A) gives yellow ppt. with K_2CrO_4 solution.

19. A Colourless crystalline compound (A) is warmed with Al and NaOH solution gives a gas which produces fumes with HCl, brown ppt. when passed through Nessler's reagent and is oxidised to a colourless gas when passed over heated CuO. The latter does not support combustion, however, Mg continues burning in it producing white solid. The compound (A) when heated alone gives a brown coloured gas and another gas (B) which is essential for living beings, leaving behind a yellow solid (C). The solid (C) gives the following reactions.

(i) It dissolves in dil. HNO_3 giving a colourless solution which gives white ppt. on addition of dilute HCl which is soluble in hot and reappears on cooling.

(ii) When heated in presence of air, the yellow solid (C) changes to red powder. Identify (A), (B) and (C), giving the equations involved.

20. An inorganic halide (A) gives the following reactions :

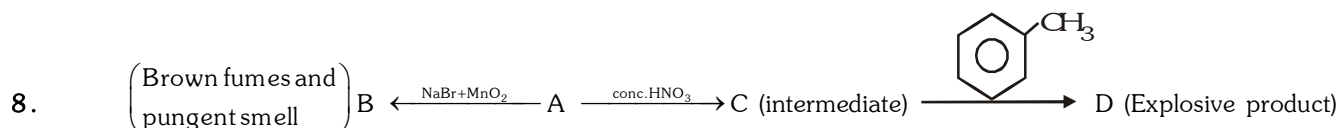
- (i) The cation of (A) on reaction with H_2S in HCl medium, gives a black ppt. of (B). (A) neither gives ppt. with HCl nor blue colour with $\text{K}_4\text{Fe}(\text{CN})_6$.
- (ii) (B) on heating with dil. HCl gives back compound (A) and a gas (C) which gives a black ppt. With lead acetate solution.
- (iii) The anion of (A) gives chromyl chloride test.
- (iv) (B) dissolves in hot dil. HNO_3 to give a solution, (D). (D) gives ring test.
- (v) When NH_4OH solution is added to (D), a white precipitate (E) is formed. (E) dissolves in minimum amount of dil. HCl to give a solution of (A). Aqueous solution of (A) on addition of water gives a whitish turbidity (F).
- (vi) Aqueous solution of (A) on warming with alkaline sodium stannite gives a black precipitate of a metal (G) and sodium stannate. The metal (G) dissolves in hydrochloric acid to give solution of A.

Identify (A) to (G) and give balanced chemical equations of reactions

CONCEPTUAL SUBJECTIVE EXERCISE			ANSWER KEY	EXERCISE -4(B)
1. Mixture consists - K_2SO_4 and MgBr_2				
2. (A) - SnCl_2	(B) - $\text{Sn}(\text{OH})\text{Cl}$	(C) - $\text{Sn}(\text{OH})_2$	3. $\text{X} \rightarrow \text{Pb}(\text{NO}_3)_2$	
4. (A) - BiCl_3	(B) - BiOCl	(C) - Bi	(D) - H_2SO_4	
5. (A) - SO_2	(B) - CO	(C) - O_2		
6. $\text{X} \rightarrow \text{AgNO}_3$				
7. $\text{X} \rightarrow \text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$				
8. (A) - $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$	(B) - Fe_2O_3	(C) - SO_2	(D) - SO_3	
(E) - FeCl_3	(F) - S	(G) - FeCl_2	(H) - $\text{Fe}(\text{CNS})_3$	
9. (A) - FeS	(B) - H_2S	(C) - HNO_3	(D) - S	
(E) - CuSO_4	(F) - CuS	(G) - $[\text{Cu}(\text{NH}_3)_4](\text{NO}_3)_2$	(H) - $\text{Cu}_2[\text{Fe}(\text{CN})_6]$	
10. (A) - NaOH	(B) - H_2	(C) - Phosphorous (P_4)	(D) - NH_4Cl	
11. $\text{X} \rightarrow \text{H}_2\text{O}_2$	12. $\text{X} \rightarrow \text{CuSO}_4$		13. $\text{X} \rightarrow \text{CaOCl}_2$	
14. (A) - FeSO_4	(B) - BaCl_2			
15. (A) - CrO_2Cl_2	(B) - $\text{Fe}_3[\text{Fe}(\text{CN})_6]_2$			
16. (A) - CO_2	(B) - H_2O	$\text{X} \rightarrow \text{Na}_2\text{CO}_3$		
17. $\text{X} \rightarrow \text{HgCl}_2$				
18. (A) - $\text{Pb}(\text{NO}_3)_2$	(B) - FeSO_4	(C) - PbSO_4	(D) - $\text{Fe}(\text{NO}_3)_2$	
19. (A) - $\text{Pb}(\text{NO}_3)_2$	(B) - O_2	(C) - Pb_3O_4		
20. (A) - BiCl_3	(B) - Bi_2S_3	(C) - H_2S	(D) - $\text{Bi}(\text{NO}_3)_3$	
(E) - $\text{Bi}(\text{OH})_3$	(F) - BiOCl	(G) - Bi		

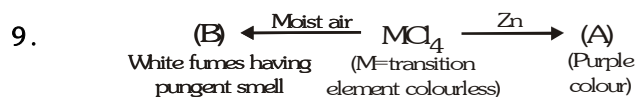
EXERCISE-05**PREVIOUS YEARS QUESTIONS****SUBJECTIVE QUESTIONS**

1. An aqueous blue coloured solution of a transition metal sulphate reacts with H_2S in acidic medium to give a black precipitate (A) which is insoluble in warm aqueous solution of KOH . The blue solution on treatment with KI in weakly acidic medium turns yellow and produces a white precipitate (B). Identify the transition metal ion. Write the chemical reaction involved in the formation of (A) and (B). [JEE 2000]
2. (i) A powdered substance (A) on treatment with fusion mixture gives a green coloured compound (B).
(ii) The solution of (B) in boiling water on acidification with dilute H_2SO_4 gives a pink coloured compound (C)
(iii) The aqueous solution of (A) on treatment with NaOH and Br_2 water gives a compound (D).
(iv) A solution of (D) in conc. HNO_3 on treatment with lead peroxide at boiling temperature produced a compound (E) which was of the same colour as that of (C).
(v) A solution of (A) in dilute HCl on treatment with a solution of barium chloride gave a white precipitate of compound (F) which was insoluble in conc. HNO_3 and conc. HCl .
Identify (A) to (F) and give balanced chemical equations for the reactions at steps (i) to (v). [JEE -2001]
3. Identify the following :
- $$\text{Na}_2\text{CO}_3 \xrightarrow{\text{SO}_2} \text{A} \xrightarrow{\text{Na}_2\text{CO}_3} \text{B} \xrightarrow[\text{A}]{\text{Elemental S}} \text{C} \xrightarrow{\text{I}_2} \text{D}$$
- Also mention the oxidation state of S in all the compounds. [JEE -2002]
4. A mixture consists A (yellow solid) and B (colourless solid) which gives Lilac colour in flame.
(a) Mixture gives black precipitate C on passing H_2S (g).
(b) C is soluble in aqua-regia and on evaporation of aqua-regia and adding SnCl_2 gives greyish black precipitate D. The salt solution with NH_4OH gives a brown precipitate.
(i) The sodium extract of the salt with $\text{CCl}_4/\text{FeCl}_3$ gives a violet layer.
(ii) The sodium extract gives yellow precipitate with AgNO_3 solution which is insoluble in NH_3 .
Identify A and B, and the precipitates C and D. [JEE -2003]
5. Dimethyl glyoxime is added to alcoholic solution of NiCl_2 . When ammonium hydroxide is slowly added to it a rosy red precipitate of a complex appears.
(i) Give the structure of complex showing hydrogen bonds
(ii) Give oxidation state and hybridization of central metal ion.
(iii) Identify whether it is paramagnetic or diamagnetic. [JEE -2004]
6. There are two ores (A_1) and (A_2) of metal (M). When ore (A_1) is calcinated a black solid (S) is obtained along with the liberation of CO_2 and water. The ore (A_1) on treatment with HCl and KI gives a precipitate (P) and iodine is liberated. Another ore (A_2) on roasting gives a gas (G) and metal (M) is set free. When gas (G) is passed through $\text{K}_2\text{Cr}_2\text{O}_7$ it turns green. Identify (M), (A_1), (A_2), (S), (P) and (G). [JEE -2004]
7. $\text{Fe}^{3+} \xrightarrow{\text{SCN}^- (\text{excess})} \text{blood red (A)} \xrightarrow{\text{F}^- (\text{excess})} \text{colourless (B)}$
Identify (A) and (B)
(a) Write IUPAC name of A and B.
(b) Find out spin only magnetic moment of B [JEE -2005]



Find A, B, C and D. Also write equations A to B and A to C.

[JEE -2001]



Identify the metal M and hence MCl_4 . Explain the difference in colours of MCl_4 and A. [JEE -2005]

10. During the qualitative analysis of a mixture containing Cu^{2+} and Zn^{2+} ions, H_2S gas is passed through an acidified solution containing these ions in order to test Cu^{2+} alone. Explain. [IIT -98, 2M]

11. Write the chemical reactions associated with the 'brown ring test'. [JEE -2000]

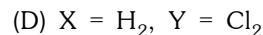
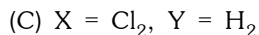
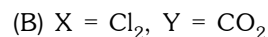
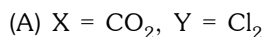
12. Write the chemical reaction associated with the 'borax bead test' of cobalt (II) oxide. [JEE -2000, 3M]

13. A white substance A reacts with dilute H_2SO_4 to produce a colourless gas B and a colourless solution C. The reaction between B and acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution produces a green solution and a slightly coloured precipitate D. The substance D burns in air to produce a gas E which reacts with B to yield D and a colourless liquid. Anhydrous copper sulphate is turned blue on addition of this colourless liquid. Addition of aqueous NH_3 or NaOH to C produces first a precipitate, which dissolves in the excess of the respective reagent to produce a clear solution in each case. Identify A, B, C, D and E. Write the equations of the reactions involved. [JEE-2001 10M]

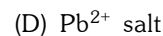
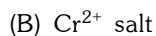
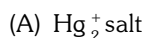
14. When a white crystalline compound X is heated with $\text{K}_2\text{Cr}_2\text{O}_7$ and concentrated H_2SO_4 , a reddish brown gas A is evolved. On passing A into caustic soda solution, a yellow coloured solution B is obtained. Neutralizing the solution of B with acetic acid and on subsequent addition of lead acetate a yellow precipitate C is obtained. When X is heated with NaOH solution, colourless gas is evolved and on passing this gas into K_2HgI_4 solution, a reddish brown precipitate D is formed. Identify A, B, C, D and X. Write the equations of reactions involved. [JEE -2002 5M]

MCQ's WITH ONE CORRECT ANSWER

15. A gas 'X' is passed through water to form a saturated solution. The aqueous solution on treatment with silver nitrate gives a white precipitate. The saturated aqueous solution also dissolves magnesium ribbon with evolution of a colourless gas 'Y'. Identify 'X' and 'Y': [JEE -2002]



16. An aqueous solution of a substance gives a white precipitate on treatment with dilute hydrochloric acid, which dissolves on heating. When hydrogen sulphide is passed through the hot acidic solution, a black precipitate is obtained. The substance is a - [JEE -2000]

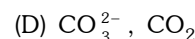
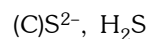
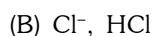
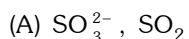


17. $[\text{X}] + \text{H}_2\text{SO}_4 \longrightarrow [\text{Y}]$ a colourless gas with irritating smell



[X] and [Y] are -

[JEE -2003]



18. A sodium salt of an unknown anion when treated with MgCl_2 gives white precipitate only on boiling. The anion is - [IIT -2004]
 (A) SO_4^{2-} (B) HCO_3^- (C) CO_3^{2-} (D) NO_3^-
19. A metal nitrate reacts with KI to give a black precipitate which on addition of excess of KI convert into orange colour solution. The cation of metal nitrate is - [IIT -2005]
 (A) Hg^{2+} (B) Bi^{3+} (C) Pb^{2+} (D) Cu^+
20. A solution when diluted with H_2O and boiled, it gives a white precipitate, On addition of excess $\text{NH}_4\text{Cl}/\text{NH}_4\text{OH}$, the volume of precipitate decreases leaving behind a white gelatinous precipitate. Identify the precipitate which dissolves in $\text{NH}_4\text{OH}/\text{NH}_4\text{Cl}$: [IIT -2006]
 (A) $\text{Zn}(\text{OH})_2$ (B) $\text{Al}(\text{OH})_3$ (C) $\text{Mg}(\text{OH})_2$ (D) $\text{Ca}(\text{OH})_2$
21. CuSO_4 decolourises on addition of KCN, the product is - [IIT -2006]
 (A) $[\text{Cu}(\text{CN})_4]^{2-}$
 (B) Cu^{2+} get reduced to form $[\text{Cu}(\text{CN})_4]^{3-}$
 (C) $\text{Cu}(\text{CN})_2$
 (D) CuCN
22. Aqueous solution of $\text{Na}_2\text{S}_2\text{O}_3$ on reaction with Cl_2 gives :- [IIT -2008]
 (A) $\text{Na}_2\text{S}_4\text{O}_6$ (B) NaHSO_4 (C) NaCl (D) NaOH

MCQ's WITH ONE OR MORE THAN ONE CORRECT ANSWER

23. The reagents, NH_4Cl and aqueous NH_3 will precipitate :- [1991, 1M]
 (A) Ca^{2+} (B) Al^{3+} (C) Bi^{3+} (D) Mg^{2+}
 (E) Zn^{2+}
24. Which of the following statement (s) is (are) correct with reference to the ferrous and ferric ions :- [1998, 2M]
 (A) Fe^{3+} gives brown colour with potassium ferricyanide
 (B) Fe^{2+} gives blue precipitate with potassium ferricyanide
 (C) Fe^{3+} gives red colour with potassium thiocyanate
 (D) Fe^{2+} gives brown colour with ammonium thiocyanate
25. The species present in solution when CO_2 is dissolved in water are :- [JEE-2006]
 (A) CO_2 , H_2CO_3 , HCO_3^- , CO_3^{2-}
 (B) H_2CO_3 , CO_3^{2-}
 (C) HCO_3^- , CO_3^{2-}
 (D) CO_2 , H_2CO_3
26. A solution of a metal ion when treated with KI gives a red precipitate which dissolves in excess KI to give a colourless solution. Moreover, the solution of metal ion on treatment with a solution of cobalt (II) thiocyanate gives rise to a deep blue crystalline precipitate. The metal ion is - [JEE -2007]
 (A) Pb^{2+} (B) Hg^{2+}
 (C) Cu^{2+} (D) Co^{+2}
27. A solution of colourless salt H on boiling with excess NaOH produces a non-flammable gas. The gas evolution ceases after sometime. Upon addition of Zn dust to the same solution, the gas evolution restarts. The colourless salt(s) H is (are) :- [IIT -2008]
 (A) NH_4NO_3 (B) NH_4NO_2 (C) NH_4Cl (D) $(\text{NH}_4)_2\text{SO}_4$

MATCH THE COLUMN

28. Match the complexes in Column I with their properties listed in Column II. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS. [JEE-2007]

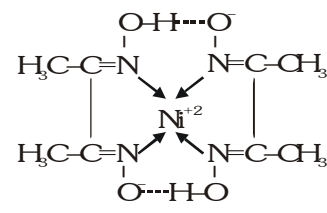
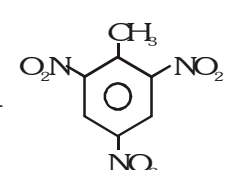
<u>Column-I</u>		<u>Column-II</u>	
(A)	$\text{O}_2^- \rightarrow \text{O}_2 \rightarrow \text{O}_2^{2-}$	(p)	redox reaction
(B)	$\text{CrO}_4^{2-} + \text{H}^+ \rightarrow$	(q)	One of the products has trigonal planar structure
(C)	$\text{MnO}_4^- + \text{NO}_2^- + \text{H}^+ \rightarrow$	(r)	Dimeric bridged tetrahedral metal ion
(D)	$\text{NO}_3^- + \text{H}_2\text{SO}_4 + \text{Fe}^{2+} \rightarrow$	(s)	disproportionation

29. **Statement-I** : $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$ is paramagnetic. [IIT -2008]

Because

Statement-II : The Fe in $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$ has three unpaired electrons.

- (A) Statement-I is True, Statement-II is True; Statement-II is a correct explanation for Statement-I
(B) Statement-I is True, Statement-II is True; Statement-II is **NOT** a correct explanation for Statement-I
(C) Statement-I is True, Statement-II is False
(D) Statement-I is False, Statement-II is True

CONCEPTUAL SUBJECTIVE EXERCISE		ANSWERS		EXERCISE - 5	
1. (A) - CuS		(B) - Cu ₂ I ₂			
2. (A) - MnSO ₄		(B) - Na ₂ MnO ₄		(C) - NaMnO ₄	
(E) - HMnO ₄		(D) - MnO ₄			
(F) - BaSO ₄					
3. A - NaHSO ₃		(B) - Na ₂ SO ₃		(C) - Na ₂ S ₂ O ₃	
				(D) - Na ₂ S ₄ O ₆	
4. (A) - HgI ₂		(B) - KI		(C) - HgS	
				(D) - Hg	
 (ii) Hybridisation - dsp², (iii) Magnetic moment = 0, Diamagnetic					
5. (i)					
6. A ₁ - CuCO ₃ ·Cu(OH) ₂ (Malachite)		A ₂ - Cu ₂ S (Copper glance)			
S - CuO		P - Cu ₂ I ₂		G - SO ₂ ↑	
7. A - [Fe(SCN)(H ₂ O) ₅] ²⁺ (Pentaaquathiocyanato-S-iron (III) ion), Magnetic moment = $\sqrt{35}$					
B - [FeF ₆] ³⁻ (hexafluoroferrate (III) ion), Magnetic moment = $\sqrt{35}$					
8. (A) - conc. H ₂ SO ₄					
(B) - Br ₂		(C) - NO ₂ ⁺		(D) -  (Explosive)	
9. (A) - [Ti(H ₂ O) ₆] ³⁺					
(B) - HCl, MCl ₄ - TiCl ₄ (Purple colour of [Ti(H ₂ O) ₆] ³⁺ is due to d-d transition)					
10. Hint :- K _{sp} (solubility product) of CuS is less than K _{sp} of ZnS.					
11. Hint :- [Fe(NO)SO ₄]					
(Brown ring) (Ferrous nitroso sulphate)					
12. Hint :- CO(BO ₂) ₂ [Cobalt metaborate (Blue colour)]					
13. (A) - ZnS		(B) - H ₂ S		(C) - ZnSO ₄	
				(D) - S	
				(E) - SO ₂	
14. X → NH ₄ Cl,		(A) - CrO ₂ Cl ₂		(B) - Na ₂ CrO ₄	
				(C) - PbCrO ₄	
				(D) - NH ₄ (HgO)HgI	
15. (C)		16. (D)		17. (A)	
				18. (B)	
				19. (B)	
20. (A)		21. (D)		22. (B)	
				23. (B), (C)	
24. (B), (C)					
25. (A)		26. (B)		27. (A,B)	
28. (A) - p,s (B) - r, (C) - p, q, (D) - p, s					
29. (A)					

EXERCISE - 01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. All samples of carbon dioxide contain carbon and oxygen in the mass ratio of 3 : 8. This is in agreement with the law of :
(A) conservation of mass (B) constant proportion
(C) multiple proportions (D) gaseous volumes
2. That the atom is indivisible was proposed by :
(A) Rutherford (B) Dalton (C) Bohr (D) Einstein
3. Atomic mass of an element is :
(A) actual mass of one atom of the element
(B) relative mass of an atom of the element
(C) average relative mass of different atoms of the element
(D) always a whole number
4. Which of the following expressions is correct (n = no. of moles of the gas, N_A = Avogadro constant, m = mass of 1 molecule of the gas, N = no. of molecules of the gas) ?
(A) $n = mN_A$ (B) $m = N_A$ (C) $N = nN_A$ (D) $m = n/N_A$
5. The volume of 1 mol of a gas at standard temperature and pressure is :
(A) 11.2 litres (B) 22.4 litres (C) 100 litres (D) none of these
6. The charge on 1 gram ions of Al^{3+} is :
(A) $\frac{1}{27} N_A e$ coulomb (B) $\frac{1}{3} N_A e$ coulomb (C) $\frac{1}{9} N_A e$ coulomb (D) $3 N_A e$ coulomb
7. In which of the following pairs do 1 g of each have an equal number of molecules :
(A) N_2O and CO (B) N_2 and C_3O_2 (C) N_2 and CO (D) N_2O and C_2O
8. A quantity of aluminium has a mass of 54.0 g. What is the mass of the same number of magnesium atoms ? (At. wt. $Al = 27$, $Mg = 24$)
(A) 12.1 g (B) 23.3 g (C) 48 g (D) 97.2 g
9. Which of the following samples contains the largest number of atoms ?
(A) 1 g of $Ni(s)$ (B) 1 g of $Ca(s)$ (C) 1 g of $N_2(g)$ (D) 1 g of $B(s)$
10. Which of the following contains greatest number of oxygen atoms :
(A) 1 g of O (B) 1 g of O_2
(C) 1 g of O_3 (D) all have the same number of atoms
11. A sample of ammonium phosphate, $(NH_4)_3PO_4$, contains 3.18 mol of hydrogen atoms. The number of moles of oxygen atoms in the sample is :
(A) 0.265 (B) 0.795 (C) 1.06 (D) 3.18
12. How many moles of electron weigh one kilogram :
(A) 6.023×10^{23} (B) $\frac{1}{9.108} \times 10^{31}$ (C) $\frac{6.023}{9.108} \times 10^{54}$ (D) $\frac{1}{9.108 \times 6.023} \times 10^8$
13. A compound was found to contain 5.37% nitrogen. What is the minimum molecular wt. of compound :
(A) 26.07 g (B) 2.607 (C) 260.7 (D) none

14. Under the same conditions, two gases have the same number of molecule. They must :
 (A) be noble gases (B) have equal volumes
 (C) have a volume of 22.4 dm³ each (D) have an equal number of atoms
15. Four 1 litre flasks are separately filled with the gases H₂, He, O₂ and O₃ at the same temperature and pressure. The ratio of total number of atoms of these gases present in different flask would be :
 (A) 1 : 1 : 1 : 1 (B) 1 : 2 : 2 : 3 (C) 2 : 1 : 2 : 3 (D) 3 : 2 : 2 : 1
16. The atomic weight of Cu is 63.546. There are only two naturally occurring isotopes of copper ⁶³Cu and ⁶⁵Cu. The natural abundance of the ⁶³Cu isotope must be approximately :
 (A) 10% (B) 30% (C) 50% (D) 70%
17. If the percentage of water of crystallization in MgSO₄·x H₂O is 13%. What is the value of x :
 (A) 1 (B) 4 (C) 5 (D) 7
18. A pure gas that is 14.3% hydrogen and 85.7% carbon by mass has a density of 2.5 g L⁻¹ at 0 C and 1 atm pressure. What is the molecular formula of the gas :
 (A) CH₂ (B) C₂H₄ (C) C₄H₈ (D) C₆H₁₂
19. A certain alkaloid has 70.8% carbon, 6.2% hydrogen, 4.1% nitrogen and the rest oxygen. What is its empirical formula :
 (A) C₂₀H₂₁NO₄ (B) C₂₀H₂₀NO₄ (C) C₂₁H₂₀NO₃ (D) C₂₀H₁₉NO₃
20. The empirical formula of a compound of molecular mass 120 is CH₂O. The molecular formula of the compound is :
 (A) C₂H₄O₂ (B) C₄H₈O₄ (C) C₃H₆O₃ (D) all of these
21. 0.250 g of an element M, reacts with excess fluorine to produce 0.547 g of the hexafluoride MF₆. What is the element : [Cr = 52, Mo = 96, S = 32, Te = 127.6]
 (A) Cr (B) Mo (C) S (D) Te
22. A 1000 gram sample of NaOH contains 3 moles of O atoms, what is the % purity of NaOH :
 (A) 14% (B) 100% (C) 12% (D) 24%
23. A 15 mL sample of 0.20 M MgCl₂ is added to 45 mL of 0.40 M AlCl₃, What is the molarity of Cl⁻ions in the final solution:
 (A) 1.0 M (B) 0.60 M (C) 0.35 M (D) 0.30 M
24. Mole fraction of ethanol in ethanol and water mixture is 0.25. Hence percentage concentration of ethanol by weight of mixture is :
 (A) 25% (B) 75% (C) 46% (D) 54%
25. How many moles of Na⁺ ions are in 20 mL of 0.40 M Na₃PO₄ :
 (A) 0.0080 (B) 0.024 (C) 0.050 (D) 0.20
26. Out of Molarity (M), Molality (m), Formality (F) and Mole fraction (x), those independent of temperature are :
 (A) M,m (B) F,x (C) m,x (D) M,x
27. The molality of a 1 L solution with x % H₂SO₄ is 9. The weight of solvent present in the solution is 910 grams. The value of x is :
 (A) 90 (B) 80.3 (C) 30.38 (D) 46.87
28. Density of ozone relative to oxygen is under the same temperature & pressure :
 (A) 1 (B) 2 (C) 1.5 (D) 2.5

29. Mole fraction of A in H_2O is 0.2. The molality of A in H_2O is :
(A) 13.8 (B) 15.5 (C) 14.5 (D) 16.8
30. The molarity of the solution containing 2.8% mass-volume solution of KOH/(Given atomic mass of K = 39) is :
(A) 0.1 M (B) 0.5 M (C) 0.2 M (D) 1 M
31. The molality of a sulphuric acid solution is 0.2 mol/kg. Calculate the total weight of the solution :
(A) 1000 g (B) 1098.6 g (C) 980.4 g (D) 1019.6 g
32. What volume of a 0.8 M solution contains 100 millimoles of the solute :
(A) 100 mL (B) 125 mL (C) 500 mL (D) 62.5 mL
33. 500 mL of a glucose solution contains 6.02×10^{22} molecules. The concentration of the solution is :
(A) 0.1 M (B) 1.0 M (C) 0.2 M (D) 2.0 M
34. 50 mL of CO is mixed with 20 mL of oxygen and sparked, After the reaction, the mixture is treated with an aqueous KOH solution. Choose the correct option :
(A) the volume of CO that reacts = 30 mL
(B) volume of CO_2 formed = 50 mL
(C) volume of CO that remains after treatment with KOH = 10 mL
(D) the volume of the CO that remains after treatment with KOH = 20 mL

[illegible]

EXERCISE-02**BRAIN TEASERS****SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)**

1. An alloy of gold and silver contains 38.5% silver by mass and has a density of 14.6 g.mL^{-1} . What is the molar concentration of silver in this alloy :
(A) 52.1 mol.L^{-1} (B) 45.6 mol.L^{-1} (C) 3.57 mol.L^{-1} (D) 2.64 mol.L^{-1}
2. "Suvarnabhasm", an ayurvedic drug, is found to contain 400 ppm of colloidal gold. Mass % of gold (atomic mass of Au = 197) will be :
(A) 0.040 % (B) 7.88 % (C) 0.0788 % (D) $4 \times 10^{-4} \%$
3. A solution containing 12.0% sodium hydroxide by mass has a density of 1.131 g/mL . What volume of this solution contains 5.00 mol of NaOH :
(A) 0.0240 L (B) 1.67 L (C) 1.47 L (D) 1.00 L
4. An aqueous solution of concentrated hydrobromic acid contains 48% HBr by mass. If the density of the solution is 1.50 g / mL , what is its concentration :
(A) 11.4 mol/L (B) 8.9 mol/L
(C) 5.9 mol/L (D) 18.5 mol/L
5. An antifreeze mixture consists of 40% ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) by weight in aqueous solution. If the density of this solution is 1.05 g/mL , what is the molar concentration :
(A) 6.77 M (B) 6.45 M (C) 0.017 M (D) 16.9 M
6. What is the molality m, of methanol in a solution prepared by dissolving 160 g of methanol, CH_3OH , in 200.0 g of water :
(A) 1.0 m (B) 5.0 m (C) 10.0 m (D) 25.0 m
7. XeF_6 fluorinates I_2 to IF_7 and liberates Xenon (g). 210 mmol of XeF_6 can yield a maximum of ___ mmol of IF_7 :
(A) 420 (B) 180 (C) 210 (D) 245
8. When 100 g of ethylene polymerises entirely to polyethene, the weight of polyethene formed as per the equation $n\text{CH}_2 = \text{CH}_2 \rightarrow (\text{CH}_2 - \text{CH}_2)_n$ is :
(A) $(n/2)\text{g}$ (B) 100g (C) $(100/n)\text{g}$ (D) $100n \text{ g}$
9. A 10 gram sample of natural gas containing CH_4 and C_2H_4 was burnt in excess of oxygen to give 29.0 grams of CO_2 and some water. How many grams of water are formed :
(A) 9.42 g (B) 18.81 g (C) 11.42 g (D) 15.31 g
10. 50 g of ZnS are strongly heated in air to effect partial oxidation and the resultant mass weighed 44 g. What is the ratio of ZnO to ZnS in the resultant mixture :
(A) 13.5 : 30.5 (B) 27 : 12.58 (C) 27 : 15.31 (D) 30.52 : 13.48
11. The % loss in weight after heating a pure sample of potassium chlorate (M. wt. 122.5) will be :
(A) 12.25 (B) 24.50
(C) 39.17 (D) 49.0
12. In a gaseous reaction of the type $a\text{A} + b\text{B} \rightarrow c\text{C} + d\text{D}$, which is wrong :
(A) a litre of A combines with b litre of B to give C & D
(B) a mole of A combines with b mole of B to give C & D
(C) a g of A combines with b g litre of B to give C & D
(D) a molecules of A combines with b molecule of B to give C & D

13. Iodobenzene (C_6H_5I) is prepared from aniline ($C_6H_5NH_2$) in a two step process as shown below
 $C_6H_5NH_2 + HNO_2 + HCl \rightarrow C_6H_5N_2^+Cl^- + 2H_2O$ $C_6H_5N_2^+Cl^- + KI \rightarrow C_6H_5I + N_2 + KCl$
 In an actual preparation 9.30 g of aniline was converted to 16.32 g of iodobenzene. The percentage yield of iodobenzene is :
 (A) 8 % (B) 50 % (C) 75 % (D) 80 %
14. In an organic compound of molar mass greater than 100 containing only C, H and N, the percentage of C is 6 times the percentage of H while the sum of the percentages of C and H is 1.5 times the percentage of N. What is the least molar mass :
 (A) 175 (B) 140 (C) 105 (D) 210
15. The simplest formula of a compound containing 50% of element X (atomic mass = 10) and 50% of the element Y (atomic mass = 20) by weight is :
 (A) XY (B) X_2Y (C) XY_2 (D) X_2Y_3
16. The sodium salt of methyl orange has 7% sodium. What is the minimum molecular weight of the compound? :
 (A) 420 (B) 375 (C) 329 (D) 295
17. In the preceding problem, if the compound contains 12.8% nitrogen 9.8% sulphur how many nitrogen and sulphur atoms are present per atom of sodium :
 (A) 2 and 1 (B) 1 and 3 (C) 1 and 2 (D) 3 and 1
18. How many grams of urea on heating yield 10^{22} molecules of biuret by the reaction :

$$2 CO (NH_2)_2 \longrightarrow H_2N - CO - NH - CO - NH_2 + NH_3 ?$$

 (A) 1.495 (B) 0.995 (C) 1.99 (D) 1.753
19. A certain compound has the molecular formula X_4O_6 . If 10 g of X_4O_6 has 5.72 g of X, atomic mass of X is :
 (A) 32 amu (B) 37 amu (C) 42 amu (D) 98 amu
20. DNA has density 1.1 g/mL and its molecular weight is 6×10^3 g/mol. Average volume occupied by its single molecule will be :
 (A) 9.1×10^{-20} cc (B) 9.1×10^{-21} cc
 (C) 9.8×10^{-21} cc (D) 9.6×10^{-20} cc
21. For an infinitely dilute aqueous solution molality will be equal to :
 (A) formality (B) molarity (C) mole fraction (D) ppm
22. If 1 g of HCl and 1 g of MnO_2 heated together the maximum weight of Cl_2 gas evolved will be :
 $[MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O]$:
 (A) 2 g (B) 0.975 g (C) 0.486 g (D) 0.972 g
23. Molarity of H_2SO_4 is 18 M. Its density is 1.8 g/cm^3 , hence molality is :
 (A) 18 (B) 100 (C) 36 (D) 500
24. If $1\frac{1}{2}$ moles of oxygen combine with Al to form Al_2O_3 , the weight of Al used in the reaction is (Al = 27):
 (A) 27 g (B) 54 g (C) 40.5 g (D) 81 g
25. $Na_2SO_4 \cdot xH_2O$ has 50% H_2O . Hence, x is :
 (A) 4 (B) 5 (C) 6 (D) 8

EXERCISE-03**MISCELLANEOUS TYPE QUESTIONS****TRUE / FALSE**

- Two elements always combine together in a fixed ratio by weight.
- There is no difference between one mole and one gram molecule.
- Equal volumes of different gases under similar conditions of temperature and pressure contain equal number of atoms.
- Empirical formula represents the actual number of atoms present in a molecule of the substance.
- A balanced equation contains equal number of atoms of each element on both sides of the equation.
- Mass of reactants is always equal to the mass of the products.

FILL IN THE BLANKS

- Law of conservation of mass was put forward by
- The formation SO_2 and SO_3 illustrates the law of
- The number of atoms present in one molecule of an elementary substance is called its
- The mass of an atom of carbon is
- The reactant which reacts completely in a reaction is called
- According to Dulong and Petit's law, Atomic wt. Specific heat = (approx)
- The SI unit of density is

MATCH THE COLUMN

1.	Column-I	Column-II
(A)	Law of conservation of mass	(p) CH_4 has carbon and hydrogen in 3 : 1 mass ratio.
(B)	Law of multiple proportion	(q) 10 mL N_2 combines with 30 mL of H_2 to form 20 mL of NH_3
(C)	Law of definite proportion	(r) S and O_2 combine to form SO_2 and SO_3
(D)	Law of reciprocal proportion	(s) In H_2S and SO_2 mass ratio of H and O w.r.t. sulphur is 1 : 16, hence in H_2O , mass ratio of H and O is 1 : 8.
(E)	Gay Lussac's Law	(t) 4.2 g MgCO_3 gives 2.0 g residue on heating.

2.	Column-I	Column-II (mass of product)
(A)	$2\text{H}_2 + \text{O}_2 \longrightarrow 2\text{H}_2\text{O}$ 1g 1g	(p) 1.028 g
(B)	$3\text{H}_2 + \text{N}_2 \longrightarrow 2\text{NH}_3$ 1g 1g	(q) 1.333 g
(C)	$\text{H}_2 + \text{Cl}_2 \longrightarrow 2\text{HCl}$ 1g 1g	(r) 1.125 g
(D)	$2\text{H}_2 + \text{C} \longrightarrow \text{CH}_4$ 1g 1g	(s) 1.214 g

ASSERTION & REASON

These questions contains, Statement I (assertion) and Statement II (reason).

(A) Statement-I is true, Statement-II is true ; Statement-II is correct explanation for Statement-I.

(B) Statement-I is true, Statement-II is true ; Statement-II is NOT a correct explanation for statement-I

(C) Statement-I is true, Statement-II is false

(D) Statement-I is false, Statement-II is true

1. **Statement-I** : 16 g each O_2 and O_3 contains $\frac{N_A}{2}$ and $\frac{N_A}{3}$ atoms respectively.

Because

Statement-II : 16 g O_2 and O_3 contains same no. of atoms.

2. **Statement-I** : 44 g of CO_2 , 28 g of CO have same volume at STP.

Because

Statement-II : Both CO_2 and CO are formed by C and oxygen.

3. **Statement-I** : Law of conservation of mass hold good for nuclear reaction.

Because

Statement-II : Law states that mass can be neither created nor destroyed in a chemical reaction.

4. **Statement-I** : A reactant that is entirely consumed when a reaction goes to completion is known as limiting reactant.

Because

Statement-II : The amount of reactant limits the amount of product formed.

5. **Statement-I** : The balancing of chemical equations is based on law of conservation of mass.

Because

Statement-II : Total mass of reactants is equal to total mass of products.

6. **Statement-I** : Pure water obtained from different sources such as, river, well, spring, sea etc. always contains hydrogen and oxygen combined in the ratio 1 : 8 by mass.

Because

Statement-II : A chemical compound always contains elements combined together in same proportion by mass, it was discovered by French chemist, Joseph Proust (1799).

7. **Statement-I** : The percentage weight of a compound A in a solution is given by

$$\% \text{ of A} = \frac{\text{Mass A}}{\text{Total mass of solution}} \times 100$$

Because

Statement-II : The mole fraction of a component A is given by,

$$\text{Mole fraction of A} = \frac{\text{No. of moles of A}}{\text{Total no. of moles of all components}}$$

8. **Statement-I** : A one molal solution prepared at 20 C will retain the same molality at 100 C, provided there is no loss of solute or solvent on heating.

Because

Statement-II : Molality is independent of temperature.

9. **Statement-I** : The molality and molarity of dilute aqueous solutions differ very little.

Because

Statement-II : The density of water is 1.0 g cm^{-3} at room temperature.

10. **Statement-I** : The average mass of one Mg atom is 24.305 amu, which is not the actual mass of one Mg atom.

Because

Statement-II : Three isotopes, ^{24}Mg , ^{25}Mg and ^{26}Mg , of Mg are found in nature.

11. **Statement-I** : A molecule of butane, C_4H_{10} has a mass of 58.12 amu.
Because
Statement-II : One mole of butane contains 6.022×10^{23} molecules and has a mass of 58.12 g.
12. **Statement-I** : Both 12 g. of carbon and 27 g. of aluminium will have 6.02×10^{23} atoms.
Because
Statement-II : Gram atomic mass of an element contains Avogadro's number of atoms.

COMPREHENSION BASED QUESTIONS

Comprehension # 1

Potash is any potassium mineral that is used for its potassium content. Most of the potash produced in the United States goes into fertilizer. The major sources of potash are potassium chloride (KCl) and potassium sulphate (K_2SO_4). Potash production is often reported as the potassium oxide (K_2O) equivalent or the amount of K_2O that could be made from a given mineral. KCl costs Rs. 50 per kg.

- What is the cost of K per mole of the KCl sample?
 (A) Rs. 13.42 mol^{-1} (B) Rs. 3.73 mol^{-1} (C) Rs. 1.00 mol^{-1} (D) Rs. 2.00 mol^{-1}
- For what price must K_2SO_4 be sold in order to supply the same amount of potassium as in KCl ?
 (A) Rs. 58.40 kg^{-1} (B) Rs. 50.00 kg^{-1} (C) Rs. 42.82 kg^{-1} (D) Rs. 25.00 kg^{-1}
- What mass (in kg) of K_2O contains the same number of moles of K atoms as 1.00 kg KCl?
 (A) 0.158 kg (B) 0.315 kg (C) 1.262 kg (D) 0.631 kg

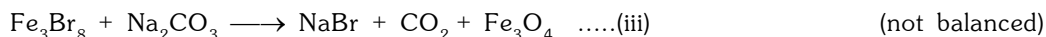
Comprehension # 2

Calcium lactate is used in the food and beverage industries. It has also been used medicinally for treatment of various allergies, for treatment of muscular leg cramps, and as an antidote for a variety of poisons, including lead, arsenicals and carbon tetrachloride. A 0.8274 g sample of anhydrous calcium lactate is found by analysis to contain 0.2732 g of C, 0.0382 g H, 0.1520 g Ca and 0.3640 g O. Each mole of calcium lactate is found to contain one mole of calcium ions. Calcium lactate can be crystallised from water as pentahydrate salt.

- Simplest formula of the calcium lactate is :
 (A) $CaO_6C_6H_{10}$ (B) $CaO_3C_3H_5$ (C) $CaO_2C_3H_3$ (D) $CaO_2C_3H_5$
- Formula weight of calcium lactate is :
 (A) 129 g mol^{-1} (B) 111 g mol^{-1} (C) 218 g mol^{-1} (D) 113 g mol^{-1}
- How many grams of calcium lactate pentahydrate would be recovered from 1 g of anhydrous salt :
 (A) 1.41 g (B) 1.00 g (C) 1.27 g (D) 1.51 g

Comprehension # 3

NaBr, used to produce AgBr for use in photography can be self prepared as follows :



How much Fe in kg is consumed to produce 2.06×10^3 kg NaBr $\dots(iv)$

- Mass of iron required to produce 2.06×10^3 kg NaBr
 (A) 420 g (B) 420 kg (C) 4.2×10^5 kg (D) 4.2×10^8 g
- If the yield of (ii) is 60% & (iii) reaction is 70% then mass of iron required to produce 2.06×10^3 kg NaBr.
 (A) 10^5 kg (B) 10^5 g (C) 10^3 kg (D) none

3. If yield of (iii) reaction is 90% then mole of CO_2 formed when $2.06 \times 10^3 \text{ kg}$ NaBr is formed.
 (A) 20 (B) 10 (C) 40 (D) none

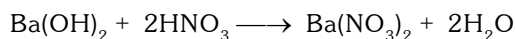
Comprehension # 4

A monobasic acid of weight 15.5 g is heated with excess of oxygen & evolved gases when passed through KOH solution increased its weight by 22 g and when passed through anhydrous CaCl_2 , increased its weight by 13.5 g. When the same mass of this organic acid is reacted with excess of silver nitrate solution form 41.75 g silver salt of the acid which on ignition gave the residue of weight 27 g.

- The molecular formula of the organic acid is.
 (A) C_2H_6 (B) $\text{C}_2\text{H}_5\text{O}_2$ (C) $\text{C}_2\text{H}_6\text{O}_2$ (D) $\text{C}_2\text{H}_4\text{O}$
- The molar masses of the acid & its silver salt respectively are:
 (A) 60, 168 (B) 167, 60 (C) 60, 167 (D) 168, 60

Comprehension # 5

342 g of 20% by mass of Ba(OH)_2 solution (sp. gr. 0.57) is reacted with 200 mL of 2 M HNO_3 according to given balanced reaction :



- The nature of the final solution is :
 (A) acidic (B) neutral (C) basic (D) can't say
- Find the molarity of the ion in resulting solution by which nature of the above solution is identified, is
 (A) 0.5 M (B) 0.8 M (C) 0.4 M (D) 1 M

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
True / False - F - T - F - F - T - T Fill in the Blanks - Lavoisier - Multiple proportions - Atomicity $1.99 \times 10^{-23} \text{ g}$ - Limiting reagent kg m^{-3} kg m^{-3} Match the Column (A) $\rightarrow$ t ; (B) $\rightarrow$ r ; (C) $\rightarrow$ p ; (D) $\rightarrow$ s ; (E) $\rightarrow$ q (A) $\rightarrow$ r ; (B) $\rightarrow$ s ; (C) $\rightarrow$ p ; (D) $\rightarrow$ q Assertion - Reason Questions - D - B - D - A - A - A - B - A - A - A - A - A Comprehension Based Questions Comprehension # 1: (B) (C) (D) Comprehension # 2: (A) (C) (A) Comprehension # 3: (B) (C) (B) Comprehension # 4: (C) (A) Comprehension # 5: (C) (A)		

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

1. 1.375 g of cupric oxide was reduced by heating in a current of hydrogen and the weight of copper obtained was 1.098 g. In another experiment, 1.179 g of copper was dissolved in nitric acid and the resulting solution was evaporated to dryness. The residue of copper nitrate when strongly heated was converted into 1.4476 g of cupric oxide. Show that the results are in agreement with the law of constant proportion.
2. Elements X and Y from two different compounds. In the first 0.324 g of X is combined with 0.471 g of Y. In second, 0.117 g of X is combined with 0.509 g of Y. Show that these data illustrate the law of multiple proportions.
3. How many g of element are present in 35.125 g atom of Si. (Given at. wt. of Si = 28.)
4. Calculate the no. of molecules in a drop of water weighing 0.07 g.
5. Calculate no. of each atom present in 106.5 g of NaClO_3 .
6. Find the no. of mole of phosphorus in 92.9 g of phosphorus assuming that molecular formula of phosphorus is P_4 . Also determine the no. of atoms and molecules of phosphorus in the sample.
7. Calculate the number of moles in 5.75 g of sodium. (Atomic mass of sodium = 23.)
8. How many grams of each of the following elements must be taken to get 1 mol of the element?
(a) Sodium (b) Chlorine (c) Copper
9. The density of liquid mercury is $13.6 / \text{cm}^3$. How many moles of mercury are there in 1 litre of the metal ? (Atomic mass of Hg = 200)
10. 50 g of CaCO_3 is allowed to react with 70 g of H_3PO_4 . Calculate :
(i) amount of $\text{Ca}_3(\text{PO}_4)_2$ formed (ii) amount of unreacted reagent
11. N_2H_4 , Hydrazine a rocket fuel can be produced according to the following reaction :
$$\text{ClNH}_2 + 2\text{NH}_3 \longrightarrow \text{N}_2\text{H}_4 + \text{NH}_4\text{Cl}$$

When 1000 g ClNH_2 is reacted with excess of NH_3 , 473 g N_2H_4 is produced. What is the % yield of the reaction.
12. Carbon disulphide ' CS_2 ', can be made from by product SO_2 . The overall reaction is
$$5\text{C} + 2\text{SO}_2 \longrightarrow \text{CS}_2 + 4\text{CO}$$

How much CS_2 can be produced from 450 kg of waste SO_2 with excess of coke if the SO_2 conversion is 82%.
13. Calculate the percent of BaO in 29.0 g of a mixture of BaO and CaO which just reacts with 100.8 mL of 6.00 M HCl.
$$\text{BaO} + 2\text{HCl} \rightarrow \text{BaCl}_2 + \text{H}_2\text{O}$$
$$\text{CaO} + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O}$$
14. Calculate the amount of 95% pure Na_2CO_3 required to prepare 5 litre of 0.5 M solution.
15. Calculate the molality of a sulphuric acid solution of specific gravity 1.2 containing 27% H_2SO_4 by weight.
16. A gaseous alkane is exploded with oxygen. The moles of O_2 for complete combustion and CO_2 formed is in the ratio 7 : 4. Deduce molecular formula of alkane.
17. When 2.86 g of a mixture of 1-butene, C_4H_8 and butane C_4H_{10} was burned in excess of oxygen, 8.80 g of CO_2 and 4.14 g of H_2O were obtained. What is percentage by mass of butane in the mixture.

18. If v mL of a gaseous hydrocarbon, after explosion with excess of oxygen, showed a contraction of $2.5 v$ mL and a further contraction of $2v$ mL with caustic potash, Find the formula of hydrocarbon.
19. The average mass of one gold atom in a sample of naturally occurring gold is 3.2707×10^{-22} g. Use this to calculate the molar mass of gold.
20. A plant virus is found to consist of uniform symmetrical particles of $150 \text{ \AA}$ in diameter and $5000 \text{ \AA}$ long. The specific volume of the virus is $0.75 \text{ cm}^3/\text{g}$. If the virus is considered to be a single particle, find its molecular weight.
21. Density of a gas relative to air is 1.17. Find the mol. mass of the gas [$M_{\text{air}} = 29 \text{ g/mol}$]
22. One type of artificial diamond (commonly called YAG for yttrium aluminium garnet) can be represented by the formula $\text{Y}_3\text{Al}_5\text{O}_{12}$.
- (a) Calculate the weight percentage composition of this compound.
- (b) What is the weight of yttrium present in a 200 – carat YAG if 1 carat = 200 mg ? (Y = 89, Al = 27)
23. A chemical commonly called "dioxin" has been very much in the news in the past few years. (It is the by product of herbicide manufacture and is thought to be quite toxic.) Its formula is $\text{C}_{12}\text{H}_4\text{Cl}_4\text{O}_2$. If you have a sample of dirt (28.3 g) that contains $1.0 \times 10^{-4} \%$ dioxin, how many moles of dioxin are in the dirt sample?
24. A chemist wants to prepare diborane by the reaction
- $$6 \text{ LiH} + 8 \text{ BF}_3 \longrightarrow 6 \text{ Li BF}_4 + \text{B}_2\text{H}_6$$
- If it starts with 2.0 moles each of LiH & BF_3 . How many moles B_2H_6 can be prepared.
25. One gram of an alloy of aluminium and magnesium when heated with excess of dil. HCl forms magnesium chloride, aluminium chloride and hydrogen. The evolved hydrogen collected over mercury at 0°C has a volume of 1.2 litres at 0.92 atm pressure. Calculate the composition of the alloy.
26. A 10 g sample of a mixture of calcium chloride and sodium chloride is treated with Na_2CO_3 to precipitate calcium as calcium carbonate. This CaCO_3 is heated to convert all the calcium to CaO and the final mass of CaO is 1.62 g. Calculate % by mass of NaCl in the original mixture.
27. By the reaction of carbon and oxygen, a mixture of CO and CO_2 is obtained. What is the composition of the mixture by mass obtained when 20 grams of O_2 reacts with 12 grams of carbon ?
28. The action of bacteria on meat and fish produces a poisonous compound called cadaverine. As its name and origin imply, it stinks ! It is 58.77% C, 13.81 % H, and 27.42 % N. Its molar mass is 102 g/mol. Determine the molecular formula of cadaverine.
29. Given the following empirical formulae and molecular weight, compute the true molecular formulae :
- | | Empirical formula | Molecular weight |
|-----|-----------------------|------------------|
| (a) | CH_2 | 84 |
| (b) | CH_2O | 150 |
| (c) | HO | 34 |
| (d) | HgCl | 472 |
| (e) | HF | 80 |
30. What is the percentage of nitrogen in an organic compound 0.14 g of which gave by Dumas method 82.1 c.c. of nitrogen collected over water at 27°C and at a barometric pressure of 774.5 mm? (aqueous tension of water at 27°C is 14.5 mm)

31. Calculate the molarity of the following solutions :
- 4g of caustic soda is dissolved in 200 mL of the solution.
 - 5.3 g of anhydrous sodium carbonate is dissolved in 100 mL of solution.
 - 0.365 g of pure HCl gas is dissolved in 50 mL of solution.
32. A mixture of ethanol and water contains 54 % water by mass. Calculate the mole fraction of alcohol in this solution.
33. 10 mL of a mixture of CO, CH₄, and N₂ exploded with excess of oxygen gave a contraction of 6.5 mL. There was a further contraction of 7 mL, when the residual gas treated with KOH. What is the composition of the original mixture?
34. When 100 mL of a O₂ – O₃ mixture was passed through turpentine, there was reduction of volume by 20 mL. If 100 mL of such a mixture is heated, what will be the increase in volume?
-

CONCEPTUAL SUBJECTIVE EXERCISE			ANSWER KEY		EXERCISE-4(A)		
3.	983.5 g of Si		4.	2.34 10 ²¹ molecules of H ₂ O			
5.	6.023 10 ²³ atom Na, 6.023 10 ²³ atom Cl, 18.06 10 ²³ atom O						
6.	0.75 mol, 4.52 10 ²³ molecules P ₄ , 18.04 10 ²³ atom P						
7.	0.25 mol		8.	(a) 23 g (b) 35.5 g (c) 63.5 g			
9.	68		10.	(i) 51.66 g (ii) 37.31 g			
11.	76.12%	12.	219.09 kg CS ₂	13.	65.65%	14.	278.94 g
15.	3.8	16.	C ₂ H ₆	17.	60.8 %	18.	C ₂ H ₆
19.	196.2	20.	7.09 10 ⁷	21.	33.9		
22.	(a) Y = 44.95%, Al = 22.73 %, O = 32.32%		(b) 17.98 g		23.	8.8 10 ⁻⁸ mol	
24.	0.250	25.	Al = 0.546 g, Mg = 0.454g		26.	67.9%	
27.	CO : CO ₂ =21 : 11	28.	C ₅ H ₁₄ N ₂	29.	(a) C ₆ H ₁₂ (b) C ₅ H ₁₀ O ₅ (c) H ₂ O ₂ (d) Hg ₂ Cl ₂ (e) H ₄ F ₄		
30.	66.7 %	31.	(a) 0.5M, (b) 0.5 M, (c) 0.2 M		32.	0.25	
33.	CO = 5mL, CH ₄ = 2 mL, N ₂ = 3 mL		34.	10 mL			

EXERCISE-04 [B]**BRAIN STORMING SUBJECTIVE EXERCISE**

1. A crystalline hydrated salt on being rendered anhydrous, loses 45.6% of its weight. The percentage composition of anhydrous salt is : Al = 10.5%, K = 15.1%, S = 24.8% and I = 49.6%. Find the empirical formula of the anhydrous and crystalline salt :
2. How much quantity of zinc will have to be reacted with excess of dilute HCl solution to produce sufficient hydrogen gas for completely reacting with the oxygen obtained by decomposing 5.104 g of potassium chlorate?
3. A 1.85 g sample of mixture of CuCl_2 and CuBr_2 was dissolved in water and mixed thoroughly with 1.8 g portion of AgCl. After reaction, the solid which now contain AgCl and AgBr was filtered, dried and weighed to be 2.052 g. What was the % by weight of CuBr_2 in the mixture?
4. 1.0 g of a sample containing NaCl, KCl and some inert impurity is dissolved in excess of water and treated with excess of AgNO_3 solution. A 2.0 g precipitate to AgCl separate out. also sample is 23% by mass in sodium. Determine mass percentage of KCl in the sample :
5. A mixture of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ and $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ was heated until all the water was driven-off. If 5.0 g of mixture gave 3 g of anhydrous salts, what was the percentage by mass of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ in the original mixture :
6. A compound containing Ca, C, N and S was subjected to quantitative analysis and formula mass determination. A 0.25 g of this compound was mixed with Na_2CO_3 to convert all Ca into 0.16 g CaCO_3 . A 0.115 g sample of compound was carried through a series of reactions until all its S was changed into SO_4^{2-} and precipitated as 0.344 g of BaSO_4 . A 0.712 g sample was processed to liberate all of its N as NH_3 and 0.155 g NH_3 was obtained. The formula mass was found to be 156. Determine the empirical and molecular formula of the compound :
7. A 0.2 g sample, which is mixture of NaCl, NaBr and NaI was dissolved in water and excess of AgNO_3 was added. The precipitate containing AgCl, AgBr and AgI was filtered, dried and weighed to be 0.412 g. The solid was placed in water and treated with excess of NaBr, which converted all AgCl into AgBr. The precipitate was then weighed to be 0.4881 g. It was then placed into water and treated with excess of NaI, which converted all AgBr into AgI. The precipitate was then weighed to be 0.5868 g. What was the percentage of NaCl, NaBr and NaI in the original mixture :
8. 2.5 g of a sample containing Na_2CO_3 ; NaHCO_3 and some non-volatile impurity on gentle heating loses 12% of its weight. Residue is dissolved in 100 mL water and its 10 mL portion required 15 mL 0.1 M aqueous solution of BaCl_2 for complete precipitation of carbonates. Determine mass percentage of Na_2CO_3 in the original sample ?
9. Based on the following information, determine value x and y :
$$\begin{array}{ccccccc} (\text{CH}_3)_x\text{AlCl}_y & \longrightarrow & x\text{CH}_4(\text{g}) + y\text{Cl}^- + \text{Al}^{3+} & \xrightarrow{\text{AgNO}_3} & \text{AgCl}(\text{s}) \\ 0.643 \text{ g} & & 0.222 \text{ g} & & 0.996 \text{ g} \end{array}$$
10. A 5.0 g sample of felspar containing Na_2O , K_2O and some inert impurity is dissolved in dilute HCl solution and NaCl and KCl formed are separated by fractional crystallization. During crystallization some less soluble impurities also comes out. Mass of NaCl, KCl and impurity accompanying these salts was found to be 6.47 g. Solid crystal was then re-dissolved and required 300 mL of 0.3 M AgNO_3 for complete precipitation of chlorides. The precipitate this, obtained was found to contain 4.23 % insoluble impurity. Determine mass percentage of Na_2O and K_2O in the original sample :
11. $\text{Pb}(\text{NO}_3)_2$ and KI reacts in aqueous solution to form an yellow precipitate of PbI_2 . In one series of experiments, the masses of two reactants varied, but the total mass of the two was held constant at 5.0 g. What maximum mass of PbI_2 can be produced in the above experiment :
12. Uranium is isolated from its ore by dissolving it as $\text{UO}_2(\text{NO}_3)_2$ and separating it as solid $\text{UO}_2(\text{C}_2\text{O}_4) \cdot x\text{H}_2\text{O}$. A 1.0 g sample of ore on treatment with nitric acid yielded 1.48 g $\text{UO}_2(\text{NO}_3)_2$ which on further treatment with 0.4 g $\text{Na}_2\text{C}_2\text{O}_4$ yielded 1.23 g $\text{UO}_2(\text{C}_2\text{O}_4) \cdot x\text{H}_2\text{O}$. Determine weight percentage of uranium in the original sample and x :

13. A mother cell disintegrate into sixty identical cells and each daughter cell further disintegrate into 24 smaller cells. The smallest cells are uniform cylindrical in shape with diameter of 120 Å and each cell is 6000 Å long. Determine molar mass of the mother cell if density of the smallest cell is 1.12 g/cm³ :
14. A sample is a mixture of Mohr's salts and (NH₄)₂SO₄. A 0.5 g sample on treatment with excess of BaCl₂ solution gave 0.75 g BaSO₄. Determine percentage composition of the salt mixture . What weight of Fe₂O₃ would be obtained if 0.2 g of the sample were ignited in air ?
15. A chloride mixture is prepared by grinding together pure BaCl₂·2H₂O, KCl and NaCl. What is the smallest and largest volume of 0.15 M AgNO₃ solution that may be used for complete precipitation of chloride from a 0.3 g sample of the mixture which may contain any one or all of the constituents ?
16. One mole of a mixture of N₂, NO₂ and N₂O₄ has a mean molar mass of 55.4. On heating to a temperature at which N₂O₄ may be dissociated : N₂O₄ → 2NO₂, the mean molar mass tends to the lower value of 39.6. What is the mole ratio of N₂ : NO₂ : N₂O₄ in the original mixture?
17. 10 mL of gaseous organic compound contain C, H and O only was mixed with 100 mL of O₂ and exploded under identical conditions and then cooled. The volume left after cooling was 90 mL. On treatment with KOH a contraction of 20 mL was observed, if vapour density of compound is 23 derive molecular formula of the compound.
18. Fluorocarbon polymers can be made by fluorinating polyethylene according to the reaction (CH₂)_n + 4nCoF₃ → (CF₂)_n + 2nHF + 4nCoF₂, where n is a large integer. The CoF₃ can be regenerated by the reaction 2 CoF₂ + F₂ → 2CoF₃. If the HF formed in the first reaction cannot be reused, how many kg of fluorine are consumed per kg of fluorocarbon produced, (CF₂)_n? If HF can be recovered and electrolyzed to hydrogen and fluorine, and if this fluorine is used for regenerating CoF₃, what is the net consumption of fluorine per kg of fluorocarbon?
19. $A_2 + 2B_2 \longrightarrow A_2B_4$
 $\frac{3}{2}A_2 + 2B_2 \longrightarrow A_3B_4$
 Two substance A₂ & B₂ react in the above manner when A₂ is limited it gives A₂B₄ in excess gives A₃B₄. A₂B₄ can be converted to A₃B₄ when reacted with A₂. Using this information calculate the composition of the final mixture when the mentioned amount of A & B are taken :c
 (a) 4 moles A₂ & 4 moles B₂
 (b) $\frac{1}{2}$ moles A₂ & 2 moles B₂
 (c) 1.25 moles A₂ & 2 moles B₂
20. In a water treatment plant, Cl₂ used for the treatment of water is produced from the following reaction 2KMnO₄ + 16HCl → 2KCl + 2MnCl₂ + 8H₂O + 5Cl₂. If during each feed 1 L KMnO₄ having 79% (w/v) KMnO₄ & 9 L HCl with d = 1.825 g/mL & 10% (w/w) HCl are entered & if that percent yield is 80% then calculate :
 (a) amount of Cl₂ produced.
 (b) amount of water that can be treated by Cl₂ if 1 litre consumes 28.4 g of Cl₂ for treatment.
 (c) calculate efficiency η of the process if $\eta = \frac{\text{vol. of water treated}}{\text{vol. of total feed}}$
21. A sea water sample has a density of 1.03 g/cm³ and 2.8% NaCl by mass. A saturated solution of NaCl in water is 5.45 M NaCl. How much water would have to be evaporated from 1.00 × 10⁶ L of the sea water before NaCl would precipitate ?
22. A sample of oleum is such that ratio of "free SO₃" by "combined SO₃" is equal to unity. Calculate its labelling in terms of percentage oleum.

23. One litre of milk weighs 1.035 kg. The butter fat is 4% (v/v) of milk has density of 875 kg/m^3 . Find the density of fat free skimmed milk.
24. A sample of fuming sulphuric acid containing H_2SO_4 , SO_3 and SO_2 weighing 1.00 g is found to require 23.47 mL of 1.00 M alkali (NaOH) for neutralisation. A separate sample shows the presence of 1.50% SO_2 . Find the percentage of "free" SO_3 , H_2SO_4 and "combined" SO_3 in the sample.
25. In one process for waterproofing, a fabric is exposed to $(\text{CH}_3)_2\text{SiCl}_2$ vapour. The vapour reacts with hydroxyl groups on the surface of the fabric or with traces of water to form the waterproofing film $[(\text{CH}_3)_2\text{SiO}]_n$, by the reaction ;
- $$n(\text{CH}_3)_2\text{SiCl}_2 + 2n\text{OH} \longrightarrow 2n\text{Cl}^- + n\text{H}_2\text{O} + [(\text{CH}_3)_2\text{SiO}]_n$$
- where n stands for a large integer. The waterproofing film is deposited on the fabric layer upon layer. Each layer is $6.0 \text{ \AA}$ thick [the thickness of the $(\text{CH}_3)_2\text{SiO}$ group]. How much $(\text{CH}_3)_2\text{SiCl}_2$ is needed to waterproof one side of a piece of fabric, 1.00 m by 3.00 m, with a film 300 layers thick ? The density of the film is 1.0 g/cm^3 .
26. 20 mL of a mixture of methane and a gaseous compound of Acetylene series were mixed with 100 mL of oxygen and exploded. The volume of the products after cooling to original room temperature and pressure, was 80 mL and on treatment with potash solution a further contracting of 40 mL was observed. Calculate (a) the molecular formula of the hydrocarbon, (b) the percentage composition of the mixture.
27. In a solution the concentration of CaCl_2 is 5 M & that of MgCl_2 is 5 m. The specific gravity of solution is 1.05, calculate the concentration of Cl^- in the solution in terms of Molarity.

BRAIN STORMING SUBJECTIVE EXERCISE	ANSWER KEY	EXERCISE-4(B)
1. KAlS_2O_8 , $\text{KAlS}_2\text{O}_8 \cdot 12 \text{ H}_2\text{O}$	2. 8.124 g Zn	3. 34.18
4. 29.28	5. 74.4	6. $\text{CaC}_2\text{N}_2\text{S}_2$, $\text{CaC}_2\text{N}_2\text{S}_2$
7. 50, 20.23, 29.77	8. 42.4 % Na_2CO_3	9. 2, 1
10. 31.16, 37.34	11. 3.464 g	12. 89.4, 3
13. $6.6 \times 10^{10} \text{ g}$	14. 18.77 mg	15. 16.38 mL, 34.18 L
16. 0.5 : 0.1 : 0.4	17. $\text{C}_2\text{H}_6\text{O}$	18. 1.52 kg, 0.76 kg
19. (a) $\text{A}_3\text{B}_4 = 2$ & $\text{A}_2 = 1$; (b) $\text{A}_2\text{B}_4 = \frac{1}{2}$ & $\text{B}_2 = 1$ (c) $\text{A}_2\text{B}_4 = 0.5$ & $\text{A}_3\text{B}_4 = 0.5$	21. 9.095 $\times 10^5 \text{ L}$	22. 110.11 %
20. (a) 10 mol, (b) L, (c) 2.5		
23. 1.041 g/mL		
24. $\text{H}_2\text{SO}_4 = 35.38\%$, Free $\text{SO}_3 = 63.1\%$, combined $\text{SO}_3 = 28.89\%$		25. 0.9413 gram
26. (b) 50	27. $[\text{Cl}^-] = 13.36 \text{ M}$	

JEE-[MAIN] : PREVIOUS YEAR QUESTIONS

- | PREVIOUS YEARS QUESTIONS | | | | | ANSWER KEY | | | EXERCISE-5 [A] | | |
|--------------------------|----|---|---|---|------------|---|---|----------------|---|----|
| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| Ans. | 1 | 1 | 4 | 1 | 1 | 4 | 3 | 1 | 1 | 3 |
| Que. | 11 | | | | | | | | | |
| Ans. | 1 | | | | | | | | | |

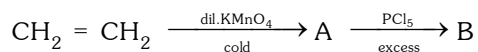
EXERCISE - 05 [B] JEE-[ADVANCED] : PREVIOUS YEAR QUESTIONS

1. How many moles of e^- weight one Kg : [JEE '2002 (Scr), 1]
- (A) 6.023×10^{23} (B) $\frac{1}{9.108} \times 10^{31}$ (C) $\frac{6.023}{9.108} \times 10^{54}$ (D) $\frac{1}{9.108 \times 6.023} \times 10^8$
2. Calculate the molarity of pure water using its density to be 1000 kg m^{-3} : [JEE' 2003]
3. One gm of charcoal adsorbs 100 mL 0.5 M CH_3COOH to form a monolayer, and there by the molarity of CH_3COOH reduces to 0.49. Calculate the surface area of the charcoal adsorbed by each molecule of acetic acid. Surface area of charcoal = $3.01 \times 10^2 \text{ m}^2/\text{g}$: [JEE' 2003]
4. Which has maximum number of atoms : [JEE 2003]
- (A) 24 g C(12) (B) 56 g Fe(56) (C) 27 g Al (27) (D) 108 g Ag(108)
5. Calculate the amount of Calcium oxide required when it reacts with 852 g of P_4O_{10} . [JEE 2005]
- $$6\text{CaO} + \text{P}_4\text{O}_{10} \longrightarrow 2 \text{Ca}_3(\text{PO}_4)_2$$
6. 20% surface sites have adsorbed N_2 . On heating N_2 gas evolved from sites and were collected at 0.001 atm and 298 K in a container of volume is 2.46 cm^3 . Density of surface sites is $6.023 \times 10^{14}/\text{cm}^2$ and surface area is 1000 cm^2 , find out the no. of surface sites occupied per molecule of N_2 : [JEE 2005]
7. Given that the abundances of isotopes ^{54}Fe , ^{56}Fe , ^{57}Fe are 5%, 90% and 5% respectively, the atomic mass of Fe is : [JEE 2009]
- (A) 55.85 u (B) 55.95 u (C) 55.75 u (D) 56.05 u

PREVIOUS YEARS QUESTIONS		ANSWER KEY		EXERCISE -5 [B]	
1.	D	2.	55.5 mol L^{-1}	3.	$5 \times 10^{-19} \text{ m}^2$
4.	A	5.	1008 g	6.	2
7.	B				

EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

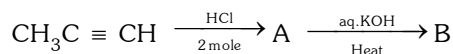
1. Consider the following reaction sequence



The products (A) and (B) are, respectively :

- (A) $\text{CH}_3\text{CH}_2\text{OH}$ and $\text{CH}_3\text{CH}_2\text{Cl}$ (B) CH_3CHO and CH_3CHCl_2
 (C) $\text{CH}_2\text{OHCH}_2\text{OH}$ and $\text{CH}_2\text{ClCH}_2\text{Cl}$ (D) $\text{CH}_2\text{OHCH}_2\text{OH}$ and $\text{CH}_2\text{OHCH}_2\text{Cl}$

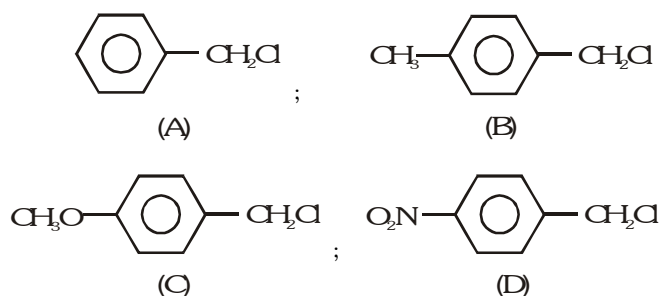
2. Consider the following reaction sequence,



The products (A) and (B) are, respectively,

- (A) $\text{CH}_3\text{CH}_2\text{CHCl}_2$ and $\text{CH}_3\text{CH}_2\text{CHO}$ (B) $\text{CH}_3\text{CCl}_2\text{CH}_3$ and CH_3COCH_3
 (C) $\text{CH}_3\text{CCl}_2\text{CH}_3$ and $\text{CH}_3\text{C} \equiv \text{CH}$ (D) $\text{CH}_3\text{CHClCH}_2\text{Cl}$ and $\text{CH}_3\text{CHOHCH}_2\text{OH}$

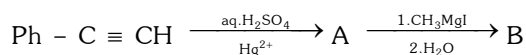
3. Consider the following chlorides :



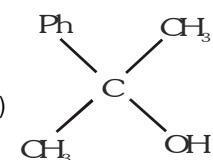
The order of reactivity of A, B, C and D towards hydrolysis by $\text{S}_{\text{N}}1$ mechanism is :

- (A) $\text{A} < \text{B} < \text{C} < \text{D}$ (B) $\text{D} < \text{C} < \text{B} < \text{A}$ (C) $\text{D} < \text{A} < \text{B} < \text{C}$ (D) $\text{C} < \text{B} < \text{A} < \text{D}$

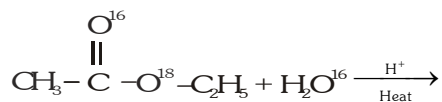
4. In the following reaction sequence



the product (B) is :

- (A) PhCOCH_3 (B) $\text{PhCH}_2\text{CHOHCH}_3$ (C)  (D) $\text{PhCH}_2\text{COCH}_3$

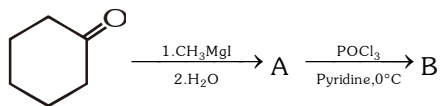
5. Consider the following reaction.



The products formed in the reaction are

- (A) $\text{CH}_3 - \overset{\overset{\text{O}^{16}}{\parallel}}{\text{C}} - \text{O}^{16} - \text{H}$ and $\text{C}_2\text{H}_5\text{O}^{18}\text{H}$ (B) $\text{CH}_3 - \overset{\overset{\text{O}^{16}}{\parallel}}{\text{C}} - \text{O}^{18} - \text{H}$ and $\text{C}_2\text{H}_5\text{O}^{16}\text{H}$
 (C) $\text{CH}_3 - \overset{\overset{\text{O}^{16}}{\parallel}}{\text{C}} - \text{O}^{16} - \text{H}$ and $\text{C}_2\text{H}_5\text{O}^{16}\text{H}$ (D) $\text{CH}_3 - \overset{\overset{\text{O}^{18}}{\parallel}}{\text{C}} - \text{O}^{18} - \text{H}$ and $\text{C}_2\text{H}_5\text{O}^{18}\text{H}$

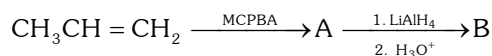
6. Consider the following reaction sequence,



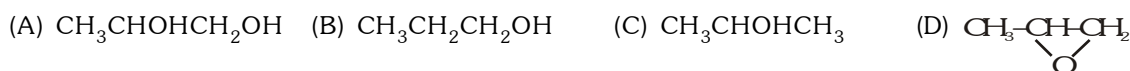
The product (B) is-



7. Propene is allowed to react with m-chloroperoxybenzoic acid. The product (A) is then reduced with LiAlH_4 in dry ether to give (B).



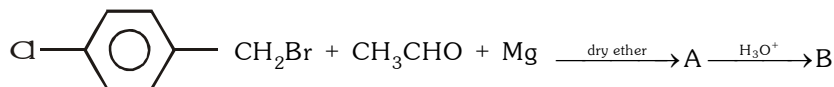
The structure of the product (B) is:



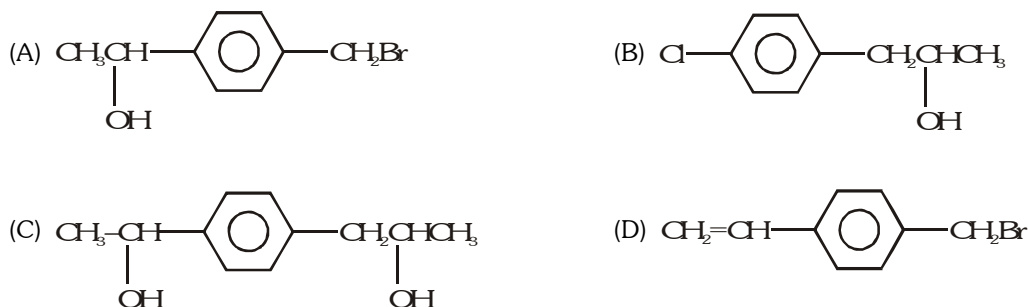
8. 2-Phenylethanol may be prepared by the reaction of phenylmagnesium bromide with :



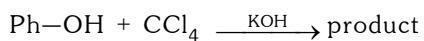
9. In the reaction



the product (B) is :

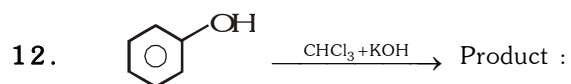


10. The final product in the reaction is :



11. In the reaction of phenol with CHCl_3 and aqueous NaOH at 70°C , the electrophile attacking the ring is :

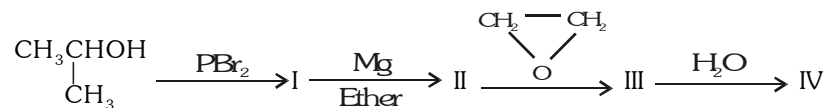




about above reaction the incorrect statement is

- (A) The name of reaction is Reimer teimann's reaction
 (B) The intermediate in the reaction is dichloro carbene
 (C) The final product is o-hydroxy benzaldehyde (D) the final product is benzyl chloride

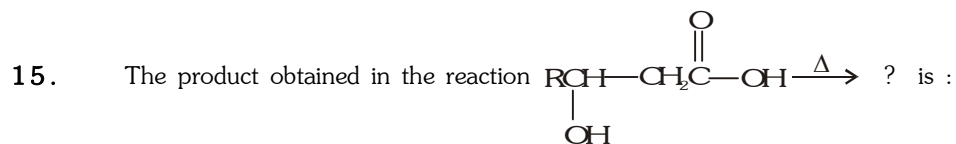
13. The final product (IV) in the sequence of reactions is :

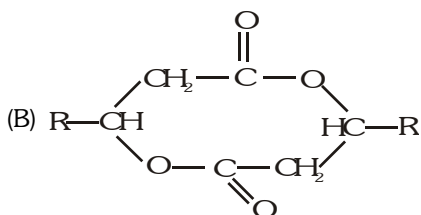


- (A) $\text{CH}_3-\text{CH}(\text{CH}_3)-\text{CH}_2\text{CH}_2\text{OH}$ (B) $\text{CH}_3-\text{CH}(\text{CH}_3)-\text{CH}_2\text{CH}_2\text{Br}$
 (C) $\text{CH}_3-\text{CH}(\text{CH}_3)-\text{CH}_2\text{CH}_2\text{OH}$ (D) $\text{CH}_3-\text{CH}(\text{CH}_3)-\text{CH}_2\text{CH}_2\text{CH}_3$

14. The reaction of 1 mol each of p-hydroxyacetophenone and methyl magnesium iodide will give :

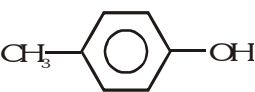
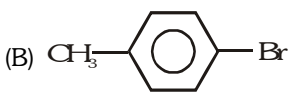
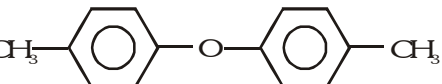
- (A) $\text{CH}_4 + \text{IMgO}-\text{C}_6\text{H}_4-\text{COOCH}_3$ (B) $\text{CH}_3\text{O}-\text{C}_6\text{H}_4-\text{COOCH}_3$
 (C) $\text{H}_3\text{C}-\text{C}(\text{OMgI})(\text{CH}_3)-\text{C}_6\text{H}_4-\text{OH}$ (D) $\text{CH}_3\text{O}-\text{C}_6\text{H}_4-\text{COOCH}_3$



- (A) $\text{RCH}(\text{OH})-\text{CH}_2-\text{C}(\text{O})-\text{OH}$ (B) 
 (C) $\text{RCH} = \text{CHCOOH}$ (D) None of these

16. The final product obtained in the reaction is :



- (A)  (B) 
 (C)  (D) None of these

17. The acidity of the compounds RCOOH , H_2CO_3 , $\text{C}_6\text{H}_5\text{OH}$, ROH decreases in the order :

- (A) $\text{RCOOH} > \text{H}_2\text{CO}_3 > \text{C}_6\text{H}_5\text{OH} > \text{ROH}$ (B) $\text{C}_6\text{H}_5\text{OH} > \text{RCOOH} > \text{H}_2\text{CO}_3 > \text{ROH}$
 (C) $\text{ROH} > \text{C}_6\text{H}_5\text{OH} > \text{RCOOH} > \text{H}_2\text{CO}_3$ (D) $\text{H}_2\text{CO}_3 > \text{RCOOH} > \text{C}_6\text{H}_5\text{OH} > \text{ROH}$

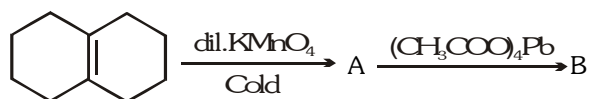
18. Arrange the following in order of decreasing acidic strength. p-nitrophenol (I), p-cresol (II), m-cresol (III), phenol (IV) :

- (A) $\text{I} > \text{II} > \text{III} > \text{IV}$ (B) $\text{IV} > \text{III} > \text{II} > \text{I}$ (C) $\text{I} > \text{III} > \text{II} > \text{IV}$ (D) $\text{III} > \text{II} > \text{I} > \text{IV}$

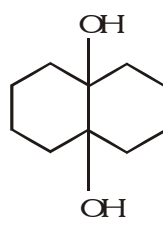
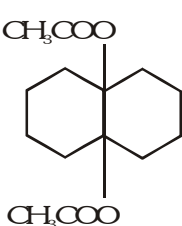
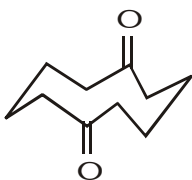
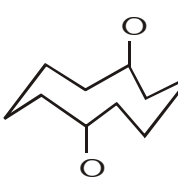
19. For the cleavage of ethers by halogen acids, the order of reactivity of halogen acids is :

- (A) $\text{HI} > \text{HBr} > \text{HCl}$ (B) $\text{HBr} > \text{HI} > \text{HCl}$
 (C) $\text{HCl} > \text{HBr} > \text{HI}$ (D) Ethers do not undergo cleavage

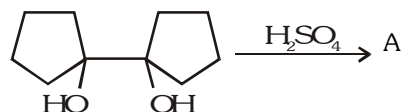
20. Consider the following reactions :



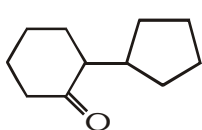
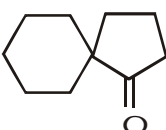
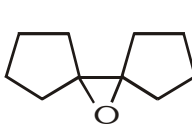
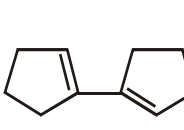
The product (B) is

- (A)  (B)  (C)  (D) 

21. The major product (A) formed in the reaction

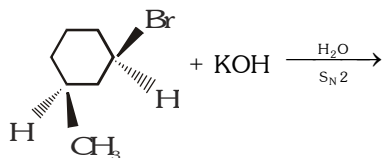


is :

- (A)  (B)  (C)  (D) 

EXERCISE-02**BRAIN TEASERS****SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)**

1. Consider the following reaction.

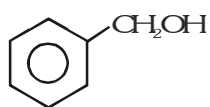


(1R, 3S)-cis-1 -Bromo -3-methylcyclohexane

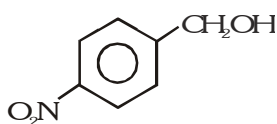
The product formed in the reaction is

- (A) (1R, 3S)-cis-3- methylcyclohexanol (B) (1S,3R)-cis-3-methylcyclohexanol
(C) (1S, 3S)-trans-3-methylcyclohexanol (D) (1R,3R)-trans-3-methylcyclohexanol

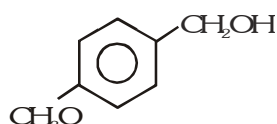
2. Consider the following alcohols



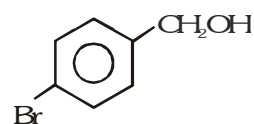
(I)



(II)



(III)

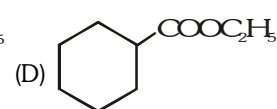
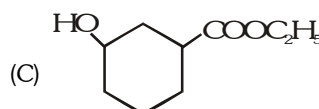
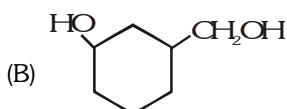
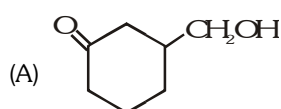
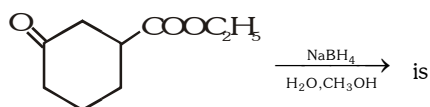


(IV)

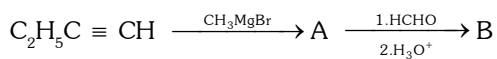
The order of decreasing reactivities of these alcohols towards nucleophilic substitution with HBr is

- (A) III > I > IV > II (B) III > I > II > IV (C) I > III > IV > II (D) I > III > II > IV

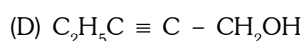
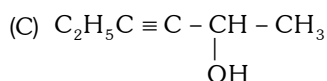
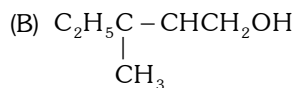
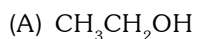
3. The product formed in the following reaction



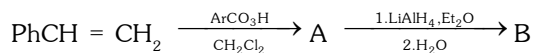
4. Consider the following sequence of reactions.



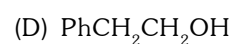
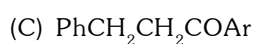
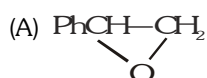
The product (B) is



5. In the transformations

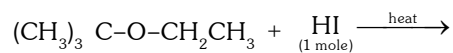


the end product (B) is



6. The reactivities of methanol (I), 1- propanol (II), 2- butanol (III) and 2- methyl-2- propanol (IV) towards sodium metal follow the order
 (A) I > II > III > IV (B) IV > III > II > I (C) I > IV > II > III (D) IV > II > III > I

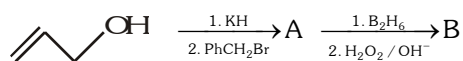
7. In the reaction



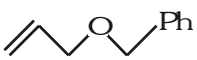
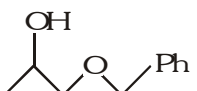
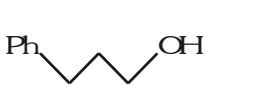
the product(s) formed is (are)

- (A) $(\text{CH}_3)_3\text{C}-\text{OH}$ and $\text{CH}_3\text{CH}_2\text{I}$ (B) $(\text{CH}_3)_3\text{C}-\text{I}$ and $\text{CH}_3\text{CH}_2\text{OH}$
 (C) $(\text{CH}_3)_3\text{C}-\text{I}$ and $\text{CH}_3\text{CH}_2\text{I}$ (D) $(\text{CH}_3)_3\text{C}-\overset{+}{\underset{\text{H}}{\text{O}}}-\text{CH}_2\text{CH}_3\text{I}^-$

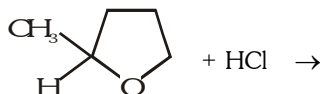
8. Consider the following sequence of reactions.



The end product (B) is

- (A)  (B) 
 (C)  (D) 

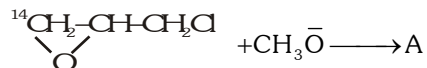
9. In the reaction



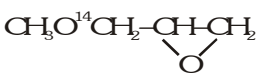
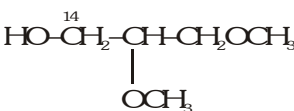
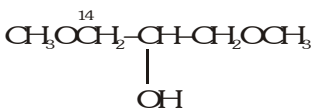
the major product formed is

- (A)  (B) 
 (C)  (D) 

10. Consider the following reaction.



The product (A) is

- (A)  (B) 
 (C)  (D) 

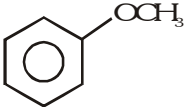
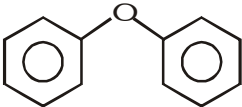
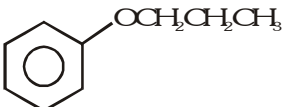
11. In the reaction $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{C}-\text{CH}_2 \\ | \quad \diagup \\ \text{O} \end{array} \xrightarrow[\text{H}^+]{\text{H}_2\text{O}^{18}} \text{A}$, the product (A) has the structure :

- (A) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{C}-\text{CH}_2\text{OH} \\ | \\ ^{18}\text{OH} \end{array}$ (B) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{C}-\text{CH}_2^{18}\text{OH} \\ | \\ \text{OH} \end{array}$ (C) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{C}-\text{CH}_3 \\ | \\ ^{18}\text{OH} \end{array}$ (D) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{CH}-\text{CH}_2^{18}\text{OH} \end{array}$

12. The compound which does not react with sodium is

- (A) $\text{CH}_3\text{CHOHCH}_3$ (B) CH_3OCH_3 (C) CH_3COOH (D) $\text{CH}_3\text{CH}_2\text{OH}$

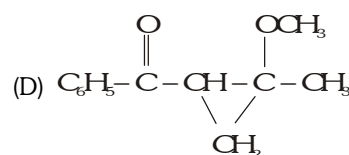
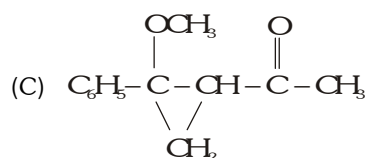
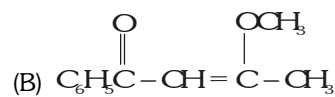
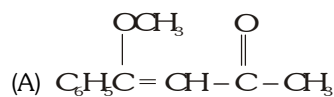
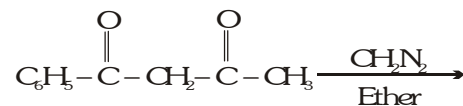
13. Which of the following ethers is not cleaved by concentrated HI even at 525 K ?

- (A)  (B) 
 (C)  (D) 

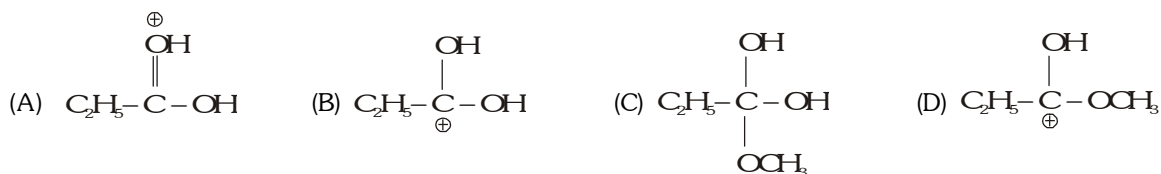
14. Which of the following reactions can be used to prepare $\begin{array}{c} \text{OH} \\ | \\ \text{CH}_3-\text{C}-\text{C}_6\text{H}_5 \\ | \\ \text{C}_2\text{H}_5 \end{array}$

- (A) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{C}_6\text{H}_5 + \text{C}_2\text{H}_5\text{MgBr} \longrightarrow [\quad] \xrightarrow{\text{H}_3\text{O}^+}$
 (B) $\text{C}_2\text{H}_5-\overset{\text{O}}{\parallel}{\text{C}}-\text{C}_6\text{H}_5 + \text{CH}_3\text{MgBr} \longrightarrow [\quad] \xrightarrow{\text{H}_3\text{O}^+}$
 (C) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{C}_2\text{H}_5 + \text{C}_6\text{H}_5\text{MgBr} \longrightarrow [\quad] \xrightarrow{\text{H}_3\text{O}^+}$
 (D) $\begin{array}{c} \text{Br} \\ | \\ \text{CH}_3-\text{C}-\text{C}_6\text{H}_5 \\ | \\ \text{C}_2\text{H}_5 \end{array} \xrightarrow[\Delta]{\text{Alc.KOH}}$

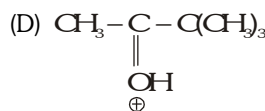
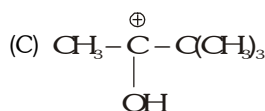
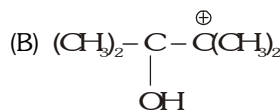
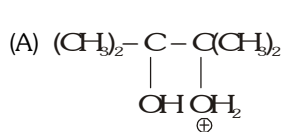
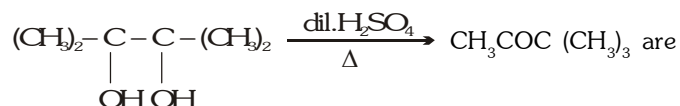
15. What are the products expected in the following reactions ?



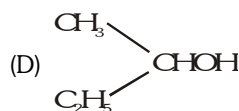
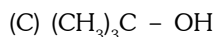
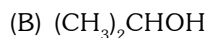
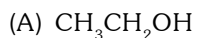
16. Which of the following groups will increase the acidity of phenol?
(A) $-\text{NO}_2$ (B) $-\text{CN}$ (C) $-\text{X}$ (halogen) (D) None of these
17. In the esterification of propanoic acid with methanol in the presence of a mineral acid, which of the following are intermediate species?



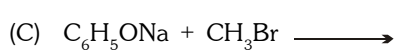
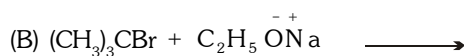
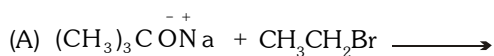
- 18.** The intermediate stages in the conversion



19. An alcohol, on treatment with $P + I_2$ followed by the reaction of the formed product first with $AgNO_2$ and then with HNO_3 and final basicification, gives a blue colour. Which of the following alcohols can it be?



20. Which of the following will result in the formation of an ether?



BRAIN TEASERS						ANSWER KEY					EXERCISE -2				
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	C	A	C	D	B	A	B	B	B	B	A	B	B	A,B,C	A,B
Que.	16	17	18	19	20										
Ans.	A,B,C	A,B,C,D	A,B,C,D	B,D	A,C										

EXERCISE-03**MISCELLANEOUS TYPE QUESTIONS****TRUE OR FALSE :**

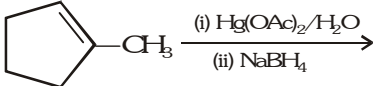
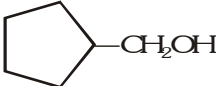
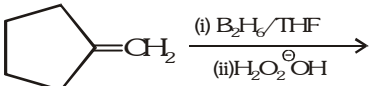
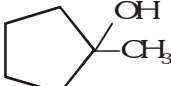
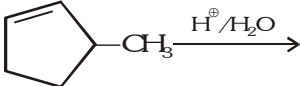
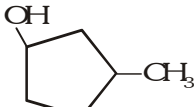
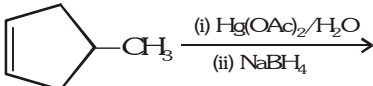
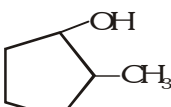
1. Lower alcohols are more soluble in water than higher alcohols.
2. Tertiary alcohols are more reactive towards halogen acids than secondary alcohols.
3. Glycerine is more viscous than ethylene glycol.
4. Tertiary alcohols are more acidic than phenol.
5. Phenol on heating with CCl_4 in presence of NaOH at 340 K followed by acetylation gives aspirin.
6. Phenol does not decompose NaHCO_3 to evolve CO_2 but picric acid does.
7. m-Methoxyphenol is a weaker acid than phenol.
8. Sodium ethoxide is prepared by the reaction of ethanol with aqueous sodium hydroxide.

FILL IN THE BLANKS :

1. Lower alcohols are highly soluble in water due to
2. The reaction of phenol with a diazonium salt in weak alkaline medium is called.....
3. Absolute alcohol can be prepared from rectified spirit by distillation.
4. A mixture of phenol and phthalic anhydride when heated with conc. H_2SO_4 formswhich is used as anin acid-base titrations.
5. Tertiary alcohols when passed over heated copper undergoto form.....
6. One mole of glycerol when heated with two moles of HIO_4 gives two moles ofand one mole of
7. The acidity of phenol is due to theof its anion.
8. A..... diol has two hydroxyl groups on.....
9. Phenol is acidic because of resonance stabilization of its conjugate base, namely.....
10. Amongst the three isomers of nitrophenols, the one that is least soluble in water is.....

MATCH THE COLUMN

1. Match the column I with column II.

Column-I (Reaction)		Column-II (Possible products)	
(A)		(p)	
(B)		(q)	
(C)		(r)	
(D)		(s)	

2. Match the column I with column II.

Column-I (substrate + RMgX)		Column-II (Product)	
(A)	HCHO	(p)	Tertiary alcohol
(B)	$\text{CH}_3-\overset{\text{O}}{\underset{\text{ }}{\text{C}}}-\text{CH}_3$	(q)	First ketone then 3° alcohol
(C)	$\text{CH}_3-\underset{\text{O}}{\underset{\text{ }}{\text{CH}}}-\text{CH}_2-\text{CH}_3$	(r)	Secondary alcohol
(D)	Ester	(s)	Primary alcohol

ASSERTION & REASON QUESTION :

These questions contain, Statement-I (assertion) and Statement-II (reason).

(A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I

(B) Statement-I is True, Statement-II is True ; Statement-II is NOT a correct explanation for Statement-I

(C) Statement-I is True, Statement-II is False.

(D) Statement-I is False, Statement-II is True.

1. **Statement-I :** Solubility of n-alcohols in water decreases with increase in molecular weight.

Because

Statement-II : The relative proportion of the hydrocarbon part in alcohols increases with increasing molecular weight which permits enhanced hydrogen bond with water.

2. **Statement-I :** p- Nitrophenol is a stronger acid than o-nitrophenol.

Because

Statement-II : Intramolecular H-bond makes o-isomer weaker than p-isomer.

3. **Statement-I :** Phenol is more reactive than benzene towards

Because

Statement-II : In the case of phenol, the intermediate carbocation is more resonance stabilized.

4. **Statement-I :** Tert-Butoxide is a stronger base than OH^- or $\text{C}_2\text{H}_5\text{O}^-$ ion but is a much poorer nucleophile.

Because

Statement-II : A negatively charged ion is always more powerful nucleophile than its conjugate acid.

5. **Statement-I** With $\text{Br}_2-\text{H}_2\text{O}$, phenol gives 2,4,6- tribromophenol but with $\text{Br}-\text{CS}_2$, it gives 4-bromophenol as the major product.

Because

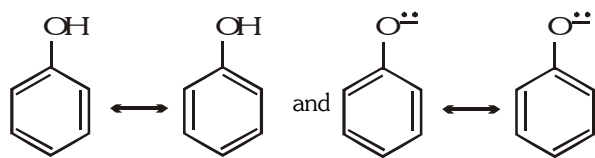
Statement-II : In water ionization of phenol is enhanced but in CS_2 , it is greatly suppressed.

COMPREHENSION BASED QUESTIONS :

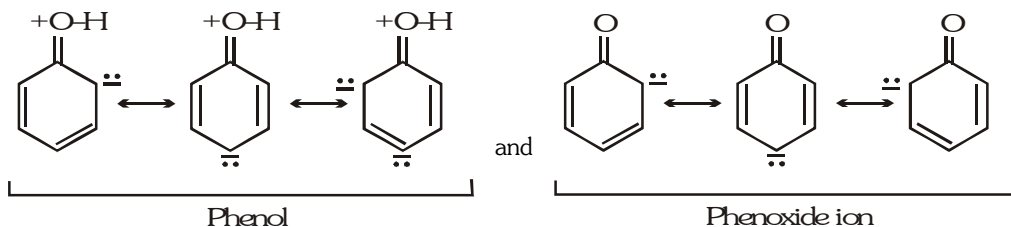
Comprehension # 1

Phenols are converted into their salts by aqueous NaOH but not by aqueous bicarbonates. The salts are converted into the free phenols by aqueous mineral acids, carboxylic acid or carbonic acids. Most phenols have K_a value of about 10^{-10} , and are tremendously more acidic than alcohols. The difference in acidity are

due to difference in stabilities of reactants and products. Phenol and phenoxide ions contain benzene ring and therefore must be hybrid of Kekuley structures

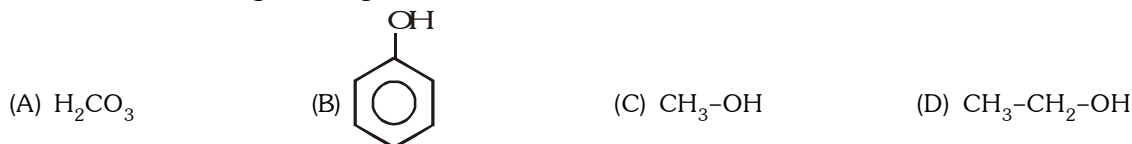


Being basic, oxygen can share more than a pair of electron with the ring.



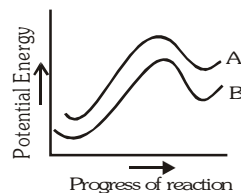
Since energy must be supplied to separate opposite charge, the structure of phenols should contain more energy. The net effect of resonance is therefore to stabilise the phenoxide ion to a greater extent than the phenol and thus to shift the equilibrium towards ionization and make K_a larger than for an alcohol.

1. Which of the following is strongest acid ?

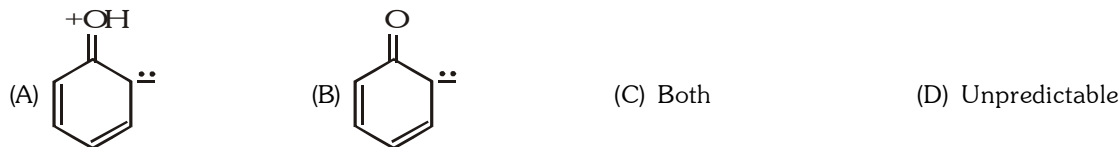


2. Consider the following curves :

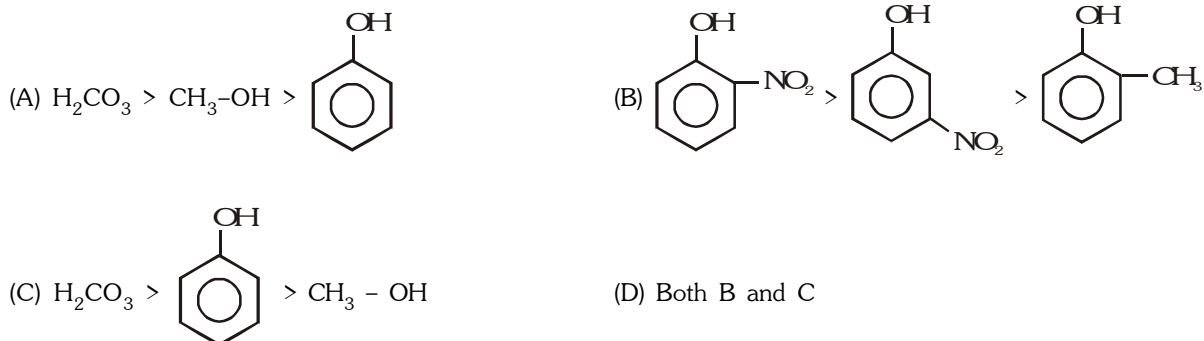
- (A) Curve A represents the ionisation of alcohol
(B) Curve B represent the ionisation of Phenols
(C) Curve A represents the ionisation of phenol
(D) None



3. Which of the following is more stable:



4. Correct order of acidity is

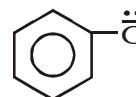


5

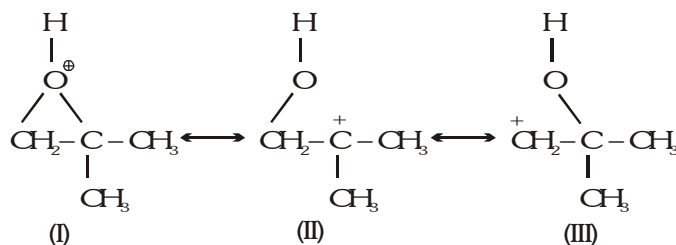
Choose the correct statement:

(A) Phenol gives effervescence with NaHCO_3

(B) Picric acid is weaker acid than carbonic acid

(C) Picric acid gives effervescence with NaHCO_3 (D) $\text{R}-\ddot{\text{O}}^-$ is more stable than**Comprehension # 2**

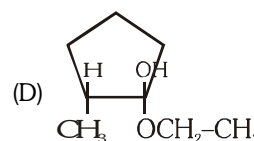
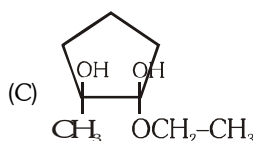
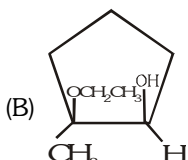
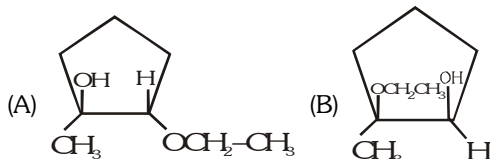
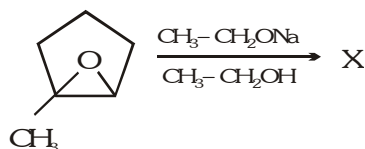
Symmetrically substituted epoxides give the same products in both the acid catalysed and base catalyzed ring opening. An unsymmetrical epoxide gives different products under acid catalysed and base catalysed conditions. Under basic conditions, the alkoxide ion simply attacks the less hindered carbon atom in an S_N^2 displacement. Under acidic conditions, the alcohol, attacks the protonated epoxide.



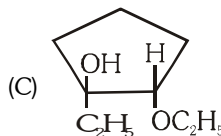
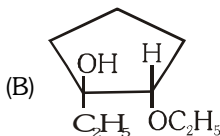
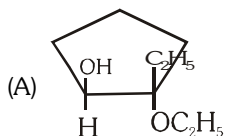
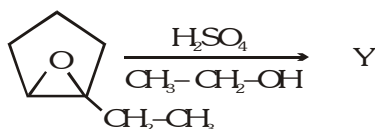
Structure II and III show that the oxirane carbon share part of positive charge. The tertiary carbon bear a larger part of positive charge and it is more strongly electrophilic. The bond between tertiary carbon and oxygen is weaker implying a lower transition state energy for attack at the tertiary carbon. Attack by the weak nucleophilic is sensitive to the strength of electrophilic is sensitive to the strength of electrophile. Centre attack takes place at more electrophilic carbon which is usually the more substituted carbon because it can better support the positive charge.

1.

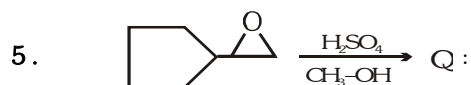
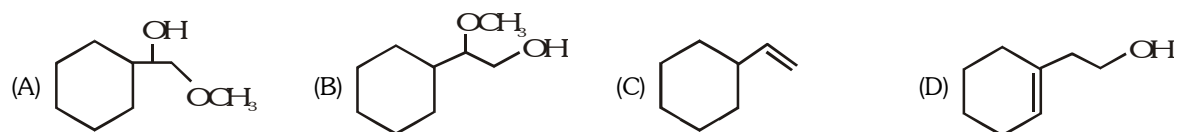
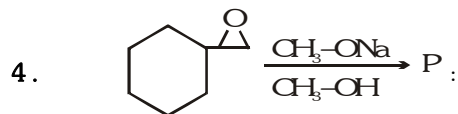
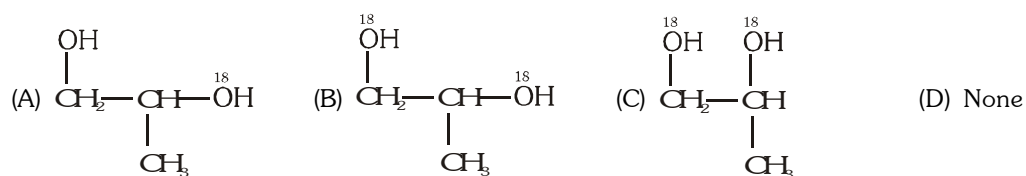
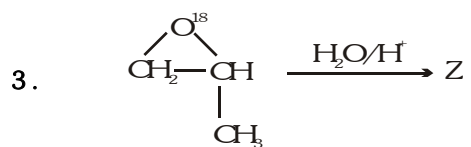
What will be the products in following reactions



2.

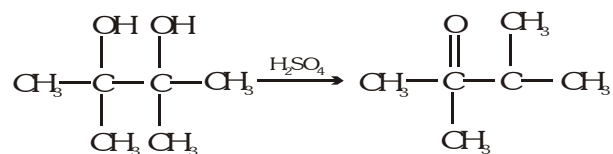


(D) None



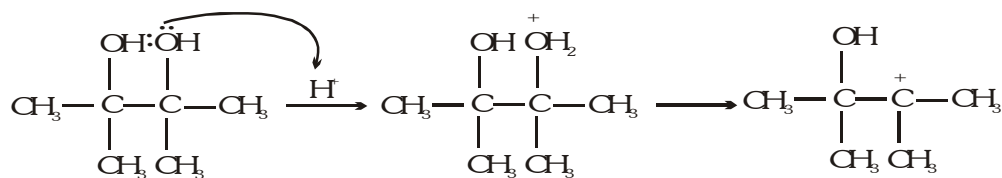
Comprehension # 3

When pinacol is treated with dilute H_2SO_4 , a re-arrangement reaction takes place which leads to the formation of a ketone.



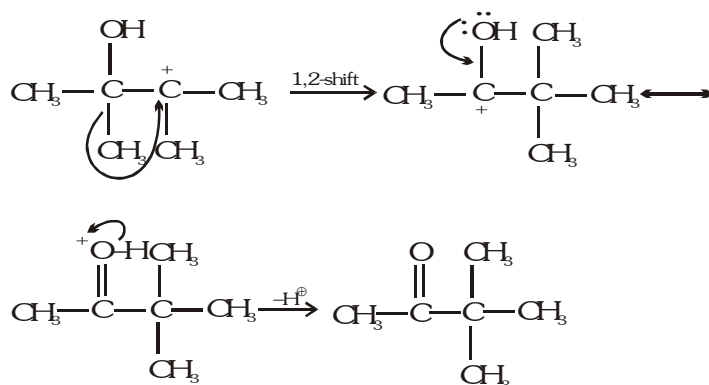
This reaction involves re-arrangement of carbocation.

Step 1:

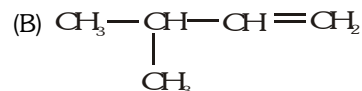
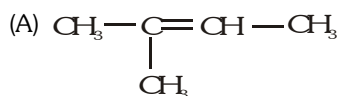


Step 2 :

Carbocation rearrange by hydride, alkyl or aryl shift to get as stable as they can. Stability is the driving force for re-arrangement migration of bond may also occur. Where by ring expansion and ring contraction takes place. The relief of strain can provide a powerful driving force for re-arrangement.



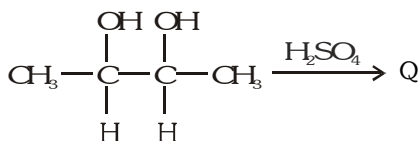
1. What will be the product of following reaction $\text{CH}_3-\text{C}(\text{CH}_3)_2-\text{CH}_2-\text{Br} \xrightarrow{\text{NaOH}}$ P, P is :



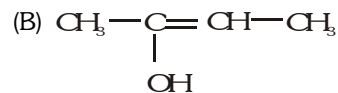
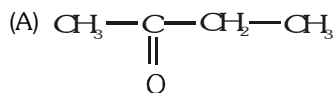
(C) Both of these

(D) None of these

2. What will be the product of following reaction



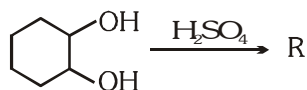
Q is :



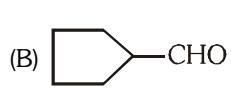
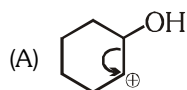
(C) Both

(D) None

3. What will be the product of following reaction

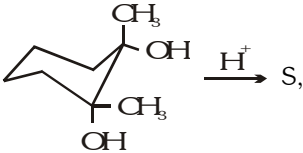
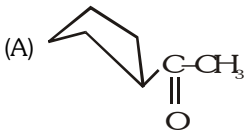
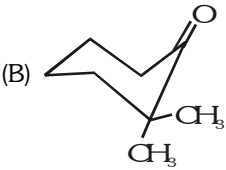
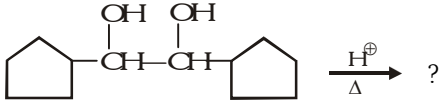
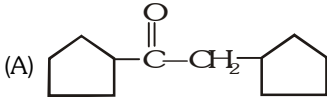
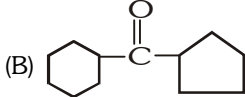


R is :



(C) Both

(D) None

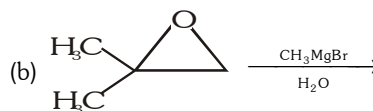
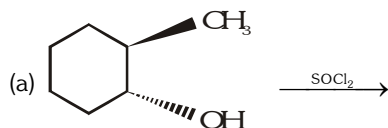
4. What will be the product of following reaction  $\xrightarrow{H^+}$ S,
S is :
- (A)  (B)  (C) Both (D) None
5. What will be the product of following reaction  $\xrightarrow[\Delta]{H^+}$?
- (A)  (B) 
(C) both (D) None

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
True / False 1. T 2. T 3. T 4. F 5. T 6. T 7. T 8. T Fill in the Blanks 1. H-bonding 2. coupling reaction 3. azeotropic 4. phenolphthalein, indicator 5. dehydration, alkene 6. formaldehyde, formic acid 7. resonance stabilisation 8. vicinal, adjacent 9. phenoxide ion 10. O-nitrophenol Match the Column 1. (A) $\rightarrow$ q ; B $\rightarrow$ p ; (C) $\rightarrow$ r ; (D) $\rightarrow$ q 2. (A) $\rightarrow$ s ; (B) $\rightarrow$ p ; (C) $\rightarrow$ r ; (D) $\rightarrow$ q Assertion - Reason Questions 1. C 2. A 3. A 4. B 5. A Comprehension Based Questions Comprehension #1 : 1. (A) 2. (B) 3. (B) 4. (D) 5. (C) Comprehension #2 : 1. (A) 2. (A) 3. (B) 4. (A) 5. (B) Comprehension #3 : 1. (A) 2. (A) 3. (B) 4. (A) 5. (B)		

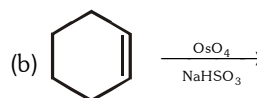
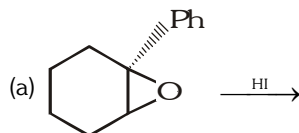
EXERCISE-04 [A]

CONCEPTUAL SUBJECTIVE EXERCISE

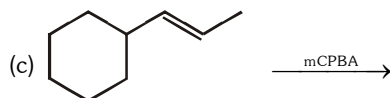
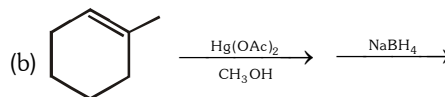
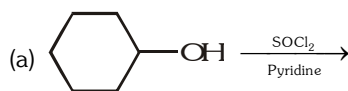
- $C_2H_5OH \xrightarrow{PCl_5} (A) \xrightarrow{KCN} (B) \xrightarrow{H_3O^+} (C) \xrightarrow{NH_3} (D) \xrightarrow{\text{heat}} (E)$
- $CH_3CH_2CH_2OH \xrightarrow{PBr_5} (A) \xrightarrow{KOH(Alc.)} (B) \xrightarrow{HBr} (C) \xrightarrow{NH_3} (D)$
- t-butyl alcohol reacts less rapidly with metallic sodium than the primary alcohol. Explain why?
- Explain why ArOR ethers are cleaved to give RI and ArOH rather than ArI and ROH.
- Complete the following reactions :



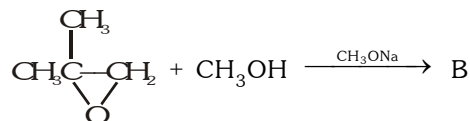
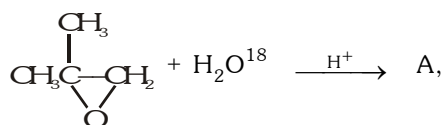
- Complete the following reactions :



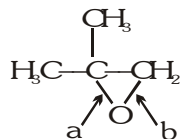
- Provide products in the following reactions :



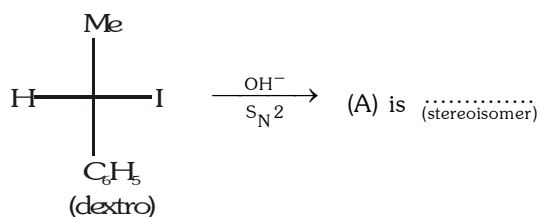
- Identify the products A and B giving proper explanation :



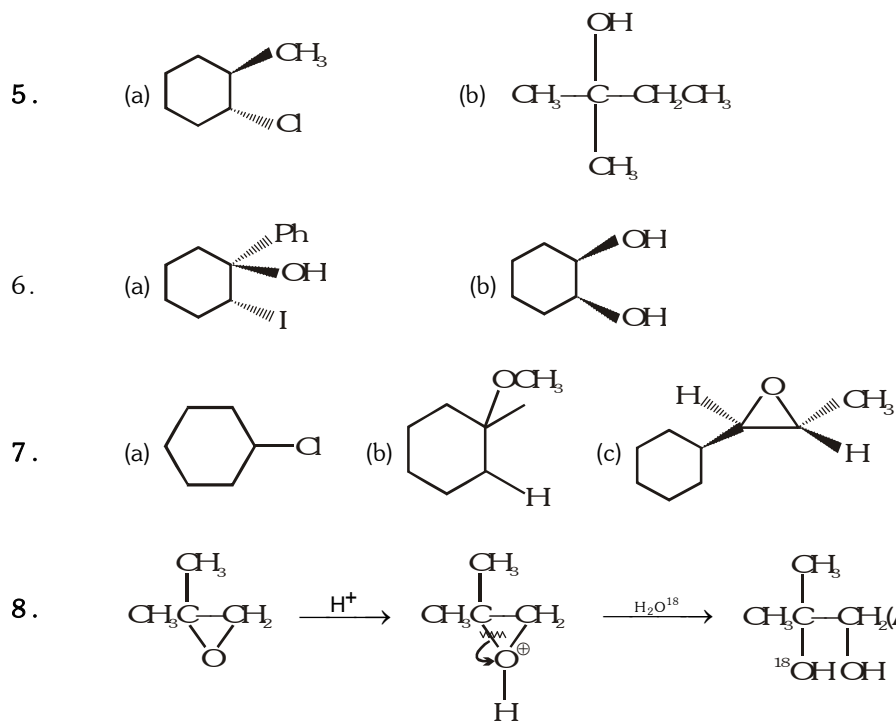
- Indicate bonds which are cleaved
I : in basic conditions II : in acidic conditions



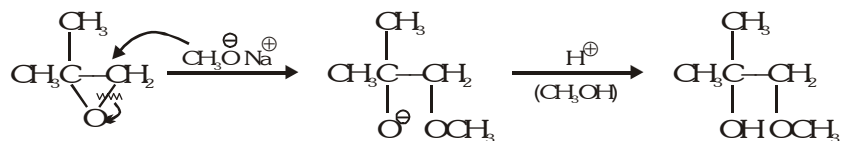
- In the following S_N2 reaction



- (A) C_2H_5Cl ; (B) C_2H_5CN ; (C) C_2H_5COOH ; (D) $C_2H_5COONH_4$;
(E) $C_2H_5CONH_2$
- (A) C_3H_7Br ; (B) $CH_3CH=CH_2$; (C) $CH_3CHBrCH_3$; (D) $CH_3CHNH_2CH_3$
- The +I.E. of three methyl groups on central C-atom of tert-butyl alcohol makes it partially negative with the result that it pushes the electron pair of $-OH$ bond towards H-atom and thus H-atom is not replaced easily.
- S_N2 attack on a carbon of benzene ring does not occur nor does the high energy $C_6H_5^+$ form by an S_N1 reaction. Hence ArI cannot be a product even in an excess of conc. HI.



(nucleophile attacks the more substituted carbon in acid-catalysed reaction)



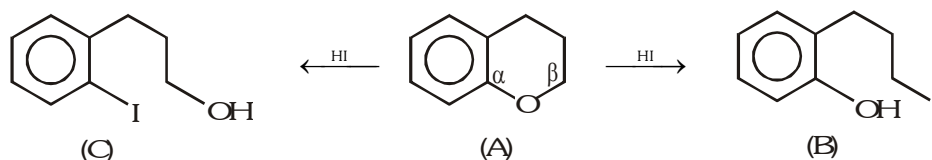
(nucleophile attacks the less substituted carbon in base-catalysed reaction)

- I : bond b ; II : bond a
- A is laevo - isomer

EXERCISE-04 [B]

BRAIN STORMING SUBJECTIVE EXERCISE

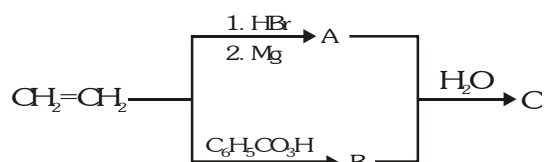
1. A compound (X) reacts with thionyl chloride to give a compound (Y). (Y) reacts with Mg to form a Grignard reagent, which is treated with acetone and the product is hydrolysed to give 2-methyl-2-butanol. What are structural formulae of (X) and (Y) ?
2. Compound (A) $C_4H_{10}O$ reacts rapidly with metallic sodium, but undergoes almost no reaction with Lucas reagent. When (A) is treated with hot concentrated sulphuric acid, a new compound (B) C_4H_8 is formed. If C_4H_8 is hydrated with sulphuric acid a new compound (C) C_4H_9OH is formed, which is almost inert to metallic sodium but reacts rapidly with Lucas reagent. What are (A), (B) and (C) ?
3. Give the product of the reaction of Ph_2CHCH_2OH with HBr and explain its formation.
4. Hydration of 3-phenyl-1-butene in dilute H_2SO_4 is not a satisfactory method for preparing 3-phenyl-2-butanol, because 2-phenyl-2-butanol is obtained instead. Explain.
5. When A (given below) reacts with HI products is B and not C. Explain.



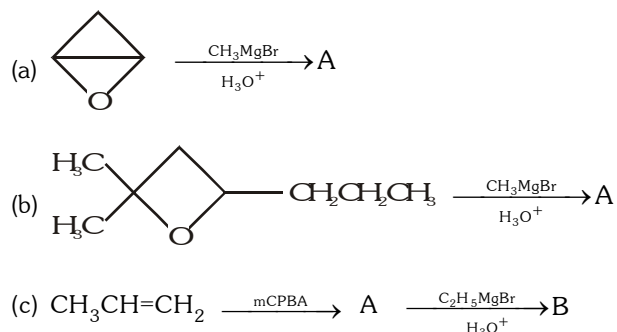
6. Isotopic carbon-14 in (A) appears at new position (as in B) when (A) reacts with CH_3ONa . Explain.



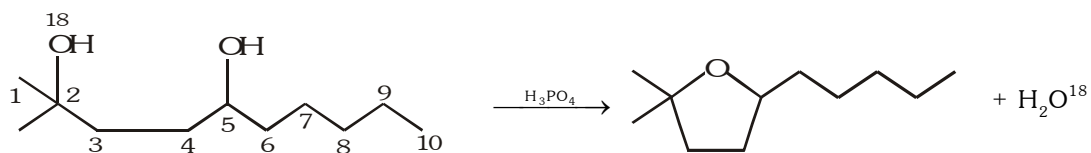
7. Complete the following reaction



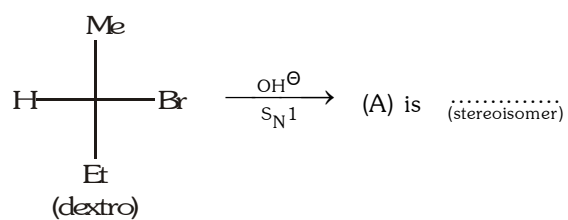
8. Complete the following reactions



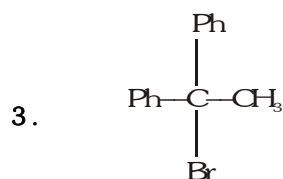
9. In the following dehydration of diol with H_3PO_4 , following product is formed such that isotopic ^{18}O goes with H_2O . Explain.



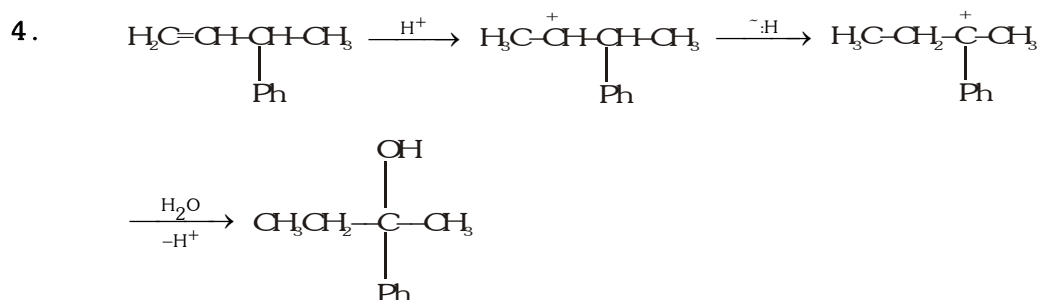
10. In the following $\text{S}_{\text{N}}1$ reaction :



1. $\text{C}_2\text{H}_5\text{OH}$ $\text{C}_2\text{H}_5\text{Cl}$
Ethyl alcohol Ethyl chloride
(X) (Y)
2. $(\text{CH}_3)_2\text{CHCH}_2\text{OH}$; $(\text{CH}_3)_2\text{C}=\text{CH}_2$; $(\text{CH}_3)_3\text{COH}$
(A) (B) (C)

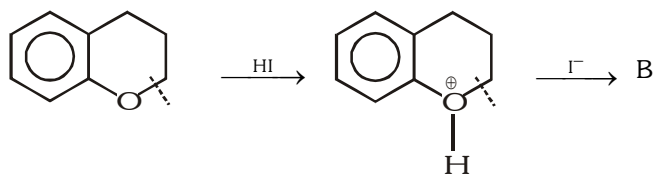


Synchronous migration of Ph provides greater assistance in the removal of H_2O from the protonated alcohol than does a methyl group.

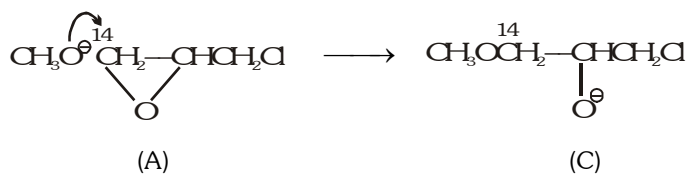


(Hydration gives an intermediate 2° C^+ , which undergoes a hydride shift. Although phenyl is a better migrator than H, migration of H occurs leading to a more stable 3° benzylic carbocation.)

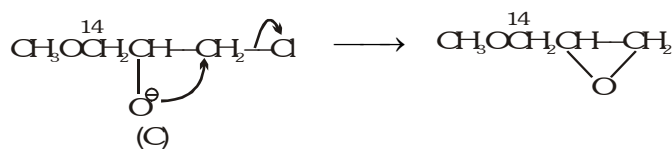
5. Bond energy of C (vinylic)—O bond (α) is greater than that of C(alkylic)—O bond (β). Hence, when A reacts with HI, bond β breaks forming B.

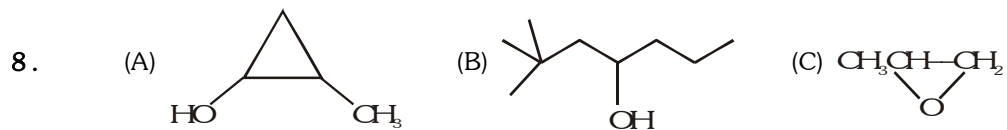
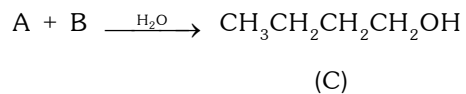
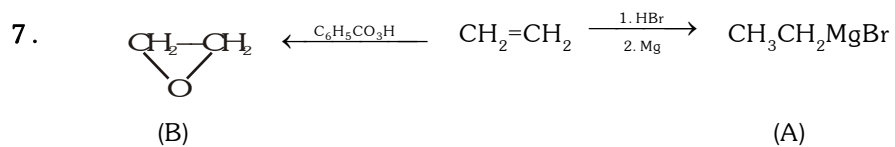


6. CH_3O^- (nucleophile) attacks less substituted carbon (which is C-14 in this case) forming intermediate (C).

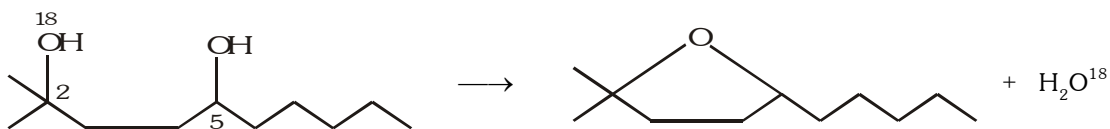


(C) will displace Cl^- forming (B)





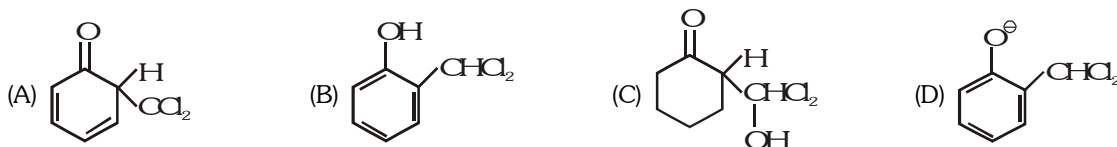
9. 3 alcohol at C_2 is more basic than 2 alcohol at C_5 . Hence, intramolecular dehydration takes place such that H of 2 alcoholic group at C_5 eliminates OH of 3 alcoholic groups at C_2 (with isotopic ^{18}O) to form cyclic ether



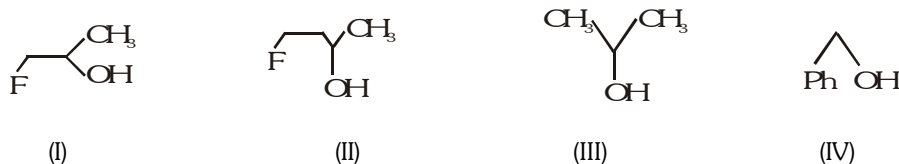
10. A is a mixture of d- and l- and thus racemic mixture.
-

EXERCISE-05**PREVIOUS YEARS QUESTIONS**

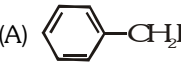
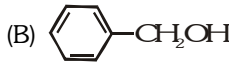
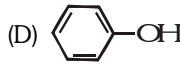
1. The products of combustion of an aliphatic thiol (RSH) at 298 K are - [IIT-92]
(A) CO_2 (l), H_2O (g) and SO_2 (g) (B) CO_2 (g), H_2O (g) and SO_2 (g)
(C) CO_2 (l), H_2O (l) and SO_2 (g) (D) CO_2 (g), H_2O (l) and SO_2 (l)
2. An organic compound $\text{C}_3\text{H}_6\text{O}$ does not give a precipitate with 2, 4-dinitrophenyl hydrazine reagent and does not react with sodium metal. It could be - [IIT-93]
(A) $\text{CH}_3 - \text{CH}_2 - \text{CHO}$ (B) $\text{CH}_3 - \text{CO} - \text{CH}_3$
(C) $\text{CH}_2 = \text{CH} - \text{CH}_2\text{OH}$ (D) $\text{CH}_2 = \text{CH} - \text{OCH}_3$
3. When phenol is reacted with CHCl_3 and NaOH followed by acidification, salicylaldehyde is obtained which of the following species are involved in the above mentioned reaction as intermediates ? [IIT-95]



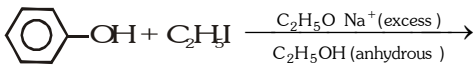
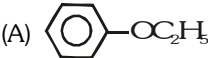
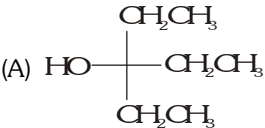
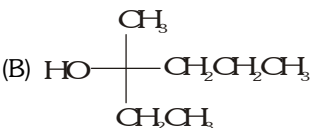
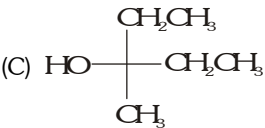
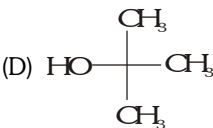
4. The reaction products of, $\text{C}_6\text{H}_5\text{OCH}_3 + \text{HI} \xrightarrow{\Delta}$ are : [IIT-95]
(A) $\text{C}_6\text{H}_5\text{OH} + \text{CH}_3\text{I}$ (B) $\text{C}_6\text{H}_5\text{I} + \text{CH}_3\text{OH}$ (C) $\text{C}_6\text{H}_5\text{CH}_3 + \text{HOI}$ (D) $\text{C}_6\text{H}_6 + \text{CH}_3\text{OI}$
5. The order of reactivity of the following alcohols - [IIT-97]

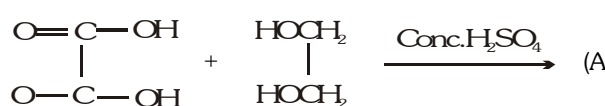
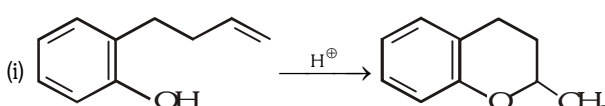
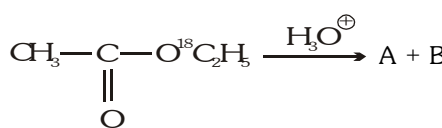
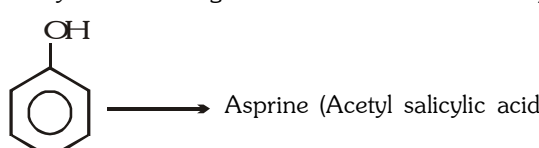
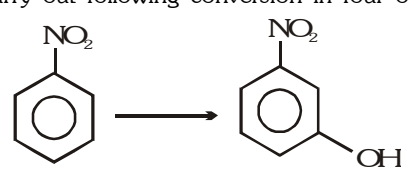


towards conc. HCl is -

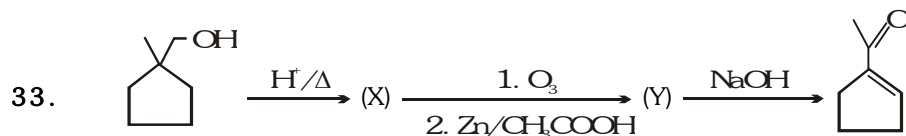
- (A) $\text{I} > \text{II} > \text{III} > \text{IV}$ (B) $\text{I} > \text{III} > \text{II} > \text{IV}$ (C) $\text{IV} > \text{III} > \text{II} > \text{I}$ (D) $\text{IV} > \text{III} > \text{I} > \text{II}$
6. Among the following compounds, the strongest acid is - [IIT-98]
(A) $\text{HC} \equiv \text{CH}$ (B) C_6H_6 (C) C_2H_6 (D) CH_3OH
7. Benzenediazonium chloride on reaction with phenol in weakly basic medium gives - [IIT-98]
(A) Diphenyl ether (B) p-hydroxyazobenzene
(C) Chlorobenzene (D) Benzene
8. The ether  when treated with HI produces [IIT-99]
(A)  (B)  (C)  (D) 
9. Which one of the following will most readily be dehydrated in acidic condition - [IIT-2000]



10. 1-propanol & 2-propanal can be best distinguished by - [IIT-01]
 (A) Oxidation with alkaline KMnO_4 followed by reaction with Fehling solution
 (B) Oxidation with acidic dichromate followed by reaction with Fehling solution
 (C) Oxidation by heating with copper followed by reaction with Fehling solution
 (D) Oxidation with concentrated H_2SO_4 followed by reaction with Fehling solution
11. Identify the correct order of boiling point of the following compounds - [IIT-02]
 (A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$; (B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$; (C) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
 (A) $A > B > C$ (B) $C > A > B$ (C) $A > C > B$ (D) $C > B > A$
12.  [IIT-03]
 (A)  (B)  (C) $\text{C}_6\text{H}_5\text{OC}_6\text{H}_5$ (D) $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$
13. Reaction of enantiomerically pure acid with 1 chiral carbon and racemic alcohol with 1 chiral carbon gives an ester which is : [IIT-03]
 (A) Meso (B) Optically active
 (C) Racemic mixture (D) Enantiomerically pure
14. $\text{CH}_3\text{MgBr} + \text{Ethyl ester} \rightarrow$ which can be formed as product (excess) : [IIT-03]
 (A)  (B) 
 (C)  (D) 
15. On acid catalysed hydration, 2-phenyl propene gives : [IIT-04]
 (A) 3-phenyl-2-propanol (B) 2-phenyl-1-propanol
 (C) 1-phenyl-3-propanol (D) 2-phenyl-2-propanol
16. Conversion of cyclohexanol into cyclohexene is most effective in : [IIT-05]
 (A) concentrated H_3PO_4 (B) concentrated HCl
 (C) concentrated $\text{HCl} / \text{ZnCl}_2$ (D) concentrated HBr
17. When t-butanol and n-butanol are separately treated with a few drops of dilute KMnO_4 in one case only, purple colour disappears and a brown precipitate is formed. Which of the two alcohols gives the above reaction and what is the brown precipitate ? [IIT-94]
18. 3,3-Dimethylbutan-2-ol loses a molecule of water in the presence of a concentrated sulphuric acid to give tetramethyl ethylene as a major product. Suggest a suitable mechanism. [IIT-96]
19. A compound $\text{D}(\text{C}_8\text{H}_{10}\text{O})$ upon treatment with alkaline solution of iodine gives a yellow precipitate. The filtrate on acidification gives a white solid (E) ($\text{C}_7\text{H}_6\text{O}_2$). Write the structures of (D) and (E) and explain the formation of (E). [IIT-96]

20. Which of the following is the correct method for synthesising methyl-t-butyl ether and why?
 (i) $(\text{CH}_3)_3\text{CBr} + \text{NaOMe} \rightarrow$ (ii) $\text{CH}_3\text{Br} + \text{tert-BuONa} \rightarrow$ [IIT- 97]
21.  [IIT-97]
22. Write the intermediate steps for the following reaction: [IIT- 98]
- (i) 
23. Discuss why o-hydroxy benzaldehyde is a liquid at room temperature while p-hydroxy benzaldehyde is a high melting solid? [IIT-99]
24. Write the structures of the product A & B
-  [IIT-2000]
25. Cyclobutyl bromide on treatment with magnesium in dry ether forms an organometallic A. The organometallic reacts with ethanal to give an alcohol B after mild acidification. Prolonged treatment of alcohol B with an equivalent amount of HBr gives 1- bromo-1-methylcyclopentane (C). Write the structures of A, B and explain how C is obtained from B. [IIT-2001]
26. How would you synthesis 4- methoxyphenol from bromobenzene in NOT more than five steps? State clearly the reagents used in each step and show the structures of the intermediate compounds in your synthetic scheme. [IIT-2001]
27. Identify X, Y and Z in the following synthetic scheme and write their structure. Explain the formation of labelled formaldehyde ($\text{H}_2\text{C}^*\text{O}$) as one of the products when compound Z is treated with HBr and subsequently ozonolysed. Mark the C^* carbon in the entire scheme.
 $\text{BaC}^*\text{O}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{X (gas)}$ [C^* denotes C^{14}]
 $\text{CH}_2 = \text{CH} - \text{Br} \xrightarrow[\text{(ii) X, (iii) H}_3\text{O}^+]{\text{(i) Mg/ ether}} \text{Y} \xrightarrow{\text{LiAlH}_4} \text{Z}$ [IIT-2001]
28. Compound X on reduction with LiAlH_4 gives a hydride Y containing 21.72% hydrogen along with other products. The compound Y reacts with air explosively resulting in boron trioxide. Identify X and Y. Give balanced reactions involved in the formation of Y and its reaction with air. Draw the structure of Y. [IIT-2001]
29. Mention two esters produced when a racemic mixture of 2- phenyl propanoic acid is treated with (+) 2- butanol. What is the stereochemical relationship between these esters? [IIT-2003]
30. Carry out following conversion in 3 or less steps.
 [IIT-2003]
31. Carry out following conversion in four or less steps. Also mention all the reagents used and reaction conditions.
 [IIT-2004]

32. An organic compound $P(C_5H_{10}O)$ reacts 10^{15} times faster than ethylene with dilute H_2SO_4 to give two products Q and R. Both Q and R give positive iodoform test. Identify P, Q and R and also give reason for very high reactivity of P. [IIT-2004]

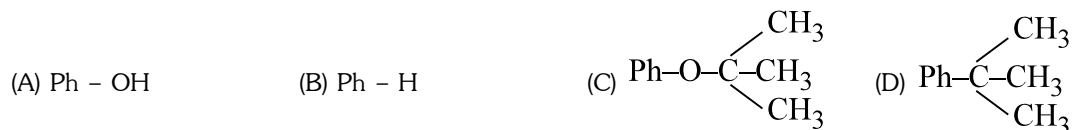


Identify (X) and (Y).

[IIT-2004]

34. Phenyl magnesium bromide reacting with t-Butyl alcohol gives

[JEE 2005]



35. **Statement-1:** p-Hydroxybenzoic acid has a lower boiling point than o-hydroxybenzoic acid.

[JEE 2007]

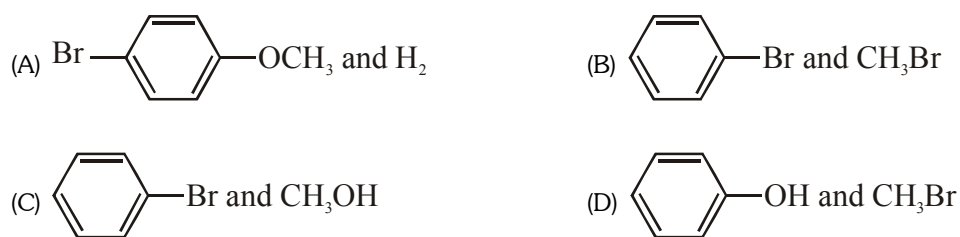
because

Statement-2 : o-Hydroxybenzoic acid has intramolecular hydrogen bonding.

- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.

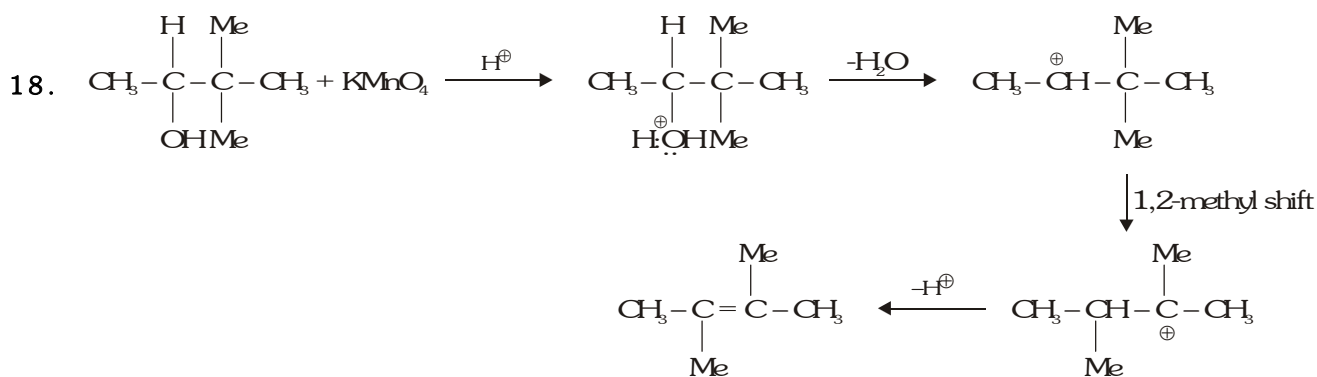
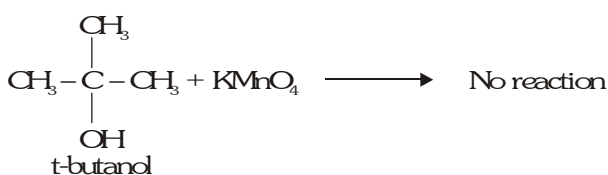
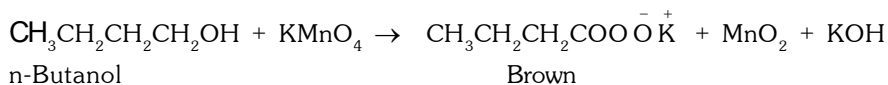
36. In the reaction  the products are

[JEE 2010]



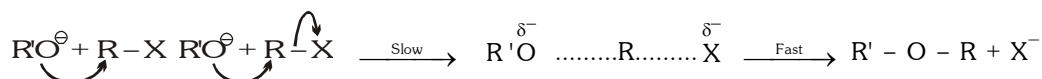
1. (B) 2. (D) 3. (D) 4.(A) 5. (C) 6. (D) 7.(B) 8. (D) 9. (A)
10. (C) 11. (B) 12. (A) 13. (C) 14. (C) 15. (D) 16. (A)

17. n-butanol is oxidised by KMnO_4 and not t-butanol as the latter does not contain H atom attached to carbinol carbon atom.

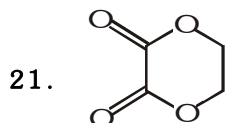


19. (D) $\text{Ph} - \underset{\text{OH}}{\text{CH}} - \text{CH}_3$ $\text{C}_6\text{H}_5\text{COOH}$
- (D) (E)

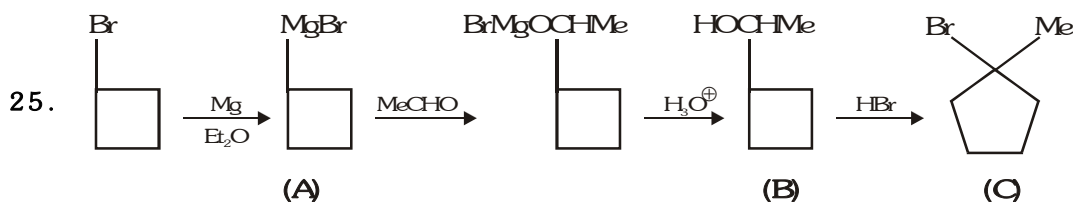
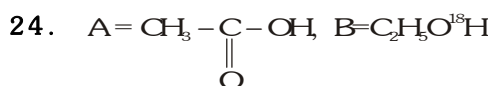
20. The ether formation involves nucleophilic substitution of alkoxide ion for halide ion.



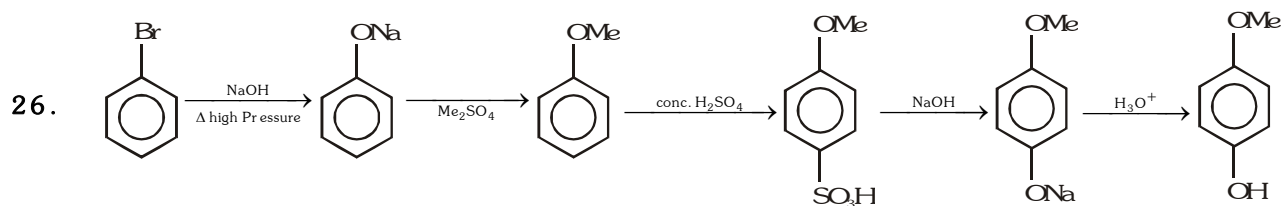
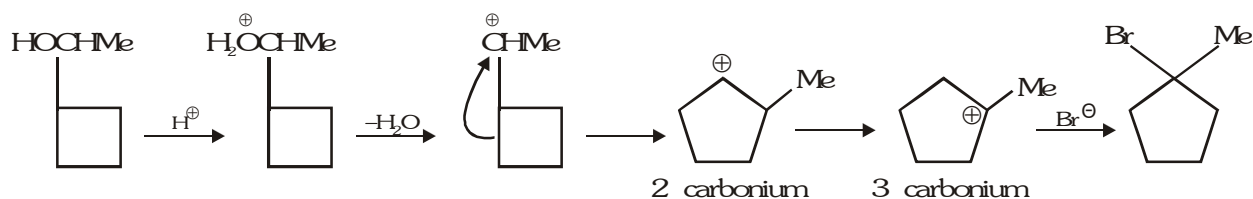
3 alkyl halide can also involve elimination of HX to give alkene in the presence of a base. So, it is better to start with 3 alkoxide and 1 alkyl halide, i.e., equation (b).



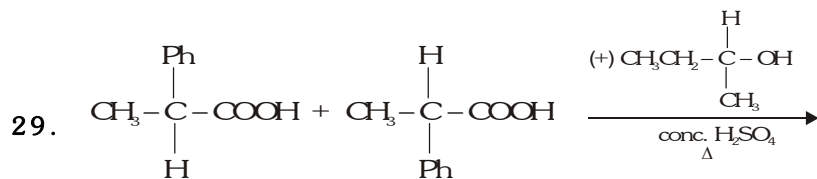
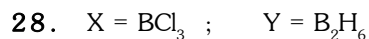
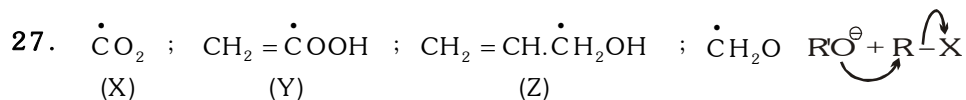
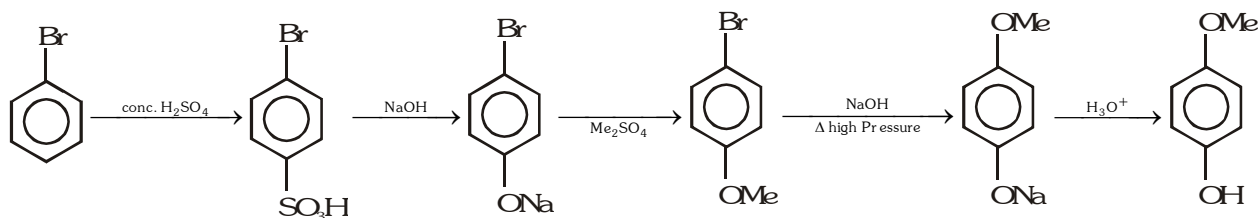
23. Due to intermolecular H-bonding



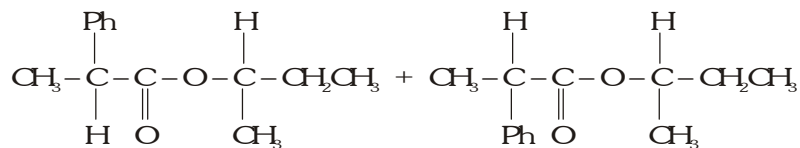
Meachnism



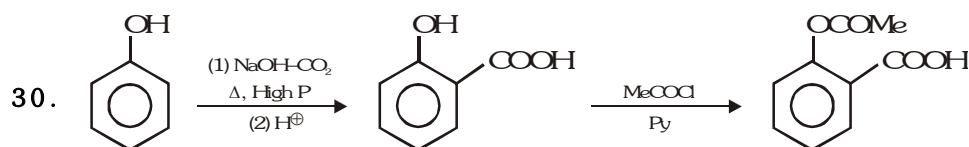
Alternative rout



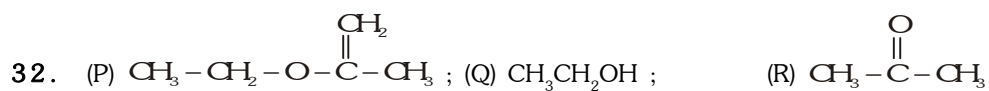
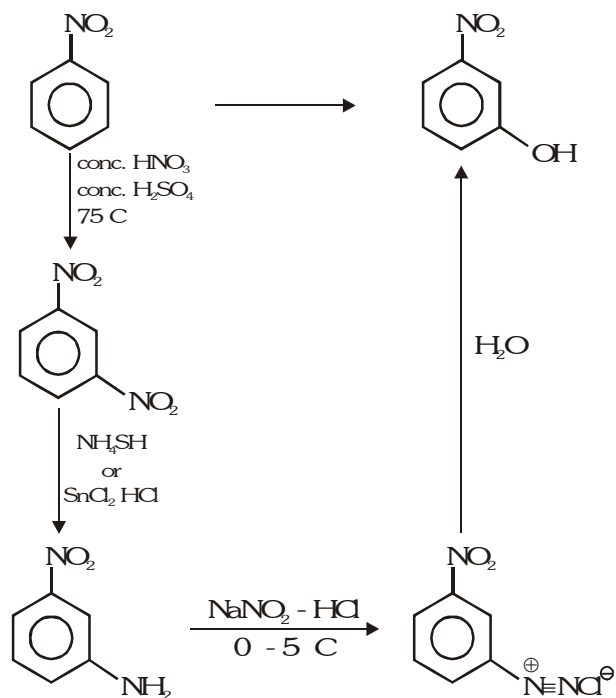
(racemic mixture)



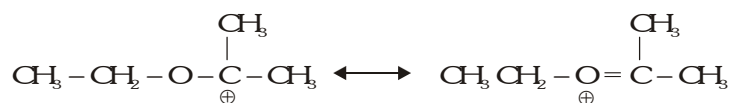
during esterification reaction only $-COOH$ and $-OH$ participates. There is no effect on structure or configuration of carbon adjacent to these groups. So when $(\pm)$ acid reacts with pure $(+)$ alcohol two esters are produced which are diastereoisomers of each other.



31.

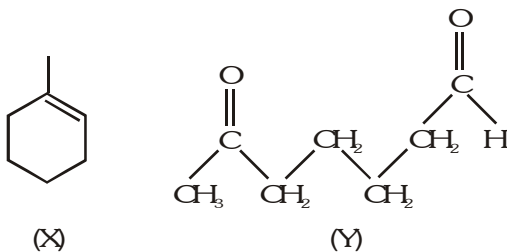


When ethylene reacts with dil. H_2SO_4 $\text{CH}_3\text{CH}_2^\oplus$ is produced during rate determining step, whereas P gives resonance stabilized intermediate.



Due to extra stability of intermediate the rate of reaction is very fast.

33.



34.

B

35.

D

36.

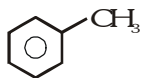
D

EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

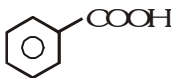
1. The correct order of reactivity towards the electrophilic substitution of the compounds aniline (I), benzene (II) and nitrobenzene (III) is :-

(A) III > II > I (B) II > III > I (C) I < II > III (D) I > II > III

2. Which of the following order is correct for the decreasing reactivity to ring monobromination of the following compounds :-



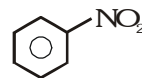
I



II



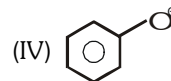
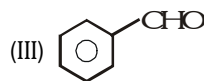
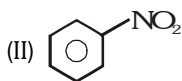
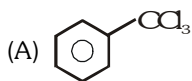
III



IV

(A) I > II > III > IV (B) II > III > IV > I (C) I > III > II > IV (D) III > I > II > IV

3. Electrophile NO_2^+ attacks the following :



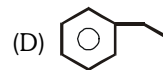
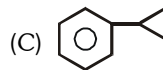
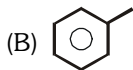
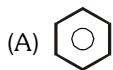
in which cases NO_2^+ will attack at meta position

(A) II and IV (B) I, II and III (C) II and IV (D) I only

4. The strongest deactivating effect on aromatic ring is

(A) $-\text{CH}_2\text{Cl}$ (B) $-\text{OCH}_3$ (C) $-\text{CH}_3$ (D) $-\text{CCl}_3$

5. Which of the following is maximum reactive for nitration is :-



6. The order of decreasing reactivity towards electrophilic reagent for the following :

(a) Benzene (b) Toluene

(c) Chloro benzene (d) Anisole

(A) b > d > a > c (B) d > c > b > a (C) d > b > a > c (D) a > b > c > d

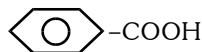
7. Increasing order of the following for electrophilic substitution reaction as -



(I)



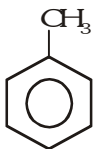
(II)



(III)

(A) I < II < III (B) III < II < I (C) II < III < I (D) I < III < II

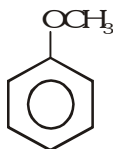
8. Among the compounds :



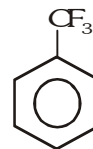
(I)



(II)



(III)



(IV)

the order of decreasing reactivity towards electrophilic substitution is :

(A) II > I > III > IV (B) III > I > II > IV (C) IV > I > II > III (D) I > II > III > IV

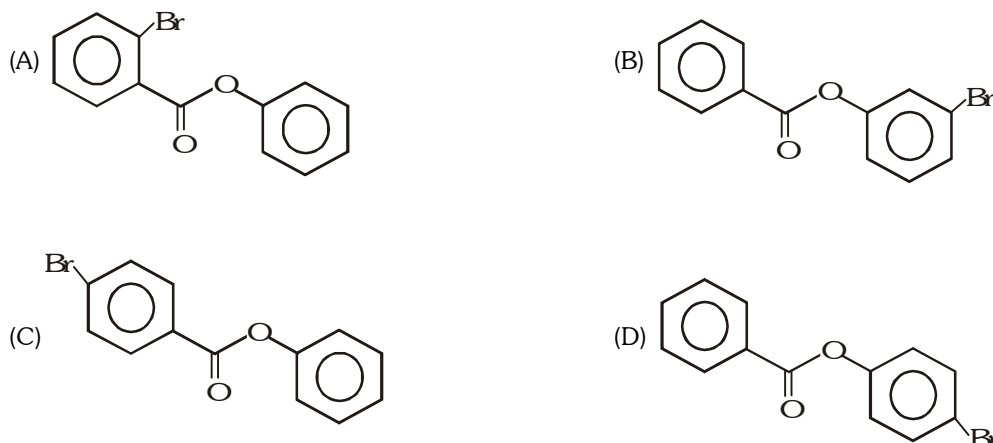
9. Choose the most reactive among the following compound :



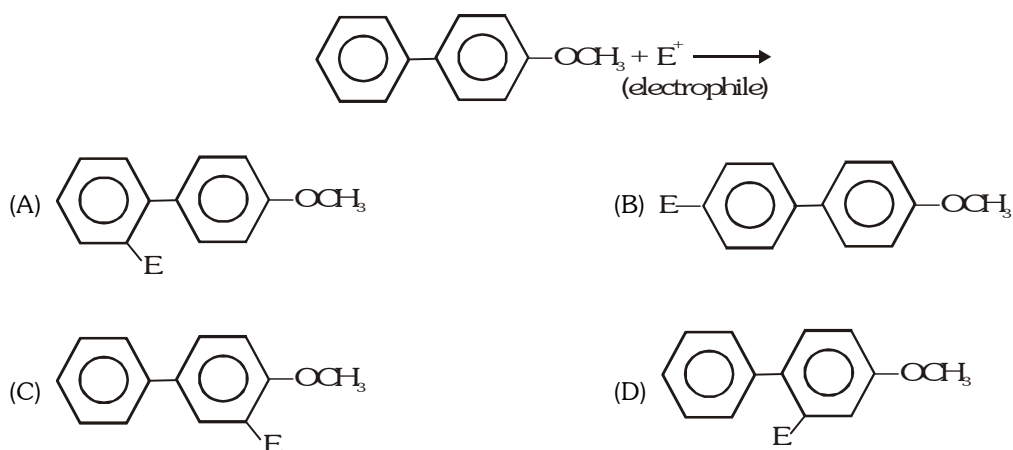
10. The number of possible dichloronitrobenzene isomers is :

- (A) 3 (B) 4 (C) 6 (D) 8

11. The major product formed on monobromination of phenylbenzoate is :



12. The major product formed in the reaction is :



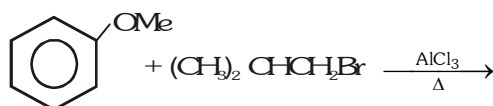
13. The electrophilic aromatic substitution of a compound C_6H_5Y produces mainly a meta-disubstituted product. Among the following which one could be the substituent Y ?

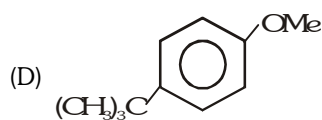
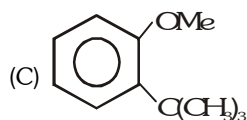
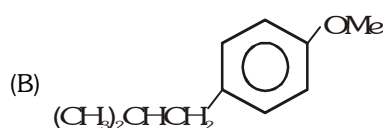
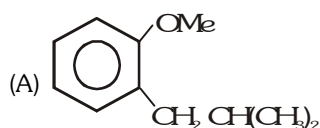
- (A) $-NH_2$ (B) $-COOH$ (C) $-CH_3$ (D) $-OCH_3$

14. Which of the following is an o-, p-directing but deactivating substituent in an electrophilic aromatic substitution :

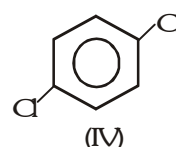
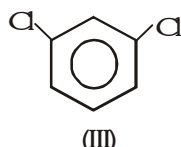
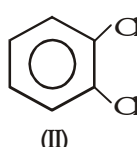
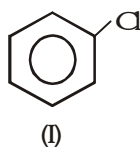
- (A) $-CCl_3$ (B) $-Cl$ (C) $-NHCOCH_3$ (D) $-OCH_3$

15. The major product formed in the reaction is :

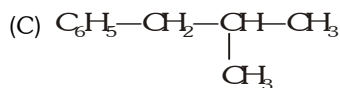
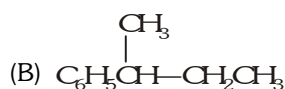
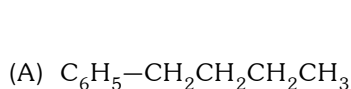




16. The dipole moment of chlorobenzene is 1.6 D. The expected dipole moment of meta-dichlorobenzene is:
 (A) 1.6 D (B) 3.2 D (C) $1.6\sqrt{2}$ D (D) 0.0 D
17. In the nitration of benzene with a mixture of concentrated HNO_3 and concentrated H_2SO_4 , the active species involved is :
 (A) NO_3^- (B) NO_2 (C) NO_2^- (D) NO_2^+
18. Which of the following substituted benzene derivatives would produce three isomeric products when one more substituent is introduced ?

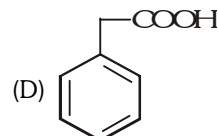
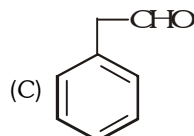
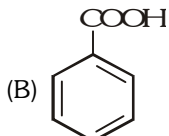
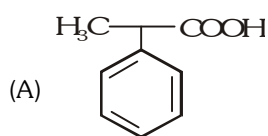


- (A) I, II and III (B) I and III (C) II and IV (D) I and IV
19. In the sulphonation of benzene, the active electrophilic species is :
 (A) SO_2 (B) SO_3 (C) SO_4^{2-} (D) HSO_4^-
20. The Friedel-Crafts reaction of benzene with n-butyl chloride at 0°C produces :

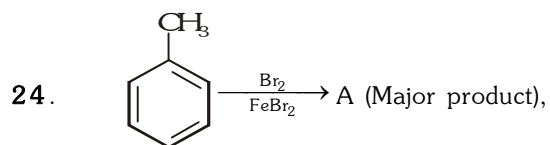


(D) all of these

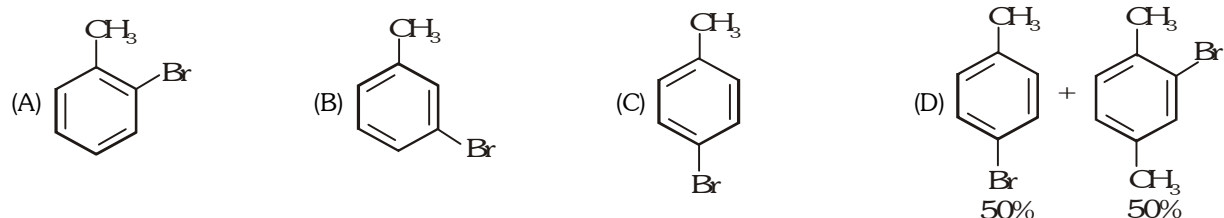
21. Sulfonation differs from most of electrophilic aromatic substitution reactions with the fact that the reaction-
 (A) is reversible (B) requires Lewis acid as catalyst
 (C) is explosive (D) takes place at high temperature
22. Cumene on treatment with KMnO_4 gives -



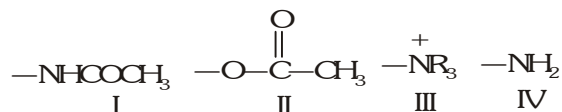
23. Benzene reacts with CH_3COCl in the presence of anhyd. AlCl_3 to give :
 (A) $\text{C}_6\text{H}_5\text{CH}_3$ (B) $\text{C}_6\text{H}_5\text{Cl}$ (C) $\text{C}_6\text{H}_5\text{O}_2\text{Cl}$ (D) $\text{C}_6\text{H}_5\text{COCH}_3$



then the major product A is -



25. Arrange the following in correct activating order towards EAS -

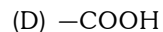
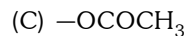
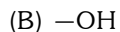
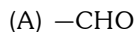


- (A) $\text{III} < \text{I} < \text{II} < \text{IV}$ (B) $\text{IV} < \text{I} < \text{II} < \text{III}$
 (C) $\text{III} < \text{II} < \text{I} < \text{IV}$ (D) $\text{II} < \text{III} < \text{I} < \text{IV}$

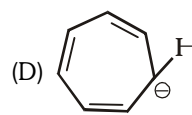
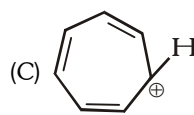
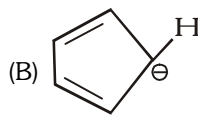
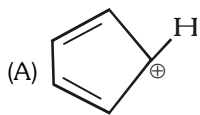
CHECK YOUR GRASP						ANSWER KEY					EXERCISE -1				
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	D	C	B	D	B	C	B	B	B	C	D	C	B	B	D
Que.	16	17	18	19	20	21	22	23	24	25					
Ans.	A	D	B	B	A	A	B	D	A	C					

EXERCISE-02**BRAIN TEASERS****SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)**

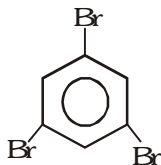
1. Which of the following groups are m-directing :



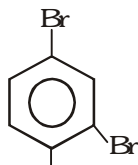
2. Amongst the ions the aromatic species are :



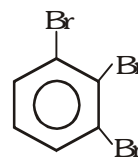
3. The following three isomeric tribromobenzenes are subjected to mononitration, which is/are given two isomers :



(I)



(II)



(III)

Which of these would produce three possible mononitrotribromobenzenes :

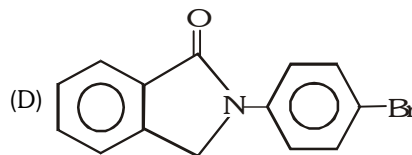
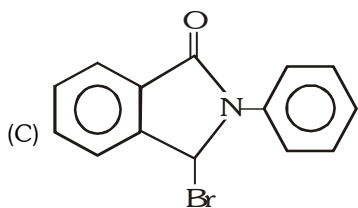
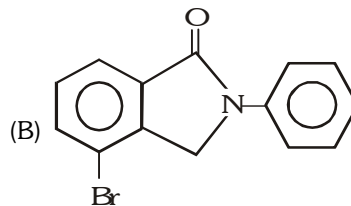
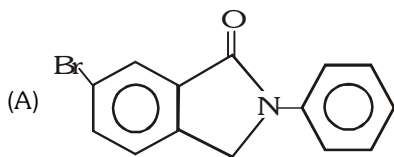
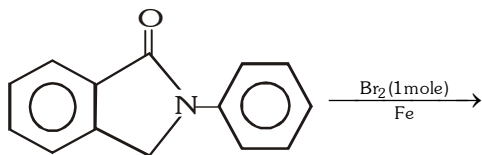
(A) II and III

(B) I and II

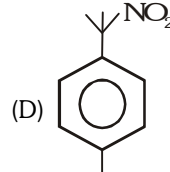
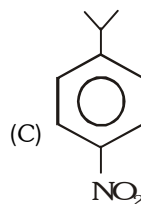
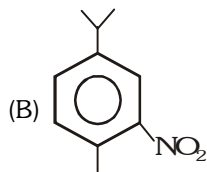
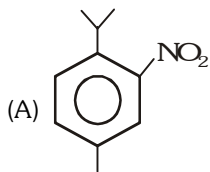
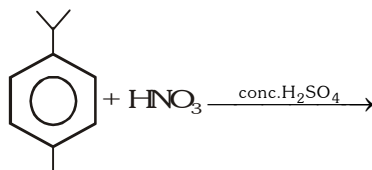
(C) II

(D) III

4. In the reaction the major product formed is :

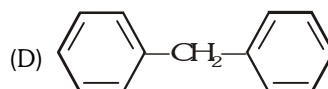
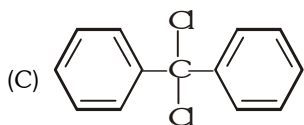
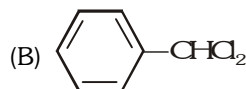
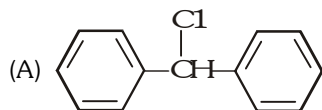


5. The major product formed in the reaction is :

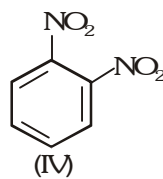
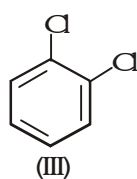
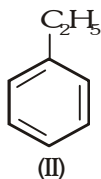
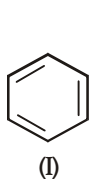


6. Nitrobenzene reacts with Br_2 in the presence of FeBr_3 to give m-bromonitrobenzene as the major product. Which of the following provides the best reason for the formation of m-bromonitrobenzene as the major product :
- The electron density at the meta position is greater than those at the ortho and para positions
 - Aromaticity is lost in the σ -complexes formed by the attack of Br^+ at the ortho and para positions but not at the meta position.
 - The σ -complex formed by the attack of Br^+ at the meta position is the least destabilized and the most stable among the three σ -complexes
 - In the final step of regeneration of benzene ring by the loss of H^+ from the σ -complexes, the meta-oriented σ -complex loses H^+ most readily
7. Isopropylbenzene can be prepared by :
- Benzene + $\text{CH}_3\text{CH}=\text{CH}_2 \xrightarrow{\text{H}_2\text{SO}_4}$
 - Benzene + $\text{CH}_3-\underset{\text{Cl}}{\underset{|}{\text{CH}}}-\text{CH}_3 \xrightarrow{\text{H}_2\text{SO}_4}$
 - Benzene + $\text{CH}_3-\underset{\text{Cl}}{\underset{|}{\text{CH}}}-\text{CH}_3 \xrightarrow{\text{AlCl}_3}$
 - Benzene + $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} \xrightarrow[\Delta]{\text{AlCl}_3}$
8. Which of the following characteristic does an aromatic compound exhibit :
- It should have $(4n + 2)$ π -electrons in the ring
 - It should be planar and conjugated
 - It should have $4n$ π -electrons in the ring
 - It should possess high resonance energy
9. Which of the following groups are meta-directing :
- $-\text{NH}_2$
 - $-\text{OH}$
 - $-\text{NO}_2$
 - $-\text{CN}$
10. Which of the following groups are ortho-and para-directing :
- $-\text{OH}$
 - $-\text{CHO}$
 - $-\text{CN}$
 - $-\text{NHCOCH}_3$
11. Which of the following statements are correct :
- An activating group is an electron-releasing group
 - An activating group activates all positions of the benzene ring
 - The effect of any group—whether activating or deactivating—is the strongest at the ortho-and para-positions in the benzene ring
 - An activating group activates only the ortho-and para-positions in the benzene ring
12. The major products formed in the reaction of toluene with chlorine in the presence of ferric chloride are:
- o-chlorotoluene
 - m-chlorotoluene
 - p-chlorotoluene
 - benzyl chloride
13. Benzene can undergo :
- substitution
 - addition
 - elimination
 - oxidation
14. Which of the following statements about the nitration of aromatic compounds are correct :
- The rate of nitration of toluene is greater than that of benzene
 - The rate of nitration of benzene is almost the same as that of hexadeutrobenezene
 - The rate of nitration of benzene is greater than that of hexadeutrobenezene
 - Nitration is an electrophilic substitution reaction

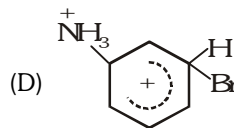
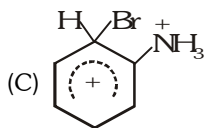
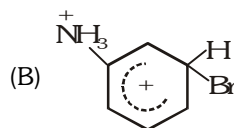
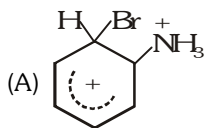
15. Halogenation of benzene in presence of AlCl_3 (anhy.) is :
 (A) nucleophilic substitution (B) nucleophilic addition
 (C) electrophilic substitution (D) free radical substitution
16. Which of the following structures correspond to the product expected, when excess of C_6H_6 reacts with CH_2Cl_2 in presence of anhydrous AlCl_3 ?



17. Select the incorrect statement among the following :
 (A) benzene undergoes predominantly electrophilic substitution reactions
 (B) toluene is more easily sulphonated than benzene
 (C) benzene reacts with CCl_4 in the presence of anhydrous AlCl_3 to give triphenyl methyl chloride
 (D) benzene reacts with chlorine (Cl_2) in presence of light to give benzyl chloride
18. Identify the correct order of reactivity in electrophilic substitution reactions of the following compounds:



- (A) $\text{I} > \text{II} > \text{III} > \text{IV}$ (B) $\text{IV} > \text{III} > \text{II} > \text{I}$
 (C) $\text{II} > \text{I} > \text{III} > \text{IV}$ (D) $\text{II} > \text{III} > \text{I} > \text{IV}$
19. The structure of Wheland intermediate obtained after the attack of Br^+ on anilinium ion is :



EXERCISE-03

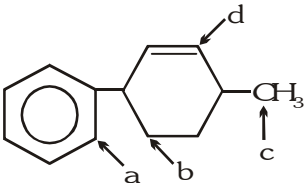
MISCELLANEOUS TYPE QUESTIONS

TRUE OR FALSE :

1. Substitution of benzene occurs through nucleophilic attack.
2. AlCl_3 (aq.) is used as catalyst in Friedel-Crafts reaction.
3. Diphenyl methane is obtained when excess benzene is treated with dichloro methane in presence of anhydrous AlCl_3 .
4.  gives ortho-methyl aniline when treated with CH_3Cl in presence of AlCl_3 (anhydrous).
5. $-\text{CCl}_3$ and $-\text{Cl}$ are the meta director.

FILL IN THE BLANKS :

1. Give the structure of the electrophile formed as per the major product :
 - (i) $\text{FeBr}_3 + \text{Br}_2 \longrightarrow \dots\dots\dots$
 - (ii) $\text{HNO}_3 + \text{H}_2\text{SO}_4 \xrightarrow[\text{conc.}]{\text{conc.}} \dots\dots\dots$
 - (iii) $\text{SO}_3 + \text{H}_2\text{SO}_4 \xrightarrow[\text{fuming sulphuric acid}]{} \dots\dots\dots$
 - (iv) $\text{CH}_3\text{Cl} + \text{AlCl}_3 \longrightarrow \dots\dots\dots$
 - (v) $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \text{CH}_2\text{Cl} + \text{AlCl}_3 \xrightarrow{\Delta} \dots\dots\dots$
2. Write the structure of σ -complex (carbocation) in the following S_E reactions :
 - (i)  + SO_3 (neutral) $\longrightarrow \dots\dots\dots$
 - (ii)  + NO_2^+ $\longrightarrow \dots\dots\dots$
3. Write hybridisation of carbon atoms as indicated :



(a)

(b)

(c)

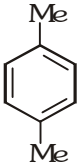
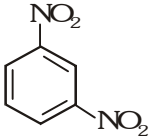
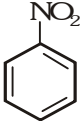
(d)

MATCH THE COLUMN

1. Match the column I with column II.

Column-I		Column-II	
(A)	Cyclic conjugated polyenes with $(4n + 2) \pi$ -electrons	(p)	Arenes and alkyl halides in presence of anhydrous AlCl_3
(B)	Dichlorobenzene	(q)	Aromatic compounds
(C)	Friedel crafts reaction	(r)	Delocalization of π -electrons
(D)	Meta directing group	(s)	Deactivates the ring towards electrophilic substitution

2. Match the column I with column II.

Column-I		Column-II	
	(Compounds)		(orientation of attacking electrophile)
(A)		(p)	Meta position with respect to $-\text{NO}_2$
(B)		(q)	ortho with respect to methyl group
(C)		(r)	only one mono substituted product
(D)		(s)	meta with respect to $-\text{CH}_3$

ASSERTION & REASON QUESTION :

These questions contains, Statement-I (assertion) and Statement-II (reason).

(A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I

(B) Statement-I is True, Statement-II is True ; Statement-II is NOT a correct explanation for Statement-I

(C) Statement-I is True, Statement-II is False.

(D) Statement-I is False, Statement-II is True.

1. **Statement-I** : In nitration H_2SO_4 is used as sulphonating agent.

Because

Statement-II : Benzene has multicentre π -bonding.

2. **Statement-I** : C_2H_5 is a meta-directing group.

Because

Statement-II : The groups which direct the incoming group to meta position are called meta-directing groups.

3. **Statement-I** : Tropylium cation is aromatic in nature.



Because

Statement-II : The only property that determines its aromatic behavior is its planar structure.

4. **Statement-I** : Benzene on heating with conc. H_2SO_4 gives benzene sulphonic acid which when heated with superheated steam under pressure gives benzene.

Because

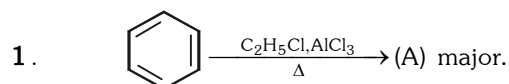
Statement-II : Sulphonation is a reversible process.

5. **Statement-I** : Chlorination of ethyl benzene with Cl_2 in presence of heat and light nearly yield 1-chloro-1-phenyl ethane as major product.
Because
Statement-II : The reaction occurs through intermediate formation of the radical $\text{C}_6\text{H}_5\dot{\text{C}}\text{H}-\text{CH}_3$.
6. **Statement-I** : Nitration of toluene is easier than benzene.
Because
Statement-II : The methyl group in toluene is electron-releasing.
7. **Statement-I** : Benzene reacts with n-propyl chloride in presence of AlCl_3/Δ to give isopropyl benzene.
Because
Statement-II : Benzene undergoes electrophilic substitution readily.
8. **Statement-I** : Most o-, p -directing substitution are deactivating.
Because
Statement-II : o-, p-directors have at least one lone pair of non-bonding electrons.
9. **Statement-I** : Nitrobenzene does not undergo Friedel-Craft's reaction.
Because
Statement-II : Nitrogroup is a deactivating group.
10. **Statement-I** : The rate of nitration of benzene is less than hexa deuterio benzene.
Because
Statement-II : Deuterium is an isotope of hydrogen.

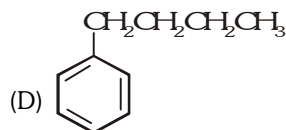
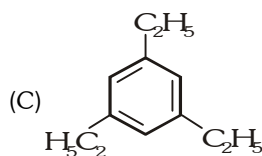
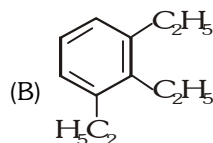
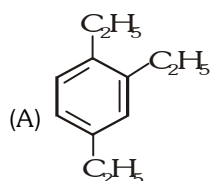
COMPREHENSION BASED QUESTIONS :

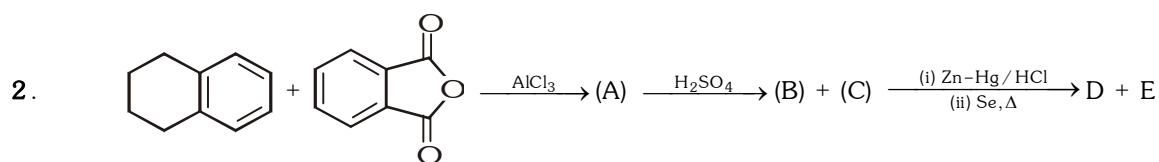
Comprehension # 1

The typical reaction of benzene and other aromatic compounds are electrophilic substitution. Presence of electron donating group activates the ring towards electrophilic substitution, while presence of electron withdrawing group deactivates the ring towards electrophilic substitution but at the same time activates the ring towards nucleophilic substitution. Some groups are predominantly meta-directing and all of these are deactivating. Except halogen, most of the o-and p-directing groups are activating groups.

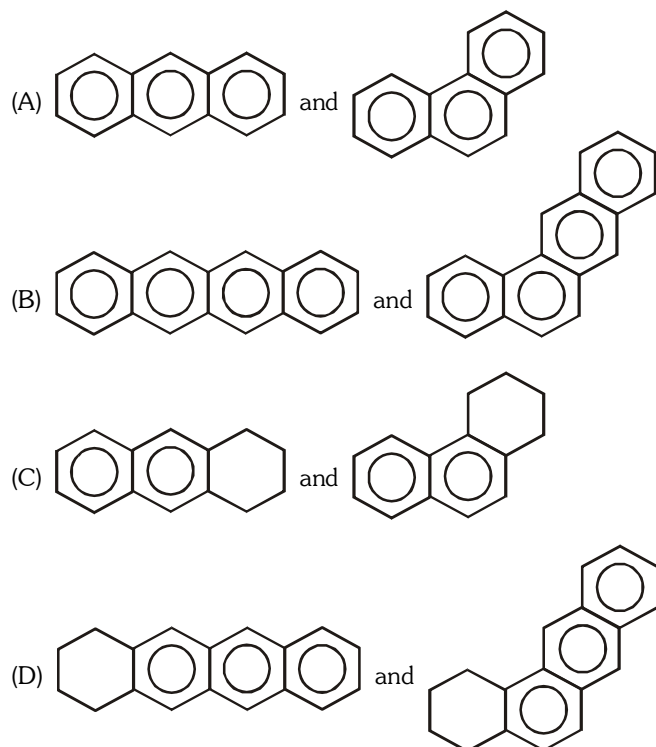


A is trisubstituted benzene. The structure of A is :

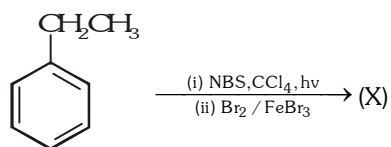




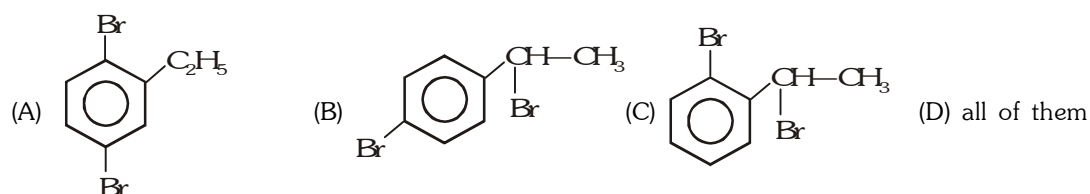
D and E are :



3. Which of the following compound is not formed.



X represents mixture of organic compounds. The mixture does not contain.



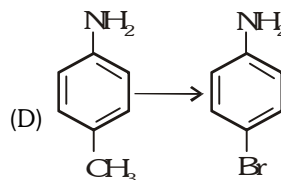
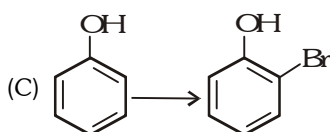
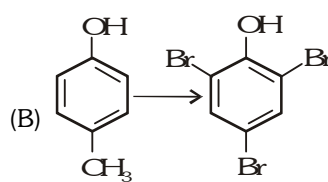
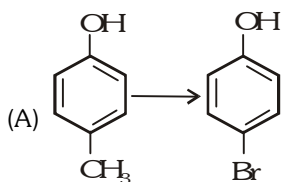
Comprehension # 2

It is not always easy to predict the position of attack on multiply substituted benzene. If the benzene ring bears different ortho/para directing group at the 1 and 4 positions, the position of further substitution is not immediately clear. Sometimes steric effects determine the outcome. In other cases, electronic factors determine the outcome, and further reaction will be at the position activated by the more strongly activating group.

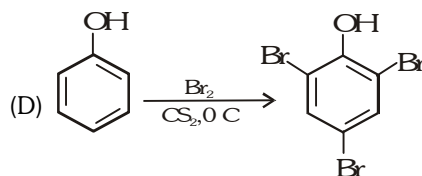
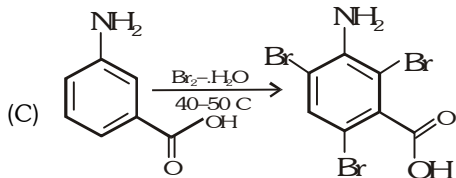
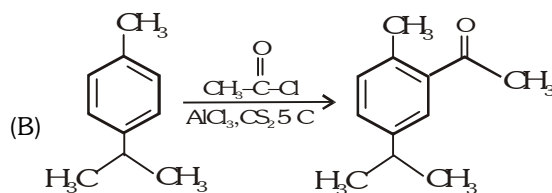
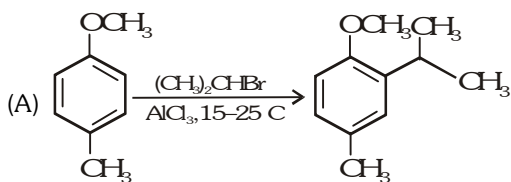
Some substituents are so strongly activating that no catalyst is needed, and it is often difficult to stop substitution after mono substitution. Mild conditions are needed to restrict the reaction to mono-substitution.

It is possible to reduce the activity such groups (by side chain reaction) so that the reaction can be stopped after mono substitution then and again by a side chain reaction the original group is restored. Effective use can sometimes be made of removable blocking groups on the ring.

1. Which of the following synthesis could be done in the ring step ?



2. Which of the following is the correct major product ?



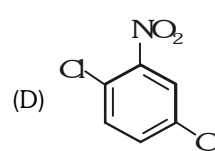
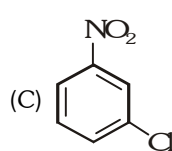
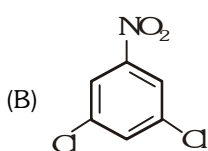
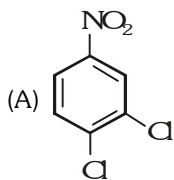
3. Which of the following side chain reaction/s can be used to reduce the activity of strongly activating groups like $-\text{OH}$?
- (A) benzylation (B) acetylation (C) both of the above (D) none of the above

Comprehension # 3

A third group is least likely to enter between two groups in the meta relationship. This is the result of steric hindrance and increases in importance with the size of the groups on the ring and with the size of the attacking species.

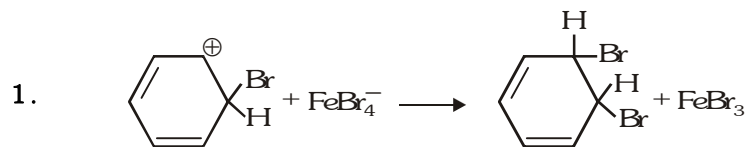
When a Meta-directing group is meta to an ortho-para directing group, the incoming group primarily goes ortho to the meta directing group rather than para.

1. Chlorination of m-chloro nitro benzene gives :



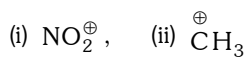
2. $\xrightarrow{E^+}$?
- (A) (B) (C) (D) all of them
3. $\xrightarrow{CH_2/CH_3COOH}$?
- (A) (B) (C) (D)
4. $\xrightarrow{HNO_3/H_2SO_4}$, unlikely to form is -
- (A) (B) (C) (D)

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
True / False 1. F 2. F 3. T 4. F 5. F Fill in the Blanks 1. (i) Br^+; (ii) NO_2^+; (iii) SO_3; (iv) $[CH_3 - \overset{\delta+}{Cl} - \overset{\delta-}{AlCl_3}]$; (v) 2. (i) (ii) 3. (a) sp^2; (b) sp^3; (c) sp^3; (d) sp^2 Match the Column 1. (A) $\rightarrow$ q, r ; B $\rightarrow$ q, r, s ; (C) $\rightarrow$ p ; (D) $\rightarrow$ s 2. (A) $\rightarrow$ q, r ; (B) $\rightarrow$ p, r ; (C) $\rightarrow$ q, p ; (D) $\rightarrow$ p Assertion - Reason Questions 1. D 2. D 3. C 4. A 5. A 6. A 7. B 8. D 9. A 10. D Comprehension Based Questions Comprehension #1 : 1. (A) 2. (B) 3. (A) Comprehension #2 : 1. (C) 2. (D) 3. (C) Comprehension #3 : 1. (A) 2. (D) 3. (B) 4. (C)		

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

Why does the given step does not occur

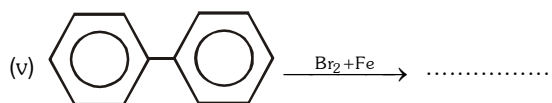
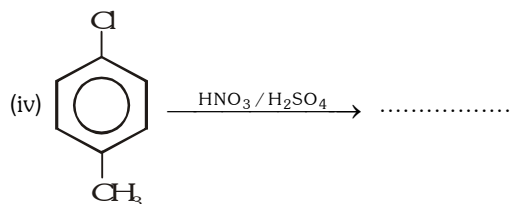
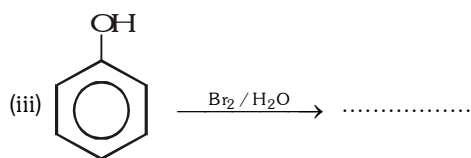
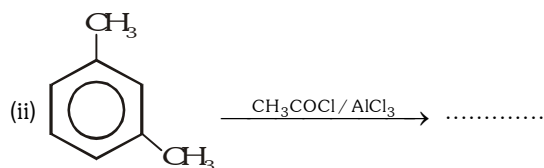
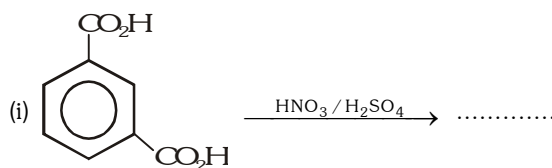
2. Write resonance structures of arenium ions with :



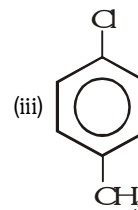
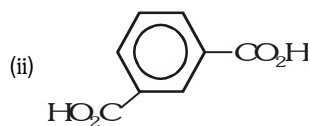
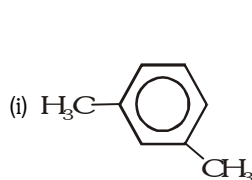
3. $\text{C}_6\text{H}_4(\text{NO}_2)\text{CH}_3$ can form different isomers. Write three isomers.

(I) (II) (III)

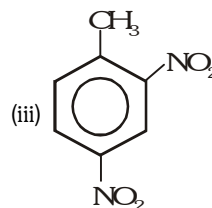
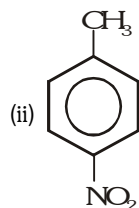
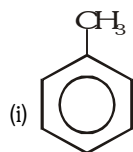
4. Write the structure of the products formed in the following S_E reactions :



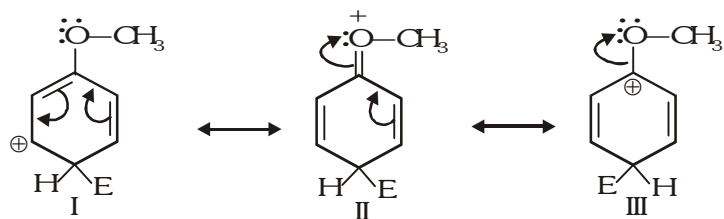
5. Indicate the position of the electrophile in the following compounds :



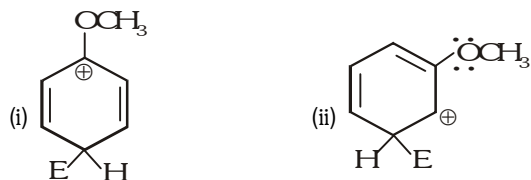
6. Indicate the compound which can be nitrated most easily :



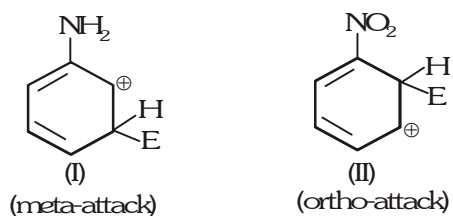
7. Indicate wrong electron-migration :



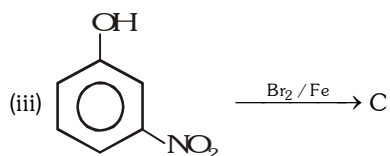
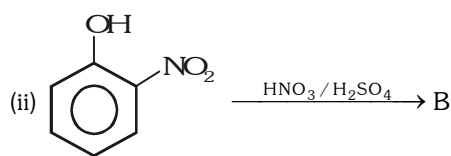
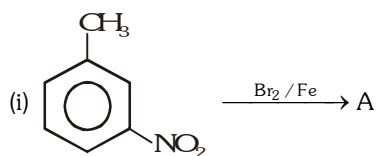
8. Indicate more stable carbocation and explain :



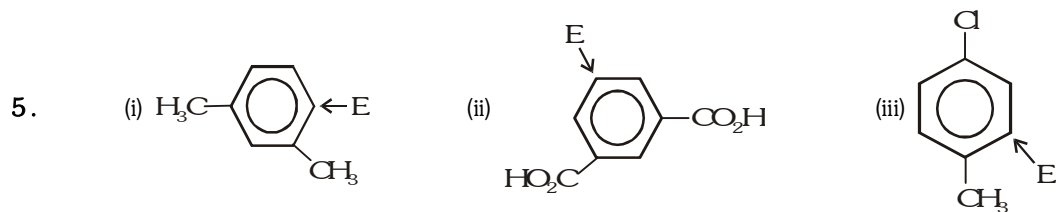
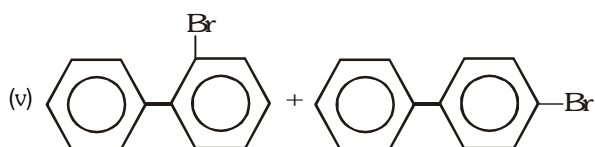
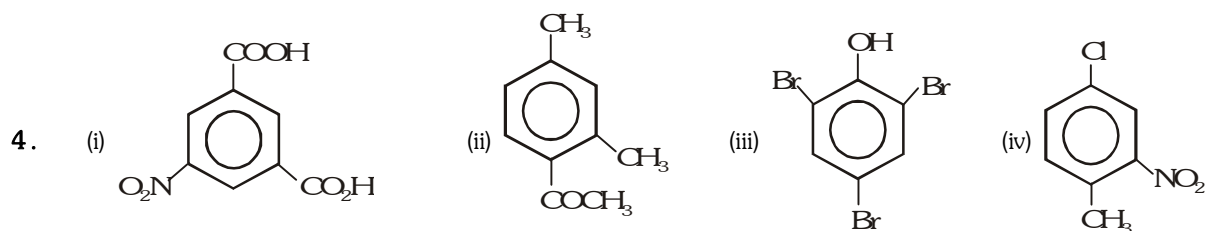
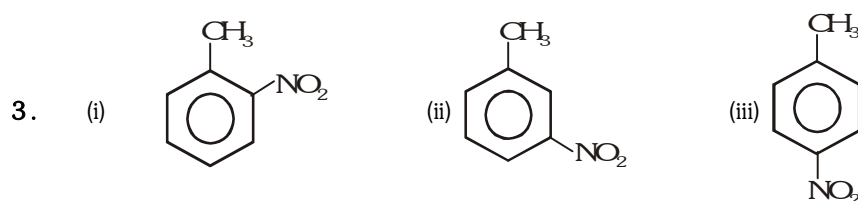
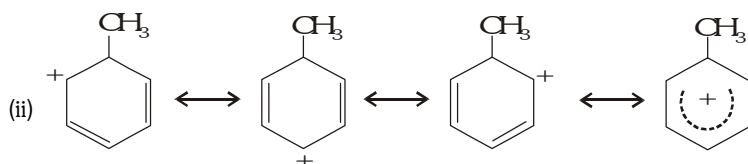
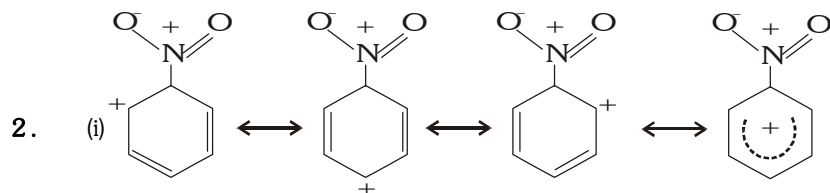
9. In case of meta- attack on aniline (where —NH_2 group is o-, p-directing) and in case of ortho-attack on nitrobenzene, resonating structures are possible. Which of them are more stable ?



10. What are the major products of the following :



1. The given reaction does not occur because on addition the ring lost its aromaticity and on deprotonation of σ -complex it retains the aromaticity hence substitution reaction takes place.

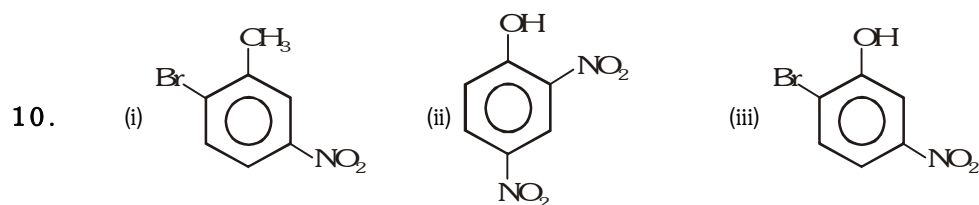


6. (i) ; because $-\text{CH}_3$ is an activating group hence toluene is most reactive.

7. (III)

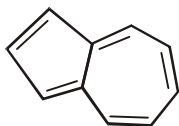
8. (I) is more stable, due to the + M effect of $-\text{OCH}_3$.

9. (I)

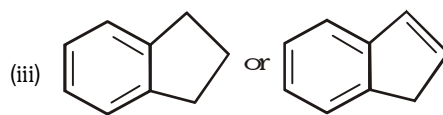
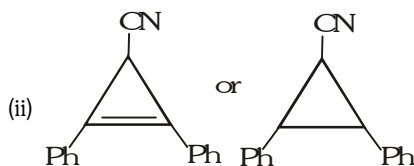
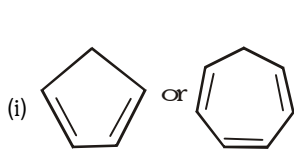


EXERCISE-04 [B]**BRAIN STORMING SUBJECTIVE EXERCISE**

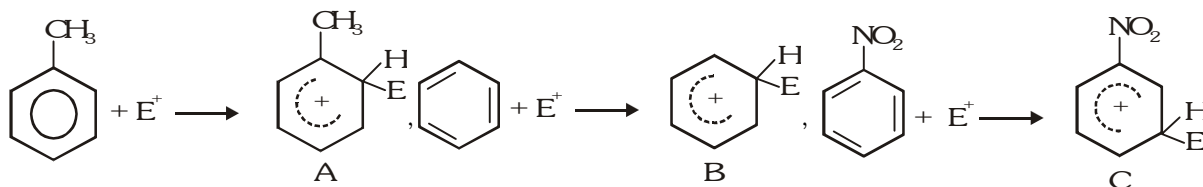
1. Unlike most hydrocarbons, azulene ($C_{10}H_8$) is highly coloured (deep blue), although its isomer, naphthalene, does not have significant zwitter-ionic character, azulene does.
- (i) Draw a resonance structure of azulene in which five-membered ring is anionic and the seven-membered ring cationic.



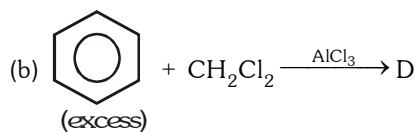
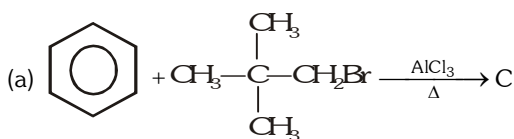
- (ii) Can azulene be considered aromatic ?
2. Explain, which of these compounds is a stronger acid. :



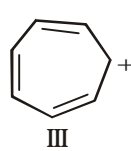
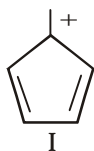
3. Arrange A, B, C in increasing potential energy :



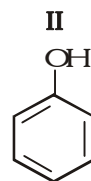
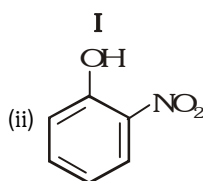
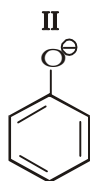
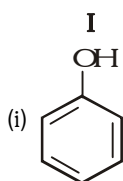
4. $C_6H_5^+$ has electron withdrawing inductive effect yet each ring of biphenyl, $C_6H_5-C_6H_5$ is more reactive than benzene towards S_E reaction and chief products are ortho and para-isomer. Show how reactivity and orientation can be accounted for on the basis of resonance.
5. What are the major products in the following reactions :

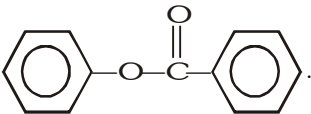


6. Arrange the following in increasing order of electrophilic strength :

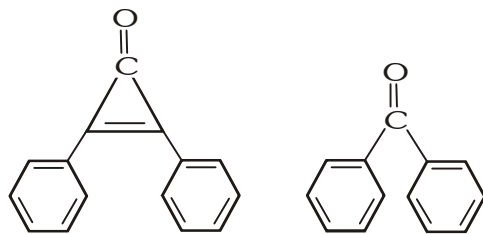


7. Which is more reactive towards S_E and explain ?

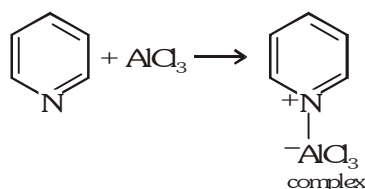
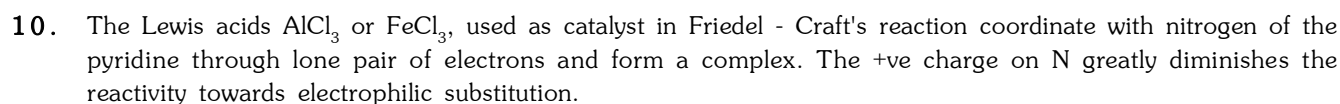
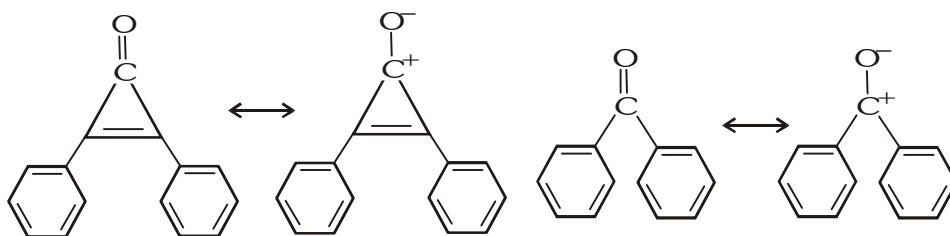
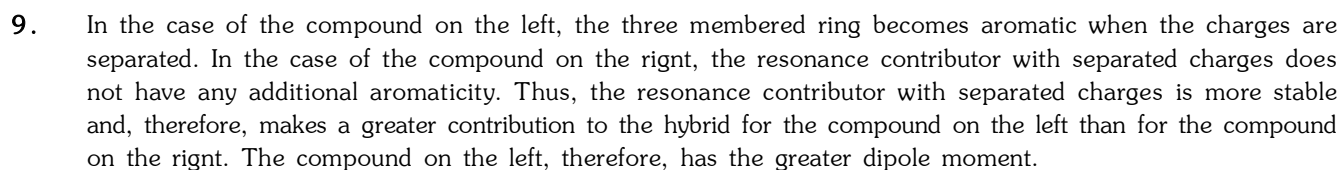
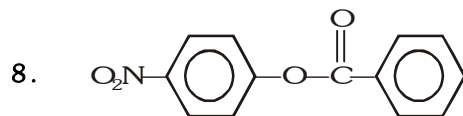
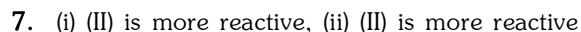
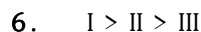
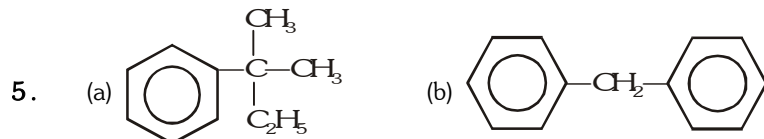
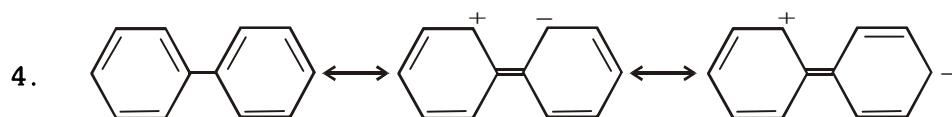
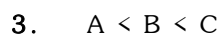
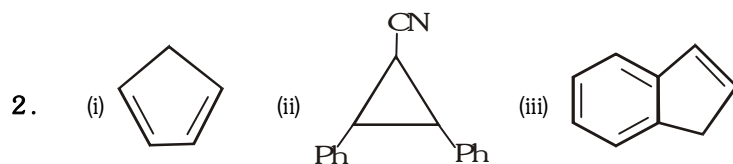
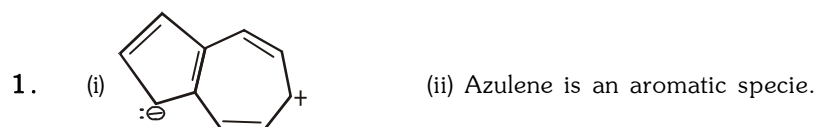


8. What are the major products when $\text{NO}_2^\oplus$ (nitronium) reacts with .

9. Which of the following compounds has the greater dipole moment ?

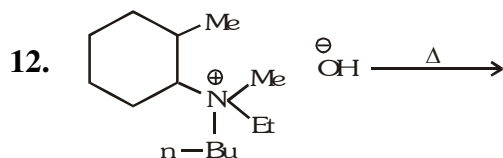


10. Explain why pyridine does not undergo Friedel-Craft's reaction ?



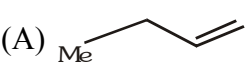
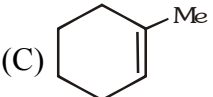
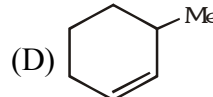
EXERCISE - 05 [A]**JEE-[MAINS] : PREVIOUS YEAR QUESTIONS**

1. Which of these will not react with acetylene - [AIEEE-2002]
(A) NaOH (B) ammonical AgNO_3 (C) Na (D) HCl
2. What is the product formed when acetylene reacts with hypochlorous acid - [AIEEE-2002]
(A) CH_3COCl (B) ClCH_2CHO (C) Cl_2CHCHO (D) ClCH_2COOH
3. 1-Butene may be converted to butane by reaction with - [AIEEE-2003]
(A) Pd/H_2 (B) $\text{Zn} - \text{HCl}$ (C) $\text{Sn} - \text{HCl}$ (D) $\text{Zn} - \text{Hg/HCl}$
4. On mixing a certain alkane with chlorine and irradiating it with ultraviolet light, it forms only one monochloroalkane. This alkane could be - [AIEEE-2003]
(A) neopentane (B) propane (C) pentane (D) isopentane
5. Which one of the following is reduced with Zn-Hg/HCl to give the corresponding hydrocarbon [AIEEE-2004]
(A) Butan-2-one (B) Acetic acid (C) Acetamide (D) Ethyl acetate
6. Which one of the following has the minimum boiling point : [AIEEE-2004]
(A) isobutane (B) 1-butyne (C) 1-butene (D) n-butane
7. 2-Methylbutane on reacting with bromine in the presence of sunlight gives mainly [AIEEE-2005]
(A) 2-bromo-2-methylbutane (B) 1-bromo-2-methylbutane
(C) 1-bromo-3-methylbutane (D) 2-bromo-3-methylbutane
8. Alkyl halides react with dialkyl copper reagent to give [AIEEE-2005]
(A) alkyl copper halides (B) alkenes
(C) alkenyl halides (D) alkanes
9. Reaction of one molecule of HBr with one molecule of 1,3-butadiene at 40°C gives predominantly [AIEEE-2005]
(A) 1-bromo-2-butene under thermodynamically controlled conditions
(B) 3-bromobutene under kinetically controlled conditions
(C) 1-bromo-2-butene under kinetically controlled conditions
(D) 3-bromobutene under thermodynamically controlled conditions
10. Acid catalyzed hydration of alkenes except ethene leads to the formation of [AIEEE-2005]
(A) secondary or tertiary alcohol
(B) primary alcohol
(C) mixture of secondary and tertiary alcohols
(D) mixture of primary and secondary alcohols
11. Elimination of bromine from 2-bromobutane results in the formation of [AIEEE-2005]
(A) predominantly 2-butene (B) equimolar mixture of 1 and 2-butene
(C) predominantly 2-butyne (D) predominantly 1-butene



The alkene formed as a major product in the above elimination reaction is-

[AIEEE-2006]

- (A)  (B) $\text{CH}_2=\text{CH}_2$ (C)  (D) 

13. Reaction of trans-2-phenyl-1-bromocyclopentane on reaction with alcoholic KOH produces-
 (A) 4-phenyl cyclopentene (B) 2-phenyl cyclopentene [AIEEE-2006]
 (C) 1-phenyl cyclopentene (D) 3-phenyl cyclopentene

14. Phenyl magnesium bromide reacts with methanol to give- [AIEEE-2006]
 (A) A mixture of anisole and $\text{Mg}(\text{OH})\text{Br}$ (B) A mixture of benzene and $\text{Mg}(\text{OMe})\text{Br}$
 (C) A mixture of toluene and $\text{Mg}(\text{OH})\text{Br}$ (D) A mixture of phenol and $\text{Mg}(\text{Me})\text{Br}$

15. Which of the following reactions will yield, 2, 2-dibromopropane [AIEEE-2007]
 (A) $\text{CH}_3-\text{C}\equiv\text{CH} + 2\text{HBr} \longrightarrow$ (B) $\text{CH}_3\text{CH}=\text{CHBr} + \text{HBr} \longrightarrow$
 (C) $\text{CH}\equiv\text{CH} + 2\text{HBr} \longrightarrow$ (D) $\text{CH}_3-\text{CH}=\text{CH}_2 + \text{HBr} \longrightarrow$

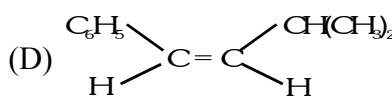
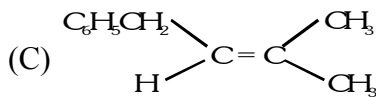
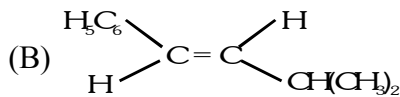
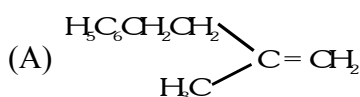
16. In the following sequence of reactions, the alkene affords the compound 'B' :- [AIEEE-2008]
 $\text{CH}_3\text{CH}=\text{CHCH}_3 \xrightarrow{\text{O}_3} \text{A} \xrightarrow[\text{Zn}]{\text{H}_2\text{O}} \text{B}.$

The compound B is

- (A) $\text{CH}_3\text{CH}_2\text{CHO}$ (B) CH_3COCH_3 (C) $\text{CH}_3\text{CH}_2\text{COCH}_3$ (D) CH_3CHO
 17. The hydrocarbon which can react with sodium in liquid ammonia is [AIEEE-2008]
 (A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{C}\equiv\text{CCH}_2\text{CH}_2\text{CH}_3$ (B) $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH}$
 (C) $\text{CH}_3\text{CH}=\text{CHCH}_3$ (D) $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CCH}_2\text{CH}_3$
 18. The treatment of CH_3MgX with $\text{CH}_3\text{C}\equiv\text{C}-\text{H}$ produces [AIEEE-2008]

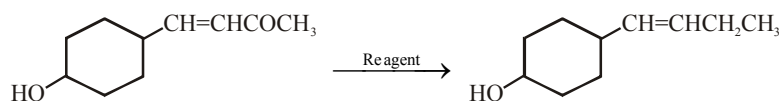
- (A) $\text{CH}_3-\text{CH}=\text{CH}_2$ (B) $\text{CH}_3\text{C}\equiv\text{C}-\text{CH}_3$ (C) $\begin{array}{c} \text{H} \quad \text{H} \\ | \quad | \\ \text{CH}_3-\text{C}=\text{C}-\text{CH}_3 \end{array}$ (D) CH_4

19. The main product of the following reaction is [AIEEE-2010]
 $\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{OH})\text{CH}(\text{CH}_3)_2 \xrightarrow{\text{Conc. H}_2\text{SO}_4} ?$



20. One mole of a symmetrical alkene on ozonolysis gives two moles of an aldehyde having a molecular mass of 44 u. The alkene is :- [AIEEE-2010]
 (A) ethene (B) propene (C) 1-butene (D) 2-butene
 21. Ozonolysis of an organic compound gives formaldehyde as one of the products. This confirms the presence of :- [AIEEE-2011]
 (A) an isopropyl group (B) an acetylenic triple bond
 (C) two ethylenic double bonds (D) a vinyl group

22. Ozonolysis of an organic compound 'A' produces acetone and propionaldehyde in equimolar mixture. Identify 'A' from the following compounds :- [AIEEE-2011]
- (A) 2 - Methyl - 1 - pentene (B) 1 - Pentene
(C) 2 - Pentene (D) 2 - Methyl - 2 - pentene
23. 2-Hexyne gives trans -2-Hexene on treatment with :- [AIEEE-2012]
- (A) LiAlH_4 (B) Pt/H_2 (C) Li/NH_3 (D) Pd/BaSO_4
24. In the given transformation, which of the following is the most appropriate reagent ? [AIEEE-2012]

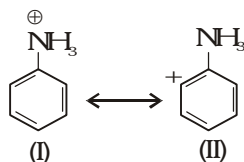


- (A) NaBH_4 (B) $\text{NH}_2\text{NH}_2, \text{OH}^-$ (C) $\text{Zn} - \text{Hg} / \text{HCl}$ (D) Na, Liq.NH_3

JEE-[MAIN] : PREVIOUS YEAR QUESTIONS							ANSWER KEY			EXERCISE -5[A]					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans	1	3	1	1	1	1	1	4	1	1	1	2	4	2	1
Que.	16	17	18	19	20	21	22	23	24						
Ans	4	2	4	2	4	4	4	3	2						

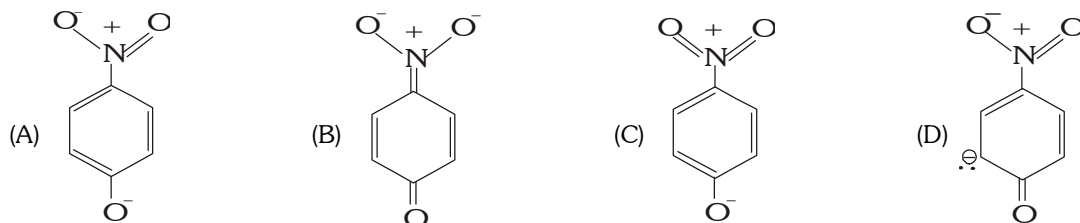
EXERCISE - 05 [B]**JEE-[ADVANCED] : PREVIOUS YEAR QUESTIONS**

1. The chlorination of toluene in presence of ferric chloride gives predominantly : [IIT-86]
(A) Benzyl chloride (B) m-chlorotoluene
(C) Benzal chloride (D) o-and-p-chlorotoluene
2. Aryl halides are less stable towards nucleophilic substitution reaction as compared to alkyl halides due to-
(A) The formation of less stable carbonium ion (B) Resonance stabilization
(C) Longer carbon-halogen bond (D) The inductive effect [IIT-90]
3. The most basic compound among the following is - [IIT-90]
(A) Benzylamine (B) Aniline (C) Acetanilide (D) p-nitro aniline
4. Excess chlorination of toluene in the presence of light and heat followed by treatment with aqueous NaOH gives :
(A) o-cresol (B) p-cresol (C) 2, 4-dihydroxytoluene (D) Benzoic acid [IIT-90]
5. Choose the correct statement from the ones given below for two anilium in - [IIT-93]

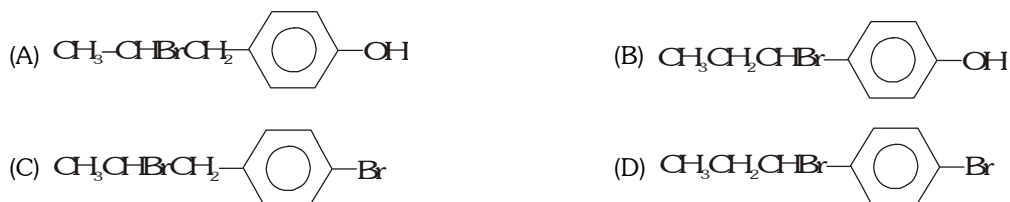


- (A) II is not an acceptable canonical structure because carbonium ions are less stable than ammonium ions
(B) II is not an acceptable canonical structure because it is aromatic
(C) II is not an acceptable canonical structure because the nitrogen has 10 valence electrons
(D) II is an acceptable canonical structure
6. The major product of nitration of benzoic acid is : [IIT-93]
(A) 3-nitrobenzoic acid (B) 4-nitrobenzoic acid
(C) 2-nitrobenzoic acid (D) 2,4-dinitrobenzoic acid
7. Most stable carbonium ion is : [IIT-95]
(A) $p\text{-NO}_2\text{-C}_6\text{H}_4\text{-}^+\text{CH}_2$ (B) $\text{C}_6\text{H}_5\text{-}^+\text{CH}_2$
(C) $p\text{-Cl-C}_6\text{H}_4\text{-}^+\text{CH}_2$ (D) $p\text{-CH}_3\text{O-C}_6\text{H}_4\text{-}^+\text{CH}_2$
8. Arrange in order of decreasing trend towards S_E reactions, [IIT-95]
Chlorobenzene, Benzene, Anilinium chloride, Toluene
I II III IV
(A) I > II > III > IV (B) III > I > II > IV (C) IV > II > I > III (D) I > II > III > IV
9. Among the following statements on the nitration of aromatic compounds, the false one is - [IIT-97]
(A) The rate of nitration of benzene is almost the same as that of hexadeuterobenzene
(B) The rate of nitration of toluene is greater than that of benzene
(C) The rate of nitration of benzene is greater than that of hexadeuterobenzene
(D) Nitration is an electrophilic substitution reaction
10. Nitrobenzene can be prepared from benzene by using a mixture of conc. HNO_3 and conc. H_2SO_4 . In the nitrating mixture HNO_3 acts as a - [IIT-98]
(A) Base (B) Acid (C) Reducing agent (D) Catalyst
11. Benzyl chloride ($\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$) can be prepared from toluene by chlorination with - [IIT-98]
(A) $\text{SO}_2\text{Cl}_2/h\nu$ (B) SOCl_2 (C) PCl_5 (D) NaOCl

12. The most unlikely representation of resonance structure of p-nitrophenoxide ions is - [IIT-98]



13. The reaction of $\text{CH}_3\text{-CH=CH-}$  -OH with HBr gives - [IIT-98]

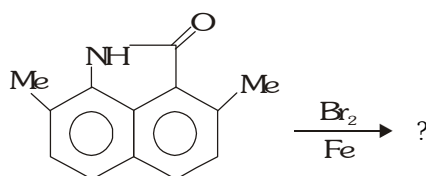


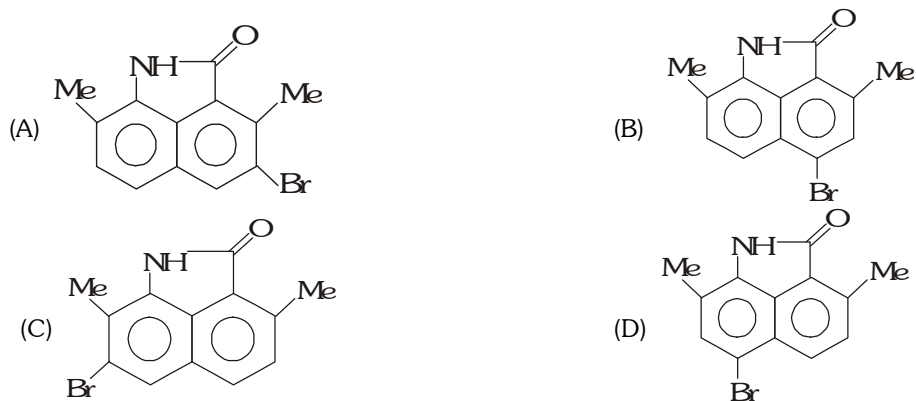
14. A new C-C bond is formed in - [IIT-98]

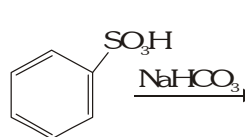
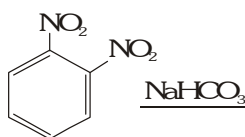
- (A) Cannizzaro' reaction (B) Friedel Crafts reaction
(C) Both (A) & (B) (D) None of these

15. Amongst the following the strongest base is - [IIT-2000]

- (A) $\text{C}_6\text{H}_5\text{NH}_2$ (B) $\text{p-O}_2\text{NC}_6\text{H}_4\text{NH}_2$ (C) $\text{m-O}_2\text{NC}_6\text{H}_4\text{NH}_2$ (D) $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$

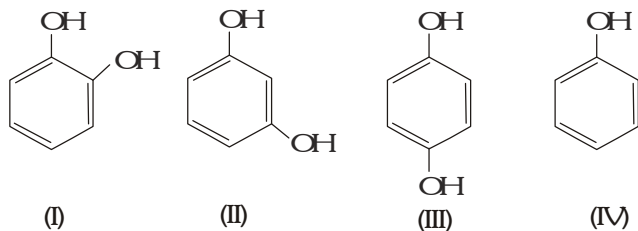
16.  Major product in this reaction is : [IIT-2004]



17. (I)  (II)  Gases released in reaction (I) and (II) are : [IIT-06]
(A) CO_2 , CO_2 (B) SO_2 , NO_2 (C) SO_2 , CO_2 (D) SO_2 , NO

18. Order of boiling point for the following compounds is -

[IIT-06]



(A) (I) < (II) < (III) < (IV)

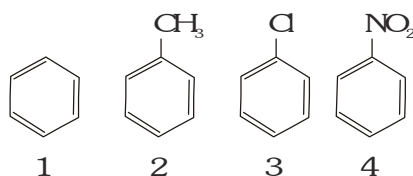
(B) (I) < (II) < (IV) < (III)

(C) (IV) < (I) < (II) < (III)

(D) (II) < (I) < (III) < (IV)

19. Identify the correct order of reactivity in electrophilic substitution reaction of the following compounds

[IIT-2002]



(A) 1 > 2 > 3 > 4

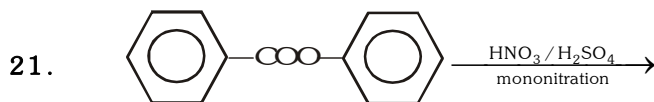
(B) 4 > 3 > 2 > 1

(C) 2 > 1 > 3 > 4

(D) 2 > 3 > 1 > 4

20. $\text{C}_6\text{H}_6 + (\text{CH}_3)_2\text{CHCH}_2\text{Cl} \xrightarrow[\Delta]{\text{AlCl}_3} ?$

[IIT 1992]

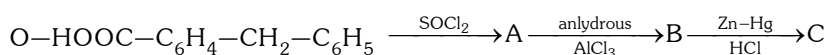


[IIT 1993]

22. $\text{C}_6\text{H}_6 + (\text{CH}_3)_2\text{CH}.\text{CH}_2\text{OH} \xrightarrow{\text{H}_2\text{SO}_4} ?$

[IIT 1994]

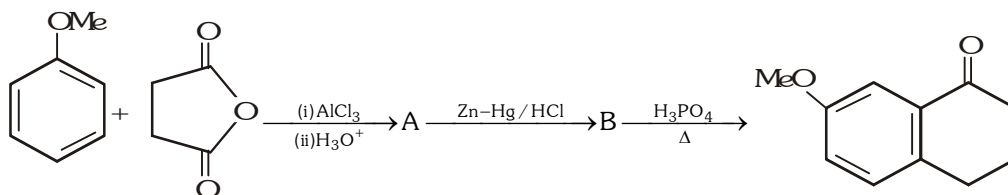
23. Fill in the blanks with appropriate structure of reaction products in the following transformation :



[IIT 1995]

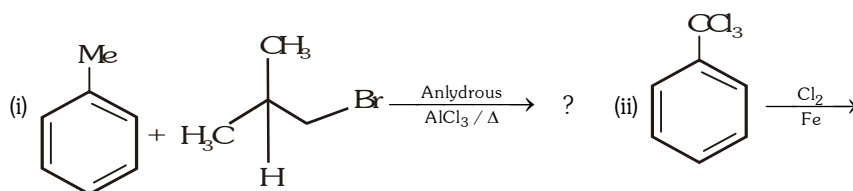
24. Predict the structures of the intermediates/products in the following reaction sequence.

[IIT 1996]



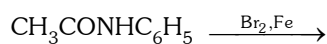
25. Predict the product

[IIT 1997]



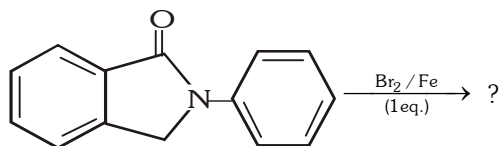
26. Each of the following reaction gives two products. Write the structure of the products :

[IIT 1998]



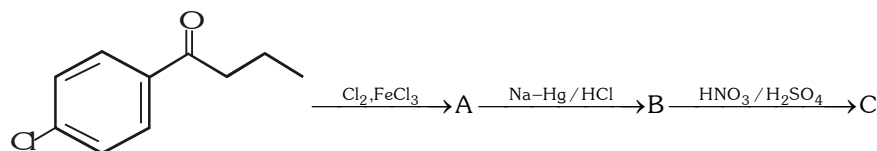
27. Predict the product

[IIT 2002]

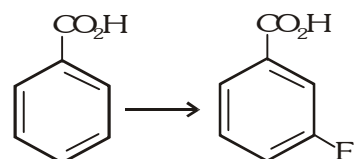


28. Predict the product

[IIT 2002]



29. Convert :

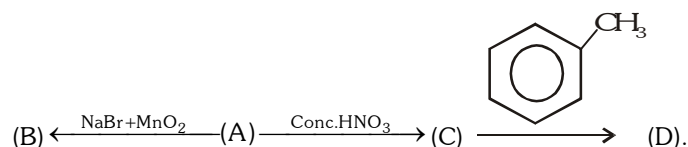


[IIT 2003]

30. 7-bromo-1, 3, 5-cycloheptatriene exists as a ionic substance whereas 5-bromo-1, 3-cyclopentadiene does not ionize even in the presence of Ag^+ ion. Explain ?

[IIT 2004]

31.



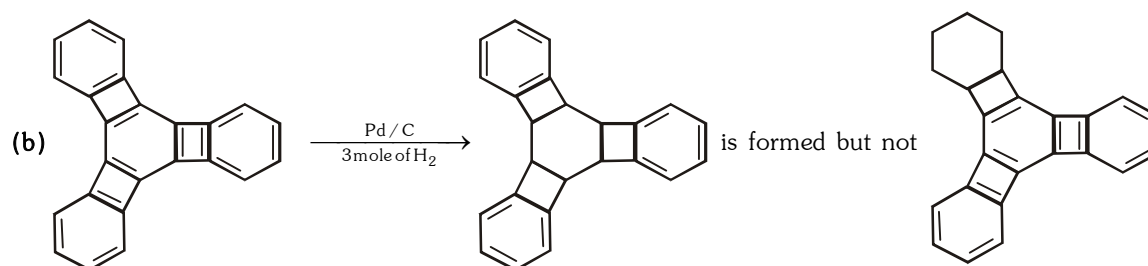
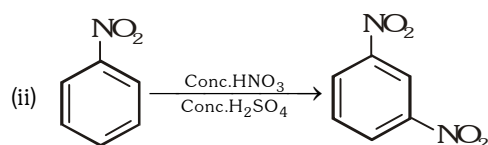
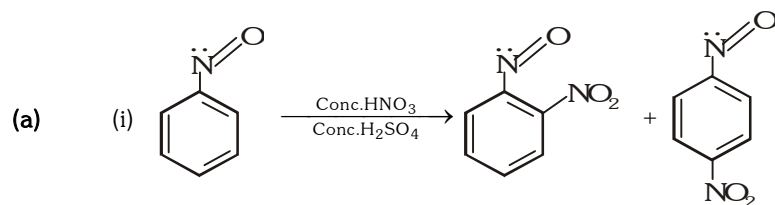
(brown fumes & Pungent smell)

(Explosive Product intermediate)

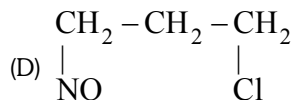
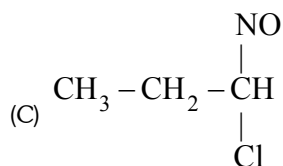
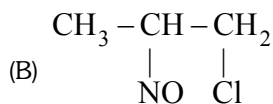
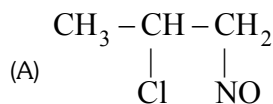
Identify the missing compounds. Give the equation from A to B and A to C.

[IIT 2005]

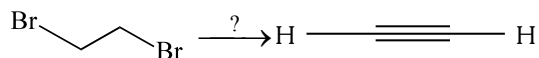
32. Give reason :



33. $\text{CH}_3\text{-CH=CH}_2 + \text{NOCl} \rightarrow \text{P}$ [IIT 2006]
Identify the adduct.



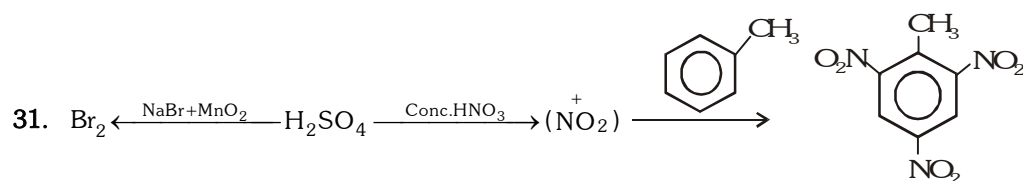
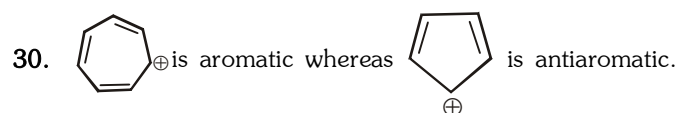
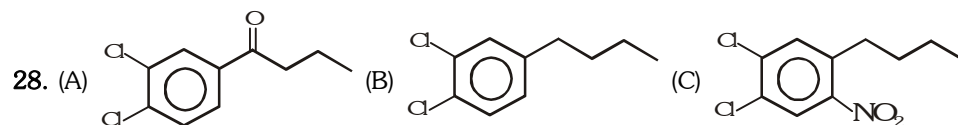
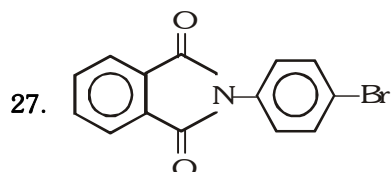
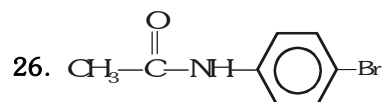
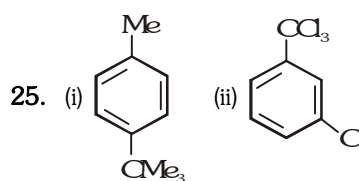
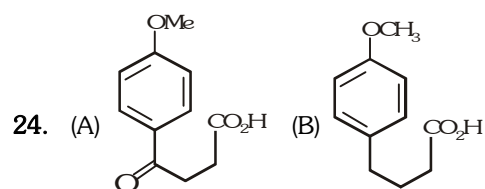
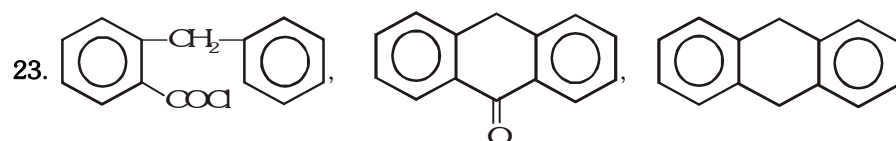
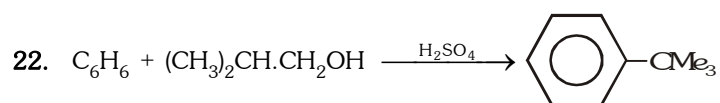
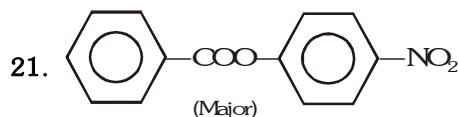
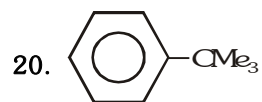
34. The number of stereoisomers obtained by bromination of trans-2-butene is [IIT 2007]
(A) 1 (B) 2 (C) 3 (D) 4
35. The number of structural isomers for C_6H_{14} is [IIT 2007]
(A) 3 (B) 4 (C) 5 (D) 6
36. The reagent(s) for the following conversion, [IIT 2007]



is / are

- (A) alcoholic KOH (B) alcoholic KOH followed by NaNH_2
(C) aqueous KOH followed by NaNH_2 (D) $\text{Zn} / \text{CH}_3\text{OH}$
37. The synthesis of 3-octyne is achieved by adding a bromoalkane into a mixture of sodium amide and an alkyne. [IIT-2010]
The bromoalkane and alkyne respectively are
(A) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH}$
(B) $\text{BrCH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{C}\equiv\text{CH}$
(C) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{C}\equiv\text{CH}$
(D) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH}$

1. (D) 2. (B) 3. (A) 4.(D) 5. (C) 6. (A) 7.(D) 8. (C)
 9. (C) 10. (A) 11. (A) 12. (C) 13. (B) 14. (B) 15. (D) 16. (D)
 17. (A) 18. (C) 19. (A)



32. (a) (i) $-\text{NO}$ is ortho and para director. (ii) $-\text{NO}_2$ is meta director.
 (b) due to high resonance energy of the product.

33. A 34. A 35. C 36. B 37. D

EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. The element having no neutron in the nucleus of its atom is -
(A) hydrogen (B) nitrogen (C) helium (D) boron
2. The particles present in the nucleus of an atom are -
(A) the proton and the electron (B) the electron and the neutron
(C) the proton and the neutron (D) none of these
3. The fraction of volume occupied by the nucleus with respect to the total volume of an atom is -
(A) 10^{-15} (B) 10^{-5} (C) 10^{-30} (D) 10^{-10}
4. Which of the following is iso-electronic with neon -
(A) O^{2-} (B) F^+ (C) Mg (D) Na
5. The approximate size of the nucleus of ${}^{64}_{28}\text{Ni}$ is -
(A) 3 fm (B) 4 fm (C) 5 fm (D) 2 fm
6. Which is true about an electron -
(A) rest mass of electron is 9.1×10^{-28} g
(B) mass of electron increases with the increase in velocity
(C) molar mass of electron is 5.48×10^{-4} g/mole
(D) e/m of electron is 1.7×10^8 coulomb/g
7. An isotone of ${}^{76}_{32}\text{Ge}$ is -
(A) ${}^{77}_{32}\text{Ge}$ (B) ${}^{77}_{33}\text{As}$ (C) ${}^{77}_{34}\text{Se}$ (D) ${}^{78}_{34}\text{Se}$
8. When alpha particles are sent through a thin metal foil, most of them go straight through the foil because -
(A) alpha particles are much heavier than electrons
(B) alpha particles are positively charged
(C) most part of the atom is empty space
(D) alpha particles move with high speed
9. Many elements have nonintegral atomic masses because -
(A) they have isotopes
(B) their isotopes have non-integral masses
(C) their isotopes have different masses
(D) the constituents, neutrons, protons and electrons combine to give fractional masses
10. The MRI (magnetic resonance imaging) body scanners used in hospitals operate with 400 MHz radio frequency energy. The wavelength corresponding to this radio frequency is -
(A) 0.75 m (B) 0.75 cm (C) 1.5 m (D) 2 cm
11. Photon of which light has maximum energy -
(A) Red (B) Blue (C) Violet (D) Green
12. The value of Planck's constant is 6.63×10^{-34} Js. The velocity of light is 3×10^8 m/sec. Which value is closest to the wavelength in nanometer of a quantum of light with frequency of $8 \times 10^{15} \text{ sec}^{-1}$ -
(A) 5×10^{-18} (B) 4×10^1 (C) 3×10^7 (D) 2×10^{-25}
13. Bohr's theory is not applicable to -
(A) He (B) Li^{2+} (C) He^{2+} (D) the H-atom

14. What is likely to be principal quantum number for a circular orbit of diameter 20 nm of the hydrogen atom if we assume Bohr orbit be the same as that represented by the principal quantum number ?
 (A) 10 (B) 14 (C) 12 (D) 16
15. Which is the correct relationship -
 (A) E_1 of H = $1/2 E_2$ of He^+ = $1/3 E_3$ of Li^{2+} = $1/4 E_4$ of Be^{3+}
 (B) $E_1(\text{H}) = E_2(\text{He}^+) = E_3(\text{Li}^{2+}) = E_4(\text{Be}^{3+})$
 (C) $E_1(\text{H}) = 2E_2(\text{He}^+) = 3E_3(\text{Li}^{2+}) = 4E_4(\text{Be}^{3+})$
 (D) No relation
16. If the value of $E = -78.4$ kcal/mole, the order of the orbit in hydrogen atom is -
 (A) 4 (B) 3 (C) 2 (D) 1
17. If velocity of an electron in 1st orbit of H atom is V, what will be the velocity of 3rd orbit of Li^{+2} -
 (A) V (B) V/3 (C) 3 V (D) 9 V
18. In a certain electronic transition in the hydrogen atoms from an initial state (1) to a final state (2), the difference in the orbit radius ($r_1 - r_2$) is 24 times the first Bohr radius. Identify the transition -
 (A) $5 \rightarrow 1$ (B) $25 \rightarrow 1$ (C) $8 \rightarrow 3$ (D) $7 \rightarrow 5$
19. Match the following -
 (a) Energy of ground state of He^+ (i) -6.04 eV
 (b) Potential energy of I orbit of H-atom (ii) -27.2 eV
 (c) Kinetic energy of II excited state of He^+ (iii) 8.7×10^{-18} J
 (d) Ionisation potential of He^+ (iv) -54.4 eV
 (A) A - (i), B - (ii), C - (iii), D - (iv) (B) A - (iv), B - (iii), C - (ii), D - (i)
 (C) A - (iv), B - (ii), C - (i), D - (iii) (D) A - (ii), B - (iii), C - (i), D - (iv)
20. The energy of hydrogen atom in its ground state is -13.6 eV. The energy of the level corresponding to $n = 5$ -
 (A) -0.54 eV (B) -5.40 eV (C) -0.85 eV (D) -2.72 eV
21. Total no. of lines in Lyman series of H spectrum will be -
 (where n = no. of orbits)
 (A) n (B) $n - 1$ (C) $n - 2$ (D) $n(n + 1)$
22. The spectrum of He^+ is expected to be similar to that of -
 (A) Li^+ (B) He (C) H (D) Na
23. What possibly can the ratio be of the de Broglie wavelengths for two electrons having the same initial energy and accelerated through 50 volts and 200 volts ?
 (A) 3 : 10 (B) 10 : 3 (C) 1 : 2 (D) 2 : 1
24. The uncertainty in the momentum of an electron is $1.0 \times 10^{-5} \text{ kg m s}^{-1}$. The uncertainty of its position will be ($h = 6.626 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$) -
 (A) $1.05 \times 10^{-28} \text{ m}$ (B) $1.05 \times 10^{-26} \text{ m}$ (C) $5.27 \times 10^{-30} \text{ m}$ (D) $5.25 \times 10^{-28} \text{ m}$
25. An α -particle is accelerated through a potential difference of V volts from rest. The de-Broglie's wavelength associated with it is -
 (A) $\sqrt{\frac{150}{V}} \text{ \AA}$ (B) $\frac{0.286}{\sqrt{V}} \text{ \AA}$ (C) $\frac{0.101}{\sqrt{V}} \text{ \AA}$ (D) $\frac{0.983}{\sqrt{V}} \text{ \AA}$
26. The orbital with zero angular momentum is -
 (A) s (B) p (C) d (D) f

27. Which of the following is electronic configuration of Cu^{2+} ($Z = 29$) -
 (A) $[\text{Ar}]4s^1 3d^8$ (B) $[\text{Ar}]4s^2 3d^{10} 4p^1$ (C) $[\text{Ar}]4s^1 3d^{10}$ (D) $[\text{Ar}] 3d^9$
28. The electronic configuration of the Mn^{4+} ion is -
 (A) $3d^4 4s^0$ (B) $3d^2 4s^1$ (C) $3d^1 4s^2$ (D) $3d^3 4s^0$
29. Which of the following ions has the maximum number of unpaired d-electrons -
 (A) Zn^{2+} (B) Fe^{2+} (C) Ni^{3+} (D) Cu^+
30. The total spin resulting from a d^7 configuration is -
 (A) 1 (B) 2 (C) $5/2$ (D) $3/2$
31. Given K L M N
 2 8 11 2
 The number of electrons present in $\ell = 2$ is -
 (A) 3 (B) 6 (C) 5 (D) 4
32. The configuration $1s^2 2s^2 2p^5 3s^1$ shows the -
 (A) ground state of the fluorine atom (B) excited state of the fluorine atom
 (C) excited state of the neon atom (D) excited state of O_2^- ion
33. The value ℓ and m for the last electron in the Cl^- ion are -
 (A) 1 and 2 (B) 2 and +1 (C) 3 and -1 (D) 1 and -1
34. In which transition, one quantum of energy is emitted -
 (A) $n = 4 \rightarrow n = 2$ (B) $n = 3 \rightarrow n = 1$ (C) $n = 4 \rightarrow n = 1$ (D) $n = 2 \rightarrow n = 1$
35. Choose the correct relation on the basis of Bohr's theory -
 (A) velocity of electron $\propto \frac{1}{n}$ (B) frequency of revolution $\propto \frac{Z^2}{n^3}$
 (C) radius of orbit $\propto n^2 Z$ (D) force on electron $\propto \frac{Z^3}{n^4}$
36. The magnitude of the spin angular momentum of an electron is given by -
 (A) $S = \sqrt{s(s+1)} \frac{h}{2\pi}$ (B) $S = s \frac{h}{2\pi}$ (C) $S = \frac{\sqrt{3}}{2} \times \frac{h}{2\pi}$ (D) $S = \pm \frac{1}{2} \times \frac{h}{2\pi}$
37. The change in orbital angular momentum corresponding to an electron transition inside a hydrogen atom can be -
 (A) $\frac{h}{4\pi}$ (B) $\frac{h}{\pi}$ (C) $\frac{h}{2\pi}$ (D) $\frac{h}{8\pi}$
38. In which of these options do both constituents of the pair have the same magnetic moment -
 (A) Zn^{2+} and Cu^+ (B) Co^{2+} and Ni^{2+} (C) Mn^{4+} and Co^{2+} (D) Mg^{2+} and Sc^+

CHECK YOUR GRASP						ANSWER KEY				EXERCISE -1					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	A	C	A	A	C	A,B,C,D	B,D	A,C	A,C	A	C	B	A,C	B	B
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	C	A	A	C	A	B	C	D	C	C	A	D	D	B	D
Que.	31	32	33	34	35	36	37	38							
Ans.	A	C	D	A,B,C,D	A,B,D	A,C	B,C	A,C							

EXERCISE-02**BRAIN TEASERS****SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)**

1. The maximum energy is present in any electron at :-
(A) Nucleus (B) Ground state
(C) First excited state (D) Infinite distance from the nucleus
2. Which electronic level would allow the hydrogen atom to absorb a photon but not to emit a photon :-
(A) 3s (B) 2p (C) 2s (D) 1s
3. The third line in Balmer series corresponds to an electronic transition between which Bohr's orbits in hydrogen :-
(A) $5 \rightarrow 3$ (B) $5 \rightarrow 2$ (C) $4 \rightarrow 3$ (D) $4 \rightarrow 2$
4. Correct set of four quantum numbers for valence electron of rubidium ($Z = 37$) is :-
(A) $5, 0, 0, +\frac{1}{2}$ (B) $5, 1, 0, +\frac{1}{2}$ (C) $5, 1, 1, +\frac{1}{2}$ (D) $6, 0, 0, +\frac{1}{2}$
5. The orbital diagram in which the Aufbau's principle is violated is :-
(A) $\begin{array}{cccc} 2s & 2p_x & 2p_y & 2p_z \\ \uparrow\downarrow & \uparrow\downarrow & \uparrow & \end{array}$ (B) $\begin{array}{cccc} 2s & 2p_x & 2p_y & 2p_z \\ \uparrow & \uparrow\downarrow & \uparrow & \uparrow \end{array}$ (C) $\begin{array}{cccc} 2s & 2p_x & 2p_y & 2p_z \\ \uparrow\downarrow & \uparrow & \uparrow & \uparrow \end{array}$ (D) $\begin{array}{cccc} 2s & 2p_x & 2p_y & 2p_z \\ \uparrow\downarrow & \uparrow\downarrow & \uparrow\downarrow & \uparrow \end{array}$
6. The total number of neutrons in dipositive zinc ions with mass number 70 is :-
(A) 34 (B) 40 (C) 36 (D) 38
7. Which of the following sets of quantum numbers represent an impossible arrangement :-

n	l	m	m_s
(A) 3	2	-2	$\frac{1}{2}$
(B) 4	0	0	$\frac{1}{2}$
(C) 3	2	-3	$\frac{1}{2}$
(D) 5	3	0	$\frac{1}{2}$
8. The explanation for the presence of three unpaired electrons in the nitrogen atom can be given by :-
(A) Pauli's exclusions principle (B) Hund's rule
(C) Aufbau's principle (D) Uncertainty principle
9. The electronic configuration of an element is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$. This represents its :-
(A) Excited state (B) Ground state (C) Cationic form (D) None
10. Which of the following has maximum number of unpaired electron (atomic number of Fe 26) :-
(A) Fe (B) Fe (II) (C) Fe (III) (D) Fe (IV)
11. Which quantum number is not related with Schrodinger equation :-
(A) Principal (B) Azimuthal (C) Magnetic (D) Spin
12. If λ_0 is the threshold wavelength for photoelectric emission, λ wavelength of light falling on the surface of metal, and m mass of electron, then de Broglie wavelength of emitted electron is :-

(A) $\left[\frac{h(\lambda\lambda_0)}{2mc(\lambda_0 - \lambda)} \right]^{\frac{1}{2}}$ (B) $\left[\frac{h(\lambda_0 - \lambda)}{2mc\lambda\lambda_0} \right]^{\frac{1}{2}}$ (C) $\left[\frac{h(\lambda - \lambda_0)}{2mc\lambda\lambda_0} \right]^{\frac{1}{2}}$ (D) $\left[\frac{h\lambda\lambda_0}{2mc} \right]^{\frac{1}{2}}$

13. It is known that atom contain protons, neutrons and electrons. If the mass of neutron is assumed to half of its original value where as that of proton is assumed to be twice of its original value then the atomic mass of $^{14}_6\text{C}$ will be :-
 (A) same (B) 25 % more
 (C) 14.28 % more (D) 28.5 % less
14. Give the correct order of initials T (true) or F (false) for following statements :-
 (I) If an ion has 2 electrons in K shell, 8 electrons in L shell and 6 electrons in M shell, then number of S electrons present in that element is 6.
 (II) The maximum number of electrons in a subshell is given by $2n^2$
 (III) If electron has magnetic quantum number -1 , then it cannot be present in s-orbital.
 (IV) Only one radial node is present in 3p orbital.
 (A) TTFF (B) FFTF (C) TFFT (D) FFTF
15. The shortest wavelength of He^+ in Balmer series is x, then longest wavelength in the Paschene series of Li^{+2} is :-
 (A) $\frac{36x}{5}$ (B) $\frac{16x}{7}$ (C) $\frac{9x}{5}$ (D) $\frac{5x}{9}$
16. An electron in a hydrogen atom in its ground state absorbs energy equal to ionisation energy of Li^{+2} . The wavelength of the emitted electron is :-
 (A) $3.32 \times 10^{-10} \text{ m}$ (B) $1.17 \text{ \AA}$ (C) $2.32 \times 10^{-9} \text{ nm}$ (D) 3.33 pm
17. In compound FeCl_2 the orbital angular momentum of last electron in its cation & magnetic moment (in Bohr Magnetron) of this compound are :-
 (A) $(\sqrt{6})\hbar, \sqrt{35}$ (B) $(\sqrt{6})\hbar, \sqrt{24}$ (C) $0, \sqrt{35}$ (D) none of these
18. An electron, a proton and an alpha particle have kinetic energy of 16 E, 4E and E respectively. What is the qualitative order of their de Broglie wavelengths :-
 (A) $\lambda_e > \lambda_p = \lambda_\alpha$ (B) $\lambda_p = \lambda_\alpha > \lambda_e$
 (C) $\lambda_p > \lambda_e > \lambda_\alpha$ (D) $\lambda_\alpha < \lambda_e \gg \lambda_p$
19. Question : Is the specie paramagnetic ?
 STATE-1 : The atomic number of specie is 29.
 STATE-2 : The charge on the specie is +1.
 (A) Statements (1) alone is sufficient but statement (2) is not sufficient
 (B) Statement (2) alone is sufficient but statement (1) is not sufficient
 (C) Both statement together are sufficient but neither statement alone is sufficient
 (D) Statement (1) & (2) together are not sufficient
20. Given ΔH for the process $\text{Li (g)} \longrightarrow \text{Li}^{+3} \text{ (g)} + 3e^-$ is 19800 kJ/mole & IE_1 for Li is 520 then IE_2 & IE_3 of Li^+ are respectively (approx value) :-
 (A) 11775, 7505 (B) 19280, 520
 (C) 11775, 19280 (D) Data insufficient
21. The ratio of difference in wavelengths of 1st and 2nd lines of Lyman series in H-like atom to difference in wavelength for 2nd and 3rd lines of same series is :-
 (A) 2.5 : 1 (B) 3.5 : 1
 (C) 4.5 : 1 (D) 5.5 : 1

22. Which of the following statement is INCORRECT.

- (A) $\frac{e}{m}$ ratio for canal rays is maximum for hydrogen ion.
 (B) $\frac{e}{m}$ ratio for cathode rays is independent of the gas taken.
 (C) The nature of canal rays is dependent on the electrode material.
 (D) The $\frac{e}{m}$ ratio for electron is expressed as $\frac{E^2}{2B^2V}$, when the cathode rays go undeflected under the influence of electric field (E), magnetic field (B) and V is potential difference applied across electrodes.

23. The quantum numbers of four electrons (e1 to e4) are given below :-

	n	ℓ	m	s
e1	3	0	0	+1/2
e2	4	0	1	1/2
e3	3	2	2	-1/2
e4	3	1	-1	1/2

The correct order of decreasing energy of these electrons is :

- (A) $e4 > e3 > e2 > e1$ (B) $e2 > e3 > e4 > e1$
 (C) $e3 > e2 > e4 > e1$ (D) none

24. If radius of second stationary orbit (in Bohr's atom) is R. Then radius of third orbit will be :-

- (A) R/3 (B) 9R (C) R/9 (D) 2.25 R

25. The wavelength associated with a gold weighing 200 g and moving at a speed of 5 m/h is of the order :-

- (A) 10^{-10} m (B) 10^{-20} m (C) 10^{-30} m (D) 10^{-40} m

26. If the nitrogen atom had electronic configuration $1s^7$, it would have energy lower than that of normal ground state configuration $1s^2 2s^2 2p^3$, because the electrons would be closer to the nucleus. Yet $1s^7$ is not observed because it violates :-

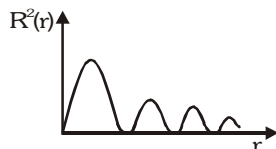
- (A) Heisenberg uncertainty principle (B) Hund's rule
 (C) Pauli's exclusion principle (D) Bohr postulate of stationary orbits

27. From the following observations predict the type of orbital :

Observation 1 : x y plane acts as nodal plane

Observation 2 : The angular function of the orbital intersects the three axes at origin only.

Observation 3 : $R^2(r)/r$ curve is obtained for the orbital is



- (A) $5p_z$ (B) $6d_{xy}$ (C) $6d_{x^2-y^2}$ (D) $6d_{yz}$

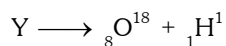
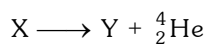
28. Question : Is the orbital of hydrogen atom $3p_x$?

STATE 1 : The radial function of the orbital is $R(r) = \frac{1}{9\sqrt{6}a_0^{3/2}} (4 - \sigma)\sigma e^{-\sigma/2}$, $\sigma = \frac{r}{2}$

STATE 2: The orbital has 1 radial node & 0 angular node.

- (A) Statement (1) alone is sufficient. (B) Statement (2) alone is sufficient
 (C) Both together are sufficient (D) Neither is sufficient

29. Consider the following nuclear reactions involving X & Y.

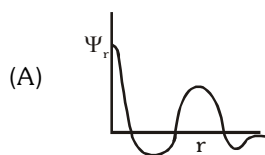


If both neutrons as well as protons in both the sides are conserved in nuclear reaction then identify period number of X & moles of neutrons in 4.6 g of X

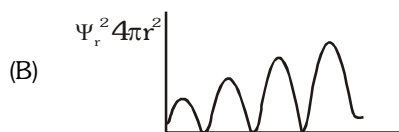
- (A) 3, $2.4 N_A$ (B) 3, 2.4 (C) 2, 4.6 (D) 3, $0.2 N_A$
30. Electromagnetic radiations having $\lambda = 310 \text{ \AA}$ are subjected to a metal sheet having work function = 12.8 eV. What will be the velocity of photoelectrons with maximum Kinetic Energy...
- (A) 0, no emission will occur (B) $2.18 \times 10^6 \text{ m/s}$
 (C) $2.18\sqrt{2} \times 10^6 \text{ m/s}$ (D) $8.72 \times 10^6 \text{ m/s}$
31. If in Bohr's model, for unielectronic atom, time period of revolution is represented as $T_{n,z}$ where n represents shell no. and z represents atomic number then the value of $T_{1,2} : T_{2,1}$ will be :-
- (A) 8 : 1 (B) 1 : 8 (C) 1 : 1 (D) None of these
32. Column I & Column II contain data on Schrodinger Wave-Mechanical model, where symbols have their usual meanings. Match the columns :-

Column I

Column II (Type of orbital)



(p) 4s



(q) 5p_x

(C) $\Psi(\theta, \phi) = K$ (independent of θ & ϕ)

(r) 3s

(D) at least one angular node is present

(s) 6d_{xy}

33. Which orbital is non-directional :-
- (A) s (B) p (C) d (D) All
34. A hydrogen - like atom has ground state binding energy 122.4 eV. Then :
- (A) its atomic number is 3
 (B) an electron of 90 eV can excite it to a higher state
 (C) an 80 eV electron cannot excite it to a higher state
 (D) an electron of 8.2 eV and a photon of 91.8 eV are emitted when a 100 eV electron interacts with it
35. Uncertainty in position is twice the uncertainty in momentum Uncertainty in velocity is :-
- (A) $\sqrt{\frac{h}{\pi}}$ (B) $\frac{1}{2m} \sqrt{\frac{h}{\pi}}$ (C) $\frac{1}{2m} \sqrt{h}$ (D) $\frac{h}{4\pi}$
36. For which orbital angular probability distribution is maximum at an angle of 45° to the axial direction :-
- (A) d_{x²-y²} (B) d_{z²} (C) d_{xy} (D) P_x

37. Which orbit would be the first to have 'g' subshell :-
 (A) 3rd (B) 4th (C) 5th (D) 6th
38. The decreasing order of energy of the 3d, 4s, 3p, 3s orbitals is :-
 (A) 3d > 3s > 4s > 3p (B) 3s > 4s > 3p > 3d (C) 3d > 4s > 3p > 3s (D) 4s > 3d > 3s > 3p
39. If n and ℓ are respectively the principle and azimuthal quantum numbers, then the expression for calculating the total number of electrons in any orbit is :-
 (A) $\sum_{\ell=1}^{\ell=n} 2(2\ell+1)$ (B) $\sum_{\ell=1}^{\ell=n-1} 2(2\ell+1)$ (C) $\sum_{\ell=0}^{\ell=n+1} 2(2\ell+1)$ (D) $\sum_{\ell=0}^{\ell=n-1} 2(2\ell+1)$
40. If wavelength is equal to the distance travelled by the electron in one second, then :-
 (A) $\lambda = \frac{h}{p}$ (B) $\lambda = \frac{h}{m}$ (C) $\lambda = \sqrt{\frac{h}{p}}$ (D) $\lambda = \sqrt{\frac{h}{m}}$
41. According to Schrodinger model nature of electron in an atom is as :-
 (A) Particles only (B) Wave only
 (C) Both simultaneously (D) Sometimes waves and sometimes particle
42. Which describes orbital :-
 (A) ψ (B) ψ^2 (C) $|\psi^2| \psi$ (D) All
43. In order to have the same wavelength for the electron (mass m_e) and the neutron (mass m_n) their velocities should be in the ratio (electron velocity/neutron velocity) :-
 (A) m_n/m_e (B) $m_n \cdot m_e$ (C) m_e/m_n (D) one
44. The quantum numbers +1/2 and -1/2 for the electron spin represent :-
 (A) Rotation of the electron in clockwise and anticlockwise direction respectively.
 (B) Rotation of the electron in anticlockwise and clockwise direction respectively.
 (C) Magnetic moment of the electron pointing up and down respectively.
 (D) Two quantum mechanical spin states which have no classical analogue.
45. Which is true about ψ :-
 (A) ψ represents the probability of finding an electron around the nucleus
 (B) ψ represent the amplitude of the electron wave
 (C) Both A and B
 (D) None of these
46. Consider an electron in the n^{th} orbit of a hydrogen atom in the Bohr model. The circumference of the orbit can be expressed in terms of the de Broglie wavelength λ of the electron as :-
 (A) $(0.529) n\lambda$ (B) $\sqrt{n}\lambda$ (C) $(13.6) \lambda$ (D) $n\lambda$
47. A particle X moving with a certain velocity has a de Broglie wave length of 1Å. If particle Y has a mass of 25% that of X and velocity 75% that of X, de Broglie's wave length of Y will be :-
 (A) 3 Å (B) 5.33 Å (C) 6.88 Å (D) 48 Å
48. What are the values of the orbital angular momentum of an electron in the orbitals 1s, 3s, 3d and 2p :-
 (A) 0, 0, $\sqrt{6}\hbar, \sqrt{2}\hbar$ (B) 1, 1, $\sqrt{4}\hbar, \sqrt{2}\hbar$ (C) 0, 1, $\sqrt{6}\hbar, \sqrt{3}\hbar$ (D) 0, 0, $\sqrt{20}\hbar, \sqrt{6}\hbar$
49. If m = magnetic quantum number and ℓ = azimuthal quantum number then :-
 (A) $m = \ell + 2$ (B) $m = 2\ell^2 + 1$ (C) $\ell = \frac{m-1}{2}$ (D) $\ell = 2m + 1$

50. The number of unpaired electrons in Mn^{4+} ($Z = 25$) is :-
(A) Four (B) Two (C) Five (D) Three
51. After np orbitals are filled, the next orbital filled will be :-
(A) $(n + 1) s$ (B) $(n + 2) p$ (C) $(n + 1) d$ (D) $(n + 2) s$
52. The value of the magnetic moment of a particular ion is 2.83 Bohr magneton. The ion is :-
(A) Fe^{2+} (B) Ni^{2+} (C) Mn^{2+} (D) Co^{3+}
53. In Bohr's model of the hydrogen atom the ratio between the period of revolution of an electron in the orbit of $n = 1$ to the period of the revolution of the electron in the orbit $n = 2$ is :-
(A) 1 : 2 (B) 2 : 1 (C) 1 : 4 (D) 1 : 8
54. Let ν_1 be the frequency of the series limit of the Lyman series, ν_2 be the frequency of the first line of the lyman series, and ν_3 be the frequency of the series limit of the Balmer series :-
(A) $\nu_1 - \nu_2 = \nu_3$ (B) $\nu_2 - \nu_1 = \nu_3$ (C) $\nu_3 = 1/2 (\nu_1 - \nu_3)$ (D) $\nu_1 + \nu_2 = \nu_3$
55. The energies of energy levels A, B and C for a given atom are in the sequence $E_A < E_B < E_C$. If the radiations of wavelengths λ_1 , λ_2 and λ_3 are emitted due to the atomic transitions C to B, B to A and C to A respectively then which of the following relations is correct :-
(A) $\lambda_1 + \lambda_2 + \lambda_3 = 0$ (B) $\lambda_3 = \lambda_1 + \lambda_2^2$ (C) $\lambda_3 = \lambda_1 + \lambda_2$ (D) $\lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$
56. The wavelengths of photons emitted by electron transition between two similar levels in H and He^+ are λ_1 and λ_2 respectively. Then :-
(A) $\lambda_2 = \lambda_1$ (B) $\lambda_2 = 2\lambda_1$ (C) $\lambda_2 = \lambda_1/2$ (D) $\lambda_2 = \lambda_1/4$
57. If first ionization potential of an atom is 16 V, then the first excitation potential will be :-
(A) 10.2 V (B) 12 V (C) 14 V (D) 16 V
58. In which transition minimum energy is emitted :-
(A) $\infty \rightarrow 1$ (B) $2 \rightarrow 1$ (C) $3 \rightarrow 2$ (D) $n \rightarrow (n - 1)$ ($n \geq 4$)
59. No. of visible lines when an electron returns from 5th orbit to ground state in H spectrum :-
(A) 5 (B) 4 (C) 3 (D) 10

[illegible]

EXERCISE-03**MISCELLANEOUS TYPE QUESTIONS****TRUE / FALSE**

- The electron density in xy plane of $3d_{x^2-y^2}$ orbital is zero.
- $3d^6$ configuration is more stable than $3d^5$.
- The potential energy of an electron in an orbit is twice in magnitude as compared to its kinetic energy.
- The increasing order for the values of e/m (charge/mass) for electron (e), proton (p), neutron (n) and alpha particle (α) is $n < \alpha < p < e$.
- The orbital $3d_{xy}$ has no probability of finding electron along x and y-axis.

FILL IN THE BLANKS

- Nitrogen has an atomic number of 7 and oxygen has an atomic number of 8. The total number of electrons in the nitrate ion (NO_3^-) is
- h/π is the angular momentum of the electron in the orbit of He^+ .
- An emission spectrum has electromagnetic radiation of definite
- The maximum number of electrons that can be accommodated in all the orbitals for which $\ell = 3$ is
- The values of n_1 and n_2 in the Paschen spectral series of hydrogen atom are and respectively.

MATCH THE COLUMN

1.	Column-I	Column-II
(A)	Aufbau principle	(p) Line spectrum in visible region
(B)	de broglie	(q) Orientation of an electron in an orbital
(C)	Angular momentum	(r) Photon
(D)	Hund's rule	(s) $\lambda = h/mv$
(E)	Balmer series	(t) Electronic configuration
(F)	Planck's law	(u) mvr

2.	Column-I	Column-II
(A)	Cathode rays	(p) Helium nuclei
(B)	dumb-bell	(q) Uncertainty principle
(C)	Alpha particles	(r) Electromagnetic radiation
(D)	Moseley	(s) p-orbital
(E)	Heisenberg	(t) Atomic number
(F)	X-ray	(u) Electrons

3. Frequency = f, Time period = T, Energy of n^{th} orbit = E_n , radius of n^{th} orbit = r_n , Atomic number = Z, Orbit number = n :

Column-I	Column-II
(A) f	(p) n^3
(B) T	(q) Z^2
(C) E_n	(r) $\frac{1}{n^2}$
(D) $\frac{1}{r_n}$	(s) Z

4.	Column-I	Column-II
(A)	Lyman series	(p) maximum number of spectral line observed = 6
(B)	Balmer series	(q) maximum number of spectral line observed = 2
(C)	In a sample $5 \rightarrow 2$	(r) 2^{nd} line has wave number $\frac{8R}{9}$
(D)	In a single isolated H-atom for $3 \rightarrow 1$ transition	(s) 2^{nd} line has wave number $\frac{3R}{4}$
		(t) total number of spectral line is 10

ASSERTION & REASON

These questions contains, Statement I (assertion) and Statement II (reason).

(A) Statement-I is true, Statement-II is true ; Statement-II is correct explanation for Statement-I.

(B) Statement-I is true, Statement-II is true ; Statement-II is NOT a correct explanation for statement-I

(C) Statement-I is true, Statement-II is false

(D) Statement-I is false, Statement-II is true

1. **Statement-I** : Nodal plane of p_x atomic orbital is yz plane.

Because

Statement-II : In p_x atomic orbital electron density is zero in the yz plane.

2. **Statement-I** : No two electrons in an atom can have the same values of four quantum numbers.

Because

Statement-II : No two electrons in an atom can be simultaneously in the same shell, same subshell, same orbitals and have same spin.

3. **Statement-I** : p-orbital has dumb-bell shape.

Because

Statement-II : Electrons present in p-orbital can have one of three values for 'm', i.e. 0, +1, -1

4. **Statement-I** : The ground state configuration of Cr is $3d^5 4s^1$.

Because

Statement-II : A set of exactly half filled orbitals containing parallel spin arrangement provide extra stability.

5. **Statement-I** : Mass numbers of most of the elements are fractional.

Because

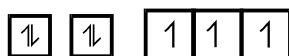
Statement-II : Mass numbers are obtained by comparing with the mass number of carbon taken as 12.

6. **Statement-I** : Limiting line in the balmer series has a wavelength of $36.4 \mu\text{m}$.

Because

Statement-II : Limiting lines is obtained for a jump of electron from $n = \infty$ to $n = 2$ for Balmer series.

7. **Statement-I** : The electronic configuration of nitrogen atom is represented as :



not as



Because

Statement-II : The configuration of ground state of an atom is the one which has the greatest multiplicity.

8. **Statement-I** : The configuration of B atom cannot be $1s^2 2s^3$.
Because
Statement-II : Hund's rule demands that the configuration should display maximum multiplicity.
9. **Statement-I** : 2p orbitals do not have spherical nodes.
Because
Statement-II : The number of spherical nodes in p-orbitals is given by $(n - 2)$.
10. **Statement-I** : In Rutherford's gold foil experiment, very few α - particles are deflected back.
Because
Statement-II : Nucleus present inside the atom is heavy.
11. **Statement-I** : Each electron in an atom has two spin quantum numbers.
Because
Statement-II : Spin quantum numbers are obtained by solving Schrodinger wave equation.
12. **Statement-I** : There are two spherical nodes in 3s-orbital.
Because
Statement-II : There is no angular node in 3s-orbital.

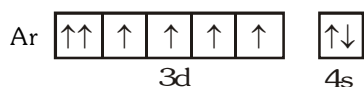
COMPREHENSION BASED QUESTIONS

Comprehension # 1

Read the following rules and answer the questions at the end of it.

- ◆ Electrons in various suborbitals of an orbit are filled in increasing order to their energies.
- ◆ Pairing of electrons in various orbitals of a suborbital takes place only after each orbital is half-filled.
- ◆ No two electrons in an atom can have the same set of quantum number.

1. Cr ($Z = 24$), Mn^+ ($Z = 25$), Fe^{2+} ($Z = 26$) and Co^{3+} ($Z = 27$) are isoelectronic each having 24 electrons. Thus,
 (A) all have configurations as $[Ar] 4s^1 3d^5$
 (B) Cr and Mn^+ have configurations as $[Ar] 4s^1 3d^5$ while Fe^{2+} and Co^{3+} have configurations as $[Ar] 3d^5$.
 (C) all have configurations as $[Ar] 3d^6$
 (D) all have configurations as $[Ar] 4s^2 3d^6$
2. A compound of vanadium has a magnetic moment of 1.73 BM. Electronic configuration of the vanadium ion in the compound is :
 (A) $[Ar] 4s^0 3d^1$ (B) $[Ar] 4s^2 3d^3$ (C) $[Ar] 4s^1 3d^0$ (D) $[Ar] 4s^0 3d^5$
3. Which of these ions are expected to be paramagnetic and coloured in aqueous solution ?
 (A) Fe^{3+} , Ti^{3+} , Co^{3+} (B) Cu^+ , Ti^{4+} , Sc^{3+} (C) Fe^{3+} , Ni^{2+} , V^{5+} (D) Cu^+ , Cu^{2+} , Fe^{2+}
4. While writing the following electronic configuration of Fe some rules have been violated :
 I : Aufbau rule,
 II : Hund's rule
 III : Pauli's exclusion principle



- (A) I, II (B) II, III (C) I, III (D) I, II, III

5. How many elements would be in the second period of the periodic table if the spin quantum number (m_s) could have the value of $-\frac{1}{2}, 0, +\frac{1}{2}$?
 (A) 8 (B) 10 (C) 12 (D) 18
6. The sub-shell that arises after f sub-shell is called g sub-shell.
 (A) it contains 18 electrons and 9 orbitals
 (B) it corresponds to $\ell = 4$ and first occurs in 5th energy level
 (C) a g -orbital can have maximum of two electrons
 (D) all the above statements are true.
-

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
<u>True / False</u> 1. F 2. F 3. T 4. T 5. T <u>Fill in the Blanks</u> 1. 32 2. 2nd 3. frequency or wavelength 4. 14 5. 3, (4, 5, 6,...) <u>Match the Column</u> 1. (A) → t; (B) → s; (C) → u; (D) → q; (E) → p; (F) → r 2. (A) → u; (B) → s; (C) → p; (D) → t; (E) → q; (F) → r 3. (A) → q; (B) → p; (C) → q,r; (D) → r,s 4. (A) → r; (B) → s; (C) → p; (D) → q <u>Assertion - Reason Questions</u> 1. A 2. A 3. B 4. A 5. E 6. A 7. A 8. B 9. A 10. B 11. E 12. B <u>Comprehension Based Questions</u> Comprehension #1 : 1. (B) 2. (A) 3. (A) 4. (D) 5. (C) 6. (D)		

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

1. How long would it take a radio wave of frequency $6 \times 10^3 \text{ sec}^{-1}$ to travel from mars to the earth, a distance of $8 \times 10^7 \text{ km}$?
2. The energy levels of hypothetical one electron atom are shown below.
 $0 \text{ eV} \text{ --- } n = \infty$
 $-0.50 \text{ eV} \text{ --- } n = 5$
 $-1.45 \text{ eV} \text{ --- } n = 4$
 $-3.08 \text{ eV} \text{ --- } n = 3$
 $-5.3 \text{ eV} \text{ --- } n = 2$
 $-15.6 \text{ eV} \text{ --- } n = 1$
 - (a) Find the ionisation potential of atom?
 - (b) Find the short wavelength limit of the series terminating at $n = 2$?
 - (c) Find the wave no. of photon emitted for the transition made by the electron from third orbit to first orbit ?
 - (d) Find the minimum energy that an electron will have after interacting with this atom in the ground state, if the initial kinetic energy of the electron is (i) 6 eV (ii) 11 eV ?
3. Suppose 10^{-17} J of light energy is needed by the interior of the human eye to see an object. How many photons of green light ($\lambda = 550 \text{ nm}$) are needed to generate this minimum amount of energy?
4. Find the number of photons of radiation of frequency $5 \times 10^{13} \text{ s}^{-1}$ that must be absorbed in order to melt one g ice when the latent heat of fusion of ice is 330 J/g .
5. The eyes of certain member of the reptile family pass a single visual signal to the brain when the visual receptors are struck by photons of wavelength 850 nm . If a total energy of $3.15 \times 10^{-14} \text{ J}$ is required to trip the signal, what is the minimum number of photons that must strike the receptor?
6. The wavelength of a certain line in the Paschen series is 1093.6 nm . What is the value of n_{high} for this line [$R_{\text{H}} = 1.0973 \times 10^7 \text{ m}^{-1}$].
7. Wavelength of the Balmer H_{α} line (first line) is $6565 \text{ \AA}$. Calculate the wavelength of H_{β} (second line).
8. Calculate the Rydberg constant R if He^+ ions are known to have the wavelength difference between the first (of the longest wavelength) lines of Balmer and Lyman series equal to 133.7 nm .
9. Calculate the energy emitted when electrons of 1.0 g atom of hydrogen undergo transition giving the spectral line of lowest energy in the visible region of its atomic spectrum.
10. A photon having $\lambda = 854 \text{ \AA}$ causes the ionization of a nitrogen atom. Give the I.E. per mole of nitrogen in KJ .
11. Calculate energy of electron which is moving in the orbit that has its radius, Sixteen times the radius of first Bohr orbit for H -atom.
12. The electron energy in hydrogen atom is given by $E_n = \frac{-21.7 \times 10^{-12}}{n^2} \text{ ergs}$. Calculate the energy required to remove an e^- completely from $n = 2$ orbit. What is the largest wavelength in cm of light that can be used to cause this transition.

13. Calculate the wavelength in angstrom of photon that is emitted when an e^- in Bohr orbit $n = 2$ returns to the orbit $n = 1$. The ionization potential of the ground state of hydrogen atom is 2.17×10^{-11} erg/atom.
14. The velocity of e^- in a certain Bohr orbit of the hydrogen atom bears the ratio 1 : 275 to the velocity of light. What is the quantum no. "n" of the orbit and the wave no. of the radiation emitted for the transition from the quantum state $(n + 1)$ to the ground state.
15. A doubly ionised lithium atom is hydrogen like with atomic number $Z = 3$. Find the wavelength of the radiation required to excite the electron in Li^{2+} from the first to the third Bohr orbit.
16. Estimate the difference in energy between I and II Bohr Orbit for a hydrogen atom. At what minimum At. no. a transition from $n = 2$ to $n = 1$ energy level would result in the emission of X-rays with $\lambda = 3.0 \times 10^{-8}$ m? Which hydrogen like species does this At. no. correspond to:
17. 1.8 g atoms of hydrogen are excited to radiations. The study of spectra indicates that 27% of the atoms are in 3rd energy level and 15% of atoms in 2nd energy level and the rest in ground state. If I.P. of H is 21.7×10^{-12} erg. Calculate.
 - (i) No. of atoms present in III & II energy level.
 - (ii) Total energy evolved when all the atoms return to ground state.
18. One mole He^+ ions are excited. Spectral analysis showed existence of 50% ions in 3rd orbit, 25% in 2nd and rest in ground state. Calculate total energy evolved when all the ions return to the ground state.
19. The energy of an excited H-atom is -3.4 eV. Calculate angular momentum of e^- .
20. The vapours of Hg absorb some electrons accelerated by a potential difference of 4.5 volt as a result of it light is emitted. If the full energy of single incident e^- is supposed to be converted into light emitted by single Hg atom, find the wave no. of the light.
21. The hydrogen atom in the ground state is excited by means of monochromatic radiation of wavelength $x \text{ \AA}$. The resulting spectrum consists of 15 different lines. Calculate the value of x.
22. If the average life time of an excited state of H atom is of order 10^{-8} sec, estimate how many orbits an e^- makes when it is in the state $n = 2$ and before it suffers a transition to $n = 1$ state.
23. Calculate the frequency of e^- in the first Bohr orbit in a H-atom.
24. A single electron orbits around a stationary nucleus of charge $+Ze$ where Z is a constant from the nucleus and e is the magnitude of the electric charge. The hydrogen like species required 47.2 eV to excite the electron from the second Bohr orbit to the third Bohr orbit. Find -
 - (i) the value of Z give the hydrogen like species formed.
 - (ii) the kinetic energy and potential energy of the electron in the first Bohr orbit.
25. A stationary He^+ ion emitted a photon corresponding to a first line of the Lyman series. The photon liberated a photoelectron from a stationary H atom in ground state. What is the velocity of photoelectron ?
26. To what series does the spectral lines of atomic hydrogen belong if its wave number is equal to the difference between the wave number of the following two lines of the Balmer series 486.1 and 410.2 nm. What is the wavelength of this ?

27. A particle of charge equal to that of an electron and mass 208 times the mass of the electron moves in a circular orbit around a nucleus of charge $+3e$. Assuming that the Bohr model of the atom is applicable to this system, (a) derive an expression for the radius of the n^{th} bohr orbit, (b) find the value of n for which the radius of the orbit is approximately the same as that of the first Bohr orbit for the hydrogen atom, and (c) find the wavelength of the radiation emitted when the revolving particle jumps from the third orbit to the first.
28. A neutrons breaks into a proton and an electron. This decay of neutron is accompanied by release of energy. Assuming that 50% of the energy is produced in the form of electromagnetic radiation, what will be the frequency of radiation produced. Will this photon be sufficient to cause ionization of Aluminium. In case it is able to do so what will be the energy of the electron ejected from the Aluminum atom. IE_1 of Al = 577 kJ/mol.
29. Calculate the threshold frequency of metal if the binding energy is $180.69 \text{ kJ mol}^{-1}$ of electron.
30. Calculate the binding energy per mole when threshold wavelength of photon is 240 nm.
31. A metal was irradiated by light of frequency $3.2 \times 10^{15} \text{ s}^{-1}$. The photoelectron produced had its KE, 2 times the KE of the photoelectron which was produced when the same metal was irradiated with a light of frequency $2.0 \times 10^{15} \text{ s}^{-1}$. What is work function ?
32. U.V. light of wavelength 800 Å & 700 Å falls on hydrogen atoms in their ground state & liberates electrons with kinetic energy 1.8 eV and 4 eV respectively. Calculate planck's constant.
33. A potential difference of 20 kV is applied across an X-ray tube. Find the minimum wavelength of X-ray generated.
34. The K.E. of an electron emitted from tungsten surface is 3.06 eV. What voltage would be required to bring the electron to rest.
35. What is de-Broglie wavelength of a He-atom in a container at room temperature. $\left(\text{Use } U_{\text{avg.}} = \sqrt{\frac{8kT}{\pi m}} \right)$
36. Through what potential difference must an electron pass to have a wavelength of 500 Å .
37. A proton is accelerated to one tenth of the velocity of light. If its velocity can be measured with a precision $\pm 1\%$. What must be its uncertainty in position ?
38. To what effective potential a proton beam be subjected to give its protons a wavelength of $1 \times 10^{-10} \text{ m}$.
39. Calculate the number of exchange pairs of electrons present in configuration of Cu according to Aufbau principle considering 3d orbitals.
40. He atom can be excited to $1s^1 2p^1$ by $\lambda = 58.44 \text{ nm}$. If lowest excited state for He lies 4857 cm^{-1} below the above. Calculate the energy for the lower excitation state.
41. A certain dye absorbs 4530 Å and fluorecence at 5080 Å these being wavelengths of maximum absorption that under given conditions 47% of the absorbed energy is emitted. Calculate the ratio of the no. of quanta emitted to the number absorbed.
42. The reaction between H_2 and Br_2 to form HBr in presence of light is initiated by the photo decomposition of Br_2 into free Br atoms (free radicals) by absorption of light. The bond dissociation energy of Br_2 is 192 kJ/mole. What is the longest wavelength of the photon that would initiate the reaction?
43. The quantum yield for decomposition of HI is 0.2. In an experiment 0.01 moles of HI are decomposed. Find the number of photons absorbed.

44. Calculate the wavelength of the radiation that would cause photo dissociation of chlorine molecule if the Cl – Cl bond energy is 243 kJ/mol.
45. The dissociation energy of H₂ is 430.53 kJ/mol. If H₂ is exposed to radiant energy of wavelength 253.7 nm, what % of radiant energy will be converted into K.E. ?
46. Iodine molecule dissociates into atoms after absorbing light of 4500 Å⁰. If one quantum of radiation is absorbed by each molecule, calculate the K.E. of iodine atoms.
(Bond energy of I₂ = 240 kJ/mol)
47. X-rays emitted from a copper target and a molybdenum target are found to contain a line of wavelength 22.85 nm attributed to the K_α line of an impurity element. The K_α lines of a copper (Z = 29) and molybdenum (Z = 42) have wavelength 15.42 nm and 7.12 nm respectively. Using Moseley's law, $\gamma^{1/2} = a(Z - b)$. Calculate the atomic number of the impurity element.
48. What is de-Broglie wavelength associated with an e⁻ accelerated through P.D. = 100 kV ?
49. Calculate the de-broglie wavelength associated with motion of earth (mass 6 × 10²⁴ kg) orbiting around the sun at a speed of 3 × 10⁶ m/s.
50. A base ball of mass 200 g is moving with velocity 30 × 10² cm/s. If we can locate the base ball with error equal in magnitude to the λ of the light used (5000 Å), how will the uncertainty in momentum compared with the total momentum of base ball ?
51. An electron has a speed of 40 m/s, accurate up 99.99 %. What is the uncertainty in locating position?

CONCEPTUAL SUBJECTIVE EXERCISE			ANSWER KEY		EXERCISE-4(A)	
1. 2.66 × 10 ² sec	2. (a) 15.6 eV		(b) 233.9 nm,	(c) 1.008 × 10 ⁷ m ⁻¹		
			(d) (i) electron will not interact	(ii) 0.7 eV		
3. 28 photons	4. 10 ²²		5. 1.35 × 10 ⁵	6. 6	7. 4863 Å	
8. 1.096 × 10 ⁷ m ⁻¹	9. 1.827 × 10 ⁵ J/mol	10. 1403 kJ/mol	11. -1.36 × 10 ⁻¹⁹ Joules			
12. 5.425 × 10 ⁻¹² ergs, 3.7 × 10 ⁻⁵ cm		13. 1220 Å	14. 2 ; 9.75 × 10 ⁴ cm ⁻¹			
15. 113.74 Å	16. 10.2 eV, Z=2	17. 292.68 × 10 ²¹ atoms, 162.60 × 10 ²¹ atoms, 832.50 kJ				
18. 331.13 × 10 ⁴ J	19. h/π	20. 3.63 × 10 ⁶ m ⁻¹	21. 938 Å			
22. 8 × 10 ⁶	23. 6530 × 10 ¹² Hz	24. (i) Z = 5, (ii) 340 eV, -680 eV				
25. 3.09 × 10 ⁸ cm/sec		26. Brackett ; 2.63 × 10 ⁻⁴ cm				
27. $r_n = \frac{n^2 h^2}{4K\pi^2 \times 3e^2 \times 208m_e}$; n = 25 ; 55.2 pm		28. 9.15 × 10 ¹⁹ Hz, yes, 58.5 × 10 ⁻¹⁵ J				
29. 4.5 × 10 ¹⁴ s ⁻¹	30. 497 kJ/mol	31. 319.2 kJ/mol	32. 6.57 × 10 ⁻³⁴ Js			
33. 0.62 Å	34. 3.06 V	35. 0.79 Å	36. 6.03 × 10 ⁻⁴ volt			
37. 1.05 × 10 ⁻¹³ m	38. 0.0826 volts	39. 16	40. 3.3 × 10 ⁻¹⁸ J			
41. 0.527	42. 6235 Å	43. 3 × 10 ²²	44. 4.9 × 10 ⁻⁷ m			
45. 8.68 %	46. 2.186 × 10 ⁻²⁰ Joules	47. 24	48. 3.88 pm			
49. 3.68 × 10 ⁻⁶⁵ m	50. 1.75 × 10 ⁻²⁹	51. 0.0144 m				

EXERCISE-04 [B]**BRAIN STORMING SUBJECTIVE EXERCISE**

1. To what series does the spectral lines of atomic hydrogen belong if its wave number is equal number is equal to the difference between the wave numbers of the following two lines of the Balmer series 486.1 and 410.2 nm? What is the wavelength of this line ?
2. Energy required for the excitation of H-atom its ground state to the 2nd excited state is 2.67 times smaller than dissociation energy of H₂(g). If H₂ (g) placed in 1.0 litre flask at 27 C and 1.0 bar is to be excited to their 2nd excited state, what will be the total energy consumption ?
3. Find the quantum number 'n' corresponding to the excited state of He⁺ ion if on transition to the ground state that ion emits two photons in succession with wavelengths 108.5 and 30.4 nm.
4. A gas of identical H-like atom has some atoms in the lowest (ground) energy level A and some atoms in a particular upper (excited) energy level B and there are no atoms in any other energy level. The atoms of the gas make transition to a higher energy level by absorbing monochromatic light of photon energy 2.7 eV. Subsequently, the atoms emit radiation of only six different photons energies. Some of the emitted photons have energy 2.7 eV. Some have more and some have less than 2.7 eV.
 - (a) Find the principal quantum number of initially excited level B.
 - (b) Find the ionisation energy for the gas atoms.
 - (c) Find the maximum and the minimum energies of the emitted photons.
5. A hydrogen like atom (atomic number Z) is in a higher excited state of quantum number n. This excited atom can make a transition to the first excited state by successively emitting two photons of energies 10.20 eV and 17.00 eV respectively. Alternatively, the atom from the same excited state can make a transition to the second excited state by successively emitting two photons of energy 4.25 eV and 5.95 eV respectively. Determine the values of n and z (ionisation energy of hydrogen atom = 13.6 eV).
6. Hydrogen atom in its ground state is excited by means of monochromatic radiation of wavelength 975 Å . How many different lines are possible in the resulting spectrum? Calculate the longest wavelength amongst them.
7. An alpha particle after passing through a potential difference of 2×10^6 volt falls on a silver foil. The atomic number of silver is 47. Calculate (i) the K.E. of the alpha-particle at the time of falling on the foil. (ii) K.E. of the α -particle at a distance of 5×10^{-14} m from the nucleus, (iii) the shortest distance from the nucleus of silver to which the α -particle reaches.
8. Suppose the potential energy between electron and proton at a distance r is given by $-\frac{ke^2}{3r^3}$. Use Bohr's theory to obtain energy of such a hypothetical atom.
9. An energy of 68 eV is required to excite a hydrogen like atom from its second Bohr orbit to the third. The nuclear charge is Ze. Find the value of Z, the kinetic energy of the electron in the first Bohr orbit and the wavelength of the radiation required to eject the electrons from the first Bohr orbit to infinity .
10. The ionisation energy of a H-like Bohr atom is 4 Rydbergs.
 - (i) What is the wavelength of radiation emitted when the e⁻ jumps from the first excited state to the ground state?
 - (ii) What is the radius of first Bohr orbit for this atom? [1 Rydberg = 2.18×10^{-18} J]
11. Photon having wavelength 12.4 nm was allowed to strike a metal plate having work function 25 eV. Calculate the -

- (a) Maximum kinetic energy of photoelectrons emitted in eV.
- (b) Wavelength of electron with maximum kinetic energy in Å .
- (c) Calculate the uncertainty in wavelength of emitted electron, if the uncertainty in the momentum is $6.62 \times 10^{-28} \text{ kg m/sec}$.
- 12.** Electron present in single electron species jumps from energy level 3 to 1. Emitted photons when passed through a sample containing excited He^+ ion causes further excitation to some higher energy level (Given $E_n = 13.6 \frac{Z^2}{n^2}$) : Determine .
- (i) Atomic number of single electron species.
- (ii) Principal quantum number of initial excited level & higher energy of He^+
- 13.** The angular momentum of an electron in a Bohr's orbit of H-atom is $3.1652 \times 10^{-34} \text{ kg-m}^2/\text{sec}$. Calculate the wave number in terms of Rydberg constant (R) of the spectral line emitted when an electron falls from this level to the ground state. (Use $h = 6.626 \times 10^{-34} \text{ Js}$).
-

BRAIN STORMING SUBJECTIVE EXERCISE		ANSWER KEY	EXERCISE-4(B)
1.	$n_1 = 4, n_2 = 6, 2.63 \times 10^{-4} \text{ cm}$	2.	21.8 kJ
3.	$n = 5$	4.	(a) $n = 2$, (b) 14.4 eV, (c) 13.5eV, 0.7eV
5.	$n = 6, Z = 3$	6.	six, 18800 Å
7.	$6.4 \times 10^{-13} \text{ J}, 2.1 \times 10^{-13} \text{ J}, 3.4 \times 10^{-14} \text{ m}$	8.	$E = \frac{n^6 h^6}{384 m^3 K^2 e^4 \pi^6}$
9.	6 ; 489.6 eV, 25.28 Å	10.	300.89 Å , $2.645 \times 10^{-9} \text{ cm}$
11.	(a) 75 eV ; (b) 1.414 Å ; (c) $2 \times 10^{-14} \text{ m}$		
12.	(i) $Z = 1$ (ii) For He^+ ion this energy corresponds to excitation from 2 to 6.		
13.	$R \left(\frac{8}{9} \right)$		

EXERCISE - 05 [A]**JEE-[MAIN] : PREVIOUS YEAR QUESTIONS**

1. An atom has a mass of 0.02 kg and uncertainty in its velocity is 9.218×10^{-6} m/s then uncertainty in position is ($h = 6.626 \times 10^{-34}$ Js) [AIEEE 2002]
(1) 2.86×10^{-28} m (2) 2.86×10^{-32} cm (3) 1.5×10^{-27} m (4) 3.9×10^{-10} m
2. Energy of H- atom in the ground state is -13.6 eV, Hence energy in the second excited state is- [AIEEE 2002]
(1) -6.8 eV (2) -3.4 eV (3) -1.51 eV (4) -4.3 eV
3. Uncertainty in position of a particle of 25 g in space is 10^{-5} m. Hence uncertainty in velocity (ms^{-1}) is (Planck's constant $h = 6.6 \times 10^{-34}$ Js) [AIEEE-2002]
(1) 2.1×10^{-28} (2) 2.1×10^{-34} (3) 0.5×10^{-34} (4) 5.0×10^{-24}
4. The orbital angular momentum for an electron revolving in an orbit is given by $\sqrt{\ell(\ell+1)} \cdot \frac{h}{2\pi}$. This momentum for an s-electron will be given by [AIEEE-2003]
(1) $\sqrt{2} \cdot \frac{h}{2\pi}$ (2) $+\frac{1}{2} \cdot \frac{h}{2\pi}$ (3) zero (4) $\frac{h}{2\pi}$
5. The number of d-electrons retained in Fe^{2+} (At. no. of Fe = 26) ion is : [AIEEE-2003]
(1) 6 (2) 3 (3) 4 (4) 5
6. The de Broglie wavelength of a tennis ball of mass 60 g moving with a velocity of 10 metres per second is approximately : [AIEEE 2003]
(1) 10^{-25} metres (2) 10^{-33} metres (3) 10^{-31} metres (4) 10^{-16} metres
7. In Balmer series of lines of hydrogen spectrum, the third line from the red end corresponds to which one of the following inter-orbit jumps of the electron for Bohr orbits in an atom of hydrogen ? [AIEEE-2003]
(1) $2 \rightarrow 5$ (2) $3 \rightarrow 2$ (3) $5 \rightarrow 2$ (4) $4 \rightarrow 1$
8. Which of the following sets of quantum number is correct for an electron in 4f orbital ? [AIEEE-2004]
(1) $n = 3, l = 2, m = -2, s = +\frac{1}{2}$ (2) $n = 4, l = 4, m = -4, s = -\frac{1}{2}$
(3) $n = 4, l = 3, m = +1, s = +\frac{1}{2}$ (4) $n = 4, l = 3, m = +4, s = +\frac{1}{2}$
9. Consider the ground state of Cr atom ($Z = 24$). The numbers of electrons with the azimuthal quantum numbers, $l = 1$ and 2 are, respectively [AIEEE-2004]
(1) 16 and 5 (2) 12 and 5 (3) 16 and 4 (4) 12 and 4
10. The wavelength of the radiation emitted, when in a hydrogen atom electron falls from infinity to stationary state 1, would be [AIEEE-2004]
(Rydberg constant = $1.097 \times 10^7 \text{ m}^{-1}$) :
(1) 9.1×10^{-8} nm (2) 192 nm (3) 406 nm (4) 91 nm
11. Which one of the following sets of ions represents the collection of isoelectronic species ? [AIEEE-2004]
(1) $\text{Na}^+, \text{Mg}^{2+}, \text{Al}^{3+}, \text{Cl}^-$ (2) $\text{Na}^+, \text{Ca}^{2+}, \text{Sc}^{3+}, \text{F}^-$
(3) $\text{K}^+, \text{Cl}^-, \text{Mg}^{2+}, \text{Sc}^{3+}$ (4) $\text{K}^+, \text{Ca}^{2+}, \text{Sc}^{3+}, \text{Cl}^-$
12. In a multi-electron atom, which of the following orbitals described by the three quantum members will have the same energy in the absence of magnetic and electric fields ? [AIEEE-2005]
(A) $n = 1, l = 0, m = 0$ (B) $n = 2, l = 0, m = 0$
(C) $n = 2, l = 1, m = 1$ (D) $n = 3, l = 2, m = 1$
(E) $n = 3, l = 2, m = 0$
(1) (D) and (E) (2) (C) and (D) (3) (B) and (C) (4) (A) and (B)

13. Of the following sets which one does not contain isoelectronic species ? [AIEEE-2005]
 (1) BO_3^{3-} , CO_3^{2-} , NO_3^- (2) SO_3^{2-} , CO_3^{2-} , NO_3^- (3) CN^- , N_2 , C_2^{2-} (4) PO_4^{3-} , SO_4^{2-} , ClO_4^-
14. Which of the following statements in relation to the hydrogen atom is correct ? [AIEEE-2005]
 (1) 3s, 3p and 3d orbitals all have the same energy
 (2) 3s and 3p orbitals are of lower energy than 3d orbitals
 (3) 3p orbital is lower in energy than 3d orbital
 (4) 3s orbitals is lower in energy than 3p orbital
15. According to Bohr's theory angular momentum of electron in 5th shell is :- [AIEEE-2006]
 (1) $1.0 \text{ h}/\pi$ (2) $10 \text{ h}/\pi$ (3) $2.5 \text{ h}/\pi$ (4) $25 \text{ h}/\pi$
16. Uncertainty in the position of an electron (mass = $9.1 \times 10^{-31} \text{ Kg}$) moving with a velocity 300 ms^{-1} , accurate upto 0.001%, will be :- ($h = 6.63 \times 10^{-34} \text{ Js}$) [AIEEE-2006]
 (1) $5.76 \times 10^{-2} \text{ m}$ (2) $1.92 \times 10^{-2} \text{ m}$ (3) $3.84 \times 10^{-2} \text{ m}$ (4) $19.2 \times 10^{-2} \text{ m}$
17. Which of the following sets of quantum numbers represents the highest energy of an atom ? [AIEEE-2007]
 (1) $n = 3, l = 1, m = 1, s = +1/2$ (2) $n = 3, l = 2, m = 1, s = +1/2$
 (3) $n = 4, l = 0, m = 0, s = +1/2$ (4) $n = 3, l = 0, m = 0, s = +1/2$
18. Which one of the following constitutes a group of the isoelectronic species? [AIEEE-2008]
 (1) C_2^{2-} , O_2^- , CO , NO (2) NO^+ , C_2^{2-} , CN^- , N_2 (3) CN^- , N_2 , O_2^{2-} , C_2^{2-} (4) N_2 , O_2^- , NO^+ , CO
19. The ionization enthalpy of hydrogen atom is $1.312 \times 10^6 \text{ J mol}^{-1}$. The energy required to excite the electron in the atom from $n = 1$ to $n = 2$ is [AIEEE-2008]
 (1) $8.51 \times 10^5 \text{ J mol}^{-1}$ (2) $6.56 \times 10^5 \text{ J mol}^{-1}$ (3) $7.56 \times 10^5 \text{ J mol}^{-1}$ (4) $9.84 \times 10^5 \text{ J mol}^{-1}$
20. In an atom, an electron is moving with a speed of 600 m/s with an accuracy of 0.005%. Certainty with which the position of the electron can be located is ($h = 6.6 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$, mass of electron, $m_e = 9.1 \times 10^{-31} \text{ kg}$):- [AIEEE-2009]
 (1) $1.92 \times 10^{-3} \text{ m}$ (2) $3.84 \times 10^{-3} \text{ m}$ (3) $1.52 \times 10^{-4} \text{ m}$ (4) $5.10 \times 10^{-3} \text{ m}$
21. Calculate the wavelength (in nanometer) associated with a proton moving at $1.0 \times 10^3 \text{ ms}^{-1}$ (Mass of proton = $1.67 \times 10^{-27} \text{ kg}$ and $h = 6.63 \times 10^{-34} \text{ Js}$) :- [AIEEE-2009]
 (1) 2.5 nm (2) 14.0 nm (3) 0.032 nm (4) 0.40 nm
22. The energy required to break one mole of Cl-Cl bonds in Cl_2 is 242 kJ mol^{-1} . The longest wavelength of light capable of breaking a single Cl-Cl bond is ($C = 3 \times 10^8 \text{ ms}^{-1}$ and $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$) [AIEEE-2010]
 (1) 494 nm (2) 594 nm (3) 640 nm (4) 700 nm
23. Ionisation energy of He^+ is $19.6 \times 10^{-18} \text{ J atom}^{-1}$. The energy of the first stationary state ($n = 1$) of Li^{2+} is:- [AIEEE-2010]
 (1) $8.82 \times 10^{-17} \text{ J atom}^{-1}$ (2) $4.41 \times 10^{-16} \text{ J atom}^{-1}$
 (3) $-4.41 \times 10^{-17} \text{ J atom}^{-1}$ (4) $-2.2 \times 10^{-15} \text{ J atom}^{-1}$
24. A gas absorbs a photon of 355 nm and emits at two wavelengths. If one of the emissions is at 680 nm, the other is at :- [AIEEE-2011]
 (1) 743 nm (2) 518 nm (3) 1035 nm (4) 325 nm
25. The frequency of light emitted for the transition $n = 4$ to $n = 2$ of He^+ is equal to the transition in H atom corresponding to which of the following :- [AIEEE-2011]
 (1) $n = 3$ to $n = 1$ (2) $n = 2$ to $n = 1$ (3) $n = 3$ to $n = 2$ (4) $n = 4$ to $n = 3$
26. The increasing order of the ionic radii of the given isoelectronic species is :- [AIEEE-2012]
 (1) K^+ , S^{2-} , Ca^{2+} , Cl^- (2) Cl^- , Ca^{2+} , K^+ , S^{2-} (3) S^{2-} , Cl^- , Ca^{2+} , K^+ (4) Ca^{2+} , K^+ , Cl^- , S^{2-}

27. The electrons identified by quantum numbers n and ℓ :-

[AIEEE-2012]

(a) $n = 4$, $\ell = 1$

(b) $n = 4$, $\ell = 0$

(c) $n = 3$, $\ell = 2$

(d) $n = 3$, $\ell = 1$

Can be placed in order of increasing energy as

(1) (a) < (c) < (b) < (d)

(2) (c) < (d) < (b) < (a)

(3) (d) < (b) < (c) < (a)

(4) (b) < (d) < (a) < (c)

PREVIOUS YEARS QUESTIONS						ANSWER KEY				EXERCISE-5 [A]					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans	1	3	1	3	1	2	3	3	2	4	4	1	2	1	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27			
Ans	2	2	2	4	1	4	1	3	1	2	4	3			

EXERCISE - 05 [B] JEE-[ADVANCED] : PREVIOUS YEAR QUESTIONS

1. Rutherford's experiment, which established the nuclear model of atom, used a beam of :-
(A) β - particles, which impinged on a metal foil and got absorbed. [JEE 2002]
(B) γ - rays, which impinged on a metal foil and ejected electron.
(C) Helium atoms, which impinged on a metal foil and got scattered.
(D) Helium nuclei, which impinged on a metal foil and got scattered.

2. The magnetic moment of cobalt of the compound $\text{Hg}[\text{Co}(\text{SCN})_4]$ is [Given : Co^{+2}] [JEE 2004]
(A) $\sqrt{3}$ (B) $\sqrt{8}$ (C) $\sqrt{15}$ (D) $\sqrt{24}$

3. The radius of which of the following orbit is same as that of the first Bohr's orbit of hydrogen atom? [JEE 2004]
(A) He^+ ($n = 2$) (B) Li^{2+} ($n = 2$) (C) Li^{2+} ($n = 3$) (D) Be^{3+} ($n = 2$)

4. (a) The Schrodinger wave equation for hydrogen atom is [IIT-2004]

$$\Psi_{2s} = \frac{1}{4(2\pi)^{1/2}} \left(\frac{1}{a_0} \right)^{3/2} \left(2 - \frac{r_0}{a_0} \right) e^{-r/a}$$

Where a_0 is Bohr's radius. Let the radial node in 2s be at r_0 . Then find r_0 in terms of a_0 .

(b) A baseball having mass 100 g moves with velocity 100 m/s. Find out the value of wavelength of baseball.

5. (a) Calculate velocity of electron in first Bohr orbit of hydrogen atom (Given $r = a_0$) [IIT-2005]
(b) Find de-Broglie wavelength of the electron in first Bohr orbit.
(c) Find the orbital angular momentum of 2p-orbital in terms of $h/2\pi$ units.

6. Given in hydrogenic atom r_n , V_n , E , K_n stand for radius, potential energy, total energy and kinetic energy in n^{th} orbit. Find the value of U, v, x, y. [JEE 2006]

- (A) $U = \frac{V_n}{K_n}$ (P) 1
(B) $\frac{1}{r_n} \propto E^x$ (Q) -2
(C) $r_n \propto Z^y$ (R) -1
(Z = Atomic number)
(D) $v =$ (Orbital angular momentum of electron in its lowest energy) (S) 0

7. Match the entries in **Column I** with the correctly related quantum number(s) in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS.

[JEE 2008]

Column I		Column II	
(A)	Orbital angular momentum of the electron in a hydrogen-like atomic orbital	(P)	Principal quantum number
(B)	A hydrogen-like one-electron wave function obeying Pauli principle	(Q)	Azimuthal quantum number
(C)	Shape, size and orientation of hydrogen like atomic orbitals	(R)	Magnetic quantum number
(D)	Probability density of electron at the nucleus in hydrogen-like atom	(S)	Electron spin quantum number
(A)			

Paragraph for questions 8 to 10

[JEE 2010]

The hydrogen-like species Li^{2+} is in a spherically symmetric state S_1 with one radial node. Upon absorbing light the ion undergoes transition to a state S_2 . The state S_2 has one radial node and its energy is equal to the ground state energy of the hydrogen atom.

8. The state S_1 is :-
 (A) 1s (B) 2s (C) 2p (D) 3s
9. Energy of the state S_1 in units of the hydrogen atom ground state energy is :-
 (A) 0.75 (B) 1.50 (C) 2.25 (D) 4.50
10. The orbital angular momentum quantum number of the state S_2 is :-
 (A) 0 (B) 1 (C) 2 (D) 3
11. The maximum number of electrons that can have principal quantum number, $n=3$, and spin quantum number, $m_s = -1/2$, is [JEE 2011]
12. The work function (ϕ) of some metals is listed below. The number of metals which will show photoelectric effect when light of 300 nm wavelength falls on the metal is :- [JEE 2011]

Metal	Li	Na	K	Mg	Cu	Ag	Fe	Pt	W
$\phi(\text{eV})$	2.4	2.3	2.2	3.7	4.8	4.3	4.7	6.3	4.75

13. The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is [a_0 is Bohr radius] [JEE 2012]

(A) $\frac{h^2}{4\pi^2 m a_0^2}$ (B) $\frac{h^2}{16\pi^2 m a_0^2}$ (C) $\frac{h^2}{32\pi^2 m a_0^2}$ (D) $\frac{h^2}{32\pi^2 m a_0^2}$

PREVIOUS YEARS QUESTIONS			ANSWER KEY		EXERCISE-5 [B]	
1. (D)	2. (C)	3. (D)	4. (a) $r_0 = 2a_0$	(b) $6.626 \times 10^{-25} \text{ Å}$		
5. (a) $2.197 \times 10^6 \text{ m/s}$	(b) 3.31 Å	(c) $\sqrt{2} \cdot \frac{h}{2\pi}$	6. (A) Q, (B) P, (C) R, (D) S			
7. (A) Q,R (B) P, Q, R,S (C) P, Q, R (D) P, Q			8. (B)	9. (C)		
10. (B)	11. 9	12. 4	13. (C)			

EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. If a spoon of copper metal is placed in a solution of ferrous sulphate :-
(A) Cu will precipitate out (B) iron will precipitate
(C) Cu and Fe will precipitate (D) no reaction will take place
2. During discharge of a lead storage cell the density of sulphuric acid in the cell :-
(A) increasing (B) decreasing
(C) remains unchanged (D) initially increases but decreases subsequently
3. Which of the following represents the reduction potential of silver wire dipped into 0.1 M AgNO_3 solution at 25 C :-
(A) E_{red} (B) $(E_{\text{red}} + 0.059)$ (C) $(E_{\text{ox}} - 0.059)$ (D) $(E_{\text{red}} - 0.059)$
4. If the solution of the CuSO_4 in which copper rod is immersed is diluted to 10 times, the electrode reduction potential. ($T = 298 \text{ K}$) :-
(A) increases by 0.030 volt (B) decreases by 0.030 volt
(C) increases by 0.059 volt (D) decreases by 0.0059 volt
5. The oxidation potential of a hydrogen electrode at $\text{pH} = 1$ is ($T = 298 \text{ K}$) :-
(A) 0.059 volt (B) 0 volt (C) -0.059 volt (D) 0.59 volt
6. The electrode potential becomes equal to standard electrode potential when reactants and products concentration ratio is :-
(A) equal to 1 (B) greater than 1 (C) less than 1 (D) none of the above
7. The reduction potential values are given below
 $\text{Al}^{3+} / \text{Al} = -1.67 \text{ volt}$, $\text{Mg}^{2+} / \text{Mg} = -2.34 \text{ volt}$ $\text{Cu}^{2+} / \text{Cu} = +0.34 \text{ volt}$
 $\text{I}_2 / 2\text{I}^- = +0.53 \text{ volt}$. Which one is the best reducing agent ?
(A) Al (B) Mg (C) Cu (D) I_2
8. When iron is rusted, it is :-
(A) reduced (B) oxidised (C) evaporated (D) decomposed
9. The reference electrode is made from which of the following :-
(A) ZnCl_2 (B) CuSO_4 (C) Hg_2Cl_2 (D) HgCl_2
10. Given standard electrode potentials :
 $\text{Fe}^{3+} + 3\text{e}^- \longrightarrow \text{Fe}$; $E = -0.036 \text{ volt}$
 $\text{Fe}^{2+} + 2\text{e}^- \longrightarrow \text{Fe}$; $E = -0.440 \text{ volt}$
The standard electrode potential E for
 $\text{Fe}^{3+} + \text{e}^- \longrightarrow \text{Fe}^{2+}$ is :-
(A) -0.476 volt (B) -0.404 volt (C) 0.440 volt (D) 0.772 volt
11. The reduction potential of a hydrogen electrode at $\text{pH} 10$ at 298 K is : ($p = 1 \text{ atm}$)
(A) 0.51 volt (B) 0 volt (C) -0.591 volt (D) 0.059 volt
12. The emf of the cell, $\text{Ni} | \text{Ni}^{2+} (1.0 \text{ M}) || \text{Ag}^+ (1.0 \text{ M}) | \text{Ag}$ [E for $\text{Ni}^{2+} / \text{Ni} = -0.25 \text{ volt}$, E for $\text{Ag}^+ / \text{Ag} = 0.80 \text{ volt}$] is given by : [E for $\text{Ag}^+ / \text{Ag} = 0.80 \text{ volt}$]
(A) $-0.25 + 0.80 = 0.55 \text{ volt}$ (B) $-0.25 - (+0.80) = -1.05 \text{ volt}$
(C) $0 + 0.80 - (-0.25) = +1.05 \text{ volt}$ (D) $-0.80 - (+0.25) = -0.55 \text{ volt}$
13. The position of some metals in the electrochemical series in decreasing electropositive character is given as $\text{Mg} > \text{Al} > \text{Zn} > \text{Cu} > \text{Ag}$. What will happen if a copper spoon is used to stir a solution of aluminium nitrate?
(A) The spoon will get coated with aluminium (B) An alloy of aluminium and copper nitrate
(C) The solution becomes blue (D) There is no reaction
14. $E (\text{Ni}^{2+} / \text{Ni}) = -0.25 \text{ volt}$, $E (\text{Au}^{3+} / \text{Au}) = 1.50 \text{ volt}$. The emf of the voltaic cell $\text{Ni} | \text{Ni}^{2+} (1.0 \text{ M}) || \text{Au}^{3+} (1.0 \text{ M}) | \text{Au}$ is :-
(A) 1.25 volt (B) -1.75 volt (C) 1.75 volt (D) 4.0 volt

15. When the electric current is passed through a cell having an electrolyte, the positive ions move towards cathode and negative ions towards the anode. If the cathode is pulled out of the solution :-
 (A) the positive and negative ions will move towards anode
 (B) the positive ions will start moving towards the anode while negative ions will stop moving
 (C) the negative ions will continue to move towards anode while positive ions will stop moving
 (D) the positive and negative ions will start moving randomly
16. The oxidation potentials of Zn, Cu, Ag, H₂ and Ni are 0.76, -0.34, -0.80, 0.00, 0.25 volt respectively. Which of the following reactions will provide maximum voltage :-
 (A) $\text{Zn} + \text{Cu}^{2+} \longrightarrow \text{Cu} + \text{Zn}^{2+}$ (B) $\text{Zn} + 2\text{Ag}^+ \longrightarrow 2\text{Ag} + \text{Zn}^{2+}$
 (C) $\text{H}_2 + \text{Cu}^{2+} \longrightarrow 2\text{H}^+ + \text{Cu}$ (D) $\text{H}_2 + \text{Ni}^{2+} \longrightarrow 2\text{H}^+ + \text{Ni}$
17. Which one of the following will increase the voltage of the cell : (T = 298 K)
 $\text{Sn} + 2\text{Ag}^+ \longrightarrow \text{Sn}^{2+} + 2\text{Ag}$
 (A) increase in the size of silver rod (B) increase in the concentration of Sn^{+2} ions
 (C) increase in the concentration of Ag^+ ions (D) none of the above
18. A chemist wants to produce Cl₂ (g) from molten NaCl. How many grams could be produced if he uses a steady current of 2 ampere for 2.5 minutes :-
 (A) 3.55 g (B) 1.775 g (C) 0.110 g (D) 0.1775 g
19. Consider the reaction : (T = 298 K)
 $\text{Cl}_2 (\text{g}) + 2\text{Br}^- (\text{aq}) \longrightarrow 2\text{Cl}^- (\text{aq}) + \text{Br}_2 (\text{aq})$
 The emf of the cell, when $[\text{Cl}^-] = [\text{Br}_2] = [\text{Br}^-] = 0.01 \text{ M}$ and Cl₂ gas is 1 atm pressure, will be (E for the above reaction is = 0.29 volt) :-
 (A) 0.54 volt (B) 0.35 volt (C) 0.24 volt (D) -0.29 volt
20. Three faradays of electricity was passed through an aqueous solution of iron (II) bromide. The mass of iron metal (at. mass 56) deposited at the cathode is :-
 (A) 56 g (B) 84 g (C) 112 g (D) 168 g
21. The standard emf for the cell reaction $\text{Zn} + \text{Cu}^{2+} \longrightarrow \text{Zn}^{2+} + \text{Cu}$ is 1.10 volt at 25 C. The emf for the cell reaction when 0.1 M Cu²⁺ and 0.1 M Zn²⁺ solutions are used at 25 C is :-
 (A) 1.10 volt (B) 0.110 volt (C) -1.10 volt (D) -0.110 volt
22. Three moles of electrons are passed through three solutions in succession containing AgNO₃, CuSO₄ and AuCl₃ respectively. The molar ratio of amounts of cations reduced at cathode will be :-
 (A) 1 : 2 : 3 (B) 2 : 1 : 3 (C) 3 : 2 : 1 (D) 6 : 3 : 2
23. The emf of the cell involving the following reaction, $2\text{Ag}^+ + \text{H}_2 \longrightarrow 2\text{Ag} + 2\text{H}^+$ is 0.80 volt. The standard oxidation potential of silver electrode is :-
 (A) -0.80 volt (B) 0.80 volt (C) 0.40 volt (D) -0.40 volt
24. For the electrochemical cell, $\text{M}|\text{M}^+||\text{X}^-|\text{X}$, $E_{\text{M}^+/\text{M}} = 0.44 \text{ V}$ and $E_{\text{X}/\text{X}^-} = 0.33 \text{ V}$. From this data we can deduce that :-
 (A) $\text{M} + \text{X} \longrightarrow \text{M}^+ + \text{X}^-$ is the spontaneous reaction
 (B) $\text{M}^+ + \text{X}^- \longrightarrow \text{M} + \text{X}$ is the spontaneous reaction
 (C) $E_{\text{cell}} = 0.77 \text{ V}$
 (D) $E_{\text{cell}} = -0.77 \text{ V}$
25. For the net cell reaction of the cell $\text{Zn}(\text{s})|\text{Zn}^{2+}||\text{Cd}^{2+}|\text{Cd}(\text{s})$ ΔG in kilojoules at 25 C is ($E_{\text{cell}} = 0.360 \text{ V}$):-
 (A) 112.5 (B) 69.47 (C) -34.73 (D) -69.47
26. How many faradays are required to reduce one mol of MnO_4^- to Mn^{2+} :-
 (A) 1 (B) 2 (C) 3 (D) 5
27. $\text{Cu}^+ + \text{e}^- \longrightarrow \text{Cu}$, $E = x_1 \text{ volt}$;
 $\text{Cu}^{2+} + 2\text{e}^- \longrightarrow \text{Cu}$, $E = x_2 \text{ volt}$, then for
 $\text{Cu}^{2+} + \text{e}^- \longrightarrow \text{Cu}^+$, E (volt) will be :
 (A) $x_1 - 2x_2$ (B) $x_1 + 2x_2$ (C) $x_1 - x_2$ (D) $2x_2 - x_1$

28. $\text{Zn}|\text{Zn}^{2+}(\text{C}_1)||\text{Zn}^{2+}(\text{C}_2)|\text{Zn}$. for this cell ΔG is negative if :-
 (A) $\text{C}_1 = \text{C}_2$ (B) $\text{C}_1 > \text{C}_2$ (C) $\text{C}_2 > \text{C}_1$ (D) None
29. $\text{Pt} \left| \begin{matrix} \text{H}_2 \\ (p_1) \end{matrix} \right| \left| \begin{matrix} \text{H}^+ \\ (1\text{M}) \end{matrix} \right| \left| \begin{matrix} \text{H}^+ \\ (1\text{M}) \end{matrix} \right| \left| \begin{matrix} \text{H}_2 \\ (p_2) \end{matrix} \right| \text{Pt}$ (where p_1 and p_2 are pressure) cell reaction will be spontaneous if :-
 (A) $p_1 = p_2$ (B) $p_1 > p_2$ (C) $p_2 > p_1$ (D) $p_1 = 1 \text{ atm}$
30. $\text{Pt} \left| \begin{matrix} \text{H}_2 \\ 1 \text{ atm} \end{matrix} \right| \text{pH} = 2 || \text{pH} = 3 \left| \begin{matrix} \text{H}_2 \\ 1 \text{ atm} \end{matrix} \right| \text{Pt}$. The cell reaction for the given cell is :-
 (A) spontaneous (B) non-spontaneous (C) equilibrium (D) none is correct
31. $\text{M}^{2+} + 2\text{e}^- \longrightarrow \text{M}$. 0.275 g of metal M is deposited at the cathode due to passage of 1A of current for 965 s. Hence atomic weight of the metal M is :-
 (A) 27.5 (B) 55.0 (C) 110.0 (D) 13.75
32. In an electrochemical cell that function as a voltaic cell :-
 (A) electrons move from the cathode to the anode
 (B) electrons move through a salt-bridge
 (C) electrons can move either from the cathode to the anode or from the anode to the cathode
 (D) reduction occurs at the cathode
33. A certain metal salt solution is electrolysed in series with a silver coulometer. The weight of silver and the metal deposited are 0.5094 g and 0.2653 g. Calculate the valency of the metal if its atomic weight is nearly that of silver.
 (A) .1 (B) 2 (C) 3 (D) 4
34. The cell $\text{Pt}(\text{H}_2)(1 \text{ atm}) | \text{H}^+(\text{pH} = ?), \Gamma(a = 1) | \text{AgI(s)}, \text{Ag}$ has emf, $E_{298 \text{ K}} = 0$. The electrode potential for the reaction $\text{AgI} + \text{e}^- \rightarrow \text{Ag} + \text{I}^-$ is -0.151 volt. Calculate the pH value :-
 (A) 3.37 (B) 5.26 (C) 2.56 (D) 4.62
35. Using the information in the preceding problem, calculate the solubility product of AgI in water at 25 C
 $[E_{(\text{Ag}^+, \text{Ag})} = +0.799 \text{ volt}]$:-
 (A) 1.97×10^{-17} (B) 7.91×10^{-17} (C) 1.79×10^{-17} (D) 9.17×10^{-17}
36. If same quantity of electricity is passed through CuCl and CuSO_4 , the ratio of the weights of Cu deposited from CuSO_4 and CuCl is :-
 (A) 2 : 1 (B) 1 : 2 (C) 1 : 1 (D) 4 : 1
37. For Zn^{2+}/Zn , $E = -0.76 \text{ V}$, for Ag^+/Ag , $E = 0.799 \text{ V}$. The correct statement is :-
 (A) the reaction Zn getting reduced Ag getting oxidized is spontaneous
 (B) Zn undergoes reduction and Ag is oxidized
 (C) Zn undergoes oxidation and Ag^+ gets reduced
 (D) No suitable answer
38. The oxidation potential of a hydrogen electrode at $\text{pH} = 10$ and $P_{\text{H}_2} = 1$ is :-
 (A) 0.51 V (B) 0.00 V (C) + 0.59 V (D) 0.059 V

CHECK YOUR GRASP						ANSWER KEY				EXERCISE -1					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	D	B	D	B	A	A	B	B	C	D	C	C	D	C	D
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	B	C	C	B	B	A	D	A	B	D	D	D	C	B	B
Que.	31	32	33	34	35	36	37	38							
Ans.	B	D	B	C	B	B	C	C							

EXERCISE-02**BRAIN TEASERS****SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)**

- What a lead storage is discharged :-
(A) PbSO_4 is formed (B) Pb is formed (C) SO_2 is consumed (D) H_2SO_4 is consumed
- The standard emf of the cell, $\text{Cd (s)}|\text{CdCl}_2\text{ (aq) (0.1 M)}||\text{AgCl (s)}|\text{Ag (s)}$ in which the cell reaction is, $\text{Cd (s)} + 2 \text{AgCl (s)} \longrightarrow 2 \text{Ag (s)} + \text{Cd}^{2+}\text{ (aq)} + 2\text{Cl}^-\text{ (aq)}$ is 0.6915 V at 0 C and 0.6753 V at 25 C. The ΔH of the reaction at 25 C is :-
(A) -176 kJ (B) -234.7 kJ (C) +123.5 kJ (D) -167.26 kJ
- A current of 2.68 A is passed for one hour through an aqueous solution of CuSO_4 using copper electrodes. Select the correct statement (s) from the following :-
(A) increase in mass of cathode = 3.174 g
(B) decrease in mass of anode = 3.174 g
(C) no change in masses of electrodes
(D) the ratio between the change of masses of cathode and anode is 1 : 2
- Any redox reaction would occur spontaneously, if :-
(A) the free energy change (ΔG) is negative (B) the (ΔG) is positive
(C) the cell e.m.f. (E) is negative (D) the cell e.m.f. is positive
- Fe is reacted with 1.0 M HCl. E for $\text{Fe/Fe}^{2+} = +0.34$ volt. The correct observation (s) regarding this reaction is / are :-
(A) Fe will be oxidised to Fe^{2+}
(B) Fe^{2+} will be reduced to Fe
(C) since e.m.f. is positive, the reaction shall occur
(D) since e.m.f. is positive, the reaction shall not occur
- An acidic solution of copper (II) sulphate containing some contaminations of zinc and iron (II) ions was electrolysed till all the copper is deposited. If electrolysis is further continued for sometime, the product liberated at cathode is :-
(A) Fe (B) Zn (C) H_2 (D) Alloy of Zn and Fe
- A current of 9.65 ampere is passed through the aqueous solution NaCl using suitable electrodes for 1000 s. The amount of NaOH formed during electrolysis is :-
(A) 2.0 g (B) 4.0 g (C) 6.0 g (D) 8.0 g
- The potential of the Daniel cell, $\text{Zn} \left| \text{ZnSO}_4 \right| (1\text{M}) \left| \text{CuSO}_4 \right| (1\text{M}) \left| \text{Cu} \right|$ was reported by Buckbee, Surdzial and Metz as $E = 1.1028 - 0.641 \cdot 10^{-3} T + 0.72 \cdot 10^{-5} T^2$, where T is the celsius temperature. Calculate ΔS for the cell reaction at 25 C :-
(A) -45.32 EU (B) -34.52 EU (C) -25.43 EU (D) -54.23 EU
- Using the data in the preceding problem, calculate the equilibrium constant of the reaction at 25 C.
$$\text{Zn} + \text{Cu}^{++} \rightleftharpoons \text{Zn}^{++} + \text{Cu}, K = \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}$$

(A) $8.314 \cdot 10^{24}$ (B) $4.831 \cdot 10^{31}$ (C) $8.314 \cdot 10^{36}$ (D) $4.831 \cdot 10^{44}$
- The standard electrode potential (reduction) of $\text{Ag}^+|\text{Ag}$ is 0.800 V at 25 C. Its electrode potential in a solution containing 10^{-3} M ion of Ag^+ ions is :-
(A) 0.623 V (B) -0.977 V (C) 0.892 V (D) 1.246 V

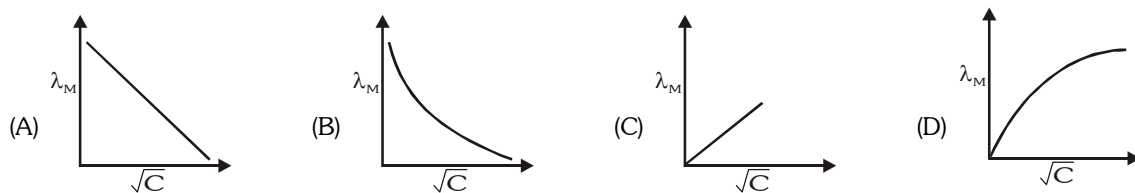
11. The number of faradays required to produce one mole of water from hydrogen - oxygen fuel cell containing aqueous alkali as electrolyte is :-
 (A) 1 (B) 3 (C) 2 (D) 4
12. The solubility product of silver iodide is 8.3×10^{-17} and the standard potential (reduction) of Ag, Ag^+ electrode is +0.800 volts at 25 C. The standard potential of Ag, AgI / I^- electrode (reduction) from these data is :-
 (A) -0.30 V (B) +0.15 V (C) +0.10 V (D) -0.15 V
13. The standard electrode potentials (reduction) of Pt/ Fe^{3+} , Fe^{2+} and Pt/ Sn^{4+} , Sn^{2+} are +0.77 V and 0.15 V respectively at 25 C. The standard EMF of the reaction $\text{Sn}^{4+} + 2 \text{Fe}^{2+} \longrightarrow \text{Sn}^{2+} + 2\text{Fe}^{3+}$ is :-
 (A) -0.62 V (B) -0.92 V (C) +0.31 V (D) +0.85 V
14. The standard reduction potential E of the following systems are :-

System	E (volts)
(i) $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \longrightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51
(ii) $\text{Sn}^{4+} + 2\text{e}^- \longrightarrow \text{Sn}^{2+}$	0.15
(iii) $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \longrightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	1.33
(iv) $\text{Ce}^{4+} + \text{e}^- \longrightarrow \text{Ce}^{3+}$	1.61

The oxidising power of the various species decreases in the order :

- (A) $\text{Ce}^{4+} > \text{Cr}_2\text{O}_7^{2-} > \text{Sn}^{4+} > \text{MnO}_4^-$ (B) $\text{Ce}^{4+} > \text{MnO}_4^- > \text{Cr}_2\text{O}_7^{2-} > \text{Sn}^{4+}$
 (C) $\text{Cr}_2\text{O}_7^{2-} > \text{Sn}^{4+} > \text{Ce}^{4+} > \text{MnO}_4^-$ (D) $\text{MnO}_4^- > \text{Ce}^{4+} > \text{Sn}^{4+} > \text{Cr}_2\text{O}_7^{2-}$
15. The e.m.f. of the following cell
 $\text{Ni(s)} / \text{NiSO}_4 (1.0 \text{ M}) \parallel \text{H}^+ (1.0 \text{ M}) | \text{H}_2 (1 \text{ atm}), \text{Pt}$ at 25 C is 0.236 V. The electrical energy which can be produced is :-
 (A) 22.73 kJ (B) 45.55 kJ (C) 19.30 kJ (D) 23.60 kJ
16. The reduction potential of a hydrogen electrode at pH = 10, at 25 C is :-
 (A) -0.59 V (B) -0.0591 V (C) zero (D) -0.591 V
17. The reduction potential of a half-cell consisting of a Pt electrode immersed in 1.5 M Fe^{2+} and 0.015 M Fe^{3+} solution at 25 C is ($E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^0 = 0.770 \text{ V}$) :
 (A) 0.652 V (B) 0.88 V (C) 0.710 V (D) 0.850 V
18. Consider the cell $\text{H}_2(\text{Pt}) \left| \text{H}_3\text{O}^+(\text{aq}) \right| \left| \text{Ag}^+ \right| \text{Ag}$. The measured EMF of the cell is 1.023 V. What is the value of x ?
 $E_{\text{Ag}^+, \text{Ag}}^0 = + 0.799 \text{ V}$. [T = 25 C]
 (A) $2 \times 10^{-2} \text{ M}$ (B) $2 \times 10^{-3} \text{ M}$ (C) $1.5 \times 10^{-3} \text{ M}$ (D) $1.5 \times 10^{-2} \text{ M}$
19. Hg_2Cl_2 is produced by the electrolytic reduction of Hg^{2+} ion in presence of Cl^- ion is $2\text{Hg}^{2+} + 2\text{Cl}^- + 2\text{e}^- \rightarrow \text{Hg}_2\text{Cl}_2$. Calculate the current required to have a rate production of 44 g per hour of Hg_2Cl_2 . [Atomic weight of Hg = 200.6] :-
 (A) 5 ampere (B) 4 ampere
 (C) 6.5 ampere (D) 3.5 ampere
20. The ionization constant of a weak electrolyte is 25×10^{-6} while the equivalent conductance of its 0.01 M solution is $19.6 \text{ S cm}^2 \text{ eq}^{-1}$. The equivalent conductance of the electrolyte at infinite dilution (in $\text{S cm}^2 \text{ eq}^{-1}$) will be :-
 (A) 250 (B) 196 (C) 392 (D) 384

21. Which of the following curve represents the variation of λ_M with $\sqrt{C}$ for AgNO_3 ?



22. Four moles of electrons were transferred from anode to cathode in an experiment on electrolysis of water. The total volume of the two gases (dry and at STP) produced will be approximately (in litres) :-

(A) 22.4 (B) 44.8 (C) 67.2 (D) 89.4

23. Equivalent conductance of BaCl_2 , H_2SO_4 & HCl at infinite dilution are A_∞^1 , A_∞^2 & A_∞^3 respectively. Equivalent conductance of BaSO_4 solution is :

(A) $A_\infty^1 + A_\infty^2 - 2 A_\infty^3$ (B) $A_\infty^1 + A_\infty^2 + A_\infty^3$
 (C) $A_\infty^1 + A_\infty^2 - A_\infty^3$ (D) $A_\infty^1 - A_\infty^2 + A_\infty^3$

24. Salts of A (atomic weight = 7), B (atomic weight = 27) and C (atomic weight = 48) were electrolysed under identical conditions using the same quantity of electricity. It was found that when 2.1 g of A was deposited, the weights of B and C deposited were 2.7g and 7.2 g. The valencies of A, B and C respectively are :-

(A) 3, 1 and 2 (B) 1, 3 and 2 (C) 3, 1 and 3 (D) 2, 3 and 2

25. During electrolysis of an aqueous solution of CuSO_4 using copper electrodes, if 2.5 g of Cu is deposited at cathode, then at anode :-

(A) 890 mL of Cl_2 at STP is liberated
 (B) 445 mL of O_2 at STP is liberated
 (C) 2.5 g of copper is deposited
 (D) a decrease of 2.5 g of mass takes place

26. The cost at 5 paise/KWH of operating an electric motor for 8 hours which takes 15 amp at 110 V is :-

(A) Rs. 66 (B) 66 paise (C) 37 paise (D) Rs. 6.60

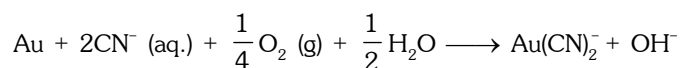
27. When an aqueous solution of lithium chloride is electrolysed using graphite electrodes :-

(A) Cl_2 is liberated at the anode (B) Li is deposited at the cathode
 (C) as the current flows, pH of the solution around the cathode remains constant
 (D) as the current flows, pH of the solution around the cathode decreases

28. A silver wire dipped in 0.1 M HCl solution saturated with AgCl develops a potential of -0.25 V. If $E_{\text{Ag}/\text{Ag}^+} = -0.799$ V, the K_{sp} of AgCl in pure water will be :-

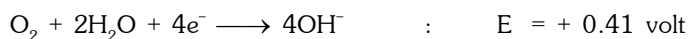
(A) 2.95×10^{-11} (B) 5.1×10^{-11} (C) 3.95×10^{-11} (D) 1.95×10^{-11}

29. Consider the reaction of extraction of gold from its ore



Use the following data to calculate ΔG for the reaction

$$K_f \{ \text{Au}(\text{CN})_2^- \} = X$$



(A) $-RT \ln X + 1.29 F$

(B) $-RT \ln X - 2.11 F$

(C) $-RT \ln \frac{1}{X} + 2.11 F$

(D) $-RT \ln X - 1.29 F$

EXERCISE-03**MISCELLANEOUS TYPE QUESTIONS****TRUE / FALSE**

- Compounds of active metals (Zn, Na, Mg) are reducible by H_2 whereas those of noble metals (Cu, Ag, Au) are not reducible.
- The mass of a substance deposited on the cathode or anode during electrolysis is given by $w = \frac{EIt}{F}$.
- Faraday's second law of electrolysis is related to the equivalent mass of the electrolyte.
- Equivalent conductance at infinite dilution of salt AB is equal to the sum of equivalent conductances of ions, A^+ and B^- at infinite dilution.
- The standard reaction potential of $Cl^-|AgCl|Ag$ half-cell is related to that of $Ag^+|Ag$ half-cell through the expression $E_{Ag^+|Ag} = E_{Cl^-|AgCl|Ag} + \frac{RT}{F} \ln K_{sp}(AgCl)$.
- The cell potential is given by $E_{cell} = E_{RP(cathode)} - E_{RP(anode)}$.
- In a galvanic cell, the half-cell with higher reduction potential acts as a reducing agent.
- In an electrode concentration cell, the cell reaction $Zn(c_1) \longrightarrow Zn(c_2)$ will be spontaneous if $c_1 > c_2$.
- The redox reaction involved in galvanic cell is a non-spontaneous process.

FILL IN THE BLANKS

- A dilute solution of sulphuric acid during electrolysis liberate gas at anode.
- Among metals is the strongest reducing agents in aqueous solutions.
- Lead is able to displace silver from $AgNO_3$ solution because its standard oxidation potential is than that of silver.
- Coulomb refers to of electricity while ampere refers to at which it flows.
- During electrolysis of aqueous solution of $CuSO_4$ using Pt electrodes the product at anode is
- The quantity of electricity required for complete reduction of 0.5 mole of MnO_4^- to Mn^{2+} is C.
- λ_{eq} normality =
- The presence of electrolytes the rate of corrosion.
- The more negative the standard potential, the is its ability to displace hydrogen from acids.
- In the calomel half-cell, the reduction reaction to be considered is

MATCH THE COLUMN

Match the items of column I to those of column I :

- | Column-I | | Column-II | |
|----------|----------------------------|-----------|--|
| (A) | Cell constant | (p) | $E_{cathode}^0 + E_{anode}^0$ |
| (B) | Anode | (q) | ℓ/A |
| (C) | Conductance | (r) | Mass of product deposited by 1 coulomb of electricity. |
| (D) | Electrochemical equivalent | (s) | $(Resistance)^{-1}$ |
| (E) | E_{cell}^0 | (u) | Involve oxidation |
- | Column-I | | Column-II | |
|----------|---------------------------------------|-----------|---|
| (A) | Cathode | (p) | Primary cell |
| (B) | 1 Coulomb | (q) | Secondary cell |
| (C) | Dry cell | (r) | 6.24×10^{18} electrones |
| (D) | Lead strong cell | (s) | Concentration cell |
| (E) | $Zn Zn^{2+}(0.01M) Zn^{2+}(0.1M) Zn$ | (u) | Positive terminal of electrochemical cell |

3.

Column-I		Column-II	
(A)	Electrolytic cell	(p)	$-\Delta G$
(B)	nFE_{Cell}^0	(q)	Concentration cell
(C)	$E_{\text{cell}} = \frac{0.059}{n} \log \frac{C_{\text{cathode}}}{C_{\text{anode}}}$	(r)	96500 Coulombs
(D)	Diffusion of ions	(s)	Device converting electrical energy into chemical energy
(E)	1 Faraday	(u)	Salt bridge

4.

Column-I		Column-II	
(A)	Conductance	(p)	Cm^{-1}
(B)	Specific conductance	(q)	$\text{Ohm}^{-1} \text{cm}^2 \text{mol}^{-1}$
(C)	Cell constant	(r)	Ohm^{-1}
(D)	Equivalent conductance	(s)	$\text{Ohm}^{-1} \text{cm}^{-1}$
(E)	Molar conductance	(u)	$\text{Ohm}^{-1} \text{cm}^2 \text{equivalent}^{-1}$

ASSERTION & REASON

These questions contains, Statement I (assertion) and Statement II (reason).

(A) Statement-I is true, Statement-II is true ; Statement-II is correct explanation for Statement-I.

(B) Statement-I is true, Statement-II is true ; Statement-II is NOT a correct explanation for statement-I

(C) Statement-I is true, Statement-II is false

(D) Statement-I is false, Statement-II is true

1. **Statement-I :** In electrolysis, the quantity of electricity needed for depositing 1 mole of silver is different from that required for 1 mole of copper.

Because

Statement-II : The molecular weights of silver and copper are different.

2. **Statement-I :** Equivalent conductance of all electrolytes decreases with increasing concentration.

Because

Statement-II : Lesser number of ions are available per gram equivalent at higher concentration.

3. **Statement-I :** If an aqueous solution of NaCl is electrolysed, the product obtained at the cathode is H_2 gas and not Na.

Because

Statement-II : Gases are liberated faster than the metals.

4. **Statement-I :** Molar conductivity of a weak electrolyte at infinite dilution cannot be determined experimentally.

Because

Statement-II : Kohlrausch law help to find the molar conductivity of a weak electrolyte at infinite dilution.

5. **Statement-I :** Gold chloride (AuCl_3) solution cannot be stored in a vessel made of copper, iron, nickel, chromium, zinc or tin.

Because

Statement-II : Gold is very precious metal.

6. **Statement-I** : In the Daniel cell, if concentration of Cu^{2+} and Zn^{2+} ions are doubled, the emf of the cell will not change.
Because
Statement-II : If the concentration of ions in contact with the metals is doubled, the electrode potential is doubled.
7. **Statement-I** : $\text{H}_2 + \text{O}_2$ fuel cell gives a constant voltage throughout its life.
Because
Statement-II : In this fuel cell, H_2 reacts with OH^- ions, yet the overall concentration of OH^- ions does not change.
8. **Statement-I** : Blocks of magnesium are often strapped to steel hulls of ocean going ships.
Because
Statement-II : Magnesium causes cathodic protection of iron.
9. **Statement-I** : Absolute value of E_{red}^0 of an electrode cannot be determined.
Because
Statement-II : Neither oxidation nor reduction can take place alone.

COMPREHENSION BASED QUESTIONS

Comprehension # 1

Copper reduces NO_3^- into NO and NO_2 depending upon conc. of HNO_3 in solution. Assuming $[\text{Cu}^{2+}] = 0.1 \text{ M}$, and $P_{\text{NO}} = P_{\text{NO}_2} = 10^{-3} \text{ atm}$ and using given data answer the following questions :

$$E_{\text{Cu}^{2+}/\text{Cu}} = +0.34 \text{ volt} ;$$

$$E_{\text{NO}_3^-/\text{NO}}^0 = +0.96 \text{ volt}$$

$$E_{\text{NO}_3^-/\text{NO}_2}^0 = +0.79 \text{ volt} ;$$

$$\text{at } 298 \text{ K } \frac{RT}{F} (2.303) = 0.06 \text{ volt}$$

1. E_{Cell} for reduction of $\text{NO}_3^- \longrightarrow \text{NO}$ by Cu(s), when $[\text{HNO}_3] = 1 \text{ M}$ is [At $T = 298$]
 (A) ~ 0.61 (B) ~ 0.71 (C) ~ 0.51 (D) ~ 0.81
2. At what HNO_3 concentration thermodynamic tendency for reduction of NO_3^- into NO and NO_2 by copper is same ?
 (A) $10^{1.23} \text{ M}$ (B) $10^{0.56} \text{ M}$ (C) $10^{0.66} \text{ M}$ (D) $10^{0.12} \text{ M}$

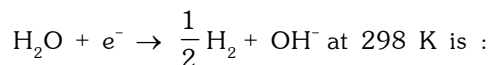
Comprehension # 2

Accidentally chewing on a stray fragment of aluminium foil can causes a sharp tooth pain if the aluminium comes in contact with an amalgam filling. The filling, an alloy of silver, tin and mercury, acts as the cathode of a tiny galvanic cell, the aluminium behaves as the anode, and saliva serves as the electrolyte. When the aluminium and the filling come in contact, an electric current passage from the aluminium to the filling which is sensed by a nerve in the tooth. Aluminium is oxidized at the anode, and O_2 gas is reduced to water at the cathode.

$$E_{\text{Al}^{3+}/\text{Al}}^0 = -1.66 \text{ V}, E_{\text{O}_2, \text{H}^+/\text{H}_2\text{O}}^0 = 1.23 \text{ V}$$

1. Net reaction taking place when amalgam is in contact with aluminium foil :
 (A) $\text{Al} + \text{O}_2 + \text{OH}^- \rightarrow \text{Al}(\text{OH}) + \text{H}_2\text{O}$
 (B) $4\text{Al} + 3\text{O}_2 + 12\text{H}^+ \rightarrow 4\text{Al}^{3+} + 6\text{H}_2\text{O}$
 (C) $4\text{Al} + 3\text{O}_2 \rightarrow 4\text{Al}_2\text{O}_3$
 (D) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
2. Standard E.M.F. experienced by the person with dental filling is :
 (A) $+2.89 \text{ V}$ (B) -2.89 V (C) -0.93 V (D) $+0.43 \text{ V}$

3. The standard reduction potential of the reaction,



(A) $E = \frac{RT}{2F} \ln K_w$

(B) $E = \frac{RT}{F} \ln [P_{\text{H}_2}]^{1/2} [\text{OH}^-]$

(C) $E = \frac{RT}{F} \ln \frac{[P_{\text{H}_2}]^{1/2}}{[\text{H}^+]}$

(D) $E = \frac{RT}{F} \ln K_w$

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
<u>True / False</u> 1. F 2. T 3. T 4. T 5. F 6. T 7. F 8. T 9. F <u>Fill in the Blanks</u> 1. O₂ 2. Lithium 3. Higher 4. Amount, rate 5. Oxygen 6. 2.5 96500 C 7. $\kappa \cdot 10^3$ 8. Increases 9. Greater 10. $\text{Hg}_2\text{Cl}_2(\text{s}) + 2e^- \rightarrow 2\text{Hg}(\ell) + 2\text{Cl}^-(\text{aq})$ <u>Match the Column</u> 1. A - (q), B - (u), C - (s), D - (r), E - (p) 2. A - (u), B - (r), C - (p), D - (q), E - (s) 3. A - (s), B - (p), C - (q), D - (u), E - (r) 4. A - (r), B - (s), C - (p), D - (u), E - (r) <u>Assertion - Reason Questions</u> 1. B 2. C 3. C 4. B 5. B 6. C 7. A 8. A 9. A <u>Comprehension Based Questions</u> Comprehension # 1: 1. (B) 2. (C) Comprehension # 2: 1. (A) 2. (C) 3. (A) Comprehension # 3: 1. (B) 2. (A) 3. (D)		

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

1. The resistance of a N/10 KCl solution is 245 ohms. Calculate the specific conductance and the equivalent conductance of the solution if the electrodes in the cell are 4 cm apart and each having an area of 7.0 sq. cm.
2. The resistance of a solution 'A' is 50 ohms and that of solution 'B' is 100 ohms, both solutions being taken in the same conductivity cell. If equal volumes of solution A and B are mixed, what will be the resistance of the mixture using the same cell. (Assume that there is no increase in the degree of dissociation of A and B on mixing.)
3. In a conductivity cell the two platinum electrodes, each of area 10 sq. cm. are fixed 1.5 cm apart. The cell contained 0.05 N solution of a salt. If the two electrodes are just half dipped into the solution which has a resistance of 50 ohms, find equivalent conductance of the salt solution.
4. A big irregular shaped vessel contained water, the sp. conductance of which was $2.56 \times 10^{-5} \text{ mho cm}^{-1}$. 500g of NaCl was then added to the water and the specific conductance after the addition of NaCl, was found to be $3.10 \times 10^{-5} \text{ mho cm}^{-1}$. Find the capacity of the vessel if it is fulfilled with water. ($\lambda_{\infty} \text{NaCl} = 149.9$)
5. The equivalent conductance of 0.10 N solution of MgCl_2 is $97.1 \text{ mho cm}^2 \text{ eq}^{-1}$. A cell with electrodes that are 1.50 cm^2 in surface area and 0.50 cm apart is filled with 0.1N MgCl_2 solution. How much current will flow when the potential difference between the electrodes is 5 volts?
6. At 18 C the mobilities of NH_4^+ and ClO_4^- ions are 6.6×10^{-4} and $5.7 \times 10^{-4} \text{ cm}^2 \text{ volt}^{-1} \text{ sec}^{-1}$ at infinite dilution. Calculate equivalent conductance of ammonium chlorate solution.
7. For H^+ and Na^+ the values of λ_{∞} are 349.8 and 50.11. Calculate the mobilities of these ions and their velocities if they are in a cell in which the electrodes are 5 cm apart and to which a potential of 2 volts is applied.
8. The equivalent conductance of an infinitely dilute solution NH_4Cl is 150 and the ionic conductances of OH^- and Cl^- ions are 198 and 76 respectively. What will be the equivalent conductance of the solution of NH_4OH at infinite dilution. If the equivalent conductance of a 0.01 N solution NH_4OH is 9.6, what will be its degree of dissociation?
9. Calculate the dissociation constant of water at 25 C from the following data.
Specific conductance of $\text{H}_2\text{O} = 5.8 \times 10^{-8} \text{ mho cm}^{-1}$, $\lambda_{\text{H}^+}^{\infty} = 350.0$ and $\lambda_{\text{OH}^-}^{\infty} = 198.0 \text{ mho cm}^2$
10. Calculate K_a of acetic acid if its 0.05 N solution has equivalent conductance of 7.36 mho cm^2 at 25 C.
($\lambda_{\text{CH}_3\text{COOH}}^{\infty} = 390.70$)
11. The sp. cond. of a saturated solution of AgCl at 25 C after subtracting the sp. conductance of conductivity of water is $2.28 \times 10^{-6} \text{ mho cm}^{-1}$. Find the solubility product of AgCl at 25 C. ($\lambda_{\text{AgCl}}^{\infty} = 138.3 \text{ mho cm}^2$)
12. The specific conductance of a N/10 KCl solution at 18 C is $1.12 \times 10^{-2} \text{ mho cm}^{-1}$. The resistance of the solution contained in the cell is found to be 65 ohms. Calculate the cell constant.
13. When a solution of conductance 1.342 mho m^{-1} was placed in a conductivity cell with parallel electrodes, the resistance was found to be 170.5 ohm. The area of the electrodes is $1.86 \times 10^{-4} \text{ sq. meter}$. Calculate the distance between the two electrodes in meter.

14. The resistance of two electrolytes X and Y were found to be 45 and 100 respectively when equal volumes of both the solutions were taken in the same cell in two different experiments. If equal volumes of these solutions are mixed in the same cell, what will be the conductance of the mixture?
15. The resistance of an aqueous solution containing 0.624 g of CuSO₄ · 5H₂O per 100 cm³ of the solution in a conductance cell of cell constant 153.7 per meter is 520 ohms at 298 K. Calculate the molar conductivity. (CuSO₄ · 5H₂O = 249.5)
16. Given the equivalent conductance of sodium butyrate, sodium chloride and hydrogen chloride as 83, 127 and 426 mho cm² at 25 C respectively. Calculate the equivalent conductance of butyric acid at infinite dilution.
17. For 0.0128 N solution of acetic acid at 25 C, equivalent conductance of the solution is 1.4 mho cm³ eq⁻¹ and λ[∞] = 391 mho cm² eq⁻¹. Calculate dissociation constant (K_a) of acetic acid.
18. The specific conductance at 25 C of a saturated solution of SrSO₄ is 1.482 × 10⁻⁴ ohm⁻¹ cm⁻¹ while that of water used is 1.5 × 10⁻⁶ mho cm⁻¹. Determine at 25 C the solubility in g per litre of SrSO₄ in water. Molar ionic conductance of Sr²⁺ and SO₄²⁻ ions at infinite dilution are 59.46 and 79.8 ohm⁻¹ cm² mole⁻¹ respectively. [Sr = 87.6, S = 32, O = 16]
19. Specific conductance of pure water at 25 C is 0.58 × 10⁻⁷ mho cm⁻¹. Calculate ionic product of water (K_w) if ionic conductances of H⁺ and OH⁻ ions at infinite dilution are 350 and 198 mho cm² respectively at 25 C.
20. How long a current of 3A has to be passed through a solution of AgNO₃, to coat a metal surface of 80 cm² with 5μm thick layer? Density of silver = 10.8 g/cm³.
21. 3A current was passed through an aqueous solution of an unknown salt of Pd for 1 hr. 2.977 g of Pdⁿ⁺ was deposited at cathode. Find n.
22. 50 mL of 0.1 M CuSO₄ solution is electrolysed with a current of 0.965 A for a period of 200 sec. The reactions at electrodes are :

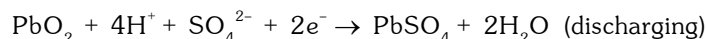
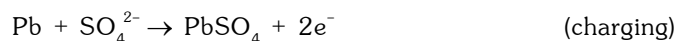
Cathode : Cu²⁺ + 2e⁻ → Cu(s) Anode : 2H₂O → O₂ + 4H⁺ + 4e⁻.

Assuming no change in volume during electrolysis, calculate the molar concentration of Cu²⁺, H⁺ and SO₄²⁻ at the end of electrolysis.
23. A metal is known to form fluoride MF₂. When 10 A of electricity is passed through a molten salt for 330 sec., 1.95 g of metal is deposited. Find the atomic weight of M. What will be the quantity electricity required to deposit the same mass of Cu from CuSO₄?
24. An electric current is passed through electrolytic cells in series one containing Ag(NO₃) (aq.) and other H₂SO₄ (aq.) What volume of O₂ measured at 25 C and 750 mm Hg pressure would be liberated from H₂SO₄ if

(a) one mole of Ag⁺ is deposited from AgNO₃ solution

(b) 8 × 10²² ions of Ag⁺ are deposited from AgNO₃ solution.
25. Cd amalgam is prepared by electrolysis of a solution CdCl₂ using a mercury cathode. How long should a current of 5A be passed in order to prepare 12 % Cd-Hg amalgam on a cathode of 2g, Hg (atomic weight of Cd = 112.4)
26. After electrolysis of NaCl solution with inert electrodes for a certain period of time, 600 mL of the solution was left. Which was found to be 1N in NaOH. During the same time, 31.75 g of Cu deposited in the copper voltameter in series with the electrolytic cell. Calculate the percentage yield of NaOH obtained.
27. During the discharge of a lead storage battery, the density of sulphuric acid fell from 1.294 to 1.139 g.mL⁻¹. H₂SO₄ of density 1.294 g.mL⁻¹ is 39% and that of density 1.139 g.mL⁻¹ is 20% by weight.

The battery holds 3.5 L of acid and the volume practically remains constant during discharge. Calculate the number of ampere hours for which the battery must have been used. The discharging and charging reactions are.

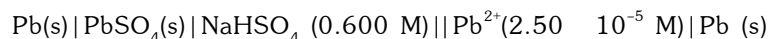


28. The e.m.f. of the cell obtained by combining Zn and Cu electrode of a Daniel cell with N calomel electrode in two different arrangements are 1.083 V and 0.018 V respectively at 25 C. If the standard reduction potential of N calomel electrode is 0.28 V find the emf of Daniel cell.
29. Same quantity of electricity is being used to liberate iodine (at anode) and a metal x (at cathode). The mass of x liberated is 0.617 g and the iodine liberated is completely reduced by 46.3 cc of 0.124 M sodium thiosulphate. Find the equivalent mass of metal x.
30. 100 mL CuSO_4 (aq) was electrolyzed using inert electrodes by passing 0.965 A till the pH of the resulting solution was 1. The solution after electrolysis was neutralized, treated with excess KI and titrated with 0.04 M $\text{Na}_2\text{S}_2\text{O}_3$. Volume of $\text{Na}_2\text{S}_2\text{O}_3$ required was 35 mL. Assuming no volume change during electrolysis, calculate :
 - (a) duration of electrolysis if current efficiency is 80 %
 - (b) initial concentration (M) of CuSO_4 .
31. A current of 3.7 A is passed for 6 hrs between Pt electrodes in 0.5 L of 2M solution of $\text{Ni}(\text{NO}_3)_2$. What will be the molarity of solution at the end of electrolysis ?
32. Calculate the EMF of a Daniel cell when the concentration of ZnSO_4 and CuSO_4 are 0.001 M and 0.1 M respectively. The standard potential of the cell is 1.1 V.
33. EMF of the cell $\text{Zn}|\text{ZnSO}_4(a = 0.2)||\text{ZnSO}_4(a_2)|\text{Zn}$ is -0.0088 V at 25 C. Calculate the value of a_2 .
34. The EMF of the cell $\text{M}|\text{M}^{n+}(0.02 \text{ M})||\text{H}^+(1 \text{ M})|\text{H}_2(\text{g}) (1 \text{ atm}), \text{Pt}$ at 25 C is 0.81 V. Calculate the valency of the metal if the standard oxidation potential of the metal is 0.76 V.
35. Equinormal solution of two weak acids, HA ($\text{pK}_a = 3$) and HB ($\text{pK}_a = 5$) are each placed in contact with standard hydrogen electrode at 25 C. When a cell is constructed by interconnecting them through a salt bridge find the e.m.f. of the cell.
36. In two vessels each containing 500 mL water, 0.5 m mol of aniline ($K_b = 10^{-9}$) and 25 m mol of HCl are added separately. Two hydrogen electrodes are constructed using these solutions. Calculate the emf of cell made by connecting them appropriately.
37. The emf of the cell $\text{Ag}|\text{AgI}|\text{KI} (0.05 \text{ M})||\text{AgNO}_3(0.05 \text{ M})|\text{Ag}$ is 0.788 V. Calculate the solubility product of AgI.
38. The cell $\text{Pt}, \text{H}_2 (1 \text{ atm})|\text{H}^+ (\text{pH} = x)|| \text{Normal calomel Electrode}$ has an EMF of 0.67 V at 25 C. Calculate the pH of the solution. The oxidation potential of the calomel electrode on hydrogen scale is -0.28 V.
39. Estimate the cell potential of a Daniel cell having 1.0 M Zn^{++} and originally having 1.0 M Cu^{++} after sufficient NH_3 has been added to the cathode compartment to make NH_3 concentration 2.0 M. Given K_f for $[\text{Cu}(\text{NH}_3)_4]^{2+} = 1 \times 10^{12}$, E for the reaction, $\text{Zn} + \text{Cu}^{2+} \longrightarrow \text{Zn}^{2+} + \text{Cu}$ is 1.1 V.
40. Consider the cell $\text{Ag}|\text{AgBr(s)}|\text{Br}^-||\text{AgCl(s)}|\text{Cl}^-|\text{Ag}$ at 25 C. The solubility product constants of AgBr & AgCl are respectively 5×10^{-13} & 1×10^{-10} . For what ratio of the concentration of Br^- & Cl^- ions would the em.f. of the cell be zero?
41. The pK_{sp} of AgI is 16.07. If the E value for Ag^+/Ag is 0.7991 V, find the E for the half cell reaction $\text{AgI (s)} + e^- \longrightarrow \text{Ag} + \text{I}^-$

42. Voltage of the cell $\text{Pt, H}_2 (1 \text{ atm}) | \text{HOCN} (1.3 \times 10^{-3} \text{ M}) || \text{Ag}^+ (0.8 \text{ M}) | \text{Ag(s)}$ is 0.982 V. Calculate the K_a for HOCN. Neglect $[\text{H}^+]$ because of oxidation of $\text{H}_2(\text{g})$.
 $\text{Ag}^+ + \text{e}^- \longrightarrow \text{Ag (s)} = 0.8 \text{ V}$
43. The standard oxidation potential of Zn referred to SHE is 0.76 V and that of Cu is -0.34 V at 25 C. When excess of Zn is added to CuSO_4 , Zn displaces Cu^{2+} till equilibrium is reached. What is the ratio of Zn^{2+} to Cu^{2+} ions at equilibrium?
44. The standard reduction potential values, $E (\text{Bi}^{3+} / \text{Bi})$ and $E (\text{Cu}^{2+} / \text{Cu})$ are 0.226 V and 0.344 V respectively. A mixture of salts of bismuth and copper at unit concentration each is electrolysed at 25 C. To what value can $[\text{Cu}^{2+}]$ be brought down before bismuth starts to deposit, in electrolysis.
45. Calculate the potential of an indicator electrode versus the standard hydrogen electrode, which originally contained 0.1 M MnO_4^- and 0.8 M H^+ and which was treated with Fe^{2+} necessary to reduce 90% of the MnO_4^- to Mn^{2+} .
 $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \longrightarrow \text{Mn}^{2+} + \text{H}_2\text{O}, E = 1.51 \text{ V},$
46. K_d for dissociation of $[\text{Ag}(\text{NH}_3)_2]^+$ into Ag^+ and NH_3 is 6×10^{-8} . Calculate E for the following half reaction;
 $\text{Ag} (\text{NH}_3)_2^+ + \text{e}^- \longrightarrow \text{Ag} + 2\text{NH}_3$
 Given $\text{Ag}^+ + \text{e}^- \longrightarrow \text{Ag}, E = 0.799 \text{ V}$
47. The overall formation constant for the reaction of 6 mole of CN^- with cobalt (II) is 1×10^{19} . The standard reduction potential for the reaction $[\text{Co}(\text{CN})_6]^{3-} + \text{e}^- \longrightarrow \text{Co}(\text{CN})_6^{4-}$ is -0.83 V. Calculate the formation constant of $[\text{Co}(\text{CN})_6]^{3-}$.
 Given $\text{Co}^{3+} + \text{e}^- \longrightarrow \text{Co}^{2+}; E = 1.82 \text{ V}$
48. Calculate the e.m.f. of the cell
 $\text{Pt} | \text{H}_2 (1.0 \text{ atm}) | \text{CH}_3\text{COOH} (0.1 \text{ M}) || \text{NH}_3 (\text{aq}, 0.01 \text{ M}) | \text{H}_2 (1.0 \text{ atm}) | \text{Pt}$
 $K_a (\text{CH}_3\text{COOH}) = 1.8 \times 10^{-5}, K_b, (\text{NH}_3) = 1.8 \times 10^{-5}.$
49. An external current source giving a current of 5.0 A was joined with Daniel cell arrangement opposing the normal current flow and was removed after 10 hrs. Before passing the current the LHE and RHE contained 1L each of 1M Zn^{2+} and Cu^{2+} respectively. Find the EMF supplied by the Daniel cell after removal of the external current source. E of Zn^{2+}/Zn and Cu^{2+}/Cu at 25 C is -0.76 and +0.34 V respectively.
50. Calculate the equilibrium constant for the reaction :
 $3\text{Sn (s)} + 2\text{Cr}_2\text{O}_7^{2-} + 28 \text{H}^+ \longrightarrow 3 \text{Sn}^{+4} + 4\text{Cr}^{3+} + 14\text{H}_2\text{O}$
 E for $\text{Sn}/\text{Sn}^{2+} = 0.136 \text{ V}$ E for $\text{Sn}^{2+}/\text{Sn}^{4+} = -0.154 \text{ V}$
 E for $\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+} = 1.33 \text{ V}$
51. Calculate the voltage E, of the cell at 25 C
 $\text{Mn(s)} | \text{Mn(OH)}_2(\text{s}) | \text{Mn}^{2+}(\text{x M}), \text{OH}^- (1.00 \times 10^{-4} \text{ M}) || \text{Cu}^{2+} (0.675 \text{ M}) | \text{Cu(s)}$
 given that $K_{sp} = 1.9 \times 10^{-13}$ for $\text{Mn(OH)}_2 (\text{s})$ E (Mn^{2+}/Mn) = -1.18 V, E (Cu^{+2}/Cu) = +0.34 V
52. Calculate the voltage E, of the cell
 $\text{Ag(s)} | \text{AgIO}_3(\text{s}) | \text{Ag}^+ (\text{x M}), \text{HIO}_3 (0.300 \text{ M}) || \text{Zn}^{2+} (0.175 \text{ M}) | \text{Zn (s)}$

if $K_{sp} = 3.02 \times 10^{-8}$ for $\text{AgIO}_3(\text{s})$ and $K_a = 0.162$ for HIO_3 , $E(\text{Zn}^{+2}/\text{Zn}) = -0.76 \text{ V}$, $E(\text{Ag}/\text{Ag}^+) = -0.8 \text{ V}$

53. The voltage of the cell



is $E = +0.061 \text{ V}$. Calculates $K_2 = [\text{H}^+][\text{SO}_4^{2-}]/[\text{HSO}_4^-]$, the dissociation constant for HSO_4^- .

Given $\text{Pb(s)} + \text{SO}_4^{2-} \rightarrow \text{PbSO}_4 + 2\text{e}^-$ ($E = 0.356 \text{ V}$), $E(\text{Pb}^{2+}/\text{Pb}) = -0.126 \text{ V}$.

54. The voltage of the cell

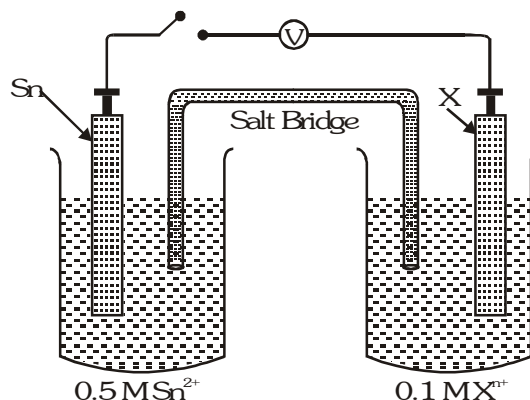
$\text{Zn(s)} | \text{Zn(CN)}_4^{2-} (0.450 \text{ M}), \text{CN}^- (2.65 \times 10^{-3} \text{ M}) || \text{Zn}^{2+} (3.84 \times 10^{-4} \text{ M}) | \text{Zn(s)}$ is $E = +0.099 \text{ V}$.

Calculate the constant K_f for $\text{Zn}^{2+} + 4\text{CN}^- \rightarrow \text{Zn(CN)}_4^{2-}$, the only $\text{Zn}^{2+} + \text{CN}^-$ complex reaction of importance.

CONCEPTUAL SUBJECTIVE EXERCISE		ANSWER KEY	EXERCISE-4(A)	
1.	$2.332 \times 10^{-3} \text{ mho cm}^{-1}$, $23.32 \text{ mho cm}^2 \text{ eq}^{-1}$	2.	$R = 66.67 \text{ ohms}$	
4.	$V = 2.34 \times 10^8 \text{ cc}$	5.	0.1456 amp	
7.	$\mu_{\text{H}^+} = 3.62 \times 10^{-3} \text{ cm}^2 \text{ volt}^{-1} \text{ sec}^{-1}$, velocity of $\text{H}^+ = 1.45 \times 10^{-3} \text{ cm sec}^{-1}$,	$\mu_{\text{Na}} = 5.20 \times 10^{-4} \text{ cm}^2 \text{ volt}^{-1} \text{ sec}^{-1}$ velocity of $\text{Na}^+ = 2.08 \times 10^{-4} \text{ cm sec}^{-1}$	3.	$120 \text{ mho cm}^2 \text{ eq}^{-1}$
8.	272, 0.0353	9.	$1.8 \times 10^{-16} \text{ mole/litre}$	
11.	$2.70 \times 10^{-10} (\text{mole/litre})^2$	12.	0.728 cm^{-1}	
14.	0.016 mho	15.	$118.2 \text{ mho cm}^2 \text{ mol}^{-1}$	
17.	1.6×10^{-7}	18.	0.1934 g/L	
20.	$t = 128.66 \text{ sec}$	21.	$n = 4$	
22.	$\text{Cu}^{2+} = 0.08\text{M}$, $\text{H}^+ = 0.04\text{M}$, $\text{SO}_4^{2-} = 0.1\text{M}$	23.	$A=114$, $Q=5926.8\text{C}$	
24.	(a) $V(\text{O}_2)=6.2 \text{ L}$, (b) $V(\text{O}_2)=0.824\text{L}$	25.	$t = 93.65 \text{ sec}$	
27.	265 Amp. hr.	26.	$\eta = 60\%$	
30.	1250 S, 0.064 M	28.	$E = 1.1 \text{ V}$	
33.	$a_2 = 0.1006 \text{ M}$	29.	$\text{Eq.wt} = 107.468$	
36.	$E = 0.395 \text{ V}$	31.	1.172 M	
39.	$E = 0.71 \text{ V}$	32.	$E = 1.159 \text{ V}$	
42.	$K_a = 6.74 \times 10^{-4}$	34.	$n = 2$	
45.	1.39 V	35.	$E = 0.059 \text{ V}$	
48.	-0.46 V	37.	$K_{\text{sp}} = 1.16 \times 10^{-16}$	
51.	1.66 V	38.	$\text{pH} = 6.61$	
54.	5.4×10^{16}	40.	$[\text{Br}^-] : [\text{Cl}^-] = 1 : 200$	
		41.	$E = -0.149 \text{ V}$	
		43.	$[\text{Zn}^{2+}]/[\text{Cu}^{2+}]=1.941 \times 10^{37}$	
		44.	$[\text{Cu}^{2+}] = 10^{-4} \text{ M}$	
		46.	0.373 V	
		47.	$K_f = 8.28 \times 10^{63}$	
		49.	1.143 V	
		50.	$K = 10^{268}$	
		52.	-1.188 V	
		53.	1.06×10^{-2}	

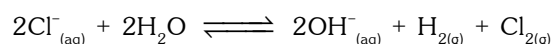
EXERCISE-04 [B]**BRAIN STORMING SUBJECTIVE EXERCISE**

1. The e.m.f. of cell : $\text{H}_2(\text{g})|\text{Buffer}||\text{Normal calomal electrode}$ is 0.6885 V at 40 C when the barometric pressure is 725 mm of Hg. What is the pH of the solution. $E_{\text{calomal}} = 0.28$.
2. A direct current of 1.25 A was passed through 200 mL of 0.25 M $\text{Fe}_2(\text{SO}_4)_3$ solution for a period of 1.1 hour. The resulting solution in cathode chamber was analyzed by titrating against acidic KMnO_4 solution. 25 mL permanganate solution was required to reach the end point. Determine molarity of KMnO_4 solution.
3. An electrochemical cell is constructed with an open switch as shown below :



When the switch is closed, mass of tin-electrode increases. If $E(\text{Sn}^{2+} / \text{Sn}) = -0.14 \text{ V}$ and for $E(\text{X}^{n+} / \text{X}) = -0.78 \text{ V}$ and initial emf of the cell is 0.65 V, determine n and indicate the direction of electron flow in the external circuit.

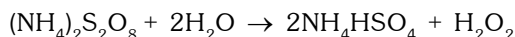
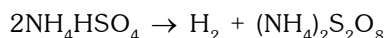
4. 10 g fairly concentrated solution of CuSO_4 is electrolyzed using 0.01 F of electricity. Calculate :
 - (a) The weight of resulting solution
 - (b) Equivalents of acid or alkali in the solution.
5. One of the methods of preparation of per disulphuric acid, $\text{H}_2\text{S}_2\text{O}_8$, involve electrolytic oxidation of H_2SO_4 at anode ($2\text{H}_2\text{SO}_4 \longrightarrow \text{H}_2\text{S}_2\text{O}_8 + 2\text{H}^+ + 2\text{e}^-$) with oxygen and hydrogen as by - products. In such an electrolysis, 9.722 L of H_2 and 2.35 L of O_2 were generated at STP. What is the weight of $\text{H}_2\text{S}_2\text{O}_8$ formed?
6. Assume that impure copper contains only iron, silver and a gold as impurities. After passage of 140 A, for 482.5 sec, of the mass of the anode decreased by 22.260 g and the cathode increased in mass by 22.011 g. Estimate the % iron and % copper originally present.
7. For the galvanic cell : $\text{Ag}|\text{AgCl}(\text{s})|\text{KCl}(0.2 \text{ M})||\text{KBr}(0.001 \text{ M})|\text{AgBr}(\text{s})|\text{Ag}$,
Calculate the EMF generated and assign correct polarity to each electrode for a spontaneous process after taking into account the cell reaction at 25 C.
 $[K_{\text{sp}}(\text{AgCl}) = 2.8 \times 10^{-10}; K_{\text{sp}}(\text{AgBr}) = 3.3 \times 10^{-13}]$
8. An aqueous solution of NaCl on electrolysis gives $\text{H}_{2(\text{g})}$, $\text{Cl}_{2(\text{g})}$ and NaOH according to the reaction :



A direct current of 25 amperes with a current efficiency of 62% is passed through 20 liters of NaCl solution (20% by weight). Write down the reactions taking place at the anode and the cathode. How long will it take to produce 1 kg of Cl_2 ? What will be the molarity of the solution with respect to hydroxide ion? (Assume no loss due to evaporation).

9. An acidic solution of Cu^{2+} salt containing 0.4 of Cu^{2+} is electrolyzed until all the copper is deposited. The electrolysis is continued for seven more minutes with the volume of solution kept at 100 mL and the current at 1.2 amp. Calculate the volume of gases evolved at NTP during the entire electrolysis.
10. In the refining of silver by electrolytic method what will be the weight of 100 g Ag anode if 5 ampere current is passed for 2 hours? Purity of silver is 95% by weight.

11. Hydrogen peroxide can be prepared by successive reactions :



The first reaction is an electrolytic reaction the second is steam distillation. What amount of current would have to be used in first reaction to produce enough intermediate to yield 100 g pure H_2O_2 per hour? Assume 50% anode current efficiency.

12. Dal lake has water 8.2×10^{12} litre approximately. A power reactor produces electricity at the rate of 1.5×10^6 coulomb per second at an appropriate voltage. How many years would it take to electrolyse the lake?
13. Determine the degree of hydrolysis and hydrolysis constant of aniline hydrochloride in M/32 solution of salt at 298 K from the following cell data at 298 K.



14. The emf of the cell, $\text{Pt}|\text{H}_2(1 \text{ atm})|\text{H}^+(0.1 \text{ M}, 30 \text{ mL})|| \text{Ag}^+(0.8 \text{ M}) \text{ Ag}$ is 0.9 V. Calculate the emf when 40 mL of 0.05 M NaOH is added.
15. A dilute aqueous solution of KCl was placed between two electrodes 10 cm apart, across which a potential of 6 volt was applied. How far would the K^+ ion move in 2 hours at 25 C? Ionic conductance of K^+ ion at infinite dilution at 25 C is $73.52 \text{ ohm}^{-1} \text{ cm}^2 \text{ mole}^{-1}$?
16. 100 mL $\text{CuSO}_4(\text{aq})$ was electrolyzed using inert electrodes by passing 0.965 A till the pH of the resulting solution was 1. The solution after electrolysis was neutralized, treated with excess KI and titrated with 0.04 M $\text{Na}_2\text{S}_2\text{O}_3$. Volume of $\text{Na}_2\text{S}_2\text{O}_3$ required was 35 mL. Assuming no volume change during electrolysis, calculate :

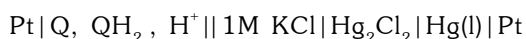
(a) duration of electrolysis if current efficiency is 80%

(b) initial concentration (M) of CuSO_4

17. Calculate the equilibrium concentration of all ions in an ideal solution prepared by mixing 25.00 mL of 0.100 M Ti^+ with 25.00 mL of 0.200 M Co^{3+} .

$$E(\text{Ti}^+/\text{Ti}^{3+}) = -1.25 \text{ V}, E(\text{Co}^{3+}/\text{Co}^{2+}) = 1.84 \text{ V}$$

18. Determine at 298 for cell :



(a) it's emf when pH = 5.0

(b) the pH when $E_{\text{cell}} = 0$

(c) the positive electrode when pH = 7.5

$$\text{given } E_{\text{RP(RHS)}} = 0.28, E_{\text{RP(LHS)}} = 0.699$$

19. At 25 C, $\Delta H_f(\text{H}_2\text{O}, \text{l}) = -56700 \text{ J/mol}$ and energy of ionization of $\text{H}_2\text{O}(\text{l}) = 19050 \text{ J/mol}$. What will be the reversible EMF at 25 C of the cell,
 $\text{Pt}|\text{H}_2(\text{g})(1 \text{ atm})|\text{H}^+||\text{OH}^-|\text{O}_2(\text{g})(1 \text{ atm})|\text{Pt}$, if at 26 C the emf increase by 0.001158 V.
20. Calculate the cell potential of a cell having reaction $\text{Ag}_2\text{S} + 2\text{e}^- \rightleftharpoons 2 \text{Ag} + \text{S}^{2-}$ in a solution buffered at $\text{pH} = 3$ and which is also saturated with 0.1 M H_2S .
 For H_2S : $K_1 = 10^{-8}$ and $K_2 = 1.1 \cdot 10^{-13}$, $K_{\text{sp}}(\text{Ag}_2\text{S}) = 2 \cdot 10^{-49}$, $E_{\text{Ag}^+/\text{Ag}} = 0.8$.
21. Calculate the solubility and solubility product of $\text{Co}_2[\text{Fe}(\text{CN})_6]$ in water at 25 C from the following data :
 Conductivity of a saturated solution of $\text{Co}_2[\text{Fe}(\text{CN})_6]$ is $2.06 \cdot 10^{-6} \Omega^{-1}\text{cm}^{-1}$ and that of water used $4.1 \cdot 10^{-7} \Omega^{-1}\text{cm}^{-1}$. The ionic molar conductivities of Co^{2+} and $\text{Fe}(\text{CN})_6^{4-}$ are $86.0 \Omega^{-1}\text{cm}^2\text{mol}^{-1}$ and $444.0 \Omega^{-1}\text{cm}^2\text{mol}^{-1}$.
22. A sample of water from a large swimming pool has a resistance of 9200Ω at 25 C when placed in a certain conductance cell. When filled with 0.02 M KCl solution, the cell has a resistance of 85Ω at 25 C. 500 g of NaCl were dissolved in the pool, which was thoroughly stirred. A sample of this solution gave a resistance of 7600Ω . Calculate the volume of water in the pool.
 Given : Molar conductance of NaCl at that concentrations is $126.5 \Omega^{-1}\text{cm}^2\text{mol}^{-1}$ and molar conductivity of KCl at 0.02 M is $138 \Omega^{-1}\text{cm}^2\text{mol}^{-1}$.

BRAIN STORMING SUBJECTIVE EXERCISE	ANSWER KEY	EXERCISE-4(B)
1. 6.6	2. 0.41 M	
3. $n = 3$ & X-electrode to Sn-electrode	4. final weight = 9.6 g, 0.01 eq. of acid	
5. 43.45 g	6. Cu = 98.88%, Fe = 0.85%	
7. - 0.037 V	8. 48.71 hour, $[\text{OH}^-] = 1.41 \text{ M}$	
9. $V(\text{O}_2) = 99.68 \text{ mL}$, $V(\text{H}_2) = 58.46 \text{ mL}$, Total vol. = 158.1 mL		
10. 57.5894 g	11. 315.36 A	
12. 1.9 million year	13. $h = 2.12 \cdot 10^{-2}$, $K_h = 1.43 \cdot 10^{-5} \text{ M}$	
14. 0.95 V	15. 3.29 cm	
16. 1250 s, 0.064 M	17. $\text{Ti}^+ = 10^{-8}$, $\text{Co}^{+3} = 2 \cdot 10^{-8}$	
18. (a) -0.124V (b) 7.1 (c) calomel electrode		
19. 0.4414 V	20. -0.167 V	
21. $K_{\text{sp}} = 7.682 \cdot 10^{-17}$	22. $2 \cdot 10^5 \text{ dm}^3$	

EXERCISE - 05 [A]**JEE-[MAINS] : PREVIOUS YEAR QUESTIONS**

1. For a cell reaction involving a two-electron change, the standard e.m.f. of the cell is found to be 0.295V at 25 °C. The equilibrium constant of the reaction at 25 °C will be : [AIEEE-2003]
(A) 10 (B) 1×10^{10} (C) 1×10^{-10} (D) 29.5×10^{-2}
2. Standard reduction electrode potentials of three metals A, B and C are respectively + 0.5V, - 3.0V and -1.2 V. The reducing powers of these metals are : [AIEEE-2003]
(A) $C > B > A$ (B) $A > C > B$ (C) $B > C > A$ (D) $A > B > C$
3. For the redox reaction :
 $\text{Zn(s)} + \text{Cu}^{2+} (0.1\text{M}) \rightarrow \text{Zn}^{2+} (1\text{M}) + \text{Cu(s)}$ taking place in a cell,
 E_{cell} is 1.10 volt. E_{cell} for the cell will be $\left(2.303 \frac{RT}{F} = 0.0591 \right)$ [AIEEE-2003]
(A) 1.07 volt (B) 0.82 volt (C) 2.14 volt (D) 1.80 volt
4. When during electrolysis of a solution of AgNO_3 9650 coulombs of charge pass through the electroplating bath, the mass of silver deposited on the cathode will be : [AIEEE-2003]
(A) 21.6g (B) 108g (C) 1.08g (D) 10.8g
5. Consider the following E° values [AIEEE-2004]
 $E^\circ_{\text{Fe}^{3+}/\text{Fe}^{2+}} = + 0.77\text{V}$ $E^\circ_{\text{Sn}^{2+}/\text{Sn}} = - 0.14\text{V}$
Under standard conditions the potential for the reaction
 $\text{Sn(s)} + 2\text{Fe}^{3+} (\text{aq}) \longrightarrow 2\text{Fe}^{2+}(\text{aq}) + \text{Sn}^{2+}(\text{aq})$ is
(A) 0.91V (B) 1.40V (C) 1.68V (D) 0.63V
6. The standard e.m.f. of a cell, involving one electron change is found to be 0.591 V at 25 °C. The equilibrium constant of the reaction is ($F = 96,500\text{C mol}^{-1}$; $R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$) [AIEEE-2004]
(A) 1.0×10^{10} (B) 1.0×10^5 (C) 1.0×10^1 (D) 1.0×10^{30}
7. The limiting molar conductivities Λ° for NaCl, KBr and KCl are 126, 152 and $150\text{S cm}^2 \text{ mol}^{-1}$ respectively. The Λ° for NaBr is : [AIEEE-2004]
(A) $278 \text{ S cm}^2 \text{ mol}^{-1}$ (B) $176 \text{ S cm}^2 \text{ mol}^{-1}$ (C) $128 \text{ S cm}^2 \text{ mol}^{-1}$ (D) $302 \text{ S cm}^2 \text{ mol}^{-1}$
8. In a cell that utilises the reaction
 $\text{Zn(s)} + 2\text{H}^+(\text{aq.}) \longrightarrow \text{Zn}^{2+}(\text{aq}) + \text{H}_2(\text{g})$ addition of H_2SO_4 to cathode compartment, will [AIEEE-2004]
(A) increase the E and shift equilibrium to the right (B) lower the E and shift equilibrium to the right
(C) lower the E and shift equilibrium to the left (D) increase the E and shift equilibrium to the left
9. The $E^\circ_{\text{M}^{3+}/\text{M}^{2+}}$ values for Cr, Mn, Fe and Co are -0.41, + 1.57, +0.77 and + 1.97V respectively. For which one of these metals the change in oxidation state from +2 to +3 is easiest ? [AIEEE-2004]
(A) Fe (B) Mn (C) Cr (D) Co

10. For a spontaneous reaction the ΔG , equilibrium constant (K) and E_{Cell}^0 will be respectively [AIEEE-2005]
 (A) -ve, < 1, -ve (B) -ve, >1, -ve (C) -ve, >1, +ve (D) +ve, >1, -ve
11. Aluminium oxide may be electrolysed at 1000 C to furnish aluminium metal (At. Mass = 27 amu; 1 Faraday = 96500 Coulombs). The cathode reaction is [AIEEE-2005]
 $\text{Al}^{3+} + 3e^- \longrightarrow \text{Al}^0$
 To prepare 5.12 kg of aluminium metal by this method would require.
 (A) 5.49 10^4 C of electric charge (B) 5.49 10^1 C of electric charge
 (C) 5.49 10^7 C of electric charge (D) 1.83 10^7 C of electric charge
12. The highest electrical conductivity of the following aqueous solution is of [AIEEE-2005]
 (A) 0.1 M fluoroacetic acid (B) 0.1 M difluoroacetic acid
 (C) 0.1 M acetic acid (D) 0.1 M chloroacetic acid
13. Electrolyte $\Lambda^\infty (\text{S cm}^2 \text{ mol}^{-1})$
 KCl 149.9
 KNO_3 145.0
 HCl 426.2
 NaOAc 91.0
 NaCl 126.5
 Calculate $\Lambda_{\text{HOAc}}^\infty$ using appropriate molar conductances of the electrolytes listed above at infinite dilution in H_2O at 25 C [AIEEE-2005]
 (A) 390.7 (B) 217.5 (C) 517.2 (D) 552.7
14. Given the data at 25 C,
 $\text{Ag} + \text{I}^- \rightarrow \text{AgI} + e^-$, $E = 0.152\text{V}$
 $\text{Ag} \rightarrow \text{Ag}^+ + e^-$, $E = -0.800\text{V}$
 What is the value of $\log K_{\text{sp}}$ for AgI ? $\left(2.303 \frac{RT}{F} = 0.059\text{V} \right)$ [AIEEE 2006]
 (A) -8.12 (B) +8.612 (C) -37.83 (D) -16.13
15. Resistance of a conductivity cell filled with a solution of an electrolyte of concentration 0.1M is 100Ω . The conductivity of this solution is 1.29 Sm^{-1} . Resistance of the same cell when filled with 0.02M of the same solution is 520Ω . The molar conductivity of 0.02M solution of the electrolyte will be. [AIEEE-2006]
 (A) $124 \cdot 10^{-4} \text{ Sm}^2 \text{ mol}^{-1}$ (B) $1240 \cdot 10^{-4} \text{ Sm}^2 \text{ mol}^{-1}$
 (C) $1.24 \cdot 10^4 \text{ Sm}^2 \text{ mol}^{-1}$ (D) $12.4 \cdot 10^{-4} \text{ Sm}^2 \text{ mol}^{-1}$
16. The molar conductivities Λ_{NaOAc}^0 and Λ_{HCl}^0 at infinite dilution in water at 25 C are 91.0 and $426.2 \text{ S cm}^2/\text{mol}$ respectively. To calculate Λ_{HOAc}^0 the additional value required is : [AIEEE-2006]
 (A) KCl (B) NaOH (C) NaCl (D) H_2O
17. Given $E_{\text{Cr}^{3+}/\text{Cr}}^0 = -0.72\text{V}$, $E_{\text{Fe}^{2+}/\text{Fe}}^0 = -0.42\text{V}$. The potential for the cell [AIEEE-2008]
 $\text{Cr} | \text{Cr}^{3+} (0.1 \text{ M}) || \text{Fe}^{2+} (0.01 \text{ M}) | \text{Fe}$ is
 (A) 0.26 V (B) 0.339 V (C) -0.339 V (D) -0.26 V

18. Given :

[AIEEE-2009]

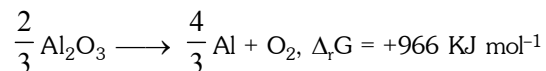
$$E_{\text{Fe}^{3+}/\text{Fe}}^0 = -0.036\text{V}, E_{\text{Fe}^{2+}/\text{Fe}}^0 = -0.439\text{V},$$

The value of standard electrode potential for the change, $\text{Fe}_{(\text{aq})}^{3+} + e^- \rightarrow \text{Fe}_{(\text{aq})}^{2+}$ will be :-

- (A) 0.770 V (B) -0.270 V (C) -0.072 V (D) 0.385 V

19. The Gibbs energy for the decomposition of Al_2O_3 at 500 °C is as follows :

[AIEEE-2010]



The potential difference needed for electrolytic reduction of Al_2O_3 at 500 °C is at least :-

- (A) 5.0 V (B) 4.5 V (C) 3.0 V (D) 2.5 V

20. The correct order of $E_{M^{2+}/M}^{\circ}$ values with negative sign for the four successive elements Cr, Mn, Fe and Co

is :-

[AIEEE-2010]

- (A) $\text{Cr} > \text{Mn} > \text{Fe} > \text{Co}$ (B) $\text{Mn} > \text{Cr} > \text{Fe} > \text{Co}$
(C) $\text{Cr} > \text{Fe} > \text{Mn} > \text{Co}$ (D) $\text{Fe} > \text{Mn} > \text{Cr} > \text{Co}$

21. The reduction potential of hydrogen half-cell will be negative if :-

[AIEEE-2011]

- (A) $p(\text{H}_2) = 2 \text{ atm}$ $[\text{H}^+] = 1.0 \text{ M}$ (B) $p(\text{H}_2) = 2 \text{ atm}$ and $[\text{H}^+] = 2.0 \text{ M}$
(C) $p(\text{H}_2) = 1 \text{ atm}$ and $[\text{H}^+] = 2.0 \text{ M}$ (D) $p(\text{H}_2) = 1 \text{ atm}$ and $[\text{H}^+] = 1.0 \text{ M}$

22. Resistance of 0.2 M solution of an electrolyte is $50\ \Omega$. The specific conductance of the solution is $1.3\ \text{S m}^{-1}$. If resistance of the 0.4M solution of the same electrolyte is $260\ \Omega$, its molar conductivity is :-

[AIEEE-2011]

- (A) $6250 \text{ S m}^2 \text{ mol}^{-1}$ (B) $6.25 \times 10^{-4} \text{ S m}^2 \text{ mol}^{-1}$
(C) $625 \times 10^{-4} \text{ S m}^2 \text{ mol}^{-1}$ (D) $62.5 \text{ S m}^2 \text{ mol}^{-1}$

23. The standard reduction potentials for $\text{Zn}^{2+} / \text{Zn}$, $\text{Ni}^{2+} / \text{Ni}$ and $\text{Fe}^{2+} / \text{Fe}$ are -0.76 , -0.23 and -0.44 V respectively. The reaction $\text{X} + \text{Y}^{2+} \rightarrow \text{X}^{2+} + \text{Y}$ will be spontaneous when **[AIEEE-2012]**

[AIEEE-2012]

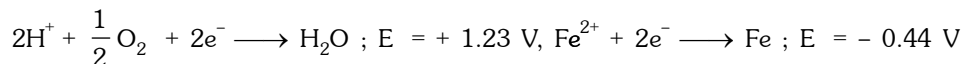
- (A) X = Zn, Y = Ni
(B) X = Ni, Y = Fe
(C) X = Ni, Y = Zn
(D) X = Fe, Y = Zn

JEE-[MAIN] : PREVIOUS YEAR QUESTIONS							ANSWER KEY			EXERCISE -5[A]					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans	B	C	A	D	A	A	C	A	C	C	C	B	A	D	A
Que.	16	17	18	19	20	21	22	23							
Ans	C	A	A	D	B	A	B	A							

EXERCISE - 05 [B]**JEE-[ADVANCED] : PREVIOUS YEAR QUESTIONS**

1. The standard reduction potentials of $\text{Cu}^{2+} / \text{Cu}$ and $\text{Cu}^{2+} / \text{Cu}^+$ are 0.337 and 0.153 V respectively. The standard electrode potential of Cu^+ / Cu half cell is :- [JEE 1997]
(A) 0.184 V (B) 0.827 V (C) 0.521 V (D) 0.490 V
2. A standard hydrogen electrode has zero electrode potential because :- [JEE 1997]
(A) hydrogen is easier to oxidise (B) this electrode potential is assumed to be zero
(C) hydrogen atom has only one electron (D) hydrogen is the lowest element
3. The standard reduction potential values of the three metallic cations X, Y, Z are 0.52, -3.03, and -1.18V respectively. The order of reducing power of the corresponding metals is :- [JEE 1998]
(A) $Y > Z > X$ (B) $X > Y > Z$ (C) $Z > Y > X$ (D) $Z > X > Y$
4. A gas X at 1 atm is bubbled through a solution containing a mixture of 1M Y and 1M Z at 25 C. If the reduction potential of $Z > Y > X$, then :- [JEE 1999]
(A) Y will oxidise X and not Z (B) Y will oxidise Z and not X
(C) Y will oxidise both X and Z (D) Y will reduce both X and Z
5. For the electrochemical cell, $\text{M} | \text{M}^+ || \text{X}^- | \text{X}$, $E^\circ(\text{M}^+/\text{M}) = 0.44 \text{ V}$ and $E^\circ(\text{X}/\text{X}^-) = 0.33 \text{ V}$. From this data, one can deduce that :- [JEE 2000]
(A) $\text{M} + \text{X} \longrightarrow \text{M}^+ + \text{X}^-$ is the spontaneous reaction
(B) $\text{M}^+ + \text{X}^- \longrightarrow \text{M} + \text{X}$ is the spontaneous reaction
(C) $E_{\text{cell}} = 0.77 \text{ V}$
(D) $E_{\text{cell}} = -0.77 \text{ V}$
6. The reaction,
 $3 \text{ClO}^- (\text{aq}) \longrightarrow \text{ClO}_3^- (\text{aq}) + 2 \text{Cl}^- (\text{aq})$
is an example of :- [JEE 2001]
(A) Oxidation reaction (B) Reduction reaction
(C) Disproportionation reaction (D) Decomposition reaction
7. The correct order of equivalent conductance at infinite dilution of LiCl, NaCl and KCl is:-[JEE 2001]
(A) $\text{LiCl} > \text{NaCl} > \text{KCl}$ (B) $\text{KCl} > \text{NaCl} > \text{LiCl}$
(C) $\text{NaCl} > \text{KCl} > \text{LiCl}$ (D) $\text{LiCl} > \text{KCl} > \text{NaCl}$
8. Saturated solution of KNO_3 is used to make salt bridge because :- [JEE 2001]
(A) velocity of K^+ is greater than that of NO_3^- (B) velocity of NO_3^- is greater than that of K^+
(C) velocities of both K^+ and NO_3^- are nearly the same (D) KNO_3 is highly soluble in water
9. Standard electrode potential data are useful for understanding the suitability of an oxidant in a redox titration. Some half cell reactions and their standard potentials are given below :- [JEE 2002]
 $\text{MnO}_4^- (\text{aq}) + 8\text{H}^+ (\text{aq}) + 5\text{e}^- \longrightarrow \text{Mn}^{2+} (\text{aq}) + 4\text{H}_2\text{O} (\text{l}) ; E^\circ = 1.51 \text{ V}$
 $\text{CrO}_7^{2-} (\text{aq}) + 14\text{H}^+ (\text{aq}) + 6\text{e}^- \longrightarrow 2\text{Cr}^{3+} (\text{aq}) + 7\text{H}_2\text{O} (\text{l}) ; E^\circ = 1.38 \text{ V}$
 $\text{Fe}^{3+} (\text{aq}) + \text{e}^- \longrightarrow \text{Fe}^{2+} (\text{aq}) ; E^\circ = 0.77 \text{ V}$
 $\text{Cl}_2 (\text{g}) + 2\text{e}^- \longrightarrow 2\text{Cl}^- (\text{aq}) ; E^\circ = 1.40 \text{ V}$
Identify the only incorrect statement regarding quantitative estimation of aqueous $\text{Fe}(\text{NO}_3)_2$.
(A) MnO_4^- can be used in aqueous HCl (B) $\text{Cr}_2\text{O}_7^{2-}$ can be used in aqueous HCl
(C) MnO_4^- can be used in aqueous H_2SO_4 (D) $\text{Cr}_2\text{O}_7^{2-}$ can be used in aqueous H_2SO_4

10. In the electrolytic cell, flow of electrons is from :- [JEE 2003]
 (A) Cathode to anode in solution
 (B) Cathode to anode through external supply
 (C) Cathode to anode through internal supply
 (D) Anode to cathode through internal supply
11. $\text{Zn}|\text{Zn}^{2+}(\text{a} = 0.1 \text{ M})||\text{Fe}^{2+}(\text{a} = 0.01 \text{ M})|\text{Fe}$. The emf of the above cell is 0.2905 V. Equilibrium constant for the cell reaction is :- [JEE 2004]
 (A) $10^{0.32/0.591}$ (B) $10^{0.32/0.0295}$ (C) $10^{0.26/0.0295}$ (D) $e^{0.32/0.295}$
12. The half cell reactions for rusting of iron are : [JEE 2000]

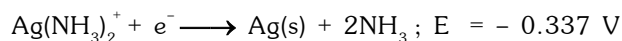
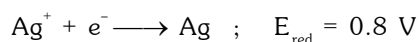


ΔG (in kJ) for the reaction is :

- (A) -76 (B) -322 (C) -122 (D) -176

Question No. 13 to 15 (3 question)

Tollen's reagent is used for the detection of aldehyde when a solution of AgNO_3 is added to glucose with NH_4OH then gluconic acid is formed.



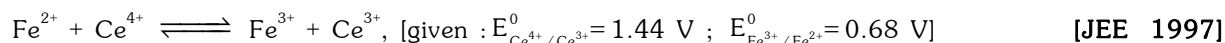
[Use $2.303 \frac{RT}{F} = 0.0592$ and $\frac{F}{RT} = 38.92$ at 298 K]

[JEE 2006]

13. $2\text{Ag}^+ + \text{C}_6\text{H}_{12}\text{O}_6 + \text{H}_2\text{O} \longrightarrow 2\text{Ag(s)} + \text{C}_6\text{H}_{12}\text{O}_7 + 2\text{H}^+$
 Find $\ell n K$ of this reaction
 (A) 66.13 (B) 58.38 (C) 28.30 (D) 46.29
14. When ammonia is added to the solution, pH is raised to 11. Which half-cell reaction is affected by pH and by how much ?
 (A) E_{oxd} will increase by a factor of 0.65 from E_{oxd}
 (B) E_{oxd} will decrease by a factor of 0.65 from E_{oxd}
 (C) E_{red} will increase by a factor of 0.65 from E_{red}
 (D) E_{red} will decrease by a factor of 0.65 from E_{red}
15. Ammonia is always added in this reaction. Which of the following must be incorrect?
 (A) NH_3 combines with Ag^+ to form a complex
 (B) $\text{Ag}(\text{NH}_3)_2^+$ is a weaker oxidising reagent than Ag^+
 (C) In absence of NH_3 silver salt of gluconic acid is formed
 (D) NH_3 has affected the standard reduction potential of glucose / gluconic acid electrode
16. Electrolysis of a solution of MnSO_4 in aqueous sulphuric acid is a method for the preparation of MnO_2 as per the reaction, $\text{Mn}_{\text{aq}}^{2+} + 2\text{H}_2\text{O} \rightarrow \text{MnO}_2(\text{s}) + 2\text{H}_{\text{aq}}^+ + \text{H}_2(\text{g})$
 Passing a current of 27 A for 24 hours gives one kg of MnO_2 . What is the value of current efficiency?
 Write the reaction taking place at the cathode and at the anode. [JEE 1997]

17. How many grams of silver could be plated out on a serving tray by electrolysis of a solution containing silver in +1 oxidation state for a period of 8.0 hours at a current of 8.46 Amperes? What is the area of the tray if the thickness of the silver plating is 0.00254 cm? Density of silver is 10.5 g/cm³ :- [JEE 1997]

18. Calculate the equilibrium constant for the reaction



19. Calculate the equilibrium constant for the reaction, $2\text{Fe}^{3+} + 3\text{I}^- \rightleftharpoons 2\text{Fe}^{2+} + \text{I}_3^-$. The standard reduction potentials in acidic condition are 0.77 and 0.54 V respectively for $\text{Fe}^{3+}/\text{Fe}^{2+}$ and I_3^-/I^- couples :- [JEE 1998]

20. Find the solubility product of a saturated solution of Ag_2CrO_4 in water at 298 K if the emf of the cell $\text{Ag}|\text{Ag}^+(\text{satd. Ag}_2\text{CrO}_4 \text{ soln})||\text{Ag}^+(0.1 \text{ M})|\text{Ag}$ is 0.164 V at 298 K :- [JEE 1998]

21. Copper sulphate solution (250 mL) was electrolysed using a platinum anode and a copper cathode. A constant current of 2 mA was passed for 16 minute it was found that after electrolysis, the concentration of the solution was reduced to 50% of its original value. Calculate the concentration of copper sulphate in the solution to begin with. [JEE 2000]

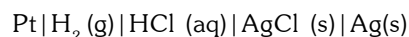
22. The following electrochemical cell has been set up :- [JEE 2000]



$$E_{\text{Fe}^{3+}/\text{Fe}^{2+}} = 0.77 \text{ V and } E_{\text{Ce}^{4+}/\text{Ce}^{3+}} = 1.61 \text{ V}$$

If an ammeter is connected between the two platinum electrodes. Predict the direction of flow of current. Will the current increase or decrease with time?

23. The standard potential of the following cell is 0.23 V at 15 C C & 0.21 V at 35 C.



(i) Write the cell reaction.

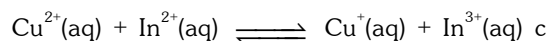
(ii) Calculate ΔH^0 , ΔS^0 for the cell reaction by assuming that these quantities remain unchanged in the range 15 C to 35 C.

(iii) Calculate the solubility of AgCl in water at 25 C. Given standard reduction potential of the Ag^+/Ag couples is 0.80 V at 25 C. [JEE 2001]

24. Two students use same stock solution of ZnSO_4 and a solution of CuSO_4 . The e.m.f of one cell is 0.03 V higher than the other. The conc. of CuSO_4 in the cell with higher e.m.f value is 0.5 M. Find out the

conc. of CuSO_4 in the other cell $\left(\frac{2.303RT}{F} = 0.06\right)$:- [JEE 2003]

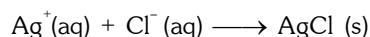
25. Find the equilibrium constant at 298 K for the reaction,



Given that :

$$E_{\text{Cu}^{2+}|\text{Cu}^+}^0 = 0.15 \text{ V}, E_{\text{In}^{3+}|\text{In}^+}^0 = -0.42 \text{ V}, E_{\text{In}^{2+}|\text{In}^+}^0 = -0.40 \text{ V} \quad [\text{JEE 2000}]$$

26. (a) Calculate ΔG_f^0 of the following reaction



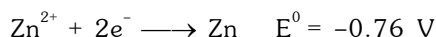
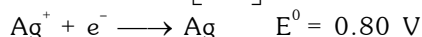
$$\text{Given} : \Delta G_f^0(\text{AgCl}) = -109 \text{ kJ/mole}, \Delta G_f^0(\text{Cl}^-) = -129 \text{ kJ/mole}, \Delta G_f^0(\text{Ag}^+) = 77 \text{ kJ/mole}$$

Represent the above reaction in form of a cell

Calculate E^0 of the cell. Find $\log_{10} K_{\text{SP}}$ of AgCl

(b) 6.593×10^{-2} g of metallic Zn (amu = 65.39) was added to 100 mL of saturated solution of AgCl.

Calculate $\log_{10} \frac{[Zn^{2+}]}{[Ag^+]^2}$, given that



Also find how many moles of Ag will be formed ?

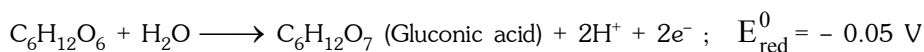
[JEE 2005]

27. We have taken a saturated solution of AgBr, K_{sp} of AgBr is 12×10^{-14} . If 10^{-7} mole of $AgNO_3$ are added to 1 litre of this solution find conductivity (specific conductance) of this solution in terms of $10^{-7} \text{ S m}^{-1} \text{ mol}^{-1}$.

Given : $\lambda_{(Ag^+)}^0 = 6 \times 10^{-3} \text{ S m}^2 \text{ mol}^{-1}$; $\lambda_{(Br^-)}^0 = 8 \times 10^{-3} \text{ S m}^2 \text{ mol}^{-1}$; $\lambda_{(NO_3^-)}^0 = 7 \times 10^{-3} \text{ S m}^2 \text{ mol}^{-1}$:-[JEE 2006]

Question No. 28 to 30 (3 questions)

Tollen's reagent is used for the detection of aldehyde when a solution of $AgNO_3$ is added to glucose with NH_4OH then gluconic acid is formed



[Use $2.303 \frac{RT}{F} = 0.0592$ and $\frac{F}{RT} = 38.92$ at 298 K]

[JEE 2006]

28. $2Ag^+ + C_6H_{12}O_6 + H_2O \longrightarrow 2Ag(s) + C_6H_{12}O_7 + 2H^+$

Find $\ln K$ of this reaction

- (A) 66.13 (B) 58.38 (C) 28.30 (D) 46.29

29. When ammonia is added to the solution, pH is raised to 11. Which half-cell reaction is affected by pH and by how much?

(A) E_{oxd} will increase by a factor of 0.65 from E_{oxd}^0

(B) E_{oxd} will decrease by a factor of 0.65 from E_{oxd}^0

(C) E_{red} will increase by a factor of 0.65 from E_{red}^0

(D) E_{red} will decrease by a factor of 0.65 from E_{red}^0

30. Ammonia is always added in this reaction. Which of the following must be incorrect?

(A) NH_3 combines with Ag^+ to form a complex.

(B) $Ag(NH_3)_2^+$ is a weaker oxidising reagent than Ag^+ .

(C) In absence of NH_3 silver salt of gluconic acid is formed.

(D) NH_3 has affected the standard reduction potential of glucose/gluconic acid electrode.

Paragraph for Question Nos. 31 to 33 (3 questions)

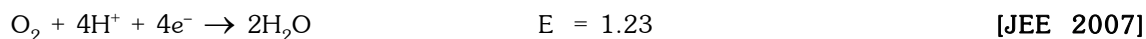
Chemical reactions involve interaction of atoms and molecules. A large number of atoms/molecules (approximately 6.023×10^{23}) are present in a few grams of any chemical compound varying with their atomic/molecular masses. To handle such large numbers conveniently, the mole concept was introduced. This concept has implications in diverse areas such as analytical chemistry, biochemistry, electrochemistry and radiochemistry. The following example illustrates a typical case, involving chemical/ electrochemical reaction, which requires a clear understanding of the mole concept.

A 4.0 molar aqueous solution of NaCl is prepared and 500 mL of this solution is electrolysed. This leads to the evolution of chlorine gas at one of the electrodes (atomic mass : Na = 23, Hg = 200; 1 Faraday = 96500 coulombs) [JEE 2007]

31. The total number of moles of chlorine gas evolved is
(A) 0.5 (B) 1.0 (C) 2.0 (D) 3.0
32. If the cathode is a Hg electrode, the maximum weight (g) of amalgam formed from this solution is
(A) 200 (B) 225 (C) 400 (D) 446
33. The total charge (coulombs) required for complete electrolysis is
(A) 24125 (B) 48250 (C) 96500 (D) 193000

Paragraph for Question Nos. 34 & 35 (2 questions)

Redox reactions play a pivoted role in chemistry and biology. The values of standard redox potential (E) of two half-cell reactions decide which way the reaction is expected to proceed. A simple example is a Daniel cell in which zinc goes into solution and copper gets deposited. Given below are a set of half-cell reactions (acidic medium) along with their E (V with respect to normal hydrogen electrode) values. Using this data obtain the correct explanations to Questions 14-16.



34. Among the following, identify the correct statement.
(A) Chloride ion is oxidised by O_2 (B) Fe^{2+} is oxidised by iodine
(C) Iodine ion is oxidised by chlorine (D) Mn^{2+} is oxidised by chlorine
35. While Fe^{3+} is stable, Mn^{3+} is not stable in acid solution because
(A) O_2 oxidises Mn^{2+} to Mn^{3+} (B) O_2 oxidises both Mn^{2+} to Mn^{3+} and Fe^{2+} to Fe^{3+}
(C) Fe^{3+} oxidises H_2O to O_2 (D) Mn^{3+} oxidises H_2O to O_2
36. Electrolysis of dilute aqueous NaCl solution was carried out by passing 10 milli ampere current. The time required to liberate 0.01 mol of H_2 gas at the cathode is (1 Faraday = 96500 C mol^{-1})
(A) $9.65 \times 10^4 \text{ sec}$ (B) $19.3 \times 10^4 \text{ sec}$
(C) $28.95 \times 10^4 \text{ sec}$ (D) $38.6 \times 10^4 \text{ sec}$ [JEE 2008]
37. For the reaction of NO_3^- ion in an aqueous solution, E is +0.96 V. Values of E for some metal ions are given below



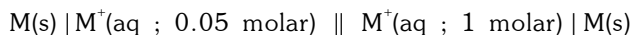


The pair(s) of metal that is(are) oxidised by NO_3^- in aqueous solution is(are) [JEE 2009]

- (A) V and Hg (B) Hg and Fe (C) Fe and Au (D) Fe and V

Paragraph for Questions 38 to 39

The concentration of potassium ions inside a biological cell is at least twenty times higher than the outside. The resulting potential difference across the cell is important in several processes such as transmission of nerve impulses and maintaining the ion balance. A simple model for such a concentration cell involving a metal M is : [JEE 2010]



For the above electrolytic cell the magnitude of the cell potential $|E_{\text{cell}}| = 70 \text{ mV}$.

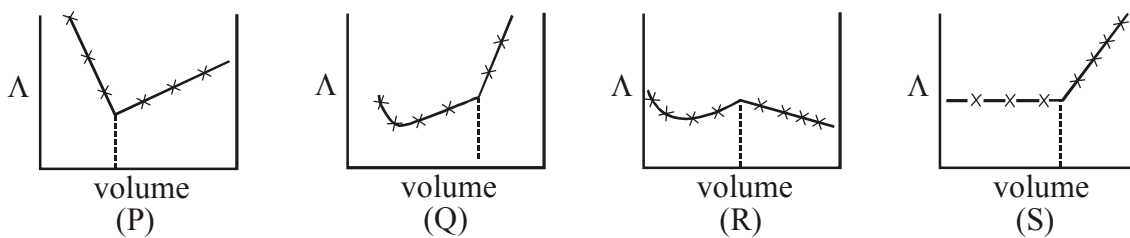
38. For the above cell :-

- (A) $E_{\text{cell}} < 0$; $\Delta G > 0$ (B) $E_{\text{cell}} > 0$; $\Delta G < 0$
 (C) $E_{\text{cell}} < 0$; $\Delta G^0 > 0$ (D) $E_{\text{cell}} > 0$; $\Delta G^0 < 0$

39. If the 0.05 molar solution of M^+ is replaced by a 0.0025 molar M^+ solution, then the magnitude of the cell potential would be :-

- (A) 35 mV (B) 70 mV (C) 140 mV (D) 700 mV

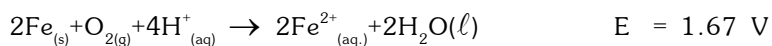
40. $\text{AgNO}_3(\text{aq.})$ was added to an aqueous KCl solution gradually and the conductivity of the solution was measured. the plot of conductance (Λ) versus the volume of AgNO_3 is - [JEE 2011]



- (A) (P) (B) (Q) (C) (R) (D) (S)

41. Consider the following cell reaction :

[JEE 2011]



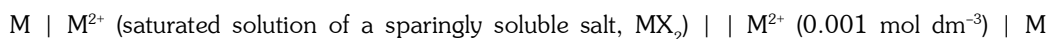
At $[\text{Fe}^{2+}] = 10^{-3} \text{ M}$, $P(\text{O}_2) = 0.1 \text{ atm}$ and $\text{pH} = 3$, the cell potential at 25°C is -

- (A) 1.47 V (B) 1.77 V (C) 1.87 V (D) 1.57 V

Paragraph for Question 42 & 43

The electrochemical cell shown below is a concentration cell.

[JEE 2012]



The emf of the cell depends on the difference in concentrations of M^{2+} ions at the two electrodes. The emf of the cell at 298 K is 0.059 V .

EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. Which of the following equations represents a reaction that provides the heat of formation of ethane (CH_3CH_3)?
- (A) $2 \text{ C(s)} + 6 \text{ H(g)} \rightarrow \text{CH}_3\text{CH}_3(\text{g})$ (B) $2 \text{ C(s)} + 3 \text{ H}_2(\text{g}) \rightarrow \text{CH}_3\text{CH}_3(\text{g})$
(C) $\text{CH}_2 = \text{CH}_2(\text{g}) + 2 \text{ H}_2(\text{g}) \rightarrow \text{CH}_3\text{CH}_3(\text{g})$ (D) $\text{CH}^-\text{CH}(\text{g}) + 2 \text{ H}_2\text{O(g)} \rightarrow \text{CH}_3\text{CH}_3(\text{g}) + \text{O}_2(\text{g})$
2. Which of the following equations represents a reaction that provides the heat of formation of CH_3Cl ?
- (A) $\text{C(s)} + \text{HCl(g)} + \text{H}_2(\text{g}) \rightarrow \text{CH}_3\text{Cl(g)}$
(B) $\text{C(s)} + 3/2 \text{ H}_2(\text{g}) + 1/2 \text{ Cl}_2(\text{g}) \rightarrow \text{CH}_3\text{Cl(g)}$
(C) $\text{C(s)} + 3 \text{ H(g)} + \text{Cl(g)} \rightarrow \text{CH}_3\text{Cl(g)}$
(D) $\text{CH}_4(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow \text{CH}_3\text{Cl(g)} + \text{HCl(g)}$
3. Use the given standard enthalpies of formation to determine the heat of reaction of the following reaction:
- $\text{TiCl}_4(\text{g}) + 2 \text{ H}_2\text{O(g)} \rightarrow \text{TiO}_2(\text{g}) + 4 \text{ HCl(g)}$
 $\Delta H_f^\circ \text{ TiCl}_4(\text{g}) = -763.2 \text{ kJ/mole}$
 $\Delta H_f^\circ \text{ TiO}_2(\text{g}) = -944.7 \text{ kJ/mole}$
 $\Delta H_f^\circ \text{ H}_2\text{O(g)} = -241.8 \text{ kJ/mole}$
 $\Delta H_f^\circ \text{ HCl(g)} = -92.3 \text{ kJ/mole}$
(A) - 278.1 (B) + 369.2 (C) + 67.1 (D) - 67.1
4. The heats of formation of $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O(l)}$ are -394 kJ/mole and -285.8 kJ/mole respectively. Using the data for the following combustion reaction, calculate the heat of formation of $\text{C}_3\text{H}_8(\text{g})$.
- $\text{C}_3\text{H}_8(\text{g}) + 5 \text{ O}_2(\text{g}) \rightarrow 3 \text{ CO}_2(\text{g}) + 4 \text{ H}_2\text{O(l)}$
 $\Delta H^\circ = -2221.6 \text{ kJ}$
(A) 212.2 (B) - 143.3 (C) 185.4 (D) - 103.6
5. The heats of formation of $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O(l)}$ are -394 kJ/mole and -285.8 kJ/mole respectively. Using the data for the following combustion reaction, calculate the heat of formation of $\text{C}_2\text{H}_2(\text{g})$.
- $2 \text{ C}_2\text{H}_2(\text{g}) + 5 \text{ O}_2(\text{g}) \rightarrow 4 \text{ CO}_2(\text{g}) + 2 \text{ H}_2\text{O(l)}$
 $\Delta H^\circ = - 2601 \text{ kJ}$
(A) - 238.6 (B) 253.2 (C) 238.7 (D) 226.7
6. Using the following information calculate the heat of formation of NaOH .
- $2 \text{ Na(s)} + 2 \text{ H}_2\text{O(l)} \rightarrow 2 \text{ NaOH(s)} + \text{H}_2(\text{g})$
 $\Delta H^\circ = - 281.9 \text{ kJ}$
 $\Delta H_f^\circ \text{ H}_2\text{O(l)} = -285.8 \text{ kJ/mole}$
(A) - 141.6 (B) - 712.6 (C) - 426.8 (D) - 650.4
7. Using the following information calculate the heat of formation of CH_4 .
- $\text{CH}_4(\text{g}) + 2 \text{ O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2 \text{ H}_2\text{O(l)}$
 $\Delta H^\circ = - 890.4 \text{ kJ}$
 $\Delta H_f^\circ \text{ CO}_2(\text{g}) = - 393.5 \text{ kJ/mole}$
 $\Delta H_f^\circ \text{ H}_2\text{O(l)} = - 285.9 \text{ kJ/mole}$
(A) - 98.6 (B) - 65.5 (C) - 74.9 (D) - 43.5

8. The heat of formation of $\text{CO}_2(\text{g})$ is -394 kJ/mole and that of $\text{H}_2\text{O}(\text{l})$ is -286 kJ/mole . The heat of combustion of C_2H_4 is -1412 kJ/mole . What is the heat of formation of C_2H_4 ?
- $$\text{C}_2\text{H}_4(\text{g}) + 3 \text{O}_2(\text{g}) \rightarrow 2 \text{CO}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{l})$$
- (A) $+1360$ (B) -108 (C) $+52$ (D) $+108$
9. What amount of heat energy (kJ) is released in the combustion of 12.0 g of C_3H_4 ?
(Atomic weights: C = 12.01 , H = 1.008 , O = 16.00).
- $$\text{C}_3\text{H}_4(\text{g}) + 4 \text{O}_2(\text{g}) \rightarrow 3 \text{CO}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{l})$$
- $$\Delta H^\circ = -1939.1 \text{ kJ}$$
- (A) 725 (B) 504 (C) 783 (D) 581
10. The standard heat of combustion of solid boron is equal to :-
- (A) $\Delta H_f(\text{B}_2\text{O}_3)$ (B) $\frac{1}{2} \Delta H_f(\text{B}_2\text{O}_3)$ (C) $2\Delta H_f(\text{B}_2\text{O}_3)$ (D) $\frac{1}{2} \Delta H_f(\text{B}_2\text{O}_3)$
11. In the reaction, $\text{CO}_2(\text{g}) + \text{H}_2(\text{g}) \longrightarrow \text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g})$; $\Delta H = 2.8 \text{ kJ}$ ΔH represents :-
- (A) heat of reaction (B) heat of combustion (C) heat of formation (D) heat of solution
12. A gas mixture 3.67 L in volume contain C_2H_4 and CH_4 is proportion of $2 : 1$ by moles and is at 25°C and 1 atm . If the $\Delta H_c(\text{C}_2\text{H}_4)$ and $\Delta H_c(\text{CH}_4)$ are -1400 and -900 kJ/mol find heat evolved on burning this mixture :-
- (A) 20.91 kJ (B) 50.88 kJ (C) 185 kJ (D) 160 kJ
13. The bond dissociation energy of gaseous H_2 , Cl_2 and HCl are 104 , 58 and $103 \text{ kcal mol}^{-1}$ respectively. The enthalpy of formation for HCl gas will be :-
- (A) -44.0 kcal (B) -22.0 kcal (C) 22.0 kcal (D) 44.0 kcal
14. The average, S - F bond energy in SF_6 if the ΔH_f value are -1100 , $+275$ and $+80 \text{ kJ/mol}$ respectively for $\text{SF}_6(\text{g})$, $\text{S}(\text{g})$ and $\text{F}(\text{g})$ is :-
- (A) 390.1 kJ/mol (B) 103.9 kJ/mol (C) 903.1 kJ/mol (D) 309.1 kJ/mol
15. The average O-H bond energy in H_2O with the help of following data :-
- (1) $\text{H}_2\text{O}_{(\text{l})} \longrightarrow \text{H}_2\text{O}_{(\text{g})}$; $\Delta H = +40.6 \text{ kJ mol}^{-1}$
 (2) $2\text{H}_{(\text{g})} \longrightarrow \text{H}_{2(\text{g})}$; $\Delta H = -435.0 \text{ kJ mol}^{-1}$
 (3) $\text{O}_{2(\text{g})} \longrightarrow 2\text{O}_{(\text{g})}$; $\Delta H = +489.6 \text{ kJ mol}^{-1}$
 (4) $2\text{H}_{2(\text{g})} + \text{O}_{2(\text{g})} \longrightarrow 2\text{H}_2\text{O}(\text{l})$; $\Delta H = -571.6 \text{ kJ mol}^{-1}$
- (A) $584.9 \text{ kJ mol}^{-1}$ (B) $279.8 \text{ kJ mol}^{-1}$ (C) $462.5 \text{ kJ mol}^{-1}$ (D) 925 kJ mol^{-1}
16. The enthalpy change for the following reaction is 514 kJ . Calculate the average Cl - F bond energy.
- $$\text{ClF}_3(\text{g}) \rightarrow \text{Cl}(\text{g}) + 3 \text{F}(\text{g})$$
- (A) 1542 (B) 88 (C) 171 (D) 514
17. Given the following equations and ΔH° values, determine the heat of reaction (kJ) at 298 K for the reaction:
- $$4 \text{H}_2\text{O}(\text{g}) + 3 \text{Fe}(\text{s}) \rightarrow \text{Fe}_3\text{O}_4(\text{s}) + 4 \text{H}_2(\text{g})$$
- $$\text{H}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{g}) \quad \Delta H^\circ/\text{kJ} = -285.8$$
- $$3 \text{FeO}(\text{s}) + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{Fe}_3\text{O}_4(\text{s}) \quad \Delta H^\circ/\text{kJ} = -302.4$$
- $$\text{FeO}(\text{s}) + \text{H}_2(\text{g}) \rightarrow \text{Fe}(\text{s}) + \text{H}_2\text{O}(\text{g}) \quad \Delta H^\circ/\text{kJ} = -13.8$$
- (A) $+391.7$ (B) -24.8 (C) -602.0 (D) $+24.8$

CHECK YOUR GRASP						ANSWER KEY				EXERCISE -1					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	B	B	D	D	D	C	C	C	D	B	A	C	B	D	C
Que.	16	17	18	19	20	21	22	23	24						
Ans.	C	D	D	B	A	A	A	D	B						

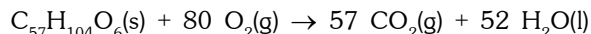
EXERCISE-02**BRAIN TEASERS****SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)**

1. The standard enthalpy of formation of propene, C_3H_6 , is +20.6 kJ/mole. Calculate the heat of combustion of one mole of C_3H_6 . The heats of formation of $CO_2(g)$ and $H_2O(l)$ are -394 kJ/mole and -285.8 kJ/mole respectively.

(A) 1721.2 (B) -1939.1 (C) 2060.0 (D) 2221.6

2. The fat, glyceryl trioleate, is metabolized via the following reaction. Given the enthalpies of formation, calculate the energy (kJ) liberated when 1.00 g of this fat reacts.

(Atomic weights: C = 12.01, H = 1.008, O = 16.00).



$$\Delta H^\circ C_{57}H_{104}O_6 = -70870 \text{ kJ/mole}$$

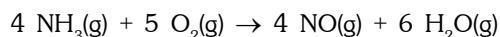
$$\Delta H^\circ H_2O(l) = -285.8 \text{ kJ/mole}$$

$$\Delta H^\circ CO_2(g) = -393.5 \text{ kJ/mole}$$

(A) 40.4 (B) 33.4 (C) 37.8 (D) 42.6

3. Using the enthalpies of formation, calculate the energy (kJ) released when 3.00 g of $NH_3(g)$ reacts according to the following equation.

(Atomic weights: N = 14.01, O = 16.00, H = 1.008).



$$\Delta H^\circ NH_3(g) = -46.1 \text{ kJ/mole}$$

$$\Delta H^\circ NO(g) = +90.2 \text{ kJ/mole}$$

$$\Delta H^\circ H_2O(g) = -241.8 \text{ kJ/mole}$$

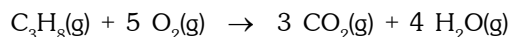
(A) 34.3 (B) 30.8 (C) 39.9 (D) 42.6

4. A sheet of 15.0 g of gold at 25.0°C is placed on a 30.0 g sheet of copper at 45.0°C. What is the final temperature of the two metals assuming that no heat is lost to the surroundings. The specific heats of gold and copper are 0.129 J/g°C and 0.385 J/g°C respectively.

(A) 42.1 (B) 40.1

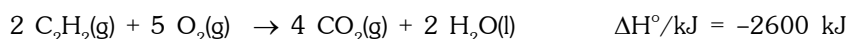
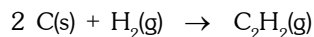
(C) 41.1 (D) 43.4

5. Calculate the heat of combustion (kJ) of propane, C_3H_8 using the listed standard enthalpy of reaction data:



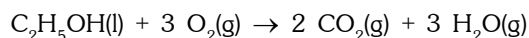
(A) -2043.9 (B) -1532.9 (C) -1021.9 (D) -739.1

6. Calculate the value of $\Delta H^\circ/kJ$ for the following reaction using the listed thermochemical equations:



(A) +184 (B) +214 (C) +202 (D) +234

7. Use the given standard enthalpies of formation to determine the heat of reaction of the following reaction:



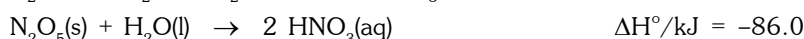
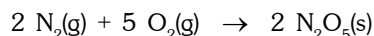
$$\Delta H_f^\circ \text{C}_2\text{H}_5\text{OH}(\text{l}) = -277.7 \text{ kJ/mole}$$

$$\Delta H_f^\circ \text{CO}_2(\text{g}) = -393.5 \text{ kJ/mole}$$

$$\Delta H_f^\circ \text{H}_2\text{O}(\text{g}) = -241.8 \text{ kJ/mole}$$

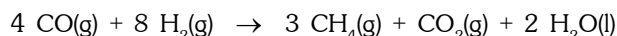
$$(A) -1456.3 \quad (B) -1234.7 \quad (C) -1034.0 \quad (D) -1119.4$$

8. Calculate $\Delta H^\circ/\text{kJ}$ for the following reaction using the listed standard enthalpy of reaction data:



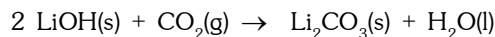
$$(A) -84.4 \quad (B) -243.6 \quad (C) -71.2 \quad (D) -121.8$$

9. Determine $\Delta H^\circ/\text{kJ}$ for the following reaction using the listed enthalpies of reaction:



$$(A) -622.4 \quad (B) -686.2 \quad (C) -747.5 \quad (D) -653.5$$

10. Use the given standard enthalpies of formation to determine the heat of reaction of the following reaction :



$$\Delta H_f^\circ \text{LiOH}(\text{s}) = -487.23 \text{ kJ/mole}$$

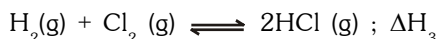
$$\Delta H_f^\circ \text{Li}_2\text{CO}_3(\text{s}) = -1215.6 \text{ kJ/mole}$$

$$\Delta H_f^\circ \text{H}_2\text{O}(\text{l}) = -285.85 \text{ kJ/mole}$$

$$\Delta H_f^\circ \text{CO}_2(\text{g}) = -393.5 \text{ kJ/mole}$$

$$(A) +303.4 \quad (B) -133.5 \quad (C) -198.6 \quad (D) +198.6$$

11. $\text{NH}_3(\text{g}) + 3\text{Cl}_2(\text{g}) \rightleftharpoons \text{NCl}_3(\text{g}) + 3\text{HCl}(\text{g}) ; -\Delta H_1$

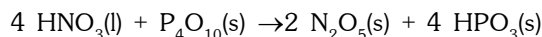


The heat of formation of $\text{NCl}_3(\text{g})$ in the terms of ΔH_1 , ΔH_2 and ΔH_3 is ?

$$(A) \Delta H_f = -\Delta H_1 + \frac{\Delta H_2}{2} - \frac{3}{2} \Delta H_3 \quad (B) \Delta H_f = \Delta H_1 + \frac{\Delta H_2}{2} - \frac{3}{2} \Delta H_3$$

$$(C) \Delta H_f = \Delta H_1 - \frac{\Delta H_2}{2} - \frac{3}{2} \Delta H_3 \quad (D) \text{None}$$

12. Determine ΔH° of the following reaction using the listed heats of formation :



$$\Delta H_f^\circ \text{HNO}_3(\text{l}) = -174.1 \text{ kJ/mole}$$

$$\Delta H_f^\circ \text{N}_2\text{O}_5(\text{s}) = -43.1 \text{ kJ/mole}$$

$$\Delta H_f^\circ \text{P}_4\text{O}_{10}(\text{s}) = -2984.0 \text{ kJ/mole}$$

$$\Delta H_f^\circ \text{HPO}_3(\text{s}) = -948.5 \text{ kJ/mole}$$

$$(A) -176.3 \quad (B) -199.8 \quad (C) +276.2 \quad (D) -242.4$$

13. If x_1 , x_2 and x_3 are enthalpies of H-H, O=O and O-H bonds respectively, and x_4 is the enthalpy of vaporisation of water, estimate the standard enthalpy of combustion of hydrogen.

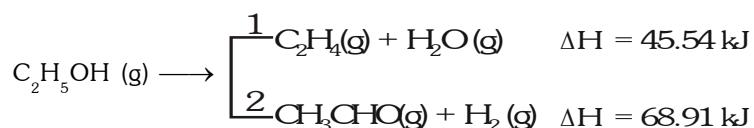
$$(A) x_1 + \frac{x_2}{2} - 2x_3 + x_4 \quad (B) x_1 + \frac{x_2}{2} - 2x_3 - x_4 \quad (C) x_1 + \frac{x_2}{2} - x_3 + x_4 \quad (D) 2x_1 - x_1 - \frac{x_2}{2} - x_4$$

14. Use the given bond enthalpy data to estimate the ΔH° (kJ) for the following reaction.
 (C – H = 414 kJ, H – Cl = 431 kJ, Cl – Cl = 243 kJ, C – Cl = 331 kJ).
 $\text{CH}_4(\text{g}) + 4 \text{Cl}_2(\text{g}) \rightarrow \text{CCl}_4(\text{g}) + 4 \text{HCl}(\text{g})$
 (A) 620 (B) 330 (C) 420 (D) 105
15. For which of the following change $\Delta H \neq \Delta E$?
 (A) $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightarrow 2\text{HI}(\text{g})$ (B) $\text{HCl}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$
 (C) $\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$ (D) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$
16. ΔH_f of which of the following reactions is zero ?
 (A) $\text{H}_2(\text{g}) \rightarrow 2\text{H}^+(\text{g}) + 2\text{e}^-$ (B) $2\text{H}(\text{g}) + \text{aq} \rightarrow 2\text{H}^+(\text{aq}) + 2\text{e}^-$
 (C) $2\text{H}(\text{g}) \rightarrow 2\text{H}^+(\text{g}) + 2\text{e}^-$ (D) $\text{H}_2(\text{g}) + \text{aq} \rightarrow 2\text{H}^+(\text{aq}) + 2\text{e}^-$
17. ΔH_f° of water is $-285.8 \text{ kJ mol}^{-1}$. If enthalpy of neutralisation of monoacid strong base is $-57.3 \text{ kJ mol}^{-1}$, ΔH_f° of OH^- ion will be ?
 (A) $-228.5 \text{ kJ mol}^{-1}$ (B) $228.5 \text{ kJ mol}^{-1}$ (C) $114.25 \text{ kJ mol}^{-1}$ (D) $-114.25 \text{ kJ mol}^{-1}$
18. 4 grams of sodium hydroxide pellets were dissolved in 100 cm^3 of water. The temperature before adding the sodium hydroxide pellets was 25 degrees C, and after adding the pellets it was 35 degrees C. Calculate the enthalpy change in kJ/mole of the reaction (Specific heat capacity of water = 4.2 J/K/g)
 (A) 42 kJ/mole (B) 4.2 kJ/mole (C) 4200 kJ/mole (D) none
19. 50.0 mL of 0.10 M HCl is mixed with 50.0 mL of 0.10 M NaOH. The solution temperature rises by 3.0 C. Calculate the enthalpy of neutralization per mol of HCl.
 (A) $-2.5 \times 10^2 \text{ kJ}$ (B) $-1.3 \times 10^2 \text{ kJ}$ (C) $-8.4 \times 10^1 \text{ kJ}$ (D) $-6.3 \times 10^1 \text{ kJ}$
20. Which of the following statements is (are) correct ?
 (A) the reaction between the strong acid and strong base takes place with the evolution of heat
 (B) $\Delta H_{\text{neut.}}$ of weak acid/strong base is less than the $\Delta H_{\text{neut.}}$ of strong acid/strong base
 (C) $\Delta H_{\text{neut.}}$ of strong acid/strong base is equal to the ΔH of formation of $\text{H}_2\text{O}(\text{l})$ from its ions in the aqueous medium
 (D) $\Delta H_{\text{neut. (weak acid/strong base)}} - \Delta H_{\text{neut. (strong acid/strong base)}} = \Delta H_{\text{ion (weak acid)}}$
21. The standard enthalpies of formation of $\text{CO}_{2(\text{g})}$, and $\text{HCOOH}(\text{l})$ are -393.7 kJ/mol and -409.2 kJ/mol respectively :-
 (A) -393.7 kJ/mol is the enthalpy change for the reaction $\text{C}_{(\text{s})} + \text{O}_{2(\text{g})} \rightarrow \text{CO}_{2(\text{g})}$
 (B) the enthalpy change for the reaction $\text{CO}_{2(\text{g})} + \text{H}_{2(\text{g})} \rightarrow \text{HCOOH}_{(\text{l})}$ would be -15.5 kJ/mol
 (C) the enthalpy change for the reaction

$$\text{H}_2\text{O} + \text{CO} \rightarrow \text{HCOOH} \text{ is } -409.2 \text{ kJ/mol}$$

 (D) the enthalpy change for the reaction

$$\text{H}_{2(\text{g})} + \text{CO}_{2(\text{g})} \rightarrow \text{H}_2\text{O}_{(\text{l})} + \text{CO}_{(\text{g})} \text{ is } -409.2 \text{ kJ/mol}$$
22. Ethanol can undergoes decomposition to form two sets of products ?



if the molar ratio of C_2H_4 to CH_3CHO is 8 : 1 in a set of product gases, then the energy involved in the decomposition of 1 mole of ethanol is ?

- (A) 65.98 kJ (B) 48.137 kJ (C) 48.46 kJ (D) 57.22 kJ

23. Reactions involving gold have been of particular interest to a chemist. Consider the following reactions,
- $$\text{Au(OH)}_3 + 4 \text{HCl} \longrightarrow \text{HAuCl}_4 + 3 \text{H}_2\text{O}, \quad \Delta H = -28 \text{ kcal}$$
- $$\text{Au(OH)}_3 + 4 \text{HBr} \longrightarrow \text{HAuBr}_4 + 3 \text{H}_2\text{O}, \quad \Delta H = -36.8 \text{ kcal}$$
- In an experiment there was an absorption of 0.44 kcal when one mole of HAuBr_4 was mixed with 4 moles of HCl. What is the percentage conversion of HAuBr_4 into HAuCl_4 ?
- (A) 0.5 % (B) 0.6 % (C) 5 % (D) 50%
24. Which of the following statement is (are) correct ?
- (A) for an exothermic reactions, $\sum \Delta H_f^\circ$ (products) < $\sum \Delta H_f^\circ$ (reactants)
- (B) $\sum \Delta H_f^\circ$ of $\text{CO}_2(\text{g})$ is same as the $\Delta H_{\text{comb.}}$ of carbon graphite
- (C) all exothermic reactions have a free energy change negative
- (D) for a reaction $\text{N}_{2(\text{g})} + \text{O}_{2(\text{g})} \longrightarrow 2\text{NO}_{(\text{g})}$, the heat at constant pressure and the heat at constant volume at a given temperature are same
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BRAIN TEASERS						ANSWER KEY				EXERCISE -2					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	C	C	C	A	A	D	B	A	C	B	A	B	B	C	D
Que.	16	17	18	19	20	21	22	23	24						
Ans.	D	A	A	A	A,B,C,D	A,B	B	C	A,B,D						

EXERCISE-03**MISCELLANEOUS TYPE QUESTIONS****TRUE / FALSE**

- The enthalpies of elements are always taken to be zero.
- Heat of neutralisation of weak acid-strong base is not constant.
- Resonance energy is always negative.
- For reaction $2A(g) + B(g) \longrightarrow 3C(g)$ $\Delta H = -x$ kJ
then for reaction $\frac{3}{2}C(g) \longrightarrow A(g) + \frac{B}{2}(g)$ $\Delta H = \frac{x}{2}$ kJ.
- ΔH_f (C, diamond) $\neq 0$
- For a particular reaction $\Delta E = \Delta H + P\Delta V$
- If BE (bond energy) of $N \equiv N$ bond is x_1 that of $H - H$ bond is x_2 and $N - H$ bond is x_3 then enthalpy change of the reaction is
$$N_2 + 3H_2 \longrightarrow 2NH_3 \qquad \Delta H_r = x_1 + 3x_2 - 2x_3$$
- Enthalpy change is dependent on temperature and on the path adopted.
- Heat of hydrogenation of ethene is x_1 and that of benzene is x_2 , hence resonance energy of benzene is $(x_1 - x_2)$.

FILL IN THE BLANKS

- The combustion of a substance is always
- If heat content of X is smaller than that of Y then the process $X \rightarrow Y$ is
- $C(\text{Diamond}) \longrightarrow C(\text{Graphite}) + x$ kJ. The heat change in this process is called
- As per reaction, $N_2(g) + 2O_2(g) \longrightarrow 2NO_2(g)$ -66 kJ the value of ΔH_f of NO_2 is
- Heats of combustion of methane, carbon and hydrogen are -212, -94, -68 kcal mol⁻¹. The heat of formation of CH_4 is
- The heat of neutralisation of 1 mole of $HClO_4$ with 1 gm-equivalent of NaOH in aqueous solutions is kJ mol⁻¹.
- The heat of combustion of graphite and carbon monoxide respectively are -393.5 kJ mol⁻¹ and -283 kJ mol⁻¹. Thus heat of formation of carbon monoxide in kJ mol⁻¹ is
- For the reaction,
$$C_3H_8(g) + 5O_2(g) \longrightarrow 3CO_2(g) + 4H_2O(l)$$

at constant temperature, $\Delta H - \Delta U$ is

MATCH THE COLUMN

- Match the reaction (In Column I) with relation between ΔH and ΔE (in Column II) :

Column-I		Column-II	
(A)	$C(s) + O_2(g) \longrightarrow CO_2(g)$	(p)	$\Delta H = \Delta E + RT$
(B)	$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$	(q)	$\Delta H = \Delta E$
(C)	$NH_4HS(s) \longrightarrow NH_3(g) + H_2S(g)$	(r)	$\Delta H = \Delta E - 2RT$
(D)	$PCl_5(g) \longrightarrow PCl_3(g) + Cl_2$	(s)	$\Delta H = \Delta E + 2RT$
(E)	$2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$	(t)	$\Delta H = \Delta E - RT$

2. List equation/law (in Column I) with statement (in Column II) :

Column-I		Column-II	
(A)	Arrhenius equation	(p)	Variation of enthalpy of a reaction with temperature
(B)	Kirchhoff equation	(q)	Variation of rate constant with temperature
(C)	Second law of thermodynamics	(r)	Entropy of an isolated system tends to increase and reach a maximum value
(D)	Hess's law of constant heat summation	(s)	Enthalpy change in a reaction is always constant and independent of the manner in which the reaction occurs.

- 3.

Column-I		Column-II	
(A)	$S(g) + O_2(g) \longrightarrow SO_2(g) ; \Delta H$	(p)	Heat of solution
(B)	$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l) ; \Delta H$	(q)	Heat of neutralisation
(C)	$NaOH(s) + aq \longrightarrow NaOH(aq) ; \Delta H$	(r)	Heat of formation
(D)	$NaOH(aq) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l) ; \Delta H$	(s)	Heat of combustion

ASSERTION & REASON

These questions contains, Statement I (assertion) and Statement II (reason).

- (A) Statement-I is true, Statement-II is true ; Statement-II is correct explanation for Statement-I.
 (B) Statement-I is true, Statement-II is true ; Statement-II is NOT a correct explanation for statement-I
 (C) Statement-I is true, Statement-II is false
 (D) Statement-I is false, Statement-II is true

1. **Statement-I** : Enthalpy of neutralization of CH_3COOH by $NaOH$ is less than that of HCl by $NaOH$.

Because

Statement-II : Enthalpy of neutralization of CH_3COOH is less because of the absorption of heat in the ionization process.

2. **Statement-I** : Enthalpy changes are positive when $Na_2SO_4 \cdot 10H_2O$, $CuSO_4 \cdot 5H_2O$ and salts like $NaCl$, KCl , etc., which do not form hydrates are dissolved in water. But enthalpy changes are negative when anhydrous salts capable of forming hydrates are dissolved in water.

Because

Statement-II : The difference in the behaviour is due to large differences in the molecular weight of hydrated and anhydrous salts. The substances with larger molecular weights usually show positive enthalpy changes on dissolution.

3. **Statement-I** : Heat of neutralisation of $HF(aq.)$, a weak acid, with $NaOH(aq.)$ is more than 13.7 kcal, in an exothermic reaction.

Because

Statement-II : Some heat is lost in the ionisation of a weak acid.

4. **Statement-I** : Enthalpy of formation of HCl is equal to bond energy of HCl.
Because
Statement-II : Enthalpy of formation and bond energy both involve the formation of one mole of HCl from the elements.
5. **Statement-I** : Enthalpy of atomization is the heat of reaction $\text{H}_2\text{O(l)} \longrightarrow \text{H}_2\text{O(g)}$.
Because
Statement-II : Gaseous molecules are far apart of each other due to less attraction.
6. **Statement-I** : The enthalpy of formation of $\text{H}_2\text{O(l)}$ is greater than that of $\text{H}_2\text{O(g)}$.
Because
Statement-II : Enthalpy change is negative for the condensation reaction, $\text{H}_2\text{O(g)} \longrightarrow \text{H}_2\text{O(l)}$.
7. **Statement-I** : As temperature increases, heat of reaction also increases for exothermic as well as for endothermic reactions.
Because
Statement-II : ΔH varies with temperature as given by $\Delta H_2 \text{ (at } T_2) = \Delta H_1 \text{ (at } T_1) + \Delta C_p(T_2 - T_1)$
8. **Statement-I** : Heat of combustion is always negative.
Because
Statement-II : Heat of combustion is used to calculate of fuels.

COMPREHENSION BASED QUESTIONS

Comprehension # 1

The industrial preparation of a polymer, PTFE, is based on the synthesis of the monomer $\text{CF}_2 = \text{CF}_2$, which is produced according to reaction (i) below :



The monomer $\text{CF}_2 = \text{CF}_2$ is also obtained by reaction (ii) below :

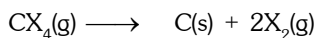


Consider the information below to answer the questions:

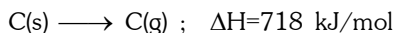
Compound	ΔH_f	Molecule X-X	$\Delta(X-X)$
HCl(g)	-92.3	F-F	154.7
CHClF ₂ (g)	-485.2	Cl-Cl	246.7
CF ₂ = CF ₂ (g)	-658.3		
CF ₄	-679.6		
CCl ₄	-106.6		

1. The enthalpy change for reaction (i).
 (A) 100.23 kJ/mol (B) 127.5 kJ/mol
 (C) -127.5 kJ/mol (D) -100.23 kJ/mol

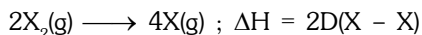
2. (i) Use the expressions



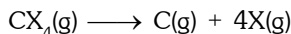
$$\Delta H = -\Delta H_f$$



and



where X = F, Cl, to the enthalpy change for the two processes



What is the average C-X bond energies for the species $\text{CX}_4(\text{g})$ (where X = F, Cl).

(A) 329.5 kJ/mol, 426.75 kJ/mol

(B) 426.75 kJ/mol, 329.5 kJ/mol

(C) -329.5 kJ/mol, -426.75 kJ/mol

(D) -426.75 kJ/mol, -329.5 kJ/mol

(ii) Given that the C-H bond energy is 416.1 kJ/mol, the order of relative chemical reactivities of C-H, C-F, and C-Cl bonds.

(A) C - H > C - Cl > C - F

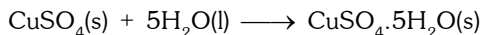
(B) C - F > C - Cl > C - H

(C) C - Cl > C - H > C - F

(D) C - Cl > C - F > C - H

Comprehension # 2

The hydration enthalpy of anhydrous copper (II) sulphate is defined as the heat absorbed or evolved when one mole of anhydrous solid is converted to one mole of crystalline hydrated solid.



It cannot be measured directly.

In an experiment to determine the hydration enthalpy indirectly, 4 g of the anhydrous solid were added to 50 g of water and the rise in temperature was 8 degrees. When 4 g of the hydrated solid was added to 50 g of water the fall in temperature was 1.3 degrees.

1. What is the heat produced when 4 g of anhydrous solid is added to 50 g of water.

(A) 400 kJ

(B) 1672 kJ

(C) 200 kJ

(D) 836 kJ

2. What is the enthalpy of solution of anhydrous copper (II) sulphate in kJ/mol.

(A) 69.9472 kJ/mol

(B) 4054.375 kJ/mol

(C) 139.8948 kJ/mol

(D) 8108.750 kJ/mol

3. Given that the enthalpy of solution of the hydrated copper (II) sulphate is +11.3 kJ/mol, what is the enthalpy of hydration of the anhydrous solid.

(A) 97.321 kJ/mol

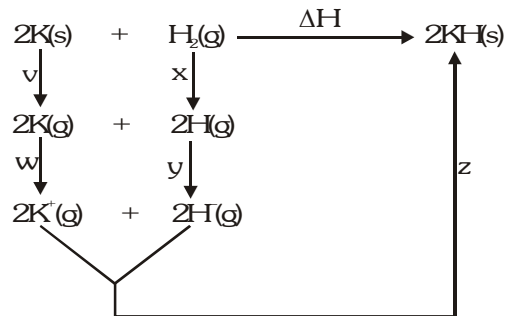
(B) -97.321 kJ/mol

(C) -77.971 kJ/mol

(D) 77.971 kJ/mol

Comprehension # 3

The Born Haber cycle below represents the energy changes occurring at 298K when KH is formed from its elements



$$\Delta H_{\text{atomisation}} \text{ K} = 90 \text{ kJ/mol}$$

$$\Delta H_{\text{ionisation}} \text{ K} = 418 \text{ kJ/mol}$$

$$\Delta H_{\text{dissociation}} \text{ H} = 436 \text{ kJ/mol}$$

$$\Delta H_{\text{electron affinity}} \text{ H} = -78 \text{ kJ/mol}$$

$$\Delta H_{\text{lattice}} \text{ KH} = -710 \text{ kJ/mol}$$

- In terms of the letters v to z write down expressions for
 - ΔH for the reaction

$$2K(s) + H_2(g) \longrightarrow 2KH(s)$$
 - ΔH_f of K
 - $\Delta H_{\text{electron affinity}}$ of H
 - $\Delta H_{\text{lattice}}$ of KH(s)
- Which of v to y is
 - The most exothermic
 - The most endothermic
- Calculate the value of ΔH showing all your working.
- Calculate the ΔH_f of KH(s).
- Write a balanced equation for the reaction of KH with water.
- On complete reaction with water, 0.1 g of KH gave a solution requiring 25 cm³ of 0.1M HCl for neutralisation. Calculate the relative atomic mass of potassium from this information.

MISCELLANEOUS TYPE QUESTION	ANSWER KEY				EXERCISE -3			
● <u>True / False</u>								
1. F	2. T	3. T	4. T	5. T	6. F	7. F	8. F	
9. F								
● <u>Fill in the Blanks</u>								
1. exothermic	2. endothermic	3. heat of transition	4. +33 kJ mol ⁻¹					
5. -18 kcal mol ⁻¹	6. -57.2	7. -110.5	8. -3 RT					
● <u>Match the Column</u>								
1. (A) - q, (B) - r, (C) - s, (D) - p, (E) - t	2. (A) - q, (B) - p, (C) - r, (D) - s							
3. (A) - r, (B) - s, (C) - p, (D) - q								
● <u>Assertion - Reason Questions</u>								
1. (A)	2. (C)	3. (C)	4. (A)	5. (D)	6. (A)	7. (D)	8. (B)	
● <u>Comprehension Based Questions</u>								
Comprehension # 1 : 1. (B) 2. (i) (B) (ii) (C)								
Comprehension #2 : 1. (B) 2. (A) 3. (C)								
Comprehension #3 : 1. (i) $\Delta H = (v + w + x + y + z)$ (ii) w/2 (iii) y/2 (iv) z/2								
2. (i) y				(ii) w				
3. -124 kJ/mol				4. -62 kJ/mol			6. 39	

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

- When 2 moles of C_2H_6 are completely burnt 3120 kJ of heat is liberated. Calculate the heat of formation, ΔH_f for C_2H_6 . Give ΔH_f for $CO_2(g)$ & $H_2O(l)$ are -395 & -286 kJ respectively.
- The standard heats of formation of $CH_4(g)$, $CO_2(g)$ and $H_2O(l)$ are -76.2 , -398.8 , -241.6 kJ mol^{-1} . Calculate amount of heat evolved by burning $1m^3$ of methane measured under normal (STP) conditions.
- Calculate the enthalpy change when infinitely dilute solution of $CaCl_2$ and Na_2CO_3 mixed ΔH_f for $Ca^{2+}(aq)$, $CO_3^{2-}(aq)$ and $CaCO_3(s)$ are -129.80 , -161.65 , -288.5 kcal mol^{-1} respectively.
- The enthalpies of neutralization of $NaOH$ & NH_4OH by HCl are -13680 calories and -12270 cal respectively. What would be the enthalpy change if one gram equivalent of $NaOH$ is added to one gram equivalent of NH_4Cl is solution ? Assume that NH_4OH and $NaCl$ are quantitatively obtained.
- The heat of solution of anhydrous $CuSO_4$ is -15.9 kcal and that of $CuSO_4 \cdot 5H_2O$ is 2.8 kcal. Calculate the heat of hydration of $CuSO_4$.
- The heat of reaction $\frac{1}{2} H_2(g) + \frac{1}{2} Cl_2(g) \longrightarrow HCl(g)$ at $27^\circ C$ is -22.1 kcal. Calculate the heat of reaction at $77^\circ C$. The molar heat capacities at constant pressure at $27^\circ C$ for hydrogen, chlorine & HCl are 6.82 , 7.70 & 6.80 cal mol^{-1} respectively.
- Calculate standard heats of formation of carbon-di-sulphide (l). Given the standard heat of combustion of carbon (s), sulphur (s) & carbon-di-sulphide (l) are : -393.3 , -293.72 and -1108.76 kJ mol^{-1} respectively.
- The standard enthalpy of neutralization of KOH with (a) HCN (b) HCl in dilute solution is -2480 cal. $g\text{eq}^{-1}$ and -13.68 kcal, $g\text{eq}^{-1}$ respectively. Find the enthalpy of dissociation of HCN at the same temperature.
- At $300K$, the standard enthalpies of formation of $C_6H_5COOH(s)$, $CO_2(g)$ & $H_2O(l)$ are; -408 , -393 & -286 kJ mol^{-1} respectively. Calculate the heat of combustion of benzoic acid at :
(i) constant pressure & (ii) constant volume.
- The heat liberated on complete combustion of 7.8 g of benzene is 327 kJ. This heat has been measured at constant volume & at $27^\circ C$. Calculate the heat of combustion of benzene at constant pressure.
- If the enthalpy of formation of $HCl(g)$ and $Cl^-(aq)$ are -92.3 kJ/mole and -167.44 kJ/mol, find the enthalpy of solution of hydrogen chloride gas.
- 0.16 g of methane was subjected to combustion at $27^\circ C$ in a bomb calorimeter. The temperature of calorimeter system (including water) was found to rise by $0.5^\circ C$. Calculate the heat of combustion of methane at
(i) constant volume (ii) constant pressure.
The thermal capacity of calorimeter system is 17.7 kJ K^{-1} . ($R = 8.313$ $mol^{-1}K^{-1}$)
- When 1.0 g of fructose $C_6H_{12}O_6(s)$ is burned in oxygen in a bomb calorimeter, the temperature of the calorimeter water increases by $1.56^\circ C$. If the heat capacity of the calorimeter and its contents is 10.0 kJ/ $^\circ C$. Calculate the enthalpy of combustion of fructose at $298 K$.
- The enthalpy of dissociation of PH_3 is 954 kJ/mol and that of P_2H_4 is 1.485 M J mol^{-1} . What is the bond enthalpy of the $P-P$ bond ?
- Using the bond enthalpy data given below, calculate the enthalpy change for the reaction,
$$C_2H_4(g) + H_2(g) \longrightarrow C_2H_6(g)$$

Data :

Bond	C - C	C = C	C - H	H - H
Bond Enthalpy	336.81 kJ/mol	606.68 kJ/mol	410.87 kJ/mol	431.79 kJ/mol

16. The enthalpy change for the following process at 25 C and under constant pressure at 1 atm are as follows:

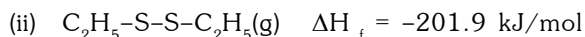


Calculate C–C bond energy in C_2H_6 & heat of formation of $\text{C}_2\text{H}_6(\text{g})$

Given : $\Delta_{\text{sub}} \text{C}(\text{s}) = 171.8 \text{ kcal/mole}$

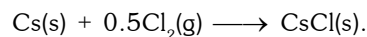
B.E. (H–H) = 104.1 kcal/mole

17. Find the enthalpy of S–S bond from the following data.



18. Calculate the electron affinity of fluorine atom using the following data. Make Born–Haber's cycle. All the values are in kJ mol^{-1} at 25 C, $\Delta H_{\text{diss}}(\text{F}_2) = 160$, $\Delta H_f^\circ (\text{NaF}(\text{s})) = -571$. I.E. $[\text{Na}(\text{g})] = 494$, $\Delta H_{\text{vap}} [\text{Na}(\text{s})] = 101$. Lattice energy of $\text{NaF}(\text{s}) = -894$.

19. Cesium chloride is formed according to the following equation :



The enthalpy of sublimation of Cs, enthalpy of dissociation of chlorine, ionization energy of Cs & electron affinity of chlorine are 81.2, 243.0, 375.7 and $-348.3 \text{ kJ mol}^{-1}$. The energy change involved in the formation of CsCl is $-388.6 \text{ kJ mol}^{-1}$. Calculate the lattice energy of CsCl .

20. The enthalpy of formation of ethane, ethylene and benzene from the gaseous atom are -2839.2 , -2275.2 and $-5506 \text{ kJ mol}^{-1}$ respectively. Calculate the resonance energy of benzene. The bond enthalpy of C–H bond is given as equal to $+410.87 \text{ kJ/mol}$.

21. Two mole of ideal diatomic gas ($C_{v,m} = 5/2 R$) at 300 K and 5 atm expanded irreversibly & adiabatically to a final pressure of 2 atm against a constant pressure of 1 atm. Calculate q , w , ΔH & ΔU .

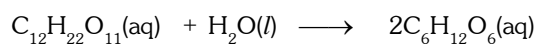
22. A bomb containing 5.4g of Al and 15.97g of Fe_2O_3 is placed in an ice calorimeter containing initially 8 kg of ice and 8 kg of water. The reaction $2\text{Al}(\text{s}) + \text{Fe}_2\text{O}_3(\text{s}) \rightarrow \text{Al}_2\text{O}_3(\text{s}) + 2\text{Fe}(\text{s})$ is set off by remote control and it is then observed that the calorimeter contains 7.746 kg of ice and 8.254 kg of water. Find the ΔH for the above reaction.

$$\Delta H_{\text{fusion}} (\text{ice}) = 1.436 \text{ kcal/mole}$$

23. A sample of the sugar D–ribose ($\text{C}_5\text{H}_{10}\text{O}_5$) of mass 0.727g was placed in a calorimeter and then ignited in the presence of excess oxygen. The temperature raise by 0.910 K. In a separate experiment in the same calorimeter, the combustion of 0.825g of benzoic acid, for which the internal energy of combustion is $-3251 \text{ kJ mol}^{-1}$, gave a temperature rise of 1.940K. Calculate the internal energy of combustion of D–ribose and its enthalpy of formation.

24. The heat of combustion of formaldehyde_(g) is $-134 \text{ kcal mole}^{-1}$ and the heat of combustion of paraformaldehyde_(s) is $-122 \text{ kcal per } (1/n) (\text{CH}_2\text{O})_n$. Calculate the heat of polymerization of formaldehyde to paraformaldehyde.

25. The disaccharide α – maltose can be hydrolysed to glucose according to the equation



Using the following values, calculate the standard enthalpy change in this reaction:

$$\begin{aligned}\Delta_f H (\text{H}_2\text{O}, \text{l}) &= -285.85 \text{ kJ mol}^{-1} \\ \Delta_f H (\text{C}_6\text{H}_{12}\text{O}_6, \text{aq}) &= -1263.1 \text{ kJ mol}^{-1} \\ \Delta_f H (\text{C}_{12}\text{H}_{22}\text{O}_{11}, \text{aq}) &= -2238.3 \text{ kJ mol}^{-1}\end{aligned}$$

CONCEPTUAL SUBJECTIVE EXERCISE			ANSWER KEY		EXERCISE-4(A)	
1. -88 kJ/mol	2. 35.973 MJ	3. 2.95 kcal	4. -1410 cal	5. -18.7 kcal		
6. -22.123 kcal	7. 128.02 kJ	8. 11.2 kcal	9. (i) -3201 kJ/mol (ii) -3199.75 kJ/mol			
10. -3273.77 kJ/mol	11. -75.14 kJ/mol	12. (i) -885 kJ/mol (ii) -889.980 kJ/mol				
13. -2808 kJ/mol	14. 213 kJ/mol	15. -120.08 J/mol				
16. B.E. (C-C) = 82 kcal/mol, $\Delta_f H [\text{C}_2\text{H}_6(\text{g})] = -20.1 \text{ kcal/mol}$	17. 277.5 kJ/mol					
18. E.A = -352 kJ mol ⁻¹	19. -618.7 kJ mol ⁻¹					
20. -23.68 kJ/mol	21. q = 0, $\Delta U = w = -1247.1 \text{ J}$, $\Delta H = -1745.94 \text{ J}$					
22. $\Delta H = -202.5 \text{ kcal}$	23. -2.13 MJ mol ⁻¹ , 1.267 MJ mol ⁻¹					
24. -12 kcal	25. -2.05 kJ mol ⁻¹					

EXERCISE-04 [B]**BRAIN STORMING SUBJECTIVE EXERCISE**

1. For the reaction $\text{cis-2-butene} \rightarrow \text{trans-2-butene}$ and $\text{cis-2-butene} \rightarrow 1\text{-butene}$, $\Delta H = -950$ and $+1771$ cal/mol respectively. The heat of combustion of 1-butene is -649.8 kcal/mol. Determine the heat of combustion of trans-2-butene. Also calculate the bond energy of C=C bond in trans-2-butene. Given B.E of C = O = 196, O—H = 110, O = O = 118, C—C = 80 and C—H = 98 kcal/mol respectively. $\Delta H_v(\text{H}_2\text{O}) = 11$ kcal/mol.
2. Using the data (all values are in kJ/mol at 25 C) given below :
 - (i) Enthalpy of polymerization of ethylene = -72 .
 - (ii) Enthalpy of formation of benzene (ℓ) = 49
 - (iii) Enthalpy of vaporization of benzene (ℓ) = 30
 - (iv) Resonance energy of benzene (ℓ) = -152
 - (v) Heat of formation of gaseous atoms from the elements in their standard states H = 218 , C = 715 . Average bond energy of C—H = 415 . Calculate the B.E. of C—C and C=C. [A : 331 and 590 kJ/mol]
3. Determine resonance energy of benzene [C_6H_6 (ℓ)] from the following information :
$$\Delta H_f \text{ of } \text{C}_6\text{H}_6 (\ell) = +49 \text{ kJ ;}$$
$$\Delta H_f \text{ of } \text{C}_2\text{H}_2 (\text{g}) = +75 \text{ kJ } \Delta H_v \text{ of } \text{C}_6\text{H}_6 (\ell) = +45 \text{ kJ}$$
B.E. C $\equiv$ C = 930 kJ/mol ; C = C = 615 kJ/mol ; C — C = 348 kJ/mol
4. Consider the following thermodynamic data :

Enthalpy of formation of CaC_2 (s) = -60 kJ/mol ;
Enthalpy of sublimation of Ca (s) = 179 kJ/mol ;
Enthalpy of sublimation of C (s) = 718 kJ/mol ;
First ionization energy of Ca (g) = 590 kJ/mol ;
Second ionization energy of Ca (g) = 1143 kJ/mol ;
Bond energy of C_2 (g) = 614 kJ/mol ;
First electron affinity of C_2 (g) = -315 kJ/mol ;
Second electron affinity of C_2 (g) = $+410$ kJ/mol ;

Draw a clear Born-Haber cycle and determine lattice energy of CaC_2 (s).
5. A swimmer breaths 20 times in one minute when swimming and inhale 200 mL of air in one breath. Inhaled air contain 20% O_2 by volume and exhaled air contain 10% O_2 by volume. If all oxygen are consumed in combustion of glucose in the body and 25% of energy obtained from combustion is available for muscular work. Determine the maximum distance this swimmer can swim in one hour if 100 kJ energy is required for 1.0 km swimming. Standard molar enthalpy of combustion of glucose is -2880 kJ/mol and body temperature is 37°C .
6. Given the following standard molar enthalpies :
$$\Delta H_f \text{ of } \text{CH}_3\text{CN} (\text{g}) = 88 \text{ kJ/mol, } \Delta H_f \text{ of } \text{C}_2\text{H}_6 = -84 \text{ kJ/mol, } \Delta H_{\text{Sublimation}} \text{ of } \text{C}(\text{gr}) = 717 \text{ kJ/mol,}$$
bond dissociation energy of N_2 (g) and H_2 (g) are 946 and 436 kJ/mole respectively, B.E. (C—H) = 410 kJ/mol. Determine C—C and C $\equiv$ N bond energies.
7. By using the following data draw an appropriate energy cycle & calculate the enthalpy change of hydration of (i) the chloride ion ; (ii) the iodide ion.

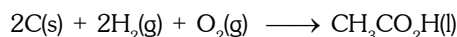
Comment on the difference in their values.

 - * enthalpy change of solution of $\text{NaCl}(\text{s}) = -2$ kJ/mol.
 - * enthalpy change of solution of $\text{NaI}(\text{s}) = +2$ kJ/mol.
 - * enthalpy change of hydration of $\text{Na}^+(\text{g}) = -390$ kJ/mol.
 - * lattice energy of $\text{NaCl} = -772$ kJ/mol.
 - * lattice energy of $\text{NaI} = -699$ kJ/mol.

8. Use the following enthalpies of combustion in the calculations that follow.

Element/Compound	Enthalpy of Combustion
C(s)	- 394
H ₂ (g)	- 286
CH ₃ CO ₂ H(l)	- 876
C ₄ H ₆ (g)	- 2542
C ₂ H ₆ (g)	- 1561
C ₂ H ₄ (g)	- 1393

- (i) Calculate the enthalpy change for the reaction :



- (ii) Calculate the enthalpy of formation of buta-1, 3- diene C₄H₆.

- (iii) Calculate the enthalpy of formation of ethene C₂H₄.

- (iv) Calculate the enthalpy change for the conversion of ethene C₂H₄ to ethane C₂H₆.

9. Becker and Roth measured the heat evolved in the following processes at 20 C: (1) 1 mole of solid (COONH₄)₂H₂O is burned in oxygen, (2) 1 mole of solid (COOH)₂(H₂O)₂ is burned in oxygen, (3) 1 mole of solid (COONH₄)₂H₂O is dissolved in a large excess of water, (4) 1 mole of solid (COOH)₂(H₂O)₂ is dissolved in a large excess of water, (5) 1 mole of oxalic acid in dilute solution is neutralized with gaseous ammonia. They found (1) 189.86 kcal, (2) 53.10 kcal, (3) -11.47 kcal, (4) -8.62 kcal, (5) 43.13 kcal; (1) and (2) were measured at constant volume, the others at constant pressure. The end products of (1) and (2) were nitrogen, carbon dioxide, and water. The heat of formation for 1 mole of water from the elements had previously been determined as 68.35 kcal at constant pressure and 20 C. Find the change in enthalpy when 1 mole of NH₃ is formed from the elements at 20 C.

10. Set up a thermodynamic cycle for determining the enthalpy of hydration of Ca²⁺ ions using the following data :

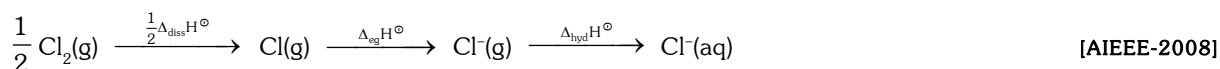
Enthalpy of sublimation of Ca(s), +178.2kJ mol⁻¹; first and second ionization enthalpies of Ca(g), 589.7 kJ mol⁻¹ and 1145 kJ mol⁻¹; enthalpy of vaporization of bromine, + 30.91kJmol⁻¹; dissociation enthalpy of Br₂(g), +192.9 kJmol⁻¹; electron gain enthalpy of Br(g), -331.0kJmol⁻¹; enthalpy of solution of CaBr₂(s), -103.1 kJ mol⁻¹; enthalpy of hydration of Br⁻(g), -337 kJmol⁻¹.

BRAIN STORMING SUBJECTIVE EXERCISE	ANSWER KEY	EXERCISE-4(B)
1. $\Delta H = -647.079 \text{ kcal}$, B.E. (C = C) = 159.921 kcal		
2. C - C = 343.67 kJ, C = C = 615.33		
3. R.E. = -132 kJ/mol		
4. L.E. = -2889 kJ		
5. 1.1328 km		
6. 366 kJ/mol, 1095 kJ/mol		
7. for Cl ⁻ - 384 kJ mol ⁻¹ , for I ⁻ - 307 kJ mol ⁻¹		
8. (i) $\Delta H = -484 \text{ kJ/mol}$ (ii) $\Delta H = 108 \text{ kJ/mol}$ (iii) $\Delta H = 33$ (iv) $\Delta H = -118 \text{ kJ/mol}$		
9. -11.01 kcal		
10. -1587kJmol ⁻¹		

EXERCISE-05(A)**PREVIOUS YEARS QUESTIONS**

1. If at 298 K the bond energies of C-H, C-C, C=C and H-H bonds are respectively 414, 347, 615 and 435 kJ mol⁻¹, the value of enthalpy change for the reaction : [AIEEE-2003]
- $$\text{H}_2\text{C}=\text{CH}_2(\text{g}) + \text{H}_2(\text{g}) \longrightarrow \text{H}_3\text{C}-\text{CH}_3(\text{g})$$
- at 298 K will be :-
-
- (1) +125 kJ (2) -125 kJ
-
- (3) +250 kJ (4) -250 kJ
2. The enthalpies of combustion of carbon and carbon monoxide are -393.5 and -283 kJ mol⁻¹ respectively. The enthalpy of formation of carbon monoxide per mole :- [AIEEE-2004]
- (1) 110.5 kJ (2) 676.5 kJ
(3) -676.5 kJ (4) -110.5 kJ
3. Consider the reaction : $\text{N}_2 + 3\text{H}_2 \longrightarrow 2\text{NH}_3$ carried out at constant temperature and pressure, if ΔH and ΔU are the enthalpy and internal energy changes for the reaction, which of the following expressions is true? [AIEEE-2005]
- (1) $\Delta H = \Delta U$ (2) $\Delta H = 0$
(3) $\Delta H > \Delta U$ (4) $\Delta H < \Delta U$
4. If the bond dissociation energies of XY, X₂ and Y₂ (all diatomic molecules) are in the ratio of 1 : 1 : 0.5 and $\Delta_f H$ for the formation of XY is -200 kJ mol⁻¹. The bond dissociation energy of X₂ will be :- [AIEEE-2005]
- (1) 200 kJ mol⁻¹ (2) 100 kJ mol⁻¹
(3) 800 kJ mol⁻¹ (4) 300 kJ mol⁻¹
5. The standard enthalpy of formation ($\Delta_f H$) at 298K for methane, CH₄(g), is -74.8 kJ mol⁻¹. The additional information required to determine the average energy for C-H bond formation would be:- [AIEEE-2006]
- (1) Latent heat of vapourization of methane
(2) The first four ionization energies of carbon and electron gain enthalpy of hydrogen
(3) The dissociation energy of hydrogen molecule H₂
(4) The dissociation energy of H₂ and enthalpy of sublimation of carbon
6. The enthalpy changes for the following processes are listed below : [AIEEE-2006]
- $$\begin{aligned}\text{Cl}_2(\text{g}) &= 2\text{Cl}(\text{g}), & 242.3 \text{ kJ mol}^{-1} \\ \text{I}_2(\text{g}) &= 2\text{I}(\text{g}), & 151.0 \text{ kJ mol}^{-1} \\ \text{ICl}(\text{g}) &= \text{I}(\text{g}) + \text{Cl}(\text{g}), & 211.3 \text{ kJ mol}^{-1} \\ \text{I}_2(\text{s}) &= \text{I}_2(\text{g}), & 62.76 \text{ kJ mol}^{-1}\end{aligned}$$
- Given that the standard states for iodine and chlorine are I₂(s) and Cl₂(g), the standard enthalpy of formation for ICl(g) is :-
- (1) -16.8 kJ mol⁻¹ (2) +16.8 kJ mol⁻¹
(3) +244.8 kJ mol⁻¹ (4) -14.6 kJ mol⁻¹

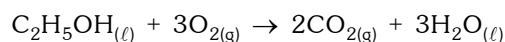
7. ($\Delta H - \Delta U$) for the formation of carbon monoxide (CO) from its elements at 298 K is
($R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$) [AIEEE-2006]
- (1) $1238.78 \text{ J mol}^{-1}$ (2) $-2477.57 \text{ J mol}^{-1}$
(3) $2477.57 \text{ J mol}^{-1}$ (4) $-1238.78 \text{ J mol}^{-1}$
8. Assuming that water vapour is an ideal gas, the internal energy change (ΔU) when 1 mol of water is vapourised at 1 bar pressure and 100 C, (Given : Molar enthalpy of vapourisation of water at 1 bar and 373 K = 41 kJ mol^{-1} and $R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$ will be) :- [AIEEE-2007]
- (1) $4.100 \text{ kJ mol}^{-1}$ (2) $3.7904 \text{ kJ mol}^{-1}$
(3) $37.904 \text{ kJ mol}^{-1}$ (4) $41.00 \text{ kJ mol}^{-1}$
9. Oxidising power of chlorine in aqueous solution can be determined by the parameters indicated below:



The energy involved in the conversion of $\frac{1}{2} \text{Cl}_2(\text{g})$ to $\text{Cl}^-(\text{aq})$

(using the data $\Delta_{\text{diss}} H_{\text{Cl}_2}^\ominus = 240 \text{ kJ mol}^{-1}$, $\Delta_{\text{eg}} H_{\text{Cl}}^\ominus = -349 \text{ kJ mol}^{-1}$, $\Delta_{\text{hyd}} H_{\text{Cl}^-}^\ominus = -381 \text{ kJ mol}^{-1}$) will be:-

- (1) -610 kJ mol^{-1} (2) -850 kJ mol^{-1}
(3) $+120 \text{ kJ mol}^{-1}$ (4) $+152 \text{ kJ mol}^{-1}$
10. On the basis of the following thermochemical data : ($\Delta G_f^\ominus H_{(\text{aq})}^+ = 0$)
- $\text{H}_2\text{O}(\ell) \rightarrow \text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) ; \Delta H = 57.32 \text{ kJ}$
- $\text{H}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\ell) ; \Delta H = -286.20 \text{ kJ}$
- The value of enthalpy of formation of OH^- ion at 25 C is :- [AIEEE-2009]
- (1) $+228.88 \text{ kJ}$ (2) -343.52 kJ
(3) -22.88 kJ (4) -228.88 kJ
11. The standard enthalpy of formation of NH_3 is $-46.0 \text{ kJ mol}^{-1}$. If the enthalpy of formation of H_2 from its atoms is -436 kJ mol^{-1} and that of N_2 is -712 kJ mol^{-1} , the average bond enthalpy of N-H bond in NH_3 is :- [AIEEE-2010]
- (1) $-1102 \text{ kJ mol}^{-1}$ (2) -964 kJ mol^{-1}
(3) $+352 \text{ kJ mol}^{-1}$ (4) $+1056 \text{ kJ mol}^{-1}$
12. The value of enthalpy change (ΔH) for the reaction



at 27 C is $-1366.5 \text{ kJ mol}^{-1}$. The value of internal energy change for the above reaction at this temperature will be :- [AIEEE-2011]

- (1) -1371.5 kJ (2) -1369.0 kJ
(3) -1364.0 kJ (4) -1361.5 kJ

13. Consider the reaction :
 $4\text{NO}_{2(g)} + \text{O}_{2(g)} \rightarrow 2\text{N}_2\text{O}_{5(g)}$, $\Delta_r H = -111\text{kJ}$.
 If $\text{N}_2\text{O}_{5(s)}$ is formed instead of $\text{N}_2\text{O}_{5(g)}$ in the above reaction, the $\Delta_r H$ value will be :-
 (given, ΔH of sublimation for N_2O_5 is 54 kJ mol^{-1}) [AIEEE-2011]
 (1) -165 kJ (2) $+54\text{ kJ}$
 (3) $+219\text{ kJ}$ (4) -219 kJ
14. The enthalpy of neutralisation of NH_4OH with HCl is $-51.46\text{ kJ mol}^{-1}$ and the enthalpy of neutralisation of NaOH with HCl is $-55.90\text{ kJ mol}^{-1}$. The enthalpy of ionisation of NH_4OH is :- [AIEEE-2012 (Online)]
 (1) $+107.36\text{ kJ mol}^{-1}$ (2) -4.44 kJ mol^{-1}
 (3) $-107.36\text{ kJ mol}^{-1}$ (4) $+4.44\text{ kJ mol}^{-1}$
15. The difference between the reaction enthalpy change ($\Delta_r H$) and reaction internal energy change ($\Delta_r U$) for the reaction :
 $2\text{C}_6\text{H}_6(\ell) + 15\text{O}_2(\text{g}) \rightarrow 12\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\ell)$ at 300K is ($R = 8.314\text{ J mol}^{-1}\text{ K}^{-1}$) [AIEEE-2012 (Online)]
 (1) 0 J mol^{-1} (2) 2490 J mol^{-1}
 (3) -2490 J mol^{-1} (4) -7482 J mol^{-1}

PREVIOUS YEAR QUESTIONS										ANSWER KEY						EXERCISE-5(A)			
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15				
Ans	2	4	4	3	4	2	1	3	1	4	3	3	4	4	4				

EXERCISE-05(B)**PREVIOUS YEARS QUESTIONS**

1. Which of the following reaction defines ΔH_f ? [IIT-JEE-2003]
- (A) $C_{(\text{diamond})} + O_2(g) \longrightarrow CO_2(g)$ (B) $\frac{1}{2} H_2(g) + \frac{1}{2} F_2(g) \longrightarrow HF(g)$
- (C) $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$ (D) $CO(g) + \frac{1}{2} O_2(g) \longrightarrow CO_2(g)$
2. ΔH_f for $CO_2(g)$, $CO(g)$ and $H_2O(g)$ are -393.5 , -110.5 and $-241.8 \text{ kJ mol}^{-1}$ respectively. The standard enthalpy change (in kJ) for the relation : [IIT-JEE-2000]
- $CO_2(g) + H_2(g) \longrightarrow CO(g) + H_2O(g)$ is :-
- (A) 524.1 (B) 41.2 (C) -262.5 (D) -41.2
3. Which of the following is not an endothermic reaction :- [IIT-JEE-1999]
- (A) Combustion of methane (B) Decomposition of water
- (C) Dehydrogenation of ethene to ethylene (D) Conversion of graphite to diamond
4. Standard molar enthalpy of formation of CO_2 is equal to :- [IIT-JEE-1997]
- (A) Zero
- (B) Standard molar enthalpy of combustion of carbon (graphite)
- (C) Standard molar enthalpy of combustion of gaseous carbon
- (D) Sum of molar enthalpies of formation of CO and O_2
5. Diborane is a potential rocket fuel which undergoes combustion according to the reaction, [IIT-JEE-2000]
- $B_2H_6(g) + 3O_2(g) \longrightarrow B_2O_3(s) + 3H_2O(g)$
- From the following data, calculate the enthalpy change for the combustion of diborane :
- $2B(s) + \frac{3}{2} O_2(g) \longrightarrow B_2O_3(s) ; \quad \Delta H = -1273 \text{ kJ}$
- $H_2(g) + \frac{1}{2} O_2(g) \longrightarrow H_2O(l) ; \quad \Delta H = -286 \text{ kJ}$
- $H_2O(l) \longrightarrow H_2O(g) ; \quad \Delta H = 44 \text{ kJ}$
- $2B(s) + 3H_2(g) \longrightarrow B_2H_6(g) ; \quad \Delta H = 36 \text{ kJ}$
6. Estimate the average S-F bond energy in SF_6 . The ΔH_f values of $SF_6(g)$, $S(g)$ and $F(g)$ are -1100 , 275 and 80 kJ/mol respectively. [IIT-JEE-1999, 2005]
7. From the following data, calculate the enthalpy change for the combustion of cyclopropane at 298 K . The enthalpy of formation of $CO_2(g)$, $H_2O(l)$ and propene (g) are -393.5 , -285.8 & 20.42 kJ/mol respectively. The enthalpy of isomerisation of cyclopropane to propene is -33.0 kJ/mol . [IIT-JEE-1998, 2005]

8. Using the data provided, calculate the multiple bond energy (kJ mol^{-1}) of a $\text{C} \equiv \text{C}$ bond in C_2H_2 . That energy is (take the bond energy of a C-H bond as 350 kJ mol^{-1} .) [JEE 2012]



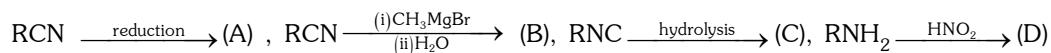
- (A) 1165 (B) 837 (C) 865 (D) 815

9. In a constant volume calorimeter, 3.5 g of a gas with molecular weight 28 was burnt in excess oxygen at 298.0 K. The temperature of the calorimeter was found to increase from 298.0 K to 298.45 K due to the combustion process. Given that the heat capacity of the calorimeter is 2.5 kJ K^{-1} , the numerical value for the enthalpy of combustion of the gas in kJ mol^{-1} is [JEE 2009]

PREVIOUS YEARS QUESTIONS		ANSWER KEY		EXERCISE -5(B)
1. (B)	2. (B)	3. (A)	4. (B)	
5. $-2035 \text{ kJ mol}^{-1}$	6. 309.16 kJ/ mol	7. $-2091.32 \text{ kJ mol}^{-1}$		
8. D	9. 9			

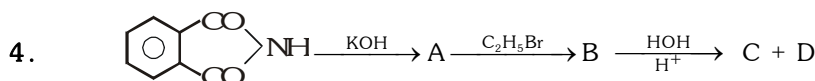
EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. Tertiary butyl amine is a
 (A) 1 Amine (B) 2 Amine
 (C) 3 Amine (D) Quaternary salt
2. The correct set of the products obtained in the following reactions



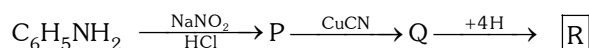
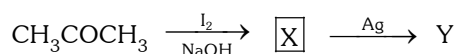
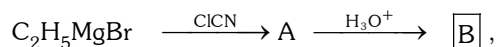
The answer is :

- | | A | B | C | D |
|-----|----------|---------------|----------|----------|
| (A) | 2 Amine | Methyl ketone | 1 Amine | Alcohol |
| (B) | 1 Amine | Methyl ketone | 1 Amine | Alcohol |
| (C) | 2 Amine | Methyl ketone | 2 Amine | Acid |
| (D) | 2 Amine | Methyl ketone | 2 Amine | Aldehyde |
3. Which one of the following tests can be used to identify primary amino group in a given organic compound
 (A) Iodoform test (B) Victor Meyer's test
 (C) Carbylamine reaction (D) Libermann's reaction

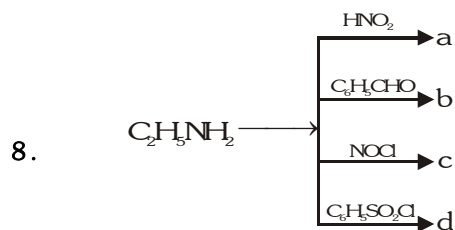


C and D in the above sequence are

- (A) Benzoic acid + aniline (B) Phthalic acid + ethylamine
 (C) Phthalic acid + aniline (D) Benzoic acid + ethylamine
5. Lowest boiling point will be of the compound
 (A) 2-propanamine (B) Ethylmethanamine
 (C) 1-propanamine (D) N, N-dimethylmethanamine
6. How many primary amines can be formulated by $\text{C}_3\text{H}_9\text{N}$ and how many 1 hydrogen are associated with carbon atoms of each compound
 (A) Two primary amines [3, 6] (B) One primary amine [3]
 (C) Three primary amines [3, 6, 6] (D) Two primary amines [5, 6]
7. Identify B, X and R respectively in the following sequence of reactions



- (A) $\text{C}_2\text{H}_5\text{COOH}$, CHI_3 , $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$
 (B) $\text{C}_2\text{H}_5\text{COOH}$, CH_3I , $\text{C}_6\text{H}_5\text{COOH}$
 (C) $\text{C}_2\text{H}_5\text{CH}_2\text{NH}_2$, CH_3I , $\text{C}_6\text{H}_5\text{COOH}$
 (D) $\text{C}_2\text{H}_5\text{COOH}$, $\text{C}_2\text{H}_5\text{I}$, $\text{C}_6\text{H}_5\text{CONH}_2$



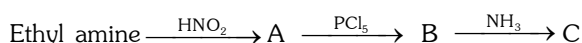
Which product is a Schiff's base :-

- (A) a (B) b (C) c (D) d

9. A + CS₂ + HgCl₂ gives C₂H₅-N=C=S. Thus compound A is :-

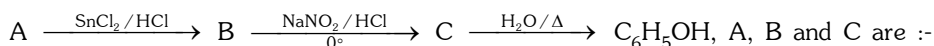
- (A) C₂H₅NH₂ (B) C₂H₅NHC₂H₅ (C) CH₃-CH=NOH (D) CH₃CH₂NO₂

10. The end-product in the reaction sequence would be :



- (A) Ethyl cyanide (B) Ethyl amine (C) Methyl amine (D) Acetamide

11. In the reaction sequence :-



- (A) Benzene, nitrobenzene, aniline (B) Nitrobenzene, aniline and azo-compound
(C) Nitrobenzene, benzene, aniline (D) Benzene, amino compound, aniline

12. Methylamine on treatment with chloroform and ethanolic caustic alkali gives foul smelling compound, the compound is

- (A) CH₃NCO (B) CH₃CNO (C) CH₃CN (D) CH₃NC

13. Which of the following is used as a solvent in the Friedel-Crafts reaction :-

- (A) Toluene (B) Nitrobenzene (C) Benzene (D) Aniline



Aqueous solution of A

- (A) Turns blue litmus to red (B) Turns red litmus to blue
(C) Does not affect the litmus (D) Decolourise the litmus

15. Hinsberg reagent is used to distinguish between

- (A) —CHO , >C=O (B) $\text{—CH}_2\text{OH}$, >CHOH , >C—OH
(C) —O— , —OH (D) —NH_2 , —NH— , >N

16. Reactants of reaction - I are : CH₃CONH₂, KOH, Br₂

Reactants of reaction-II are : CH₃NH₂, CHCl₃, KOH

The intermediate species of reaction-I and reaction-II are respectively

- (A) Carbonium ion, carbene (B) Alkyl isocyanate, carbene
(C) Carbene, nitrene (D) Carbocation, carbanion

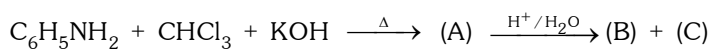
17. Which of the following amine does not respond to carbylamine reaction :-

- (A) Ethylamine (B) (CH₃)₂NH (C) CH₃NH₂ (D) Phenylamine

SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)

1. Carbylamine test is performed in alcoholic KOH by heating a mixture of :-
(A) Chloroform and silver powder
(B) Trihalogenated methane and a primary amine
(C) An alkyl halide and a primary amine
(D) An alkyl cyanide and a primary amine
2. Acetamide is treated separately with following reagents. Which one of these would give methylamine ?
(A) PCl_5 (B) Sodalime
(C) $\text{NaOH} + \text{Br}_2$ (D) Hot concentrated H_2SO_4
3. Maximum basic in gas phase is ?
(A) NH_3 (B) $\text{CH}_3\text{CH}_2\text{NH}_2$ (C) $(\text{CH}_3\text{CH}_2)\text{NH}$ (D) $(\text{CH}_3\text{CH}_2)_3\text{N}$
4. Benzenediazonium chloride on reaction with aniline in weakly basic medium gives
(A) diphenyl ether (B) p-aminoazobenzene (C) Chlorobenzene (D) benzene
5. Which of the following statements is correct ?
(A) Replacement of halogen by NH_2 in alkyl halide is a nucleophilic substitution reaction
(B) Aryl halides show more reactivity as compared to alkyl halides in the replacements of halogen by the NH_2 group
(C) During the replacement of halogen by $-\text{NH}_2$ group, ammonia is taken in large excess so as to avoid the formation of 2 and 3 amines
(D) Tertiary alkyl halide generally produces alkene instead of the replacement of halogen by NH_2 group
6. Which of the following statements is correct ?
(A) Primary amines show intermolecular hydrogen bonding
(B) Secondary amines show intermolecular hydrogen bonding
(C) Tertiary amines show intermolecular hydrogen bonding
(D) Amines have lower boiling points as compared to those of alcohols and carboxylic acid of comparable molar masses.
7. Which of the following statements is correct ?
(A) Aliphatic amines are stronger bases than ammonia
(B) Aromatic amines are stronger bases than ammonia
(C) The alkyl group in alkyl ammonium ion more stabilizes the ion relative to the amine
(D) The aryl group in aryl ammonium ion less stabilizes the ion relative to the amine
8. Which of the following orders regarding the basic strength of substituted aniline is correct ?
(A) p-methylaniline > p-chloroaniline > p-aminoacetophenone
(B) p-methylaniline > p-aminoacetophenone > p-chloroaniline
(C) p-aminoacetophenone > p-methylaniline > p-chloroaniline
(D) p-aminoacetophenone > p-chloroaniline > p-methylaniline

9. Consider the following reaction,



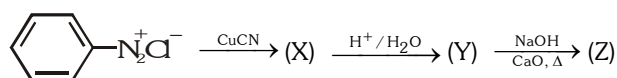
The compounds (B) and (C) are :-

- (A) $\text{C}_6\text{H}_5\text{COOH}$ and NH_3 respectively (B) $\text{C}_6\text{H}_5\text{NH}_2$ and HCOOH respectively
(C) $\text{C}_6\text{H}_5\text{NH}_2$ and H_2O respectively (D) None of these

10. $\text{CH}_3\text{CH}_2\text{NH}_2$ is soluble in :-

- (A) Dilute HCl (B) CuSO_4 solution (C) AgNO_3 (D) None of these

11. The end product (Z) of the following reaction is :-



- (A) A cyanide (B) A carboxylic acid (C) An amine (D) An arene

12. The increasing order of basicity of RCN , $\text{RCH} = \text{NR}$ and RNH_2 is :-

- (A) $\text{RCN} < \text{RCH} = \text{NR} < \text{RNH}_2$ (B) $\text{RNH}_2 < \text{RCN} < \text{RCH} = \text{NR}$
(C) $\text{RCH} = \text{NR} < \text{RNH}_2 < \text{RCN}$ (D) $\text{RH}_2\text{N} < \text{RCH} = \text{NR} < \text{RCN}$

13. Amongst the following, the most basic compound is :-

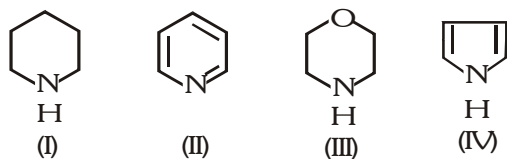
- (A) Benzylamine (B) Aniline (C) Acetanilide (D) p-nitroaniline

14. isocyanide test can be given by :

- (A) $\text{C}_2\text{H}_5\text{NH}_2$ (B) $\text{C}_6\text{H}_5\text{NH}_2$ (C) $\text{C}_6\text{H}_5\text{NHC}_2\text{H}_5$ (D) $\text{C}_6\text{H}_5\text{CH}(\text{NH}_2)\text{C}_2\text{H}_5$

(D) II is an acceptable canonical structure

15. In the following compounds



the order of basicity is

- (A) $\text{IV} > \text{I} > \text{III} > \text{II}$ (B) $\text{III} > \text{I} > \text{IV} > \text{II}$
(C) $\text{II} > \text{I} > \text{III} > \text{IV}$ (D) $\text{I} > \text{III} > \text{II} > \text{IV}$

16. A positive carbylamine test is not given by :

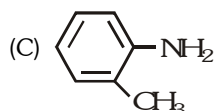
- (A) N, N-dimethylaniline (B) 2, 4-hydroxyazobenzene
(C) N-methyl-o-methylaniline (D) p-methylbenzylamine

17. Among the following, the strongest base is

- (A) $\text{C}_6\text{H}_5\text{NH}_2$ (B) $\text{p-NO}_2\text{C}_6\text{H}_4\text{NH}_2$
(C) $\text{m-NO}_2\text{C}_6\text{H}_4\text{NH}_2$ (D) $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$

18. Phenyl cyanide on reduction with $\text{Na}/\text{C}_2\text{H}_5\text{OH}$ yields

- (A) $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$ (B) $\text{C}_6\text{H}_5\text{NHCH}_3$



- (D) $\text{C}_6\text{H}_5\text{NH}_2$

EXERCISE-03**MISCELLANEOUS TYPE QUESTIONS****TRUE OR FALSE :**

1. Secondary amines show carbylamine reaction.
2. The aqueous solution of ethylamine precipitates iron as ferric hydroxide when added to ferric chloride solution.
3. Nitrogen is evolved when ethylamine is treated with nitrous acid.
4. Primary, secondary and tertiary amines can be separated from their mixture by using ethyl oxalate.
5. Tertiary amines react with Grignard reagents to form hydrocarbon.
6. Primary amines are less soluble than tertiary amines.
7. α -Amino acids on heating with Ba(OH)_2 shows decarboxylation to produce primary amines.

FILL IN THE BLANKS :

1. $(\text{CH}_3)_2\text{CHNH}_2$ is a amine.
2. Methyl cyanide on reaction with sodium and $\text{C}_2\text{H}_5\text{OH}$ forms
3. Acetamide on treatment with forms ethyl amine.
4. Tertiary amines do not react with acetyl chloride since they do not have on nitrogen.
5. The reaction $\text{RCN} \xrightarrow{\text{SnCl}_2/\text{HCl}} (\text{A}) \xrightarrow{\text{H}_2\text{O}} \text{RCHO} + \text{NH}_4\text{Cl}$, is known as
6. nitroalkanes when reduced with zinc dust and ammonium chloride form
7. Primary amines on heating with and form alkyl isocyanides.

MATCH THE COLUMN

1. Match the column I with column II.

Column-I		Column-II	
(A)	$\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$	(p)	Secondary amine
(B)	Conversion of amide to amine	(q)	Hofmann's bromamide reaction
(C)	Conversion of primary amine to isocyanide	(r)	Hinsberg's reagent
(D)	Dimethylamine	(s)	Carbylamine reaction

2. Match the column I with column II.

Column-I		Column-II	
(A)	Tetraethyl ammonium iodide	(p)	Quaternary salt
(B)	>C-NH_2	(q)	Tertiary amine
(C)	$\text{RCOOH} + \text{N}_3\text{H} + \text{Conc. H}_2\text{SO}_4$	(r)	Schmidt reaction
(D)	R_3N	(s)	Primary amine

3. Match list I with II and choose the correct answer from the codes given below :-

Column-I		Column-II	
(A)	Aniline	(p)	Used in making azodyes
(B)	Nitrobenzene	(q)	Sulpha drug
(C)	Sulphanilamide	(r)	Solvent in the Friedel Crafts reaction
(D)	Trinitrotoluene	(s)	Used as explosive

ASSERTION & REASON QUESTION :

These questions contains, Statement-I (assertion) and Statement-II (reason).

- (A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I
 (B) Statement-I is True, Statement-II is True ; Statement-II is NOT a correct explanation for Statement-I
 (C) Statement-I is True, Statement-II is False.
 (D) Statement-I is False, Statement-II is True.

- Statement-I** : Alkyl isocyanide can be prepared by carbyl amine reaction.
Because
Statement-II : Ethyl amine when heated with chloroform in presence of alcoholic KOH, alkanamide is formed.
- Statement-I** : CN^- ion is an ambident nucleophile.
Because
Statement-II : Nucleophiles are electron rich species.
- Statement-I** : Conversion of alkyl halides into alkyl cyanide or isocyanides is a nucleophilic substitution reaction.
Because
Statement-II : When an alkyl halide is treated with alcoholic solution of AgCN , alkyl cyanides are formed as major product.
- Statement-I** : Nitrobenzene undergoes electrophilic substitution at m-position, while nucleophilic substitution occurs at o-and p-position.
Because
Statement-II : Nitro group in benzene ring significantly lowers the electron density at o- and p-positions, while at m-position, the electron density is only slightly lowered.
- Statement-I** : Nitrobenzene is used as a solvent in Friedel-Craft's reaction.
Because
Statement-II : Fusion of p-chloronitrobenzene with KOH gives a low yield of a mixture of p-nitro phenoles.
- Statement-I** : In Hoffmann bromide reaction, the amine formed has one carbon atom less than the parent 1 amide.
Because
Statement-II : N-methyl acetamide undergoes Hoffmann bromamide reaction.
- Statement-I** : Nitrobenzene does not undergo Friedel-Craft's alkylation.
Because
Statement-II : Nitrobenzene is used as solvent in laboratory and industry.
- Statement-I** : A reaction between a diazo salt and an aromatic amine or a phenol, giving an amino-azo or hydroxy-azo compound is called coupling reaction.
Because
Statement-II : Condensation of diazonium with phenol is carried out in weakly acidic medium.

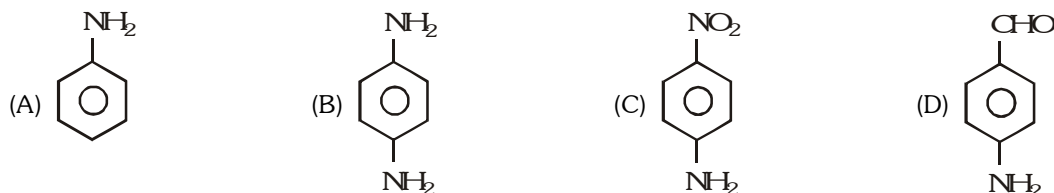
9. **Statement-I** : Aniline can be prepared by the reaction of chlorobenzene with $\text{NaNH}_2/\text{Liq. NH}_3$.
Because
Statement-II : NH_2^- ion is a stronger nucleophile.
10. **Statement-I** : Carbylamine reaction involves the reaction between 1 amine and chloroform in basic medium.
Because
Statement-II : In carbylamine reaction, $-\text{NH}_2$ group is converted into $-\text{NC}$ group via reaction with carbene intermediate.

COMPREHENSION BASED QUESTIONS :

Comprehension # 1

The origin of acidity and basicity in organic compound is great interest and provides an extensive comparison. Among hydrocarbons % s character is taken in account while to decide the acidity, in simple aliphatic acids, more the number of alkyl groups, (+I effect) less is the acidity & more the (-I effect) alkyl groups larger the acidity and vice-versa in the case of simple aliphatic bases. Benzoic acid is more stronger than carboxylic acid as benzoate ion is stabilised more by resonance. Aromatic amines are less basic than aliphatic amine as the electron pair is less available in case of aromatic amines. The presence of solvent also plays a very important role and at times governs the order too.

1. Amongst the following the most basic compound is -



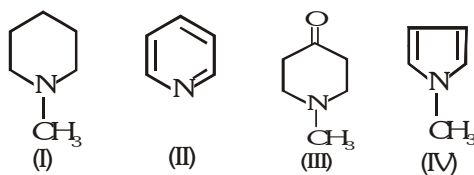
2. Among the following compounds the strongest acid is



3. Which of the following orders regarding acid strength is correct ?



4. In the following compounds, the order of basicity

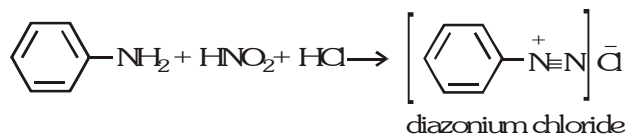


5. Which of the following is most acidic

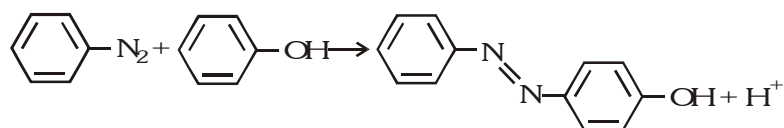


Comprehension # 2

Diazonium salt formation and coupling reactions : When a reaction mixture of phenyl amine and nitrous acid is kept below 10°C , a diazonium salt is formed. This reaction is called diazotization reaction.

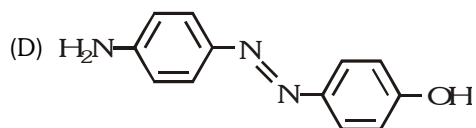
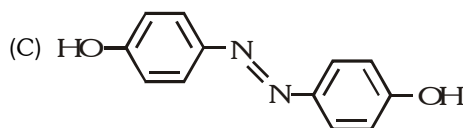
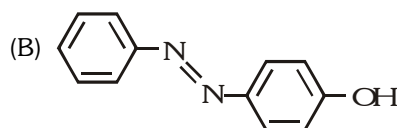
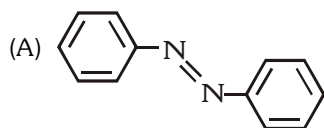


The diazonium ion, $-\text{N}_2^+$, is rather unstable and decomposes readily to nitrogen. However, delocalization of the diazonium from π -bond electron over a benzene ring stabilizes phenyl diazonium sufficiently for it to form at low temperatures. The phenyl diazonium ion behaves as an electrophile, and will attack another arene molecule such as phenol. Electrophilic substitution takes place at the 4 position, producing 4-hydroxy phenyl azobenzene. The reaction is known as coupling reaction.



The compound formed is an energetically stable, yellow azo dye (the azo group is $-\text{N}=\text{N}-$). The stability is due to extensive delocalisation of electrons via the nitrogen-nitrogen double bonds.

1. The azo dye obtained on reacting 4-aminophenol with nitrous acid (in dilute hydrochloric acid) below 10°C and coupling the resulting diazonium salt with phenol is :



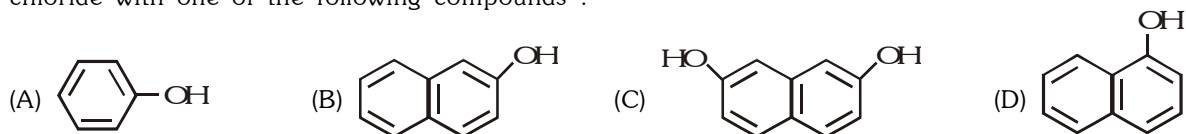
2. Benzene diazonium chloride on reaction with phenol in weakly basic medium gives :

- (A) diphenyl ether
(C) chlorobenzene

- (B) p-hydroxy azobenzene
(D) benzene

3. The product which is a red azodye obtained on reacting benzene diazonium

chloride with one of the following compounds :



Comprehension # 3

Amines are derivatives of ammonia and are classified as 1° , 2° , and 3° . Primary and secondary (but not tertiary amines) form intermolecular hydrogen bonds and thus they boil at higher temperatures than expected. Like ammonia, all amines are basic, although they differ in their basic nature.

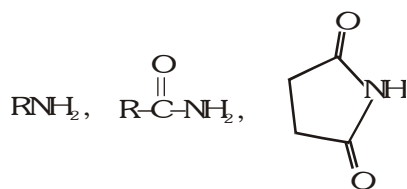
As amines are considered as derivatives of ammonia, quaternary ammonium salts are considered as derivatives of ammonium salts. Only the quaternary ammonium salts can show optical activity.

- Which of the following statement is correct ?
 (A) All classes of amines form hydrogen bonds with each other
 (B) Only primary and secondary amines form hydrogen bonds with water
 (C) All classes of amines can form hydrogen bonds with water
 (D) All amines are completely soluble in water
- When nitrogen is bonded to three different groups
 (A) The molecule is optically inactive
 (B) The molecule is tetrahedral
 (C) The molecule is not superimposable on its mirror image
 (D) The amine boils at nearly similar b.p. as the 1°, and 2° amine of comparable molecular weight.
- Which of the following shows configurational isomerism ?
 (A) Diethyldimethylammonium iodide (B) Dimethylpropylamine
 (C) Methylallylphenylbenzyl ammonium bromide (D) None of these
- Methylethylpropyl amine is optically inactive because
 (i) It is not tetrahedral
 (ii) Its molecule is superimposable on its mirror image
 (iii) The enantiomers are rapidly interconverted
 (iv) The nitrogen is sp^2 hybridized
 Which of the above statement(s) is (are) true ?
 (A) Only (i) (B) Only (iii) (C) (i) and (iii) (D) All the four

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
True / False 1. F 2. T 3. T 4. T 5. F 6. F 7. T Fill in the Blanks 1. Primary 2. ethyl amine 3. $LiAlH_4$ 4. replaceable hydrogen 5. Stephen's reaction 6. hydroxyl amine 7. $CHCl_3$, KOH or NaOH Match the Column 1. (A) → r ; (B) → q ; (C) → s ; (D) → p 2. (A) → p ; B → s ; (C) → r ; (D) → q 3. (A) → p ; (B) → r ; (C) → q ; (D) → s Assertion - Reason Questions 1. C 2. B 3. C 4. B 5. B 6. C 7. B 8. C 9. D 10. A Comprehension Based Questions Comprehension #1 : 1. (B) 2. (A) 3. (B) 4. (D) 5. (B) Comprehension #2 : 1. (C) 2. (B) 3. (B) Comprehension #3 : 1. (C) 2. (C) 3. (C) 4. (B)		

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

1. Arrange the following in decreasing basic order with proper reasoning.



2. Sulphanilic acid is not soluble in organic solvents. Explain ?

3. What is the order of basicity of the following compounds ?

(I) CH_3NH_2 , $(\text{CH}_3)_2\text{NH}$, $(\text{CH}_3)_3\text{N}$ (in aprotic solvent)

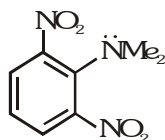
(II) $\text{C}_2\text{H}_5\text{NH}_2$, $(\text{C}_2\text{H}_5)_2\text{NH}$, $(\text{C}_2\text{H}_5)_3\text{N}$ (in aprotic solvent)

4. What is the order of basicity of the following compounds ?

(I) CH_3NH_2 , $(\text{CH}_3)_2\text{NH}$, $(\text{CH}_3)_3\text{N}$ (in protic solvent)

(II) $\text{C}_2\text{H}_5\text{NH}_2$, $(\text{C}_2\text{H}_5)_2\text{NH}$, $(\text{C}_2\text{H}_5)_3\text{N}$ (in protic solvent)

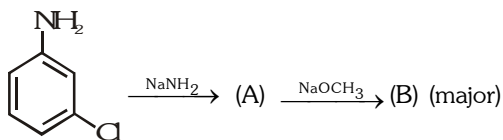
5. Unlike other aromatic amines, why is the following amine strongly basic ?



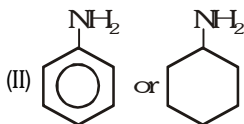
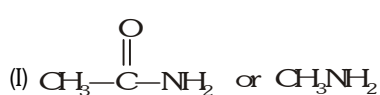
6. In this compound $\text{OH}-\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$ which site acts as an acid and which as a base ?

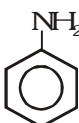
7. Alkyl cyanides (CH_3CN) when treated with hydrogen in presence of Pt or with LiAlH_4 produces same carbon number compound. What is the formula of that compound ?

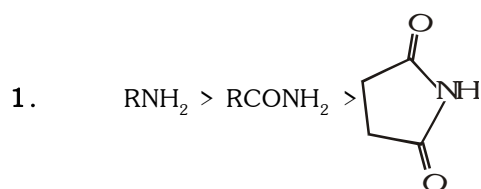
8. Write the compound (A) and (B) formed in this



9. Identify the stronger base in each of the following pairs :



10.  $\xrightarrow[0-5^\circ\text{C}]{\text{NaNO}_2/\text{HCl}}$ A $\xrightarrow[\text{H}_2\text{O}]{\text{warm}}$ B ; what are A & B ?



the lone pair of electrons on N is accommodated by sp^3 hybrid atomic orbital in RNH_2 and does not involve in resonance, whereas in RCONH_2 and succinamide the lone pair of $-\text{NH}_2$ is delocalised.

2. Sulphanilic acid exist as Zwitterion

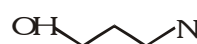


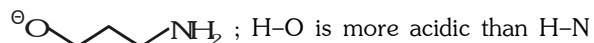
Due to its ionic character it is insoluble in organic solvents.

3. (I) $(\text{CH}_3)_3\text{N} > (\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2$; (II) $(\text{C}_2\text{H}_5)_3\text{N} > (\text{C}_2\text{H}_5)_2\text{NH} > \text{C}_2\text{H}_5\text{NH}_2$

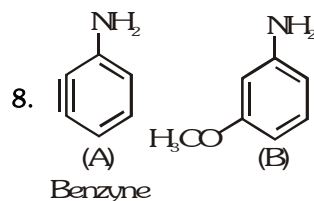
4. (I) $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N}$; (II) $(\text{C}_2\text{H}_5)_2\text{NH} > (\text{C}_2\text{H}_5)_3\text{N} > \text{C}_2\text{H}_5\text{NH}_2$

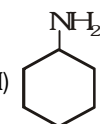
5. Due to the presence of bulky $-\text{NO}_2$ groups on its ortho positions, the $-\text{NMe}_2$ group goes outside the plane of resonance to avoid steric repulsion. The C-N bond rotates and hence the lone pair of N goes perpendicular to the plane of benzene ring. As a result the resonance is stopped and hence the lone pair is readily available as a base.

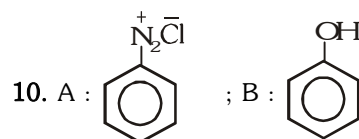
6.  ; $-\text{NH}_2$ is a more basic site than OH ;



7. $\text{CH}_3\text{CH}_2\text{NH}_2$



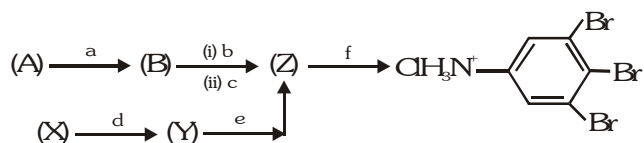
9. (I) CH_3NH_2 ; (II) 



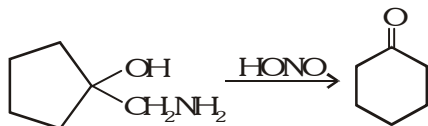
EXERCISE-04 [B]

BRAIN STORMING SUBJECTIVE EXERCISE

- An organic compound (A) composed of C, H and O gives characteristic colour with ceric ammonium nitrate. Treatment of (A) with PCl_5 gives (B), which reacts with KCN to form (C). The reduction of (C) with warm $\text{Na}/\text{C}_2\text{H}_5\text{OH}$ produces (D), which on heating gives (E) with evolution of ammonia. Pyridine is obtained on treatment of (E) with nitrobenzene. Give structure of compounds (A) to (E) with proper reasoning.
- Compound (A) having M.F. $\text{C}_8\text{H}_8\text{O}$ on treatment with $\text{NH}_2\text{OH} \cdot \text{HCl}$ gives (B) and (C). (B) and (C) rearrange to give (D) and (E), respectively on treatment with acid. Compounds (B), (C), (D) and (E) are all isomers of molecular formula $\text{C}_8\text{H}_9\text{NO}$. When (D) is boiled with alcoholic KOH, an oil (F) $\text{C}_6\text{H}_7\text{N}$ separated out. (F) reacts rapidly with CH_3COCl to give back (D). On the other hand, (E) on boiling with alkali followed by acidification gives a white solid (G), $\text{C}_7\text{H}_6\text{O}_2$. Identify the compounds (A) to (G).
- Two isomeric compounds (A) and (B) have $\text{C}_4\text{H}_{11}\text{N}$ as molecular formula. Both on separately treating with HNO_2 lose their N_2 producing two isomeric alcohols (C) and (D) respectively of molecular formula $\text{C}_4\text{H}_{10}\text{O}$. (C) reacts with Lucas reagent immediately and undergoes oxidation. (D) does not react with Lucas reagent in cold but can be easily oxidized. Complete methylation of either (A) or (B) is made which on decomposition does not produce 1-butene. Identify A to D.
- A mixture of two aromatic compounds A and B was separated by dissolving in chloroform followed by extraction with aqueous KOH solution. The organic layer containing compound A, when heated with alcoholic solution of KOH produced a compound C ($\text{C}_7\text{H}_5\text{N}$) associated with an unpleasant odour. The alkaline aqueous layer on the other hand, when heated with chloroform and then acidified gave a mixture of two isomeric compounds D and E of molecular formula $\text{C}_7\text{H}_6\text{O}_2$. Identify the compound A, B, C, D and E write their structures.
- Give structures of (A), (B), (X), (Y) in the given scheme of reactions, if reagents a, b, c, d, e and f are $\text{Br}_2/\text{CH}_3\text{COOH}$, $\text{NaNO}_2/\text{H}_2\text{SO}_4$, CuBr , $\text{HNO}_3/\text{H}_2\text{SO}_4$, Br_2/Fe and Sn/HCl respectively.



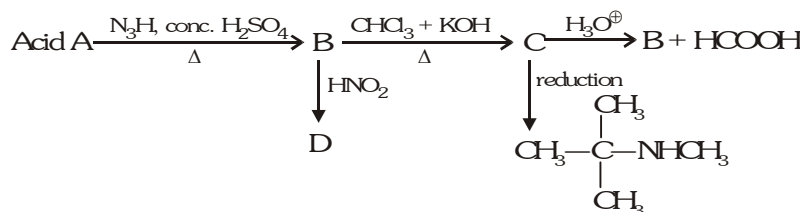
- Explain the mechanism of the following change :



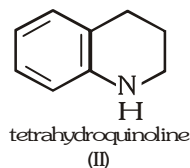
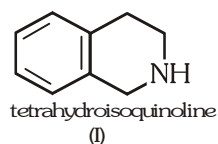
- $$\text{CH}_3\text{CH}_2\text{N}^+(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_3 \xrightarrow{\Delta} \text{A} + \text{B} + \text{C}$$
 Identify A, B (both alkene) and C.

- Write isomeric amines of the formula $\text{C}_3\text{H}_9\text{N}$.

- Identify A, B, C and D.



10. The two amines shown differ by a factor of about 40,000 in their ionisation constants. Which is stronger base? Explain:

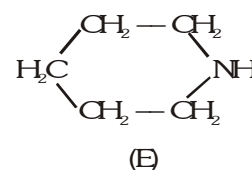
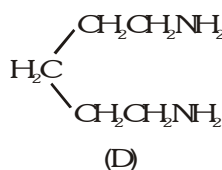
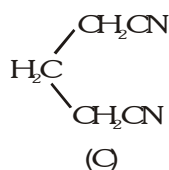
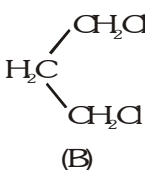
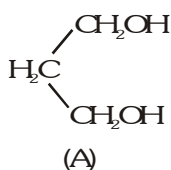


BRAIN STORMING SUBJECTIVE EXERCISE

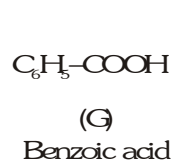
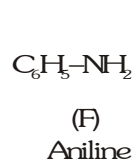
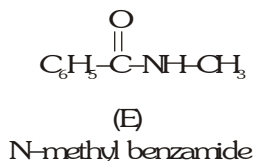
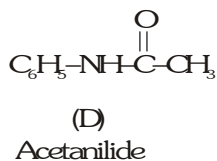
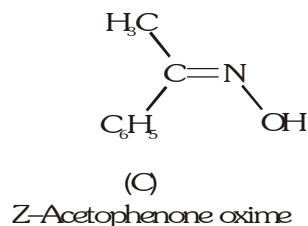
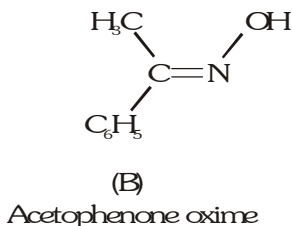
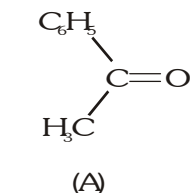
ANSWER KEY

EXERCISE -4(B)

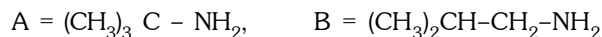
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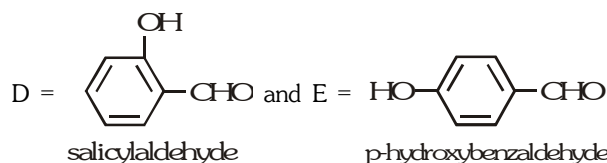
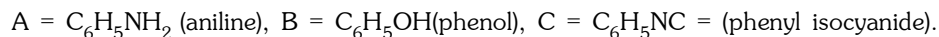
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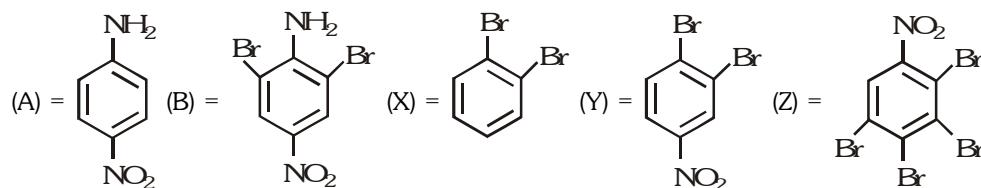
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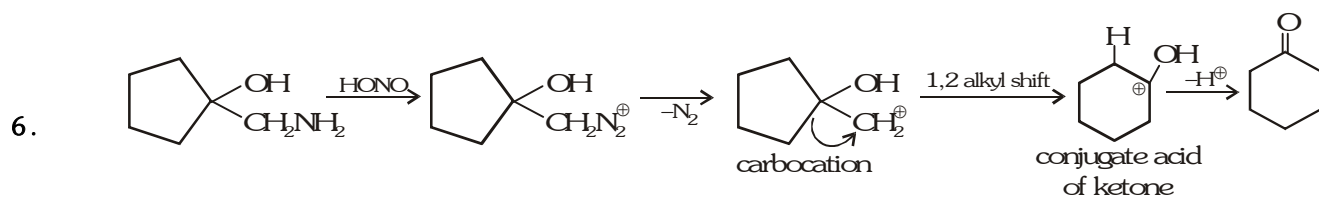


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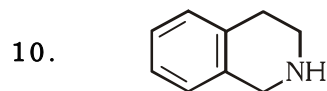
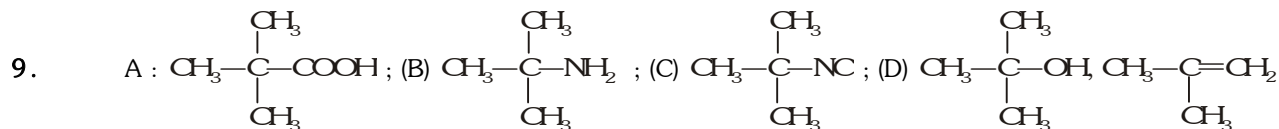
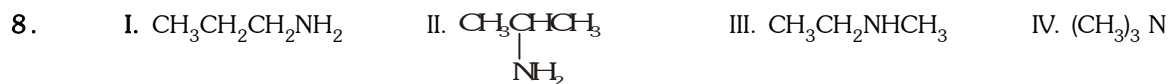
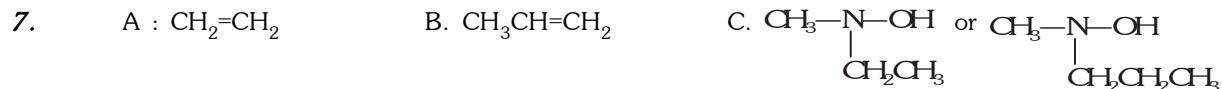


5.





Thus, it involves pinacol-type ring expansion via the carbocation.

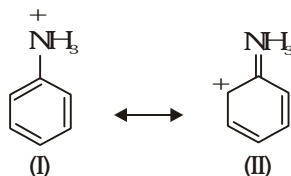


tetrahydroisoquinoline

is a stronger base. In tetrahydroquinoline (II), lone-pair on nitrogen is used in delocalisation of π -electrons of benzene ring hence, basicity of II is decreased. I resembles aliphatic amine II resembles aniline.

EXERCISE-05[A]**PREVIOUS YEARS QUESTIONS**

1. Among the following compounds, which will react with acetone to give a product containing $>C=N-$?
(A) $C_6H_5NH_2$ (B) $(CH_3)_3N$ [IIT-90]
(C) $C_6H_5NHC_6H_5$ (D) $C_6H_5NHNH_2$
2. The compound that will react most readily with NaOH to form methanol is [IIT-90]
(A) $(CH_3)_4N^+I^-$ (B) CH_3OCH_3 (C) $(CH_3)_3S^+I^-$ (D) $(CH_3)_3CCl$
3. Examine the following two structures for the anilinium ion and choose the correct statement from the ones given below : [IIT-93]



- (A) II is not an acceptable canonical structure because carbonium ions are less stable than ammonium ions
(B) II is not an acceptable canonical structure because it is non-aromatic
(C) II is not an acceptable canonical structure because the nitrogen has ten valence electrons
(D) Both (A) & (C)
4. Allyl isocyanide contains σ and π bonds are - [IIT-95]
(A) 9σ and 3π (B) 9σ and 9π (C) 3σ and 4π (D) 5σ and 7π
5. Nitrobenzene can be prepared from benzene by using a mixture of conc. HNO_3 and conc. H_2SO_4 . In the nitrating mixture, nitric acid acts as - [IIT-97]
(A) base (B) acid (C) reducing agent (D) catalyst
6. Among following statements on the nitration of aromatic compounds, the false one is - [IIT-97]
(A) the rate of nitration of benzene is almost the same as that of hexadeuterobenzene
(B) the rate of nitration of toluene is greater than that of benzene
(C) the rate of nitration of benzene is greater than that of hexadeuterobenzene
(D) nitration is an electrophilic substitution reaction.
7. p-Chloroaniline and anilinium hydrochloride can not be distinguished by [IIT-98]
(A) Sandmeyer reaction (B) $NaHCO_3$ (C) $AgNO_3$ (D) Carbylamine test
8. Read the following statement and explanation and answer as per the option given below : [IIT-98]
Statement I : Benzonitrile is prepared by the reaction of chlorobenzene with potassium cyanide
Because
Statement II : Cyanide (CN^-) is a strong nucleophile
(A) If both assertion and reason are correct, and reason is the correct explanation of the assertion
(B) If both assertion and reason are correct, but reason is not correct explanation of the assertion
(C) If assertion is correct but reason is incorrect
(D) If assertion is incorrect but reason is correct
9. Benzenediazonium chloride on reaction with phenol in weakly basic medium gives [IIT-98]
(A) Diphenyl ether (B) p-Hydroxyazobenzene
(C) Chlorobenzene (D) Benzene

10. A positive carbylamine test is given by [IIT-99]
 (A) N, N-Dimethylaniline (B) 2, 4-Dimethylaniline
 (C) N-Methyl-o-methylaniline (D) o-methylbenzylamine

11. Among the following, the strongest base is : [IIT-2000]
 (A) $C_6H_5NH_2$ (B) $p\text{-NO}_2\text{-C}_6\text{H}_4\text{NH}_2$ (C) $m\text{-NO}_2\text{-C}_6\text{H}_4\text{NH}_2$ (D) $C_6H_5CH_2NH_2$

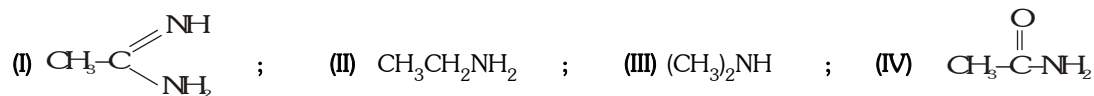
12. Read the following statement and explanation and answer as per the option given below : [IIT-01]

Statement I : In strongly acidic solutions, aniline becomes less reactive towards electrophilic reagents.

Beacuse

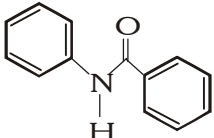
Statement II : The amino group being completely protonated in strongly acidic solution, the lone pair of electrons on the nitrogen is no longer available for resonance.

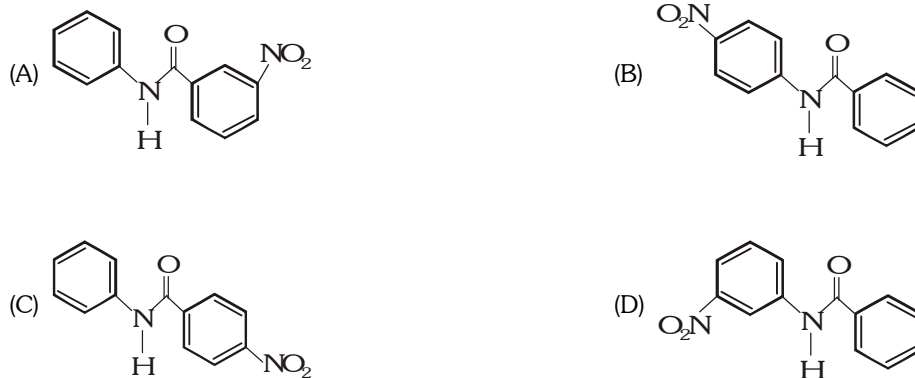
- (A) If both assertion and reason are correct, and reason is the correct explanation of the assertion
 (B) If both assertion and reason are correct, but reason is not correct explanation of the assertion
 (C) If assertion is correct but reason is incorrect
 (D) If assertion is incorrect but reason is correct
13. The correct order of basicities of the following compounds is - [IIT-01]



- (A) II > I > III > IV (B) I > III > II > IV (C) III > I > II > IV (D) I > II > III > IV
14. When benzamide is treated with POCl_3 , the product formed is : [IIT-04]
 (A) Benzonitrile (B) Aniline (C) Chlorobenzene (D) Benzylamine

15. $\text{CH}_3\text{NH}_2 + \text{CHCl}_3 \xrightarrow{\text{KOH}}$ Product, Product is- [IIT-06]
 (A) $\text{CH}_3\text{-N}^{\oplus}\equiv\text{C}^{\ominus}$ (B) $\text{CH}_3\text{-N}^{\ominus}\equiv\text{C}^{\oplus}$ (C) $\text{CH}_3\text{-NH-CH}_3$ (D) $\text{CH}_3\text{-C}\equiv\text{N}$

16. In the following reaction,  $\xrightarrow[\text{conc. H}_2\text{SO}_4]{\text{conc. HNO}_3}$ X, the structure of the major product 'X' is -



[IIT-07]

17. An aromatic compound contains 69.4% carbon and 5.8% hydrogen. A sample of 0.303 g of this compound was analysed for nitrogen by Kjeldahl's method. The ammonia evolved was absorbed in 50ml of 0.05 M sulphuric acid. The excess of acid required 25 ml of 0.1 M sodium hydroxide for neutralization. Determine the molecular formula of the compound if its molecular weight is 121. Draw two possible structures for this compound. [IIT-82]

18. Compound 'X' containing chlorine, on treatment with NH_3 gives a solid 'Y' which is free from chlorine. (Y) analysed as C = 49.31%, H = 9.59% and N = 19.18% and reacts with Br_2 and caustic soda to give a basic compound (Z), (Z) reacts with HNO_2 to give ethanol. Suggest structure for (X), (Y) and (Z).

[IIT-88]

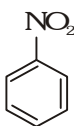
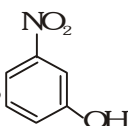
19. Show with equations how the following compounds are prepared (equations need not be balanced) :

(i) 4-nitroaniline to 1, 2,3-tribromobenzene.

[IIT-90]

(ii) Aniline $\longrightarrow$ Benzylamine (in 3 steps)

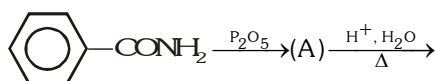
[IIT-2000]

(iii) Convert  to  in not more than four steps.

[IIT-04]

20. Complete the following with appropriate structures :

[IIT-92]

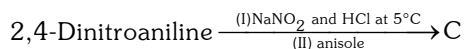


21. The aqueous solution of a nitrogen and chlorine containing compound (A) is acidic to litmus. (A) on treatment with aqueous NaOH gives a compound (B) containing nitrogen, but not chlorine. Compound (B) on treatment with $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$ in presence of NaOH gives an insoluble product (C), $\text{C}_{13}\text{H}_{13}\text{NO}_2\text{S}$. Give structure of (A) and (B).

[IIT-93]

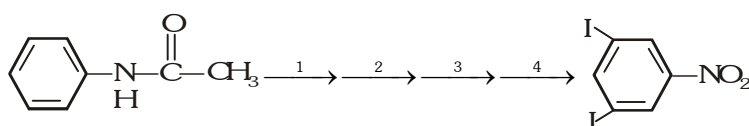
22. Complete the following with appropriate structures :

[IIT-95]



23. Complete the following reaction with appropriate reagents :

[IIT-99]

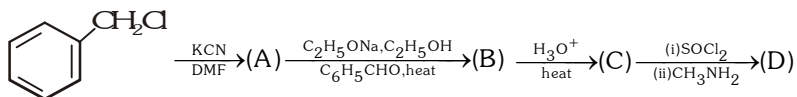


24. Compound (A), $\text{C}_5\text{H}_{11}\text{NO}$ is not soluble in cold alkaline or acidic solutions. When (A) is refluxed in NaOH solution, a gas (B) is evolved and salt (C) is formed. Acetyl chloride reacts with (B) to give (D), ($\text{C}_4\text{H}_9\text{NO}$). (B) reacts with HNO_2 to give a yellow oil (E). Give structures of (A) to (E) with reason.

[IIT-99]

25. Identify (A) to (D) in the following series of reactions :

[IIT - 04]



26. $\text{C}_5\text{H}_{13}\text{N}$ (optically active) $\xrightarrow[\text{-N}_2]{\text{aq. NaNO}_2 / \text{HCl}}$ Y (Tertiary alcohol) + Some other products

[IIT-05]

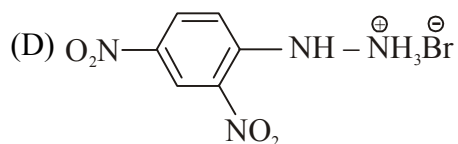
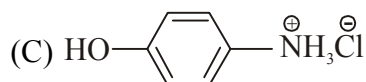
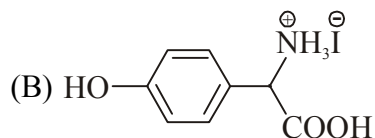
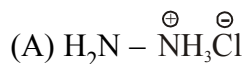
(i) Identify (X) and (Y)

(ii) Is (Y) optically active ?

(iii) Give structure (s) of intermediate (s), if any, in the formation of (Y) from (X).

27. Match the compounds in **Column I** with their characteristic test(s)/reaction(s) given in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS.

Column I



Column II

[IIT 2008]

(P) sodium fusion extract of the compound gives Prussian blue colour with FeSO_4

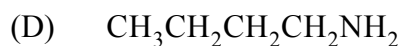
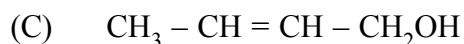
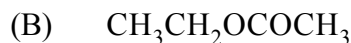
(Q) gives positive FeCl_3 test

(R) gives white precipitate with AgNO_3

(S) reacts with aldehydes to form the corresponding hydrazone derivative

28. Match each of the compound in Column I with its characteristic reaction(s) in Column II.

Column I



Column II

[IIT 2009]

(P) Reduction with $\text{Pd-C} / \text{H}_2$

(Q) Reduction with $\text{SnCl}_2 / \text{HCl}$

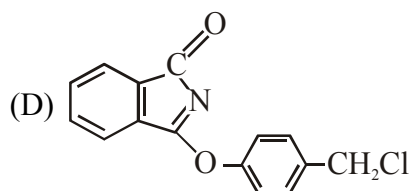
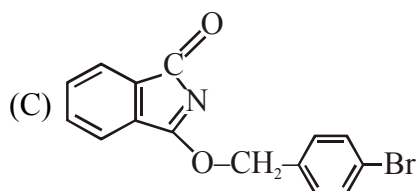
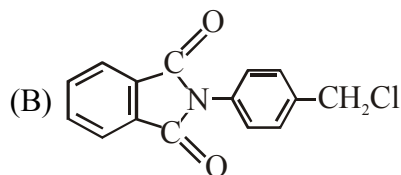
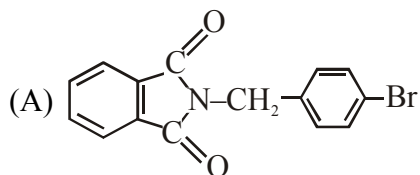
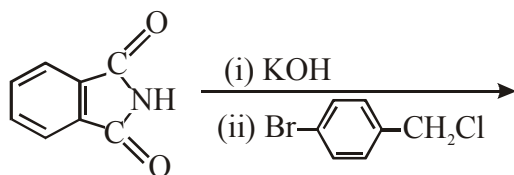
(R) Development of foul smell on treatment with chloroform and alcoholic KOH

(S) Reduction with diisobutylaluminium hydride (DIBAL - H)

(T) Alkaline hydrolysis

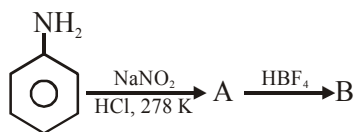
29. The major product of the following reaction is

[IIT 2011]



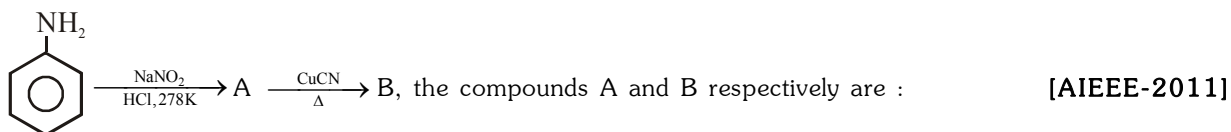
EXERCISE-05[B]**PREVIOUS YEARS QUESTIONS**

1. Reaction - [AIEEE-2002]
Primary amine + $\text{CHCl}_3 + \text{KOH} \rightarrow$ product, here product will be -
(1) Cyanide (2) Isocyanide (3) Amine (4) Alcohol
2. The compound formed in the positive test for nitrogen with the Lassaigne solution of an organic compound is- [AIEEE-2004]
(1) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ (2) $\text{Na}_3[\text{Fe}(\text{CN})_6]$ (3) $\text{Fe}(\text{CN})_3$ (4) $\text{Na}_4[\text{Fe}(\text{CN})_5]\text{NOS}$
3. Which one of the following methods is neither meant for the synthesis nor for separation of amines- [AIEEE-2005]
(1) Hofmann method (2) Hinsberg method (3) Curtius reaction (4) Wurtz reaction
4. In the chemical reaction, $\text{CH}_3\text{CH}_2\text{NH}_2 + \text{CHCl}_3 + 3\text{KOH} \rightarrow (\text{A}) + (\text{B}) + 3\text{H}_2\text{O}$, the compounds (A) and (B) are respectively - [AIEEE-2007]
(1) $\text{C}_2\text{H}_5\text{CN}$ and 3KCl (2) $\text{CH}_3\text{CH}_2\text{CONH}_2$ and 3KCl
(3) $\text{C}_2\text{H}_5\text{NC}$ and K_2CO_3 (4) $\text{C}_2\text{H}_5\text{NC}$ and 3KCl
5. In the chemical reactions, [AIEEE-2010]



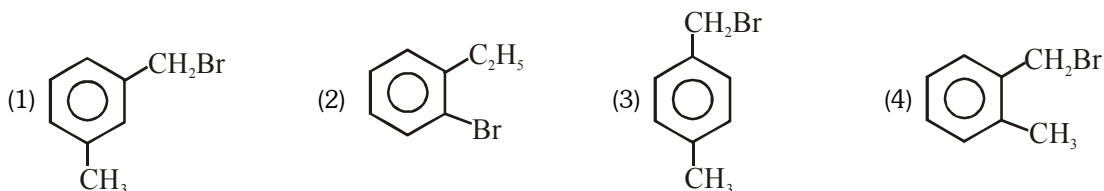
the compounds 'A' and 'B' respectively are :-

- (1) Nitrobenzene and chlorobenzene (2) Nitrobenzene and fluorobenzene
(3) Phenol and benzene (4) Benzene diazonium chloride and fluorobenzene
6. In the chemical reactions



- (1) Fluorobenzene and phenol (2) Benzene diazonium chloride and benzonitrile
(3) Nitrobenzene and chlorobenzene (4) Phenol and bromobenzene

7. Compound (A), $\text{C}_8\text{H}_9\text{Br}$, gives a white precipitate when warmed with alcoholic AgNO_3 . Oxidation of (A) gives an acid (B), $\text{C}_8\text{H}_6\text{O}_4$. (B) easily forms anhydride on heating. Identify the compound (A): [AIEEE-2013]



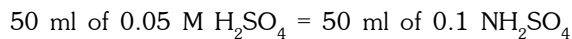
8. An organic compound A upon reacting with NH_3 gives B. On heating, B gives C. C in presence of KOH reacts with Br_2 to give $\text{CH}_3\text{CH}_2\text{NH}_2$. A is :- [AIEEE-2013]

- (1) CH_3COOH (2) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$ (3) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{COOH}$ (4) $\text{CH}_3\text{CH}_2\text{COOH}$

1. (D) 2. (C) 3. (C) 4.(A) 5. (A) 6. (C) 7.(C) 8. (D) 9. (B)

10. (B,D) 11. (D) 12. (A) 13. (B) 14. (A) 15. (A) 16. (B)

17. Calculation of % nitrogen



($\therefore$ Normality of H_2SO_4) = 2 molarity)

Excess of acid requires 25 ml of 0.1 M or 0.1 N NaOH

($\therefore$ Normality of NaOH = molarity of NaOH)

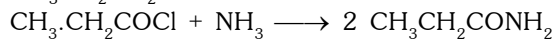
25 ml of 0.1 N NaOH $\equiv$ 25 ml of 0.1 N H_2SO_4

$\therefore$ vol. of 0.1 NH_2SO_4 used for the neutralisation of NH_3 = 50 - 25 = 25 ml

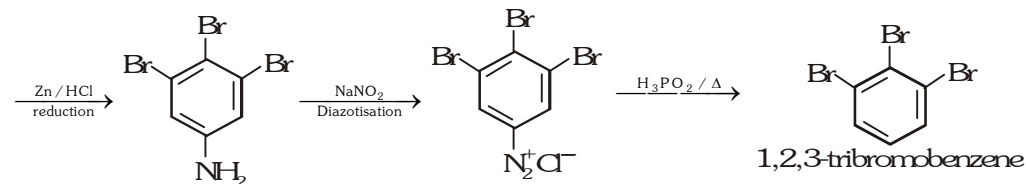
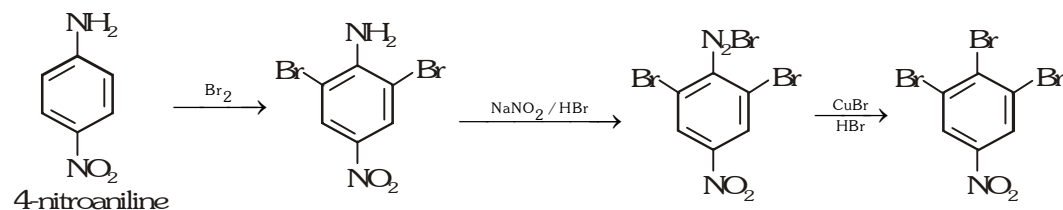
$$\text{Now we know that, \% of nitrogen} = \frac{1.4 \times \text{Normality of acid} \times \text{Vol. of acid}}{\text{Wt. of compound}}$$

$$= \frac{1.4 \times 0.1 \times 25}{0.303} = 11.55\% \quad ; \text{ Hence \% of oxygen} = 100 - (69.4 + 5.8 + 11.55) = 13.25$$

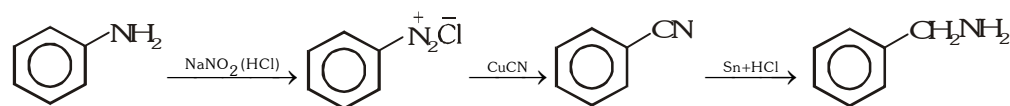
18. Empirical formula of (Y) is $\text{C}_3\text{H}_7\text{NO}$



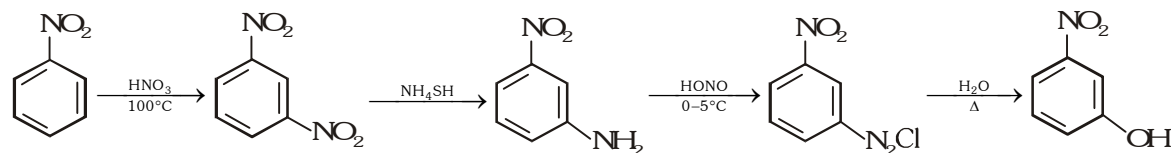
19. (i)



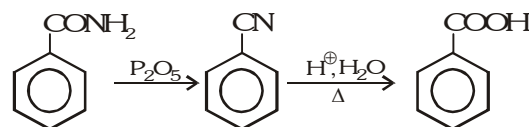
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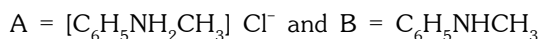
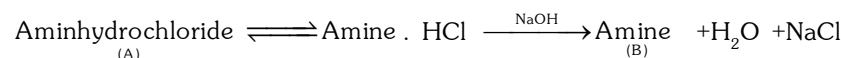
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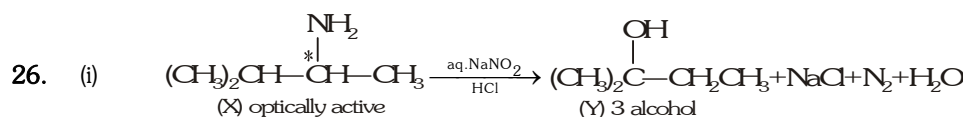
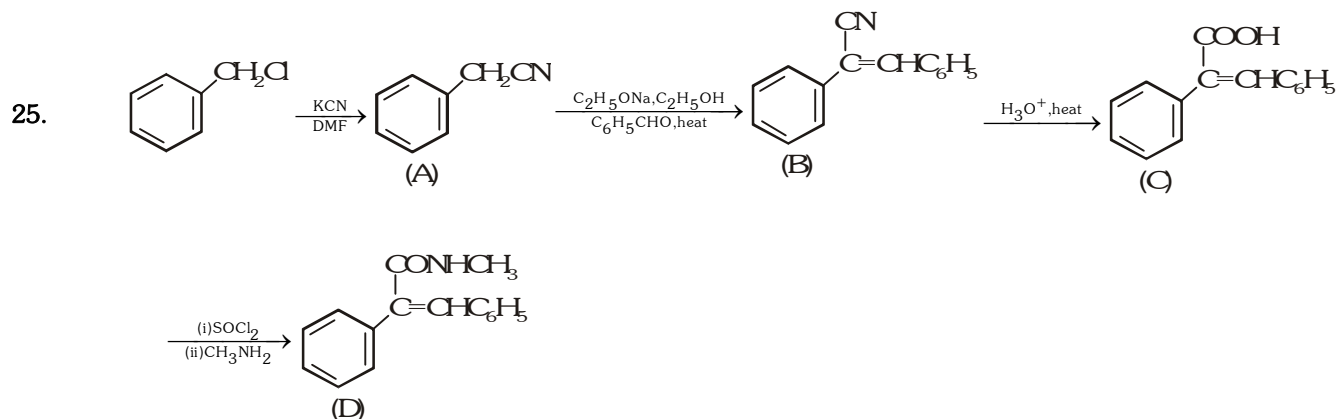
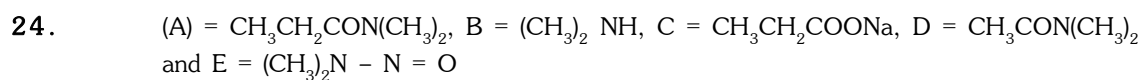
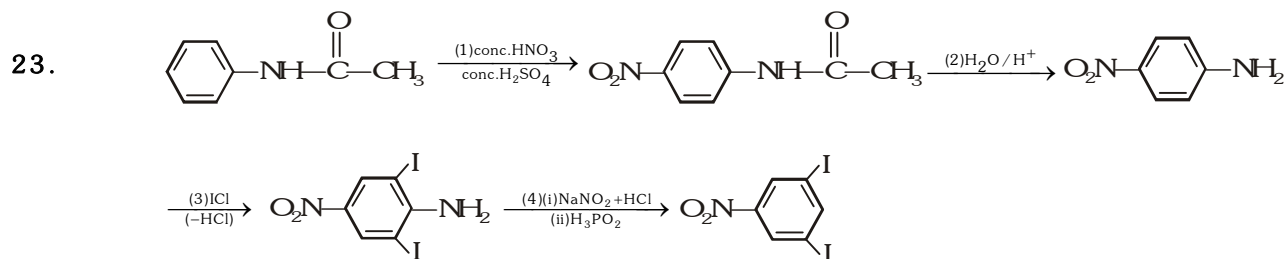
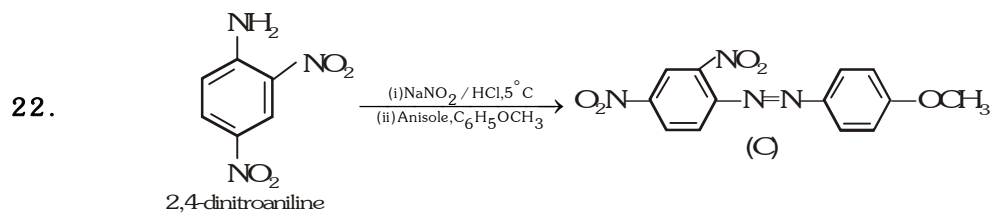


20.



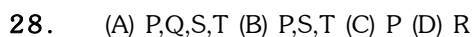
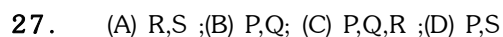
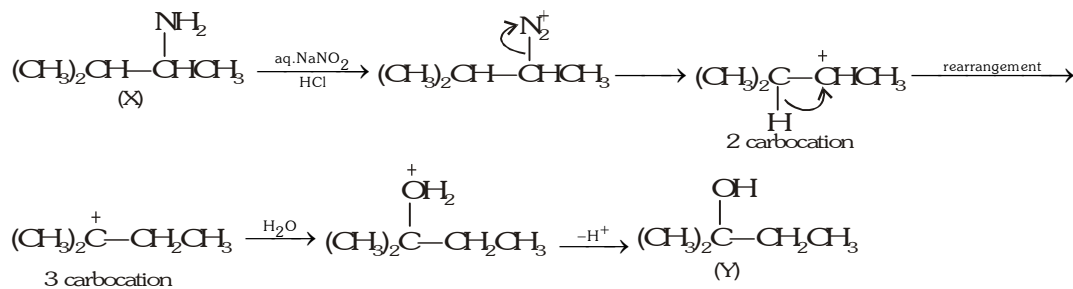
21.





(ii) [Y], a 3 alcohol is optically inactive

(iii) Formation of [Y] from [X]

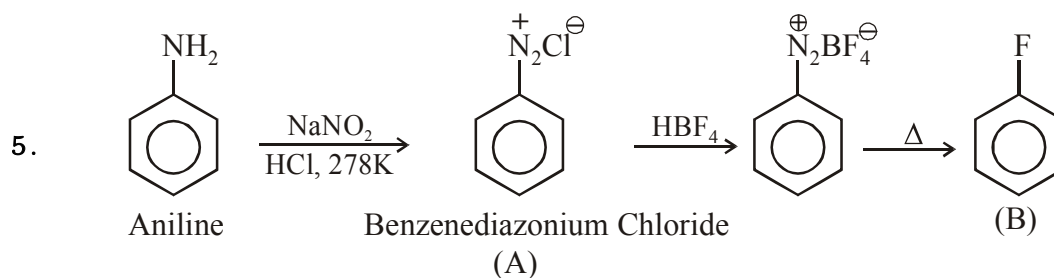


1. 2 2. 1 3. 4 4. 4 5. 4 6. 2
 7. 4 8. 4

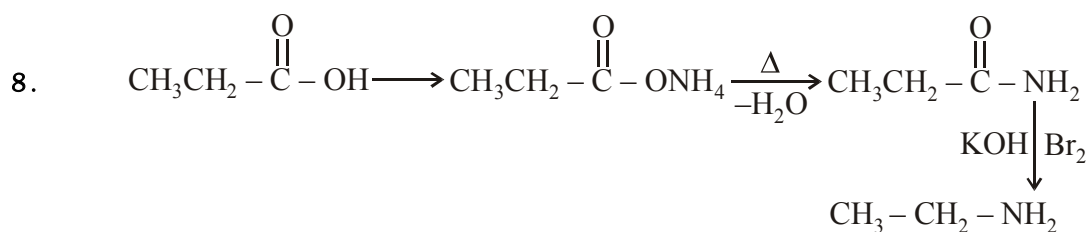
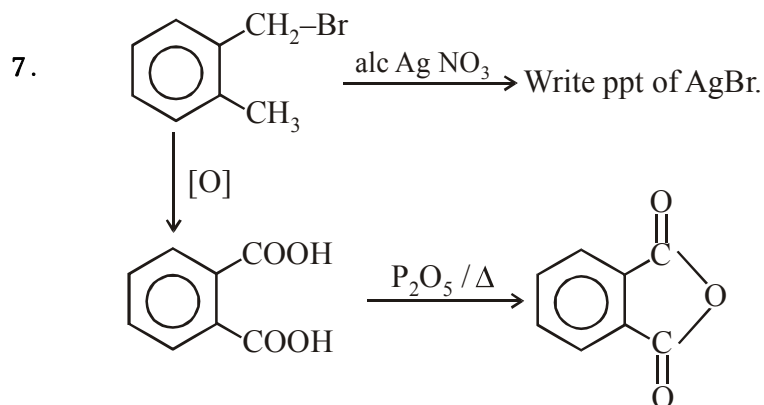
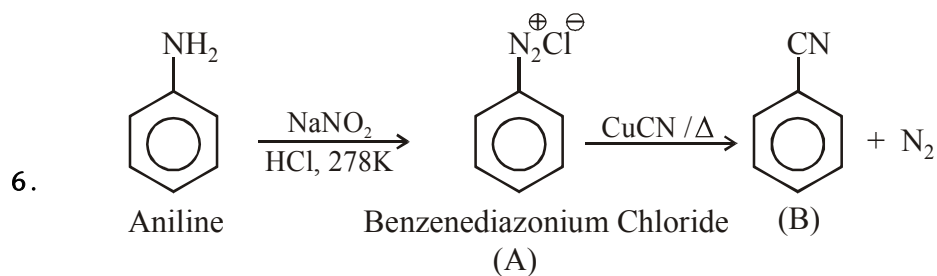
1. Carbylamine Test

3. Wurtz reaction $R \xrightarrow[\text{dry ether}]{Na} R-R + NaX$

4. Isocyanide test (carbylamine test). Pungent smelling isocyanide is formed as a major product.



Sandmeyer's Reaction



EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. The molarity of a glucose solution containing 36 g of glucose per 400 mL of the solution is:
(A) 1.0 (B) 0.5 (C) 2.0 (4) 0.05
2. 1 kg of NaOH solution contains 4g of NaOH. The approximate concentration of the solution is:
(A) 0.1 molar (B) 0.1 molal (C) Decinormal (D) About 0.1 N
3. To prepare 0.1 M KMnO_4 solution in 250 mL flask, the weight of KMnO_4 required is:
(A) 4.80g (B) 3.95g (C) 39.5g (D) 0.48 g
4. The number of moles present in 2 litre of 0.5 M NaOH is:
(A) 2 (B) 1 (C) 0.1 (D) 0.5
5. The weight of solute present in 200 mL of 0.1 M H_2SO_4 :
(A) 2.45g (B) 4.9g (C) 1.96g (D) 3.92 g
6. The nature of mixture obtained by mixing 50 mL of 0.1 M H_2SO_4 and 50 mL of 0.1 M NaOH is:
(A) Acidic (B) Basic (C) Neutral (D) Amphoteric
7. If 250 mL of a solution contains 24.5g H_2SO_4 the molarity and normality respectively are:
(A) 1M, 2N (B) 1M, 0.5M (C) 0.5M, 1N (D) 2M, 1N
8. The volume strength of H_2O_2 solution is 10. What does it mean :
(A) at S.T.P. 10 g solution of H_2O_2 gives 10 mL of O_2
(B) at S.T.P. 1 g equivalent of H_2O_2 gives 10 mL of O_2
(C) at S.T.P. 10 litre solution of H_2O_2 gives 10 mL of O_2
(D) at S.T.P. 1 mL solution of H_2O_2 gives 10 mL of O_2
9. The normality of 0.3 M phosphorus acid (H_3PO_3) is:
(A) 0.1 (B) 0.9 (C) 0.3 (D) 0.6
10. The normality of 4% (wt./vol.) NaOH is:
(A) 0.1 (B) 1.0 (C) 0.05 (D) 0.01
11. The density of NH_4OH solution is 0.6 g/mL. It contains 34% by weight of NH_4OH . Calculate the normality of the solution :
(A) 4.8 N (B) 10 N (C) 0.5 N (D) 5.8 N
12. A molal solution is one that contains one mole of a solute in :
(A) 1000 g of the solution (B) 1000 c.c. of the solution
(C) 1000 c.c of the solvent (D) 1000 g of the solvent
13. Out of molarity (M), molality (m), formality (F) and mole fraction (x) those independent of temperature are :
(A) M, m (B) F, x (C) m, x (D) M, x
14. 3.0 molal NaOH solution has a density of 1.110 g/mL. The molarity of the solution is:
(A) 2.9732 (B) 3.05 (C) 3.64 (D) 3.0504
15. 1000 gram aqueous solution of CaCO_3 contains 10 gram of carbonate. Concentration of solution is:
(A) 10ppm (B) 100ppm (C) 1000ppm (D) 10,000 ppm
16. When 5.0 gram of BaCl_2 is dissolved in water to have 10^6 gram of solution. The concentration of solution is :
(A) 2.5 ppm (B) 5 ppm (C) 5M (D) 5 g L^{-1}
17. How many grams of glucose be dissolved to make one litre solution of 10% glucose :
(A) 10g (B) 180g (C) 100g (D) 1.8g

18. Vapour pressure of a solvent containing nonvolatile solute is :
 (A) more than the vapour pressure of a solvent (B) less than the vapour pressure of solvent
 (C) equal to the vapour pressure of solvent (D) none
19. The relative lowering in vapour pressure is:
 (A) $\propto X_{\text{solute}}$ (B) $\propto \frac{1}{X_{\text{solute}}}$ (C) $= X_{\text{solute}}$ (D) $\propto m$
20. The vapour pressure of a dilute solution of a solute is not influenced by :
 (A) temperature of solution (B) melting point of solute
 (C) mole fraction of solute (D) degree of dissociation of solute
21. An aqueous solution of methanol in water has vapour pressure :
 (A) equal to that of water (B) equal to that of methanol
 (C) more than that of water (D) less than that of water
22. When a substance is dissolved in a solvent, the vapour pressure of solvent decreases. This brings:
 (A) an increase in b.pt. of the solution (B) a decrease in b.pt of a solution
 (C) an increase in f.pt of the solvent (D) none
23. Solute when dissolved in water:
 (A) increases the vapour pressure of water (B) decreases the boiling point of water
 (C) decreases the freezing point of water (D) all of the above
24. If the vapour pressure of solutions of two liquids are less than those expected from ideal solution they are said to have :
 (A) negative deviation from ideal behaviour
 (B) positive deviations from ideal behaviour
 (C) ideal behaviour
 (D) positive deviation for lower concentration and negative deviations for higher concentration
25. A 5.8% solution of NaCl has vapour pressure closest to :
 (A) 5.8 % solution of urea (B) 2 m solution of glucose
 (C) 1 m solution of urea (D) 5.8 % solution of glucose
26. The boiling point of C_6H_6 , CH_3OH , $C_6H_5NH_2$ and $C_6H_5NO_2$ are 80 C, 65 C, 184 C and 212 C respectively. Which will show highest vapour pressure at room temperature:
 (A) C_6H_6 (B) CH_3OH (C) $C_6H_5NH_2$ (D) $C_6H_5NO_2$
27. Boiling point of water is defined as the temperature at which :
 (A) vapour pressure of water equal to that of atmospheric pressure
 (B) bubbles are formed
 (C) steam comes out
 (D) none of the above
28. Which solution will show maximum elevation in b.pt:
 (A) 0.1 M KCl (B) 0.1 M $BaCl_2$ (C) 0.1 M $FeCl_3$ (D) 0.1 M $Fe_2(SO_4)_3$
29. The correct relationship between the boiling points of very dilute solutions of $AlCl_3$ (t_1) and $CaCl_2$ (t_2) having the same molar concentration is :
 (A) $t_1 = t_2$ (B) $t_1 > t_2$ (C) $t_2 > t_1$ (D) $t_2 \geq t_1$
30. Cryoscopic constant of a liquid is:
 (A) decrease in freezing point when 1 gram of solute is dissolved per kg of the solvent
 (B) decrease in the freezing point when 1 mole of solute is dissolved per kg of the solvent
 (C) the elevation for 1 molar solution
 (D) a factor used for calculation of elevation in boiling point

31. At certain Hill-station pure water boils at 99.725 C. If K_b for water is 0.513 C kg mol⁻¹, the boiling point of 0.69 m solution of urea will be :
 (A) 100.079 C (B) 103 C (C) 100. 359 C (D) un predictable
32. The freezing point of 1 molal NaCl solution assuming NaCl to be 100% dissociated in water is :
 (A) -1.86 C (B) -3.72 C (C) +1.86 C (D) +3.72 C
33. 10 gram of solute with molecular mass 100 gram mol⁻¹ is dissolved in 100 gram solvent to show 0.3 C elevation in boiling point. The value of molal ebullioscopic constant will be :
 (A) 10 (B) 3 (C) 0.3 (D) un predictable
34. Depression in freezing point of solution of electrolytes are generally:
 (A) lower
 (B) higher than what should be normally
 (C) low or high depending upon nature of electrolyte
 (D) what it should be normally
35. A liquid is in equilibrium with its vapour at its boiling point. On the average the molecules in the two phase have equal:
 (A) inter-molecular forces (B) potential energy (C) total energy (D) kinetic energy
36. Which salt may show the same value of vant Hoff factor (i) as that of $K_4Fe(CN)_6$ in very dilute solution state :
 (A) $Al_2(SO_4)_3$ (B) NaCl (C) $Al(NO_3)_3$ (D) Na_2SO_4
37. Which compound corresponds vant Hoff factor (i) to be equal to 2 in dilute solution:
 (A) K_2SO_4 (B) $NaHSO_4$ (C) Sugar (D) $MgSO_4$
38. In which of the following, the vant Hoff factor (i) is equal to one:
 (A) NaCl (B) KNO_3 (C) Urea (D) all
39. If the observed and theoretical molecular mass of NaCl is found to be 31.80 and 58.50, then the degree of dissociation of NaCl is :
 (A) 83.96% (B) 8.39% (C) 90% (D) 100%
40. The substance A when dissolved in solvent B shows the molecular mass corresponding to A_3 . The vant Hoff's factor will be:
 (A) 1 (B) 2 (C) 3 (D) 1/3
41. Which of the following conditions is not correct for ideal solution :
 (A) no change in volume on mixing (B) no change in enthalpy on mixing
 (C) it obey's Raoult's law
 (D) lonisation of solute should occurs to a small extent
42. Solutions distilled without change in composition at a temperature are called :
 (A) Amorphous (B) Azeotropic mixture
 (C) Ideal solution (D) Super saturated solution
43. If mole fraction of the solvent in a solution decreases then :
 (A) vapour pressure of solution increases (B) b.pt decreases
 (C) osmotic pressure increases (D) all are correct
44. An azeotropic solution of two liquids has boiling point lower than either of them when it:
 (A) shows a negative deviation from Raoult's law (B) shows no deviation from Raoult's law
 (C) shows positive deviation from Raoult's law (D) is saturated
45. The passing of particles through semipermeable membrane is called :
 (A) osmosis (B) electrodialysis (C) electrophoresis (D) electroplating

46. From the colligative properties of solution which one is the best method for the determination of mol. wt of proteins & polymers :
- (A) osmotic pressure (B) lowering in vapour pressure
(C) lowering in freezing point (D) elevation in B.Pt
47. As a result of osmosis, the volume of the concentrated solution :
- (A) gradually decreases (B) gradually increases
(C) suddenly increases (D) none
48. The osmotic pressure of a solution of benzoic acid dissolved in benzene is less than expected because:
- (A) benzoic acid is an organic solute (B) benzene is a non-polar solvent
(C) benzoic acid dissociates in benzene (D) benzoic acid gets associated in benzene
49. Two solutions have different osmotic pressures. The solution of higher osmotic pressure is called:
- (A) isotonic solution (B) hypotonic solution
(C) isotopic solution (D) hypertonic solution
50. Blood is isotonic with :
- (A) 0.16 M NaCl (B) Conc.NaCl (C) 30% NaCl (D) 50% NaCl
51. Which one of the following pairs of solution can we expect to be isotonic at the same temperature:
- (A) 0.1 M urea and 0.1 M NaCl (B) 0.1 M urea and 0.2 M MgCl_2
(C) 0.1 M NaCl and 0.1 M Na_2SO_4 (D) 0.1 M $\text{Ca}(\text{NO}_3)_2$ and 0.1 M Na_2SO_4
52. A 5% solution of cane sugar is isotonic with 0.877 % of X. The molecular weight of substance X is:
- (A) 58.98 (B) 119.96 (C) 95.58 (D) 126.98
53. Which statement is incorrect about osmotic pressure (π), volume (V) and temperature (T):
- (A) $\pi \propto \frac{1}{V}$ if T is constant (B) $\pi \propto T$ if V is constant
(C) $\pi \propto V$ if T is constant (D) πV is constant if T is constant
54. The osmotic pressure of equimolar solutions of urea, BaCl_2 and AlCl_3 will be in the order:
- (A) $\text{AlCl}_3 > \text{BaCl}_2 > \text{Urea}$ (B) $\text{BaCl}_2 > \text{AlCl}_3 > \text{Urea}$
(C) $\text{Urea} > \text{BaCl}_2 > \text{AlCl}_3$ (D) $\text{BaCl}_2 > \text{Urea} > \text{AlCl}_3$

CHECK YOUR GRASP						ANSWER KEY				EXERCISE -1					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	B	B	B	B	C	A	A	D	D	B	D	D	C	A	D
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	B	C	B	C	B	C	A	C	A	B	B	A	D	B	B
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	A	B	C	C	D	A	D	C	A	D	D	B	C	C	A
Que.	46	47	48	49	50	51	52	53	54						
Ans.	A	B	D	D	A	D	A	C	A						

EXERCISE-02**BRAIN TEASERS****SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)**

- When 0.6 g of urea dissolved in 100 g of water, the water will boil at (K_b for water = $0.52 \text{ kJ. mol}^{-1}$ and normal boiling point of water = 100°C) :
 (A) 373.052 K (B) 273.52 K (C) 372.48 K (D) 273.052 K
- Solutions having the same osmotic pressure are called :
 (A) isotonic solution (B) molar solutions (C) hypotonic solutions (D) ideal solutions
- Consider 1 M solutions of the following salts. State which solution will have the lowest freezing point.
 (A) Na_2SO_4 (B) BaCl_2 (C) NaCl (D) $\text{Al}_2(\text{SO}_4)_3$
- A solution prepared by dissolving a 2.50 g sample of an unknown compound dissolved in 34.0 g of benzene, C_6H_6 boils at 1.38°C higher than pure benzene. Which expression gives the molar mass of the unknown compound ?

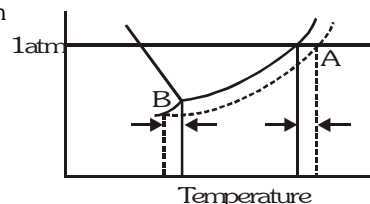
Compound	K_b
C_6H_6	$2.53^\circ\text{C.m}^{-1}$

 (A) $2.53 \frac{2.50}{1.38}$ (B) $1.38 \frac{34.0}{2.53} \times 2.50$
 (C) $2.50 \cdot 10^3 \frac{2.53}{34.0} \times \frac{1}{1.38}$ (D) $2.50 \cdot 10^3 \frac{1.38}{34.0} \cdot 2.53$
- When 1.20 g of sulphur is melted with 15.00 g of naphthalene, the solution freezes at 77.2°C . What is the molar mass of this from of sulphur. Data for Naphthalene
 Melting point, m.p 80°C
 Freezing point depression constant, $k_f = 6.80^\circ\text{C m}^{-1}$
 (A) 180 g mol^{-1} (B) 194 g mol^{-1} (C) 260 g mol^{-1} (D) 450 g mol^{-1}
- 12.2 gm benzoic acid ($M = 122$) in 100 g H_2O has elevation of boiling point of 0.27°C , $K_b = 0.54^\circ\text{C kg/mol}$. If there is 100% dimerization, the no. of molecules of benzoic acid in associated state is :
 (A) 1 (B) 2 (C) 3 (D) 4
- The Van't Hoff factor 0.1 M $\text{La}(\text{NO}_3)_3$ solution is found to be 2.74 the percentage dissociation of the salt is :
 (A) 85 % (B) 58 % (C) 65.8% (D) 56.8%
- Maximum freezing point will be for 1 molal solution of (assuming equal ionisation in each case) :
 (A) $[\text{Fe}(\text{H}_2\text{O})_6 \text{Cl}_3]$ (B) $[\text{Fe}(\text{H}_2\text{O})_5 \text{Cl}] \text{Cl}_2 \cdot \text{H}_2\text{O}$ (C) $[\text{Fe}(\text{H}_2\text{O})_4 \text{Cl}_2] \text{Cl} \cdot 2\text{H}_2\text{O}$ (D) $[\text{Fe}(\text{H}_2\text{O})_3 \text{Cl}_3] \cdot 3\text{H}_2\text{O}$
- 1.0 molal aqueous solution of an electrolyte X_3Y_2 is 25% ionized. The boiling point of the solution is (K_b for $\text{H}_2\text{O} = 0.52^\circ\text{C kg/mol}$) :
 (A) 375.5 K (B) 374.04 K (C) 377.12 K (D) 373.25 K
- Which one of the following aqueous solution has the highest freezing point at 1 atm :
 (A) 0.1 M urea (B) 0.1 M acetic acid (C) 0.1 M NaCl (D) 0.1 M BaCl_2
- If in solvent, n simple molecules of solute combine to form an associated molecule, X is degree of association the Van't Hoff's factor 'i' is equal to :

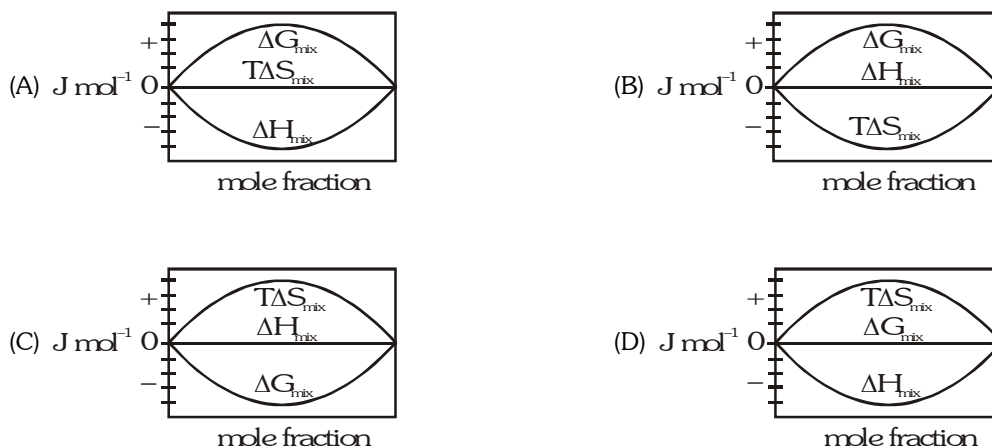
(A) $\frac{1}{1-nx}$	(B) $\frac{1-x+nx}{1}$	(C) $\frac{1-x+\frac{x}{n}}{1}$	(D) $\frac{\frac{x}{n}-1+x}{1}$
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- The decrease in the freezing point of an aqueous solution of a substance is 1.395 K and that in the freezing point of benzene solution of the same substance is 1.280 K . Explain the difference in ΔT . The substance :
 (A) dissociates in the aqueous solution as well as in the benzene solution
 (B) forms complexes in solution
 (C) associates in the benzene solution
 (D) dissociates in the aqueous solution and not in the benzene solution

13. The molal boiling point constant of water is $0.573\text{ }^{\circ}\text{C kg mole}^{-1}$. When 0.1 mole of glucose is dissolved in 1000 g of water, the solution boils under atmospheric pressure at :
 (A) 100.513 C (B) 100.0573 C (C) 100.256 C (D) 101.025 C
14. A 0.2 molal aqueous solution of a weak acid HX is 20% ionized. The freezing point of the solutions is ($k_f = 1.86\text{ K kg mole}^{-1}$ for water) :
 (A) $-0.45\text{ }^{\circ}\text{C}$ (B) $-0.9\text{ }^{\circ}\text{C}$ (C) $-0.31\text{ }^{\circ}\text{C}$ (D) $-0.53\text{ }^{\circ}\text{C}$
15. The boiling point of an azeotropic mixture of water and ethyl alcohol is less than that of theoretical value of water and alcohol mixture. Hence the mixture shows :
 (A) that solution is highly saturated (B) positive deviation from Raoult's law
 (C) negative deviation from Raoult's law (D) none of these
16. The boiling point of an aqueous solution of a non-volatile solute is $100.15\text{ }^{\circ}\text{C}$. What is the freezing point of an aqueous solution obtained by diluting the above solution with an equal volume of water. The values of K_b and K_f for water are 0.512 and $1.86\text{ }^{\circ}\text{C mol}^{-1}$:
 (A) $-0.544\text{ }^{\circ}\text{C}$ (B) $-0.512\text{ }^{\circ}\text{C}$ (C) $-0.272\text{ }^{\circ}\text{C}$ (D) $-1.86\text{ }^{\circ}\text{C}$
17. An aqueous solution of NaCl freezes at $-0.186\text{ }^{\circ}\text{C}$. Given that $K_{b(\text{H}_2\text{O})} = 0.512\text{ }^{\circ}\text{C kg mol}^{-1}$ and $K_{f(\text{H}_2\text{O})} = 1.86\text{ }^{\circ}\text{C kg mol}^{-1}$, the elevation in boiling point of this solution is :
 (A) $0.0585\text{ }^{\circ}\text{C}$ (B) $0.0512\text{ }^{\circ}\text{C}$ (C) $1.864\text{ }^{\circ}\text{C}$ (D) $0.0265\text{ }^{\circ}\text{C}$
18. The Van't Hoff factors i for an electrolyte which undergoes dissociation and association in solvents are respectively :
 (A) greater than one and less than one (B) less than one and greater than one
 (C) less than one and less than one (D) greater than one and greater than one
19. A solution of 0.450 g of urea (mol. wt. 60) in 22.5 g of water showed $0.170\text{ }^{\circ}\text{C}$ of elevation in boiling point, the molal elevation constant of water :
 (A) 0.51 (B) 0.95 (C) 0.25 (D) 2.25
20. Colligative properties of the solution depend on :
 (A) Nature of solute (B) Nature of solvent
 (C) Number of particles present in the solution (D) Number of moles of solvent only
21. Which of the following solutions will have highest boiling point :
 (A) 1% glucose in water (B) 1% sucrose in water (C) 1% NaCl in water (D) 1% urea in water
22. In cold countries, ethylene glycol is added to water in the radiators of cars during winters. It results in :
 (A) reducing viscosity (B) reducing specific heat
 (C) reducing freezing point (D) reducing boiling point
23. An azeotropic solution of two liquids has a boiling point lower than either of them when it :
 (A) shows negative deviation from Raoult's law (B) shows no deviation from Raoult's law
 (C) shows positive deviation from Raoult's law (D) is saturated
24. When mercuric iodide is added to an aqueous solution of potassium iodide the :
 (A) freezing point is raised (B) freezing point is lowered
 (C) freezing point does not change (D) boiling point does not change
25. For an ideal solution containing a nonvolatile solute, which of the following expressions represents the vapour pressure of the solution? ($x_1 \rightarrow$ mole fraction of solvent)
 (A) $p = x_2 p_2^*$ (B) $p = x_1 p_2^*$ (C) $p = x_1 p_1^*$ (D) $p_1^* - p = x_2$
26. For a dilute solution containing a nonvolatile solute, the molar mass of solute evaluated from the elevation of boiling point is given by the expression :
 (A) $M_2 = \frac{\Delta T_b \cdot m_1}{K_b \cdot m_2}$ (B) $M_2 = \frac{\Delta T_b \cdot m_2}{K_b \cdot m_1}$ (C) $M_2 = \frac{K_b \cdot m_2}{\Delta T_b \cdot m_1}$ (D) $M_2 = \frac{K_b \cdot m_1}{\Delta T_b \cdot m_2}$

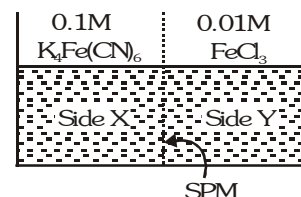
27. For a dilute solution containing a nonvolatile solute, the molar mass of solute evaluated from the osmotic pressure measurement is given as :
- (A) $M_2 = \frac{m_2 RT}{V \pi}$ (B) $M_2 = \frac{m_2 \pi}{V RT}$ (C) $M_2 = m_2 \frac{RT}{\pi}$ (D) $M_2 = m_2 \frac{\pi}{RT}$
28. An aqueous solution of acetone, CH_3COCH_3 , is 10.00% acetone by weight. What is the mole percentage of acetone in this solution :
- (A) 3.332 % (B) 5.000 % (C) 10.00 % (D) 11.11 %
29. The freezing point of an aqueous solution of a non-electrolyte is -0.14°C . The molarity of this solution is $[K_f(\text{H}_2\text{O}) = 1.86 \text{ K kg mol}^{-1}]$:
- (A) 1.86 m (B) 1.00 m (C) 0.15 m (D) 0.075 m
30. The boiling point of a 0.1 M solution of CaCl_2 should be elevated by :
- (A) exactly 0.51 (B) somewhat less than 1.02
(C) exactly 1.02 (D) some what less than 1.53
31. Of the following measurements the one most suitable for the determination of the molecular weight of oxyhaemoglobin, a molecule with a molecular weight of many thousand, is :
- (A) the vapour pressure lowering (B) the elevation of the boiling point
(C) the depression of the freezing point (D) the osmotic pressure
32. The vapour pressure of pure benzene at 50°C is 268 torr. How many mol of non-volatile solute per mol of benzene is required to prepare a solution of benzene having a vapour pressure of 167 torr at 50°C :
- (A) 0.377 (B) 0.605 (C) 0.623 (D) 0.395
33. If P° the vapour pressure of a pure solvent and P is the vapour pressure of the solution prepared by dissolving a non volatile solute in it. The mole fraction of the solvent X_A is given by :
- (A) $\frac{P^\circ - P}{P^\circ} = X_A$ (B) $\frac{P^\circ - P}{P} = X_A$ (C) $\frac{P}{P^\circ} = X_A$ (D) $P^\circ - P = X_A$
34. Dry air was passed successively through a solution of 5 g of a solute in 180 g of water and then through pure water. The loss in weight of solution was 2.5 g and that of pure solvent 0.04 g. The molecular weight of the solute is :
- (A) 31.25 (B) 3.125 (C) 312.5 (D) None of these
35. The relative lowering of the vapour pressure is equal to the ratio between the number :
- (A) solute molecules to the solvent molecules
(B) solute molecules to the total molecules in the solution
(C) solvent molecules to the total molecules in the solution
(D) solvent molecules to the total number of ions of the solute
36. The vapour pressure of pure liquid solvent A is 0.80 atm when a non volatile solute B is added to the solvent its vapours pressure falls to 0.60 atm. Mole fraction of solute B in the solution is :
- (A) 0.50 (B) 0.25
(C) 0.75 (D) given data is not sufficient
37. Which of the following plots represents an ideal binary mixture ?
- (A) plot of P_{total} v/s $1/X_B$ is linear (X_B = mole fraction of 'B' in liquid phase)
(B) plot of P_{total} v/s Y_A is linear (Y_A = mole fraction of 'A' in vapour phase)
(C) plot of $\frac{1}{P_{\text{total}}}$ v/s Y_A is linear
(D) plot of $\frac{1}{P_{\text{total}}}$ v/s Y_B is non linear



38. The lowering of vapour pressure in a saturated aq. solution of salt AB is found to be 0.108 torr. If vapour pressure of pure solvent at the same temperature is 300 torr, find the solubility product of salt AB:
 (A) 10^{-8} (B) 10^{-6} (C) 10^{-4} (D) 10^{-5}
39. Which of the following represents correctly the changes in thermodynamic properties during the formation of 1 mol of an ideal binary solution :



40. FeCl_3 on reaction with $\text{K}_4[\text{Fe}(\text{CN})_6]$ in aqueous solution gives blue colour. These are separated by a semipermeable membrane AB as shown. Due to osmosis there is :
 (A) blue colour formation in side X.
 (B) blue colour formation in side Y.
 (C) blue colour formation in both of the sides X and Y.
 (D) no blue colour formation.



41. $P_A = (235y - 125xy)$ mm of Hg. P_A is partial pressure of A, x is mole fraction of B in liquid phase in the mixture of two liquids A and B and y is the mole fraction of A in vapour phase, then P_B in mm of Hg is :
 (A) 235 (B) 0 (C) 110 (D) 125

BRAIN TEASERS						ANSWER KEY				EXERCISE -2					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	A	A	D	C	B	B	B	D	B	A	C	D	B	A	B
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	C	B	A	A	C	C	C	C	A	C	C	A	A	D	D
Que.	31	32	33	34	35	36	37	38	39	40	41				
Ans.	D	B	C	A	B	B	C	C	C	D	C				

EXERCISE-03

MISCELLANEOUS TYPE QUESTIONS

TRUE / FALSE

1. Relative lowering of vapour pressure is a colligative property.
2. The components of an azeotropic solution can be separated by simple distillation.
3. Addition of non-volatile solute to water always lowers its vapour pressure.
4. Reverse osmosis is generally used to make saline water fit for domestic use.
5. A 6% solution of NaCl should be isotonic with 6% solution of sucrose.
6. On diluting solution, its normality and molarity changes but molality remains constant.
7. The unit of k_b is $\text{kg K}^{-1} \text{mol}^{-1}$.
8. The value of K_b or K_f depends only on the type of solvent & not solute dissolved in it.
9. Limiting value of van't Hoff factor of $[\text{K}_4\text{Fe}(\text{CN})_6]$ is 11.
10. The increasing order of osmotic pressure of 0.1 M aqueous solution containing different electrolyte is as follows
 $0.1 \text{ M Glucose} < 0.1 \text{ M sodium chloride} < 0.1 \text{ M magnesium chloride}$.

FILL IN THE BLANKS

1. Lowering of vapour pressure is to the mole fraction of the solute.
2. The ratio of the value of any colligative property for NaCl solution of that of equimolal solution of sugar is nearly
3. Semipermeable membrane allows the passage of through it.
4. A binary solution which has same composition in liquid as well as vapour phase is called
5. The number of urea molecules in 1 litre of 0.5 M solution
6. 0.1 M solution of urea would be with 0.1 M solution of NaCl. (hypotonic / hypertonic / isotonic)
7. The value of k_f depends on nature of
8. Among 0.1 M solution of NaCl, CaCl_2 and $\text{Al}_2(\text{SO}_4)_3$ the one with highest vapour
9. For a non-ideal solution exhibiting positive deviation from Raoult's law, $\Delta_{\text{mix}} H$ has a (nonzero) value.
10. If in a binary solution, forces of attraction between like molecules are weaker than those prevailing between unlike molecules, the solution is expected to exhibit deviations from Raoult's law.
11. The van't Hoff factor of a weak electrolyte AB in a solution is 1.1. Its degree of dissociation would be
12. The density of a solution expressed in g cm^{-3} and kg dm^{-3} have values.

MATCH THE COLUMN

1.	Column-I (Properties)		Column-II (Affecting factors)	
	(A)	Relative lowering of vapour pressure	(p)	Directly proportional to van't Hoff factor, i
	(B)	Elevation in boiling point	(q)	Directly proportional to molality
	(C)	Freezing point	(r)	Directly proportional to molarity
	(D)	Osmotic pressure	(s)	Indirectly proportional to lowering of vapour pressure

ASSERTION & REASON

These questions contains, Statement I (assertion) and Statement II (reason).

(A) Statement-I is true, Statement-II is true ; Statement-II is correct explanation for Statement-I.

(B) Statement-I is true, Statement-II is true ; Statement-II is NOT a correct explanation for statement-I

(C) Statement-I is true, Statement-II is false

(D) Statement-I is false, Statement-II is true

1. **Statement-I** : 0.1 M solution of NaCl has greater osmotic pressure than 0.1 M solution of glucose at same temperature.

Because

Statement-II : In solution, NaCl dissociates to produce more number of particles.

2. **Statement-I** : Relative lowering of vapour pressure is equal to mole fraction of the solvent.

Because

Statement-II : Relative lowering of vapour pressure is a colligative property.

3. **Statement-I** : Molal elevation constant depends on the nature of solvent.

Because

Statement-II : Molal elevation constant is the elevation in boiling point when 1 mole of the solute is dissolved in 1 kg of solvent.

4. **Statement-I** : 0.02 m solutions of urea and sucrose will freeze at same temperature.

Because

Statement-II : Freezing point of a solution is inversely proportional to the conc. of solution.

5. **Statement-I** : When mercuric iodide is added to the aqueous solution of KI, the freezing point is raised.

Because

Statement-II : HgI_2 reacts with KI forming complex ion $[\text{HgI}_4]^{2-}$.

6. **Statement-I** : 1 M solution of Glauber's salt is isotonic with 1 M solution of KNO_3 .

Because

Statement-II : Solutions having same molar concentration of solute may or may not have same osmotic pressure.

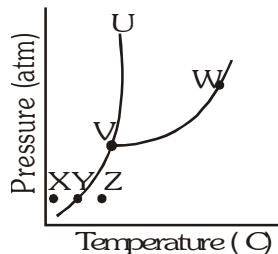
7. **Statement-I** : If decimolal solution of sodium chloride boils at 101.2°C , then decimolal solution of calcium chloride will also boil at the same temperature.

Because

Statement-II : For same molal concentration of aqueous solutions of electrolytes, the elevation of boiling point may not be same.

COMPREHENSION BASED QUESTIONS

Comprehension # 1



The phase diagram for a pure substance is shown above. Use this diagram and your knowledge about changes of phase to answer the following questions.

1. What does point V represent :
(A) point of equilibrium (B) point of fusion (C) point of vaporisation (D) Triple point
2. What characteristics are specific to the system only at point V?
(A) Liquid $\rightleftharpoons$ Solid (B) Solid $\rightleftharpoons$ Vapour
(C) Liquid $\rightleftharpoons$ Vapour (D) Solid $\rightleftharpoons$ Liquid $\rightleftharpoons$ Vapour

3. What happens if temperature is increased from X to Y at 1.0 atm ?
 (A) solid is completely vaporised (B) solid and vapour are in equilibrium
 (C) solid and liquid are in equilibrium (D) liquid and vapour are in equilibrium
4. Select correct statement (s) :
 (A) curve VU is solid-liquid equilibrium curve
 (B) curve VU has a positive slope
 (C) curve VW is vapour pressure curve for liquid substance
 (D) In the solid - liquid mixture of the substance, solid will float
5. If the given substance is water then :
 (A) curve VU would have negative slope
 (B) in ice $\rightleftharpoons$ water liquid mixture, ice will float
 (C) as the temperature increases, pressure at which solid and liquid are in equilibrium, decreases
 (D) increase in pressure at constant temperature causes ice to be converted to liquid water
6. If the triple point pressure of a substance is greater than 1 atm, we expect :
 (A) the solid to sublime without melting
 (B) the boiling point temperature to be lower than the triple point temperature
 (C) the melting point of the solid to come at a lower temperature than the triple point
 (D) that the substance cannot exist as a liquid
7. In a phase change (say solid to liquid or liquid to solid) $\Delta G = \Delta H - T\Delta S$ where :
 (A) ΔH is the enthalpy change associated with making or breaking the intermolecular attractions that hold solid and liquid together and ΔS is associated with change in disorder between the various phases.
 (B) ΔH is associated with change in disorder while ΔS is associated with energy change
 (C) both are associated with change in disorder
 (D) both are associated with change in energy

Comprehension # 2

Following passage explains effect of temperature on the vapour pressure of liquid. Answer the questions given at the end.

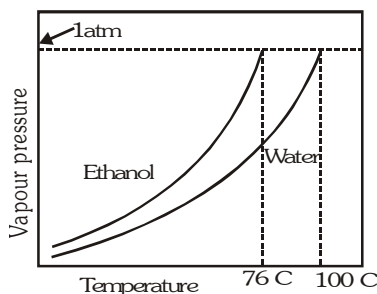
Effect of temperature on Vapour pressure

The quantity of heat required to evaporate a given liquid at constant temperature is defined as the heat of vaporisation. Variation of vapour pressure with temperature is given by

Clausius-Clapeyron equation.

$$\log_e P = -\frac{\Delta H_{\text{vap}}}{RT} + \log_e A$$

A liquid is said to be at its boiling temperature if its vapour pressure is equal to external pressure. Therefore, the boiling point of water in particular and of liquids in general decreases as altitude of a place increases where the external pressure is less than 1 atmosphere (normal b.p. of water is 373.15 K at 1 atmosphere)



On top of Mount Everest, for example, where the atmospheric pressure is only about 260 mm Hg, water boils at approximately 71 C. Conversely, if the external pressure on a liquid is greater than 1 atm., the vapour pressure necessary for boiling than normal boiling is reached later, and the liquid boils at a temperature greater than normal boiling point.

1. Clausius-Clapeyron equation can be written in the following form :

$$(A) P = Ae^{-\Delta H_{\text{vap}}/RT} \quad (B) \frac{d \log_{10} P}{dT} = \frac{\Delta H_{\text{vap}}}{2.303RT^2}$$

$$(C) \frac{d \log_e P}{dT} = -\frac{\Delta H_{\text{vap}}}{RT^2} \quad (D) P = Ae^{\Delta H_{\text{vap}}/RT}$$

2. For a given liquid at a given temperature vapour pressure is given by :

$$\log_{10} P \text{ (mm)} = -\frac{400(K)}{T} + 10$$

Vapour pressure of the liquid at 400 K is :

- (A) 9 mm (B) -9 mm (C) 10^9 mm (D) 10^{-9} mm

3. Latent heat of vaporisation of the above case in the given temperature range is :

- (A) -400 R (B) 400 R
(C) -400 2.303 R (D) 400 2.303 R

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
<u>True / False</u> 1. T 2. F 3. T 4. T 5. F 6. F 7. F 8. T 9. F 10. T <u>Fill in the Blanks</u> 1. proportional 2. 2 : 1 3. solvent molecule 4. azeotropic mixture 5. 3.01×10^{23} 6. hypotonic 7. solvent 8. 0.1 M NaCl 9. positive 10. negative 11. 0.1 12. same <u>Match the Column</u> 1. (A) → p ; (B) → p, q ; (C) → s ; (D) → p, r <u>Assertion - Reason Questions</u> 1. A 2. D 3. B 4. B 5. A 6. D 7. D <u>Comprehension Based Questions</u> Comprehension #1 : 1. (D) 2. (D) 3. (B) 4. (A,B,C) 5. (A,B,C,D) 6. (A) 7. (A) Comprehension #2 : 1. (A,B) 2. (C) 3. (D)		

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

1. The vapour pressure of ethanol and methanol are 44.5 mm Hg and 88.7 mm Hg, respectively. An ideal solution is formed at the same temperature by mixing 60 g of ethanol and 40 g of methanol. Calculate the total vapour pressure of the solution and the mole fraction of methanol in the vapour.
2. The vapour pressure of pure benzene at a certain temperature is 640 mm Hg. A non-volatile solid weighing 2.175 g is added to 39.0 g of benzene. The vapour pressure of the solution is 600 mm Hg. What is the molar mass of the solid substance?
3. Addition of 0.643 g of a compound to 50 mL of benzene (density : 0.879 g mL^{-1}) lower the freezing point from 5.51 C to 5.03 C. If K_f for benzene is $5.12 \text{ K kg mol}^{-1}$, calculate the molar mass of the compound.
4. (a) The vapour pressure of n-hexane and n-heptane at 273 K are 45.5 mm Hg and 11.4 mm Hg, respectively. What is the composition of a solution of these two liquids if its vapour pressure at 273 K is 37.3 mm Hg.
(b) The mole fraction of n-hexane in the vapour above a solution of n-hexane and n-heptane is 0.75 at 273 K. What is the composition of the liquid solution.
5. A solution containing 30 g of a nonvolatile solute in exactly 90 g water has a vapour pressure of 21.85 mm Hg at 25 C. Further 18 g of water is then added to the solution. The resulting solution has vapour pressure of 22.18 mm Hg at 25 C. Calculate (a) molar mass of the solute, and (b) vapour pressure of water at 25 C.
6. The freezing point of ether was lowered by 0.60 C on dissolving 2.0 g of phenol in 100 g of ether. Calculate the molar mass of phenol and comment on the result. Given : $K_f(\text{ether}) = 5.12 \text{ K kg mol}^{-1}$.
7. A solution contains 3.22 g of HClO_2 in 47.0 g of water. The freezing point of the solution is 271.10 K. Calculate the fraction of HClO_2 that undergoes dissociation to H^+ and ClO_2^- . Given : $K_f(\text{water}) = 1.86 \text{ K kg mol}^{-1}$.
8. A 0.1 molar solution of NaCl is found to be isotonic with 1% urea solution. Calculate (a) Van't Hoff factor, and (b) degree of dissociation of sodium chloride. Assume density of 1% urea equal to 1 g cm^{-3} .
9. The addition of 3 g of a substance to 100 g CCl_4 ($M = 154 \text{ g mol}^{-1}$) raises the boiling point of CCl_4 by 0.60 C. If $K_b(\text{CCl}_4)$ is $5 \text{ K mol}^{-1} \text{ kg}$, calculate (a) the freezing point depression (b) the relative lowering of vapour pressure (c) the osmotic pressure at 298 K and (d) the molar mass of the substance. Given : $K_f(\text{CCl}_4) = 31.8 \text{ K kg mol}^{-1}$ and $\rho(\text{solution}) = 1.64 \text{ g cm}^{-3}$.
10. To 500 cm^3 of water 3.0 $\times 10^{-3} \text{ kg}$ of acetic acid is added. If 23% of acetic acid is dissociated, what will be the depression of freezing point? K_f and density of water are $1.86 \text{ K kg mol}^{-1}$ and 0.997 g cm^{-3} , respectively.
11. A 0.01 m aqueous solution of $\text{K}_3[\text{Fe}(\text{CN})_6]$ freezes at -0.062 C . What is the apparent percentage of dissociation? [K_f for water = 1.86]
12. The degree of dissociation of $\text{Ca}(\text{NO}_3)_2$ in a dilute aqueous solution containing 7 g of the salt per 100 g of water at 100 C is 70%. If the vapour pressure of water at 100 C is 760 mm, calculate the vapour pressure of the solution.
13. A solution containing 0.122 kg of benzoic acid in 1 kg of benzene (b. pt. 353 K) boils at 354.5 K. Determine the apparent molar mass of benzoic acid (which dimerizes) in the solution and the degree of dimerization. Given : $\Delta_{\text{vap}}H_{1\text{m}}(\text{benzene}) = 394.57 \text{ J g}^{-1}$.
14. A solution containing 0.011 kg of barium nitrate in 0.1 kg of water boils at 100.46 C. Calculate the degree of ionization of the salt. $K_b(\text{water}) = 0.52 \text{ K kg mol}^{-1}$.

15. When 3.24 g of mercuric nitrate $\text{Hg}(\text{NO}_3)_2$ dissolved in 1 kg of water, the freezing point of the solution is found to be -0.0558°C . When 10.84 g of mercuric chloride HgCl_2 is dissolved in 1 kg of water, the freezing point of the solution is -0.0744°C . $K_f = 1.86^\circ\text{C kg mol}^{-1}$. Will either of these dissociate into ions in an aqueous solution?
16. The vapour pressure of solution containing 6.69 g of $\text{Mg}(\text{NO}_3)_2$ dissolved in 100 g of water is 747 Torr at 373 K. Calculate the degree of dissociation of the salt in the solution.
17. At 353 K, the vapour pressure of pure ethylene bromide and propylene bromide are 22.93 and 16.93 k Nm^{-2} , respectively, and these compounds form a nearly ideal solution. 3 mol of ethylene bromide and 2 mole of propylene bromide are equilibrated at 553 K and a total pressure of 20.4 k Nm^{-2} .
 - (a) What is the composition of the liquid phase?
 - (b) What amount of each compound is present in the vapour phase?
18. The vapour pressure of two pure liquids, A and B, that form an ideal solution are 300 and 800 torr, respectively, at temperature T. A mixture of the vapour of A and B for which the amount fraction of A is 0.25 is slowly compressed at temperature T. Calculate :
 - (a) The composition of the first drop of the condensate,
 - (b) The total pressure when this drop is formed,
 - (c) The composition of the solution whose normal boiling point is T.
 - (d) The pressure when only the last bubble of vapour remains.
 - (e) The composition of the last bubble.
19. Sea water is found to contain 5.85% NaCl and 9.50% MgCl_2 by weight of solution. Calculate its normal boiling point assuming 80% ionisation for NaCl and 50% ionisation of MgCl_2 [$K_b(\text{H}_2\text{O}) = 0.51^\circ\text{C kg mol}^{-1}$].
20. Find the freezing point of a glucose solution whose osmotic pressure at 25 $^\circ\text{C}$ is found to be 30 atm. $K_f(\text{water}) = 1.86^\circ\text{C kg mol}^{-1}$.
21. The latent heat of fusion of ice is 80 calories per gram at 0 $^\circ\text{C}$. What is the freezing point of a solution of KCl in water containing 7.45 grams of solute 500 grams of water, assuming that the salt is dissociated to the extent of 95%?
22. A certain mass of a substance, when dissolved in 100 g C_6H_6 , lowers the freezing point by 1.28 $^\circ\text{C}$. The same mass of solute dissolved in 100 g water lowers the freezing point by 1.40 $^\circ\text{C}$. If the substance has normal molecular weight in benzene and is completely ionised in water, into how many ions does it dissociate in water? K_f for H_2O and C_6H_6 are 1.86 and 5.12 $^\circ\text{C kg mol}^{-1}$.
23. The cryoscopic constant for acetic acid is 3.6 $^\circ\text{C kg mol}^{-1}$. A solution of 1 g of a hydrocarbon in 100 g of acetic acid freezes at 16.14 $^\circ\text{C}$ instead of the usual 16.60 $^\circ\text{C}$. The hydrocarbon contains 92.3% carbon. What is the molecular formula?
24. A radiator was filled with 10 L of water to which 2.5 L of methanol (density = 0.8 g mL^{-1}) were added. At 9 : 00 pm, the vehicle is parked outdoors where the temperature is 0 $^\circ\text{C}$. The temperature is decreasing at a uniform rate of 0.5 $^\circ\text{C} / \text{min}$. Upto what time will there be no danger to the radiator of the car. $K_f(\text{water}) = 1.86^\circ\text{C kg mol}^{-1}$. Assume methanol to be non-volatile.
25. At 300 K, two solutions of glucose in water of concentration 0.01 M and 0.001 M are separated by semipermeable membrane. Pressure needs to be applied on which solution, to prevent osmosis? Calculate the magnitude of this applied pressure?
26. At 10 $^\circ\text{C}$, the osmotic pressure of urea solution is 500 mm. The solution is diluted and the temperature is raised to 25 $^\circ\text{C}$, when the osmotic pressure is found to be 105.3 mm. Determine extent of dilution.

27. When cells of the skeletal vacuole of a frog were placed in a series of NaCl solutions of different concentration at 25 C, it was observed microscopically that they remained unchanged in 0.7% NaCl solution, shrank in more concentrated solutions, and swelled in more dilute solutions. Water freezes from the 0.7% salt solution at -0.406°C . What is the osmotic pressure of the cell cytoplasm at 25 C ? $K_f = 1.86 \text{ kg mol}^{-1} \text{ K}$.
28. A 0.1 M solution of potassium ferrocyanide is 46% dissociated at 18 C. What will be its osmotic pressure?
29. At 100 C, benzene & toluene have vapour pressure of 1375 & 558 Torr respectively. Assuming these two form an ideal binary solution that boils at 1 atm & 100 C. What is the composition of vapour issuing at these conditions?
30. An ideal solution of two volatile liquid A and B has a vapour pressure of 402.5 mmHg, the mole fraction of A in vapour & liquid state being 0.35 & 0.65 respectively. What are the vapour pressure of the two liquid at this temperature.
31. Dry air was drawn through bulbs containing a solution of 40 grams of urea in 300 grams of water, then through bulbs containing pure water at the same temperature and finally through a tube in which pumice moistened with strong H_2SO_4 was kept. The water bulbs lost 0.0870 grams and the sulphuric acid tube gained 2.036 grams. Calculate the molecular weight of urea.

CONCEPTUAL SUBJECTIVE EXERCISE			ANSWER KEY		EXERCISE-4(A)		
1.	66.11mmHg, 0.656	2.	69.6 g/mole	3.	156.06 g/mol		
4.	(a) 0.76, 0.34, (b) 0.61	5.	(a) 61.21 g mol ⁻¹ , (b) 23.99 mmHg	6.	170.7 g mol ⁻¹		
7.	$\alpha = 0.102$	8.	(a) 1.667, (b) 0.667				
9.	(a) 3.816 k (b) .01814	(c) 4.669 atm (d) 250 g mol ⁻¹		10.	0.23 k		
11.	0.78 %	12.	746.10 mm	13.	0.214 kg mol ⁻¹ , 0.86		
14.	0.55	16.	0.56	17.	(a) 0.578, 0.422, (b) 0.9967 mol, 0.5374 mol		
18.	(a) 0.4706 (b) 564.7 Torr	(c) 0.08, 0.92	(d) 675 Torr (e) .111, .889				
19.	T _b = 102.3 C	20.	T _f = - 2.28 C	21.	T _f = - 0.73 C	22.	3 ions
23.	C ₆ H ₆	24.	23.25 min	25.	P = 0.2217 atm should be applied		
26.	(V _{final} = 5 V _{original})	27.	5.34 atm	28.	p = 6.785 atm		
29.	x _b = 0.2472, y _b = 0.4473	30.	p _A = 216.7 mm Hg, p _B = 747.5 mm Hg	31.	M = 53.8		

EXERCISE-04 [B]**BRAIN STORMING SUBJECTIVE EXERCISE**

1. A one litre solution is prepared by dissolving some solid lead-nitrate in water. The solution was found to boil at 100.15 C. To the resulting solution 0.2 mole NaCl was added. The resulting solution was found to freeze at -0.83 C. Determine solubility product of PbCl_2 . Given $K_b = 0.5$ and $K_f = 1.86$. Assume molality to be equal to molarity in all case.
2. A protein has been isolated as sodium salt with their molecular formula Na_xP (this notation means that $x\text{Na}^+$ ions are associated with a negatively charged protein P^{x-}). A solution of this salt was prepared by dissolving 0.25 g of this sodium salt of protein in 10 g of water and ebulliscopic analysis revealed that solution boils at temperature 5.93×10^{-3} C higher than the normal boiling point of pure water. K_b of water 0.52 kg mol^{-1} . Also elemental analysis revealed that the salt contain 1% sodium metal by weight. Deduce molecular formula and determine molecular weight of acidic form of protein H_xP .
3. The vapour pressure of two miscible liquids (A) and (B) are 300 and 500 mm of Hg respectively. In a flask 10 mole of (A) is mixed with 12 mole of (B). However, as soon as (B) is added, (A) starts polymerising into a completely insoluble solid. The polymerisation follows first-order kinetics. After 100 minute, 0.525 mole of a solute is dissolved which arrests the polymerisation completely. The final vapour pressure of the solution is 400 mm of Hg. Estimate the rate constant of the polymerisation reaction. Assume negligible volume change on mixing and polymerisation, and ideal behaviour for the final solution.
4. Two beaker A and B present in a closed vessel. Beaker A contains 152.4 g aqueous solution of urea, containing 12 g of urea. Beaker B contains 196.2 g glucose solution, containing 18 g of glucose. Both solutions allowed to attain the equilibrium. Determine wt. % of glucose in it's solution at equilibrium :
5. The vapour pressure of two pure liquids A and B, that form an ideal solution are 100 and 900 mm Hg respectively at temperature T. This liquid solution of A and B is composed of 1 mole of A and 1 mole of B. What will be the pressure, when 1 mole of mixture has been vaporized ?
6. The addition of 3 g of substance to 100 g CCl_4 ($M = 154 \text{ g mol}^{-1}$) raises the boiling point of CCl_4 by 0.60 C of K_b (CCl_4) is $5.03 \text{ kg mol}^{-1} \text{ K}$. Calculate :
 - (a) the freezing point depression
 - (b) the relative lowering of vapour pressure
 - (c) the osmotic pressure at 298 K
 - (d) the molar mass of the substanceGiven $K_f(\text{CCl}_4) = 31.8 \text{ kg mol}^{-1} \text{ K}$ and ρ (density) of solution = 1.64 g/cm^3 .
7. If 20 mL of ethanol (density = 0.7893 g/mL) is mixed with 40 mL water (density = 0.9971 g/mL) at 25 C, the final solution has density of 0.9571 g/mL . Calculate the percentage change in total volume of mixing. Also calculate the molality of alcohol in the final solution.
8. Mixture of two liquids A and B is placed in cylinder containing piston. Piston is pulled out isothermally so that volume of liquid decreases but that of vapour increases. When negligibly small amount of liquid was remaining, the mole fraction of A in vapour is 0.4. Given $P_A = 0.4 \text{ atm}$ and $P_B = 1.2 \text{ atm}$ at the experimental temperature. Calculate the total pressure at which the liquid has almost evaporated. (Assume ideal behaviour)
9. 1.5 g of monobasic acid when dissolved in 150 g of water lowers the freezing point by 0.165 C. 0.5 g of the same acid when titrated, after dissolution in water, requires 37.5 mL of N/10 alkali. Calculate the degree of dissociation of the acid (K_f for water = 1.86 C mol^{-1}).

10. The molar volume of liquid benzene (density = 0.877 g mL^{-1}) increase by a factor of 2750 as it vaporizes at 20 C and that of liquid toluene (density = 0.867 g mL^{-1}) increases by a factor of 7720 at 20 C solution has a vapour pressure of 46.0 torr. Find the mole fraction of benzene in the vapour above the solution.
11. Calculate the boiling point of a solution containing 0.61 g of benzoic acid in 50 g of carbon disulphide assuming 84% dimerization of the acid. The boiling point and K_b of CS_2 are 46.2 C and $2.3 \text{ K kg mol}^{-1}$, respectively.
12. At 25 C, 1 mol of A having a vapour pressure of 100 torr and 1 mol of B having a vapour pressure of 300 torr were mixed. The vapour at equilibrium is removed, condensed and the condensate is heated back to 25 C. The vapour now formed are again removed, recondensed and analyzed. What is the mole fraction of A in this condensate ?
13. 30 mL of CH_3OH ($d = 0.7980 \text{ g cm}^{-3}$) and 70 mL of H_2O ($d = 0.9984 \text{ g cm}^{-3}$) are mixed at 25 C to form a solution of density 0.9575 g cm^{-3} . Calculate the freezing point of the solution. $K_f(\text{H}_2\text{O})$ is $1.86 \text{ kg mol}^{-1} \text{ K}$. Also calculate its molarity.
14. Vapour pressure of C_6H_6 and C_7H_8 mixture at 50 C is given by $P (\text{mm Hg}) = 179 X_B + 92$, where X_B is the mole fraction of C_6H_6 . A solution is prepared by mixing 936 g benzene and 736 g toluene and if the vapours over this solution are removed and condensed into liquid and again brought to the temperature of 50 C, what would be mole fraction of C_6H_6 in the vapour state ?
15. When the mixture of two immiscible liquids (water and nitrobenzene) boils at 372 K and the vapour pressure at this temperature are 97.7 kPa (H_2O) and 3.6 kPa ($\text{C}_6\text{H}_5\text{NO}_2$). Calculate the weight % of nitrobenzene in the vapour.
16. The vapour pressure of a certain liquid is given by the equation :

$\text{Log}_{10} P = 3.54595 - \frac{313.7}{T} + 1.40655 \log_{10} T$ where P is the vapour pressure in mm and T = Kelvin Temperature. Determine the molar latent heat of vaporisation as a function of temperature. Calculate the its value at 80 K.

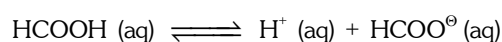
17. A very dilute saturated solution of a sparingly soluble salt A_3B_4 has a vapour pressure of 20 mm of Hg at temperature T, while pure water exerts a pressure of 20.0126 mm Hg at the same temperature. Calculate the solubility product constant of A_3B_4 at the same temperature.
18. The molar volume of liquid benzene (density = 0.877 g mL^{-1}) increases by a factor of 2750 as it vaporises at 20 C while in equilibrium with liquid benzene. At 27 C when a non - volatile solute (that does not dissociate) is dissolved in 54.6 cm^3 of benzene vapour pressure of this solution, is found to be 98.88 mm Hg. Calculate the freezing point of the solution.

Given : Enthalpy of vaporization of benzene (l) = 394.57 J/g

Molal depression constant for benzene = $5.12 \text{ K kg. mol}^{-1}$

Freezing point of benzene = 278.5 K.

19. An ideal solution was prepared by dissolving some amount of cane sugar (non-volatile) in 0.9 moles of water. The solution was then cooled just below its freezing temperature (271 K), where some ice get separated out. The remaining aqueous solution registered a vapour pressure of 700 torr at 373 K. Calculate the mass of ice separated out, if the molar heat of fusion of water is 96 kJ.
20. The freezing point depression of a 0.109 M aq. solution of formic acid is -0.21 C . Calculate the equilibrium constant for the reaction,



K_f for water = $1.86 \text{ kg mol}^{-1} \text{ K}$

21. 10 g of NH_4Cl (mol. weight = 53.5) when dissolved in 1000 g of water lowered the freezing point by 0.637 C. Calculate the degree of hydrolysis of the salt if its degree of dissociation of 0.75. The molal depression constant of water is $1.86 \text{ kg mol}^{-1} \text{ K}$.
22. The freezing point of 0.02 mol fraction solution of acetic acid (A) in benzene (B) is 277.4 K. Acetic acid exists partly as a dimer $2\text{A} = \text{A}_2$. Calculate equilibrium constant for the dimerisation. Freezing point of benzene is 278.4 K and its heat of fusion ΔH_f is $10.042 \text{ kJ mol}^{-1}$.
23. Tritium, T (an isotope of H) combines with fluorine to form weak acid TF, which ionizes to give T^+ . Tritium is radioactive and is a β -emitter. A freshly prepared aqueous solution of TF has pT (equivalent of pH) of 1.5 and freezes at -0.372 C . If 600 mL of freshly prepared solution were allowed to stand for 24.8 years, calculate (i) ionization constant of TF. (ii) Number of β -particles emitted.
(Given K_f for water = $1.86 \text{ kg mol K}^{-1}$, $t_{1/2}$ for tritium = 12.4 years.)

BRAIN STORMING SUBJECTIVE EXERCISE		ANSWER KEY		EXERCISE-4(B)	
1. 1.46×10^{-5}	2. H_{20}P , 45563 amu	3. 1.0×10^{-4}	4. 14.49 %		
5. 300 mm Hg	6. (a) 3.793 C, (b) 0.018, (c) 4.65 atm, (d) 251.5				
7. 3.1 %, 8.6	8. 0.66 atm	9. 18.27%	10. 0.732		
11. 46.31 C	12. 0.1	13. -19.91 C , 7.63 M			
14. 0.9286	15. 20.11%				
16. $\Delta H = 1659.9 \text{ Cal. at } 80 \text{ K}$, $\Delta H = R[313.7 - 2.303 + 1.40655 \text{ T}]$			17. 5.4×10^{-13}		
18. $T_f = 277.51 \text{ C}$	19. 12.54 g	20. 1.44×10^{-4}	21. $h = 0.109$		
22. 3.225	23. (i) $k_a = 7.3 \times 10^{-3}$ (ii) 3.7 10^{22}				

EXERCISE - 05 [A]**JEE-[MAINS] : PREVIOUS YEAR QUESTIONS**

1. Which of the following concentration factor is affected by change in temperature? [AIEEE-2002]
(A) Molarity (B) Molality (C) Mol fraction (D) Weight fraction
2. For an aqueous solution, freezing point is -0.186°C . The boiling point of the same solution is ($K_f = 1.86^\circ\text{C mol}^{-1}\text{ kg}$) and ($K_b = 0.512^\circ\text{C mol}^{-1}\text{ kg}$) [AIEEE-2002]
(A) 0.186 (B) 100.0512 (C) 1.86 (D) 5.12
3. In a mixture of A and B, components show negative deviation when - [AIEEE-2003]
(A) A - B interaction is stronger than A - A and B - B interaction
(B) A - B interaction is weaker than A - A and B - B interaction
(C) $\Delta V_{\text{mix}} > 0$, $\Delta S_{\text{mix}} > 0$
(D) $\Delta V_{\text{mix}} = 0$, $\Delta S_{\text{mix}} > 0$
4. A pressure cooker reduces cooking time for food because - [AIEEE-2003]
(A) The higher pressure inside the cooker crushes the food material
(B) Cooking involves chemical changes helped by a rise in temperature
(C) Heat is more evenly distributed in the cooking space
(D) Boiling point of water involved in cooking is increased
5. If liquids A and B form an ideal solution - [AIEEE-2003]
(A) The free energy of mixing is zero
(B) The free energy as well as the entropy of mixing are each zero
(C) The enthalpy of mixing is zero
(D) The entropy of mixing is zero
6. In a 0.2 molal aqueous solution of a weak acid HX the degree of ionization is 0.3 . Taking K_f for water as 1.85 , the freezing point of the solution will be nearest to - [AIEEE-2003]
(A) -260°C (B) $+0.480^\circ\text{C}$ (C) -0.480°C (D) -0.360°C
7. Which one of the following aqueous solutions will exhibit highest boiling point? [AIEEE-2004]
(A) $0.01\text{M Na}_2\text{SO}_4$ (B) 0.01M KNO_3 (C) 0.015M urea (D) 0.015M glucose
8. Which of the following liquid pairs shows a positive deviation from Raoult's law? [AIEEE-2004]
(A) Water-hydrochloric acid (B) Benzene-methanol
(C) Water-nitric acid (D) Acetone-chloroform
9. Which one of the following statement is False? [AIEEE-2004]
(A) Raoult's law states that the vapour pressure of a component over a solution is proportional to its mole fraction
(B) The osmotic pressure (π) of a solution is given by the equation $\pi = MRT$ where M is the molarity of the solution
(C) The correct order of osmotic pressure for 0.01M aqueous solution of each compound is $\text{BaCl}_2 > \text{KCl} > \text{CH}_3\text{COOH} > \text{Sucrose}$
(D) Two sucrose solutions of same molality prepared in different solvent will have the same freezing point depression
10. If α is the degree of dissociation of Na_2SO_4 , the vant of Hoff's factor (i) used for calculating the molecular mass is - [AIEEE-2005]
(A) $1 - \alpha$ (B) $1 + \alpha$ (C) $1 - 2\alpha$ (D) $1 + 2\alpha$

11. Benzene and toluene form nearly ideal solutions. At 20 C, the vapour pressure of benzene is 75 torr and that of toluene is 22 torr. The partial vapour pressure of benzene at 20 C for a solution containing 78 g of benzene and 46 g of toluene in torr is - [AIEEE-2005]
(A) 25 (B) 50 (C) 53.5 (D) 37.5
12. Two solutions of a substance (non electrolyte) are mixed in the following manner. 480 ml of 1.5 M first solution + 520 mL of 1.2 M second solution. What is the molarity of the final mixture? [AIEEE-2005]
(A) 1.50 M (B) 1.20 M (C) 2.70 M (D) 1.344 M
13. Equimolar solutions in the same solvent have - [AIEEE-2006]
(A) Same freezing point but different boiling point
(B) Same boiling point but different freezing point
(C) Different boiling and different freezing point
(D) Same boiling and same freezing points
14. 18 g of glucose ($C_6H_{12}O_6$) is added to 178.2g of water. The vapour pressure of water for this aqueous solution at 100 C is - [AIEEE-2006]
(A) 7.60 Torr (B) 76.00 Torr (C) 752.40 Torr (D) 759.00 Torr
15. Density of a 2.05 M solution of acetic acid in water is 1.02 g/mL. The molality of the solution is - [AIEEE-2006]
(A) 3.28 mol kg^{-1} (B) 2.28 mol kg^{-1} (C) 0.44 mol kg^{-1} (D) 1.14 mol kg^{-1}
16. A mixture of ethyl alcohol and propyl alcohol has a vapour pressure of 290 mm at 300 K. The vapour pressure of propyl alcohol is 200 mm. If the mole fraction of ethyl alcohol is 0.6, its vapour pressure (in mm) at the same temperature will be [AIEEE-2007]
(A) 350 (B) 300 (C) 700 (D) 360
17. A 5.25% solution of a substance is isotonic with a 1.5% solution of urea (molar mass = 60 g mol^{-1}) in the same solvent. If the densities of both the solutions are assumed to be equal to 1.0 g cm^{-3} , molar mass of the substance will be- [AIEEE-2007]
(A) 90.0 g mol^{-1} (B) 115.0 g mol^{-1} (C) 105.0 g mol^{-1} (D) 210.0 g mol^{-1}
18. The density (in g mL^{-1}) of a 3.60 M sulphuric acid solution that is 29% H_2SO_4 (Molar mass = 98 g mol^{-1}) by mass will be - [AIEEE-2007]
(A) 1.64 (B) 1.88 (C) 1.22 (D) 1.45
19. At 80 C, the vapour pressure of pure liquid 'A' is 520 mm Hg and that of pure liquid 'B' is 1000 mm Hg. If a mixture solution of 'A' and 'B' boils at 80 C and 1 atm pressure, the amount of 'A' in the mixture is (1 atm = 760 mm Hg) [AIEEE-2008]
(A) 52 mol % (B) 34 mol % (C) 48 mol % (D) 50 mol %
20. The vapour pressure of water at 20 C is 17.5 mm Hg. If 18 g of glucose ($C_6H_{12}O_6$) is added to 178.2 g of water at 20 C, the vapour pressure of the resulting solution will be [AIEEE-2008]
(A) 17.675 mm Hg (B) 15.750 mm Hg (C) 16.500 mm Hg (D) 17.325 mm Hg
21. Two liquids X and Y form an ideal solution. At 300K, vapour pressure of the solution containing 1 mol of X and 3 mol of Y is 550 mm Hg. At the same temperature, if 1 mol of Y is further added to this solution, vapour pressure of the solution increases by 10 mm Hg. Vapour pressure (in mmHg) of X and Y in their pure states will be, respectively [AIEEE-2009]
(A) 400 and 600 (B) 500 and 600 (C) 200 and 300 (D) 300 and 400

22. A binary liquid solution is prepared by mixing n-heptane and ethanol. Which one of the following statements is correct regarding the behaviour of the solution ? [AIEEE-2009]
 (A) The solution is non-ideal, showing -ve deviation from Raoult's law
 (B) n-heptane shows +ve deviation while ethanol shows -ve deviation from Raoult's law
 (C) The solution formed is an ideal solution.
 (D) The solution is non-ideal, showing +ve deviation from Raoult's law
23. If sodium sulphate is considered to be completely dissociated into cations and anions in aqueous solution, the change in freezing point of water (ΔT_f), when 0.01 mol of sodium sulphate is dissolved in 1 kg of water, is ($K_f = 1.86 \text{ K kg mol}^{-1}$) :- [AIEEE-2010]
 (A) 0.0186 K (B) 0.0372 K (C) 0.0558 K (D) 0.0744 K
24. On mixing, heptane and octane form an ideal solution. At 373 K, the vapour pressures of the two liquid components (heptane and octane) are 105 kPa and 45 kPa respectively. Vapour pressure of the solution obtained by mixing 25.0 g of heptane and 35 g of octane will be (molar mass of heptane = 100 g mol^{-1} and of octane = 114 g mol^{-1}) :- [AIEEE-2010]
 (A) 144.5 kPa (B) 72.0 kPa (C) 36.1 kPa (D) 96.2 kPa
25. The degree of dissociation (α) of a weak electrolyte, A_xB_y is related to van't Hoff factor (i) by the expression [AIEEE-2011]
 (A) $\alpha = \frac{x+y-1}{i-1}$ (B) $\alpha = \frac{x+y+1}{i-1}$ (C) $\alpha = \frac{i-1}{(x+y-1)}$ (D) $\alpha = \frac{i-1}{x+y+1}$
26. Ethylene glycol is used as an antifreeze in a cold climate. Mass of ethylene glycol which should be added to 4 kg of water to prevent it from freezing at -6°C will be :
 (K_f for water = $1.86 \text{ K kg mol}^{-1}$, and molar mass of ethylene glycol = 62 g mol^{-1}) [AIEEE-2011]
 (A) 400.00 g (B) 304.60 g (C) 804.32 g (D) 204.30 g
27. A 5% solution of cane sugar (molar mass 342) is isotonic with 1% of a solution of an unknown solute. The molar mass of unknown solute in g/mol is :- [AIEEE-2011]
 (A) 136.2 (B) 171.2 (C) 68.4 (D) 34.2
28. The molality of a urea solution in which 0.0100g of urea, $[(\text{NH}_2)_2\text{CO}]$ is added to 0.3000 dm^3 of water at STP is :- [AIEEE-2011]
 (A) 0.555 m (B) $5.55 \times 10^{-4} \text{ m}$ (C) 33.3 m (D) $3.33 \times 10^{-2} \text{ m}$
29. K_f for water is $1.86 \text{ K kg mol}^{-1}$. If your automobile radiator holds 1.0 kg of water, how many grams of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) must you add to get the freezing point of the solution lowered to -2.8°C ? [AIEEE-2012]
 (A) 27 g (B) 72 g (C) 93 g (D) 39 g

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans	A	B	A	D	C	C	A	B	D	D	B	D	D	C	B
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	
Ans	A	D	C	D	D	A	D	C	B	C	C	C	B	C	

EXERCISE - 05 [B]**JEE-[ADVANCED] : PREVIOUS YEAR QUESTIONS**

1. An azeotropic solution of two liquids has boiling point lower than either of them when it : [JEE 1981]
(A) shows negative deviation from Raoult's law (B) shows no deviation from Raoult's law
(C) shows positive deviation from Raoult's law (D) is saturated
2. For a dilute solution, Raoult's law state that : [JEE 1985]
(A) the lowering of vapour pressure is equal to mole fraction of solute
(B) the relative lowering of vapour pressure is equal to the mole fraction of solute
(C) the relative lowering of vapour pressure is proportional to the amount of solute in solution
(D) the vapour pressure of the solution is equal to the mole fraction of solvent
3. Which of the following 0.1 M aqueous solutions will have the lowest freezing point ? [JEE 1989]
(A) Potassium sulphate (B) Sodium chloride (C) Urea (D) Glucose
4. The freezing point of equimolal aqueous solution is highest for :
(A) $C_6H_5N^+H_3Cl^-$ (aniline hydrochloride) (B) $Ca(NO_3)_2$
(C) $La(NO_3)_3$ (D) $C_6H_{12}O_6$ (glucose)
5. Osmotic pressure of a solution is 0.0821 atm at a temperature of 300 K. The concentration in mole/litre will be : [JEE 1990]
(A) 0.33 (B) 0.066 (C) 0.3×10^{-2} (D) 3
6. Increasing the temperature of an aqueous solution will cause : [JEE 1983]
(A) decrease in molality (B) decrease in molarity
(C) decrease in the mole fraction (D) decrease in % (w/w)
7. How many moles of Fe^{2+} ions are formed when excess iron is treated with 500 mL of 0.4 M HCl under inert atmosphere ? Assume no change in volume : [JEE 1993]
(A) 0.4 (B) 0.1 (C) 0.2 (D) 0.8
8. A 0.2 molal aqueous solution of a weak acid (HX) is 20 percent ionised. The freezing point of this solution is (given $k_f = 1.86 \text{ } ^\circ\text{C kg mol}^{-1}$ for water) : [JEE 1995]
(A) $-0.45 \text{ } ^\circ\text{C}$ (B) $-0.90 \text{ } ^\circ\text{C}$ (C) $-0.31 \text{ } ^\circ\text{C}$ (D) $-0.53 \text{ } ^\circ\text{C}$
9. The molecular weight of benzoic acid in benzene as determined by depression in freezing point of the solution is : [JEE 1996]
(A) ionization of benzoic acid (B) dimerization of benzoic acid
(C) trimerization of benzoic acid (D) solvation of benzoic acid
10. The van't Hoff factor for 0.1 M $Ba(NO_3)_2$ solution is 2.74. The degree of dissociation is : [JEE 1996]
(A) 91.3% (B) 87% (C) 100% (D) 74%
11. In the depression of freezing point experiment, it is found that : [JEE 1999]
(i) The vapour pressure of the solution is less than that of pure solvent.
(ii) The vapour pressure of the solution is more than that of pure solvent.
(iii) Only solute molecules solidify at the freezing point.
(iv) Only solvent molecules solidify at the freezing point.
(A) (i), (ii) (B) (ii), (iii) (C) (i), (iv) (D) (i), (ii), (iii)
12. To 500 cm^3 of water, $3.0 \times 10^{-3} \text{ kg}$ of acetic acid is added. If 23% of acetic acid is dissociated, what will be the depression in freezing point? k_f and density of water are $1.86 \text{ } ^\circ\text{C kg mol}^{-1}$ and 0.997 g cm^{-3} respectively : [JEE 2000]
(A) 0.186 K (B) 0.228 K (C) 0.372 K (D) 0.556 K
13. An aqueous solution of 6.3 g oxalic acid dihydrate is made up to 250 mL. The volume of 0.1 N NaOH required to completely neutralise 10 mL of this solution is : [JEE 2001]
(A) 40 mL (B) 20 mL (C) 10 mL (D) 4 mL

14. During depression of freezing point in a solution, the following are in equilibrium : [JEE 2003]
 (A) Liquid solvent - solid solvent (B) Liquid solvent - solid solute
 (C) Liquid solute - solid solute (D) Liquid solute - solid solvent
15. The elevation in boiling point of a solution of 13.44 g of CuCl_2 in 1 kg of water using the following information will be (Molecular weight of $\text{CuCl}_2 = 134.4$ and $K_b = 0.52 \text{ K molal}^{-1}$) : [JEE 2005]
 (A) 0.16 (B) 0.05 (C) 0.1 (D) 0.2

SUBJECTIVE QUESTIONS :

16. A very small amount of a nonvolatile solute (that does not dissociate) is dissolved in 56.8 cm^3 of benzene (density 0.889 g cm^{-3}). At room temperature, vapour pressure of this solution is 98.88 mm Hg while that of benzene is 100 mm Hg. Find the molality of this solution If the freezing point depression constant of benzene ? [JEE 1997]
17. A solution of a nonvolatile solute in water freezes at -0.30°C . The vapour pressure of pure water at 298 K is 23.51 mm Hg and K_f for water is $1.86^\circ\text{C molal}^{-1}$. Calculate the vapour pressure of this solution at 298 K. [JEE 1998]
18. The vapour pressure of two miscible liquids (A) and (B) are 300 and 500 mm of Hg respectively. In a flask 10 mole of (A) is mixed with 12 mole of (B). However, as soon as (B) is added, (A) starts polymerising into a completely insoluble solid. The polymerisation follows first-order kinetics. After 100 minute, 0.525 mole of a solute is dissolved which arrests the polymerisation complete. The final vapour pressure of the solution is 400 mm of Hg. Estimate the rate constant of the polymerisation reaction. Assume negligible volume change on mixing and polymerisation and ideal behaviour for the final solution. [JEE 2001]
19. Match the boiling point with K_b for x, y and z, if molecular weight of x, y are same : [JEE 2003]
- | | b.pt | K_b |
|---|------|-------|
| x | 100 | 0.68 |
| y | 27 | 0.53 |
| z | 253 | 0.98 |
20. 1.22 g of benzoic acid is dissolved in (i) 100 g acetone (K_b for acetone = 1.7) and (ii) 100 g benzene (K_b for benzene = 2.6). The elevation in boiling points T_b is 0.17°C and 0.13°C respectively : [JEE 2004]
 (a) What are the molecular weight of benzoic acid in both the solutions?
 (b) What do you deduce out of it in terms of structure of benzoic acid?
21. 72.5 g of phenol is dissolved in 1 kg of a solvent ($k_f = 14$) which leads to dimerisation of phenol and freezing point is lowered by 7 kelvin. What percent of total phenol is present in dimeric form: [JEE 2006]
22. When 20 g of naphtholic acid ($\text{C}_{11}\text{H}_8\text{O}_2$) is dissolved in 50 g of benzene ($K_f = 1.72 \text{ K kg mol}^{-1}$), a freezing point depression of 2 K is observed. The van't Hoff factor (i) is [JEE 2007]
 (A) 0.5 (B) 1 (C) 2 (D) 3

Paragraph for Question No. Q.23 to Q.24

Properties such as boiling point, freezing point and vapour pressure of a pure solvent change when solute molecules are added to get homogeneous solution. These are called colligative properties. Applications of colligative properties are very useful in day-to-day life. One of its examples is the use of ethylene glycol and water mixture as anti-freezing liquid in the radiator of automobiles.

A solution **M** is prepared by mixing ethanol and water. The mole fraction of ethanol in the mixture is 0.9.

Given : Freezing point depression constant of water (K_f^{water}) = $1.86 \text{ K kg mol}^{-1}$

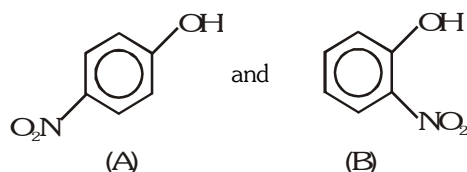
Freezing point depression constant of ethanol (K_f^{ethanol}) = $2.0 \text{ K kg mol}^{-1}$

Boiling point elevation constant of water (K_b^{water}) = $0.52 \text{ K kg mol}^{-1}$

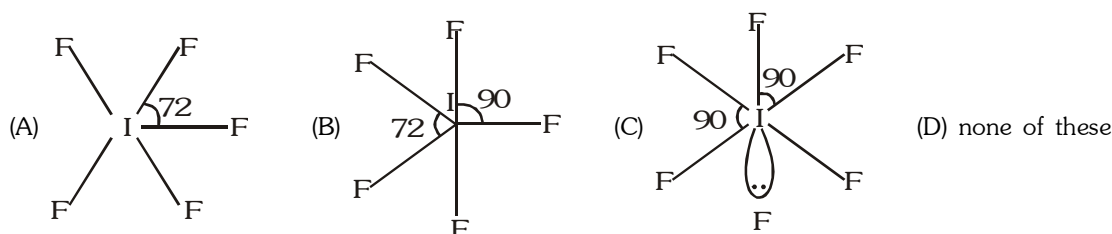
JEE-[ADVANCED] : PREVIOUS YEAR QUESTIONS										ANSWER KEY						EXERCISE -5[B]					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15						
Ans.	C	B	A	D	C	B	B	A	B	B	C	B	A	A	A						

EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. Out of the two compounds shown below, the vapour pressure of B at a particular temperature is expected to be



- (A) higher than that of A
(B) lower than that of A
(C) same as that of A
(D) can be higher or lower depending upon the size of the vessel
2. The structure of IF_5 can be best described as :-



3. The correct order of the bond angles is :-

- (A) $\text{NH}_3 > \text{H}_2\text{O} > \text{PH}_3 > \text{H}_2\text{S}$ (B) $\text{NH}_3 > \text{PH}_3 > \text{H}_2\text{O} > \text{H}_2\text{S}$
(C) $\text{NH}_3 > \text{H}_2\text{S} > \text{PH}_3 > \text{H}_2\text{O}$ (D) $\text{PH}_3 > \text{H}_2\text{S} > \text{NH}_3 > \text{H}_2\text{O}$

4. The correct increasing bond angle among BF_3 , PF_3 and ClF_3 follows the order :-

- (A) $\text{BF}_3 < \text{PF}_3 < \text{ClF}_3$ (B) $\text{PF}_3 < \text{BF}_3 < \text{ClF}_3$ (C) $\text{ClF}_3 < \text{PF}_3 < \text{BF}_3$ (D) $\text{BF}_3 = \text{PF}_3 = \text{ClF}_3$

5. How many sigma and pi bonds are present in tetracyanoethylene ?

- (A) nine σ and nine π (B) five π and nine σ (C) nine σ and seven π (D) eight σ and eight π

6. The types of bond present in N_2O_5 are :-

- (A) only covalent (B) only ionic (C) ionic and covalent (D) covalent & coordinate

7. How many bonded electron pairs are present in IF_7 molecule :-

- (A) 6 (B) 7 (C) 5 (D) 8

8. When $2s - 2s$, $2p - 2p$ and $2p - 2s$ orbitals overlap, the bond strength decreases in the order :-

- (A) $p - p > s - s > p - s$ (B) $p - p > p - s > s - s$
(C) $s - s > p - p > p - s$ (D) $s - s > p - s > p - p$

9. The volatility of HF is low as compare to other Hydra acid of Halogen because of :-

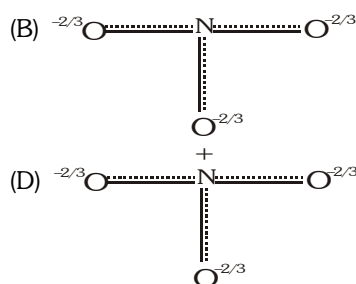
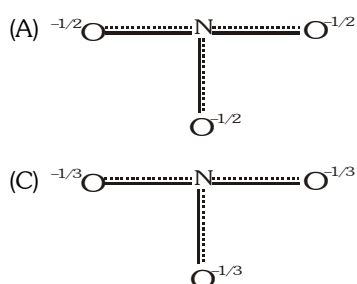
- (A) its low polarizability
(B) the weak dispersion interaction between the molecules
(C) its small molecular mass
(D) its strong hydrogen bonding

10. The shapes of PCl_4^+ , PCl_4^- and AsCl_5 are respectively :-

- (A) square planar, tetrahedral and see-saw
(B) tetrahedral, see-saw and trigonal bipyramidal
(C) tetrahedral, square planar and pentagonal bipyramidal
(D) trigonal bipyramidal, tetrahedral and square pyramidal

11. The experimental value of the dipole moment of HCl is 1.03 D. The length of the H – Cl bond is 1.275 Å. The percentage of ionic character in HCl is :-
 (A) 43 (B) 21 (C) 17 (D) 7
12. The shapes of IF_5 and IF_7 are respectively :-
 (A) tetragonal pyramidal and pentagonal bipyramidal
 (B) octahedral and pyramidal
 (C) trigonal bipyramidal and square antiprismatic
 (D) distorted square planar and distorted octahedral
13. Amongst LiCl, RbCl, BeCl_2 and MgCl_2 , the compounds with the greatest and the least ionic character, respectively are :-
 (A) LiCl and RbCl (B) RbCl and BeCl_2 (C) RbCl and MgCl_2 (D) MgCl_2 and BeCl_2
14. PCl_5 , exists but NCl_5 does not because :-
 (A) Nitrogen has no vacant 2d-orbitals (B) NCl_5 is unstable
 (C) Nitrogen atom is much smaller than P (D) Nitrogen is highly inert
15. (C – Cl) bond in $\text{CH}_2 = \text{CH} - \text{Cl}$ (vinyl chloride) is stabilised in the same way as in :-
 (A) benzyl chloride (B) ethyl chloride (C) chlorobenzene (D) allyl chloride
16. In ICl_4^- , the shape is square planar. The number of bond pair-lone pair repulsion at 90° are :-
 (A) 6 (B) 8 (C) 12 (D) 4
17. The structure of diborane (B_2H_6) contains :-
 (A) four ($2c - 2e^-$) bonds and two ($2c - 3e^-$) bonds (B) two ($2c - 2e^-$) bonds and two ($3c - 2e^-$) bonds
 (C) four ($2c - 2e^-$) bonds and four ($3c - 2e^-$) bonds (D) none
18. Among the following species, identify the isostructural pairs : NF_3 , NO_3^- , BF_3 , H_3O^+ , HN_3 :-
 (A) $[\text{NF}_3, \text{NO}_3^-]$ and $[\text{BF}_3, \text{H}_3\text{O}^+]$ (B) $[\text{NF}_3, \text{HN}_3]$ and $[\text{NO}_3^-, \text{BF}_3]$
 (C) $[\text{NF}_3, \text{H}_3\text{O}^+]$ and $[\text{NO}_3^-, \text{BF}_3]$ (D) $[\text{NF}_3, \text{H}_3\text{O}^+]$ and $[\text{HN}_3, \text{BF}_3]$
19. Which of the following exhibit/s H-bonding ?
 (A) CH_4 (B) H_2Se (C) N_2H_4 (D) H_2S
20. Valency expresses generally :
 (A) total e^- in an atom (B) atomicity of an element
 (C) oxidation number of an element (D) combining capacity of an element
21. Which element do not have valency equals to its group no. :
 (A) sodium (B) aluminium (C) oxygen (D) carbon
22. Which condition favours the bond formation:-
 (A) maximum attraction and maximum potential energy
 (B) minimum attraction and minimum potential energy
 (C) minimum potential energy and maximum attraction
 (D) none of the above
23. Number of σ and π bonds present in :
 $\text{CH}_3 - \text{CH} = \text{CH} - \text{C} \equiv \text{CH}$ are -
 (A) 10 σ , 3 π (B) 10 σ , 2 π (C) 9 σ , 2 π (D) 8 σ , 3 π
24. Which is not characteristic of π -bond:-
 (A) π - bond is formed when a sigma bond already formed
 (B) π - bond are formed from hybrid orbitals
 (C) π - bond may be formed by the overlapping of p-orbitals
 (D) π -bond results from lateral overlap of atomic orbitals
25. When sodium and chlorine reacts then energy is :-
 (A) released and ionic bond is formed (B) released and covalent bond is formed
 (C) absorbed and ionic bond is formed (D) absorbed and covalent bond is formed

26. The electron pair which forms a bond between two similar non-metallic atoms will be :-
 (A) dissimilar shared between the two (B) by complete transfer from one atom to other
 (C) in a similar spin condition (D) equally shared in between the two
27. In N_2 molecule, the atoms are bonded by :-
 (A) 1 σ and 2 π - bonds, 2 L.P. (B) 1 σ and 1 π - bonds, 1 L.P.
 (C) 2 σ and 1 π - bonds, No L.P. (D) 1 σ , 2 π & No L.P.
28. An atom of element A has three electrons in its outer shell and B has six electrons in its outermost shell. The formula of the compound formed between these two will be :-
 (A) A_6B_6 (B) A_2B_3 (C) A_3B_2 (D) A_2B
29. Two element X and Y have following electronic configuration :-
 X $1s^2 ; 2s^2, 2p^6 ; 3s^2, 3p^6 ; 4s^2$
 Y $1s^2 ; 2s^2, 2p^6 ; 3s^2, 3p^5$
 The expected compound formed by combination of X and Y will be expressed as :
 (A) XY_2 (B) X_5Y_2 (C) X_2Y_5 (D) XY_5
30. Polarizability of halide ions increases in the order :-
 (A) F^- , I^- , Br^- , Cl^- (B) Cl^- , Br^- , I^- , F^- (C) I^- , Br^- , Cl^- , F^- (D) F^- , Cl^- , Br^- , I^-
31. The correct order of the O-O bond length in O_2 , H_2O_2 and O_3 is :-
 (A) $O_2 > O_3 > H_2O_2$ (B) $O_3 > H_2O_2 > O_2$ (C) $H_2O_2 > O_3 > O_2$ (D) $H_2O_2 > O_2 > O_3$
32. In which of the following the central atom does not use sp^3 hybrid orbitals in its bonding :-
 (A) BeF_3^- (B) OH_3^+ (C) NH_2^- (D) NF_3
33. According to Fajjan's rule, covalent bond is favoured by :-
 (A) large cation and small anion (B) large cation and large anion
 (C) Small cation and large anion (D) Small cation and small anion
34. Resonance hybrid of nitrate ion is :-



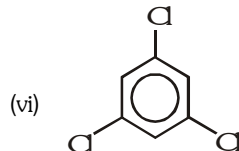
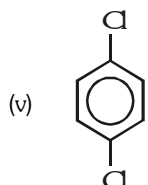
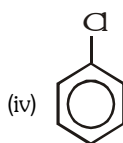
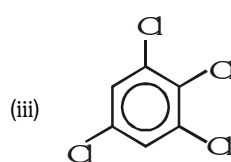
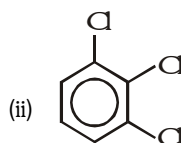
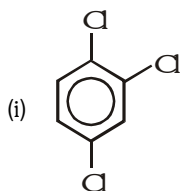
35. The correct order of bond angle (smallest first) in H_2S , NH_3 , BF_3 and SiH_4 is :-
 (A) $H_2S < NH_3 < BF_3 < SiH_4$ (B) $NH_3 < H_2S < SiH_4 < BF_3$
 (C) $H_2S < NH_3 < SiH_4 < BF_3$ (D) $H_2S < SiH_4 < NH_3 < BF_3$

CHECK YOUR GRASP								ANSWER KEY								EXERCISE -1				
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	A	C	A	C	A	D	B	B	D	B	C	A	B	A	C	B	D	C	C	D
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35					
Ans.	C	C	A	B	A	D	A	B	A	D	C	A	C	C	C					

EXERCISE-02**BRAIN TEASERS****SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)**

1. Which of the following does not have same shape :-
(A) $\text{SO}_4^{2-}, \text{S}_5^{2-}$ (B) $\text{ICl}_4^\ominus, \text{I}_5^\ominus$ (C) CO_2, SO_2 (D) $\text{NO}_2^\oplus, \text{NO}_2^\ominus$
2. $\text{B}_3\text{N}_3\text{H}_6 + \text{solution of hydrochloric acid} \longrightarrow ?$
Select correct about above reaction :
(A) no reaction
(B) $\text{B}_3\text{N}_3\text{H}_6$ show substitution reaction & produce $\text{B}_3\text{N}_3\text{Cl}_6$
(C) $\text{B}_3\text{N}_3\text{H}_6$ show addition reaction and produce $\text{B}_3\text{N}_3\text{H}_9\text{Cl}_3$ in which Cl is bonded to boron
(D) $\text{B}_3\text{N}_3\text{H}_6$ show addition reaction and produce $\text{B}_3\text{N}_3\text{H}_9\text{Cl}_3$ in which Cl is bonded to nitrogen
3. Nodal planes of π bonds in benzene are located in :
(A) all are in molecular plane
(B) one in molecular plane and two in plane perpendicular to molecular plane which contain C – C σ bonds.
(C) two in molecular plane and one in plane perpendicular to molecular plane which contain C – C σ bond and C – H σ bond
(D) perpendicular to molecular plane which bisect benzene ring in two equal half
4. Which of the following has fractional bond order :-
(A) O_2^{2+} (B) O_2^{2-} (C) F_2^{2-} (D) H_2^-
5. Which is correct statement ?
As the s-character of a hybrid orbital decreases
(I) The bond angle decreases (II) The bond strength increases
(III) The bond length increases (IV) Size of orbitals increases
(A) (I), (III) and (IV) (B) (II), (III) and (IV) (C) (I) and (II) (D) all are correct
6. Which of the following compounds have the same no. of lone pairs with their central atom :-
(I) XeF_5^- (II) BrF_3 (III) XeF_2 (IV) Triple methylene
(A) (IV) and (V) (B) (I) and (III) (C) (I) and (II) (D) (II), (IV) and (V)
7. Select pair of compounds in which both have different hybridization but have same molecular geometry :-
(A) $\text{BF}_3, \text{BrF}_3$ (B) $\text{ICl}_2^\ominus, \text{BeCl}_2$ (C) $\text{BeCl}_3, \text{PCl}_3$ (D) $\text{PCl}_3, \text{NCl}_3$
8. The states of hybridization of boron and oxygen atoms in boric acid (H_3BO_3) are respectively.
(A) sp^3 and sp^2 (B) sp^2 and sp^3 (C) sp^2 and sp^2 (D) sp^3 and sp^3
9. Which of the following option w.r.t. increasing bond order is correct ?
(A) $\text{NO} < \text{C}_2 < \text{O}_2^- < \text{He}_2^+$ (B) $\text{C}_2 < \text{NO} < \text{He}_2^+ < \text{O}_2^-$
(C) $\text{He}_2^+ < \text{O}_2^- < \text{NO} < \text{C}_2$ (D) $\text{He}_2^+ < \text{O}_2^- < \text{C}_2 < \text{NO}$
10. Which is most ionic :
(A) P_2O_5 (B) MnO (C) CrO_3 (D) Mn_2O_7
11. The molecular orbital configuration of CN^+ is :-
(A) $\text{KK } \sigma(2s)^2, \sigma^*(2s)^2, \pi(2p_x)^2, \pi(2p_y)^2$ (B) $\text{KK } \sigma(2s)^2, \sigma^*(2s)^2, \sigma(2p_z)^2, \pi(2p_x)^1, \pi(2p_y)^1$
(C) $\text{KK } \sigma(2s)^2, \sigma^*(2s)^2, \sigma(2p_z)^2, \pi(2p_x)^2, \pi(2p_y)^1$ (D) $\text{KK } \sigma(2s)^2, \sigma^*(2s)^2, \sigma(2p_z)^2, \pi(2p_x)^2, \pi(2p_y)^2$
12. Among the following orbital bonds, the angle is minimum between :-
(A) sp^3 bonds (B) p_x and p_y orbitals
(C) H–O–H in water (D) sp bonds

13. Molecule having dipole moment is :-
 (A) 2, 2-dimethylpropane (B) trans-2-pentene
 (C) neopentane (D) 2,2,3,3-tetramethylbutane.
14. The AsF_5 molecule is trigonal bipyramidal. The hybrid orbitals used by the As atoms for bonding are :-
 (A) $dx^2 - y^2$, dz^2 , s, p_x , p_y (B) dxy , s, p_x , p_y , p_z
 (C) s, p_x , p_y , p_z , dz^2 (D) $dx^2 - y^2$, s, p_x , p_y
15. Polarisation is the distortion of the shape of an anion by an adjacently placed cation. Which of the following statements is correct :-
 (A) maximum polarisation is brought about by a cation of high charge
 (B) minimum polarisation is brought about by a cation of low radius
 (C) a large cation is likely to bring about a large degree of polarisation
 (D) polarising power of a cation is less than that of anion
16. Amongst NO_3^- , AsO_3^{3-} , CO_3^{2-} , ClO_3^- , SO_3^{2-} and BO_3^{3-} , the non-planar species are :-
 (A) CO_3^{2-} , SO_3^{2-} and BO_3^{3-} (B) AsO_3^{3-} , ClO_3^- and SO_3^{2-}
 (C) NO_3^- , CO_3^{2-} and BO_3^{3-} (D) SO_3^{2-} , NO_3^- and BO_3^{3-}
17. The nature of π -bonds in perchlorate ion is :-
 (A) O (d π) - Cl (p π) (B) O (p π) - Cl (d π) (C) O (d π) - Cl (d π) (D) O (p π) - Cl (p π)
18. CaO and NaCl have the same crystal structure and approximately the same ionic radii. If U is the lattice energy of NaCl, the approximate lattice energy of CaO is :-
 (A) U/2 (B) U (C) 2 U (D) 4 U
19. The ease of hydrolysis of trichlorides of group 15 elements decreases in the order :-
 (A) $\text{NCl}_3 > \text{PCl}_3 > \text{AsCl}_3 > \text{SbCl}_3 > \text{BiCl}_3$ (B) $\text{PCl}_3 > \text{NCl}_3 > \text{AsCl}_3 > \text{SbCl}_3 > \text{BiCl}_3$
 (C) $\text{AsCl}_3 > \text{NCl}_3 > \text{PCl}_3 > \text{SbCl}_3 > \text{BiCl}_3$ (D) $\text{SbCl}_3 > \text{BiCl}_3 > \text{PCl}_3 > \text{NCl}_3 > \text{AsCl}_3$
20. Which of the following solid have highest value of K_p when heated in closed vessel :-
 (A) Li_2CO_3 (s) (B) BeCO_3 (s) (C) Na_2CO_3 (s) (D) BaCO_3 (s)
21. Type of bonds between calcium and carbon in CaC_2 are :-
 (A) σ , π (B) only σ (C) only π (D) ionic bond
22. Ethanol has a higher boiling point than dimethyl ether though they have the same molecular weight. This is due to :-
 (A) resonance (B) coordinate bonding (C) hydrogen bonding (D) ionic bonding
23. Write order of dipole moment of following compounds :-



- (A) (iii) > (ii) > (i) > (iv) > (v) > (vi) (B) (iii) > (i) > (ii) = (vi) > (iv) > (v)
 (C) (ii) > (i) = (iii) = (iv) > (v) = (vi) (D) (iii) > (i) > (iv) > (v) > (ii) > (vi)

24. The correct order of increasing X – O – X bond angle is (X = H, F or Cl) :-
 (A) $\text{H}_2\text{O} > \text{Cl}_2\text{O} > \text{F}_2\text{O}$ (B) $\text{Cl}_2\text{O} > \text{H}_2\text{O} > \text{F}_2\text{O}$
 (C) $\text{F}_2\text{O} > \text{Cl}_2\text{O} > \text{H}_2\text{O}$ (D) $\text{F}_2\text{O} > \text{H}_2\text{O} > \text{Cl}_2\text{O}$
25. Out of given reaction which show change in hybridisation of central atom :-
 (A) H_2BO_3 dissolve in water (B) H_2SO_4 dissolve in water
 (C) $\text{N}_2\text{O}_{5(\text{g})} \longrightarrow \text{N}_2\text{O}_{5(\text{s})}$ (D) $\text{PBr}_{5(\text{g})} \longrightarrow \text{PBr}_{5(\text{s})}$
 (E) $\text{C}_2\text{H}_6 \xrightarrow[\text{bond cleavage of C-C bond}]{\text{Homolytic}} \longrightarrow$
26. In the cyclo- S_8 molecule of rhombic sulphur, all the S-S bond lengths and all the S-S-S bond angles are respectively (give approximate values) :-
 (A) 204 pm and 105 (B) 102 pm and 120
 (C) 204 pm and 180 (D) 102 pm and 60
27. The structure of the SO_3 molecule in the gaseous phase contains :-
 (A) only σ -bonds between sulphur and oxygen
 (B) σ -bonds and a $(p\pi-p\pi)$ bonds between sulphur and oxygen
 (C) σ -bonds and a $(d\pi-p\pi)$ bonds between sulphur and oxygen
 (D) σ -bonds, and a $(p\pi-p\pi)$ and a $(p\pi-d\pi)$ bonds between sulphur and oxygen
28. Which of the following allotropic forms of sulphur is the most stable thermodynamically :-
 (A) Orthorhombic (B) β -monoclinic
 (C) γ -monoclinic (D) Plastic sulphur
29. Which of the following acids is not a peroxo acid :-
 (A) $\text{CF}_3\text{CO}_3\text{H}$ (B) $\text{H}_2\text{S}_2\text{O}_8$ (C) $\text{H}_2\text{S}_2\text{O}_7$ (D) $\text{H}_2\text{N}_2\text{O}_2$
30. The hydrolysis of 1 mol of peroxodisulphuric acid produces :-
 (A) 2 mol of sulphuric acid
 (B) 2 mol of peroxomonosulphuric acid
 (C) 1 mol each of sulphuric acid and peroxomonosulphuric acid
 (D) 1 mol each of sulphuric acid, peroxomonosulphuric acid and hydrogen peroxide
31. Which of the following statements is correct :-
 (A) SF_6 does not react with water (B) OF_6 is $d^2\text{sp}^3$ -hybridized
 (C) $\text{S}_2\text{O}_3^{2-}$ is a linear ion (D) There is no π -bonding in SO_4^{2-}
32. In H_2O , the bond angle H-O-H is $104^\circ 28'$ but in H_2S , H_2Se and H_2Te the bond angles are pretty close to 90° . This suggests that :-
 (A) oxygen uses sp^2 -hybrid orbitals while S, Se and Te use sp^3 -hybrid orbitals for bonding with the hydrogen atoms
 (B) oxygen uses sp^3 -hybrid orbitals to bond with the two hydrogen atoms while S, Se and Te use almost pure p orbitals
 (C) oxygen uses sp^3 -hybrid orbitals while S, Se and Te utilize d orbitals for bonding with the hydrogen atoms
 (D) all the atoms use pure p orbitals to bond with the hydrogen atoms
33. Which of the following statements are correct for the SO_4^{2-} ion ?
 (A) it is tetrahedral
 (B) all the S-O bond length are equal, and shorter than expected
 (C) it contains four σ -bonds between the S and the O atoms, two π -bonds delocalized over the S and the four O atoms, and all the S-O bonds have a bond order of 1.5
 (D) Oxidation state of sulphur is +6 and all oxygen in -2

34. The colour of halogens progressively deepens from fluorine to iodine because :-
 (A) halogens of higher atomic number absorb light of longer wavelength since the difference in energy between the ground state and excited state decreases as the atomic number increases.
 (B) fluorescence and phosphorescence become more intense as the atomic numbers of halogen increases
 (C) the standard electrode potential increases from I_2 to F_2
 (D) halogens of higher atomic number absorb light of shorter wavelength since the difference in energy between the ground state and excited state increases as the atomic number increases.
35. Which of the following pairs of halogens have approximately identical bond energy ?
 (A) F_2 and Br_2 (B) F_2 and I_2 (C) F_2 and Cl_2 (D) Cl_2 and I_2
36. Which of the following is arranged in order of increasing ionic character :-
 (A) $PbCl_2 < SnCl_4 < KCl < MgCl_2$ (B) $SnCl_4 < PbCl_2 < KCl < MgCl_2$
 (C) $SnCl_4 < PbCl_2 < MgCl_2 < KCl$ (D) $PbCl_2 < SnCl_4 < MgCl_2 < KCl$
37. $XeOF_4$ contains :-
 (A) six electron pairs forming an octahedron with two positions occupied by lone pairs
 (B) two π -bonds and the remaining six electron pairs, forming an octahedron
 (C) three π -bonds and the remaining four electron pairs forming an tetrahedron
 (D) one π -bonds and the remaining six electron pairs forming an octahedron with one position occupied by a lone pair
38. The azide ion has :-
 (A) 20 outer electrons and is isoelectronic with Br_2O (B) 18 outer electrons and is isoelectronic with NO_2^-
 (C) 16 outer electrons and is isoelectronic with CO_2 (D) 14 outer electrons and is isoelectronic with H_2O_2
39. Which of the following pairs of ions do not represent cyclic and chain silicates ?
 (A) $Si_2O_7^{2-}$ and $(SiO_3)_n^{2n-}$ (B) $Si_3O_9^{6-}$ and $(Si_4O_{11})_n^{6n-}$
 (C) $Si_2O_7^{2-}$ and $(Si_2O_5)_n^{2n-}$ (D) $Si_2O_7^{7-}$ and $(SiO_3)_n^{2n-}$
40. Which of the following statements is/are incorrect ?
 (A) B_2H_6 is not an electron-deficient molecule.
 (B) the dipole moment of BF_3 is zero
 (C) $B(OH)_3$ partially reacts with water to form H_3O^+ and $[B(OH)_4]^-$, and behaves like a weak acid.
 (D) BF_3 and BrF_3 molecules have different shapes.
41. Rotation around the bond (between the underlined atoms) is restricted in :
 (A) $\underline{C}C_2H_4$ (B) $H_2\underline{O}O_2$ (C) $\underline{C}C_2H_2$ (D) $\underline{C}C_2H_6$
42. The H bond in solid HF can be best represented as :
 (A) $H - F \cdots H - F \cdots H - F$
 (B) $\begin{array}{c} H & & H & & H & & H \\ & \diagdown & & \diagdown & & \diagdown & \\ & F & \cdots & H & \cdots & F & \cdots & H \end{array}$
 (C) $\begin{array}{c} & F & & & F & & \\ & \diagdown & & \diagup & \diagdown & & \diagup \\ H & & H & & H & & H \\ & \diagup & & \diagdown & \diagup & & \diagdown \\ & F & \cdots & H & \cdots & F & \cdots & H \end{array}$
 (D) $\begin{array}{c} & H & & & H & & \\ & \diagdown & & \diagup & \diagdown & & \diagup \\ & F & \cdots & H & \cdots & F & \cdots & H \\ & \diagup & & \diagdown & \diagup & & \diagdown \\ & H & & H & & H & & H \end{array}$
43. Which of the following statements is/are correct ?
 (A) NH_2^+ shows sp^2 - hybridisation whereas NH_2^- shows sp^3 - hybridisation
 (B) $Al(OH)_4^-$ has a regular tetrahedral geometry
 (C) sp^2 -hybridized orbitals have equal s-and p-character
 (D) usually hybridized orbitals form σ -bonds

EXERCISE-03**MISCELLANEOUS TYPE QUESTIONS****TRUE OR FALSE :**

1. The polarising power of a cation is directly proportional to its size.
2. The polarisability of an anion is directly proportional to its charge.
3. For a given cation, greater the polarisability of the anion, more the covalent character.
4. An element with low ionization potential is most likely to form a covalent bond with an other element having a high electron affinity.
5. Ionic interactions are stronger than covalent bonds.
6. Two non-metal atoms are likely to form covalent bonds on combination.
7. Ionic interactions are directional.
8. All molecules having polar bonds are polar.
9. The CH_2Cl_2 molecule may be polar or nonpolar depending on its geometry.
10. Two isomers of $\text{C}_2\text{H}_2\text{Cl}_2$ are polar.
11. The net dipole in the water molecules is the resultant of its bond dipoles.
12. SO_2 is polar whereas CO_2 is non-polar.
13. If all bonds in a molecule are polar, the molecule as a whole must be polar.
14. The bond angle around B in BCl_3 and BF_3 is same.
15. NH_3 molecule involve sp^3 hybridisation of N-atom.
16. The bond length decreases with increase in multiplicity of bonds between two atoms.
17. The geometry of NH_3 and BH_3 is same.
18. Dipole moment of NF_3 is less than that of NH_3 .
19. A non-polar molecule can have a polar bond in it.
20. In ionic bond formation octet is completed.
21. π -bond is formed by the colateral overlapping of atomic orbitals.
22. Normally the covalent bond is non-directional.
23. He_2 having no existence because its bond order is zero.
24. O_2 , O_2^- and O_2^+ all are paramagnetic.
25. When bond order increases, bond length decreases.
26. Bond order is the measurement of bond strength.
27. In hybridisation, only orbitals are involved not electrons.
28. SF_6 molecule is octahedral.
29. Ethyne is a linear molecule.
30. CO_2 is a polar molecule but not have polar bond.
31. C-C sigma bond in ethyne is sp^2-sp^2 .
32. XeF_4 molecule is square planar in shape.
33. Hybridised orbitals have identical shape.
34. The bond angle in Cl_2O is equal to that of OF_2 .
35. The density of water is more than ice.
36. HCl is a gas while HF is high boiling point liquid because H-F bond is stronger.
37. Liquid NH_3 does not contain H-bond.

38. Dipole moment is completely based on ionic nature of bond.
39. BeF_2 containing dipole moment while H_2O having zero dipole moment.
40. In CO_2 molecule, C–O bond is polar but CO_2 molecule is non-polar because the vector sum of two C–O bond is zero.
41. Odd electron molecule is paramagnetic.
42. The reason for resonance, is delocalisation of π -electrons.
43. During the formation of covalent bond both shared electron having opposite spin.
44. The bond Hg-Cl is more ionic in HgCl_2 than Hg_2Cl_2 .
45. The solubility of ionic solids decreases when hydration energy of ions increases.
46. Between layers of graphite, bond will be covalent.
47. The polarising power of Zn^{2+} is greater than Ca^{2+} ions.
48. The colour of PbI_2 is yellow. The reason for this is large size of Pb^{+2} ion.
49. The H – N – H bond angle in NH_3 is greater than H – As – H bond angle in AsH_3 .
50. Linear overlap of atomic p-orbitals leads to a sigma bond.
51. The dipole moment of CH_3F is greater than that of CH_3Cl .
52. sp^2 hybrid orbitals have equal s- and p-character.
53. All the Al – Cl bonds in Al_2Cl_6 are not equivalent.

FILL IN THE BLANKS :

1. Hydrogen bond energy is around
2. The valence atomic orbitals on carbon in silver acetylide are hybridised.
3. The hybridisation state of oxygen in water molecule is
4. When N_2 goes to N_2^+ , then N–N bond distance and when O_2 goes to O_2^+ , then O–O bond distance
5. Among N_2O , SO_2 , I_3^+ and I_3^- , the linear species are and

MATCH THE COLUMN :

1.

Column-I		Column-II	
(A)	O_3	(p)	covalent bond
(B)	$\text{H}_2\text{O}_{(l)}$	(q)	Co-ordinate bond
(C)	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$	(r)	Hydrogen bond
(D)	$(\text{NH}_4)_2\text{SO}_4$	(s)	Ionic bond

2.

Column-I		Column-II	
(A)	$\text{B}_3\text{N}_3\text{H}_6$	(p)	planar geometry
(B)	H_2O_2	(q)	non planar geometry
(C)	B_2H_6	(r)	no lone pair
(D)	I_2Cl_6	(s)	non polar molecule

3.

Column-I	
(A)	$B_2H_6 + 2NaH \longrightarrow$
(B)	$H_3BO_3 + \text{water} \longrightarrow$
(C)	$BeH_2 \longrightarrow (BeH_2)_{(s)}$
(D)	$BF_3 + NaF \longrightarrow$

Column-II	
(p)	no change in hybridisation of under line atom
(q)	$sp^2 \longrightarrow sp^3$ (change in hybridisation)
(r)	breaking of $3C - 2e^-$ bond
(s)	formation of $3C - 2e^-$ bond

4.

Column-I	
(A)	N_2^+ is stable than N_2^-
(B)	NO can easily loss its electron than N_2
(C)	NO have large bond length than NO^+
(D)	He_2^+ exist but less stable than H_2^+

Column-II	
(p)	due to one have higher electrons in antibonding than other
(q)	one have B.O. 3 and other have 2.5
(r)	both are paramagnetic with same bond order
(s)	one paramagnetic and other diamagnetic

ASSERTION & REASON QUESTION :

These questions contains, Statement-I (assertion) and Statement-II (reason).

(A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I

(B) Statement-I is True, Statement-II is True ; Statement-II is NOT a correct explanation for Statement-I

(C) Statement-I is True, Statement-II is False.

(D) Statement-I is False, Statement-II is True.

1. **Statement-I** : FeI_3 cannot exist in an aqueous solution.

Because

Statement-II : Fe^{3+} oxidizes I^- to I_2 easily.

2. **Statement-I** : SF_6 exists but SH_6 does not.

Because

Statement-II : $d\pi-p\pi$ bonding cannot take place in SH_6

3. **Statement-I** : The stability of peroxides and superoxides increases in passing from Li to Cs.

Because

Statement-II : The electropositive character of the elements in the periodic table increases on moving down a group.

4. **Statement-I** : Borazole is aromatic in nature.

Because

Statement-II : Nitrogen contributes π -electrons to the system.

5. **Statement-I** : The first ionization energy of Be is greater than that of B.

Because

Statement-II : The 2p orbital is lower in energy than the 2s.

6. **Statement-I** : Bond order of O_2 and BN is same.

Because

Statement-II : O_2 and BN are isoelectronic

7. **Statement-I** : CO_2 is non polar while SO_2 is polar molecule.

Because

Statement-II : S-O bonds are polar while C-O non polar.

8. **Statement-I** : CO_2 and SiO_2 has same physical state at room temp.

Because

Statement-II : Carbon has more electronegativity than silicon atom.

9. **Statement-I** : In NF_3 molecule lone pair resides in sp^3 hybrid orbital.

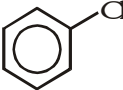
Because

Statement-II : NF_3 has pyramidal shape.

10. **Statement-I** : N_2O , CO_2 & I_3^- are isostructural.

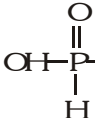
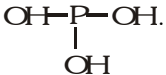
Because

Statement-II : All three have same hybridise central atom.

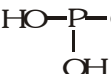
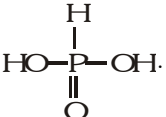
11. **Statement-I** : Dipole moment of  is similar as bromo benzene.

Because

Statement-II : Dipole moment of $\text{Cl} - \text{C}$ bond is greater than $\text{Br} - \text{C}$ bond.

12. **Statement-I** : PCl_3 on hydrolysis gives  and .

Because

Statement-II : H_3PO_3 exists in two tautomeric forms :  $\rightleftharpoons$ .

13. **Statement-I** : Super oxide ion is paramagnetic whereas peroxide ion is diamagnetic.

Because

Statement-II : Super oxide ion has one unpaired electron whereas per oxide ion has no unpaired electron.

14. **Statement-I** : Although PF_5 , PCl_5 and PBr_5 are known, the penta halide of nitrogen have not been observed.

Because

Statement-II : Phosphorus has lower electronegativity than nitrogen.

15. **Statement-I** : Among alkali metal cations, $\text{Li}^+(\text{aq.})$ has highest electrical conductance.

Because

Statement-II : $\text{Li}^+(\text{aq.})$ is largest alkali metal cation because of greater degree of hydration.

16. **Statement-I** : HNO_3 is a stronger acid than HNO_2 .

Because

Statement-II : In HNO_3 , there are two $\text{N}-\text{O}$ linkage whereas in HNO_2 there is only one.

17. **Statement-I** : $\text{Al}(\text{OH})_3$ is amphoteric in nature.

Because

Statement-II : $\text{Al}-\text{O}$ and $\text{O}-\text{H}$ bonds can be broken with equal ease in $\text{Al}(\text{OH})_3$.

18. **Statement-I** : H_2SO_4 is more viscous than water.

Because

Statement-II : In H_2SO_4 , S has highest oxidation state.

19. **Statement-I** : F_3^- ion is known, which has same geometry as X_3^- ($\text{X} = \text{Cl}, \text{Br}, \text{I}$)

Because

Statement-II : F is most electronegative element.

20. **Statement-I** : The p-isomer of dichlorobenzene has higher melting point than o-and m-isomer.
Because
Statement-II : p-isomer is symmetrical and thus shows more closely packed structure.
21. **Statement-I** : Boron does not show univalent nature but unipositive nature of thallium is quite stable.
Because
Statement-II : Inert pair effect predominates in thallium.
22. **Statement-I** : H_3BO_3 is monobasic acid in water.
Because
Statement-II : In water ionise as $\text{H}_3\text{BO}_3 \rightleftharpoons \text{H}_2\text{BO}_3^- + \text{H}^+$.

COMPREHENSION BASED QUESTIONS :

Comprehension # 1

In order to explain the shape and geometry of molecules. The valence bond theory was supplemented by the concept of Hybridization. This is a Hypothetical concept and has been introduced by Pauling and Slater. According to this concept any number of atomic orbitals of an atom which differ in energy slightly may mix with each other to form new orbitals called hybrid orbitals. The process of mixing or amalgamation of atomic orbitals of nearly same energy to produced a set of entirely new orbitals of equivalent energy is known as Hybridization.

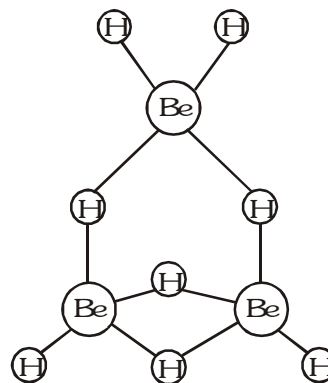
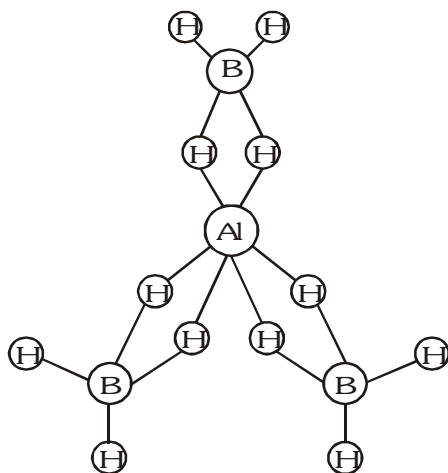
- The hybridization of carbon atoms in $\text{C} - \text{C}$ single bond of $\text{H} - \text{C} \equiv \text{C} - \text{CH} = \text{CH}_2$ is :
 (A) $\text{sp}^3 - \text{sp}^3$ (B) $\text{sp}^2 - \text{sp}^3$ (C) $\text{sp} - \text{sp}^2$ (D) $\text{sp}^3 - \text{sp}$
- In XeF_2 , XeF_4 and XeF_6 the number of lone pairs on Xe is respectively :
 (A) 2, 3, 1 (B) 1, 2, 3 (C) 4, 1, 2 (D) 3, 2, 1
- Which of the following is the correct set :
 (A) H_2O ; sp^3 , angular (B) H_2O ; sp^2 , linear
 (C) NH_3 ; sp^2 , pyramidal (D) BF_3 ; sp^3 , trigonal planar
- In NO_3^- ion, number of bond pair and lone pairs of electrons on nitrogen atom are :
 (A) 2, 2 (B) 3, 1 (C) 1, 3 (D) 4, 0
- T-shape is exhibited by molecule :
 (A) ClF_3 (B) CHCl_3 (C) CCl_4 (D) PCl_5

Comprehension # 2

$\text{Na}[\text{BH}_4]$ is ionic compound contain $\text{BH}_4^\ominus$ tetrahydrido borate ion and in solid state NaBH_4 has sodium chloride structure. Not all tetrahydridoborates are ionic. The beryllium, aluminium and transition metal borohydrides become increasingly covalent and volatile. In these type of tetrahydrido borate, the $\text{BH}_4^\ominus$ form covalent bond with metal ion. One or more H atoms in a $\text{BH}_4^\ominus$ act as a bridging and bond to metal, forming a three centre bond with two electrons shared by three atoms.

The $\text{BH}_4^\ominus$ is usually in that it may form one two or three such three centre bonds to the metal ion when forms covalent bond.

$\text{Be}(\text{BH}_4)_2$ and $\text{Al}(\text{BH}_4)_3$ are covalent and structures are given below.



1. $\text{Li}[\text{AlH}_4]$ is used as a reducing agent in many reaction & it is prepared by excess LiH and AlCl_3 . Select incorrect statement about $\text{Li}[\text{AlH}_4]$.

(A) hybridisation of Al is same as B in $\text{Na}[\text{BH}_4]$ (B) geometry of around Al is same as $\text{AlCl}_4^\ominus$

(C) $\text{AlH}_4^\ominus$, $\text{BH}_4^\ominus$, $\text{AlCl}_4^\ominus$ are iso-structural (D) $\text{AlH}_4^\ominus$, $\text{AlCl}_4^\ominus$, $\text{BH}_4^\ominus$ are iso electronic.
2. Select correct about $\text{Al}(\text{BH}_4)_3$:

(A) all three tetrahydride borate form two hydrogen bridges

(B) two $\text{BH}_4^\ominus$ form 2 hydrogen bridges and one form one hydrogen bridge.

(C) one $\text{BH}_4^\ominus$ form 2 hydrogen bridge and two form one hydrogen bridge

(D) B form only $2c-2e^-$ bond
3. Total no. of $2c-2e^-$ bond and $3c-2e^-$ bond in $\text{Al}(\text{BH}_4)_3$ are respectively :

(A) 6 , 12 (B) 6 , 6 (C) 12 , 12 (D) 12 , 6
4. Total $2c-2e$ and $3c-2e$ bonds in $\text{Be}(\text{BH}_4)_2$ are respectively :

(A) 8, 4 (B) 4, 8 (C) 4, 4 (D) 8, 8

Comprehension # 3

The molecular orbital with the lowest energy is filled first. Thus σ (1s) is filled first where as σ^* (2p) is filled in the last, also the maximum number of electron in bonding and antibonding molecular orbitals are according to Pauli and Hund's rule. As an electron in an antibonding molecular orbital cancels out the stability introduced by the electron in a bonding molecular orbital, it means that in order for bonding of atoms to occur there should be an excess of bonding electrons over antibonding electrons. In case where the number of bonding and antibonding electrons are equal, no bond will be formed between the atoms. With the help of above discussion, we can define easily bond order, relative bond length, relative stability and magnetic properties for a molecule.

Read the above paragraph carefully and give the answer of following questions :

- In an antibonding molecular orbital, there is a point between the two probability contours of hydrogen atoms. This is called.
 - antinode
 - node
 - a plane where electron charge density is maximum
 - A and C both are correct
- According to MOT which statement is correct about Boron molecule ?
 - it is diamagnetic in nature
 - it is paramagnetic in nature having magnetic moment 2.8 B.M. by using spin only formula
 - it is paramagnetic but having magnetic moment 1.7 B.M.
 - its bond order is 2
- Which of the following are paramagnetic in nature ?

(A) B_2 , N_2 and C_2	(B) O_2 , B_2 , N_2^- and O_2^-
(C) O_2^- , N_2^+ and CO	(D) B_2 , C_2 and F_2

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -5
True / False 1. F 2. T 3. T 4. F 5. F 6. T 7. F 8. F 9. F 10. T 11. F 12. T 13. F 14. T 15. T 16. T 17. F 18. T 19. T 20. T 21. T 22. F 23. T 24. T 25. T 26. T 27. T 28. T 29. T 30. F 31. F 32. T 33. T 34. F 35. T 36. F 37. F 38. F 39. F 40. T 41. T 42. T 43. T 44. F 45. F 46. F 47. T 48. F 49. T 50. T 51. F 52. F 53. T		
Fill in the Blanks 1. $4.2 - 8.4 \text{ kJ mol}^{-1}$ 2. sp 3. sp^3 4. increases, decreases 5. N_2O , I_3^-		
Match the Column 1. A - (p, q), B - (p, r), C - (p, q, r, s), D - (p, q, s) 2. A - (p, r, s), B - (q), C - (q, r, s), D - (p, s) 3. A - (p, r), B - (q), C - (s), D - (q) 4. A - (p, r), B - (p, q, s), C - (p, q, s), D - (p, r)		
Assertion - Reason Questions 1. A 2. B 3. A 4. B 5. C 6. C 7. C 8. D 9. A 10. C 11. D 12. A 13. A 14. B 15. D 16. C 17. C 18. B 19. D 20. A 21. A 22. C		
Comprehension Based Questions Comprehension #1 : 1. (C) 2. (D) 3. (A) 4. (D) 5. (A) Comprehension #2 : 1. (D) 2. (A) 3. (B) 4. (C) Comprehension #3 : 1. (B) 2. (B) 3. (B)		

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE****SUBJECTIVE QUESTIONS :**

1. Give reasons for the following :
 - (a) KHF_2 is known whereas KHCl_2 is unknown.
 - (b) $(\text{CH}_3)_3\text{N}$ is pyramidal but $(\text{SiH}_3)_3\text{N}$ is trigonal planar.
 - (c) CO_2 is a gas but SiO_2 is a solid.
2.
 - (a) The structures of N_2O_3 and P_2O_3 are different. Explain.
 - (b) Among H-I, H-Br and H-Cl bonds which is weakest?
3. Suppose that the stability of carbonates when heated depends on the ability of the metal cation to polarize the carbonate ion and remove an oxide ion from it, so releasing CO_2 .
 - (a) Predict the order of thermal stability of the group 1 and 2 metal carbonates ?
 - (b) Comment on the likely stability of aluminium carbonate ?
4.
 - (a) BF_3 has less lewis acidic property than BBr_3 . Why ?
 - (b) $(\text{CH}_3)_3\text{C}-\text{OH}$ is less acid than $(\text{CH}_3)_3\text{Si}-\text{OH}$. Why ?
 - (c) Why in the presence of ethyne HF exhibit more acidic character.
 - (d) Why SF_4 can be hydrolysed easily but SF_6 can not ?
5. In what hybridization state is the beryllium atom in BeCl_2 molecule? How will the type of hybridization change when BeCl_2 transform to the solid state.
6. Give a suitable reason for the high mobility of H^+ ions through the ice than through liquid water.
7.
 - (a) Example the following :
 - (i) Boron trichloride has triangular planar geometry whereas aluminium trichloride has tetrahedral dimeric structure.
 - (ii) Barium sulphate is sparingly soluble in water whereas beryllium sulphate is freely soluble.
 - (iii) Dioxygen (O_2) molecule is stable whereas disulphur (S_2) is unstable at the room temperature.
8. Assuming Z-axis as molecular axis, label the molecular orbitals formed by the following combination of atomic orbitals :
 - (i) $1s + 1s$
 - (ii) $2p_y - 2p_y$
 - (iii) $2p_z + 2p_z$
 - (iv) $2s + 2s$
 - (v) $2p_x + 2p_x$
9. When a magnet is dipped in a jar of liquid O_2 , some O_2 clings to it. Why ?
10.
 - (a) What type of bonding holds the layer in graphite ?

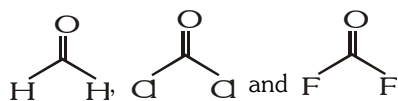
Why will graphite conduct electricity well in a direction parallel to the planes of hexagons, but not at all well in a direction perpendicular to the planes ?
 - (b) PCl_5 exists as solid in the form of $[\text{PCl}_4]^+ [\text{PCl}_6]^-$, yet it is a non conductor of electricity. Why?
11. A diatomic molecule has a dipole moment of 1.2 D. If the bond distance is 0.1 Å, what fraction of an electron charge, e exist on each atom?
12. Through the electronegativities of nitrogen and chlorine are same, NH_3 exists as liquid whereas HCl as gas. Why?

13. The percent ionic character in HCl is 18.08. The observed dipole moment is 1.08 D. Find the internuclear distance in HCl.
14. HBr has dipole moment 2.6×10^{-30} CM. If the ionic character of the bond is 11.5 %, calculate the interatomic spacing.
15. Dipole moment of LiF was experimentally determined and was found to be 6.32 D. Calculate percentage ionic character in LiF molecule Li – F bond length is 0.156 nm.
16. Based upon M.O. theory state reason for the paramagnetic character of CN, the diamagnetic character of CN^- , the stability of CN^- and calculate their respective bond orders.
17. Draw the structure of following compound
- (i) S_8
 - (ii) N_2H_4
 - (iii) P_4H_{10}
 - (iv) POCl_3
 - (v) XeOF_4
 - (vi) C_3O_2
 - (vii) BrF_5
-

CONCEPTUAL	SUBJECTIVE	EXERCISE	ANSWER KEY	EXERCISE -4(A)
14.	1.4 Å	15.	84.35 %	16. $2\frac{1}{2}$, 3

EXERCISE-04 [B]**BRAIN STORMING SUBJECTIVE EXERCISE****SUBJECTIVE QUESTIONS :**

- Draw the structure of following compound
 - $\text{Na}[\text{B}_3\text{O}_3(\text{OH})_4]$
 - $\text{Na}[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$
 - $\text{Na}_6\text{P}_6\text{O}_{18}$
 - $\text{S}_3\text{O}_6^{-2}$
 - S_3O_9
 - $(\text{CN})_2$
- What is the increasing order of the bond angle for the following compounds ?



- How many S – S linkage present in $\text{H}_2\text{S}_n\text{O}_6$
- Draw the geometry of following three molecules and explain with proper reasoning :
 - $\text{N}(\text{CH}_3)_3$
 - $\text{N}(\text{SiH}_3)_3$
 - $\text{P}(\text{SiH}_3)_3$
- The dipole moment of LiH is 1.964×10^{-29} CM and the intermolecular distance between Li and H in this molecule is $1.596 \text{ \AA}$. What is percent ionic character in molecule ?
- The dipole moment of KCl is 3.336×10^{-29} coulomb metre which indicates that it is a highly polar molecule. The interatomic distance between K^+ and Cl^- in this molecule is 2.6×10^{-10} m. Calculate the dipole moment of KCl molecule, if these were opposite charges of one fundamental unit located at each nucleus. Calculate percentage ionic character of KCl.
- Assuming covalent radii to be additive property ; calculate the iodine - iodine distance in o-, m-, p- di-iodobenzene. The benzene ring is regular hexagon and each C – I bond lies on a line passing through the centre of hexagon. The C – C bond length C_6H_6 are $1.40 \text{ \AA}$ and covalent radius of iodine and carbon atom are $1.33 \text{ \AA}$ and $0.77 \text{ \AA}$. Also neglect different overlapping effect.
- Assuming that all the four valency of carbon atom in propane pointing towards the corners of a regular tetrahedron. Calculate the distance between the terminal carbon atoms in propane. Given, C – C single bond length is $1.54 \text{ \AA}$.

CONCEPTUAL SUBJECTIVE EXERCISE**ANSWER KEY****EXERCISE -4(B)**

5. 76.82%

6. 4.165×10^{-29} coulomb metre.7. $3.50 \text{ \AA}$, $6.06 \text{ \AA}$, $7.0 \text{ \AA}$ 8. $2.514 \text{ \AA}$

EXERCISE-05[A]**PREVIOUS YEARS QUESTIONS**

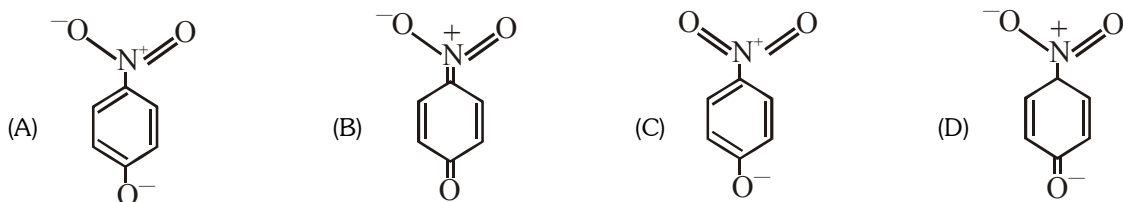
1. In which of the following species is the underlined carbon having sp^3 - hybridisation ? [AIEEE 2002]
(1) $CH_3-\underline{C}OOH$ (2) $CH_3\underline{C}H_2OH$ (3) $CH_3\underline{C}OCH_3$ (4) $CH_2=\underline{C}H-CH_3$
2. Which of the following statements is true ? [AIEEE 2002]
(1) HF is less polar than HBr
(2) Water does not contain any ions
(3) Chemical bond formation takes place when forces of attraction overcome the forces of repulsion
(4) In covalent bond, transfer of electrons takes place
3. A square planar complex is formed by hybridisation of which atomic orbital ? [AIEEE 2002]
(1) s, p_x, p_y, d_{yz} (2) $s, p_x, p_y, d_{x^2-y^2}$ (3) s, p_x, p_y, d_{z^2} (4) s, p_x, p_y, d_{xy}
4. The reason for double helical structure of DNA is operation of : [AIEEE 03]
(1) dipole-dipole interaction (2) hydrogen bonding
(3) electrostatic attractions (4) vander Walls' forces
5. Which one of the following pairs of molecules will have permanent dipole moments for both members [AIEEE 03]
(1) NO_2 and CO_2 (2) NO_2 and O_3 (3) SiF_4 and CO_2 (4) SiF_4 and NO_2
6. The pair of species having identical shapes for molecules of both species is [AIEEE 03]
(1) XeF_2, CO_2 (2) BF_3, PCl_3 (3) PF_5, IF_5 (4) CF_4, SF_4
7. The correct order of bond angles (smallest first) in H_2S, NH_3, BF_3 and SiH_4 is :- [AIEEE-2004]
(1) $H_2S < NH_3 < SiH_4 < BF_3$ (2) $NH_3 < H_2S < SiH_4 < BF_3$
(3) $H_2S < SiH_4 < NH_3 < BF_3$ (4) $H_2S < NH_3 < BF_3 < SiH_4$
8. The bond order in NO is 2.5 while that in NO^+ is 3. Which of the following statements is true for these two species ? [AIEEE-2004]
(1) Bond length in NO^+ is equal to that in NO (2) Bond length in NO is greater than in NO^+
(3) Bond length in NO^+ is greater than in NO (4) Bond length is unpredictable
9. The states of hybridization of boron and oxygen atoms in boric acid (H_3BO_3) are respectively [AIEEE-2004]
(1) sp^3 and sp^2 (2) sp^2 and sp^3 (3) sp^2 and sp^2 (4) sp^3 and sp^3
10. Which one of the following has the regular tetrahedral structure ? [AIEEE-2004]
(1) BF_4^- (2) SF_4 (3) XeF_4 (4) $[Ni(CN)_4]^{2-}$
(Atomic nos.: B = 5, S = 16, Ni = 28, Xe = 54)
11. The maximum number of 90° angles between bond pair-bond pair of electrons is observed in :- [AIEEE-2004]
(1) dsp^2 hybridization (2) sp^3d hybridization (3) dsp^3 hybridization (4) sp^3d^2 hybridization
12. Beryllium and aluminium exhibit many properties which are similar. But, the two elements differ in [AIEEE-2004]
(1) Forming covalent halides (2) Forming polymeric hydrides
(3) Exhibiting maximum covalency in compounds (4) Exhibiting amphoteric nature in their oxides

13. Which one of the following species is diamagnetic in nature ? [AIEEE-2005]
- (1) He_2^+ (2) H_2 (3) H_2^+ (4) H_2^-
14. lattice energy of an ionic compound depends upon [AIEEE-2005]
- (1) charge on the ion only (2) size of the ion only
(3) packing of the ion only (4) charge and size of the ion
15. The molecular shapes of SF_4 , CF_4 and XeF_4 are [AIEEE-2005]
- (1) the same with 2, 0 and 1 lone pair of electrons on the central atom, respectively
(2) the same with 1, 1 and 1 lone pair of electrons on the central atoms, respectively
(3) different with 0, 1 and 2 lone pair of electrons on the central atoms, respectively
(4) different with 1, 0 and 2 lone pair of electrons on the central atoms, respectively
16. Of the following sets which one does not contain isoelectronic species ? [AIEEE-2005]
- (1) PO_4^{3-} , SO_4^{2-} , ClO_4^- (2) CN^- , N_2 , C_2^{2-}
(3) SO_3^{2-} , CO_3^{2-} , NO_3^- (4) BO_3^{3-} , CO_3^{2-} , NO_3^-
17. Which of the following molecules\ions does not contain unpaired electrons? [AIEEE-2006]
- (1) N_2^+ (2) O_2 (3) O_2^{2-} (4) B_2
18. Among the following mixtures, dipole-dipole as the major interaction, is present in [AIEEE-2006]
- (1) KCl and water (2) benzene and carbon tetrachloride
(3) benzene and ethanol (4) acetonitrile and acetone
19. A metal, M forms chlorides in its +2 and +4 oxidation states. Which of the following statements about these chlorides is correct? [AIEEE-2006]
- (1) MCl_2 is more ionic than MCl_4 (2) MCl_2 is more easily hydrolysed than MCl_4
(3) MCl_2 is more volatile than MCl_4 (4) MCl_2 is more soluble in anhydrous ethanol than MCl_4
20. In which of the following molecules/ions are all the bonds **not** equal? [AIEEE-2006]
- (1) XeF_4 (2) BF_4^- (3) SF_4 (4) SiF_4
21. The decreasing values of bond angles from NH_3 (106°) to SbH_3 (91°) down group-15 of the periodic table is due to [AIEEE-2006]
- (1) decreasing lp – bp repulsion (2) increasing electronegativity
(3) increasing bp – bp repulsion (4) increasing p-orbital character in sp^3
22. In which of the following ionization processes, the bond order has increased and the magnetic behaviour has changed [AIEEE-2007]
- (1) $\text{NO} \rightarrow \text{NO}^+$ (2) $\text{O}_2 \rightarrow \text{O}_2^+$ (3) $\text{N}_2 \rightarrow \text{N}_2^+$ (4) $\text{C}_2 \rightarrow \text{C}_2^+$

EXERCISE-05[B]**PREVIOUS YEARS QUESTIONS**

Choose the correct alternative (only one correct answer).

1. The geometry & the type of hybrid orbitals present about the central atom in BF_3 is : [JEE '98]
(A) linear, sp (B) trigonal planar, sp^2 (C) tetrahedral, sp^3 (D) pyramidal, sp^3
2. The correct order of increasing C–O bond length of, CO , CO_3^{2-} , CO_2 is [JEE '99]
(A) $\text{CO}_3^{2-} < \text{CO}_2 < \text{CO}$ (B) $\text{CO}_2 < \text{CO}_3^{2-} < \text{CO}$ (C) $\text{CO} < \text{CO}_3^{2-} < \text{CO}_2$ (D) $\text{CO} < \text{CO}_2 < \text{CO}_3^{2-}$
3. In the dichromate anion [JEE '99]
(A) 4Cr – O bonds are equivalent
(B) 6Cr – O bonds are equivalent
(C) all Cr – O bonds are equivalent
(D) all Cr – O bonds are non equivalent
4. The geometry of H_2S and its dipole moment are [JEE '99]
(A) angular & non zero (B) angular & zero (C) linear & non zero (D) linear & zero
5. In compounds type ECl_3 , where E = B, P, As or Bi, the angles Cl – E – Cl for different E are in the order [JEE '99]
(A) $\text{B} > \text{P} = \text{As} = \text{Bi}$ (B) $\text{B} > \text{P} > \text{As} > \text{Bi}$ (C) $\text{B} < \text{P} = \text{As} = \text{Bi}$ (D) $\text{B} < \text{P} < \text{As} < \text{Bi}$
6. The most likely representation of resonance structure of p-nitrophenoxide is: [JEE '99]



7. Amongst H_2O , H_2S , H_2Se and H_2Te , the one with the highest boiling point is [JEE 2000]
(A) H_2O because of hydrogen bonding (B) H_2Te because of higher molecular weight
(C) H_2S because of hydrogen bonding (D) H_2Se because of lower molecular weight
8. The hybridization of atomic orbitals of nitrogen in NO_2^+ , NO_3^- and NH_4^+ are [JEE 2000]
(A) sp^2 , sp^3 and sp^2 respectively (B) sp , sp^2 and sp^3 respectively
(C) sp^2 , sp , and sp^3 respectively (D) sp^2 , sp^3 and sp respectively
9. The correct order of hybridization of the central atom in the following species NH_3 , PtCl_4^{2-} , PCl_5 and BCl_3 is [JEE 2001]
(A) dsp^2 , sp^3d , sp^2 and sp^3 (B) sp^3 , dsp^2 , sp^3d , sp^2
(C) dsp^2 , sp^2 , sp^3 , sp^3d (D) dsp^2 , sp^3 , sp^2 , sp^3d
10. The common features among the species CN^- , CO and NO^+ are [JEE 2001]
(A) Bond order three and isoelectronic
(B) Bond order three and weak field ligands
(C) Bond order two and π -acceptors
(D) Isoelectronic and weak field ligands
11. Specify hybridization of N and B atoms in a 1 : 1 complex of BF_3 and NH_3 [JEE 2002]
(A) N : tetrahedral, sp^3 ; B : tetrahedral, sp^3 (B) N : pyramidal, sp^3 ; B : pyramidal, sp^3
(C) N : pyramidal, sp^3 ; B : planar, sp^2 (D) N : pyramidal, sp^3 ; B : tetrahedral, sp^3

12. The nodal plane in the π -bond of ethene is located in [JEE 2002]
 (A) the molecular plane
 (B) a plane parallel to the molecular plane
 (C) a plane perpendicular to the molecular plane which bisects, the carbon-carbon a bond at right angle.
 (D) a plane perpendicular to the molecular plane which contains, the carbon-carbon bond.
13. Identify the least stable ion amongst the following : [JEE 2002]
 (A) Li^- (B) Be^- (C) B^- (D) C^-
14. Which of the following molecular species has unpaired electron(s) ? [JEE 2002]
 (A) N_2 (B) F_2 (C) O_2^- (D) O_2^{2-}
15. Which of the following are isoelectronic and isostructural ? [JEE 2003]
 NO_3^- , CO_3^{2-} , ClO_3^- , SO_3
 (A) NO_3^- , CO_3^{2-} (B) SO_3 , NO_3^- (C) ClO_3^- , CO_3^{2-} (D) CO_3^{2-} , SO_3
16. According to molecular orbital theory which of the following statement about the magnetic character and bond order is correct regarding O_2^+ [JEE 2004]
 (A) Paramagnetic and Bond order $< \text{O}_2$ (B) Paramagnetic and Bond order $> \text{O}_2$
 (C) Diamagnetic and Bond order $< \text{O}_2$ (D) Diamagnetic and Bond order $> \text{O}_2$
17. Which species has the maximum number of lone pair of electrons on the central atom? [JEE 2005]
 (A) ClO_3^- (B) XeF_4 (C) SF_4 (D) I_3^-
18. The percentage of p-character in the orbitals forming P-P bonds in P_4 is
 (A) 25 (B) 33 (C) 50 (D) 75
19. Among the following, the paramagnetic compound is [JEE 2007]
 (A) Na_2O_2 (B) O_3 (C) N_2O (D) KO_2
20. The species having bond order different from that in CO is [JEE 2007]
 (A) NO^- (B) NO^+ (C) CN^- (D) N_2
21. The structure of XeO_3 is [JEE 2007]
 (A) linear (B) planar (C) pyramidal (D) T-shaped
22. **Statement-1** : p-Hydroxybenzoic acid has a lower boiling point than o-hydroxybenzoic acid. [JEE 2007]
and
Statement-2 : o-Hydroxybenzoic acid has intramolecular hydrogen bonding.
 (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.
23. **Statement-1** : In water, orthoboric acid behaves as a weak monobasic acid. [JEE 2007]
and
Statement-2 : In water, orthoboric, acid acts as a proton donor.
 (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.

24. **Statement-1** : Pb^{+4} compounds are stronger oxidizing agents than Sn^{4+} compounds [JEE 2008]
and
Statement-2 : The higher oxidation states for the group 14 elements are more stable for the heavier members of the group due to 'inert pair effect' .
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement-2 is False.
(D) Statement-1 is False, Statement-2 is True.
25. Match each of the diatomic molecules in Column I with its property / properties in Column II. [JEE 2009]
- | Column I | Column II |
|--------------------|------------------------------------|
| (A) B_2 | (P) Paramagnetic |
| (B) N_2 | (Q) undergoes oxidation |
| (C) O_2^- | (R) Undergoes reduction |
| (D) O_2 | (S) Bond order ≥ 2 |
| | (T) Mixing of 's' and 'p' orbitals |
26. The nitrogen oxide(s) that contain(s) N–N bond(s) is (are) [JEE 2009]
(A) N_2O (B) N_2O_3 (C) N_2O_4 (D) N_2O_5
27. In the reaction

$$2\text{X} + \text{B}_2\text{H}_6 \longrightarrow [\text{BH}_2(\text{X})_2]^+ [\text{BH}_4]^-$$
the amine(s) X is (are) [JEE 2009]
(A) NH_3 (B) CH_3NH_2 (C) $(\text{CH}_3)_2\text{NH}$ (D) $(\text{CH}_3)_3\text{N}$
28. The species having pyramidal shape is [JEE 2010]
(A) SO_3 (B) BrF_3 (C) SiO_3^{2-} (D) OSF_2
29. Assuming that Hund's rule is violated, the bond order and magnetic nature of the diatomic molecule B_2 is [JEE 2010]
(A) 1 and diamagnetic (B) 0 and diamagnetic
(C) 1 and paramagnetic (D) 0 and paramagnetic
30. In allene (C_3H_4), the type(s) of hybridisation of the carbon atoms is (are) [JEE 2012]
(A) sp and sp^3 (B) sp and sp^2
(C) only sp^2 (D) sp^2 and sp^3
31. Which ordering of compounds is according to the decreasing order of the oxidation state of nitrogen- [JEE 2012]
(A) HNO_3 , NO , NH_4Cl , N_2 (B) HNO_3 , NO , N_2 , NH_4Cl
(C) HNO_3 , NH_4Cl , NO , N_2 (D) NO , HNO_3 , NH_4Cl , N_2
32. The shape of XeO_2F_2 molecule is : [JEE 2012]
(A) Trigonal bipyramidal (B) Square planar (C) tetrahedral (D) see-saw

Explain the following.

- Give reason carbon oxygen bond lengths in formic acid are $1.23\text{\AA}$ & $1.36\text{\AA}$ and both the carbon oxygen bonds in sodium formate have the same value i.e. $1.27\text{\AA}$. [JEE '88]
- Explain the first I.E. of carbon atom is greater than that of boron atom whereas the reverse is true for the second I.E. [JEE '89]

- The experimentally determined N-F bond length in NF_3 is greater than the sum of single bond covalent radii of N & F. Explain. [JEE '95]
- Explain the difference in the nature of bonding in LiF & LiI. [JEE '96]
- Give reasons for the following in one or two sentences only. [JEE '99]
 - BeCl_2 can be easily hydrolysed
 - CrO_3 is an acid anhydride

Arrange as directed.

- N_2 , O_2 , F_2 , Cl_2 in increasing order of bond dissociation energy. [JEE '88]
- CO_2 , N_2O_5 , SiO_2 , SO_3 is the increasing order of acidic character. [JEE '88]
- HOCl , HClO_2 , HClO_3 , HClO in increasing order of thermal stability. [JEE '88]
- The decreasing order of acid strength of ClOH , BrOH , IOH . [JEE '97]
- Arrange in order of increasing radii, Li^+ , Mg^{2+} , K^+ , Al^{3+} [JEE '97]
- Arrange BeSO_4 , MgSO_4 , CaSO_4 , SrSO_4 in order of decreasing thermal stability. [JEE '97]
- Decreasing order of the O-O bond length present in them [JEE 2004]
 O_2 , KO_2 and $\text{O}_2 [\text{AsF}_4]$

Subjective

- The number of water molecule(s) directly bonded to the metal centre in $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is [JEE 2009]
- Based on VSEPR theory, the number of 90 degree F-Br-F angles in BrF_5 is [JEE 2010]
- The value of n in the molecular formula $\text{Be}_n\text{Al}_2\text{Si}_6\text{O}_{18}$ is [JEE 2010]
- The total number of diprotic acids among the following is [JEE 2010]
 H_3PO_4 , H_2SO_4 , H_3PO_3 , H_2CO_3 , $\text{H}_2\text{S}_2\text{O}_7$
 H_3BO_3 , H_3PO_2 , H_2CrO_4 , H_2SO_3
- Among the following, the number of elements showing only one non-zero oxidation state is [JEE 2010]
 O, Cl, F, N, P, Sn, Tl, Na, Ti
- The difference in the oxidation numbers of the two types of sulphur atoms in $\text{Na}_2\text{S}_4\text{O}_6$ is. [JEE 2011]

PREVIOUS YEARS QUESTIONS			ANSWER KEY		EXERCISE-5 [B]	
1. (B)	2. (D)	3. (B)	4. (A)	5. (B)	6. (A)	7. (A)
8. (B)	9. (B)	10. (A)	11. (A)	12. (A)	13. (B)	14. (C)
15. (A)	16. (B)	17. (D)	18. (D)	19. (D)	20. (A)	21. (C)
22. (D)	23. (C)	24. (C)				
25. (A) → P,Q,R,T ; B → Q,R,S,T ; (C) → P,Q,R ; (D) → P,Q,R,S						
26. A,B,C	27. B,C	28. (D)	29. (A)	30. (B)	31. (B)	32. (D)
●	<u>Arrange as directed :</u>					
1. $\text{F}_2 < \text{Cl}_2 < \text{O}_2 < \text{N}_2$			2. $\text{SiO}_2 < \text{CO}_2 < \text{N}_2\text{O}_5 < \text{SO}_3$			
3. $\text{HOCl} < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4$			4. $\text{HOCl} > \text{HOBr} > \text{HOI}$			
5. $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Li}^+ < \text{K}^+$			6. $\text{SrSO}_4 > \text{CaSO}_4 > \text{MgSO}_4 > \text{BeSO}_4$			
7. $\text{KO}_2 > \text{O}_2 > \text{O}_2 [\text{AsF}_4]$						
●	<u>Subjective Questions</u>					
1. 4	2. 0	3. 3	4. 6	5. 2	6. 5	

EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. The final product obtained when boric acid is heated to red heat is –
(A) Metaboric acid (B) Tetraboric acid (C) Boron oxide (D) Pyroboric acid
2. Which of the following can be detected by the borax-bead test ?
(A) Ni^{2+} (B) Co^{2+} (C) Pb^{+2} (D) Both (A) & (B)
3. Boric acid polymerizes due to –
(A) The presence of hydrogen bonds (B) Its acidic nature
(C) Its geometry (D) Its monobasic nature
4. Aluminium is obtained by–
(A) Reduction of Al_2O_3 with coke
(B) Electrolysis of Al_2O_3 dissolved in Na_3AlF_6
(C) Reduction of Al_2O_3 with chromium
(D) Heating cryolite and alumina
5. In thermite welding, aluminium acts as –
(A) A solder (B) A flux (C) An oxidising agent (D) A reducing agent
6. From B_2H_6 , all the following can be prepared except –
(A) B_2O_3 (B) H_3BO_3 (C) $\text{B}_2(\text{CH}_3)_6$ (D) NaBH_4
7. The product formed in the reaction,
 $\text{BCl}_3 + \text{H}_2\text{O} \longrightarrow$ Product is –
(A) $\text{H}_3\text{BO}_3 + \text{HCl}$ (B) $\text{B}_2\text{O}_3 + \text{HOCl}$ (C) $\text{B}_2\text{H}_6 + \text{HCl}$ (D) No reaction
8. The hydrides of boron are called
(A) Boron hydrogen compounds (B) Hydrogen borides
(C) Boranes (D) Hydroboric acids
9. Which one of the following mixed sulphates is not an alum ?
(A) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ (B) $\text{K}_2\text{SO}_4 \cdot \text{Cr}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
(C) $\text{Na}_2\text{SO}_4 \cdot \text{Fe}(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ (D) $\text{CuSO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
10. Higher percentage of carbon is found in –
(A) Anthracite (B) Lignite (C) Bituminous (D) Peat
11. Silicones have the general formula –
(A) SiO_4^{4-} (B) $\text{Si}_2\text{O}_7^{6-}$ (C) $(\text{R}_2\text{SiO})_n$ (D) $(\text{SiO}_{3/2})_n^{2-}$
12. In which of the following there exists a $p\pi - d\pi$ bonding –
(A) Diamond (B) Graphite (C) Dimethylamine (D) Trisilylamines
13. Glass or silica soluble in –
(A) HClO_4 (B) HF (C) Aqua-regia (D) H_2SO_4

14. Producer gas is a mixture of –
 (A) CO and N₂ (B) CO₂ and H₂ (C) CO and H₂ (D) CO₂ and N₂
15. Which variety of glass is used for manufacture of optical glasses ?
 (A) Sodium glass (B) Flint glass (C) Ground glass (D) Quartz
16. The colour imparted by Co(II) compounds to glass is –
 (A) Green (B) Deep-Blue (C) Yellow (D) Red
17. The species present in solution when CO₂ is dissolved in water are –
 (A) CO₂, H₂CO₃, HCO₃⁻, CO₃²⁻ (B) H₂CO₃, CO₃²⁻
 (C) CO₃²⁻, HCO₃⁻ (D) CO₂, H₂CO₃
18. P₂O₅ is used extensively as a –
 (A) Dehydrating agent (B) Catalytic agent
 (C) Reducing agent (D) Preservative
19. The number of molecules of water needed to convert one molecules of P₂O₅ into orthophosphoric acid is –
 (A) 2 (B) 3 (C) 4 (D) 5
20. In warfare smoke screens are prepared from –
 (A) PH₃ (B) CaC₂ (C) P₂O₅ (D) COCl₂
21. In Haber's process for the manufacture of ammonia, the catalyst used is –
 (A) Finely divided nickel (B) Finely divided molybdenum
 (C) Finely divided iron (D) Finely divided platinum
22. Which one of the following nitrogen oxides is the anhydride of nitrous acid ?
 (A) N₂O (B) N₂O₃ (C) N₂O₄ (D) NO
23. Which acts both an oxidising as well as reducing agent –
 (A) HNO₃ (B) HNO₂ (C) H₂SO₄ (D) HCl
24. NO₂ is released by heating –
 (A) Pb(NO₃)₂ (B) KNO₃ (C) NaNO₂ (D) NaNO₃
25. A deep brown gas is formed by mixing two colourless gases which are –
 (A) NO₂ and O₂ (B) N₂O and NO (C) NO and O₂ (D) NH₃ and HCl
26. A metal X on heating in nitrogen gas gives Y. Y on treatment with H₂O gives a colourless gas which when passed through CuSO₄ solution gives a blue colour. Y is –
 (A) Mg(NO₃)₂ (B) Mg₃N₂ (C) NH₃ (D) MgO
27. Oil of vitriol is –
 (A) H₂SO₄ (B) H₂SO₃ (C) H₂S₂O₉ (D) H₂S₂O₈
28. The compound which gives off oxygen on moderate heating is –
 (A) Cupric oxide (B) Mercuric oxide (C) Zinc oxide (D) Aluminium oxide

29. When conc. H_2SO_4 comes in contact with sugar, it becomes black due to –
 (A) Hydrolysis (B) Hydration (C) Decolourisation (D) Dehydration
30. Which one of the following reacts with conc. H_2SO_4 ?
 (A) Au (B) Ag (C) Pt (D) All
31. HCOOH reacts with conc. H_2SO_4 to produce –
 (A) CO (B) CO_2 (C) NO (D) NO_2
32. Chlorine is manufactured by –
 (A) Birkland and Eyde's process (B) Deacon's process
 (C) Bosch process (D) Solvay's process
33. When chlorine water is kept in sunlight oxygen is evolved therefore –
 (A) Affinity of hydrogen for oxygen is less
 (B) Affinity of hydrogen for oxygen is more
 (C) Affinity of hydrogen for chlorine is more
 (D) Hydrogen is a reducing agent
34. The following acids have been arranged in the order of decreasing acid strength. Identify the correct order-
 ClOH(I) BrOH(II) IOH(III)
 (A) $\text{I} > \text{II} > \text{III}$ (B) $\text{II} > \text{I} > \text{III}$ (C) $\text{III} > \text{II} > \text{I}$ (D) $\text{I} > \text{III} > \text{II}$
35. Sea weed are important source of –
 (A) Iron (B) Chlorine (C) Iodine (D) Bromine
36. Euchlorine is a mixture of –
 (A) Cl_2 and SO_2 (B) Cl_2 and ClO_2 (C) Cl_2 and CO (D) None of these
37. Which of the following represents the correct order of increasing pK_a values of the given acids –
 (A) $\text{HClO}_4 < \text{HNO}_3 < \text{H}_2\text{CO}_3 < \text{B(OH)}_3$ (B) $\text{HNO}_3 < \text{HClO}_4 < \text{B(OH)}_3 < \text{H}_2\text{CO}_3$
 (C) $\text{B(OH)}_3 < \text{H}_2\text{CO}_3 < \text{HClO}_4 < \text{HNO}_3$ (D) $\text{HClO}_4 < \text{HNO}_3 < \text{B(OH)}_3 < \text{H}_2\text{CO}_3$
38. The word Argon means –
 (A) Noble (B) Now
 (C) Strange (D) Lazy
39. Iodine and hypo react to produce –
 (A) Na_2S (B) Na_2SO_3 (C) Na_2SO_4 (D) $\text{Na}_2\text{S}_4\text{O}_6$

CHECK YOUR GRASP					ANSWER KEY					EXERCISE-1					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	C	D	A	B	D	C	A	C	D	A	C	D	B	A	B
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	B	A	A	B	A	C	B	B	A	C	B	A	B	D	B
Que.	31	32	33	34	35	36	37	38	39						
Ans.	A	B	C	A	C	B	A	D	D						

SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)

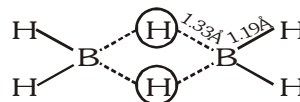
1. A mixture of boric acid with ethyl alcohol burns with green edged flame due to the formation of –
(A) Ethyl borax (B) Ethyl borate (C) Methyl borax (D) Methyl borate
2. AlCl_3 on hydrolysis gives –
(A) $\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$ (B) $\text{Al}(\text{OH})_3$ (C) Al_2O_3 (D) $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$
3. When a solution of sodium hydroxides is added in excess to the solution of potash alum, we obtain –
(A) A white precipitate (B) Bluish white precipitate
(C) A clear solution (D) A crystalline mass
4. BCl_3 does not exist as dimer but BH_3 exist as dimer (B_2H_6) because –
(A) Chlorine is more electronegative than hydrogen
(B) There is $p\pi - p\pi$ back bonding in BCl_3 but BH_3 does not contain such multiple bonding
(C) Large sized chlorine atoms do not fit in between the small boron atoms whereas small sized hydrogen atoms get fitted in between boron atoms
(D) None of the above
5. Amorphous boron on burning in air forms –
(A) $\text{B}(\text{OH})_3$ (B) Mixture of B_2O_3 and BN
(C) Only B_2O_3 (D) Only BN
6. Which of the following statements is correct ?
(A) BCl_3 and AlCl_3 are both Lewis acids and BCl_3 is stronger than AlCl_3
(B) BCl_3 and AlCl_3 both Lewis acids and AlCl_3 is stronger than BCl_3
(C) BCl_3 and AlCl_3 are both equally strong Lewis acids
(D) Both BCl_3 and AlCl_3 are not Lewis acids.
7. Which one of the following does not exist in the free form ?
(A) BF_3 (B) BCl_3 (C) BBr_3 (D) BH_3
8. Thermite is a mixture of –
(A) 3 Parts of powdered Al and 1 part of Fe_2O_3 (B) 1 part of powdered Al and 3 parts of Fe_2O_3
(C) 1 part of powdered Al and 1 part of Fe_2O_3 (D) 2 Parts of powdered Al and 1 part of Fe_2O_3
9. Borax is used as cleansing agent because on dissolving in water it gives –
(A) Alkaline solution (B) Acidic solution (C) Bleaching solution (D) Colloidal solution
10. Trisilylamine $[\ddot{\text{N}}(\text{SiH}_3)_3]$ has a –
(A) Planar geometry (B) Tetrahedral geometry
(C) Pyramidal geometry (D) None of these

11. The halide that is not hydrolysed is –
 (A) SiCl_4 (B) SiF_4 (C) CCl_4 (D) PbCl_4
12. What is false about N_2O_5 ?
 (A) It is anhydride of HNO_3 (B) It is a powerful oxidizing agent
 (C) Solid N_2O_5 is called nitronium nitrate (D) Structure of N_2O_5 contains no $[\text{N} \rightarrow \text{O}]$ bond
13. SbCl_3 and BiCl_3 on hydrolysis gives –
 (A) Sb^{+3} and Bi^{+3} (B) $\text{Sb}(\text{OH})_3$ and $\text{Bi}(\text{OH})_3$
 (C) SbOCl and BiOCl (D) None
14. The percentage of nitrogen in urea is about –
 (A) 70 (B) 63 (C) 47 (D) 28
15. Sequence of acidic character is –
 (A) $\text{SO}_2 > \text{CO}_2 > \text{CO} > \text{N}_2\text{O}_5$ (B) $\text{SO}_2 > \text{N}_2\text{O}_5 > \text{CO} > \text{CO}_2$
 (C) $\text{N}_2\text{O}_5 > \text{SO}_2 > \text{CO} > \text{CO}_2$ (D) $\text{N}_2\text{O}_5 > \text{SO}_2 > \text{CO}_2 > \text{CO}$
16. Tip of saftymatch stick are made up of –
 (A) Sulphur and potassium (B) Sulphur
 (C) Sulphur, dichromate and phosphorus (D) Sulphur, dichromate and potassium
17. Of the following, which has three electron bond in its structure ?
 (A) Nitrous oxide (B) Nitric oxide
 (C) Dinitrogen trioxide (D) Nitrogen pentoxide
18. Which of the following leaves no residue on heating ?
 (A) $\text{Pb}(\text{NO}_3)_2$ (B) NH_4NO_3 (C) $\text{Cu}(\text{NO}_3)_2$ (D) NaNO_3
19. H_2SO_4 has very high corrosive action on skin because –
 (A) it reacts with proteins
 (B) it acts as an oxidising agent
 (C) it acts as a dehydrating agent
 (D) it acts as dehydrating agent and absorption of water is highly exothermic
20. A black sulphide when treated with ozone becomes white. The white compound is –
 (A) ZnSO_4 (B) CaSO_4 (C) BaSO_4 (D) PbSO_4
21. Which of the following does not react with AgCl –
 (A) $\text{Na}_2\text{S}_2\text{O}_3$ (B) NH_4OH (C) NaNO_3 (D) NH_3

22. By passing H_2S gas in acidified KMnO_4 solution, we get –
 (A) K_2S (B) S (C) K_2SO_3 (D) MnO_2
23. Chloride of lime is –
 (A) CaOCl_2 (B) $\text{Ca}(\text{OCl})_2$ (C) CaCl_2 (D) $(\text{CaO})_2 \text{Cl}$
24. Which one of the following oxy acid of fluorine exists ?
 (A) HOF (B) HFO_3 (C) HFO_4 (D) HFO_2
25. Chromyl chloride test is performed for the confirmation of the presence of the following in mixture –
 (A) SO_4^{2-} (B) Cr^{+++} (C) Cl^- (D) Cr^{+++} and Cl^-
26. Iodine gas turns starch iodide paper –
 (A) Blue (B) Red (C) Colourless (D) Yellow
27. Essential trace element involved in physiology of thyroid glands –
 (A) K (B) Mg (C) Ni (D) I_2
28. When chlorine is passed over dry slaked lime at room temperature, the main reaction product is –
 (A) $\text{Ca}(\text{ClO}_2)_2$ (B) CaCl_2 (C) CaOCl_2 (D) $\text{Ca}(\text{OCl}_2)_2$
29. Iodine is formed when KI reacts with a solution of –
 (A) ZnSO_4 (B) CuSO_4 (C) FeSO_4 (D) $(\text{NH}_4)_2\text{SO}_4$
30. Which amongst the following reactions cannot be used for the preparation of the halogen acid ?
 (A) $2\text{KBr} + \text{H}_2\text{SO}_4(\text{Conc.}) \longrightarrow \text{K}_2\text{SO}_4 + 2\text{HBr}$
 (B) $2\text{NaCl} + \text{H}_2\text{SO}_4(\text{Conc.}) \longrightarrow \text{NaHSO}_4 + \text{HCl}$
 (C) $\text{NaHSO}_4 + \text{NaCl} \longrightarrow \text{Na}_2\text{SO}_4 + \text{HCl}$
 (D) $\text{CaF}_2 + \text{H}_2\text{SO}_4(\text{conc.}) \longrightarrow \text{CaSO}_4 + 2\text{HF}$
31. HI can be prepared by all the following methods except –
 (A) $\text{PI}_3 + \text{H}_2\text{O}$ (B) $\text{KI} + \text{H}_2\text{SO}_4$
 (C) $\text{H}_2 + \text{I}_2 \longrightarrow$ (D) $\text{I}_2 + \text{H}_2\text{S}$
32. When I_2 is passed through KCl , KF , KBr solution –
 (A) Cl_2 and Br_2 are evolved (B) Cl_2 is evolved
 (C) Cl_2 , Br_2 , F_2 are evolved (D) None of these
33. Which two of the following salts are used for preparing iodized salt–
 (i) KIO_3 (ii) KI (iii) I_2 (iv) HI
 (A) (i) and (ii) (B) (i) and (iii) (C) (ii) and (iv) (D) (iii) and (iv)
34. Helium is obtained from which of the following.?
 (A) Natural gases trapped under rock formations. (B) Liquid air
 (C) Radioactive decay (D) Gasoline
35. The statement, which prompted Neil Bartlett to prepare the first noble gas compound was –
 (A) Xe-F bond has high bond energy
 (B) F_2 has exceptionally low bond energy
 (C) PtF_6 is a strong oxidant
 (D) O_2 molecule and Xe atom have very similar ionization energies.

36. Pick out the correct statement for XeF_6
- (A) XeF_6 is hydrolysed partially to form XeOF_4
- (B) It react with SiO_2 to form XeOF_4
- (C) On complete hydrolysis, it forms XeO_3
- (D) All

37. The molecular shapes of diborane is shown:
Consider the following statements for diborane



1. Boron is approximately sp^3 hybridised
2. B-H-B angle is 180°
3. There are two terminal B-H bonds for each boron atom
4. There are only 12 bonding electrons available

Of these statements –

- (A) 1, 3 and 4 are correct
- (B) 1, 2 and 3 are correct
- (C) 2, 3 and 4 are correct
- (D) 1, 2 and 4 are correct
38. Borax is actually made of two tetrahedral and two triangular units joined together and should be written as : $\text{Na}_2 [\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$.

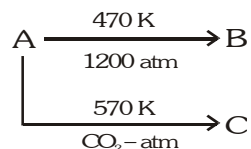
Consider the following statements about borax:

1. Each boron atom has four B-O bonds
2. Each boron atom has three B-O bonds
3. Two boron atoms have four B-O bonds while other two have three B-O bonds
4. Each boron atom has one-OH groups

Select correct statement(s) –

- (A) 1, 2 (B) 2, 3 (C) 3, 4 (D) 1, 3

39. Three allotropes (A), (B) and (C) of phosphorous in the following change are respectively –



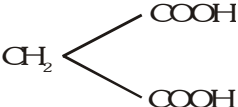
- (A) White, black, red (B) Black, white, red (C) Red, black, white (D) Red, violet, black
40. One mole of calcium phosphide on reaction with excess of water gives –
- (A) One mole of phosphine (B) Two moles of phosphoric acid
- (C) Two moles of phosphine (D) One mole of phosphorus penta-oxide

41. $\text{Ca} + \text{C}_2 \longrightarrow \text{CaC}_2 \xrightarrow{\text{N}_2} \text{A}$

Compound (A) is used as a/an –

- (A) Fertilizer (B) Dehydrating agent (C) Oxidising agent (D) Reducing agent
42. Which one of the following statements is not true regarding diborane?
- (A) It has two bridging hydrogens and four perpendicular to the rest.
- (B) When methylated, the product is $\text{Me}_4\text{B}_2\text{H}_2$
- (C) The bridging hydrogens are in a plane perpendicular to the rest.
- (D) All the B-H bond distances are equal.

43. A red coloured mixed oxide (X) on treatment with concentrate HNO_3 gives a compound (Y). (Y) with HCl, produces a chloride compound (Z) which can also be produced by treating (X) with concentrate HCl. Compounds (X), (Y) and (Z) will be –
- (A) Mn_3O_4 , MnO_2 , MnCl_2 (B) Pb_3O_4 , PbO_2 , PbCl_2
 (C) Fe_3O_4 , Fe_2O_3 , FeCl_2 (D) Fe_3O_4 , Fe_2O_3 , FeCl_3
44. Match List-I with List-II
- | List-I Chemical reaction | List-II Name of process |
|--|-------------------------|
| (1) $4\text{NH}_3 + 5\text{O}_2 \xrightarrow{800^\circ\text{C}/\text{Pt}} 4\text{NO} + 6\text{H}_2\text{O}$ | (a) Contact process |
| (2) $4\text{HCl} + \text{O}_2 \xrightarrow[450-500^\circ/\text{V}_2\text{O}_5]{3230^\circ\text{C}/\text{CuCl}_2} 2\text{Cl}_2 + 2\text{H}_2\text{O}$ | (b) Ostwald's process |
| (3) $2\text{SO}_2 + \text{O}_2 \longrightarrow 2\text{SO}_3$ | (c) Deacon's process |
| (4) $2\text{N}_2 + 3\text{H}_2 \xrightarrow{\text{Fe}+\text{Mo}} 2\text{NH}_3$ | (d) Haber's process |
- (A) 1-a, 2-b, 3-d, 4-c (B) 1-b, 2-c, 3-a, 4-d (C) 1-a, 2-d, 3-c, 4-b (D) 1-a, 2-c, 3-b, 4-d
45. A gas which exists in three allotropic forms α , β and γ is –
- (A) SO_2 (B) SO_3 (C) CO_2 (D) NH_3
46. $\text{HNO}_3 + \text{P}_4\text{O}_{10} \longrightarrow \text{HPO}_3 + \text{A}$; The product A is –
- (A) N_2O (B) N_2O_3 (C) NO_2 (D) N_2O_5
47. Which of the following statements is/are correct regarding B_2H_6 ?–
- (A) Banana bonds are longer but stronger than normal B–H bonds
 (B) B_2H_6 is also known as 3c-2e compound
 (C) The hybrid state of B in B_2H_6 is sp^3 while that of sp^2 in BH_3
 (D) It cannot be prepared by reacting BF_3 with LiBH_3 in the presence of dry ether
48. Which of the following pairs of nitrates gives the same gaseous products on thermal decomposition?
- (A) KNO_3 and $\text{Pb}(\text{NO}_3)_2$ (B) KNO_3 and NaNO_3
 (C) $\text{Pb}(\text{NO}_3)_2$ and $\text{Cu}(\text{NO}_3)_2$ (D) NaNO_3 and $\text{Ca}(\text{NO}_3)_2$
49. When an inorganic compound (X) having 3c - 2e as well as 2c -2e bonds reacts with ammonia gas at a certain temperature, gives a compound (Y) iso-structural with benzene. Compound (X) with ammonia at a high temperature, produces a hard substance (Z). Then –
- (A) (X) is B_2H_6 (B) (Z) is known as inorganic graphite
 (C) (Z) having structure similar to graphite (D) (Z) having structure similar to (X)
50. Boric acid –
- (A) Exists in polymeric form due to inter-molecular hydrogen bonding
 (B) Is used in manufacturing of optical glasses
 (C) Is a tri-basic acid
 (D) With borax, it is used in the preparation of a buffer solution.
51. Select correct statement(s) –
- (A) Borax is used as a buffer
 (B) 1 M borax solution reacts with equal volumes of 2 M HCl solution
 (C) Titration of borax can be made using methyl orange as the indicator
 (D) Coloured bead obtained in borax-bead test contains metaborate.

52. Select correct statement about B_2H_6 –
- (A) Bridging groups are electron-deficient with 12 valence electrons
 (B) It has 2c-2e B-H bonds
 (C) It has 3c-2e B-H-B bonds
 (D) All of above are correct statements
53. Which of the following is/are correct for group 14 elements –
- (A) The stability of dihalides are in the order $CX_2 < SiX_2 < GeX_2 < SnX_2 < PbX_2$
 (B) The ability to form $p\pi-p\pi$ multiple bonds among themselves increases down the group
 (C) The tendency for catenation decreases down the group
 (D) They all form oxides with the formula MO_2
54. The solubility of anhydrous $AlCl_3$ and hydrous $AlCl_3$ in diethyl ether are S_1 and S_2 respectively. Then –
- (A) $S_1 = S_2$ (B) $S_1 > S_2$ (C) $S_1 < S_2$ (D) $S_1 < S_2$ but not $S_1 = S_2$
55. Concentrated HNO_3 reacts with iodine to give –
- (A) HI (B) HOI (C) $HOIO_2$ (D) $HOIO_3$
56. Conc. H_2SO_4 cannot be used to prepare HBr from NaBr because it –
- (A) Reacts slowly with NaBr (B) Oxidises HBr
 (C) Reduces HBr (D) Disproportionates HBr
57. CH_2  $\xrightarrow{P_4O_{10}, 150^\circ C}$ X. Compound (X) is –
- (A) Malonic acid (B) Carbon suboxide (C) Tartaric acid (D) Acetic acid
58. $H_3BO_3 \xrightarrow{T_1} X \xrightarrow{T_2} Y \xrightarrow{\text{red hot}} B_2O_3$ if $T_1 < T_2$ then X and Y respectively are –
- (A) X = Metaboric acid and Y = Tetraboric acid
 (B) X = Tetraboric acid and Y = Metaboric acid
 (C) X = Borax and Y = Metaboric acid
 (D) X = Tetraboric acid and Y = Borax
59. In a molecule of phosphorus (V) oxide, there are –
- (A) 4P – P, 10P – O and 4P = O bonds (B) 12P – O and 4P = O bonds
 (C) 2P – O and 4P = P bonds (D) 6P – P, 12P – O and 4P = P bonds
60. Conc. HNO_3 is yellow coloured liquid due to –
- (A) Dissolution of NO in conc. HNO_3
 (B) Dissolution of NO_2 in conc. HNO_3
 (C) Dissolution of N_2O in conc. HNO_3
 (D) Dissolution of N_2O_3 in conc. HNO_3
61. When chlorine gas is passed through an aqueous solution of a potassium halide in the presence of chloroform, a violet colouration is obtained. On passing more of chlorine water, the violet colour is disappeared and solution becomes colourless. This test confirms the presence of in aqueous solution. –
- (A) Chlorine (B) Fluorine (C) Bromine (D) Iodine

62. An inorganic salt (A) is decomposed at about 523 K to give products (B) and (C). compound (C) is a liquid at room temperature and is neutral to litmus paper while oxide (B) on burning with white phosphorous, given a dehydrating agent (D). compounds (A), (B), (C) and (D) will be identified as –
- (A) NH_4NO_3 , N_2O , H_2O , P_2O_5
- (B) NH_4NO_2 , N_2O , H_2O , P_2O_5
- (C) CaCO_3 , CaO , H_2O , CaCl_2
- (D) CaCO_3 , CaO , H_2O , $\text{Ca}(\text{OH})_2$
63. Aqueous solution of borax reacts with 2 mol of acids. This is because of –
- (A) Formation of 2 mol of $\text{B}(\text{OH})_3$ only.
- (B) Formation of 2 mol of $[\text{B}(\text{OH})_4]^-$ only.
- (C) Formation of 1 mol each of $\text{B}(\text{OH})_3$ and $[\text{B}(\text{OH})_4]^-$
- (D) Formation of 2 mol each of $[\text{B}(\text{OH})_4]^-$ and $\text{B}(\text{OH})_3$, of which only $[\text{B}(\text{OH})_4]^-$ reacts with acid
64. Borax is used as a buffer since –
- (A) Its aqueous solution contains equal amount of weak acid and its salt
- (B) It is easily available
- (C) Its aqueous solution contains equal amount of strong acid and its salt
- (D) Statement that borax is a buffer, is wrong.
65. When fluoride is heated with conc. H_2SO_4 and MnO_2 the gas evolved is –
- (A) HF
- (B) F_2
- (C) SF_4
- (D) None

Brain Teasers					Answer Key						Exercise-2				
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	B	B	C	C	B	A	D	B	A	A	C	D	C	C	D
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	C	B	B	D	D	C	B	A	A	C	A	D	C	B	A
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	B	D	A	A,C	D	D	A	C	A	C	A	D	B	B	B
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	D	A,B,C	B,C	A,B,C	A,B,D	A,B,C,D	B,C	A,C,D	B	C	B	B	A	B	B
Que.	61	62	63	64	65										
Ans.	D	A	D	A	A										

EXERCISE-03**MISCELLANEOUS TYPE QUESTIONS****TRUE / FALSE**

1. Goldschmidt thermite process is used in the extraction of aluminium.
2. Cryolite is added to alumina as to lower the fusion temperature and make the mass good conductor of electricity.
3. Nitrous acid acts both as an oxidising and a reducing agent.
4. NF_3 is stable but NCl_3 and NI_3 are explosive in nature.
5. Bleaching action of SO_2 is due to reduction and temporary.
6. Mercury in presence of ozone is oxidised to mercuric oxide.
7. Hydrofluoric acid cannot be kept in glass vessels.
8. The amount of chlorine obtained from a sample of bleaching powder by treatment with excess of dilute acid or CO_2 is called available chlorine.

FILL IN THE BLANKS

1. The increase in solubility of iodine in an aqueous solution of KI is due to the formation of
2. Silver fluoride is fairly.....in water.
3. HI cannot be prepared by the action of conc. H_2SO_4 on KI because H_2SO_4 is anagent.
4. Antichlor is a compound which convertsinto
5. Iodine reacts with hot NaOH solution giving the products as NaI and
6. In the known interhalogen compounds, the maximum number of halogen atom is
7. Conc. HNO_3 reacts with iodine to give
8. The only halogen that is not found in nature is
9. Tincture of iodine contains I_2 ,.....and.....
10. Iodine deficiency in diet is known to cause.....

MATCH THE COLUMN

1.

<u>Column-I</u>		<u>Column-II</u>	
(A)	Hypo phosphoric acid	(p)	All hydrogen are ionizable in water
(B)	Pyro phosphoric acid	(q)	Lewis acid
(C)	Boric acid	(r)	Monobasic in water
(D)	Hypo phosphorus acid	(s)	sp^3 hybridized central atom

ASSERTION & REASON QUESTIONS

These questions contains, Statement I (assertion) and Statement II (reason).

- (A) Statement-I is true, Statement-II is true ; Statement-II is correct explanation for Statement-I.
(B) Statement-I is true, Statement-II is true ; Statement-II is not a correct explanation for statement-I
(C) Statement-I is true, Statement-II is false
(D) Statement-I is false, Statement-II is true

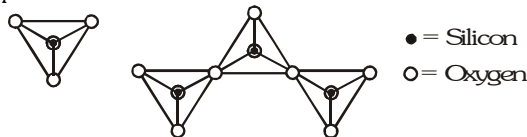
1. **Statement -I** : Silicones are very inert polymers.
Because
Statement -II : Both Si—O and Si—C bond energies are very high.
2. **Statement -I** : Chlorine gas disproportionates in hot & conc. NaOH solution
Because
Statement -II : NaCl and NaOCl are formed in the above reaction.
3. **Statement -I** : $\text{Al}(\text{OH})_3$ is amphoteric in nature.
Because
Statement -II : It can not be used as an antacid.
4. **Statement -I** : Oxygen is more electronegative than sulphur, yet H_2S is acidic, while H_2O is neutral.
Because
Statement -II : H-S bond is weaker than O-H bond.
5. **Statement -I** : Conc. H_2SO_4 can not be used to prepare pure HBr from NaBr.
Because
Statement -II : It reacts slowly with NaBr.
6. **Statement -I** : Aluminium and zinc metal evolve H_2 gas from NaOH solution.
Because
Statement -II : Several non-metals such as P, S, Cl, etc. yield a hydride instead of H_2 gas from NaOH
7. **Statement -I** : Borax bead test is applicable only to coloured salt.
Because
Statement -II : In borax bead test, coloured salts are decomposed to give coloured metal metaborates.
8. **Statement -I** : Cl_2 gas bleaches the articles permanently.
Because
Statement -II : Cl_2 is a strong reducing agent.
9. **Statement -I** : HClO is stronger acid than HBrO
Because
Statement -II : Greater is the electronegativity of the halogen, greater will be attraction of electron pair towards it and hence more easily the H^+ ion will be released.

COMPREHENSION BASED QUESTIONS

Comprehension # 1

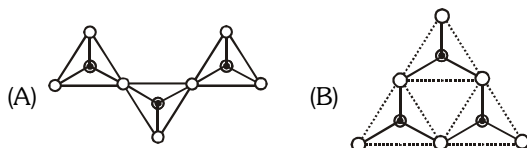
The name 'Silica' covers an entire group of minerals, which have the general formula SiO_2 , The most common of which is quartz. Quartz is a framework silicate with SiO_4 tetrahedral arranged in spirals. the spirals can turn in a clockwise or anticlockwise direction – a feature that results in there being two mirror images, optically active, varieties of quartz.

1. The following pictures represent various silicate anions. Their formulae are respectively –



- (A) SiO_3^{2-} $\text{Si}_3\text{O}_7^{2-}$ (B) SiO_4^{4-} $\text{Si}_3\text{O}_{10}^{8-}$
 (C) SiO_4^{2-} $\text{Si}_3\text{O}_9^{2-}$ (D) SiO_3^{4-} $\text{Si}_3\text{O}_7^{8-}$

2. $\text{Si}_3\text{O}_9^{6-}$ (having three tetrahedral) is represented as –



- (C) Both (D) None of these

3. The silicate anion in the mineral kaolinite is a chain of three SiO_4 tetrahedral that share corners with adjacent tetrahedral. The mineral also contains Ca^{2+} ions, Cu^{2+} ions, and water molecules in a 1:1:1 ratio. Mineral is represented as –

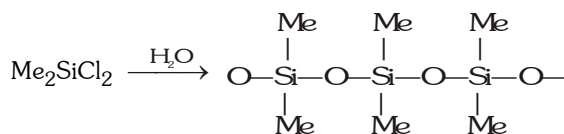
- (A) $\text{CaCuSi}_3\text{O}_{10} \cdot \text{H}_2\text{O}$ (B) $\text{CaCuSi}_3\text{O}_{10} \cdot 2\text{H}_2\text{O}$
 (C) $\text{Ca}_2\text{Cu}_2\text{Si}_3\text{O}_{10} \cdot 2\text{H}_2\text{O}$ (D) None of these

Comprehension # 2

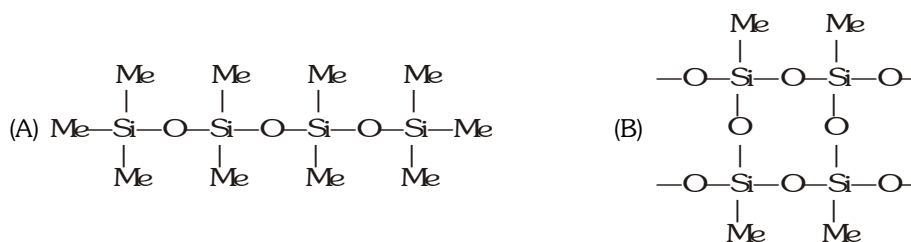
Silicones are synthetic polymers containing repeating R_2SiO units. Since, the empirical formula is that of a ketone (R_2CO), the name silicone has been given to these materials. Silicones can be made into oils, rubbery elastomers and resins. they find a variety of applications because of their chemical inertness, water repelling nature, heat-resistance and good electrical insulating property.

Commercial silicon polymers are usually methyl derivatives and to a lesser extent phenyl derivatives and are synthesised by the hydrolysis of

R_2SiCl_2 [R = methyl (Me) or phenyl (ϕ)]

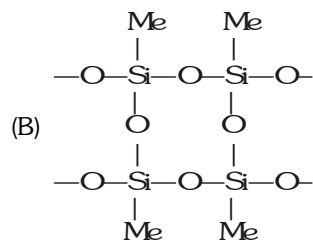
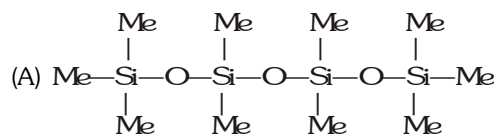


1. If we mix Me_3SiCl with Me_2SiCl_2 , we get silicones of the type –



- (C) Both of the above (D) None of the above

2. If we start with MeSiCl_3 as the starting material, silicones formed is -



- (C) Both of the above
(D) None of the above

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
<u>True / False</u> 1. F 2. T 3. T 4. T 5. T 6. F 7. T 8. T <u>Fill in the Blanks</u> 1. KI_3 2. Soluble 3. Oxidising 4. Cl_2, Hydrochloric acid 5. NaIO_3 6. Eight 7. HIO_3 8. At 9. KI, Rectified spirit 10. Goiter <u>Match the Column</u> 1. (A) $\rightarrow$ p,s ; (B) $\rightarrow$ s ; (C) $\rightarrow$ q,r ; (D) $\rightarrow$ r,s <u>Assertion - Reason Questions</u> 1. A 2. C 3. C 4. A 5. C 6. B 7. A 8. C 9. A <u>Comprehension Based Questions</u> Comprehension # 1 : 1. B 2. B 3. C Comprehension # 2 : 1. A 2. B		

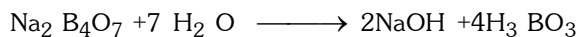
EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

- 1 Colourless salt (A) + NaOH (excess) $\xrightarrow{\Delta}$ gas (B) giving white fumes with HCl + alkaline solution (C)
(C) + Zn $\longrightarrow$ gas (B)
(A) $\xrightarrow{\Delta}$ gas (D) + liquid (E)
D,E Both triatomic
Identify (A,B,C,D) and (E)
- 2 Complete and balance the following reactions-
- (a) $\text{Cu} + \text{dil. HNO}_3 \longrightarrow \text{NO} + \dots + \dots$
- (b) $\text{Pb}(\text{NO}_3)_2 \xrightarrow{\text{heat}} \text{PbO} + \dots + \dots$
- (c) $\text{CuSO}_4 + \text{NH}_4\text{OH}(\text{excess}) \longrightarrow \dots$
- (d) $\text{AgCl} + \text{NH}_4\text{OH} \longrightarrow \dots$
- 3 Aqua-regia dissolves gold. Write reaction.
- 4 Precipitation of second group sulphides in qualitative analysis is carried out with H_2S in presence of HCl and not nitric acid. Why?
- 5 KMnO_4 should not be dissolved in conc. H_2SO_4 . Why?
- 6 What happens when H_2S gas is passed through nitric acid ?
- 7 What happens when –
- (1) Hydrogen sulphide is bubbled through an aqueous solution of sulphur dioxide.
- (2) Hydrogen sulphide is passed through acidified ferric chloride solution.
- (3) Sulphur is boiled with caustic soda solution.
- 8 Sodium salt (A) of a dibasic acid $\xrightarrow{\text{HCl}}$ gas (B) and clear solution gas (B) and clear solution turns $\text{K}_2\text{Cr}_2\text{O}_7$ to green and also lime water milky. identify (A) and (B).
- 9 A certain salt 'X' gives the following test.
- (i) Its aqueous solution is alkaline to litmus.
- (ii) On strongly heating it swells to give glassy bead.
- (iii) When concentrated H_2SO_4 is added to solution of 'X', white crystals of a weak acid. separate out.
Identify 'X' and write down all reaction.
- 10 SnCl_2 gives white precipitate with HgCl_2 which turns grey later on, but SnCl_4 does not. Explain why ?
- 11 Identify A and B (Compound/reaction condition)
- $$\text{PbS} \xrightarrow[\text{in air}]{\text{Heat}} \text{A} + \text{PbS} \xrightarrow{\text{B}} \text{Pb} + \text{SO}_2$$
- 12 A bottle of liquor ammonia should be cooled before opening. Why ?
- 13 Iodine is liberated in the reaction between KI and Cu^{2+} but chlorine is not liberated when KCl is added to Cu^{2+} . Why?
- 14 What happens when an aqueous solution of hydrazine reacts with:
- (i) An aqueous solution of I_2
- (ii) An alkaline solution of copper sulphate
- (iii) An aqueous alkaline solution of potassium ferricyanide
- (iv) an ammoniacal solution of silver nitrate.

- 15 An acid (A) is pale-blue. The potassium salt of this acid does not give any reaction with BaCl_2 but gives white crystalline precipitate (B) with Ag^+ ions. The acid (A) reacts with urea to liberate two gases (C) and (D). Gas (D) is used in synthesis of Urea also. On adding thiourea in acid (A) followed by addition of FeCl_3 /dilute HCl red coloured substance (E) is obtained. Identify substances (A) to (E).

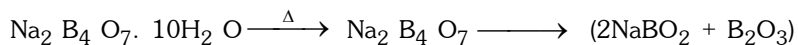
CONCEPTUAL	SUBJECTIVE	EXERCISE	ANSWER KEY	EXERCISE -4(A)
1.	(A) $\text{NH}_4 \text{NO}_3$ H_2O	(B) NH_3	(C) $(\text{NaNO}_3 + \text{NaOH})$	(D) N_2O (E)
2.	(a) $2\text{HNO}_3 \rightarrow \text{H}_2\text{O} + 2\text{NO} + 3(\text{O})$ $3\text{Cu} + 3(\text{O}) + 6\text{HNO}_3 \rightarrow 3\text{Cu}(\text{NO}_3)_2 + 3\text{H}_2\text{O}$ $3\text{Cu} + 8\text{HNO}_3 \rightarrow 3\text{Cu}(\text{NO}_3)_2 + 2\text{NO} + 4\text{H}_2\text{O}$ (b) $2\text{Pb}(\text{NO}_3)_2 \rightarrow 2\text{PbO} + 4\text{NO}_2 + \text{O}_2$ (c) $\text{CuSO}_4 + 2\text{NH}_4\text{OH} \rightarrow \text{Cu}(\text{OH}_2) + (\text{NH}_4)_2\text{SO}_4$ $\text{Cu}(\text{OH}_2) + (\text{NH}_4)_2\text{SO}_4 + 2\text{NH}_4\text{OH} \rightarrow [\text{Cu}(\text{NH}_3)_4]\text{SO}_4 + 4\text{H}_2\text{O}$ (g) $\text{AgCl} + \text{NH}_4\text{OH} \rightarrow \text{AgOH} + \text{NH}_4\text{Cl}$ $\text{AgOH} + \text{NH}_4\text{Cl} + \text{NH}_4\text{OH} \rightarrow \text{Ag}(\text{NH}_3)_2\text{Cl} + 2\text{H}_2\text{O}$ $\text{AgCl} + 2\text{NH}_4\text{OH} \rightarrow \text{Ag}(\text{NH}_3)_2\text{Cl} + 2\text{H}_2\text{O}$ 			
3.	Noble metals like gold, platinum, iridium, rhodium etc., do not react with nitric acid. However, these metals dissolve in aqua-regia (3 parts conc. HCl and one part conc. HNO_3). Aqua-regia gives nascent chlorine which attacks these metals. Gold: $[\text{HNO}_3 + 3\text{HCl} \rightarrow \text{NOCl} + 2\text{H}_2\text{O} + 2\text{Cl}] \quad 3$ Nitrosyl chloride $[\text{Au} + 3\text{Cl} \rightarrow \text{AuCl}_3] \quad 2$ $[\text{AuCl}_3 + \text{HCl} \rightarrow \text{HAuCl}_4] \quad 2$ $2\text{Au} + 3\text{HNO}_3 + 11\text{HCl} \rightarrow 2\text{HAuCl}_4 + 3\text{NOCl} + 6\text{H}_2\text{O}$ Chloroauric acid			
4.	HNO_3 , a strong oxidant, will oxidise H_2S to give colloidal sulphur. Thus precipitation of second group sulphide will not occur. $\text{H}_2\text{S} \xrightarrow[\text{HNO}_3]{[\text{O}]} \text{H}_2\text{O} + \text{S}$			
5.	KMnO_4 forms explosive covalent compound, Mn_2O_7 with conc. H_2SO_4 . $2\text{KMnO}_4 + \text{H}_2\text{SO}_4 \text{ conc.} \longrightarrow \text{K}_2\text{SO}_4 + \text{Mn}_2\text{O}_7 + \text{H}_2\text{O}$ However, it can be dissolved in dil. H_2SO_4 to give nascent oxygen if it is to be used as oxidant.			
6.	Yellow colloidal sulphur is formed. $2\text{HNO}_3 + \text{H}_2\text{S} \longrightarrow 2\text{NO}_2 + 2\text{H}_2\text{O} + \text{S}$			
7.	(1) (Moist) $\text{SO}_2 + 2\text{H}_2\text{S} \longrightarrow 3\text{S} + 2\text{H}_2\text{O}$ (2) $2\text{FeCl}_3 + \text{H}_2\text{S} \longrightarrow 2\text{FeCl}_2 + 2\text{HCl} + \text{S}$ (3) $4\text{S} + 6\text{NaOH} \longrightarrow \text{Na}_2\text{S}_2\text{O}_3 + 2\text{Na}_2\text{S} + 3\text{H}_2\text{O} \xrightarrow{\text{excess sulphur}} \text{Na}_2\text{S}_5$			

9. $\text{Na}_2 \text{B}_4 \text{O}_7 \cdot 10\text{H}_2 \text{O}$

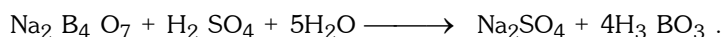


Weak

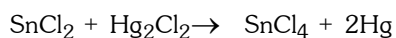
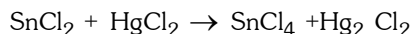
Due to presence of NaOH. the aqueous solution is alkaline to litmus.



Glassy Bead



10. SnCl_2 is strong reducing agent and reduces HgCl_2 to Hg_2Cl_2 (white) which is further reduced to Hg (grey).

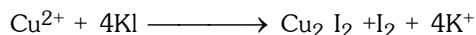


11. A = PbO or PbSO_4

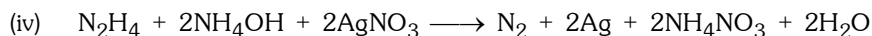
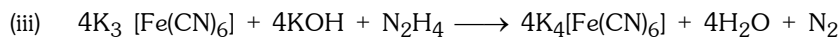
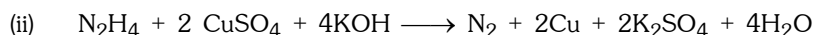
B = high temperature

12. It should be cooled to lower the pressure of NH_3 inside the bottle, otherwise NH_3 will bump out of the bottle.

13. The I^- ion acts as strong reducing agent where as Cl^- acts as reductant only in presence of strong oxidant and therefore KI reduces Cu^{2+} to Cu^+



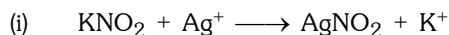
14. (i) $\text{N}_2\text{H}_4 + 2\text{I}_2 \longrightarrow \text{N}_2 + 4\text{HI}$



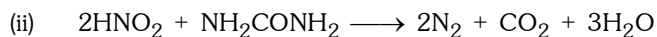
15. (A) = HNO_2 (Nitrous acid) ; (B) = AgNO_2 (Silver nitrite)

(C) = N_2 ; (D) = CO_2 ;

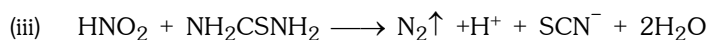
(E) = $\text{Fe}(\text{SCN})_3$ (Ferric thiocyanate)



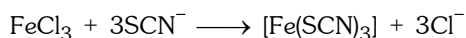
(B) Silver nitrite



(A) (Urea) (C) (D)



(A) Thiourea



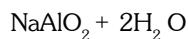
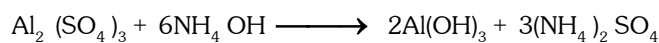
(E) Ferric thiocyanate

BRAIN STORMING SUBJECTIVE EXERCISE

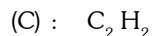
- $$\text{Cyclohexylidenemethane} \xrightarrow[\text{(ii) H}_2\text{O}_2/\text{OH}^-]{\text{(i) B}_2\text{H}_6} \text{A}$$

6. An inorganic compound (A) of S, Cl and oxygen has vapour density 67.5. It reacts with water to form two acids (B) and (C). (A) also reacts with $\text{KOH}_{(\text{aq})}$ to form two salts (D) and (E). (D) and (E) gives white precipitate with AgNO_3 and BaCl_2 solutions respectively. What are (A) to (E)?
7. A certain compound (X) gives brick red flame on performing the flame test and shows the following reactions.
- (i) When KI is added to an aqueous suspension of (X) containing acetic acid, iodine is liberated.
 - (ii) When CO_2 is passed through an aqueous suspension of (X) the turbidity transforms to a ppt.
 - (iii) When a paste of (X) in water is heated with ethyl alcohol, a product of anaesthetic use is obtained.
- Identify (X) and write down chemical equations for reactions at step (i), (ii) and (iii).

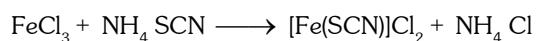
1. A : K_2SO_4 $Al_2(SO_4)_3$ 24 H_2O



2. (A) : CaC_2 (Ca^{2+} , C_2^{2-} binary)



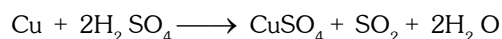
3. $FeCl_3$ on reacting with NH_4SCN gives complex ion $[Fe(SCN)]^{2+}$ which is of red colour.



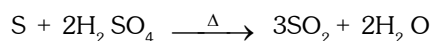
blood red colour

$FeCl_3$ while reacting with H_2S , gives $FeCl_2$ which is not reacting with NH_4SCN . So no colour is obtained.

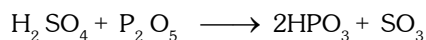
4. (a) SO_2 is obtained by heating copper with conc. H_2SO_4



It can also be obtained by boiling sulphur with conc. H_2SO_4 .

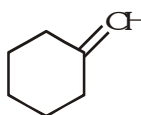


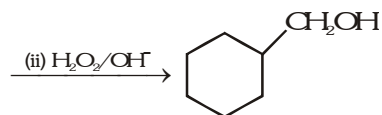
(b) When treated with P_2O_5 H_2SO_4 loses water and forms SO_3



(c) SO_2Cl_2 is formed when conc. H_2SO_4 is treated with excess of PCl_5 .



5.  $\xrightarrow{(i) B_2H_6} (C_6H_{12}CH_2)_3B_2$

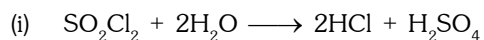


6. Vapour density of S, Cl and oxygen compound = 67.5

$\therefore$ Molecular weight of compound = 135

the molecular, weight suggest it may be SO_2Cl_2

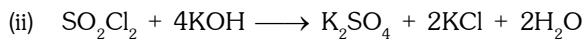
The give reaction are



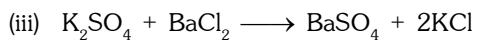
(A)

(B)

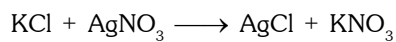
(C)



(D) (E)

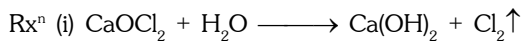


(D) Insoluble

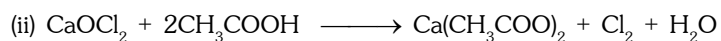


(E) Insoluble

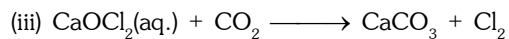
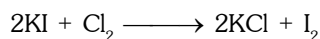
7. (X) gives brick red flame so, it contains Ca^{2+} . Reactions (i), (ii) and (iii) suggest that the probable compound is bleaching powder (CaOCl_2).



smell

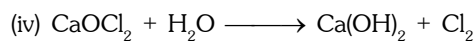


(X)

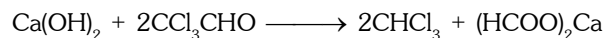
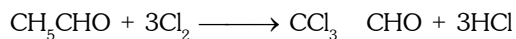
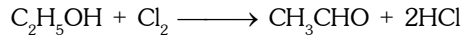


Turbid

white ppt



(X)



(Anaesthetic)

EXERCISE-05 [A]**PREVIOUS YEARS QUESTIONS**

1. Aluminium is industrially prepared by: [AIEEE-2002]
(1) Fused cryolite (2) Bauxite ore (3) Alunite (4) Borax
2. In case of nitrogen NCl_3 is possible but not NCl_5 while in case of phosphorous, PCl_3 as well as PCl_5 are possible. This is due to :- [AIEEE-2002]
(1) Availability of vacant d-orbitals in P but not in N
(2) Lower electronegativity of P than N
(3) Lower tendency of H-bond formation in P than N
(4) Occurrence of P in solid while N in gaseous state at room temperature
3. Which products are expected from the disproportionation of hypochlorous acid : [AIEEE-2002]
(1) HClO_3 and Cl_2O (2) HClO_2 and HClO
(3) HCl and Cl_2O (4) HCl and HClO_3
4. Identify the incorrect statement among the following [AIEEE-2002]
(1) Ozone reacts with SO_2 to give SO_3
(2) Silicon reacts with NaOH(aq.) in the presence of air to give Na_2SiO_3 and H_2O
(3) Cl_2 reacts with excess of NH_3 to give N_2 and HCl
(4) Br_2 reacts with hot and strong NaOH solution to give NaBr , NaBrO_4 and H_2O
5. In XeF_2 , XeF_4 , XeF_6 the number of lone pairs on Xe is respectively : [AIEEE-2002]
(1) 2,3,1 (2) 1,2,3 (3) 4, 1, 2 (4) 3,2,1
6. What may be expected when phosphine gas is mixed with chlorine gas: [AIEEE-2003]
(1) PCl_5 and HCl are formed and mixture cools down
(2) $\text{PH}_3 \cdot \text{Cl}_2$ is formed with warming up
(3) The mixture only cools down
(4) PCl_3 and HCl are formed and the mixture warms up
7. For making good quality mirrors, plates of float glass are used. These are obtained by floating molten glass over a liquid metal which does not solidify before glass. The metal used can be : [AIEEE-2003]
(1) Sodium (2) Magnesium (3) Mercury (4) Tin
8. Graphite is a soft solid lubricant extremely difficult to melt. The reason for this anomalous behaviour is that graphite : [AIEEE-2003]
(1) Has molecules of variable molecular masses like polymers
(2) Has carbon atoms arranged in large plates of rings of strongly bonded carbon atoms with weak interplate bonds
(3) Is a non crystalline substance
(4) Is an allotropic form of diamond
9. Concentrated hydrochloric acid when kept in open air sometimes produces a cloud of white fumes. This is due to : [AIEEE-2003]
(1) Strong affinity of HCl gas for moisture in air results in forming of droplets of liquid solution which appears like a cloudy smoke
(2) Due to strong affinity for water, conc. HCl pulls moisture of air towards self. The moisture forms droplets of water and hence the cloud
(3) Conc. HCl emits strongly smelling HCl gas all the time
(4) Oxygen in air reacts with emitted HCl gas to form a cloud of Cl_2 gas
10. Aluminium chloride exists as dimer, Al_2Cl_6 in solid state as well as in solution of non-polar solvents such as benzene. When dissolved in water, it gives- [AIEEE-2004]
(1) $\text{Al}^{3+} + 3\text{Cl}^-$ (2) $[\text{Al}(\text{H}_2\text{O})_6]^{3+} + 3\text{Cl}^-$
(3) $[\text{Al}(\text{OH})_6]^{3-} + 3\text{HCl}$ (4) $\text{Al}_2\text{O}_3 + 6\text{HCl}$

11. The soldiers of Napoleon army while at Alps during freezing winter suffered a serious problem as regards to the tin buttons of their uniforms. White Metallic tin buttons get converted to grey powder. This transformation is related to:- [AIEEE-2004]
- (1) An interaction with water vapour contained in humid air
 - (2) A change in crystalline structure of tin
 - (3) A change in the partial pressure of O_2 in air
 - (4) An interaction with N_2 of air at low temperature
12. Which one of the following statements regarding helium is incorrect [AIEEE-2004]
- (1) It is used to produce and sustain powerful superconducting magnets
 - (2) It is used as a cryogenic agent for carrying out experiments at low temperatures
 - (3) It is used to fill gas balloons instead of hydrogen because it is lighter than hydrogen and non-inflammable
 - (4) It is used in gas-cooled nuclear reactors
13. The structure of diborane contains [AIEEE- 2005]
- (1) Four 2c-2e bonds and four 3c-2e bonds
 - (2) Two 2c-2e bonds and two 3c-2e bonds
 - (3) Two 2c-2e bonds and four 3c-2e bonds
 - (4) Four 2c-2e bonds and two 3c-2e bonds
14. Heating an aqueous solution of aluminium chloride to dryness will give :- [AIEEE-2005]
- (1) $AlCl_3$
 - (2) Al_2Cl_6
 - (3) Al_2O_3
 - (4) $Al(OH)Cl_2$
15. Which one of the following is the correct statement [AIEEE-2005]
- (1) Boric acid is a protonic acid
 - (2) Beryllium exhibits coordination number of six
 - (3) Chlorides of both beryllium and aluminium have bridged chloride structures in solid phase
 - (4) B_2H_6 , $2NH_3$ is known as "inorganic benzene"
16. In silicon dioxide : [AIEEE-2005]
- (1) Each silicon atom is surrounded by four oxygen atoms and each oxygen atom is bonded to two silicon atoms
 - (2) Each silicon atom is surrounded by two oxygen atoms and each oxygen atom is bonded to two silicon atoms
 - (3) Silicon atom is bonded to two oxygen atoms
 - (4) There are double bonds between silicon and oxygen atoms
17. The number of hydrogen atoms attached to phosphorus atom in hypophosphorous acid is : [AIEEE-2005]
- (1) Zero
 - (2) Two
 - (3) One
 - (4) Three
18. The correct order of the thermal stability of hydrogen halide (H-X) is : [AIEEE-2005]
- (1) $HI > HBr > HCl > HF$
 - (2) $HF > HCl > HBr > HI$
 - (3) $HCl < HF > HBr < HI$
 - (4) $HI > HCl < HF > HBr$
19. The stability of dihalides of Si, Ge, Sn and Pb increases steadily in the sequence: [AIEEE-2007]
- (1) $GeX_2 \ll SiX_2 \ll SnX_2 \ll PbX_2$
 - (2) $SiX_2 \ll GeX_2 \ll PbX_2 \ll SnX_2$
 - (3) $SiX_2 \ll GeX_2 \ll SnX_2 \ll PbX_2$
 - (4) $PbX_2 \ll SnX_2 \ll GeX_2 \ll SiX_2$

20. Regular use of which of the following fertilizer increases the acidity of soil : [AIEEE-2007]
 (1) Potassium nitrate
 (2) Urea
 (3) Superphosphate of lime
 (4) Ammonium sulphate
21. Four species are listed below : [AIEEE-2008]
 (i) HCO_3^- (ii) H_3O^+ (iii) HSO_4^- (iv) HSO_3F
 Which one of the following is the correct sequence of their acid strength?
 (1) $\text{iv} < \text{ii} < \text{iii} < \text{i}$ (2) $\text{ii} < \text{iii} < \text{i} < \text{iv}$
 (3) $\text{i} < \text{iii} < \text{ii} < \text{iv}$ (4) $\text{iii} < \text{i} < \text{iv} < \text{ii}$
22. Among the following substituted silanes the one which will give rise to cross linked silicone polymer on hydrolysis is [AIEEE-2008]
 (1) R_4Si (2) RSiCl_3 (3) R_2SiCl_2 (4) R_3SiCl
23. Which one of the following reactions of Xenon compounds is not feasible ? [AIEEE-2009]
 (1) $2\text{XeF}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{Xe} + 4\text{HF} + \text{O}_2$
 (2) $\text{XeF}_6 + \text{RbF} \rightarrow \text{Rb}[\text{XeF}_7]$
 (3) $\text{XeO}_3 + 6\text{HF} \rightarrow \text{XeF}_6 + 3\text{H}_2\text{O}$
 (4) $3\text{XeF}_4 + 6\text{H}_2\text{O} \rightarrow 2\text{Xe} + \text{XeO}_3 + 12\text{HF} + 1.5\text{O}_2$
24. Boron cannot form which one of the following anions ? [AIEEE-2011]
 (1) $\text{B}(\text{OH})_4^-$ (2) BO_2^- (3) BF_6^{3-} (4) BH_4^-
25. In view of the signs of $\Delta_r G$ for the following reactions
 $\text{PbO}_2 + \text{Pb} \rightarrow 2 \text{PbO}$, $\Delta_r G < 0$
 $\text{SnO}_2 + \text{Sn} \rightarrow 2 \text{SnO}$, $\Delta_r G > 0$,
 Which oxidation states are more characteristic for lead and tin ? [AIEEE-2011]
 (1) For lead + 4, for tin + 2
 (2) For lead + 2, for tin + 2
 (3) For lead + 4, for tin + 4
 (4) For lead + 2, for tin + 4
26. Which of the following statement is wrong ? [AIEEE-2011]
 (1) Single N-N bond is weaker than the single P-P bond
 (2) N_2O_4 has two resonance structures
 (3) The stability of hydrides increases from NH_3 to BiH_3 in group 15 of the periodic table
 (4) Nitrogen cannot form $d\pi\text{-p}\pi$ bond
27. Which of the following statements regarding sulphur is incorrect ? [AIEEE-2011]
 (1) At 600 C the gas mainly consists of S_2 molecules
 (2) The oxidation state of sulphur is never less than +4 in its compounds
 (3) S_2 molecule is paramagnetic
 (4) The vapour at 200 C consists mostly of S_8 rings

PREVIOUS YEARS QUESTIONS						ANSWER KEY						EXERCISE -5[A]				
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
Ans	2	1	4	4	4	1	3	2	3	2	2	3	4	3	3	
Que.	16	17	18	19	20	21	22	23	24	25	26	27				
Ans	1	2	2	3	4	3	2	3	3	4	3	2				

EXERCISE-05 [B]**PREVIOUS YEARS QUESTIONS**

1. The number of P—O—P bonds in cyclic metaphosphoric acid is – [IIT-2000]
(A) Zero (B) Two (C) Three (D) Four
2. The correct order of acidic strength is – [IIT- 2000]
(A) $\text{Cl}_2\text{O}_7 > \text{SO}_2 < \text{P}_4\text{O}_{10}$ (B) $\text{CO}_2 > \text{N}_2\text{O}_5 > \text{SO}_3$
(C) $\text{Na}_2\text{O} > \text{MgO} > \text{Al}_2\text{O}_3$ (D) $\text{K}_2\text{O} > \text{CaO} > \text{MgO}$
3. Amongst H_2O , H_2S , H_2Se and H_2Te , the one with the highest boiling point is – [IIT- 2000]
(A) H_2O because of hydrogen bonding (B) H_2Te because of higher molecular weight
(C) H_2S because of hydrogen bonding (D) H_2Se because of lower molecular weight.
4. Ammonia can be dried by – [IIT- 2000]
(A) Conc. H_2SO_4 (B) P_4O_{10} (C) CaO (D) Anhydrous CaCl_2
5. Which of the following are hydrolysed – [REE 2000]
(A) NCl_3 (B) BCl_3 (C) CCl_4 (D) SiCl_4
6. The set with correct order of acidity is – [IIT- 2001]
(A) $\text{HClO} < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4$ (B) $\text{HClO}_4 < \text{HClO}_3 < \text{HClO}_2 < \text{HClO}$
(C) $\text{HClO} < \text{HClO}_4 < \text{HClO}_3 < \text{HClO}_2$ (D) $\text{HClO}_4 < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}$
7. The reaction, $3\text{ClO}^- (\text{aq}) \longrightarrow \text{ClO}_3^- (\text{aq}) + 2\text{Cl}^- (\text{aq})$ is an example of – [IIT- 2001]
(A) Oxidation reaction (B) reduction reaction
(C) Disproportionation reaction (D) Decomposition reaction
8. The number of S—S bonds in sulphur trioxide trimer, (S_3O_9) is – [IIT- 2001]
(A) Three (B) Two (C) One (D) Zero
9. **Statement-I** : Between SiCl_4 and CCl_4 , only SiCl_4 reacts with water [IIT- 2001]
Because :
Statement-II : SiCl_4 is ionic and CCl_4 is covalent
(A) If both assertion and reason are correct and reason is the correct explanation of the assertion
(B) If both assertion and reason are correct, but reason is not the correct explanation of the assertion
(C) If assertion is correct, but reason is incorrect
(D) If assertion is incorrect, but reason is correct.
10. Specify the coordination geometry around and hybridisation of N and B atoms in a 1 : 1 complex of BF_3 and NH_3 – [IIT 2001]
(A) N: tetrahedral sp^3 ; B : tetrahedral sp^3
(B) N: pyramidal sp^3 ; B : pyramidal sp^3
(C) N: pyramidal sp^3 ; B : planar sp^2
(D) N: pyramidal sp^3 ; B : tetrahedral sp^3 d
11. Polyphosphates are used as water softening agents because they – [IIT- 2002]
(A) Form soluble complexes with anionic species
(B) Precipitate anionic species
(C) Form soluble complexes with cationic species
(D) Precipitate cationic species

12. Identify, the correct order of acidic strength of CO_2 , CuO , CaO , H_2O – [IIT- 2002]
 (A) $\text{CaO} < \text{CuO} < \text{H}_2\text{O} < \text{CO}_2$ (B) $\text{H}_2\text{O} < \text{CuO} < \text{CaO} < \text{CO}_2$
 (C) $\text{CaO} < \text{H}_2\text{O} < \text{CuO} < \text{CO}_2$ (D) $\text{H}_2\text{O} < \text{CO}_2 < \text{CaO} < \text{CuO}$
13. Identify the correct order of solubility of Na_2S , CuS , and ZnS in aqueous medium – [IIT- 2002]
 (A) $\text{CuS} > \text{ZnS} > \text{Na}_2\text{S}$ (B) $\text{ZnS} > \text{Na}_2\text{S} > \text{CuS}$
 (C) $\text{Na}_2\text{S} > \text{CuS} > \text{ZnS}$ (D) $\text{Na}_2\text{S} > \text{ZnS} > \text{CuS}$
14. H_3BO_3 is – [IIT- 2002]
 (A) Monobasic acid and weak Lewis acid (B) Monobasic and weak Bronsted acid
 (C) Monobasic and strong Lewis acid (D) Tribasic and weak Bronsted acid
15. $(\text{Me})_2\text{SiCl}_2$ on hydrolysis will produce – [IIT- 2003]
 (A) $(\text{Me})_2\text{Si}(\text{OH})_2$ (B) $(\text{Me})_2\text{Si} = \text{O}$
 (C) $[-\text{O}-(\text{Me})_2\text{Si}-\text{O}-]_n$ (D) $\text{Me}_2\text{SiCl}(\text{OH})$
16. When I^- is oxidised by MnO_4^- in alkaline medium, I^- converts into – [IIT- 2003]
 (A) IO_3^- (B) I_2 (C) IO_4^- (D) IO^-
17. Which is the most thermodynamically stable allotropic form of phosphorus? [IIT- 2004]
 (A) Red (B) White (C) Black (D) Yellow
18. Which of the following is not oxidised by O_3 ? [IIT- 2005]
 (A) KI (B) FeSO_4 (C) KMnO_4 (D) K_2MnO_4
19. Which blue-liquid is obtained on reacting equimolar amounts of two gases at -30°C ? [IIT- 2005]
 (A) N_2O (B) N_2O_3 (C) N_2O_4 (D) N_2O_5
20. Name of the structure of silicates in which three oxygen atoms of $[\text{SiO}_4]^{4-}$ are shared is – [IIT- 2005]
 (A) Pyrosilicate (B) Sheet silicate
 (C) Linear chain silicate (D) Three dimensional silicate
21. $\text{B}(\text{OH})_3 + \text{NaOH} \rightleftharpoons \text{NaBO}_2 + \text{Na}[\text{B}(\text{OH})_4] + \text{H}_2\text{O}$ how can this reaction is made to proceed in forward direction? [IIT- 2006]
 (A) Addition of cis 1, 2 diol (B) Addition of borax
 (C) Addition of trans 1, 2 diol (D) Addition of Na_2HPO_4
22. The percentage of p-character in the orbitals forming P-P bonds in P_4 is – [IIT- 2007]
 (A) 25 (B) 33 (C) 50 (D) 75
23. Among the following, the paramagnetic compound is – [IIT- 2007]
 (A) Na_2O_2 (B) O_3 (C) N_2O (D) KO_2
24. Draw the structure of a cyclic silicate, $(\text{Si}_3\text{O}_9)^{6-}$ with proper labelling – [IIT-1998, 4 M]
25. Complete and balance the following chemical equations – [IIT-1998, 2 M]
 (i) $\text{P}_4\text{O}_{10} + \text{PCl}_5 \longrightarrow$ (ii) $\text{SnCl}_4 + \text{C}_2\text{H}_5\text{Cl} + \text{Na} \longrightarrow$
26. Work out the following using chemical equations [IIT- 1998, 2M]
 "Chlorination of calcium hydroxide produces bleaching powder"

27. Hydrogen peroxide acts both as an oxidizing and as a reducing agent in alkaline solution towards certain first row transition metal ion. Illustrate both these properties of H_2O_2 using chemical equations – [IIT- 1998, 4 M]
28. Give reasons for the following in one or two sentences only. [IIT- 1999]
 BeCl_2 can be easily hydrolysed."
29. In the contact process for industrial manufacture of sulphuric acid, some amount of sulphuric acid is used as a starting material. Explain briefly. What is the catalyst used in the oxidation of SO_2 ? [IIT- 1998, 4 M]
30. Give an example of oxidation of one halide by another halogen. Explain the feasibility of the reaction. [IIT- 2000]
31. Write the balanced chemical equation for developing photographic films. [IIT- 2001]
32. Draw the molecular structures of XeF_2 , XeF_4 and XeO_2F_2 , indicating the location of lone pair (s) of electrons. [IIT- 2000]
33. Give reason [IIT- 2000]
 Why elemental nitrogen exists as a diatomic molecule whereas elemental phosphorus is a tetra atomic molecule.
34. Compounds X on reduction with LiAlH_4 gives a hydride Y containing 21.72% hydrogen along with other products. The compound Y reacts with air explosively resulting in boron trioxide. Identify X and Y. Give balanced reactions involved in the formation of Y and its reaction with air Draw the structure of Y. [IIT- 2001]
35. Starting from SiCl_4 , prepare the following in steps not exceeding the number given in parenthesis (reactions only) [IIT- 2001]
 (i) Silicon (1) (ii) Linear silicon containing methyl group only (4) (iii) Na_2SiO_3 (3)
36. Identify (X) in the following synthetic scheme and write their structures. [IIT- 2001]
 $\text{Ba}^*\text{CO}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{X (gas)}$ (C^* denotes C^{14})
37. Write the balanced equations for the reactions of the following compounds with water: [IIT- 2002]
 (i) Al_4C_3 (ii) CaNCN (iii) BF_3 (iv) NCl_3 (v) XeF_4
38. Identify the following:

$$\text{Na}_2\text{CO}_3 \xrightarrow[\text{(aq)}]{\text{SO}_2} \text{A} \xrightarrow{\text{Na}_2\text{CO}_3} \text{B} \xrightarrow[\Delta]{\text{elemental S}} \text{C} \xrightarrow{\text{I}_2} \text{D}$$
 [IIT- 2003]
 Also mention the oxidation state of S in all the compounds.
39. Arrange the following oxides in the increasing order of Bronsted basicity. [IIT- 2004]
 Cl_2O_7 , BaO , SO_3 , CO_2 , B_2O_3

MATCH THE COLUMN

[IIT- 2003]

40.	Column-I	Column-II
(A)	$\text{Bi}^{3+} \longrightarrow (\text{BiO})^+$	(p)
(B)	$[\text{AlO}_2]^- \longrightarrow \text{Al}(\text{OH})_3$	(q)
(C)	$\text{SiO}_4^{4-} \longrightarrow \text{Si}_2\text{O}_7^{6-}$	(r)
(D)	$(\text{B}_4\text{O}_7^{2-}) \longrightarrow [\text{Bi}(\text{OH})_3]$	(s)
		Dilution by water

41. **Statement-I** : In water, orthoboric acid behaves as a weak monobasic acid. [IIT- 2007]

Because :

Statement-II : In water, orthoboric acid acts as a proton donor.

- (A) Statement-I is True, Statement-II is True; Statement-II is a correct explanation for Statement-I
(B) Statement-I is true, Statement-II is True ; Statement-II is not a correct explanation for Statement-I
(C) Statement-I is True, Statement-II is false
(D) Statement-I is False, Statement-II is True.

Comprehension # 1 (Q-42, 43, 44)

The noble gases have closed-shell electronic configuration and are monoatomic gases under normal conditions. The low boiling point of the lighter noble gases are due to weak dispersion forces between the atoms and the absence of other interatomic interactions. The direct reaction of xenon with fluorine leads to a series of compounds with oxidation number + 2, + 4 and + 6. XeF_4 reacts violently with water to give XeO_3 . The compounds of xenon exhibit rich stereochemistry and their geometries can be deduced considering the total number of electron pairs in the valence shell. [IIT- 2007]

42. Argon is used in arc welding because of its –

- (A) Low reactivity with metal (B) Ability to lower the melting point of metal
(C) Flammability (D) High calorific value

43. The structure of XeO_3 is –

- (A) Linear (B) Planar (C) Pyramidal (D) T-shaped

44. XeF_4 and XeF_6 are expected to be –

- (A) Oxidising (B) Reducing (C) Unreactive (D) Strongly basic

Comprehension # 2 (Q-45, 46, 47)

There are some deposits of nitrates and phosphates in earth's crust. Nitrates are more soluble in water. Nitrates are difficult to reduce under the laboratory conditions but microbes do it easily. Ammonia forms large number of complexes with transition metal ions. Hybridization easily explains the ease of sigma donation capability of NH_3 and PH_3 . Phosphine is a flammable gas and is prepared from white phosphorous. [IIT- 2008]

45. Among the following, the correct statement is :-

- (A) Phosphates have no biological significance in humans
(B) Between nitrates and phosphates, phosphates are less abundant in earth's crust
(C) Between nitrates and phosphates, nitrates are less abundant in earth's crust
(D) Oxidation of nitrates is possible in soil

46. Among the following, the correct statement is :-

- (A) Between NH_3 and PH_3 , NH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional
(B) Between NH_3 and PH_3 , PH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional
(C) Between NH_3 and PH_3 , NH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional
(D) Between NH_3 and PH_3 , PH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional.

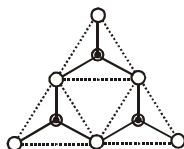
47. White phosphorus on reaction with NaOH gives PH_3 as one of the products. This is a :-
 (A) dimerization reaction (B) disproportionation reaction
 (C) condensation reaction (D) precipitation reaction
48. The reaction of P_4 with X leads selectively to P_4O_6 . The X is [JEE 2009]
 (A) Dry O_2 (B) A mixture of O_2 , and N_2
 (C) Moist O_2 (D) O_2 in the presence of aqueous NaOH
49. The nitrogen oxide(s) that contain(s) N-N bond(s) is (are) [JEE 2009]
 (A) N_2O (B) N_2O_3 (C) N_2O_4 (D) N_2O_5
50. In the reaction [JEE 2009]

$$2\text{X} + \text{B}_2\text{H}_6 \longrightarrow [\text{BH}_2(\text{X})_2]^+[\text{BH}_4]^-$$
 the amine(s) X is (are)
 (A) NH_3 (B) CH_3NH_2 (C) $(\text{CH}_3)_2\text{NH}$ (D) $(\text{CH}_3)_3\text{N}$
51. The reaction of white phosphorus with aqueous NaOH gives phosphine along with another phosphorus containing compound. The reaction type ; the oxidation states of phosphorus in phosphine and the other product are respectively [JEE 2012]
 (A) redox reaction ; -3 and -5
 (B) redox reaction ; +3 and +5
 (C) disproportionation reaction ; -3 and +1
 (D) disproportionation reaction ; -3 and +3
52. Bleaching powder contains a salt of an oxoacid as one of its components. The anhydride of that oxoacid is : [JEE 2012]
 (A) Cl_2O (B) Cl_2O_7 (C) ClO_2 (D) Cl_2O_6
53. With respect to graphite and diamond, which of the statement(s) given below is (are) correct ? [JEE 2012]
 (A) Graphite is harder than diamond.
 (B) Graphite has higher electrical conductivity than diamond.
 (C) Graphite has higher thermal conductivity than diamond.
 (D) Graphite has higher C-C bond order than diamond.

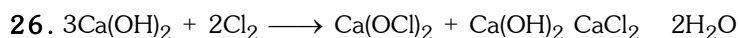
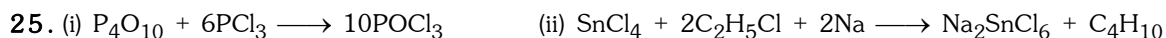
- 1. C 2. A 3. A 4. C 5. B,D 6. A 7. C
 8. D 9. C 10. A 11. C 12. A 13. D 14. A
 15. C 16. A 17. C 18. C 19. B 20. B 21. A
 22. D 23. D

24. In cyclic $(\text{Si}_3\text{O}_9)^{6-}$, three tetrahedral of SiO_4 are joined together by sharing of two oxygen atoms per tetrahedral.

Structure of $(\text{Si}_3\text{O}_9)^{6-}$

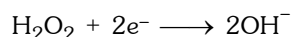


In it dark circles () represent Si and open circles (O) represent oxygen atom or iron.

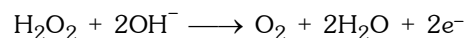


Bleaching powder is a mixture of CaOCl_2
and hydrated basic calcium chloride.

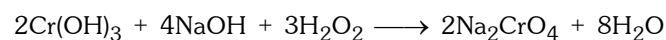
27. When H_2O_2 acts as oxidizing agent, therefore, following reaction takes place:



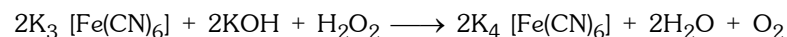
while, regarding its action on reducing agent, the following reaction takes place :



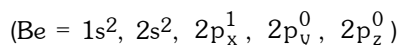
Oxidizing character :



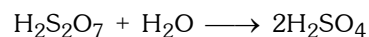
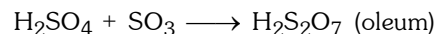
Reducing character:



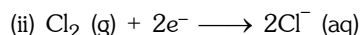
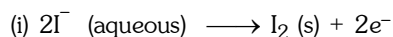
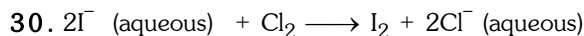
28. BeCl_2 is hydrolysed due to high polarising power and presence of vacant p-orbitals in Be atom.



29. In $\text{SO}_3 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4$ reaction, H_2SO_4 is obtained in misty form and reaction is explosive (highly exothermic). By adding H_2SO_4 the above reaction is prevented.

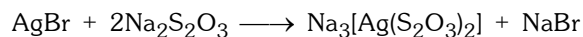


The catalyst used is V_2O_5 and K_2O is used as promotor for the oxidation of SO_2 into SO_3 .

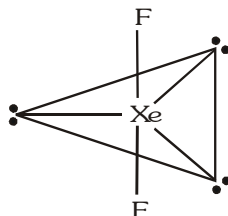


Thus, I^- is oxidised into I_2 by Cl_2 due to higher oxidised potential of Cl_2 than I_2

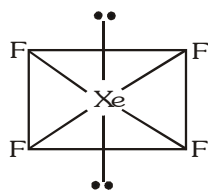
31. Unreacted AgBr is removed by hypo ($\text{Na}_2\text{S}_2\text{O}_3$)



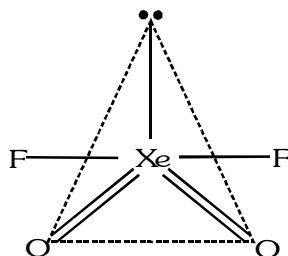
32.



In it Xe is sp^3d -hybrid but its shape is linear due to involvement of VSEPR theory. (i.e., due to presence of three free pair of electrons, geometry of XeF_2 is distorted from trigonal bipyramidal to linear).



In it Xe is sp^3d^2 -hybrid but its shape is square planar due to involvement of VSEPR theory. (i.e., due to presence of two free pair of electrons, geometry of XeF_4 is distorted from octahedral to square planar).



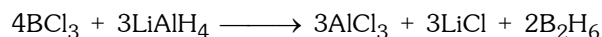
In it Xe is sp^3 d-hybrid but its geometry is planar due to involvement of VSEPR theory. (i.e., due to presence of a free pair of electron, its geometry is distorted from trigonal bipyramidal to planar).

33. In nitrogen, d-orbitals are not present, so in it the possibility of intramolecular multiplicity exists which leads to the completion of octet through π -bond between two nitrogen atoms.

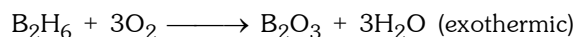
In phosphorus, d-orbitals are present, so in it due to large size of P, the P-P bonds are longer and hence intramolecular multiplicity is ruled out. So, for the completion of octet, it forms the bonds with three other 'P' atoms. Hence due to this reason it shows molecular formula as P_4 .

34. X : BCl_3

Y : B_2H_6

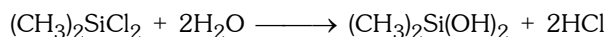
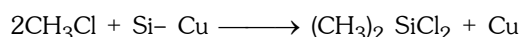
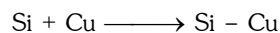
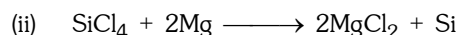


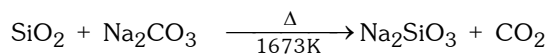
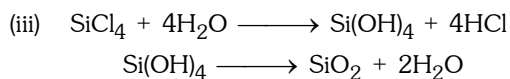
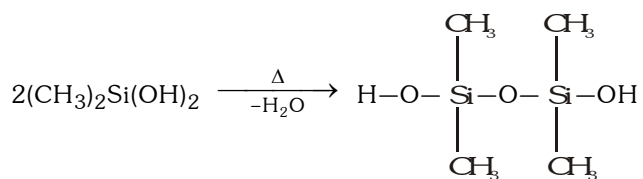
X



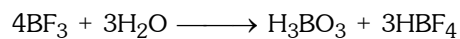
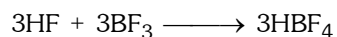
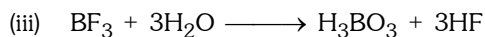
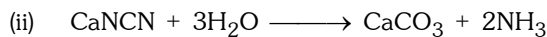
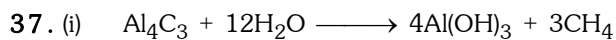
Y

35. (i) $3\text{SiCl}_4 + 4\text{Al} \longrightarrow 3\text{Si} + 4\text{AlCl}_3$ (in one step)

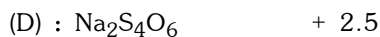
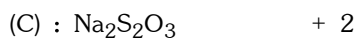
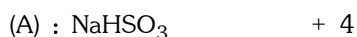




36. $^{14}\text{CO}_2$

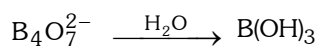
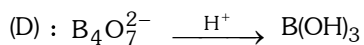
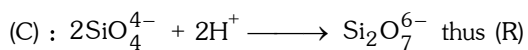
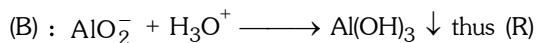


38. Oxidation state



39. $\text{Cl}_2\text{O}_7 < \text{SO}_3 < \text{CO}_2 < \text{B}_2\text{O}_3 < \text{BaO}$

40. Thus, Q and R



Thus, (Q) and (R)

41. (C)

Comprehension # 1

42.. (A)

43. (C)

44. (A)

Comprehension # 2

45. (C)

46. (C)

47. (B)

48. (B)

49. (A, B, C)

50. (B, C)

51. (C)

52. (A)

53. (B, D)

EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. Alkaline earth metals (group 2 or IIA elements) differ from group 12 (or IIB) elements in the electronic configuration of their :
(A) Antipenultimate shell (B) Innermost shell
(C) Outermost shell (D) Penultimate shell
2. The first ionization enthalpy of magnesium is lower than the first ionization enthalpy of :
(A) Lithium (B) Sodium (C) Calcium (D) Beryllium
3. Chemical A is used for water softening to remove temporary hardness. A reacts with Na_2CO_3 to generate caustic soda. When CO_2 is bubbled through A, it turns cloudy. What is the chemical formula of A :
(A) CaCO_3 (B) CaO (C) Ca(OH)_2 (D) $\text{Ca(HCO}_3)_2$
4. The substance not likely to contain CaCO_3 is :
(A) Calcined gypsum (B) Sea shells (C) Dolomite (D) A marble statue
5. A metal M readily forms water soluble sulphate MSO_4 , water insoluble hydroxide M(OH)_2 and oxide MO which becomes inert on heating. The hydroxide is soluble in NaOH. The M is :
(A) Be (B) Mg (C) Ca (D) Sr
6. A chloride dissolves appreciably in cold water. When placed on a Pt wire in Bunsen flame, no distinctive colour is noted. Which cation could be present?
(A) Be^{2+} (B) Ba^{2+} (C) Pb^{2+} (D) Ca^{2+}
7. The hydroxide which is best soluble in water is :
(A) Ba(OH)_2 (B) Mg(OH)_2 (C) Sr(OH)_2 (D) Ca(OH)_2
8. What is X in the following reaction?
 $\text{MgCl}_2 + 2 \text{H}_2\text{O} \longrightarrow \text{X} + 2 \text{HCl} + \text{H}_2\text{O}$
(A) MgO (B) Mg (C) Mg(OH)_2 (D) Mg(OH)Cl
9. (Yellow ppt) $\text{T} \xleftarrow{\text{K}_2\text{CrO}_4} \text{X} \xrightarrow{\text{dil. HCl}} \text{Y (Yellow ppt)} + \text{Z} \uparrow$ (pungent smelling gas) If X gives green flame test. Then, X is :
(A) MgSO_4 (B) BaS_2O_3 (C) CuSO_4 (D) PbS_2O_3
10. The correct statement is/are :
(A) BeCl_2 is a covalent compound (B) BeCl_2 is an electron deficient molecule
(C) BeCl_2 can form dimer (D) The hybrid state of Be in BeCl_2 is sp^2
11. The reaction of an element A with water produces combustible gas B and an aqueous solution of C. When another substance D reacts with this solution C also produces the same gas B. D also produces the same gas even on reaction with dilute H_2SO_4 at room temperature. Element A imparts golden yellow colour to Bunsen flame. Then A, B, C and D may be identified as :
(A) Na, H_2 , NaOH and Zn (B) K, H_2 , KOH and Zn
(C) K, H_2 , NaOH and Zn (D) Ca, H_2 , CaCO_3 and Zn
12. An alkaline earth metal (M) gives a salt with chlorine, which is insoluble in water at room temperature but soluble in boiling water. It also forms an insoluble sulphate whose mixture with a sulphide of a transition metal is called 'lithopone'a white pigment. Metal M is :
(A) Ca (B) Mg (C) Ba (D) Sr
13. In electrolysis of NaCl when Pt electrode is taken then H_2 is liberated at cathode while with Hg cathode it forms sodium amalgam:
(A) Hg is more inert than Pt
(B) More voltage is required to reduce H^+ at Hg than at Pt
(C) Na is dissolved in Hg while it does not dissolve in Pt
(D) Conc. of H^+ ions is larger when Pt electrode is taken

14. The correct sequence of increasing covalent character is represented by –
 (A) $\text{BeCl}_2 < \text{NaCl} < \text{LiCl}$ (B) $\text{NaCl} < \text{LiCl} < \text{BeCl}_2$
 (C) $\text{BeCl}_2 < \text{LiCl} < \text{NaCl}$ (D) $\text{LiCl} < \text{NaCl} < \text{BeCl}_2$
15. The paramagnetic species is :
 (A) KO_2 (B) SiO_2 (C) TiO_2 (D) BaO_2
16. The pair of amphoteric hydroxides is :-
 (A) Al(OH)_3 , LiOH (B) Be(OH)_2 , Mg(OH)_2
 (C) B(OH)_3 , Be(OH)_2 (D) Be(OH)_2 , Zn(OH)_2
17. Maximum thermal stability is shown by
 (A) MgCO_3 (B) CaCO_3 (C) SrCO_3 (D) BaCO_3
18. Stable oxide is obtained by heating the carbonate of the element
 (A) Li (B) K (C) Na (D) Rb
19. The stable superoxide is formed by the element
 (A) Li (B) Na (C) K (D) Ca
20. The metallic lustre exhibited by sodium is explained by
 (A) diffusion of sodium ions
 (B) oscillation of loose electrons
 (C) excitation of free protons
 (D) existence of body centred cubic lattice
21. A solution of sodium sulphate in water is electrolysed using inert electrodes. The products at the cathode and anode are respectively
 (A) H_2 , O_2 (B) O_2 , H_2 (C) O_2 , Na (D) O_2 , SO_2
22. The hydration energy of Mg^{2+} is greater than that of
 (A) Al^{3+} (B) Na^+ (C) Be^{2+} (D) Mg^{3+}
23. Calcium is obtained by the
 (A) electrolysis of molten calcium chloride
 (B) electrolysis of a solution of CaCl_2 in water
 (C) reduction of CaCl_2 with carbon
 (D) roasting of limestone
24. The material used in photoelectric cells contains –
 (A) Cs (B) Si (C) Sn (D) Ti
25. Four alkali metals A, B, C and D are having respectively standard reduction potentials as -3.05 , -1.66 , -0.40 and 0.80 V. Which one will be the most reducing agent ?
 (A) A (B) B (C) C (D) D
26. Which of the following imparts violet colouration to the Bunsen burner non-luminous flame
 (A) NaCl (B) BaCl_2 (C) CaCl_2 (D) KCl
27. Which one of the following is most basic ?
 (A) Al_2O_3 (B) MgO (C) SiO_2 (D) P_2O_5
28. Molten sodium is used in nuclear reactors to
 (A) absorb neutrons in order to control the chain reaction
 (B) slow down the fast neutrons
 (C) absorb the heat generated by nuclear fission
 (D) extract radio-isotopes produced in the reactor

EXERCISE-02**BRAIN TEASERS****SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT NSWERS)**

1. Which of the following is incorrect?
(A) Mg burns in air releasing dazzling light rich in UV rays.
(B) $\text{CaCl}_2 + 6 \text{H}_2\text{O}$ when mixed with ice gives freezing mixture.
(C) Mg cannot form complexes
(D) Be can form complexes due to its very small size.
2. On dissolving moderate amount of sodium metal in liquid NH_3 at low temperature, which one of the following does not occur
(A) Blue coloured solution is obtained.
(B) Na^+ ions are formed in the solution.
(C) Liquid NH_3 becomes good conductor of electricity.
(D) Liquid ammonia remains diamagnetic.
3. The minimum equivalent conductance in fused state is shown by –
(A) MgCl_2 (B) BeCl_2 (C) CaCl_2 (D) SrCl_2
4. The metal which cannot be produced on reduction of its oxide by aluminium is
(A) K (B) Mn (C) Cr (D) Fe
5. Magnesium on reaction with very dilute HNO_3 gives
(A) NO (B) N_2O (C) H_2 (D) NO_2
6. The alkali metal that reacts with nitrogen directly to form nitride is
(A) Li (B) Na (C) K (D) Rb
7. Which of the following statement is/are false for alkali metals ?
(A) Lithium is the strongest reducing agent
(B) Na is amphoteric in nature
(C) Li^+ is exceptionally small
(D) All alkali metals give blue solution in liquid ammonia
8. Amongst LiCl, RbCl, BeCl_2 and MgCl_2 , the compounds with the greatest and least ionic character respectively are :-
(A) LiCl, RbCl (B) RbCl, BeCl_2 (C) RbCl, MgCl_2 (D) MgCl_2 , BeCl_2
9. K_2CS_3 can be called potassium
(A) sulphocyanide (B) thiocarbide (C) thiocarbonate (D) thiocyanate
10. Anhydrous MgCl_2 can be prepared by heating $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$
(A) in a current of dry HCl gas (B) with carbon
(C) until it fuses (D) with lime
11. Oxygen ions structure in its peroxide, superoxide, ozonide :
(A) O_2^- , O_2^{2-} , O_3^{2-} (B) O_2^{2-} , O_2^- , O_3^- (C) O_2^{2-} , O^{2-} , O_3^- (D) O_2^- , O_2^{3-} , O_3^{2-}

12. In presence of iron, alkali metal react with liquid ammonia and form
 (A) Metal mixture + H_2 (B) Iron metal mixture + H_2
 (C) Metal mixture (D) Metal amide + H_2
13. The ionic conductance of following cation in a given concentration are in the order
 (A) $Li^+ < Na^+ < K^+ < Rb^+$ (B) $Li^+ > Na^+ > K^+ > Rb^+$
 (C) $Li^+ < Na^+ > K^+ > Rb^+$ (D) $Li^+ = Na^+ < K^+ < Rb^+$
14. Which of the following does not give an oxide on heating –
 (A) $MgCO_3$ (B) Li_2CO_3 (C) $ZnCO_3$ (D) K_2CO_3
15. On heating sodium metal in the current of dry ammonia leads to the formation of which gas–
 (A) $NaNH_2$ (B) NaN_3 (C) NH_3 (D) H_2
16. On allowing ammonia solution of s-block metals to stand for a long time, blue colour becomes fade. The reason is:-
 (A) Formation of NH_3 gas (B) Formation of metal amide
 (C) Cluster formation of metal ions (D) Formation of metal nitrate
17. When Na and Li placed in dry air we get :-
 (A) $NaOH$, Na_2O , Li_2O (B) Na_2CO_3 , Na_2O_2 , Li_2O
 (C) Na_2O , Li_3N , NH_3 (D) Na_2O , Li_2O , Li_3N
18. The hydride ion H^- is stronger base than its hydroxide ion OH^- . Which of the following reaction will occur if sodium hydride is dissolved in water:-
 (A) $H_{(aq)}^- + H_2O \rightarrow H_3O^+$ (B) $H_{(aq)}^- + H_2O \rightarrow OH^- + H_2$
 (C) $H^- + H_2O \rightarrow H_2 + O_2$ (D) $H^- + H_2O \rightarrow$ No reaction
19. Which can not be used to generate H_2 :-
 (A) $Al + NaOH$ (B) $Zn + NaOH$ (C) $Mg + NaOH$ (D) $LiH + H_2O$
20. Only those elements of s-block can produce superoxides which have :-
 (A) High ionisation energy (B) High electronegativity
 (C) High charge density (D) Low ionisation potential
21. Alum is the name used for all double salts having the composition $M_2^I SO_4 \cdot M_2^{III} (SO_4)_3 \cdot 24H_2O$. Where M^{III} stands for Al^{+3} , Cr^{+3} , Fe^{+3} , while M^I stands for:-
 (A) Li^+ , Cu^+ , Ag^+ (B) Li^+ , NH_4^+ , Na^+ (C) Na^+ , K^+ , Rb^+ (D) Ca^{+2} , Mg^{+2} , Sr^{+2}
22. Identify the correct statement -
 (A) Gypsum contains a lower percentage of Ca than plaster of paris
 (B) Gypsum is obtained by heating plaster of paris
 (C) Plaster of paris can be obtained by hydration of gypsum
 (D) Plaster of paris is obtained by partial oxidation of gypsum
23. In the reaction $M + O_2 \longrightarrow MO_2$ (super oxide) the metal is
 (A) Li (B) Na
 (C) K (D) Ba

24. Na^+ and Ag^+ differ in
- (A) Na_2CO_3 is thermally stable while Ag_2CO_3 decomposes into Ag, CO_2 and O_2
- (B) Ag^+ forms complexes, Na^+ does not
- (C) NaCl is water soluble, AgCl is insoluble
- (D) NaBr-yellow and AgBr pale yellow
25. The stability order of oxide, peroxide and superoxide of alkali metal is
- (A) Normal oxide > super oxide > per oxide
- (B) Normal oxide > per oxide > super oxide
- (C) super oxide > per oxide > normal oxide
- (D) per oxide > normal oxide > super oxide
26. Match list I with list II and choose the correct answer from the codes given below

List I	List II
(A) NaNO_3	(a) Baking soda
(B) $\text{Na}(\text{NH}_4)\text{HPO}_4$	(b) Chile salt peter
(C) NaHCO_3	(c) Microcosmic salt
(D) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$	(d) Washing soda

Codes is :

	A	B	C	D
(A)	a	b	c	d
(B)	b	c	a	d
(C)	c	a	b	d
(D)	d	a	b	c

27. Which of the following statement is not correct
- (A) LiOH is amphoteric in nature
- (B) LiCl is soluble in pyridine
- (C) Li_3N is stable while Na_3N doesn't exist even at room temperature
- (D) BeO is amphoteric in nature
28. Which of the following statement is correct for s-block elements :-
- (A) Be has smallest atomic size in II A group
- (B) Li is most metallic
- (C) Mg impart red colour to the flame
- (D) Cs is most reducing in water
29. Which of the following are ionic carbides?
- (A) CaC_2 (B) Al_4C_3 (C) SiC (D) Be_2C
30. Which of the following groups of elements have chemical properties that are most similar :
- (A) Na, K, Ca (B) Mg, Sr, Ba
- (C) Be, Al, Ca (D) Be, Ra, Cs

31. Which of the following statements are false?
- (A) BeCl_2 is a linear molecule in the vapour state but it is polymeric in the solid state
- (B) Calcium hydride is called hydrolith
- (C) Carbides of both Be and Ca react with water to form acetylene
- (D) Oxides of both Be and Ca are amphoteric.
32. The incorrect statement(s) is/are :
- (A) Mg cannot form complexes
- (B) Be can form complexes due to a very small atomic size
- (C) The first ionisation potential of Be is higher than that of Mg.
- (D) Mg forms an alkaline hydroxide while Be forms amphoteric oxides.
33. Na_2SO_4 is water soluble but BaSO_4 is insoluble because :
- (A) The hydration energy of Na_2SO_4 is higher than that of its lattice energy
- (B) The hydration energy of Na_2SO_4 is less than that of its lattice energy
- (C) The hydration energy of BaSO_4 is less than that of its lattice energy
- (D) The hydration energy of BaSO_4 is higher than that of its lattice energy
34. $\text{BeCl}_2 + \text{LiAlH}_4 \longrightarrow \text{X} + \text{LiCl} + \text{AlCl}_3$
- (A) X is lithium hydride
- (B) X is BeH_2
- (C) X is $\text{BeCl}_2 \cdot 2\text{H}_2\text{O}$
- (D) X is LiH
35. $\text{X} \xrightarrow{\text{CaCl}_2} \text{CaCl}_2 + \text{Y} \uparrow$; the effective ingredient of X is :
- (A) OCl^-
- (B) Cl^-
- (C) OCl^+
- (D) OCl_2^-
36. Which of the following substance(s) is/are used in laboratory for drying purposes?
- (A) Anhydrous P_2O_5
- (B) Graphite
- (C) Anhydrous CaCl_2
- (D) Na_3PO_4
37. If X and Y are the second ionisation potentials of alkali and alkaline earth metals of same period, then :
- (A) $X > Y$
- (B) $X < Y$
- (C) $X = Y$
- (D) $X \ll Y$
38. $\text{X} \xrightarrow{\text{N}_2, \Delta} \text{Y} \xrightarrow{\text{H}_2\text{O}} \text{Z (colourless gas)} \xrightarrow{\text{CuSO}_4} \text{T (blue colour)}$:
- Then, substances Y and T are -
- (A) $\text{Y} = \text{Mg}_3\text{N}_2$ and $\text{T} = \text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
- (B) $\text{Y} = \text{Mg}_3\text{N}_2$ and $\text{T} = \text{CuSO}_4 \cdot 4\text{NH}_3$
- (C) $\text{Y} = \text{Mg}(\text{NO}_3)_2$ and $\text{T} = \text{CuO}$
- (D) $\text{Y} = \text{MgO}$ and $\text{T} = \text{CuSO}_4 \cdot 4\text{NH}_3$
39. When K_2O is added to water, the solution becomes basic in nature because it contains a significant concentration of :
- (A) K^+
- (B) O^{2-}
- (C) OH^-
- (D) O_2^{2-}

EXERCISE-03**MISCELLANEOUS TYPE QUESTIONS****TRUE / FALSE**

1. Magnesium is an essential constituent of chlorophyll, the green colouring matter of plants.
2. Setting of cement is an endothermic process.
3. Calcium bicarbonate is known in solid state.
4. BeH_2 is an ionic hydride.
5. BeCO_3 is thermally stable compounds.
6. In the electrolysis of fused calcium hydride, hydrogen is liberated at cathode.
7. $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ on heating forms MgCl_2 .
8. Sodium when heated in excess of oxygen gives sodium oxide.
9. In group IA of alkali metals, the ionisation potential decreases down the group. Therefore, lithium is a poorer reducing agent in gaseous medium.
10. The softness of group IA metals increases down the group with increasing atomic number.

FILL IN THE BLANKS

1. Anhydrous magnesium chloride is obtained by heating the hydrated salt with.....
2. Ca(OH)_2 is..... basic than Mg(OH)_2 .
3. CaH_2 is commercially known as.....
4. Magnesium burn in air forming and
5. Ba react with cold water Mg reacts with..... while Be has..... with boiling water.
6. A standard solution of sodium hydroxide cannot be prepared by direct weighing because.....
7. Potassium bicarbonate cannot be prepared by solvay process because.....
8. Solution of alkali metals in liquid ammonia conducts electricity due to.....

MATCH THE COLUMN

1.

<u>Column-I</u>		<u>Column-II</u>	
(A)	Hydrolith	(p)	Contain Ca
(B)	Nitrolium	(q)	Used as a fertilizer
(C)	Dolomite	(r)	Used to prepare H_2
(D)	Pearl's ash	(s)	Contain potassium

2.

<u>Column-I</u>		<u>Column-II</u>	
(A)	Solvay process	(p)	NaCl
(B)	Evolve $\text{CO}_2 \uparrow$ on heating	(q)	Na_2O_2
(C)	aq. soln. is neutral towards litmus	(r)	NaHCO_3
(D)	Oxone	(s)	Na_2CO_3

3.

Column-I		Column-II	
(A)	Metal sulphate $\xrightarrow{\Delta}$ metal oxide + SO_2 + O_2	(p)	Ba
(B)	Metal cation + $\text{K}_2\text{CrO}_4 \longrightarrow$ yellow ppt	(q)	Sr
(C)	Metal + $\text{NH}_3 \xrightarrow{\text{(liquid)}}$ blue solution	(r)	Na
(D)	MCl_2 + conc. $\text{H}_2\text{SO}_4 \longrightarrow$ white ppt.	(s)	Mg

ASSERTION & REASON QUESTIONS

These questions contains, Statement I (assertion) and Statement II (reason).

- (A) Statement-I is true, Statement-II is true ; Statement-II is correct explanation for Statement-I.
 (B) Statement-I is true, Statement-II is true ; Statement-II is not a correct explanation for statement-I
 (C) Statement-I is true, Statement-II is false
 (D) Statement-I is false, Statement-II is true

1. **Statement-I** : Li_2SO_4 do not form double salt like alum.

Because

Statement-II : Atomic size of Li is too small.

2. **Statement-I** : NaCl when exposed in air it becomes wet.

Because

Statement-II : NaCl contains hygroscopic impurities like CaCl_2 , MgCl_2 etc.

3. **Statement-I** : Lithium is the weakest reducing agent among alkali metals.

Because

Statement-II : In alkali metals I.P. decreases down the group.

4. **Statement-I** : BaCO_3 is more soluble in HNO_3 than in plain water.

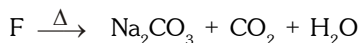
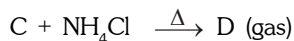
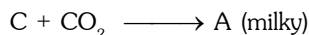
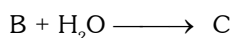
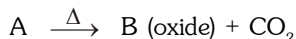
Because

Statement-II : Carbonate is a weak base and reacts with the H^+ from the strong acid causing the barium salt to dissociate.

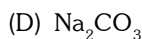
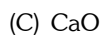
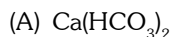
5. **Statement-I** : BeCl_2 fumes in moist air.

Because

Statement-II : BeCl_2 reacts with moisture of form HCl gas.

COMPREHENSION BASED QUESTIONS**Comprehension # 1**

1. A is :



2. B and C are :
- (A) CaO , Ca(OH)_2 (B) Ca(OH)_2 , CaCO_3
 (C) CaCO_3 , Ca(OH)_2 (D) Ca(OH)_2 , CaO
3. D, E and F are :
- (A) NH_3 , NH_4Cl , NH_4HCO_3
 (B) NH_3 , NH_4HCO_3 , NaHCO_3
 (C) NH_4HCO_3 , Na_2CO_3 , NaHCO_3
 (D) None

Comprehension # 2

Following given passage the five observation regarding alkali metals are mentioned.

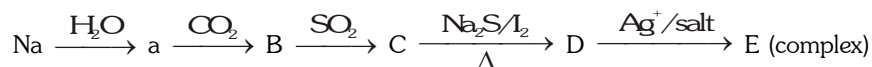
- On exposure to air, sodium hydroxide becomes liquid and after some time it changes to white powder.
- In water LiF is least soluble fluorides among fluorides of alkali metal, but its solubility increases as HF is added in aqueous solution.
- LiH more stable than NaH when heated separately
- When excess of $\text{Na}_2\text{S}_2\text{O}_3$ solution is added to the FeCl_3 solution an intense violet colouration is produced, but violet colour disappeared shortly
- Between Na^+ and Ag^+ , Ag^+ is stronger Lewis acid.

- The explanation of observation (v) is :
 - Because Na^+ has inert gas configuration which has greater polarisation power
 - Because Ag^+ has inert gas configuration which has greater polarisation power
 - Because Ag^+ has pseudo inert gas configuration which has lesser polarisation power
 - Because Ag^+ has pseudo inert gas configuration which has greater polarisation power
- The explanation for the observation (iv) is :
 - Initially with FeCl_3 , $\text{Na}_2\text{S}_2\text{O}_3$ produce an intense violet colour substance $\text{Fe}_2(\text{S}_2\text{O}_3)_3$. But $\text{Fe}_2(\text{S}_2\text{O}_3)_3$ changes to Fe^{2+} & $\text{S}_4\text{O}_6^{2-}$ on standing
 - Initially with FeCl_3 , is reduced to FeCl_2 by $\text{Na}_2\text{S}_2\text{O}_3$. FeCl_2 so produced undergo unstable complex formation $\text{Fe}(\text{S}_2\text{O}_3)_3^{4-}$ which is violet in colour.
 - Initially with FeCl_3 , $\text{Na}_2\text{S}_2\text{O}_3$ produce colloidal Fe which is violet in colour. After sometime, the colloidal suspension changes to the ppt of Fe .
 - There is no reaction
- As per observation (iii) LiH is more stable than NaH , because :
 - Due to small size of Li^+ , the lattice energy of LiH is greater
 - Due to greater size of H^- , the lattice energy of LiH is greater
 - LiH is more covalent than NaH
 - Due to greater size of Na^+ , the lattice energy of NaH is greater.
- As per observation (ii) the solubility of LiF increases in the presence of HF , because :
 - The HF further ionises to H^+ & F^- (B) In the presence of HF , there will be a common ion effect
 - In the presence HF , F^- is converted to HF_2^- (D) All of the above

5. The reaction for observation (i) can be explained as –

- (A) $\text{NaOH (S)} \xrightarrow{\text{H}_2\text{O}} \text{NaOH(aq)} \xrightarrow{\text{H}_2\text{O}} \text{N}$
- (B) $\text{NaOH (S)} \xrightarrow{\text{H}_2\text{O}} \text{NaOH(aq)} \xrightarrow{-\text{H}_2\text{O}} \text{Na}_2\text{O (S)}$
- (C) $\text{NaOH (S)} \xrightarrow{\text{H}_2\text{O}} \text{NaOH(aq)} \xrightarrow{\text{O}_2 / \text{air}} \text{Na}_2\text{O (S)}$
- (D) $\text{NaOH (S)} \xrightarrow{\text{H}_2\text{O}} \text{NaOH(aq)} \xrightarrow{\text{CO}_2} \text{Na}_2\text{CO (S)}$

Comprehension # 3



1. The compound B and C are :

- (A) Na_2CO_3 , Na_2SO_4
- (B) NaHCO_3 , Na_2SO_4
- (C) Na_2CO_3 , Na_2SO_3
- (D) None of these

2. The compound D is :

- (A) Na_2SO_4
- (B) $\text{Na}_2\text{S}_4\text{O}_6$
- (C) $\text{Na}_2\text{S}_2\text{O}_5$
- (D) $\text{Na}_2\text{S}_2\text{O}_3$

3. Oxidation number of each 'S' atom in compound D :

- (A) + 2, + 2
- (B) + 4, 0
- (C) + 6, - 2
- (D) + 5, - 1

Comprehension # 4

Alkali metals readily react with oxyacids forming corresponding salts like M_2CO_3 , MHCO_3 , MNO_3 , M_2SO_4 etc. with evolution of hydrogen. They also dissolve in liquid NH_3 but without the evolution of hydrogen. The colour of its dilute solution is blue but when it is heated and concentrated then its colour becomes bronze.

1. Among the nitrate of alkali metals which one can be decomposed to its oxide?

- (A) NaNO_3
- (B) KNO_3
- (C) LiNO_3
- (D) All of these

2. Among the carbonates of alkali metals which one has highest stability?

- (A) Cs_2CO_3
- (B) Rb_2CO_3
- (C) K_2CO_3
- (D) Na_2CO_3

3. Which of the following statement about the sulphate of alkali metal is correct?

- (A) Except Li_2SO_4 all sulphate of other alkali metals are soluble in water
- (B) All sulphates of alkali metals except lithium sulphate forms alum.
- (C) The sulphates of alkali metals cannot be hydrolysed.
- (D) All of these

4. Which of the following statement about solution of alkali metals in liquid ammonia is correct?
- (A) The solution have strong oxidizing properties.
- (B) Both the dilute solution as well as concentrated solution are paramagnetic in nature
- (C) Charge transfer is the responsible for the colour of the solution
- (D) None of these
5. Which metal bicarbonates does not exist in solid state?
- (i) LiHCO_3 (ii) $\text{Ca}(\text{HCO}_3)_2$ (iii) $\text{Zn}(\text{HCO}_3)_2$ (iv) NaHCO_3 (v) AgHCO_3
- (A) (ii), (iii), (v) (B) (i), (ii), (iii) (C) (i), (ii), (v) (D) (ii), (iii), (iv)
-

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
• <u>True / False</u>		
1. T 2. F 3. F 4. F 5. F 6. F 7. F 8. F 9. T 10. T		
• <u>Fill in the Blanks</u>		
1. Dry HCl 2. More 3. Hydrolyth 4. Mgo , Mg_3N_2 5. Vigorously, slowly, no action		
6. Absorb moisture & CO_2 from atmosphere 7. Ammonated e^-		
• <u>Match the Column</u>		
1. (A) $\rightarrow$ p,r ; (B) $\rightarrow$ p,q ; (C) $\rightarrow$ p ; (D) $\rightarrow$ s 2. (A) $\rightarrow$ r,s ; (B) $\rightarrow$ r,s ; (C) $\rightarrow$ p ; (D) $\rightarrow$ q		
3. (A) $\rightarrow$ p,q,s ; (B) $\rightarrow$ p,q; (C) $\rightarrow$ p,q,r,s ; (D) $\rightarrow$ p,q		
• <u>Assertion - Reason Questions</u>		
1. A 2. A 3. D 4. C 5. A		
• <u>Comprehension Based Questions</u>		
Comprehension #1 : 1. (B) 2. (A) 3. (B)		
Comprehension #2 : 1. (D) 2. (A) 3. (A) 4. (C) 5. (D)		
Comprehension #3 : 1. (C) 2. (D) 3. (C)		
Comprehension #4 : 1. (C) 2. (A) 3. (D) 4. (D) 5. (A)		

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

1. Explain the following :
 - (i) The reaction between marble and dilute H_2SO_4 is not used to prepare carbon dioxide.
 - (ii) Lime water becomes turbid on passing CO_2 though it, but becomes clear when more CO_2 is passed.
 - (iii) Alkaline earth metals have higher melting points than alkali metals.
 - (iv) Beryllium does not exhibit a covalency beyond 4.
2. PbO_2 is soluble in NaOH and also in HCl . What does it reflect about the nature of PbO_2 ?
3. What happens when :
 - (i) Hot and concentrated caustic soda solution reacts with iodine.
 - (ii) White phosphorus is heated with caustic soda.
 - (iii) Excess of caustic soda reacts with zinc sulphate solution.
 - (iv) Excess of NaOH is added to AlCl_3 solution.
4. Write balanced equation for reaction between
 - (i) Na_2O_2 and water
 - (ii) KO_2 and water
 - (iii) Na_2O_2 and CO_2
5. Element A burns in nitrogen to give an ionic compound B. Compound B reacts with water to give C and D. A solution of C becomes milky on bubbling carbon dioxide . Identify A, B, C and D.
6. In water LiF is least soluble fluoride among fluorides of alkali metals, but its solubility increases as HF is added in aqueous solution, why?
7. What happens when $\text{CuSO}_4(\text{aq.})$ is treated with excess of $\text{Na}_2\text{S}_2\text{O}_3$ solution?
8. Arrange the following in order of increasing
 - (i) Thermal stability $\text{BeSO}_4, \text{MgSO}_4, \text{CaSO}_4$
 - (ii) Polarising power $\text{Be}^{2+}, \text{Mg}^{2+}, \text{Ca}^{2+}$
 - (iii) Solubility in H_2O $\text{Be}(\text{OH})_2, \text{Mg}(\text{OH})_2, \text{Ca}(\text{OH})_2$
 - (iv) Covalent nature $\text{BeCl}_2, \text{MgCl}_2, \text{CaCl}_2$
 - (v) Hydrolysis nature $\text{BeCl}_2, \text{MgCl}_2, \text{CaCl}_2$
 - (vi) Lattice energy $\text{CaF}_2, \text{MgF}_2, \text{BaF}_2$
 - (vii) Hydration energy $\text{Be}^{2+}, \text{Mg}^{2+}, \text{Ba}^{2+}$
 - (viii) Solubility in water $\text{MgF}_2, \text{BaF}_2, \text{BeF}_2$
 - (ix) Basic nature $\text{Be}, \text{Mg}, \text{Ca}, \text{Sr}$
9. Hydrogen reacts with a metal (A) to give an ionic hydride (B). The metal (A) gives brick red colour with bunsen flame. The hydride formed is commonly known by its trade name. The compound (B) on treating with water gives back H_2 and (C). Identify (A), (B) and (C).

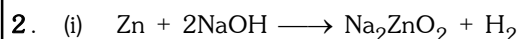
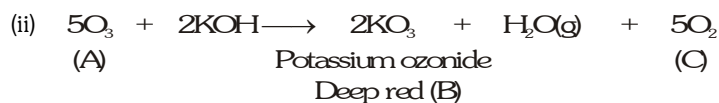
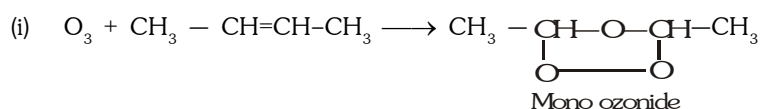
1. (i) Insoluble CaSO_4 is formed which deposits on the surface of marble and prevents further action of dilute H_2SO_4 , so the evolution of CO_2 ceases after sometime.
- (ii) Insoluble CaCO_3 is first precipitated which dissolves in excess of CO_2 due to the form of $\text{Ca}(\text{HCO}_3)_2$.
- $$\text{Ca}(\text{OH})_2 + \text{CO}_2 \longrightarrow \text{CaCO}_3 + \text{H}_2\text{O};$$
- (Insoluble)
- $$\text{CaCO}_3 + \text{H}_2\text{O} + \text{CO}_2 \longrightarrow \text{Ca}(\text{HCO}_3)_2$$
- (Soluble)
- (iii) Metallic bonding is much stronger in alkaline earth metals as two electrons are present in valence shell.
- (iv) The outermost energy shell in beryllium is the second. It cannot accommodate more than 8 electrons and hence a covalency limit 4 cannot be exceeded.
2. PbO_2 are amphoteric nature
- $$\text{PbO}_2 + 4\text{HCl} \longrightarrow \text{PbCl}_4 + 2\text{H}_2\text{O}$$
- $$\text{PbO}_2 + 2\text{NaOH} \longrightarrow \text{Na}_2\text{PbO}_3 + \text{H}_2\text{O}$$
3. (i) $3\text{I}_2 + 6\text{NaOH} \rightarrow 5\text{NaI} + \text{NaIO}_3 + 3\text{H}_2\text{O}$ (ii) $\text{P}_4 + 3\text{NaOH} + 3\text{H}_2\text{O} \rightarrow \text{PH}_3 + 3\text{NaH}_2\text{PO}_2$
- (iii) $\text{ZnSO}_4 + 2\text{NaOH} \longrightarrow \text{Zn}(\text{OH})_2 + \text{Na}_2\text{SO}_4$ (iv) $\text{AlCl}_3 + 3\text{NaOH} \longrightarrow \text{Al}(\text{OH})_3 + 3\text{NaCl}$
- ppt
- $$\text{Zn}(\text{OH})_2 + 2\text{NaOH} \longrightarrow \text{Na}_2[\text{Zn}(\text{OH})_4]$$
- ppt
- $$\text{Al}(\text{OH})_3 + \text{NaOH} \longrightarrow \text{Na}[\text{Al}(\text{OH})_4]$$
- or
- NaAlO_2
4. (i) $\text{Na}_2\text{O}_2 + 2\text{H}_2\text{O} \longrightarrow 2\text{NaOH} + \text{H}_2\text{O}_2$
- (ii) $\text{KO}_2 + \text{H}_2\text{O} \longrightarrow \text{KOH} + \text{H}_2\text{O}_2 + \text{O}_2$
- (iii) $2\text{Na}_2\text{O}_2 + 2\text{CO}_2 \longrightarrow 2\text{Na}_2\text{CO}_3 + \text{O}_2$
5. A = Ca, B = Ca_3N_2 C = $\text{Ca}(\text{OH})_2$ D = NH_3
6. In presence of HF, F^- is converted into bifluoride ion HF_2^- , allowing further dissolution of solid LiF.
7. $\text{CuSO}_4 + \text{Na}_2\text{S}_2\text{O}_3 \longrightarrow \text{CuS}_2\text{O}_3 + \text{Na}_2\text{SO}_4$
- $$2\text{CuS}_2\text{O}_3 + \text{Na}_2\text{S}_2\text{O}_3 \longrightarrow \text{CuS}_2\text{O}_3 + \text{Na}_2\text{S}_4\text{O}_6$$
- Cupric thiosulphate
- $$3\text{CuS}_2\text{O}_3 + 2\text{Na}_2\text{S}_2\text{O}_3 \longrightarrow \text{Na}_4[\text{Cu}_6(\text{S}_2\text{O}_3)_5]$$
- Sodium cuprothiosulphate.
8. (i) $\text{BeSO}_4 < \text{MgSO}_4 < \text{CaSO}_4$ (ii) $\text{Ca}^{2+} < \text{Mg}^{2+} < \text{Be}^{2+}$
- (iii) $\text{Be}(\text{OH})_2 < \text{Mg}(\text{OH})_2 < \text{Ca}(\text{OH})_2$ (iv) $\text{CaCl}_2 < \text{MgCl}_2 < \text{BeCl}_2$
- (v) $\text{CaCl}_2 < \text{MgCl}_2 < \text{BeCl}_2$ (vi) $\text{BaF}_2 < \text{CaF}_2 < \text{MgF}_2$
- (vii) $\text{Ba}^{2+} < \text{Mg}^{2+} < \text{Be}^{2+}$ (viii) $\text{BaF}_2 < \text{MgF}_2 < \text{BeF}_2$
- (ix) $\text{Be} < \text{Mg} < \text{Ca} < \text{Sr}$
9. (i) Ca gives brick red colour to flame (ii) $\text{Ca} + \text{H}_2 \longrightarrow \text{CaH}_2$ (hydrolith, trade name)
- (A) (B)
- (iii) $\text{CaH}_2 + 2\text{H}_2\text{O} \longrightarrow \text{Ca}(\text{OH})_2 + 2\text{H}_2$
- (B) (C)

BRAIN STORMING SUBJECTIVE EXERCISE

- When a gas (A) is passed through dry KOH at low temperature, a deep red coloured compound (B) and a gas (C) are obtained. The gas (A) on reaction with but-2-ene followed by treatment with Zn/H₂O yields acetaldehyde. Identify (A), (B) and (C)
- A compound (X) imparts a golden yellow flame and shows the following reactions:
 - Zinc powder when boiled with a concentrated aqueous solution of (X) dissolves and hydrogen is evolved.
 - When an aqueous solution of (X) is added to an aqueous solution of stannous chloride, a white precipitate is obtained first which dissolves in excess of solution of (X). Identify (X) and write equations at step (i) and (ii).
- A white solid is either Na₂O or Na₂O₂. A piece of red litmus paper turns white when it is dipped into a freshly made aqueous solution of the white solid.
 - Identify the substance and explain with balanced equation
 - Explain what would happen to the red litmus if the white solid were the other compound
- (A) is binary compound of a univalent metal. 1.422 g of (A) reacts completely with 0.321 g of sulphur in an evacuated and sealed tube to give 1.743 g of a white crystalline solid (B) that formed a hydrated double salt (C) with Al₂(SO₄). Identify (A), (B) and (C).
- Element (M) is a shiny and highly reactive metal (melting point 63°C) and element (X) is a highly reactive non-metal (melting point – 7.2°C). They react to form a compound with the empirical formula MX, a colourless, brittle solid that melts at 734°C. When dissolved in water or when in the molten state, the substance conducts electricity. When chlorine gas is bubbled through an aqueous solution containing (MX), a reddish-brown liquid appears and Cl[–] are formed. From these observations, identify M and X.
- Name an element which is invariably bivalent and whose oxide is soluble in excess of NaOH and its dipositive ion has a noble gas core.
- Out of the elements marked A, B, C, D, E, F, G and H:
 - Which form superoxide?
 - Which form thermally stable carbonate?
 - Which forms strongest base?
 - Which show diagonal relationship?
 - Which forms amphoteric oxide?

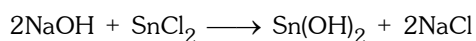
[illegible][illegible]

1. The gas (A) on treatment with but-2-ene followed by treatment with $\text{Zn}/\text{H}_2\text{O}$ yields acetaldehyde and thus (A) is ozone



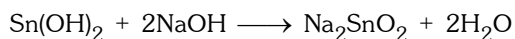
(X)

- (ii) (X) is also justified by step 2 reactions:



(X)

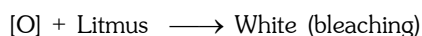
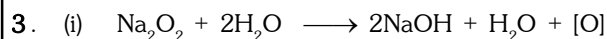
White ppt.



(Excess)

Soluble

(X)



Red

- (ii) The other compound Na_2O will give NaOH on dissolution in water. The red litmus will turn to blue.

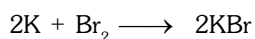


(A)

(B)

(C)

5. The given facts suggest M to be potassium (K) and (X) to be bromine (Br_2).



(Ionic solid with m. pt. 734°C)

6. Be

7. (a) E and G

(b) C, E and G

(c) G

(d) A and D

(e) B

EXERCISE-05 (A)**PREVIOUS YEARS QUESTIONS**

- A metal M readily forms its sulphate MSO_4 which is water soluble. It forms oxide MO which becomes inert on heating. It forms insoluble hydroxide which is soluble in NaOH. The metal M is:- [AIEEE-2002]
 (1) Mg (2) Ba (3) Ca (4) Be
- KO_2 is used in space and submarines because it [AIEEE-2002]
 (1) Absorbs CO_2 and increase O_2 concentration
 (2) Absorbs moisture
 (3) Absorbs CO_2
 (4) Produces ozone
- In curing cement plasters, water is sprinkled from time to time. This helps in :- [AIEEE-2003]
 (1) Hydrating sand and gravel mixed with cement
 (2) Converting sand into silicate
 (3) Developing interlocking needle like crystals of hydrated silicates
 (4) Keeping it cool
- The solubilities of carbonates decreases down the magnesium group due to decrease in- [AIEEE-2003]
 (1) Inter-ionic attraction
 (2) Entropy of solution formation
 (3) Lattice energy of solids
 (4) Hydration energy of cations
- The substance not likely to contain CaCO_3 is :- [AIEEE-2003]
 (1) Sea shells (2) Dolomite
 (3) A marble statue (4) Calcined gypsum
- One mole of magnesium nitride on reaction with excess of water gives :- [AIEEE-2004]
 (1) Two mole of HNO_3 (2) Two mole of NH_3
 (3) 1 mole of NH_3 (4) 1 mole of HNO_3
- The ionic mobility of alkali metal ions in aqueous solution is maximum for :- [AIEEE-2006]
 (1) Rb^+ (2) Li^+ (3) Na^+ (4) K^+
- Which of the following on thermal-decomposition yields a basic as well as an acidic oxide ? [AIEEE-2012]
 (1) NH_4NO_3 (2) NaNO_3 (3) KClO_3 (4) CaCO_3
- Fire extinguishers contain H_2SO_4 and which one of the following :- [AIEEE-2012 (Online)]
 (1) CaCO_3 (2) NaHCO_3 and Na_2CO_3
 (3) Na_2CO_3 (4) NaHCO_3
- Which one of the following will react most vigorously with water ? [AIEEE-2012 (Online)]
 (1) Li (2) K (3) Rb (4) Na
- A metal M on heating in nitrogen gas gives Y. Y on treatment with H_2O gives a colourless gas which when passed through CuSO_4 solution gives a blue colour, Y is :- [AIEEE-2012 (Online)]
 (1) NH_3 (2) MgO (3) Mg_3N_2 (4) $\text{Mg}(\text{NO}_3)_2$

PREVIOUS YEAR QUESTIONS											s-BLOCK		EXERCISE-05(A)					
Q.	1	2	3	4	5	6	7	8	9	10	11							
A.	4	1	3	4	4	2	1	4	3	3	3							

EXERCISE-05 (B)**PREVIOUS YEARS QUESTIONS**

1. Which process is used in the extractive metallurgy of Mg :
(A) Fused salt electrolysis (B) Self reduction
(C) Aqueous solution electrolysis (D) Thermite reduction
2. A sodium salt on treatment with MgCl_2 gives white precipitate only on heating. The anion of sodium salt is :
(A) HCO_3^- (B) CO_3^{2-} (C) NO_3^- (D) SO_4^{2-}
3. The following compounds have been arranged in order of their increasing thermal stabilities. Identify the correct order: (IIT 96)
(I) K_2CO_3 (II) MgCO_3 (III) CaCO_3 (IV) BeCO_3
(A) $\text{I} < \text{II} < \text{III} < \text{IV}$ (B) $\text{IV} < \text{II} < \text{III} < \text{I}$
(C) $\text{IV} < \text{II} < \text{I} < \text{III}$ (D) $\text{II} < \text{IV} < \text{III} < \text{I}$
4. Property of the alkaline earth metals that increases with their atomic number is – (IIT 97)
(A) Ionisation energy
(B) Solubility of their hydroxides
(C) Solubility of their sulphates
(D) Electronegativity
5. The characteristics of solid sodium chloride are (REE 96)
(1) Brittle (2) Ionic (3) Covalent (4) Non-conductor
(A) 1 & 2 (B) 3 & 4 (C) 1, 2, & 4 (D) 1, 3, & 4
6. Which of the following are not amphoteric – (REE 97)
(1) $\text{Be}(\text{OH})_2$ (2) $\text{Sr}(\text{OH})_2$ (3) $\text{Ca}(\text{OH})_2$ (4) $\text{Al}(\text{OH})_3$
(A) 1 & 3 (B) 2 & 3 (C) 1 & 4 (D) 2 & 4
7. Highly dilute solution of sodium in liquid ammonia :
(i) Shows blue colour (ii) Exhibits electrical conductivity
(iii) Produces sodium amide (iv) Produces hydrogen gas
(A) (i), (ii), (iii) (B) (i), (ii) (C) (iii), (iv) (D) Only (ii)
8. Which of the following hydrides is not ionic
(A) CaH_2 (B) BaH_2 (C) SrH_2 (D) BeH_2
9. The compound(s) formed upon combustion of sodium metal in excess air is (are) [JEE 2009]
(A) Na_2O_2 (B) Na_2O (C) NaO_2 (D) NaOH

ASSERTION & REASON QUESTIONS

- (A) Statement-I is true, statement-II is true ; statement-II is a correct explanation for statement-I
(B) Statement-I is true, statement-II is true ; statement-II is NOT a correct explanation for statement-I
(C) Statement-I is true, statement-II is false
(D) Statement-I is false, statement-II is true

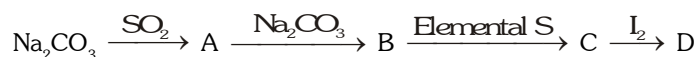
1. **Statement-I** : Alkali metals dissolve in liquid ammonia to give blue solutions. (IIT 2007)

Because :

Statement-II : Alkali metals in liquid ammonia give solvated species of the type $[\text{M}(\text{NH}_3)_n]^+$ (M = alkali metals)

SUBJECTIVE QUESTIONS

- 1.** Identify the following:



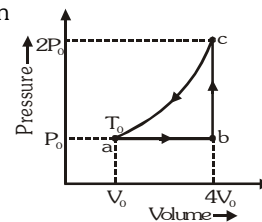
Also mention the oxidation state of S in all the compounds.

2. Beryllium chloride shows acidic nature in water or why BeCl_2 is easily hydrolysed?
3. The crystalline salts of alkaline earth metals contain more water of crystallisation than the corresponding alkali metal salts, why?
4. Arrange the following sulphates of alkaline earth metals in order of their decreasing thermal stability. BeSO_4 , MgSO_4 , CaSO_4 , SrSO_4 .
5. Why the solubility of calcium acetate decreases while that of lead nitrate increases with increase in temperature.
6. Why magnesium is not precipitated from a solution of its salt by NH_4OH In the presence of NH_4Cl .

PREVIOUS YEARS QUESTIONS	ANSWER KEY						EXERCISE -05(B)
• 1. (A) 2. (A) 3. (B) 4. B 5. C 6. B 7. B 8. D 9. A, B							
• <u>Assertion - Reason Questions</u>							
1. B							
• <u>Subjective Questions</u>							
1. A = NaHSO ₃ , Oxidation state of S = + 4 B = Na ₂ SO ₃ , Oxidation state of S = + 4 C = Na ₂ S ₂ O ₃ , Oxidation state of S = + 6 & - 2 D = Na ₂ S ₄ O ₆ , Oxidation state of S = + 5 & 0							
2. BeCl ₂ is a salt of weak base Be(OH) ₂ and strong acid HCl and thus undergoes hydrolysis to result in an acidic solution in water.							
$\text{BeCl}_2 + 4\text{H}_2\text{O} \xrightarrow{\text{Hydration}} [\text{Be}(\text{H}_2\text{O})_4]^{2+} + 2\text{Cl}^-$							
3. Alkaline earth metals have smaller size and more nuclear charge.							
4. SrSO ₄ > CaSO ₄ > MgSO ₄ > BeSO ₄							
5. (CH ₃ COO) ₂ Ca shows exothermic dissolution whereas Pb(NO ₃) ₂ show endothermic dissolution.							
6. The dissociation of NH ₄ OH (a weak electrolyte) is suppressed in presence of NH ₄ Cl due to common ion effect. Thus, [OH ⁻] in solution becomes low. The ionic product of concentrations of Mg ²⁺ and OH ⁻ ions does not exceed the solubility product of Mg(OH) ₂ and thus Mg(OH) ₂ is not precipitated.							

EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. The piece of zinc at a temperature of 20.0 C weighing 65.38 g is dropped into 180 g of boiling water ($T = 100\text{ C}$). The specific heat of zinc is $0.400\text{ J g}^{-1}\text{ C}^{-1}$ and that of water is $4.20\text{ J g}^{-1}\text{ C}^{-1}$. What is the final common temperature reached by both the zinc and water :-
(A) 97.3 C (B) 33.4 C (C) 80.1 C (D) 60.0 C
2. What is the change in internal energy ΔU , for a system that does 70 joules of work as it absorbs 45 joules of heat ?
(A) 115 J (B) 25 J (C) -25 J (D) -115 J
3. If the internal energy of an ideal gas decreases by the same amount as the work done by the system, the process is :-
(A) cyclic (B) isothermal (C) adiabatic (D) isolated
4. When 1 mol gas is heated at constant volume, temperature is raised from 298 to 308 K. Heat supplied to the gas is 500 J. Then which statement is correct :-
(A) $q = W = 500\text{ J}$, $\Delta U = 0$ (B) $q = \Delta U = 500\text{ J}$, $W = 0$
(C) $q = W = 500\text{ J}$, $\Delta U = 0$ (D) $\Delta U = 0$, $q = W = -500\text{ J}$,
5. Internal energy does not include
(A) vibrational energy (B) rotational energy
(C) nuclear energy (D) energy arising by gravitational pull
6. Which one of the following quantity is dependent on path ?
(A) molar internal energy (B) volume (C) w (D) $q + w$
7. For a monatomic gas, the value of the ratio of $C_{p,m}$ and $C_{v,m}$ is :-
(A) $\frac{5}{3}$ (B) $\frac{7}{5}$ (C) $\frac{9}{7}$ (D) $\frac{9}{11}$
8. How much heat, in joules, must be added to 0.250 mol of Ar(g) to raise its temperature from 20 C to 36.0 C at constant pressure ?
(A) 50.0 (B) 83.2 (C) 187 (D) 200
9. In thermodynamics, a process is called reversible when :-
(A) surroundings and system change into each other
(B) there is no boundary between system and surroundings
(C) the surrounding are always in equilibrium with the system
(D) the system changes into the surroundings spontaneously
10. One mole of an ideal monoatomic gas is caused to go through the cycle shown in figure. Then the change in the internal energy in expanding the gas from a to c along the path abc is :-
(A) $3P_0V_0$ (B) $6RT_0$
(C) $4.5RT_0$ (D) $10.5RT_0$
11. An adiabatic process is one in which there is no transfer of heat across the boundary between system and surroundings. For such a process.
(A) $P_{\text{ext}}\Delta V = 0$ (B) $q = w$ (C) $\Delta U = w$ (D) $\Delta U = 0$



12. Two moles of an ideal monoatomic gas are allowed to expand adiabatically and reversibly from 300 K and 200 K. The work done in the system is ($C_v = 12.5 \text{ J/K/mol}$)
- (A) -12.5 kJ (B) -2.5 kJ (C) -6.25 kJ (D) 500 kJ

13. One mole of an ideal monoatomic gas at temperature T and volume 1L expands to 2L against a constant external pressure of one atm under adiabatic conditions, then final temperature of gas will be :-

(A) $T + \frac{2}{3 \times 0.0821}$ (B) $T - \frac{2}{3 \times 0.0821}$ (C) $\frac{T}{2^{5/3-1}}$ (D) $\frac{T}{2^{5/3+1}}$

14. Enthalpy of the system is given as :-

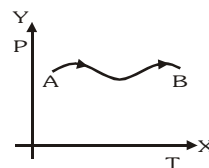
(A) $H + PV$ (B) $U + PV$ (C) $U - PV$ (D) $H - PV$

15. The P-T graph as given below was observed for a process of an ideal gas, which of the following statement is true :

(A) $w = +ve, \Delta H = +ve$ (B) $w = -ve, \Delta H = -ve$
 (C) $w = -ve, \Delta H = +ve$ (D) $w = +ve, \Delta H = -ve$

16. The difference between heats of reaction at constant pressure and constant volume for the reaction $2C_6H_6(l) + 15O_2(g) \rightarrow 12CO_2(g) + 6H_2O(l)$ at 25 in kJ mol^{-1} is :-

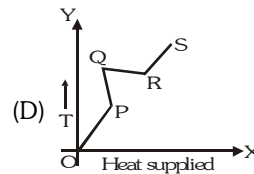
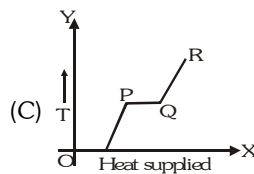
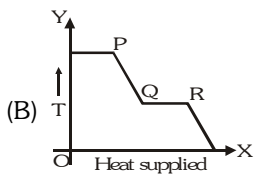
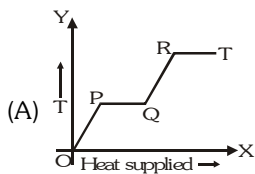
(A) -7.43 (B) +3.72
 (C) -3.72 (D) +7.43



17. The reaction $NH_2CN(s) + \frac{3}{2} O_2(g) \rightarrow N_2(g) + CO_2(g) + H_2O(l)$ was carried out at 300 K in a bomb calorimeter. The heat released was 743 kJ mol^{-1} . The value of ΔH_{300K} for this reaction would be :-

(A) $-740.5 \text{ kJ mol}^{-1}$ (B) $-741.75 \text{ kJ mol}^{-1}$ (C) $-743.0 \text{ kJ mol}^{-1}$ (D) $-744.25 \text{ kJ mol}^{-1}$

18. A block of ice at -10°C is slowly heated and converted into steam at 100°C . Which of the following curves represents the phenomenon qualitatively ?



19. How much energy must be supplied to change 36 g of ice at 0°C to water at room temperature 25°C

Data for water, H_2O $\Delta H_{\text{fusion}} = 6.01 \text{ kJ mol}^{-1}$ $C_{p,\text{liquid}} = 4.18 \text{ J.K}^{-1} \text{ g}^{-1}$

(A) 12 kJ (B) 16 kJ (C) 19 kJ (D) 22 kJ

20. Molar heat capacity of water in equilibrium with ice at constant pressure is :-

(A) zero (B) infinity
 (C) $40.45 \text{ kJ K}^{-1} \text{ mol}^{-1}$ (D) $75.48 \text{ kJ K}^{-1} \text{ mol}^{-1}$

21. A vessel contains 100 litres of a liquid x. Heat is supplied to the liquid in such a fashion that, Heat given = change in enthalpy. The volume of the liquid increases by 2 litres. If the external pressure is one atm, and 202.6 Joules of heat were supplied then, $[U \rightarrow \text{total internal energy}]$:-

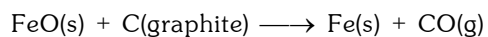
(A) $\Delta U = 0, \Delta H = 0$ (B) $\Delta U = +202.6 \text{ J}, \Delta H = +202.6 \text{ J}$
 (C) $\Delta U = -202.6 \text{ J}, \Delta H = -202.6 \text{ J}$ (D) $\Delta U = 0, \Delta H = +202.6 \text{ J}$

22. 1 mole of an ideal gas at 25 C is subjected to expand reversibly and adiabatically to ten times of its initial volume. Calculate the change in entropy during expansion (in $\text{J K}^{-1} \text{mol}^{-1}$) :-
 (A) 19.15 (B) -19.15 (C) 4.7 (D) Zero
23. One mole of an ideal diatomic gas ($C_v = 5 \text{ cal}$) was transformed from initial 25 C and 1 L to the state when temperature is 100 C and volume 10L. The entropy change of the process can be expressed as ($R = 2 \text{ calories/mol/K}$) :-
 (A) $3 \ln \frac{298}{373} + 2 \ln 10$ (B) $5 \ln \frac{373}{298} + 2 \ln 10$
 (C) $7 \ln \frac{373}{298} + 2 \ln \frac{1}{10}$ (D) $5 \ln \frac{373}{298} + 2 \ln \frac{1}{10}$
24. Calculate the total entropy change for the transition at 368 K of 1 mol of sulphur from the monoclinic to the rhombic solid state, is $\Delta H = -401.7 \text{ J mol}^{-1}$ for the transition. Assume the surrounding to be an ice-water both at 0 C :-
 (A) -1.09 JK^{-1} (B) 1.47 JK^{-1} (C) 0.38 JK^{-1} (D) None of these
25. One mole of an ideal monoatomic gas expands isothermally against constant external pressure of 1 atm from initial volume of 1L to a state where its final pressure becomes equal to external pressure. If initial temperature of gas is 300 K then total entropy change of system in the above process is :-
 ($R = 0.082 \text{ L atm mol}^{-1} \text{ K}^{-1} = 8.3 \text{ mol}^{-1} \text{ K}^{-1}$)
 (A) 0 (B) $R \ln(24.6)$ (C) $R \ln(2490)$ (D) $\frac{3}{2} R \ln(24.6)$
26. The enthalpy change for a given reaction at 298 K is $-x \text{ J mol}^{-1}$ (x being positive). If the reaction occurs spontaneously at 298 K, the entropy change at that temperature :
 (A) Can be negative but numerically larger than $\frac{x}{298}$
 (B) Can be negative but numerically smaller than $\frac{x}{298}$
 (C) Cannot be negative
 (D) Cannot be positive
27. For the gas - phase decomposition, $\text{PCl}_5(\text{g}) \xrightleftharpoons{\Delta} \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$:-
 (A) $\Delta H < 0, \Delta S < 0$ (B) $\Delta H > 0, \Delta S > 0$ (C) $\Delta H > 0, \Delta S < 0$ (D) $\Delta H < 0, \Delta S > 0$
28. A reaction has $\Delta H = -33 \text{ kJ}$ and $\Delta S = -58 \text{ J/K}$. This reaction would be :-
 (A) Spontaneous at all temperature
 (B) Non-spontaneous at all temperature
 (C) Spontaneous above a certain temperature only
 (D) Spontaneous below a certain temperature only
29. If $\Delta G = -177 \text{ K cal}$ for (1) $2\text{Fe}(\text{s}) + \frac{3}{2}\text{O}_2(\text{g}) \longrightarrow \text{Fe}_2\text{O}_3(\text{s})$
 and $\Delta G = -19 \text{ K cal}$ for (2) $4\text{Fe}_2\text{O}_3(\text{s}) + \text{Fe}(\text{s}) \longrightarrow 3\text{Fe}_3\text{O}_4(\text{s})$
 What is the Gibbs free energy of formation of Fe_3O_4 ?
 (A) $+229.6 \frac{\text{kcal}}{\text{mol}}$ (B) $-242.3 \frac{\text{kcal}}{\text{mol}}$ (C) $-727 \frac{\text{kcal}}{\text{mol}}$ (D) $-229.6 \frac{\text{kcal}}{\text{mol}}$

30. Given the following data :

Substance	ΔH_f (kJ/mol)	S (J/mol K)
FeO(s)	-266.3	57.49
C(Graphite)	0	5.74
Fe(s)	0	27.28
CO(g)	-110.5	197.6

Determine at what temperature the following reaction is spontaneous ?



- (A) 298 K
 (B) 668 K
 (C) 966 K
 (D) ΔG is +ve, hence the reaction will never be spontaneous
-

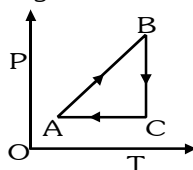
CHECK YOUR GRASP						ANSWER KEY				EXERCISE -1					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	A	C	C	B	D	C	A	B	C	D	C	B	B	B	C
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	A	B	A	B	B	D	D	B	C	B	B	B	D	B	C

EXERCISE-02

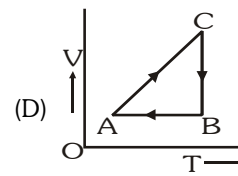
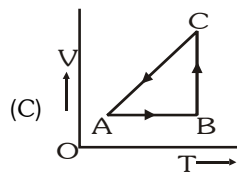
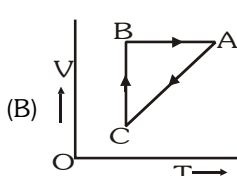
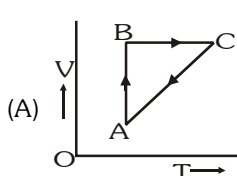
BRAIN TEASERS

SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)

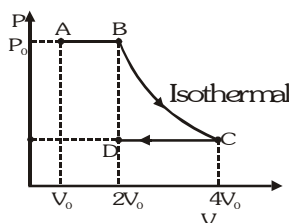
- Out of boiling point (I), entropy (II), pH (III) and e.m.f. of a cell (IV) Intensive properties are :
 (A) I, II (B) I, II, III (C) I, III, IV (D) all of these
- Which has maximum internal energy at 298 K ?
 (A) helium gas (B) oxygen gas (C) ozone gas (D) equal
- One mole of ideal gas is allowed to expand reversibly and adiabatically from a temperature of 27 C. If the work done by the gas in the process is 3kJ, the final temperature will be equal to ($C_v = 20 \text{ J/K mol}$)
 (A) 100 K (B) 450 K (C) 150K (D) 400 K
- Two moles of an ideal gas ($C_v = \frac{5}{2} R$) was compressed adiabatically against constant pressure of 2 atm. which was initially at 350 K and 1 atm pressure. The work involve in the process is equal to ?
 (A) 250 R (B) 300 R (C) 400 R (D) 500 R
- A cyclic process is shown in the P-T diagram.



Which of the curves show the same process on V-T diagram ?



- q, w, ΔE and ΔH for the following process ABCD on a monoatomic gas are :-

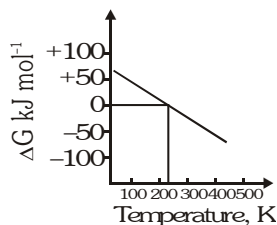


- (A) $w = -2P_0 V_0 \ln 2$, $q = 2P_0 V_0 \ln 2$, $\Delta E = 0$, $\Delta H = 0$
 (B) $w = -2P_0 V_0 \ln 2$, $q = 2P_0 V_0 \ln 2$, $\Delta E = 0$, $\Delta H = 2P_0 V_0 \ln 2$
 (C) $w = -P_0 V_0 (1 + \ln 2)$, $q = P_0 V_0 (1 + \ln 2)$, $\Delta E = 0$, $\Delta H = 0$
 (D) $w = -P_0 V_0 \ln 2$, $q = P_0 V_0 \ln 2$, $\Delta E = 0$, $\Delta H = 0$
- One mole of a real gas is subjected to heating at constant volume from (P_1, V_1, T_1) state to (P_2, V_1, T_2) state. Then it is subjected to irrversible adiabatic compression against constant external pressure P_3 atm till system reaches final state (P_3, V_2, T_3). If the constant volume molar heat capacity of real gas is C_v . Find out correct expression for ΔH from state 1 to state 3.
- (A) $C_v (T_3 - T_1) + (P_3 V_1 - P_1 V_1)$ (B) $C_v (T_2 - T_1) + (P_3 V_2 - P_1 V_1)$
 (C) $C_v (T_2 - T_1) + (P_3 V_1 - P_1 V_1)$ (D) $C_p (T_2 - T_1) + (P_3 V_1 - P_1 V_1)$

8. The maximum efficiency of a heat engine operating between 100 C and 25 C is ?
 (A) 20% (B) 22.2% (C) 25% (D) None
9. A heat engine operating between 227 C and 27 C absorbs 2 kcal of heat from the 227 C reservoir reversibly per cycle. The amount of work done in one cycle is ?
 (A) 0.4 kcal (B) 0.8 kcal (C) 4 kcal (D) 8 kcal
10. A system is changed from an initial state to a final state by a manner such that $\Delta H = q$. If the same change from the initial state to the final state were made by a different path, which of the following statements are correct.
 (A) ΔH remain the same
 (B) ΔH will depend upon the type of path
 (C) Heat exchanged q will be same if the path is isobaric
 (D) Heat exchanged q will be different if the path is not isobaric
11. For which of the following reactions. ΔS will the maximum :-



12. The entropy change when two moles of ideal monoatomic gas is heated from 200 to 300 C reversibly and isochorically ?
 (A) $\frac{3}{2} R \ln \left(\frac{300}{200} \right)$ (B) $\frac{5}{2} R \ln \left(\frac{573}{273} \right)$ (C) $3R \ln \left(\frac{573}{473} \right)$ (D) $\frac{3}{2} R \ln \left(\frac{573}{473} \right)$
13. When two equal sized pieces of the same metal at different temperatures T_h (hot piece) and T_c (cold piece) are brought into contact into thermal contact and isolated from it's surrounding. The total change in entropy of system is given by ? [C_v (J/K) = heat capacity of metal]
 (A) $C_v \ln \frac{T_c + T_h}{2T_c}$ (B) $C_v \ln \frac{T_2}{T_1}$ (C) $C_v \ln \frac{(T_c + T_h)^2}{2T_h \cdot T_c}$ (D) $C_v \ln \frac{(T_c + T_h)^2}{4T_h \cdot T_c}$
14. What can be concluded about the values of ΔH and ΔS from this graph ?



- (A) $\Delta H > 0, \Delta S > 0$ (B) $\Delta H > 0, \Delta S < 0$ (C) $\Delta H < 0, \Delta S > 0$ (D) $\Delta H < 0, \Delta S < 0$
15. For which of the following processes, total entropy of universe increases.
 (A) Melting one mole of ice to water at 0 C
 (B) Freezing one mole of water to ice at 0 C
 (C) Freezing one mole of water to ice at - 10 C
 (D) Melting one mole of ice at 10 C into water
16. 9.0 g ice at 0 C is mixed with 36 g of water at 50 C in a thermally insulated container. Using the following data, answer the question that follow ?
 $C_p (H_2O) = 4.18 J g^{-1} K^{-1}$; $\Delta H_{fusion} (ice) = 335 J g^{-1}$
 (i) final temperature of water is
 (A) 304.43 K (B) 296.97 K (C) 303.93 K (D) 287 K

- (ii) ΔS_{ice} is
 (A) 11.04 JK^{-1} (B) 3.16 JK^{-1} (C) 14.2 JK^{-1} (D) 7.84 JK^{-1}
- (iii) ΔS_{water} is
 (A) -12.64 JK^{-1} (B) -0.34 JK^{-1} (C) -5.42 JK^{-1} (D) 12.64 JK^{-1}
- (iv) What is the total entropy change in the process ?
 (A) -1.56 JK^{-1} (B) -1.60 JK^{-1} (C) 1.56 JK^{-1} (D) 1.60 JK^{-1}
17. 1 mole of an ideal gas at 25 C is subjected to expand reversibly ten times of its initial volume. The change in entropy due to expansion is :-
 (A) $19.15 \text{ JK}^{-1} \text{ mole}^{-1}$ (B) $16.15 \text{ JK}^{-1} \text{ mole}^{-1}$
 (C) $22.15 \text{ JK}^{-1} \text{ mole}^{-1}$ (D) none
18. Following reaction occurs at 25 C :
 $2\text{NO} (\text{g}, 1 \times 10^{-5} \text{ atm}) + \text{Cl}_2 (\text{g}, 1 \times 10^{-2} \text{ atm}) \rightleftharpoons 2\text{NOCl} (\text{g}, 1 \times 10^{-2} \text{ atm}).$
 ΔG is :-
 (A) -45.65 kJ (B) -28.53 kJ (C) -22.82 kJ (D) -57.06 kJ
19. For the reaction at 300 K
 $\text{A}(\text{g}) + \text{B}(\text{g}) \longrightarrow \text{C}(\text{g})$
 $\Delta E = -3.0 \text{ kcal}$; $\Delta S = -10.0 \text{ cal/K}$ value of ΔG is ?
 (A) -600 cal (B) -6600 cal (C) -6000 cal (D) none
20. What is the free energy change (ΔG) when 1.0 mole of water at 100 C and 1 atm pressure is converted into steam at 100 C and 1 atm pressure ?
 (A) 80 cal (B) 540 cal (C) 620 cal (D) zero
21. What is the free energy change (ΔG) when 1.0 mole of water at 100 C and 1 atm pressure is converted into steam at 100 C and 2 atm pressure ?
 (A) zero cal (B) 540 cal (C) 515.4 cal (D) none

BRAIN TEASERS						ANSWER KEY				EXERCISE -2					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	C	C	C	D	C	A	C	A	B	A,C	B	C	D	A	C,D
Que.	16 (i)	(ii)	(iii)	(iv)	17	18	19	20	21						
Ans.	B	C	A	C	A	A	A	D	C						

EXERCISE-03**MISCELLANEOUS TYPE QUESTIONS****TRUE / FALSE**

1. Pressure is an intensive property.
2. Like U and H, S is also a state function.
3. When a system undergoes a change at constant pressure, it is referred to an isothermal process.
4. The work done by a gas during free expansion is equal to zero.
5. First law of T.D. is applicable to all processes irrespective to whether they are reversible or irreversible.
6. All spontaneous process proceed in one direction only.
7. Positive value of ΔS_{system} during the process can be taken as sole criterion of spontaneity.
8. The ΔH of a reaction is independent of temperature.

FILL IN THE BLANKS

1. According to IUPAC conventions work done on the surroundings is
2. A system is said to be if it can neither exchange matter nor energy with surrounding.
3. The efficiency of a carnot engine can be increased by sink temperature when the source temperature is held constant.
4. Entropy change of a system is determine by the and states only, irrespective of how the system has changed its states.
5. Solidification of liquid shows in entropy.
6. When Fe(s) is dissolved in a aqueous HCl in a closed rigid vessel, the work done is
7. For Non-spontaneous process at constant T & P ΔG is

MATCH THE COLUMN

1.	Column-I (Process of reaction)	Column-II (Positive, negative)
(A)	For the process $\text{H}_2\text{O}_{(l)} \rightleftharpoons \text{H}_2\text{O}_{(s)}$, ΔH & ΔS are	(p) -ve, +ve
(B)	For the endothermic reaction $2\text{A}_{(s)} + \frac{1}{2}\text{O}_{2(g)} \rightleftharpoons \text{A}_2\text{O}_{(s)}$ at 298 K, ΔS & ΔG are	(q) +ve, -ve
(C)	$\text{C(diamond)} \rightleftharpoons \text{C(graphite)}$, favourable conditions for formation of diamond are high pressure and high temperature then ΔH_f of diamond and ΔS_f of graphite from diamond are	(r) +ve, +ve
(D)	For the given reaction $\text{N}_2\text{O}_{4(g)} \rightleftharpoons 2\text{NO}_{2(g)}$, $E_{a(\text{forward})} = 57.2 \text{ kJ}$ and $E_{a(\text{backward})} = 3.2 \text{ kJ}$, ΔH & ΔS for the given reaction	(s) -ve, -ve

2.	Column-I (Process)	Column-II (Entropy change)
(A)	Reversible isothermal ideal gas expansion	(p) $\Delta S_{\text{surrounding}} = 0$
(B)	Reversible adiabatic ideal gas compression	(q) $\Delta S_{\text{surrounding}} < 0$
(C)	Adiabatic free expansion ($P_{\text{ext}} = 0$) of an ideal gas	(r) $\Delta S_{\text{surrounding}} > 0$
(D)	Irreversible isothermal ideal gas compression	(s) $\Delta S_{\text{system}} = 0$

ASSERTION & REASON

These questions contains, Statement I (assertion) and Statement II (reason).

- (A) Statement-I is true, Statement-II is true ; Statement-II is correct explanation for Statement-I.
 (B) Statement-I is true, Statement-II is true ; Statement-II is NOT a correct explanation for Statement-I
 (C) Statement-I is true, Statement-II is false.
 (D) Statement-I is false, Statement-II is true.

- Statement-I** : The heat absorbed during the isothermal expansion of an ideal gas against vacuum is zero.

Because

Statement-II : The volume occupied by the molecules of an ideal gas is zero.
- Statement-I** : The magnitude of the work involved in an isothermal expansion is greater than that involved in an adiabatic expansion.

Because

Statement-II : P-V curve (P on y-axis and V on x-axis) decrease more rapidly for reversible adiabatic expansion compared to reversible isothermal expansion starting from same initial state.
- Statement-I** : Entropy change in reversible adiabatic expansion of an ideal gas is zero.

Because

Statement-II : The increase in entropy due to volume increase just compensate the decrease in entropy due to fall in temperature.
- Statement-I** : The standard free energy changes of all spontaneously occurring reactions are negative.

Because

Statement-II : The standard free energies of the elements in their standard states at 1 bar and 298 K are taken as zero.
- Statement-I** : Enthalpy and entropy of any elementary substance in the standard states are taken as zero.

Because

Statement-II : At absolute zero, particles of the perfectly crystalline substance become completely ordered.
- Statement-I** : A reaction which is spontaneous and accompanied by decrease of randomness must be exothermic.

Because

Statement-II : All exothermic reactions are accompanied by decrease of randomness.
- Statement-I** : Decrease of free energy during the process under constant temperature and pressure provides a measure of its spontaneity.

Because

Statement-II : A spontaneous change must have +ve sign of ΔS_{system} .

COMPREHENSION BASED QUESTIONS

Comprehension # 1

An ideal gas, having ratio of specific heat γ undergoes a process in which its internal energy relates to the volume as $U = \alpha\sqrt{V}$, where α is a constant. If the gas is expanded from volume V_1 to V_2 .

1. The work performed by gas is :

(A) $2\alpha(\gamma - 1)[\sqrt{V_2} - \sqrt{V_1}]$

(B) $\alpha(\gamma - 1)[\sqrt{V_2} - \sqrt{V_1}]$

(C) $2\alpha(\gamma - 1)[V_2 - V_1]$

(D) $\alpha(\gamma - 1)[V_2 - V_1]$

2. If the ideal gas is diatomic and its increase in internal energy is 100 J then the work performed by gas is : (Ignore vibrational degree of freedom)

(A) 80 J

(B) 180 J

(C) 100 J

(D) 20 J

3. In the above question, the heat supplied to gas is :

(A) 80 J

(B) 180 J

(C) 100 J

(D) 20 J

Comprehension # 2

Standard Gibb's energy of reaction ($\Delta_r G$) at a certain temperature can be computed as $\Delta_r G = \Delta_r H - T \Delta_r S$ and the change in the value of $\Delta_r H$ and $\Delta_r S$ for a reaction with temperature can be computed as follows :

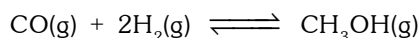
$$\Delta_r H_{T_2}^\circ - \Delta_r H_{T_1}^\circ = \Delta_r C_p^\circ (T_2 - T_1)$$

$$\Delta_r S_{T_2}^\circ - \Delta_r S_{T_1}^\circ = \Delta_r C_p^\circ \ln\left(\frac{T_2}{T_1}\right)$$

$$\Delta_r G = \Delta_r H - T \Delta_r S$$

and by $\Delta_r G = -RT \ln K_{eq}$

Consider the following reaction :



Given : $\Delta_f H$ ($\text{CH}_3\text{OH, g}$) = -201 kJ/mol ; $\Delta_f H$ (CO, g) = - 114 kJ/mol

S ($\text{CH}_3\text{OH, g}$) = 240 J/K-mol ;

S (H_2, g) = 29 JK⁻¹ mol⁻¹

S (CO, g) = 198 J/mol-K ;

$C_{p, m}$ (H_2) = 28.8 J/mol-K

$C_{p, m}$ (CO) = 29.4 J/mol-K ;

$C_{p, m}$ (CH_3OH) = 44 J/mol-K

and $\ln\left(\frac{320}{300}\right) = 0.06$, all data at 300 K

1. $\Delta_r S$ at 300 K for the reaction is :

(A) 152.6 J/K-mol

(B) 181.6 J/K-mol

(C) -16 J/K-mol

(D) none of these

2. $\Delta_r H$ at 300 K for the reaction is :

(A) -87 kJ/mol

(B) 87 kJ/mol

(C) -315 kJ/mol

(D) -288 kJ/mol

3. $\Delta_r S$ at 320 K is :

(A) 155.18 J/mol-K

(B) 150.02 J/mol-K

(C) 172 J/mol-K

(D) none of these

4. $\Delta_f H$ at 320 K is :
 (A) -288.86 kJ/mol (B) -289.1 kJ/mol
 (C) -87.86 kJ/mol (D) none of these
5. $\Delta_f G$ at 320 K is :
 (A) -48295.2 kJ/mol (B) -240.85 kJ/mol
 (C) 240.85 kJ/mol (D) -81.91 kJ/mol
-

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
<u>True / False</u> 1. T 2. T 3. F 4. T 5. T 6. T 7. F 8. F <u>Fill in the Blanks</u> 1. Negative 2. isolated 3. decreasing 4. (i) initial (ii) final 5. decrease 6. zero 7. positive <u>Match the Column</u> 1. (A) $\rightarrow$ s ; (B) $\rightarrow$ p ; (C) $\rightarrow$ r ; (D) $\rightarrow$ r 2. (A) $\rightarrow$ q ; (B) $\rightarrow$ p,s ; (C) $\rightarrow$ p ; (D) $\rightarrow$ r <u>Assertion - Reason Questions</u> 1. B 2. A 3. A 4. B 5. D 6. C 7. C <u>Comprehension Based Questions</u> - Comprehension #1 : 1. (A) 2. (A) 3. (B) - Comprehension #2 : 1. (C) 2. (A) 3. (D) 4. (C) 5. (D)		

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

1. In which of the following changes at constant pressure is work done by system on surrounding ? By the surrounding on system ?

Initial state		Final state
(i) $\text{H}_2\text{O} (\text{g})$	$\longrightarrow$	$\text{H}_2\text{O}(\ell)$
(ii) $\text{H}_2\text{O} (\text{s})$	$\longrightarrow$	$\text{H}_2\text{O}(\text{g})$
(iii) $\text{H}_2\text{O} (\ell)$	$\longrightarrow$	$\text{H}_2\text{O}(\text{s})$
(iv) $3\text{H}_2 (\text{g}) + \text{N}_2(\text{g})$	$\longrightarrow$	$2\text{NH}_3(\text{g})$
(v) $\text{CaCO}_3 (\text{s})$	$\longrightarrow$	$\text{CaO} (\text{s}) + \text{CO}_2(\text{g})$

2. The gas is cooled and loses 65 J of heat. The gas contracts as it cools and work done on the system equal to 20 J which is exchanged with the surroundings. What are q, w and ΔE ?
3. The enthalpy of combustion of glucose is $-2808 \text{ kJ mol}^{-1}$ at 25 C. How many grams of glucose do you need to consume [Assume wt = 62.5 Kg].
- (a) To climb a flight of stairs rising through 3m.
- (b) To climb a mountain of altitude 3000 m ?

Assume that 25% of enthalpy can be converted to useful work.

4. What is ΔE when 2.0 mole of liquid water vaporises at 100 C ? The heat of vaporisation, ($\Delta H_{\text{vap.}}$) of water at 100 C is $40.66 \text{ kJ mol}^{-1}$.
5. If 1.0 kcal of heat is added to 1.2 L of O_2 in a cylinder of constant pressure of 1 atm, the volume increases to 1.5 L. Calculate ΔE and ΔH of the process.
6. When the following reaction was carried out in a bomb calorimeter, ΔE is found to be -742.7 kJ/mol of $\text{NH}_2\text{CN} (\text{s})$ at 298 K.
- $$\text{NH}_2\text{CN}(\text{s}) + \frac{3}{2} \text{O}_2(\text{g}) \longrightarrow \text{N}_2(\text{g}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\ell)$$
- Calculate ΔH_{298} for the reaction.
7. When 1 mole of ice melt at 0 C and at constant pressure of 1 atm. 1440 calories of heat are absorbed by the system. The molar volumes of ice and water are 0.0196 and 0.0180 litre respectively. Calculate ΔH and ΔE for the reaction.
8. Water expands when it freezes. Determine amount of work in joules, done when a system consisting of 1.0 L of liquid water freezes under a constant pressure of 1.0 atm and forms 1.1 L of ice.
9. One mole of solid Zn is placed in excess of dilute H_2SO_4 at 27 C in a cylinder fitted with a piston. Find the value of ΔE , q and w for the process if the area of piston is 500 cm^2 and it moves out by 50 cm against a pressure of 1 atm during the reaction. The heat given to surrounding is 36.5 kJ.



10. Five moles of an ideal gas at 300K, expanded isothermally from an initial pressure of 4 atm to a final pressure of 1 atm against a cont. ext. pressure of 1 atm. Calculate q, w, ΔU & ΔH . Calculate the corresponding value of all if the above process is carried out reversibly.
11. 1 mole of CO_2 gas at 300 K is expanded under reversible adiabatic condition such that its volume becomes 27 times.

(a) What is final temperature

(b) What is work done.

Given $\gamma = 1.33$ and $C_V = 25.08 \text{ J mol}^{-1} \text{ K}^{-1}$ for CO_2 .

12. A sample of a fluorocarbon was allowed to expand reversibly and adiabatically to twice its volume. In the expansion the temperature dropped from 298.15 to 248.44 K. Assume the gas behaves perfectly. Estimate the value of $C_{V, m}$.

13. Find the work done when one mole of the gas is expanded reversibly and isothermally from 5 atm to 1 atm at 25 C ?

14. One mole of an ideal monoatomic gas is carried through the cycle of the given figure consisting of step A, B and C and involving state 1, 2 and 3. Fill in the blank space in the table given below assuming reversible steps.

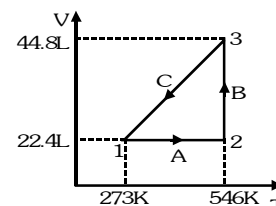


Table-1			
State	P	V	T
1			
2			
3			

Step	Name of process	q	w	ΔE	ΔH
A					
B					
C					
overall					

15. The standard enthalpy of formation of water liquid is -285.76 kJ at 298 K. Calculate the value at 373K. The molar heat capacities at constant pressure (C_p) in the given temperature range of $\text{H}_2(\text{g})$, $\text{O}_2(\text{g})$ and $\text{H}_2\text{O}(\text{l})$ are respectively 38.83, 29.16 and $75.312 \text{ JK}^{-1}\text{mol}^{-1}$.

16. Methane (Considered to be an ideal gas) initially at 25 C and 1 bar pressure is heated at constant pressure until the volume has doubled. The variation of the molar heat capacity with absolute temperature is given by :

$C_p = 22.34 + 48.1 \times 10^{-3} T$, where C_p is in $\text{JK}^{-1} \text{ mol}^{-1}$. Calculate molar (a) ΔH (b) ΔU .

17. One mole of $\text{NaCl}(\text{s})$ on melting absorbed 30.5 kJ of heat and its entropy is increased by 28.8 JK^{-1} . What is the melting point of sodium chloride ?

18. Oxygen is heated from 300 K to 600 K at a constant pressure of 1 bar. What is the increases in molar entropy? The molar heat capacity in $\text{JK}^{-1} \text{ mol}^{-1}$ for the O_2 is.

$$C_p = 25.5 + 13.6 \times 10^{-3} T - 42.5 \times 10^{-7} T^2$$

19. Calculate the free energy change at 298 K for the reaction :

$\text{Br}_2(\text{l}) + \text{Cl}_2(\text{g}) \longrightarrow 2\text{BrCl}(\text{g})$. For the reaction $\Delta H = 29.3 \text{ kJ}$ & the entropies of $\text{Br}_2(\text{l})$, $\text{Cl}_2(\text{g})$ & $\text{BrCl}(\text{g})$ at the 298 K are 152.3, 223.0, $239.7 \text{ J mol}^{-1} \text{ K}^{-1}$ respectively.

20. Using the data given below, establish that the vaporization of $\text{CCl}_4(l)$ at 298K to produce $\text{CCl}_4(g)$ at 1 atm pressure does not occur spontaneously.

Given : $\text{CCl}_4(l, 1 \text{ atm}) \longrightarrow \text{CCl}_4(g, 1 \text{ atm})$; $\Delta S = 94.98 \text{ JK}^{-1} \text{ mol}^{-1}$

$$\Delta H_f(\text{CCl}_4, g) = -106.7 \text{ kJ mol}^{-1} \text{ \& } \Delta H_f(\text{CCl}_4, l) = -139.3 \text{ kJ mol}^{-1}$$

21. From the given table answer the following questions :

Reaction : $\text{H}_2\text{O}(g) + \text{CO}(g) \rightleftharpoons \text{H}_2(g) + \text{CO}_2(g)$

- (i) Calculate $\Delta_r H_{298}$ (ii) Calculate $\Delta_r G_{298}$
 (iii) Calculate $\Delta_r S_{298}$ (iv) Calculate $\Delta_r E_{298}$
 (v) Calculate $S_{298}[\text{H}_2\text{O}(g)]$

	$\text{CO}(g)$	$\text{CO}_2(g)$	$\text{H}_2\text{O}(g)$	$\text{H}_2(g)$
$\Delta H_{298} \text{ (-kJ/mol)}$	-26.42	-94.05	-57.8	0
$\Delta G_{298} \text{ (-kJ/mol)}$	-32.79	-94.24	-54.64	0
$S_{298} \text{ (-Cal/kmole)}$	47.3	51.1	?	31.2

CONCEPTUAL SUBJECTIVE EXERCISE

ANSWER KEY

EXERCISE-4(A)

1. (i) + w, (ii) - w, (iii) + w, (iv) + w, (v) - w

2. q = -65J, w = 20 J, ΔE = -45 J

3. (A) 0.47 g , (B) 0.47 kg

4. ΔE = 75.11 kJ

5. ΔE = 0.993 kcal, ΔH = 1 kcal

6. -741.5 kJ

7. ΔH ≅ ΔE = 1440 calories

8. -10 J

9. ΔE = -39.03 kJ mol⁻¹, q = -36.5 kJ, w = -2.53 J

10. w_{irr} = -9356.25 J, w_{rev} = -17291.6 J, ΔU = ΔH = 0

11. T₂ = 100 k, w = -5.016 kJ

12. C_{v,m} = 31.6 Jk⁻¹ mol⁻¹

13. w = -3.988 kJ

14.

Table-1			
State	P	V	T
1	1 atm	22.4	273
2	2 atm	22.4	546
3	1 atm	44.8	546

Step	Name of process	q	w	ΔE	ΔH
A	Isochoric	3/2 R(273)	0	3/2 R(273)	5/2 R (273)
B	Isothermal	546 R ln 2	-546 R ln 2	0	0
C	Isobaric	-5/2 R (273)	R (273)	-3/2 R(273)	-5/2 R (273)
overall	cyclic	546Rln2-273R	273R-546Rln2	0	0

15. ΔH₃₇₃^o (H₂O, (ℓ)) = -284.11 kJ

16. (a) 13.064 kJ mol⁻¹ (b) 10.587 kJ mol⁻¹

17. T = 1059 k

18. 21.18 Jk⁻¹ mol⁻¹

19. -1721.8 J

20. ΔG = 4.3 kJ mol⁻¹ > 0

21. (i) -9,83 kcal mol⁻¹ (ii) -6.81 kcal mol⁻¹

(iii) -10.13 cal/K mol (iv) -9.83 kcal/mol

(v) +45.13 cal/K mole

EXERCISE-04 [B]

BRAIN STORMING SUBJECTIVE EXERCISE

- The increase in entropy of 1 kg of ice at 200 K which is heated to 400 K (super heated steam) at constant atmospheric pressure, will be :
Given that $C_p(\text{ice}) = 2.09 \times 10^3 \text{ J/kg degree}$; $C_p(\text{water}) = 4.18 \times 10^3 \text{ J/kg degree}$
 $C_p(\text{steam}) = 2.09 \times 10^3 \text{ J/kg degree}$; $L_f(\text{C}, 273 \text{ K}) = 3.34 \times 10^5 \text{ J/kg}$;
 $L_v(\text{water}, 273 \text{ K}) = 22.6 \times 10^5 \text{ J/kg}$
- Calculate workdone in adiabatic compression of one mole of an ideal gas (monoatomic) from an initial pressure of 1 atm to final pressure of 2 atm. Initial temperature = 300 K
 - If process is carried out reversibly.
 - If process is carried out irreversible against 2 atm external pressure.Compute the final volume reached by gas in two cases.
- 20.0 dm³ of an ideal gas (diatomic $C_{v,m} = 5 R/2$) at 673 K and 0.7 MPa expands until pressure of the gas is 0.2 MPa. Calculate q, w, ΔU and ΔH for the process if the expansion is :
 - Isothermal and reversible
 - Adiabatic and reversible
 - Isothermal and adiabatic
 - Against 0.2 MPa and adiabatic
 - Against 0.2 MPa and isothermal.
- One mole of an ideal gas is expanded isothermally at 298 K until its volume is tripled. Find the values of ΔS_{gas} and ΔS_{total} under the following conditions.
 - Expansion is carried out reversibly.
 - Expansion is carried out irreversibly where 836.6 J of heat is less absorbed than in (i)
 - Expansion is free.
- 10 g of neon initially at a pressure of 506.625 kPa and temperature of 473 K expand adiabatically to a pressure of 202.65 kPa. Calculate entropy change of the system and total entropy change for the following ways of carrying out this expansion.
 - Expansion is carried out reversibly.
 - Expansion occurs against a constant external pressure of 202.65 kPa.
 - Expansion is free expansion.
- Pressure over 1000 mL of a liquid is gradually increases from 1 bar to 1001 bar under adiabatic conditions. If the final volume of the liquid is 990 mL, calculate ΔU and ΔH of the process, assuming linear variation of volume with pressure.
- One mole of ideal monoatomic gas was taken through reversible isochoric heating from 100 K to 1000 K. Calculate ΔS_{system} , ΔS_{surr} , and ΔS_{total} in
 - when the process carried out reversibly
 - when the process carried out irreversibly (one step)
- Fixed amount of an ideal gas contained in a sealed rigid vessel ($V = 24.6$ litre) at 1.0 bar is heated reversibly from 27 C to 127 C. Determine change in Gibb's energy (in Joule) if entropy of gas $S = 10 + 10^{-2} T$ (J/K).
- A 32 g sample of CH_4 gas initially at 101.325 kPa and 300 K is heated to 550 K. $C_{p,m}/\text{JK}^{-1} \text{ mol}^{-1} = 12.552 + 8.368 \times 10^{-2} T/\text{K}$. Assuming CH_4 behaves ideally, compute w, q, ΔU and ΔH for (a) an isobaric reversible process and (b) an isochoric reversible process.
- At 298 K, $\Delta H_{\text{combustion}}(\text{sucrose}) = -5737 \text{ kJ/mol}$ & $\Delta G_{\text{combustion}}(\text{sucrose}) = -6333 \text{ kJ/mol}$. Estimate additional non-PV work that is obtained by raising temperature to 310 K . Assume $\Delta_f C_p = 0$ for this temperature change.

11. Compute $\Delta_r G$ for the reaction $\text{H}_2\text{O}(\ell, 1 \text{ atm}, 323 \text{ K}) \rightarrow \text{H}_2\text{O}(\text{g}, 1 \text{ atm}, 323 \text{ K})$
 Given that : $\Delta_{\text{vap}} H$ at $373 \text{ K} = 40.639 \text{ kJ mol}^{-1}$, $C_p(\text{H}_2\text{O}, \ell) = 75.312 \text{ J K}^{-1} \text{ mol}^{-1}$,
 $C_p(\text{H}_2\text{O}, \text{g}) = 33.305 \text{ J K}^{-1} \text{ mol}^{-1}$.
12. (a) An ideal gas undergoes a single state expansion against a constant opposing pressure from (P_1, V_1, T) to (P_2, V_2, T) . What is the largest mass m which can be lifted through a height h in this expansion ?
 (b) The system (a) restored to its initial state by a single state compression. What is the smallest mass m' which must fall through the height h to restore system?
 (c) What is the net mass lowered through height h in the cyclic transformation in (a) and (b) ?

BRAIN STORMING SUBJECTIVE EXERCISE	ANSWER KEY	EXERCISE-4(B)
1.	9390 J kg^{-1}	
2.	(a) $T_2 = 395.8$; $V_2 = 16.24 \text{ L}$; $w_{\text{rev}} = 1194.72 \text{ J}$, (b) $V_2^1 = 17.24 \text{ L}$; $T_2^1 = 420 \text{ K}$; $w_{\text{irrev}} = 1496.52 \text{ J}$	
3.	(i) $q = -w = 17.54 \text{ kJ}$, $\Delta U = 0$ and $\Delta H = 0$; (ii) $q = 0$, $w = \Delta U = -10.536 \text{ kJ}$ and $\Delta H = -14.75 \text{ kJ}$ (iii) $q = 0$, $w = 0$, $\Delta U = 0$ and $\Delta H = 0$; (iv) $q = 0$; $\Delta U = w = -7.14 \text{ kJ}$; $\Delta H = -9.996 \text{ kJ}$ (v) $q = -w = 9.94 \text{ kJ}$, $\Delta U = \Delta H = 0$	
4.	$\Delta S_{\text{gas}} = -\Delta S_{\text{surr}}$ and $\Delta S_{\text{total}} = 0$, (ii) $\Delta S_{\text{total}} = 2.808 \text{ J K}^{-1}$, (iii) $\Delta S_{\text{total}} = \Delta S_{\text{sys}} = 9.134 \text{ J K}^{-1}$	
5.	(i) $\Delta S_{\text{sys}} = 0$; $\Delta S_{\text{surr}} = 0$ and $\Delta S_{\text{total}} = 0$, (ii) $\Delta S_{\text{surr}} = 0$; $\Delta S_{\text{total}} = \Delta S_{\text{sys}} = 0.957 \text{ JK}^{-1}$ (iii) $\Delta S_{\text{sys}} = \Delta S_{\text{total}} = 3.81 \text{ JK}^{-1}$	
6.	$\Delta U = 501 \text{ J}$; $\Delta H = 99.5 \text{ kJ}$	
7.	(i) Rev. process $\Delta S_{\text{sys}} = \frac{3}{2} R \ln 10$; $\Delta S_{\text{surr}} = -\frac{3}{2} R \ln 10$, (ii) Irr. process $\Delta S_{\text{sys}} = \frac{3}{2} R \ln 10$; $\Delta S_{\text{surr}} = -\frac{3}{2} (0.9)$; $\Delta S_{\text{total}} = \frac{3}{2} R (1.403)$	
8.	-530 J	
9.	(a) $q_p = \Delta H = 24.058 \text{ kJ}$, $w = -4.157 \text{ kJ}$, $\Delta U = 19.90 \text{ kJ}$; (b) $\Delta U = 19.90 \text{ kJ}$, $\Delta H = 24.058$; $w = 0$	
10.	24 kJ/mol	11. $\Delta_r G = 5.59 \text{ kJ mol}^{-1}$
12.	(a) $m = m = \frac{nRT}{gh} \left(1 - \frac{P_2}{P_1} \right)$, (b) $m' = \frac{nRT}{gh} \left(\frac{P_1}{P_2} - 1 \right)$, (c) $m' - m = \frac{nRT}{gh} \left(\frac{(P_1 - P_2)^2}{P_1 P_2} \right)$	

EXERCISE-05(A)**PREVIOUS YEARS QUESTIONS**

1. In a irreversible process taking place at constant T and P and in which only pressure-volume work is being done the change in Gibbs free energy (ΔG) and change in entropy (ΔS) satisfy the criteria :-
[AIEEE-2003]
(1) $(\Delta S) = 0, (\Delta G) = 0$
(2) $(\Delta S) = 0, (\Delta G) > 0$
(3) $(\Delta S) < 0, (\Delta G) < 0$
(4) $(\Delta S) > 0, (\Delta G) < 0$
2. The internal energy change when a system goes from state A to B is 40 kJ/mole. If the system goes from A to B by a reversible path and returns to state A by an irreversible path what would be the net change in internal energy ?
[AIEEE-2003]
(1) < 40 kJ
(2) Zero
(3) 40 kJ
(4) > 40 kJ
3. The enthalpy change for a reaction does not depend upon :-
[AIEEE-2003]
(1) The nature of intermediate reaction steps
(2) The differences in initial or final temperatures of involved substances
(3) The physical states of reactants and products
(4) Use of different reactants for the same product
4. The correct relationship between free energy change in a reaction and the corresponding equilibrium constant K_c is :-
[AIEEE-2003]
(1) $\Delta G = RT \ln K_c$
(2) $-\Delta G = RT \ln K_c$
(3) $\Delta G = RT \ln K_c$
(4) $-\Delta G = RT \ln K_c$
5. In conversion of lime-stone to lime,
 $\text{CaCO}_3(\text{s}) \longrightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$
the values of ΔH and ΔS are $+179.1 \text{ kJ mol}^{-1}$ and 160.2 J/K respectively at 298 K and 1 bar. Assuming that ΔH and ΔS do not change with temperature, temperature above which conversion of limestone to lime will be spontaneous is :-
[AIEEE-2007]
(1) 1008 K
(2) 1200 K
(3) 845 K
(4) 1118 K
6. Identify the correct statement regarding a spontaneous process :-
[AIEEE-2007]
(1) For a spontaneous process in an isolated system, the change in entropy is positive
(2) Endothermic processes are never spontaneous
(3) Exothermic processes are always spontaneous
(4) Lowering of energy in the reaction process is the only criterion for spontaneity
7. Standard entropy of X_2 , Y_2 and XY_3 are 60, 40 and $50 \text{ JK}^{-1} \text{ mol}^{-1}$, respectively. For the reaction,
 $\frac{1}{2} X_2 + \frac{3}{2} Y_2 \rightarrow XY_3$, $\Delta H = -30 \text{ kJ}$, to be at equilibrium, the temperature will be
[AIEEE-2008]
(1) 1250 K
(2) 500 K
(3) 750 K
(4) 1000 K

8. The entropy change involved in the isothermal reversible expansion of 2 moles of an ideal gas from a volume of 10 dm³ to a volume of 100 dm³ at 27 C is :- [AIEEE-2011]
- (1) 32.3 J mol⁻¹ K⁻¹ (2) 42.3 J mol⁻¹ K⁻¹
 (3) 38.3 J mol⁻¹ K⁻¹ (4) 35.8 J mol⁻¹ K⁻¹
9. The incorrect expression among the following is :- [AIEEE-2012]
- (1) $K = e^{-\Delta G / RT}$
 (2) $\frac{\Delta G_{\text{system}}}{\Delta S_{\text{total}}} = - T$
 (3) In isothermal process, $W_{\text{reversible}} = - nRT \ln \frac{V_f}{V_i}$
 (4) $\ln K = \frac{\Delta H^\circ - T\Delta S^\circ}{RT}$
10. One mole of an ideal gas is expanded isothermally and reversibly to half its initial pressure. ΔS for the process in J K⁻¹ mol⁻¹ is [$\ln 2 = 0.693$ and $R = 8.314$, J/(mol/K)] : [AIEEE-2012 (Online)]
- (1) 10.76 (2) 6.76 (3) 8.03 (4) 5.76
11. The entropy of a sample of a certain substance increases by 0.836 JK⁻¹ on adding reversibly 0.3344 J of heat at constant temperature. The temperature of the sample is :- [AIEEE-2012 (Online)]
- (1) 0.4 K (2) 2.5 K (3) 0.016 K (4) 0.3 K
12. A piston filled with 0.04 mol of an ideal gas expands reversibly from 50.0 mL to 375 mL at a constant temperature of 37.0 C. As it does so, it absorbs 208J of heat. The values of q and w for the process will be :- [JEE(Main)-2013]
- ($R = 8.314$ J/mol K) ($\ln 7.5 = 2.01$)
- (1) q = + 208 J, w = - 208 J (2) q = - 208 J, w = - 208 J
 (3) q = - 208 J, w = + 208 J (4) q = + 208 J, w = + 208 J

PREVIOUS YEAR QUESTIONS ANSWER KEY EXERCISE-5(A)													
Que.	1	2	3	4	5	6	7	8	9	10	11	12	
Ans	4	2	1	2	4	1	3	3	4	4	2	1	

EXERCISE-05(B)**PREVIOUS YEARS QUESTIONS**

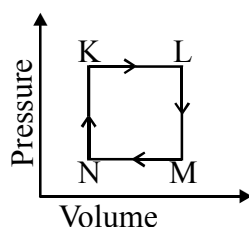
1. A process $A \longrightarrow B$ is difficult to occur directly instead it takes place in three successive steps :-
- $\begin{array}{c} C \longrightarrow D \\ \uparrow \qquad \downarrow \\ A \qquad B \end{array}$

$\Delta S(A \longrightarrow C) = 50 \text{ e.u.}$
 $\Delta S(C \longrightarrow D) = 30 \text{ e.u.}$
 $\Delta S(B \longrightarrow D) = 20 \text{ e.u.}$
- where e.u. is entropy unit.
- Then the entropy change ΔS for the process $(A \longrightarrow B)$ is [IIT-JEE-2006]
- (A) +100 e.u. (B) -60 e.u. (C) -100 e.u. (D) +60 e.u.
2. The molar heat capacity of a monoatomic gas for which the ratio of pressure and volume is one :- [IIT-JEE-2006]
- (A) $4/2 R$ (B) $3/2 R$ (C) $5/2 R$ (D) zero
3. One mole of monoatomic ideal gas expands adiabatically at initial temp. T against a constant external pressure of 1 atm from one litre to two litre. Find out the final temp. ($R = 0.0821 \text{ litre. atm K}^{-1} \text{ mol}^{-1}$) [IIT-JEE-2005]
- (A) T (B) $\frac{T}{(2)^{\frac{5}{3}-1}}$ (C) $T - \frac{2}{3 \times 0.0821}$ (D) $T + \frac{2}{3 \times 0.0821}$
4. Two mole of an ideal gas is expanded isothermally and reversibly from 1 litre to 10 litre at 300 K. The enthalpy change (in kJ) for the process is :- [IIT-JEE-2004]
- (A) 11.4 kJ (B) -11.4 kJ (C) 0 kJ (D) 4.8 kJ
5. The enthalpy of vapourization of a liquid is 30 kJ mol^{-1} and entropy of vapourization is $75 \text{ J mol}^{-1} \text{ K}$. The boiling point of the liquid at 1 atm is :- [IIT-JEE-2004]
- (A) 250 K (B) 400 K (C) 450 K (D) 600 K
6. One mole of non-ideal gas undergoes a change of state (2.0 atm, 3.0 L, 95 K) to (4.0 atm, 5.0 L, 245 K) with a change in internal energy (ΔU) = 30.0 L-atm. The change in enthalpy (ΔH) of the process in L-atm :- [IIT-JEE-2002]
- (A) 40.0
(B) 42.3
(C) 44.0
(D) not defined, because pressure is not constant
7. Which of the following statement is false ? [IIT-JEE-2001]
- (A) Work is a state function
(B) temperature is a state function
(C) Change of state is completely defined when initial and final states are specified
(D) Work appears at the boundary of the system
8. Molar heat capacity of water in equilibrium with ice at constant pressure is :- [IIT-JEE-1997]
- (A) zero (B) ∞
(C) $40.45 \text{ kJ K}^{-1} \text{ mol}^{-1}$ (D) $75.48 \text{ JK}^{-1} \text{ mol}^{-1}$
9. For the reaction, [IIT-JEE-2006]
- $$2\text{CO(g)} + \text{O}_2\text{(g)} \longrightarrow 2\text{CO}_2\text{(g)} ; \Delta H = -560 \text{ kJ mol}^{-1}$$
- In one litre vessel at 500 K the initial pressur is 70 atm and after the reaction it becomes 40 atm at constant volume of one litre. Calculate change in internal energy. All the above gases show significant deviation from ideal behaviour. (1 Latm = 0.1 kJ)

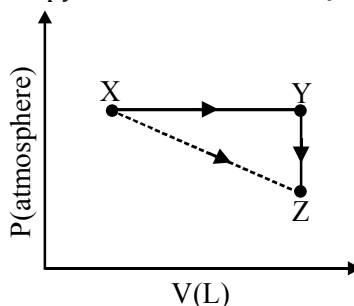
10. One mole of a liquid (1 bar, 100 mL) is taken in an adiabatic container and the pressure increases steeply to 100 bar. Then at a constant pressure of 100 bar, volume decreases by 1 mL. Find ΔU and ΔH [IIT-JEE-2004]
11. Two moles of a perfect gas undergoes the following processes : [IIT-JEE-2002]
- a reversible isobaric expansion from (1.0 atm, 20.0 L) to (1.0 atm, 40.0 L) ;
 - a reversible isochoric change of state from (1.0 atm, 40.0 L) to (0.5 atm, 40.0 L) ;
 - a reversible isothermal compression from (0.5 atm, 40.0 L) to (1.0 atm, 20.0 L) ;
- Sketch with labels each of the processes on the same P-V diagram.
 - Calculate the total work (w) and the total heat change (q) involved in the above processes.
 - What will be the values of ΔU , ΔH and ΔS for the overall process ?
12. Show that the reaction $\text{CO(g)} + \frac{1}{2} \text{O}_2 \text{(g)} \longrightarrow \text{CO}_2 \text{(g)}$ at 300 K is spontaneous and exothermic, when the standard entropy is $-0.094 \text{ kJ mol}^{-1} \text{ K}^{-1}$. The standard Gibbs free energies of formation for CO_2 and CO are -394.4 and $-137.2 \text{ kJ mol}^{-1}$, respectively. [IIT-JEE-2001]
13. A sample of argon gas at 1 atm pressure and 27°C expands reversibly and adiabatically from 1.25 dm^3 to 2.50 dm^3 . Calculate the enthalpy change in this process. $C_{v,m}$ for argon is $12.48 \text{ JK}^{-1} \text{ mol}^{-1}$. [IIT-JEE-2000]

Paragraph for Question 14 and 15

A fixed mass 'm' of a gas is subjected to transformation of states from K to L to M to N and back to K as shown in the figure. [JEE 2013]



14. The pair of isochoric processes among the transformation of states is
- K to L and L to M
 - L to M and N to K
 - L to M and M to N
 - M to N and N to K
15. The succeeding operations that enable this transformation of states are
- Heating, cooling, heating, cooling
 - cooling, heating, cooling, heating
 - Heating, cooling, cooling, heating
 - Cooling, heating, heating, cooling
16. For an ideal gas, consider only P-V work in going from an initial state X to the final state Z. The final state Z can be reached by either of the two paths shown in the figure. Which of the following choice(s) is (are) correct ? [take ΔS as change in entropy and w as work done] [JEE 2012]



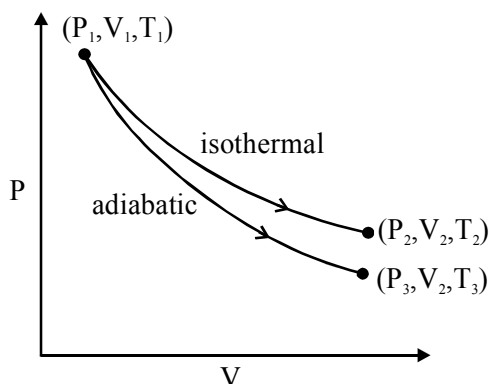
(A) $\Delta S_{x \rightarrow z} = \Delta S_{x \rightarrow y} + \Delta S_{y \rightarrow z}$

(B) $W_{x \rightarrow z} = W_{x \rightarrow y} + W_{y \rightarrow z}$

(C) $W_{x \rightarrow y \rightarrow z} = W_{x \rightarrow y}$

(D) $\Delta S_{x \rightarrow y \rightarrow z} = \Delta S_{x \rightarrow y}$

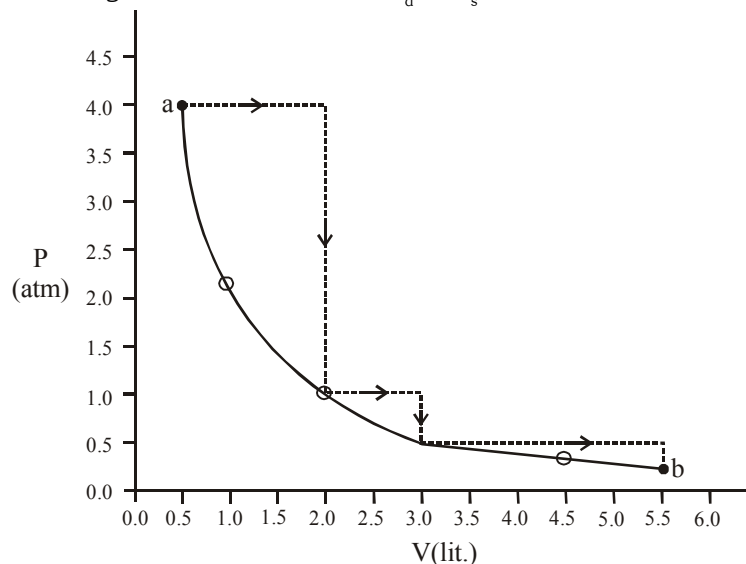
17. The reversible expansion of an ideal gas under adiabatic and isothermal conditions is shown in the figure. Which of the following statement(s) is (are) correct ? [JEE 2012]



- (A) $T_1 = T_2$ (B) $T_3 > T_1$ (C) $w_{\text{isothermal}} > w_{\text{adiabatic}}$ (D) $\Delta U_{\text{isothermal}} > \Delta U_{\text{adiabatic}}$
18. Match the transformations in **Column-I** with appropriate option in **Column-II** [JEE 2011]

Column-I	Column-II
(A) $\text{CO}_2(\text{s}) \rightarrow \text{CO}_2(\text{g})$	(p) phase transition
(B) $\text{CaCO}_3(\text{g}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$	(q) allotropic change
(C) $2\text{H}\cdot \rightarrow \text{H}_2(\text{g})$	(r) ΔH is positive
(D) $\text{P}_{(\text{white, solid})} \rightarrow \text{P}_{(\text{red, solid})}$	(s) ΔS is positive
	(t) ΔS is negative

19. One mole of an ideal gas is taken from a to b along two paths denoted by the solid and the dashed lines as shown in the graph below. If the work done along the solid line path is w_s and that along the dotted line path is w_d , then the integer closed to the ratio w_d / w_s is- [JEE 2010]



20. Among the following, the state function(s) is (are) [JEE 2009]
 (A) Internal energy (B) Irreversible expansion work
 (C) Reversible expansion (D) Molar enthalpy
21. **Statement-1** : For every chemical reaction at equilibrium, standard Gibbs energy of reaction is zero.
Statement-2 : At constant temperature and pressure, chemical reactions are spontaneous in the direction of decreasing Gibbs energy. [JEE 2008]
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
22. **Statement-1** : There is a natural asymmetry between converting work to heat and converting heat to work.
Statement-2 : No process is possible in which the sole result is the absorption of heat from a reservoir and its complete conversion into work. [JEE 2008]
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
23. For the process $\text{H}_2\text{O}(\text{l})$ (1 bar, 373 K) $\longrightarrow$ $\text{H}_2\text{O}(\text{g})$ (1 bar, 373 K), the correct set of thermodynamic parameters is :- [JEE 2007]
 (A) $\Delta G = 0$, $\Delta S = +ve$ (B) $\Delta G = 0$, $\Delta S = -ve$
 (C) $\Delta G = +ve$, $\Delta S = 0$ (D) $\Delta G = -ve$, $\Delta S = +ve$

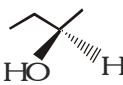
PREVIOUS YEARS QUESTIONS				ANSWER KEY		EXERCISE -5(B)
1. (D)	2. (A)	3. (C)	4. (C)			
5. (B)	6. (C)	7. (A)	8. (B)			
9. -557 kJ/mol						
10. $\Delta U = 0.1$ litre atm, $\Delta H = 9.9$ litre atm						
11. (ii) $-w = q = 620.77$ J, (iii) $\Delta H = 0$, $\Delta U = 0$, $\Delta S = 0$						
12. $\Delta H = -285.4$ kJ/mol, $\Delta G = -257.2$ kJ/mol	0					
13. $\Delta H = -114.52$ J						
14. (B)	15. (C)	16. (A), (C)	17. (A), (D)			
18. (A) $\rightarrow$ (p, r, s) ; (B) $\rightarrow$ (r, s) ; (C) $\rightarrow$ (t) ; (D) $\rightarrow$ (p, q, t)						
19. (2)	20. (A), (D)	21. (D)	22. (A)			
23. (A)						

EXERCISE-01**CHECK YOUR GRASP**

SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)

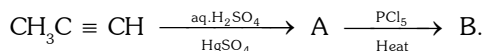
1. The product formed in the reaction  + $\text{SOCl}_2 \rightarrow$ is-



2. The reaction  + $\text{SOCl}_2 \xrightarrow{\text{pyridine}}$  + SO_2 + 

proceeds by the mechanism

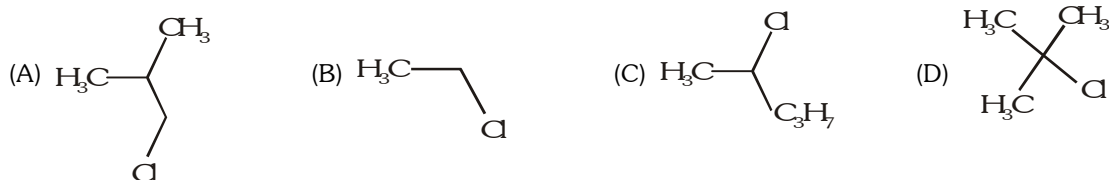
- (A) $\text{S}_{\text{N}}1$ (B) $\text{S}_{\text{N}}2$ (C) $\text{S}_{\text{N}}\text{i}$ (D) $\text{S}_{\text{E}}2$
3. 1, 3- Dibromopropane reacts with metallic zinc to form
(A) propene (B) cyclopropane (C) propane (D) hexane
4. Consider the following reaction sequence,



The products (A) and (B) are, respectively,

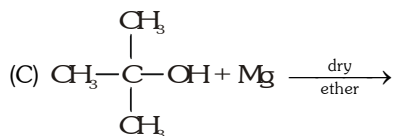
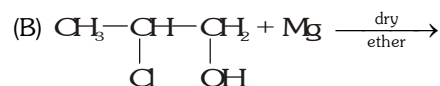
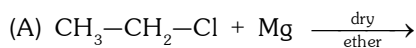
- (A) CH_3COCH_3 and $\text{CH}_3\text{CCl}_2\text{CH}_3$ (B) $\text{CH}_3\text{CH}_2\text{CHO}$ and $\text{CH}_3\text{CH}_2\text{CHCl}_2$
(C) $\text{CH}_3\text{CHOHCH}_3$ and $\text{CH}_3\text{CHClCH}_3$ (D) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$

5. Which of the following has highest dipole moment:
(A) CH_3Cl (B) CH_3F (C) CH_3Br (D) CH_3I
6. In $\text{S}_{\text{N}}1$ the first step involves the formation of
(A) free radical (B) carbanion (C) carbocation (D) final product
7. To form alkane isonitrile, alkyl halide is reacted with:
(A) KCN (B) AgCN (C) HCN (D) NH_4CN
8. Which one of the following compounds most readily undergoes substitution by $\text{S}_{\text{N}}2$ mechanism ?



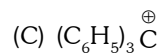
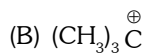
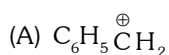
9. Sec. Butyl chloride undergo alkaline hydrolysis in the polar solvent by
(A) $\text{S}_{\text{N}}2$ (B) $\text{S}_{\text{N}}1$ (C) $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ (D) none of these

10. Grignard reagent can be prepared by

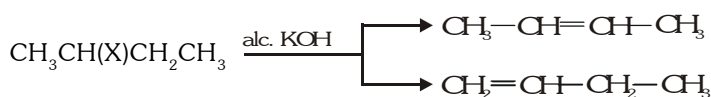


(D) All of them

11. Most stable carbocation formed from $(\text{CH}_3)_3\text{C—Br}$, $(\text{C}_6\text{H}_5)_3\text{CBr}$, $(\text{C}_6\text{H}_5)_2\text{CHBr}$ and $\text{C}_6\text{H}_5\text{CH}_2\text{Br}$ would be



12. For the reaction



(A) $\text{CH}_3\text{—CH=CH—CH}_3$ predominates

(B) $\text{CH}_2=\text{CH—CH}_2\text{—CH}_3$ predominates

(C) Both are formed in equal amounts

(D) The product ratio depends on the halogen

13. The products of reaction of alcoholic silver nitrite with ethyl bromide are

(A) Ethane

(B) Ethene

(C) Ethyl alcohol

(D) Nitro ethane

14. The reaction $\text{CH}_3\text{Br} + \text{OH}^- \longrightarrow \text{CH}_3\text{OH} + \text{Br}^-$ obeys the mechanism

(A) $\text{S}_{\text{N}}1$

(B) $\text{S}_{\text{N}}2$

(C) E_1

(D) E_2

15. Ethylidene chloride can be prepared by the reaction of HCl and

(A) Ethane

(B) Ethylene

(C) Acetylene

(D) Ethylene glycol

16. 1-phenyl-2-chloropropane on treating with alc. KOH gives mainly

(A) 1-phenylpropene

(B) 2-phenylpropene

(C) 1-phenylpropan-2-ol

(D) 1-phenylpropan-1-ol

17. Grignard reagent is obtained when magnesium is treated with

(A) Alkyl halide in presence of alcohol

(B) Alkyl halide in presence of phenol

(C) Alkyl halide in presence of dry ether

(D) Alkyl halide in presence of alcoholated ether

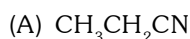
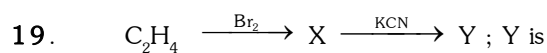
18. Ethylene reacts with bromine to form -

(A) Chloroethane

(B) Ethylene dibromide

(C) Cyclohexane

(D) 1-bromo propane



20. Reactivity order of halides for dehydrohalogenation is

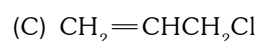
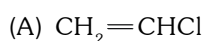
(A) $\text{R—F} > \text{R—Cl} > \text{R—Br} > \text{R—I}$

(B) $\text{R—I} > \text{R—Br} > \text{R—Cl} > \text{R—F}$

(C) $\text{R—I} > \text{R—Cl} > \text{R—Br} > \text{R—F}$

(D) $\text{R—F} > \text{R—I} > \text{R—Br} > \text{R—Cl}$

21. Which of the following is least reactive in a nucleophilic substitution reaction



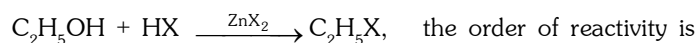
22. The correct reactivity order of alcohols towards H-X will be

- (I) $\text{CH}_2=\text{CH}-\text{OH}$ (II) $\text{H}_3\text{C}-\text{C}(\text{OH})=\text{CH}_2$ (III) $\text{CH}_3-\text{CH}_2-\text{OH}$ (IV) $\text{CH}_3-\text{CH}(\text{OH})-\text{CH}_3$
- (A) $\text{II} > \text{I} > \text{III} > \text{IV}$ (B) $\text{IV} > \text{III} > \text{II} > \text{I}$ (C) $\text{II} > \text{IV} > \text{I} > \text{III}$ (D) $\text{II} > \text{IV} > \text{III} > \text{I}$

23. Identify 'Z' in the following reaction series, $\text{CH}_3\cdot\text{CH}_2\text{CH}_2\text{Br} \xrightarrow{\text{aq. NaOH}} (\text{X}) \xrightarrow[\text{Heat}]{\text{Al}_2\text{O}_3} (\text{Y}) \xrightarrow{\text{HOCl}} (\text{Z})$:

- (A) Mixture of $\text{CH}_3\text{CH}(\text{Cl})\text{CH}_2\text{Cl}$ and $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{OH}$ (B) $\text{CH}_3-\text{CH}(\text{OH})-\text{CH}_2\text{Cl}$
- (C) $\text{CH}_3-\text{CH}(\text{Cl})-\text{CH}_2\text{OH}$ (D) $\text{CH}_3-\text{CH}(\text{Cl})-\text{CH}_2\text{Cl}$

24. For the reaction,



- (A) $\text{HI} > \text{HCl} > \text{HBr}$ (B) $\text{HI} > \text{HBr} > \text{HCl}$ (C) $\text{HCl} > \text{HBr} > \text{HI}$ (D) $\text{HBr} > \text{HI} > \text{HCl}$

25. Ethyl alcohol reacts at a faster rate with HI than with HCl in forming the corresponding ethyl halides under identical conditions mainly because -

- (A) HI, being a stronger acid, protonates ethyl alcohol at oxygen much better and helps substitution
 (B) the bond length in HI is much shorter than that in HCl
 (C) I^- is a much better leaving group
 (D) I^- is a much better nucleophile than Cl^-

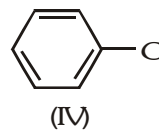
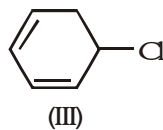
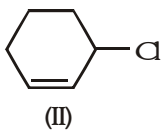
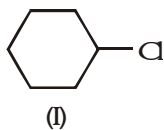
CHECK YOUR GRASP						ANSWER KEY					EXERCISE -1				
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	B	B	B	A	A	C	B	B	C	A	C	D	D	B	C
Que.	16	17	18	19	20	21	22	23	24	25					
Ans.	A	C	B	B	B	A	D	B	B	D					

EXERCISE-02**BRAIN TEASERS****SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)**

1. Which of the following does/do produce a white precipitate of AgCl on warming with alcoholic silver nitrate?

(A) Allyl chloride (B) t-Butyl chloride (C) Benzyl chloride (D) Vinyl chloride

2. What is the order of reactivity of the following compounds towards nucleophilic substitution?

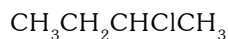
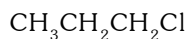


(A) I < II < III < IV (B) IV < III < II < I (C) IV < I < II < III (D) IV < II < I < III

3. The order of decreasing nucleophilicity of the following is

(A) $\text{H}_2\text{O} > \text{OH}^- > \text{CH}_3\text{COO}^- > \text{CH}_3\text{O}^-$ (B) $\text{CH}_3\text{O}^- > \text{OH}^- > \text{CH}_3\text{COO}^- > \text{H}_2\text{O}$
(C) $\text{CH}_3\text{COO}^- > \text{CH}_3\text{O}^- > \text{OH}^- > \text{H}_2\text{O}$ (D) $\text{HO}^- > \text{CH}_3\text{O}^- > \text{CH}_3\text{COO}^- > \text{H}_2\text{O}$

4. The order of decreasing $\text{S}_{\text{N}}1$ reactivities of the halides



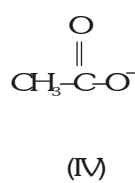
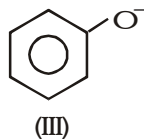
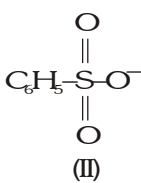
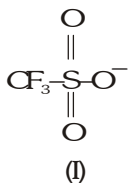
I

II

III

(A) I > II > III (B) II > I > III (C) II > III > I (D) III > II > I

5. Consider the following anions.



When attached to sp^3 - hybridized carbon, their leaving group ability in nucleophilic substitution reactions decreases in the order

(A) I > II > III > IV (B) I > II > IV > III (C) IV > I > II > III (D) IV > III > II > I

6. The basicity of RO^- , HO^- , RCOO^- , ROH , and H_2O are of the order -

(A) $\text{HO}^- > \text{RO}^- > \text{H}_2\text{O} > \text{ROH} > \text{RCOO}^-$
(B) $\text{RO}^- > \text{HO}^- > \text{RCOO}^- > \text{ROH} > \text{H}_2\text{O}$
(C) $\text{H}_2\text{O} > \text{ROH} > \text{RCOO}^- > \text{HO}^- > \text{RO}^-$
(D) $\text{ROH} > \text{H}_2\text{O} > \text{HO}^- > \text{RCOO}^- > \text{RO}^-$

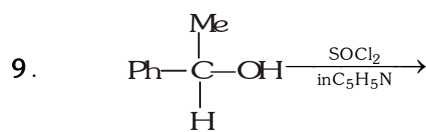
7. Which of the following are aprotic solvents :

(A) DMSO (B) DMF (C) H_2O (D) CH_3COOH

8. Which is/are true statements (s) :

(A) Protonation increases electrophilic nature of carbonyl group
(B) CF_3SO_3^- is better leaving group than CH_3SO_3^-
(C) Benzyl carbonium ion is stabilised by resonance

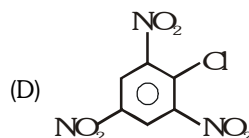
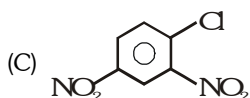
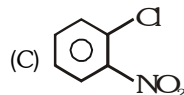
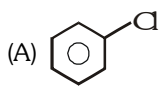
(D) $\text{CCl}_3\text{CH} \begin{matrix} \text{OH} \\ \diagup \\ \text{OH} \end{matrix}$ is stable, due to H-Bonding



Which statement is true for the above reaction ?

- (A) Retention of configuration (B) Inversion of configuration
(C) Inversion and Retention both (D) None

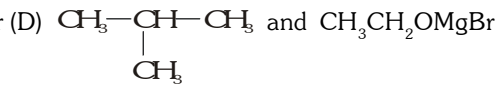
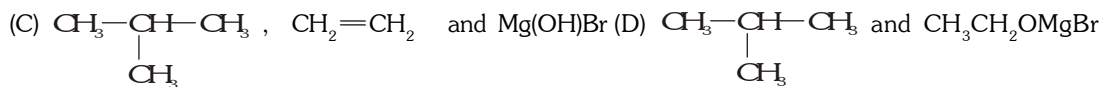
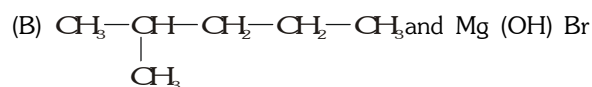
10. Which of the following undergoes hydrolysis most easily



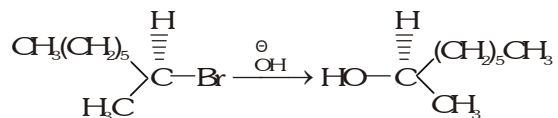
11. A compound 'A' formula of $\text{C}_3\text{H}_6\text{Cl}_2$ on reaction with alkali can give 'B' of formula $\text{C}_3\text{H}_6\text{O}$ or 'C' of formula C_3H_4 . 'B' on oxidation gave a compound of the formula $\text{C}_3\text{H}_6\text{O}_2$. 'C' with dilute H_2SO_4 containing Hg^{2+} ion gave 'D' of formula $\text{C}_3\text{H}_6\text{O}$, which with bromine and alkali gave the sodium salt of $\text{C}_2\text{H}_4\text{O}_2$. Then 'A' is

- (A) $\text{CH}_3\text{CH}_2\text{CHCl}_2$ (B) $\text{CH}_3\text{CCl}_2\text{CH}_3$ (C) $\text{CH}_2\text{ClCH}_2\text{CH}_2\text{Cl}$ (D) $\text{CH}_3\text{CHClCH}_2\text{Cl}$

12. Isobutyl magnesium bromide with dry ether and absolute alcohol gives



13. Following reaction is



- (A) E_1 (B) $\text{S}_{\text{N}}1$ (C) E_2 (D) $\text{S}_{\text{N}}2$

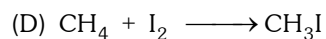
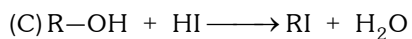
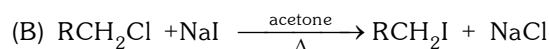
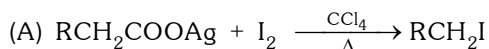
14. On treatment with chlorine in presence of sunlight, toluene gives the product -

- (A) o-chloro toluene (B) 2, 5-dichloro toluene
(C) p-chloro toluene (D) Benzyl chloride

15. In $\text{S}_{\text{N}}1$ reaction an optically active substrates mainly gives :

- (A) Retention in configuration (B) Inversion in configuration
(C) Racemic product (D) No product

16. Alkyl iodides can be prepared by :-



17. Which of the following reagents can be used to prepare an alkyl halide :-
 (A) NaCl (B) HCl + ZnCl₂ (C) SOCl₂ (D) PCl₅
18. Which of the following reactions depict the nucleophilic substitution of C₂H₅Br :-
 (A) C₂H₅Br + C₂H₅SNa $\longrightarrow$ C₂H₅SC₂H₅ + NaBr
 (B) C₂H₅Br $\xrightarrow{\text{Na} + \text{C}_2\text{H}_5\text{OH}}$ C₂H₆ + HBr
 (C) C₂H₅Br + AgCN $\longrightarrow$ C₂H₅NC + AgBr
 (D) C₂H₅Br + KOH $\longrightarrow$ C₂H₅OH + KBr
19. For an S_N² reaction, which of the following statements are true :-
 (A) The rate of reaction is independent of the concentration of the nucleophile
 (B) The nucleophile attacks the C-atom on the side of the molecule opposite to the group being displaced
 (C) The reaction proceeds with simultaneous bond formation and rupture
 (D) None of these
20. Which of the following is an S_N² reaction :-
 (A) CH₃CH₂Br + KOH $\longrightarrow$ CH₃CH₂OH + KBr
 (B) CH₃CH₂Br + CH₃CH₂ONa $\longrightarrow$ CH₃CH₂OCH₂CH₃ + NaBr
 (C) (CH₃)₃CBr + KOH $\longrightarrow$ (CH₃)₃COH + KBr
 (D)
$$\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{CH}_2 - \text{C} - \text{Br} \\ | \\ \text{CH}_3 \end{array} + \text{KOH} \longrightarrow \begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3\text{CH}_2 - \text{C} - \text{OH} \\ | \\ \text{CH}_3 \end{array} + \text{KBr}$$

BRAIN TEASERS						ANSWER KEY				EXERCISE -2					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	A,B,C	C	B	C	B	B	A,B	A,B,C,D	B	D	A	D	D	D	C
Que.	16	17	18	19	20										
Ans.	B,C	B,C,D	A,C,D	B,C	A,B										

EXERCISE-03

MISCELLANEOUS TYPE QUESTIONS

TRUE OR FALSE :

- Alkyl halides follow the reactivity sequence,
 $R-I > R-Br > R-Cl > R-F$
- Vinyl chloride reacts with dilute NaOH to form vinyl alcohol.
- Allyl chloride is more reactive than vinyl chloride.
- Tertiary butyl bromide undergoes S_N1 reactions.
- Both vic- and gem-dihalides on heating with zinc dust in presence of alcohol form same alkene.

FILL IN THE BLANKS :

- The interaction of elemental sulphur with Grignard reagent gives
- An alkyl halide may be converted into alcohol byreaction
- Diethyl ether is obtained from ethyl bromide by treating it with and the name of the reaction is
- Allyl chloride is a compound while vinyl chloride is inert towards nucleophilic substitution.
- The dihalides in which halogen atoms are attached to adjacent carbon atoms are termed as.....
- Alkyl halides are formed when thionyl chloride and are refluxed in presence of pyridine.
- Ethylene chloride on hydrolysis with aq. KOH forms.....

MATCH THE COLUMN

- Match the column I with column II.

Column-I (reaction)		Column-II (Mechanism)	
(A)	$\begin{array}{c} \text{CH}_3 \\ \\ \text{Ph}-\text{C}-\text{OH} \\ \\ \text{H} \end{array} + \text{SOCl}_2 \longrightarrow \begin{array}{c} \text{CH}_3 \\ \\ \text{Ph}-\text{C}-\text{Cl} \\ \\ \text{H} \end{array}$	(p)	S_N1
(B)	$\begin{array}{c} \text{CH}_3 \\ \\ \text{Ph}-\text{C}-\text{OH} \\ \\ \text{H} \end{array} + \text{SOCl}_2 \xrightarrow{\text{Pyridine}} \begin{array}{c} \text{CH}_3 \\ \\ \text{H}_5\text{C}_6-\text{C}-\text{Cl} \\ \\ \text{H} \end{array}$	(q)	S_N2
(C)	$\text{CH}_3-\text{CH}(\text{CH}_3)-\text{Br} + \text{OC}_2\text{H}_5/\text{EtOH} \longrightarrow$	(r)	S_Ni
(D)	$\begin{array}{c} \text{CH}_3 \\ \\ \text{H}_5\text{C}_6-\text{C}-\text{OH} \\ \\ \text{H} \end{array} + \text{HCl} \longrightarrow \begin{array}{c} \text{CH}_3 \\ \\ \text{C}_6\text{H}_5-\text{C}-\text{Cl} \\ \\ \text{H} \end{array}$	(s)	E_2

- Match the column I with column II.

Column-I (Substrate)		Column-II (Relative rate of solvolysis in 50% aqueous ethanol at 45 C)	
(A)		(p)	7700
(B)		(q)	1
(C)		(r)	91
(D)		(s)	1,30,000

ASSERTION & REASON QUESTION :

These questions contains, Statement-I (assertion) and Statement-II (reason).

- (A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I
(B) Statement-I is True, Statement-II is True ; Statement-II is NOT a correct explanation for Statement-I
(C) Statement-I is True, Statement-II is False.
(D) Statement-I is False, Statement-II is True.

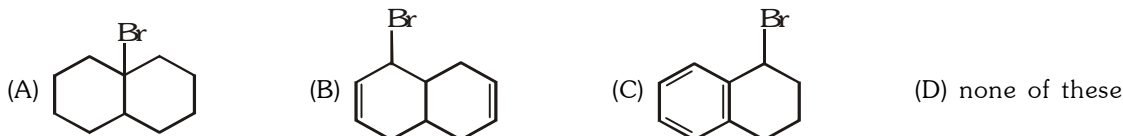
1. **Statement-I** : Iodination of alkanes is carried out by heat in presence of reducing agent.
Because
Statement-II : Iodination of alkanes takes place explosively.
2. **Statement-I** : Chloropropane has higher boiling point than chloroethane.
Because
Statement-II : Haloalkanes are polar molecules.
3. **Statement-I** : Polar solvent slows down S_N2 reaction.
Because
Statement-II : $\text{CH}_3\text{-Br}$ is less reactive than CH_3Cl .
4. **Statement-I** : Primary benzylic halides are more reactive than primary alkyl halides towards S_N1 reaction.
Because
Statement-II : Reactivity depends upon the nature of the nucleophile and the solvent.
5. **Statement-I** : Vinylic halides are reactive towards nucleophilic substitution reaction.
Because
Statement-II : Reactivity is due to the polarity of carbon-halogen bond.
6. **Statement-I** : Aryl halides undergo electrophilic substitution less readily than benzene.
Because
Statement-II : Aryl halide gives only meta product w.r.t. electrophilic substitution.
7. **Statement-I** : Optically active 2-iodobutane on treatment with NaI in acetone undergoes racemisation.
Because
Statement-II : Repeated Walden inversions on the reactant and its product eventually gives a racemic mixture.
8. **Statement-I** : Free radical chlorination of n-butane gives 72% of 2-chlorobutane and 28% of 1-chlorobutane though it has six primary and four secondary hydrogens.
Because
Statement-II : A secondary hydrogen is abstracted more easily than the primary hydrogen.
9. **Statement-I** : Nucleophilic substitution reaction on an optically active alkyl halide gives a mixture of enantiomers.
Because
Statement-II : The reaction occurs by S_N1 mechanism.
10. **Statement-I** : Boiling point of alkyl halide increases with increase in molecular weight.
Because
Statement-II : Boiling point of alkylhalides are in the order $\text{RI} > \text{RBr} > \text{RCI} > \text{RF}$.

COMPREHENSION BASED QUESTIONS :

Comprehension # 1

An organic compound A has molecular formula $C_{10}H_{17}Br$ and it is non-resolvable. A does not decolourize brown colour of bromine water solution. A on treatment with $(CH_3)_2COK/(CH_3)_3COH$ yields B as major product. B on treatment with H_2/Pt yields $(C_{10}H_{16})$ which on treatment with $Cl_2/h\nu$ yields three monochloro derivative. Also B on boiling with acidic permanganate solution yields $C(C_{10}H_{16}O_3)$. C on heating with sodalime yields D ($C_9H_{16}O$). D on reducing with $LiAlH_4$ followed by heating the product with concentrated H_2SO_4 yields E (C_9H_{16}) as major product. E on treatment with ozone followed by work-up with $Zn-H_2O$ yields 6-Ketnonanal.

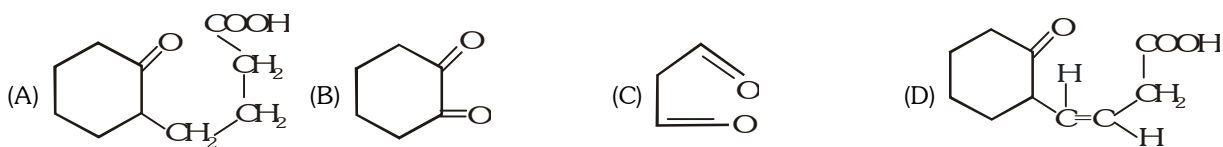
1. Compound A is :



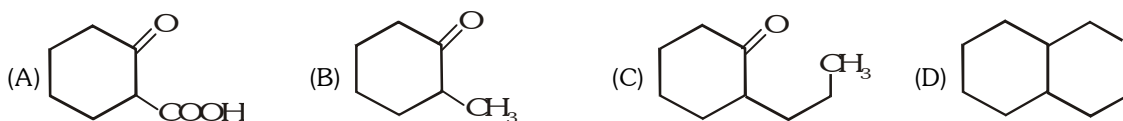
2. Compound B is :



3. Compound C is :

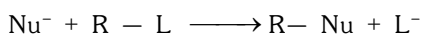


4. Compound D is :



Comprehension # 2

Nucleophilic substitution reactions generally expressed as

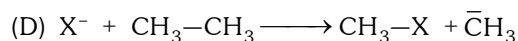
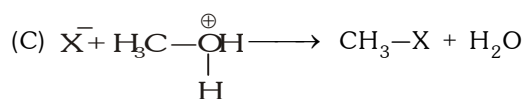
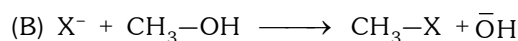
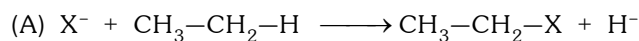


Where $Nu^- \longrightarrow$ Nucleophile ; $R-L \longrightarrow$ substrate ; $L \longrightarrow$ leaving group

The best leaving groups are those that become the most stable ions after they depart. Since most leaving groups leave as a negative ion, the best leaving groups are those ions that stabilize a negative charge most effectively. A good leaving group should be

- (a) electron-withdrawing to polarize the carbon
- (b) stable once it has left (not a strong base)
- (c) polarisable- to maintain partial bonding with the carbon in the transition state (both S_N1 and S_N2). This bonding helps to stabilise the transition state and reduces the activation energy.

1. Among the following which is feasible ?



2. Among the following which is false statement ?

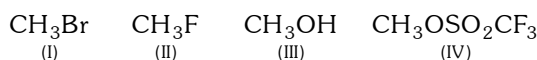
(A) The weaker the base after the group departs, the better the leaving group

(B) A reactive leaving group would raise the energy of the product, driving the equilibrium towards the reactants

(C) Relative leaving group ability may vary with change of solvent

(D) Better leaving group only increases S_N2 rate, not S_N1 .

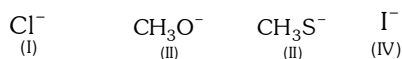
3.



The correct order of decreasing reactivity of the above compounds towards CH_3O^- in an S_N2 reaction is :

(A) $I > IV > II > III$ (B) $IV > I > II > III$ (C) $IV > I > III > II$ (D) $IV > II > I > III$

4.



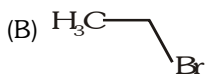
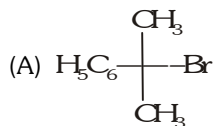
The correct order of increasing leaving group capability of above anions

(A) $III < IV < II < I$ (B) $II < III < I < IV$ (C) $II < IV < III < I$ (D) $I < III < II < IV$

Comprehension # 3

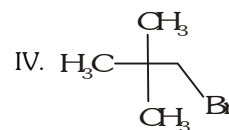
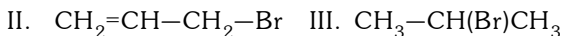
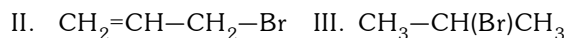
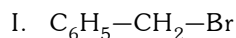
Nucleophilic aliphatic substitution reaction is mainly of two types : S_N1 and S_N2 . The S_N1 mechanism is a two step process. Reaction velocity of S_N1 reaction depends only on the concentration of the substrate. Since product formation takes place by the formation of carbocation, optically active substrate gives (+) and (-) forms of the product. In most of the cases the product usually consists of 5-20% inverted product and 80-95% racemised species. The more stable the carbocation, the greater is the proportion of racemisation. In solvolysis reaction, the more nucleophilic the solvent, the greater is the proportion of inversion.

1. Which one of the following compound will give S_N1 reaction predominantly ?



(D) All of these

2. Which of the following compounds will give S_N1 and S_N2 reactions with considerable rate ?



Select the correct answer from the codes given below

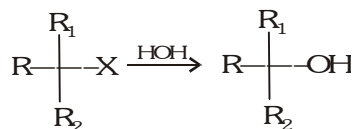
(A) I, II and III

(B) I, II and IV

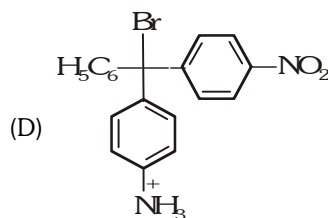
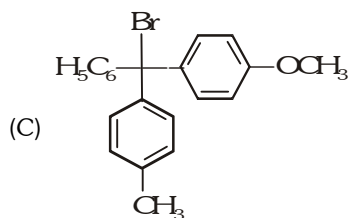
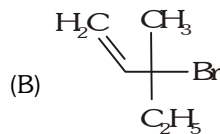
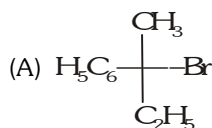
(C) II, III and IV

(D) I, III and IV

3. For the given reaction



Which substrate will give maximum racemisation ?

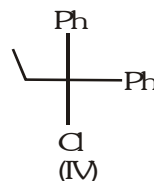
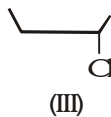
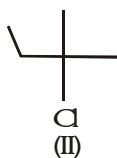
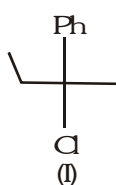


MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
True / False 1. T 2. F 3. T 4. T 5. F Fill in the Blanks 1. thioalcohol 2. nucleophilic substitution 3. sodium ethoxide, williamson's synthesis 4. reactive 5. vic-dihalides or alkylene halides 6. alcohols 7. glycol Match the Column 1. (A) → r ; B → q ; (C) → s ; (D) → p 2. (A) → q ; (B) → r ; (C) → s ; (D) → p Assertion - Reason Questions 1. D 2. B 3. C 4. B 5. D 6. C 7. A 8. A 9. A 10. B Comprehension Based Questions Comprehension #1 : 1. (A) 2. (C) 3. (A) 4. (C) Comprehension #2 : 1. (C) 2. (D) 3. (B) 4. (B) Comprehension #3 : 1. (A) 2. (A) 3. (C)		

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

1. Arrange the following compounds in order of :

Decreasing S_N1 reaction rate :

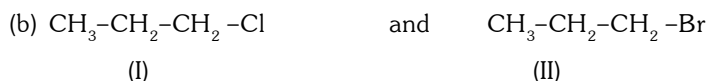


2. Select the member of each pair that shows faster rate of S_N2 reaction with KI in acetone.



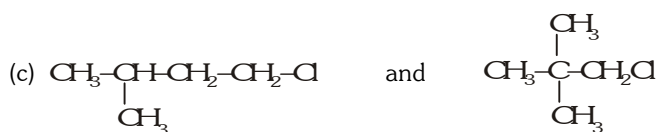
(I)

(II)



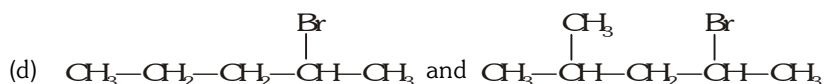
(I)

(II)



(I)

(II)



(I)

(II)

3. Of the following statements which are true for S_N1 reaction.

- (a) Tertiary alkyl halides react faster than secondary.
- (b) The absolute configuration of the product is opposite to that of the reactant when an optical active substrate is used.
- (c) The reaction shows first order kinetics.
- (d) The rate of reaction depends markedly on the nucleophilicity of the nucleophile.
- (e) The mechanism is two step.
- (f) Carbocations are intermediate.
- (g) $\text{Rate} \propto [\text{Alkyl halides}]$
- (h) The rate of the reaction depends on the nature of the leaving group.

4. Of the following statements, which are true for S_N2 reaction.

- (a) Tertiary alkyl halides react faster than secondary.
- (b) The absolute configuration of product is opposite to that of the reactant when an optically active substrate is used.
- (c) The reaction shows first order kinetics.
- (d) The rate of the reaction depends markedly on the nucleophilicity of the attacking reagent.
- (e) The mechanism is one step.

(f) Carbocations are intermediate.

(g) $\text{Rate} \propto [\text{Alkyl halides}]$

(h) The rate of the reaction depends on the nature of the leaving group.

5. Arrange the isomers of molecular formula $\text{C}_4\text{H}_9\text{Cl}$ in order of decreasing rate of reaction with sodium iodide in acetone.

6. There is an overall 29-fold difference in reactivity of 1-chlorohexane, 2-chlorohexane towards potassium iodide in acetone. Which one is the most reactive? why?

7. Identify the product when A reacts with 
(A)

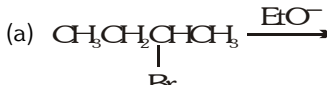
(a) Br_2/Fe

(b) Br_2/CCl_4

(c) NBS

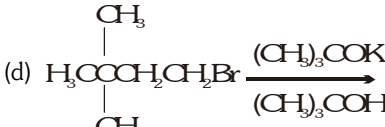
(d) HBr

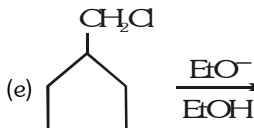
8. Identify major product in the following : $\text{ClCH}_2\text{CHCl}_2 \xrightarrow{\text{OH}^-} \text{CH}_2=\text{CCl}_2$

(a) 

(b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br} \xrightarrow{\text{EtO}^-}$

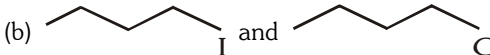
(c) 

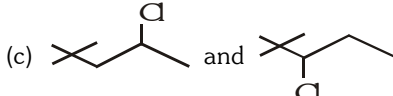
(d) 

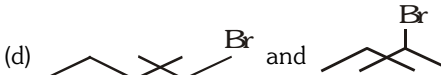
(e) 

9. Which is faster in the following pairs of halogen compounds undergoing $\text{S}_{\text{N}}2$ reactions?

(a) 

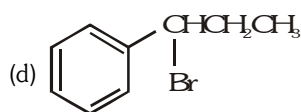
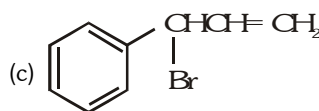
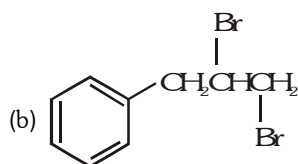
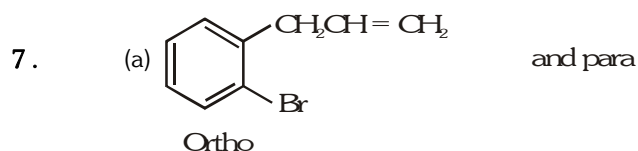
(b) 

(c) 

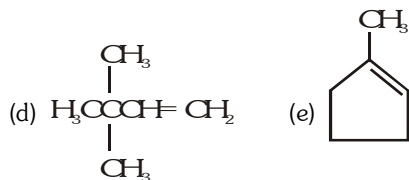
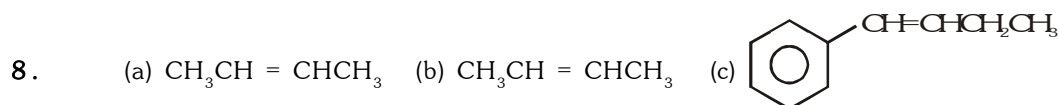
(d) 

10. $\text{R}-\text{Mg}-\text{Br}$ (A) on reaction with H_2O forms a gas (B), which occupied 1.4 L/g at NTP. What is product when $\text{R}-\text{Br}$ reacts with benzene in presence of AlCl_3 ?

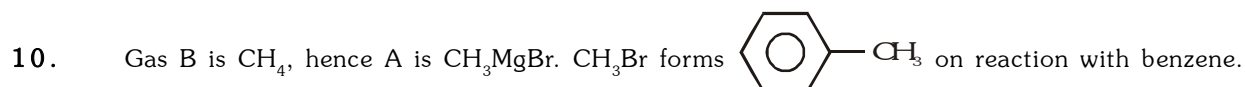
- IV > I > II > III
- (a) I (b) II (c) I (d) I
- (a) T (b) F (c) T (d) F (e) T (f) T (g) T (h) T
- (a) F (b) T (c) F (d) T (e) T (f) F (g) F (h) T
- 1 > 2 > 3 Anion of acetic acid is more stabilised by resonance than phenoxide ion.
- 1-chlorohexane
Because it follows S_N2 path.



(1, 2-H⁺ shift gives more stable benzylic carbocation)



9. Ease of backside attack (less steric hindrance) decides which undergoes S_N2 faster (except in (b) in which iodide is better leaving group). In all cases first one is faster than the other for S_N2 reaction.

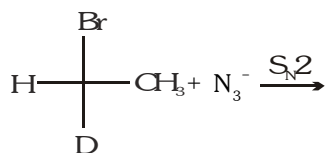


EXERCISE-04 [B]**BRAIN STORMING SUBJECTIVE EXERCISE**

1. Explain the following observations:

(a) Azide ion (N_3^-) react with 2-bromopentane thousand times faster than with neopentyl bromide in a $\text{S}_\text{N}2$ reaction though former is a secondary halide while latter is primary.

(b) What will happen to the stereochemistry of product of the following reaction:



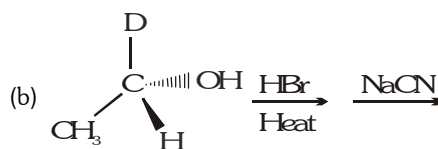
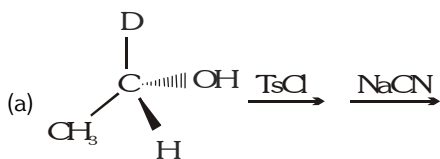
(c) What will happen to the rate if the concentration of alkyl bromide in (b) is doubled?

(d) What will happen to the rate if the concentration of azide ion in (b) is doubled?

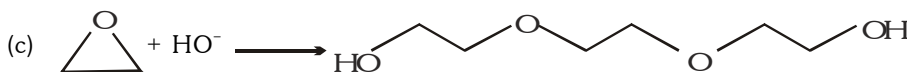
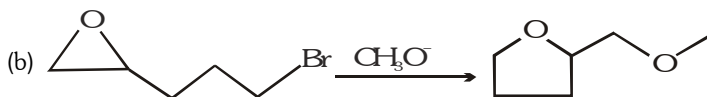
(e) How the sign of optical rotation of reactant and product are related in (b)

(f) When allowed to stand in dilute H_2SO_4 , laevo-rotatory 2-butanol slowly loses optical activity.

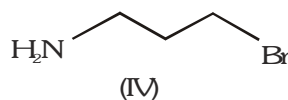
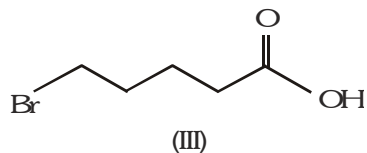
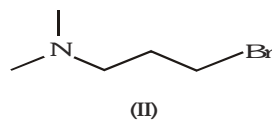
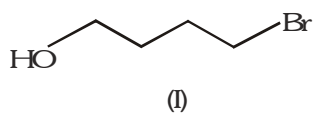
2. Provide structure of major product in the following reaction indicating stereochemistry where appropriate:



3. Propose mechanism of the following reactions:

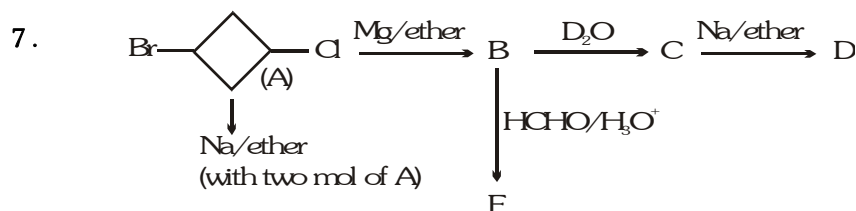
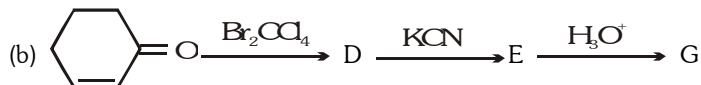
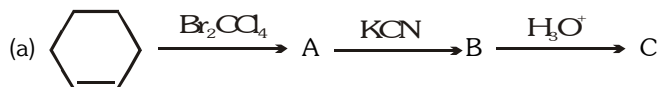


4. Which of the following alkyl halide could be successfully used to synthesize Grignard reagent and why other fail?



5. An alkyl bromide A has molecular formula $C_8H_{17}Br$ and four different structures can be drawn for it, all optically active. A on refluxing with ethanolic KOH solution yields only one elimination product $B(C_8H_{16})$ which is still enantiomeric. B on treatment with H_2/Pt yields $C(C_8H_{18})$ which does not rotate the plane polarized light, B on ozonolysis followed by work-up with H_2O_2 yields $D(C_7H_{14}O)$ as one product which is still resolvable. Deduce structures of A to D.

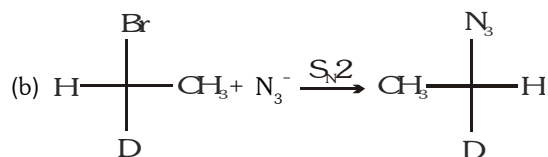
6. Identify A to G in the following.



Identify B to F

8. Vinyl chloride does not give S_N reaction but allyl chloride gives. Explain.
9. Arrange the following in the increasing order of their ability as a leaving group:
- (a) CH_3S^- , CH_3O^- , CF_3^- and F^-
- (b) $CF_3SO_3^-$, $CH_3SO_3^-$ and CH_3COO^-
10. RBr when treated with $AgCN$ in a highly polar solvent gives RNC whereas when it is treated with $NaCN$ it gives RCN . Explain.
-

1. (a) Though neopentylbromide is primary, bulky tertiary butyl group possess very large steric hindrance to the attack of bulky nucleophile N_3^- .



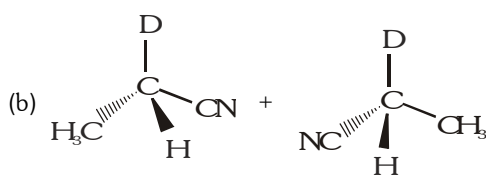
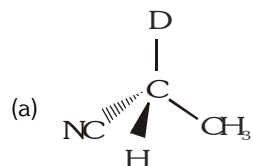
(c) Rate will double

(d) Rate will double

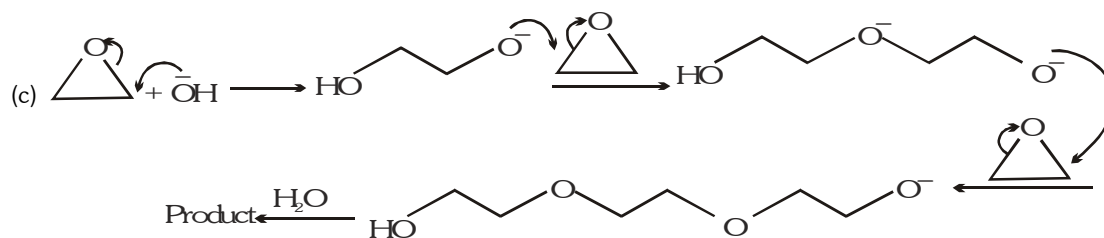
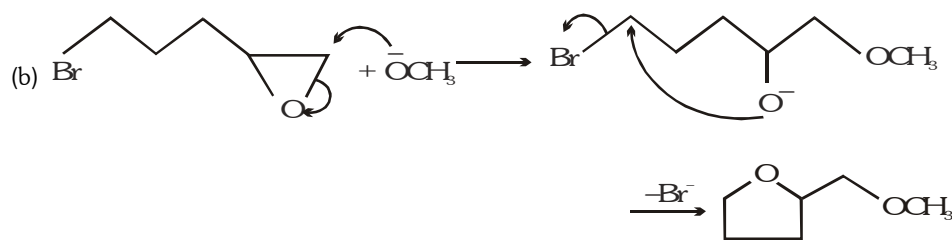
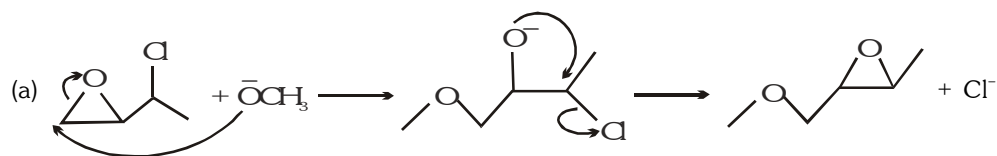
(e) not related

(f) Racemization occur through carbocation intermediate

2.

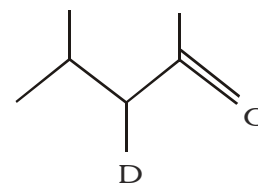
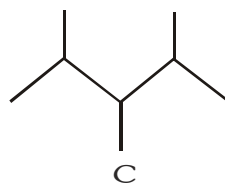
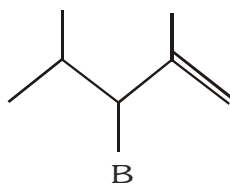
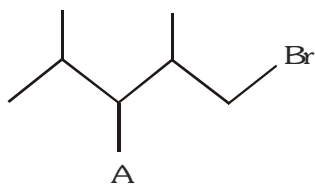


3.

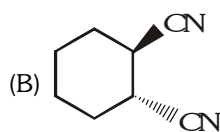
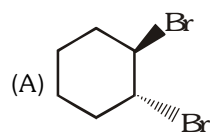


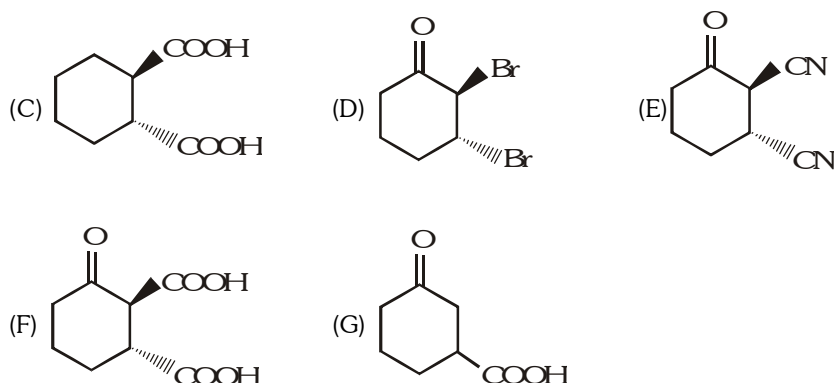
4. Only II can be used for successful synthesis of Grignard reagent, rest all contain acidic proton and will react with R^- (from Grignard reagent) forming alkane.

5.

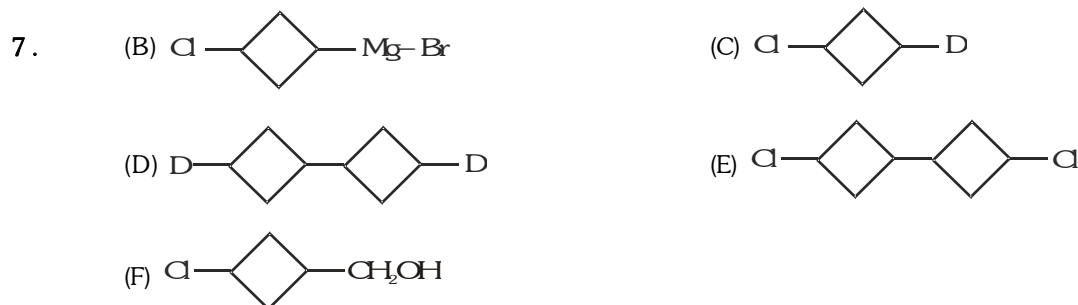


6.

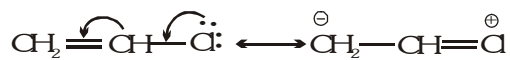




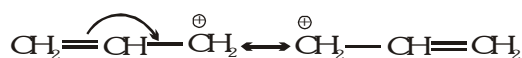
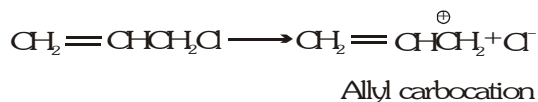
(decarboxylation takes place on heating when there is a keto group at β - position)



8. In Vinyl chloride, C - Cl bond is stable due to resonance (as in chlorobenzene)

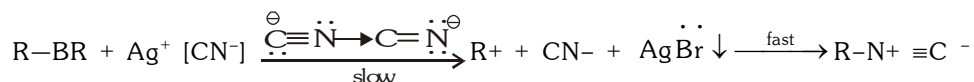


Hence S_N reaction in which Cl is replaced by nucleophile is not possible. In addition to this, sp^2 - hybridised carbon is more acidic than sp^3 - carbon, hence removal of proton (H^+) is easier than removal of halide (Cl^-) In allyl chloride, S_N reaction is easier since allyl carbocation formed after removal of Cl^- is stabilised by resonance.



9. (a) $\text{CF}_3^- < \text{CH}_3\text{O}^- < \text{CH}_3\text{S}^-$; (b) $\text{CH}_3\text{COO}^- < \text{CH}_3\text{SO}_3^- < \text{CF}_3\text{SO}_3^-$

10. As $[\text{CN}]^-$ is an ambident nucleophile which have two nucleophile which have two nucleophilic sites and can attack from either side. In a highly polar solvent, AgCN promotes the formation of carbocation R^+ , precipitation of AgBr .



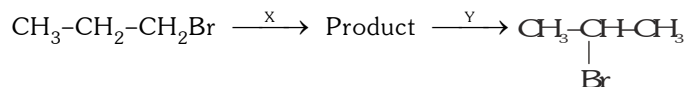
In the absence of such promotion by Ag^+ , with $\text{Na}^+[\text{CN}]^-$, the resulting S_N2 reaction is found to proceed with preferential attack on the atom in the nucleophile which is more polarisable i.e. C.



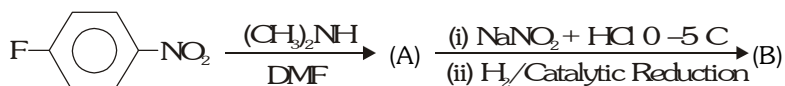
EXERCISE-05**PREVIOUS YEARS QUESTIONS**

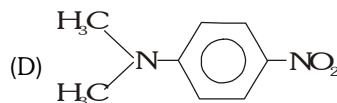
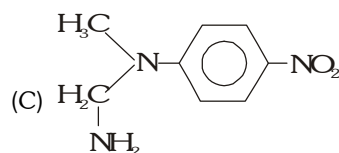
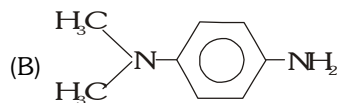
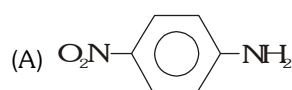
1. Chlorination of toluene in the presence of light and heat followed by treatment with aqueous NaOH gives-
(A) o-cresol (B) p-cresol [IIT-90]
(C) 2, 4 dihydroxytoluene (D) Benzoic acid
2. Aryl halides are less reactive towards nucleophilic substitution reaction as compared to alkyl halides due to-
(A) The formation of less stable carbonium ion (B) Resonance stabilization [IIT-90]
(C) The inductive effect (D) sp^2 hybridised carbon attached to the halogen
3. 1-Chlorobutane on reaction with alcoholic potash gives - [IIT-91]
(A) 1-butene (B) 1-butanol (C) 2-butene (D) 2-butanol
4. In the addition of HBr to propene in the absence of peroxides, the first step involves the addition of -
(A) H^+ (B) Br^- (C) $H^\bullet$ (D) $Br^\bullet$ [IIT-93]
5. Arrange the following compounds in order of increasing dipole moment : [IIT-96]
(I) Toluene ; (II) m-dichlorobenzene ; (III) o-dichlorobenzene ; (IV) p-dichlorobenzene
(A) $I < IV < II < III$ (B) $IV < I < II < III$ (C) $I < IV < II < III$ (D) $IV < II < I < III$
6. In the reaction of p-chloro toluene with KNH_2 in liq. NH_3 , the major product is - [IIT-97]
(A) o-toluidine (B) m-toluidine (C) m-chloroaniline (D) p-chloroaniline
7. $(CH_3)_3CMgCl$ reaction with D_2O produces : [IIT-97]
(A) $(CH_3)_3CD$ (B) $(CH_3)_3OD$ (C) $(CD_3)_3CD$ (D) $(CH_3)_3OD$
8. The intermediate during the addition of HCl to propene in presence of peroxide is - [IIT-97]
(A) $CH_3\dot{C}HCH_2Cl$ (B) $CH_3\overset{+}{C}HCH_3$ (C) $CH_3\dot{C}HCH_2$ (D) $CH_3CH_2\dot{C}H_2$
9. The number of possible enantiomeric pairs that can be produced during monochlorination of isopentane-
(A) 2 (B) 3 (C) 4 (D) 1 [IIT-97]
10. During debromination of meso-2,3-dibromobutane, with Zn dust/ CH_3COOH the major compound formed is - [IIT-97]
(A) n-butane (B) 1-butene (C) cis-2-butene (D) trans-2-butene
11. Benzyl chloride ($C_6H_5CH_2Cl$) can be prepared from toluene by chlorination with [IIT-98]
(A) SO_2Cl_2 (B) $SOCl_2$ (C) PCl_5 (D) $NaOCl$
12. Toluene, when treated with Br_2/Fe , gives o and p-bromotoluene, because the CH_3 group - [IIT-99]
(A) is ortho and para directing (B) is meta directing
(C) deactivates the ring by hyperconjugation (D) deactivates the ring
13. A solution of (+) 2-chloro-2phenylethane in toluene racemises slowly in the presence of small amount of $SbCl_5$ due to the formation of - [IIT-99]
(A) Carbanion (B) Carbene (C) Free-radical (D) Carbocation
14. The order of reactivity of the following alkyl halides for a S_N2 reaction is - [IIT-2000]
(A) $RF > RCl > R-Br > R-I$ (B) $R-F > R-Br > R-Cl > R-I$
(C) $R-Cl > R-Br > R-F > R-I$ (D) $R-I > R-Br > R-Cl > R-F$
15. Which of the following has the highest nucleophilicity : [IIT-2000]
(A) F^- (B) OH^- (C) CH_3^- (D) NH_2^-
16. An S_N2 reaction at an asymmetric carbon of a compound always gives : [IIT-2001]
(A) an enantiomer of the substrate (B) a product with opposite optical rotation
(C) a mixture of diastereomers (D) a single stereoisomer

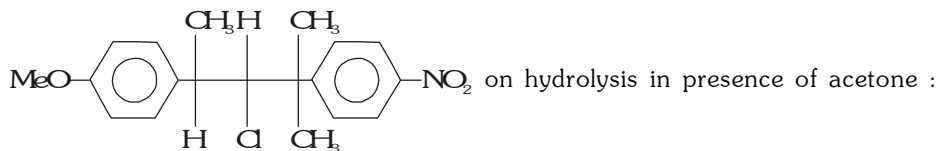
17. The number of isomer for the compound with molecular formula $C_2BrClFI$ is - [IIT-01]
 (A) 3 (B) 4 (C) 5 (D) 6
18. In the presence of peroxide, hydrogen chloride and hydrogen iodide do not give anti-Markovnikov's addition to alkenes because : [IIT-01]
 (A) both are highly ionic
 (B) one is oxidising and the other is reducing
 (C) one of the steps in endothermic in both the cases
 (D) all the steps are exothermic in both the reactions
19. Identify the set of reagents/reaction conditions 'X' and 'Y' in the following set of transformations - [IIT-02]

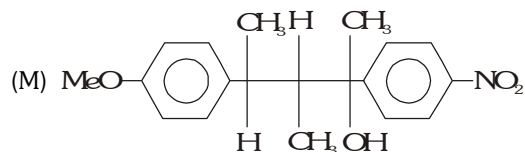
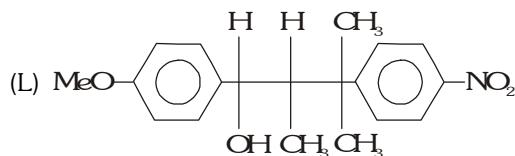
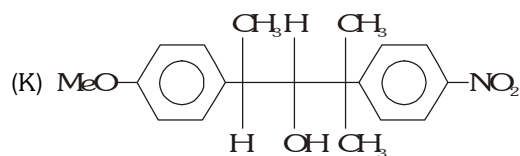


- (A) X = dilute aqueous NaOH, 20 C, Y = HBr/acetic acid, 20 C
 (B) X = concentrated alcoholic NaOH, 80 C; Y = HBr/acetic acid, 20
 (C) X = dilute aqueous NaOH 20 , Y = $Br_2/CHCl_3$, 0 C
 (D) X = concentrated alcoholic NaOH, 80 C; Y = $Br_2/CHCl_3$, 0 C

20.  [IIT-03]



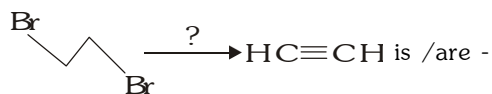
21.  on hydrolysis in presence of acetone : [IIT-05]



- (A) K & L (B) only L (C) M only (D) K & M

22. The reagent (s) for the following conversion :

[IIT-07]



- (A) alcoholic KOH (B) Alcoholic KOH followed by NaNH_2
(C) aqueous KOH followed by NaNH_2 (D) $\text{Zn}/\text{CH}_3\text{OH}$

23. In the following groups :

[IIT-97]

(I) $-\text{OAc}$; (II) $-\text{OMe}$; (III) $-\text{OSO}_2\text{Me}$; (IV) OSO_2CF_3

the order of leaving group ability is -

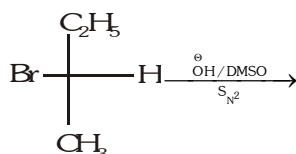
- (A) $\text{I} > \text{II} > \text{III} > \text{IV}$ (B) $\text{IV} > \text{III} > \text{I} > \text{II}$ (C) $\text{III} > \text{II} > \text{I} > \text{IV}$ (D) $\text{II} > \text{III} > \text{IV} > \text{I}$

24. $\text{CH}_3\text{CH}_2\text{CHCl}_2 \xrightarrow{\text{Aq. KOH}}$?

[IIT 1991]

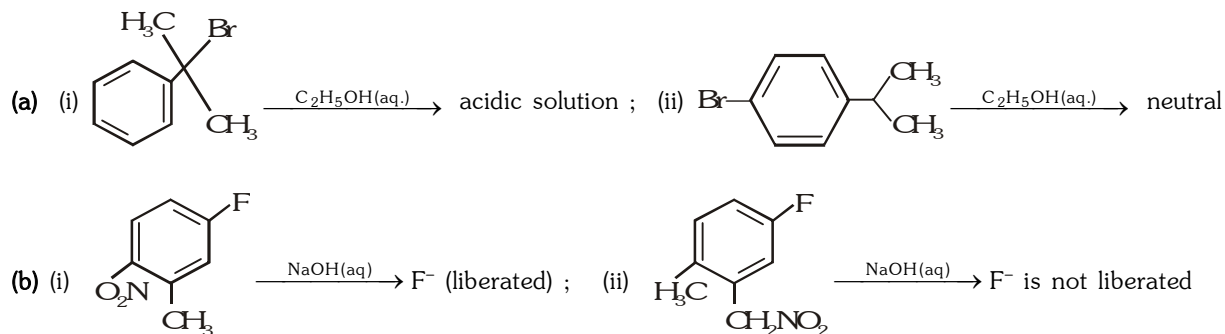
25. Draw the stereochemical structure of the products in the following reaction.

[IIT 1994]



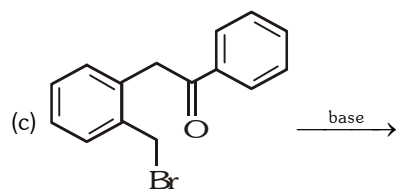
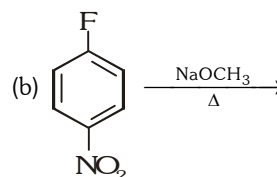
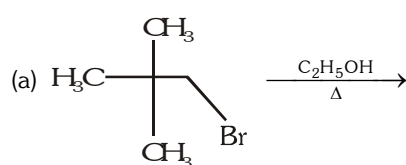
26. Give reasons :

[IIT 2005]



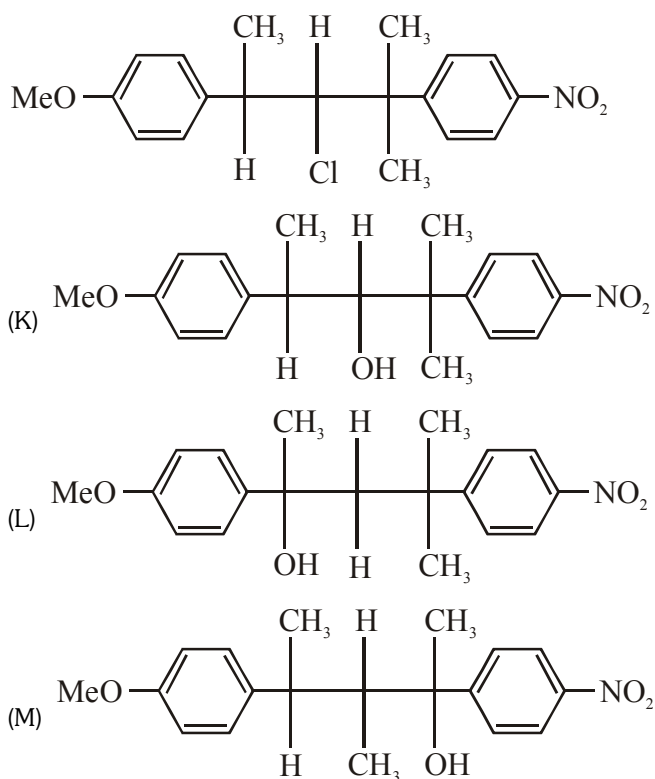
27. An alkyl halide (A) of formula $\text{C}_6\text{H}_{11}\text{Cl}$ on treatment with potassium tertiary butoxide gives two isomeric alkenes (B) and (C) C_6H_{10} . Both alkene on hydrogenation give methycyclopentane. Predict the structure of (A), (B) and (C).

28. Which would be the major product in each of the following reactions ?



29. The following compound on hydrolysis in aqueous acetone will give:

[IIT 2005]



It mainly gives

(A) K and L

(B) Only K

(C) L and M

(D) Only M

30. Match the following:

[IIT 2006]

Column I

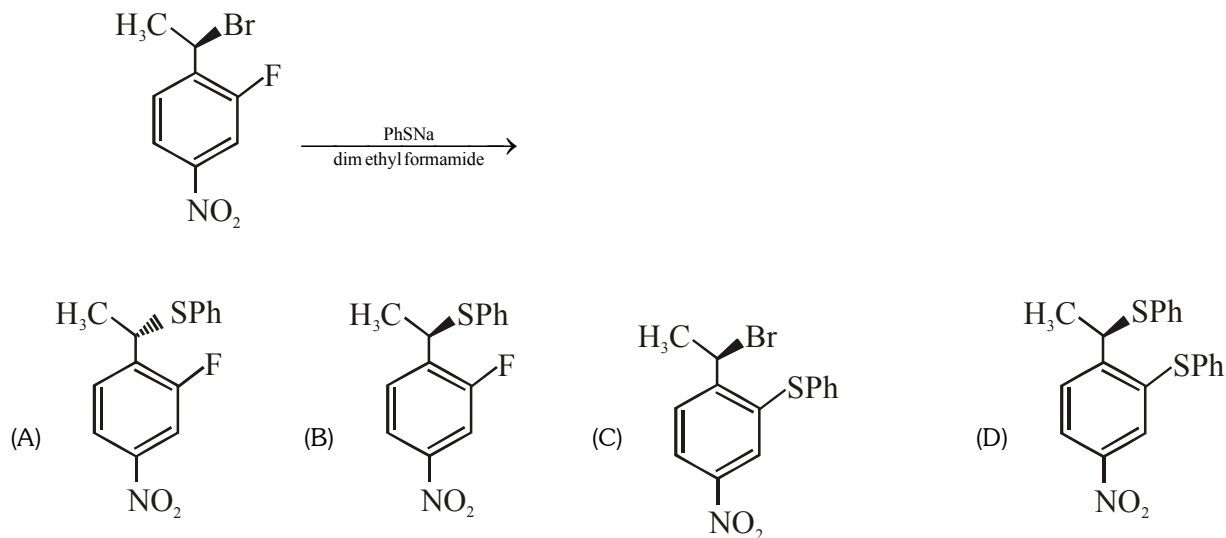
- (A) $\text{CH}_3\text{-CHBr-CD}_3$ on treatment with alc. KOH gives $\text{CH}_2=\text{CH-CD}_3$ as a major product.
- (B) Ph-CHBr-CH_3 reacts faster than Ph-CHBr-CD_3 .
- (C) $\text{Ph-CD}_2\text{-CH}_2\text{Br}$ on treatment with $\text{C}_2\text{H}_5\text{OD/C}_2\text{H}_5\text{O}^-$ gives Ph-CD=CH_2 as the major product.
- (D) $\text{PhCH}_2\text{CH}_2\text{Br}$ and $\text{PhCD}_2\text{CH}_2\text{Br}$ react with same rate.

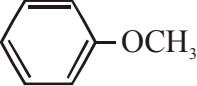
Column II

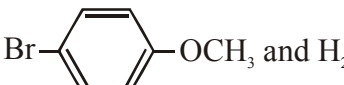
- (P) E1 reaction
- (Q) E2 reaction
- (R) E1cb reaction
- (S) First order reaction

31. The major product of the following reaction is

[IIT 2008]

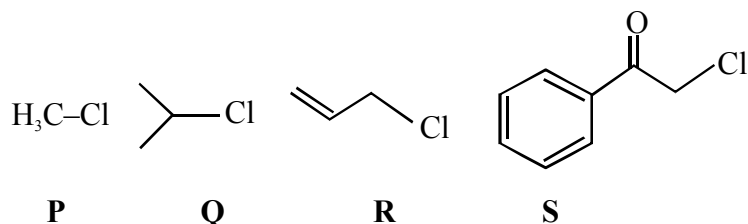


32. In the reaction  $\xrightarrow{\text{HBr}}$ the products are [IIT 2010]

- (A)  and H_2 (B)  and CH_3Br
- (C)  and CH_3OH (D)  and CH_3Br

33. The total number of alkenes possible by dehydrobromination of 3-bromo-3-cyclopentylhexane using alcoholic KOH is [IIT-2011]

34. KI in acetone, undergoes $\text{S}_{\text{N}}2$ reaction with each of P, Q, R and S. The rates of the reaction vary as - [IIT 2013]

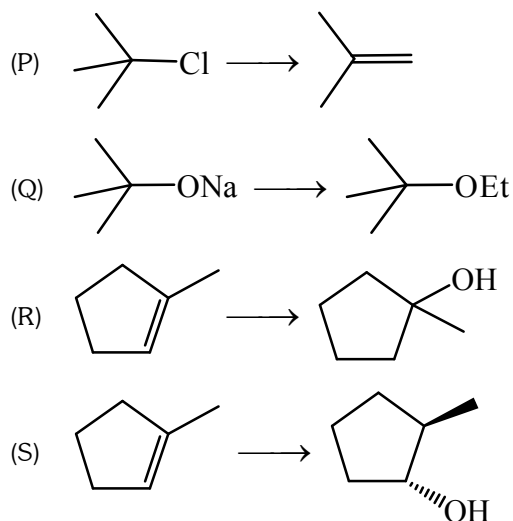


- (A) $\text{P} > \text{Q} > \text{R} > \text{S}$ (B) $\text{S} > \text{P} > \text{R} > \text{Q}$
- (C) $\text{P} > \text{R} > \text{Q} > \text{S}$ (D) $\text{R} > \text{P} > \text{S} > \text{Q}$

35. Match the chemical conversions in List-I with the appropriate reagents in List-II and select the correct answer using the code given below this lists - [IIT 2013]

List-I

List-II



(1) (i) $\text{Hg}(\text{OAc})_2$ (ii) NaBH_4

(2) NaOEt

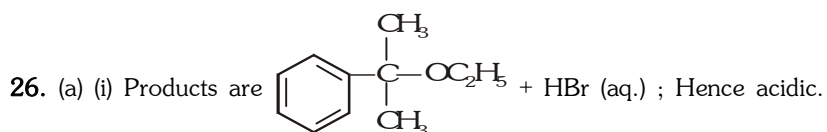
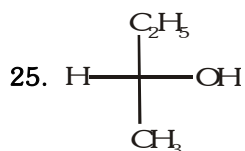
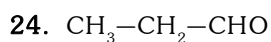
(3) Et-Br

(4) (i) BH_3 (ii) $\text{H}_2\text{O}_2 / \text{NaOH}$

Codes :

- | | P | Q | R | S |
|-----|---|---|---|---|
| (A) | 2 | 3 | 1 | 4 |
| (B) | 3 | 2 | 1 | 4 |
| (C) | 2 | 3 | 4 | 1 |
| (D) | 3 | 2 | 4 | 1 |

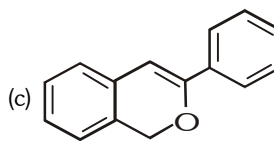
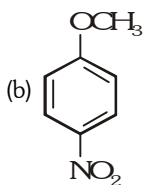
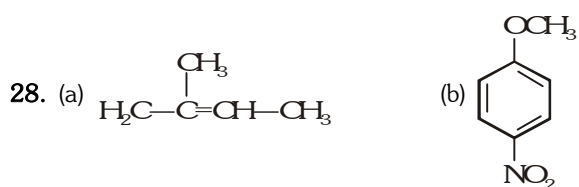
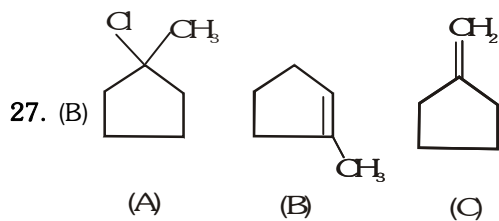
1. (D) 2. (B) 3. (A) 4. (A) 5. (B) 6. (D) 7. (A) 8. (B)
 9. (A) 10. (C) 11. (A) 12. (A) 13. (D) 14. (D) 15. (C) 16. (D)
 17. (D) 18. (C) 19. (B) 20. (B) 21. (A) 22. (B) 23. (B)



(ii) No reaction takes place, as aryl halide is inert to nucleophilic substitution.

(b) (i) —NO_2 group facilitates the nucleophilic attack because it stabilised the anion formed when attack of OH^- takes place.

(ii) Does not show reaction.



29. A, C 30. D 31. C 32. D 33. A
 34. B 35. A

EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. For a catalyst which condition is not essential :
(A) Variable valency (B) High ionisation energy
(C) Empty orbitals (D) Free valency on the surface
2. Which element is having lowest melting and boiling point :
(A) Ti (B) Cu (C) Zn (D) Mn
3. Density of which of the following element is highest :
(A) Pt (B) Hg (C) Mn (D) Cu
4. The adsorption of hydrogen by platinum black is called :
(A) hydrogenation (B) Reduction (C) Occlusion (D) Hydration
5. To which of the following series the transition element from $Z = 39$ to $Z = 48$ belong :
(A) 3d series (B) 4d series (C) 5d series (D) 6d series
6. Permanent magnets are generally made of alloys of :
(A) Co (B) Zn (C) Mn (D) Pb
7. The reaction of O_2 and CO with haemoglobin gives :
(A) Only oxygen-haeme complex
(B) Only CO-haeme complex
(C) Both but oxygen-haeme-complex is more stable
(D) Both but CO-haeme-complex is more stable
8. Which of the following statement is not correct :
(A) Fe, Ni, Co form interstitial compound
(B) $CuSO_4 + Ca(OH)_2$ is called Bordeaux mixture
(C) Verdigris is basic copper acetate $[Cu(COOCH_3)_2Cu(OH)_2]$
(D) 24 carat gold is an alloy of Au and Cu
9. Chromium forms most stable compound in the following oxidation state :
(A) Cr (I) (B) Cr (II) (C) Cr (III) (D) Cr (iv)
10. Not more than one oxidation state is show by :
(A) Mn (B) Cr (C) Fe (D) Sc
11. Which pair of ions is colourless :
(A) Mn^{+3} , CO^{+3} (B) Fe^{+3} , Cr^{+3} (C) Zn^{2+} , Sc^{3+} (D) Ti^{2+} , Cu^{2+}
12. Which of the following set of metals can form alloy :
(A) Cu – Au (B) Li–Na (C) Fe–Hg (D) All
13. Which of the following is ferromagnetic :
(A) Cu, Ag, Au (B) Fe, Co, Ni (C) Zn, Cd, Hg (D) Ca, Sr, Ba
14. The number of moles of acidified $KMnO_4$ required to convert sulphite ion into sulphate ion is :
(A) $2/5$ (B) $3/5$ (C) $4/5$ (D) 1

15. $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \xrightleftharpoons{\text{Fe} + \text{Mo}} 2\text{NH}_3(\text{g})$; Haber's process, Mo is used as :
 (A) A catalyst (B) A catalytic promoter
 (C) An oxidising agent (D) As a catalytic poison
16. $\text{Cr}_2\text{O}_7^{2-} \xrightleftharpoons[\text{Y}]{\text{X}} 2\text{CrO}_4^{2-}$, X and Y are respectively :
 (A) $\text{X} = \text{OH}^-$, $\text{Y} = \text{H}^+$ (B) $\text{X} = \text{H}^+$, $\text{Y} = \text{OH}^-$
 (C) $\text{X} = \text{OH}^-$, $\text{Y} = \text{H}_2\text{O}_2$ (D) $\text{X} = \text{H}_2\text{O}_2$, $\text{Y} = \text{OH}^-$
17. The magnetism of Ni^{2+} ion is :
 (A) $\frac{eh}{4\pi m}$ (B) $\sqrt{6} \left(\frac{eh}{4\pi m} \right)$
 (C) $\sqrt{8} \left(\frac{eh}{4\pi m} \right)$ (D) $\sqrt{10} \left(\frac{eh}{4\pi m} \right)$
18. During estimation of oxalic acid Vs KMnO_4 , self indicator is :
 (A) KMnO_4 (B) Oxalic acid (C) K_2SO_4 (D) MnSO_4
19. Bayer's reagent used to detect olifinic double bond is :
 (A) Acidified KMnO_4 (B) Aqueous KMnO_4
 (C) 1% alkaline KMnO_4 solution (D) KMnO_4 in benzene
20. $\text{Cu} + \text{conc. HNO}_3 \xrightarrow{\text{(hot)}} \text{Cu}(\text{NO}_3)_2 + \text{X}$ (oxide of nitrogen); then X is :
 (A) N_2O (B) NO_2 (C) NO (D) N_2O_3
21. The higher oxidation states of transition elements are found to be the combination with A and B, which are:
 (A) F, O (B) O, N (C) O, Cl (D) F, Cl
22. Manganese steel is used for making railways because :
 (A) It is hard with high percentage of Mn
 (B) It is soft with high percentage of Mn
 (C) It is hard with small concentration of manganese with the impurities
 (D) It is soft with small concentration of manganese with impurities.
23. 'Bordeaux mixture' is used as a fungicide. It is a mixture of :
 (A) $\text{CaSO}_4 + \text{Cu}(\text{OH})_2$ (B) $\text{CuSO}_4 + \text{Ca}(\text{OH})_2$
 (C) $\text{CuSO}_4 + \text{CaO}$ (D) $\text{CuO} + \text{CaO}$
24. Anhydrous ferric chloride is prepared by :
 (A) heating hydrated ferric chloride at a high temperature in a stream of air
 (B) heating metallic iron in a stream of dry chlorine gas
 (C) reaction of ferric oxide with HCl (aq)
 (D) reaction of metallic iron with HCl (aq)

25. The product of oxidation of I^- with MnO_4^- in alkaline medium is :
 (A) IO_3^- (B) I_2 (C) IO^- (D) IO_4^-
26. Copper becomes green when exposed to moist air for longer period.
 (A) Because of the formation of a layer of cupric oxide on the surface of copper
 (B) Because of the formation of a layer of basic carbonate of copper on the surface of copper
 (C) Because of the formation of a layer of cupric hydroxide on the surface of copper.
 (D) (A) and (C) both
27. $\text{CuFeS}_2 + \text{O}_2 \text{ (excess)} \xrightarrow{\Delta} \text{X(s)} + \text{Y(s)}$ (Unbalanced Equation) :
 Which is correct choice for (X) and (Y) ?
 (A) (X) liberates iodine from soluble iodides like KI
 (B) (Y) on heating liberates SO_3 only
 (C) (X) forms bluish white ppt. with sodium hydroxide which redissolves in excess of sodium hydroxide.
 (D) (Y) on reaction with potassium ferricyanide gives brown ppt.
28. $\text{FeCl}_3 \cdot 6\text{H}_2\text{O} + \text{C}(\text{CH}_3)_2 (\text{CH}_3\text{O})_2 \rightarrow \text{Products}$
 Reaction products are
 (A) FeCl_3 , CH_3OH and CH_3COCH_3 (B) $(\text{CH}_3\text{O})_3\text{Fe}$, HCl and H_2O
 (C) FeCl_2 , HCl and CH_3COCH_3 (D) $\text{Fe}(\text{OH})_3$, FeCl_3 and CH_3COCH_3
29. Which of the following statements is incorrect ?
 (A) Transition element exhibit higher enthalpies of atomization as they have stronger interatomic interaction
 (B) IE_2 of ${}_{23}\text{V} < {}_{24}\text{Cr} > {}_{25}\text{Mn}$ and ${}_{28}\text{Ni} < {}_{29}\text{Cu} > {}_{30}\text{Zn}$
 (C) Ni (II) compounds are more stable than pt(II) where as pt(IV) compounds are more stable than Ni (IV)
 (D) The elements which gives the greatest number of oxidation states does not occur in or near the middle of the series
30. Stability of Cu^+ and Ag^+ halide complexes are in order :
 (A) $\text{I} > \text{Br} > \text{Cl} > \text{F}$ (B) $\text{F} > \text{Cl} > \text{Br} > \text{I}$ (C) $\text{Cl} > \text{F} > \text{I} > \text{Br}$ (D) $\text{Br} > \text{I} > \text{Cl} > \text{F}$
31. When KI (excess) is added to :
 I : CuSO_4 II. HgCl_2 III. $\text{Pb}(\text{NO}_3)_2$
 (A) A white ppt. of CuI in I, an orange ppt. HgI_2 in II and a yellow ppt. of PbI_2 in III
 (B) A white ppt. of CuI in I, an orange ppt. dissolving to HgI_4^{2-} in II, and a yellow ppt. of PbI_2 in III
 (C) A white ppt. of CuI , HgI_2 and PbI_2 in each case
 (D) none is correct
32. CuSO_4 can be estimated volumetrically :
 (A) by reaction with KI followed by reaction with $\text{Na}_2\text{S}_2\text{O}_3$
 (B) by reaction with BaCl_2
 (C) by reaction with $\text{K}_4[\text{Fe}(\text{CN})_6]$
 (D) none is correct

33. Cl_2 gas is obtained by various reactions but not by :
- (A) $\text{KMnO}_4 + \text{conc. HCl} \xrightarrow{\Delta}$ (B) $\text{KCl} + \text{K}_2\text{Cr}_2\text{O}_7 + \text{conc. H}_2\text{SO}_4 \xrightarrow{\Delta}$
- (C) $\text{MnO}_2 + \text{conc. HCl} \xrightarrow{\Delta}$ (D) $\text{KCl} + \text{F}_2 \xrightarrow{\Delta}$
34. Maximum magnetic moment is shown by :
- (A) d^5 (B) d^6 (C) d^7 (D) d^8
35. A jeweller is selling 22-carat gold articles with 95% purity, it is approximately :
- (A) exact (B) 3.5% higher (C) 3.5% lower (D) 5% lower
36. The basic character of the transition metal monoxides follows the order :
- (Atomic no. Ti = 22, V = 23, Cr = 24, Fe = 26)
- (A) $\text{TiO} > \text{FeO} > \text{VO} > \text{CrO}$ (B) $\text{TiO} > \text{VO} > \text{CrO} > \text{FeO}$
- (C) $\text{VO} > \text{CrO} > \text{TiO} > \text{FeO}$ (D) $\text{CrO} > \text{VO} > \text{FeO} > \text{TiO}$
37. AgCl on fusion with sodium carbonate, gives :
- (A) Ag_2CO_3 (B) Ag_2O (C) Ag (D) Ag_2C_2
38. Which of the following reactions is used to estimate copper volumetrically ?
- (A) $2\text{Cu}^{2+} + 4\text{CN}^- \longrightarrow \text{Cu}_2(\text{CN})_2 + (\text{CN})_2$
- (B) $\text{Cu}^{2+} + 4\text{NH}_3 \longrightarrow [\text{Cu}(\text{NH}_3)_4]^{2+}$
- (C) $2\text{Cu}^{2+} + 2\text{CNS}^- + \text{SO}_2 + 2\text{H}_2\text{O} \longrightarrow \text{Cu}_2(\text{CNS})_2 + \text{H}_2\text{SO}_4 + 2\text{H}^+$
- (D) $2\text{Cu}^{2+} + 4\text{I}^- \longrightarrow \text{Cu}_2\text{I}_2 + \text{I}_2$
39. Atomic size of gold is almost the same as that of silver. It is due to :
- (A) the same crystal structure of silver and gold
- (B) almost the same electropositive character of the two metals
- (C) transition metals contraction in a series
- (D) the effect of lanthanide contraction
40. Which of the following compounds is most sensitive of light ?
- (A) AgCl (B) AgCN (C) AgI (D) AgBr

CHECK YOUR GRASP					ANSWER KEY						EXERCISE-1				
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	B	C	A	C	B	A	D	D	C	D	C	A	B	A	B
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	A	C	D	C	B	A	A	B	B	A	B	A	A	A	A
Que.	31	32	33	34	35	36	37	38	39	40					
Ans.	B	A	B	A	B	B	C	D	D	D					

EXERCISE-02**BRAIN TEASERS****SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)**

1. An inorganic salt is lemon yellow in colour. It becomes orange in colour like methyl orange when it is acidic and again becomes yellow when it is alkaline. The inorganic salt will be :
(A) Copper nitrate (B) Ferric chloride
(C) Potassium chromate (D) Potassium ferri cyanide
2. Magnetic moment $\sqrt{35}$ is true for which of the following pair :
(A) Co^{+2} , Fe^{+2} (B) Fe^{+3} , Mn^{+2} (C) Co^{+3} , Cr^{+2} (D) Fe^{+2} , Mn^{+2}
3. Fe^{+3} is more stable than Fe^{+2} , the reason is/are :
(A) 1st and 2nd I.P. difference is less than 11.0 eV
(B) Core of Fe^{+3} is more stable
(C) 2nd and 3rd IP difference is less than 11.0 eV
(D) IP of Fe^{+3} is high
4. Addition of non-metals like B and C to the interstitial sites of a transition metal results the metal :
(A) of more ductability (B) of less ductability
(C) Less malleable (D) of more hardness
5. The metal(s) which does/do not form amalgam is/are :
(A) Fe (B) Pt (C) Zn (D) Ag
6. Correct statement(s) is/are :
(A) An acidified solution of $\text{K}_2\text{Cr}_2\text{O}_7$ liberates iodine from KI
(B) $\text{K}_2\text{Cr}_2\text{O}_7$ is used as a standard solution for estimation of Fe^{2+} ions
(C) In acidic medium, $N = M/6$ for $\text{K}_2\text{Cr}_2\text{O}_7$
(D) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ on heating decomposes to yield Cr_2O_3 through an endothermic reaction
7. The highest oxidation state shown by transition element is :
(A) + 7 by Mn (B) + 8 by Os (C) + 8 by Ru (D) + 7 by Fe
8. A compound of mercury used in cosmetics, in Ayurvedic and Yunani medicines and known as Vermilion is :
(A) HgCl_2 (B) HgS (C) Hg_2Cl_2 (D) HgI
9. Acidified chromic acid + $\text{H}_2\text{O}_2 \longrightarrow \text{X} + \text{Y}$, X and Y are :
(Blue colour)
(A) CrO_5 and H_2O (B) Cr_2O_3 and H_2O
(C) CrO_2 and H_2O (D) CrO and H_2O
10. $\uparrow \text{Y(g)} \xleftarrow{\text{KI}} \text{CuSO}_4 \xrightarrow{\text{dil H}_2\text{SO}_4} \text{X(Blue colour)}$, X and Y are :
(A) $\text{X} = \text{I}_2$, $\text{Y} = [\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ (B) $\text{X} = [\text{Cu}(\text{H}_2\text{O})_4]^{2+}$, $\text{Y} = \text{I}_2$
(C) $\text{X} = [\text{Cu}(\text{H}_2\text{O})_4]^+$, $\text{Y} = \text{I}_2$ (D) $\text{X} = [\text{Cu}(\text{H}_2\text{O})_5]^{2+}$, $\text{Y} = \text{I}_2$
11. $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ (Ammonium dichromate) is used in fire works. The green coloured powder blown in air is :
(A) Cr_2O_3 (B) CrO_2 (C) Cr_2O_4 (D) CrO_3

12. Iron becomes passive by.....due to formation of..... :

- (A) dil. HCl, Fe₂O₃ (B) Aqua-regia, Fe₃O₄
 (C) conc. H₂SO₄, Fe₃O₄ (D) conc. HCl, Fe₃O₄



13. $\begin{array}{l} \xrightarrow{+ ye^- \text{ (Acidic medium)}} \text{Mn}^{+2} \\ \xrightarrow{+ ze^- \text{ (Natural medium)}} \text{MnO}_2 \end{array}$ x, y and z are respectively :

- (A) 1, 2, 3 (B) 1, 5, 3 (C) 1, 3, 5 (D) 5, 3, 1

14. In the equation : $\text{M} + 8 \text{CN}^- + 2\text{H}_2\text{O} + \text{O}_2 \longrightarrow 4[\text{M}(\text{CN})_2]^- + 4\text{OH}^-$, metal M is :

- (A) Ag (B) Au (C) Cu (D) Hg

15. Amongst CuF₂, CuCl₂ and CuBr₂ :

- (A) Only CuF₂ is ionic
 (B) Both CuCl₂ and CuBr₂ are covalent
 (C) CuF₂ and CuCl₂ are ionic but CuBr₂ is covalent
 (D) CuF₂, CuCl₂ as well as CuBr₂ are ionic

16. A metal M which is not affected by strong acids like conc. HNO₃, conc. H₂SO₄ and conc. solution of alkalis like NaOH, KOH forms MCl₃ which finds use for toning in photography. The metal M is :

- (A) Ag (B) Hg (C) Au (D) Cu

17. $\text{CuSO}_4(\text{aq.}) + 4\text{NH}_3 \longrightarrow \text{X}$, then X is :

- (A) [Cu(NH₃)₄]²⁺ (B) Paramagnetic
 (C) Coloured (D) Of a magnetic moment of 1.73 BM

18. Number of moles of SnCl₂ required for the reduction of K₂Cr₂O₇ into Cr₂O₃ is (in acidic medium) :

- (A) 3 (B) 2 (C) 1 (D) 1/3

19. Acidified KMnO₄ can be decolourised by :

- (A) SO₂ (B) H₂O₂ (C) FeSO₄ (D) FeCl₃

20. The Ziegler-Natta catalyst used for polymerisation of ethene and styrene is TiCl₄ + (C₂H₅)₃Al, the catalysing species (active species) involved in the polymerisation is :

- (A) TiCl₄ (B) TiCl₃ (C) TiCl₂ (D) TiCl

21. Which of the following reaction is possible at anode?

- (A) $2\text{Cr}^{3+} + 7\text{H}_2\text{O} \longrightarrow \text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+$ (B) $\text{F}_2 \longrightarrow 2\text{F}^-$
 (C) $\frac{1}{2}\text{O}_2 + 2\text{H}^+ \longrightarrow \text{H}_2\text{O}$ (D) None of these

22. Colourless solution of the following four salts are placed separately in four different test tubes and a strip of copper is dipped in each one of these. Which solution will turn blue?

- (A) KNO₃ (B) AgNO₃ (C) Zn(NO₃)₂ (D) ZnSO₄

23. Cuprous chloride is obtained from cupric chloride :

- (A) By heating cupric chloride with chlorine
 (B) By the electrolysis of cupric chloride containing HCl
 (C) By heating cupric chloride with conc. HCl and copper turnings
 (D) By passing H₂ over CuCl₂

24. When excess of sodium thiosulphate is added to dil. AgNO_3 solution a soluble compound X is formed. However, when dil. $\text{Na}_2\text{S}_2\text{O}_3$ solution is added to conc. AgNO_3 solution a white ppt. turning yellow and finally black ppt. of Y is obtained. Which is correct pair. :

(A) X is Ag_2S and Y is $\text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2]$
 (B) X is $\text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2]$ and Y is Ag_2S
 (C) X is $\text{Ag}_2\text{S}_2\text{O}_3$ and Y is Ag_2S
 (D) X is $\text{Ag}_2\text{S}_2\text{O}_3$ and Y is $\text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2]$

25. Carat is a measure of :

(A) Impure gold (B) Purity of gold
 (C) Pure gold (D) None of these

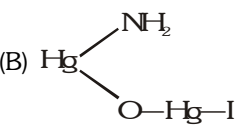
26. Fulminating gold is :

(A) CuFeS_2 (B) FeS_2
 (C) $\text{Au}(\text{NH}_2) = \text{NH}$ or AuN_2H_3 (D) AuCl_3

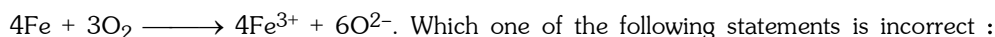
27. In the reaction $\text{HgCl}_2 + 4\text{KI} \longrightarrow \text{A} + 2\text{KCl}$, A is :

(A) HgI_2 (B) K_2HgI_3 (C) K_2HgI_4 (D) KHgI_3

28. Iodide of Millon's base is :

(A) $\text{K}_2[\text{HgI}_4]$ (B) 
 (C) $[\text{Hg}_2\text{O} \cdot \text{NH}_2\text{OH}] \cdot \text{H}_2\text{O}$ (D) $\text{Hg}(\text{NH}_2)\text{I} + \text{Hg}$

29. The following reaction describes the rusting of iron,



(A) This is an example of a redox reaction
 (B) Metallic iron is reduced to Fe^{2+}
 (C) Fe^{3+} is an oxidising agent
 (D) Metallic iron is a reducing agent

30. Which of the following chemical reaction(s) is (are) involved in the developing of photographic plate ?

(A) $\text{C}_6\text{H}_4(\text{OH})_2 + 2\text{AgBr} \longrightarrow 2\text{Ag} + \text{C}_6\text{H}_4\text{O}_2 + 2\text{HBr}$
 (B) $\text{AgBr} + 2\text{Na}_2\text{S}_2\text{O}_3 \longrightarrow \text{Na}_3[\text{AgS}_2\text{O}_3] + \text{NaBr}$
 (C) $\text{AgBr} + 2\text{NH}_3(\text{aq}) \longrightarrow [\text{Ag}(\text{NH}_3)_2] \text{Br}$
 (D) $2\text{AgBr} + \text{Na}_2\text{S}_2\text{O}_3 \longrightarrow \text{Ag}_2\text{S}_2\text{O}_3 + 2\text{NaBr}$

31. Which of the following statement(s) is (are) not correct with reference to ferrous and ferric ions

(A) Fe^{3+} gives brown colour with potassium ferricyanide
 (B) Fe^{2+} gives blue ppt with potassium ferricyanide
 (C) Fe^{3+} gives red colour with potassium sulphocyanide
 (D) Fe^{2+} gives brown colour with potassium sulphocyanide

32. KMnO_4 on treatment with conc. H_2SO_4 forms a compound (X) which decomposes explosively on heating forming (Y). The (X) and (Y) are respectively :
- (A) Mn_2O_7 , MnO_2 (B) Mn_2O , Mn_2O_3 (C) MnSO_4 , Mn_2O_3 (D) Mn_2O_3 , MnO_2
33. The hydrated cupric chloride is strongly heated. Which of the following statement(s) is (are) correct for this :
- (A) It is reduced to Cu_2Cl_2 (B) Cupric oxide is formed along with Cu_2Cl_2
 (C) Only Cl_2 is liberated (D) Cl_2 and HCl both are liberated
34. Select the incorrect statement(s) :
- (A) In K_2MnO_4 and CrO_2Cl_2 , The central transition metals/ions have the same oxidation state.
 (B) Both sodium and potassium dichromate can be used as primary standard in volumetric estimations.
 (C) Potassium dichromate on strong heating evolves oxygen gas and forms green powder.
 (D) Potassium permanganate on heating with solid KOH evolves oxygen gas and forms a black powder.
35. Cuprous chloride can not be prepared :
- (A) by passing SO_2 through the solution containing CuSO_4 and NaCl
 (B) by heating excess of copper with conc. HCl in presence of a little KClO_3
 (C) by boiling copper sulphate solution with excess of copper turnings in presence of hydrochloric acid
 (D) by dissolving cupric oxide or copper carbonate in conc. HCl
36. Which of the following ion is not coloured ?
- (A) $\text{Ni}(\text{DMG})_2$ (B) $[\text{Co}(\text{SCN})_4]^{2-}$ (C) $[\text{Fe}(\text{H}_2\text{O})_5\text{SCN}]^{2+}$ (D) $[\text{Al}(\text{OH})_4]^-$
37. The total spin and paramagnetism (B.M.) of ferrocyanide ion are respectively :
- (A) $0, 2\sqrt{6}$ (B) $5/2, \sqrt{35}$ (C) $2, \sqrt{24}$ (D) $1, 2\sqrt{2}$
38. Which is not correct about $\text{FeO}_{0.94}$?
- (A) It is non-stoichiometric compound
 (B) Some of Fe^{2+} ions get replaced by as many two third Fe^{3+} ions
 (C) It is metal excess solid
 (D) It is metal deficient solid.
39. The transition metal used in X-rays tube is :
- (A) Mo (B) Ta (C) Tc (D) Pm
40. The reagent used in the gravimetric analysis of Co^{2+} ion is :
- (A) DMG (B) α -nitro- β -naphthol (C) β -nitro- α -naphthol (D) $\text{K}_4[\text{Fe}(\text{CN})_6]$

BRAIN TEASERS					ANSWER KEY						EXERCISE-2				
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	C	B	B, C	B,C,D	A,B	A,B,C	B,C	B	B	B	A	B	B	B	A,B
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	C	A,B,C,D	A	A,B,C	B	A	B	C	B	B	C	C	B	B	A,B
Que.	31	32	33	34	35	36	37	38	39	40					
Ans.	D	A	A,B,D	B,D	D	D	C	C	A	B					

EXERCISE-03**MISCELLANEOUS TYPE QUESTIONS****TRUE / FALSE**

1. Fe_3O_4 is mixed oxide of FeO and Fe_2O_3 .
2. Acidic potassium permanganate is used in organic chemistry under the name Baeyer's reagent.
3. Cast iron has the maximum percentage of carbon.
4. No rusting takes place in absence of moisture.
5. Silver is found in nature both in native and combined state.
6. Copper dissolves in hydrochloric acid.

FILL IN THE BLANKS

1.ion show maximum paramagnetic character among the bivalent ions of first transition on series.
2. The salts and are isostructural ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, $\text{MnSO}_4 \cdot 4\text{H}_2\text{O}$, $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$)
3. Which one of the following CuCl_2 , CaCl_2 or CdCl_2 is paramagnetic in character
4. The chromate ion in acidic medium changes to
5. Among the ion, Sc^{3+} , V^{3+} and Cr^{3+} , the ion that gives colourless compound is.....

MATCH THE COLUMN

1.

Column-I (Metals)		Column-II (Ores)	
(A)	Zn	(p)	Cyanide process
(B)	Cu	(q)	hydrometallurgical process
(C)	Ag	(r)	roasting
(D)	Pt	(s)	brass.

2.

Column-I (Alloys)		Column-II (Constituents)	
(A)	TiCl ₄	(p)	Adams catalyst in reduction
(B)	PdCl ₂	(q)	In preparation of (CH ₃) ₂ SiCl ₂
(C)	Pt/PtO	(r)	Used as the Natta catalyst in polythene production
(D)	Cu	(s)	Wake process for converting C ₂ H ₄ to CH ₃ CHO

ASSERTION & REASON QUESTIONS

These questions contains, Statement-I (assertion) and Statement-II (reason).

- (A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I
 (B) Statement-I is True, Statement-II is True ; Statement-II is NOT a correct explanation for Statement-I
 (C) Statement-I is True, Statement-II is False.
 (D) Statement-I is False, Statement-II is True.

1. **Statement-I** : General formula of transition element is $(n-1)d^{1-10} ns^{1-2}$
Because
Statement-II : Transition element have 10 electrons in $(n-1)d$ orbitals.

2. **Statement-I** : 1st ionisation potential of mercury is greater than cadmium.
Because
Statement-II : Hg has stable electronic configuration ($5d^{10} 6s^2$).
3. **Statement-I** : Cu^+ is more stable than Cu^{+2} .
Because
Statement-II : ΔIP is greater than 16 eV
4. **Statement-I** : AgI is coloured while AgF is colourless.
Because
Statement-II : Unpaired electron is present in AgI
5. **Statement-I** : Change in colour of acidic solution of potassium dichromate by breath is used to test drunk drivers.
Because
Statement-II : Change in colour is due to the complexation of alcohol with potassium dichromate.
6. **Statement-I** : K_2CrO_4 has yellow colour due to charge transfer.
Because
Statement-II : CrO_4^{2-} ion is tetrahedral in shape.
7. **Statement-I** : CrO_3 reacts with HCl to form chromyl chloride gas.
Because
Statement-II : Chromyl chloride (CrO_2Cl_2) has tetrahedral shape.
8. **Statement-I** : Zinc does not show characteristic properties of transition metals.
Because
Statement-II : In zinc outermost shell is completely filled
9. **Statement-I** : Equivalent mass of KMnO_4 is equal to one-third of its molecular mass when it acts as an oxidising agent in an alkaline medium.
Because
Statement-II : Oxidation number of Mn is +7 in KMnO_4

COMPREHENSION BASED QUESTIONS

Comprehension # 1

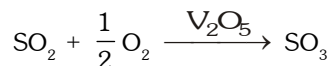
Hg shows two oxidation states (I) and (II) in its compounds. Chemistry of Hg(I) is interesting as Hg^+ does not exist. It has been proved on the basis of magnetic moment determination and equilibrium studies on Hg(I) and Hg(II) salts. It is interesting to note that Hg(I) salts are diamagnetic, yet Hg^+ contains unpaired electron. In solutions Hg_2^{2+} disproportionates as : $\text{Hg}_2^{2+} \rightleftharpoons \text{Hg}_{(l)} + \text{Hg}^{2+}$, but equilibrium constant is very low. But when anions like S^{2-} are added the formation of insoluble HgS promotes the disproportionation. The same reason is used to explain non-existence of some mercurous salts.

1. The equilibrium established when HgCl_2 is shaken with excess of Hg is :
 (A) $\text{Hg}^{2+} + \text{Hg} \rightleftharpoons 2\text{Hg}^+$ (B) $\text{Hg}^{2+} + \text{Hg}_{(l)} \rightleftharpoons \text{Hg}_2^{2+}$
 (C) $\text{Hg} + 2\text{Hg}^{2+} \rightleftharpoons \text{Hg}_2^{2+} + 2\text{Hg}^+$ (D) No equilibrium is established
2. In an experiment when placed in weak magnetic field, calomel was slightly repelled by the magnetic field. This experimental observation suggests that :
 (A) Hg^+ ion has no unpaired electron
 (B) Mercurous ion has formula Hg_2^{2+} instead of Hg^+
 (C) This experimental observation is not correct and actually mercurous salts are paramagnetic due to 6s unpaired electron
 (D) Sometimes mercurous ion may exist as Hg_2^{2+}
3. Which is well known compound :
 (A) Hg_2S (B) Hg_2O (C) Hg_2Cl_2 (D) All of the above

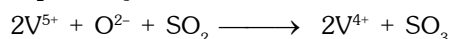
4. Hg_2Cl_2 on treatment with alkali gives red solid which is :
 (A) Hg_2O (B) HgO (C) $\text{Hg}_2(\text{OH})_2$ (D) $\text{Hg}(\text{OH})_2$

Comprehension # 2

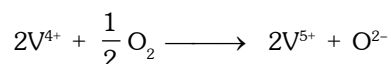
Transition metal and their compounds are used as catalysts in industry and in biological system. For example, in the Contact process, vanadium compounds in the +5 state (V_2O_5 or VO_3^-) are used to oxidise SO_2 to SO_3 :



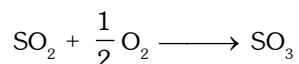
It is thought that the actual oxidation process takes place in two stages. In the first step, V^{5+} in the presence of oxide ions converts SO_2 to SO_3 . At the same time, V^{5+} is reduced to V^{4+} .



In the second step, V^{5+} is regenerated from V^{4+} by oxygen :



The overall process is, of course, the sum of these two steps:



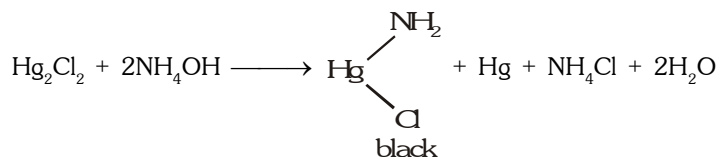
- Transition metals and their compounds catalyse reactions because :
 (A) They have completely filled s-subshell
 (B) They have a comparable size due to poor shielding of d-subshell
 (C) They introduce an entirely new reaction mechanism with a lower activation energy
 (D) They have variable oxidation states differ by two units
- During the course of the reaction :
 (A) Catalyst undergoes changes in oxidation state
 (B) Catalyst increases the rate constant
 (C) Catalyst is regenerated in its original form when the reactants form the products
 (D) All are correct.
- Catalytic activity of transition metals depends on :
 (A) Their ability to exist in different oxidation states
 (B) The size of the metal atoms
 (C) The number of empty atomic orbitals available
 (D) None of these
- Which of the following ion involved in the above process will show paramagnetism?
 (A) V^{5+} (B) V^{4+} (C) O^{2-} (D) VO_3^-

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
True / False 1. T 2. F 3. F 4. T 5. T 6. F Fill in the Blanks 1. Mn^{+2} 2. $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$, $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ 3. CuCl_2 4. $\text{Cr}_2\text{O}_7^{-2}$ 5. Sc^{+3} Match the Column 1. (A) $\rightarrow$ r,s; (B) $\rightarrow$ r,s ; (C) $\rightarrow$ p,q ; (D) $\rightarrow$ p 2. (A) $\rightarrow$ r ; (B) $\rightarrow$ s ; (C) $\rightarrow$ p ; (D) $\rightarrow$ q Assertion - Reason Questions 1.C 2.B 3.D 4. C 5. C 6.B 7.B 8.C 8.B Comprehension Based Questions Comprehension #1 : 1. B 2. B 3. C 4. B Comprehension #2 : 1. C 2. D 3. A 4. B		

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

1. What happens when excess of ammonia is added to CuSO_4 solution ?
 2. Why does AgCl dissolve in ammonia solution ?
 3. Why is that orange solution of $\text{K}_2\text{Cr}_2\text{O}_7$ turns yellow on adding NaOH to it ?
 4. Name one ore of manganese and chromium.
 5. The electronic configuration of Co^{2+} and Cu^{2+} is d^7 and d^9 respectively, which of these ions is expected to be more paramagnetic ?
 6. Write the highest oxidation state shown by element with atomic number 23
 7. One unpaired electron in atom contributes a magnetic moment of 1.1 BM. Calculate the magnetic moment of Cr (At. No. 24)
 8. Why first ionisation energy of Cu is higher than that of sodium ?
 9. Which is more stable Fe^{2+} or Fe^{3+} and why ?
 10. Why is KMnO_4 kept in dark bottles ?
 11. What is Baeyer's reagent ?
 12. Account for the following that enthalpies of atomisation of transition elements are quite high.
 13. Which d-block element is not normally considered as transition elements ?
 14. Transition metals are weak reducing agents. Why?
 15. Zinc but not copper is used for the recovery of silver from the complex $[\text{Ag}(\text{CN})_2]$. Why?
 16. Explain : Blue colour of CuSO_4 solution is discharged slowly when an iron rod is dipped into it.
 17. Explain : Green solution of potassium manganate (VI) K_2MnO_4 turns purple and a brown solid is precipitated when CO_2 is bubbled into the solution.
 18. Explain : When $\text{Mn}(\text{OH})_2$ is made by adding an alkali to a solution containing Mn^{2+} ions, the ppt quickly darkens, and eventually goes black.
 19. Explain : A deep blue colour is produced when Fe^{2+} reacts with $\text{K}_3[\text{Fe}(\text{CN})_6]$
 20. Explain : The colour of mercurous chloride changes from white to black when treated with ammonia solution.
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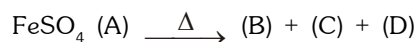
1. $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$
2. Complex formation $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$
4. Pyrolusite = MnO_2
Chromite = $\text{FeO} \cdot \text{Cr}_2\text{O}_3$
 $\text{Fe}(\text{CrO}_2)_2$
 FeCr_2O_4
5. Co^{+2} = more paramagnetic
 $d^7 = n = 3$ $\text{Cu}^{+2} = d^9 = n = 1$
6. + 5
7. $\text{Cr} = 3d^5 4s^1$
 $n = 6$
 $\mu = 6.6$
8. Small size and maximum zeff of Cu than Na
9. $\text{Fe}^{+3} > \text{Fe}^{+2}$
 $3d^5$ $3d^6$
Half filled Incomplete
10. KMnO_4 is photosensitive.
12. Strong metallic bond strength.
13. IIB or 12 or volatile metal (Zn, Cd, Hg)
14. Due to high heat of sublimation, high ionisation energy and low $E_{\text{R.P.}}^0$.
15. Zinc is cheaper as well as strong reducing agent than Cu and replaces Ag from complex.
16. Iron is above copper in electrochemical series.
17. CO_2 in aqueous solution gives bicarbonate and is acidic.
 $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3 \rightleftharpoons \text{H}^+ + \text{HCO}_3^-$
(purple by oxidation) and MnO_2 (brown solid, by reduction)
18. The black colour is due to the manganese (VI) oxide MnO_2 . It is made by the $\text{Mn}(\text{OH})_2$ being oxidized by oxygen in air
19. $\text{Fe}^{\text{II}}\text{SO}_4 + \text{K}_3[\text{Fe}^{\text{III}}(\text{CN})_6] \longrightarrow \text{KFe}^{\text{II}}[\text{Fe}^{\text{III}}(\text{CN})_6]$
Turn bull's blue (deep blue colour)
20. Hg_2Cl_2 absorbs NH_3 to form a mixture of mercury and mercuric qminochloride, black substance.



EXERCISE-04 [B]**BRAIN STORMING SUBJECTIVE EXERCISE**

1. What happens when silver sulphide is shaken with NaCN solution ?
2. Give the formula of the complex used in electroplating of object by silver.
3. What is the coordination entity formed when excess of aqueous KCN is added to an aqueous solution of copper sulphate? Why is it that no precipitate of copper sulphide is obtained when $\text{H}_2\text{S}(\text{g})$ is passed through this solution?
4. Why are the compounds of transition metal generally coloured ?
5. Why HCl not used to acidify a KMnO_4 solutions in volumetric estimations of Fe^{2+} or $\text{C}_2\text{O}_4^{2-}$?
6. Why is $\text{K}_2\text{Cr}_2\text{O}_7$ generally preferred over $\text{Na}_2\text{Cr}_2\text{O}_7$ in volumetric analysis although both are oxidising agents ?
7. K_2PtCl_6 is well known compound whereas corresponding Ni compound is not known .
Explain
8. Complete the following :
$$\text{TiCl}_4 + \text{H}_2\text{O} \rightarrow$$
9. What is the percentage of gold in 22 carat gold ?
10. Why do the transition metals exhibit higher enthalpies of atomisation ?
11. (a) Of the ions Ag^+ , Co^{2+} , Ti^{4+} , which one will be coloured in aqueous solutions.
[Atomic no : Ag = 47, Co = 27, Ti = 22]
(b) If each one of the above ionic species is in turn kept in a magnetic field, how will it respond and why ?
12. Write balanced equations for the reaction between $\text{K}_2\text{Cr}_2\text{O}_7$ and acidified solution :
(i) Ferrous sulphate
(ii) KI solution.
13. Describe the preparation of potassium permanganate. How does the acidified permanganate solution re-acts with
(a) iron (II) ions
(b) SO_2
(c) oxalic acid ?
Write the ionic equation for the reactions.
14. What happens when aqueous ammonia reacts with :
(a) Silver chloride (b) Mercury (I) chloride (c) Mercury (II) chloride ?
15. What are the equivalent weights of oxalic acid and potassium permanganate in their titration in acidic medium? (K=39, Mn=55)
16. Complete the following chemical equations :
(i) $\text{MnO}_2 + \text{KOH} + \text{O}_2 \xrightarrow{\text{Heat}}$
(ii) $\text{Cr}_2\text{O}_7^{2-} + \text{H}_2\text{S} + \text{H}^+ \longrightarrow$
(iii) $\text{I}^- + 2\text{MnO}_4^- + \text{H}_2\text{O} \longrightarrow \text{MnO}_2 + 2\text{OH}^- + \text{IO}_3^{-1}$
17. When H_2S is passed into FeCl_3 solution yellow colour of FeCl_3 changes to light green ? Explain.

18. Identify (A) to (F) in the following:



(C) and (D) are gases, (B) is red-brown solid, (C) can be oxidised to (D). (C) also turns $\text{Cr}_2\text{O}_7^{2-}/\text{H}^+$ solution green. (B) dissolves in HCl forming deep yellow solution (E). (E) gives blue colour (F) with $\text{K}_4[\text{Fe}(\text{CN})_6]$. (D) dissolves in H_2O forming a well known mineral acid called "king of acids".

19. 2.78 g of $\text{FeSO}_4 \cdot x\text{H}_2\text{O}$ is treated with excess of BaCl_2 then 2.33 g of white precipitate are obtained. Calculate the value of x. (Ba = 137, S = 32, Fe = 56, H = 1, O = 16).

20. Suggest a series of reactions, using common chemicals, by which each of the following synthesis can be performed:

- (a) $\text{Fe}(\text{OH})_3$ from FeS
- (b) BaCrO_4 from BaCO_3 and $\text{K}_2\text{Cr}_2\text{O}_7$
- (c) CrCl_3 from $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$
- (d) MnCO_3 from MnO_2

21. A white amorphous powder (A) on heating yields a colourless, non combustible gas (B) and a solid (C). The later compound assumes a yellow colour on heating and changes to white on cooling. (C) dissolves in dilute acid and the resulting solution gives a white ppt. on adding $\text{K}_4[\text{Fe}(\text{CN})_6]$ solution. (A) dissolves in dilute HCl with evolution of gas, which is identical in all respect to (B). The gas (B) turns lime water milky, but the milkiness disappears with the continuous passage of gas. The solution of (A) as obtained above gives a white precipitate (D) on the addition of excess NH_4OH and passing H_2S . Another portion of the solution initially gives a white precipitate (E) on the addition of NaOH solution, but dissolves on further addition of the base. Identify (A) to (E).

22. A compound (A) is greenish crystalline salt which gave the following results:

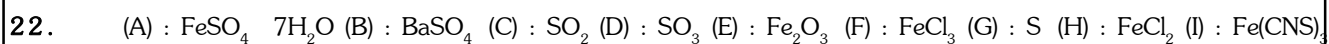
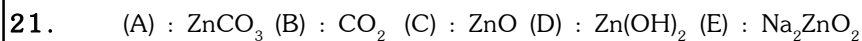
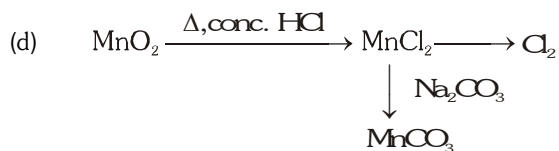
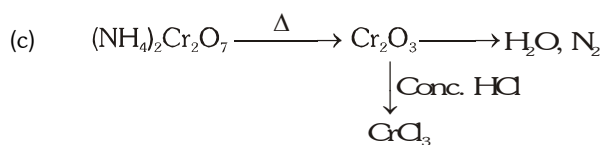
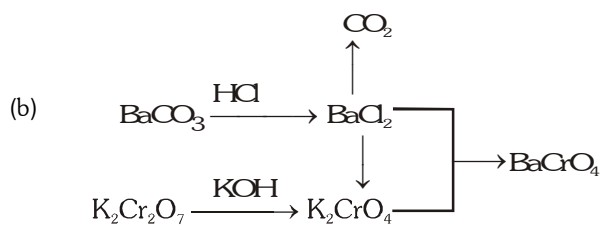
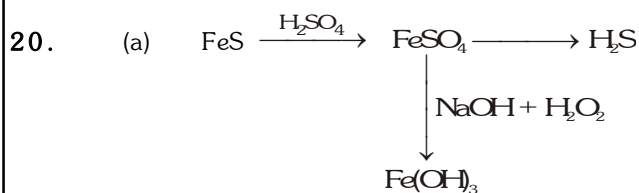
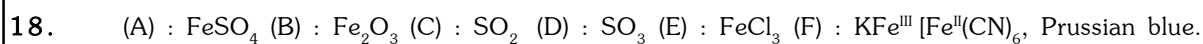
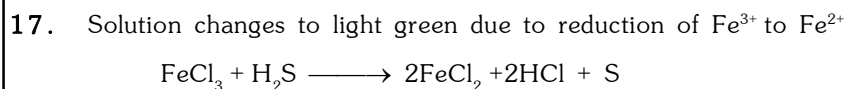
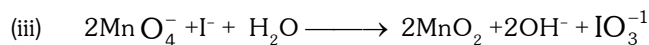
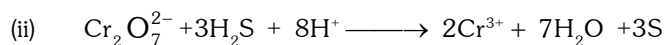
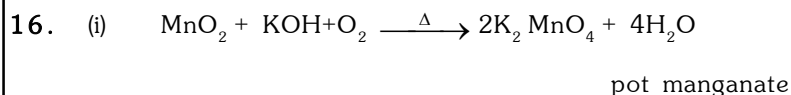
- (i) Addition of BaCl_2 solution to solution of (A) results in the formation of a white ppt. (B), which is insoluble in dil. HCl.
- (ii) On heating (A), water vapours and two oxides of sulphur, (C) and (D) are liberated leaving a red brown residue (E).
- (iii) (E) dissolves in warm conc. HCl to give a yellow solution (F).
- (iv) With H_2S , the solution (F) yields a pale yellow ppt. (G), which when filtered, leaves a greenish filtrate (H).
- (v) Solution (F) on treatment with thiocyanate ions gives blood red coloured compound (I).

Identify the substances from (A) to (I)

1. $\text{Ag}_2\text{S} + 4\text{NaCN} \rightarrow 2\text{Na}[\text{Ag}(\text{CN})_2] + \text{Na}_2\text{S}$
2. $\text{Na}[\text{Ag}(\text{CN})_2]$
3. $\text{CuSO}_4 + \text{KCN} \rightarrow \text{Cu}(\text{CN})_2$
 $2\text{Cu}(\text{CN})_2 \rightarrow 2\text{CuCN} + \text{NC-CN}$ (Cyanogen)
 $\text{CuCN} + \text{KCN} \rightarrow \text{K}_3[\text{Cu}(\text{CN})_4]$
 $\text{K}_3[\text{Cu}(\text{CN})_4] \rightarrow 3\text{K}^+ + [\text{Cu}(\text{CN})_4]^{-3} \xrightarrow{\text{H}_2\text{S}} \text{no ppt}$
4. Due to presence of unpaired electron and d-d transition
5. KMnO_4 oxidizes HCl to Cl_2
6. $\text{Na}_2\text{Cr}_2\text{O}_7$ is deliquescent in nature
7. Pt^{+4} is thermo dynamically more stable than Ni^{+4} .
8. $\text{TiCl}_4 + \text{H}_2\text{O} \longrightarrow \text{Ti}(\text{OH})_4 + 4\text{HCl}$
10. Due to strong metallic bond strength
11. (a) $\text{Ag}^+ = 5\text{d}^{10}$ $n = 0$
 $\text{Ti}^{+4} = 3\text{s}^2 3\text{p}^6$ $n = 0$
 $\text{Co}^{+2} = [\text{Ar}] 3\text{d}^7$ $n = 3$
 (b) Co^{+2} attracted by a magnetic field due to presence of unpaired electron.
12. (i) $\text{Cr}_2\text{O}_7^{-2} + 6\text{Fe}^{+2} + 14\text{H}^+ \longrightarrow 2\text{Cr}^{+3} + 6\text{Fe}^{+3} + 7\text{H}_2\text{O}$
 (ii) $\text{Cr}_2\text{O}_7^{-2} + 6\text{I}^- + 14\text{H}^+ \longrightarrow 2\text{Cr}^{+3} + 3\text{I}_2 + 7\text{H}_2\text{O}$
13. (a) $2\text{MnO}_4^- + 16\text{H}^+ + 10\text{Fe}^{+2} \longrightarrow 2\text{Mn}^{+2} + 8\text{H}_2\text{O} + 10\text{Fe}^{+3}$
 (b) $2\text{MnO}_4^- + 5\text{SO}_2 + 2\text{H}_2\text{O} \longrightarrow 5\text{SO}_4^{-2} + 2\text{Mn}^{+2} + 4\text{H}^+$
 (c) $2\text{MnO}_4^- + 16\text{H}^+ + 5\text{C}_2\text{O}_4^{-2} \longrightarrow 2\text{Mn}^{+2} + 8\text{H}_2\text{O} + 10\text{CO}_2$
14. (a) $\text{AgCl} + 2\text{NH}_4\text{OH} \longrightarrow [\text{Ag}(\text{NH}_3)_2] + 2\text{H}_2\text{O}$
 (b) $\text{Hg}_2\text{Cl}_2 + \text{NH}_3 \longrightarrow \text{Hg} + \text{HgNH}_2\text{Cl} + \text{NH}_4\text{Cl}$
 (c) $\text{HgCl}_2 + 2\text{NH}_3 \xrightarrow{\text{aq.}} \text{Hg}(\text{NH}_2)\text{Cl} + \text{NH}_4\text{Cl}$
 Mercury aminochloride
15. In acidic medium

$$\text{Equivalent weight of oxalic acid} = \frac{90}{2} = 45$$

$$\text{Equivalent weight of KMnO}_4 = \frac{158}{5} = 31.6$$



EXERCISE-05 [A]**PREVIOUS YEARS QUESTIONS**

- The radius of La^{3+} is $1.06\text{\AA}$, which of the following given values will be closest to the radius of Lu^{3+} (At no. of Lu = 71, La = 57)- [AIEEE-2003]
 - $1.6\text{\AA}$
 - $1.4\text{\AA}$
 - $1.06\text{\AA}$
 - $0.85\text{\AA}$
- What would happen when a solution of potassium chromate is treated with an excess of dilute nitric acid - [AIEEE-2003]
 - Cr^{3+} and $\text{Cr}_2\text{O}_7^{2-}$ are formed
 - $\text{Cr}_2\text{O}_7^{2-}$ and H_2O are formed
 - $\text{Cr}_2\text{O}_7^{2-}$ is reduced to +3 state of Cr
 - $\text{Cr}_2\text{O}_7^{2-}$ is oxidised to +7 state of Cr
- Cerium ($Z = 58$) is an important member of the lanthanoids. Which of the following statements about cerium is incorrect - [AIEEE-2004]
 - Cerium (IV) acts as an oxidising agent
 - The +3 oxidation state of cerium is more stable than the +4 oxidation state
 - The +4 oxidation state of cerium is not known in solutions
 - The common oxidation states of cerium are +3 and +4
- Excess of KI reacts with CuSO_4 solution and then $\text{Na}_2\text{S}_2\text{O}_3$ solution is added to it. Which of the statements is incorrect for this reaction : [AIEEE-2004]
 - Evolved I_2 is reduced
 - CuI_2 is formed
 - $\text{Na}_2\text{S}_2\text{O}_3$ is oxidised
 - Cu_2I_2 is formed
- Calomel on reaction with NH_4OH gives [AIEEE-2004]
 - HgNH_2Cl
 - $\text{NH}_2\text{-Hg-Hg-Cl}$
 - Hg_2O
 - HgO
- The lanthanoid contraction is responsible for the fact that - [AIEEE-2005]
 - Zr and Y have about the same radius
 - Zr and Nb have similar oxidation state
 - Zr and Hf have about the same radius
 - Zr and Zn have similar oxidation state
- Lanthanoid contraction is caused due to [AIEEE-2006]
 - the same effective nuclear charge from Ce to Lu
 - the imperfect shielding on outer electrons by 4f electrons from the nuclear charge
 - the appreciable shielding on outer electrons by 4f electrons from the nuclear charge
 - the appreciable shielding on outer electrons by 5d electrons from the nuclear charge
- Identify the incorrect statement among the following- [AIEEE-2007]
 - d-block elements show irregular and erratic chemical properties among themselves
 - La and Lu have partially filled d-orbitals and no other partially filled orbitals
 - The chemistry of various lanthanoids is very similar
 - 4f and 5f-orbitals are equally shielded
- The actinoids exhibit more number of oxidation states in general than the lanthanoids. This is because - [AIEEE-2007]
 - The 5f-orbitals are more buried than the 4f-orbitals
 - There is a similarity between 4f and 5f in their angular part of the wave function
 - The actinoids are more reactive than the lanthanoids
 - The 5f-orbitals extend further from the nucleus than the 4f-orbitals
- Larger number of oxidation states are exhibited by the actinides than those by the lanthanides, the main reason being [AIEEE-2008]
 - 4f orbitals more diffused than the 5f orbitals
 - lesser energy difference between 5f and 6d than between 4f and 5d orbitals
 - more energy difference between 5f and 6d than between 4f and 5d orbitals
 - more reactive nature of the actinides than the lanthanides

11. In context with the transition elements, which of the following statements is incorrect ? [AIEEE-2009]
 (1) In the highest oxidation states of the first five transition elements (Sc to Mn), all the 4s and 3d electrons are used for bonding.
 (2) Once the d^5 configuration is exceeded, the tendency to involve all the 3d electrons in bonding decreases.
 (3) In addition to the normal oxidation states, the zero oxidation state is also shown by these elements in complexes.
 (4) In the highest oxidation states, the transition metal show basic character and form cationic complexes.
12. Knowing that the chemistry of lanthanoids (Ln) is dominated by its +3 oxidation state, which of the following statements is incorrect ? [AIEEE-2009]
 (1) Ln(III) compounds are generally colourless
 (2) Ln(III) hydroxides are mainly basic in character
 (3) Because of the large size of the Ln(III) ions the bonding in its compounds is predominantly ionic in character
 (4) The ionic sizes of Ln(III) decrease in general with increasing atomic number
13. In context of the lanthanoids, which of the following statements is not correct ? [AIEEE-2011]
 (1) Because of similar properties the separation of lanthanoids is not easy
 (2) Availability of 4f electrons results in the formation of compounds in +4 state for all the members of the series
 (3) There is a gradual decrease in the radii of the members with increasing atomic number in the series
 (4) All the members exhibit +3 oxidation state
14. Iron exhibits +2 and +3 oxidation states. Which of the following statements about iron is **incorrect** ? [AIEEE-2012]
 (1) Ferrous compounds are more easily hydrolysed than the corresponding ferric compounds.
 (2) Ferrous oxide is more basic in nature than the ferric oxide.
 (3) Ferrous compounds are relatively more ionic than the corresponding ferric compounds.
 (4) Ferrous compounds are less volatile than the corresponding ferric compounds.
15. Which of the following arrangements does not represent the correct order of the property stated against it ? [JEE MAINS-2013]
 (1) $V^{2+} < Cr^{2+} < Mn^{2+} < Fe^{2+}$: paramagnetic behaviour
 (2) $Ni^{2+} < Co^{2+} < Fe^{2+} < Mn^{2+}$: ionic size
 (3) $Co^{3+} < Fe^{3+} < Cr^{3+} < Sc^{3+}$: stability in aqueous solution
 (4) $Sc < Ti < Cr < Mn$: number of oxidation states
16. Four successive members of the first row transition elements are listed below with atomic numbers. Which one of them is expected to have the highest $E_{M^{3+}/M^{2+}}^0$ value ? [JEE MAINS-2013]
 (1) Cr(Z = 24) (2) Mn(Z = 25)
 (3) Fe (Z = 26) (4) Co(Z = 27)
17. Experimentally it was found that a metal oxide has formula $M_{0.98}O$. Metal M, is present as M^{2+} and M^{3+} in its oxide. Fraction of the metal which exists as M^{3+} would be :- [JEE MAINS-2013]
 (1) 7.01% (2) 4.08%
 (3) 6.05% (4) 5.08

PREVIOUS YEARS QUESTIONS							ANSWER KEY				EXERCISE -5 [A]						
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Ans	4	2	3	2	1	3	2	4	4	2	4	1	2	1	1	4	2

EXERCISE-05 [B]**PREVIOUS YEARS QUESTIONS**

1. Anhydrous ferric chloride is prepared by – [IIT 2002]
(A) Heating hydrated ferric chloride at a high temperature in a steam of air
(B) Heating metallic iron in a steam of dry chlorine gas
(C) Reaction of ferric oxide with HCl
(D) Reaction of metallic iron with HCl
2. When MnO_2 is fused with KOH and KNO_3 , a coloured compound is formed, the product and its colour is – [IIT 2003]
(A) K_2MnO_4 , green (B) KMnO_4 , purple (C) Mn_2O_3 , brown (D) Mn_3O_4 , black
3. $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ on heating gives a gas, which is also given by – [IIT- 2004]
(A) Heating NH_4NO_2 (B) Heating NH_4NO_3 (C) $\text{Mg}_3\text{N}_2 + \text{H}_2\text{O}$ (D) $\text{Na} + \text{H}_2\text{O}_2$
4. The pair of compounds having metals in their highest oxidation state is – [IIT-2004]
(A) MnO_2 , FeCl_3 (B) $[\text{MnO}_4]^-$, CrO_2Cl_2 (C) $[\text{Fe}(\text{CN})_6]^{3-}$, $[\text{Co}(\text{CN})_3]$ (D) $[\text{NiCl}_4]^{2-}$, $[\text{CoCl}_4]^-$
5. The compound having tetrahedral geometry is – [IIT- 2004]
(A) $[\text{Ni}(\text{CN})_4]^{2-}$ (B) $[\text{Pd}(\text{CN})_4]^{2-}$ (C) $[\text{PdCl}_4]^{2-}$ (D) $[\text{NiCl}_4]^{2-}$
6. Spin only magnetic moment of the compound $\text{Hg}[\text{Co}(\text{SCN})_4]$ is – [IIT- 2004]
(A) $\sqrt{3}$ (B) $\sqrt{15}$ (C) $\sqrt{24}$ (D) $\sqrt{8}$
7. Which pair of compounds is expected to show similar colour in aqueous medium? [IIT- 2004]
(A) FeCl_3 and CuCl_2 (B) VOCl_2 and CuCl_2 (C) VOCl_2 and FeCl_2 (D) FeCl_2 and MnCl_2
8. NiCl_2 in the presence of dimethyl glyoxime (DMG) gives a complex which precipitates in the presence of NH_4OH , giving a bright red colour. [IIT- 2004]
(a) Draw its structure and show H-bonding
(b) Give oxidation state of Ni and its hybridisation
(c) Predict whether it is paramagnetic or diamagnetic.
9. A solution of a metal ion when treated with KI gives a red precipitate which dissolves in excess KI to give a colourless solution. Moreover, the solution of metal ion on treatment with a solution of cobalt(II) thiocyanate gives rise to a deep blue crystalline precipitate. The metal ion is – [IIT- 2005]
(A) Pb^{2+} (B) Hg^{2+} (C) Cu^{2+} (D) Co^{2+}
10. Write balanced chemical equation for developing a black and white photographic film. Also give reason why the solution of sodium thiosulphate on acidification turns milky white and give balance equation of this reaction – [IIT- 2005]
11. $\text{Fe}^{3+} \xrightarrow{\text{SNC}^-(\text{excess})} \text{A} \xrightarrow{\text{F}^-(\text{excess})} \text{B}$ [IIT- 2005]
Blood red Colourless
Identify A and B.
(i) Write IUPAC name of A and B.
(ii) Find out spin only magnetic moment of B
12. Consider a titration of potassium dichromate solution with acidified Mohr's salt solution using diphenylamine as indicator. The number of moles of Mohr's salt required per mole of dichromate is – [IIT- 2007]
(A) 3 (B) 4 (C) 5 (D) 6

13. Match the column

[IIT- 2007]

Column-I		Column-II	
(A)	$\text{O}_2^- \longrightarrow \text{O}_2 + \text{O}_2^{2-}$	(p)	redox reaction
(B)	$\text{CrO}_4^{2-} + \text{H}^+ \longrightarrow$	(q)	one of the products has trigonal planar structure
(C)	$\text{MnO}_4^- + \text{NO}_2^- + \text{H}^+ \longrightarrow$	(r)	dimeric bridged tetrahedral metal ion
(D)	$\text{NO}_3^- + \text{H}_2\text{SO}_4 + \text{Fe}^{2+} \longrightarrow$	(s)	disproportionation

14. Upon treatment with ammonical H_2S , the metal ion that precipitates as a sulphide is - [IIT-2013]

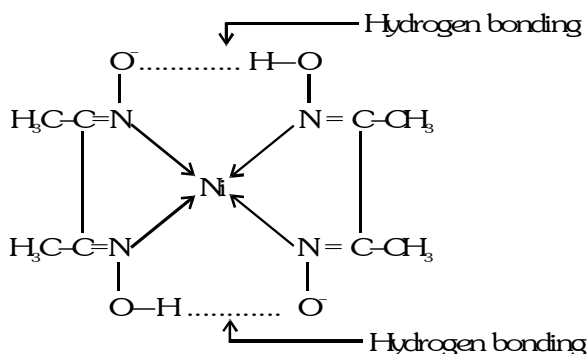
- (A) Fe(III) (B) Al(III)
(C) Mg(II) (D) Zn (II)

PREVIOUS YEARS QUESTIONS

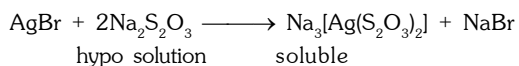
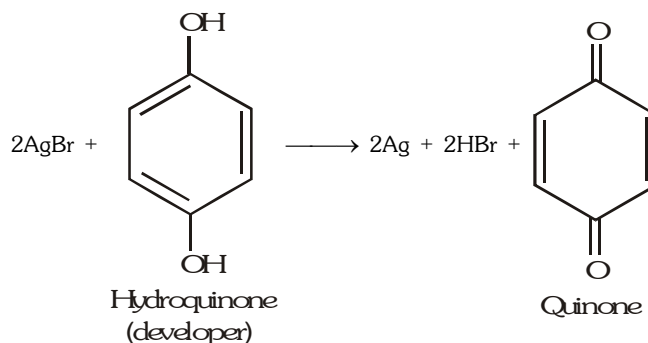
ANSWER KEY

EXERCISE -5 [B]

1. B 2. A 3. A 4. B 5. D 6. B 7. B
8. (a) Structure of the complex is



- (b) The oxidation state of Ni in this compound is +2 and it shows dsp^2 hybridisation.
(c) Since the coordination number of Ni in this complex is 4. According to crystal field theory it is diamagnetic in nature
9. (B)
10. (i) The following reactions are used in developing the photographic film.



- (ii) Aqueous solution of sodium thiosulphate ($\text{Na}_2\text{S}_2\text{O}_3$) on acidification gives milky white turbidity of colloidal sulphur. $\text{Na}_2\text{S}_2\text{O}_3 + 2\text{H}^+_{(\text{aq})} \longrightarrow 2\text{Na}^+ + \text{H}_2\text{SO}_3 + \text{S}$ (colloidal sulphur)

11. (a) A : $[\text{Fe}(\text{SCN})(\text{H}_2\text{O})_5]^{2+}$ (Pentaaquathiocyanatoferrate(III)ion) B : $[\text{FeF}_6]^{3-}$ (Hexafluoroferrate(III) ion)
(b) 5.92 BM

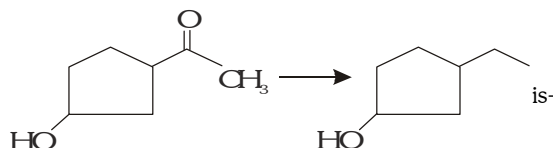
12. D

13. (A) $\rightarrow$ p,s ; (B) $\rightarrow$ r ; (C) $\rightarrow$ p,q ; (D) $\rightarrow$ p

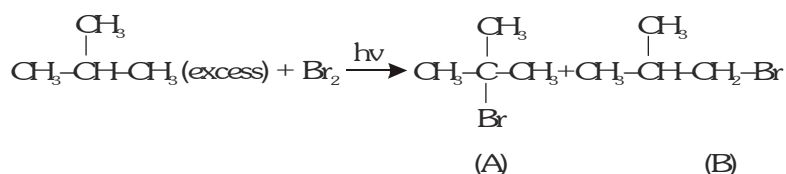
14. (D)

EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. The smallest alkane which can show optical isomerism possesses -
(A) 5 carbons (B) 6 carbons (C) 7 carbons (D) 8 carbons
2. The appropriate reagent for the transformation



- (A) Zn(Hg)/HCl (B) $\text{NH}_2\text{NH}_2, \text{OH}^-$ (C) H_2/Ni (D) NaBH_4
3. The relative reactivity of 1°H , 2°H and 3°H in bromination reaction has been found to be 1 : 82 : 1600 respectively. In the reaction -



the percentage yields of the products (A) and (B) are expected to be -

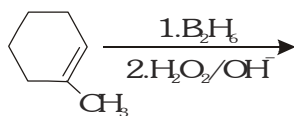
- (A) 99.4%, 0.6% (B) 50%, 50% (C) 0.6%, 99.4% (D) 80%, 20%
4. The least reactive alkane towards free-radical substitution reactions is -
(A) CH_4 (B) $(\text{CH}_3)_3\text{CH}$ (C) CH_3CH_3 (D) $\text{CH}_3\text{CH}_2\text{CH}_3$
5. For the reaction



the major product is :



6. Propene is allowed to react with B_2D_6 and the product is treated with acetic acid. The final product obtained is -
(A) 1-deuteriopropene (B) 2-deuteriopropene
(C) 1-deuteriopropene (D) 2-deuteriopropene
7. 1-Methylcyclohexene is allowed to react with B_2H_6 . The product is then treated with H_2O_2 and NaOH . The reaction is -

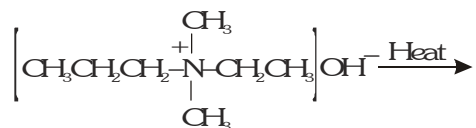


The product formed is

- (A) 1-methylcyclohexanol (B) 2-methylcyclohexanol
(C) methylcyclohexane (D) cyclohexanol

8. Propene on reaction with ICl produces mainly -
 (A) 1-chloro-2-iodopropane (B) 2-chloro-1-iodopropane
 (C) ($\pm$)-2-chloro-1-iodopropane (D) ($\pm$)-1-chloro-2-iodopropane

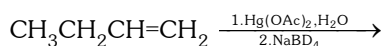
9. Consider the reaction



Which of the following is formed in major amount

- (A) $\text{CH}_2=\text{CH}_2$ (B) $\text{CH}_3\text{CH}=\text{CH}_2$
 (C) Both (A) and (B) in equal amount (D) None, as no reaction takes place
10. In the addition of HBr to propene in the absence of a peroxide, the first step involves the addition of -
 (A) H^+ (B) Br^- (C) $\text{H}^\bullet$ (D) $\text{Br}^\bullet$

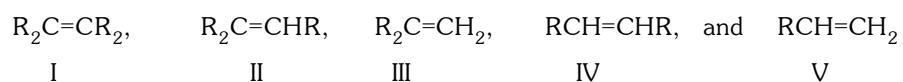
11. In the reaction



the product obtained is -

- (A) $\text{CH}_3\text{CH}_2\text{CHOHCH}_2\text{D}$ (B) $\text{CH}_3\text{CH}_2\text{CHDCH}_2\text{OH}$
 (C) $\text{CH}_3\text{CH}_2\text{CD}(\text{OH})\text{CH}_3$ (D) $\text{CH}_3\text{CH}_2\text{CD}_2\text{CH}_2\text{OH}$
12. The major product obtained in the reaction of 1,3-Butadiene with HCl (1 mole) at a higher temperature (100°C or above) is
 (A) 3,4-dichloro-1-butene (B) 3-chloro-1-butene
 (C) 1-chloro-2-butene (D) 2-chloro-2-butene
13. An optically active hydrocarbon (X) on catalytic hydrogenation gives an optically inactive compound (Y), C_6H_{14} . The hydrocarbon (X) is-
 (A) 3-methyl-1-pentene (B) 3-methyl-2-pentene
 (C) 2-ethyl-1-butene (D) 3-methylcyclopentene
14. The addition of HCl to 1-phenylpropene gives-
 (A) $\text{C}_6\text{H}_5\text{CHClCH}_2\text{CH}_3$ (B) $\text{C}_6\text{H}_5\text{CH}_2\text{CHClCH}_3$
 (C) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$ (D) $\text{C}_6\text{H}_5\text{CH}(\text{CH}_3)\text{CH}_2\text{Cl}$
15. The reduction of 4-octyne with H_2 in the presence of Pd/CaCO_3 - quinoline gives-
 (A) trans-4-octene (B) cis-4-octene
 (C) a mixture of cis-and trans-4-octene (D) a completely reduced product C_8H_{18}
16. The ease of formation of free radicals follows the order -
 (A) $3^\circ > 2^\circ > 1^\circ > \dot{\text{C}}\text{H}_3$ (B) $\dot{\text{C}}\text{H}_3 > 1^\circ > 2^\circ > 3^\circ$
 (C) $1^\circ > 2^\circ > 3^\circ > \dot{\text{C}}\text{H}_3$ (D) $2^\circ > 1^\circ > 3^\circ > \dot{\text{C}}\text{H}_3$
17. Which of the following has the lowest heat of hydrogenation per mole -
 (A) cis-2-Butene (B) trans-2-Butene (C) 1-Butene (D) 1,3-Butadiene
18. The intermediate formed during the addition of HCl to propene in the presence of peroxide is-
 (A) $\text{CH}_3\dot{\text{C}}\text{HCH}_2\text{Cl}$ (B) $\text{CH}_3\text{CH}_2\dot{\text{C}}\text{H}_2$ (C) $\text{CH}_3\overset{+}{\text{C}}\text{HCH}_3$ (D) $\text{CH}_3\text{CH}_2\overset{+}{\text{C}}\text{H}_2$

19. The order of stability of the alkenes



is -

(A) $\text{I} > \text{II} > \text{III} > \text{IV} > \text{V}$

(B) $\text{I} = \text{II} > \text{III} > \text{IV} > \text{V}$

(C) $\text{II} > \text{I} > \text{IV} > \text{III} > \text{V}$

(D) $\text{V} > \text{IV} > \text{III} > \text{I} > \text{II}$

20. Which of the following will not react with an ammonical silver nitrate solution :

(A) $\text{CH}_3\text{C}\equiv\text{CH}$

(B) $(\text{CH}_3)_2\text{CH}-\text{C}\equiv\text{H}$

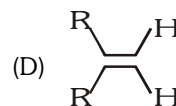
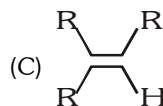
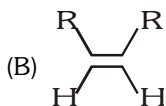
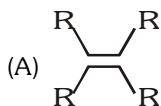
(C) $\text{CH}_3\text{C}\equiv\text{CCH}_3$

(D) $\text{HC}\equiv\text{CH}$

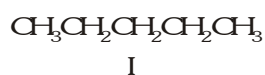
CHECK YOUR GRASP								ANSWER KEY					EXERCISE -1							
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	A	A	C	B	B	B	A	A	A	C	A	A	B	A	D	C	A	C

SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)

- Which of the following reactions will result in the formation of a chiral centre in the product -
 (A) $\text{CH}_3\text{CH}=\text{CHCH}_3+\text{HBr} \longrightarrow$ (B) $\text{CH}_3\text{CH}=\text{CH}_2+\text{HOBr} \longrightarrow$
 (C) $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2+\text{HBr} \xrightarrow{\text{H}_2\text{O}_2}$ (D) $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2+\text{HBr} \longrightarrow$
- Propene on reaction with N-bromosuccinimide in CCl_4 produces -
 (A) 1, 2-dibromopropane (B) 3-bromopropene
 (C) 1-bromopropene (D) 2-bromopropene
- cis-2-Butene on reaction with Br_2 in CCl_4 produces mainly -
 (A) 1-bromo-2-butene (B) 2,3-dibromobutane
 (C) meso-2,3-dibromobutane (D) ($\pm$) 2,3-dibromobutane
- The bond dissociation energies of the following
 $\text{CH}_3\text{-H}$ $\text{CH}_3\text{CH}_2\text{-H}$ $\text{CH}_2=\text{CH-CH}_2\text{-H}$ $\text{C}_6\text{H}_5\text{-H}$
 I II III IV
 vary in the order :
 (A) $\text{I} > \text{II} > \text{III} > \text{IV}$ (B) $\text{IV} > \text{III} > \text{II} > \text{I}$ (C) $\text{IV} > \text{I} > \text{II} > \text{III}$ (D) $\text{II} > \text{I} > \text{IV} > \text{III}$
- Which of the following decolourises alkaline KMnO_4 solution
 (A) C_3H_8 (B) C_2H_4 (C) CH_4 (D) CCl_4
- Compounds capable of reacting with ammonical AgNO_3 solution are
 (A) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{C}\equiv\text{CH}$ (B) $\text{HC}\equiv\text{CH}$ (C) 1- Butyne (D) all the above
- A hydrocarbon which decolourises KMnO_4 but does not give any precipitate with ammoniated AgNO_3
 (A) Benzene (B) Acetylene (C) Butyne (D) 2-Butene
- Compound 'A' on chlorination gives compound 'B', compound 'B' reacts with alc. KOH gives gas 'C', which decolourise Baeyer reagent. ozonolysis of compound 'C' gives only HCHO compound. 'A' is :
 (A) C_2H_6 (B) C_2H_4 (C) C_4H_{10} (D) $\text{C}_2\text{H}_5\text{Cl}$
- Which reagent converts propene to 1-propanol
 (A) H_2O , H_2SO_4 (B) B_2H_6 , H_2O_2 , OH^-
 (C) $\text{Hg}(\text{OAc})_2$, $\text{NaBH}_4/\text{H}_2\text{O}$ (D) Aq. KOH
- Which one of the following alkenes will react faster with H_2 under catalytic hydrogenation conditions :
 [R = Alkyl Substituent]



11. Arrange the following in order of increase/decrease in boiling point.

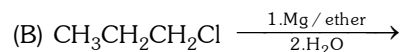
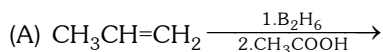


- (A) I > II > III (B) II > I > III (C) III > I > II (D) III < II < I

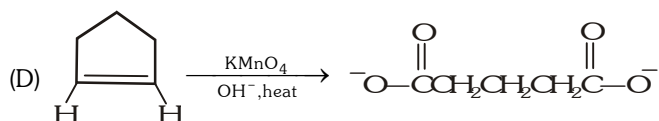
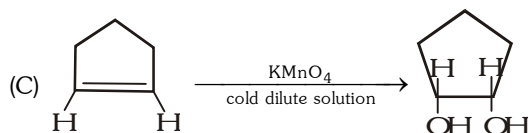
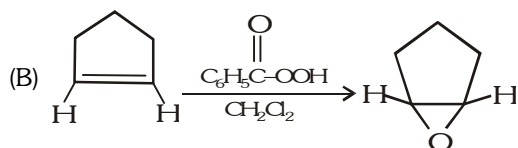
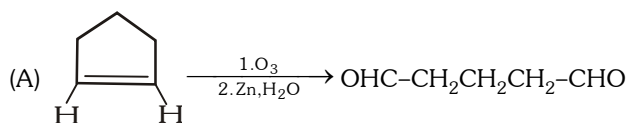
12. What are the products obtained upon the ozonolysis of 2-pentene ?

- (A) $\text{CH}_3\text{CH}_2\text{CHO}$ (B) CH_3CHO (C) CH_3COCH_3 (D) $\text{CH}_3\text{COCH}_2\text{CH}_3$

13. Which of the following can be used for the preparation of propane ?



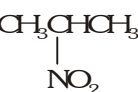
14. Which of the following are correct :



15. 2-Bromo-3-phenylpropane can be synthesised by

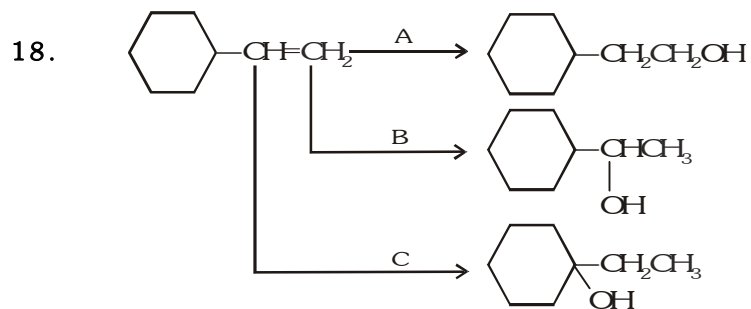
- (A) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{OH})\text{CH}_3 + \text{PBr}_3 \longrightarrow$
 (B) $\text{C}_6\text{H}_5\text{CH}=\text{CHCH}_3 + \text{HBr} + \text{benzoyl peroxide} \longrightarrow$
 (C) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{CH}_3 + \text{Br}_2 + \text{light} \longrightarrow$
 (D) none of these

16. The nitration of propane with concentrated HNO_3 gives :

- (A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NO}_2$ (B)  (C) $\text{CH}_3\text{CH}_2\text{NO}_2$ (D) CH_3NO_2

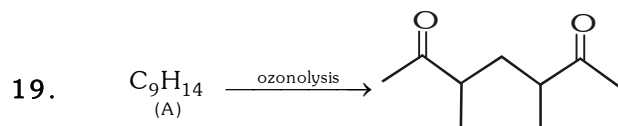
17. Which of the following will react with sodium metal :

- (A) Ethyne (B) 1-Butyne (C) 2-Butyne (D) Ethane

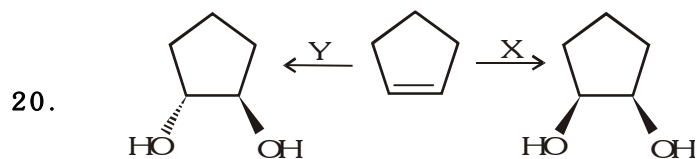
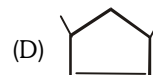
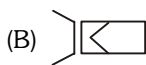
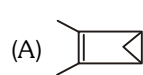


A, B and C are :

- (A) simple hydration
(B) hydroboration, mercuration-demercuration, hydration
(C) hydration, hydroboration, mercuration-demercuration
(D) mercuration-demercuration, hydration, hydroboration



Hence A is :



Select X and Y out of :

I : $\text{MnO}_4^- / \text{OH}^-$ II : HCO_3H

- (A) X -I, Y- II (B) X -II, Y- I (C) X -I, Y- I (D) X -II, Y- II

BRAIN TEASERS						ANSWER KEY				EXERCISE -2					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	A, B,D	B	D	C	B	D	D	A	B	B	A, D	A, B	A,BC,D	A,BC,D	A,B
Que.	16	17	18	19	20										
Ans.	A,BC,D	A, B	B	B	A										

EXERCISE-03

MISCELLANEOUS TYPE QUESTIONS

TRUE OR FALSE :

- Although acetylene acidic in nature it does not react with NaOH/KOH.
- Although C—H bond in acetylene has greatest bond energy of all C—H bond, yet it is most acidic.
- $\text{CH}_2=\ddot{\text{C}}\text{H}$ is less basic than $\text{HC}\equiv\ddot{\text{C}}$
- $-\text{C}\equiv\text{C}-$ has two π bond yet it is less reactive than $-\text{C}=\text{C}-$ towards electrophilic addition reaction.
- Partial reduction of alkynes is either syn or anti.

FILL IN THE BLANKS :

- Out of cis-2 butene and trans-2-butene, has the lower melting point.
- A four-carbon alkyne having weakly acidic character is
- Alkanes undergo reactions whereas alkynes give reaction.
- is a versatile method for locating the position of the double bond in an alkene.
- The valence atomic orbital on carbon in silver acetylide is hybridized.

MATCH THE COLUMN

- Match the column I with column II.

Column-I		Column-II	
(A)	Wurtz reaction	(p)	Electrophilic substitution reaction
(B)	Hydration of alkenes	(q)	Free radical substitution
(C)	Nitration of alkane	(r)	Electrophilic addition reaction
(D)	Reaction of alkene with NBS	(s)	Nucleophilic substitution

- Match the column I with column II.

Column-I		Column-II	
$\text{CH}_3-\text{CH}=\text{CH}_2$	(A) $\rightarrow \text{CH}_3-\text{CH}_2-\text{CH}_2\text{Br}$	(p)	HBr
	(B) $\rightarrow \text{CH}_3-\text{CHBr}-\text{CH}_3$	(q)	HBr + peroxide
	(C) $\rightarrow \text{CH}_3-\text{CHBr}-\text{CH}_2\text{Br}$	(r)	NBS
	(D) $\rightarrow \text{BrCH}_2-\text{CH}=\text{CH}_2$	(s)	Br_2 , low temp., dark

- Match the column I with column II.

Column-I		Column-II	
(A)	Dehydrohalogenation of alkyl halides	(p)	Kolbe reaction
(B)	Electrolysis of sodium salt	(q)	Alc. KOH
(C)	Ozonolysis	(r)	Addition product of ethylene
(D)	Dichloro ethylene	(s)	Sodalime
(E)	Decarboxylation	(t)	Alkene

ASSERTION & REASON QUESTION :

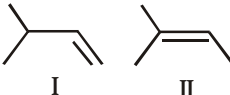
These questions contains, Statement-I (assertion) and Statement-II (reason).

(A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I

(B) Statement-I is True, Statement-II is True ; Statement-II is NOT a correct explanation for Statement-I

(C) Statement-I is True, Statement-II is False.

(D) Statement-I is False, Statement-II is True.

1. **Statement-I** : Of two isomeric alkenes shown,  II is more stable than I.

Because

Statement-II : The alkene (II) has sp^2 -hybridized carbons.

2. **Statement-I** : Addition of HBr to 1-butene gives two optical isomers.

Because

Statement-II : The product contains one asymmetric carbon.

3. **Statement-I** : 1-Butene on reaction with HBr in the presence of a peroxide produces 1-bromobutane.

Because

Statement-II : It involves the formation of a primary radical.

4. **Statement-I** : Addition of Br_2 - water containing dissolved NaCl to ethylene gives a mixture of 1, 2-dibromoethane, 1-bromo-2-chloroethane and 2-bromoethanol.

Because

Statement-II : Addition occurs through a carbocation intermediate.

5. **Statement-I** : Addition of bromine to trans-2-butene yields meso-2,3-dibromobutane.

Because

Statement-II : Bromine addition to an alkene is an electrophilic addition.

6. **Statement-I** : Alkynes are more reactive than alkenes towards electrophilic reagents like H^+ .

Because

Statement-II : The alkyl carbocation formed from alkene is more stable than the vinyl carbocation formed from alkyne.

7. **Statement-I** : Neopentane forms only one monochlorinated product.

Because

Statement-II : Neopentane has four identical methyl group attached to a quaternary carbon.

8. **Statement-I** : Buta-1,3-diene is less stable than Penta-1, 4-diene.

Because

Statement-II : Buta-1,3-diene has greater number of resonating structures and delocalised electron cloud,

9. **Statement-I** : Iodination of alkanes is carried out in the presence of iodic acid.

Because

Statement-II : Iodic acid removes I_2 gas from the reaction mixture.

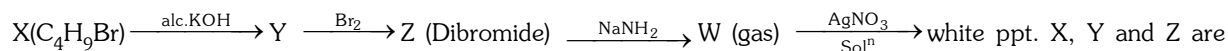
10. **Statement-I** : Propene is more reactive than ethene with HCl.

Because

Statement-II : Propene is more stable than ethene.

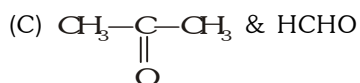
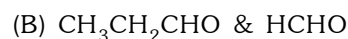
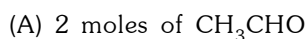
COMPREHENSION BASED QUESTIONS :

Comprehension # 1



1.	X	Y	Z
(A)	$CH_3-CH_2CH_2CH_2Br$	$CH_3-CH=CH-CH_3$	$\begin{array}{c} CH_3-CH-CH_2-CH_2 \\ \quad \quad \\ Br \quad \quad Br \end{array}$
(B)	$CH_3-CH_2CH_2CH_2Br$	$CH_3-CH_2-CH=CH_2$	$\begin{array}{c} CH_3-CH_2-CH-CH_2 \\ \quad \quad \\ Br \quad \quad Br \end{array}$
(C)	$CH_3-CH_2-CH_2-CH_3$	$CH_3-CH=CH-CH_3$	$\begin{array}{c} CH_3-CH-CH-CH_3 \\ \quad \quad \\ Br \quad \quad Br \end{array}$
(D)	$\begin{array}{c} CH_3-CH_2-CH-CH_3 \\ \\ Br \end{array}$	$CH_3-CH=CH-CH_3$	$\begin{array}{c} CH_3-CH_2-CH-CH_2 \\ \quad \quad \\ Br \quad \quad Br \end{array}$

2. Reductive ozonolysis of Y yields :



3. Which of the following statement (s) is/are correct :

(A) Compound W has 2 DU

(B) Y & W are functional isomers

(C) W can be converted into Y with Lindlar catalyst

(D) W can be converted into Y with Ni/Pt

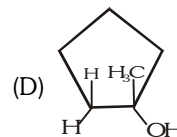
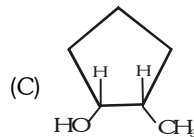
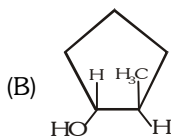
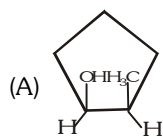
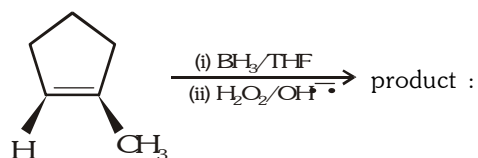
Comprehension # 2

Borane is an electron deficient compound. It has only six valence electrons, so the boron atom lacks an octet. Acquiring an octet is the driving force for the unusual bonding structure found in boron compounds. As an electron deficient compound, BH_3 is a strong electrophile, capable of adding to a double bond. This hydroboration of double bond is thought to occur in one step, with the boron atom adding to the less highly substituted end of the double bond. In transition state, the boron atom withdraws electrons from the pi bond and the carbon at the other end of the double bond acquires a partial positive charge. This positive charge is more stable on the more highly substituted carbon atom. The second step is the oxidation of boron atom, removing it from carbon and replacing it with a hydroxyl group by using H_2O_2/OH^- .

The simultaneous addition of boron and hydrogen to the double bond leads to a syn addition. Oxidation of the trialkyl borane replaces boron with a hydroxyl group in the same stereochemical position. Thus, hydroboration of alkene is an example of stereospecific reaction, in which different stereoisomers of starting compounds react to give different stereoisomers of the product.

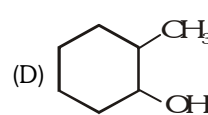
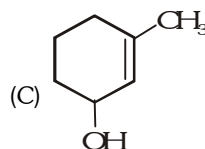
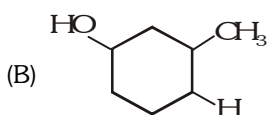
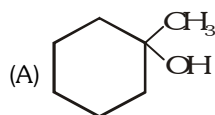
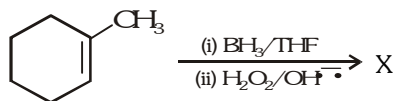
1.

What will be the product of following reaction

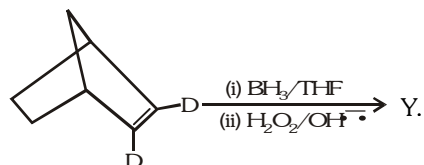


2.

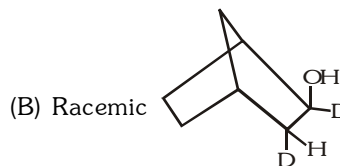
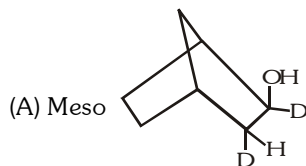
Find the product of following reaction



3.



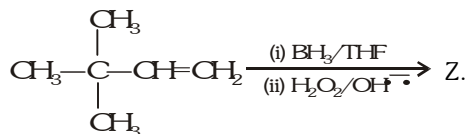
Y is :



(C) both

(D) None of these

4.

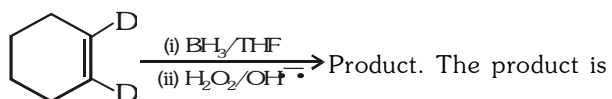


Z is :

(A) Optically active 1 -alcohol
 (C) Optically inactive 1 -alcohol

(B) Optically active 2 -alcohol
 (D) Optically inactive 3 -alcohol

5.



(A) Threo cyclic alcohol
 (C) Optically active alcohol

(B) Erythro cyclic alcohol
 (D) Both (B) and (C)

Comprehension # 3

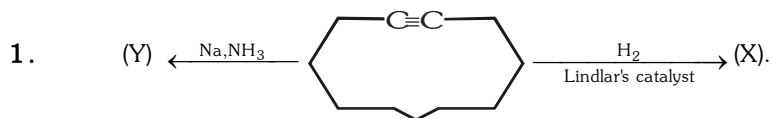
Hydrocarbon (A) C_6H_{10} on treatment with H_2/Ni , $H_2/Lindlar's$ catalyst or $Na/liq. NH_3$ forms three different reduction products (B), (C), (D) respectively. (A) does not form any salt with ammonical $AgNO_3$ solution, but (E) forms salt on heating with $NaNH_2$ in an inert solvent. Compound (E) reacts with CH_3I to give (F). Compound (D) on oxidative ozonolysis gives n-butanoic acid along with other product.

- (D) and (C) are
 (A) Isomeric alkane (B) Isomeric alkene
 (C) Not isomers (D) (C) is an alkene and (D) is salt
- If (E) is reacted with acetaldehyde followed by acidification, product is
 (A) acid (B) ketone (C) ether (D) alcohol
- (F) on ozonolysis will produce -
 (A) acetic acid (B) formic acid (C) propanoic acid (D) formaldehyde

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
<u>True / False</u> 1. T 2. T 3. F 4. T 5. T <u>Fill in the Blanks</u> 1. Cis-2-butene 2. 1-Butyne 3. Free radical substitution, electrophilic substitution 4. Ozonolysis 5. sp <u>Match the Column</u> 1. (A) $\rightarrow$ s ; B $\rightarrow$ r ; (C) $\rightarrow$ p ; (D) $\rightarrow$ q 2. (A) $\rightarrow$ q ; (B) $\rightarrow$ p ; (C) $\rightarrow$ s ; (D) $\rightarrow$ r 3. (A) $\rightarrow$ q ; (B) $\rightarrow$ p ; (C) $\rightarrow$ t ; (D) $\rightarrow$ r ; (E) $\rightarrow$ s <u>Assertion - Reason Questions</u> 1. C 2. A 3. C 4. A 5. B 6. D 7. A 8. D 9. C 10. B <u>Comprehension Based Questions</u> Comprehension #1 : 1. (B) 2. (B) 3. (A, C) Comprehension #2 : 1. (B) 2. (D) 3. (B) 4. (C) 5. (D) Comprehension #3 : 1. (B) 2. (D) 3. (A)		

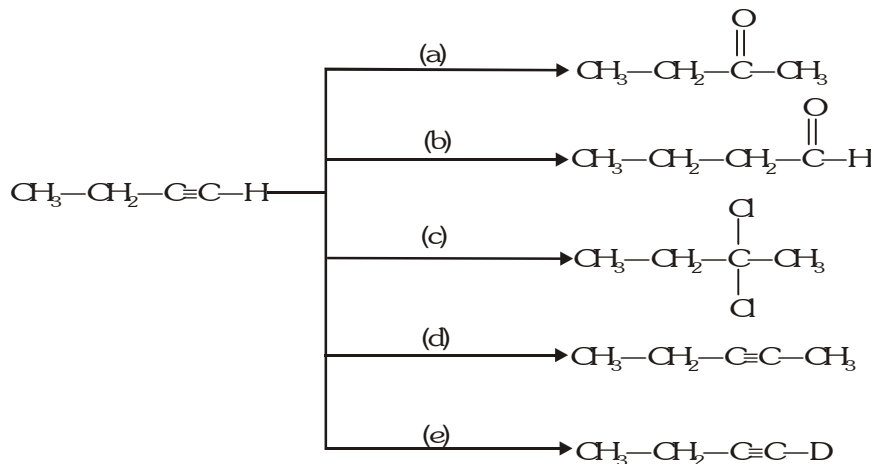
EXERCISE-04 [A]

CONCEPTUAL SUBJECTIVE EXERCISE

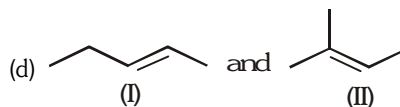
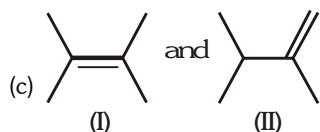
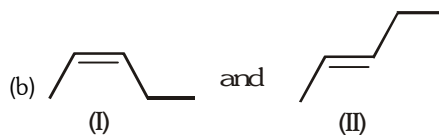
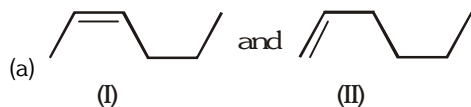


What are X and Y in above reaction.

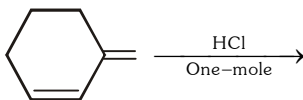
2. Find out reagents involved in following conversions :



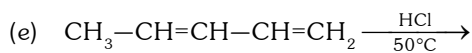
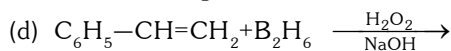
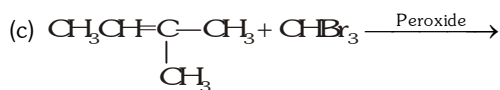
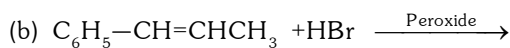
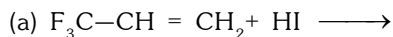
3. Select one from the following pair of isomer that has higher heat of combustion, justifying your choice.



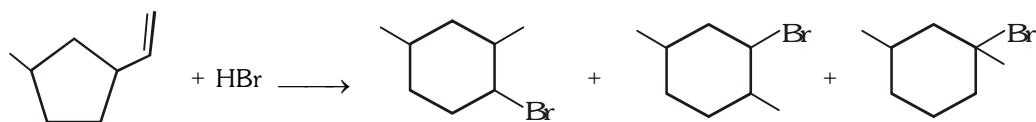
4. The reaction of the diene shown below with dry HCl can lead to four products. Provide structural formula of all the products.



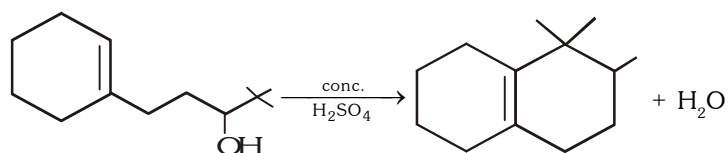
5. Write major product in the following reactions :



6. Propose mechanism :



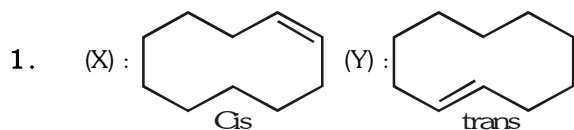
7. A hydrocarbon (A) was found to have vapour density 36. It forms only single mono chlorosubstitution product. Suggest (A).
8. An olefin was treated with ozone and the resulting product on reductive hydrolysis gave 2-pentanone and acetaldehyde. What is the structure of olefin ? Write reactions.
9. Why n-Pentane has higher boiling point than neopentane ?
10. Write a plausible mechanism for the following transformation.



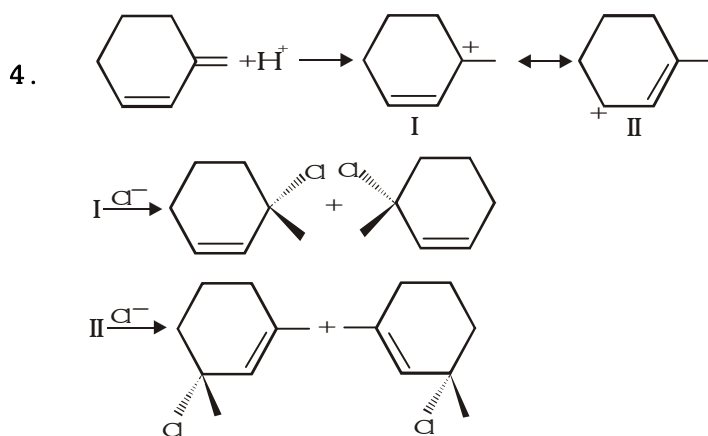
CONCEPTUAL SUBJECTIVE EXERCISE

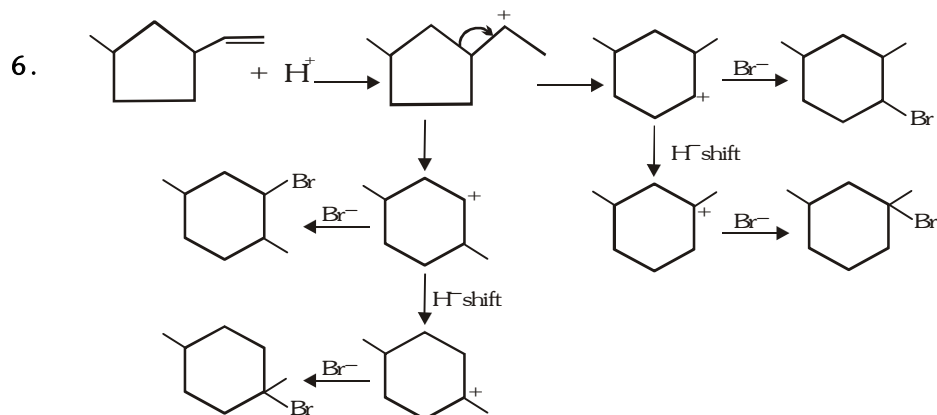
ANSWER KEY

EXERCISE -4(A)

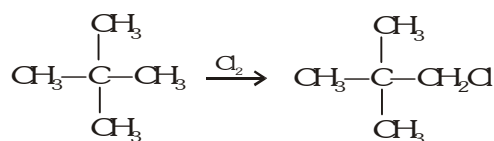


2. (a) $\text{H}_2\text{O}/\text{Hg}^{2+}/\text{H}^+$, (b) $(\text{Sia})_2\text{BH}/\text{H}_2\text{O}_2$, OH^- , (c) 2HCl ,
 (d) Na , $\text{CH}_3 - \text{I}$, (e) Na , D-O-D
3. (a) II ; (b) I ; (c) II ; (d) I

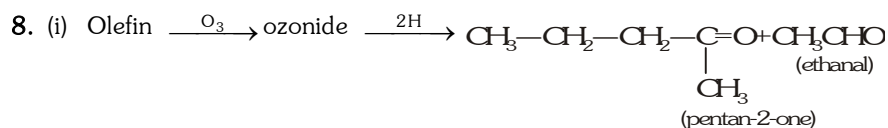




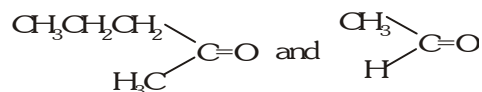
7. Mol.wt of alkane = Vapour density $\times 2 = 36 \times 2 = 72$
 Alkane is $\text{C}_n\text{H}_{2n+2}$
 $\therefore 12n + 2n + 2 = 72$
 $\therefore n = 5$ $\therefore$ Alkane is C_5H_{12}
 Thus alkane C_5H_{12} showing only one mono substitution product is



(A) (2,2-dimethylpropane)

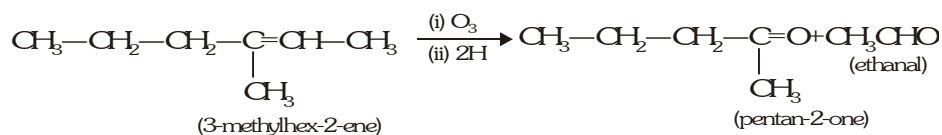


(ii) Thus olefin has 7 carbon atoms with two units round double bond as



(iii) The olefin is therefore $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{C}(\text{CH}_3)=\text{CH}-\text{CH}_3$

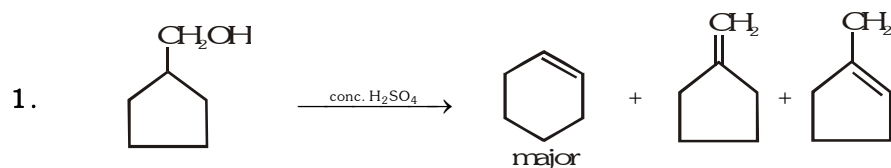
(iv) The reaction is



9. n-Pentane has higher b.p. (36.2 C) than neopentane (9.5 C). It is due to vander waals forces. n-Pentane has a rod-like shape, while neopentane is sphere-like. Rods can touch along their entire length, while the spheres touch only at a point. The more the contact between molecules, the greater the vander waals forces and hence higher the b.p.

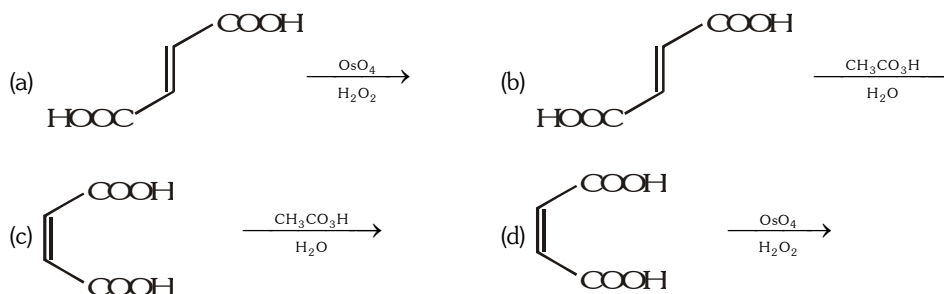
EXERCISE-04 [B]

BRAIN STORMING SUBJECTIVE EXERCISE



Give mechanism of this reaction.

2. Identify the products in the following, giving their configurations. (if any)

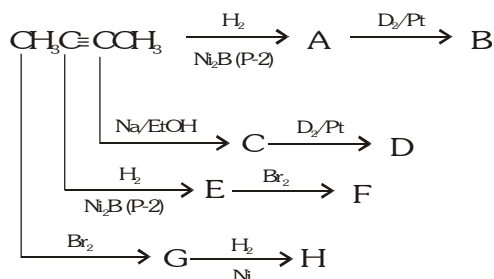


3. Write one reagent in each case to make distinction between

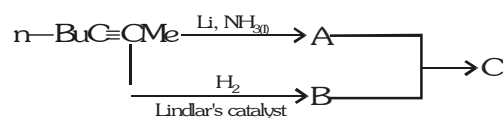
(a) 1-butyne and 2-butyne (b) 1-butene and 1-butyne

(c) ethene and ethane

4. 2-Butyne undergoes following reactions in steps as indicated. Identify A to H.

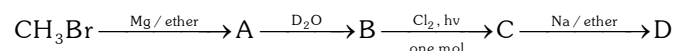


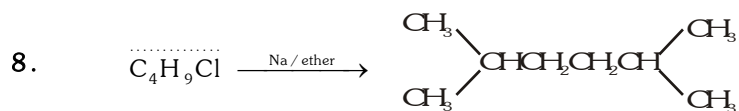
5. Identify A, B, C with proper explanation.



6. A hydrocarbon A(C₁₀H₁₂) has no chiral carbon. A gives a white precipitate with ammonical solution of silver nitrate. A on treatment with H₂/Pt gives B(C₁₀H₂₀). A on ozonolysis gives C(C₈H₁₂O₄) as one product which on heating with soda lime gives D(C₆H₁₂). D on monochlorination with Cl₂/hν gives C₆H₁₁Cl as sole isomer. Identify A to D.

7. Identify A to D in the following scheme.





Write structure of $\text{C}_4\text{H}_9\text{Cl}$.

9. Specify the reagents in which addition to alkene is syn or anti

(i) Br_2 water

(ii) H_2/pd

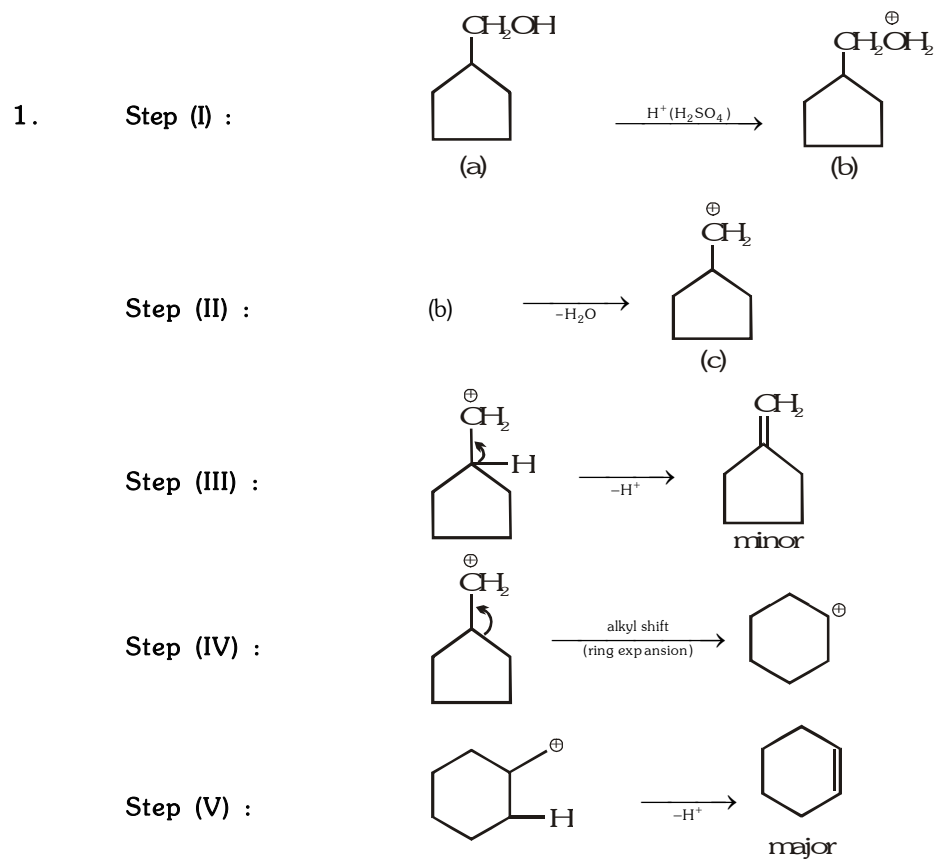
(iii) $\text{MnO}_4^-/\text{OH}^-$

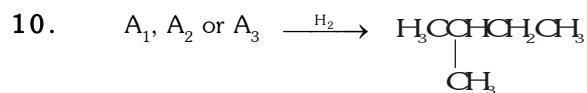
(iv) HCO_3H

(v) MCPBA

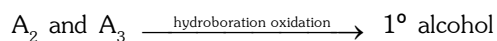
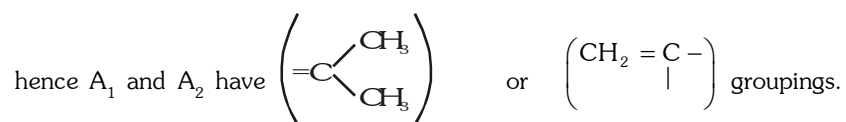
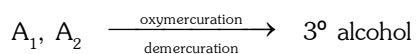
(vi) OsO_4 in ether, followed by $\text{OH}^-/\text{H}_2\text{O}$

10. C_5H_{10} represents three isomeric alkenes A_1 , A_2 and A_3 . Each on hydrogenation gives 2-methylbutane. A_1 and A_2 on oxymercuration-demercuration give the same 3° alcohol. A_2 and A_3 on hydroboration oxidation give different 1° alcohol. Assign structures to A_1 , A_2 and A_3 and explain the reactions.

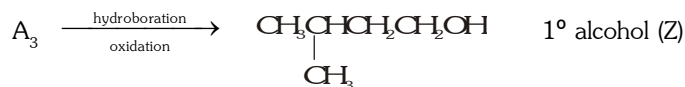
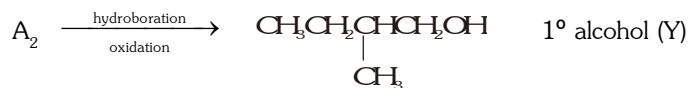
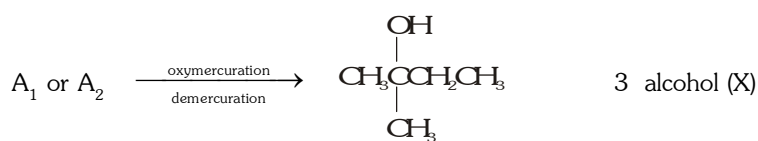
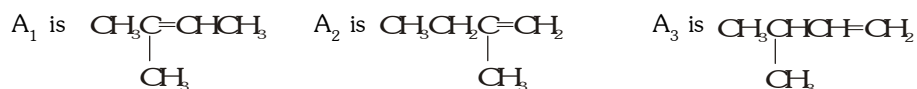




This shows that each of A_1 , A_2 and A_3 have same C-chain; only position of (C=C) is to be decided.



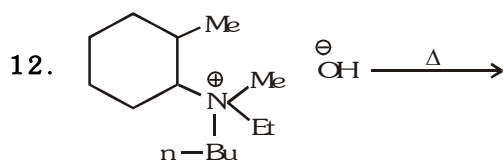
This indicates presence of $(CH_2=)$ grouping at the terminal. Hence,



Y and Z are different 1° alcohols.

EXERCISE - 05 [A]**JEE-[MAINS] : PREVIOUS YEAR QUESTIONS**

1. Which of these will not react with acetylene - [AIEEE-2002]
(A) NaOH (B) ammonical AgNO_3 (C) Na (D) HCl
2. What is the product formed when acetylene reacts with hypochlorous acid - [AIEEE-2002]
(A) CH_3COCl (B) ClCH_2CHO (C) Cl_2CHCHO (D) ClCH_2COOH
3. 1-Butene may be converted to butane by reaction with - [AIEEE-2003]
(A) Pd/H_2 (B) $\text{Zn} - \text{HCl}$ (C) $\text{Sn} - \text{HCl}$ (D) $\text{Zn} - \text{Hg}/\text{HCl}$
4. On mixing a certain alkane with chlorine and irradiating it with ultraviolet light, it forms only one monochloroalkane. This alkane could be - [AIEEE-2003]
(A) neopentane (B) propane (C) pentane (D) isopentane
5. Which one of the following is reduced with $\text{Zn-Hg}/\text{HCl}$ to give the corresponding hydrocarbon [AIEEE-2004]
(A) Butan-2-one (B) Acetic acid (C) Acetamide (D) Ethyl acetate
6. Which one of the following has the minimum boiling point : [AIEEE-2004]
(A) isobutane (B) 1-butyne (C) 1-butene (D) n-butane
7. 2-Methylbutane on reacting with bromine in the presence of sunlight gives mainly [AIEEE-2005]
(A) 2-bromo-2-methylbutane (B) 1-bromo-2-methylbutane
(C) 1-bromo-3-methylbutane (D) 2-bromo-3-methylbutane
8. Alkyl halides react with dialkyl copper reagent to give [AIEEE-2005]
(A) alkyl copper halides (B) alkenes
(C) alkenyl halides (D) alkanes
9. Reaction of one molecule of HBr with one molecule of 1,3-butadiene at 40 C gives predominantly [AIEEE-2005]
(A) 1-bromo-2-butene under thermodynamically controlled conditions
(B) 3-bromobutene under kinetically controlled conditions
(C) 1-bromo-2-butene under kinetically controlled conditions
(D) 3-bromobutene under thermodynamically controlled conditions
10. Acid catalyzed hydration of alkenes except ethene leads to the formation of [AIEEE-2005]
(A) secondary or tertiary alcohol
(B) primary alcohol
(C) mixture of secondary and tertiary alcohols
(D) mixture of primary and secondary alcohols
11. Elimination of bromine from 2-bromobutane results in the formation of [AIEEE-2005]
(A) predominantly 2-butene (B) equimolar mixture of 1 and 2-butene
(C) predominantly 2-butyne (D) predominantly 1-butene



The alkene formed as a major product in the above elimination reaction is-

[AIEEE-2006]

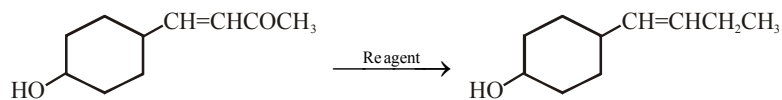


13. Reaction of trans-2-phenyl-1-bromocyclopentane on reaction with alcoholic KOH produces- [AIEEE-2006]
(A) 4-phenyl cyclopentene (B) 2-phenyl cyclopentene
(C) 1-phenyl cyclopentene (D) 3-phenyl cyclopentene
14. Phenyl magnesium bromide reacts with methanol to give- [AIEEE-2006]
(A) A mixture of anisole and $\text{Mg}(\text{OH})\text{Br}$ (B) A mixture of benzene and $\text{Mg}(\text{OMe})\text{Br}$
(C) A mixture of toluene and $\text{Mg}(\text{OH})\text{Br}$ (D) A mixture of phenol and $\text{Mg}(\text{Me})\text{Br}$

15. Which of the following reactions will yield, 2, 2-dibromopropane [AIEEE-2007]
- (A) $\text{CH}_3-\text{C}\equiv\text{CH} + 2\text{HBr} \longrightarrow$ (B) $\text{CH}_3\text{CH}=\text{CHBr} + \text{HBr} \longrightarrow$
- (C) $\text{CH}\equiv\text{CH} + 2\text{HBr} \longrightarrow$ (D) $\text{CH}_3-\text{CH}=\text{CH}_2 + \text{HBr} \longrightarrow$
16. In the following sequence of reactions, the alkene affords the compound 'B' :- [AIEEE-2008]
- $\text{CH}_3\text{CH}=\text{CHCH}_3 \xrightarrow{\text{O}_3} \text{A} \xrightarrow[\text{Zn}]{\text{H}_2\text{O}} \text{B}$.
- The compound B is
- (A) $\text{CH}_3\text{CH}_2\text{CHO}$ (B) CH_3COCH_3 (C) $\text{CH}_3\text{CH}_2\text{COCH}_3$ (D) CH_3CHO
17. The hydrocarbon which can react with sodium in liquid ammonia is [AIEEE-2008]
- (A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{C}\equiv\text{CCH}_2\text{CH}_2\text{CH}_3$ (B) $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH}$
- (C) $\text{CH}_3\text{CH}=\text{CHCH}_3$ (D) $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CCH}_2\text{CH}_3$
18. The treatment of CH_3MgX with $\text{CH}_3\text{C}\equiv\text{C}-\text{H}$ produces [AIEEE-2008]
- (A) $\text{CH}_3-\text{CH}=\text{CH}_2$ (B) $\text{CH}_3\text{C}\equiv\text{C}-\text{CH}_3$ (C) $\begin{array}{c} \text{H} \quad \text{H} \\ | \quad | \\ \text{CH}_3-\text{C}=\text{C}-\text{CH}_3 \end{array}$ (D) CH_4
19. The main product of the following reaction is [AIEEE-2010]
- $\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{OH})\text{CH}(\text{CH}_3)_2 \xrightarrow{\text{Conc. H}_2\text{SO}_4} ?$
- (A) $\begin{array}{c} \text{H}_5\text{C}_6\text{CH}_2\text{CH}_2 \\ | \quad \diagdown \\ \text{H}_3\text{C} \quad \text{C}=\text{CH}_2 \end{array}$ (B) $\begin{array}{c} \text{H}_5\text{C}_6 \quad \text{H} \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{H} \quad \text{CH}(\text{CH}_3)_2 \end{array}$
- (C) $\begin{array}{c} \text{C}_6\text{H}_5\text{CH}_2 \quad \text{CH}_3 \\ \diagdown \quad \diagup \\ \text{H} \quad \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \quad \quad \text{CH}_3 \end{array}$ (D) $\begin{array}{c} \text{C}_6\text{H}_5 \quad \text{CH}(\text{CH}_3)_2 \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{H} \quad \text{H} \end{array}$
20. One mole of a symmetrical alkene on ozonolysis gives two moles of an aldehyde having a molecular mass of 44 u. The alkene is :- [AIEEE-2010]
- (A) ethene (B) propene (C) 1-butene (D) 2-butene
21. Ozonolysis of an organic compound gives formaldehyde as one of the products. This confirms the presence of :- [AIEEE-2011]
- (A) an isopropyl group (B) an acetylenic triple bond
- (C) two ethylenic double bonds (D) a vinyl group
22. Ozonolysis of an organic compound 'A' produces acetone and propionaldehyde in equimolar mixture. Identify 'A' from the following compounds :- [AIEEE-2011]
- (A) 2 - Methyl - 1- pentene (B) 1 - Pentene
- (C) 2 - Pentene (D) 2 - Methyl - 2 - pentene
23. 2-Hexyne gives trans -2-Hexene on treatment with :- [AIEEE-2012]
- (A) Li AlH_4 (B) Pt/H_2 (C) Li/NH_3 (D) Pd/BaSO_4

24. In the given transformation, which of the following is the most appropriate reagent ?

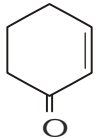
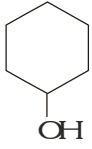
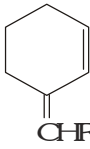
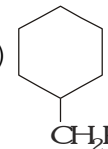
[AIEEE-2012]



- (A) NaBH_4 (B) $\text{NH}_2\text{NH}_2 \cdot \overset{\ominus}{\text{OH}}$ (C) $\text{Zn} - \text{Hg} / \text{HCl}$ (D) $\text{Na}, \text{Liq.NH}_3$

JEE-[MAIN] : PREVIOUS YEAR QUESTIONS							ANSWER KEY			EXERCISE -5[A]					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans	1	3	1	1	1	1	1	4	1	1	1	2	4	2	1
Que.	16	17	18	19	20	21	22	23	24						
Ans	4	2	4	2	4	4	4	3	2						

EXERCISE - 05 [B]**JEE-[ADVANCED] : PREVIOUS YEAR QUESTIONS**

1. Alcoholic solution of KOH is a specific reagent for - [IIT-90]
(A) Dehydration (B) Dehydrogenation (C) Dehydro halogenation (D) Dehalogenation
2. Of the following, unsaturated hydrocarbons are - [IIT-90]
(A) Ethyne (B) Cyclohexane (C) n-propane (D) Ethane
3. 1-chlorobutane on reaction with alcoholic potash gives [IIT-91]
(A) 1-butene (B) 1-butanol (C) 2-butene (D) 2-butanol
4. The hybridisation of carbon atoms in C-C single bond of $\text{HC} \equiv \text{C} - \text{CH}=\text{CH}_2$ [IIT-93]
(A) sp^3-sp^3 (B) sp^2-sp^3 (C) $\text{sp}-\text{sp}^2$ (D) sp^2-sp^2
5. The product (s) obtained via oxymercuration ($\text{HgSO}_4 + \text{H}_2\text{SO}_4$) of 1-butyne would be - [IIT-92]
(A) $\text{CH}_3-\text{CH}_2-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{CH}_3$ (B) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CHO}$
(C) $\text{CH}_3-\text{CH}_2-\text{CHO}-\text{HCHO}$ (D) $\text{CH}_3-\text{CH}_2-\text{COOH}+\text{HCOOH}$
6. Which is the decreasing order of strength of bases : [IIT-93]
 OH^- , NH_2^- , $\text{HC} \equiv \text{C}^-$ and $\text{CH}_3-\text{CH}_2^-$
(A) $\text{CH}_3-\text{CH}_2^- > \text{NH}_2^- > \text{HC} \equiv \text{C}^- > \text{OH}^-$ (B) $\text{HC} \equiv \text{C}^- > \text{CH}_3-\text{CH}_2^- > \text{NH}_2^- > \text{OH}^-$
(C) $\text{OH}^- > \text{NH}_2^- > \text{HC} \equiv \text{C}^- > \text{CH}_3-\text{CH}_2^-$ (D) $\text{NH}_2^- > \text{HC} \equiv \text{C}^- > \text{OH}^- > \text{CH}_3-\text{CH}_2^-$
7. The chief reaction product of reaction between n-butane and bromine at 130°C is - [IIT-95]
(A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$ (B) $\text{CH}_3\text{CH}_2\underset{\text{CH}_3}{\text{CH}}\text{Br}$
(C) $\text{CH}_3\underset{\text{CH}_3}{\text{CH}}\text{CH}_2\text{Br}$ (D) $\text{CH}_3\underset{\text{CH}_3}{\text{C}}\text{Br}_2$
8. When cyclohexane is poured on water, it floats, because - [IIT-97]
(A) Cyclohexane is in 'boat' form (B) Cyclohexane is in 'chair' form
(C) Cyclohexane is in 'crown' form (D) Cyclohexane is less dense than water
9. $(\text{CH}_3)_3\text{CMgCl}$ on reaction with D_2O produces [IIT-97]
(A) $(\text{CH}_3)_3\text{CD}$ (B) $(\text{CH}_3)_3\text{OD}$ (C) $(\text{CD}_3)_3\text{CD}$ (D) $(\text{CD}_3)_3\text{OD}$
10. When  reacts with $\text{Ph}_3\text{P}^{\oplus}-\text{CHR}^{\ominus}$, the product is - [IIT-97]
(A)  (B)  (C)  (D) 
11. The intermediate during the addition of HCl to propene in the presence of peroxide is : [IIT-97]
(A) $\text{CH}_3\dot{\text{C}}\text{HCH}_2\text{Cl}$ (B) $\text{CH}_3\overset{\oplus}{\text{C}}\text{HCH}_3$ (C) $\text{CH}_3\text{CH}_2\dot{\text{C}}\text{H}_2$ (D) $\text{CH}_3\text{CH}_2\overset{\oplus}{\text{C}}\text{H}_2$

12. Read the following statement and explanation and answer as per the option given below : [IIT-98]

Assertion : Addition of Br_2 to 1-butene gives two optical isomers.

Reason : The product contains one asymmetric carbon.

- (A) If both assertion and reason are correct, and reason is the correct explanation of the assertion
(B) If both assertion and reason are correct, but reason is not correct explanation of the assertion
(C) If assertion is correct but reason is incorrect
(D) If assertion is incorrect but reason is correct

13. The reaction of $\text{CH}_3\text{CH}=\text{CH}-\text{C}_6\text{H}_4-\text{OH}$ with HBr gives - [IIT-98]

- (A) $\text{CH}_3\text{CHBrCH}_2-\text{C}_6\text{H}_4-\text{OH}$ (B) $\text{CH}_3\text{CH}_2\text{CHBr}-\text{C}_6\text{H}_4-\text{OH}$
(C) $\text{CH}_3\text{CHBrCH}_2-\text{C}_6\text{H}_4-\text{Br}$ (D) $\text{CH}_3\text{CH}_2\text{CHBr}-\text{C}_6\text{H}_4-\text{Br}$

14. In the compound $\text{CH}_2=\text{CH}-\text{CH}_2-\text{C}\equiv\text{CH}$, the C_2-C_3 bond is of the type [IIT-99]

- (A) $\text{sp}-\text{sp}^2$ (B) sp^2-sp^3
(C) $\text{sp}-\text{sp}^3$ (D) sp^3-sp^3

15. The product obtained via oxymercuration ($\text{HgSO}_4 + \text{H}_2\text{SO}_4$) of 1-Butyne would give - [IIT-99]

- (A) $\text{CH}_3\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$ (B) $\text{CH}_3\text{CH}_2\text{CH}_2-\text{CHO}$
(C) $\text{CH}_3\text{CH}_2\text{CHO} + \text{HCHO}$ (D) $\text{CH}_3\text{CH}_2\text{COOH} + \text{HCOOH}$

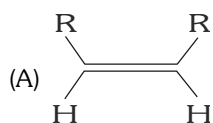
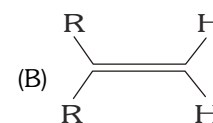
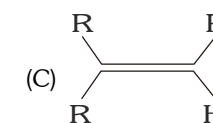
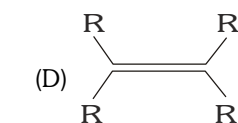
16. Read the following statement and explanation and answer as per the option given below : [IIT-2000]

Assertion : 1-Butene on reaction with HBr in the presence of a peroxide produces 1-bromobutane.

Reason : It involves the formation of a primary radical.

- (A) If both assertion and reason are correct, and reason is the correct explanation of the assertion
(B) If both assertion and reason are correct, but reason is not correct explanation of the assertion
(C) If assertion is correct but reason is incorrect (D) If assertion is incorrect but reason is correct

17. Which one of the following alkenes will react fastest with H_2 under catalytic hydrogenation condition - [IIT-2000]

- (A)  (B)  (C)  (D) 

18. Propyne and propene can be distinguished by - [IIT-2000]

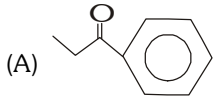
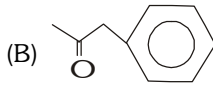
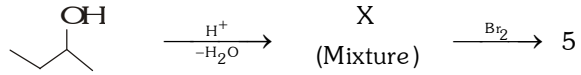
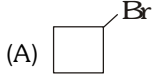
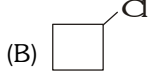
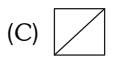
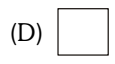
- (A) conc. H_2SO_4 (B) Br_2 in CCl_2 (C) dil. KMnO_4 (D) AgNO_3 in ammonia

19. Read the following statement and explanation and answer as per the option given below : [IIT-2001]

Assertion : Addition of bromine to trans-2-butene yields meso-2, 3-dibromobutane.

Reason : Bromine addition to an alkene is an electrophilic addition.

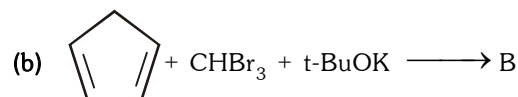
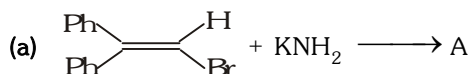
- (A) If both assertion and reason are correct, and reason is the correct explanation of the assertion
(B) If both assertion and reason are correct, but reason is not correct explanation of the assertion
(C) If assertion is correct but reason is incorrect
(D) If assertion is incorrect but reason is correct

20. In the presence of peroxide, hydrogen chloride and hydrogen iodide do not give anti-Markovnikov addition to alkene because - [IIT-2001]
 (A) Both are highly ionic
 (B) One is oxidising and the other is reducing
 (C) One of the step is endothermic in both the cases
 (D) All the steps are exothermic in both cases
21. The reaction of propene with HOCl proceeds via the addition of - [IIT-2001]
 (A) H^+ in the first step
 (B) Cl^+ in first step
 (C) OH^- in first step
 (D) Cl^+ and OH^- in single step
22. The nodal plane in the π -bond of ethene is located in - [IIT-2001]
 (A) the molecular plane
 (B) a plane parallel to the molecular plane
 (C) a plane perpendicular to the molecular plane which contains the carbon-carbon σ -bond at right angle
 (D) a plane perpendicular to the molecular plane which contains the carbon-carbon σ -bond
23. Consider the following reaction $H_3C-\underset{\substack{| \\ D}}{CH}-\underset{\substack{| \\ CH_3}}{CH}-CH_3 + \dot{Br} \rightarrow X + HBr$. Identify the structure of major product X : [IIT-2002]
 (A) $H_3C-\underset{\substack{| \\ D}}{CH}-\underset{\substack{| \\ CH_3}}{CH}-\dot{C}H_2$ (B) $H_3C-\underset{\substack{| \\ D}}{CH}-\dot{C}-CH_3$ (C) $H_3C-\underset{\substack{| \\ D}}{\dot{C}}-\underset{\substack{| \\ CH_3}}{CH}-CH_3$ (D) $H_3C-\dot{C}H-\underset{\substack{| \\ CH_3}}{CH}-CH_3$
24. Identify a reagent from the following list which can easily distinguish between 1-butyne and 2-butyne : [IIT-2002]
 (A) Bromine, CCl_4
 (B) H_2 , Lindlar catalyst
 (C) dilute H_2SO_4 , $HgSO_4$
 (D) ammonical Cu_2Cl_2 solution
25. $C_6H_5-C\equiv C-CH_3 \xrightarrow[H_2SO_4]{HgSO_4} A$ [IIT-2003]
 (A)  (B)  (C) $C_6H_5-C(OH)=CH-CH_3$ (D) $C_6H_5-CH=C(OH)-CH_3$
26.  [IIT-2003]
 compounds of molecular formula $C_4H_8Br_2$. No. of compounds X will be
 (A) 2 (B) 3 (C) 4 (D) 5
27. 2-hexyne can be converted into trans -2-hexene by the reaction of - [IIT-2004]
 (A) $H_2-Pd-BaSO_4$ (B) Li in Liquid NH_3 (C) H_2-PtO_2 (D) $NaBH_4$
28. 1-Bromo-3-chloro cyclobutane on reaction with 2-equivalent of sodium in ether gives - [IIT-2005]
 (A)  (B)  (C)  (D) 
29. $CH_3-CH=CH_2 \xrightarrow{NOCl}$ Product, product is : [IIT-2006]
 (A) $CH_3-\underset{\substack{| \\ Cl}}{CH}-CH_2-NO$ (B) $CH_3-\underset{\substack{| \\ NO}}{CH}-CH_2-Cl$
 (C) $CH_3-CH_2-CH_2-Cl$ (D) $NO-CH_2-CH_2-CH_2-Cl$

30. Cyclohexene on ozonolysis followed by reaction with zinc dust and water gives compound E. Compound E on further treatment with aqueous KOH yields compound F. Compound F is : [IIT-2007]

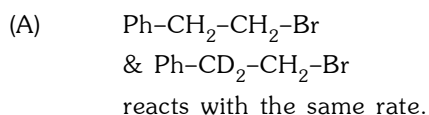


31. Complete the following, giving the structures of the principal organic products. [IIT 1997]

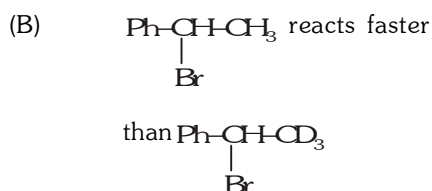


32. Match the following :

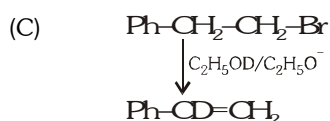
[IIT 2006]



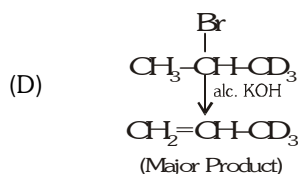
(p) E^1 reaction



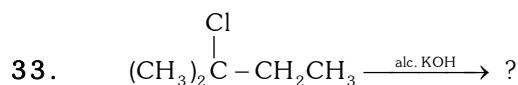
(q) E^2 reaction



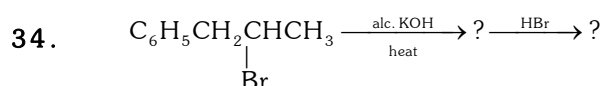
(r) E_{cb}^1 reaction



(s) 1st order reaction



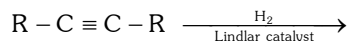
[IIT-1992]



[IIT-1993]

35. $\text{C}(\text{C}_6\text{H}_{12})$, an optically active hydrocarbon which on catalytic hydrogenation gives an optically inactive compound, C_6H_{14} . [IIT-1993]

36. Draw the stereochemical structure of the product in the following reactions. [IIT-1994]

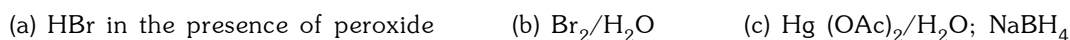


37. Write down the structure of the stereoisomers formed when cis-2-butene is reacted with bromine.

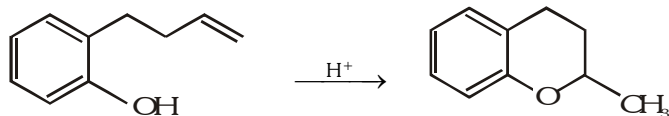
[IIT-1995]

38. An organic compound $\text{E}(\text{C}_5\text{H}_8)$ on hydrogenation gives compound $\text{F}(\text{C}_5\text{H}_{12})$. Compound E on ozonolysis gives formaldehyde and 2-ketopropanal. Deduce the structure of compound E. [IIT-1995]

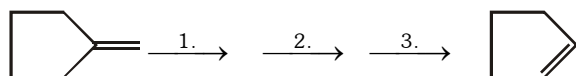
39. Give the structures of the major organic products from 3-ethyl-2-pentene under each of the following reaction conditions. [IIT-1996]



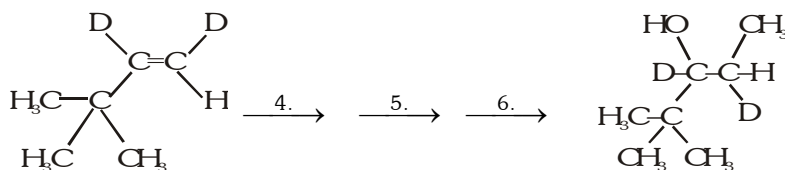
40. An alkyl halide, (X) of formula $C_6H_{13}Cl$ on treatment with potassium tertiary butoxide gives two isomeric alkenes (Y) and (Z) (C_6H_{12}). Both alkenes on hydrogenation give 2, 3-dimethylbutane. Predict the structures of (X), (Y) and (Z). [IIT-1996]
41. 3, 3-Dimethyl-butan-2-ol loses a molecule of water in the presence of concentrated sulphuric acid to give tetramethylethylene as a major product. Suggest a suitable mechanism. [IIT-1996]
42. One mole of the compound A (molecular formula C_8H_{12}), incapable of showing stereoisomerism, reacts with only one mole of H_2 on hydrogenation over Pd. A undergoes ozonolysis to give a symmetrical diketone B ($C_8H_{12}O_2$). What are the structure of A and B ? [IIT-1997]
43. Compound (A) C_6H_{12} gives a positive test with bromine in carbon tetrachloride. Reaction of (A) with alkaline $KMnO_4$ yields only (B) which is the potassium salt of an acid. Write structure formulae and IUPAC name of (A) and (B). [IIT-1997]
44. The central carbon-carbon bond in 1, 3-butadiene is shorter than that of n-butane. Why ? [IIT-1998]
45. Write the intermediate steps for the following reaction. [IIT-1998]
 $C_6H_5CH(OH)C\equiv CH \rightarrow C_6H_5CH=CHCHO$
46. Write the intermediate steps for the following reaction. [IIT-1998]



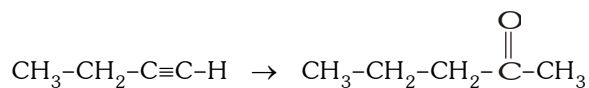
47. Complete the following - [IIT-1999]



48. Complete the following- [IIT-1999]

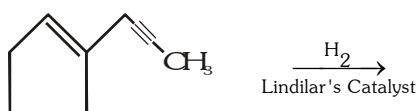


49. Carry out the following transformation in not more than three steps. [IIT 1999]



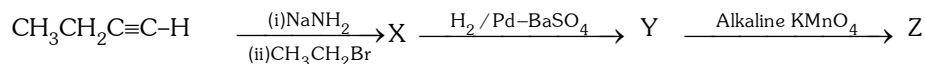
50. $CH_2=CH^-$ is more basic than $HC\equiv C^-$. Why ? [IIT 2000]

51. What would be the major product in each of the following reaction ? [IIT 2000]



52. On reaction with 4N alcoholic KOH at $175^\circ C$ 1-pentyne is slowly converted into equilibrium mixture of 1.3% 1-pentyne (A), 95.2% 2-pentyne (B) and 3.5% 1, 2-pentadiene (C). Give the suitable mechanism of formation of A, B and C with all intermediates. [IIT-2001]

53. Identify X, Y and Z in the following synthetic scheme and write their structure. Is the compound Z optically active? Justify your answer. [IIT-2002]



54. HCHO and $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$ are the products obtained on ozonolysis of a monomer (A) of a polymer. [IIT-2005]

(a) Give the structure of (A)

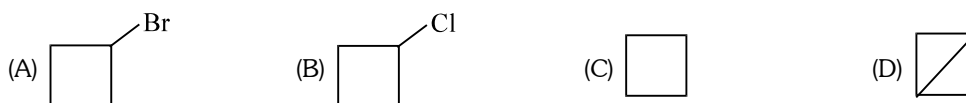
(b) Draw the all "cis" form of a polymer of a monomer (A)

- Q.55 When Phenyl Magnesium Bromide reacts with tert. butanol, which of the following is formed?

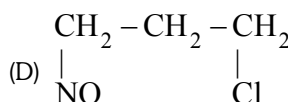
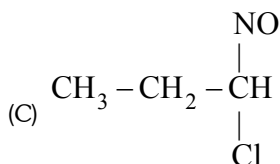
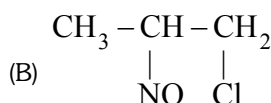
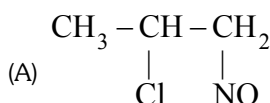
(A) Tert. butyl methyl ether (B) Benzene
(C) Tert. butyl benzene (D) Phenol

[IIT '2005]

- Q.56 1-bromo-3-chlorocyclobutane when treated with two equivalents of Na, in the presence of ether which of the following will be formed? [IIT '2005]



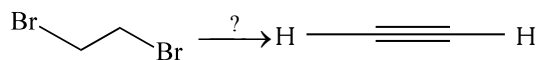
- Q.57 $\text{CH}_3-\text{CH}=\text{CH}_2 + \text{NOCl} \rightarrow \text{P}$
Identify the adduct. [IIT 2006]



- Q.58 The number of stereoisomers obtained by bromination of trans-2-butene is [IIT 2007]
(A) 1 (B) 2 (C) 3 (D) 4

- Q.59 The number of structural isomers for C_6H_{14} is [IIT 2007]
(A) 3 (B) 4 (C) 5 (D) 6

- Q.60 The reagent(s) for the following conversion, [IIT 2007]



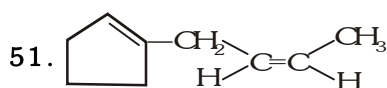
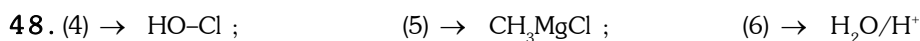
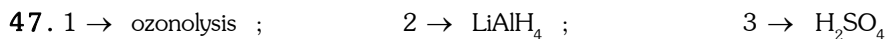
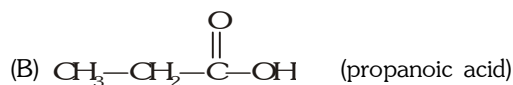
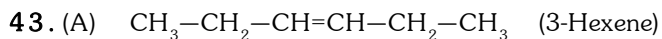
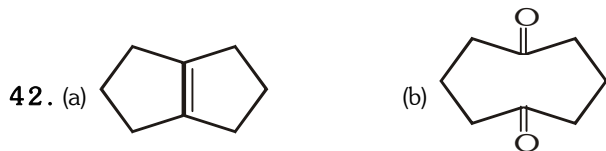
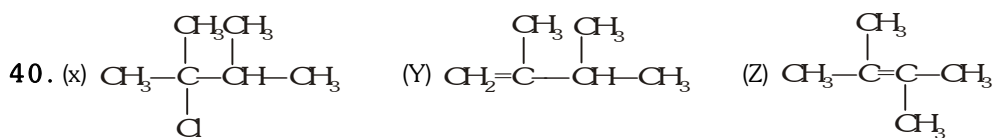
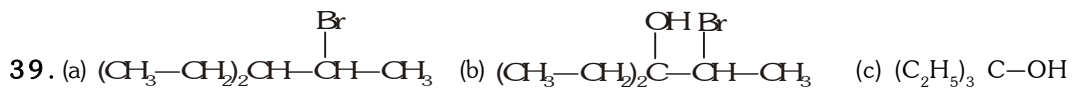
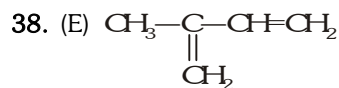
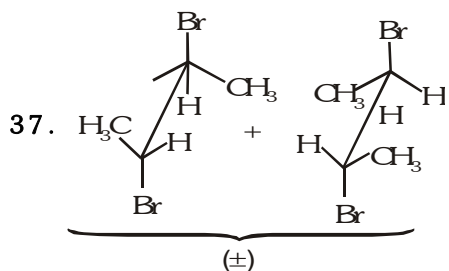
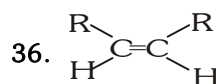
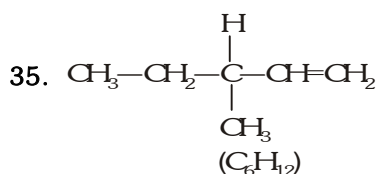
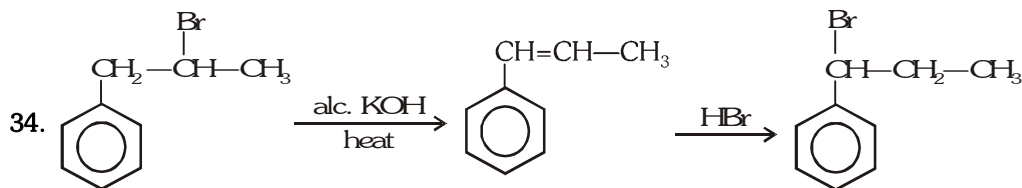
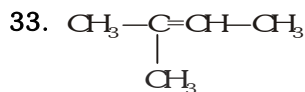
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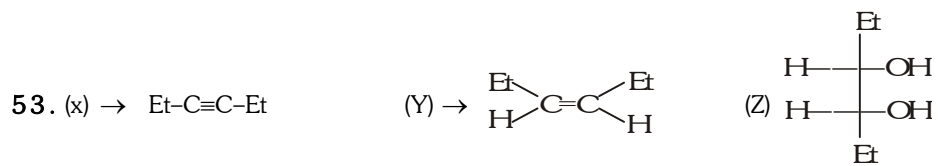
(A) alcoholic KOH (B) alcoholic KOH followed by NaNH_2
(C) aqueous KOH followed by NaNH_2 (D) $\text{Zn} / \text{CH}_3\text{OH}$

- Q.61 The synthesis of 3-octyne is achieved by adding a bromoalkane into a mixture of sodium amide and an alkyne. The bromoalkane and alkyne respectively are [IIT-2010]

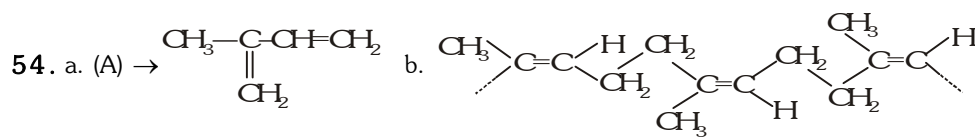
(A) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH}$ (B) $\text{BrCH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{C}\equiv\text{CH}$
(C) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{C}\equiv\text{CH}$ (D) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH}$

1. (C) 2. (A) 3. (A) 4. (C) 5. (A) 6. (A) 7. (B) 8. (D)
 9. (A) 10. (C) 11. (B) 12. (A) 13. (B) 14. (B) 15. (A) 16. (C)
 17. (A) 18. (D) 19. (B) 20. (C) 21. (B) 22. (C) 23. (B) 24. (D)
 25. (A) 26. (B) 27. (B) 28. (C) 29. (A) 30. (B)





(Z) is meso compound so optically inactive.



55. (B)

56. (D)

57. (A)

58. (A)

59. (C)

60. (B)

61. (D)

EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. Which of the following factors affects the adsorption of a gas on solid ?
(A) T_c (critical temp.) (B) Temperature of gas (C) Pressure of gas (D) All of them
2. The volume of gases NH_3 , CO_2 and CH_4 adsorbed by one gram of charcoal at 298 K are in
(A) $\text{CH}_4 > \text{CO}_2 > \text{NH}_3$ (B) $\text{NH}_3 > \text{CH}_4 > \text{CO}_2$ (C) $\text{NH}_3 > \text{CO}_2 > \text{CH}_4$ (D) $\text{CO}_2 > \text{NH}_3 > \text{CH}_4$
3. The heat of physisorption lie in the range of
(A) 1 to 10 kJ mol^{-1} (B) 20 to 40 kJ mol^{-1} (D) 40 to 200 kJ mol^{-1} (D) 200 to 400 kJ mol^{-1}
4. Which of the following is not a gel ?
(A) Cheese (B) Jellies (C) Curd (D) Milk
5. Which of the following is used to adsorb water
(A) Silica gel (B) Calcium acetate (C) Hair gel (D) Cheese
6. An emulsion is a colloidal system of
(A) two solids (B) two liquids
(C) one gas and one solid (D) one gas and one liquid
7. Which of the following is a lyophobic colloid ?
(A) Gelatin (B) Sulphur (C) Starch (D) Gum arabic
8. The nature of bonding forces in adsorption
(A) purely physical such as Vander Waal's forces (B) purely chemical
(C) both chemical and physical always (D) none of these
9. The Tyndall effect associated with colloidal particles is due to
(A) presence of electrical charges (B) scattering of light
(C) adsorption of light (D) reflection of light
10. Which one of the following is not applicable to chemisorption ?
(A) Its heat of adsorption is high (B) It takes place at high temperature
(C) It is reversible (D) It forms mono-molecular layers
11. In the colloidal state the particle size ranges
(A) below 1 nm (B) between 1 nm to 100 nm
(C) more than 100 nm (D) none of the above
12. All colloids
(A) are suspensions of one phase in another
(B) are two phase systems
(C) contain only water soluble particles
(D) are true solutions
13. Colloids can be purified by
(A) condensation (B) peptization (C) coagulation (D) dialysis
14. Milk is an example of
(A) emulsion (B) suspension (C) foam (D) sol
15. Fog is a colloidal system of
(A) gas in liquid (B) liquid in gas (C) gas in gas (D) gas in solid

16. When a colloidal solution is observed under ultramicroscope, we can see
 (A) light scattered by colloidal particle (B) size of the colloidal particle
 (C) shape of the colloidal particle (D) relative size of the colloidal particle
17. Colloidal solutions are classified on the basis of
 (A) molecular size (B) organic or inorganic (C) surface tension value (D) pH value
18. The electrical charge on a colloidal particle is indicated by
 (A) Brownian movement (B) electrophoresis (C) ultra microscope (D) molecular sieves
19. Crystalloids differ from colloids mainly in respect of
 (A) electrical behaviour (B) particle nature (C) particle size (D) solubility
20. Which one of the following is lyophilic colloid?
 (A) Milk (B) Gum (C) Fog (D) Blood
21. Small liquid droplets dispersed in another liquid is called
 (A) suspension (B) emulsion (C) gel (D) true solution
22. The process which is catalysed by one of the product is called
 (A) acid-base catalysis (B) autocatalysis
 (C) negative catalysis (D) homogeneous catalysis
23. Tyndall effect would be observed in a
 (A) solution (B) solvent (C) precipitate (D) colloidal sol
24. Adsorption is multilayer in case of
 (A) physical adsorption (B) chemisorption (C) in both (D) none of these
25. A liquid is found to scatter a beam of light but leaves no residue when passed through the filter paper. The liquid can be described as
 (A) a suspension (B) oil (C) a colloidal sol (D) a true solution
26. The ability of an ion to bring about coagulation of a given colloid depends upon
 (A) its charge (B) the sign of the charge alone
 (C) the magnitude of the charge (D) both magnitude and sign of charge
27. An emulsifier is a substance
 (A) which stabilises an emulsion
 (B) which breaks the emulsion into its constituent liquids
 (C) which can convert liquid into an emulsion
 (D) which bring about coagulation of an emulsion

CHECK YOUR GRASP						ANSWER KEY				EXERCISE -1					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	D	C	B	D	A	B	B	D	B	C	B	B	D	A	B
Que.	16	17	18	19	20	21	22	23	24	25	26	27			
Ans.	A	A	B	C	B	B	B	D	A	C	D	A			

EXERCISE-02**BRAIN TEASERS****SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)**

1. Which gas will be adsorbed on a solid to greater extent.
(A) A gas having non polar molecule (B) A gas having highest critical temperature (T_c)
(C) A gas having lowest critical temperature. (D) A gas having highest critical pressure.
2. Colloidal particles in a sol. can be coagulated by
(A) heating (B) adding an electrolyte
(C) adding oppositely charged sol (D) any of the above methods
3. Emulsifier is an agent which
(A) accelerates the dispersion (B) homogenizes an emulsion
(C) stabilizes an emulsion (D) aids the flocculation of an emulsion
4. Given below are a few electrolytes, indicate which one among them will bring about the coagulation of a gold sol. quickest and in the least of concentration?
(A) NaCl (B) $MgSO_4$ (C) $Al_2(SO_4)_3$ (D) $K_4[Fe(CN)_6]$
5. The minimum concentration of an electrolyte required to cause coagulation of a sol is called
(A) flocculation value (B) gold number (C) protective value (D) none of these
6. Smoke precipitator works on the principle of
(A) distribution law (B) neutralization of charge on colloids
(C) Le-Chatelier's principle (D) addition of electrolytes
7. Which one of following statements is not correct in respect of lyophilic sols ?
(A) There is a considerable interaction between the dispersed phase and dispersion medium
(B) These are quite stable and are not easily coagulated
(C) They carry charge
(D) The particle are hydrated
8. As_2S_3 sol is
(A) positive colloid (B) negative colloid (C) neutral colloid (D) none of the above
9. Which of the following electrolyte will be most effective in coagulation of gold sol ?
(A) $NaNO_3$ (B) $K_4[Fe(CN)_6]$ (C) Na_3PO_4 (D) $MgCl_2$
10. At the critical micelle concentration (CMC) the surfactant molecules
(A) decompose (B) dissociate
(C) associate (D) become completely soluble
11. Alums purify muddy water by
(A) dialysis (B) absorption (C) coagulation (D) forming true solution
12. Solute dispersed in ethanol is called
(A) emulsion (B) micelle (C) hydrophilic sol. (D) alcosols
13. An arsenious sulphide sol. carries a negative charge. The maximum precipitating power of this sol. is possessed by
(A) K_2SO_4 (B) $CaCl_2$ (C) Na_3PO_4 (D) $AlCl_3$
14. Reversible adsorption is
(A) chemical adsorption (B) physical adsorption (C) both (D) none

15. The function of gum arabic in the preparation of Indian ink is
 (A) coagulation (B) peptisation (C) protective action (D) adsorption
16. Which of the following is an example of associated colloid?
 (A) Protein + water (B) Soap + water (C) Rubber + benzene (D) $\text{As}_2\text{O}_3 + \text{Fe}(\text{OH})_3$
17. Adsorption of gases on solid surface is generally exothermic because
 (A) enthalpy is positive (B) entropy decreases (C) entropy increases (D) free energy increases
18. Which of the following is/are correct statements
 (A) Hardy Schulz rule is related to coagulation
 (B) Brownian movement and Tyndall effect are shown by colloids
 (C) When liquid is dispersed in liquid, it is called gel.
 (D) Gold number is a measure of protective power of lyophillic colloid.
19. Which statement is/are correct ?
 (A) Physical adsorption is multilayer non-directional and non-specific
 (B) Chemical adsorption is generally monolayer and specific in nature
 (C) Physical adsorption is due to free valence of atoms
 (D) Chemical adsorption is stronger than physical adsorption
20. Which is the following is/are correct for lyophillic sols ?
 (A) Its surface tension is lower than that of H_2O
 (B) Its viscosity is higher than that of water
 (C) Its surface tension is higher than that of water
 (D) Its viscosity is equal to that of water
21. Which statement(s) is/are correct
 (A) A solution is prepared by addition of excess of AgNO_3 solution in KI solution. The charge likely to develop on colloidal particle is positive.
 (B) The effects of pressure on physical adsorption is high if temperature is low.
 (C) Ultracentrifugation process is used for preparation of lyophobic colloids.
 (D) Gold number is the index for extent of gold plating done.
22. Colloidal solution can be purified by
 (A) Dialysis (B) Electrodialysis (C) Electrophoresis (D) ultrafiltration
23. Coagulation of colloids can be achieved by
 (A) Centrifugation (B) Adding electrolyte (C) Change in pH (D) Adding water
24. When -vely charged colloid like As_2S_3 sol is added to +vely charged $\text{Fe}(\text{OH})_3$ sol in suitable amounts
 (A) Both the sols are precipitated simultaneously
 (B) This process is called mutual coagulation
 (C) They become +vely charged colloid
 (D) They become -vely charged colloid
25. Which of the following is not lyophillic
 (A) Gelatin sol (B) Silver sol (C) Sulphur sol (D) As_2S_3 sol
26. Colloidal Gold can be prepared by
 (A) Bredig's are method (B) Reduction of AuCl_3 (C) Hydrolysis (D) Peptization

27. The coagulation of sol particles may be brought about by
(A) heating (B) addition oppositely charged sol
(C) addition electrolyte (D) persistent dialysis
28. Which one is not lyophobic in nature ?
(A) Gelatin (B) Sulphur (C) Starch (D) Protein
29. Which of the following are colloids?
(A) Milk (B) Ice cream (C) Urea solution (D) Blood
30. Which are the properties of sols.
(A) Adsorption (B) Tyndall effect (C) Flocculation (D) Paramagnetism
31. The migration of colloidal particles under the influence of an electrical field is known as
(A) electro osmosis (B) electrophoresis (C) electrodialysis (D) None

[illegible]

EXERCISE-03

MISCELLANEOUS TYPE QUESTIONS

TRUE / FALSE

1. Physisorption is non-specific.
2. Chemisorption needs activation energy.
3. A graph of x/m vs temperature at constant pressure is called adsorption isotherm.
4. Suspensions have solute particles with size less than 1 nm.
5. $\text{Fe}(\text{OH})_3$ sol contains positively charged colloidal particles.
6. Chemisorption is irreversible.
7. Adsorption isobars of chemisorption and physisorption are of the same type.
8. Milk is an example of water in oil emulsions.
9. Gold sol can be prepared by Bredig's arc method.
10. Gel is a system in which liquid is the dispersed phase and solid is the dispersion medium.

FILL IN THE BLANKS

1. The substance on whose surface adsorption takes place is called an
2. Removal of adsorbate from the surface of adsorbent is called
3. Migration of colloidal particles under the effect of electric field is called
4. The heat of adsorption in case of physisorption is approximately
5. The phenomenon of zig-zag motion of colloidal particles is known as
6. Lyophilic sols are stable than lyophobic sols.
7. Electrical properties of a colloidal solution are demonstrated by
8. Tyndall effect takes place due to of light by colloidal particles.
9. The liquid-liquid colloidal dispersions are called
10. The movement of dispersion medium under the influence of an electric field is called
11. Smoke is a colloidal solution of in
12. The adhering of the molecules of a gas on the surface of a solid is called
13. The protective action of different colloids is compared in term of
14. The colloidal dispersion of a liquid in a liquid is called
15. The colloidal dispersions of liquids in solid media are called

ASSERTION & REASON

These questions contains, Statement I (assertion) and Statement II (reason).

- (A) Statement-I is true, Statement-II is true ; Statement-II is correct explanation for Statement-I.
(B) Statement-I is true, Statement-II is true ; Statement-II is NOT a correct explanation for statement-I
(C) Statement-I is true, Statement-II is false
(D) Statement-I is false, Statement-II is true

1. **Statement-I** : Isoelectric point is pH at which colloidal can move towards either of electrode
Because
Statement-II : At isoelectric point, colloidal particles become electrically neutral.
2. **Statement-I** : When AgNO_3 is treated with excess of potassium iodide, colloidal particles gets attracted towards anode.
Because
Statement-II : Precipitate adsorb common ions (excess) and thus become charged.

3. **Statement-I** : For adsorption ΔG , ΔH , ΔS all have -ve values
Because
Statement-II : Adsorption is a spontaneous exothermic process in which randomness decreases due to force of attraction between adsorbent and adsorbate.
4. **Statement-I** : A gas with higher critical temperature gets adsorbed to more extent than a gas with lower critical temperature.
Because
Statement-II : The easily liquifiable gases get adsorbed to more extent.
5. **Statement-I** : Micelles are formed by surfactant molecules above the critical micellar concentration (CMC).
Because
Statement-II : The conductivity of a solution having surfactant molecules decreases sharply at the CMC.

COMPREHENSION BASED QUESTIONS

Comprehension # 1

Question No. 46 to 50 (5 questions)

Whenever a mixture of gases is allowed to come in contact with a particular adsorbent under the same conditions, the more strong adsorbate is adsorbed to greater extent irrespective of its amount present, e.g., H_2O is adsorbed to more extent on silica gel than N_2 and O_2 . This shows that some adsorbates are preferentially adsorbed. It is also observed that preferentially adsorbable adsorbents can displace a weakly adsorbed substance from the surface of an adsorbent.

- Which of the following gases is adsorbed to maximum extent :
 (A) He (B) Ne (C) Ar (D) Xe
- Which of the gas can displace remaining all the gases
 (A) O_2 (B) N_2 (C) CO (D) H_2
- When temperature is increased
 (A) extent of adsorption increases (B) extent of adsorption decreases
 (C) no effect on adsorption (D) extent of adsorption first decreases, then increases
- Chromatographic separations are based on
 (A) differential solubility (B) differential adsorption (C) differential absorption (D) None of these
- Activated charcoal is prepared by
 (A) heated charcoal with steam so that it becomes more porous
 (B) addition $Ca_3(PO_4)_2$ to charcoal
 (C) addition impurity to charcoal
 (D) reacted with conc. HNO_3

Comprehension # 2

Question No. 51 to 54 (4 questions)

The clouds consist of charged particles of water dispersed in air. Some of them are +vely charged, others are negatively charged. When +vely charged clouds come closer they have cause lightening and thundering whereas when +ve and -ve charged colloids come closer they cause heavy rain by aggregation of minute particles. It is possible to cause artificial rain by throwing electrified sand or silver iodide from an aeroplane and thus coagulation the mist hanging in air.

- When excess of AgNO_3 is treated with KI solution, AgI forms
(A) +ve charged sol (B) -vely charged sol (C) neutral sol (D) true solution
- Clouds are colloidal solution of
(A) liquid in gas (B) gas in liquid (C) liquid in liquid (D) solid in liquid
- AgI helps in artificial rain because
(A) It helps in coagulation (B) It helps in dispersion process
(C) Both (D) None
- Electrical chimneys are made on the principle of
(A) Electroosmosis (B) Electrophoresis (C) Coagulation (D) All of these

Comprehension # 3

Question No. 55 to 58 (4 questions)

In macromolecular type of colloids, the dispersed particles are themselves large molecules (usually polymers). Since these molecules have dimensions comparable to those of colloidal particles, their dispersions are called macromolecular colloids. Most lyophilic sols belong to this category. There are certain colloids which behave as normal strong electrolytes at low concentrations, but exhibit colloidal properties at higher concentrations due to the formation of aggregate particles. These are known as micelles or associated colloids. Surface active agents like soaps and synthetic detergent belong to this class.

- ◆ Critical micelle concentration (CMC) is the lowest concentration at which micelle formation appears. CMC increases with the total surfactant concentration. At concentration higher than CMC, they form extended parallel sheets known as lamellar micelles which resemble biological membranes. With two molecules thick, the individual molecule is perpendicular to the sheets such that hydrophilic groups are on the outside in aqueous solution and on the inside in a non-polar medium
- ◆ In concentrated solution, micelles take the form of long cylinders packed in hexagonal arrays and are called lyotropic mesomorphs.
- ◆ In an aqueous solution (polar medium), the polar group points towards the periphery and the hydrophobic hydrocarbon chains point towards the centre forming the core of the micelle.
- ◆ Micelles from the ionic surfactants can be formed only above a certain temperature called the **Kraft temperature**.
- ◆ They are capable of forming ions.
- ◆ Molecules of soaps and detergents consist of lyophilic as well as lyophobic parts which associate together to form micelles.
- ◆ Micelles may contain as many as 100 molecules or more.

- Select incorrect statement(s) :
(A) Surface active agent like soaps and synthetic detergents are micelles
(B) Soaps are emulsifying agents
(C) $\text{C}_{17}\text{H}_{35}$ (hydrocarbon part) and $-\text{COO}^-$ (carboxylate part) of stearate ion ($\text{C}_{17}\text{H}_{35}\text{COO}^-$) both are hydrophobic
(D) All are incorrect statements
- Which part of the soap (RCOO^-) dissolved grease and forms micelle ?
(A) R part (called tail of the anion) (B) $-\text{COO}^-$ part (called head of the anion)
(C) both (A) and (B) (D) none of these
- In multimolecular colloidal sols, atoms or molecules are held together by:
(A) H-bonding (B) vander Waals forces
(C) ionic bonding (D) polar covalent bonding

4. Cleansing action of soap occurs because :
- (A) oil and grease can be absorbed into the hydrophobic centres of soap micelles and washed away
- (B) oil and grease can be absorbed into hydrophilic centres of soap micelles and washed away
- (C) oil and grease can be absorbed into both hydrophilic and hydrophobic centres but not washed away
- (D) cleansing action is not related to micelles

Comprehension # 4

Question No. 59 to 61 (3 questions)

The protective power of the lyophilic colloids is expressed in terms of gold number a term introduced by Zsigmondy. Gold number is the number of milligram of the protective colloid which prevent the coagulation of 10 mL of red gold sol. when 1 mL of a 10 percent solution of sodium chloride is added to it. Thus, smaller the gold number of lyophilic colloid, the greater is the protective power.

1. On addition of one mL of solution of 10% NaCl to 10 mL of red gold sol in presence of 0.025 g of starch, the coagulation is just prevented. The gold number of starch is
 (A) 0.025 (B) 0.25 (C) 2.5 (D) 25
2. Which of the following statement(s) is/are correct
 (A) Higher the gold number, more protective power of colloid
 (B) Lower the gold number, more the protective power
 (C) Higher the coagulation value, more the coagulation power
 (D) Lower the coagulation value, higher the coagulation power
3. Gold number gives an indication of
 (A) protective nature of colloids (B) purity of gold in suspension
 (C) the charge on a colloidal solution of gold (D) g-mole of gold per litre

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
<u>True / False</u> 1. T 2. T 3. F 4. F 5. T 6. T 7. F 8. F 9. T 10. T <u>Fill in the Blanks</u> 1. absorbent 2. desorption 3. electrophoresis 4. 20–40 kJ·mol⁻¹ 5. Brownian movement 6. more 7. electrophoresis 8. scattering 9. emulsions 10. electro-osmosis 11. solid in gas 12. adsorption 13. gold number 14. emulsion 15. gel <u>Assertion - Reason Questions</u> 1. B 2. C 3. C 4. C 5. D <u>Comprehension Based Questions</u> Comprehension # 1: 1. (D) 2. (C) 3. (B) 4. (B) 5. (A) Comprehension # 2: 1. (A) 2. (A) 3. (C) 4. (B) Comprehension # 3: 1. (C,D) 2. (C) 3. (B) 4. (A) Comprehension # 4: 1. (D) 2. (B,D) 3. (A)		

JEE-[MAIN] : PREVIOUS YEAR QUESTIONS

- $$(1) \frac{V_c}{V_s} \simeq 1 \qquad (2) \frac{V_c}{V_s} \simeq 10^{23} \qquad (3) \frac{V_c}{V_s} \simeq 10^{-3} \qquad (4) \frac{V_c}{V_s} \simeq 10^3$$

- (1) Magnesium chloride solution coagulates, the gold sol more readily than the iron (III) hydroxide sol
- (2) Sodium sulphate solution cause coagulation in both sols
- (3) Mixing the sols has no effect
- (4) Coagulation in both sols can be brought about by electrophoresis

- (1) Under high pressure it results into multi molecular layer on adsorbent surface
- (2) Enthalpy of adsorption ($\Delta H_{\text{adsorption}}$) is low and positive
- (3) It occurs because of Van der Waal's forces
- (4) More easily liquefiable gases are adsorbed readily

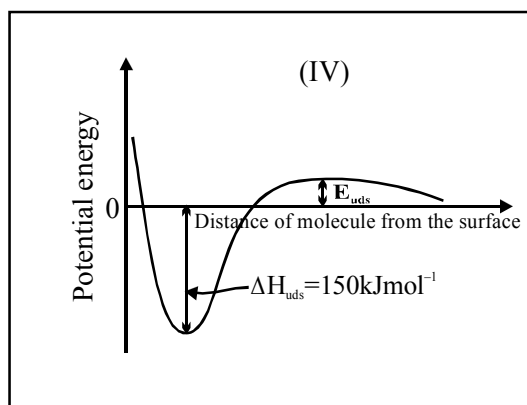
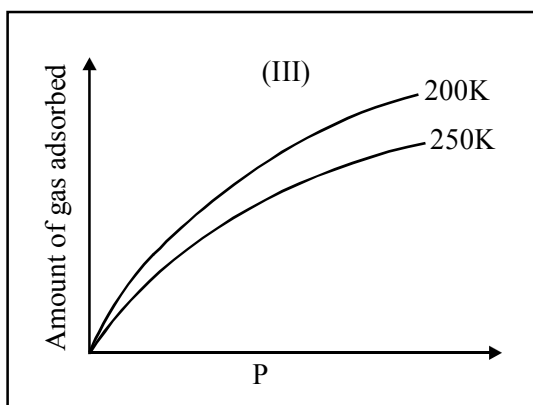
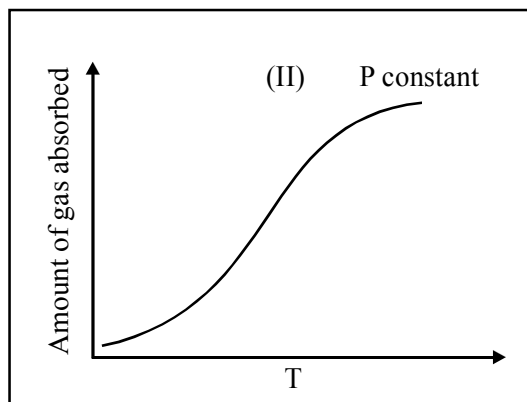
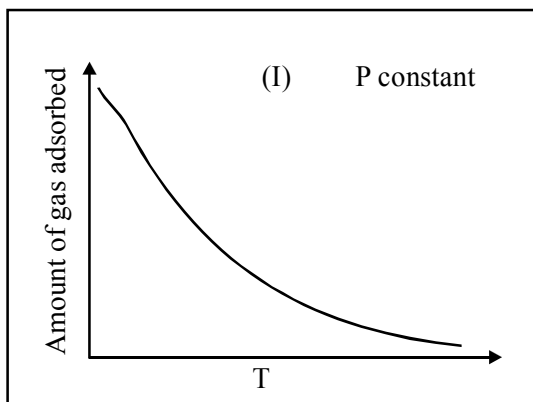
- $$(1) \frac{x}{m} \propto p^0 \qquad (2) \frac{x}{m} \propto p^1 \qquad (3) \frac{x}{m} \propto p^{1/n}$$

- (4) All the above are correct for different ranges of pressure

MISCELLANEOUS TYPE QUESTION	ANSWER KEY				EXERCISE -4(A)
Que.	1	2	3	4	
Ans	4	3	2	4	

EXERCISE - 04 [B]**JEE-[ADVANCED] : PREVIOUS YEAR QUESTIONS**

- Q.1 Among the following, the surfactant that will form micelles in aqueous solution at the lowest molar concentration at ambient conditions is [JEE 2008]
(A) $\text{CH}_3(\text{CH}_2)_{15}\text{N}^+(\text{CH}_3)_3\text{Br}^-$ (B) $\text{CH}_3(\text{CH}_2)_{11}\text{OSO}_3^-\text{Na}^+$
(C) $\text{CH}_3(\text{CH}_2)_6\text{COO}^-\text{Na}^+$ (D) $\text{CH}_3(\text{CH}_2)_{11}\text{N}^+(\text{CH}_3)_3\text{Br}^-$
- Q.2 Among the electrolytes Na_2SO_4 , CaCl_2 , $\text{Al}_2(\text{SO}_4)_3$ and NH_4Cl , the most effective coagulation agent for Sb_2S_3 sol is [JEE 2009]
(A) Na_2SO_4 (B) CaCl_2 (C) $\text{Al}_2(\text{SO}_4)_3$ (D) NH_4Cl
- Q.3 The correct statement(s) pertaining to the adsorption of a gas on a solid surface is (are) - [JEE 2011]
(A) Adsorption is always exothermic
(B) Physisorption may transform into chemisorption at high temperature
(C) Physisorption increases with increasing temperature but chemisorption decreases with increasing temperature
(D) Chemisorption is more exothermic than physisorption, however it is very slow due to higher energy of activation
- Q.4 The given graphs / data **I, II, III and IV** represent general trends observed for different physisorption and chemisorption processes under mild conditions of temperature and pressure. Which of the following choice(s) about **I, II, III and IV** is (are) correct ? [JEE 2012]



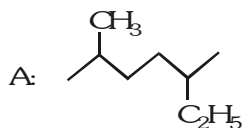
- (A) **I** is physisorption and **II** is chemisorption
(B) **I** is physisorption and **III** is chemisorption
(C) **IV** is chemisorption and **II** is chemisorption
(D) **IV** is chemisorption and **III** is chemisorption

- Q.5 Choose the correct reason(s) for the stability of the **lyophobic** colloidal particle. [JEE 2012]
- (A) Preferential adsorption of ions on their surface from the solution
 - (B) Preferential adsorption of solvent on their surface from the solution
 - (C) Attraction between different particles having opposite charges on their surface
 - (D) Potential difference between the fixed layer and the diffused layer of opposite charges around the colloidal particles

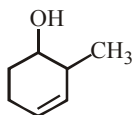
MISCELLANEOUS TYPE QUESTION		ANSWER KEY		EXERCISE -4(B)	
Q.1	A	Q.2	C	Q.3	A,B,D
Q.4	A, C	Q.5	A, D		

EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

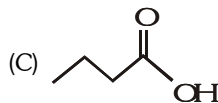
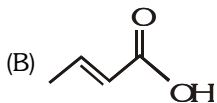
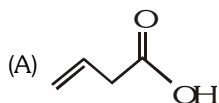
1. IUPAC name of the hydrocarbon (A) is :



- (A) 2-ethyl-5methylhexane (B) 5-ethyl-2methylhexane
(C) 2, 5-dimethylheptane (D) 5-ethyl-2,5-dimethylpentane
2. IUPAC name of neopentyl group is :
(A) 2,2-dimethylbutyl (B) 2, 2-dimethylpropyl (C) 1, 1-dimethylbutyl (D) 1, 1-dimethylpropyl
3. IUPAC name of the following compound is :



- (A) 2-methyl-3-cyclohexenol (B) 3-methyl-1-cyclohexen-4-ol
(C) 4-hydroxy-3-methyl-1-cyclohexene (D) 2-hydroxy-1-methylcyclohexene
4. Which compound is tertiary alcohol ?
(A) 1-propanol (B) 2-methyl-1-hexanol (C) 3-methyl-2-hexanol (D) 2-methyl-2-hexanol
5. 3-Butenoic acid (or But-3-enoic acid) is represented as :



(D) none of these

6. In the nomenclature of priority order is -

- (A) $-\text{CH}_3 < -\text{OH} < \text{C}=\text{C}$ (first) (B) $-\text{OH} < -\text{CH}_3 < \text{C}=\text{C}$ (first)
(C) $\text{C}=\text{C} < -\text{CH}_3 < -\text{OH}$ (first) (D) $-\text{CH}_3 < \text{C}=\text{C} < -\text{OH}$ (first)

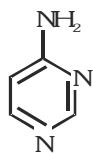
7. $\text{C}_4\text{H}_8\text{O}_2$ represents :-

- (A) An acid only (B) An ester only
(C) An ketone only (D) An acid and an ester also

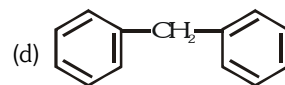
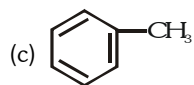
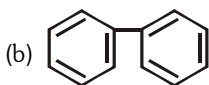
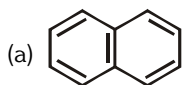
8. The higher homologue of dimethylamine ($\text{CH}_3\text{—NH—CH}_3$) has the structure :-

- (A) $(\text{CH}_3)_3\text{N}$ (B) $\text{CH}_3\text{—CH}_2\text{—CH}_2\text{—NH}_2$
(C) $\text{CH}_3\text{—NH—CH}_2\text{—CH}_3$ (D)

9. The third member of the family of alkenynes has the molecular formula :-
 (A) C_3H_2 (B) C_5H_6 (C) C_6H_8 (D) C_4H_4
10. The hetero atoms present in the following compound is/are :

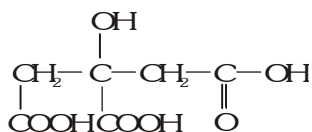


- (A) 2 (B) 3 (C) 1 (D) 4
11. Which of the following have only 2 H-atom :



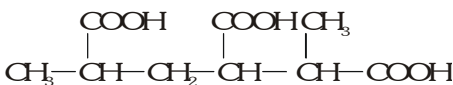
Correct code is :

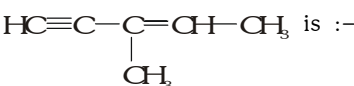
- (A) only a and b (B) a, b and d (C) a, c and d (D) All of them
12. Common name of $CH_2=CH-CN$ is :-
 (a) acrylonitrile (b) vinyl cyanide
 (c) allyl cyanide (d) allyl nitrile
 (A) a, b and d (B) a and b (C) only b (D) a, b and c
13. Which of the following names is correct :-
 (A) 4-Isopropyl-3-methyl hexane (B) 2-Ethyl-3-isopropyl pentane
 (C) 3-Isopropyl-4-methyl hexane (D) 3-Ethyl-2,4-dimethyl hexane
14. The correct systematic IUPAC name of the given compound is :

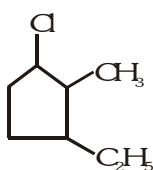


- (A) 3-Carboxy-3-hydroxy butane dioic acid (B) 2-Hydroxy propane-1,2,3-tricarboxylic acid
 (C) 3-Hydroxy butane dioic acid (D) 2-Bis(carboxymethyl)-2-hydroxy ethanoic acid
15. The IUPAC name of $CH_3-CH_2-NH-CH_3$ is :-
 (A) Methyl ethyl amine (B) 1-methyl amino ethane
 (C) N-methyl ethanamine (D) N-ethyl methanamine
16. The correct IUPAC name from the incorrect name 4-Amino-3-hydroxy-2-butene is :-
 (A) 1-Amino-2-hydroxy-2-butene (B) 4-Amino-2-buten-3-ol
 (C) 1-Amino-2-buten-2-ol (D) 1-Amino-2-butenol
17. The correct name of 2-chlorobutan-3-ol is :-
 (A) 3-Chloro-2-hydroxy butane (B) 3-Chloro-2-butanol
 (C) 3-Hydroxy-2-chloro butane (D) 2-Chloro-3-hydroxy butane

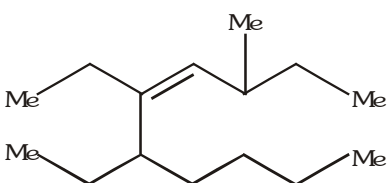
SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)

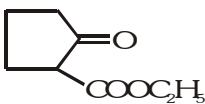
1. The IUPAC name of  is :-
 (A) 2,4,5-Hexane tricarboxylic acid (B) 2,3,5-Hexane tricarboxylic acid
 (C) 2-(1-carboxyethyl)-4-methyl pentanedioic acid (D) 3,5-Dicarboxy-2-methyl hexanoic acid

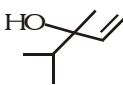
2. The name for  is :-
 (A) 3-Methyl-2-penten-4-yne (B) 3-Methyl-3-penten-1-yne
 (C) 3-Methyl-4-pentyn-1-ene (D) 3-Methyl pentenyne

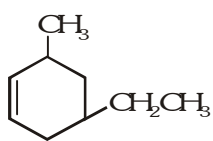
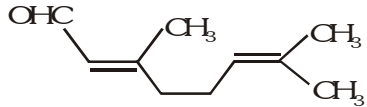
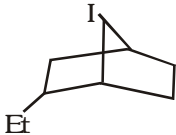
3.  has the IUPAC name as :-
 (A) 3-Chloro-1-ethyl-2-methylcyclopentane (B) 1-Chloro-3-ethyl-2-methylcyclopentane
 (C) 4-Chloro-1-ethyl-5-methylcyclopentane (D) None of above

4. The IUPAC name of $\text{CH}_3\text{CH}_2\text{NHCHO}$ is :
 (A) N-formyl ethanamine (B) Ethyl amino methanal
 (C) N-ethyl methanamide (D) None of them

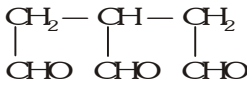
5. The IUPAC name of the  structure is :-
 (A) 2,4,5-Triethyl-3-nonene (B) 5,6-Diethyl-3-methyl-4-decene
 (C) 2,4,5-Triethyl-3-octene (D) 3-Ethyl-5-methyl-3-heptene

6.  has the IUPAC name :
 (A) Ethyl-2-oxo cyclopentane carboxylate
 (B) 2-Cyclopentanone-1-carbethoxy
 (C) 2-Ethylcarbonate cyclopentanone
 (D) 1-Keto-2-carbethoxy cyclopentanone

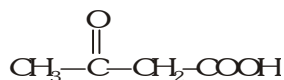
7.  has the IUPAC name :-
 (A) 3,4-Dimethyl-1-penten-3-ol (B) Isopropyl-3-methyl vinyl carbinol
 (C) 2,3-Dimethyl-4-penten-3-ol (D) None of the above

8. The IUPAC name of  is :-
- (A) 1-Methyl-5-ethyl cyclohex-2-ene (B) 5-Ethyl-3-methyl cyclohex-1-ene
(C) 4-Ethyl-6-methyl cyclohex-1-ene (D) 1-Ethyl-5-methyl cyclohex-3-ene
9. In the compound $\text{HC}\equiv\text{C}-\text{CH}_2-\text{CH}=\text{CH}-\text{CH}_3$, the C_2-C_3 bond is the type of :-
- (A) $\text{sp} - \text{sp}^2$ (B) $\text{sp}^3 - \text{sp}^3$
(C) $\text{sp} - \text{sp}^3$ (D) $\text{sp}^2 - \text{sp}^2$
10. The number of all primary amines possible with the molecular formula $\text{C}_3\text{H}_9\text{N}$ and the primary amine with amino group on a primary carbon atom is given by the set :-
- (A) 4, 1 (B) 2, 1 (C) 3, 2 (D) 2, 2
11. The number of primary alkanols, secondary alkanols and tertiary alkanols possible with the formula $\text{C}_4\text{H}_{10}\text{O}$ is given by the set :-
- (A) 2, 1, 0 (B) 1, 2, 1 (C) 2, 1, 1 (D) 2, 1, 2
12. IUPAC name of $\text{CH}_2=\text{CH}-\text{CH}_2\text{NH}_2$ is :
- (A) 3-amino propenamine (B) 2-Propen-1-amine
(C) 3-Amino-1-propenamine (D) Allyl amine
13. Which name is correct :
- (A) 
3, 7-Dimethyloct-2,6-dienal
- (B)  Spiro(3,6) decane
- (C) 
3-Ethyl-7-iodobicyclo (2.2.1) heptane
- (D)  N-Methyl-N-Ethylethanamine

SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)

14. The compound $\text{CH}_3-\text{CH}_2-\text{C}(\text{CH}_3)(\text{CH}_2\text{OCH}_2)-\text{CH}_2$ may be named as :
- (A) 2-ethyl-2-methyl oxirane (B) 1, 2-epoxy-2-methylbutane
(C) 1,2-oxapentane (D) 2-methyl-2-butoxide
15. Which of the following names are not correct for the given compound :
- 
- (A) 3-Formyl pentane-1,5-dial (B) 1,2,3-Triformylpropane
(C) 2-Formylbutane-1,4-dial (D) Propane-1,2,3-tricarbaldehyde

16. Which of the following names are correct for the compound :



- (A) 3-ketobutan-1-oic acid (B) 4-Carboxy butan -2-one
(C) 3-oxo butan-1-oic acid (D) 3-Carboxy acetone

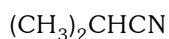
17. The compound $\text{C}_6\text{H}_5\text{—CH=CH—COOH}$ may be called as :

- (A) Succinic acid (B) 3-phenylprop-2-en-1-oic acid
(C) Mandelic acid (D) Cinnamic acid

18. Which of the following are the names of cyclic ether :

- (A) Oxirane (B) Epoxycyclohexane (C) Alkene oxide (D) Carbonyl

19. The name (s) of the following compound is :



- (A) 2-Methyl propane nitrile (B) Isobutyro nitrile
(C) Isopropyl cyanide (D) None of these

20. The name(s) of the following compound is :



- (A) tartaric acid (B) 2,3-dihydroxy butane-1,4-dioic acid
(C) $\alpha\alpha'$ -dihydroxy succinic acid (D) None of these

BRAIN TEASERS						ANSWER KEY				EXERCISE -2					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	B	B	B	C	B	A	A	B	C	B	C	B	B	A,B	B,C
Que.	16	17	18	19	20										
Ans.	A,C	B,D	A,B,C	A,B,C	A,B,C										

EXERCISE-03**MISCELLANEOUS TYPE QUESTIONS****MATCH THE COLUMN**

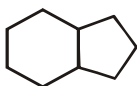
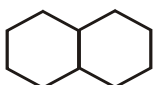
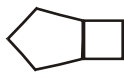
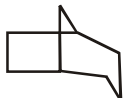
1. Match the following the compounds of column I with column II.

Column-I		Column-II	
(A)	Wood spirit	(p)	2-Butyne
(B)	Acetone	(q)	Trichloromethane
(C)	Dimethyl acetylene	(r)	Methanol
(D)	Chloroform	(s)	Propanone

2. Match the following the compounds of column I with column II.

Column-I		Column-II	
(A)	C_nH_{2n+2}	(p)	Alkynes
(B)	C_nH_{2n}	(q)	Alkenes
(C)	C_nH_{2n-2}	(r)	Cyclohexane
(D)	C_6H_{12}	(s)	Paraffins or alkanes

3. Match column I with column II and select the correct answer from the given codes :

Column-I (compounds)		Column-II (number of carbons in the bridges)	
(A)		(p)	[3.2.1]
(B)		(q)	[4.3.0]
(C)		(r)	[4.4.0]
(D)		(s)	[3.2.0]

COMPREHENSION BASED QUESTIONS :**Comprehension # 1**

If the organic compound contains more than two similar terminal groups and all of them are directly attached to the principal chain, then none of them forms a part of the principal chain. Special suffixes are used to name these :-

Functional group**Suffix**

Carboxamide

Carbonitrile

Carbaldehyde

Carboxylic acid

Carbon atoms of these terminal groups are not counted in the principal chain. If any one of these terminal groups is not directly attached to the parent chain and forms the part of side chain, then the longest chain is selected containing two such similar groups at its two ends. The groups present in the side chain are treated as substituents and are indicated by suitable prefixes.

Indicate whether the following IUPAC names are true (a) or false (b)



True (a)

False (b)



True (a)

False (b)



True (a)

False (b)



True (a)

False (b)



True (a)

False (b)

Comprehension # 2

In addition to the standard ring systems (such as cyclohexane), cyclic compounds can also be bicyclic, tricyclic, etc. or they can be spirocyclic, bicyclic or bridge head carbons. The point of attachment of two rings are called bridge head atoms.

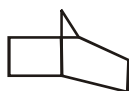
The formal names of bicyclic and related ring systems are based on

(a) Total number of atoms in the molecule.

(b) The number of atoms in each bridge connecting the bridge head atoms. These numbers are written in square bracket in decreasing order.

Spirocyclic compounds have two fused rings, but only one bridge head atom. Spirocyclic compounds are named like bicyclic compounds, but have the prefix spirocyclo. Answer the following question :

1.



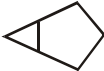
what is the IUPAC name of the above compound ?

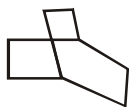
(A) cyclo [1.2.2] heptane

(B) Bicyclo [1.2.2] heptane

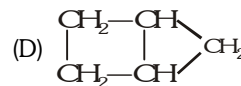
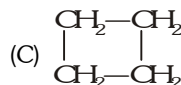
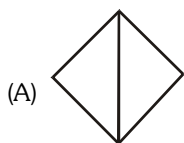
(C) Bicyclo [2.2.1] heptane

(D) cyclo [2.2.1] heptane

2.  The number of atoms in each bridge are :
 (A) [3.2.1] (B) [3.1.0] (C) [1.3.0] (D) [2.1.0]
3. Select the correct statements about the following compounds :



- (A) It is a tricyclic compound (B) It is bicyclo compound
 (C) It is spiro compound (D) Its IUPAC name is bicyclo [2.2.2] hexane
4. Which of the following is the correct structure of bicyclo [1.1.0] butane ?



Comprehension # 3

Branched- chain alkanes are named according to the following rules.

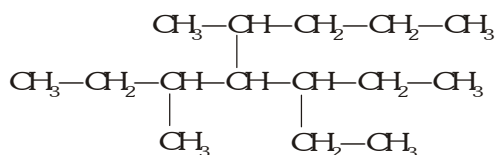
- (1) Longest chain Rule - Locate the longest continuous chain of carbon atoms. This chain determines the parent name of the alkane.
- (2) Lowest set of locants - The longest continuous chain are numbered by arabic numerals 1, 2, 3, 4, from one end of chain to the other, in such a manner that carbon atom carrying first substituent gets the lowest number.
- (3) Name of the branched chain alkane - The substituent name and the parent alkane are joined in one word and there is a hyphen between the number and the substituent name.
- (4) Alphabetical order of the side chains - When two or more substituents are present, give each substituent a number corresponding to its position on the longest chain, the substituent group should be listed alphabetically.
- (5) Numbering of different alkyl groups at equivalent positions- If two different alkyl groups are present at equivalent positions the numbering of the parent chain is done in such a way that alkyl group which comes first in the alphabetical order gets the lower number.
- (6) Naming of same alkyl groups at different positions - When two or more substituents are identical, indicate this by the use of prefixes di, tri, tetra and soon. Commas are used to separate numbers from each other.
- (7) Rule for larger number of substituents - If a compound has two or more chains of the same length, the parent hydrocarbon is the chain with the greater number of substituents.
- (8) Numbering the complex substituent - Name such as iso-propyl, sec-butyl and tert-butyl are acceptable substituent name in the IUPAC system of nomenclature but systematic name are preferable. Systematic substituent names are obtained by numbering the substituent starting at the carbon that is attached to the parent hydrocarbon. This means that the carbon that is attached to the parent hydrocarbon is always the number -1 carbon of the substituent.

1. In following compound $\begin{array}{ccccccc} & & \text{CH}_3 & & \text{CH}_3 & & \\ & & | & & | & & \\ \text{CH}_3 - & \text{CH}_2 - & \text{C} & - & \text{CH} & - & \text{CH} - \text{CH}_2 - \text{CH}_3 \\ & & | & & | & & \\ & & \text{CH}_3 & & \text{CH}_3 & & \end{array}$

The correct lowest set of locant is :-

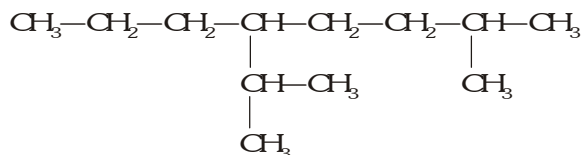
- (A) 3, 3, 4, 5 (B) 3, 4, 5, 5 (C) 4, 5, 3, 3 (D) 5, 5, 4, 3

2. The IUPAC name of the compound is ?



- (A) 5-Ethyl-3-Methyl-4-(1-Methylpropyl) octane (B) 4-Ethyl-6-Methyl-5-(1-Methylpropyl) octane
(C) 3-Ethyl-5-Methyl-4-(1-Methylpropyl) octane (D) 4-Sec-butyl-5-Ethyl-5-Methylheptane

3. The correct IUPAC name of the following compound is -

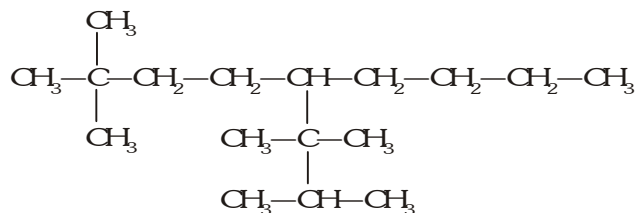


- (A) 2-Methyl-5-isopropyloctane (B) 2, 6-Dimethyl-5-propylheptane
(C) 5-isopropyl-2-Methyloctane (D) 4-(1-Methylethyl)-7-methyloctane

4. The molecular weight of following compound is 3, 7 - Diethyl -2, 2-dimethyl-4-propylnonane :-

- (A) 230 (B) 236 (C) 254 (D) 240

5. The correct IUPAC name of following compound is -

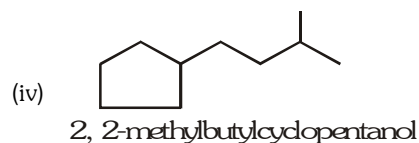
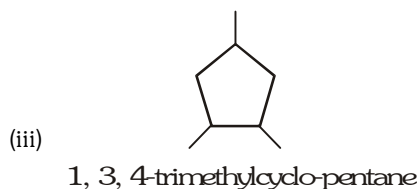
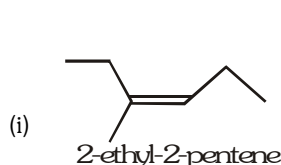


- (A) 2, 3, 3, 7, 7 Pentamethyl-4-butyloctane
(B) 4-Butyl-2, 3, 3, 7, 7 pentamethylnonane
(C) 2, 2-Dimethyl-5-(1', 1', 2'-trimethylpropyl)nonane
(D) 5-(1', 1', 2'-trimethylpropyl)-2, 2-dimethylnonane

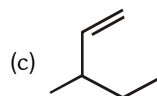
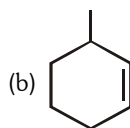
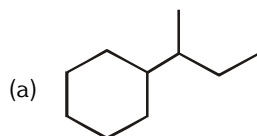
MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
Match the Column - A - (r), B - (s), C - (p), D - (q) - A - (s), B - (q, r), C - (p), D - (q, r) - A - (q), B - (r), C - (s), D - (p) Comprehension Based Questions Comprehension #1 : 1. T 2. F 3. F 4. T 5. T Comprehension #2 : 1. (C) 2. (B) 3. (B) 4. (A) Comprehension #3 : 1. (A) 2. (C) 3. (A) 4. (C) 5. (C)		

EXERCISE-04**CONCEPTUAL SUBJECTIVE EXERCISE**

1. A certain substance contains only carbon and hydrogen and has a molecular weight of 70. Photochemical chlorination gave only one monochloride. Write the structure and IUPAC name of the hydrocarbon and its monochloride.
2. A hydrocarbon of molecular weight 72 g mol^{-1} has a 2-methyl group. What is the IUPAC name? Also draw its bond-line structure?
3. Write the structure and give IUPAC systematic name of an alkane or cycloalkane with the formula :
(a) C_8H_{18} that has only primary hydrogen atoms
(b) C_6H_{12} that has only secondary hydrogen atoms.
4. What is wrong with the names given for these compounds provide the correct name for each :



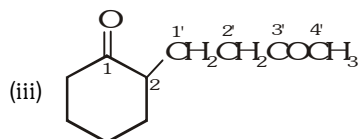
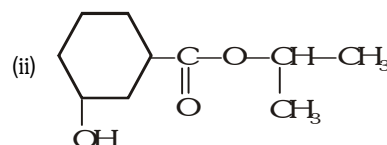
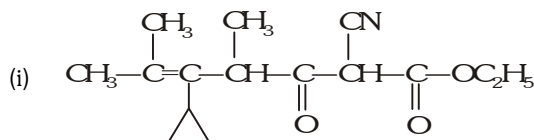
5. Write the IUPAC name for each of the following structures :



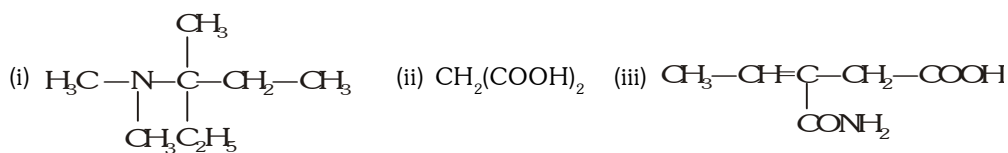
6. Write down the correct priority for citation as principal groups :



7. Write down the correct IUPAC name of the following compounds :



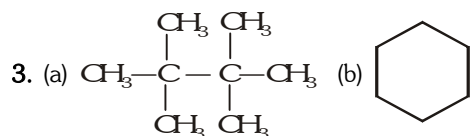
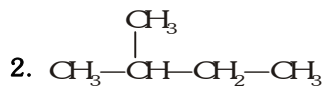
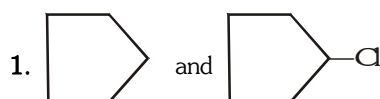
8. Write down the structure of the given compounds :
- (i) Bicyclo [4.3.1] decane
 (ii) 1-(3'-methylcyclopentyl) benzene
 (iii) 4-ethyl-2-methyl-1-propylcyclohexane
9. Answer the following :
- (i) What would be the molecular formula for a straight chain hydrocarbon having 8 carbon atoms with
 (a) all C—C single bond,
 (b) Three C—C double bond,
 (c) one C—C triple bond and one C—C double bond.
- (ii) What is the minimum number of carbon atoms in
 (a) a branched alkane,
 (b) cyclo-alkane
10. Give the IUPAC names of the following compounds :



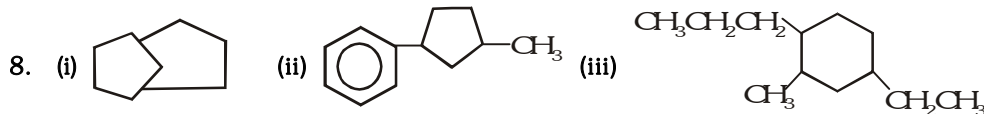
CONCEPTUAL SUBJECTIVE EXERCISE

ANSWER KEY

EXERCISE -4



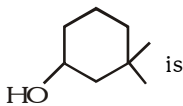
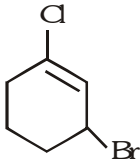
7. (a) Ethyl-2-cyano-5-cyclopropyl-4, 6-dimethyl-3-oxohept-5-en-1-oate
 (b) Isopropyl-3-hydroxycyclohexane carboxylate.
 (c) 2-(3'-oxobutyl) cyclohexan-1-one.

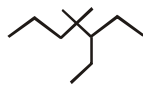
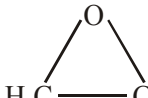
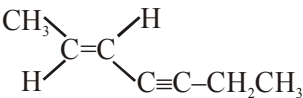


9. (i) (a) C_8H_{18} (b) C_8H_{12} (c) C_8H_{12} (ii) (a) 4 (b) 3

10. (i) 3-(N, N-dimethylamino) -3-methyl pentane;
 (ii) propan-1, 3-dioic acid (iii) 3-Carbamoylpent-3-enoic acid

EXERCISE-05(A)**PREVIOUS YEARS QUESTIONS**

1. Underlined carbon is sp hybridised in- [AIEEE -2002]
(1) $\text{CH}_3\text{CH}=\text{CH}_2$ (2) $\text{CH}_3\text{CH}_2-\text{NH}_2$
(3) CH_3CONH_2 (4) $\text{CH}_3\text{CH}_2\text{CN}$
2. Which of the following compound has wrong IUPAC name? [AIEEE -2002]
(1) $\text{CH}_3\text{CH}_2-\text{CH}_2\text{COO}-\text{CH}_2\text{CH}_3$ (Ethyl butanoate)
(2) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_2-\text{CHO}$ (3-Methylbutanal)
(3) $\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_3$ (2-Methyl-3-butanol)
(4) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\underset{\text{O}}{\text{C}}-\text{CH}_2-\text{CH}_3$ (2-Methyl-3-pentanone)
3. The IUPAC name of $\text{CH}_3\text{COCH}(\text{CH}_3)_2$ is [AIEEE-2003]
(1) isopropyl methyl ketone
(2) 2-methyl-3-butanone
(3) 4-methylisopropyl ketone
(4) 3-methyl-2-butanone
4. The general formula of $\text{C}_n\text{H}_{2n}\text{O}_2$ could be for open chain [AIEEE-2003]
(1) diketones (2) carboxylic acids
(3) diols (4) dialdehydes
5. Which one of the following does not have sp^2 hybridised carbon ? [AIEEE -2004]
(1) Acetone (2) Acetic acid
(3) Acetonitrile (4) Acetamide
6. The IUPAC name of the compound  is [AIEEE-2004]
(1) 1, 1-dimethyl-3-cyclohexanol
(2) 1, 1-dimethyl-3-hydroxy cyclohexane
(3) 3, 3-dimethyl-1-cyclohexanol
(4) 3, 3-dimethyl-1-hydroxy cyclohexane
7. The IUPAC name of the compound is  [AIEEE-2006]
(1) 6-bromo-2-chlorocyclohexene
(2) 3-bromo-1-chlorocyclohexene
(3) 1-bromo-3-chlorocyclohexene
(4) 2-bromo-6-chlorocyclohex-1-en

8. The IUPAC name of  is- [AIEEE-2007]
- (1) 1, 1-diethyl-2, 2-dimethylpentane
 - (2) 4, 4-dimethyl-5, 5-diethylpentane
 - (3) 5, 5-diethyl-4, 4-dimethylpentane
 - (4) 3-ethyl-4, 4-dimethylheptane
9. The correct decreasing order of priority for the functional groups of organic compounds in the IUPAC system of nomenclature is [AIEEE -2008]
- (1) $-\text{COOH}$, $-\text{SO}_3\text{H}$, $-\text{CONH}_2$, $-\text{CHO}$
 - (2) $-\text{SO}_3\text{H}$, $-\text{COOH}$, $-\text{CONH}_2$, $-\text{CHO}$
 - (3) $-\text{CHO}$, $-\text{COOH}$, $-\text{SO}_3\text{H}$, $-\text{CONH}_2$
 - (4) $-\text{CONH}_2$, $-\text{CHO}$, $-\text{SO}_3\text{H}$, $-\text{COOH}$
10. The IUPAC name of neopentane is :- [AIEEE -2009]
- (1) 2-methylpropane
 - (2) 2, 2-dimethylbutane
 - (3) 2-methylbutane
 - (4) 2, 2-dimethylpropane
11. The IUPAC name of the compound  is :- [AIEEE-2012 (Online)]
- (1) 1, 2-Epoxy propane
 - (2) Propylene oxide
 - (3) 1, 2-Oxo propane
 - (4) 1, 2-Propoxide
12. The IUPAC name of the following compounds is : [AIEEE-2012 (Online)]
- 
- (1) (Z) - 5 - hepten - 3 - yne
 - (2) (Z) - 2 - hepten - 4 - yne
 - (3) (E) - 5 - hepten - 3 - yne
 - (4) (E) - 2 - hepten - 4 - yne

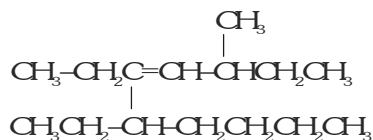
PREVIOUS YEAR QUESTIONS						ANSWER KEY			EXERCISE-05(A)			
Que.	1	2	3	4	5	6	7	8	9	10	11	12
Ans	4	3	4	2	3	3	2	4	1	4	1	4

EXERCISE-05(B)**PREVIOUS YEARS QUESTIONS**

1. The IUPAC name of the compound, $\text{CH}_2=\text{CH}-\text{CH}(\text{CH}_3)_2$ is - [IIT-88]

- (A) 1,1-Dimethylprop-2-ene (B) 3-Methylbut-1-ene
(C) 2-Vinylpropane (D) 1-Isopropylethylene

2. The correct IUPAC name of the compound is : [IIT-90]



- (A) 5,6-diethyl-8-methyl dec-6-ene (B) 5,6-diethyl-3-methyl dec-4-ene
(C) 6-butyl-5-ethyl-3-methyl oct-4-ene (D) 2,4,5-triethyl-3-ene

3. The IUPAC name of - [IIT-91]



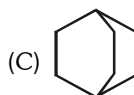
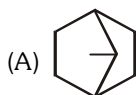
- (A) 5-Vinyloct-3-en-1-al (B) 4-Butylhexa-2,5-dien-1-al
(C) 5-Vinyloct-5-en-8-al (D) 3-Butylhexa-1,4-dien-8-al

4. Choose the correct IUPAC name for $\text{CH}_3-\text{CH}-\text{CHO}$ is - [IIT-93]



- (A) Butan-2-aldehyde (B) 2-Methylbutanal
(C) 3-Methyl iso butyraldehyde (D) 2-Ethylpropanal

5. A compound with molecular formula C_8H_{14} , contains 12 secondary and two tertiary H atoms. The compound is : [IIT-93]



6. IUPAC name of $\text{C}_6\text{H}_5-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl}$ is - [IIT-06]

- (A) Benzoylchloride (B) Benzenecarbonylchloride
(C) Chlorophenyl ketone (D) Phenylchloroketone

PREVIOUS YEARS QUESTIONS	ANSWER KEY	EXERCISE -5 (B)
1. B	2. B	3. B
4. B	5. B,C,D	6. B

EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. Alkaline earth metals (group 2 or IIA elements) differ from group 12 (or IIB) elements in the electronic configuration of their :
(A) Antipenultimate shell (B) Innermost shell
(C) Outermost shell (D) Penultimate shell
2. The first ionization enthalpy of magnesium is lower than the first ionization enthalpy of :
(A) Lithium (B) Sodium (C) Calcium (D) Beryllium
3. Chemical A is used for water softening to remove temporary hardness. A reacts with Na_2CO_3 to generate caustic soda. When CO_2 is bubbled through A, it turns cloudy. What is the chemical formula of A :
(A) CaCO_3 (B) CaO (C) Ca(OH)_2 (D) $\text{Ca(HCO}_3)_2$
4. The substance not likely to contain CaCO_3 is :
(A) Calcined gypsum (B) Sea shells (C) Dolomite (D) A marble statue
5. A metal M readily forms water soluble sulphate MSO_4 , water insoluble hydroxide M(OH)_2 and oxide MO which becomes inert on heating. The hydroxide is soluble in NaOH. The M is :
(A) Be (B) Mg (C) Ca (D) Sr
6. A chloride dissolves appreciably in cold water. When placed on a Pt wire in Bunsen flame, no distinctive colour is noted. Which cation could be present?
(A) Be^{2+} (B) Ba^{2+} (C) Pb^{2+} (D) Ca^{2+}
7. The hydroxide which is best soluble in water is :
(A) Ba(OH)_2 (B) Mg(OH)_2 (C) Sr(OH)_2 (D) Ca(OH)_2
8. What is X in the following reaction?
 $\text{MgCl}_2 + 2 \text{H}_2\text{O} \longrightarrow \text{X} + 2 \text{HCl} + \text{H}_2\text{O}$
(A) MgO (B) Mg (C) Mg(OH)_2 (D) Mg(OH)Cl
9. (Yellow ppt) $\text{T} \xleftarrow{\text{K}_2\text{CrO}_4} \text{X} \xrightarrow{\text{dil. HCl}} \text{Y (Yellow ppt)} + \text{Z} \uparrow$ (pungent smelling gas) If X gives green flame test. Then, X is :
(A) MgSO_4 (B) BaS_2O_3 (C) CuSO_4 (D) PbS_2O_3
10. The correct statement is/are :
(A) BeCl_2 is a covalent compound (B) BeCl_2 is an electron deficient molecule
(C) BeCl_2 can form dimer (D) The hybrid state of Be in BeCl_2 is sp^2
11. The reaction of an element A with water produces combustible gas B and an aqueous solution of C. When another substance D reacts with this solution C also produces the same gas B. D also produces the same gas even on reaction with dilute H_2SO_4 at room temperature. Element A imparts golden yellow colour to Bunsen flame. Then A, B, C and D may be identified as :
(A) Na, H_2 , NaOH and Zn (B) K, H_2 , KOH and Zn
(C) K, H_2 , NaOH and Zn (D) Ca, H_2 , CaCO_3 and Zn
12. An alkaline earth metal (M) gives a salt with chlorine, which is insoluble in water at room temperature but soluble in boiling water. It also forms an insoluble sulphate whose mixture with a sulphide of a transition metal is called 'lithopone'a white pigment. Metal M is :
(A) Ca (B) Mg (C) Ba (D) Sr
13. In electrolysis of NaCl when Pt electrode is taken then H_2 is liberated at cathode while with Hg cathode it forms sodium amalgam:
(A) Hg is more inert than Pt
(B) More voltage is required to reduce H^+ at Hg than at Pt
(C) Na is dissolved in Hg while it does not dissolve in Pt
(D) Conc. of H^+ ions is larger when Pt electrode is taken

14. The correct sequence of increasing covalent character is represented by –
 (A) $\text{BeCl}_2 < \text{NaCl} < \text{LiCl}$ (B) $\text{NaCl} < \text{LiCl} < \text{BeCl}_2$
 (C) $\text{BeCl}_2 < \text{LiCl} < \text{NaCl}$ (D) $\text{LiCl} < \text{NaCl} < \text{BeCl}_2$
15. The paramagnetic species is :
 (A) KO_2 (B) SiO_2 (C) TiO_2 (D) BaO_2
16. The pair of amphoteric hydroxides is :-
 (A) Al(OH)_3 , LiOH (B) Be(OH)_2 , Mg(OH)_2
 (C) B(OH)_3 , Be(OH)_2 (D) Be(OH)_2 , Zn(OH)_2
17. Maximum thermal stability is shown by
 (A) MgCO_3 (B) CaCO_3 (C) SrCO_3 (D) BaCO_3
18. Stable oxide is obtained by heating the carbonate of the element
 (A) Li (B) K (C) Na (D) Rb
19. The stable superoxide is formed by the element
 (A) Li (B) Na (C) K (D) Ca
20. The metallic lustre exhibited by sodium is explained by
 (A) diffusion of sodium ions
 (B) oscillation of loose electrons
 (C) excitation of free protons
 (D) existence of body centred cubic lattice
21. A solution of sodium sulphate in water is electrolysed using inert electrodes. The products at the cathode and anode are respectively
 (A) H_2 , O_2 (B) O_2 , H_2 (C) O_2 , Na (D) O_2 , SO_2
22. The hydration energy of Mg^{2+} is greater than that of
 (A) Al^{3+} (B) Na^+ (C) Be^{2+} (D) Mg^{3+}
23. Calcium is obtained by the
 (A) electrolysis of molten calcium chloride
 (B) electrolysis of a solution of CaCl_2 in water
 (C) reduction of CaCl_2 with carbon
 (D) roasting of limestone
24. The material used in photoelectric cells contains –
 (A) Cs (B) Si (C) Sn (D) Ti
25. Four alkali metals A, B, C and D are having respectively standard reduction potentials as -3.05 , -1.66 , -0.40 and 0.80 V. Which one will be the most reducing agent ?
 (A) A (B) B (C) C (D) D
26. Which of the following imparts violet colouration to the Bunsen burner non-luminous flame
 (A) NaCl (B) BaCl_2 (C) CaCl_2 (D) KCl
27. Which one of the following is most basic ?
 (A) Al_2O_3 (B) MgO (C) SiO_2 (D) P_2O_5
28. Molten sodium is used in nuclear reactors to
 (A) absorb neutrons in order to control the chain reaction
 (B) slow down the fast neutrons
 (C) absorb the heat generated by nuclear fission
 (D) extract radio-isotopes produced in the reactor

EXERCISE-02**BRAIN TEASERS****SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT NSWERS)**

1. Which of the following is incorrect?
(A) Mg burns in air releasing dazzling light rich in UV rays.
(B) $\text{CaCl}_2 + 6 \text{H}_2\text{O}$ when mixed with ice gives freezing mixture.
(C) Mg cannot form complexes
(D) Be can form complexes due to its very small size.
2. On dissolving moderate amount of sodium metal in liquid NH_3 at low temperature, which one of the following does not occur
(A) Blue coloured solution is obtained.
(B) Na^+ ions are formed in the solution.
(C) Liquid NH_3 becomes good conductor of electricity.
(D) Liquid ammonia remains diamagnetic.
3. The minimum equivalent conductance in fused state is shown by –
(A) MgCl_2 (B) BeCl_2 (C) CaCl_2 (D) SrCl_2
4. The metal which cannot be produced on reduction of its oxide by aluminium is
(A) K (B) Mn (C) Cr (D) Fe
5. Magnesium on reaction with very dilute HNO_3 gives
(A) NO (B) N_2O (C) H_2 (D) NO_2
6. The alkali metal that reacts with nitrogen directly to form nitride is
(A) Li (B) Na (C) K (D) Rb
7. Which of the following statement is/are false for alkali metals ?
(A) Lithium is the strongest reducing agent
(B) Na is amphoteric in nature
(C) Li^+ is exceptionally small
(D) All alkali metals give blue solution in liquid ammonia
8. Amongst LiCl, RbCl, BeCl_2 and MgCl_2 , the compounds with the greatest and least ionic character respectively are :-
(A) LiCl, RbCl (B) RbCl, BeCl_2 (C) RbCl, MgCl_2 (D) MgCl_2 , BeCl_2
9. K_2CS_3 can be called potassium
(A) sulphocyanide (B) thiocarbide (C) thiocarbonate (D) thiocyanate
10. Anhydrous MgCl_2 can be prepared by heating $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$
(A) in a current of dry HCl gas (B) with carbon
(C) until it fuses (D) with lime
11. Oxygen ions structure in its peroxide, superoxide, ozonide :
(A) O_2^- , O_2^{2-} , O_3^{2-} (B) O_2^{2-} , O_2^- , O_3^- (C) O_2^{2-} , O^{2-} , O_3^- (D) O_2^- , O_2^{3-} , O_3^{2-}

12. In presence of iron, alkali metal react with liquid ammonia and form
 (A) Metal mixture + H_2 (B) Iron metal mixture + H_2
 (C) Metal mixture (D) Metal amide + H_2
13. The ionic conductance of following cation in a given concentration are in the order
 (A) $Li^+ < Na^+ < K^+ < Rb^+$ (B) $Li^+ > Na^+ > K^+ > Rb^+$
 (C) $Li^+ < Na^+ > K^+ > Rb^+$ (D) $Li^+ = Na^+ < K^+ < Rb^+$
14. Which of the following does not give an oxide on heating –
 (A) $MgCO_3$ (B) Li_2CO_3 (C) $ZnCO_3$ (D) K_2CO_3
15. On heating sodium metal in the current of dry ammonia leads to the formation of which gas–
 (A) $NaNH_2$ (B) NaN_3 (C) NH_3 (D) H_2
16. On allowing ammonia solution of s-block metals to stand for a long time, blue colour becomes fade. The reason is:-
 (A) Formation of NH_3 gas (B) Formation of metal amide
 (C) Cluster formation of metal ions (D) Formation of metal nitrate
17. When Na and Li placed in dry air we get :-
 (A) $NaOH$, Na_2O , Li_2O (B) Na_2CO_3 , Na_2O_2 , Li_2O
 (C) Na_2O , Li_3N , NH_3 (D) Na_2O , Li_2O , Li_3N
18. The hydride ion H^- is stronger base than its hydroxide ion OH^- . Which of the following reaction will occur if sodium hydride is dissolved in water:-
 (A) $H_{(aq)}^- + H_2O \rightarrow H_3O^+$ (B) $H_{(aq)}^- + H_2O \rightarrow OH^- + H_2$
 (C) $H^- + H_2O \rightarrow H_2 + O_2$ (D) $H^- + H_2O \rightarrow$ No reaction
19. Which can not be used to generate H_2 :-
 (A) $Al + NaOH$ (B) $Zn + NaOH$ (C) $Mg + NaOH$ (D) $LiH + H_2O$
20. Only those elements of s-block can produce superoxides which have :-
 (A) High ionisation energy (B) High electronegativity
 (C) High charge density (D) Low ionisation potential
21. Alum is the name used for all double salts having the composition $M_2^I SO_4 \cdot M_2^{III} (SO_4)_3 \cdot 24H_2O$. Where M^{III} stands for Al^{+3} , Cr^{+3} , Fe^{+3} , while M^I stands for:-
 (A) Li^+ , Cu^+ , Ag^+ (B) Li^+ , NH_4^+ , Na^+ (C) Na^+ , K^+ , Rb^+ (D) Ca^{+2} , Mg^{+2} , Sr^{+2}
22. Identify the correct statement -
 (A) Gypsum contains a lower percentage of Ca than plaster of paris
 (B) Gypsum is obtained by heating plaster of paris
 (C) Plaster of paris can be obtained by hydration of gypsum
 (D) Plaster of paris is obtained by partial oxidation of gypsum
23. In the reaction $M + O_2 \longrightarrow MO_2$ (super oxide) the metal is
 (A) Li (B) Na
 (C) K (D) Ba

24. Na^+ and Ag^+ differ in
- (A) Na_2CO_3 is thermally stable while Ag_2CO_3 decomposes into Ag, CO_2 and O_2
- (B) Ag^+ forms complexes, Na^+ does not
- (C) NaCl is water soluble, AgCl is insoluble
- (D) NaBr-yellow and AgBr pale yellow
25. The stability order of oxide, peroxide and superoxide of alkali metal is
- (A) Normal oxide > super oxide > per oxide
- (B) Normal oxide > per oxide > super oxide
- (C) super oxide > per oxide > normal oxide
- (D) per oxide > normal oxide > super oxide
26. Match list I with list II and choose the correct answer from the codes given below

List I	List II
(A) NaNO_3	(a) Baking soda
(B) $\text{Na}(\text{NH}_4)\text{HPO}_4$	(b) Chile salt peter
(C) NaHCO_3	(c) Microcosmic salt
(D) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$	(d) Washing soda

Codes is :

	A	B	C	D
(A)	a	b	c	d
(B)	b	c	a	d
(C)	c	a	b	d
(D)	d	a	b	c

27. Which of the following statement is not correct
- (A) LiOH is amphoteric in nature
- (B) LiCl is soluble in pyridine
- (C) Li_3N is stable while Na_3N doesn't exist even at room temperature
- (D) BeO is amphoteric in nature
28. Which of the following statement is correct for s-block elements :-
- (A) Be has smallest atomic size in II A group
- (B) Li is most metallic
- (C) Mg impart red colour to the flame
- (D) Cs is most reducing in water
29. Which of the following are ionic carbides?
- (A) CaC_2 (B) Al_4C_3 (C) SiC (D) Be_2C
30. Which of the following groups of elements have chemical properties that are most similar :
- (A) Na, K, Ca (B) Mg, Sr, Ba
- (C) Be, Al, Ca (D) Be, Ra, Cs

31. Which of the following statements are false?
- (A) BeCl_2 is a linear molecule in the vapour state but it is polymeric in the solid state
- (B) Calcium hydride is called hydrolith
- (C) Carbides of both Be and Ca react with water to form acetylene
- (D) Oxides of both Be and Ca are amphoteric.
32. The incorrect statement(s) is/are :
- (A) Mg cannot form complexes
- (B) Be can form complexes due to a very small atomic size
- (C) The first ionisation potential of Be is higher than that of Mg.
- (D) Mg forms an alkaline hydroxide while Be forms amphoteric oxides.
33. Na_2SO_4 is water soluble but BaSO_4 is insoluble because :
- (A) The hydration energy of Na_2SO_4 is higher than that of its lattice energy
- (B) The hydration energy of Na_2SO_4 is less than that of its lattice energy
- (C) The hydration energy of BaSO_4 is less than that of its lattice energy
- (D) The hydration energy of BaSO_4 is higher than that of its lattice energy
34. $\text{BeCl}_2 + \text{LiAlH}_4 \longrightarrow \text{X} + \text{LiCl} + \text{AlCl}_3$
- (A) X is lithium hydride
- (B) X is BeH_2
- (C) X is $\text{BeCl}_2 \cdot 2\text{H}_2\text{O}$
- (D) X is LiH
35. $\text{X} \xrightarrow{\text{CaCl}_2} \text{CaCl}_2 + \text{Y} \uparrow$; the effective ingredient of X is :
- (A) OCl^-
- (B) Cl^-
- (C) OCl^+
- (D) OCl_2^-
36. Which of the following substance(s) is/are used in laboratory for drying purposes?
- (A) Anhydrous P_2O_5
- (B) Graphite
- (C) Anhydrous CaCl_2
- (D) Na_3PO_4
37. If X and Y are the second ionisation potentials of alkali and alkaline earth metals of same period, then :
- (A) $X > Y$
- (B) $X < Y$
- (C) $X = Y$
- (D) $X \ll Y$
38. $\text{X} \xrightarrow{\text{N}_2, \Delta} \text{Y} \xrightarrow{\text{H}_2\text{O}} \text{Z (colourless gas)} \xrightarrow{\text{CuSO}_4} \text{T (blue colour)}$:
- Then, substances Y and T are -
- (A) $\text{Y} = \text{Mg}_3\text{N}_2$ and $\text{T} = \text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
- (B) $\text{Y} = \text{Mg}_3\text{N}_2$ and $\text{T} = \text{CuSO}_4 \cdot 4\text{NH}_3$
- (C) $\text{Y} = \text{Mg}(\text{NO}_3)_2$ and $\text{T} = \text{CuO}$
- (D) $\text{Y} = \text{MgO}$ and $\text{T} = \text{CuSO}_4 \cdot 4\text{NH}_3$
39. When K_2O is added to water, the solution becomes basic in nature because it contains a significant concentration of :
- (A) K^+
- (B) O^{2-}
- (C) OH^-
- (D) O_2^{2-}

EXERCISE-03**MISCELLANEOUS TYPE QUESTIONS****TRUE / FALSE**

1. Magnesium is an essential constituent of chlorophyll, the green colouring matter of plants.
2. Setting of cement is an endothermic process.
3. Calcium bicarbonate is known in solid state.
4. BeH_2 is an ionic hydride.
5. BeCO_3 is thermally stable compounds.
6. In the electrolysis of fused calcium hydride, hydrogen is liberated at cathode.
7. $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ on heating forms MgCl_2 .
8. Sodium when heated in excess of oxygen gives sodium oxide.
9. In group IA of alkali metals, the ionisation potential decreases down the group. Therefore, lithium is a poorer reducing agent in gaseous medium.
10. The softness of group IA metals increases down the group with increasing atomic number.

FILL IN THE BLANKS

1. Anhydrous magnesium chloride is obtained by heating the hydrated salt with.....
2. Ca(OH)_2 is..... basic than Mg(OH)_2 .
3. CaH_2 is commercially known as.....
4. Magnesium burn in air forming and
5. Ba react with cold water Mg reacts with..... while Be has..... with boiling water.
6. A standard solution of sodium hydroxide cannot be prepared by direct weighing because.....
7. Potassium bicarbonate cannot be prepared by solvay process because.....
8. Solution of alkali metals in liquid ammonia conducts electricity due to.....

MATCH THE COLUMN

1.

<u>Column-I</u>		<u>Column-II</u>	
(A)	Hydrolith	(p)	Contain Ca
(B)	Nitrolium	(q)	Used as a fertilizer
(C)	Dolomite	(r)	Used to prepare H_2
(D)	Pearl's ash	(s)	Contain potassium

2.

<u>Column-I</u>		<u>Column-II</u>	
(A)	Solvay process	(p)	NaCl
(B)	Evolve $\text{CO}_2 \uparrow$ on heating	(q)	Na_2O_2
(C)	aq. soln. is neutral towards litmus	(r)	NaHCO_3
(D)	Oxone	(s)	Na_2CO_3

3.

Column-I		Column-II	
(A)	Metal sulphate $\xrightarrow{\Delta}$ metal oxide + SO_2 + O_2	(p)	Ba
(B)	Metal cation + $\text{K}_2\text{CrO}_4 \longrightarrow$ yellow ppt	(q)	Sr
(C)	Metal + $\text{NH}_3 \xrightarrow{\text{(liquid)}}$ blue solution	(r)	Na
(D)	MCl_2 + conc. $\text{H}_2\text{SO}_4 \longrightarrow$ white ppt.	(s)	Mg

ASSERTION & REASON QUESTIONS

These questions contains, Statement I (assertion) and Statement II (reason).

- (A) Statement-I is true, Statement-II is true ; Statement-II is correct explanation for Statement-I.
 (B) Statement-I is true, Statement-II is true ; Statement-II is not a correct explanation for statement-I
 (C) Statement-I is true, Statement-II is false
 (D) Statement-I is false, Statement-II is true

1. **Statement-I** : Li_2SO_4 do not form double salt like alum.

Because

Statement-II : Atomic size of Li is too small.

2. **Statement-I** : NaCl when exposed in air it becomes wet.

Because

Statement-II : NaCl contains hygroscopic impurities like CaCl_2 , MgCl_2 etc.

3. **Statement-I** : Lithium is the weakest reducing agent among alkali metals.

Because

Statement-II : In alkali metals I.P. decreases down the group.

4. **Statement-I** : BaCO_3 is more soluble in HNO_3 than in plain water.

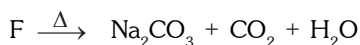
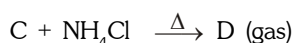
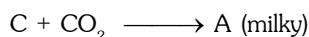
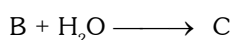
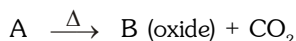
Because

Statement-II : Carbonate is a weak base and reacts with the H^+ from the strong acid causing the barium salt to dissociate.

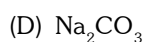
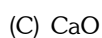
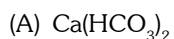
5. **Statement-I** : BeCl_2 fumes in moist air.

Because

Statement-II : BeCl_2 reacts with moisture of form HCl gas.

COMPREHENSION BASED QUESTIONS**Comprehension # 1**

1. A is :



2. B and C are :
- (A) CaO , Ca(OH)_2 (B) Ca(OH)_2 , CaCO_3
 (C) CaCO_3 , Ca(OH)_2 (D) Ca(OH)_2 , CaO
3. D, E and F are :
- (A) NH_3 , NH_4Cl , NH_4HCO_3
 (B) NH_3 , NH_4HCO_3 , NaHCO_3
 (C) NH_4HCO_3 , Na_2CO_3 , NaHCO_3
 (D) None

Comprehension # 2

Following given passage the five observation regarding alkali metals are mentioned.

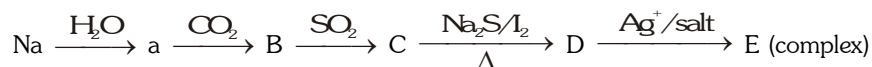
- On exposure to air, sodium hydroxide becomes liquid and after some time it changes to white powder.
- In water LiF is least soluble fluorides among fluorides of alkali metal, but its solubility increases as HF is added in aqueous solution.
- LiH more stable than NaH when heated separately
- When excess of $\text{Na}_2\text{S}_2\text{O}_3$ solution is added to the FeCl_3 solution an intense violet colouration is produced, but violet colour disappeared shortly
- Between Na^+ and Ag^+ , Ag^+ is stronger Lewis acid.

- The explanation of observation (v) is :
 - Because Na^+ has inert gas configuration which has greater polarisation power
 - Because Ag^+ has inert gas configuration which has greater polarisation power
 - Because Ag^+ has pseudo inert gas configuration which has lesser polarisation power
 - Because Ag^+ has pseudo inert gas configuration which has greater polarisation power
- The explanation for the observation (iv) is :
 - Initially with FeCl_3 , $\text{Na}_2\text{S}_2\text{O}_3$ produce an intense violet colour substance $\text{Fe}_2(\text{S}_2\text{O}_3)_3$. But $\text{Fe}_2(\text{S}_2\text{O}_3)_3$ changes to Fe^{2+} & $\text{S}_4\text{O}_6^{2-}$ on standing
 - Initially with FeCl_3 , is reduced to FeCl_2 by $\text{Na}_2\text{S}_2\text{O}_3$. FeCl_2 so produced undergo unstable complex formation $\text{Fe}(\text{S}_2\text{O}_3)_3^{4-}$ which is violet in colour.
 - Initially with FeCl_3 , $\text{Na}_2\text{S}_2\text{O}_3$ produce colloidal Fe which is violet in colour. After sometime, the colloidal suspension changes to the ppt of Fe .
 - There is no reaction
- As per observation (iii) LiH is more stable than NaH , because :
 - Due to small size of Li^+ , the lattice energy of LiH is greater
 - Due to greater size of H^- , the lattice energy of LiH is greater
 - LiH is more covalent than NaH
 - Due to greater size of Na^+ , the lattice energy of NaH is greater.
- As per observation (ii) the solubility of LiF increases in the presence of HF , because :
 - The HF further ionises to H^+ & F^- (B) In the presence of HF , there will be a common ion effect
 - In the presence HF , F^- is converted to HF_2^- (D) All of the above

5. The reaction for observation (i) can be explained as –

- (A) $\text{NaOH (S)} \xrightarrow{\text{H}_2\text{O}} \text{NaOH(aq)} \xrightarrow{\text{H}_2\text{O}} \text{N}$
- (B) $\text{NaOH (S)} \xrightarrow{\text{H}_2\text{O}} \text{NaOH(aq)} \xrightarrow{-\text{H}_2\text{O}} \text{Na}_2\text{O (S)}$
- (C) $\text{NaOH (S)} \xrightarrow{\text{H}_2\text{O}} \text{NaOH(aq)} \xrightarrow{\text{O}_2 / \text{air}} \text{Na}_2\text{O (S)}$
- (D) $\text{NaOH (S)} \xrightarrow{\text{H}_2\text{O}} \text{NaOH(aq)} \xrightarrow{\text{CO}_2} \text{Na}_2\text{CO (S)}$

Comprehension # 3



1. The compound B and C are :

- (A) Na_2CO_3 , Na_2SO_4
- (B) NaHCO_3 , Na_2SO_4
- (C) Na_2CO_3 , Na_2SO_3
- (D) None of these

2. The compound D is :

- (A) Na_2SO_4
- (B) $\text{Na}_2\text{S}_4\text{O}_6$
- (C) $\text{Na}_2\text{S}_2\text{O}_5$
- (D) $\text{Na}_2\text{S}_2\text{O}_3$

3. Oxidation number of each 'S' atom in compound D :

- (A) + 2, + 2
- (B) + 4, 0
- (C) + 6, - 2
- (D) + 5, - 1

Comprehension # 4

Alkali metals readily react with oxyacids forming corresponding salts like M_2CO_3 , MHCO_3 , MNO_3 , M_2SO_4 etc. with evolution of hydrogen. They also dissolve in liquid NH_3 but without the evolution of hydrogen. The colour of its dilute solution is blue but when it is heated and concentrated then its colour becomes bronze.

1. Among the nitrate of alkali metals which one can be decomposed to its oxide?

- (A) NaNO_3
- (B) KNO_3
- (C) LiNO_3
- (D) All of these

2. Among the carbonates of alkali metals which one has highest stability?

- (A) Cs_2CO_3
- (B) Rb_2CO_3
- (C) K_2CO_3
- (D) Na_2CO_3

3. Which of the following statement about the sulphate of alkali metal is correct?

- (A) Except Li_2SO_4 all sulphate of other alkali metals are soluble in water
- (B) All sulphates of alkali metals except lithium sulphate forms alum.
- (C) The sulphates of alkali metals cannot be hydrolysed.
- (D) All of these

4. Which of the following statement about solution of alkali metals in liquid ammonia is correct?
- (A) The solution have strong oxidizing properties.
- (B) Both the dilute solution as well as concentrated solution are paramagnetic in nature
- (C) Charge transfer is the responsible for the colour of the solution
- (D) None of these
5. Which metal bicarbonates does not exist in solid state?
- (i) LiHCO_3 (ii) $\text{Ca}(\text{HCO}_3)_2$ (iii) $\text{Zn}(\text{HCO}_3)_2$ (iv) NaHCO_3 (v) AgHCO_3
- (A) (ii), (iii), (v) (B) (i), (ii), (iii) (C) (i), (ii), (v) (D) (ii), (iii), (iv)
-

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
<u>True / False</u> 1. T 2. F 3. F 4. F 5. F 6. F 7. F 8. F 9. T 10. T <u>Fill in the Blanks</u> 1. Dry HCl 2. More 3. Hydrolyth 4. Mgo, Mg_3N_2 5. Vigrowly, slowly, no action 6. Absorb moisture & CO_2 from atmosphere 7. Ammonated e^- <u>Match the Column</u> 1. (A) $\rightarrow$ p,r ; (B) $\rightarrow$ p,q ; (C) $\rightarrow$ p ; (D) $\rightarrow$ s 2. (A) $\rightarrow$ r,s ; (B) $\rightarrow$ r,s ; (C) $\rightarrow$ p ; (D) $\rightarrow$ q 3. (A) $\rightarrow$ p,q,s ; (B) $\rightarrow$ p,q; (C) $\rightarrow$ p,q,r,s ; (D) $\rightarrow$ p,q <u>Assertion - Reason Questions</u> 1. A 2. A 3. D 4. C 5. A <u>Comprehension Based Questions</u> Comprehension #1 : 1. (B) 2. (A) 3. (B) Comprehension #2 : 1. (D) 2. (A) 3. (A) 4. (C) 5. (D) Comprehension #3 : 1. (C) 2. (D) 3. (C) Comprehension #4 : 1. (C) 2. (A) 3. (D) 4. (D) 5. (A)		

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

1. Explain the following :
 - (i) The reaction between marble and dilute H_2SO_4 is not used to prepare carbon dioxide.
 - (ii) Lime water becomes turbid on passing CO_2 through it, but becomes clear when more CO_2 is passed.
 - (iii) Alkaline earth metals have higher melting points than alkali metals.
 - (iv) Beryllium does not exhibit a covalency beyond 4.
2. PbO_2 is soluble in NaOH and also in HCl . What does it reflect about the nature of PbO_2 ?
3. What happens when :
 - (i) Hot and concentrated caustic soda solution reacts with iodine.
 - (ii) White phosphorus is heated with caustic soda.
 - (iii) Excess of caustic soda reacts with zinc sulphate solution.
 - (iv) Excess of NaOH is added to AlCl_3 solution.
4. Write balanced equation for reaction between
 - (i) Na_2O_2 and water
 - (ii) KO_2 and water
 - (iii) Na_2O_2 and CO_2
5. Element A burns in nitrogen to give an ionic compound B. Compound B reacts with water to give C and D. A solution of C becomes milky on bubbling carbon dioxide. Identify A, B, C and D.
6. In water LiF is least soluble fluoride among fluorides of alkali metals, but its solubility increases as HF is added in aqueous solution, why?
7. What happens when $\text{CuSO}_4(\text{aq.})$ is treated with excess of $\text{Na}_2\text{S}_2\text{O}_3$ solution?
8. Arrange the following in order of increasing
 - (i) Thermal stability $\text{BeSO}_4, \text{MgSO}_4, \text{CaSO}_4$
 - (ii) Polarising power $\text{Be}^{2+}, \text{Mg}^{2+}, \text{Ca}^{2+}$
 - (iii) Solubility in H_2O $\text{Be}(\text{OH})_2, \text{Mg}(\text{OH})_2, \text{Ca}(\text{OH})_2$
 - (iv) Covalent nature $\text{BeCl}_2, \text{MgCl}_2, \text{CaCl}_2$
 - (v) Hydrolysis nature $\text{BeCl}_2, \text{MgCl}_2, \text{CaCl}_2$
 - (vi) Lattice energy $\text{CaF}_2, \text{MgF}_2, \text{BaF}_2$
 - (vii) Hydration energy $\text{Be}^{2+}, \text{Mg}^{2+}, \text{Ba}^{2+}$
 - (viii) Solubility in water $\text{MgF}_2, \text{BaF}_2, \text{BeF}_2$
 - (ix) Basic nature $\text{Be}, \text{Mg}, \text{Ca}, \text{Sr}$
9. Hydrogen reacts with a metal (A) to give an ionic hydride (B). The metal (A) gives brick red colour with bunsen flame. The hydride formed is commonly known by its trade name. The compound (B) on treating with water gives back H_2 and (C). Identify (A), (B) and (C).

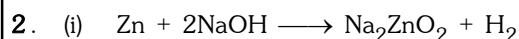
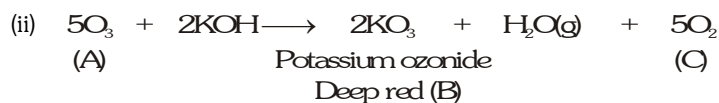
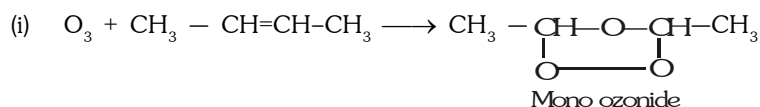
1. (i) Insoluble CaSO_4 is formed which deposits on the surface of marble and prevents further action of dilute H_2SO_4 , so the evolution of CO_2 ceases after sometime.
- (ii) Insoluble CaCO_3 is first precipitated which dissolves in excess of CO_2 due to the form of $\text{Ca}(\text{HCO}_3)_2$.
- $$\text{Ca}(\text{OH})_2 + \text{CO}_2 \longrightarrow \text{CaCO}_3 + \text{H}_2\text{O};$$
- (Insoluble)
- $$\text{CaCO}_3 + \text{H}_2\text{O} + \text{CO}_2 \longrightarrow \text{Ca}(\text{HCO}_3)_2$$
- (Soluble)
- (iii) Metallic bonding is much stronger in alkaline earth metals as two electrons are present in valence shell.
- (iv) The outermost energy shell in beryllium is the second. It cannot accommodate more than 8 electrons and hence a covalency limit 4 cannot be exceeded.
2. PbO_2 are amphoteric nature
- $$\text{PbO}_2 + 4\text{HCl} \longrightarrow \text{PbCl}_4 + 2\text{H}_2\text{O}$$
- $$\text{PbO}_2 + 2\text{NaOH} \longrightarrow \text{Na}_2\text{PbO}_3 + \text{H}_2\text{O}$$
3. (i) $3\text{I}_2 + 6\text{NaOH} \rightarrow 5\text{NaI} + \text{NaIO}_3 + 3\text{H}_2\text{O}$ (ii) $\text{P}_4 + 3\text{NaOH} + 3\text{H}_2\text{O} \rightarrow \text{PH}_3 + 3\text{NaH}_2\text{PO}_2$
- (iii) $\text{ZnSO}_4 + 2\text{NaOH} \longrightarrow \text{Zn}(\text{OH})_2 + \text{Na}_2\text{SO}_4$ (iv) $\text{AlCl}_3 + 3\text{NaOH} \longrightarrow \text{Al}(\text{OH})_3 + 3\text{NaCl}$
- ppt ppt
- $$\text{Zn}(\text{OH})_2 + 2\text{NaOH} \longrightarrow \text{Na}_2[\text{Zn}(\text{OH})_4] \quad \text{Al}(\text{OH})_3 + \text{NaOH} \longrightarrow \text{Na}[\text{Al}(\text{OH})_4]$$
- or
- NaAlO_2
4. (i) $\text{Na}_2\text{O}_2 + 2\text{H}_2\text{O} \longrightarrow 2\text{NaOH} + \text{H}_2\text{O}_2$
- (ii) $\text{KO}_2 + \text{H}_2\text{O} \longrightarrow \text{KOH} + \text{H}_2\text{O}_2 + \text{O}_2$
- (iii) $2\text{Na}_2\text{O}_2 + 2\text{CO}_2 \longrightarrow 2\text{Na}_2\text{CO}_3 + \text{O}_2$
5. A = Ca, B = Ca_3N_2 C = $\text{Ca}(\text{OH})_2$ D = NH_3
6. In presence of HF, F^- is converted into bifluoride ion HF_2^- , allowing further dissolution of solid LiF.
7. $\text{CuSO}_4 + \text{Na}_2\text{S}_2\text{O}_3 \longrightarrow \text{CuS}_2\text{O}_3 + \text{Na}_2\text{SO}_4$
- $$2\text{CuS}_2\text{O}_3 + \text{Na}_2\text{S}_2\text{O}_3 \longrightarrow \text{CuS}_2\text{O}_3 + \text{Na}_2\text{S}_4\text{O}_6$$
- Cupric thiosulphate
- $$3\text{CuS}_2\text{O}_3 + 2\text{Na}_2\text{S}_2\text{O}_3 \longrightarrow \text{Na}_4[\text{Cu}_6(\text{S}_2\text{O}_3)_5]$$
- Sodium cuprothiosulphate.
8. (i) $\text{BeSO}_4 < \text{MgSO}_4 < \text{CaSO}_4$ (ii) $\text{Ca}^{2+} < \text{Mg}^{2+} < \text{Be}^{2+}$
- (iii) $\text{Be}(\text{OH})_2 < \text{Mg}(\text{OH})_2 < \text{Ca}(\text{OH})_2$ (iv) $\text{CaCl}_2 < \text{MgCl}_2 < \text{BeCl}_2$
- (v) $\text{CaCl}_2 < \text{MgCl}_2 < \text{BeCl}_2$ (vi) $\text{BaF}_2 < \text{CaF}_2 < \text{MgF}_2$
- (vii) $\text{Ba}^{2+} < \text{Mg}^{2+} < \text{Be}^{2+}$ (viii) $\text{BaF}_2 < \text{MgF}_2 < \text{BeF}_2$
- (ix) $\text{Be} < \text{Mg} < \text{Ca} < \text{Sr}$
9. (i) Ca gives brick red colour to flame (ii) $\text{Ca} + \text{H}_2 \longrightarrow \text{CaH}_2$ (hydrolith, trade name)
- (A) (B)
- (iii) $\text{CaH}_2 + 2\text{H}_2\text{O} \longrightarrow \text{Ca}(\text{OH})_2 + 2\text{H}_2$
- (B) (C)

BRAIN STORMING SUBJECTIVE EXERCISE

1. When a gas (A) is passed through dry KOH at low temperature, a deep red coloured compound (B) and a gas (C) are obtained. The gas (A) on reaction with but-2-ene followed by treatment with $\text{Zn}/\text{H}_2\text{O}$ yields acetaldehyde. Identify (A), (B) and (C)
2. A compound (X) imparts a golden yellow flame and shows the following reactions:
 - (i) Zinc powder when boiled with a concentrated aqueous solution of (X) dissolves and hydrogen is evolved.
 - (ii) When an aqueous solution of (X) is added to an aqueous solution of stannous chloride, a white precipitate is obtained first which dissolves in excess of solution of (X). Identify (X) and write equations at step (i) and (ii).
3. A white solid is either Na_2O or Na_2O_2 . A piece of red litmus paper turns white when it is dipped into a freshly made aqueous solution of the white solid.
 - (i) Identify the substance and explain with balanced equation
 - (ii) Explain what would happen to the red litmus if the white solid were the other compound
4. (A) is binary compound of a univalent metal. 1.422 g of (A) reacts completely with 0.321 g of sulphur in an evacuated and sealed tube to give 1.743 g of a white crystalline solid (B) that formed a hydrated double salt (C) with $\text{Al}_2(\text{SO}_4)_3$. Identify (A), (B) and (C).
5. Element (M) is a shiny and highly reactive metal (melting point 63°C) and element (X) is a highly reactive non-metal (melting point -7.2°C). They react to form a compound with the empirical formula MX, a colourless, brittle solid that melts at 734°C . When dissolved in water or when in the molten state, the substance conducts electricity. When chlorine gas is bubbled through an aqueous solution containing (MX), a reddish-brown liquid appears and Cl^- are formed. From these observations, identify M and X.
6. Name an element which is invariably bivalent and whose oxide is soluble in excess of NaOH and its dipositive ion has a noble gas core.
7. Out of the elements marked A, B, C, D, E, F, G and H:
 - (a) Which form superoxide?
 - (b) Which form thermally stable carbonate?
 - (c) Which forms strongest base?
 - (d) Which show diagonal relationship?
 - (e) Which forms amphoteric oxide?

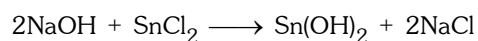
[illegible][illegible]

1. The gas (A) on treatment with but-2-ene followed by treatment with $\text{Zn}/\text{H}_2\text{O}$ yields acetaldehyde and thus (A) is ozone



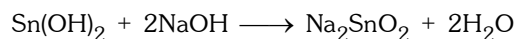
(X)

- (ii) (X) is also justified by step 2 reactions:



(X)

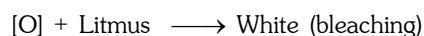
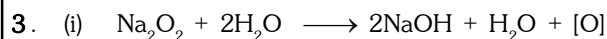
White ppt.



(Excess)

Soluble

(X)



Red

- (ii) The other compound Na_2O will give NaOH on dissolution in water. The red litmus will turn to blue.

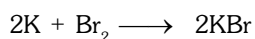


(A)

(B)

(C)

5. The given facts suggest M to be potassium (K) and (X) to be bromine (Br_2).



(Ionic solid with m. pt. 734°C)

6. Be

7. (a) E and G

(b) C, E and G

(c) G

(d) A and D

(e) B

EXERCISE-05 (A)**PREVIOUS YEARS QUESTIONS**

- A metal M readily forms its sulphate MSO_4 which is water soluble. It forms oxide MO which becomes inert on heating. It forms insoluble hydroxide which is soluble in NaOH. The metal M is:- [AIEEE-2002]
 (1) Mg (2) Ba (3) Ca (4) Be
- KO_2 is used in space and submarines because it [AIEEE-2002]
 (1) Absorbs CO_2 and increase O_2 concentration
 (2) Absorbs moisture
 (3) Absorbs CO_2
 (4) Produces ozone
- In curing cement plasters, water is sprinkled from time to time. This helps in :- [AIEEE-2003]
 (1) Hydrating sand and gravel mixed with cement
 (2) Converting sand into silicate
 (3) Developing interlocking needle like crystals of hydrated silicates
 (4) Keeping it cool
- The solubilities of carbonates decreases down the magnesium group due to decrease in- [AIEEE-2003]
 (1) Inter-ionic attraction
 (2) Entropy of solution formation
 (3) Lattice energy of solids
 (4) Hydration energy of cations
- The substance not likely to contain CaCO_3 is :- [AIEEE-2003]
 (1) Sea shells (2) Dolomite
 (3) A marble statue (4) Calcined gypsum
- One mole of magnesium nitride on reaction with excess of water gives :- [AIEEE-2004]
 (1) Two mole of HNO_3 (2) Two mole of NH_3
 (3) 1 mole of NH_3 (4) 1 mole of HNO_3
- The ionic mobility of alkali metal ions in aqueous solution is maximum for :- [AIEEE-2006]
 (1) Rb^+ (2) Li^+ (3) Na^+ (4) K^+
- Which of the following on thermal-decomposition yields a basic as well as an acidic oxide ? [AIEEE-2012]
 (1) NH_4NO_3 (2) NaNO_3 (3) KClO_3 (4) CaCO_3
- Fire extinguishers contain H_2SO_4 and which one of the following :- [AIEEE-2012 (Online)]
 (1) CaCO_3 (2) NaHCO_3 and Na_2CO_3
 (3) Na_2CO_3 (4) NaHCO_3
- Which one of the following will react most vigorously with water ? [AIEEE-2012 (Online)]
 (1) Li (2) K (3) Rb (4) Na
- A metal M on heating in nitrogen gas gives Y. Y on treatment with H_2O gives a colourless gas which when passed through CuSO_4 solution gives a blue colour, Y is :- [AIEEE-2012 (Online)]
 (1) NH_3 (2) MgO (3) Mg_3N_2 (4) $\text{Mg}(\text{NO}_3)_2$

PREVIOUS YEAR QUESTIONS											s-BLOCK		EXERCISE-05(A)					
Q.	1	2	3	4	5	6	7	8	9	10	11							
A.	4	1	3	4	4	2	1	4	3	3	3							

EXERCISE-05 (B)**PREVIOUS YEARS QUESTIONS**

1. Which process is used in the extractive metallurgy of Mg :
(A) Fused salt electrolysis (B) Self reduction
(C) Aqueous solution electrolysis (D) Thermite reduction
2. A sodium salt on treatment with MgCl_2 gives white precipitate only on heating. The anion of sodium salt is :
(A) HCO_3^- (B) CO_3^{2-} (C) NO_3^- (D) SO_4^{2-}
3. The following compounds have been arranged in order of their increasing thermal stabilities. Identify the correct order: (IIT 96)
(I) K_2CO_3 (II) MgCO_3 (III) CaCO_3 (IV) BeCO_3
(A) $\text{I} < \text{II} < \text{III} < \text{IV}$ (B) $\text{IV} < \text{II} < \text{III} < \text{I}$
(C) $\text{IV} < \text{II} < \text{I} < \text{III}$ (D) $\text{II} < \text{IV} < \text{III} < \text{I}$
4. Property of the alkaline earth metals that increases with their atomic number is – (IIT 97)
(A) Ionisation energy
(B) Solubility of their hydroxides
(C) Solubility of their sulphates
(D) Electronegativity
5. The characteristics of solid sodium chloride are (REE 96)
(1) Brittle (2) Ionic (3) Covalent (4) Non-conductor
(A) 1 & 2 (B) 3 & 4 (C) 1, 2, & 4 (D) 1, 3, & 4
6. Which of the following are not amphoteric – (REE 97)
(1) $\text{Be}(\text{OH})_2$ (2) $\text{Sr}(\text{OH})_2$ (3) $\text{Ca}(\text{OH})_2$ (4) $\text{Al}(\text{OH})_3$
(A) 1 & 3 (B) 2 & 3 (C) 1 & 4 (D) 2 & 4
7. Highly dilute solution of sodium in liquid ammonia :
(i) Shows blue colour (ii) Exhibits electrical conductivity
(iii) Produces sodium amide (iv) Produces hydrogen gas
(A) (i), (ii), (iii) (B) (i), (ii) (C) (iii), (iv) (D) Only (ii)
8. Which of the following hydrides is not ionic
(A) CaH_2 (B) BaH_2 (C) SrH_2 (D) BeH_2
9. The compound(s) formed upon combustion of sodium metal in excess air is (are) [JEE 2009]
(A) Na_2O_2 (B) Na_2O (C) NaO_2 (D) NaOH

ASSERTION & REASON QUESTIONS

- (A) Statement-I is true, statement-II is true ; statement-II is a correct explanation for statement-I
(B) Statement-I is true, statement-II is true ; statement-II is NOT a correct explanation for statement-I
(C) Statement-I is true, statement-II is false
(D) Statement-I is false, statement-II is true

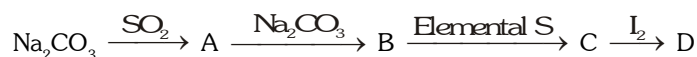
1. **Statement-I** : Alkali metals dissolve in liquid ammonia to give blue solutions. (IIT 2007)

Because :

Statement-II : Alkali metals in liquid ammonia give solvated species of the type $[\text{M}(\text{NH}_3)_n]^+$ (M = alkali metals)

SUBJECTIVE QUESTIONS

- 1.** Identify the following:



Also mention the oxidation state of S in all the compounds.

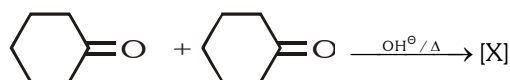
2. Beryllium chloride shows acidic nature in water or why BeCl_2 is easily hydrolysed?
3. The crystalline salts of alkaline earth metals contain more water of crystallisation than the corresponding alkali metal salts, why?
4. Arrange the following sulphates of alkaline earth metals in order of their decreasing thermal stability. BeSO_4 , MgSO_4 , CaSO_4 , SrSO_4 .
5. Why the solubility of calcium acetate decreases while that of lead nitrate increases with increase in temperature.
6. Why magnesium is not precipitated from a solution of its salt by NH_4OH In the presence of NH_4Cl .

PREVIOUS YEARS QUESTIONS	ANSWER KEY	EXERCISE -05(B)
1. (A) 2. (A) 3. (B) 4. B 5. C 6. B 7. B 8. D 9. A, B <u>Assertion - Reason Questions</u> 1. B <u>Subjective Questions</u> 1. A = NaHSO₃, Oxidation state of S = + 4 B = Na₂SO₃, Oxidation state of S = + 4 C = Na₂S₂O₃, Oxidation state of S = + 6 & - 2 D = Na₂S₄O₆, Oxidation state of S = + 5 & 0 2. BeCl₂ is a salt of weak base Be(OH)₂ and strong acid HCl and thus undergoes hydrolysis to result in an acidic solution in water.	$\text{BeCl}_2 + 4\text{H}_2\text{O} \xrightarrow{\text{Hydration}} [\text{Be}(\text{H}_2\text{O})_4]^{2+} + 2\text{Cl}^-$	
3. Alkaline earth metals have smaller size and more nuclear charge. 4. $\text{SrSO}_4 > \text{CaSO}_4 > \text{MgSO}_4 > \text{BeSO}_4$ 5. (CH₃COO)₂ Ca shows exothermic dissolution whereas Pb(NO₃)₂ show endothermic dissolution. 6. The dissociation of NH₄OH (a weak electrolyte) is suppressed in presence of NH₄Cl due to common ion effect. Thus, [OH⁻] in solution becomes low. The ionic product of concentrations of Mg²⁺ and OH⁻ ions does not exceed the solubility product of Mg(OH)₂ and thus Mg(OH)₂ is not precipitated.		

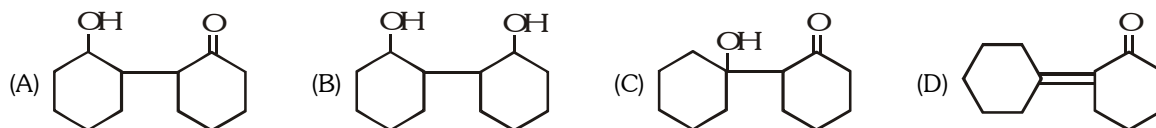
EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. Which of the following reacts with water to form very stable product ?
(A) CH_3Cl (B) CCl_4 (C) CCl_3CHO (D) $\text{CH}_2\text{ClCH}_2\text{Cl}$
2. The compound that gives a positive iodoform test is :-
(A) 1-pentanol (B) 2-pentanone (C) 3-pentanone (D) pentanal
3. Which of the following does not undergo aldol condensation ?
(A) HCHO (B) CH_3CHO (C) CH_3COCH_3 (D) $\text{CH}_3\text{CH}_2\text{CHO}$
4. Cannizzaro reaction is not given by :
(A) Trimethylacetaldehyde (B) Acetaldehyde (C) Benzaldehyde (D) Formaldehyde

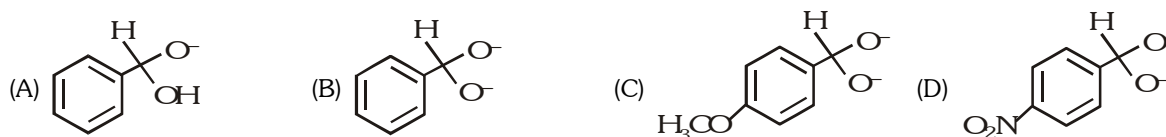
5. In the reaction :



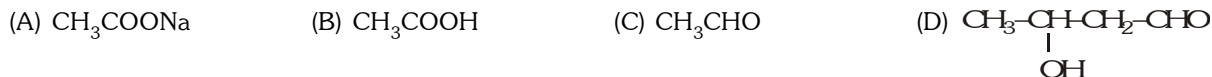
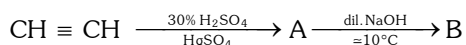
[X] will be :-



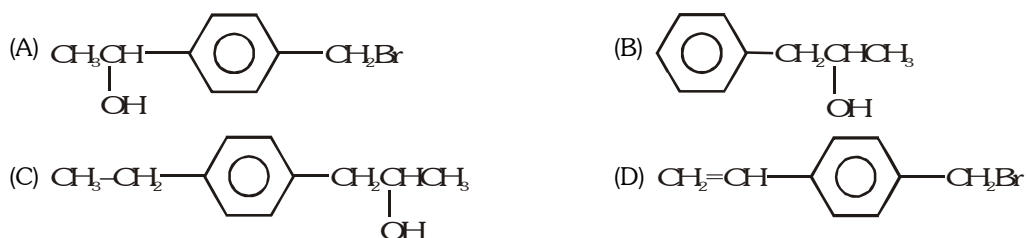
6. In a Cannizzaro reaction, the intermediate that will be the best hydride donor is :



7. If 3-hexanone is reacted with NaBH_4 followed by hydrolysis with D_2O , the product will be :
(A) $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_3$ (B) $\text{CH}_3\text{CH}_2\text{CD}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_3$
(C) $\text{CH}_3\text{CH}_2\text{CH}(\text{OD})\text{CH}_2\text{CH}_2\text{CH}_3$ (D) $\text{CH}_3\text{CH}_2\text{CD}(\text{OD})\text{CH}_2\text{CH}_2\text{CH}_3$
8. Predict the product 'B' in the sequence of reaction



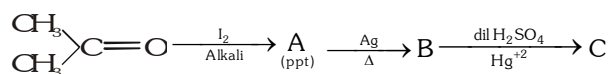
9. In the reaction $\text{CH}_2\text{Br} + \text{Mg} \xrightarrow{\text{dry ether}} \text{A} + \text{CH}_3\text{CHO} \longrightarrow \text{adduct} \xrightarrow{\text{H}_3\text{O}^+} \text{B}$.
The product (B) is :



10. Which one of the following reagents is suitable for the conversion of 2-cyclohexenone into 3-methylcyclohexanone ?
 (A) CH_3MgI (B) $(\text{CH}_3)_2\text{CuLi}$ (C) CH_3I_2 and Zn (D) CH_3AlCl_2
11. Acetaldehyde reacts with NaOH to form :-



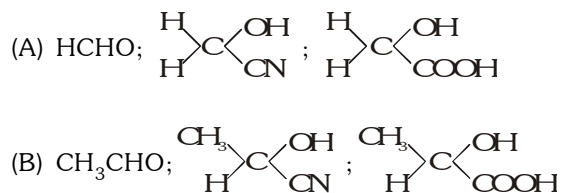
12. Benzaldehyde reacts with formaldehyde in the presence of alkali to form :-
 (A) Methyl alcohol and sodium benzoate (B) Benzyl alcohol and sodium formate
 (C) Benzoic acid and ethanol (D) Formic acid and benzyl alcohol
13. The compounds A, B and C in the reaction sequence

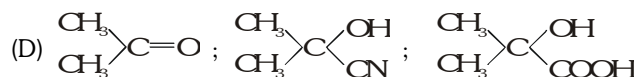


are given by the set :-

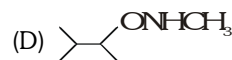
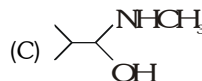
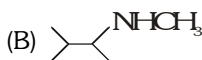
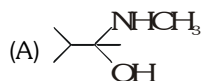
- (A) Iodoform, ethylene, ethyl alcohol (B) Iodoform, acetylene, acetaldehyde
 (C) Iodoform, propyne, acetone (D) Iodoform, 2-propanol, propanone
14. In the reaction sequence
- $$\text{RCOCl} + \text{H}_2 \xrightarrow{\text{Pd}+\text{BaSO}_4} \text{A} \xrightarrow{\text{HCN}} \text{B} \xrightarrow{\text{H}_3\text{O}^+} \text{C}$$
- A, B and C are given by the set :-
- (A) RCHO , RCH(OH)CN , $\text{RCH(OH)CH}_2\text{NH}_2$ (B) RCHO , RCH(OH)CN , RCH(OH)COOH
 (C) RCHO , $\text{R}-\overset{\text{O}}{\underset{||}{\text{C}}}-\overset{\text{CN}}{\underset{|}{\text{CH}_2}}$, $\text{R}-\overset{\text{O}}{\underset{||}{\text{C}}}-\overset{\text{CN}}{\underset{|}{\text{CH}_2}}-\overset{\text{O}}{\underset{||}{\text{C}}}-\text{OH}$ (D) RCHO , $\text{R}-\text{CH}_2-\text{CN}$, $\text{R}-\text{CH}_2-\text{COOH}$

15. Oxidation of 2-methyl propane-1,2-diol with periodic acid gives :-
 (A) Propionic acid and formaldehyde (B) Acetone and formaldehyde
 (C) Acetone and acetic acid (D) Acetone and propionic acid
16. A carbonyl compound gives a positive iodoform test but does not reduce Tollen's reagent or Fehling's solution. It forms a cyanohydrin with HCN, which on hydrolysis gives a hydroxy acid with a methyl side chain. The compound is :-
 (A) Acetaldehyde (B) Propionaldehyde (C) Acetone (D) Crotonaldehyde
17. A carbonyl compound 'A' reacts with hydrogen cyanide to form a cyanohydrin 'B' which on hydrolysis gives an optically active alpha hydroxy acid 'C'. 'A' gives a positive iodoform test 'A', 'B' and 'C' are given by the set :-

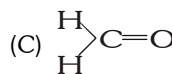
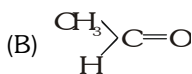




- $$\text{CH}_3\text{CH}(\text{CH}_3)\text{C}(=\text{O})\text{CH}_3 \xrightarrow[\text{(ii) LiAlH}_4, \text{(iii) H}_2\text{O}]{\text{(i) CH}_3\text{NH}_2}$$

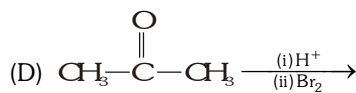
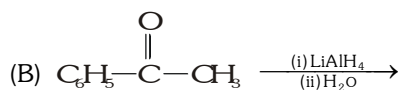


- (A) $\begin{array}{c} \text{CH}_3 \\ \text{CH}_3 \end{array} \rangle \text{C}=\text{O}$

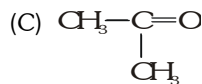
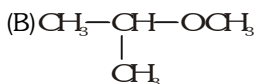


- (D) All of these

- $$(A) \text{ C}_6\text{H}_5-\overset{\text{O}}{\parallel}\text{C}-\text{CH}_3 \xrightarrow[\text{(ii) H}^+]{\text{(i) CH}_3\text{MgBr}}$$



- (A) CH_3CHO



- (D) HCHO

- (A) formaldehyde

- (B) Ethyl alcohol

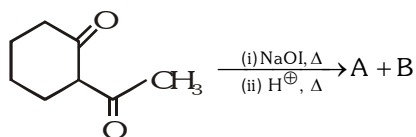
- (C) Methyl cyanide

- (D) Methyl iodide

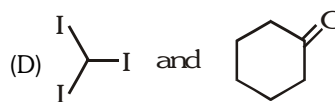
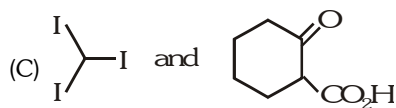
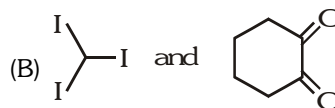
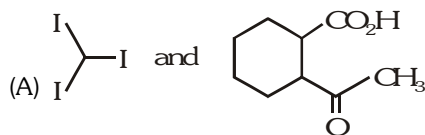
CHECK YOUR GRASP								ANSWER KEY					EXERCISE -1							
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	A	B	D	C	C	D	B	B	B	B	B	B	B	C	B	D	B	B
Que.	21	22	23	24	25															
Ans.	A	D	B	B	C															

EXERCISE-02**BRAIN TEASERS****SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)**

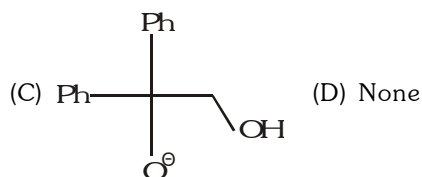
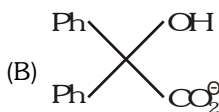
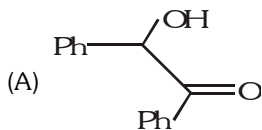
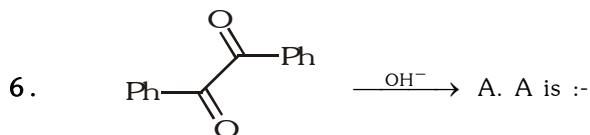
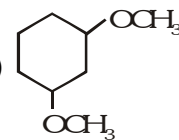
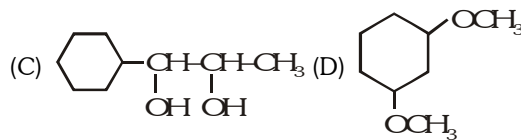
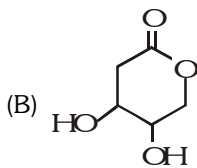
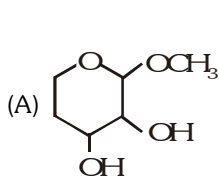
1. End products of the following sequence of reaction is



A & B are :



2. Which of the following compound on treatment with LiAlH_4 will give a product that will give positive iodoform test?
- (A) $\text{CH}_3\text{CH}_2\text{CHO}$ (B) $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3$ (C) $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$ (D) CH_3COCH_3
3. Among the given compounds, the most susceptible to nucleophilic attack at the carbonyl group is
- (A) CH_3COCl (B) CH_3CHO (C) $\text{CH}_3\text{COOCH}_3$ (D) $\text{CH}_3\text{COOCOCH}_3$
4. Which of the following does not give iodoform reaction ?
- (A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (B) CH_3OH (C) CH_3CHO (D) PhCOCH_3
5. Which of the following compounds undergo periodic oxidation

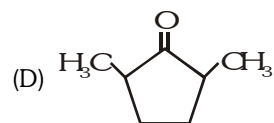
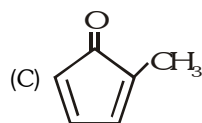
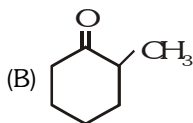
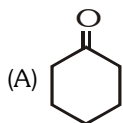


(D) None

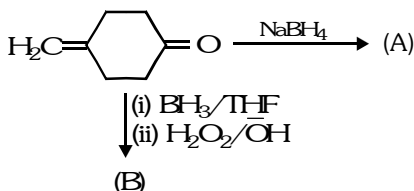
7. In the reaction sequence, [X] is ketone :



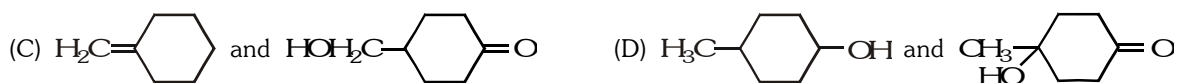
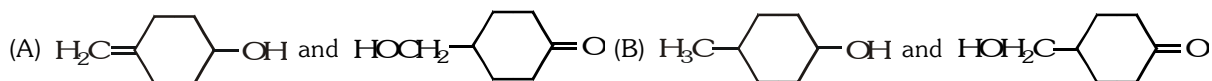
[X] will be :-



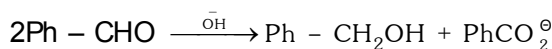
8. In the given reaction :



(A) and (B) are :

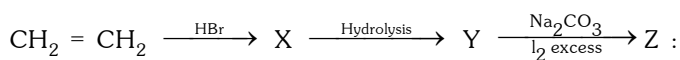


9. In the Cannizzaro reaction given below,



the slowest step is :

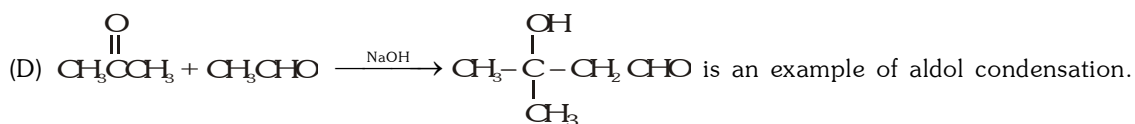
- (A) the attack of OH^- at the carbonyl group
 (B) the transfer of hydride to the carbonyl group
 (C) the abstraction of proton from the carboxylic acid
 (D) the deprotonation of $\text{Ph} - \text{CH}_2\text{OH}$
10. Which of the possible compound will be formed in the following sequence of reaction.

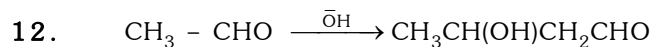


- (A) $\text{C}_2\text{H}_5\text{Br}$ (B) $\text{C}_2\text{H}_5\text{OH}$ (C) CHI_3 (D) CH_3CHO

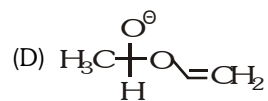
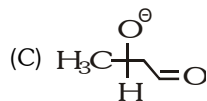
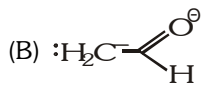
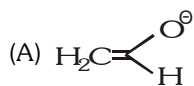
11. Which of the following statements are correct ?

- (A) Benzaldehyde reduces Fehling's solution
 (B) $\text{C}_6\text{H}_5\text{CHO} + \text{C}_6\text{H}_5\text{CHO} \xrightarrow{\text{NaOH}} \text{C}_6\text{H}_5\text{CH} = \text{CHC}_6\text{H}_5 + \text{O}_2$ is a Claisen-Schmidt reaction.
 (C) pK_a (formic acid) is less than pK_a (acetic acid)

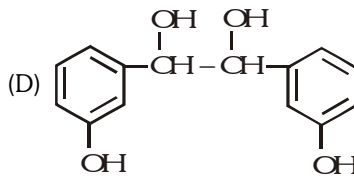
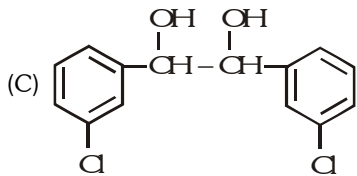
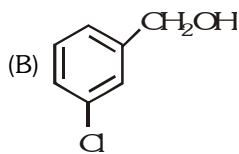
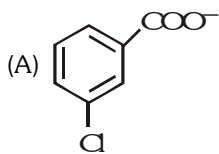




In the aldol condensation of acetaldehyde represented above, which of the following intermediate species are obtained ?

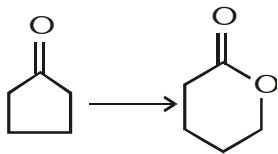


13. When m-Chlorobenzaldehyde is treated with 50% KOH solution, the products obtained is (are)

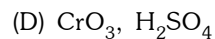
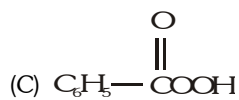
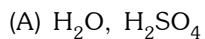


14.

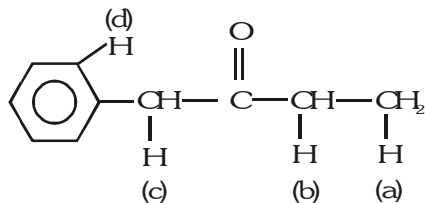
The conversion :



can be effected by using the reagent



15. Which of the following hydrogens will be the most acidic?



(A) a

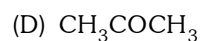
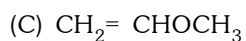
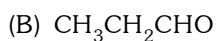
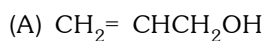
(B) b

(C) c

(D) d

16.

An organic compound with the molecular formula $\text{C}_3\text{H}_6\text{O}$ does not respond positively to the silver mirror test with Tollens reagent but produces an oxime. The compound is



EXERCISE-03

MISCELLANEOUS TYPE QUESTIONS

TRUE OR FALSE :

1. The reaction of methyl magnesium bromide with acetone followed by hydrolysis gives secondary alcohol.
2. Aldehydes are more reactive than ketones.
3. The yield of ketone, when a secondary alcohol is oxidised, is more than the yield of aldehyde when primary alcohol is oxidised by $K_2Cr_2O_7/H^+$
4. Both aldehydes and ketones reduce Tollen's reagent.
5. Aldol condensation is given by all carbonyl compounds.
6. Acetaldehyde and acetone can be distinguished by iodoform test.
7. $LiAlH_4$ converts ketones into secondary alcohols.
8. Methanol can be distinguished from ethanol by haloform reaction.
9. Propanone does not show tautomerism.
10. Ketones restore pink colour of Schiff's reagent.

FILL IN THE BLANKS :

1. To prepare ethanol, CH_3MgI is treated with the other reagent
2. Urotropine is formed by the action of with
3. The conversion of acid chlorides into aldehydes by reduction is termed
4. Aldehyde show reducing properties due to their ready conversion into
5. Hydrazone of an aldehyde when heated with sodium ethoxide forms This is known as reaction.
6. Cannizzaro's reaction is followed by those aldehydes which α -hydrogen atom.
7. Two separate solutions, Fehling's solution A(.....) and Fehling's solution B($NaOH + \dots$) are at first mixed up together and is then heated with the aldehyde. A precipitate is formed.
8. Tollen's reagent gives with acetaldehyde.
9. Aldehydes have boiling points lower than those of and higher than those of of comparable molecular masses.
10. When calcium acetate is distilled alone is formed.

MATCH THE COLUMN

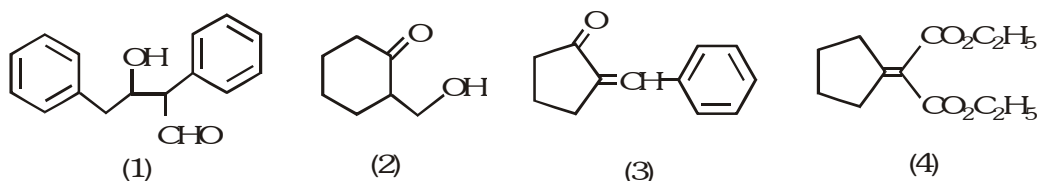
1. Match the compounds/ions in Column I with their properties/reactions in Column. II

Column -I		Column -II	
(A)	C_6H_5CHO	(p)	gives precipitate with 2,4-dinitrophenylhydrazine
(B)	$CH_3C \equiv CH$	(q)	give precipitate with $AgNO_3$
(C)	CN^-	(r)	is a nucleophile
(D)	I^-	(s)	is involved in cyanohydrin formation

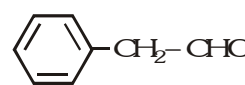
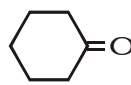
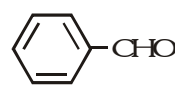
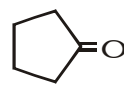
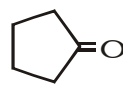
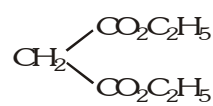
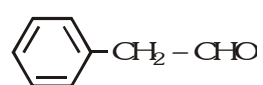
2. Match the column I with column II.

Column-I		Column-II	
(A)	$\text{RMgX}_{n\text{-carbon}} + \text{HCHO} \longrightarrow$ Adduct $\xrightarrow{\text{H}_3\text{O}^+}$	(p)	Ketone
(B)	$\text{RMgX}_{n\text{-carbon}} + (\text{CH}_2)_2\text{O} \longrightarrow$ Adduct $\xrightarrow{\text{H}_3\text{O}^+}$	(q)	1°Alcohol (n + 1) carbon
(C)	$\text{RMgX}_{n\text{-carbon}} + \text{CO}_2 \longrightarrow$ Adduct $\xrightarrow{\text{H}_3\text{O}^+}$	(r)	Acid (n + 1) carbon
(D)	$\text{RMgX}_{n\text{-carbon}} + \text{Ph-C}\equiv\text{N} \longrightarrow$ Adduct $\xrightarrow{\text{H}_3\text{O}^+}$	(s)	1°Alcohol (n + 2) carbon

3. Aldol condensation proceeds by carbon-carbon bond formation between an enolate donor and a carbonyl acceptor. For each of the following aldol products (1 through 4)



match the donor and acceptor compound.

Column -I [Donor]		Column -II [Acceptor]	
(A)		(p)	$\text{CH}_2 = \text{O}$
(B)		(q)	
(C)		(r)	
(D)		(s)	

ASSERTION & REASON QUESTION :

These questions contains, Statement-I (assertion) and Statement-II (reason).

- (A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I
 (B) Statement-I is True, Statement-II is True ; Statement-II is NOT a correct explanation for Statement-I
 (C) Statement-I is True, Statement-II is False.
 (D) Statement-I is False, Statement-II is True.

1. **Statement-I** : 2, 2,-dimethyl propanol undergoes Cannizaro reaction with concentrated NaOH.
Because
Statement-II : Cannizarro reaction is a disproportionation reaction.

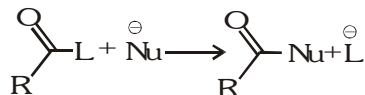
2. **Statement-I** : Benzaldehyde does not undergoes aldol condensation.
Because
Statement-II : Benzaldehyde does not contains acidic α -Hydrogen.
3. **Statement-I** : Acetaldehyde is less reactive than trichloro acetaldehyde.
Because
Statement-II : Chlorine atom exhibit -I effect in trichloro acetaldehyde.
4. **Statement-I** : Benzaldehyde gives a positive test with Benedict's and Fehling's solution.
Because
Statement-II : Benzaldehyde gives silver mirror with Tollen's reagent.
5. **Statement-I** : $R-C\equiv O^+$ is more stable than $R-C^+=O$.
Because
Statement-II : Resonance in carbonyl compound provides $\text{>}\overset{\oplus}{C}-$ and O^- .
6. **Statement-I** : Rate of addition of HCN on carbonyl compounds increases in presence of NaCN.
Because
Statement-II : Reaction involves the addition of CN^- in rate determining step.
7. **Statement-I** : Fehling's solution can be used to distinguish benzaldehyde from acetaldehyde.
Because
Statement-II : Both benzaldehyde and acetaldehyde reduces tollen's reagent.
8. **Statement-I** : Ketones are less reactive than aldehydes.
Because
Statement-II : Ketones do not give Schiff's test.
9. **Statement-I** : Benzaldehyde is more reactive than ethanol towards nucleophilic attack.
Because
Statement-II : The overall effect of -I and +R effect of phenyl group decreases the electron density on the carbon atom of $>C=O$ group in benzaldehydes.
10. **Statement-I** : In formaldehyde all the four atoms are in same plane.
Because
Statement-II : The carbon atom in formaldehyde is sp^2 hybridised.

COMPREHENSION BASED QUESTIONS :

Comprehension # 1

Aldehyde, ketone, acid and acid derivatives contain $>C=O$ group. Aldehyde and ketones gives nucleophilic addition reactions where as acid and acid derivatives gives nucleophilic addition followed by elimination reactions. Nucleophilic addition reactions followed by elimination of acid derivatives is known as acyl substitution reaction. This substitution reaction takes places by formation of tetrahedral intermediate.

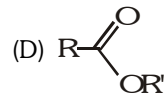
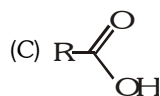
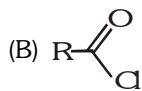
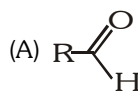
1. For the given reaction



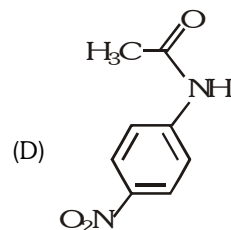
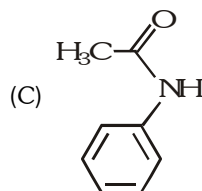
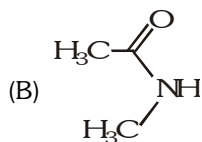
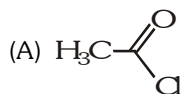
which of these is correct ?

- (A) L must be better leaving group than Nu
- (B) Nu^- must be strong enough nucleophile to attack carbonyl carbon
- (C) Carbonyl carbon must be enough electrophilic to react with Nu^-
- (D) All of these

2. Which of the following compounds has very poor leaving group ?



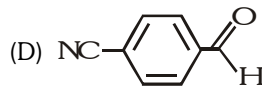
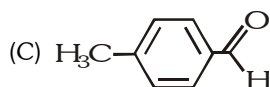
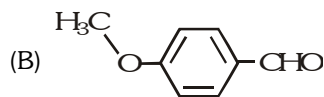
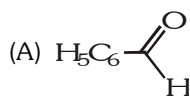
3. Which one of the following is least reactive compound for nucleophilic acyl substitution.



Comprehension # 2

Aldehydes and ketones are amphoteric. Thus they can act both as acids and bases. Under acidic conditions, the carbon of the protonated carbonyl group is much more electrophilic, reacting even with weak nucleophile. Carbonyl compound gives nucleophilic addition reaction. In this reaction the nucleophilic attack precedes the electrophilic attack.

- Which of these statements are correct ?
 (A) Carbonyl compound is amphoteric in character
 (B) Acid catalyst makes the carbonyl carbon more electrophilic
 (C) basic catalyst makes the nucleophilic attack more faster.
 (D) All of these
- Which of the statements are/is correct ?
 (A) The rate determining step of addition reaction is the addition of nucleophile
 (B) The rate-determining step is addition of electrophile
 (C) The reaction intermediate of the reaction is alkoxide ion
 (D) both (A) and (C)
- Which one of the carbonyl compounds is more reactive towards NaCN/H^+ ?



4. Carbonyl compounds gives nucleophilic addition with
- carbon nucleophile
 - oxygen nucleophile
 - Nitrogen nucleophile
 - All of these

Comprehension # 3

Aldehyde and ketones are specially susceptible to nucleophile addition because carbonyl group >C=O is polar (due to electronegativity different between carbon and oxygen).



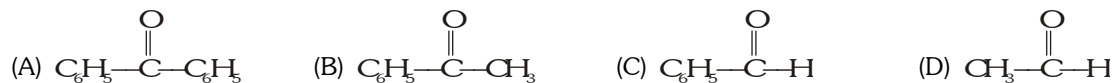
Positive charge on carbon makes it reactive towards the nucleophile. This addition is catalysed by acid.

Reactivity of carbonyl compound towards nucleophilic addition increases with increase in the electron deficiency at carbonyl carbon. Thus, (-I.E.) group increase while (+I.E.) groups decrease the reactivity of carbonyl compound.

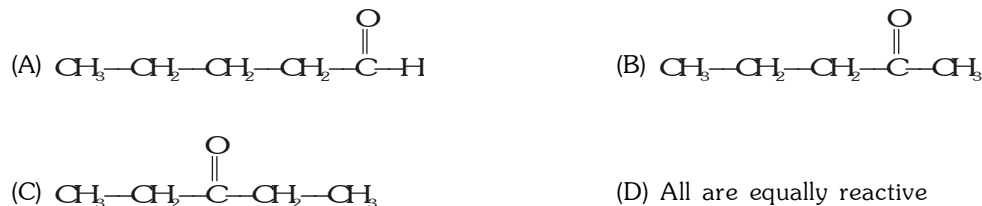
- Which of the following is most reactive to give nucleophilic addition?
(A) FCH_2CHO (B) ClCH_2CHO (C) BrCH_2CHO (D) ICH_2CHO
- Carbonyl compounds show nucleophilic addition with :
(A) HCN (B) NaHSO_3 (C) $(\text{CH}_3\text{OH} + \text{HCl})$ (D) all of these
- Which among the following carbonyl compounds is most polar?



- Select the least reactive carbonyl compound for nucleophilic addition:



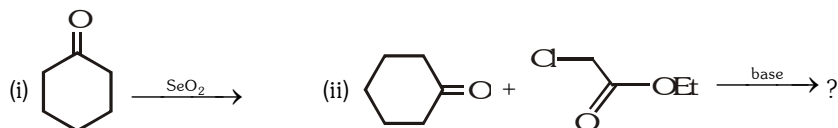
- Which among the following isomeric compound is most reactive?



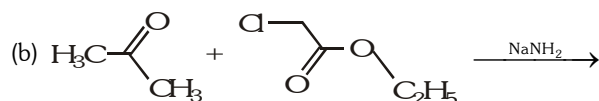
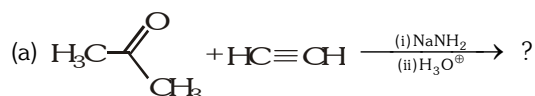
MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
True / False 1. F 2. T 3. T 4. F 5. F 6. F 7. T 8. T 9. F 10. F Fill in the Blanks 1. HCHO 2. HCHO, NH_3 3. Rosenmund reduction 4. acids 5. alkane, wolf-kishner 6. do not have 7. CuSO_4, roscheil salt, Cu_2O 8. silver mirror 9. alcohol, alkane 10. acetone Match the Column 1. (A) $\rightarrow$ p,s ; (B) $\rightarrow$ q,r ; (C) $\rightarrow$ q,r,s ; (D) $\rightarrow$ q,r 2. (A) $\rightarrow$ q ; B $\rightarrow$ s ; (C) $\rightarrow$ r ; (D) $\rightarrow$ p 3. (A) $\rightarrow$ s ; (B) $\rightarrow$ p ; (C) $\rightarrow$ q ; (D) $\rightarrow$ r Assertion - Reason Questions 1. D 2. A 3. A 4. D 5. B 6. A 7. B 8. B 9. A 10. A Comprehension Based Questions Comprehension #1 : 1. (D) 2. (A) 3. (B) Comprehension #2 : 1. (D) 2. (D) 3. (D) 4. (D) Comprehension #3 : 1. (A) 2. (D) 3. (A) 4. (A) 5. (A)		

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

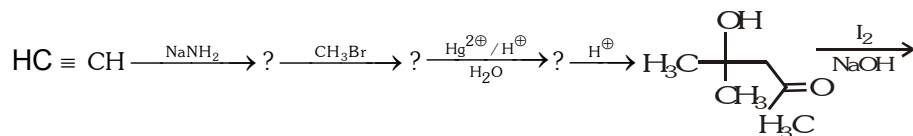
- Which of the carbonyl groups in $p\text{-MeOC}_6\text{H}_4\text{COMe}$ and $p\text{-NO}_2\text{C}_6\text{H}_4\text{COMe}$ protonates more readily in acid solution and why ?
- Identify (A) and (B) in the given sequence of reaction $\text{PhCH}_2\text{CHO} \xrightarrow{\text{SeO}_2} \text{(A)} \xrightarrow[\text{(ii)H}^+]{\text{(i)con.OH}^-} \text{(B)}$
- Distinguish between PhCOEt and $p\text{-MeC}_6\text{H}_4\text{COMe}$ by a chemical method.
 - Arrange the following in the order of increasing reactivity towards nucleophilic addition reactions.
 - CH_3CHO , $\text{C}_6\text{H}_5\text{COC}_6\text{H}_5$, $\text{CH}_3\text{COC}_6\text{H}_5$, CH_3COCH_3
 - CH_3CHO , CF_3CHO , $\text{CH}_2 = \text{CHCHO}$
- Compound X with molecular formula $\text{C}_9\text{H}_{10}\text{O}$ form a semicarbazone and give negative Tollen's and Iodoform tests. Upon reduction it gives n-propyl benzene. Deduce the structure of X.
- The double bond in aldehydes and ketones is reactive towards nucleophilic reagents like CN^- whereas that in an alkene is not.
 - Alkenes undergo electrophilic addition whereas aldehydes and ketones undergo nucleophilic addition.
- Complete the following equations giving the structures of the major organic product.



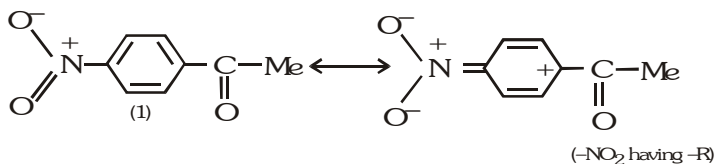
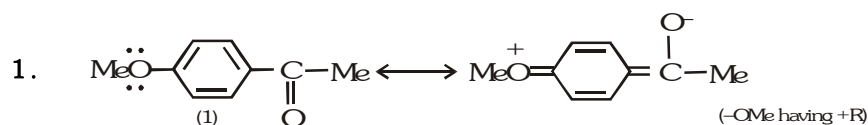
7. Complete the following equation :



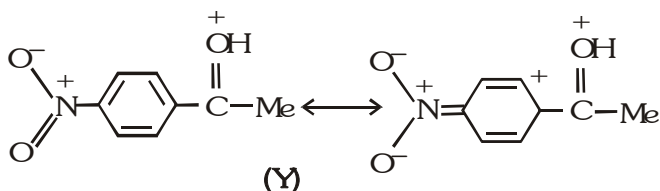
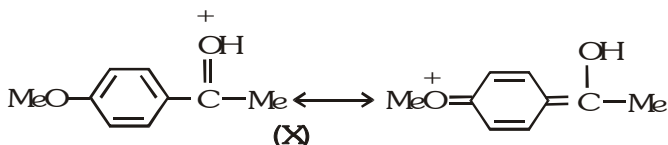
- Two organic compounds (A) and (B) have same empirical formula CH_2O . Vapour density of (B) is twice the vapour density of (A). (A) reduces Fehling solution but does not react with NaHCO_3 . Compound (B) neither reacts with NaHCO_3 nor reduces Fehling solution. What are (A) and (B) ? Also report an isomer of (B) if it reacts with NaHCO_3 .
- Complete the following equation :



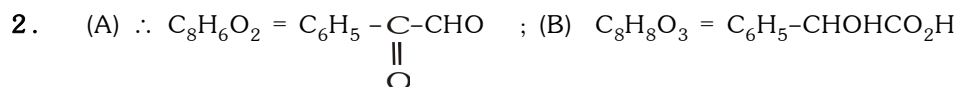
- A compound has two isomers (A) and (B) of formula $\text{C}_5\text{H}_{10}\text{O}$. Isomer (A) on treating with NaOH (aq.) give 2, 2-dimethylpropan-1-ol and 2, 2-dimethylpropanoic acid salt. The isomer (B) on treating with NaOH (aq.) gives 3-hydroxy-2-propylheptanal. What are A and B ?



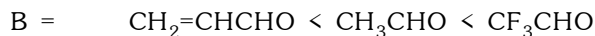
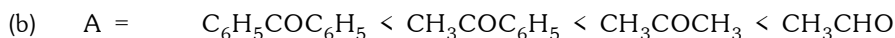
I will be more readily protonated than (II). Alternative protonated (I) is more stabilised by resonance than protonated (II).



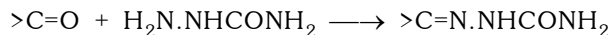
In (X) there is extended conjugation and only one charge is involved. In (Y) there is not this extended conjugation and the relative close proximity of two positive charges is a destabilising factor. Hence (X) is more stable than (Y).



3. (a) I does not undergo haloform reaction whereas II does.



4. (i) X forms semicarbazone and thus possesses carbonyl group.



(ii) It does not give Tollen's reagent test and thus it is ketone.

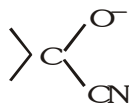
(iii) It does not give iodoform test and thus it is not methyl ketone.

(iv) Keeping in view of the above facts and molecular formula ; X is

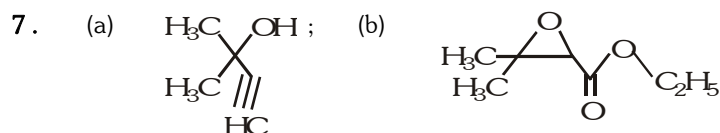
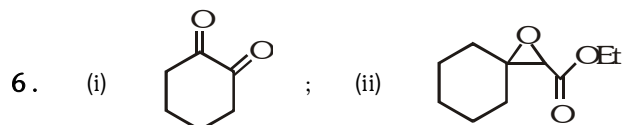


(v) This on reduction will give n-propyl benzene.

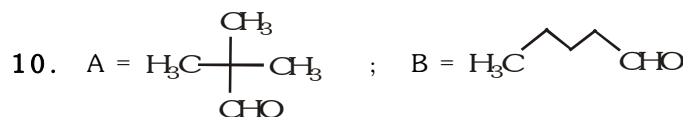
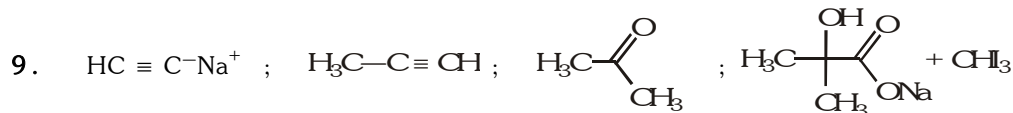
5. (a) The carbonyl group in aldehydes and ketones add on CN^- resulting in the formation of an anion where the negative charge resides on oxygen. However if a nucleophile adds on to an alkene the negative charge resides on carbon. Since carbon is much less strongly electron attracting than oxygen, this species is less stable and hence not readily formed.



- (b) In alkenes the double bond joins two carbon atoms and there is no resultant polarity. In carbonyl compounds, the carbonyl group is highly polar and the high partial positive charge on the C atom makes it susceptible to nucleophilic attack.

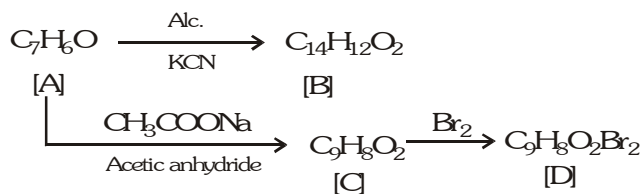


8. A = HCHO (Formaldehyde) B = HCOOCH_3 (Methyl methanoate)

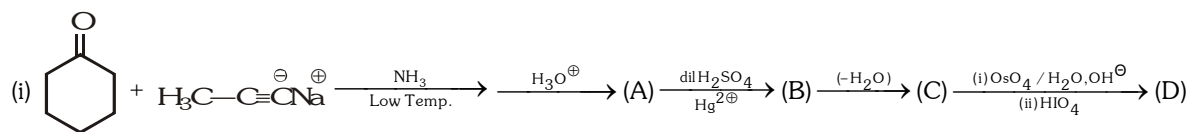


EXERCISE-04 [B]**BRAIN STORMING SUBJECTIVE EXERCISE**

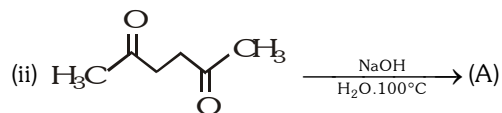
1. Identify (A) to (D) as reactant, reagent, product as name of the reaction in following :
- (i) $3(\text{CH}_3)_2\text{C}=\text{O} \xrightarrow{\text{HCl}}$ (A) Aldol condensation
- (ii) $\text{CH}_3\text{COCl} + \text{H}_2 \xrightarrow{\text{(B)}}$ (C) Rosenmund's reaction
- (iii) (D) $\xrightarrow[\text{(ii) } \text{C}_2\text{H}_5\text{ONa}]{\text{(i) } \text{NH}_2\text{NH}_2} \text{CH}_3\text{CH}_2\text{CH}_3$
- (iv) (D) $\xrightarrow{\ominus\text{OBr}} \text{CHBr}_3 + \text{CH}_3\text{COO}^\ominus$
2. Two different Grignard reagents, (X) and (Y) produce $\text{C}_6\text{H}_5\text{CH}_2\text{C}(\text{CH}_3)_2\text{OH}$ on reaction with (P) and (Q) respectively. Give structures of (X), (Y) and (Q).
3. (a) 2-methyl-1,3-cyclohexanedione is more acidic than cyclohexanone - explain with reason
(b) Explain why HCN will add to the double bond in $\text{CH}_2=\text{CHCOOH}$ but not in $\text{RCH}=\text{CHR}$
4. A compound $\text{C}_5\text{H}_{10}\text{O}$ does not reduce Fehling's solution, form a phenyl hydrazone, shows the haloform reaction, and can be converted into n-pentane by Zn-Hg and conc. HCl. What is this compound ?
5. (a) Convert PhCHO into $\text{PhCH}=\text{CHCOPh}$
(b) Identify A, B, C and D in the following reaction.



6. A ketone (A) which undergoes haloform reaction gives compound B on reduction. B on heating with sulphuric acid gives compound C, which form mono ozonide D. D on hydrolysis in presence of Zn dust gives only acetaldehyde. Identify A, B and C. Write down the reactions involved.
7. Complete the following reactions :

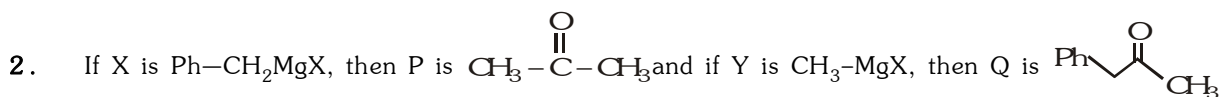
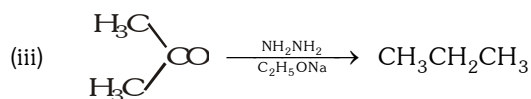
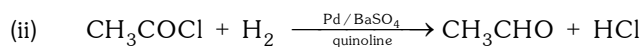
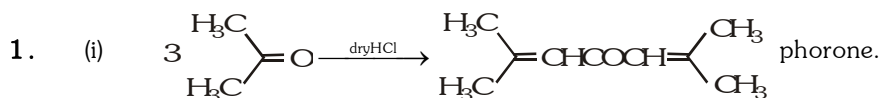
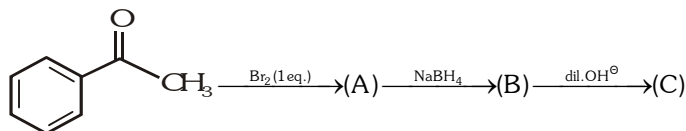


Write the structures of A to D and give the IUPAC name of (D).

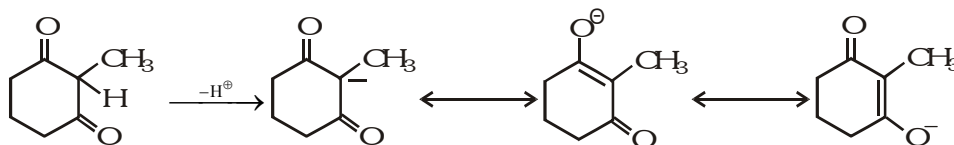


8. The sodium salt of a carboxylic acid, A, was produced by passing a gas, B, into an aqueous solution of caustic alkali at an elevated temperature and pressure. A on heating in presence of sodium hydroxide followed by treatment with sulphuric acid gave a dibasic acid C. A sample of 0.4 g of C on combustion gave 0.08 g of water and 0.39 g of carbon dioxide. The silver salt of acid weighin 1g on ignition yielded 0.71 g of silver as residue. Identify A, B and C.

9. Give reasons for the following :
- Ketones are less electrophilic than aldehydes.
 - Aldehydes are reducing agents and ketones are not.
 - HBr fails to give addition products with carbonyl compounds.
10. Identify A, B and C ?

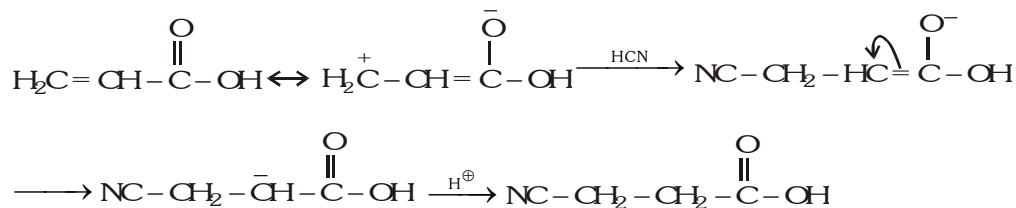


3. (a) 2-methyl-1, 3-cyclohexanedione is more acidic because its enolate ion is stabilized by an additional resonance structure.



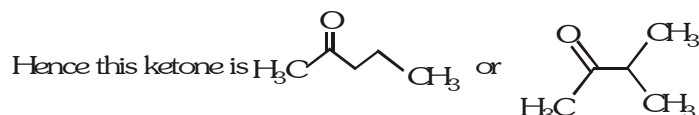
Resonance stabilized conjugate base.

- (b) In the case of the acid the double bond is activated due to the presence of a -COOH group



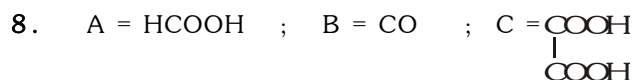
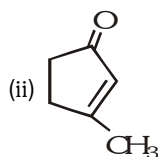
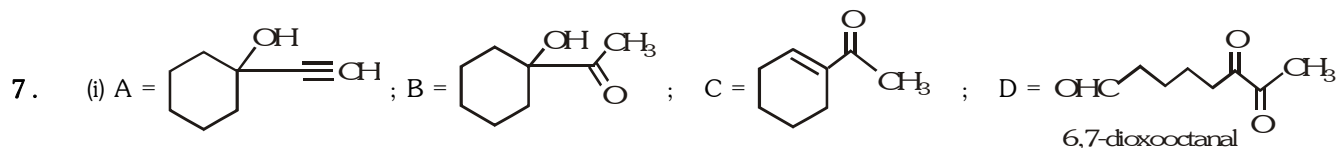
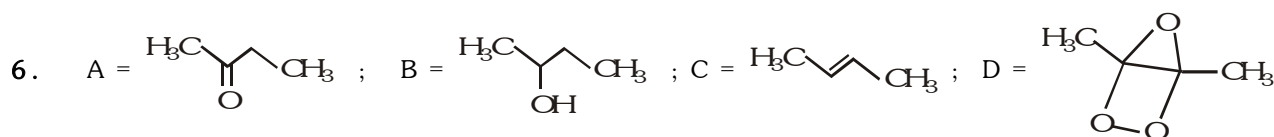
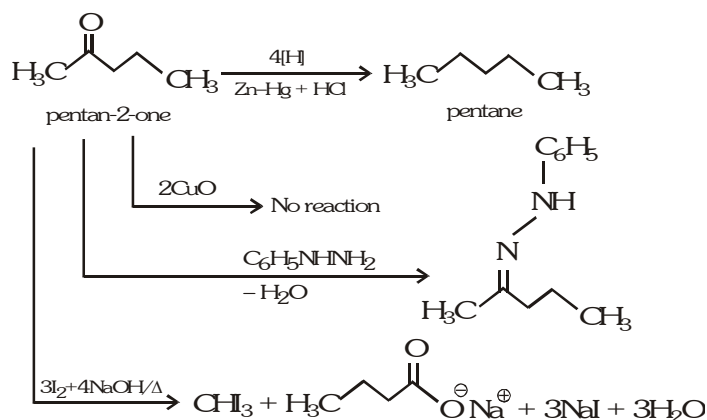
4. (i) Since $\text{C}_5\text{H}_{10}\text{O}$ forms a phenyl hydrazone, hence it is an aldehyde or ketone.
 (ii) As it does not reduce Fehling's solution, hence it is not aldehyde but a ketone.

- (iii) Since it undergoes haloform reaction, therefore it is $\text{CH}_3\text{-C(=O)-R}$ type of ketone. The value of R - can be derived as follows : $\text{R} = \text{-C}_3\text{H}_7$

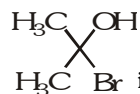


(iv) Ketone on Clemmensen reduction yields n-pentane; hence R is n-propyl and not isopropyl.

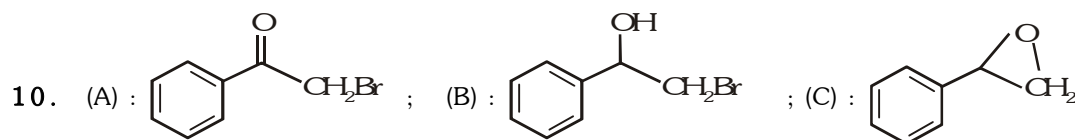
5. (a) CH_3MgI , H_3O^+ , $\text{H}^+/\text{K}_2\text{Cr}_2\text{O}_7$, $\text{PhCHO}/\text{OH}^-/\Delta$
 (b) $\text{A} = \text{C}_6\text{H}_5\text{CHO}$; $\text{B} = \text{PhCH}(\text{OH})\text{COPh}$;
 $\text{C} = \text{C}_6\text{H}_5\text{CH}=\text{CH}-\text{COOH}$; $\text{D} = \text{C}_6\text{H}_5\text{CHBr}-\text{CHBr}-\text{COOH}$



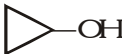
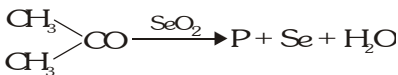
9. (a) The positive inductive effect of the second alkyl radical reinforces that of the first one decreasing still further the partial positive charge on the carbonyl carbon atom. This reduces the attraction of the atom for nucleophilic reagents. Hence ketones are less electrophilic.
 (b) The $>\text{C}=\text{O}$ group in aldehydes activates the H atom attached to the carbonyl group. This is due to the relaying of the $-I$ effect of the oxygen atom to the $\text{C}-\text{H}$ bond so that partial positive charge is created on the H atom. The result of this activation is that the H atom of the $-\text{CHO}$ group can be oxidised readily to a (OH) group. Thus aldehydes are reducers.
 (c) HBr is strongly polar and is hence readily added to the polarized $>\text{C}=\text{O}$ group. The addition product



is however unstable and decomposes to give the original carbonyl compound and HBr .



EXERCISE-05**PREVIOUS YEARS QUESTIONS**

1. The formation of cyanohydrin from a ketone is an example of [IIT-90]
(A) Electrophilic addition (B) Nucleophilic addition
(C) Nucleophilic substitution (D) Electrophilic substitution
2. The enolic form of acetone contains - [IIT-90]
(A) 9 sigma bonds, 1 pi bond and 2 lone pairs (B) 8 sigma bonds, 2 pi bond and 2 lone pairs
(C) 10 sigma bonds, 1 pi bond and 1 lone pairs (D) 9 sigma bonds, 2 pi bond and 1 lone pairs
3. m-chlorobenzaldehyde on reaction with conc. KOH at room temperature gives [IIT-91]
(A) Potassium m-chlorobenzoate and m-hydroxybenzaldehyde
(B) m-hydroxy benzaldehyde and m-chlorobenzyl alcohol
(C) m-chlorobenzyl alcohol and m-hydroxybenzyl alcohol
(D) Potassium m-chlorobenzoate and m-chlorobenzyl alcohol
4. Hydrogenation of benzoyl chloride in the presence of Pd on BaSO₄ gives [IIT-92]
(A) Benzyl alcohol (B) Benzaldehyde
(C) Benzoic acid (D) Phenol
5. An organic compound C₃H₆O does not give a precipitate with 2,4-Dinitrophenyl hydrazine reagent- [IIT-93]
(A) CH₃CH₂CHO (B) CH₃COCH₃
(C) CH₂=CH-CH₂OH (D) 
6. Under Wolff Kishner reduction conditions, the conversion which may be through about is - [IIT-95]
(A) Benzaldehyde into Benzyl alcohol (B) Cyclohexanol into Cyclohexane
(C) Cyclohexanol into Cyclohexanol (D) Benzophenone into Diphenylmethane
7. In the reaction, P is, [IIT-95]

(A) CH₃COCHO (B) CH₃COOCH₃ (C) CH₃COCH₂OH (D) None
8. In the Cannizzaro reaction given below $2 \text{ Ph-CHO} \xrightarrow{\text{OH}^-} \text{Ph-CH}_2\text{OH} + \text{PhCO}_2^-$ the slowest step is- [IIT-96]
(A) the attack of OH⁻ at the carbonyl group
(B) the transfer of hydride to the carbonyl group
(C) the abstraction of proton from the carboxylic acid
(D) the deprotonation of Ph-CH₂OH
9. Which of the following will give yellow precipitate with I₂/NaOH- [IIT-97]
(A) ICH₂COCH₂CH₂ (B) CH₃COOCOCH₃
(C) CH₃CONH₂ (D) CH₃CH(OH)CH₂CH₃
10. Among the given compounds, the most susceptible to nucleophilic attack at the carbonyl group is - [IIT-97]
(A) MeCOCl (B) MeCHO
(C) MeCOOMe (D) MeCOOCOMe

11. In a Cannizzaro reaction the intermediate which is the best hydride donor is - [IIT-97]



12. $\text{CH}_3\text{CHO} + \text{H}_2\text{NOH} \rightarrow \text{CH}_3-\text{CH}=\text{N}-\text{OH}$. The above reaction occurs at - [IIT-97]
 (A) pH = 1 (B) pH $\approx$ 4.5 (C) Any value of pH (D) pH = 12

13. Among the following compounds, which will react with acetone to give a product containing $>\text{C}=\text{N}-$ [IIT-98]



14. The product obtained via oxymercuration ($\text{HgSO}_4 + \text{H}_2\text{SO}_4$) of 1-butyne would be - [IIT-98]



15. Which of the following will not undergo aldol condensation - [IIT-98]



16. Which of the following will react with water to form a stable hydrate - [IIT-98]



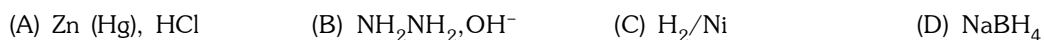
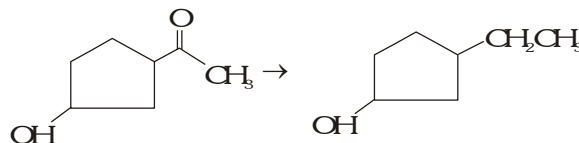
17. The enol form of acetone, after treatment with D_2O gives - [IIT-99]



18. Which of the following has the most acidic hydrogen- [IIT-2000]



19. The appropriate reagent for the following transformation - [IIT-2000]



20. Read the following statement and explanation and answer as per the option given below : [IIT-01]

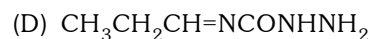
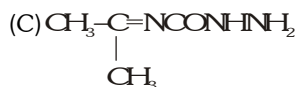
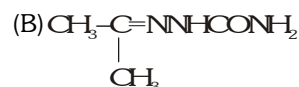
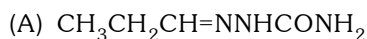
Statment-I : Dimethylsulphide is commonly used for the reduction of an ozonide of an alkene to get the carbonyl compounds.

Statment-II : It reduces the ozonide giving water soluble dimethyl sulphoxide and excess of it evaporates.

- (A) If both assertion and reason are correct, and reason is the correct explanation of the assertion
 (B) If both assertion and reason are correct, but reason is not correct explanation of the assertion
 (C) If assertion is correct but reason is incorrect
 (D) If assertion is incorrect but reason is correct

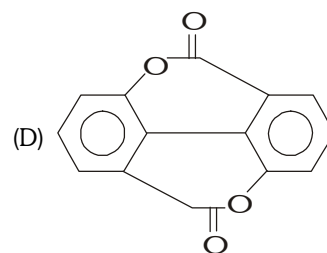
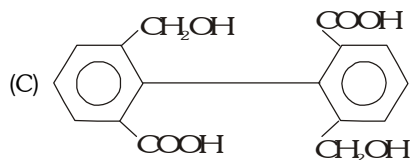
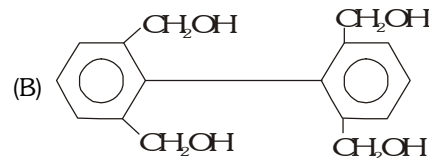
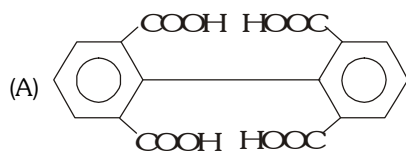
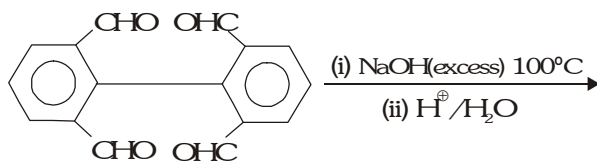
21. A mixture of benzaldehyde and formaldehyde on heating with aqueous NaOH solution gives - [IIT-01]
 (A) benzyl alcohol and sodium formate (B) sodium benzoate and methyl alcohol
 (C) sodium benzoate and sodium formate (D) benzyl alcohol and methyl alcohol

22. Compound A (molecular formula C_3H_8O) is treated with acidified potassium dichromate to form a product B (molecular formula C_3H_6O). B forms a shining silver mirror on warming with ammoniacal silver nitrate. B when treated with an aqueous solution of $H_2NCONHNH_2$, HCl and sodium acetate gives a product C. Identify the structure of C- [IIT-02]

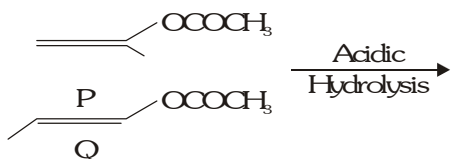


23. Any one of the product formed is :

[IIT-02]



- 24.



[IIT-03]

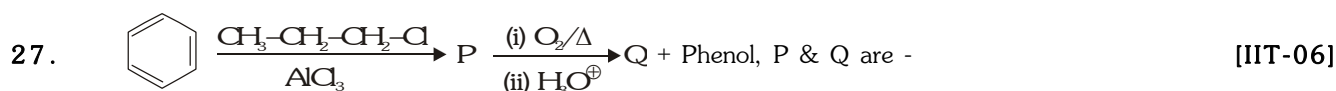
Product formed by P & Q can be differentiated by :

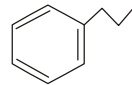
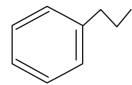
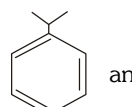
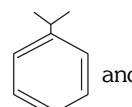
- (A) 2, 4-DNP (B) Lucas reagent ($ZnCl_2 + \text{conc. HCl}$)
 (C) $NaHSO_3$ (D) Fehling's solution
25. Which of the reagent is used to convert 2-Butanone into propanoic acid - [IIT-05]
 (A) $NaOH, I_2/H^+$ (B) Tollen's reagent (C) Fehling solution (D) $NaOH, NaI/H^+$

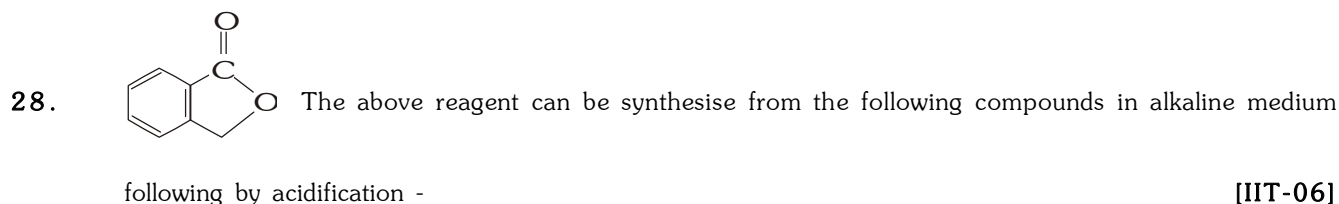


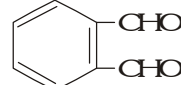
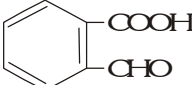
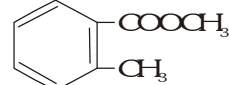
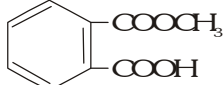
Following statements is/are correct about mixture :

- (A) mixture is 3-types of oximes (B) mixture is 2-types of oximes
 (C) all are optically active (D) one is optically active



- (A)  and $\text{CH}_3-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$ (B)  and $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$
 (C)  and $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$ (D)  and $\text{CH}_3-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$



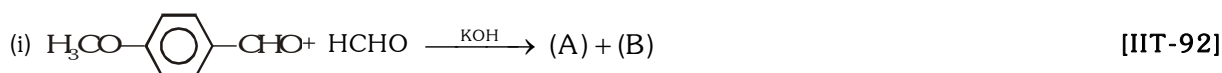
- (A)  (B)  (C)  (D) 

29. An unknown compound of carbon, hydrogen and oxygen contains 69.77% C and 11.63% H, and has a molecular weight of 86. It does not reduce Fehling solution but forms a bisulphate addition compound and gives a positive iodoform test. What are the possible structures ? [IIT-87]

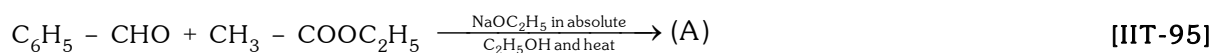
30. Iodoform is obtained by the reaction of acetone with hypoiodite but not with iodide. Why ? [IIT-91]

31. An organic compound 'A' on treatment with ethyl alcohol gives a carboxylic acid 'B' and compound 'C'. Hydrolysis of 'C' under acidic conditions gives 'B' and 'D'. Oxidation of 'D' with KMnO_4 also gives 'B'. 'B' on heating with Ca(OH)_2 gives 'E' ($\text{C}_3\text{H}_6\text{O}$). E does not give Tollen's test and does not reduce Fehling's solution but form a 2,4-dinitrophenyl hydrazone. Identify (A), (B), (C), (D) and (E). [IIT-92]

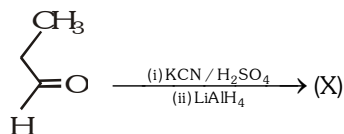
32. Complete the following :



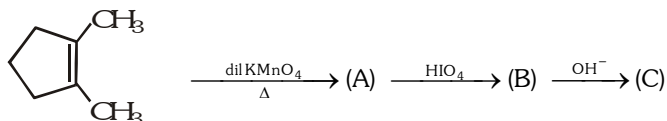
33. Complete the following :



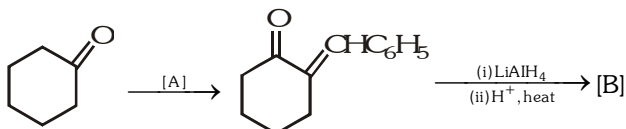
34. Complete the following reaction with appropriate structure : [IIT-96]



35. Suggest appropriate structures for the missing compounds. (The number of carbon atoms remains the same throughout the reactions). [IIT-96]

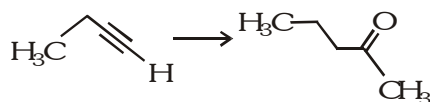


36. Complete the following : [IIT-98]



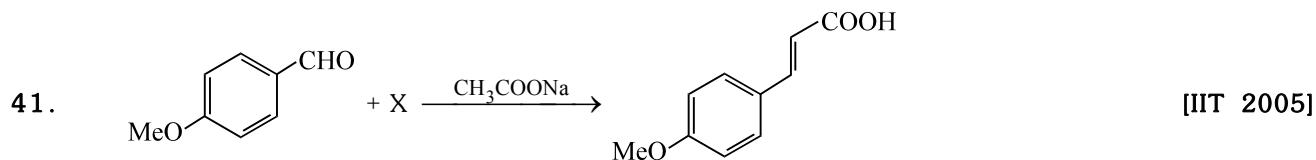
37. Compound 'A' ($\text{C}_8\text{H}_8\text{O}$) on treatment with $\text{NH}_2\text{OH} \cdot \text{HCl}$ gives 'B' and 'C'. 'B' and 'C' rearrange to give 'D' and 'E', respectively, on treatment with acid. B, C, D and E are all isomers of molecular formula ($\text{C}_8\text{H}_9\text{NO}$). When 'D' is boiled with alcoholic KOH an oil 'F' ($\text{C}_6\text{H}_7\text{N}$) separates out. 'F' reacts rapidly with CH_3COCl to give back 'D'. On the other hand, 'E' on boiling with alkali followed by acidification gives a white solid 'G' ($\text{C}_7\text{H}_6\text{O}_2$). Identify 'A' to 'G'. [IIT-99]

38. Carry out the following transformation in not more than three steps. [IIT-99]



39. An organic compound (A), $\text{C}_6\text{H}_{10}\text{O}$, on reaction with CH_3MgBr followed by acid treatment gives compound (B). The compound (B) on ozonolysis gives compound (C), which in presence of a base give 1-acetyl cyclopentene (D). The compound (B) on reaction with HBr gives compound (E). Write the structures of (A), (B), (C) and (E). Show how (D) is formed from (C). [IIT-2000]

40. An organic compound (A) reacts with H_2 to give (B) and (C) successively. On ozonolysis of (A), two aldehydes (D) $\text{C}_2\text{H}_4\text{O}$ and (E) $\text{C}_2\text{H}_2\text{O}_2$ are formed. On ozonolysis of (B) only propanal is formed. What are (A) to (E)? [IIT-01]

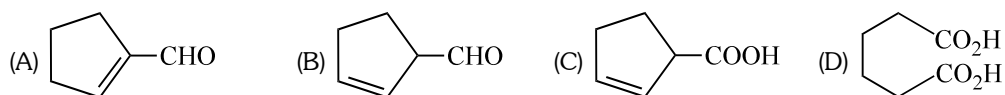


What is X?

- (A) CH_3COOH (B) BrCH_2COOH (C) $(\text{CH}_3\text{CO})_2\text{O}$ (D) $\text{CHO}-\text{COOH}$

42. The smallest ketone and its next homologue are reacted with NH_2OH to form oxime. (A) Two different oximes are formed (B) Three different oximes are formed (C) Two oximes are optically active (D) All oximes are optically active [JEE 2006]

43. Cyclohexene on ozonolysis followed by reaction with zinc dust and water gives compound E. Compound E on further treatment with aqueous KOH yields compound F. Compound F is [JEE 2007]



44. **Statement-1** : Glucose gives a reddish-brown precipitate with Fehling's solution.

because

Statement-2 : Reaction of glucose with Fehling's solution gives CuO and gluconic acid.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True.

[JEE 2007]

45. Match the compounds/ion in column I with their properties/ reaction in Column II. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS.

[JEE 2007]

Column I

(A) C_6H_5CHO

(B) $CH_3C\equiv CH$

(C) CN^-

(D) I^-

Column II

(P) gives precipitate with 2,4-dinitrophenylhydrazine

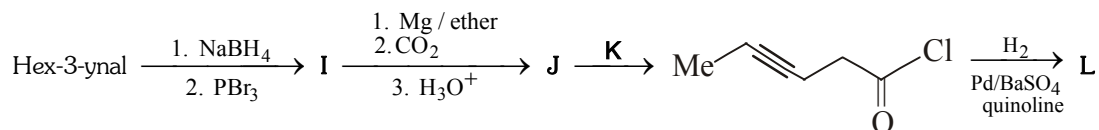
(Q) gives precipitate with $AgNO_3$

(R) is a nucleophile

(S) is involved in cyanohydrin formation

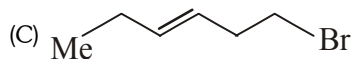
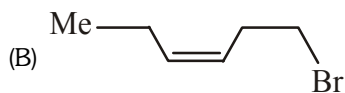
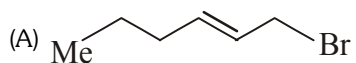
Paragraph for Question No. 46 to 48

In the following reaction sequence, products **I**, **J** and **L** are formed. **K** represents a reagent.



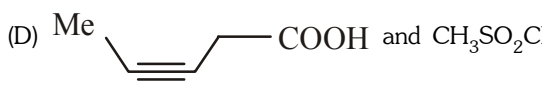
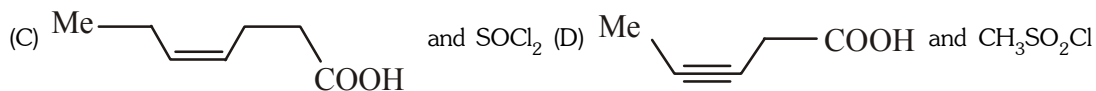
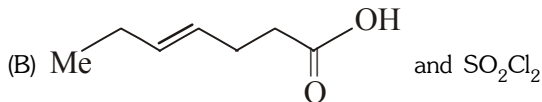
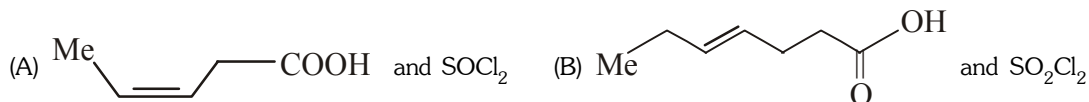
46. The structure of the product **I** is

[JEE 2008]



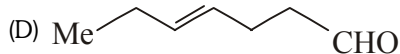
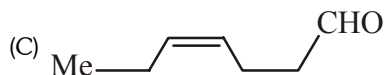
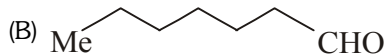
47. The structures of compounds **J** and **K**, respectively, are

[JEE 2008]



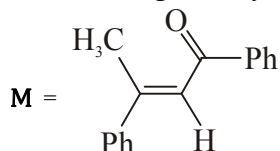
48. The structure of product **L** is

[JEE 2008]



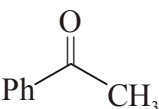
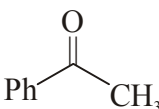
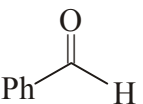
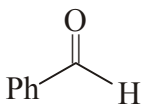
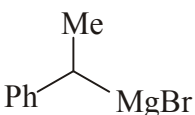
Paragraph for Question No. 49 to 51

A tertiary alcohol **H** upon acid catalysed dehydration gives a product **I**. Ozonolysis of **I** leads to compounds **J** and **K**. Compound **J** upon reaction with KOH gives benzyl alcohol and a compound **L**, whereas **K** on reaction with KOH gives only **M**.



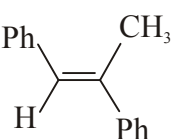
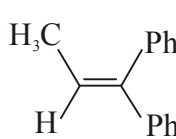
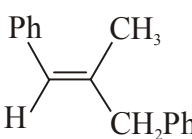
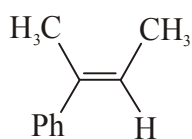
49. Compound **H** is formed by the reaction of

[JEE 2008]

- (A)  + PhMgBr (B)  + PhCH₂MgBr
- (C)  + PhCH₂MgBr (D)  + 

50. The structure of compound **I** is

[JEE 2008]

- (A)  (B)  (C)  (D) 

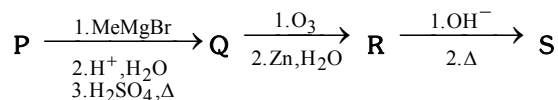
51. The structures of compounds **J**, **K** and **L**, respectively, are

[JEE 2008]

- (A) PhCOCH₃, PhCH₂COCH₃ and PhCH₂COO K⁺
 (B) PhCHO, PhCH₂CHO and PhCOO K⁺
 (C) PhCOCH₃, PhCH₂CHO and CH₃COO K⁺
 (D) PhCHO, PhCOCH₃ and PhCOO K⁺

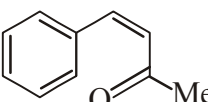
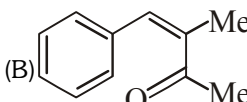
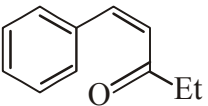
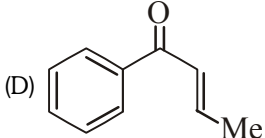
Paragraph for Question Nos. 52 to 54

A carbonyl compound **P**, which gives positive iodoform test, undergoes reaction with MeMgBr followed by dehydration to give an olefin **Q**. Ozonolysis of **Q** leads to a dicarbonyl compound **R**, which undergoes intramolecular aldol reaction to give predominantly **S**.



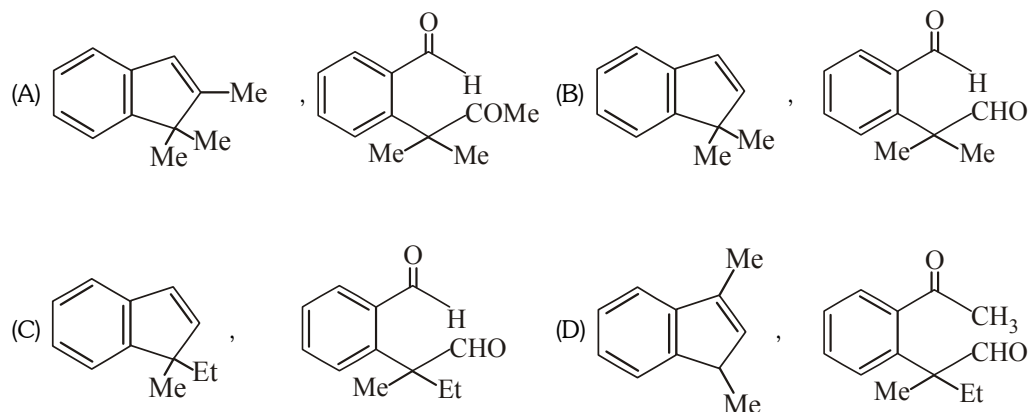
52. The structure of the carbonyl compound **P** is

[JEE 2009]

- (A)  (B) 
- (C)  (D) 

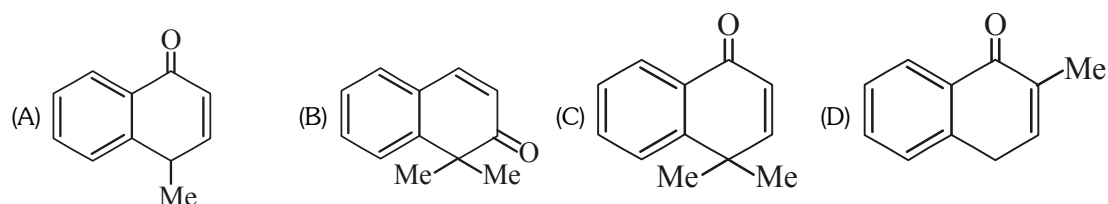
53. The structure of the products **Q** and **R**, respectively, are

[JEE 2009]



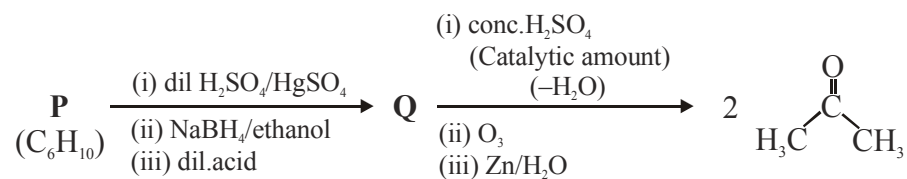
54. The structure of the product **S** is

[JEE 2009]



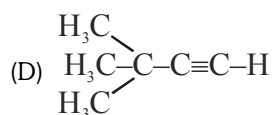
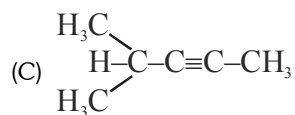
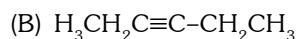
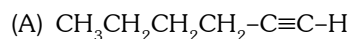
Paragraph for Questions Nos. 55 to 56

An acyclic hydrocarbon **P**, having molecular formula C_6H_{10} , gave acetone as the only organic product through the following sequence of reactions, in the which **Q** is an intermediate organic compound.

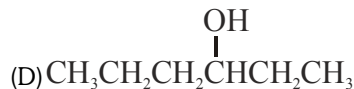
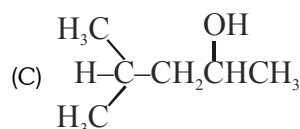
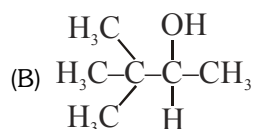
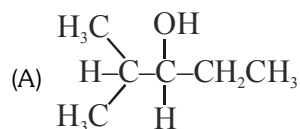


[JEE 2011]

55. The structure of compound **P** is -

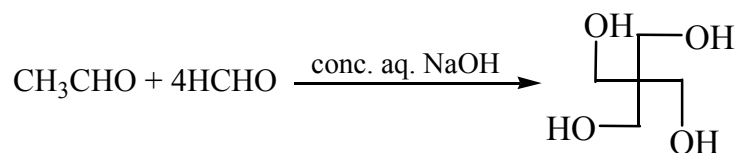


56. The structure of the compound **Q** is -



57. The number of aldol reaction(s) that occurs in the given transformation is

[JEE 2012]



(A) 1 (B) 2 (C) 3 (D) 4

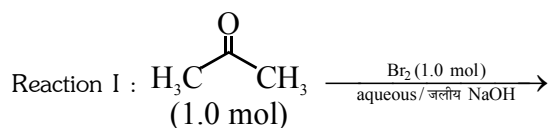
58. Among P, Q, R and S, the aromatic compound(s) is / are :

[JEE 2013]

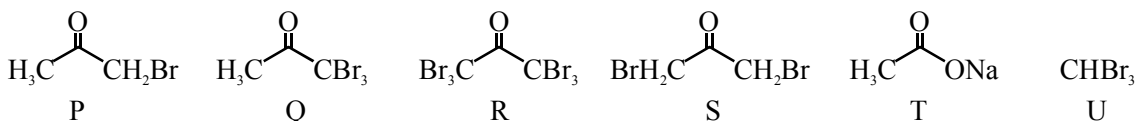
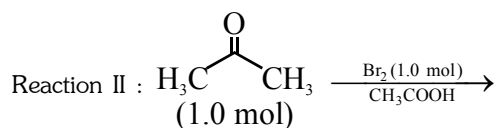


(A) P (B) Q (C) R (D) S

59. After completion of the reactions (I and II), the organic compound(s) in the reaction mixtures is(are)



[JEE 2013]



- (A) Reaction I : P and Reaction II : P
 (B) Reaction I : U, acetone and Reaction II : Q acetone
 (C) Reaction I : T, U, acetone and Reaction II : P
 (D) Reaction I : R, acetone and Reaction II : S acetone

PREVIOUS YEARS QUESTIONS				ANSWER KEY			EXERCISE -5	
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1. (B)	2. (A)	3. (D)	4.(B)	5. (C, D)	6. (D)	7.(A)	8. (B)	9. (A)
10. (A)	11. (D)	12. (B)	13. (A)	14. (A)	15. (C)	16. (B)	17. (C)	18. (B)
19. (B)	20. (A)	21. (A)	22. (A)	23. (C)	24. (D)	25. (A)	26. (A)	27. (C)

28. (A)

29. (i) For empirical formula

Element	%	Relative no. of atoms	Simplest ratio
C	69.77	5.76	5
H	11.63	11.63	10
O	19.20	1.2	1

∴ Empirical formula of compound is $C_5H_{10}O$.

and Empirical formula wt. = 86

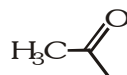
Also Molecular wt. = 86

∴ Molecular formula of compound is $C_5H_{10}O$.

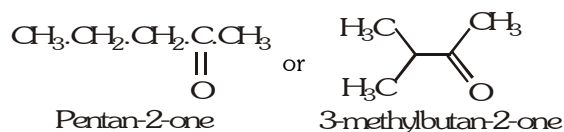
(ii) Compound form bisulphate addition compound and thus, has carbonyl group. i.e., aldehyde or ketone.

(iii) It does not reduce Fehling solution and thus, it is not aldehyde but it is ketone.

(iv) It gives positive iodoform test and thus, it has



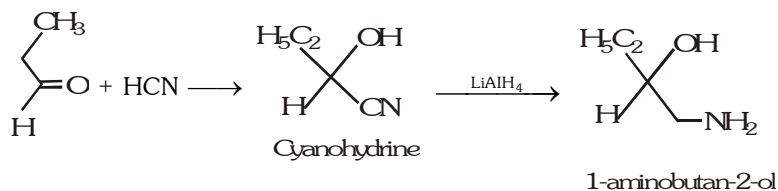
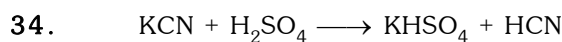
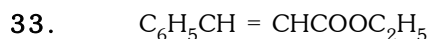
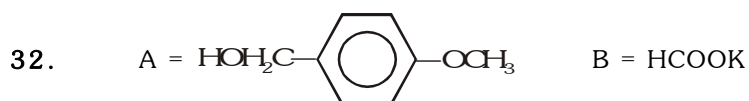
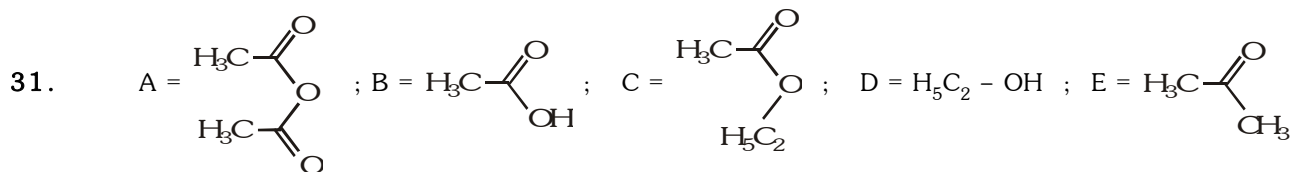
(v) Above facts reveals that compound is

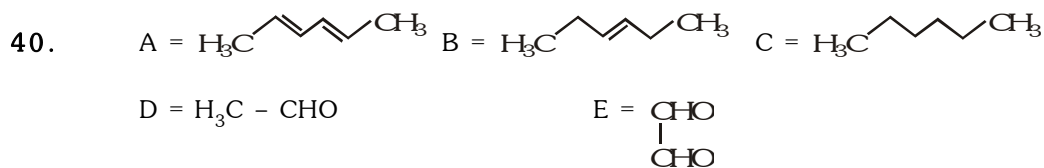
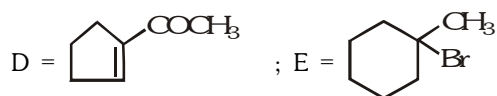
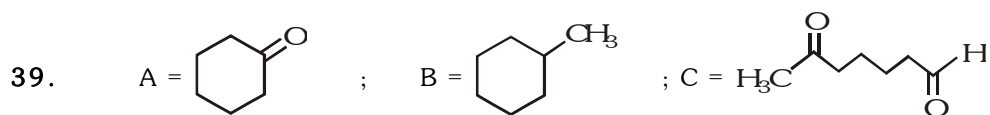
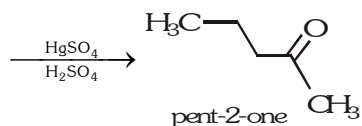
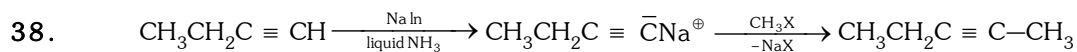
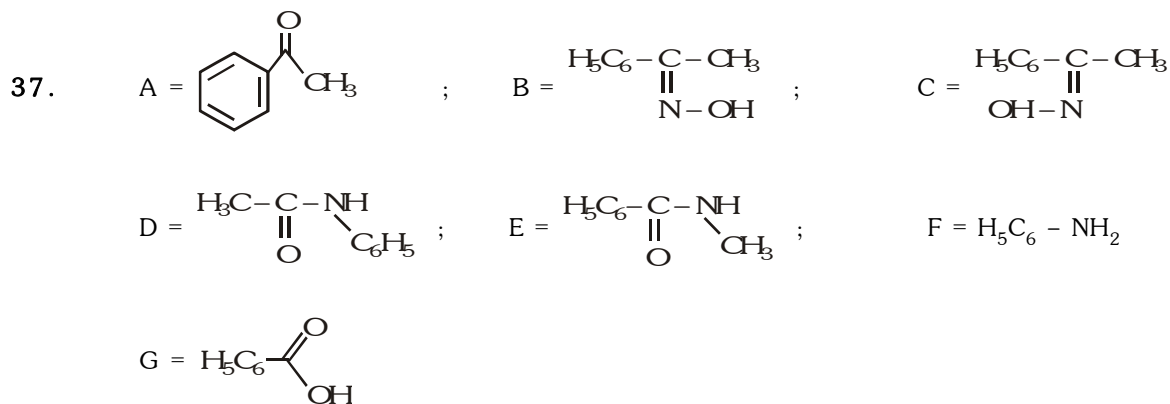
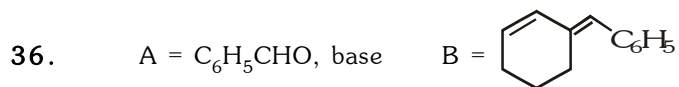
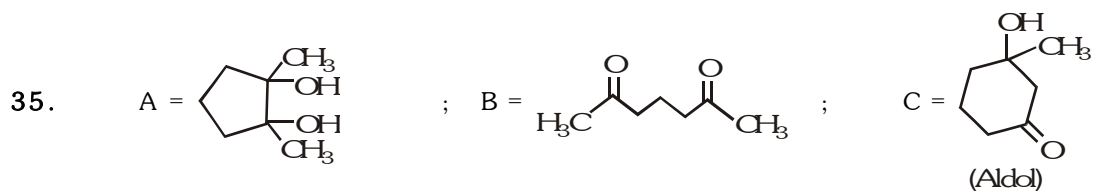


30. The formation of CHI_3 takes place as follows :



I^- cannot bring about this reaction. The active species is OI^- , OI^- is an oxidant and an iodinating agent.

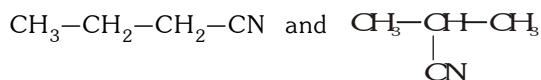




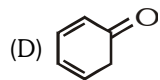
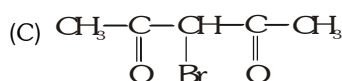
41. C 42. B 43. A 44. C 45. (A) P, S; (B) Q; (C) Q, R, S; (D) Q, R
46. D 47. A 48. C 49. B 50. A 51. D 52. B
53. A 54. B 55. D 56. B 57. C 58. A, B, C, D 59. C

EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. Type of isomerism exists between :-

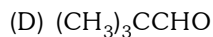
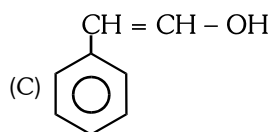
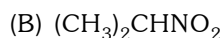
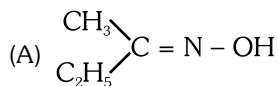


- (A) Position (B) Chain (C) Both the above (D) None of these
2. The number of primary alcohols possible with the formula $\text{C}_4\text{H}_{10}\text{O}$ is -
(A) 2 (B) 3 (C) 4 (D) 5
3. The total number of benzene derivatives having the molecular formula $\text{C}_7\text{H}_7\text{Cl}$ is -
(A) 2 (B) 3 (C) 4 (D) 5
4. Which of the following will lead to maximum enolisation :-
(A) CH_3COCH_3 (B) $\text{CH}_3\text{COCH}_2\text{CHO}$

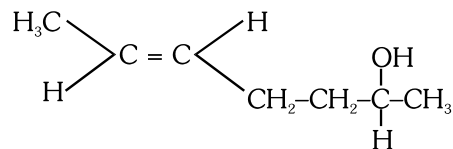


5. $\begin{array}{c} \text{H}_3\text{C} \quad \text{H} \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{H}_3\text{C} \quad \text{H} \end{array} \begin{array}{c} \text{H} \\ | \\ \text{C} \\ | \\ \text{COOH} \end{array}$ Exhibits :-

- (A) Tautomerism (B) Optical isomerism
(C) Geometrical isomerism (D) Geometrical and optical isomerism
6. Tautomerism is not exhibited by :-



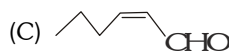
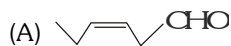
7. Given compound exhibits x geometrical isomers and y optical isomers



The value of x and y respectively are :-

- (A) 4 and 4 (B) 2 and 2 (C) 2 and 4 (D) 4 and 2
8. The total number of cyclic compounds (neglecting stereoisomers) with the molecular formula C_5H_{10} is -
(A) 4 (B) 5 (C) 6 (D) 7
9. Which of the following exhibits tautomerism ?
(A) $(\text{CH}_3)_2\text{NH}$ (B) $(\text{CH}_3)_3\text{CNO}_2$ (C) R_3CNO_2 (D) RCH_2NO_2

10. The correct structure of trans-2-hexenal is -



11. The total number of benzene derivatives having the molecular formula C_7H_8O is -

- (A) 3 (B) 4 (C) 5 (D) 6

12. Which of the following compounds does not exhibit tautomerism

- (A) CH_3NO_2 (B) $CH_3CH_2NO_2$
(C) $C_6H_5CH=CH-OH$ (D) CH_3CH_2OH

13. The total number of benzene derivatives with the molecular formula $C_6H_3Cl_3$ is -

- (A) 2 (B) 3 (C) 4 (D) 5

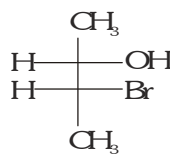
14. Which of the following compounds is not chiral ?

- (A) $DCH_2CH_2CH_2Cl$ (B) CH_3CHDCH_2Cl (C) $CH_3CHClCH_2D$ (D) CH_3CH_2CHDCI

15. The total number of stereoisomers of 2,3-dibromobutane is -

- (A) 2 (B) 3 (C) 4 (D) 5

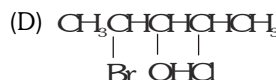
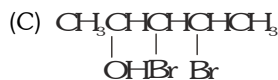
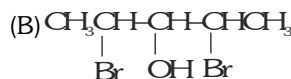
16. In the structure :



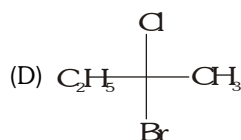
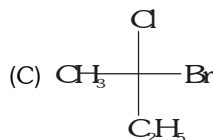
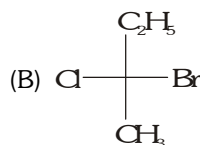
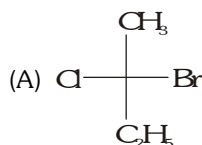
the configurations at the chiral centres are :

- (A) 2R, 3R (B) 2S, 3R (C) 2R, 3S (D) 2S, 3S

17. Which of the following compound contains a pseudo-asymmetric carbon atom



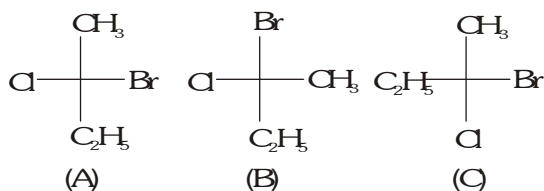
18. Consider the following structures (A), (B), (C) and (D) -



Which of the following statements is not correct

- (A) B and C are identical (B) A and B are enantiomers
(C) A and C are enantiomers (D) B and D are enantiomers

19. The interchange of two groups (Br and CH₃) at the chiral centre of the projection formula (A) yields the formula (B), while the interchange of another set of two groups (C₂H₅ and Cl) of (A) yields the projection formula (C) -



Which of the following statements is not correct about the structures (A), (B) and (C) -

- (A) B and C are identical (B) A and C are enantiomers
(C) B and C are enantiomers (D) A and B are enantiomers
20. How many meso stereoisomers are possible for 2, 3, 4-pentanetriol -
(A) 1 (B) 2 (C) 3 (D) none of these
21. The total number of stereoisomers of the compound

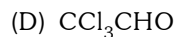
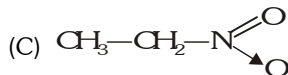
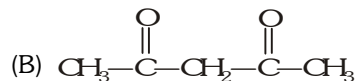


- (A) 2 (B) 3 (C) 4 (D) 8
22. The total number of aldehydes and ketones with the molecular formula C_4H_8O is -
(A) 2 (B) 3 (C) 4 (D) 5
23. In which of the following properties do enantiomers differ from each other
(A) Solubility in an achiral solvent (B) Reactivity with an achiral reagent
(C) Melting point (D) Optical rotation
24. (+) - Mandelic acid has a specific rotation of $+158^\circ$. What would be the observed specific rotation of a mixture of 25% (-) -mandelic acid and 75% (+) -mandelic acid :
(A) $+118.5$ (B) -118.5 (C) -79 (D) $+79$
25. When $C_6H_4Cl_2$ is converted into $C_6H_3Cl_3$, o-isomer will give m types of $C_6H_3Cl_3$, and p-isomer will give q types of $C_6H_3Cl_3$. m, n, q are respectively :
(A) 1, 2, 3 (B) 2, 1, 3 (C) 1, 3, 2 (D) 2, 3, 1

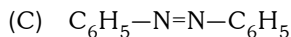
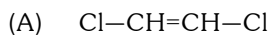
CHECK YOUR GRASP								ANSWER KEY					EXERCISE -1							
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	B	A	C	D	B	D	B	A	D	B	C	D	B	A	B	B	B	D	C	D
Que.	21	22	23	24	25															
Ans.	C	B	B	D	B															

SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)

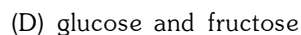
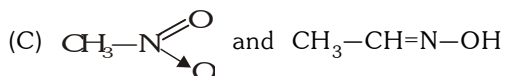
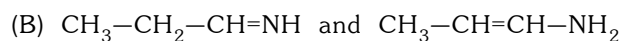
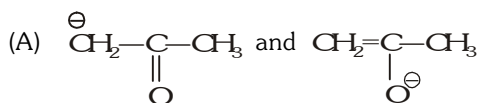
1. Tautomerism is shown by :-



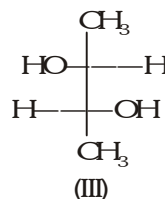
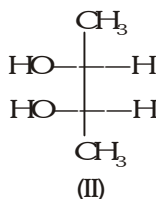
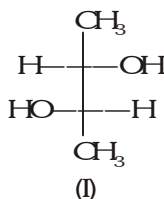
2. In which of the following cases, cis-trans nomenclature can not be used :-



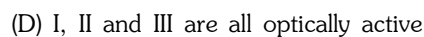
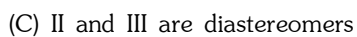
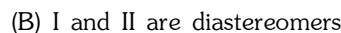
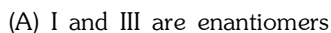
3. Among the following, which are tautomers :-



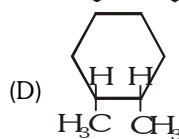
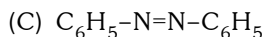
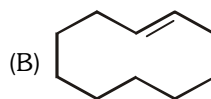
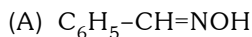
4. Consider the following compounds :



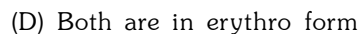
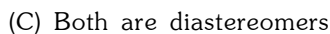
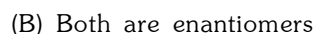
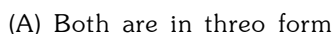
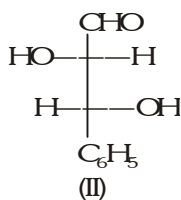
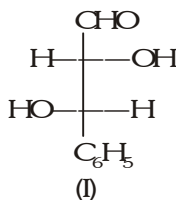
Choose the correct statements :



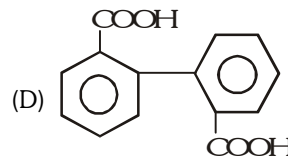
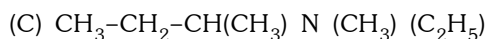
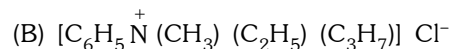
5. Which will show geometrical isomerism :-



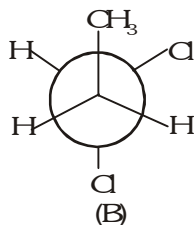
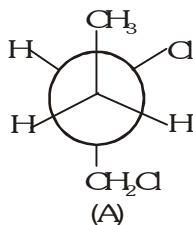
6. Which statement (s) is/are correct for :-



7. Which of the following compounds are chiral and resolvable :-

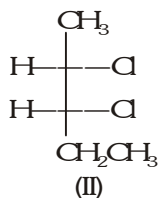
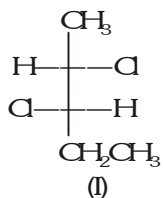


8. Observe the following structures and pick up the correct option (s) mentioned below :-



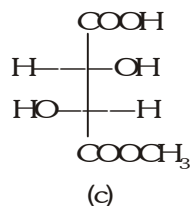
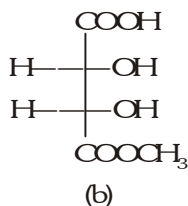
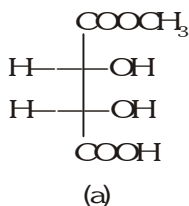
- (A) The two are position isomers
 (B) None of the two shows optical isomerism
 (C) Only A shows optical isomerism
 (D) The two are not related to each other regarding isomerism

9.



- (A) I and II are enantiomers
 (B) I is 2S, 3S; while II is 2S, 3R
 (C) I is 2R, 3R; While II is 2R, 3S
 (D) I and II are diastereomers

10. Which of the following statements are true regarding following structures :-

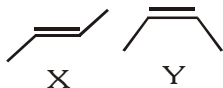


- (A) A and B are diastereomers
 (B) A and C are diastereomers
 (C) B and C are diastereomers
 (D) A and B are enantiomers

11. The R and S enantiomers of an optically active compound differ in :-

- (A) their reactivity with chiral reagents
 (B) their melting points
 (C) their optical rotation of plane polarized light
 (D) their solubility in achiral reagents

12. Which of the following statements (s) is (are) incorrect :-

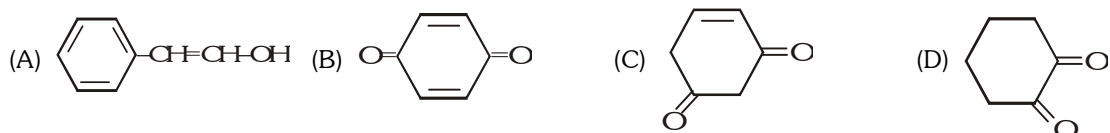


- (A) X is cis- and Y is trans (B) X is Z and Y is E
 (C) X is trans and Y is cis (D) X and Y are diastereomers

13. Which of the following compounds will show geometrical isomerism :-

- (A) 2-butene (B) propene (C) 1-phenylpropene (D) 2-methyl-2-butene

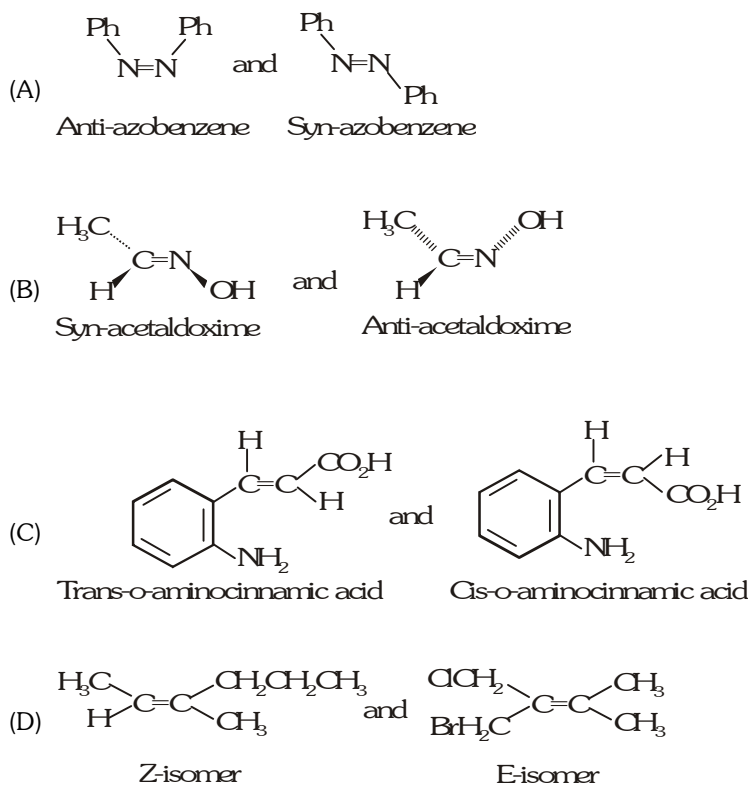
14. Tautomerism is exhibited by :-



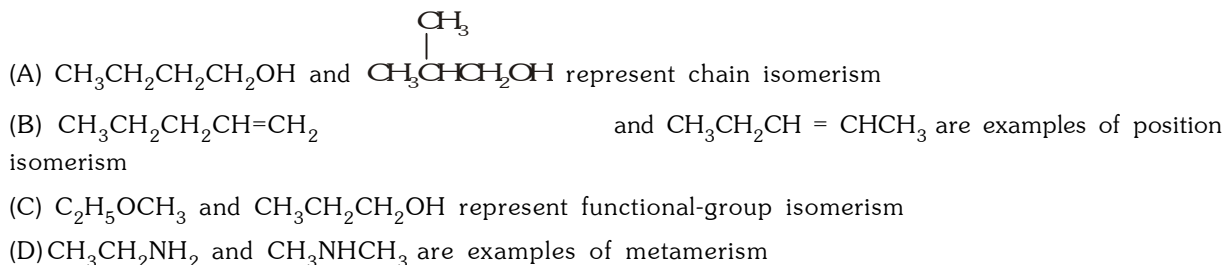
15. Which of the following notations are correct :



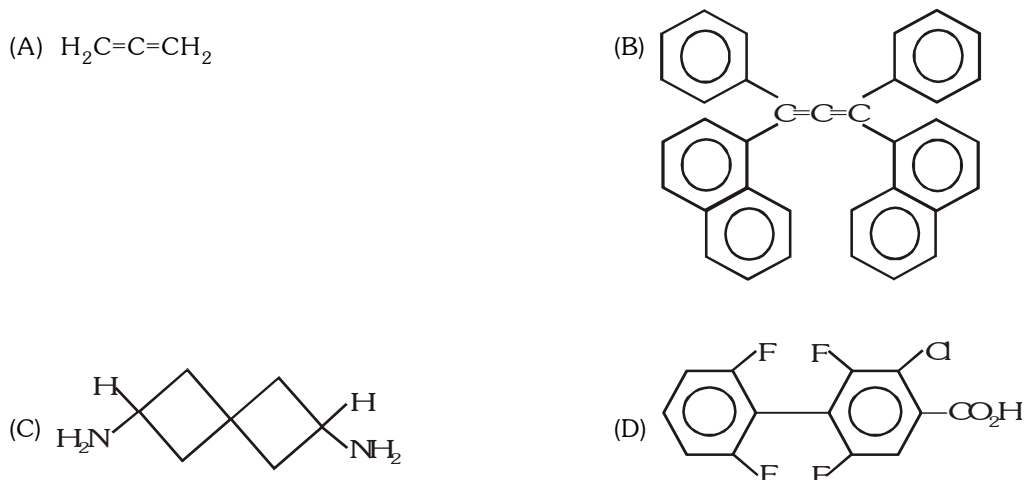
16. For which of the following pairs of compounds are the correct notation given :-



17. Which of the following statements are correct :-



18. Which of the following are optically active :-



19. An enantiomerically pure acid is treated with racemic mixture of an alcohol having one chiral carbon. The ester formed will be :

(A) optically active mixture
(B) pure enantiomer
(C) meso compound
(D) racemic mixture

20. Tartaric acid molecule contains two asymmetric carbon atoms. The number of optical isomers of tartaric acid is :-

(A) 2 (B) 3 (C) 4 (D) 5

BRAIN TEASERS						ANSWER KEY				EXERCISE -2					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	A,B,C	B,C,D	B,C	A,B,C	A,B,C,D	A,B	B,C	A,C	B,D	B,C,D	A,C	A,B	A,C	A,C,D	A,C
Que.	16	17	18	19	20										
Ans.	B,C,D	A,B,C,D	B,C	A	B										

EXERCISE-03

MISCELLANEOUS TYPE QUESTIONS

TRUE OR FALSE :

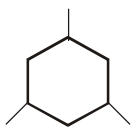
1. Stereo-isomers which are not mirror image of each other are known as diastereoisomers.
2. In every case, a pair of enantiomers have a mirror-image relationship.
3. If a compound has an enantiomer it must be chiral.
4. If a compound has a diastereomer it must be chiral
5. Any molecule containing a stereocentre must be chiral.
6. Any chiral compound with a single asymmetric carbon must have a positive optical rotation if the compound has the R configuration.

FILL IN THE BLANKS :

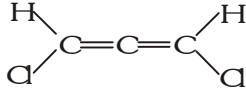
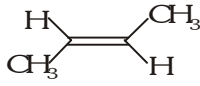
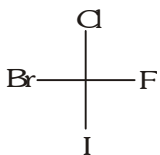
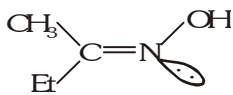
1. The possible number of dichloro derivatives of propane are
2. Ethyl benzene is isomer to xylenes.
3. The compound $\text{CHCl}=\text{CHCl}$ can show isomerism.
4. d- and l- lactic acid are known as
5. Maleic and fumaric acids are a pair of

MATCH THE COLUMN

1. Match the column I with column II.

Column-I (reaction)		Column-II (stereoisomers)	
(A)	$\text{CH}_3-\text{CH}=\text{CH}-\text{CH}=\text{N}-\text{OH}$	(p)	2
(B)		(q)	4
(C)	$\text{CH}_3-\text{CH}=\text{CH}-\text{CH}=\text{CH}-\text{CH}=\text{CH}-\text{CH}_3$	(r)	6
(D)	$\text{CH}_3-\text{CH}=\text{CH}-\text{CH}=\text{CH}-\text{CH}=\text{CH}-\text{Ph}$	(s)	8

2. Match the following compounds of column I with column II.

Column-I (Molecule)		Column-II (Property)	
(A)		(p)	Chiral compound
(B)		(q)	Presence of stereocenter
(C)		(r)	Optically active compound
(D)		(s)	Compound containing plane of symmetry

3. Match the column I with column II.

Column-I (reaction)		Column-II (stereoisomers)	
(A)		(p)	Homologs
(B)		(q)	Functional isomers
(C)		(r)	Metamer
(D)		(s)	Chain isomers

ASSERTION & REASON QUESTION :

These questions contains, Statement-I (assertion) and Statement-II (reason).

- (A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I
 (B) Statement-I is True, Statement-II is True ; Statement-II is NOT a correct explanation for Statement-I
 (C) Statement-I is True, Statement-II is False.
 (D) Statement-I is False, Statement-II is True.

1. **Statement-I** : Staggered and eclipsed ethane can not be separated.

Because

Statement-II : Energy barrier between staggered and eclipsed form of ethane is 12.6 kJ/mole.

2. **Statement-I** : All double bond containing compounds show geometrical isomerism.

Because

Statement-II : Alkenes have restricted rotation about the double bond.

3. **Statement-I** : Meso-tartaric acid is optically active.

Because

Statement-II : Optically active molecule is a molecule that cannot be superimposed on its mirror image.

4. **Statement-I** : Cyclohexanone exhibits keto-enol tautomerism.

Because

Statement-II : In cyclohexanone, one form contains the keto group (C=O) while other contains enolic group (-C=C-OH).

5. **Statement-I** : Staggered form is less stable than the eclipsed form.

Because

Statement-II : The conformation in which the bond pairs of two central atoms are very far from one another is called staggered from.

6. **Statement-I** : Trans-isomers are more stable than cis-isomer.

Because

Statement-II : The cis-isomer is the one in which two similar groups are on the same side of double bond.

7. **Statement-I** : Propadiene is optically inactive.

Because

Statement-II : Propadiene has a plane of symmetry.

COMPREHENSION BASED QUESTIONS :

Comprehension # 1

Geometrical isomerism is a kind of stereoisomerism which is present in the compounds containing a double bond ($C=C$, $C=N$, $N=N$) and arise due to the restricted or frozen rotation about the double bond. The atoms or groups attached to the doubly bonded carbons must be different. In aldoximes, the isomer is named as syn if hydrogen and hydroxyl groups are on the same side of $C=N$ bond and if these are on opposite sides, the isomer is named as anti. In ketoximes, the prefixes syn and anti indicate which group of ketoxime is syn or anti to hydroxyl group.

1. Which of the following does not show geometrical isomerism ?

- (A) 1,2-Dichloropent-1-ene (B) 1,3-Dichloropent-2-ene
(C) 1,1-Dichloropent-1-ene (D) 1,4-Dichloropent-2-ene

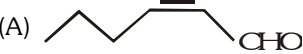
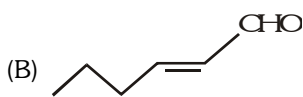
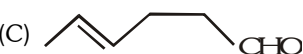
2. On treating with NH_2OH , which can form two products ?

- (A) Acetaldehyde (B) Acetone
(C) Formaldehyde (D) Benzophenone

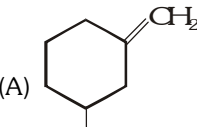
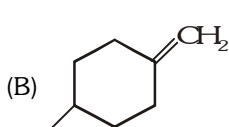
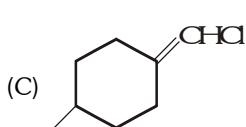
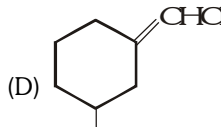
3. Number of stereoisomers of the compound 2-chloro-4-methylhex-2-ene is/are

- (A) 1 (B) 2 (C) 4 (D) 16

4. The correct structure of trans-2-hexenal is -

- (A)  (B) 
(C)  (D) 

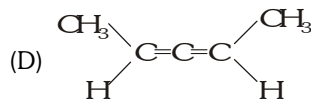
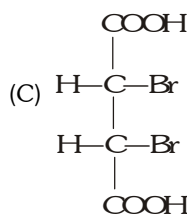
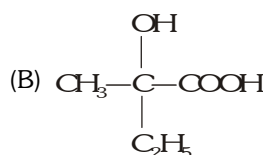
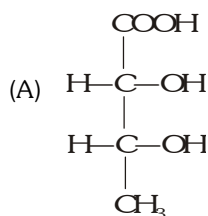
5. The geometrical isomerism is shown by :

- (A)  (B) 
(C)  (D) 

Comprehension # 2

The optical isomers rotate the plane of plane-polarised light. A sp^3 -hybridised carbon atom attached to four different atoms or groups is called an asymmetrical centre or chiral centre. Chiral molecules do not possess any of the elements of symmetry. A chiral molecule cannot be superimposed on its mirror image. These stereoisomers are called enantiomers. Molecules having a plane of symmetry or centre of symmetry are superimposable on their mirror images and are achiral. The stereoisomers that are not mirror images of each other are called diastereomers. A mesoisomer has a plane of symmetry and is optically inactive due to internal compensation.

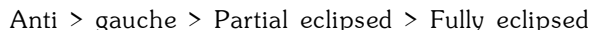
- Which of the following has a meso isomer also ?
 (A) 2-Chlorobutane (B) 2,3-Dichlorobutane
 (C) 2,3-Dichloropentane (D) 2-Hydroxypentanoic acid
- Which of the following compounds is not chiral ?
 (A) $\text{DCH}_2\text{CH}_2\text{CH}_2\text{Cl}$ (B) $\text{CH}_3\text{CH}_2\text{CHDCl}$
 (C) $\text{CH}_3\text{CHDCH}_2\text{Cl}$ (D) $\text{CH}_3\text{CHClCH}_2\text{D}$
- The total number of acyclic isomers including the stereoisomers (geometrical and optical) possible with the molecular formula $\text{C}_4\text{H}_7\text{Cl}$ is :
 (A) 12 (B) 11 (C) 10 (D) 9
- Which among the following compounds will be dissymmetric but not asymmetric :



- Two isomeric alkenes A and B have molecular formula $\text{C}_5\text{H}_9\text{Cl}$. On adding H_2 , A gives optically inactive compound while B gives chiral compound. The two isomers are :
 (A) A is 3-Chloropent-1-ene and B is 1-chloropent-2-ene
 (B) A is 2-Chloro-3-methylbut-2-ene while B is 1-Chloro-3-methylbut-1-ene
 (C) A is 3-Chloropent-2-ene and B is 2-Chloropent-2-ene
 (D) A is 4-Chloropent-2-ene and B is 4-Chloropent-1-ene

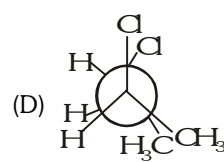
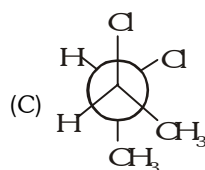
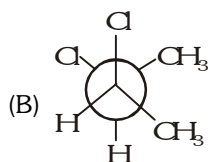
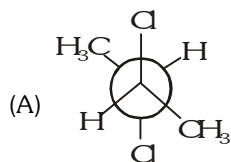
Comprehension # 3

Different spatial arrangements of the atoms that result from rotation about a single bond are conformers. n-Butane has four conformers eclipsed, fully eclipsed, gauche and anti. The stability order of these conformers are as follows :

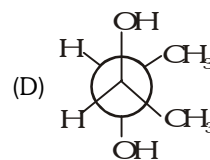
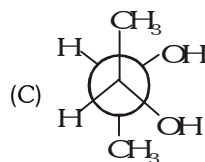
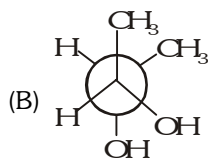
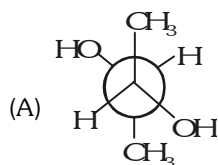


Although anti is more stable than gauche but in some cases gauche is more stable than anti.

- Which one of the following is most stable conformer :



2. Which one of the following is the most stable conformer ?



3. Number of possible conformers of n-butane is :

(A) 2

(B) 4

(C) 6

(D) infinite

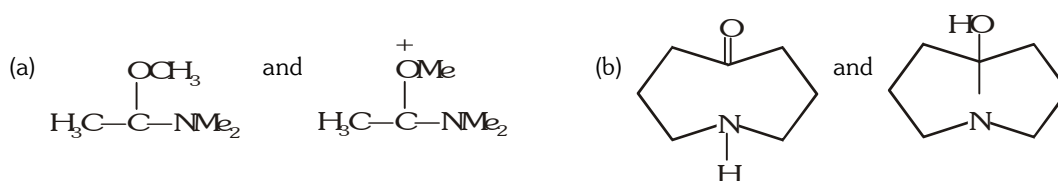
MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
• <u>True / False</u>		
1. T	2. T	3. T
4. F	5. T	6. F
• <u>Fill in the Blanks</u>		
1. four	2. chain	3. geometrical
4. enantiomer	5. geometrical	
• <u>Match the Column</u>		
1. (A) → q ; B → p ; (C) → r ; (D) → s	2. (A) → p, q, r ; (B) → q, s ; (C) → p, q, r ; (D) → q, s	
3. (A) → p ; (B) → r ; (C) → q ; (D) → r, s		
• <u>Assertion - Reason Questions</u>		
1. A	2. D	3. D
4. B	5. D	6. B
7. A		
• <u>Comprehension Based Questions</u>		
Comprehension #1 : 1. (C)	2. (A)	3. (C)
4. (B)	5. (D)	
Comprehension #2 : 1. (B)	2. (A)	3. (C)
4. (D)	5. (C)	
Comprehension #3 : 1. (A)	2. (C)	3. (D)

EXERCISE-04 [A]

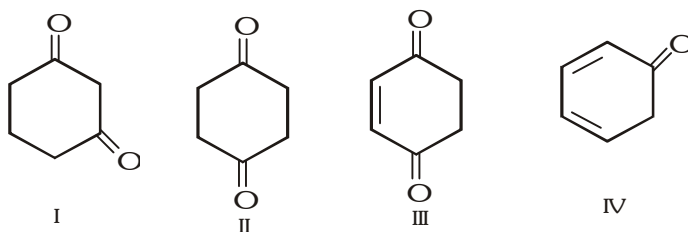
CONCEPTUAL SUBJECTIVE EXERCISE

SUBJECTIVE QUESTIONS :

1. How many isomers are there corresponding to the formula $C_4H_{10}O$?
2. If the bonds in dichloro benzene, $C_6H_4Cl_2$, were localized between specific carbon atoms, how many isomers of this compound would exist ? How many isomers actually exists.
3. Which of the following compounds can exist as geometric isomers ?
 CH_2Cl_2 , CH_2Cl-CH_2Cl , $CHBr = CHCl$, CH_2Cl-CH_2Br .
4. How will you distinguish between Maleic acid and Fumaric acid ?
5. Why does cyclopentene not exhibit geometric isomerism though it has a double bond.
6. Why does 2-butene exhibit cis-trans isomerism but 2-yne does not ?
7. Which of the following pairs show tautomerism.



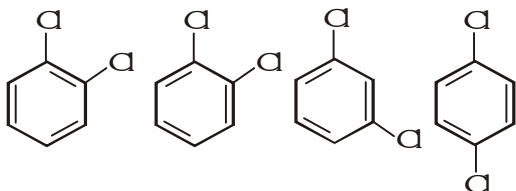
8. Write structural isomer of C_6H_{14} . What is relation between them ?
9. Arrange the following in the order of their enolic content :



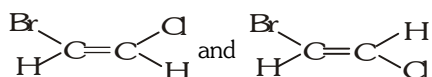
10. Which of the following does/do not exhibits tautomerism.



- There are 7—1-butanol, 2-butanol, 2-methyl-1-propanol, 2-methyl-2-propanol, diethyl ether, methyl propyl ether, and methylisopropyl ether.
- If the bonds were localized, there would be 4 isomers ; actually there are only 3 of the following, the first two are identical, because the bonds are not localized.

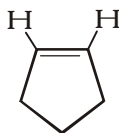
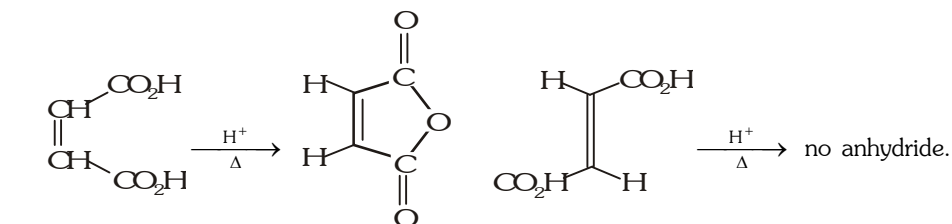


- Only $\text{CHBr}=\text{CHCl}$ can exist as geometric isomers :



In $\text{CH}_2\text{Cl}-\text{CH}_2\text{Cl}$ and $\text{CH}_2\text{Cl}-\text{CH}_2\text{Br}$, the carbon atoms are connected by a single bond about which the groups can rotate relatively freely. Thus any conformation of the halogen atoms may be converted into any other simply by rotation about the single bond. In CH_2Cl_2 , the configuration of the molecule is tetrahedral and all interchanges of atoms yield exactly equivalent configurations.

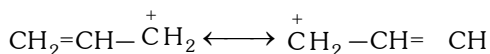
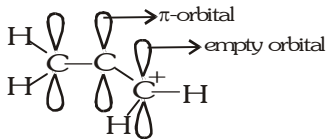
- Maleic acid forms an anhydride where as fumaric acid does not.



- This is cis form. Two H atoms on the same side. To get trans, ring must be twisted.

Double bond becomes severely twisted-destabilized. Effective overlap of P orbitals is missing, so does not exist.

- The P_z orbitals forming π -bonds and the empty P_z orbital of the carbon with +ve charge are parallel. So the electrons may be delocalized. The +ve charge is effectively spread out over two carbons; delocalized.



In n-propyl cation, + I effect of R increases the stability.

In allyl + M effect increases the stability. But + M effect in allyl cation is more effective. So allyl > propyl.

A group with + M effect stabilizes cation; destabilizes anion.

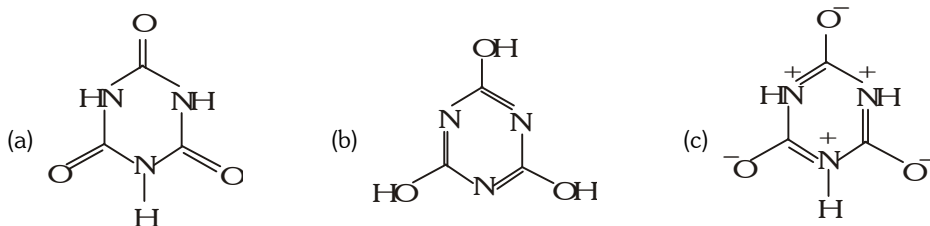
- Lone pair - $p\pi$ conjugation between fluorine and carbon will be more effective than between chlorine and carbon.
- III > IV > I > II

EXERCISE-04 [B]

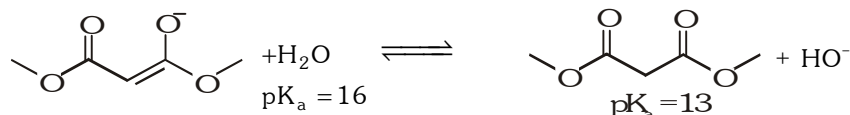
BRAIN STORMING SUBJECTIVE EXERCISE

SUBJECTIVE QUESTIONS :

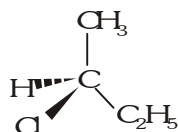
1. What is relation between (a), (b), (c) ?



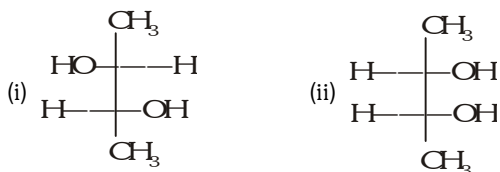
2. Which side is favoured at equilibrium, provide quantitative explanation :



3. (+) 2-butanol has specific rotation of + 13.9 when measured in pure form. A sample of 2-butanol was found to have an optical rotation of -3 . What is the stereomeric composition of this mixture ?
4. N-methylethanamine as such does not show any stereoisomerism but one of its resonance form exhibit stereoisomerism. Explain.
5. Assign Cahn-ingold prelog priorities to the following sets of substituents :
- (i) -H, -Br, -CH₂CH₃, -CH₂CH₂OH (ii) -COOH, -COOCH₃, -CH₂OH, -OH
- (iii) -CN, -CH₂NH₂, -CH₂NHCH₃, -NH₂ (iv) -Br, -CH₂Br, -Cl, -CH₂Cl
6. Identify whether the stereogenic centre is present or not :
- (i) 2-Cyclo penten-1-ol (ii) 3-cyclo penten-1-ol (iii) 2-bromopentane (iii) 3-bromopentane
7. Discuss the optical activity of tertiary amines of the type R₁R₂R₃N :
8. Draw the enantiomer of the following structure :



9. 2,4-Hexadiene has three geometrical isomers. Draw their structures.
10. Assign R and S configuration to the chiral carbons in the following :

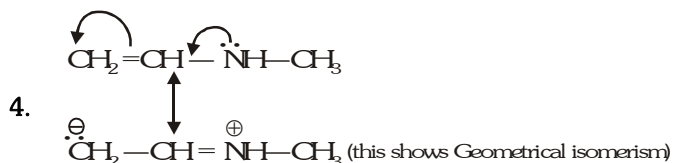


1. a & b are tautomers and a & c are resonating structures.

3. Let x is the % of (+) 2-butanol.

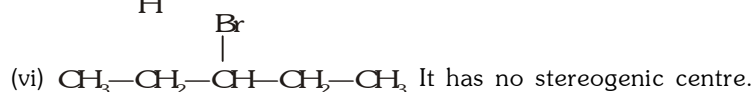
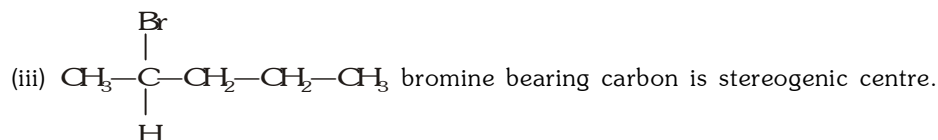
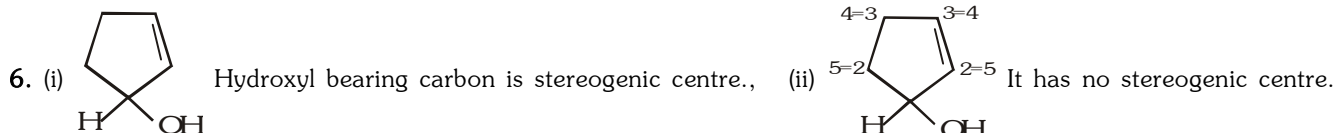
$$13.9x - 13.9(100 - x) = -300.$$

$$x = 39.2, \% \text{ of d form} = 39.2, \% \text{ of } \ell \text{ form} = 60.8.$$



5. (a) $-\text{Br} > -\text{CH}_2\text{CH}_2\text{OH} > -\text{CH}_2\text{CH}_3 > -\text{H}$, (b) $-\text{OH} > -\text{COOCH}_3 > -\text{COOH} > -\text{CH}_2\text{OH}$

(c) $-\text{NH}_2 > -\text{CN} > -\text{CH}_2\text{NHCH}_3 > -\text{CH}_2\text{NH}_2$ (d) $-\text{Br} > -\text{Cl} > -\text{CH}_2\text{Br} > -\text{CH}_2\text{Cl}$

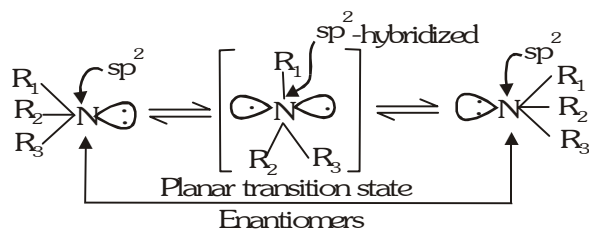
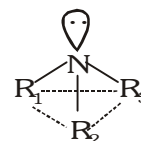


7. Tertiary amines have pyramidal geometry with sp^3 -hybridization at nitrogen.

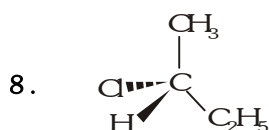
It should be a chiral molecule (assuming lone pair to be a substituent). Thus,

tertiary amines exist as racemic mixture but they cannot be resolved.

This is due to the reason that the energy difference between the isomer is very small (25 kJ mol^{-1}). Hence, rapid nitrogen or amine inversion takes place.



Tertiary amine N-oxide has four group hence nitrogen inversion is not possible, thus tertiary amine N-oxide can be resolved.



10. (i) 2R, 3R (ii) 2S, 3R

EXERCISE-05(A)**PREVIOUS YEARS QUESTIONS**

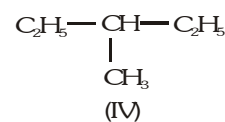
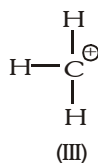
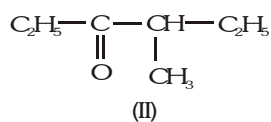
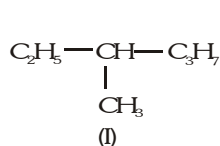
1. Racemic mixture is formed by mixing two- [AIEEE-2002]
(1) isomeric compounds (2) chiral compounds
(3) meso compounds (4) enantiomers with chiral carbon

2. Geometrical isomerism is not shown by- [AIEEE-2002]
(1) 1,1-dichloro-1-pentene (2) 1,2-dichloro-1-pentene
(3) 1,3-dichloro-2-pentene (4) 1,4-dichloro-2-pentene

3. Following types of compounds I and II [AIEEE-2002]
(I) $\text{CH}_3\text{CH}=\text{CHCH}_3$
(II) $\text{CH}_3-\underset{\text{CH}_2\text{CH}_3}{\underset{|}{\text{CH}}}-\text{OH}$, are studied

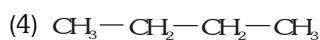
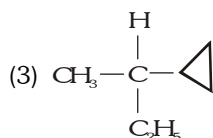
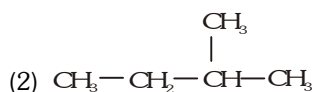
in terms of isomerism in-

- (1) chain isomerism (2) position isomerism
(3) conformers (4) stereo isomerism
4. Among the following four structures I to IV [AIEEE-2003]

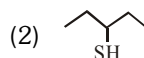
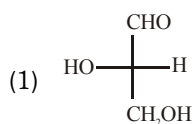


It is true that-

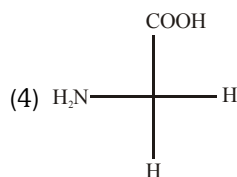
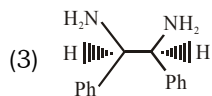
- (1) All four are chiral compounds
(2) only I and II are chiral compounds
(3) only III is a chiral compound
(4) only II and IV are chiral compounds
5. Which of the following will have a meso-isomer also- [AIEEE-2004]
(1) 2-chlorobutane (2) 2,3-dichlorobutane
(3) 2,3-dichloropentene (4) 2-hydroxy propanoic acid
6. Amongst the following compounds, the optically active alkane having lowest molecular mass is [AIEEE-2004]
(1) $\text{CH}_3-\text{CH}_2-\text{C}\equiv\text{CH}$

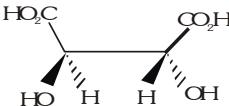


7. Which of following compounds is not chiral [AIEEE-2005]
 (1) 1-chloropentane
 (2) 2-chloropentane
 (3) 1-chloro-2-methyl pentane
 (4) 3-chloro-2-methyl pentane
8. Of the five isomeric hexanes, the isomer which can give two monochlorinated compounds is- [AIEEE-2005]
 (1) 2-methyl pentane
 (2) 2,2-dimethyl butane
 (3) 2,3-dimethyl butane
 (4) n-hexane
9. Which types of isomerism is shown by 2,3-dichloro butane- [AIEEE-2005]
 (1) structural (2) geometric
 (3) optical (4) diastereo
10. Increasing order of stability among the three main conformations (i.e. Eclipse, Anti, Gauche) of 2-fluoroethanol is [AIEEE-2006]
 (1) Gauche, Eclipse, Anti
 (2) Eclipse, Anti, Gauche
 (3) Anti, Gauche, Eclipse
 (4) Eclipse, Gauche, Anti
11. Which one of the following conformations of cyclohexane is chiral ? [AIEEE - 2007]
 (1) Twist boat (2) Rigid
 (3) Chair (4) Boat
12. Which of the following molecules is expected to rotated the plane of plane-polarised light ?



[AIEEE - 2007]



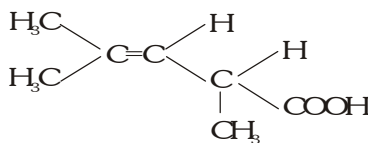
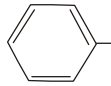
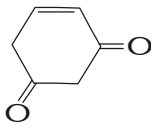
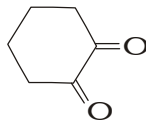
13. The absolute configuration of  is [AIEEE - 2008]
 (1) S, S (2) R, R (3) R, S (4) S, R

14. α -D-(+)-glucose and β -D-(+)-glucose are [AIEEE - 2008]
 (1) conformers (2) epimers
 (3) anomers (4) enantiomers

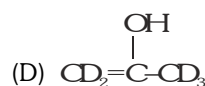
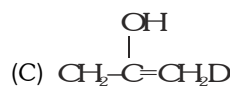
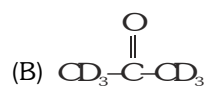
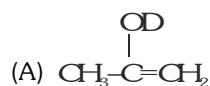
15. The alkene that exhibits geometrical isomerism is :- [AIEEE - 2009]
 (1) 2-butene (2) 2-methyl-2-butene
 (3) Propene (4) 2-methyl propene
16. The number of stereoisomers possible for a compound of the molecular formula $\text{CH}_3\text{-CH=CH-CH(OH)-Me}$ is :- [AIEEE - 2009]
 (1) 4 (2) 6 (3) 3 (4) 2
17. Out of the following, the alkene that exhibits optical isomerism is :- [AIEEE-2010]
 (1) 2-methyl-2-pentene (2) 3-methyl-2-pentene
 (3) 4-methyl-1-pentene (4) 3-methyl-1-pentene
18. Identify the compound that exhibits tautomerism :- [AIEEE-2011]
 (1) 2-Pentanone (2) Phenol
 (3) 2-Butene (4) Lactic acid
19. How many chiral compounds are possible on monochlorination of 2-methyl butane ? [AIEEE-2012]
 (1) 6 (2) 8 (3) 2 (4) 4
20. Which branched chain isomer of the hydrocarbon with molecular mass 72 u gives only one isomer of mono substituted alkyl halide ? [AIEEE-2012]
 (1) Neohexane (2) Tertiary butyl chloride
 (3) Neopentane (4) Isohexane
21. How many cyclic structures are possible for C_4H_6 :- [AIEEE-2012(Online)]
 (1) 3 (2) 5 (3) 4 (4) 6
22. Maleic acid and fumaric acids are :- [AIEEE-2012(Online)]
 (1) Tautomers (2) Chain isomers
 (3) Geometrical isomers (4) Functional isomers

PREVIOUS YEAR QUESTIONS ANSWER KEY															EXERCISE-5(A)	
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
Ans	4	1	4	2	2	3	1	3	3	2	1	1	2	3	1	
Que.	16	17	18	19	20	21	22									
Ans	1	4	1	4	3	3	3									

EXERCISE-05(B)**PREVIOUS YEARS QUESTIONS****MCQ's WITH ONE CORRECT ANSWER**

1. The enolic form of acetone contains : [IIT-90]
(A) 9σ bonds, 1π bond and 2 lone pairs (B) 8σ bonds, 2π bonds and 2 lone pairs
(C) 10σ bonds, 1π bonds and 1 lone pair (D) 9σ bonds, 2π bond and 1 lone pair
2. An organic molecule necessarily shows optical activity if it : [IIT-93]
(A) Contains asymmetric carbon atoms (B) is non polar
(C) is non superimposable on its mirror image (D) is superimposable on its mirror image
3. The compound which is not isomeric with diethyl ether is : [IIT-93]
(A) butan-1-ol (B) butanone
(C) 2-methyl propan-2-ol (D) n-propyl methyl ether
4. Ordinary light can be converted into plane polarized light with the help of a : [IIT-93]
(A) Nickel prism (B) Nicol prism (C) Diffraction grating (D) Quartz cell
5. The structure shows : [IIT-95]

(A) Geometrical isomerism (B) Optical isomerism
(C) Geometrical & optical isomerism (D) tautomerism
6. How many optically active stereoisomers are possible for butane -2,3-diol : [IIT-97]
(A) 1 (B) 2 (C) 3 (D) 4
7. Isomers which can be interconverted through rotation around of single bond are - [IIT-97]
(A) Conformers (B) Diastereomers (C) Enantiomers (D) Positional isomers
8. The number of possible enantiomeric pairs that can be produced during monochlorination of 2-methyl butane is : [IIT-97]
(A) 2 (B) 3 (C) 4 (D) 1
9. Tautomerism is not exhibited by : [IIT-98]
(A)  (B)  (C)  (D) 
10. Rotation of polarised light can be measured by : [IIT-98]
(A) Monometer (B) Galvanometer (C) Polarimeter (D) Viscometer
11. The optically active tartaric acid is named as D-(+) tartaric acid because it has a positive : [IIT-99]
(A) optical rotation and is derived from D-glucose
(B) pH in an organic solvent
(C) optical rotation and is derived from D-(+)- glyceraldehyde
(D) optical rotation only when substituted by deuterium

12. The enol form of acetone, after treatment with D_2O gives [IIT-99]



13. Which of the following compound will exhibits geometrical isomerism : [IIT-2000]

(A) 1-phenyl-2-butene

(B) 3-phenyl-1-butene

(C) 2-phenyl-1-butene

(D) 1, 1-diphenyl-1-propene

14. The number of isomers for the compound with molecular formula $C_2BrClFI$ is : [IIT-2000]

(A) 3

(B) 4

(C) 5

(D) 6

15. Which of the following exhibits stereoisomerism- [IIT-2000]

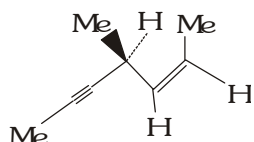
(A) 2-Methylbutene-1

(B) 3-Methylbutyne-1

(C) 3-Methylbutanoic acid

(D) 2-Methylbutanoic acid

16. Hydrogen of the following compound in the presence of poisoned palladium catalyst gives : [IIT-2002]



(A) optically active compound

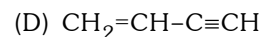
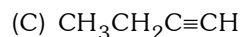
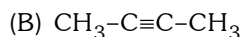
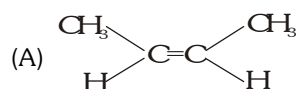
(B) an optically inactive compound

(C) a racemic mixture

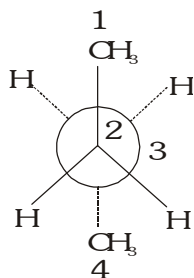
(D) a diastereomeric mixture

17. Which of the following has the lowest dipole moment :

[IIT-2002]



18. If C_2 in below compound is rotated by 120° angle in anticlockwise direction along C_2-C_3 , which of the following form will be produced : [IIT-2004]



(A) Partial eclipsed

(B) Perfectly eclipsed

(C) Perfectly staggered

(D) Gauche conformation

19. $\begin{array}{c} CH_3 \\ | \\ CH_3-CH-CH_2-CH_3 \end{array} \xrightarrow{Cl_2/h\nu} N \text{ (no. of isomers)} \xrightarrow{\text{Fractional distillation}} (F), (N) \text{ and } (F) \text{ are :}$

[IIT-2006]

(A) 6, 4

(B) 4, 4

(C) 6, 6

(D) 3, 3

20. The number of structural isomers of C_6H_{14} is : [IIT-2007]
 (A) 3 (B) 4 (C) 5 (D) 6
21. The number of stereoisomers obtained by bromination of trans-2-butene is : [IIT-2007]
 (A) 1 (B) 2 (C) 3 (D) 4

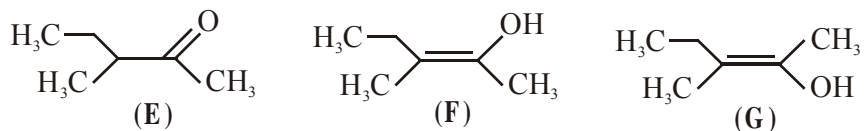
22. **Statement-I** : Molecules that are not superimposable on their mirror images are chiral
Because

Statement-II : All chiral molecules have chiral centres.

[IIT-2007]

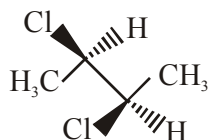
- (A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.

23. The correct statement(s) concerning the structures E, F and G is (are) [IIT-2008]



- (A) E, F and G are resonance structures
 (B) E, F and E, G are tautomers
 (C) F and G are geometrical isomers
 (D) F and G are diastereomers

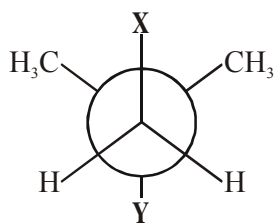
24. The correct statement(s) about the compound given below is (are) :- [IIT-2008]



- (A) The compound is optically active
 (B) The compound possesses centre of symmetry
 (C) The compound possesses plane of symmetry
 (D) The compound possesses axis of symmetry
25. The correct statement(s) about the compound $H_3C(HO)HC - CH = CH - CH(OH)CH_3$ (X) is (are) : [IIT-2009]
 (A) The total number of stereoisomers possible for X is 6
 (B) The total number of diastereomers possible for X is 3
 (C) If the stereochemistry about the double bond in X is trans, the number of enantiomers possible for X is 4
 (D) If the stereochemistry about the double bond in X is cis, the number of enantiomers possible for X is 2
26. The bond energy (in $kcal\ mol^{-1}$) of a C-C single bond is approximately : [IIT-2010]
 (A) 1 (B) 10
 (C) 100 (D) 1000

27. In the Newman projection for 2,2-dimethylbutane

[IIT-2010]



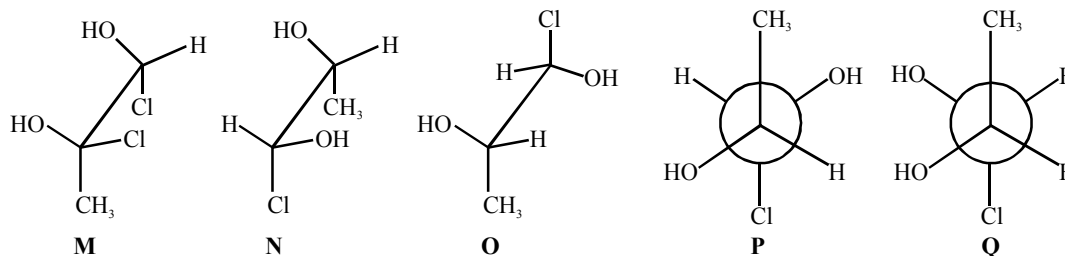
X and Y can respectively be -

- (A) H and H (B) H and C_2H_5 (C) C_2H_5 and H (D) CH_3 and CH_3

28. Amongst the given option, the compound(s) in which all the atoms are in one plane in all the possible conformations (if any), is (are) - [IIT-2011]

- (A) (B) $H-C \equiv C-C \equiv CH$ (C) $H_2C=C=O$ (D) $H_2C=C=CH_2$

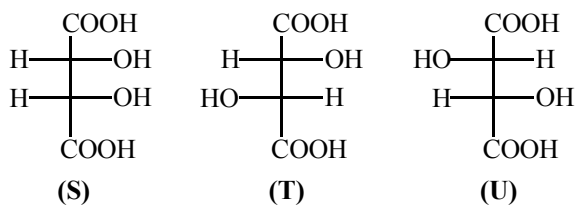
29. Which of the given statement(s) about N,O,P and Q with respect to M is (are) correct ? [IIT-2012]



- (A) M and N are non-mirror image stereoisomers (B) M and O are identical
(C) M and P are enantiomers (D) M and Q are identical

P & Q are isomers of dicarboxylic acid $C_4H_4O_4$. Both decolorize Br_2/H_2O . On heating P forms the cyclic anhydride.

Upon treatment with dilute alkaline $KMnO_4$, P as well as Q could produce one or more than one from S, T and U. [JEE ADVANCED-2013]



30. Compounds formed from P and Q are respectively
(A) Optically active S and optically active pair (T, U)
(B) Optically inactive S and optically inactive pair (T, U)
(C) Optically active pair (T, U) and optically active S
(D) Optically inactive pair (T, U) and optically inactive S

PREVIOUS YEARS QUESTIONS				ANSWER KEY		EXERCISE -5(B)	
1. (A)	2. (C)	3. (B)	4.(B)	5. (B)	6. (B)	7.(A)	8. (A)
9. (B)	10. (C)	11. (C)	12. (A)	13.(A)	14. (D)	15.(D)	16. (B)
17. (B)	18. (D)	19. (B)	20. (C)	21. (A)	22. (C)	23. (B, C, D)	
24. (A, D)	25. (A, D)	26. (C)	27. (B, D)	28. (B, C)	29. (A, B, C)	30. (B)	

EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. The atom having the valence shell electronic configuration $4s^2 4p^2$ would be in:-
(A) Group II A and period 3 (B) Group II B and period 4
(C) Group IV A and period 4 (D) Group IV A and period 3
2. An element with atomic number 106 has been discovered recently. Which of the following electronic configuration will it possess :-
(A) $[Rn] 5f^{14} 6d^5 7s^1$ (B) $[Rn] 5f^{14} 6d^5 7s^2$ (C) $[Rn] 5f^{14} 6d^6 7s^0$ (D) $[Rn] 5f^{14} 6d^1 7s^2 7p^3$
3. The electronic configuration of transition elements is exhibited by :-
(A) $ns^{1-2}(n-1)d^{1-10}$ (B) $ns^2 (n-1) d^{10}$ (C) $(n-1)d^{10}s^2$ (D) $ns^2 np^5$
4. Which of the following electronic configurations in the outermost shell is characteristic of alkali metals :-
(A) $(n-1) s^2 p^6 ns^2 p^1$ (B) $(n-1) s^2 p^6 d^{10} ns^1$ (C) $(n-1) s^2 p^6 ns^1$ (D) $ns^2 np^6 (n-1)d^{10}$
5. The chemistry of lithium is very similar to that of magnesium even though they are placed in different groups. Its reason is :-
(A) Both are found together in nature (B) Both have nearly the same size
(C) Both have similar electronic configurations (D) The ratio of their charge to size is nearly the same
6. Configuration of Br^- is : $[Ar] 3d^{10} 4s^2 4p^6$. The electronic configuration of Br^{+2} would be identical with the element :-
(A) Se (B) As (C) Ga (D) Ge
7. $4d^3 5s^2$ configuration belongs to which group :-
(A) IIA (B) IIB (C) V B (D) III B
8. The ionic radii of N^{3-} , O^{2-} and F^- are respectively given by :-
(A) 1.36, 1.40, 1.71 (B) 1.36, 1.71, 1.40 (C) 1.71, 1.40, 1.36 (D) 1.71, 1.36, 1.40
9. From the given set of species, point out the species from each set having least atomic radius:-
(a) O^{2-} , F^- , Na^+ (b) Ni, Cu, Zn (c) Li, Be, Mg (d) He, Li^+ , H^-
correct answer is -
(A) O^{2-} , Cu, Li, H^- (B) Na^+ , Ni, Be, Li^+ (C) F^- , Zn, Mg, He (D) Na^+ , Cu, Be, He
10. K^+ , Ar, Ca^{2+} and S^{2-} contains -
(A) Same electronic configuration and atomic volume
(B) Different electronic configuration but same IP.
(C) Same electronic configuration but different atomic volume
(D) None
11. Which of the following is not isoelectronic series :-
(A) Cl^- , P^{3-} , Ar (B) N^{3-} , Ne, Mg^{+2} (C) B^{+3} , He, Li^+ (D) N^{3-} , S^{2-} , Cl^-
12. Atomic radii of Fluorine and Neon in Angstrom units are given by :-
(A) 0.72, 1.60 (B) 1.60, 1.60
(C) 0.72, 0.72 (D) None of these
13. In the isoelectronic species the ionic radii (Å) of N^{3-} , Ne and Al^{+3} are respectively given by:-
(A) 1.36, 1.40, 1.71 (B) 1.36, 1.71, 1.40
(C) 1.71, 1.40, 1.36 (D) 1.71, 1.36, 1.40
14. S^{2-} is not isoelectronic with :-
(A) Ar (B) Cl^- (C) HS^- (D) Ti^{+3}
15. The IP_1 , IP_2 , IP_3 , IP_4 and IP_5 of an element are 7.1, 14.3, 34.5, 46.8, 162.2 eV respectively. The element is likely to be:-
(A) Na (B) Si (C) F (D) Ca

16. In which case the energy released is minimum:-
 (A) $\text{Cl} \rightarrow \text{Cl}^-$ (B) $\text{P} \rightarrow \text{P}^-$ (C) $\text{N} \rightarrow \text{N}^-$ (D) $\text{C} \rightarrow \text{C}^-$
17. The electron affinity values for the halogens shows the following trend :-
 (A) $\text{F} < \text{Cl} > \text{Br} > \text{I}$ (B) $\text{F} < \text{Cl} < \text{Br} < \text{I}$
 (C) $\text{F} > \text{Cl} > \text{Br} > \text{I}$ (D) $\text{F} < \text{Cl} > \text{Br} < \text{I}$
18. The process requiring the absorption of energy is.
 (A) $\text{F} \rightarrow \text{F}^-$ (B) $\text{Cl} \rightarrow \text{Cl}^-$ (C) $\text{O} \rightarrow \text{O}^{2-}$ (D) $\text{H} \rightarrow \text{H}^-$
19. The X - X bond length is 1.00 Å and C - C bond length is 1.54 Å. If electronegativities of 'X' and 'C' are 3.0 and 2.0 respectively, the C - X bond length is likely to be :-
 (A) 1.27 Å (B) 1.18 Å (C) 1.08 Å (D) 1.28 Å
20. Correct order of electronegativity of N, P, C and Si is :-
 (A) $\text{N} < \text{P} < \text{C} < \text{Si}$ (B) $\text{N} > \text{C} > \text{Si} > \text{P}$
 (C) $\text{N} = \text{P} > \text{C} = \text{Si}$ (D) $\text{N} > \text{C} > \text{P} > \text{Si}$
21. Mulliken scale of electronegativity uses the concept of :-
 (A) E. A. and EN of pauling (B) E. A. and atomic size
 (C) E.A. and I.P. (D) E.A. and bond energy
22. Which of the following general electronic configuration for transition elements is not correct :-
 (A) $(n + 1) s^{1-2} nd^{1-10}$ (B) $ns^{1-2} (n - 1)d^{1-10}$ (Where $n = 2, 3, 4, \dots$)
 (C) $ns^{0,1,2} (n - 1)s^2 p^6 d^{1-10}$ (D) $(n - 1)d^{1-10} ns^{0-2}$
23. Be and Mg have zero value of electron affinity, because :-
 (A) Be and Mg have $[\text{He}] 2s^2$ and $[\text{Ne}] 3s^2$ configuration respectively.
 (B) 2s and 3s orbitals are filled to their capacity
 (C) Be and Mg are unable to accept electron.
 (D) All the above are correct.
24. The pair with minimum difference in electronegativity is :-
 (A) F, Cl (B) C, H (C) P, H (D) Na, Cs
25. In the following electronic configuration :
 $ns^2 (n - 1) d^{0-1} (n - 2)f^{1-14}$
 If value of $(n - 1) = 6$ the configuration will be of :-
 (A) Lanthanides (B) d - block (C) Actinides (D) s - block
26. Which of the following match is correct :-
 (A) Base of mendeleef periodic table - Number of protons
 (B) Dobernier's triad - Na, K, Rb
 (C) Newland's octave rule is obeyed by H, F, Cl
 (D) Lothar meyer curve plotted between - Atomic number V/S Atomic weight
27. True statement is :-
 (A) All the transuranic elements are synthetic elements
 (B) Elements of third group are called bridge elements
 (C) Element of $1s^2$ configuration is placed in IIA group
 (D) Electronic configuration of elements of a group is same
28. The screening effect of s orbital electron is :-
 (A) Greater than p but lesser than d and f electrons
 (B) Less than p, d and f electrons
 (C) Greater than p, d and f electrons
 (D) Is equal to p, d and f electrons

29. In the first 100 elements, number of s-block elements are :-
 (A) 88 (B) 12 (C) 14 (D) 22
30. The radius of isoelectronic series :-
 (A) Decreases with decreasing nuclear charge
 (B) Decreases with increasing effective nuclear charge
 (C) Same for all
 (D) First increases then decreases
31. In a period, the elements having least melting point are :-
 (A) Noble gas (B) Alkali metals (C) Chalcogens (D) Pnictogens
32. Which set of elements has strong tendency to form cations :-
 (A) N, O, P (B) F, Cl, Br (C) Be, He, Mg (D) Cs, Ba, K
33. A neutral atom (Ar) is converted to (Ar^{+3}) by the following process

$$Ar \xrightarrow[-e]{E_1} Ar^+ \xrightarrow[-e]{E_2} Ar^{+2} \xrightarrow[-e]{E_3} Ar^{+3}$$
 The correct order of E_1 , E_2 and E_3 energies is:-
 (A) $E_1 < E_2 < E_3$ (B) $E_1 > E_2 > E_3$ (C) $E_1 = E_2 = E_3$ (D) $E_1 > E_2 < E_3$
34. The maximum energy will be released in the following process :-
 (A) $B + e^- \longrightarrow B^-$ (B) $C + e^- \longrightarrow C^-$ (C) $N + e^- \longrightarrow N^-$ (D) $O + e^- \longrightarrow O^-$
35. Which of the following represents a correct sequence of electronegativity values :-
 (A) $F > N > O > C$ (B) $F > N < O > C$
 (C) $F > N > C > O$ (D) $F < N < O < C$
36. An element with the electronic configuration $[Xe] 4f^7 5d^1 6s^2$ lies in the :-
 (A) s-block II-A group (B) d-block III-B group
 (C) f-block III-B group (D) d-block VIII group
37. In which of the following compounds cation and anion ratio is minimum :-
 (A) CsF (B) LiI (C) LiF (D) CsI
38. In which of the following the energy change corresponds to first ionisation potential :-
 (A) $X_{(g)} \rightarrow X^+_{(g)} + e$ (B) $2X_{(g)} \rightarrow 2X^+_{(g)} + 2e$
 (C) $X_{(s)} \rightarrow X^+_{(g)} + e$ (D) $X_{(aq)} \rightarrow X^+_{(aq)} + e$
39. Set of elements having one electron in their valence shell are :-
 (A) Cl, Br, I (B) Na, Mg, Al
 (C) B, Al, Ga (D) K, Rb, Cs
40. The covalent and vander Waal's radii of hydrogen respectively are :-
 (A) 0.37 Å, 0.8 Å (B) 0.37 Å, 0.37 Å (C) 0.8 Å, 0.8 Å (D) 0.8 Å, 0.37 Å
41. The electronic configuration of two neutral elements A and B are
 $A = 1s^2 2s^2 2p^6 3s^1$ and $B = 1s^2 2s^2 2p^5$
 (A) $A^+ B^-$ (B) $A^- B^+$ (C) $A - B$ (D) $A^{2+} (B^-)_2$
42. If the ionic radii of $K^{\oplus}$ and $F^{\ominus}$ are nearly the same (i.e. 1.34 Å) then the atomic radii of K and F respectively are :-
 (A) 1.34 Å, 1.34 Å (B) 0.72 Å, 1.96 Å (C) 1.96 Å, 0.72 Å (D) 1.96 Å, 1.34 Å
43. The electronegativities of the following elements H, O, F, S and Cl increase in the order:-
 (A) $H < O < F < S < Cl$ (B) $S < H < Cl < O < F$
 (C) $H < S < O < Cl < F$ (D) $H < S < Cl < O < F$
44. The correct order of size for iodine, species I, I^- , I^+ is :
 (A) $I > I^- > I^+$ (B) $I > I^+ > I^-$ (C) $I^+ > I^- > I$ (D) $I^- > I > I^+$

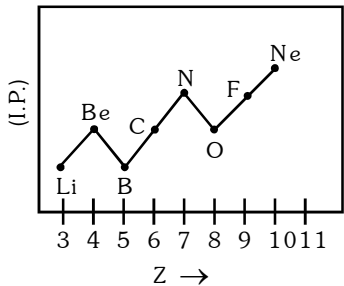
EXERCISE-02**BRAIN TEASERS****SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)**

1. If the difference in atomic size of :
Na - Li = x Rb - K = y Fr - Cs = z
Then correct order will be:-
(A) $x = y = z$ (B) $x > y > z$ (C) $x < y < z$ (D) $x < y < z$
2. In the ions P^{3-} , S^{2-} and Cl^{-} the increasing order of size is:-
(A) Cl^{-} , S^{2-} , P^{3-} (B) P^{3-} , S^{2-} , Cl^{-} (C) S^{2-} , Cl^{-} , P^{3-} (D) S^{2-} , P^{3-} , Cl^{-}
3. Which of the following order of atomic/ionic radius is not correct :-
(A) $I^{-} > I > I^{+}$ (B) $Mg^{+2} > Na^{+} > F^{-}$ (C) $P^{+5} < P^{+3}$ (D) $Li > Be > B$
4. Ionic radii of :-
(A) $Ti^{4+} < Mn^{7+}$ (B) $^{37}Cl^{-} < ^{35}Cl^{-}$ (C) $K^{+} > Cl^{-}$ (D) $P^{3+} > P^{5+}$
5. The best reason to account for the general tendency of atomic diameters to decrease as the atomic numbers increase within a period of the periodic table is the fact that
(A) Outer electrons repel inner electrons
(B) Closer packing among the nuclear particles is achieved
(C) The number of neutrons increases
(D) The increasing nuclear charge exerts a greater attractive force on the electrons
6. Correct orders of 1st Ionisation Potential are :-
(a) $Li < B < Be < C$ (b) $O < N < F$ (c) $Be < N < Ne$
(A) a, b (B) b, c (C) a, c (D) a, b, c
7. The second ionisation potentials in electron volts of oxygen and fluorine atoms are respectively given by :-
(A) 35.1, 38.3 (B) 38.3, 38.3 (C) 38.3, 35.1 (D) 35.1, 35.1
8. In which of the following pairs, the ionisation energy of the first species is less than that of the second :-
(A) O^{-} , O^{2-} (B) S, P (C) N, P (D) Be^{+} , Be
9. The correct order of stability of Al^{+} , Al^{+2} , Al^{+3} is :-
(A) $Al^{+3} > Al^{+2} > Al^{+}$ (B) $Al^{+2} > Al^{+3} > Al^{+}$ (C) $Al^{+2} < Al^{+} > Al^{+3}$ (D) $Al^{+3} > Al^{+} > Al^{+2}$
10. Mg forms $Mg(II)$ because of :-
(A) The oxidation state of Mg is + 2
(B) Difference between $I.P_1$ and $I.P_2$ is greater than 16.0 eV
(C) There are only two electrons in the outermost energy level of Mg
(D) Difference between $I.P_1$ and $I.P_2$ is less than 11 eV
11. IP_1 and IP_2 of Mg are 178 and 348 K. cal mol^{-1} . The enthalpy required for the reaction $Mg \rightarrow Mg^{2+} + 2e^{-}$ is :-
(A) + 170 K.cal (B) + 526 K.cal (C) - 170 K.cal (D) - 526 K.cal
12. Which of the following decreases in going down the halogen group :-
(A) Ionic radius (B) Atomic radius (C) Ionisation potential (D) Boiling point
13. Successive ionisation energies of an element 'X' are given below (in K. Cal)
- | IP_1 | IP_2 | IP_3 | IP_4 |
|--------|--------|--------|--------|
| 165 | 195 | 556 | 595 |
- Electronic configuration of the element 'X' is:-
(A) $1s^2, 2s^2 2p^6, 3s^2 3p^2$ (B) $1s^2, 2s^1$
(C) $1s^2, 2s^2 2p^2$ (D) $1s^2, 2s^2 2p^6, 3s^2$
14. The energy needed to remove one electron from unipositive ion is abbreviated as :-
(A) 1st I.P. (B) 3rd I.P. (C) 2nd I.P. (D) 1st E.A.

15. Which of the following has 2nd IP < 1st IP
 (A) Mg (B) Ne (C) C (D) None
16. The correct order of decreasing first ionization energy is :-
 (A) Si > Al > Mg > Na (B) Si > Mg > Al > Na
 (C) Al > Si > Mg > Na (D) Mg > Li > Al > Si
17. Which of the following transitions involves maximum amount of energy.
 (A) $M_{(g)}^- \rightarrow M_{(g)}$ (B) $M_{(g)} \rightarrow M_{(g)}^+$ (C) $M_{(g)}^+ \rightarrow M_{(g)}^{2+}$ (D) $M_{(g)}^{2+} \rightarrow M_{(g)}^{3+}$
18. Out of Na^+ , Mg^{+2} , O^{-2} and N^{-3} , the pair of species showing minimum and maximum IP would be.
 (A) Na^+ , Mg^{+2} (B) Mg^{+2} , N^{-3} (C) N^{-3} , Mg^{+2} (D) O^{-2} , N^{-3}
19. In the formation of a chloride ion, from an isolated gaseous chlorine atom, 3.8 eV energy is released, which would be equal to :-
 (A) Electron affinity of Cl^- (B) Ionisation potential of Cl
 (C) Electronegativity of Cl (D) Ionisation potential of Cl^-
20. The correct order of electron affinity is :-
 (A) Be < B < C < N (B) Be < N < B < C
 (C) N < Be < C < B (D) N < C < B < Be
21. Electron addition would be easier in :-
 (A) O (B) O^+ (C) O^- (D) O^{+2}
22. Process $Na^+ \xrightarrow{I} Na_{(g)} \xrightarrow{II} Na_{(s)}$
 (A) In (I) energy released, (II) energy absorbed (B) In both (I) and (II) energy is absorbed
 (C) In both (I) and (II) energy is released (D) In (I) energy absorbed, (II) energy released
23. In the process $Cl_{(g)} + e^- \xrightarrow{\Delta H} Cl_{(g)}^-$, ΔH is
 (A) Positive (B) Negative (C) Zero (D) None
24. $O_{(g)} + 2e^- \rightarrow O_{(g)}^{2-}$ $\Delta H_{eg} = 744.7$ KJ/mole. The positive value of ΔH_{eg} is due to :-
 (A) Energy is released to add on 1 e^- to O^{-1}
 (B) Energy is required to add on 1 e^- to O^{-1}
 (C) Energy is needed to add on 1 e^- to O
 (D) None of the above is correct
25. Second electron affinity of an element is :-
 (A) Always exothermic (B) Endothermic for few elements
 (C) Exothermic for few elements (D) Always endothermic
26. The electron affinity
 (A) Of carbon is greater than oxygen (B) Of fluorine is less than iodine
 (C) Of Cl^- is less than Cl (D) Of S is less than oxygen
27. Which of the following statement is false :-
 (A) Elements of ns^2np^6 electronic configuration lies in 1st to 6th period
 (B) Typical elements lies in 3rd period
 (C) The seventh period will accommodate thirty two elements
 (D) Boron and silicon are diagonally related
28. In boron atom screening is due to :-
 (A) Electrons of K shell only (B) All the electrons of K and L shell
 (C) Two electrons of 1s and 2s each (D) Only by electrons of L shell
29. Which will have the maximum value of electron affinity O^x , O^y and O^z [x, y and z respectively are 0, - 1 and - 2] :-
 (A) O^x (B) O^y (C) O^z (D) All have equal

30. The electron affinity of fluorine is less than that of chlorine because :-
 (A) The incoming electron enters the 3p orbital in fluorine
 (B) The incoming electron enters the 2p orbitals in fluorine and 3p orbital in chlorine
 (C) The electron density in fluorine is higher
 (D) Fluorine has lower ionisation potential than chlorine
31. If the atomic number of an element is 58, it will be placed in the periodic table in the -
 (A) III B gp and 6th period (B) IV B gp and 6th period
 (C) VB gp and 7th period (D) None of the above
32. Which of the following ion has largest size :-
 (A) F⁻ (B) Al⁺³ (C) Cs⁺ (D) O⁻²
33. Which ionisation potential (IP) in the following equations involves the greatest ammount of energy:-
 (A) $K^+ \rightarrow K^{+2} + e^-$ (B) $Li^+ \rightarrow Li^{+2} + e^-$ (C) $Fe \rightarrow Fe^+ + e^-$ (D) $Ca^+ \rightarrow Ca^{+2} + e^-$
34. Which order is wrong :-
 (A) Electronegativity - P < N < O < F
 (B) 1st ionisation potential - B < Be < O < N
 (C) Basic property - MgO > CaO > FeO > Fe₂O₃
 (D) Reactivity - Be < Li < K < Cs
35. The correct electron affinity order of N, O, S, Cl is:-
 (A) O < N < Cl < S (B) Cl > O > S > N
 (C) N < O < S < Cl (D) N = Cl > O = S
36.
$$\begin{array}{cccc} 1 & 2 & 3 & 4 \\ H_3C & -CH & =C & =CH_2 \end{array}$$

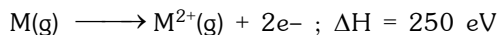
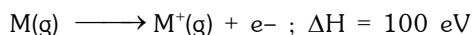
 In the given compound which carbon atom will show maximum electronegativity -
 (A) Fourth (B) First
 (C) Third (D) EN of all the carbon atoms is same
37. Decreasing order of size of ions is :-
 (A) Br⁻ > S⁻² > Cl⁻ > N⁻³ (B) N⁻³ > S⁻² > Cl⁻ > Br⁻
 (C) Br⁻ > Cl⁻ > S⁻² > N⁻³ (D) N⁻³ > Cl⁻ > S⁻² > Br⁻
38. In which case the maximum energy is needed in the formation of monpositive gaseous ion :
 (A) 1 mole of Li atoms (B) 1 mole of Na atoms
 (C) 1 mole of Cs atoms (D) 1 mole of Be atoms
39. (a) $M^+_{(g)} \rightarrow M_{(g)}$ (b) $M_{(g)} \rightarrow M^+_{(g)}$ (c) $M^+_{(g)} \rightarrow M^{+2}_{(g)}$ (d) $M^{+2}_{(g)} \rightarrow M^{+3}_{(g)}$
 Minimum and maximum I.P. would be of :-
 (A) a, d (B) b, c (C) c, d (D) d, a
40. Correct order of ionic size of elements :-
 (A) Mn⁺⁷ > Mn⁺⁶ > Mn⁺⁴ (B) C⁺ > C > C⁻
 (C) Fe⁺³ > Fe⁺² > Fe (D) All are incorrect
41. If the ionisation potential is IP, electron affinity is EA and electronegativity is x then which of the following relation is correct :-
 (A) 2X - EA - IP = 0 (B) 2EA - X - IP = 0
 (C) 2IP - X - EA = 0 (D) All of the above
42. Which are correct match :-
 (A) O < C < S < Se — Atomic size
 (B) Na < Al < Mg < Si — 1st I.P
 (C) MgO < SrO < Cs₂O < K₂O — Basic character
 (D) P₄O₁₀ > SO₃ > Cl₂O₇ - Acidic character

43. Which are correct match :-
 (A) $O > F > N > C$ — 2nd I.P.
 (B) $S^{2-} > Cl^- > K^+ > Ca^{+2}$ — Ionic radius
 (C) $N > C > P > Si$ — E. N.
 (D) $F > Na > Ne$ — 1st I.P.
44. In the third period Na to Cl seven element is/are called:-
 (A) Lanthanides (B) Typical elements
 (C) Halogen elements (D) Metalloids
45. Which of the following statement is/are not correct:-
 (A) I.P. increases down the group
 (B) IP of s-block elements is less than corresponding d- block elements
 (C) If $\Delta IP > 16$ eV higher oxidation state is more stable
 (D) IP of halogen elements is maximum in their respective period
46. Out of the following statements which is/are correct :-
 (A) H is an element of minimum atomic radius (B) He is an element of highest I.P.
 (C) Cl is an element of highest EA (D) Li is an element of lowest I.P.
47. Triad - I [N^{3-} , O^- , Na^+]
 Triad - II [N^+ , C^+ , O^+]
 Choose the species of lowest IP from triad-I and highest IP from triad-II respectively
 (A) N^{3-} , O^+ (B) Na^+ , C^+ (C) N^{3-} , N^+ (D) O^- , C^+
48. The correct values of ionization energies (in kJ mol^{-1}) of Be, Ne, He and N respectively are
 (A) 899, 2080, 1403, 2372 (B) 2080, 899, 1403, 2372
 (C) 899, 2080, 2372, 1403 (D) 899, 1403, 2080, 2372
49. Which of the following processes involve absorption of energy :-
 (A) $S(g) + e^- \rightarrow S^-(g)$ (B) $O^-(g) + e^- \rightarrow O^{2-}(g)$
 (C) $Cl(g) + e^- \rightarrow Cl^-(g)$ (D) $O(g) + e^- \rightarrow O^-(g)$
50. Following graph shows variation of I.P. with atomic number in second period (Li – Ne). Value of I.P. of Na (11) will be :-
 (A) Above Ne
 (B) Below Ne but above O
 (C) Below Li
 (D) Between N and O
- 
51. $M(g) \rightarrow M^+(g) + e^-$, $\Delta H = 100$ eV, $M(g) \rightarrow M^{2+}(g) + 2e^-$, $\Delta H = 250$ eV which is/are incorrect statement(s) :-
 (A) IP_1 of $M(g)$ is 100 eV (B) IP_1 of $M^+(g)$ is 150 eV
 (C) IP_2 of $M(g)$ is 250 eV (D) IP_2 of $M(g)$ is 150 eV
52. AB is predominantly ionic as $A^+ B^-$ if :-
 (A) $(IP)_A < (IP)_B$ (B) $(EA)_A < (EA)_B$ (C) $(EN)_A < (EN)_B$ (D) Size of A < size of B
53. Which is correct order of size of O, O^{2-} , F^- and F :-
 (A) $O^{2-} > O > F^- > F$ (B) $O > O^{2-} > F > F^-$
 (C) $O^{2-} > F^- > F > O$ (D) $O^{2-} > F^- > O > F$
54. Both metals and non-metals are found among.....elements in the periodic table :-
 (A) p-block (B) d-block (C) Transition (D) Inner transition

55. In the plot of the first ionization energy against atomic number the peaks are occupied by :-
 (A) Inert gases (B) Alkali metals
 (C) Halogens (D) Transition elements
56. The correct values of ionization energies (in KJ mole⁻¹) of Si, P, Cl and S respectively are
 (A) 786, 1012, 999, 1256 (B) 1012, 786, 999, 1256
 (C) 786, 1012, 1256, 999 (D) 786, 999, 1012, 1256
57. Which of the following sequence is correct for decreasing order of ionic radius :-
 (A) Se⁻², I⁻, Br⁻, O⁻², F⁻ (B) I⁻, Se⁻², O⁻², Br⁻, F⁻
 (C) Se⁻², I⁻, Br⁻, F⁻, O⁻² (D) I⁻, Se⁻², Br⁻, O⁻², F⁻
58. Which of the following orders for electron affinity is /are correct :-
 (a) S > O < Se (b) Cl > F (c) S > O (d) O > S
 (e) N > P (f) C > N
 (A) a, b, c, e (B) a, b, c, f (C) b, c, d, e (D) b, c, f
59. The electronic configuration of some neutral atoms are given below :-
 (1) 1s² 2s¹ (2) 1s² 2s² 2p³ (3) 1s² 2s² 2p⁴ (4) 1s² 2s² 2p⁶ 3s¹
 In which of these electronic configuration would you expect to have highest :-
 (i) IE₁ (ii) IE₂
 (A) 3, 1 (B) 2, 1 (C) 3, 2 (D) 2, 4
60. Which of the following pairs has elements containing same number of electrons in outer most orbit
 (A) Sc, Cu (B) Na, Ca (C) Pb, Sb (D) As, Bi
61. Zinc does not show the variable valency as elements of d-block, because :
 (A) This is soft metal (B) d-orbital is full
 (C) Its melting point is low (D) Volatile Metal
62. The right order of ionization potential of Li, Be, B & C is :
 (A) C > Be > B > Li (B) C > B > Be > Li (C) C > B > Li > Be (D) B > C > Be > Li
63. Increasing order of metallic characteristic of C, Sb, As, Bi, Si is shown by :
 (A) C, Si, As, Sb, Bi (B) C, Si, Bi, Sb, As
 (C) C, Si, Sb, Bi, As (D) C, Si, As, Bi, Sb
64. The correct order of second ionization potential of C, N, O and F is
 (A) C > N > O > F (B) O > N > F > C (C) O > F > N > C (D) F > O > N > C
65. The correct sequence of the electron affinity of C, N, O and F is :
 (A) C > N < O < F (B) O > N > C > F (C) C < N > O < F (D) C > N > O > F
66. The correct order of ionization energies of F⁻, Cl⁻, F and Cl is:
 (A) Cl < F < Cl⁻ < F⁻ (B) Cl⁻ < F⁻ < Cl < F
 (C) F⁻ < Cl⁻ < Cl < F (D) Cl⁻ < Cl < F⁻ < F
67. Atomic radii of alkali metals (M) follow the order Li < Na < K < Rb but ionic radii in aqueous solution follow the reverse order Li⁺ > Na⁺ > K⁺ > Rb⁺. The reason of the reverse order is :
 (A) Increase in the ionisation energy (B) Decrease in the metallic bond character
 (C) Increase in the electropositive character (D) Decrease in the amount of hydration
68. The first ionization potentials (eV) of Be and B respectively are :
 (A) 8.29eV, 9.32 eV (B) 9.32 eV, 9.32 eV
 (C) 8.29 eV, 8.29 eV (D) 9.32 eV, 8.29 eV
69. The decreasing order of the ionization potential of the following elements is :
 (A) Ne > Cl > P > S > Al > Mg (B) Ne > Cl > P > S > Mg > Al
 (C) Ne > Cl > S > P > Mg > Al (D) Ne > Cl > S > P > Al > Mg

70. One element has atomic weight 39. Its electronic configuration is $1s^2, 2s^2 2p^6, 3s^2 3p^6 4s^1$. The true statement for that element is : [IIT 95]
 (A) More (IE) (B) Transition element
 (C) Isotone with $^{36}_{18}\text{Ar}$. (D) Stable oxide M_2O
71. In which of the following arrangements the order is not according to the property indicated against it ?
 (A) $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Na}^+ < \text{F}^-$ - increasing ionic size
 (B) $\text{B} < \text{C} < \text{N} < \text{O}$ - increasing first ionization potential
 (C) $\text{I} < \text{Br} < \text{F} < \text{Cl}$ - increasing electron gain enthalpy (with negative sign)
 (D) $\text{Li} < \text{Na} < \text{K} < \text{Rb}$ - increasing metallic radius
72. Lanthanoid contraction is caused due to :
 (A) the same effective nuclear charge from Ce to Lu
 (B) the imperfect shielding on outer electrons by 4f electrons from the nuclear charge
 (C) the appreciable shielding on outer electrons by 4f electrons from the nuclear charge
 (D) the appreciable shielding on outer electrons by 5d electrons from the nuclear charge
73. Is a data sufficiency problem in which it is to be decided on the basis of given statements whether the given question can be answered or not. (No matter whether the answer is yes or no)
 If $\Delta H_{\text{solution}} < 0$ then compound acts as ionic in aqueous solution. Is $\text{AlCl}_3(\text{s})$ ionic in aqueous solution.
Statement 1 : L.E. of AlCl_3 is 5137 kJ/mol
Statement 2 : ΔH_{HE} of Al^{+3} ion is $-4665 \text{ kJ/mol}^{-1}$ & ΔH_{HE} of Cl^- is -381 kJ/mol^{-1}
 (A) Statments (A) alone is sufficient but statement (B) is not sufficient
 (B) Statments (B) alone is sufficient but statement (A) is not sufficient
 (C) Both statement together are sufficient but neither statement alone is sufficient
 (D) Statement (A) & (B) together are not sufficient
74. The properties which are common to the elements belonging to groups 1 and 17 of periodic tables are-
 (A) Electropositive character increases down the group
 (B) Reactivity decreases from top to bottom
 (C) Atomic radii increases as atomic number increases
 (D) Electronegativity decreases on moving down a group
75. The number of which subatomic particle is same in case of chlorine atom and chloride ion :
 (A) Electron (B) Proton (C) Neutrons (D) All of the above
76. Which of the following show amphoteric behaviour :
 (A) $\text{Zn}(\text{OH})_2$ (B) BeO (C) Al_2O_3 (D) $\text{Pb}(\text{OH})_2$
77. Fluorine is stronger oxidizing agent than chlorine in aqueous solution. This can be attributed to the property :
 (A) Heat of dissociation (B) Electron affinity
 (C) Ionization potential (D) Heat of hydration
78. Electron affinify of the elements or ions shown correct :
 (A) $\text{S} > \text{O}^-$ (B) $\text{O} > \text{S}^-$ (C) $\text{O}^- > \text{S}^-$ (D) $\text{N}^- > \text{S}$
79. Ionization energy of an element is :
 (A) Equal in magnitude but opposite in sign to the electron gain enthalpy of the cation of the element
 (B) Same as electron affinity of the element
 (C) Energy required to remove one valence electron from an isolated gaseous atom in its ground state
 (D) Equal in magnitude but opposite in sign to the electron gain enthalpy of the anion of the element
80. Select equations having endothermic step :
 (A) $\text{S}^-(\text{g}) \longrightarrow \text{S}^{2-}(\text{g})$
 (B) $\text{Na}^+(\text{g}) + \text{Cl}^-(\text{g}) \longrightarrow \text{NaCl}(\text{s})$
 (C) $\text{N}(\text{g}) \longrightarrow \text{N}^-(\text{g})$
 (D) $\text{Al}^{2+}(\text{g}) \longrightarrow \text{Al}^{3+}(\text{g})$

81. Consider the following ionization steps :



select correct statement(s) :

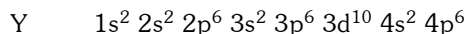
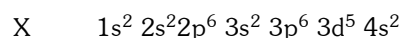
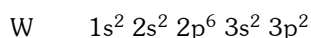
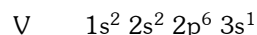
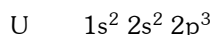
(A) I.E.₁ of M(g) is 100 eV

(B) I.E.₁ of M⁺(g) is 150 eV

(C) I.E.₂ of M(g) is 250 eV

(D) I.E.₂ of M(g) is 150 eV

82. The ground state electronic configurations of the elements, U, V, W, X and Y (these symbols do not have any chemical significance) are as follows :



Determine which sequence of elements satisfy the following statements :

(i) Element forms a carbonate which is not decomposed by heating

(ii) Element is most likely to form coloured ionic compounds

(iii) Element has largest atomic radius

(iv) Element forms only acidic oxide

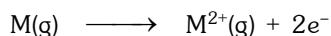
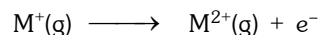
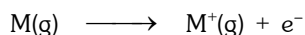
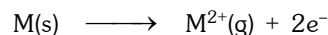
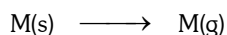
(A) V W Y U

(B) V X Y W

(C) V W Y X

(D) V X W U

83. Consider the following changes :



The second ionization energy of M could be calculated from the energy values associated with :

(A) 1 + 3 + 4

(B) 2 - 1 + 3

(C) 1 + 5

(D) 5 - 3

84. Which of the following statements are correct :

(A) F is the most electronegative and Cs is the most electropositive element.

(B) The electronegativity of halogens decreases from F to I

(C) The electron affinity of Cl is higher than that of F though their electronegativities are in the reverse order

(D) The electron affinity of noble gases is almost zero.

85. Diagonal relationships are shown by :

(A) Be and Al

(B) Li and Mg

(C) Mg and Al

(D) B and P

86. Match List I with List II and select the correct answer using the codes given below :

List I

List II

A. $1s^2, 2s^2 2p^6, 3s^2 3p^6, 4s^2$

1. In

B. $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^1$

2. Pd

C. $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2 4p^6 4d^{10}$

3. Ca

D. $1s^2, 2s^2 2p^6, 3d^{10}, 4s^2 4p^6 4d^{10}, 5s^2 5p^1$

4. Cu

Code : A

B

C

D

(A) 1

2

3

4

(B) 1

3

2

4

(C) 3

4

2

1

(D) 1

4

3

2

87. Match List I (Atomic Number of Element) with List II (Block to which the Element Belongs) and select the correct answer using the codes given below :

List I		List II	
(Atomic Number of Element)		(Block to which the element belongs)	
A. 24		1. p	
B. 38		2. f	
C. 49		3. s	
D. 59		4. d	
Code : A	B	C	D
(A) 2	1	3	4
(B) 4	3	1	2
(C) 2	3	1	4
(D) 4	1	3	2

88. Match List I (Element) with List II (Electronegativity on Pauling Scale) and select the correct answer using the codes given below :

List I		List II	
(Element)		(Electronegativity on Pauling scale)	
A. Carbon		1. 0.8	
B. Nitrogen		2. 1.6	
C. Aluminium		3. 2.5	
D. Cesium		4. 3.0	
		5. 4.0	
Code : A	B	C	D
(A) 2	4	5	1
(B) 3	1	2	4
(C) 2	1	5	4
(D) 3	4	2	1

BRAIN TEASERS						ANSWER KEY				EXERCISE -2					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	B	A	B	D	D	D	C	B	D	D	B	C	D	C	D
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	B	D	C	D	B	D	C	B	B	D	C	A	C	A	C
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	A	C	B	C	C	C	A	D	A	D	A	A,B	A, B, C	B	A, C, D
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	A, B, C	A	C	B	C	C	A,B,C	D	A	A	C	D	B	B	D
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	B	A	A	C	A	C	D	D	B	D	B	B	C	A, C, D	B, C
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88		
Ans.	A,B,C,D	A, B, D	A, B	A, C	A, C,D	A, B, D	B	D	A,B,C,D	A, B	C	B	D		

EXERCISE-03**MISCELLANEOUS TYPE QUESTIONS****TRUE / FALSE**

1. Al_2O_3 is an amphoteric oxide.
2. Third group of periodic table accommodates maximum number of elements.
3. First ionisation potential of Mg is higher than that of Al.
4. The ionic radii of trivalent lanthanides (La^{3+} , Ce^{3+} , Pr^{3+}) decreases with the increasing atomic number.
5. Successive ionisation potentials are lower.
6. The alkali metals show increasing electronegativities from Li to Cs.
7. In group I of alkali metals, the ionization potential decreases down the group. Therefore lithium is a poor reducing agent in gaseous state.
8. The decreasing order of electron affinity of F, Cl, Br is $\text{F} > \text{Cl} > \text{Br}$ [IIT-1993]
9. The basic nature of the hydroxides of Group 13 (Gr. III B) decreases progressively down the group. [IIT-1992]

FILL IN THE BLANKS

1. Most electropositive elements belong to group.
2. Most electronegative elements belong to group.
3. Transition elements are characterised by valency.
4. The second ionisation energy of calcium is than the ionisation energy of calcium.
5. The electronegativity of the elements C, N, Si and P increases in the order of
6. Total number of inner transition elements are
7. Two elements of equal electronegative values they form bond.
8. Among Na, Mg, Al & Si elements element has zero electron affinity.
9. Elements of group have greater tendency to form positive ions than elements of group IIA.
10. In aqueous solution is the best reducing agent among the alkali metals.
11. Ca^{2+} has a smaller ionic radius than K^+ because it has [IIT-1993]
12. Compounds that formally contain Pb^{4+} are easily reduced to Pb^{2+} . The stability of the lower oxidation state is due to [IIT-1997]

MATCH THE COLUMN

1. Match Column-I (atomic number of elements) with Column-II (position of element in periodic table) and select the correct answer using the codes given below :

<u>Column-I</u>		<u>Column-II</u>	
(A)	19	(p)	p-block
(B)	22	(q)	f-block
(C)	32	(r)	d-block
(D)	64	(s)	s-block

2. Match Column-I (Elements) with Column-II (configuration of elements) and select the correct answer using the codes given below :

<u>Column-I</u>		<u>Column-II</u>	
(A)	The third alkali metal	(p)	$1s^2 2s^2 2p^6 3s^2 3p^5$
(B)	The second transition element	(q)	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$
(C)	The fourth noble gas element	(r)	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^2 4s^2$
(D)	The second halogen element	(s)	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$

3.

Column-I	
(A)	Increasing ionisation potential
(B)	Increasing electronegativity
(C)	Decreasing Z_{eff}
(D)	Decreasing electron affinity

Column-II	
(p)	$N > O > F$
(q)	$N < O < F$
(r)	$O < N < F$
(s)	$O > C > N$

4.

Column-I	
(A)	Metalloid
(B)	Radioactive
(C)	Transition
(D)	Chalcogen

Column-II	
(p)	Selenium
(q)	Silver
(r)	Arsenic
(s)	Uranium

5.

Column-I	
(A)	Increasing atomic size
(B)	Decreasing atomic radius
(C)	Increasing electronegativity
(D)	Increasing effective nuclear charge

Column-II	
(p)	$Cl < O < F$
(q)	$Li < Be < B$
(r)	$Si < Al < Mg$
(s)	$N > O > F$

ASSERTION & REASON QUESTIONS

These questions contains, Statement I (assertion) and Statement II (reason).

(A) Statement-I is true, Statement-II is true ; Statement-II is correct explanation for Statement-I.

(B) Statement-I is true, Statement-II is true ; Statement-II is NOT a correct explanation for statement-I

(C) Statement-I is true, Statement-II is false

(D) Statement-I is false, Statement-II is true

- Statement -1** : Two successive ionisation energies of Argon are 56.8 eV and 36.8 eV respectively.
because

Statement -2 : Zeff of Ar ($3s^23p^6$) is greater than Ar^+ ($3s^23p^5$).
- Statement -1** : Electron affinity of fluorine is greater than chlorine.
because

Statement -2 : Ionisation potential of fluorine is less than chlorine.
- Statement -1** : Size of anion is larger than their parent atom.
because

Statement -2 : Zeff of anion is greater than that of their parent atom.
- Statement -1** : Atomic radius of inert gases is largest in the period
because

Statement -2 : Effective nuclear charge of inert gases is minimum
- Statement -1** : 2nd IP of alkali metals is maximum in the period.
because

Statement -2 : Alkali metals has smallest atomic size in the period.
- Statement -1** : First ionization energy of nitrogen is lower than oxygen.
because

Statement -2 : Across the period effective nuclear charge decreases.
- Statement -1** : The third period contains only 8 elements and not 18 like 4th period.
because

Statement -2 : In III period filling starts from $3s^1$ and complete at $3p^6$ whereas in IV period it starts from $4s^1$ and complete after $3d^{10}$ and $4s^2$.

COMPREHENSION BASED QUESTIONS

Comprehension # 1

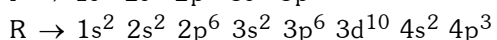
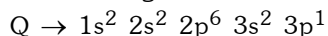
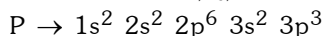
Ionization energies of five elements in kcal/mol are given below :

Atom	I	II	III
P	300	549	920
Q	99	734	1100
R	118	1091	1652
S	176	347	1848
T	497	947	1500

- Which element is a noble gas ?
(A) P (B) T (C) R (D) S
- Which element form stable unipositive ion :
(A) P (B) Q (C) R (D) S
- The element having most stable oxidation state +2 is :
(A) Q (B) R (C) S (D) T
- Which is a non-metal (excluding noble gas) :
(A) P (B) Q (C) R (D) S
- If Q reacts with fluorine and oxygen, the molecular formula of fluoride and oxide will be respectively :
(A) QF_3 , Q_2O_3 (B) QF , Q_2O (C) QF_2 , QO (D) None of these
- Which of the following pair represents elements of same group :
(A) Q, R (B) P, Q (C) P, S (D) Q, S

Comprehension # 2

Four elements P,Q,R & S have ground state electronic configuration as :



- Which of the following option represent the correct order of true (T) and False (F) Statement:
I. size of P < size of Q II. size of R < size of S
III. size of P < size of R (appreciable difference) IV. size of Q < size of S (appreciable difference)
(A) TTTT (B) TTTF (C) FFTT (D) TTFF
- Order of IE_1 values among the following is :
(A) $\text{P} > \text{R} > \text{S} > \text{Q}$ (B) $\text{P} < \text{R} < \text{S} < \text{Q}$ (C) $\text{R} > \text{S} > \text{P} > \text{Q}$ (D) $\text{P} > \text{S} > \text{R} > \text{Q}$

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
• <u>True / False</u>		
1. T 2. T 3. T 4. T 5. F 6. F 7. T 8. F 9. T		
• <u>Fill in the Blanks</u>		
1. IA group /1 2. VIIA/17 3. Variable 4. higher, first		
5. Si, P, C, N 6. 28 7. (non polar) covalent 8. Mg		
9. IA 10. Lithium 11. Higher effective nuclear charge 12. Iner pair effect		
• <u>Match the Column</u>		
1. (A)-s (B)-r (C)-p (D)-q 2. (A)-s (B)-r (C)-q (d)-p 3. (A)-r (B)-q (C)-p (D)-s		
4. (A)-r (B)-s (C)-q (D)-p 5. (A)-r (B)-s (C)-p (D)-q		
• <u>Assertion - Reason Questions</u>		
1. D 2. D 3. C 4. C 5. C 6. D 7. A		
• <u>Comprehension Based Questions</u>		
Comprehension #1 : 1. B 2. B,C 3. C 4.A 5.B 6.A		
Comprehension #2 : 1. B 2. A		

CONCEPTUAL SUBJECTIVE EXERCISE

- Can an element with atomic number 126, if discovered, be accommodated in the present long form of periodic table ?
- Third period of the periodic table contains 8 and not 18 elements. Justify.
- If scientist succeed in obtaining element with atomic number 114, which well known element would you expect it to resemble ?
- 1st and 2nd IE of few elements have been given below –

IE ₁ (KJ/mole)	IE ₂ (KJ/mole)
(A) 520	7300
(B) 1680	3380
(C) 2370	5250
(D) 900	1760

 - Which is reactive metal ?
 - Which is reactive non metal ?
 - Which is inert gas ?
 - A metal that form a stable binary halide of formulae AX₂ (X = Halogen)
- Among the elements B, Al, C and Si,
 - which has the highest first ionization enthalpy?
 - which has the most negative electron gain enthalpy ?
 - which has the largest atomic radius ?
 - which has the most metallic character ?
- Which of the elements Na, Mg, Si and P would have the greatest difference between the first and second ionization enthalpies. Briefly explain your answer.
- The diagram below shows part of the skeleton of the periodic table in which element are indicated by letters which are not their usual symbols :

[illegible]

Answer the following on the basis of periodic table :

- (I) Alkali metal(s)
- (II) An elements with the outer configuration of d^8s^2
- (III) Lanthanoids
- (IV) Representative elements(s)
- (V) Elements with incomplete f-subshell
- (VI) Halogen(s)
- (VII) s-block element(s)

(X) Non-transition element (s)

[illegible]

11. Ionisation potential and electron affinity of fluorine are 17.42 and 3.45 eV respectively. Calculate the electronegativity of fluorine on Mulliken scale and Pauling scale :
12. Addition of an electron to $\text{Na}_{(\text{g})}$ is slightly exothermic process, whereas addition of electron to $\text{Mg}_{(\text{g})}$ is strongly endothermic. Explain.

EXERCISE -4(A)

1. No. because there no provision for filling of g-block element in periodic table.
3. 14th group, carbon family
4. (i) Most reactive metal will be an alkali metal of 1st group with its $IE_2 > IE_1$. Thus most reactive metal is (a).
(ii) Most reactive non-metal will be a halogen of 17th group. Its IE_1 will be quite high. Thus most reactive non metal is (b).
(iii) A noble gas will have very, very high IE_1 . Thus (c) is a noble gas.
(iv) A metal that forms a stable binary halide will be an alkaline earth metal of 2nd group. Its IE_2 will not be much higher than IE_1 . Thus (d) is such a metal that forms a stable binary halide of formula AX_2 .
5. (a) C (b) Si (c) Al (d) Al
6. Na, because during IP_2 electron is removed from stable octet configuration ($ns^2 np^6$).
7. (i) B, (ii) H, (iii) A, (iv) B, C, F, J, I, (v) A, (vi) I (vii) B, J (viii) E, H, K (ix) G (x) L
8. (i) H, (ii) J^{2+} , (iii) H (iv) T
9. 11.2 10. 3.87 11. 10.435, 3.726

EXERCISE-04 [B]**BRAIN STORMING SUBJECTIVE EXERCISE**

1. Arrange the following ions in increasing order of their radius ?
 V^{+5} , K^+ , S^{2-} , Mn^{+7} , Ca^{+2} , Cl^- , P^{3-}
2. The ionic radii of S^{2-} and Te^{2-} are 1.84 and 2.2 Å respectively. What would you predict for the ionic radius of Se^{2-} .
3. Out of Li^+ , Be^{+2} and B^{+3} ions, which has the smallest ionic radius and why ?
4. A student reported the radii of Cu, Cu^+ and Cu^{2+} as 122 pm, 96 pm and 72 pm. Do you agree with the reported values. Justify the answer. Explain why ?
5. How many chlorine atoms will be ionised ($Cl \rightarrow Cl^+ + e^-$) by the energy released from the process $Cl + e^- \rightarrow Cl^-$ for 6.023×10^{23} atom (IP for Cl = 1250 kJ mole⁻¹ and EA = 350 KJ mole⁻¹)
6. Na and Mg^+ have same number of electrons. But removal of electron from Mg^+ requires more energy. Explain.
7. The first ionisation energy of beryllium is greater than that of lithium but reverse is true for the second ionisation energy.
8. Based on location in P.T., which of the following would you expect to be acidic & which basic.
(A) CsOH (B) IOH (C) $Sr(OH)_2$ (D) $Se(OH)_2$
(E) FrOH (F) BrOH
9. From among the elements, choose the following : Cl, Br, F, Al, C, Li, Cs & Xe.
(i) The element with highest electron affinity.
(ii) The element with lowest ionisation potential.
(iii) The element whose oxide is amphoteric.
(iv) The element which has smallest radii.
(v) The element whose atom has 8 electrons in the outermost shell.
10. For the gaseous reaction,
 $K + F \longrightarrow K^+ F^-$, ΔH was calculated to be 19 kcal under conditions where the cations and anions were prevented by electrostatic separation from combining with each other. The ionisation potential of K is 4.3 eV atom. What is the electron affinity of F ?
11. The ionisation potentials of atoms A and B are 400 and 300 kcal mol⁻¹ respectively. The electron affinities of these atoms are 80.0 and 85.0 kcal mol⁻¹ respectively. Prove that which of the atoms has higher electronegativity.
12. (a) If internuclear distance between Cl atoms in Cl_2 is 10 Å & between H atoms in H_2 is 2 Å, then calculate internuclear distance between H & Cl (Electronegativity of H = 2.1 & Cl = 3.0)
(b) Compare the following giving reasons
Acidic nature of oxides : CaO, CO, CO_2 , N_2O_5 , SO_3
13. With the help of EN values [$EN_A = 1.8$, $EN_B = 2.6$, $EN_C = 1.6$, $EN_D = 2.8$] answer the following questions for the compounds
HAO, HBO, HCO, HDO
(a) Compounds whose aqueous solution is acidic and order of their acidic strength
(b) Compounds whose aqueous solution is basic and order of their basic strength
(c) Comment on the chances of being coloured on the basis of percent ionic character for the compounds CD & AB.

1. $Mn^{+7} < V^{+5} < Ca^{+2} < K^{+} < Cl^{-} < S^{2-} < P^{3-}$

2. Ionic radius of Se^{2-} is expected to be in between the ionic radii of S^{2-} and Te^{2-} . Thus

$$\text{Ionic radius of } Se^{2-} = \frac{1.84 + 2.21}{2} = 2.025 \text{ \AA}$$

3. B^{+3} due to more z_{eff} .

4. Cu, Cu^{+} and Cu^{2+} have same number of protons but different number of electrons. In moving from Cu to Cu^{+} to Cu^{2+} , the number of electrons decreases thus effective nuclear charge and force of attraction between the nucleus and valence electron increases and hence size decreases. Thus the correct order is $cu (0.122 \text{ nm}) > Cu^{+} (0.096 \text{ nm}) > Cu^{2+} (0.072 \text{ nm})$.

5. Since $1250 \text{ kJ mole}^{-1}$ energy is required to ionise 6.023×10^{23} atoms. But 350 kJ mol^{-1} energy is released hence the no. of ionised atoms –

$$= \frac{6.023 \times 10^{23} \times 350 \text{ kJ mole}^{-1}}{1250 \text{ kJ mole}^{-1}} = 1.686 \times 10^{23}$$

6. Mg^{+} has more Z_{eff} .

7. The electronic configuration of Li and Be are $1s^2 2s^1$ and $1s^2 2s^2$ respectively.

Since in beryllium 2s orbital is complete while in lithium it is incomplete, it requires more energy to pull out an electron from beryllium than from lithium. Moreover beryllium has higher nuclear charge.

After removal of one electron, Li^{+} and Be^{+} ions have electronic configuration $1s^2$ and $1s^2 2s^1$ respectively. Now it will be easier to remove $2s^1$ electron rather than $1s^2$. Thus IE_2 of Li higher.

8. (A) Basic, (B) Acidic, (C), Basic, (D) Acidic, (E) Basic, (F) Acidic.

9. (i) Cl, (ii) Cs, (iii) Al, (iv) F, (v) Xe.

10. 3.476 eV .

11. $EN_1 > EN_2$

12. (a) $5.919 \text{ \AA}$ (b) $CaO < CO < CO_2 < N_2O_5 < SO_3$

13. (a) Acidic - HBO, HDO acidic strength - $HDO > HBO$

(b) Basic - HAO, HCO Basic strength - $HCO > HAO$

(c) % Ionic character = $16 |X_A - X_B| + 3.5 (X_A - X_B)^2$

$$\text{for CD} = 16 (1.2) + 3.5 (1.2)^2$$

$$= 24.24 \%$$

$\therefore$ Colourless.

EXERCISE - 05 [A]**JEE-[MAIN] : PREVIOUS YEAR QUESTIONS**

1. According to the Periodic law of elements, the variation in properties of elements is related to their :- [AIEEE-2003]
(1) Nuclear masses (2) Atomic numbers
(3) Nuclear neutron-proton number ratio (4) Atomic masses
2. The reduction in atomic size with increase in atomic number is a characteristic of elements of :- [AIEEE-2003]
(1) d-block (2) f-block (3) Radioactive series (4) High atomic masses
3. Which of the following groupings represent a collection of isoelectronic species ? [AIEEE-2003]
(At. no. Cs = 55, Br = 35)
(1) N^{3-} , F^- , Na^+ (2) Be , Al^{3+} , Cl^- (3) Ca^{2+} , Cs^+ , Br (4) Na^+ , Ca^{2+} , Mg^{2+}
4. The atomic numbers of vanadium (V), chromium (Cr), manganese (Mn) and iron (Fe) are respectively 23, 24, 25 and 26. Which one of these may be expected to have the highest second ionization enthalpy ? [AIEEE-2003]
(1) Cr (2) Mn (3) Fe (4) V
5. Which one of the following sets of ions represents the collection of isoelectronic species ? [AIEEE-2004]
(1) K^+ , Cl^- , Mg^{2+} , Sc^{3+} (2) Na^+ , Ca^{2+} , Sc^{3+} , F^- (3) K^+ , Ca^{2+} , Sc^{3+} , Cl^- (4) Na^+ , Mg^{2+} , Al^{3+} , Cl^-
6. Which of the following ions has the highest value of ionic radius ? [AIEEE-2004]
(1) O^{2-} (2) B^{3+} (3) Li^+ (4) F^-
7. Among Al_2O_3 , SiO_2 , P_2O_3 and SO_2 , the correct order of acid strength is :- [AIEEE-2004]
(1) $\text{Al}_2\text{O}_3 < \text{SiO}_2 < \text{SO}_2 < \text{P}_2\text{O}_3$ (2) $\text{SiO}_2 < \text{SO}_2 < \text{Al}_2\text{O}_3 < \text{P}_2\text{O}_3$
(3) $\text{SO}_2 < \text{P}_2\text{O}_3 < \text{SiO}_2 < \text{Al}_2\text{O}_3$ (4) $\text{Al}_2\text{O}_3 < \text{SiO}_2 < \text{P}_2\text{O}_3 < \text{SO}_2$
8. The formation of the oxide ion $\text{O}^{2-}(\text{g})$ requires first an exothermic and then an endothermic step as shown below :- [AIEEE-2004]
 $\text{O}(\text{g}) + \text{e}^- = \text{O}^-(\text{g}), \Delta H = -142 \text{ kJ mol}^{-1}$
 $\text{O}^-(\text{g}) + \text{e}^- = \text{O}^{2-}(\text{g}), \Delta H = 844 \text{ kJ mol}^{-1}$
This is because :-
(1) O^- ion will tend to resist the addition of another electron
(2) Oxygen has high electron affinity
(3) Oxygen is more electronegative
(4) O^- ion has comparatively larger size than oxygen atom
9. In which of the following arrangements the order is NOT according to the property indicated against it ? [AIEEE-2005]
(1) $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Na}^+ < \text{F}^-$ - increasing ionic size
(2) $\text{B} < \text{C} < \text{N} < \text{O}$ - increasing first ionization enthalpy
(3) $\text{I} < \text{Br} < \text{F} < \text{Cl}$ - increasing electron gain enthalpy (with negative sign)
(4) $\text{Li} < \text{Na} < \text{K} < \text{Rb}$ - increasing metallic radius
10. Which of the following oxides is amphoteric in character ? [AIEEE-2005]
(1) SnO_2 (2) SiO_2 (3) CO_2 (4) CaO
11. Pick out the isoelectronic structure from the following : [AIEEE-2005]
I. $^+\text{CH}_3$ II. H_3O^+ III. NH_3 IV. CH_3^-
(1) I and II (2) III and IV (3) I and III (4) II, III and IV
12. The lanthanide contraction is responsible for the fact that [AIEEE-2005]
(1) Zr and Y have about the same radius (2) Zr and Nb have similar oxidation state
(3) Zr and Hf have about the same radius (4) Zr and Zn have the same oxidation state

EXERCISE - 05 [B]**JEE-[ADVANCED] : PREVIOUS YEAR QUESTIONS**

1. Moving from right to left in a periodic table, the atomic size is: [JEE 1995]
(A) increased (B) decreased (C) remains constant (D) none of these
2. The increasing order of electronegativity in the following elements: [JEE 1995]
(A) C, N, Si, P (B) N, Si, C, P (C) Si, P, C, N (D) P, Si, N, C
3. One element has atomic weight 39. Its electronic configuration is $1s^2, 2s^2 2p^6, 3s^2 3p^6 4s^1$. The true statement for that element is: [JEE 1995]
(A) High value of IE (B) Transition element
(C) Isotone with ${}_{18}\text{Ar}^{38}$ (D) None
4. The number of paired electrons in oxygen atom is: [JEE 1995]
(A) 6 (B) 16 (C) 8 (D) 32
5. The decreasing size of K^+ , Ca^{2+} , Cl^- & S^{2-} follows the order: [JEE 1995]
(A) $\text{K}^+ > \text{Ca}^{+2} > \text{S}^{-2} > \text{Cl}^-$ (B) $\text{K}^+ > \text{Ca}^{+2} > \text{Cl}^- > \text{S}^{-2}$
(C) $\text{Ca}^{+2} > \text{K}^+ > \text{Cl}^- > \text{S}^{-2}$ (D) $\text{S}^{-2} > \text{Cl}^- > \text{K}^+ > \text{Ca}^{+2}$
6. Which of the following has the maximum number of unpaired electrons [JEE 1996]
(A) Mg^{2+} (B) Ti^{3+} (C) V^{3+} (D) Fe^{2+}
7. The incorrect statement among the following is: [JEE 1997]
(A) the first ionisation potential of Al is less than the first ionisation potential of Mg
(B) the second ionisation potential of Mg is greater than the second ionisation potential of Na
(C) the first ionisation potential of Na is less than the first ionisation potential of Mg
(D) the third ionisation potential of Mg is greater than the third ionisation potential of Al
8. Li^+ , Mg^{2+} , K^+ , Al^{3+} (Arrange in increasing order of radii) [JEE 1997]
9. Which one of the following statement (s) is (are) correct? [JEE 1998]
(A) The electronic configuration of Cr is $[\text{Ar}] 3d^5 4s^1$. (Atomic No. of Cr = 24)
(B) The magnetic quantum number may have a negative value
(C) In silver atom, 23 electrons have a spin of one type and 24 of the opposite type. (Atomic No. of Ag = 47)
(D) The oxidation state of nitrogen in HN_3 is -3.
10. The electrons, identified by n & l ; [JEE 1999]
(i) $n = 4, l = 1$ (ii) $n = 4, l = 0$ (iii) $n = 3, l = 2$
(iv) $n = 3, l = 1$ can be placed in order of increasing energy, from the lowest to highest as :
(A) (iv) < (ii) < (iii) < (i) (B) (iii) < (ii) < (iv) < (i)
(C) (i) < (iii) < (ii) < (iv) (D) (iii) < (i) < (iv) < (ii)
11. Gaseous state electronic configuration of nitrogen atom can be represented as : [JEE 1999]
(A) $\uparrow\downarrow\uparrow\downarrow\uparrow\uparrow\uparrow$ (B) $\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow\uparrow$ (C) $\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow\downarrow$ (D) $\uparrow\downarrow\uparrow\downarrow\downarrow\downarrow\downarrow$
12. The electronic configuration of an element is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$. This represents its : [JEE 2000]
(A) excited state (B) ground state (C) cationic form (D) none

13. **Assertion:** F atom has a less negative electron gain enthalpy than Cl atom. [JEE 2000]
Reason: Additional electron is repelled more efficiently by 3p electron in Cl atom than by 2p electron in F atom.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
14. The correct order of radii is: [JEE 2000]
 (A) $N < Be < B$ (B) $F^- < O^{2-} < N^{3-}$ (C) $Na < Li < K$ (D) $Fe^{3+} < Fe^{2+} < Fe^{4+}$
15. The IE_1 of Be is greater than that of B. [T/F] [JEE 2001]
16. The set representing correct order of IP_1 is [JEE 2001]
 (A) $K > Na > Li$ (B) $Be > Mg > Ca$ (C) $B > C > N$ (D) $Fe > Si > C$
17. Identify the least stable ion amongst the following: [JEE 2002]
 (A) Li^- (B) Be^- (C) B^- (D) C^-
18. The maximum number of electrons that can have principal quantum number $n=3$, and spin quantum number, $m_s = -1/2$, is [JEE 2011]

PREVIOUS YEARS QUESTIONS			ANSWER KEY		EXERCISE-5 [B]	
1. A	2. C	3. C	4. A	5. D		
6. D	7. B					
8. $Mg^{2+} < Li^+ < K^+$	Q.9 A,B,C	10. A				
11. A,D	12. B,C	13. C	14. B	15. True	16. B	
17. B	18. 9					

EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. A solid has a structure in which W atoms are located at the corners of a cubic lattice, O atom at the centre of the edges and Na atom at centre of the cube. The formula for the compound is :-
(A) NaWO_2 (B) NaWO_3 (C) Na_2WO_3 (D) NaWO_4
2. Which of the following statements is correct in the rock-salt structure of an ionic compounds?
(A) coordination number of cation is four whereas that of anion is six.
(B) coordination number of cation is six whereas that of anion is four.
(C) coordination number of each cation and anion is four.
(D) coordination number of each cation and anion is six.
3. The coordination number of cation and anion in Fluorite CaF_2 and CsCl are respectively :-
(A) 8 : 4 and 6 : 3 (B) 6 : 3 and 4 : 4 (C) 8 : 4 and 8 : 8 (D) 4 : 2 and 2 : 4
4. The interstitial hole is called tetrahedral because :-
(A) it is formed by four spheres.
(B) partly same and partly different
(C) it is formed by four spheres the centres of which form a regular tetrahedron.
(D) none of the above three
5. The tetrahedral voids formed by ccp arrangement of Cl^- ions in rock salt structure are :-
(A) occupied by Na^+ ions (B) occupied by Cl^- ions
(C) occupied by either Na^+ or Cl^- ions (D) vacant
6. The number of nearest neighbours around each particle in a face-centred cubic lattice is :-
(A) 4 (B) 6 (C) 8 (D) 12
7. The intermetallic compound LiAg crystallizes in cubic lattice in which both lithium and silver have coordination number of eight. The crystal class is :-
(A) simple cubic (B) body centred cubic (C) face centred cubic (D) none
8. The mass of a unit cell of CsCl corresponds to :-
(A) 1 Cs^+ and 1 Cl^- (B) 1 Cs^+ and 6 Cl^- (C) 4 Cs^+ and 4 Cl^- (D) 8 Cs^+ and 1 Cl^-
9. In the closest packing of atoms A (radius : r_a), the radius of atom B that can be fitted into tetrahedral voids is :-
(A) $0.155 r_a$ (B) $0.225 r_a$ (C) $0.414 r_a$ (D) $0.732 r_a$
10. Which of the following will show schottky defect :-
(A) CaF_2 (B) ZnS (C) AgCl (D) CaCl
11. A cubic solid is made up of two elements A and B. Atoms B are at the corners of the cube and A at the body centre. What is the formula of compound.
(A) AB (B) AB_2 (C) A_2B (D) None
12. A compound alloy of gold and copper crystallizes in a cubic lattice in which gold occupy that lattice point at corners of the cube and copper atom occupy the centres of each of the cube faces. What is the formula of this compound.
(A) AuCu_6 (B) AuCu (C) AuCu_3 (D) None

13. KF crystallizes in the NaCl type structure. If the radius of K^+ ions 132 pm and that of F^- ion is 135 pm, what is the closet K-K distance ?
(A) Can not say (B) 534 pm (C) 755 pm (D) 378 pm
14. AgCl has the same structure as that of NaOH. The edge length of unit cell of AgCl is found to be 555 pm and the density of AgCl is 5.561 g cm^{-3} . Find the percentage of sites that are unoccupied.
(A) 0.24% (B) 2.4% (C) 24% (D) None
15. The two ions A^+ and B^- have radii 88 and 200 pm respectively. In the closed packed crystal of compound AB, predict the co-ordination number of A^+ .
(A) 6 (B) 8 (C) 4 (D) None
16. The effective radius of the iron atom is $1.42 \text{ \AA}$. It has FCC structure. Calculate its density (Fe = 56 amu).
(A) 2.87 gm/cm^3 (B) 11.48 gm/cm^3 (C) 1.435 gm/cm^3 (D) 5.74 gm/cm^3

CHECK YOUR GRASP						ANSWER KEY				EXERCISE -1					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	B	D	C	C	D	D	B	A	B	D	A	C	D	A	A
Que.	16														
Ans.	D														

EXERCISE-02**BRAIN TEASERS****SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)**

1. The density of CaF_2 (fluorite structure) is 3.18 g/cm^3 . The length of the side of the unit cell is :-
(A) 253 pm (B) 344 pm (C) 546 pm (D) 273 pm
2. If the anions (A) form hexagonal closest packing and cations (C) occupy only $2/3$ octahedral voids in it, then the general formula of the compound is :-
(A) CA (B) CA_2 (C) C_2A_3 (D) C_3A_2
3. A solid is formed and it has three types of atoms X, Y, Z. X forms a FCC lattice with Y atoms occupying all the tetrahedral voids and Z atoms occupying half the octahedral voids. The formula of the solid is :-
(A) $\text{X}_2\text{Y}_4\text{Z}$ (B) XY_2Z_4 (C) $\text{X}_4\text{Y}_2\text{Z}$ (D) X_4YZ_2
4. A compound XY crystallizes in BCC lattice with unit cell edge length of 480 pm. If the radius of Y^- is 225 pm, then the radius of X^+ is :-
(A) 127.5 pm (B) 190.68 pm (C) 225 pm (D) 255 pm
5. Which one of the following schemes of ordering closed packed sheets of equal sized spheres do not generate close packed lattice.
(A) ABCABC (B) ABACABAC (C) ABBAABBA (D) ABCBCABCBC
6. An ionic compound AB has ZnS type structure. If the radius A^+ is 22.5 pm, then the ideal radius of B^- would be :-
(A) 54.35 pm (B) 100 pm. (C) 145.16 pm (D) none of these
7. NH_4Cl crystallizes in a body - centred cubic type lattice with a unit cell edge length of 387 pm. The distance between the oppositely charged ions in the lattice is :-
(A) 335.1 pm (B) 83.77 pm (C) 274.46 pm (D) 137.23 pm
8. $r_{\text{Na}^+} = 95 \text{ pm}$ and $r_{\text{Cl}^-} = 181 \text{ pm}$ in NaCl (rock salt) structure. What is the shortest distance between Na^+ ions ?
(A) 778.3 pm (B) 276 pm (C) 195.7 pm (D) 390.3 pm
9. In diamond, carbon atom occupy FCC lattice points as well as alternate tetrahedral voids. If edge length of the unit cell is 356 pm, then radius of carbon atom is :-
(A) 77.07 pm (B) 154.14 pm (C) 251.7 pm (D) 89 pm
10. Give the correct order of initials T (true) or F (False) for following statements :
I. In an anti-fluorite structure anions form FCC lattice and cations occupy all tetrahedral voids.
II. If the radius of cations and anions are $0.2\text{\AA}$ and $0.95\text{\AA}$ then coordinate number of cation in the crystal is 4.
III. An atom/ion is transferred from a lattice site to an interstitial position in Frenkel defect.
IV. Density of crystal always increases due to substitutinal impurity defect.
(A) TFFF (B) FTTF (C) TFFT (D) TFTF
11. A cubic solid is made by atoms A forming close pack arrangement, B occupying one-fourth of tetrahedral void and C occupying half of the octahedral voids. What is the formula of compound :-
(A) $\text{A}_4\text{B}_4\text{C}_2$ (B) $\text{A}_4\text{B}_2\text{C}_4$ (C) A_4BC (D) $\text{A}_4\text{B}_2\text{C}_2$
12. What is the percent by mass of titanium in rutile, a mineral that contain titanium and oxygen, if structure can be described as a closet packed array of oxide ions, with titanium in one half of the octahedral holes. What is the oxidation number of titanium ?
(A) $\approx 30\%$, +4 (B) $\approx 60\%$, +2 (C) $\approx 60\%$, +4 (D) $\approx 30\%$, +2

13. Spinel is an important class of oxides consisting of two types of metal ions with the oxide ions arranged in CCP pattern. The normal spinel has one - eight of the tetrahedral holes occupied by one type of metal ion and one half of the octahedral hole occupied by another type of metal ion. Such a spinel is formed by Zn^{2+} , Al^{3+} , O^{2-} , with Zn^{2+} in the tetrahedral holes. Given the formula of spinel.
(A) ZnAl_2O_4 (B) Zn_2AlO_4 (C) $\text{Zn}_2\text{Al}_2\text{O}_4$ (D) None
14. Calculate the density of diamond from the fact that it has face centered cubic structure with two atoms per lattice point and unit cell edge length of $3.569 \text{ \AA}$.
(A) 7 gm/cm^3 (B) 1.75 gm/cm^3 (C) 3.5 gm/cm^3 (D) None
15. An element crystallizes into a structure which may be described by a cubic type of unit cell having one atom on each corner of the cube and two atoms on one of its body diagonals. If the volume of this unit cell is $24 \times 10^{24} \text{ cm}^3$ and density of element is 7.2 g cm^{-3} , calculate the number of atoms present in 200 g of element.
(A) 1.1513×10^{24} (B) 3.472×10^{24} (C) 10.416×10^{24} (D) None
16. Silver has an atomic radius of 144 pm and the density of silver is 10.6 cm^{-3} . To which type of cubic crystal, silver belongs ?
(A) HCP (B) BCC (C) Simple cubic (D) FCC
17. Xenon crystallises in the face-centred cubic lattice and the edge of the unit cell is 620 pm . What is the next nearest neighbour distance ?
(A) 738.5 pm (B) 620 pm (C) 438.5 pm (D) 310 pm
18. CsCl has the bcc arrangement and its unit cell edge length is 400 pm . Calculate the interionic distance in CsCl.
(A) 141.42 pm (B) 282.84 pm (C) 346.4 pm (D) 173.2 pm
19. An element crystallizes in a structure having FCC unit cell of an edge 200 pm . Calculate the density, if 200 g of this element contains 5×10^{24} atoms.
(A) 5 m/cm^3 (B) 30 gm/cm^3 (C) 10 gm/cm^3 (D) 20 gm/cm^3
20. If the length of the body diagonal for CsCl which crystallises into a cubic structure with Cl^- ions at the corners and Cs^+ ions at the centre of the unit cells is $7 \text{ \AA}$ and the radius of the Cs^+ ion is $1.69 \text{ \AA}$. What is the radii for Cl^- ion ?
(A) $1.81 \text{ \AA}$ (B) $5.31 \text{ \AA}$ (C) $3.62 \text{ \AA}$ (D) none

BRAIN TEASERS						ANSWER KEY				EXERCISE -2					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	B	C	A	B	C	B	A	D	A	D	D	C	A	C	B
Que.	16	17	18	19	20										
Ans.	D	B	C	D	A										

EXERCISE-03

MISCELLANEOUS TYPE QUESTIONS

TRUE / FALSE

1. Crystalline solids are isotropic.
2. Rhombohedral, triclinic and hexagonal are the unit cells, which have only primitive arrangement possible.
3. Packing fraction of FCC and HCP units cells are same.
4. The minimum void fraction for any unit cell in any shape having only one type of atom and all voids unfilled is 0.26.
5. Packing fraction of a lattice structure depends on the radius of the atom crystallizing in it.
6. The location of tetrahedral voids in FCC units cell are the centres of 8 minicubes forming a large cube.
7. Effective number of octahedral voids in a unit cell is equal to the effective number of atoms in the unit cell.
8. Radius ratio for co-ordination number 4 having tetrahedral and square planar geometry is same.
9. The radius ratio value for co-ordination number 4 having square planar geometry and co-ordination number 6 having octahedral geometry is same.
10. A metallic element crystallises into a lattice containing a sequence of layers AB AB AB Any packing of spheres leaves out voids in the lattice. 26% percent by volume of this lattice is empty space.

FILL IN THE BLANKS

1. The relation between edge length (a) and radius of atom (r) for BCC lattice is
2. The relation between edge length (a) and radius of atom (r) for FCC lattice is
3. ABCABC layering pattern is called packing, found in lattice.
4. ABABAB..... layering pattern is called packing, found in lattice.
5. Height (c) of the hexagonal primitive unit cell in term of radius of atom (r) is
6. Anions would be in contact with each other only if the cation to anion radius for a given co-ordination number is
7. The number of tetrahedral voids in hexagonal primitive unit cell is
8. The limiting radius for co-ordination number 8 is
9. For cesium chloride structure, the interionic distance (in terms of edge length, a) is equal to
10. Density of a crystal due to Schottky defect and due to Frankel defect.

MATCH THE COLUMN

Column-I		Column-II	
(A)	68% occupy of space	(p)	Simple cubic lattice
(B)	CsCl	(q)	Diamond
(C)	Hexagonal close packing in three dimensions	(r)	Na ₂ O
(D)	Antifluorite structure	(s)	AB AB type of close packing
(E)	Covalent crystal	(t)	Body centred cubic lattice.

2.	Column-I		Column-II	
	(A)	Spinel structure	(p)	Framework silicate
	(B)	Glass	(q)	ZnFe ₂ O ₄
	(C)	Quartz	(r)	NaCl crystal
	(D)	Metallic crystal	(s)	Pseudo solid
	(E)	Co-ordination number 6	(t)	Malleable and ductile

ASSERTION & REASON

These questions contains, Statement I (assertion) and Statement II (reason).

(A) Statement-I is true, Statement-II is true ; Statement-II is correct explanation for Statement-I.

(B) Statement-I is true, Statement-II is true ; Statement-II is NOT a correct explanation for statement-I

(C) Statement-I is true, Statement-II is false

(D) Statement-I is false, Statement-II is true

1. **Statement-I** : Crystalline solids can cause X-rays to diffract.

Because

Statement-II : Interatomic distance in crystalline solids is of the order of 0.1 nm.

2. **Statement-I** : Graphite is an example of tetragonal crystal system.

Because

Statement-II : For a tetragonal system $a = b \neq c$, $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$

COMPREHENSION BASED QUESTIONS

Comprehension # 1

Packing refers to the arrangement of constituent units in such a way that the forces of attraction among the constituent particles is maximum and the constituents occupy the maximum available space. In two dimensions, there are square close packing and hexagonal close packing. In three dimensions, however, there are hexagonal close packing, cubic close packing and body-centred cubic packing.

(i) hcp : AB AB AB AB . . . arrangement

Coordination no. = 12

% occupied space = 74

(ii) ccp : ABC ABC . . . arrangement

Coordination no. = 12

% occupied space = 74

(iii) bcc : 68% space is occupied

Coordination no. = 8

Answer the following questions :

1. The empty space left in hcp in three dimensions is :

(A) 26%

(B) 74%

(C) 52.4%

(D) 80%

2. In closed packed lattice containing 'n' particles, the numbers of tetrahedral and octahedral voids are :

(A) n, 2n

(B) n, n

(C) 2n, n

(D) 2n, n/2

3. The pattern of successive layers of ccp arrangement can be designated as :

(A) AB AB AB . . .

(B) AB ABC AB ABC . . .

(C) ABC ABC ABC . . .

(D) AB BA AB BA . . .

4. The space occupied by spheres in bcc arrangement is :

(A) 74%

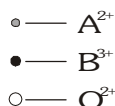
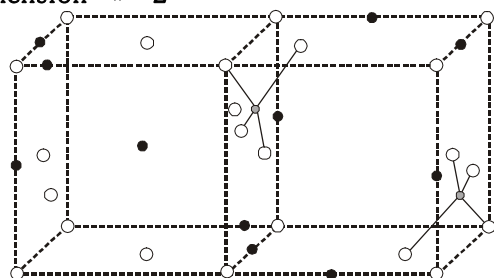
(B) 70%

(C) 68%

(D) 60.4%

5. A certain oxide of metal M crystallises in such a way that O^{2-} ions occupy hcp arrangement following AB AB . . . pattern. The metal ions, however, occupy $\frac{2}{3}$ rd of the octahedral voids. The formula of the compound is
- (A) M_2O_3 (B) M_3O (C) $M_{8/3}O_3$ (D) MO_2

Comprehension # 2



Answer the following questions :

- The space lattice given in the figure refers to :
 (A) fluoride structure (B) rock salt structure (C) spinel structure (D) inverse spinel structure
- O^{2-} ions are present in :
 (A) bcc arrangement (B) fcc arrangement
 (C) simple cubic arrangement (D) hcp arrangement
- The formula of the compound is :
 (A) ABO_2 (B) A_2BO_3 (C) AB_2O_4 (D) A_2BO_4
- Fraction of the total octahedral voids occupied will be :
 (A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{1}{8}$ (D) $\frac{1}{6}$
- B^{3+} and A^{2+} ions are present in :
 (A) tetrahedral voids (B) octahedral, tetrahedral voids
 (C) tetrahedral, octahedral voids (D) octahedral cubic voids

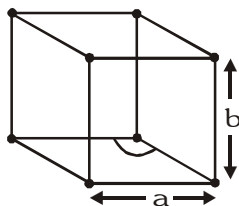
MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
True / False - F - T - T - T - F - T - T - F - T - T Fill in the Blanks $\sqrt{3}a = 4r$ $\sqrt{2}a = 4r$ - cubic close, FCC - Hexagonal close, HCP $c = 4r\sqrt{\frac{2}{3}}$ - least or minimum 12 0.732 $\frac{\sqrt{3}a}{2}$ - decreases, remains constant Match the Column (A) $\rightarrow$ t; (B) $\rightarrow$ p; (C) $\rightarrow$ s; (D) $\rightarrow$ r; (E) $\rightarrow$ q (A) $\rightarrow$ q; (B) $\rightarrow$ s; (C) $\rightarrow$ p; (D) $\rightarrow$ t; (E) $\rightarrow$ r Assertion - Reason Questions - C - D Comprehension Based Questions Comprehension # 1 : 1. (A) 2. (C) 3. (C) 4. (C) 5. (A) Comprehension #2 : 1. (C) 2. (B) 3. (C) 4. (A) 5. (B)		

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

1. A closed packed structure of uniform spheres has the edge length of 534 pm. Calculate the radius of sphere, if it exist in :-
(a) simple cubic lattice (b) BCC lattice (c) FCC lattice
2. Gold crystallizes in a face centred cubic lattice. If the length of the edge of the unit cell is 407 pm, calculate the density of gold as well as its atomic radius assuming it to be spherical. Atomic mass of gold = 197 amu:-
3. The density of KBr is 2.75 g cm^{-3} . The length of the edge of the unit cell is 654 pm. Show the KBr has face centered cubic structure.
4. A crystal of lead (II) sulphide has NaCl structure. In this crystal the shortest distance between Pb^{+2} ion and S^{2-} ions is 297 pm. What is the length of the edge of the unit cell in lead sulphide ? Also calculate the unit cell volume.
5. Iron has body centred cubic lattice structure. The edge length of the unit cell is found to be 286 pm. What is the radius of an iron atom ?
6. Cesium chloride forms a body centred cubic lattice. Cesium and chloride ions are in contact along the body diagonal of the unit cell. The length of the side of the unit cell is 412 pm and Cl^- ion has a radius of 181 pm. Calculate the radius of Cs^+ ion.
7. In a cubic closed packed structure of mixed oxides the lattice is made up of oxide ions, one eight of tetrahedral voids are occupied by divalent ions (A^{2+}) while one half of the octahedral voids occupied trivalent ions (B^{3+}). What is the formula of the oxide ?
8. A solid A^+ and B^- has NaCl type closed packed structure. If the anion has a radius of 250 pm, what should be the ideal radius of the cation ? Can a cation C^+ having a radius of 180 pm be slipped into the tetrahedral site of the crystal of A^+B^- ? Give reasons for your answer.
9. Calculate the value of Avogadro's number from the following data :
Density of NaCl = 2.165 cm^{-3}
Distance between Na^+ and Cl^- in NaCl = 281 pm.
10. If the radius of Mg^{2+} ion, Cs^+ ion, O^{2-} ion, S^{2-} ion and Cl^- ion are 0.65 Å, 1.69 Å, 1.40 Å, 1.84 Å, and 1.81 Å respectively. Calculate the co-ordination numbers of the cations in the crystals of MgS, MgO and CsCl.
11. Iron occurs as bcc as well as fcc unit cell. If the effective radius of an atom of iron is 124 pm. compute the density of iron in both these structures.
12. KCl crystallizes in the same type of lattice as does NaCl. Given that $\frac{r_{\text{Na}^+}}{r_{\text{Cl}^-}} = 0.5$ and $\frac{r_{\text{Na}^+}}{r_{\text{K}^+}} = 0.7$. Calculate :
(a) The ratio of the sides of unit cell for KCl to that for NaCl and
(b) The ratio of densities of NaCl to that for KCl.
13. An element A(Atomic weight = 100) having bcc structure has unit cell edge length 400 pm. Calculate the density of A and number of unit cells and number of atoms in 10 gm of A.
14. Prove that the void space percentage in zinc blende structure is 25%.
15. A unit cell of sodium chloride has four formula units. The edge of length of the unit cell is 0.564 nm. What is the density of sodium chloride.
16. In a cubic crystal of CsCl (density = 3.97 gm/cm^3) the eight corner are occupied by Cl^- ions with Cs^+ ions at the centre. Calculate the distance between the neighbouring Cs^+ and Cl^- ions.
17. The composition of a sample of wustite is $\text{Fe}_{0.93}\text{O}_{1.0}$. What percentage of iron is present in the form of Fe(II) ?
18. RbI crystallizes in bcc structure in which each Rb^+ is surrounded by eight iodide ions each of radius 2.17 Å.

Find the length of one side of RbI unit cell.

19. If NaCl is doped with 10^{-3} mol % SrCl_2 , what is the number of cation vacancies?
20. NaH crystallizes in the same structure as that NaCl. The edge length of the cubic unit cell of NaH is $4.88 \text{ \AA}$.
 (a) Calculate the ionic radius of H^- , provided the ionic radius of Na^+ is $0.95 \text{ \AA}$.
 (b) Calculate the density of NaH.
21. Ice crystallizes in a hexagonal lattice. At the low temperature at which the structure was determined, the lattice constants were $a = 4.53 \text{ \AA}$, and $b = 7.60 \text{ \AA}$ (see figure). How many molecules are contained in a given unit cell? [density (ice) = 0.92 gm/cm^3]



22. Using the data given below, find the type of cubic lattice to which the crystal belongs.
- | | Fe | V | Pb |
|-------------------------------|-----|------|------------|
| a in pm | 286 | 301 | 388 |
| ρ in gm cm^{-3} | | 7.86 | 5.96 12.16 |
23. Prove that void space in fluoride structure per unit volume of unit cell is 0.243.
24. A compound formed by elements X & Y. Crystallizes in a cubic structure, where X is the at corners of the cube and Y is at six face centres. What is the formula of the compound ? If side length is $5 \text{ \AA}$, estimate the density of the solid assuming atomic weight of X and Y as 60 and 90 respectively.
25. The metal nickel crystallizes in a face centred cubic structure. Its density is 8.9 gm/cm^3 . Calculate
 (a) the length of the edge of the unit cell.
 (b) the radius of the nickel atom. [Atomic weight of Ni = 58.89]

CONCEPTUAL	SUBJECTIVE	EXERCISE	ANSWER KEY	EXERCISE-4(A)
1.	267 pm, 231.2 pm, 188.8 pm	2.	19.4 g/cm^3 , 143.9 pm	
3.	$N = 6.023 \times 10^{23} \text{ mol}^{-1}$, At. mass : K = 39, Br = 80			
4.	$a = 5.94 \times 10^{-8} \text{ cm}$, $V = 2.096 \times 10^{-22} \text{ cm}^3$			
5.	123.84 pm	6.	175.8 pm	
7.	AB_2O_4	8.	103.4 pm, No	
9.	6.01×10^{23}	10.	4, 6, 8	
11.	7.887 g/cc , 8.59 gm/cm^3	12.	(a) 1.143, (b) 1.172	
13.	5.188 gm/cm^3 , 6.023×10^{22} atoms of A, 3.0115×10^{22} unit cells			
15.	2.16 gm/cm^3	16.	$3.57 \text{ \AA}$	
17.	15.053	18.	$4.34 \text{ \AA}$	
19.	$6.02 \times 10^{18} \text{ mol}^{-1}$	20.	(a) $1.49 \text{ \AA}$, (b) 1.37 g/cm^3	
21.	4 molecules of H_2O	22.	for Fe is bcc, for V is bcc, for Pd is face centred	
24.	XY_3 , 4.38 g/cm^3	25.	(a) $3.52 \text{ \AA}$, (b) $1.24 \text{ \AA}$	

EXERCISE-04 [B]**BRAIN STORMING SUBJECTIVE EXERCISE**

1. The element chromium exists as a bcc lattice whose unit cell edge is 2.88 Å. The density of chromium is 7.20 g/cc. How many atom does 52.0 g of chromium contain ?
2. The edge length of the unit cell of KCl (NaCl like structure ; fcc) is 6.28 Å. Assuming anion-cation contact along the cell edge, calculate the radius of the potassium ion. ($r_{\text{Cl}^-} = 1.8173 \text{ Å}$).
3. A cubic unit cell contains manganese ions at the corners and fluoride ions at the centre of each edge.
 - (a) What is the empirical formula ?
 - (b) What is the C.N. of the Mn ion?
 - (c) Calculate the edge length of the unit cell if the radius of a Mn ions is 0.65 Å and that of F^- ion is 1.36 Å.
 - (d) Calculate the density of the compound (Mn = 55, F = 19).
4. Silver crystallises in fcc lattice. If edge length of the cell is $4.077 \times 10^{-8} \text{ cm}$ and density is 10.5 g cm^{-3} , calculate the atomic mass of silver.
5. Formula mass of NaCl is 58.45 g mol^{-1} and density of its pure form is 2.167 g cm^{-3} . The average distance between adjacent sodium and chloride ions in the crystal is $2.814 \times 10^{-8} \text{ cm}$. Calculate Avogadro's constant.
6. Thallium chloride, TlCl (240 g mol^{-1}) crystallises in either a simple cubic lattice or a face centered cubic lattice Cl^- ions with Tl^+ ions in the holes. If the density of the solid is 9.00 g cm^{-3} and edge of the unit cell is $3.85 \times 10^{-8} \text{ cm}$, what is the unit geometry ?
7. KF has NaCl structure. What is the distance between K^+ and F^- in KF, if the density is 2.48 g cm^{-3} ?
8. BaTiO_3 crystallizes in the perovskite structure. The structure may be described as a cubic lattice with barium ions occupying the corner of the unit cell, oxide ions occupying the face-centers and titanium ion occupying the center of the unit cell.
 - (a) If titanium is described as occupying holes in BaO lattice, what type of holes does it occupy ?
 - (b) What fraction of this type hole does it occupy ?
9. Find the size of largest sphere that will fit in octahedral void in an ideal FCC crystal as a function of atomic radius 'r'. The insertion of this sphere into void does not distort the FCC lattice. Calculate the packing fraction of FCC lattice when all the octahedral voids are filled by this sphere.
10. Metallic gold crystallises in fcc lattice. The length of the cubic unit cell is $a = 4.07 \text{ Å}$.
 - (a) What is the closest distance between gold atoms.
 - (b) How many "nearest neighbours" does each gold atom have at the distance calculated in (a).
 - (c) What is the density of gold?
 - (d) Prove that the packing fraction of gold is 0.74.
11. Potassium crystallizes in a body-centered cubic lattice with edge length, $a = 5.2 \text{ Å}$.
 - (a) What is the distance between nearest neighbours?
 - (b) What is the distance between next-nearest neighbours?
 - (c) How many nearest neighbours does each K atom have?
 - (d) How many next-neighbours does each K atom have?
 - (e) What is the calculated density of crystalline potassium?
12. The olivine series of minerals consists of crystals in which Fe and Mg ions many substitute for each other causing substitutional impurity defect without changing the volume of the unit cell. In olivine series of minerals, oxide ion exist as FCC with Si^{4+} occupying $\frac{1}{4}$ th of octahedral voids and divalent ions occupying $\frac{1}{4}$ th of tetrahedral voids. The density of forsterite (magnesium silicate) is 3.21 g/cc and that of fayalite (ferrous silicate) is 4.43 g/cc. Find the formula of forsterite and fayalite minerals and the percentage of fayalite in an olivine with a density of 3.88 g/cc.

13. The mineral hawleyite, one form of CdS, crystallizes in one of the cubic lattices, with edge length 5.87 Å. The density of hawleyite is 4.63 g cm^{-3} .
- (i) In which cubic lattice does hawleyite crystallize?
- (ii) Find the Schottky defect in g cm^{-3} .
14. A strong current of trivalent gaseous boron passed through a germanium crystal decreases the density of the crystal due to part replacement of germanium by boron and due to interstitial vacancies created by missing Ge atoms. In one such experiment, one gram of germanium is taken and the boron atoms are found to be 150 ppm by weight, when the density of the Ge crystal decreases by 4%. Calculate the percentage of missing vacancies due to germanium, which are filled up by boron atoms.

Atomic wt. Ge = 72.6, B = 11

BRAIN STORMING SUBJECTIVE EXERCISE	ANSWER KEY	EXERCISE-4(B)
1. $6.05 \times 10^{23} \text{ atoms mol}^{-1}$	2. $r_{k^+} = 1.3227 \text{ Å}$	
3. (a) MnF_3 ($\text{Mn} = 8 \frac{1}{8}$; $\text{F} = 6 \frac{1}{2}$)	(b) 6(fcc) (c) 4.02 Å (d) 2.86 g/cm^3	
4. $107.09 \text{ g mol}^{-1}$	5. $6.05 \times 10^{23} \text{ mol}^{-1}$	
6. simple cubic battice	7. 269 pm	
8. (a) octahedral	(b) $1/4$	
9. 0.414 r, 79.3%	10. (a) 2.88 Å (b) 12 (c) 19.4 g/cc	
11. (a) 4.5 Å (b) 5.2 Å (c) 8 (d)	6 (e) 0.92 g/cm^3	
12. Mg_2SiO_4 , Fe_2SiO_4 , 59%	13. (i) 3.90 (ii) 0.120 g/cc	
14. 2.376 %		

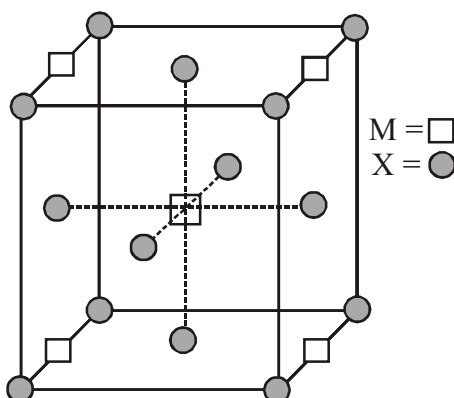
EXERCISE - 05 [A]**JEE-[MAIN] : PREVIOUS YEAR QUESTIONS**

- Q.1 Lithium forms body centred cubic structure. The length of the side of its unit cell is 351 pm. Atomic radius of the lithium will be :- **[AIEEE-2012]**
(A) 152 pm (B) 75 pm (C) 300 pm (D) 240 pm
- Q.2 In a face centred cubic lattice, atom A occupies the corner positions and atom B occupies the face centre positions. If one atom of B is missing from one of the face centred points, the formula of the compound is :- **[AIEEE-2011]**
(A) A_2B_3 (B) A_2B_5 (C) A_2B (D) AB_2
- Q.3 Copper crystallises in fcc lattice with a unit cell edge of 361 pm. The radius of copper atom is:- **[AIEEE-2011]**
(A) 181 pm (B) 108 pm (C) 128 pm (D) 157 pm
- Q.4 Percentages of free space in cubic close packed structure and in body centered packed structure are respectively :- **[AIEEE-2010]**
(A) 48% and 26% (B) 30% and 26% (C) 26% and 32% (D) 32% and 48%
- Q.5 The edge length of a face centered cubic cell of an ionic substance is 508 pm. If the radius of the cation is 110 pm, the radius of the anion is :- **[AIEEE-2010]**
(A) 144 pm (B) 288 pm (C) 398 pm (D) 618 pm
- Q.6 In a compound, atoms of element Y form ccp lattice and those of element X occupy $\frac{2}{3}$ rd of tetrahedral voids. The formula of the compound will be - **[AIEEE-2008]**
(A) X_4Y_3 (B) X_2Y_3 (C) X_2Y (D) X_3Y_4
- Q.7 Total volume of atoms present in a face-centred cubic unit cell of a metal is (r is atomic radius) : **[AIEEE-2006]**
(A) $\frac{24}{3}\pi r^3$ (B) $\frac{12}{3}\pi r^3$ (C) $\frac{16}{3}\pi r^3$ (D) $\frac{20}{3}\pi r^3$
- Q.8 Lattice energy of an ionic compound depends upon - **[AIEEE-2005]**
(A) Size of the ion only (B) Charge on the ion only
(C) Charge on the ion and size of the ion (D) Packing of ions only
- Q.9 An ionic compound has a unit cell consisting of A ions at the corners of a cube and B ions on the centres of the faces of the cube. The empirical formula of this compound would be-**[AIEEE-2005]**
(A) A_2B (B) AB (C) A_3B (D) AB_3
- Q.10 What type of crystal defect is indicated in the diagram below ? **[AIEEE-2004]**
- | | | | | | |
|-----------------|-----------------|------------------------------|-----------------|-----------------|-----------------|
| Na ⁺ | Cl ⁻ | Na ⁺ | Cl ⁻ | Na ⁺ | Cl ⁻ |
| Cl ⁻ | | Cl ⁻ | Na ⁺ | | Na ⁺ |
| Na ⁺ | Cl ⁻ | | Cl ⁻ | Na ⁺ | Cl ⁻ |
| Cl ⁻ | Na | ⁺ Cl ⁻ | Na ⁺ | | Na ⁺ |
- (A) Frenkel defect (B) Schottky defect
(C) Interstitial defect (D) Frenkel and Schottky defects
- Q.11 How many unit cells are present in a cube-shaped ideal crystal of NaCl of mass 1.00g ? **[AIEEE-2003]**
(A) 1.28×10^{21} unit cells (B) 1.71×10^{21} unit cells
(C) 2.57×10^{21} unit cells (D) 5.14×10^{21} unit cells
- Q.12 The no. of atoms per unit cell in B.C.C. & F.C.C. is respectively : **[AIEEE-2002]**
(A) 8, 10 (B) 2, 4 (C) 1, 2 (D) 1, 3

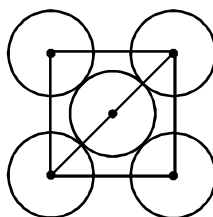
JEE-[MAIN] : PREVIOUS YEAR QUESTIONS		ANSWER KEY		EXERCISE -5[A]	
Q.1	(A)	Q.2	(B)	Q.3	(C)
Q.5	(A)	Q.6	(A)	Q.7	(C)
Q.9	(D)	Q.10	(B)	Q.11	(C)
				Q.12	(B)

EXERCISE - 05 [B]**JEE-[ADVANCED] : PREVIOUS YEAR QUESTIONS**

- Q.1 A compound M_pX_q has cubic close packing (ccp) arrangement of X. Its unit cell structure is shown below. The empirical formula of the compound is : [JEE-2012]



- (A) MX (B) MX_2 (C) M_2X (D) M_5X_{14}
- Q.2 The number of hexagonal faces that present in a truncated octahedron is. [JEE-2011]
- Q.3 The packing efficiency of the two-dimensional square unit cell shown below is [JEE-2010]



- (A) 39.27% (2) 68.02% (C) 74.05% (D) 78.54%
- Q.4 The correct statement(s) regarding defects in solid is (are) [JEE 2009]
- (A) Frenkel defect is usually favoured by a very small difference in the sizes of cation and anion.
- (B) Frenkel defect is a dislocation defect
- (C) Trapping of an electron in the lattice leads to the formation of F-center.
- (D) Schottky defects have no effect on the physical properties of solids.

Paragraph for Question No. 5 to 7

In hexagonal systems of crystals, a frequently encountered arrangement of atoms is described as a hexagonal prism. Here, the top and bottom of the cell are regular hexagons and three atoms are sandwiched in between them. A space-filling model of this structure, called hexagonal close-packed (HCP), is constituted of a sphere on a flat surface surrounded in the same plane by six identical spheres as closely as possible. Three spheres are then placed over the first layer so that they touch each other and represent the second layer. Each one of these three spheres touches three spheres of the bottom layer. Finally, the second layer is covered with a third layer that is identical to the bottom layer in relative position. Assume radius of every sphere to be 'r'.

- Q.5 The number of atoms in this HCP unit cells is [JEE 2008]
- (A) 4 (B) 6 (C) 12 (D) 17

Q.6 The volume of this HCP unit cell is [JEE 2008]

- (A) $24\sqrt{2} r^3$ (B) $16\sqrt{2} r^3$ (C) $12\sqrt{2} r^3$ (D) $\frac{64}{3\sqrt{3}} r^3$

Q.7 The empty space in this HCP unit cell is [JEE 2008]

- (A) 74% (B) 47.6 % (C) 32% (D) 26%

Q.8 Match the crystal system / unit cells mentioned in Column I with their characteristic features mentioned in Column II. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS.

Column I

- (A) simple cubic and face-centred cubic
(B) cubic and rhombohedral
(C) cubic and tetragonal
(D) hexagonal and monoclinic

Column II

- (P) have these cell parameters $a = b = c$
and $\alpha = \beta = \gamma$
(Q) are two crystal systems
(R) have only two crystallographic angles of 90°
(S) belong to same crystal system.

[JEE 2007]

Q.9 The edge length of unit cell of a metal having atomic weight 75 g/mol is 5 Å which crystallizes in cubic lattice. If the density is 2 g/cc then find the radius of metal atom. ($N_A = 6 \times 10^{23}$). Give the answer in pm. [JEE 2006]

Q.10 An element crystallises in FCC lattice having edge length 400 pm. Calculate the maximum diameter which can be placed in interstitial sites without disturbing the structure. [JEE 2005]

Q.11 Which of the following FCC structure contains cations in alternate tetrahedral voids?

- (A) NaCl (B) ZnS (C) Na_2O (D) CaF_2 [JEE 2005]

Q.12 (i) AB crystallizes in a rock salt structure with $A : B = 1 : 1$. The shortest distance between A and B is $Y^{1/3}$ nm. The formula mass of AB is 6.023 Y amu where Y is any arbitrary constant. Find the density in kg m^{-3} .

(ii) If measured density is 20 kg m^{-3} . Identify the type of point defect. [JEE-2004]

Q.13 Marbles of diameter 10 mm each are to be arranged on a flat surface so that their centres lie within the area enclosed by four lines of length each 40 mm. Sketch the arrangement that will give the maximum number of marbles per unit area, that can be enclosed in this manner and deduce the expression to calculate it.

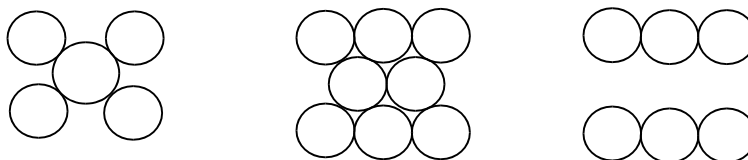
[JEE 2003]

Q.14 A substance A_xB_y crystallises in a FCC lattice in which atoms “A” occupy each corner of the cube and atoms “B” occupy the centres of each face of the cube. Identify the correct composition of the substance A_xB_y .

- (A) AB_3 (B) A_4B_3
(C) A_3B (D) composition cannot be specified [JEE-2002]

- Q.15 The figures given below show the location of atoms in three crystallographic planes in FCC lattice. Draw the unit cell for the corresponding structure and identify these planes in your diagram.

[JEE-2000]



- Q.16 In a solid “AB” having NaCl structure “A” atoms occupy the corners of the cubic unit cell. If all the face-centred atoms along one of the axes are removed, then the resultant stoichiometry of the solid is

(A) AB_2 (B) A_2B (C) A_4B_3 (D) A_3B_4 [JEE-2000]

- Q.17 In any ionic solid $[MX]$ with schottky defects, the number of positive and negative ions are same. [T/F] [JEE-2000]

- Q.18 The coordination number of a metal crystallising in a hcp structure is [JEE-2000]

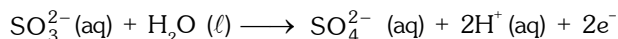
(A) 12 (B) 4 (C) 8 (D) 6

- Q.19 A metal crystallises into two cubic phases, FCC and BCC whose unit cell lengths are 3.5 and 3.0 Å respectively. Calculate the ratio of densities of FCC and BCC. [JEE-1999]

PREVIOUS YEARS QUESTIONS		ANSWER KEY		EXERCISE -5[B]	
Q.1	(B)	Q.2	(8)	Q.3	(D)
Q.5	(B)	Q.6	(A)	Q.7	(D)
Q.8	(A) P, S ; (B) -P,Q ; (C) - Q ; (D) - Q,R				
Q.9	216.5 pm	Q.10	117.1 pm	Q.11	(B)
Q.12	(i) = 5 kg m ⁻³				
	(ii) <u>There is huge difference in theoretically calculated density and observed density. It is only possible if some foreign species occupies interstitial space i.e. substitution defect.</u>				
Q.13		Q.14	(A)	Q.15	
Q.17	True	Q.18	(A)	Q.19	1.259
				Q.16	(D)

EXERCISE-01**CHECK YOUR GRASP****SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)**

1. 50 mL of 0.1 M solution of a salt reacted with 25 mL of 0.1 M solution of sodium sulphite. The half reaction for the oxidation of sulphite ion is :-



If the oxidation number of metal in the salt was 3, what would be the new oxidation number of metal :

- (A) zero (B) 1 (C) 2 (D) 4
2. An element A in a compound ABD has oxidation number A^n . It is oxidised by $\text{Cr}_2\text{O}_7^{2-}$ in acid medium. In the experiment 1.68×10^{-3} moles of $\text{K}_2\text{Cr}_2\text{O}_7$ were used for 3.26×10^{-3} moles of ABD. The new oxidation number of A after oxidation is :-
- (A) 3 (B) $3 - n$ (C) $n - 3$ (D) $+n$
3. The incorrect order of decreasing oxidation number of S in compounds is :-
- (A) $\text{H}_2\text{S}_2\text{O}_7 > \text{Na}_2\text{S}_4\text{O}_6 > \text{Na}_2\text{S}_2\text{O}_3 > \text{S}_8$ (B) $\text{H}_2\text{SO}_5 > \text{H}_2\text{SO}_3 > \text{SCl}_2 > \text{H}_2\text{S}$
 (C) $\text{SO}_3 > \text{SO}_2 > \text{H}_2\text{S} > \text{S}_8$ (D) $\text{H}_2\text{SO}_4 > \text{SO}_2 > \text{H}_2\text{S} > \text{H}_2\text{S}_2\text{O}_8$
4. Which reaction does not represent autoredox or disproportionation :-
- (A) $\text{Cl}_2 + \text{OH}^- \longrightarrow \text{Cl}^- + \text{ClO}_3^- + \text{H}_2\text{O}$ (B) $2\text{H}_2\text{O}_2 \longrightarrow \text{H}_2\text{O} + \text{O}_2$
 (C) $2\text{Cu}^+ \longrightarrow \text{Cu}^{+2} + \text{Cu}$ (D) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \longrightarrow \text{N}_2 + \text{Cr}_2\text{O}_3 + 4\text{H}_2\text{O}$
5. Match List-I (Compounds) with List-II (Oxidation states of nitrogen) and select answer using the codes given below the lists :-

List-I		List-II	
(a)	NaN_3	1.	+5
(b)	N_2H_2	2.	+2
(c)	NO	3.	-1/3
(d)	N_2O_5	4.	-1
Code :			
(A)	3	4	2
(B)	4	3	2
(C)	3	4	1
(D)	4	3	1

6. Which of the following is a redox reaction :-
- (A) $2\text{CrO}_4^{2-} + 2\text{H}^+ \rightarrow \text{Cr}_2\text{O}_7^{2-} + \text{H}_2\text{O}$ (B) $\text{CuSO}_4 + 4\text{NH}_3 \rightarrow [\text{Cu}(\text{NH}_3)_4]\text{SO}_4$
 (C) $\text{Na}_2\text{S}_2\text{O}_3 + \text{I}_2 \rightarrow \text{Na}_2\text{S}_4\text{O}_6 + \text{NaI}$ (D) $\text{Cr}_2\text{O}_7^{2-} + 2\text{OH}^- \rightarrow 2\text{CrO}_4^{2-} + \text{H}_2\text{O}$
7. In which of the following reaction is there a change in the oxidation number of nitrogen atoms :-
- (A) $2\text{NO}_2 \rightarrow \text{N}_2\text{O}_4$ (B) $\text{NH}_3 + \text{H}_2\text{O} \rightarrow \text{NH}_4^+ + \text{OH}^-$
 (C) $\text{N}_2\text{O}_5 + \text{H}_2\text{O} \rightarrow 2\text{HNO}_3$ (D) none
8. In the reaction $x\text{HI} + y\text{HNO}_3 \longrightarrow \text{NO} + \text{I}_2 + \text{H}_2\text{O}$
- (A) $x = 3, y = 2$ (B) $x = 2, y = 3$ (C) $x = 6, y = 2$ (D) $x = 6, y = 1$
9. For the redox reaction : $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \longrightarrow \text{Mn}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$
- the correct stoichiometric coefficients of MnO_4^- , $\text{C}_2\text{O}_4^{2-}$ and H^+ are respectively
- (A) 2,5,16 (B) 16,5,2 (C) 5,16,2 (D) 2,16,5

10. Which of the following relations is incorrect :-
 (A) $3 \text{ N Al}_2(\text{SO}_4)_3 = 0.5 \text{ M Al}_2(\text{SO}_4)_3$ (B) $3 \text{ M H}_2\text{SO}_4 = 6 \text{ N H}_2\text{SO}_4$
 (C) $1 \text{ M H}_3\text{PO}_4 = 1/3 \text{ N H}_3\text{PO}_4$ (D) $1 \text{ M Al}_2(\text{SO}_4)_3 = 6 \text{ N Al}_2(\text{SO}_4)_3$
11. The mass of oxalic acid crystals ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$) required to prepare 50 mL of a 0.2 N solution is :-
 (A) 4.5 g (B) 6.3 g (C) 0.63 g (D) 0.45 g
12. 125 mL of 63% (w/v) $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ is made to react with 125 mL of a 40% (w/v) NaOH solution. The resulting solution is :-
 (A) neutral (B) acidic (C) strongly acidic (D) alkaline
13. A certain weight of pure CaCO_3 is made to react completely with 200 mL of an HCl solution to give 224 mL of CO_2 gas at STP. The normality of the HCl is :-
 (A) 0.05 N (B) 0.1 N (C) 1.0 N (D) 0.2 N
14. The volume of 1.5 M H_3PO_4 solution required to neutralize exactly 90 mL of a 0.5 M $\text{Ba}(\text{OH})_2$ solution is :-
 (A) 10 mL (B) 30 mL (C) 20 mL (D) 60 mL
15. Volume V_1 mL of 0.1 M $\text{K}_2\text{Cr}_2\text{O}_7$ is needed for complete oxidation of 0.678 g N_2H_4 in acidic medium. The volume of 0.3 M KMnO_4 needed for same oxidation in acidic medium will be :-
 (A) $\frac{2}{5} V_1$ (B) $\frac{5}{2} V_1$ (C) $113 V_1$ (D) can't say
16. If equal volumes of 0.1 M KMnO_4 and 0.1 M $\text{K}_2\text{Cr}_2\text{O}_7$ solutions are allowed to oxidise Fe^{2+} to Fe^{3+} in acidic medium, then Fe^{2+} oxidised will be :-
 (A) more by KMnO_4 (B) more by $\text{K}_2\text{Cr}_2\text{O}_7$
 (C) equal in both cases (D) can't be determined
17. If 10 g of V_2O_5 is dissolved in acid and is reduced to V^{2+} by zinc metal, how many mole I_2 could be reduced by the resulting solution if it is further oxidised to VO^{2+} ions ?
 [Assume no change in state of Zn^{2+} ions] ($V = 51, O = 16, I = 127$) :
 (A) 0.11 mole of I_2 (B) 0.22 mole of I_2 (C) 0.055 mole of I_2 (D) 0.44 mole of I_2
18. Given that 50.0 mL of 0.01 M $\text{Na}_2\text{S}_2\text{O}_3$ solution and 5×10^{-4} mole of Cl_2 react according to equation,

$$\text{Cl}_2 (\text{g}) + \text{S}_2\text{O}_3^{2-} \longrightarrow \text{SO}_4^{2-} + \text{Cl}^- + \text{S}$$

 Answer the following :
 (i) The balanced molecular equation is :
 (A) $\text{Cl}_2 + \text{H}_2\text{O} + \text{Na}_2\text{S}_2\text{O}_3 \longrightarrow \text{Na}_2\text{SO}_4 + \text{S} + 2\text{HCl}$ (B) $\text{Cl}_2 + \text{Na}_2\text{S}_2\text{O}_3 \longrightarrow 2\text{NaCl} + \text{Na}_2\text{SO}_4$
 (C) $\text{Cl}_2 + \text{S}_2\text{O}_3^{2-} \longrightarrow \text{SO}_4^{2-} + \text{S} + \text{Cl}^-$ (D) none of these
 (ii) How many moles of $\text{S}_2\text{O}_3^{2-}$ are in the above sample :-
 (A) 0.00050 (B) 0.0025 (C) 0.01 (D) 0.02
 (iii) How many equivalents of oxidising agents are in this sample for the above reaction :-
 (A) 0.001 (B) 0.080 (C) 0.020 (D) 0.010
 (iv) What is the molarity of Na_2SO_4 in this solution :-
 (A) 0.080 M (B) 0.040 M (C) 0.020 M (D) 0.010 M
19. 0.3 g of an oxalate salt was dissolved in 100 mL solution. The solution required 90 mL of N/20 KMnO_4 for complete oxidation. The % of oxalate ion in salt is :-
 (A) 33% (B) 66% (C) 70% (D) 40%
20. A 0.518 g sample of limestone is dissolved in HCl and then the calcium is precipitated as CaC_2O_4 . After filtering and washing the precipitate, it requires 40.0 mL of 0.250 N KMnO_4 solution acidified with H_2SO_4 to titrate it as. The percentage of CaO in the sample is :-

$$\text{MnO}_4^- + \text{H}^+ + \text{C}_2\text{O}_4^{2-} \longrightarrow \text{Mn}^{2+} + \text{CO}_2 + 2\text{H}_2\text{O}$$

 (A) 54.0 % (B) 27.1 % (C) 42 % (D) 84 %

EXERCISE-02**BRAIN TEASERS****SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THEN ONE CORRECT ANSWERS)**

1. 1 mol of iron (Fe) reacts completely with 0.65 mol O_2 to give a mixture of only FeO and Fe_2O_3 . Mole ratio of ferrous oxide to ferric oxide is :-
(A) 3 : 2 (B) 4 : 3 (C) 20 : 13 (D) none of these
2. The molar ratio of Fe^{++} to Fe^{+++} in a mixture of $FeSO_4$ and $Fe_2(SO_4)_3$ having equal number of sulphate ion in both ferrous and ferric sulphate is :-
(A) 1 : 2 (B) 3 : 2 (C) 2 : 3 (D) can't be determined
3. If a piece of iron gains 10% of its weight due to partial rusting into Fe_2O_3 . The percentage of total iron that has rusted is :-
(A) 23 (B) 13 (C) 23.3 (D) 25.67
4. An ore of iron, Wustite has the formula $Fe_{0.93}O_{1.00}$. The mole fraction of total iron present in the form of Fe(II) is :-
(A) 0.82 (B) 0.85 (C) 0.15 (D) 0.37
5. HNO_3 oxidises NH_4^+ ions to nitrogen and itself gets reduced to NO_2 . The moles of HNO_3 required by 1 mol of $(NH_4)_2SO_4$ is :-
(A) 4 (B) 5 (C) 6 (D) 2
6. 25 mL of a 0.1 M solution of a stable cation of transition metal Z reacts exactly with 25 mL of 0.04 M acidified $KMnO_4$ solution. Which of the following is most likely to represent the change in oxidation state of Z correctly :-
(A) $Z^+ \rightarrow Z^{2+}$ (B) $Z^{2+} \rightarrow Z^{3+}$ (C) $Z^{3+} \rightarrow Z^{4+}$ (D) $Z^{2+} \rightarrow Z^{4+}$
7. How many litres of Cl_2 at S.T.P. will be liberated by oxidation of NaCl with 10 g $KMnO_4$:-
(A) 3.54 litres (B) 7.08 litres (C) 1.77 litres (D) none of these
8. During the disproportionation of iodine to iodide and iodate ions, the ratio of iodate and iodide ions formed in alkaline medium is :-
(A) 1 : 5 (B) 5 : 1 (C) 3 : 1 (D) 1 : 3
9. $28 NO_3^- + 3As_2S_3 + 4H_2O \rightarrow 6AsO_4^{3-} + 28 NO + 9SO_4^{2-} + H^+$
What will be the equivalent mass of As_2S_3 in above reaction
(A) $\frac{M.wt.}{2}$ (B) $\frac{M.wt.}{4}$ (C) $\frac{M.wt.}{24}$ (D) $\frac{M.wt.}{28}$
10. When ZnS is boiled with strong nitric acid, the products are zinc nitrate, sulphuric acid and nitrogen dioxide. What are the changes in the oxidation numbers of Zn, S and N.
(A) +2, +4, -1 (B) +2, +6, -2 (C) 0, +4, -2 (D) 0, +8, -1
11. When arsenic sulphide is boiled with NaOH, sodium arsenite and sodium thioarsenite are formed
 $x As_2S_3 + y NaOH \longrightarrow Na_3AsO_3 + x Na_3AsS_3 + \frac{y}{2} H_2O$. What are the values of x and y ?
(A) 1, 6 (B) 2, 8 (C) 2, 6 (D) 1, 4
12. An element forms two different sulphates in which its weight % is 28 and 37. What is the ratio of oxidation numbers of the element in these sulphates ?
(A) 1 : 2 (B) 1 : 3 (C) 2 : 1 (D) 3 : 2
13. CN^- is oxidised by NO_3^- in presence of acid :
$$a CN^- + b NO_3^- + c H^+ \longrightarrow (a + b) NO + a CO_2 + \frac{c}{2} H_2O$$

What are the values of a, b, c in that order :
(A) 3, 7, 7 (B) 3, 10, 7 (C) 3, 10, 10 (D) 3, 7, 10
14. Which of the following solutions will exactly oxidize 25 mL of an acid solution of 0.1 M Fe (II) oxalate :-
(A) 25 mL of 0.1 M $KMnO_4$ (B) 25 mL of 0.2 M $KMnO_4$
(C) 25 mL of 0.6 M $KMnO_4$ (D) 15 mL of 0.1 M $KMnO_4$

15. 4.9 gm of $K_2Cr_2O_7$ is taken to prepare 0.1 L of the solution. 10 mL of this solution is further taken to oxidise Sn^{2+} ion into Sn^{4+} ion Sn^{4+} so produced is used in 2nd reaction to prepare Fe^{3+} ion then the millimoles of Fe^{3+} ion formed will be (assume all other components are in sufficient amount) [Molar mass of $K_2Cr_2O_7 = 294$ g].
(A) 5 (B) 20 (C) 10 (D) none of these
16. The following equations are balanced atomwise and chargewise.
(i) $Cr_2O_7^{2-} + 8H^+ + 3H_2O_2 \longrightarrow 2Cr^{3+} + 7H_2O + 3O_2$
(ii) $Cr_2O_7^{2-} + 8H^+ + 5H_2O_2 \longrightarrow 2Cr^{3+} + 9H_2O + 4O_2$
(iii) $Cr_2O_7^{2-} + 8H^+ + 7H_2O_2 \longrightarrow 2Cr^{3+} + 11H_2O + 5O_2$
The precise equation/equations representing the oxidation of H_2O_2 is /are :
(A) (i) only (B) (ii) only (C) (iii) only (D) all the three
17. 35 mL sample of hydrogen peroxide gives of 500 mL of O_2 at 27 C and 1 atm pressure. Volume strength of H_2O_2 sample will be :-
(A) 10 volume (B) 13 volumes (C) 11 volume (D) 12 volume
18. 20 mL of 0.1 M solution of compound $Na_2CO_3 \cdot NaHCO_3 \cdot 2H_2O$ is titrated against 0.05 M HCl, x mL of HCl is used when phenolphthalein is used as an indicator and y mL of HCl is used when methyl orange is the indicator in two separate titrations. Hence (y - x) is :-
(A) 40 mL (B) 80 mL (C) 120 mL (D) none of these
19. 0.10 g of a sample containing $CuCO_3$ and some inert impurity was dissolved in dilute sulphuric acid and volume made up to 50 mL. This solution was added into 50 mL of 0.04 M KI solution where copper precipitates as CuI and I^- is oxidized into I_3^- . A 10 mL portion of this solution is taken for analysis, filtered and made up free I_3^- and then treated with excess of acidic permanganate solution. Liberated iodine required 20 mL of 2.5 mM sodium thiosulphate solution to reach the end point.
Determine weight percentage of $CuCO_3$ in the original sample.
(A) 7.41 (B) 74.1 (C) 61.75 (D) none of these
20. A 150 mL of solution of I_2 is divided into two unequal parts. I part reacts with hypo solution in acidic medium. 15 mL of 0.4 M hypo was consumed. II part was added with 100 mL of 0.3 M NaOH solution. Residual base required 10 mL of 0.3 M H_2SO_4 solution for complete neutralization. What was the initial concentration of I_2 ?
(A) 0.08 M (B) 0.1 M (C) 0.2 M (D) none of these
21. A mixture of H_2SO_4 and $H_2C_2O_4$ (oxalic acid) and some inert impurity weighing 3.185 g was dissolved in water and the solution made up to 1 litre, 10 mL of this solution required 3 mL of 0.1 N NaOH for complete neutralization. In another experiment 100 mL of the same solution in hot condition required 4 mL of 0.02M $KMnO_4$ solution for complete reaction. The wt. % of H_2SO_4 in the mixture was :-
(A) 40 (B) 50 (C) 60 (D) 80
22. 0.80 g of sample of impure potassium dichromate was dissolved in water and made upto 500 mL solution. 25 mL of this solution treated with excess of KI in acidic medium and I_2 liberated required 24 mL of a sodium thiosulphate solution. 30 mL of this sodium thiosulphate solution required 15 mL of N/20 solution of pure potassium dichromate. What was the percentage of $K_2Cr_2O_7$ in given sample?
(A) 73.5 % (B) 75.3 % (C) 36.75 % (D) none of these

BRAIN TEASERS						ANSWER KEY				EXERCISE -2					
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	B	B	C	B	C	D	A	A	D	D	A	D	D	D	C
Que.	16	17	18	19	20	21	22								
Ans.	A	B	B	B	B	A	A								

EXERCISE-03**MISCELLANEOUS TYPE QUESTIONS****TRUE / FALSE**

1. In a compound, all the atoms of a particular element have the same oxidation number.
2. In H_2O_2 , both oxygen atoms have same oxidation number but in $\text{Na}_2\text{S}_2\text{O}_3$, the two S-atoms do not have same oxidation number.
3. In the reaction :

$$3 \text{Cl}_2 + 6 \text{NaOH} \longrightarrow 5 \text{NaCl} + \text{NaClO}_3 + 3\text{H}_2\text{O}$$
 Cl_2 acts purely as an oxidizing agent.
4. In a redox reaction, the oxidation number of an element can either increase or decrease but both cannot happen simultaneously.
5. In CaOCl_2 both the chlorine atom are in same oxidation state.

FILL IN THE BLANKS

1. Oxidizing agent (or oxidant) is a substance in which oxidation number of one of the atoms
2. Reducing agent (or reductant) is a substance which electrons.
3. In the reaction $2 \text{H}_2\text{O}_2 \longrightarrow 2 \text{H}_2\text{O} + \text{O}_2$, hydrogen peroxide is
4. In the reaction $2 \text{KClO}_3 \longrightarrow 2 \text{KCl} + 3 \text{O}_2$, the element which has been oxidised is and the element which has been reduced is
5. The compound $\text{YbBa}_2\text{Cu}_3\text{O}_7$ which shows superconductivity, has copper in oxidation state Assume that the rare earth element ytterbium is in the usual + 3 oxidation state.
6. In HCN oxidation number of carbon is
7. The reaction $\text{NH}_4\text{NO}_2 \rightarrow \text{N}_2 + 2\text{H}_2\text{O}$ disproportionation reaction.

MATCH THE COLUMN

1.

Column-I	Column-II
(A) When Bi_2S_3 converted into Bi^{5+} and S	(p) 18
(B) When $\text{Al}_2(\text{Cr}_2\text{O}_7)_3$ reduced into Cr^{3+} in acidic medium	(q) 11
(C) When FeS_2 converted into Fe_2O_3 and SO_2	(r) 2
(D) When $\text{Mn}(\text{NO}_3)_2$ converted into MnO_4^{2-} and NO	(s) 10
2.

Column-I	Column-II
(A) Eq. wt. = $\frac{\text{Molecular weight}}{33}$	(p) When CrI_3 oxidises into $\text{Cr}_2\text{O}_7^{2-}$ and IO_4^-
(B) Eq. wt. = $\frac{\text{Molecular weight}}{27}$	(q) When $\text{Fe}(\text{SCN})_2$ oxidises into Fe^{3+} , SO_4^{2-} , CO_3^{2-} and NO_3^-
(C) Eq. wt. = $\frac{\text{Molecular weight}}{28}$	(r) When NH_4SCN oxidizes into SO_4^{2-} , CO_3^{2-} and NO_3^-
(D) Eq. wt. = $\frac{\text{Molecular weight}}{24}$	(s) When As_2S_3 oxidises into AsO_3^- and SO_4^{2-}

3.

Column-I		Column-II	
(A)	$\underline{\text{P}_2\text{H}_4} \longrightarrow \text{PH}_3 + \text{P}_4\text{H}_2$	(p)	$E = \frac{3M}{4}$
(B)	$\underline{\text{I}_2} \longrightarrow \text{I}^- + \text{IO}_3^-$	(q)	$E = \frac{3M}{5}$
(C)	$\text{MnO}_4^- + \text{Mn}^{2+} + \text{H}_2\text{O} \longrightarrow \underline{\text{Mn}_3\text{O}_4} + \text{H}^+$	(r)	$E = \frac{15M}{26}$
(D)	$\underline{\text{H}_3\text{PO}_2} \longrightarrow \text{PH}_3 + \text{H}_3\text{PO}_3$	(s)	$E = \frac{5M}{6}$

ASSERTION & REASON

These questions contains, Statement I (assertion) and Statement II (reason).

(A) Statement-I is true, Statement-II is true ; Statement-II is correct explanation for Statement-I.

(B) Statement-I is true, Statement-II is true ; Statement-II is NOT a correct explanation for statement-I

(C) Statement-I is true, Statement-II is false

(D) Statement-I is false, Statement-II is true

1. **Statement-I** : Oxidation involves loss of electrons and reduction involves gain of electrons.

Because

Statement-II : The overall reaction in which oxidation and reduction occur simultaneously is called redox reaction.

2. **Statement-I** : H_2SO_4 cannot act as reducing agent.

Because

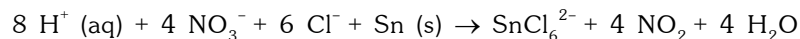
Statement-II : Sulphur cannot increase its oxidation number beyond +6.

3. **Statement-I** : The oxidation state of superoxide ion in KO_2 , CsO_2 and RbO_2 is $-1/2$.

Because

Statement-II : Since the oxidation state of an alkali metal in any compound is always +1, the oxidation state of oxygen is $-1/2$ in the O_2^- ion.

4. **Statement-I** : In the redox reaction



the reducing agent is Sn (s),

Because

Statement-II : In balancing half reaction, $\text{S}_2\text{O}_3^{2-} \rightarrow \text{S}(\text{s})$, the number of electrons added on the left is 4.

5. **Statement-I** : Among Br^- , O_2^{2-} , H^- and NO_3^- , the ions that could not act as oxidising agents are Br^- and H^- .

Because

Statement-II : $\text{Br}^\ominus$ and H^- could not be reduced.

COMPREHENSION BASED QUESTIONS**Comprehension # 1**

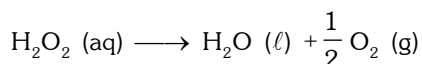
Oleum is considered as a solution of SO_3 in H_2SO_4 , which is obtained by passing SO_3 in solution of H_2SO_4 . When 100 g sample of oleum is diluted with desired weight of H_2O then the total mass of H_2SO_4 obtained after dilution is known as % labelling in oleum.

For example, a oleum bottle labelled as '109% H_2SO_4 ' means the 109 g total mass of pure H_2SO_4 will be formed when 100 g of oleum is diluted by 9 g of H_2O which combines with all the free SO_3 present in oleum to form H_2SO_4 as $\text{SO}_3 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4$

- What is the % of free SO_3 in an oleum that is labelled as '104.5 % H_2SO_4 '?
(A) 10 (B) 20 (C) 40 (D) none of these
- 9.0 g water is added into oleum sample labelled as '112% H_2SO_4 ' then the amount of free SO_3 remaining in the solution is :
(A) 14.93 L at STP (B) 7.46 L at STP (C) 3.73 L at STP (D) 11.2 L at STP
- If excess water is added into a bottle sample labelled as '112 % H_2SO_4 ' and is reacted with 5.3 g Na_2CO_3 , then find the volume of CO_2 evolved at 1 atm pressure and 300 K temperature after the completion of the reaction :
(A) 2.46 L (B) 24.6 L (C) 1.23 L (D) 12.3 L
- 1 g of oleum sample is diluted with water. The solution required 54 mL of 0.4 N NaOH for complete neutralization. The % of free SO_3 in the sample is :
(A) 74 (B) 26 (C) 20 (D) none of these

Comprehension # 2

The strength of H_2O_2 is expressed in several ways like molarity, normality, % (w/V), volume strength, etc. The strength of "10 V" means 1 volume of H_2O_2 on decomposition gives 10 volumes of oxygen at STP or 1 litre of H_2O_2 gives 10 litre of O_2 at STP. The decomposition of H_2O_2 is shown as under :



H_2O_2 can act as oxidising as well as reducing agent, as oxidizing agent H_2O_2 converted into H_2O and as reducing agent H_2O_2 converted into O_2 , both cases its n-factor is 2.

$\therefore$ Normality of H_2O_2 solution = 2 Molarity of H_2O_2 solution

- What is the molarity of "11.2 V" of H_2O_2 ?
(A) 1 M (B) 2 M (C) 5.6 M (D) 11.2 M
- What is the percentage strength (% w/V) of "11.2 V" H_2O_2 ?
(A) 1.7 (B) 3.4 (C) 34 (D) none of these
- 20 mL of H_2O_2 solution is reacted with 80 mL of 0.05 M KMnO_4 in acidic medium then what is the volume strength of H_2O_2 ?
(A) 2.8 (B) 5.6 (C) 11.2 (D) none of these
- 40 g $\text{Ba}(\text{MnO}_4)_2$ (mol. wt. = 375) sample containing some inert impurities in acidic medium is completely reacted with 125 mL of "33.6 V" of H_2O_2 . What is the percentage purity of the sample ?
(A) 28.12 % (B) 70.31 % (C) 85 % (D) none of these

Comprehension # 3

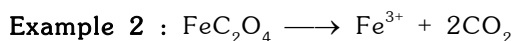
$$\text{Equivalent weight} = \frac{\text{Molecular weight} / \text{Atomic weight}}{n - \text{factor}}$$

n-factor is very important in redox as well as non-redox reactions. With the help of n-factor we can predict the molar ratio of the reactant species taking part in reactions. The reciprocal of n-factor's ratio of the reactants is the molar ratio of the reactants.

In general n-factor of acid/base is number of moles of H^+ / OH^- furnished per mole of acid/base. n-factor of a reactant is no. of moles of electrons lost or gained per mole of reactant.

Example 1 :

- In acidic medium : KMnO_4 ($n = 5$) $\longrightarrow \text{Mn}^{2+}$
- In basic medium : KMnO_4 ($n = 3$) $\longrightarrow \text{Mn}^{2+}$
- In neutral medium : KMnO_4 ($n = 1$) $\longrightarrow \text{Mn}^{6+}$



Total no. of moles of e^- lost by 1 mole of $\text{FeC}_2\text{O}_4 = 1 + 1 \times 2 \Rightarrow 3$

$\therefore$ n-factor of $\text{FeC}_2\text{O}_4 = 3$

- n-factor of $\text{Ba}(\text{MnO}_4)_2$ in acidic medium is :
(A) 2 (B) 6 (C) 10 (D) none of these
- For the reaction,
 $\text{H}_3\text{PO}_2 + \text{NaOH} \longrightarrow \text{NaH}_2\text{PO}_2 + \text{H}_2\text{O}$
What is the equivalent weight of H_3PO_2 ? (mol. wt. is M)
(A) M (B) M/2 (C) M/3 (D) none of these
- For the reaction, $\text{Fe}_{0.95}\text{O}$ (molar mass : M) $\longrightarrow \text{Fe}_2\text{O}_3$. What is the eq. wt. of $\text{Fe}_{0.95}\text{O}$?
(A) $\frac{M}{0.85}$ (B) $\frac{M}{0.95}$ (C) $\frac{M}{0.8075}$ (D) none of these
- In the reaction, $x\text{VO} + y\text{Fe}_2\text{O}_3 \longrightarrow \text{FeO} + \text{V}_2\text{O}_5$. What is the value of x and y respectively ?
(A) 1, 1 (B) 2, 3 (C) 3, 2 (D) none of these

MISCELLANEOUS TYPE QUESTION	ANSWER KEY	EXERCISE -3
True / False 1. F 2. T 3. F 4. F 5. F Fill in the Blanks 1. decrease 2. loses 3. oxidized as well as reduced 4. oxygen has been oxidised ($\text{O}^{2-} \longrightarrow \text{O}_2$) ; chlorine has been reduced ($\text{Cl}^{+5} \longrightarrow \text{Cl}^{-1}$) 5. $x = +7/3$ 6. +2 7. is not Match the Column 1. (A) $\rightarrow$ s ; (B) $\rightarrow$ p ; (C) $\rightarrow$ q ; (D) $\rightarrow$ r 2. (A) $\rightarrow$ q ; (B) $\rightarrow$ p ; (C) $\rightarrow$ s ; (D) $\rightarrow$ r 3. (A) $\rightarrow$ s ; (B) $\rightarrow$ q ; (C) $\rightarrow$ r ; (D) $\rightarrow$ p Assertion - Reason Questions 1. B 2. A 3. A 4. B 5. A Comprehension Based Questions Comprehension #1 : 1. (B) 2. (C) 3. (C) 4. (B) Comprehension #2 : 1. (A) 2. (B) 3. (B) 4. (B) Comprehension #3 : 1. (C) 2. (A) 3. (C) 4. (B)		

EXERCISE-04 [A]**CONCEPTUAL SUBJECTIVE EXERCISE**

- Calculate the oxidation number of underlined elements in the following compounds :
(a) $K[\underline{Co}(C_2O_4)_2 \cdot (NH_3)_2]$ (b) $K_4\underline{P}_2O_7$ (c) $\underline{Cr}O_2Cl_2$
(d) $Na_2[\underline{Fe}(CN)_5NO^+]$ (e) $\underline{Mn}_3O_4$ (f) $Ca(\underline{Cl}O_2)_2$
(g) $[\underline{Fe}(NO)(H_2O)_5]SO_4$ (h) $\underline{Zn}O_2^{2-}$ (c) $\underline{Fe}_{0.93}O$
- Write balanced net ionic equation for the following reactions in acidic solution.
(a) $S_4O_6^{2-}(aq) + Al(s) \longrightarrow H_2S(aq) + Al^{3+}(aq)$
(b) $S_2O_3^{2-}(aq) + Cr_2O_7^{2-}(aq) \longrightarrow S_4O_6^{2-}(aq) + Cr^{3+}(aq)$
(c) $ClO_3^-(aq) + As_2S_3(s) \longrightarrow Cl^-(aq) + H_2AsO_4^-(aq) + HSO_4^-(aq)$
(d) $IO_3^-(aq) + Re(s) \longrightarrow ReO_4^-(aq) + I^-(aq)$
(e) $HSO_4^-(aq) + As_4(s) + Pb_3O_4(s) \longrightarrow PbSO_4(s) + H_2AsO_4^-(aq)$
(f) $HNO_2(aq) \longrightarrow NO_3^- + NO(g)$
- Write balanced net ionic equations for the following reactions in basic solution :
(a) $C_4H_4O_6^{2-}(aq) + ClO_3^-(aq) \longrightarrow CO_3^{2-}(aq) + Cl^-(aq)$
(b) $Al(s) + BiONO_3(s) \longrightarrow Bi(s) + NH_3(aq) + Al(OH)_4^-(aq)$
(c) $H_2O_2(aq) + Cl_2O_7(aq) \longrightarrow ClO_2^-(aq) + O_2(g)$
(d) $Tl_2O_3(s) + NH_2OH(aq) \longrightarrow TlOH(s) + N_2(g)$
(e) $Cu(NH_3)_4^{2+}(aq) + S_2O_4^{2-}(aq) \longrightarrow SO_3^{2-}(aq) + Cu(s) + NH_3(aq)$
(f) $Mn(OH)_2(s) + MnO_4^-(aq) \longrightarrow MnO_2(s)$
- $KMnO_4$ oxidizes X^{n+} ion to XO_3^- , itself changing to Mn^{2+} in acid medium. 2.68×10^{-3} mole of X^{n+} requires 1.61×10^{-3} mole of MnO_4^- . What is the value of n? Also calculate the atomic mass of X, if the weight of 1g equivalent of XCl_n is 56.
- In a quantitative determination of iron in an ore, an analyst converted 0.40 g, of the ore into its ferrous. This required 40.00 mL of 0.1 N solution of $KMnO_4$ for titration.
(i) How many milliequivalents of $KMnO_4$ does 40.00 mL of 0.1 N solution represent?
(ii) How many equivalents of iron were present in the sample of the ore taken for analysis?
(iii) How many grams of iron were present in the sample?
(iv) What is the percentage of iron in the ore?
(v) What is the molarity of $KMnO_4$ solution used?
(vi) How many moles of $KMnO_4$ were used for titration ? (Fe = 56)
- The mixture of CuS (molar weight = M_1) and Cu_2S (molecular weight = M_2) oxidised by $KMnO_4$ (molecular weight = M_3) in acidic medium, the product obtained are Cu^{2+} , SO_2 . Find the equivalent weight of CuS , Cu_2S and $KMnO_4$ respectively.
- Consider the reaction $H^+ + IO_4^- + I^- \rightarrow I_2 + H_2O$. Find the ratio of coefficients of IO_4^- , I^- and I_2 .
- A dilute solution of H_2SO_4 is made by adding 5 mL of 3N H_2SO_4 to 245 mL of water. Find the normality and molarity of the solution.
- What volume at NTP of gaseous ammonia will be required to be passed into 30 cc of N – H_2SO_4 solution to bring down the acid strength of the latter to 0.2 N.

10. A solution containing 4.2 g of KOH and $\text{Ca}(\text{OH})_2$ is neutralized by an acid. It consumes 0.1 equivalent of acid, calculate the percentage composition of the sample.
11. How many mL of 0.1 N HCl are required to react completely with 1 g mixture of Na_2CO_3 and NaHCO_3 containing equimolar amounts of two?
12. 0.5 g of fuming H_2SO_4 (oleum) is diluted with water. The solution requires 26.7 mL of 0.4N NaOH for complete neutralization. Find the % of free SO_3 in the sample of oleum.
13. 10 g CaCO_3 were dissolved in 250 mL of M HCl and the solution was boiled. What volume of 2 M KOH would be required to equivalence point after boiling? Assume no change in volume during boiling.
14. H_3PO_4 is a tri basic acid and one of its salt is NaH_2PO_4 . What volume of 1 M NaOH solution should be added to 12 g of NaH_2PO_4 to convert it into Na_3PO_4 ?
15. 1.64 g of mixture of CaCO_3 and MgCO_3 was dissolved in 50 mL of 0.8 M HCl. The excess of acid required 16 mL of 0.25 M NaOH for neutralization. Calculate the percentage of CaCO_3 and MgCO_3 in the sample.
16. 1.5 g of chalk were treated with 10 mL of 4N – HCl. The chalk was dissolved and the solution made to 100 mL, 25 mL of this solution required 18.75 mL of 0.2 N – NaOH solution for complete neutralisation. Calculate the percentage of pure CaCO_3 in the sample of chalk?
17. A solution contains Na_2CO_3 and NaHCO_3 . 20 mL of this solution required 4 mL of 1N – HCl for titration with Ph indicator. The titration was repeated with the same volume of the solution but with MeOH. 10.5 mL of 1 – N HCl was required this time. Calculate the amount of Na_2CO_3 & NaHCO_3 .
18. A solution contains a mix of Na_2CO_3 and NaOH. Using Ph as indicator 25 mL of mixture required 19.5 mL of 0.995 N HCl for the end point. With MeOH, 25 mL of the solution required 25 mL of the same HCl for the end point. Calculate g/L of each substance in the mixture.
19. 200 mL of a solution of mixture of NaOH and Na_2CO_3 was first titrated with Ph and $\frac{N}{10}$ HCl. 17.5 mL of HCl was required for end point. After this MeOH was added and 2.5 mL of same HCl was again required for next end point. Find out amounts of NaOH and Na_2CO_3 in the mix.
20. A solution contains Na_2CO_3 and NaHCO_3 . 10 mL of this requires 2 mL of 0.1 M H_2SO_4 for neutralisation using Ph indicator. MeOH is then added when a further 2.5 mL of 0.2 M H_2SO_4 was needed. Calculate strength of Na_2CO_3 and NaHCO_3 .
21. A sample containing Na_2CO_3 & NaOH is dissolved in 100 mL solution. 10 mL of this solution requires 25 mL of 0.1 N HCl when Ph is used as indicator. If MeOH is used as indicator 10 mL of same solution requires 30 mL of same HCl. Calculate % of Na_2CO_3 and NaOH in the sample.
22. It required 40.05 mL of 1 M Ce^{4+} to titrate 20 mL of 1 M Sn^{2+} to Sn^{4+} . What is the oxidation state of the cerium in the product.
23. A volume of 12.53 mL of 0.05093 M SeO_2 reacted with exactly 25.52 mL of 0.1 M CrSO_4 . In the reaction, Cr^{2+} was oxidized to Cr^{3+} . To what oxidation state was selenium converted by the reaction.
24. Pottasium acid oxalate $\text{K}_2\text{C}_2\text{O}_4 \cdot 3\text{H}_2\text{C}_2\text{O}_4 \cdot 4\text{H}_2\text{O}$ can be oxidized by MnO_4^- in acid medium. Calculate the volume of 0.1 M KMnO_4 reacting in acid solution with one gram of the acid oxalate.
25. A 1.0 g sample of H_2O_2 solution containing x% H_2O_2 by mass requires $x \text{ cm}^3$ of a KMnO_4 solution for complete oxidation under acidic conditions. Calculate the normality of KMnO_4 solution.

26. Metallic tin in the presence of HCl is oxidized by $\text{K}_2\text{Cr}_2\text{O}_7$ to stannic chloride, SnCl_4 . What volume of deci-normal dichromate solution would be reduce by 1 g of tin.
27. 5 g sample of brass was dissolved in one litre dil. H_2SO_4 . 20 mL of this solution were mixed with KI, liberating I_2 and Cu^+ and the I_2 required 20 mL of 0.0327 N hypo solution for complete titration. Calculate the percentage of Cu in the alloy.
28. 0.84 g iron ore containing x percent of iron was taken in a solution containing all the iron in ferrous condition. The solution required x mL of a dichromatic solution for oxidizing the iron content to ferric state. Calculate the strength of dichromatic solution.
29. The neutralization of a solution of 1.2 g of a substance containing a mixture of $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$, $\text{KHC}_2\text{O}_4 \cdot \text{H}_2\text{O}$ and different impurities of a neutral salt consumed 18.9 mL of 0.5 N NaOH solution. On titration with KMnO_4 solution, 0.4 g, of the same substance needed 21.55 mL of 0.25 N KMnO_4 . Calculate the % composition of the substance.
30. 50 g of a sample of Ca(OH)_2 is dissolved in 50 mL of 0.5 N HCl solution. The excess of HCl was titrated with 0.3 N – NaOH. The volume of NaOH used was 20cc. Calculate % purity of Ca(OH)_2 .
31. One g of impure sodium carbonate is dissolved in water and the solution is made up to 250 mL. To 50 mL of this made up solution, 50 mL of 0.1 N – HCl is added and the mix after shaking well required 10 mL of 0.16 N – NaOH solution for complete titration. Calculate the % purity of the sample.
32. What amount of substance containing 60% NaCl, 37% KCl should be weighed out for analysis so that after the action of 25 mL of 0.1 N AgNO_3 solution, excess of Ag^+ is back titrates with 5 mL of NH_4SCN solution. Given that 1 mL of NH_4SCN = 1.1. mL of AgNO_3 .
33. A bottle labelled with "12 V H_2O_2 " contain 700 mL solution. If a student mix 300 mL water in it what is the g/litre strength & normality and volume strength of final solution.
34. 50 mL of an aqueous solution of H_2O_2 were treated with an excess of KI solution and dilute H_2SO_4 , the liberated iodine required 20 mL of 0.1 N $\text{Na}_2\text{S}_2\text{O}_3$ solution for complete interaction. Calculate the concentration of H_2O_2 in g/ℓ.
35. 100 kg hard water contains 5 g MgSO_4 . Find hardness.
36. One litre hard water contains 1 mg CaCl_2 and 1 mg MgSO_4 . Find hardness.
37. Calculate the hardness of water sample which contains 0.001 mol MgSO_4 per litre of water.
38. A solution of a 0.4 g sample of H_2O_2 reacted with 0.632 g of KMnO_4 in the presence of sulphuric acid. Calculate the percentage purity of the sample of H_2O_2 .
39. 5 litre of a solution of H_2O_2 with x N strength is diluted to 5.5 litre. This 5.5 litre H_2O_2 solution gives 28 litre O_2 at NTP. Find the value of x.
40. Calculate the amount of lime Ca(OH)_2 required to remove the hardness in 60 litre of pond water containing 1.62 mg of calcium bicarbonate per 100 mL of water.
41. 10 g sample of bleaching powder was dissolved into water to make the solution one litre. To this solution 35 mL of 1.0 M Mohr salt solution was added containing enough H_2SO_4 . After the reaction was complete, the excess Mohr salt required 30 mL of 0.1 M KMnO_4 for oxidation. Find out the % of available Cl_2 approximately is (mol wt. 71).

42. Calculate the amount (in milligrams) of SeO_3^{-2} in solution on the basis of following data 20 mL of M/60 solution of KBrO_3 was added to a definite volume of SeO_3^{-2} solution. The bromine evolved was removed by boiling and excess of KBrO_3 was back titrated with 5 mL of M/25 solution of NaAsO_2 . The reactions are given below. (Atomic mass of K = 39, Br = 80, As = 75, Na = 23, O = 16, Se = 79)
- (a) $\text{SeO}_3^{-2} + \text{BrO}_3^- + \text{H}^+ \longrightarrow \text{SeO}_4^{-2} + \text{Br}_2 + \text{H}_2\text{O}$
 (b) $\text{BrO}_3^- + \text{AsO}_2^- + \text{H}_2\text{O} \longrightarrow \text{Br}^- + \text{AsO}_4^{-3} + \text{H}^+$
43. A 1.0 g sample of Fe_2O_3 solid of 55.2% purity is dissolved in acid and reduced by heating the solution with zinc dust. The resultant solution is cooled and made upto 100.0 mL. An aliquot of 25.0 mL of this solution requires 17.0 mL of 0.0167 M solution of an oxidant for titration. Calculate the number of moles of electrons taken up by the oxidant in the reaction of the above titration.

CONCEPTUAL SUBJECTIVE EXERCISE			ANSWER KEY		EXERCISE-4(A)
1.	(a) +3	(b) +5	(c) +6	(d) +2	(e) 8/3 or (2 and 3)
	(f) +3	(g) +2	(h) +2	(i) 200/93 = 2.15	
2.	(a) $\text{S}_4\text{O}_6^{2-}(\text{aq}) + 6\text{Al}(\text{s}) + 20\text{H}^+ \longrightarrow 4\text{H}_2\text{S}(\text{aq}) + 6\text{Al}^{3+}(\text{aq}) + 6\text{H}_2\text{O}$				
	(b) $6\text{S}_2\text{O}_3^{2-}(\text{aq}) + \text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14\text{H}^+ \longrightarrow 3\text{S}_4\text{O}_6^{2-}(\text{aq}) + 2\text{Cr}^{3+}(\text{aq}) + 7\text{H}_2\text{O}$				
	(c) $14\text{ClO}_3^-(\text{aq}) + 3\text{As}_2\text{S}_3(\text{s}) + 18\text{H}_2\text{O} \longrightarrow 14\text{Cl}^-(\text{aq}) + 6\text{H}_2\text{AsO}_4^-(\text{aq}) + 9\text{HSO}_4^-(\text{aq}) + 15\text{H}^+$				
	(d) $7\text{IO}_3^-(\text{aq}) + 6\text{Re}(\text{s}) + 3\text{H}_2\text{O} \longrightarrow 6\text{ReO}_4^-(\text{aq}) + 7\text{I}^-(\text{aq}) + 6\text{H}^+$				
	(e) $30\text{HSO}_4^-(\text{aq}) + \text{As}_4(\text{s}) + 10\text{Pb}_3\text{O}_4(\text{s}) + 26\text{H}^+ \longrightarrow 30\text{PbSO}_4(\text{s}) + 4\text{H}_2\text{AsO}_4^-(\text{aq}) + 24\text{H}_2\text{O}$				
	(f) $3\text{HNO}_2(\text{aq}) \longrightarrow \text{HNO}_3 + 2\text{NO}(\text{g}) + \text{H}_2\text{O}$				
3.	(a) $3\text{C}_4\text{H}_4\text{O}_6^{2-}(\text{aq}) + 5\text{ClO}_3^-(\text{aq}) + 18\text{OH}^- \longrightarrow 12\text{CO}_3^{2-}(\text{aq}) + 5\text{Cl}^-(\text{aq}) + 15\text{H}_2\text{O}$				
	(b) $11\text{Al}(\text{s}) + 3\text{BiONO}_3(\text{s}) + 21\text{H}_2\text{O} + 11\text{OH}^- \longrightarrow 3\text{Bi}(\text{s}) + 3\text{NH}_3(\text{aq}) + 11\text{Al}(\text{OH})_4^-(\text{aq})$				
	(c) $4\text{H}_2\text{O}_2(\text{aq}) + \text{Cl}_2\text{O}_7(\text{aq}) + 2\text{OH}^- \longrightarrow 2\text{ClO}_2^-(\text{aq}) + 4\text{O}_2(\text{g}) + 5\text{H}_2\text{O}$				
	(d) $\text{Ti}_2\text{O}_3(\text{s}) + 4\text{NH}_2\text{OH}(\text{aq}) \longrightarrow 2\text{TlOH}(\text{s}) + 2\text{N}_2(\text{g}) + 5\text{H}_2\text{O}$				
	(e) $\text{Cu}(\text{NH}_3)_4^{2+}(\text{aq}) + \text{S}_2\text{O}_4^{2-}(\text{aq}) + 4\text{OH}^- \longrightarrow 2\text{SO}_3^{2-}(\text{aq}) + \text{Cu}(\text{s}) + 4\text{NH}_3(\text{aq}) + 2\text{H}_2\text{O}$				
	(f) $3\text{Mn}(\text{OH})_2(\text{s}) + 2\text{MnO}_4^-(\text{aq}) \longrightarrow 5\text{MnO}_2(\text{s}) + 2\text{H}_2\text{O} + 2\text{OH}^-$				
4.	2, 41				
5.	(i) 4.0, (ii) 0.0040, (iii) 0.224, (iv) 56.00%, (v) 0.02M, (vi) 0.0008 mol				
6.	$\frac{M_1}{6}, \frac{M_2}{8}, \frac{M_3}{5}$	7. 1 : 7 : 4	8. 0.06 N and 0.03 M	9. 537.6 mL	
10.	$\text{KOH} = 35\%, \text{Ca}(\text{OH})_2 = 65\%$		11. $V = 157.89\text{ mL}$	12. 20.72 %	
13.	$V = 25\text{ mL}$		14. 200 mL	15. $\text{MgCO}_3 = 51.22\%, \text{CaCO}_3 = 48.78\%$	
16.	83.33		17. 0.424 g ; 0.21g	18. 23.2 g, 22.28 g	19. 0.06 g ; 0.0265 g
20.	4.24 g/L ; 5.04 g/L		21. 39.85% ; 60.15%	22. +3	
23.	zero		24. $V = 31.68\text{ mL}$	25. 0.588 N	26. 337 mL
27.	41.53 %		28. 0.15 N	29. $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O} = 14.36\%, \text{KH}_2\text{O}_4 \cdot \text{H}_2\text{O} = 81.71\%$	
30.	1.406 %		31. 90.1%	32. 0.1281 g	
33.	25.5 g/L, 1.5 N, 8.4 V		34. 0.68 g/L	35. 41.66 ppm	
36.	1.734 ppm		37. 100 ppm	38. 85%	39. $x = 1$
40.	0.444 g		41. 7.1%	42. 84mg	43. 6

EXERCISE-04 [B]**BRAIN STORMING SUBJECTIVE EXERCISE**

1. 1.2475 g of crystalline copper sulphate was dissolved in water and excess of KI was added. The liberated iodine consumed 50 mL N/10 $\text{Na}_2\text{S}_2\text{O}_3$ solution to reach the end point of the titration. Calculate the number of water molecules of hydration in crystalline copper sulphate salt.
2. A 1g sample of $\text{K}_2\text{Cr}_2\text{O}_7$ containing some inert material was entirely reduced with conc. HCl. The chlorine liberated was passed through hot solution of NaOH at 80°C , and it completely disproportionates to form ClO_3^- and Cl^- . This NaClO_3 was isolated and its reduction with KI (aq) liberated iodine, giving Cl^- . The iodine thus liberated required 100 mL of decinormal hypo solution for complete titration. What is the percentage purity of the dichromate sample?
3. 2.5g of mixture of crystalline oxalic acid ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$) and sodium oxalate ($\text{Na}_2\text{C}_2\text{O}_4$) was dissolved in 100 mL of water. 50 mL of this solution was titrated against N/10 NaOH solution when 119.05 mL of the base was found necessary to reach the end point with phenolphthalein as the indicator. 1g of the mixture was dissolved in water and the solution titrated against N/10 KMnO_4 in the presence of dil. H_2SO_4 . What is the volume of KMnO_4 needed for getting the end point with 0.5g of the mixture?
4. 25 mL of a solution containing HCl was treated with excess of M/5 KIO_3 and KI solution of unknown concentration where I_2 liberated is titrated against a standard solution of 0.021 M $\text{Na}_2\text{S}_2\text{O}_3$ solution whose 24 mL were used up. Find the strength of HCl and volume of KIO_3 solution consumed :
5. 0.6213 g of sample contains an unknown amount of As_2O_3 . The sample was treated with HCl resulting information of AsCl_3 (g) which was distilled into a beaker of water. The hydrolysis reaction is as follows :
$$\text{AsCl}_3 + 2\text{H}_2\text{O} \rightarrow \text{HAsO}_2 + 3\text{H}^+ + 3\text{Cl}^-$$
The amount of HAsO_2 was determined by titration with 0.04134 M I_2 , requiring 23.04 mL to reach the equivalence point. The redox products in the titration were H_3AsO_4 and I^- . Find the amount of KMnO_4 needed to oxidize As in As_2O_3 to its maximum possible oxidation state in acidic medium.
6. A sample of steel weighing 0.6 g and containing S as an impurity was burnt in a stream of O_2 , when S was converted to its oxide SO_2 . SO_2 was then oxidized to SO_4^{2-} by using H_2O_2 solution containing 30 mL of 0.04 M NaOH. 22.48 mL of 0.024 M HCl was required to neutralize the base remaining after oxidation. Calculate the % of S in the sample :
7. In the presence of fluoride ion, Mn^{2+} can be titrated with MnO_4^- , both reactants being converted to a complex of Mn(III). A 0.545 g sample containing Mn_3O_4 was dissolved and all manganese was converted to Mn^{2+} . Titration in the presence of fluoride ion consumed 31.1 mL of KMnO_4 that was 0.177 N against oxalate.
 - (a) write a balanced chemical equation for the reaction, assuming that the complex is MnF_4^- .
 - (b) what was the % of Mn_3O_4 in the sample ?
8. A mixture of two gases, H_2S and SO_2 is passed through three beakers successively. The first beaker contains Pb^{2+} ions, which absorbs S^{2-} forming PbS. The second beaker contains 25 mL of 0.0396 N I_2 to oxidize SO_2 to SO_4^{2-} . The third contains 10 mL of 0.0345 N thiosulphate solution to retain any I_2 carried over from the second absorber. A 25 L gas sample was passed through the apparatus followed by an additional amount of N_2 to sweep last traces of SO_2 from first and second absorber. The solution from the first absorber was made acidic and treated with 20 mL of 0.0066 M $\text{K}_2\text{Cr}_2\text{O}_7$ which converted S^{2-} to SO_2 . The excess dichromate was reacted with solid KI and the liberated iodine required 7.45 mL of 0.0345 N $\text{Na}_2\text{S}_2\text{O}_3$ solution. The solutions in the second and third absorbers were combined and the resultant iodine was titrated with 2.44 mL of the same thiosulphate solution. Calculate the concentrations of SO_2 and H_2S in mg/L of the sample :

9. 1 g of a moist sample of a mixture of KClO_3 and KCl was dissolved in water and made upto 250 mL. 25 mL of this solution was treated with SO_2 to reduced chlorate into chloride and the excess SO_2 was boiled off. When the total chloride was precipitated, 0.1435 g of AgCl was obtained. In another experiment 25 mL of the original solution was treated with 30 mL of 0.2 N solution of FeSO_4 and unreacted FeSO_4 required 37.5 mL of 0.08 N solution of an oxidizing agent for complete oxidation. Calculate the molar ratio of chlorate and chloride in the given mixture. Fe^{2+} reacts with ClO_3^- according to equation :
- $$\text{ClO}_3^- + 6\text{Fe}^{2+} + 6\text{H}^+ \rightarrow \text{Cl}^- + 6\text{Fe}^{3+} + 3\text{H}_2\text{O}$$
- Also calculate the mass percent of moisture present in the moist sample.
10. A steel sample is to be analysed for Cr and Mn simultaneously. By suitable treatment the Cr is oxidized to $\text{Cr}_2\text{O}_7^{2-}$ and the Mn to MnO_4^- . A 10.00 g sample of steel is used to produce 250.0 mL of a solution containing $\text{Cr}_2\text{O}_7^{2-}$ and MnO_4^- . A 10.00 mL portion of this solution is added to a BaCl_2 solution and by proper adjustment of the acidity, the chromium is completely precipitated as BaCrO_4 ; 0.0549 g is obtained. A second 10.00 mL portion of this solution requires exactly 15.95 mL of 0.0750 M standard Fe^{2+} solution for its titration (in acid solution). Calculate the % of Mn and % of Cr in the steel sample.
11. 1.16 g $\text{CH}_3(\text{CH}_2)_n\text{COOH}$ was burnt in excess air and the resultant gases (CO_2 and H_2O) were passed through excess NaOH solution. The resulting solution was divided in two equal parts. One part required 50 mL of 1 N HCl for neutralization using phenolphthalein as indicator. Another part required 80 mL of 1 N HCl for neutralization using methyl orange as indicator. Find the value of n and the amount of excess NaOH solution taken initially.
12. A 1.5 g sample containing oxalic acid and some inert impurity was dissolved in enough water and volume made up to 250 mL. A 20 mL portion of this solution was then mixed with 30 mL of an alkali solution. The resulting solution was then treated with stoichiometric amount of CaCl_2 just needed for precipitation of oxalate as CaC_2O_4 . Solution was filtered off and filtrate was finally titrated against 0.1 M HCl solution. 8.0 mL of acid was required to reach the equivalence point. At last, the above neutral solution was treated with excess of AgNO_3 solution and AgCl obtained was washed, dried and weighed to be 0.4305 g. Determine mass percentage of oxalic acid in the original sample :
13. A 1 g sample containing NaOH as the only basic substance and some inert impurity was left exposed to atmosphere for a very long time so that part of NaOH got converted into Na_2CO_3 by absorbing CO_2 from atmosphere. The resulting sample was dissolved in water and volume made upto 100 mL. A 20 mL portion of this solution required 16 mL 0.25 M HCl solution to reach the equivalence point when methyl orange was used as indicator. In a separate analysis, 20 mL portion of the same solution was taken along with phenolphthalein indicator and mixed with 50 mL of 0.1 M HCl solution. An additional 9.00 mL 0.1 M Ba(OH)_2 solution was required to just restore the pink colour of solution. Determine mass percentage of NaOH in the original sample and mass percentage of Na_2CO_3 in the sample after exposure to atmosphere.

BRAIN STORMING SUBJECTIVE EXERCISE	ANSWER KEY	EXERCISE-4(B)
1. 5	2. 58.8%	3. 77.45 mL
4. $V_{\text{KIO}_3} = 0.42 \text{ mL}$, $[\text{HCl}] = 0.0168 \text{ N}$	5. 0.06 g	6. 1.76 %
7. 40.77%	8. 0.12 mg $\text{H}_2\text{S}/\text{L}$, 0.718 mg SO_2/L	
9. $\text{ClO}_3^-/\text{Cl}^- = 1$, 1.5% moisture by mass	10. Cr = 2.821% , Mn = 1.498%	
11. n = 4, $\text{NaOH} = 6.4 \text{ g}$	12. 82.5	13. 80, 36.05 %

EXERCISE-05 [A]**PREVIOUS YEARS QUESTIONS**

1. MnO_4^- is good oxidising agent in different medium changing to - [AIEEE-02]
 $\text{MnO}_4^- \longrightarrow \text{Mn}^{2+}$
 $\longrightarrow \text{MnO}_4^{2-}$
 $\longrightarrow \text{MnO}_2$
 $\longrightarrow \text{Mn}_2\text{O}_3$
Changes in oxidation number respectively are -
(1) 1, 3, 4, 5 (2) 5, 4, 3, 2 (3) 5, 1, 3, 4 (4) 2, 6, 4, 3
2. Oxidation number of Cl in CaOCl_2 (bleaching powder is) [AIEEE-02]
(1) Zero, since it contains Cl_2
(2) -1, since it contains Cl^-
(3) +1, since it contains ClO^-
(4) +1 and -1 since it contains ClO^- and Cl^-
3. Which of the following is a redox [AIEEE-02]
(1) $2\text{NaAg}(\text{CN})_2 + \text{Zn} \longrightarrow \text{Na}_2\text{Zn}(\text{CN})_4 + 2\text{Ag}$ (2) $\text{BaO}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{BaSO}_4 + \text{H}_2\text{O}_2$
(3) $\text{N}_2\text{O}_5 + \text{H}_2\text{O} \longrightarrow 2\text{HNO}_3$ (4) $\text{AgNO}_3 + \text{KI} \longrightarrow \text{AgI} + \text{KNO}_3$
4. In the coordination compound, $\text{K}_4[\text{Ni}(\text{CN})_6]$, the oxidation state of nickel is [AIEEE-03]
(1) +1 (2) +2 (3) -1 (4) 0
5. The oxidation state of Cr in $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]^+$ is - [AIEEE-05]
(1) +2 (2) +3 (3) 0 (4) +1
6. The oxidation state of chromium in the final product formed by the reaction between KI and acidified potassium dichromate solution is - [AIEEE-05]
(1) +6 (2) +4 (3) +3 (4) +2
7. Which of the following chemical reaction depicts the oxidizing behaviour of H_2SO_4 ? [AIEEE-06]
(1) $\text{Ca}(\text{OH})_2 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + 2\text{H}_2\text{O}$
(2) $\text{NaCl} + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{HCl}$
(3) $2\text{PCl}_5 + \text{H}_2\text{SO}_4 \rightarrow 2\text{POCl}_3 + 2\text{HCl} + \text{SO}_2\text{Cl}_2$
(4) $2\text{HI} + \text{H}_2\text{SO}_4 \rightarrow \text{I}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$

Que.	1	2	3	4	5	6	7
Ans	3	4	1	2	2	3	4

EXERCISE-05 [B]**PREVIOUS YEARS QUESTIONS**

1. The oxidation number of phosphorus in $\text{Ba}(\text{H}_2\text{PO}_2)_2$ is : [JEE 1990]
(A) +3 (B) +2 (C) +1 (D) -1
2. The number of electrons to balance the following equation :- [JEE 1991]
 $\text{NO}_3^- + 4\text{H}^+ + e^- \rightarrow 2\text{H}_2\text{O} + \text{NO}$ is
(A) 5 (B) 4 (C) 3 (D) 2
3. What is the volume strength of 1.5 N H_2O_2 : [JEE 1991]
(A) 4.8 (B) 8.4 (C) 3.0 (D) 8.0
4. The oxidation states of the most electronegative element in the products of the reaction of BaO_2 with dilute H_2SO_4 . [JEE 1991]
(A) 0 and -1 (B) -1 and -2 (C) -2 and 0 (D) -2 and +2
5. For the redox reaction, [JEE 1992]
 $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \rightarrow \text{Mn}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$
the correct coefficients of the reactants for the balanced reaction are :

MnO_4^-	$\text{C}_2\text{O}_4^{2-}$	H^+
(A) 2	5	16
(B) 16	5	2
(C) 5	16	2
(D) 2	16	5
6. The number of mole of KMnO_4 that will need to react completely with one mole ferrous oxalate in acidic solution is : [JEE 1997]
(A) 2/5 (B) 3/5 (C) 4/5 (D) 1
7. The number of mole of KMnO_4 that will be needed to react with one mole of sulphite ion in acidic solution is : [JEE 1997]
(A) 2/5 (B) 3/5 (C) 4/5 (D) 1
8. The equivalent mass of MnSO_4 is half its molecular mass when it is converted to : [JEE 1998]
(A) Mn_2O_3 (B) MnO_2 (C) MnO_4^- (D) MnO_4^{2-}
9. The oxidation number of sulphur in S_8 , S_2F_2 and H_2S respectively are : [JEE 1999]
(A) 0, +1 and -2 (B) +2, +1 and -2
(C) 0, +1 and +2 (D) -2, +1 and -2
10. The normality of 0.3 M phosphorus acid (H_3PO_3) is : [JEE 1999]
(A) 0.1 (B) 0.9 (C) 0.3 (D) 0.6
11. Among the following species in which oxidation state of the element is +6 : [JEE 2000]
(A) MnO_4^- (B) $\text{Cr}(\text{CN})_6^{3-}$ (C) NiF_6^{2-} (D) CrO_2Cl_2
12. Oxidation number of iron in $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}^\oplus]$ is : [JEE 2001]
(A) +2 (B) +3 (C) +8/3 (D) none of these
13. An aqueous solution of 6.3 g of oxalic acid dihydrate is made upto 250 mL. The volume of 0.1 N NaOH required to completely neutralise 10 mL of this solution is : [JEE 2001]
(A) 40 mL (B) 20 mL (C) 10 mL (D) 4 mL
14. How many moles of electron weigh one kilogram : [JEE 2002]
(A) 6.023×10^{23} (B) $\frac{1}{9.108} \times 10^{31}$ (C) $\frac{6.023}{9.108} \times 10^{54}$ (D) $\frac{1}{9.108 \times 6.023} \times 10^8$
15. Which has maximum number of atoms : [JEE 2003]
(A) 24 g of C (12) (B) 56 g of Fe (56) (C) 27 g of Al (27) (D) 108 g of Ag (108)
16. In basic medium I^- oxidises by MnO_4^- . In this process I^- replaces by : [JEE 2004]
(A) IO_3^- (B) I_2 (C) IO_4^- (D) IO^-

